MATE6301 • NANOMATERIALS • THE UNIVERSITY OF QUEENSLAND

课件合集 + 每周知识点总结

Lecture compilation with weekly knowledge summaries • Weeks 1–10, 12

每一周的课件前都附有该周的知识点总结（中文讲解，保留英文术语、公式和考试重点）。第 11 周为展示周，无课件。
期末考试：120 分钟，简答 + 计算。可通过左侧书签在各周之间跳转。
共 590 页。

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WEEK 1 · 知识点总结 · A/Prof. Shahriar Hossain

第 1 周　课程概览与纳米材料导论

Course overview & Introduction to Nanomaterials

主线：纳米材料的定义 → 表面原子与表面能带来的新性质 → 为什么需要先进表征

1. 基本定义

- **纳米材料**：至少一个维度在 **1–100 nm**；按维度分 **0D / 1D / 2D**（颗粒 / 线、管 / 薄膜、片）。
- **纳米结构材料**：体材料内部在 0–3 维上有纳米调制结构；至少一维 < 100 nm；**界面体积分数 > 1%**。
- **Nanoscience = know why**（研究尺寸接近纳米时出现的独特性质）；**Nanotechnology = know how**（在 1–100 nm 尺度控制物质，制造功能材料与器件）。
- 纳米材料的特点：小、轻、新性质、多功能、多级结构、智能。

2. "Nano is not new"

- 彩色玻璃中的 **Au 纳米颗粒**（颜色来自 LSPR，见 W7）；中世纪陶釉中的金属纳米颗粒。
- 工业早已商业化：催化剂（CeO_2）、炭黑、溶胶、气相 SiO_2、颜料；合金中的**纳米析出相强化**（如 $Ni_{65}Al_{35}$）。

3. 核心：表面原子与表面能

- 尺寸 ↓ → **表面原子比例 ↑**（表面原子缺少 1–3 个近邻，处于高能态）。
- 例：1 cm 立方体 → 切成纳米颗粒后比表面积成数量级增加；介孔 SiO_2 约 **400 m²/g**。
- 表面能：$\gamma = (\partial G/\partial A)$；断键模型 $\gamma = \tfrac{1}{2} N_b \varepsilon \rho_a$（$N_b$ 断键数，ε 键能，ρ_a 单位面积原子数）。
- **纳米颗粒的特征**：比表面积大、表面能高；表面倾向形成**刻面**（密排面）；活性高、**易氧化、易团聚**；金属纳米粉**易爆**，需小心包装运输。

4. 多孔材料（贯穿全课程的线索）

- 多孔碳：导电、孔可调、比表面积大、强度高、化学惰性、易功能化 → 储能（电池、超级电容）、能量转换、水处理、传感、生物医学、复合材料。
- **为什么需要先进表征**：合成 → 用表征了解成分、形貌、晶体结构 → 建立"结构–性能–应用"关系。

5. 实验课：电沉积介孔金薄膜

- 原理：$M^{n+} + ne^- \rightarrow M$（金属离子在阴极还原）。
- 过程：① 迁移（电场 + 扩散 + 对流）→ ② 进入双电层、脱去水合壳 → ③ 中和并吸附在阴极 → ④ 表面扩散到生长点。
- **多孔膜 vs 常规膜**：粗糙度高、比表面积大、表面缺陷多、折射率可调、功能可调、用料少、表面灵敏度高、纳米晶。
- 造孔方法：**模板法**（template-assisted）与**无模板法**。

6. 课程信息

- 考核：实验参与 + 实验报告（20%）、展示；**期末 120 分钟，简答 + 计算**，期末须 ⩾ 40% 才能通过。
- 主线：纳米材料（介孔金、磁性复合材料、多孔碳、纳米催化剂、纳米纤维素）+ 表征（XRD、SEM、TEM）+ 应用（能源、医疗、环境）。

考试重点：纳米材料 / 纳米结构材料的定义；尺寸减小 → 表面原子与表面能增大 → 活性、团聚、氧化；比表面积估算；多孔薄膜与常规薄膜的对比。

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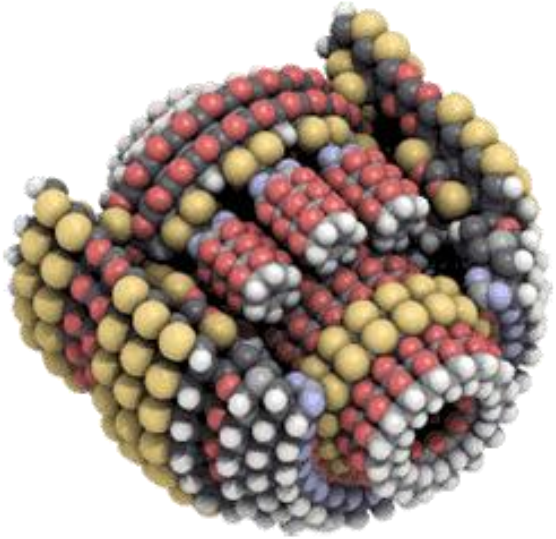
CREATE CHANGE

MATE6301 Nanomaterials



Lecture 1: Course overview and Introduction to Nanomaterials

A/Prof. Shahriar Hossain, SoMME/EAIT
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Acknowledgement of **Country**

The University of Queensland (UQ) acknowledges the Traditional Owners and their custodianship of the lands on which we meet.

We pay our respects to their Ancestors and their descendants, who continue cultural and spiritual connections to Country.

We recognise their valuable contributions to Australian and global society.

https://www.youtube.com/watch?

Mandatory Respect at UQ module

- Commencing students enrolled at UQ in Semester 2 2025 must complete the module by **5pm on Sunday 16 August 2026.**
- Students who do not complete the module by the deadline will have their access to Blackboard removed.
- Support and adjustments to the module are available. You can find a link to the module, along with information about what's included in the module and support available at respect.uq.edu.au/respect-at-uq.

Contact details

- Lecture
  - Tuesday 14:00 – 15:50 through ECHO360 +47A-141 - Sir James Foots Building
  - Laboratory/Project/Portfolio/Presentation

Friday 8:00 – 12:00 Laboratory through recording + AIBN + CMM+ ANFF and 46-914(Seminar/discussion)

Electronic Course Profile (ECP)

https://course-profiles.uq.edu.au/course-profiles/MATE6301-60587-7660#course-overview

Teaching Staffs

Course Co-Ordinator/ Magnetic nano structure synthesis, characterisation and applications

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Nanomaterials characterisation by electron microscopy: XRD, SEM and TEM

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Magneto-optical Nanomaterials applications: biomedical and agriculture

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Teaching Staffs

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Nanocatalysts fabrication and characterisation for waste water treatment

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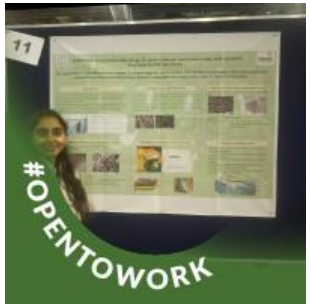


Ms Lintang

Topics/contents of the course

- Nanomaterials
  - Properties, Fabrications and Applications
  - Type of Nanomaterials
    - mesoporous gold nanostructures
    - Magnetic nanocomposites
    - Nanoporous carbon materials
    - Nanocatalysts
    - Nanocellulose
    - Metallic nano-composites
    - Semiconductor nanostructures
- Characterisation techniques
  - X-ray diffraction
  - Scanning electron microscopy
  - Transmission electron microscopy
- Applications
  - Energy, medical and environment

Highlights:

Lab pracs at AIBN, CMM, ANFF laboratories.

Weekly drop-in session, if requires. by communicating with teaching staff

Textbooks

- Textbook
  - No particular textbook recommended
- Concepts to be learnt
  - All concepts to be learnt will be outlined in the lecture notes
- References
  - Journals: *Nature*, *Science*, *Advanced Materials*, *Nano Letters*, *Materials Today* (all these journals have on-line versions at UQ)
  - Listed Reference Books
  - Recommend Internet Websites

Assessments

Assessment summary

Category	Assessment task	Weight	Due date
Portfolio	Active class/Laboratory engagement and progress portfolio	20%	13/10/2026 4:00 pm
Presentation	Critical analysis on selected published literature on nanomaterials, their characterisations and applications In-person	20%	Upload the presentation slide 21/10/2026 10:00 am Presentation 23/10/2026 8:00 am
Paper/ Report/ Annotation, Project	Development and characterisation of nanoporous gold film	20%	14/09/2026 4:00 pm
Examination	Final Exam Hurdle Identity Verified	40%	End of Semester Exam Period 7/11/2026 - 21/11/2026

Quiz to test the engagement during prac + portfolio/reflection submission (submission date has been requested to change)

Upload slide on 14/10/2026 at 10am

Presentation on 16/10/2026 at 8am to 12pm

The final exam will comprehensively test knowledge learned during the course. *A minimum achievement of 40% in the final exam is required to achieve a passing grade for the course*.

Duration: 120 minutes

Format: Short essay, Problem solving

Applications for extensions

- Individuals working in the engineering, architecture, computing, IT and design sectors are often required to submit work to strict deadlines where late submissions are not accepted.
- For this reason, and as part of preparing you to enter the professional workforce, the late penalty in this course is as follows:
- Please keep a record of your submission time.
- However, if there are medical or other acceptable reasons that will affect your ability to complete an assessment by the due date and time, then you can apply for an extension or deferred examination.

Submission guidelines

Submit via TurnItIn on Blackboard.

Deferral or extension

You may be able to apply for an extension.

The maximum extension allowed is 14 days. Extensions are given in multiples of 24 hours.

Feedback is provided to students following 14 calendar days.

A Student Access Plan (SAP) can only be used for a first extension. Extensions based on an SAP may be granted for up to seven (7) days, or the maximum number of days specified in the Electronic Course Profile (ECP), if it is less than seven (7) days. Any further extensions will require additional supporting documentation, such as a medical certificate.

Late submission

A penalty of 10% of the maximum possible mark will be deducted per 24 hours from time submission is due for up to 7 days. After 7 days, you will receive a mark of 0.

https://course-profiles.uq.edu.au/course-profiles/MATE6301-60667-7460#assessment

For example:

4 marks (2 days late x 10% x 20 marks).

2 hours over 24 hrs (26 hours) delay counts as 2 days

Academic integrity and plagiarism

- Academic Integrity
  - It is the University's task to encourage ethical scholarship and to inform students and staff about the institutional standards of academic behaviour expected of them in learning, teaching and research. Students have a responsibility to maintain the highest standards of academic integrity in their work. Students must not cheat in examinations or other forms of assessment and must ensure they do not plagiarise.
- Plagiarism
  - Plagiarism is the act of misrepresenting as one's own original work the ideas, interpretations, words or creative works of another. These include published and unpublished documents, designs, music, sounds, images, photographs, computer codes and ideas gained through working in a group. These ideas, interpretations, words or works may be found in print and/or electronic media.
  - Students are encouraged to read the UQ Student Integrity and Misconduct policy (http://ppl.app.uq.edu.au/content/3.60.04-student-integrity-and-misconduct) which makes a comprehensive statement about the University's approach to plagiarism, including the approved use of plagiarism detection software, the consequences of plagiarism and the principles associated with preventing plagiarism.
  - https://web.library.uq.edu.au/library-services/it/learnuq-blackboard-help/academic-integrity-modules

Assessments (Grades v Percentages)

Grade 7 = 85-100

Grade 6 = 75-84

Grade 5 = 65-74

Grade 4 = 50-64

Consider 40% hurdle

Grade 3 = 45-49

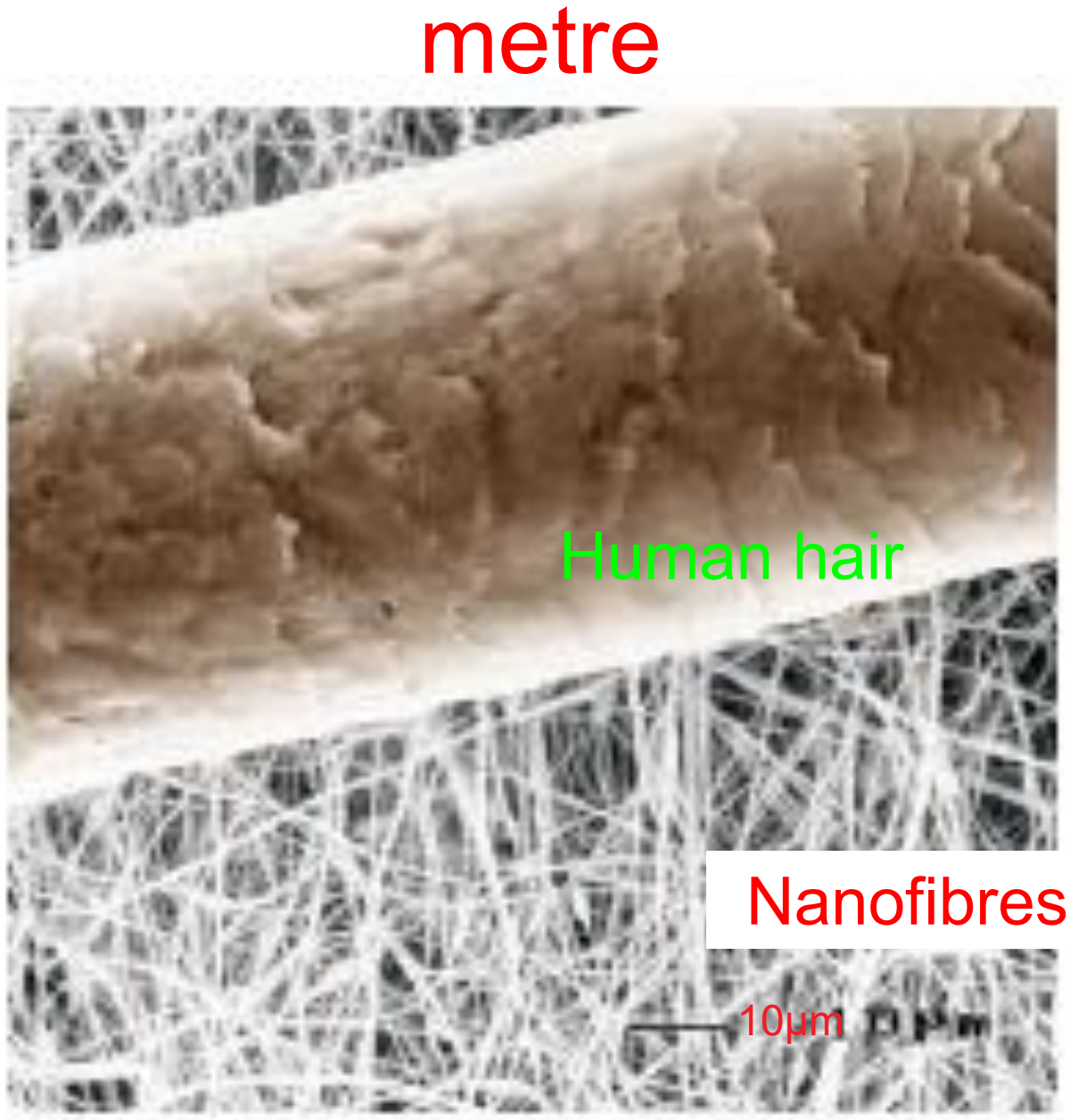
Grade 2 = 20-44

Grade 1 = 1-19

Failed

The prefixes for a given unit

Exa-	10^{18}	quintillion
Peta-	10^{15}	quadrillion
Tera-	10^{12}	trillion
Giga-	10^{9}	billion
Mega-	10^{6}	million
Kilo-	10^{3}	thousand
	1	
Milli-	10^{-3}	thousandth
Micro-	10^{-6}	millionth
Nano-	10^{-9}	billionth
Pico-	10^{-12}	trillionth
Femto-	10^{-15}	quadrillionth
Atto-	10^{-18}	quintillionth



Size matters

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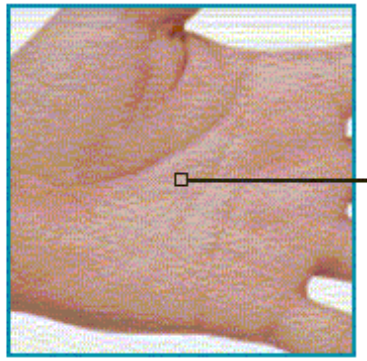
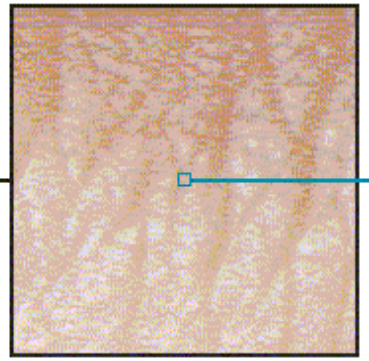
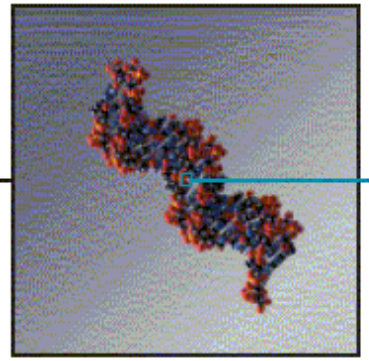
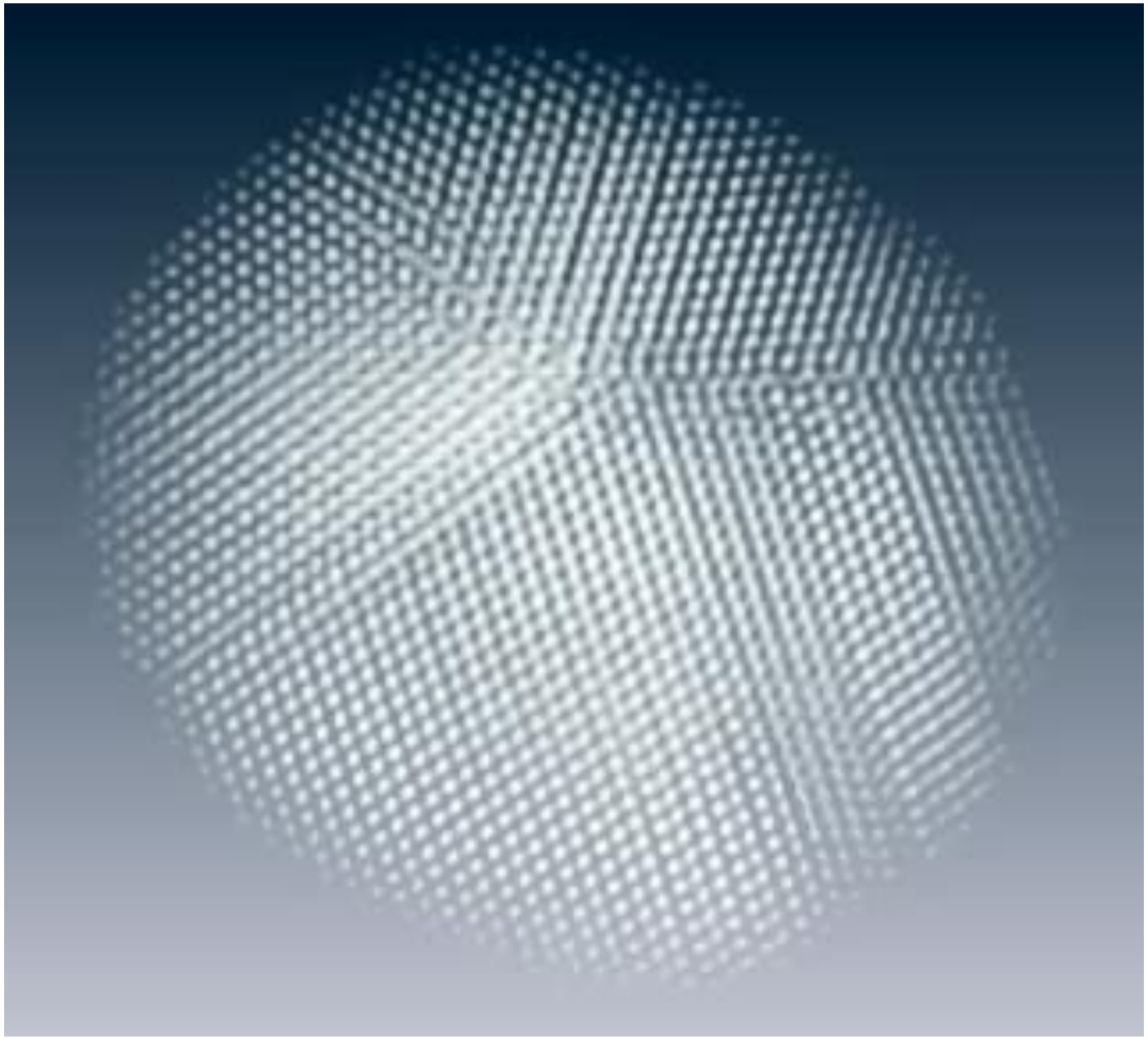
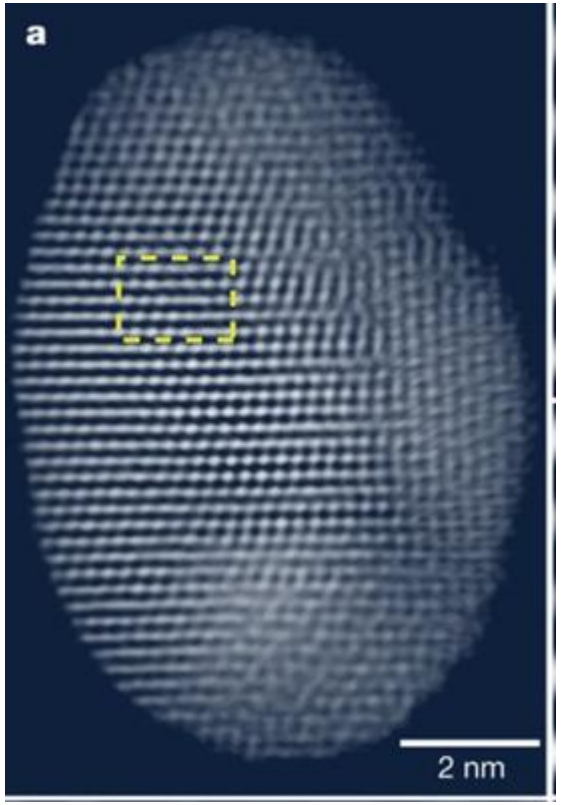
Hand	Skin	White Blood Cell	DNA	Atoms
				
10 cm	1 mm	10 microns	100 nanometers	1 nanometer

Size matters





A 7.9-Å-thick internal slice of the Pt nanoparticle. The lattice structure on the left and at the bottom parts of the slice is not well defined, mainly because this decahedral multiply twinned nanoparticle consists of five grains with different orientations.

Chen, CC., Zhu, C., White, E. *et al.* Three-dimensional imaging of dislocations in a nanoparticle at atomic resolution. *Nature* **496,** 74–77 (2013). https://doi.org/10.1038/nature12009

Predictions

Nano-Pioneers

Richard Feynman *(1918-1988) who won the 1965 Nobel Prize in Physics for his fundamental work in quantum electro-dynamics, in particular in 1948, he introduced so called Feynman diagrams, a graphical representation of interactions between different particles.*

The science of building small was first introduced in **1959** by Richard P. Feynman in a lecture titled ***There's Plenty of Room at the Bottom:***

"… ultimately---in the great future---we can arrange the atoms the way we want; the very atoms, all the way down! What would happen if we could arrange the atoms one by one the way we want them …"

Richard Smalley *(1943 - 2005) who won the 1996 Nobel Prize in Chemistry for his discovery of fullerenes in 1985.*

In a lecture *Nanotechnology and the Next 50 Years*, Smalley predicted: With the right advances in technology, a nanometre-sized solar cell could be built. Such devices could potentially provide for the world's energy needs in the year 2050.

Policy Maker

Neal Lane *(1938 -), a physicist, is the Science Advisor to former President Clinton and a former director of the National Science Foundation.*

"If I were asked for an area of science and engineering that will most likely produce the breakthroughs of tomorrow (21st Century), I would point to nanoscale science and engineering."

Global Challenges: New solutions for new challenges

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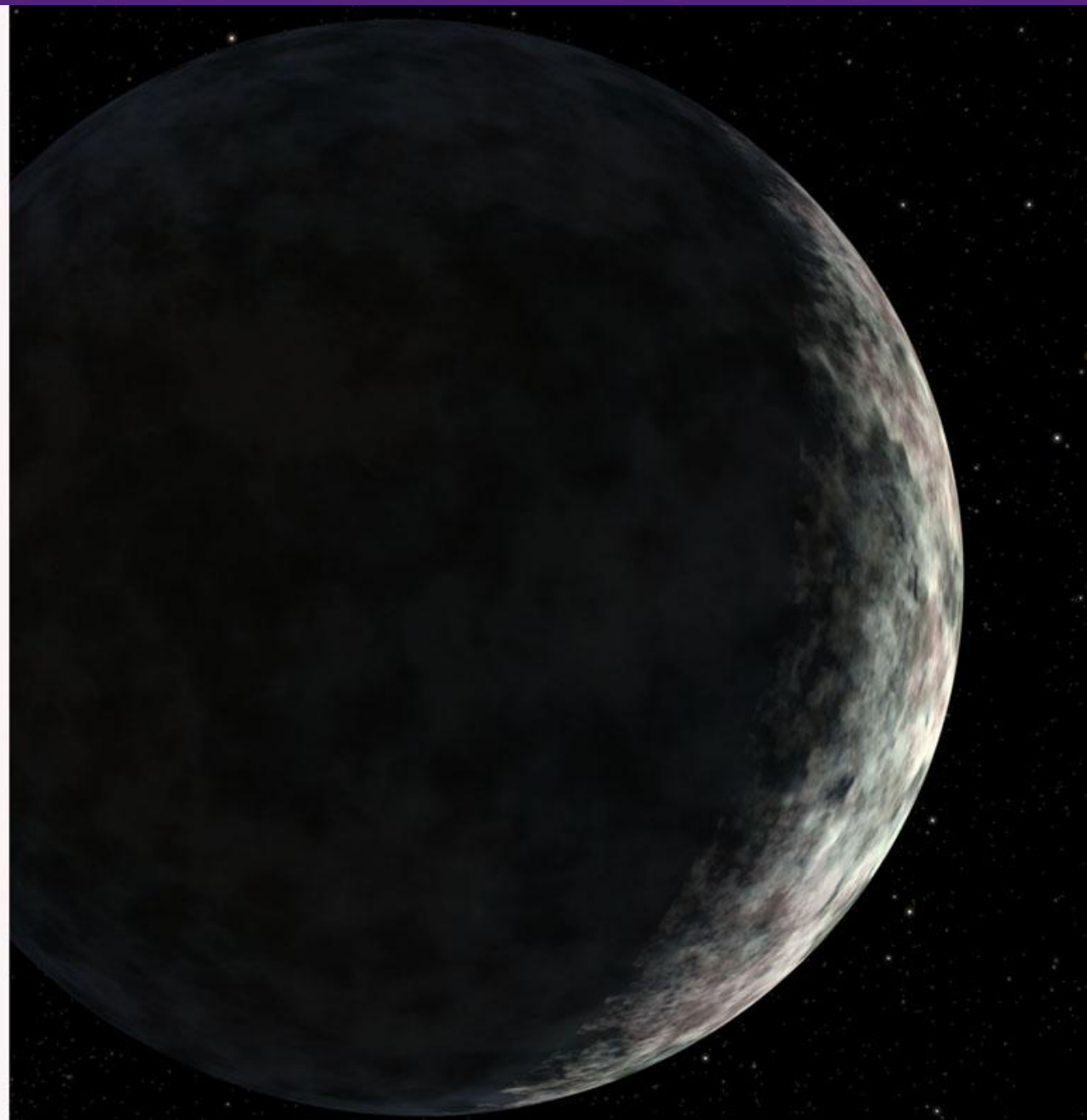
- Energy
- Water
- Food
- Environment
- Disease
- Terrorism/war
- Education
- Crime
- Poverty
- Population



Nanoscience and nanotechnology

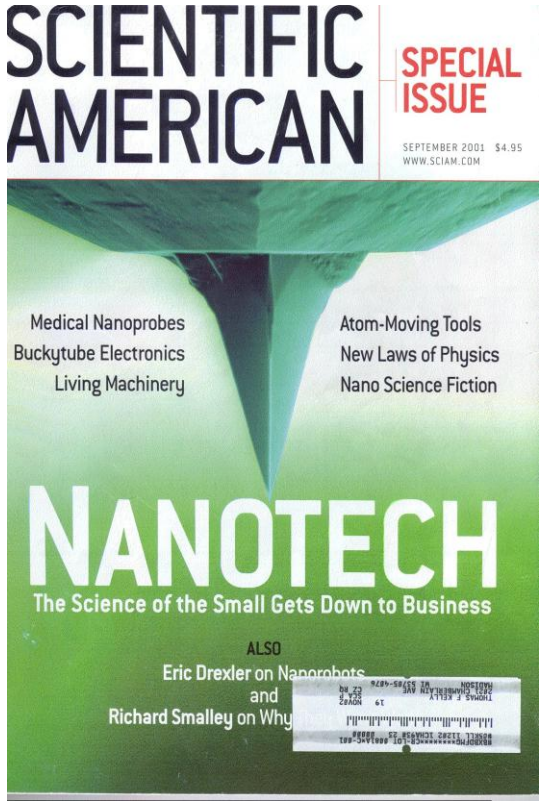
- **Nanoscience (know why)**: Study of the unique properties that emerge when system sizes approach nanometre-scale dimensions.
- **Nanotechnology(know how)**: The creation of functional/smart materials, devices, and systems through control of matter at the scale of 1 to 100 nanometres, and the exploitation of phenomena at this scale.
- **Nanoscience and nanotechnology encompasses:**
  - Sciences (Chemistry, Physics, Biology, Materials, Medicine)
  - Engineering (Mechanical, Chemical, Materials, Environmental, Metallurgical, Electrical, Biological, …)

Why nanoscience

- Small structures exhibit novel and unique properties, due to their significant fractions of surfaces/interfaces.
- Nanoscience is to understand their fundamentals
  - How they form?
  - Why do they have certain properties?
- Explore material properties and device functions beyond those that we presently know or even consider possible.
- Nanoscience is therefore an exciting new field that brings together the traditional disciplines of physics, chemistry, biology, materials science and engineering.

Why nanomaterials

- Ability to manipulate, control and build materials at the atomic and molecular level
- Provide novel affinity, capacity, and selectivity because of their unique physical, chemical and biological properties.



Nanomaterials

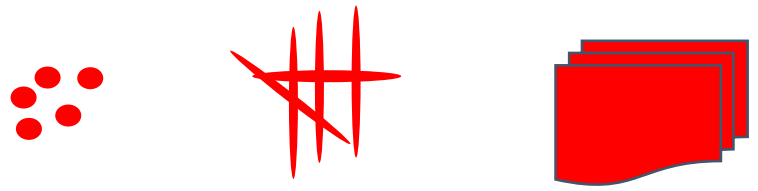
Nanomaterials stands for the materials that have a size between 1 – 100 nm.

Geometrically, nanomaterials may be considered as low (< 3) dimensional materials.

Because of their tiny in size, nanomaterials has their own characteristics.

- *Small*
- *Lightweight*
- *Novel properties*
- *Multifunctional*
- *Hierarchical*
- *Smart*

Classification



0D 1D 2D

Nanomaterials and nanostructured materials

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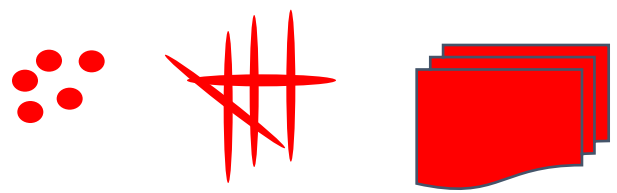
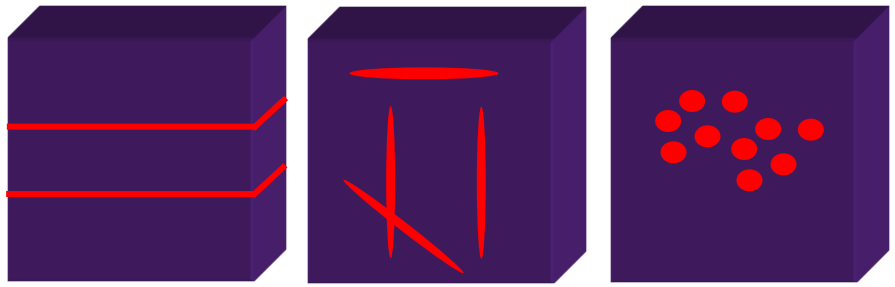
Nanomaterials



0D 1D 2D

Nanostructured Materials



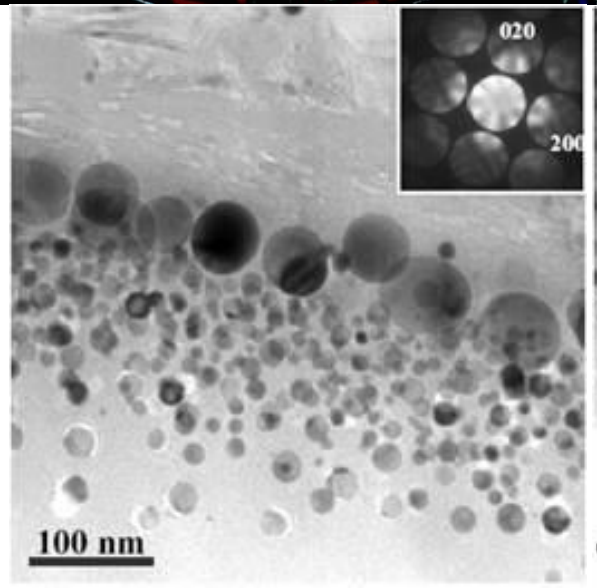
2D 1D 0D

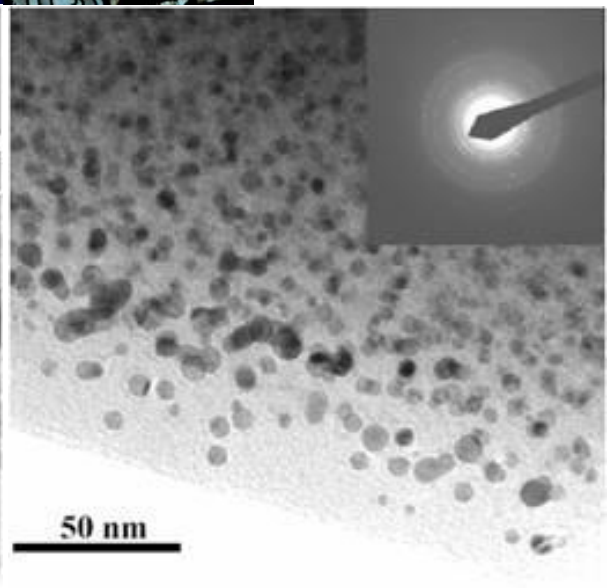
- Nanostructured materials are bulk materials with modulated structures in 0 to 3 dimensions...
- Size constraint (*confinement*) < 100nm at least in one dimension...
- Significant volume fraction (>1%) of interfaces.

Nano is not new!

The decorative glaze, found on some medieval pottery, contains metallic nanoparticles, which gives rise to its special optical properties.

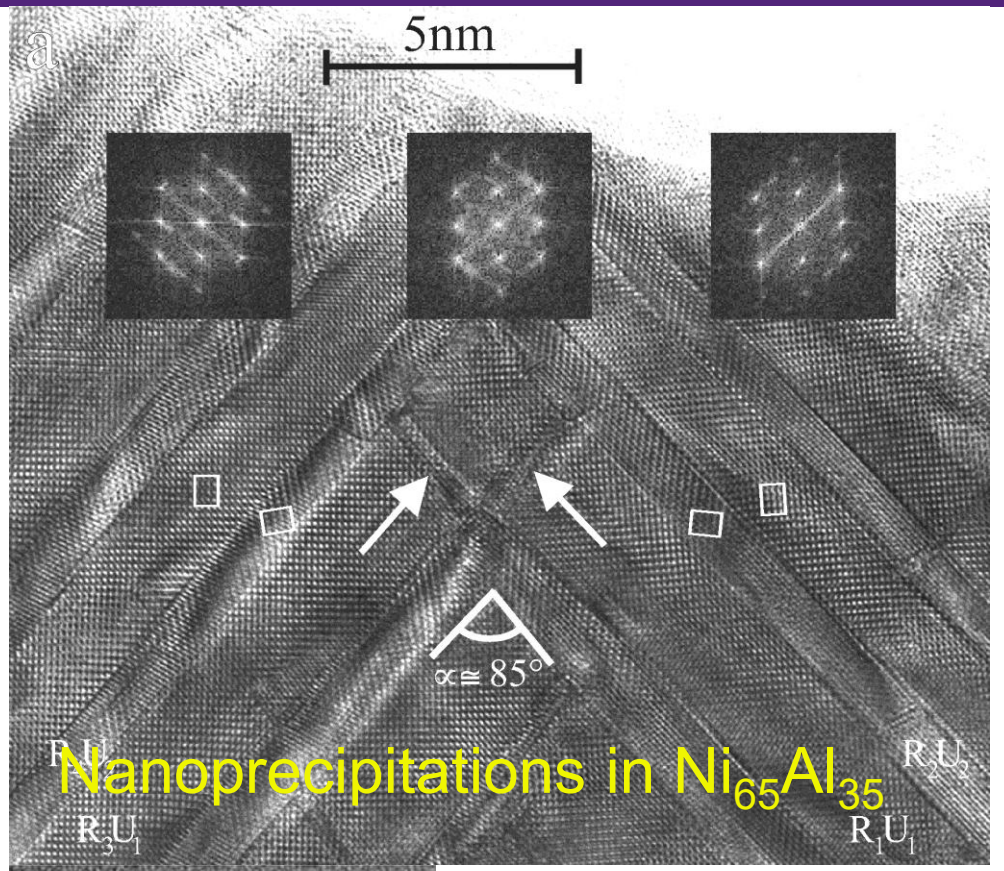
Traditionally, stained glass was made from gold nanoparticles.



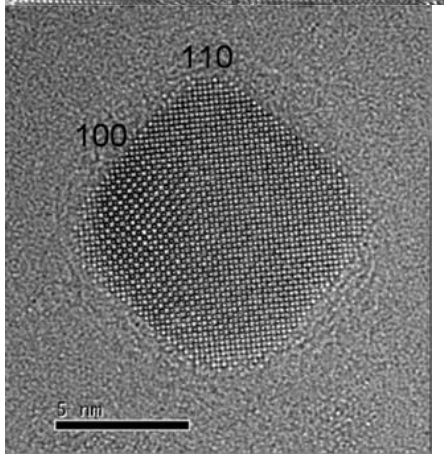


Nanomaterials are not new

- Catalysts, carbon black, sols, fumed silica and pigment industries have already commercialized several simple oxide nanopowders, nanoscale dispersion products, and nanoparticle manufacturing processes.
- Using nanosized precipitates in alloys to increase mechanical strength was developed over half a century ago.



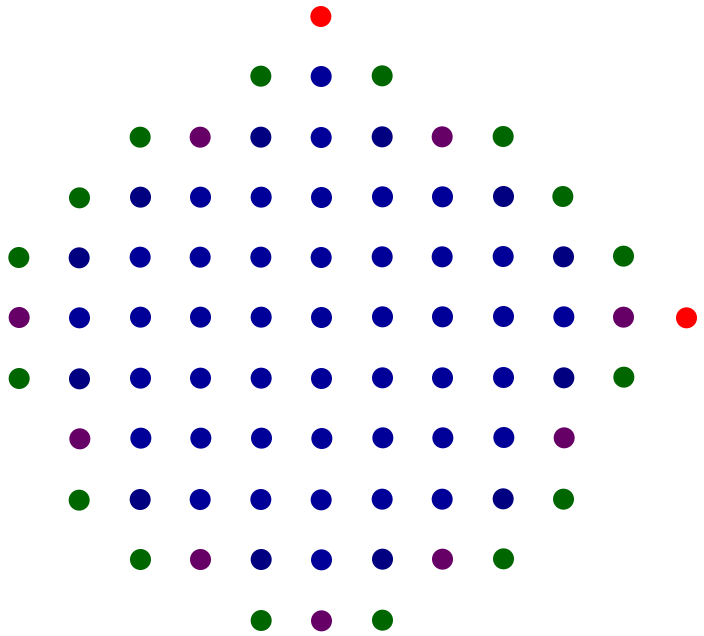
Nanoprecipitations in $Ni_{65}Al_{35}$



CeO_2 catalysts nanoparticle

Key characteristic of nanomaterials

Significant surface atoms

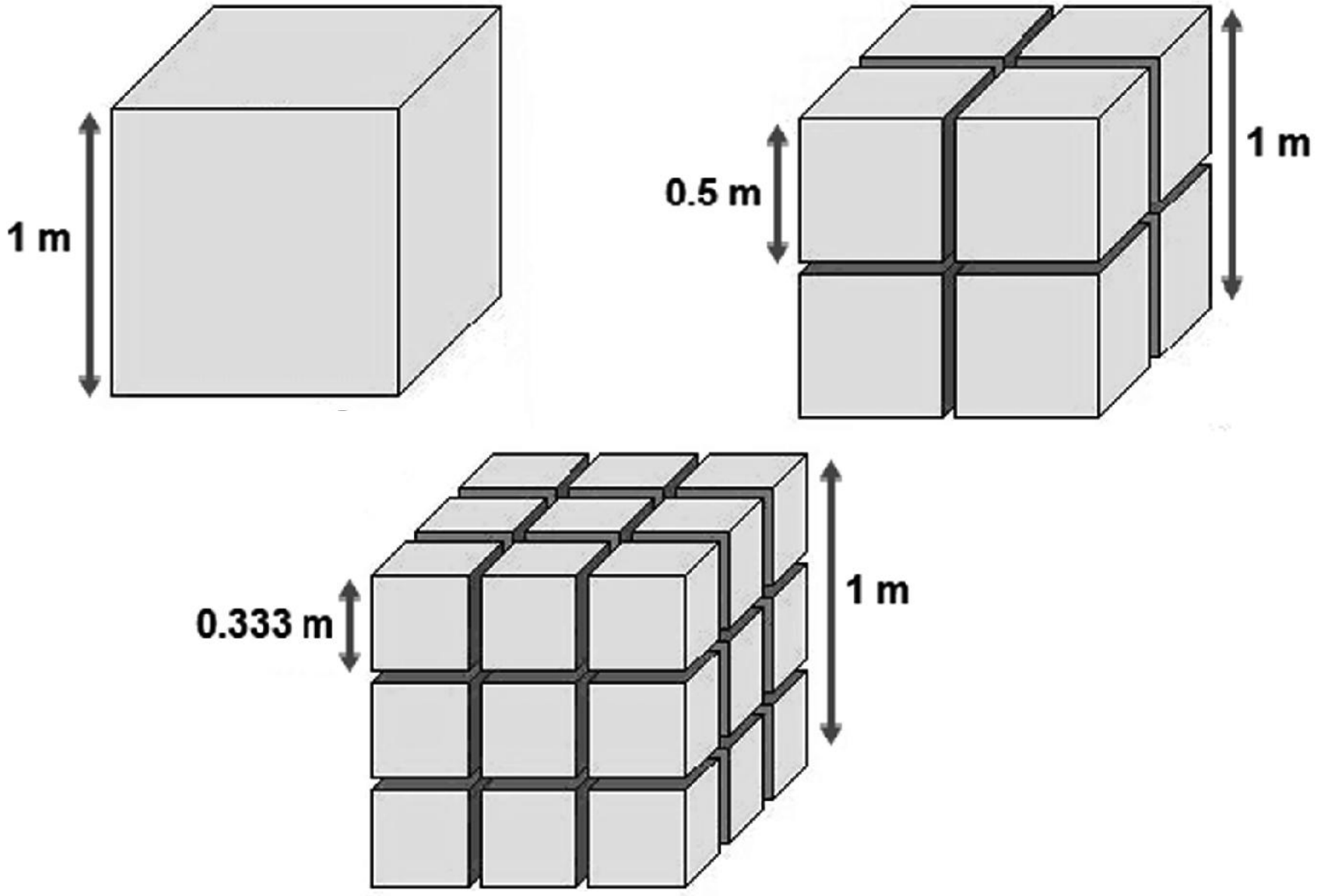


- Inner atoms

Surface atoms:

- lack of 1 nearest neighbour
- lack of 2 nearest neighbours
- lack of 3 nearest neighbours

Key characteristic of nanomaterials



SiO_2 mesoporous structure

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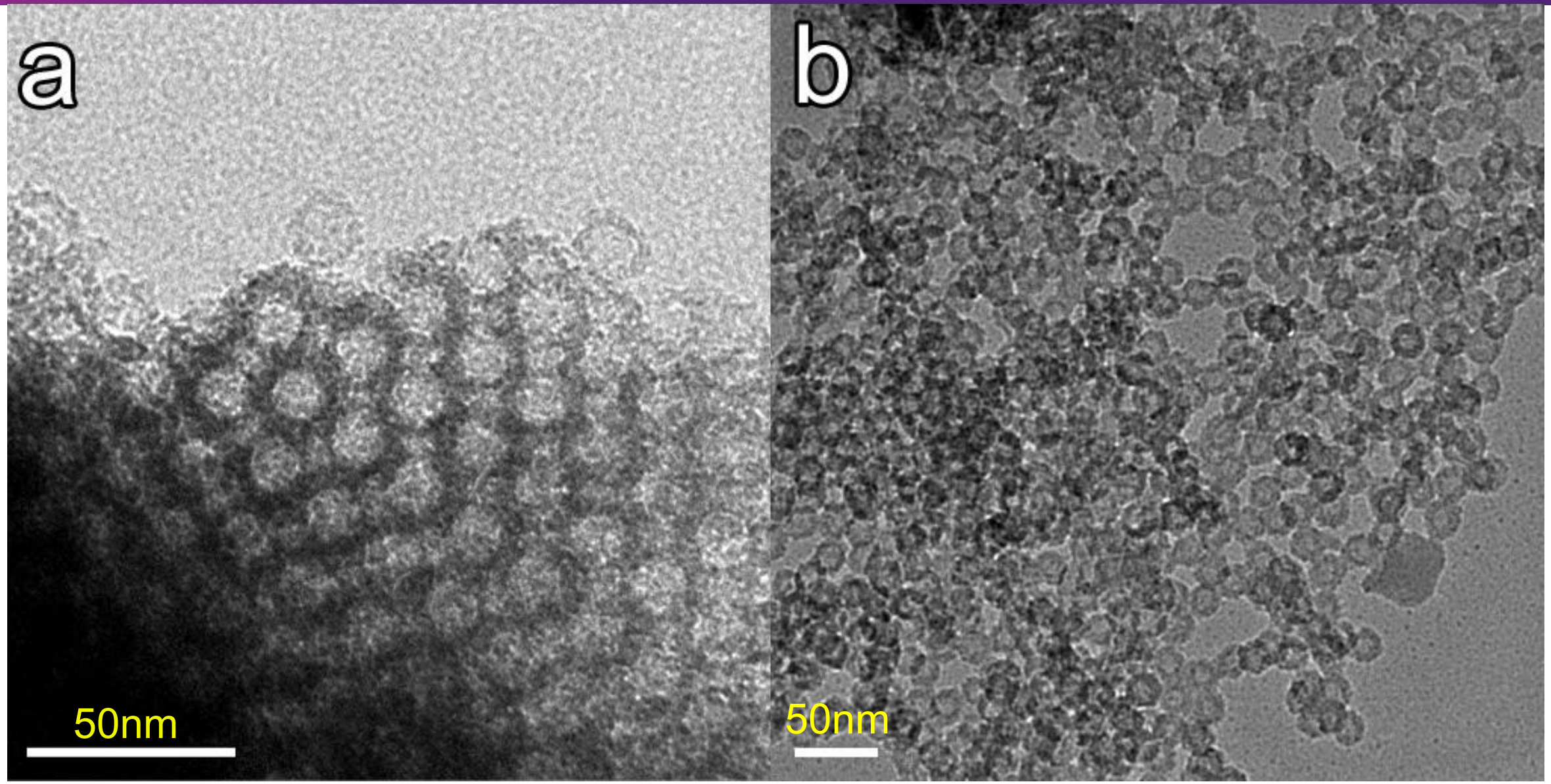
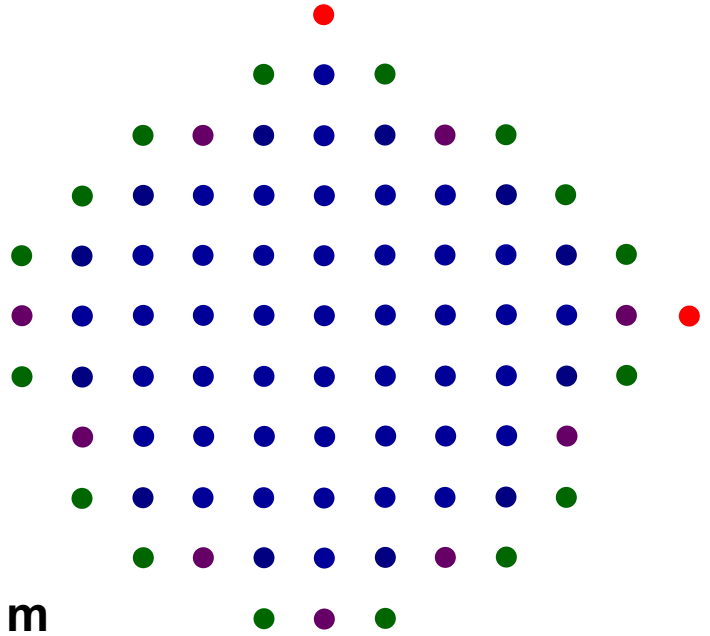


The surface area of the mesoporous structure is 400 m^2/g.

Key characteristic of nanomaterials

Significant surface atoms

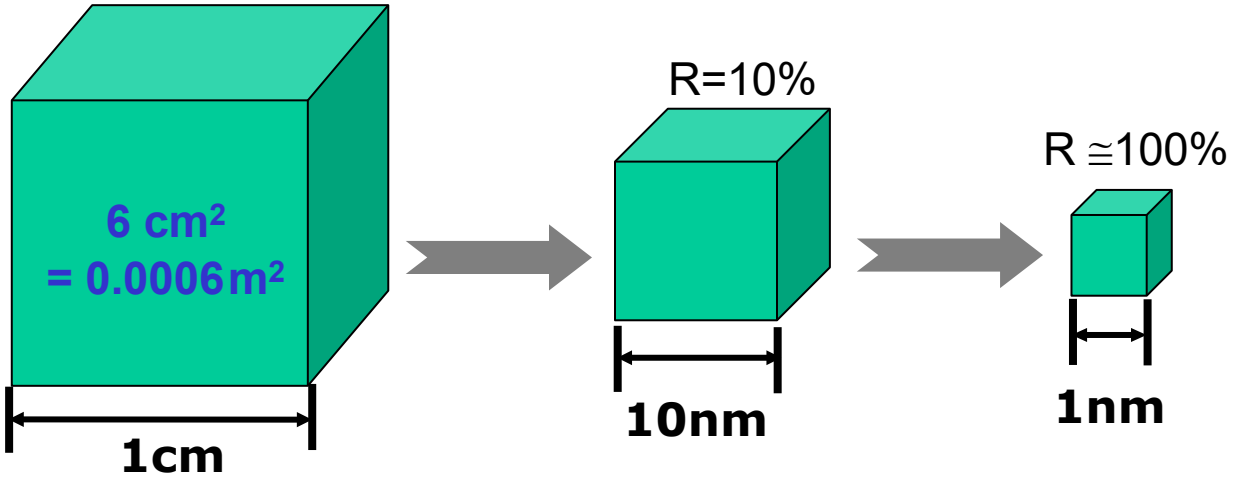


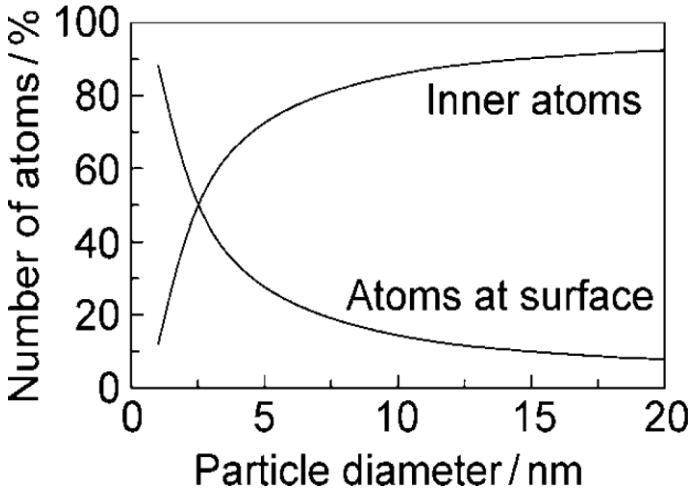
- Inner atoms

Surface atoms:

- lack of 1 nearest neighbour
- lack of 2 nearest neighbours
- lack of 3 nearest neighbours

Nanomaterials: nanometer → 10^{-9} m





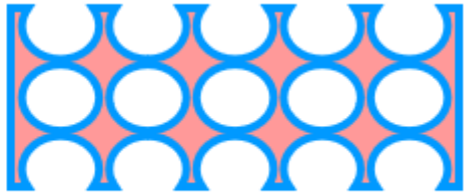
The surface area of materials dramatically increases by nanostructure and nanosize.

What is a Porous Materials?

Nonporous material

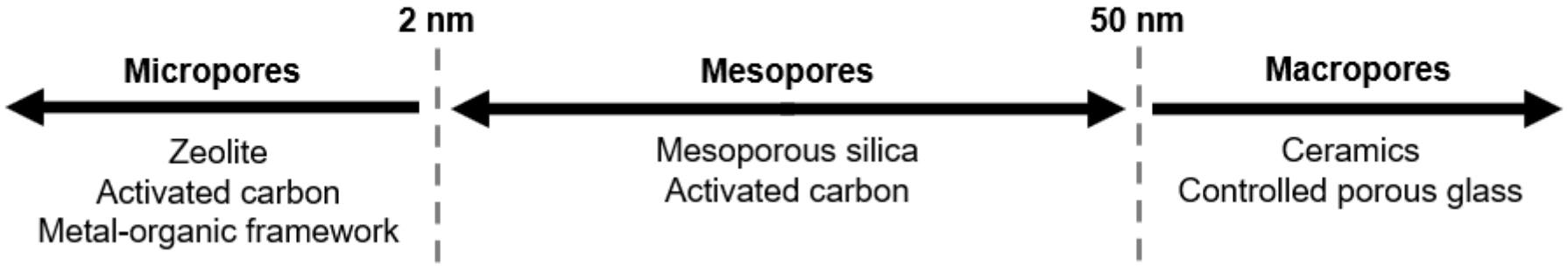
Porous material is a material containing pores

- High specific surface area & Tunable pore size

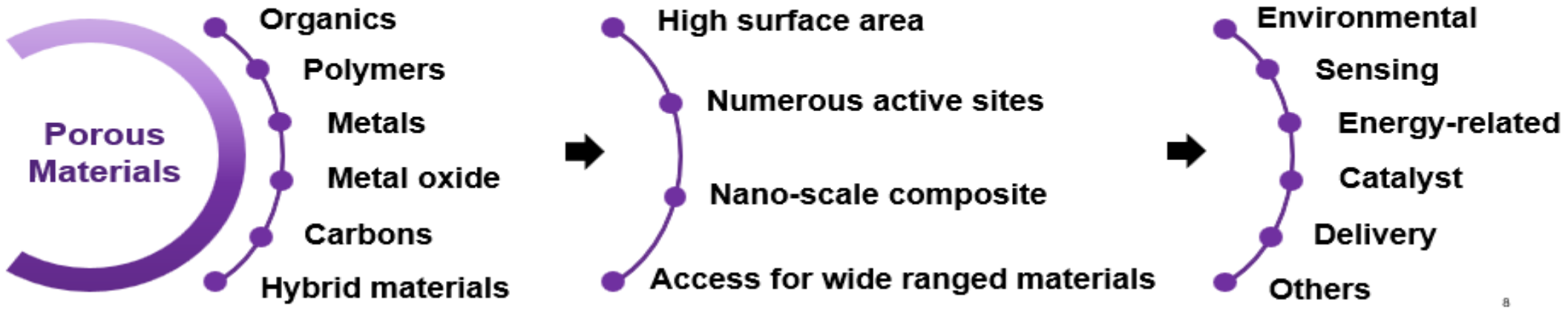


Porous material

Size of Pores (IUPAC standard)



Porous material & applications

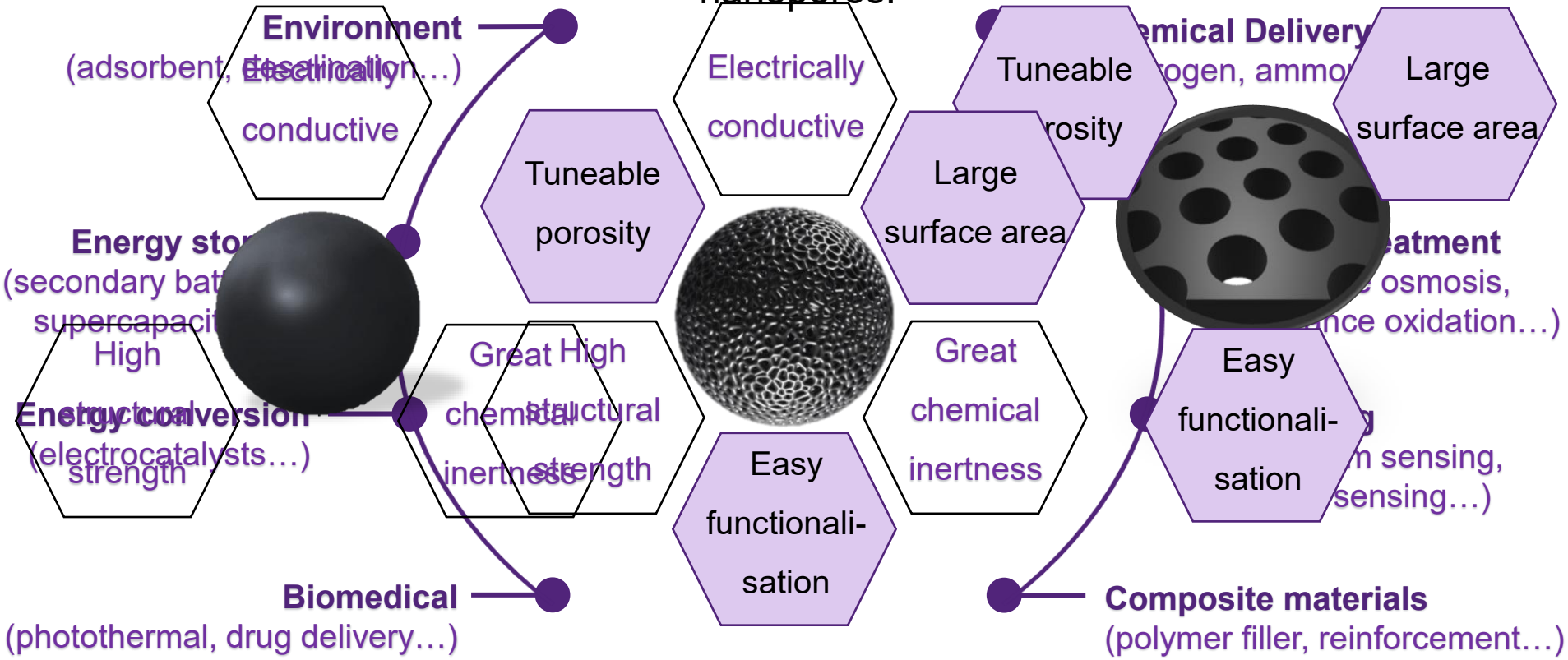


Nano-architectonics

Carbon materials are mainly composed of carbon atoms.

Porous carbon materials are mainly composed of carbon atoms and possess nanopores.

Porous materials are materials that possess pores/voids in nano-scale.



Surface energy

The surface energy (surface tension), γ, is the energy required to create a unit area of *new* surface.

$$\gamma = (\partial G/\partial A)$$

where G is the Gibbon free energy and A is the surface area.

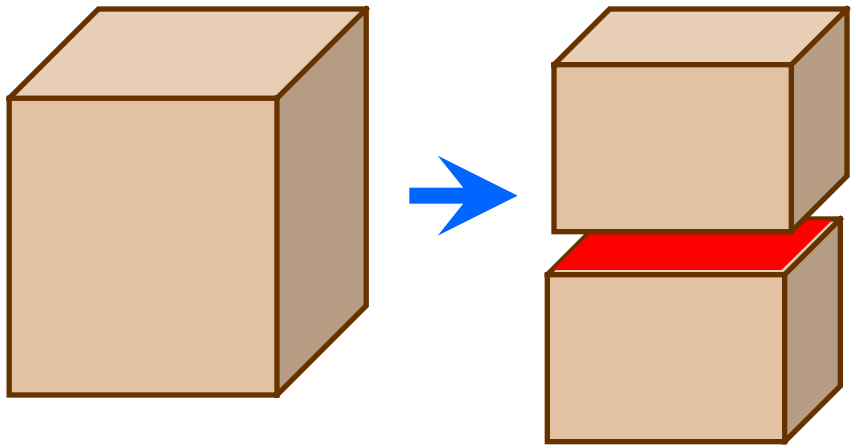
- As particle size goes down, the number of surface atoms increases
- The atoms at corners, edges and surfaces are in the high energy states compared to those in the bulk
- The nanoparticles have high surface energy and high reaction activity

Surface energy

Considering separation of a rectangular material into two pieces.

Assume N_b is the number of broken bonds, e is the bond strength, and ρ is the number of atoms per unit area on the new surface, the surface energy has the form

$$\gamma = (1/2)N_b e\rho$$



Characteristics of nanoparticles

- large surface area with large amount of atoms in surface
- high surface energy and high surface tension.
- The surface of a nano-sized crystalline particle tends to be faceted
- The faceted planes tends to be the dense plane
- The surface becomes very reactive
  - nanoparticles can be oxidised very easily when exposed in air
  - they tend to be aggregated
- Warning: Metallic nano-particles are very explosive, so that carefully packaging is critical for storage and shipping metallic nanoparticles.

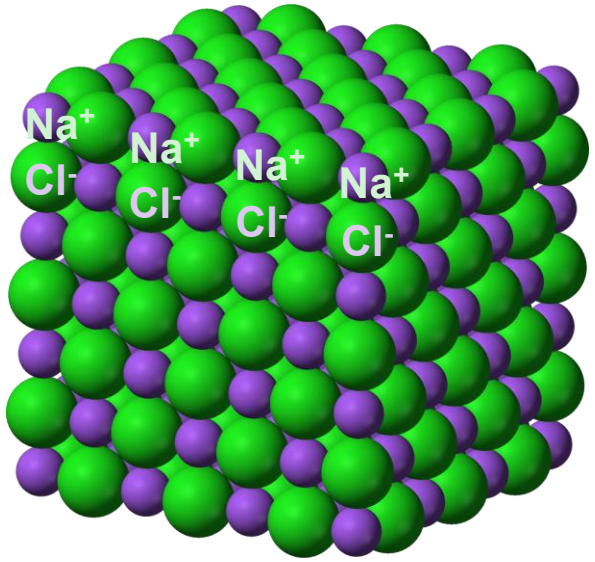
Why Advanced Characterisations?

Synthesising nanomaterials to solve problems that we are facing
e.g., energy, environmental, biosecurity, semiconductors, superconductors…

Understanding of the nanomaterials requires advanced characterisations
e.g., atomic compositions, surface topology, crystal structure…

Based on the gained understanding of the nanomaterials by advanced characterisations, we draw some conclusions on materials properties and their roles in certain applications.

Week 2 - What is crystal and crystallography?



Crystal – A crystalline solid material whose constituents (e.g., atoms, molecules or ions) are arranged in a highly ordered microscopic structure, forming a crystal lattice that extends in all directions.

Crystallography – the experimental science of determining the atomic arrangement in crystals.

General tools of crystallography – X-ray diffraction spectroscopy (XRD).

Week 2 - X-ray diffraction (XRD) crystallography

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1. Crystal structures
    1. *Lattices/unit cell/7 crystal systems/14 Bravais lattices*
    2. Symmetries
    3. Crystal directions [ ] and their families < >, as well as their determination
    4. Miller index ( ) of crystal planes and their families { }, as well as their determination
    5. Spacing of crystal planes
    6. Concept of zone axis - Weiss zone law, Interplanar angles, angles between two directions
2. Principle of diffraction
    1. the Bragg law
    2. the structure factor
3. X-ray diffraction
    1. principle of X-ray diffraction
    2. powder X-ray diffraction
    3. applications

- Lab Demo- XRD operation principals and sample characterisation

Week 2 - X-ray diffraction (XRD) crystallography

What information we can get from an XRD analysis?

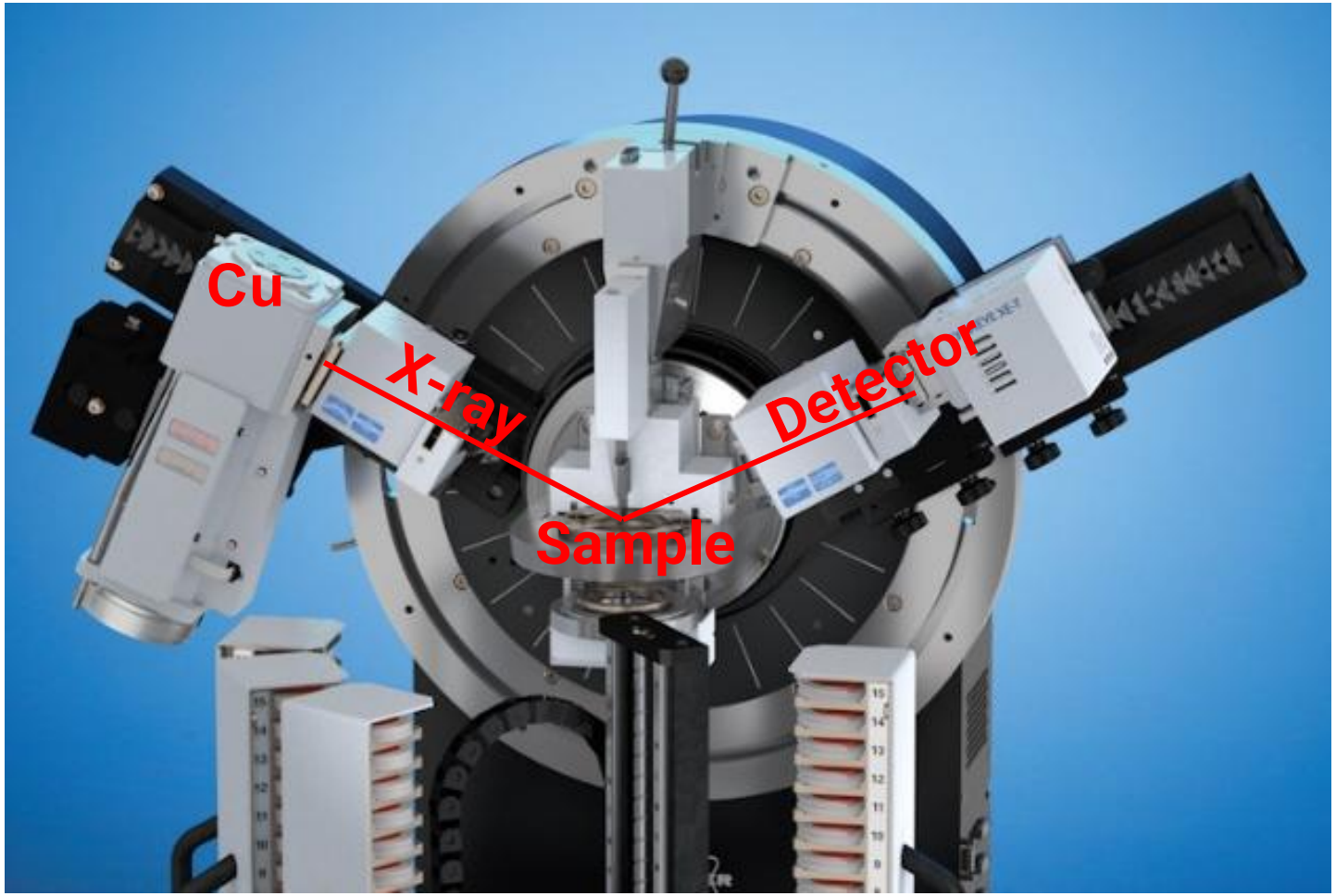
- Lattice parameters
- Phase identity
- Phase purity
- Crystallinity
- Crystal structure
- Percent phase composition

What information we can not get from an XRD analysis?

- Elemental analysis -
  - How much lithium is in this sample?
  - Is there iron in this sample
  - What elements are in this sample
- Tell me what this sample is ????
  - Unless you know something about this sample, powder XRD won't have answers !!!

Week 2 - Generation of X-rays

Metal-Ceramic Sealed Tube (Bruker)



D8 Advance XRD (Bruker)

XRD Simulation: https://myscope.training/XRD_simulator.html

Week 3 - Light Microscope vs Electron Microscope

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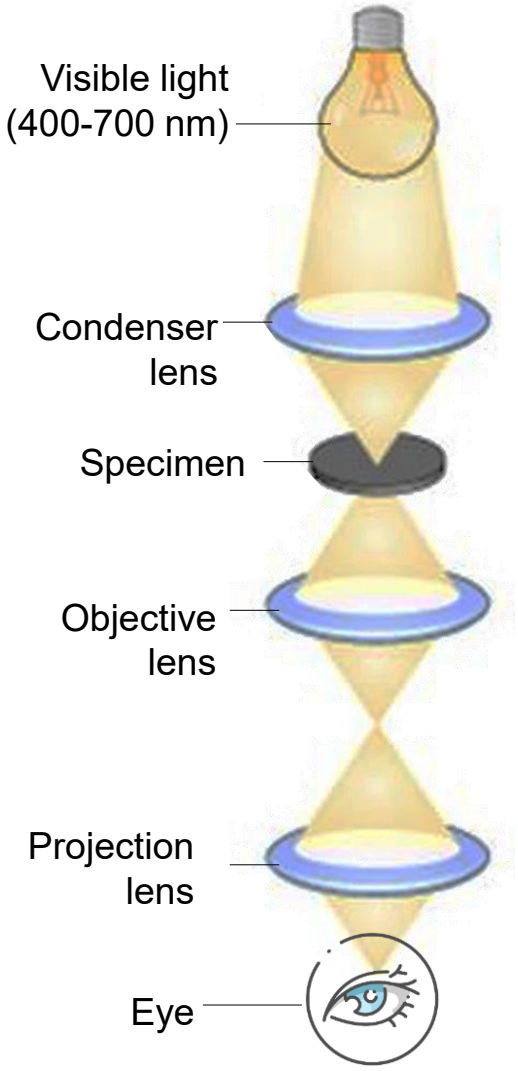
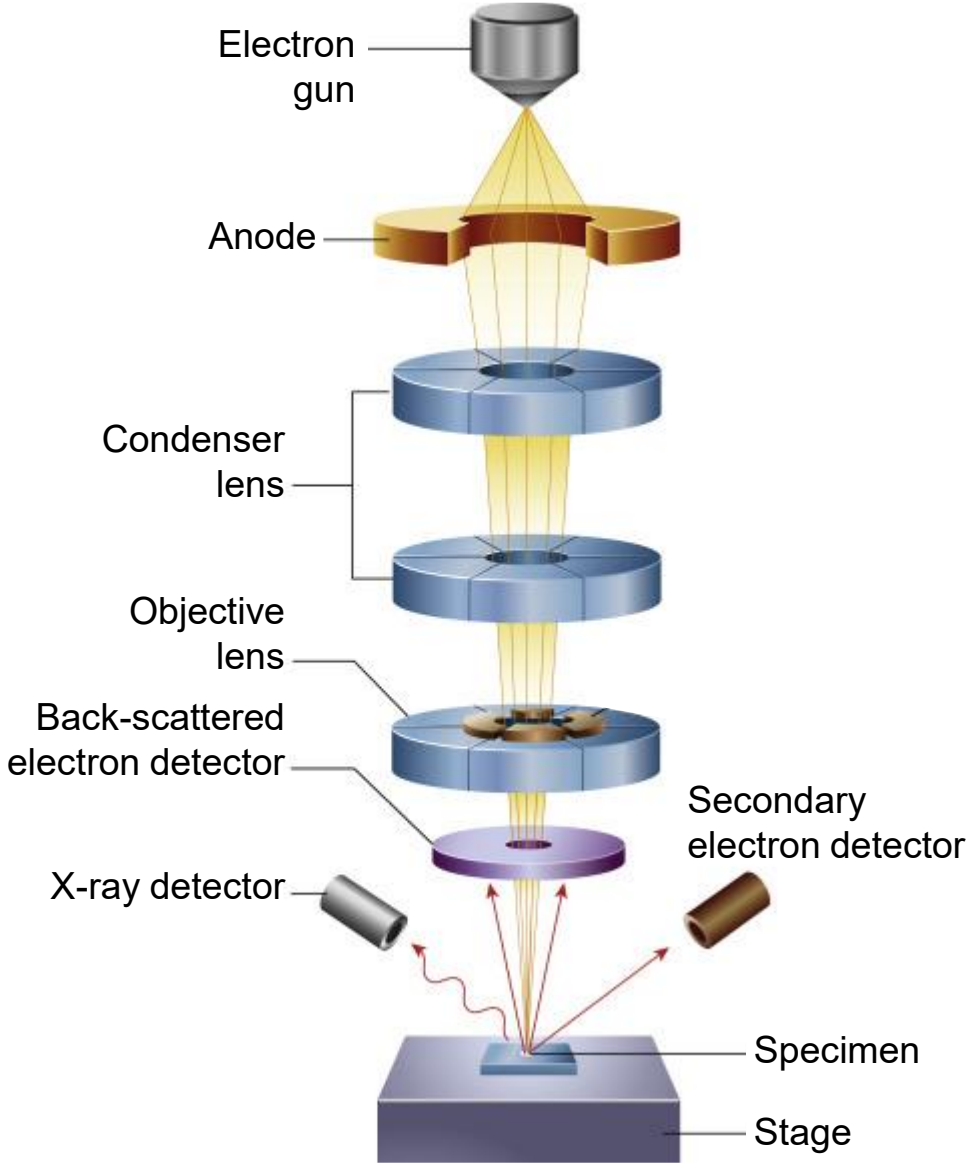
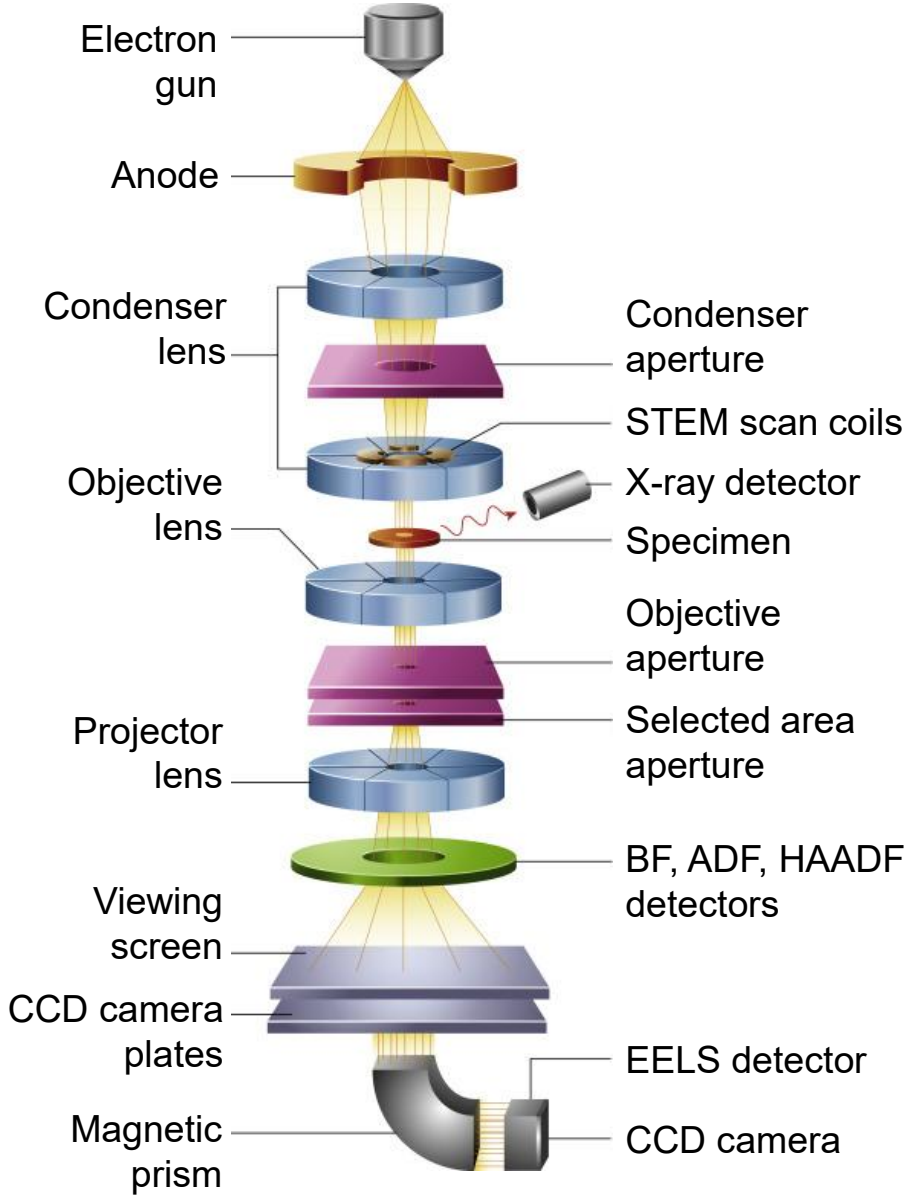
Light microscope



Scanning EM



Transmission EM



Week 3 - Light Microscope vs. Electron Microscopy

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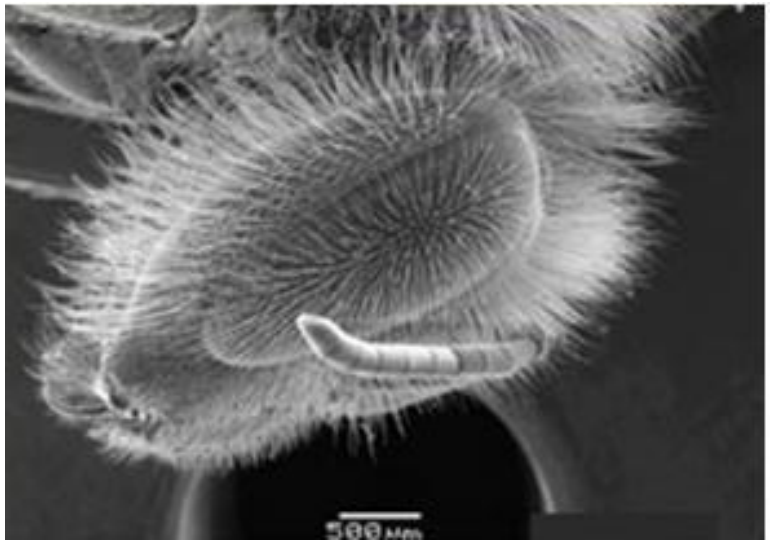
Light microscope (Colored)



SEM image (Grayscale)

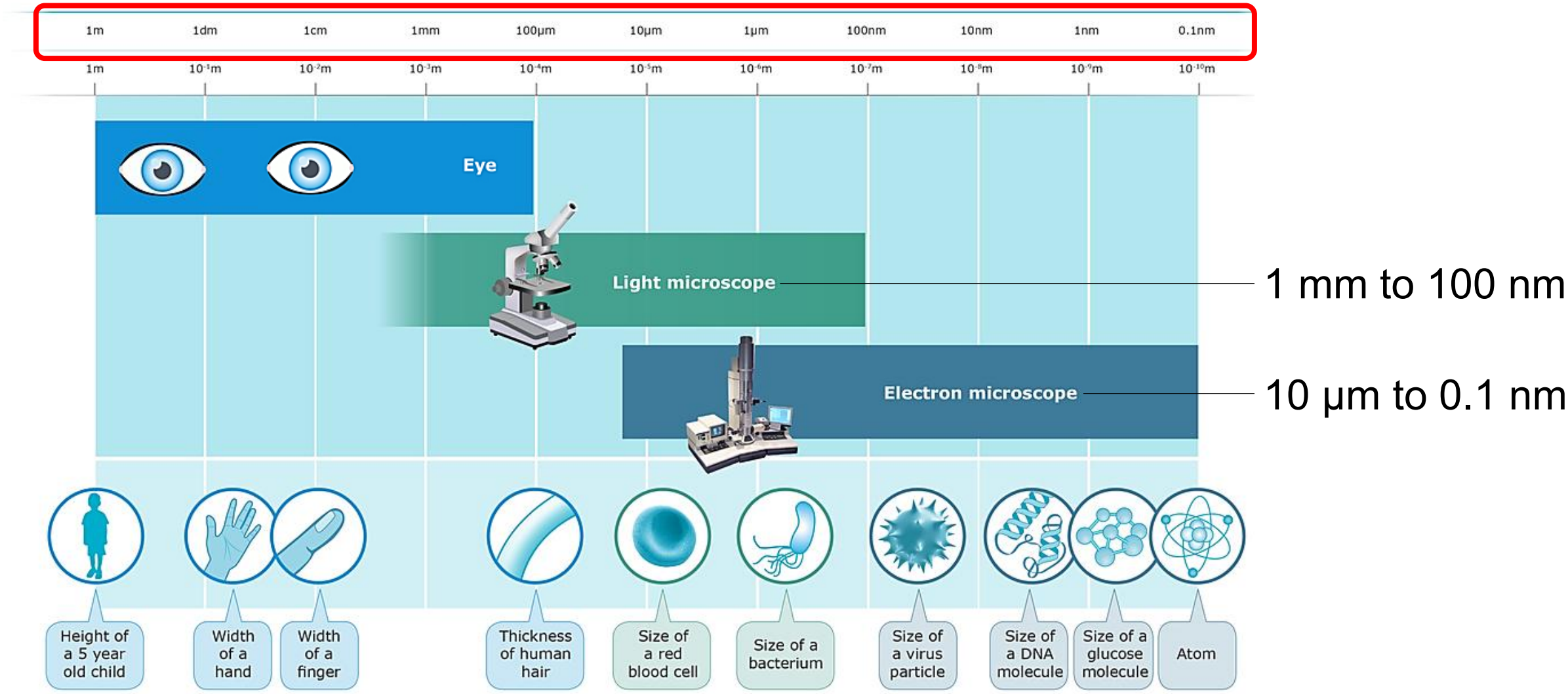
SEM offers…

- Much more topographical information.
- Much greater magnification.
- Much higher depth of field.

Why?

Week 3 - Limit of resolution (resolving power)

Limit of resolution (resolving power):
The limit or ability of a microscope to distinguish details of a specimen or sample.



1 mm to 100 nm

10 µm to 0.1 nm

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Week 3 - Electron Microscopy (EM) characterisation

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1. Interactions between electron and specimen - signals
2. Working principles of SEM signals and their applications

SE (Secondary electron)

BSE (Backscattered electrons)

EDS (Electron dispersive X-ray spectroscopy)

1. Operation setting of SEM parameters

Accelerating voltage

Spot size

Aperture

Working distance

1. Applications of SEM
2. Advantages and limitations of SEM

https://myscope.training/SEM_SEM_Basics
https://myscope.training/EDS_Welcome

- Lab Demo- SEM operation principals and sample characterisation

Week 4 - Light Microscope vs Electron Microscope

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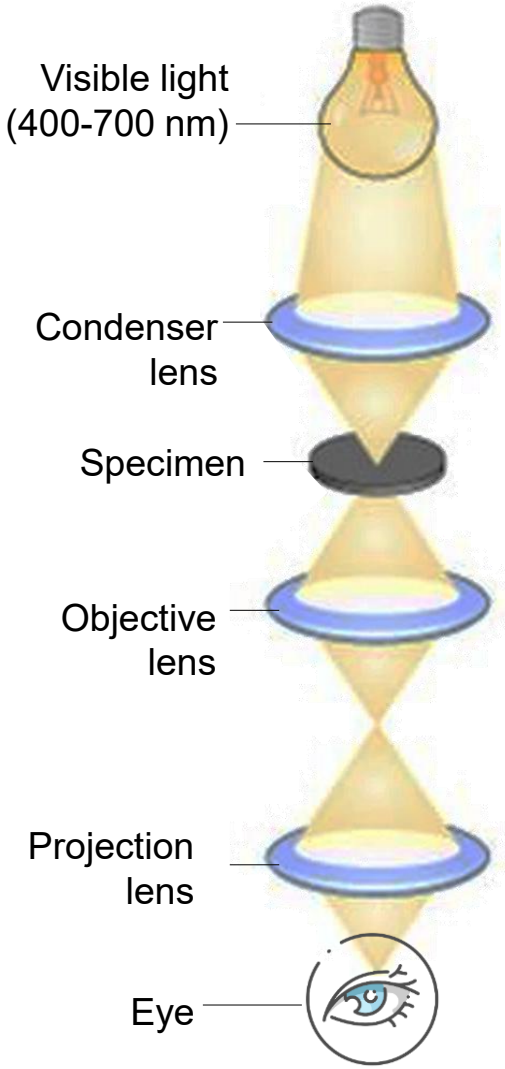
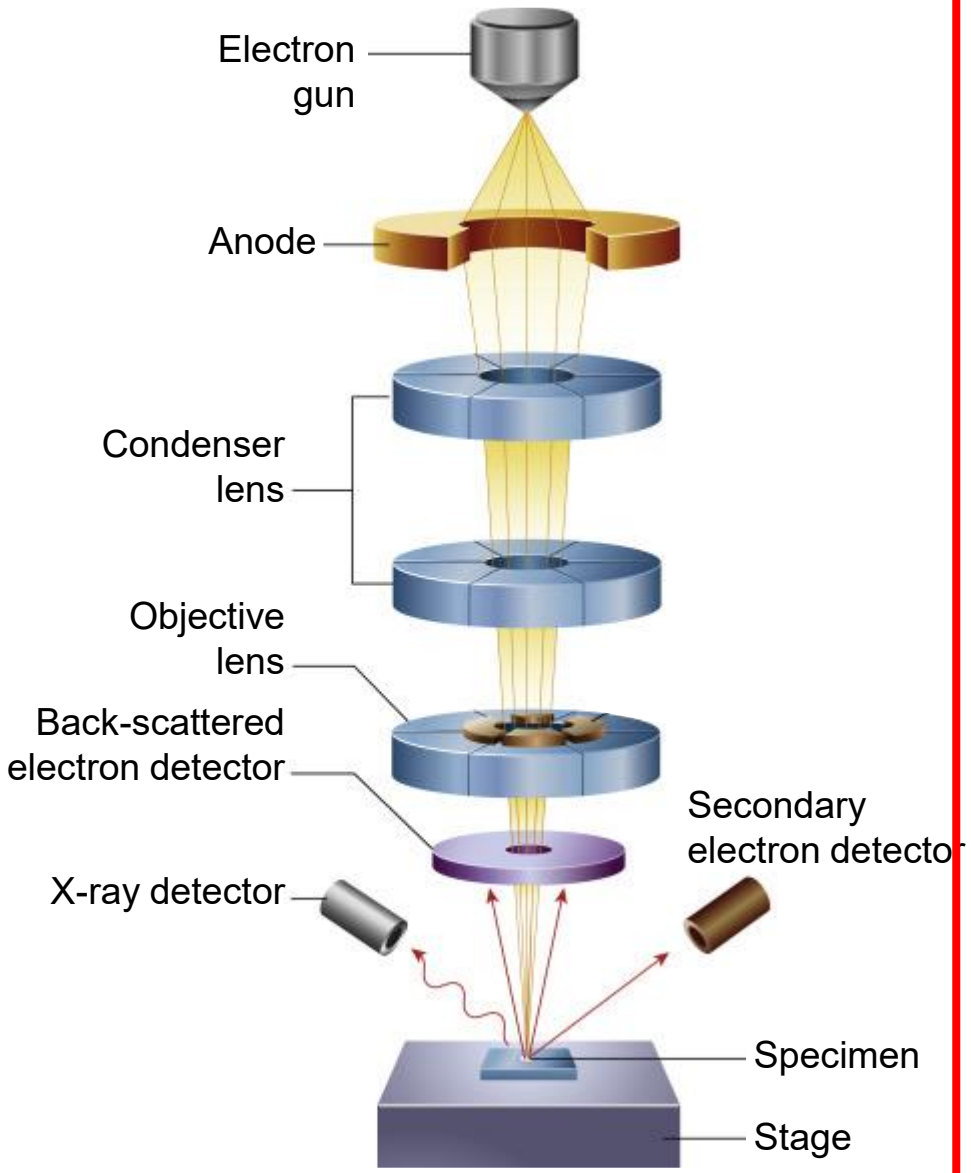
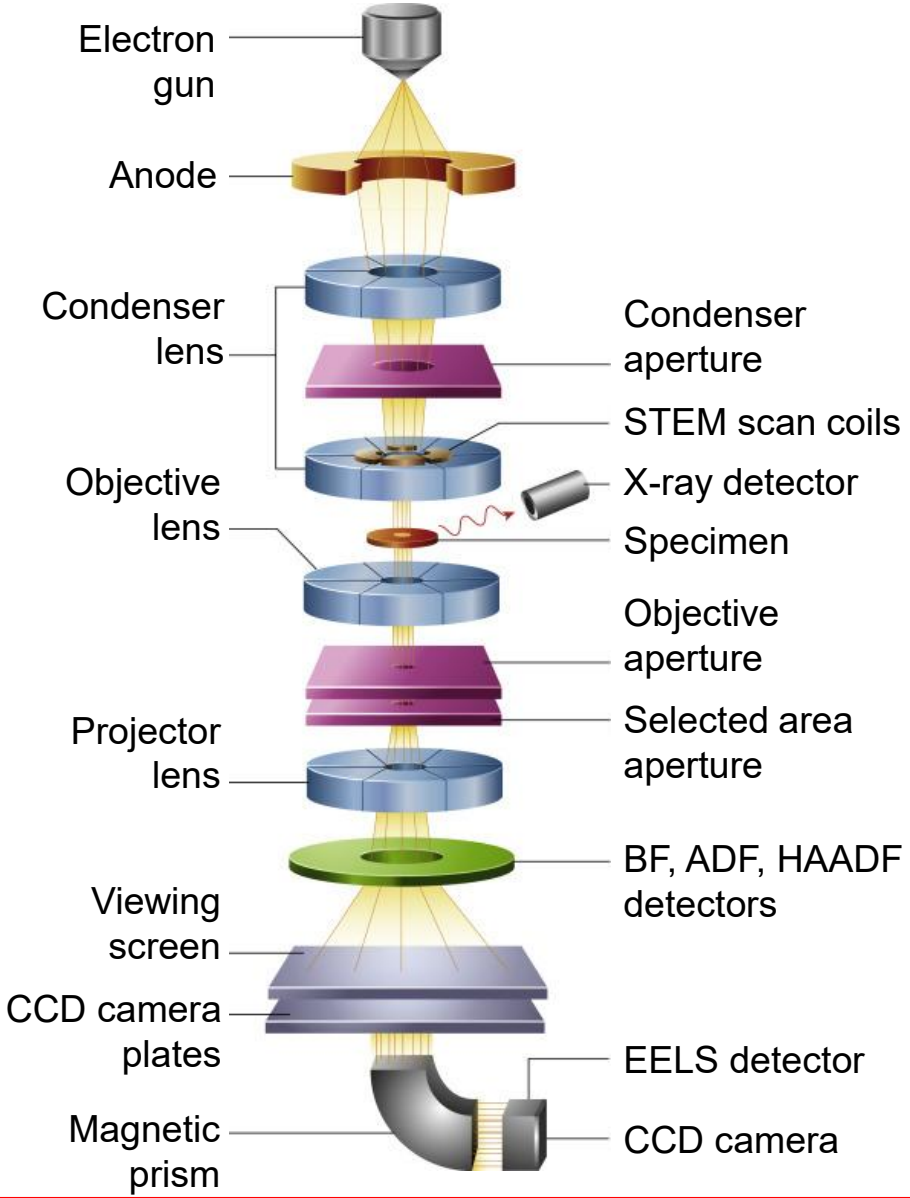
Light microscope



Scanning EM



Transmission EM



Week 4 -How does TEM work? - The electron gun

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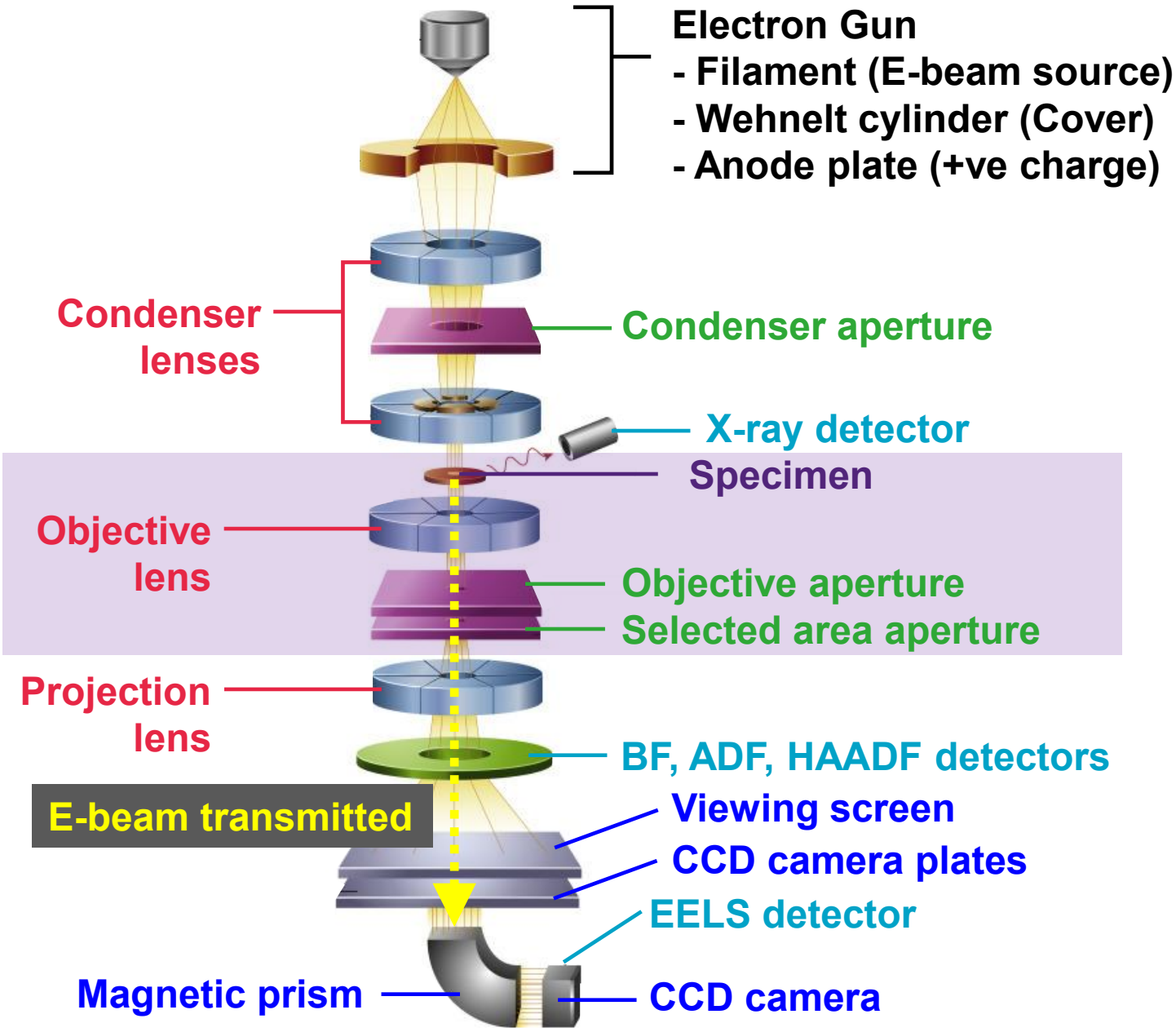
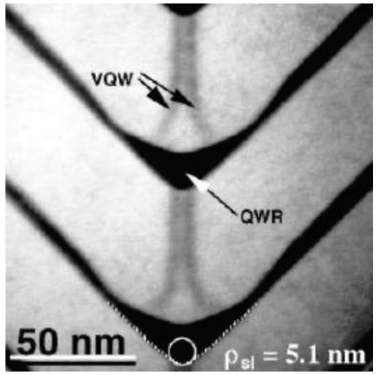
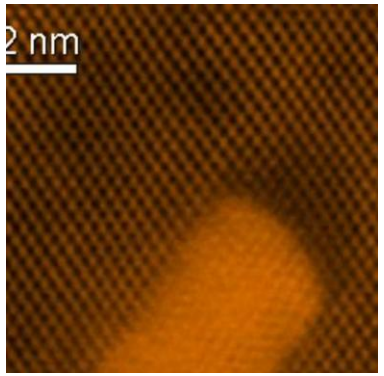
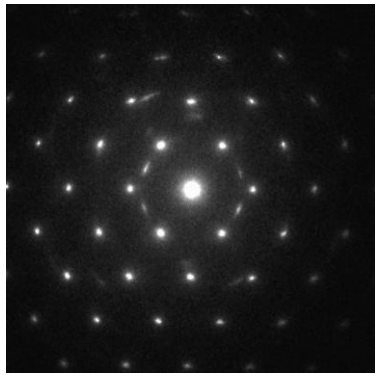




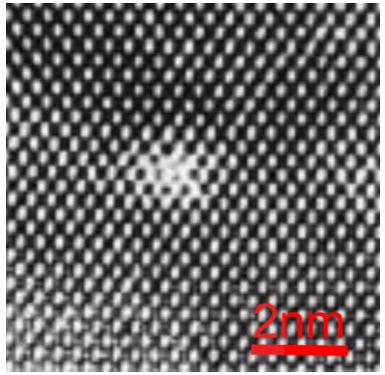
Diffraction contrast



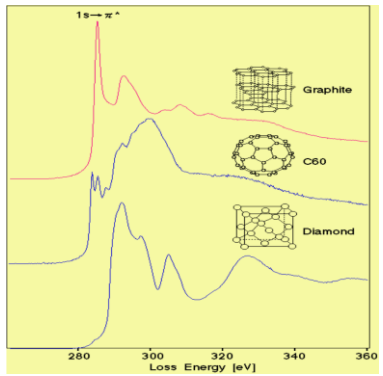
STEM imaging



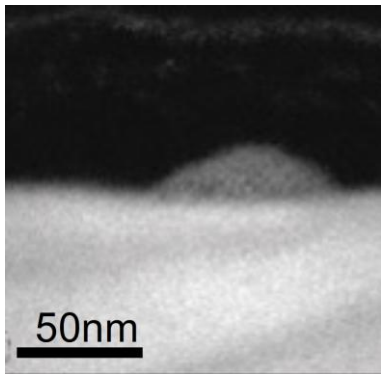
Electron diffraction



HRTEM imaging



EELS spectroscopy



Energy filtered TEM

Week 4 - Electron Microscopy (EM) characterisation

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1. Structure and Principle of a TEM
2. Electron diffraction
3. Diffraction contrast (Imaging formed by single diffraction spot)
4. Phase contract (HRTEM) (Imaging formed by multiple diffraction spots)

   TEM specimens must be thin and well orientated

1. Analytical TEM

Characteristic X-rays

Electron energy lose spectroscopy (not only chemistry, but also bonding)

STEM (Z contrast)

1. Applications of analytic S/TEM

- Lab Demo- TEM operation principals and sample characterisation

Lecture contents and Related Assessments

X-ray diffraction (XRD) - Week 1, 2

- Lecture 2
- 2Lab demonstration

Scanning electron microscope (SEM) **) - Week 3, 4**

- Lecture 3
- 2Lab demonstration

Transmission electron microscope (TEM) **) - Week 5, 6**

- Lecture 4
- 2Lab demonstration

Final examination

To be consolidated to Lab Engagement + Lab report (20%)

Week 5 - Nanomagnetism and mag nanocomposite

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- What are magnets and how they work
- Origin and fundamentals of magnetism
- Magnetic materials and their types
- Temperature effect and magnetic domain
- Magnetic hysteresis
- Magnetism and super-paramagnetism
- Synthesis of Magnetic Nanoparticles.

▪ Lab demo on the synthesis ofn magnetic nanostructures

Week 6 - Characterisation/analysis and application of mag nanocompo

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Parameters influence magnetic properties of nanoparticles/nanocomposites

Effect of annealing temperatures on magnetic phase transition and magnetic properties of nanoparticles

The particle size effect on magnetic properties and determination of critical size.

The particle shape effect on magnetic properties.

The density/porosity effect on magnetic properties of nanoparticles.

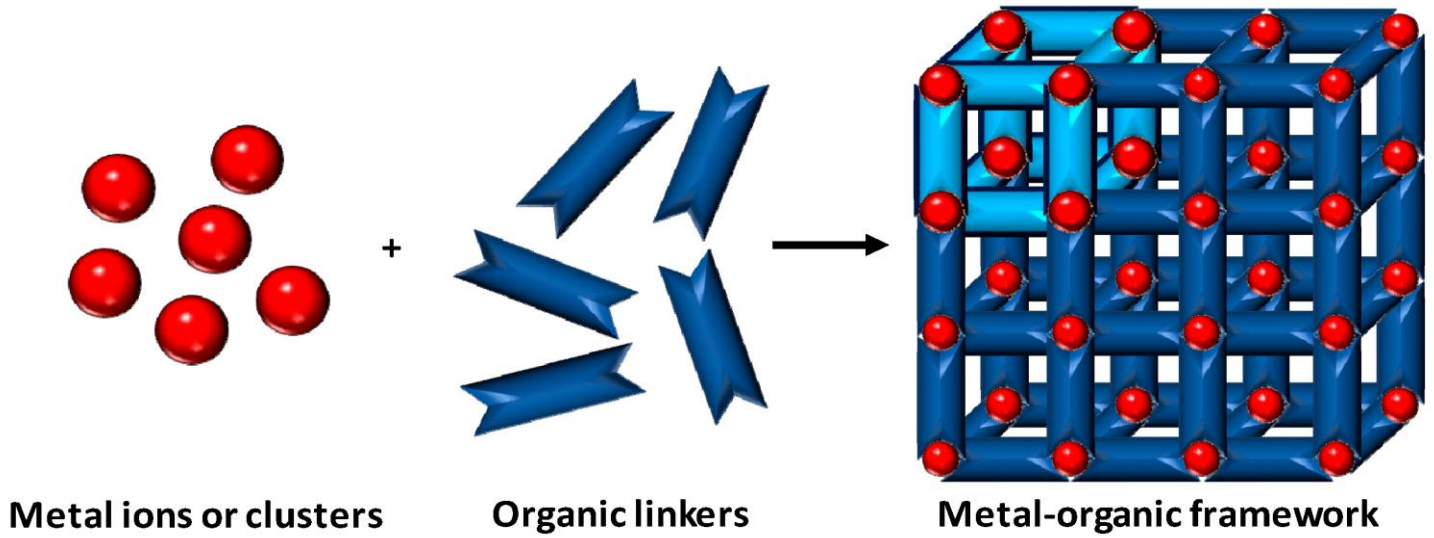
Nano Machineries – Magnetic drug delivery applications

- Laboratory – Electrochemical characterisation of magnetic nanocomposites/analysis

Week 7 - Porous Nanocatalysts Waste water treatmen

METAL-ORGANIC FRAMEWORKS (MOFs)

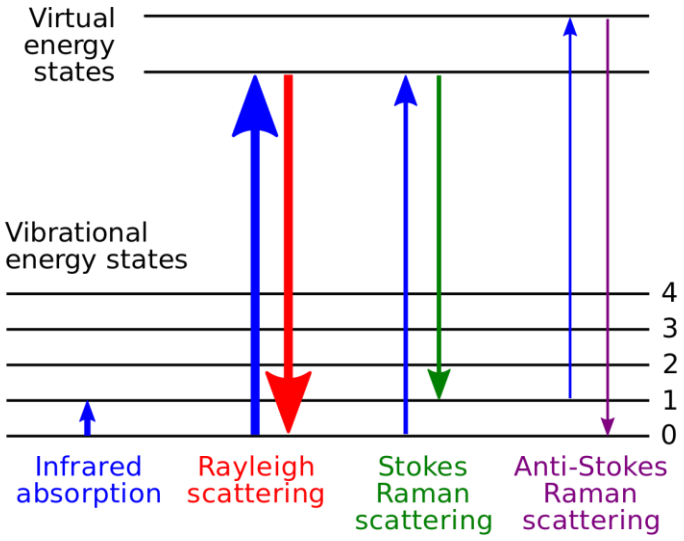
Crystalline, porous materials formed by the simple coordination of metal centres and organic linkers/ligands

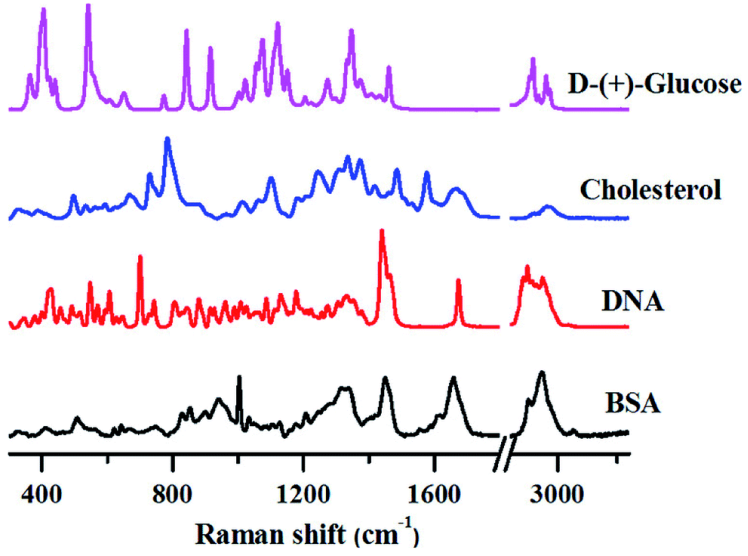


- Demo- Nanocatalysts (MOF) for waste water treatment

Week 8 -Optical properties of nanocomposite and their applications

Principles of Raman Spectroscopy: Raman spectroscopy is a type of spectroscopic techniques that determines the vibrational modes of the molecules. It relies on inelastic scattering of photons.





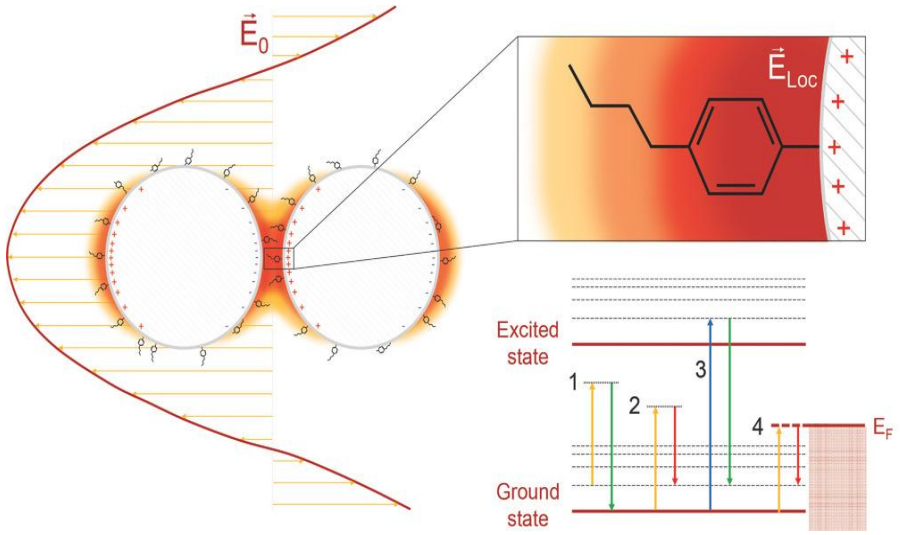
Vibrational and rotational frequencies of a molecule are characteristics of a molecule.

Kuhar *et. al. RSC Adv.* **8**, 25888-25908 (2018)

- Laboratory – metallic nanocomposites with optical properties for SERS based biosensing for pesticide detection

Week 9 –Bi-tri metallic nanocomposite film and their optical applications

Signal Enhancement mechanism of Raman Spectroscopy using nanomaterials



Electromagnetic enhancement:

- Incident light on the nanomaterials creates localised surface plasmon resonance
- Plasmon oscillates
- Scattering of the light increases with perpendicular oscillations

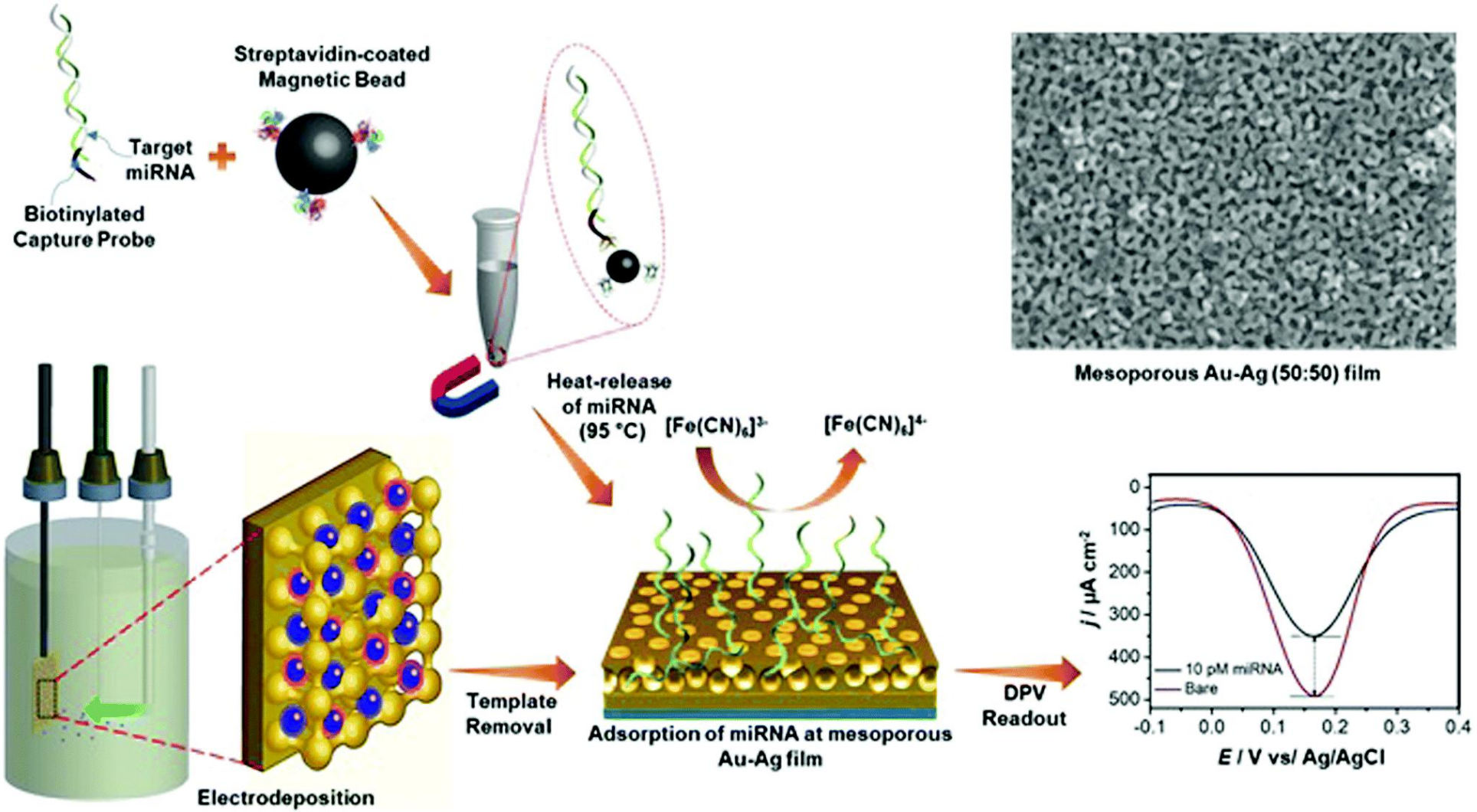
Chemical Enhancement:

- Incident light on the nanomaterials creates chemisorption of the molecule on the nanomaterial surface
- Charge transfer occurs
- Scattering of the light increases due to the charge transfer phenomena

Balcytis *et. al. Adv.Optical Mater.* 6, 1800292 (2018)

- Laboratory – metallic nanocomposites with optical properties for SERS based biosensing for pesticide detection

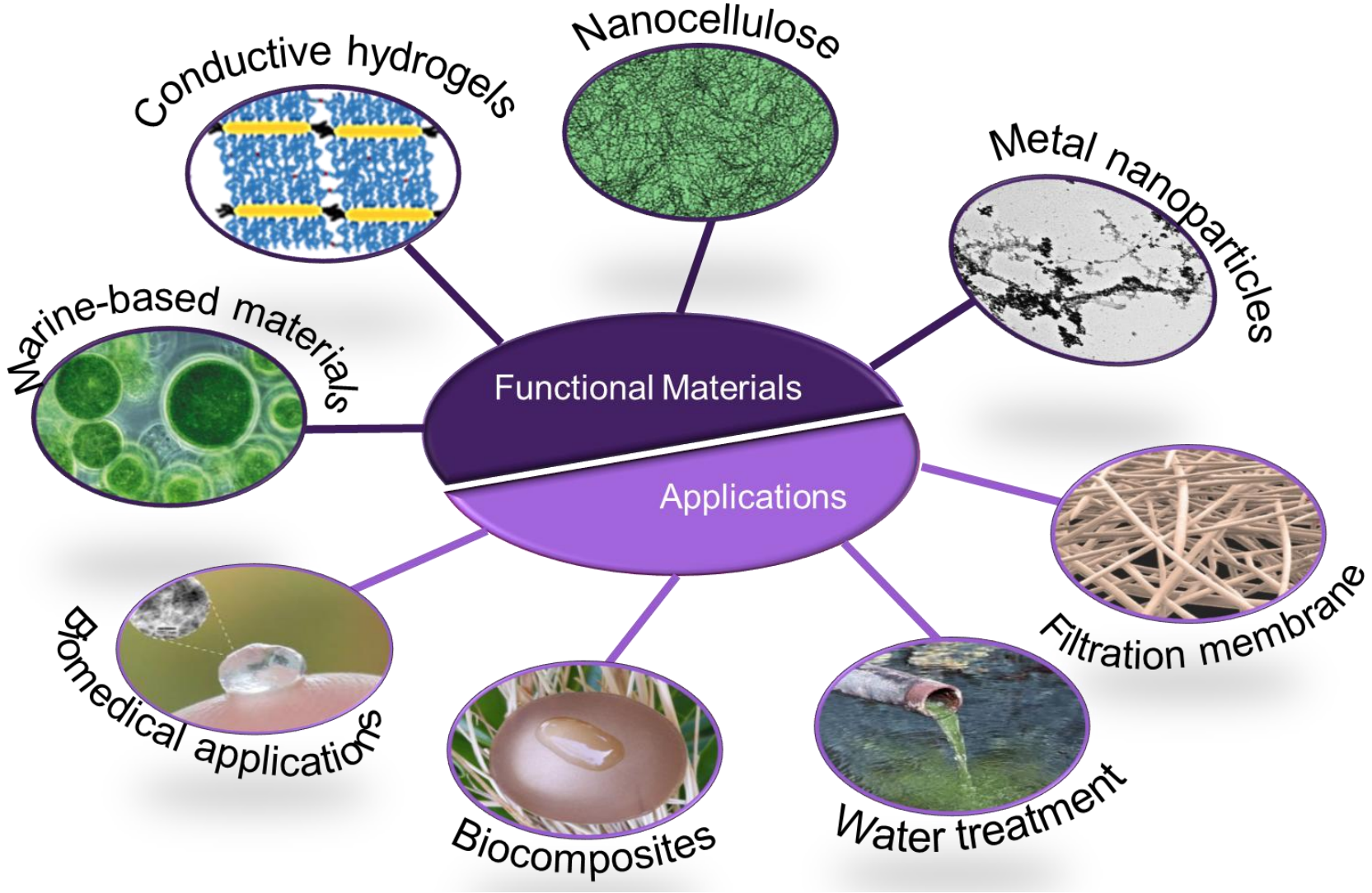
Week 9 –Bi-tri metallic nanocomposite film and their optical applications



https://pubs.rsc.org/en/content/articlelanding/2020/tb/d0tb02003f

- Laboratory – metallic nanocomposites with optical properties for SERS based biosensing for pesticide detection

Week 10 - Australian biomass derived nanocellulose



Online class and lab prac due to the public holiday

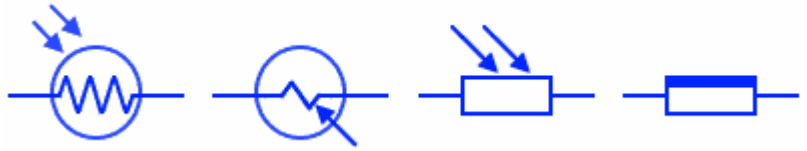
Week 11 – Presentation preparation and assessment

Week 12 – Semiconductor based photodetector application

What are photodetectors?

A photodetector is a variable resistor whose resistance depends on the light intensity of incident light. So, this is also called as *Light Dependent Resistor* (LDR)/*photoresistor*. It works on the principle of *photoconductivity*.

Symbol:



Example:

CdSe and CdS, used in *light sensing circuits* such as automatic street lights, solar lights, fire alarms, burglar alarms, etc.

Advantages:

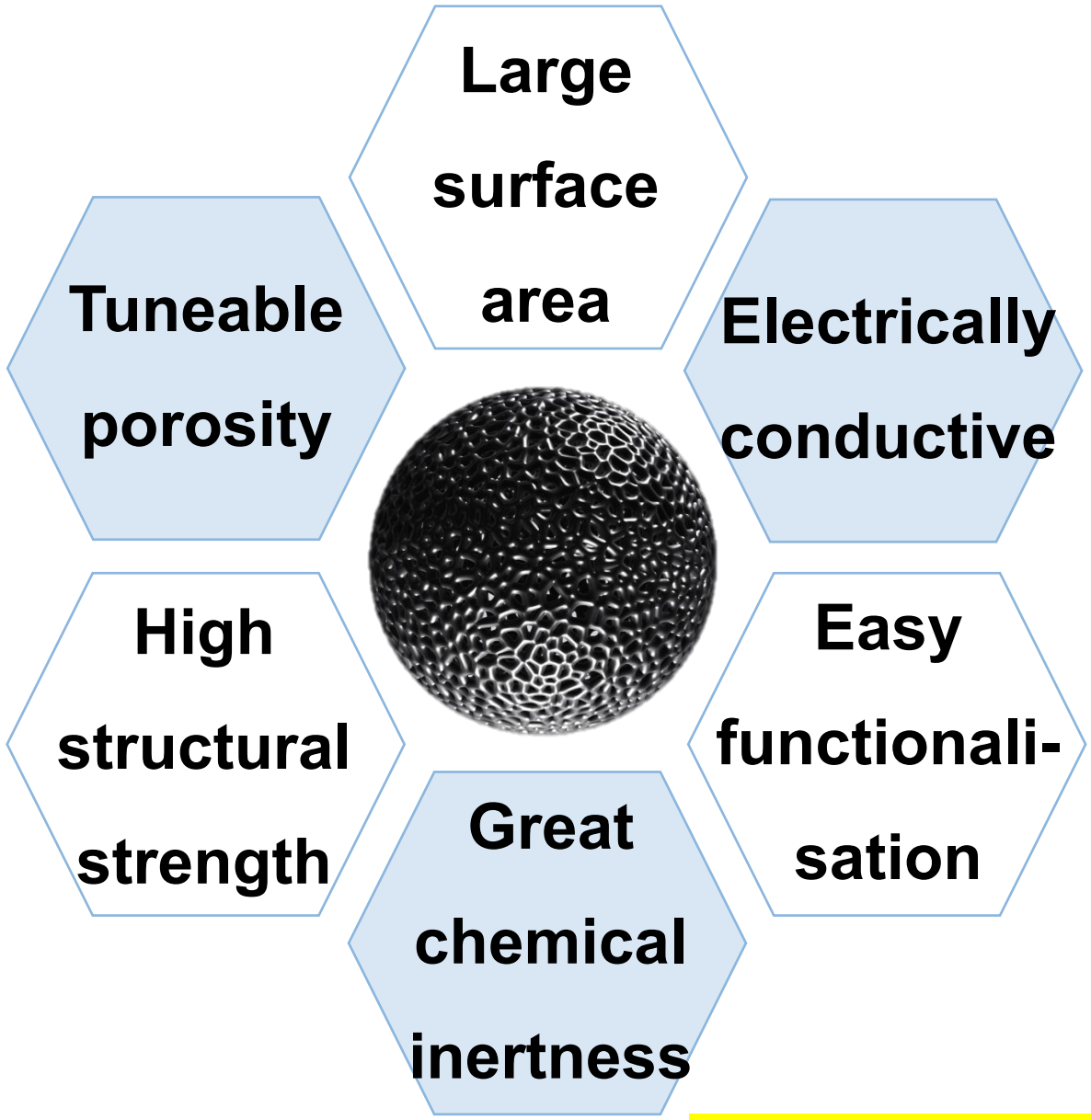
- Non-linear response to the intensity of light.
- It operates in either biasing. (It can conduct in both directions.)
- It is made of a single N-type semiconductor.
- Cost-effective and durable.
- Has a simple design and is easy to manufacture.

Disadvantages:

- It has a slower response time.

- Laboratory – photodetector demonstration

Week 13 - NANOPOROUS CARBONS



- Highly suitable for energy–related applications.
- Examples are secondary battery, supercapacitor, water-splitting electrocatalysis, hydrogen fuel cell, etc.
- It can be obtained from various materials such as biomass, porous polymers, by-products of oil refinery, etc.

▪Online Lab prac demo porous carbon synthesis and application in supercapacitors.

Week 13 – Final Exam Discussion

Q/A on final exam and discussion

MATE 6301 (Laboratory)

Deposition and Characterization of Mesoporous Gold Thin Films

Electrochemical Deposition

Electrochemical deposition is a process in which metallic ions can solidify and deposit on the cathode surface if an adequate amount of electric current flows through the electrolyte solution. The charged ions in the electrolyte are produced by dissolving metallic salt in water. Using the electrodes, an electric current is introduced into the electrolyte during the electrochemical deposition procedure.

Operating Principle

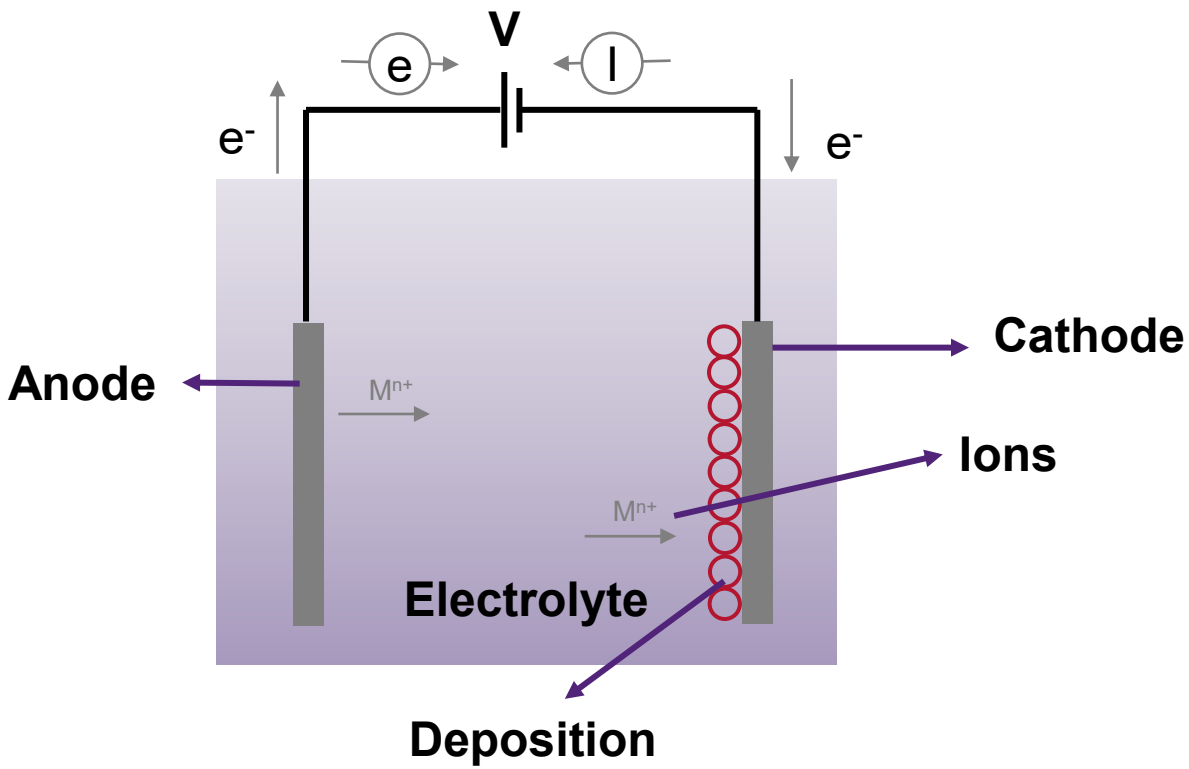
- Reduction of metal ions in an electrolyte solution onto a conductive surface through the application of an electric potential.

- The general equation for the electrochemical deposition of a metal (M) from an electrolyte can be represented as follows:

$$M^{n+} + ne^- \rightarrow M$$

where,

n: oxidation state, e: electron.



The electrochemical deposition process follows:

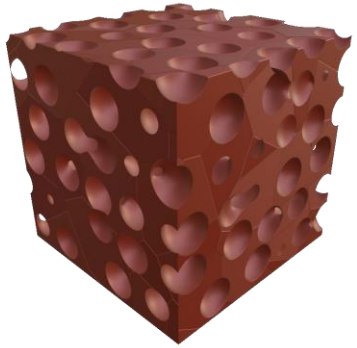
1. **Migration** – the hydrated metal ions in the solution migrate towards the cathode under the influence of applied current as well as by diffusion and convection.
2. **Electron transfer** – at the cathode surface, a hydrated metal ion enters the diffused double layer where the water molecules of the hydrated ion are aligned. Then the metal ions enter the Helmholtz double layer where it is deprived of their hydrate envelope.
3. The dehydrated ion is **neutralized and adsorbed** on the cathode surface.
4. The absorbed atom then migrates or diffuses to the **growth** point on the cathode surface.

[1]

The difference:

Conventional thin films

- Low surface roughness
- Low surface area
- Less surface defects
- No change in the refractive index
- No tunable functionality
- More material is used
- Low surface sensitivity
- Micro to nano crystalline grains

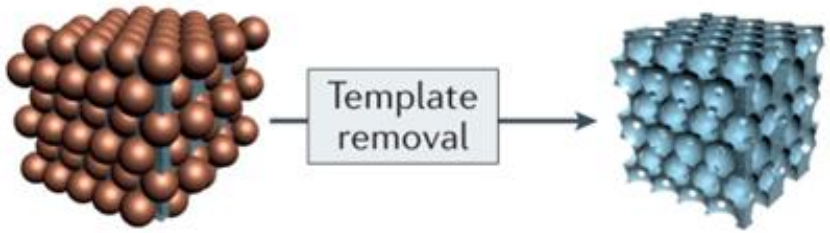


Porous thin films

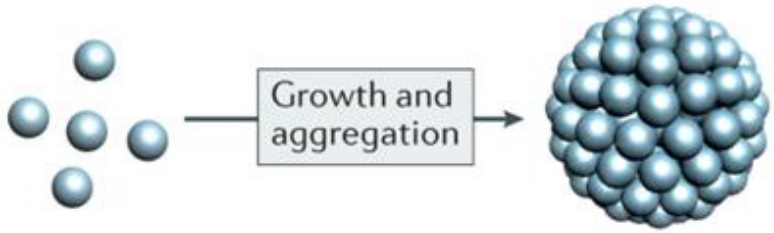
- High surface roughness
- High surface area
- More surface defects
- Tunable refractive index
- Tunable functionality
- Less material is used
- High surface sensitivity
- Nano crystalline grains

How are these porous nanostructures formed?

Template-assisted



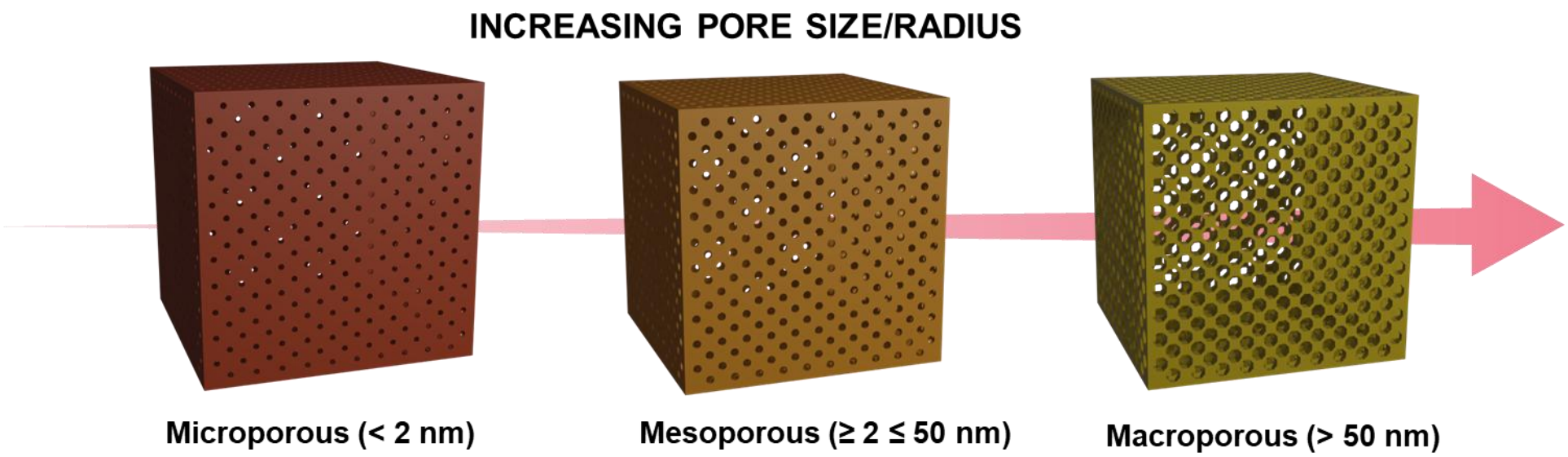
Template-free



Porous thin films?

Introducing defined void spaces also referred to as “pores” at the bulk of the film distributed symmetrically or asymmetry in a material system.

Classifications of porous materials:



Importance of mesoporous thin film?

High surface area, versatile surface functionalization, and bicontinuous percolation paths through a broad library of pore arrangements which results in improved sensitivity and detectivity. Surfaces in porous materials are more diffusive than specular. This can minimise the net reflective loss in functional materials.

Check the announcement and group formation for the lab demo starting from week 1 Friday 8-12pm

Thank you and all the best!

WEEK 2 · 知识点总结 · Dr Aditya Ashok

第 2 周　X 射线衍射

X-ray Diffraction (XRD)

主线：晶体学基础（晶胞、Miller 指数、d 间距）→ 布拉格定律 → 物相鉴定与 Scherrer 晶粒尺寸

1. 固体的三种结构

- **单晶**（长程有序）、**多晶**（微晶 / 纳米晶）、**非晶**（如石英玻璃）。
- **晶体结构 = 晶格 (lattice) + 基元 (basis)**；晶格是无限周期排列的点阵，每个点环境相同。

2. 晶胞与点阵

- 晶胞由 a, b, c 和 α, β, γ 定义（最多 6 个参数）；重复平移可填满整个晶格。
- **原胞**只在顶点有格点（含 1 个格点）；**非原胞**在面心或体心有额外格点。
- **7 大晶系**：立方、四方、正交、六方、单斜、三斜、菱方 → **14 种布拉维点阵**。
- 每个晶胞的原子数：**SC = 1，BCC = 2，FCC = 4**。

3. 晶向与晶面

- 晶向 **[uvw]**，等价晶向族 **<uvw>**；负值写成上横线。
- 晶面用 **Miller 指数 (hkl)**，晶面族 **{hkl}**；一组平行等距晶面共用同一指数。
- **求 Miller 指数**：① 求截距 → ② 取倒数 → ③ 化为最小整数（不能有分数，不写逗号）；平行于某轴 → 截距 ∞ → 指数 0。
  例：截距 1, 1/2, 3 → 倒数 1, 2, 1/3 → ×3 → **(3 6 1)**。
- **d 间距**：相邻平行晶面的垂直距离；**指数越高，d 越小**。立方晶系：$d = a / \sqrt{(h^2+k^2+l^2)}$。

4. X 射线的产生

- 高速电子轰击阳极：**连续谱**（韧致辐射）+ **特征谱**（Kα、Kβ，内层电子跃迁）。
- XRD 用 Kα：**Cu Kα λ = 0.15406 nm**；Mo Kα 0.0711 nm。不同阳极 λ 不同 → 同一晶面的 2θ 不同。

5. 布拉格定律

- $n\lambda = 2d\sin\theta$：相邻晶面散射的光程差 = 2d sinθ，等于整数倍波长时相长干涉 → 出现衍射峰。
- 衍射 = 大量散射事件的干涉；还有电子衍射、中子衍射。

6. 从 XRD 图谱读什么

- **峰位** → d 间距 → 与 **PDF/ICDD 卡**比对鉴定物相（例：Au，PDF 00-004-0784，(111)(200)(220)(311)）。
- **峰宽** → 晶粒尺寸，**Scherrer 公式**：$D = K\lambda / (\beta\cos\theta)$，K ≈ 0.9–0.94，β 为半高宽（**弧度**），θ 为布拉格角（2θ 的一半）。
- 晶粒越小，峰越宽（Pt 纳米颗粒例）；晶态 → 尖峰，非晶 → 宽包（石墨化碳 vs 无定形碳）。
- 能得到：物相及含量、晶格参数、**晶粒尺寸**、应变、择优取向（织构）、热膨胀。
- 样品要求：表面平整；抛光引入的应力需退火消除；晶粒 < 100 nm 会导致峰展宽。
- **单晶**只出现一族平行晶面的峰（如 (100)、(200)）；**多晶粉末**有成千上万个晶粒，各晶面都有合适取向的晶粒 → 出现所有峰。

考试重点：Miller 指数求法；布拉格方程计算 d 或 θ；Scherrer 计算晶粒尺寸（注意弧度、θ 而非 2θ）；单晶 vs 多晶的衍射差异；XRD 能 / 不能告诉你什么。**后续联系**：W5 案例 2、W6 案例 3、W8、W12 都用 Scherrer 公式。

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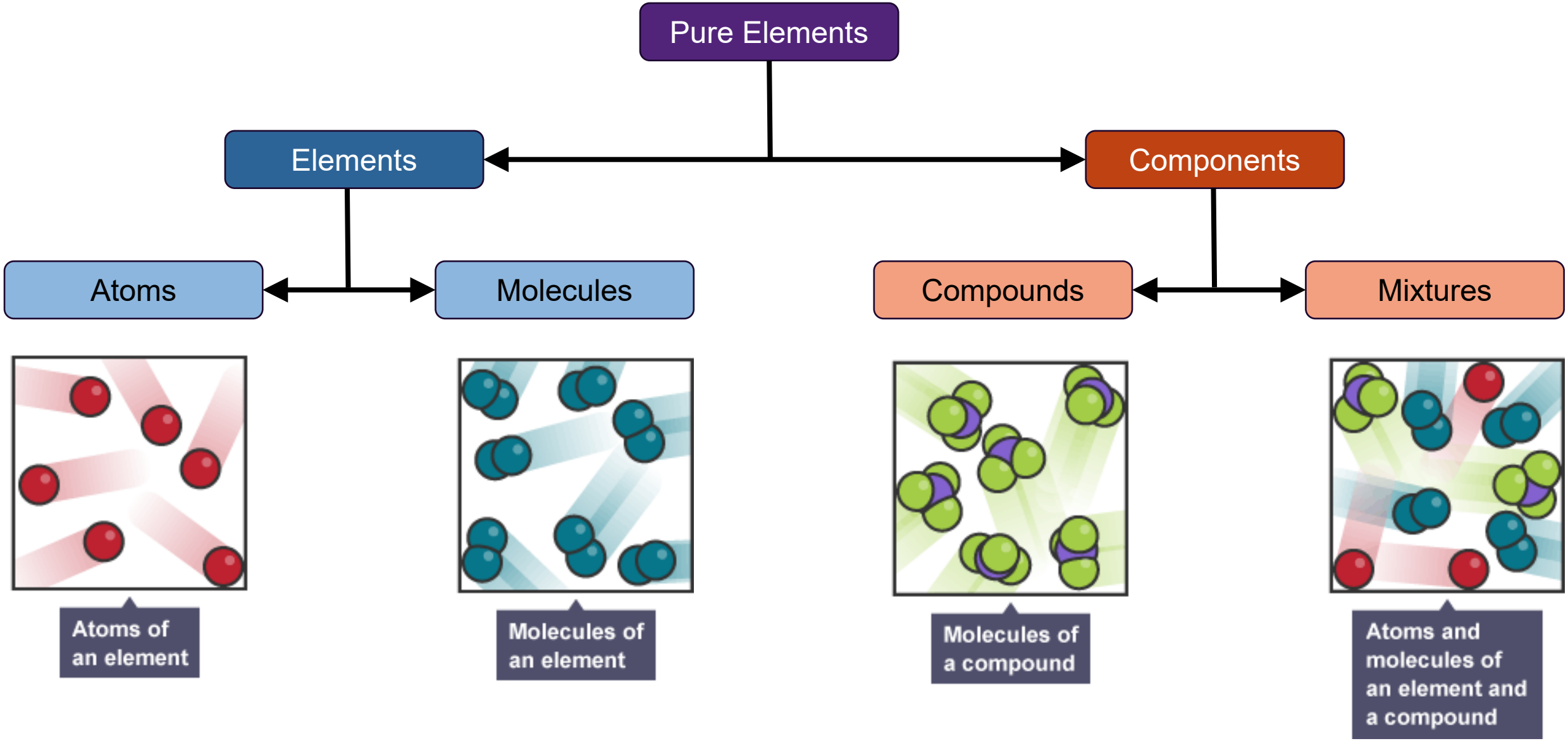
MATE 6301

X-ray Diffraction (XRD)

Lecture by,
Dr. Aditya Ashok (UQ-AIBN)

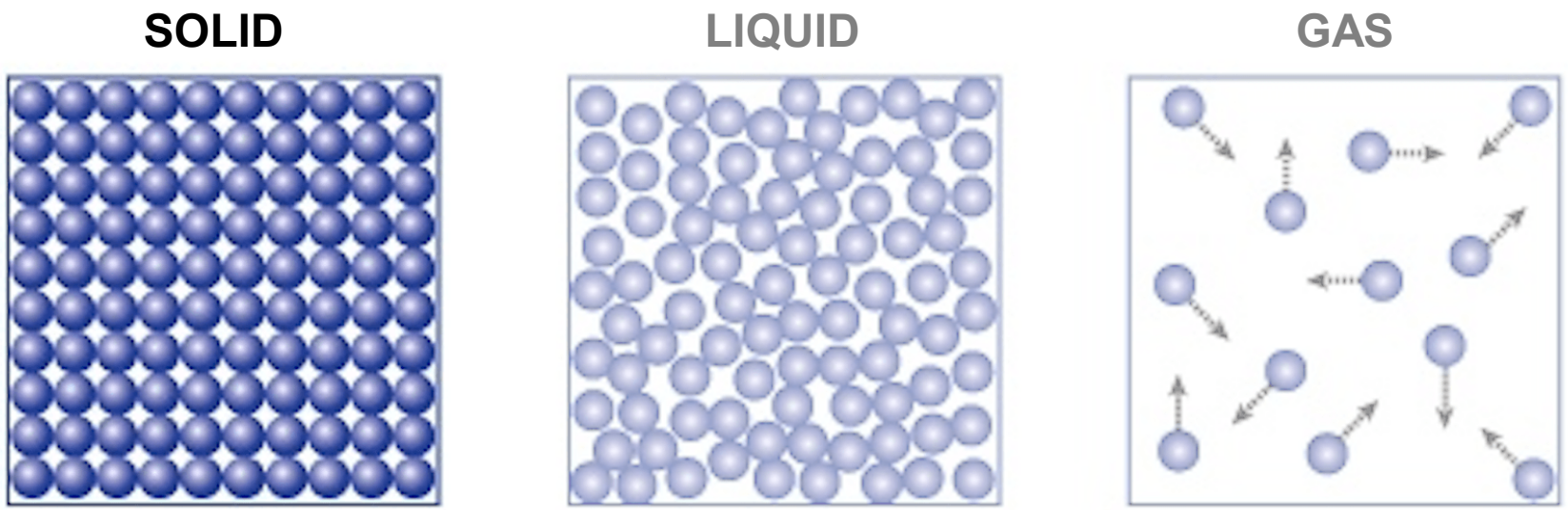

Pure Elements
Elements
Components
Atoms
Molecules
Compounds
Mixtures
Atoms of an element
Molecules of an element
Molecules of a compound
Atoms and molecules of an element and a compound

There are 3 primary states of matter,

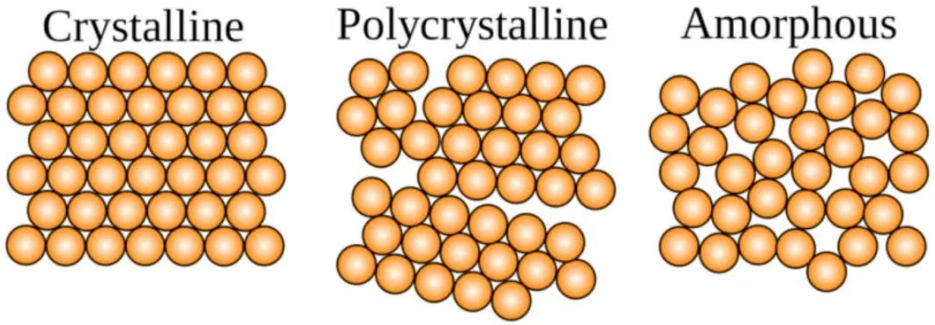
Solid – Matter with a fixed shape and volume. Particles are tightly packed and only vibrate in place (e.g., ice, rock).

Liquid – Matter with a fixed volume but no fixed shape. It takes the shape of its container. Particles are close but can move past each other (e.g., water, oil).

Gas – Matter with neither fixed shape nor fixed volume. Particles are far apart and move freely (e.g., air, steam).



Based on the arrangement of atoms,



i. Single Crystalline (long-range order)

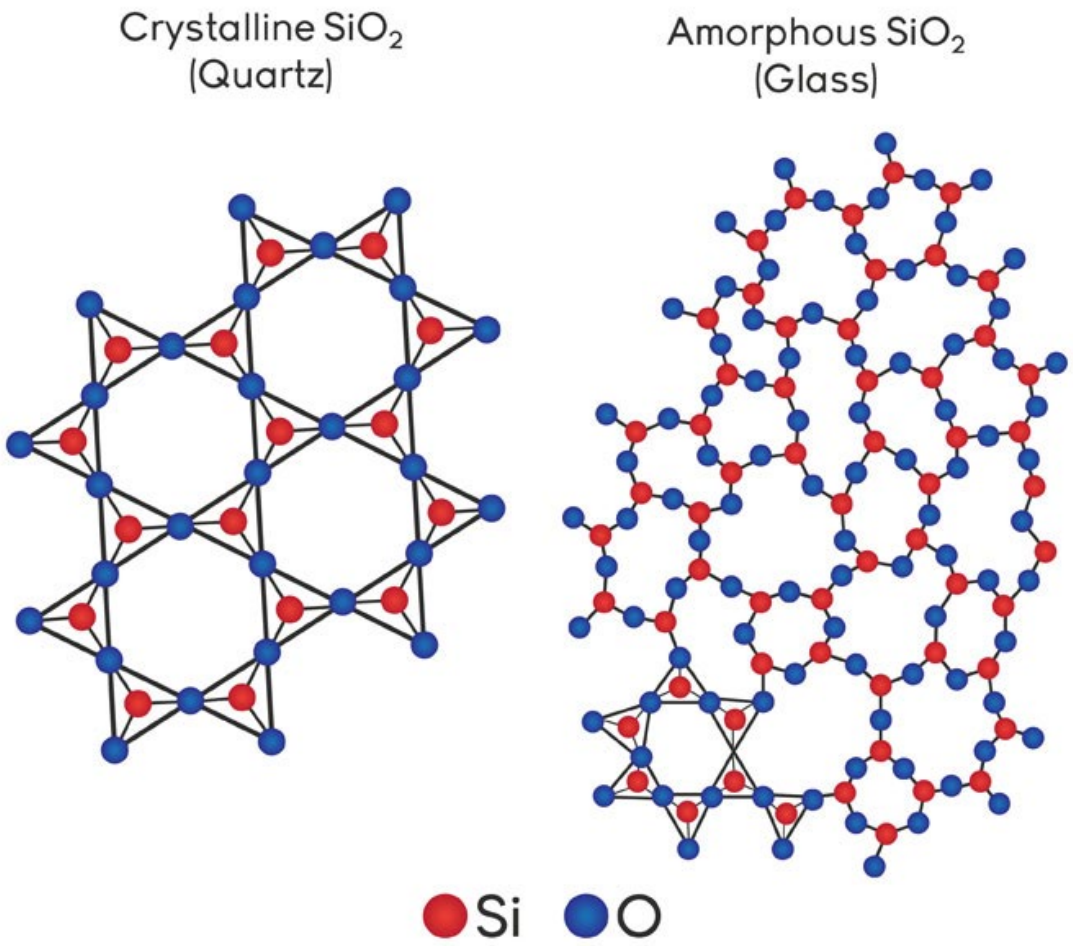
ii. Poly-crystalline (short-range order)

   a. Micro-crystalline

   b. Nano-crystalline

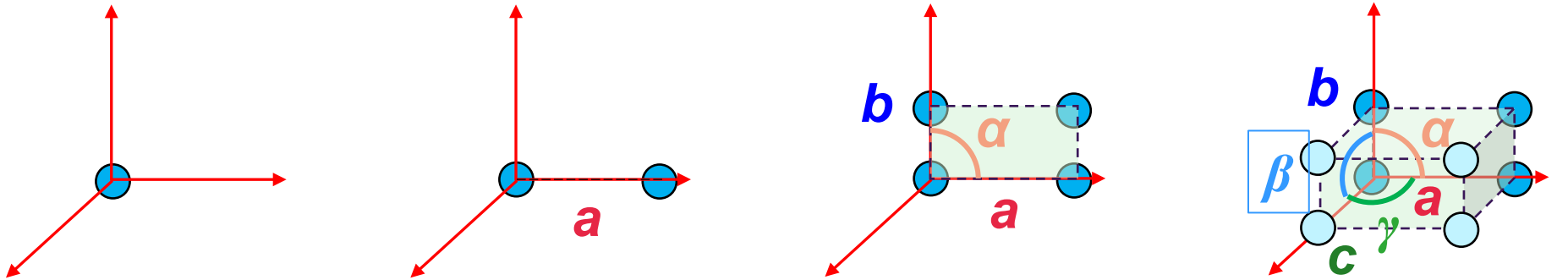
iii. Amorphous

Best example is silica glass,

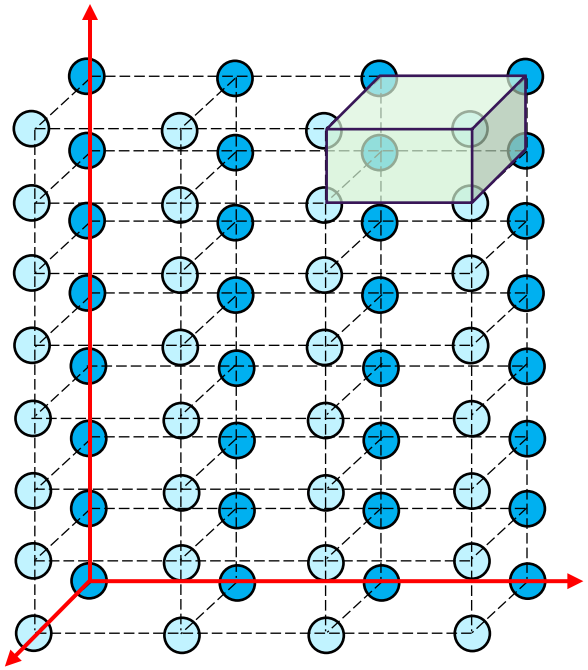


What are crystalline state?

A crystal consists of identical *structural units*, consisting of one or more atoms, which are **regularly** and **periodically** arranged in three dimensions. This regularity extends over the entire crystal.



Lattice point → **1D lattice line** → **2D lattice plane** → **3D lattice space**

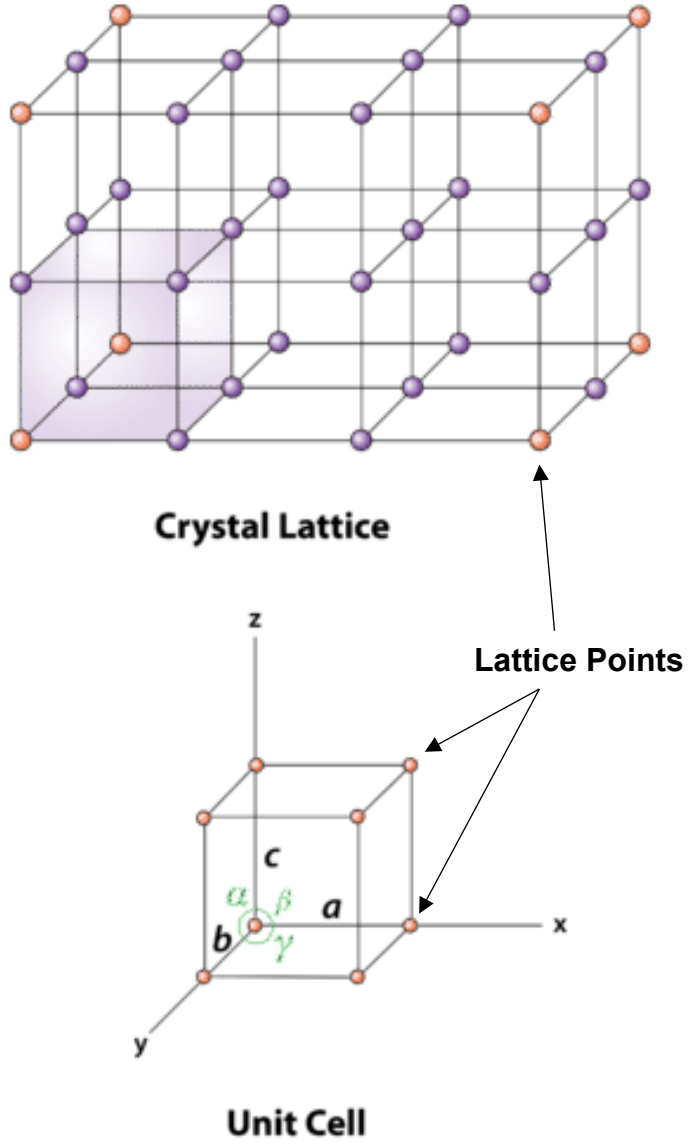


Note:
The smallest building blocks of a crystal are referred to as the **basis**,
the periodic arrangement is described by the **lattice**.

Crystal lattices and unit cells

Lattice is infinite periodic arrangement of points in a space.

The lattice is a 3D array of points (lattice points), each of them has identical surroundings.



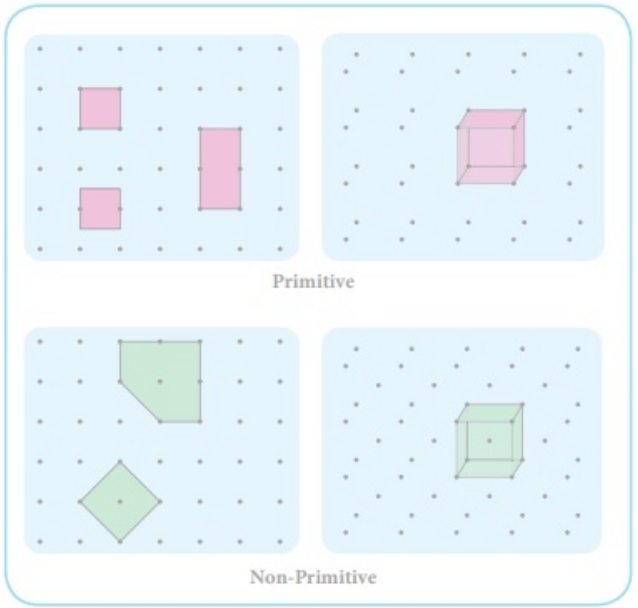
Notes on Unit Cells:

- A **Unit Cell** represents a small volume of a lattice.
- The entire lattice can be reconstructed by repeating this unit cell **without empty space in-between.**
- The unit cell is defined by three independent unit vectors, **a**, **b**, and **c.**
- The direction of the unit vectors **a**, **b**, and **c** defines the direction of the *x*, *y* and *z* axes, called the **crystal axes.**
- α, β, γ are angles between **b** & **c**, **c** & **a**, and **a** & **b.**
- Maximum 6 variables.

Primitive and non-primitive un

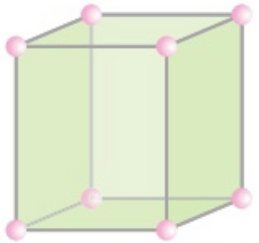
There are two types of unit c
primitive. A unit cell that conta
is called a primitive unit cell, wh
lattice points at each of the c
present translational symme

In case of non-primitive unit c
lattice points, either on a face
the unit cell.

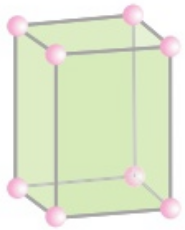


cubic	primitive	body centered	face centered	
tetragonal	primitive	body centered		
hexagonal				
orthorhombic	primitive	body centered	face centered	basis face centered
monoclinic	primitive		basis face centered	
trigonal				
triclinic				

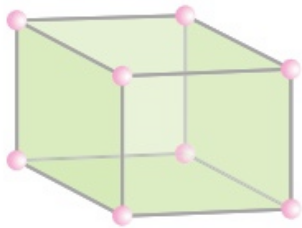
Table 6.2 14 Bravais Lattices



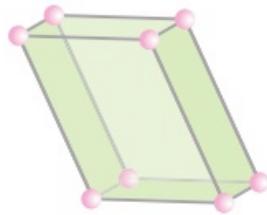
Hexagonal
$a = b \neq c$
$\alpha = \beta = 90°, \gamma = 120°$



Tetragonal
$a = b \neq c$
$\alpha = \beta = \gamma = 90°$



Monoclinic
$a \neq b \neq c$
$\alpha = \gamma = 90°, \beta \neq 90°$



Triclinic
$a \neq b \neq c$
$\alpha \neq \beta \neq \gamma \neq 90°$

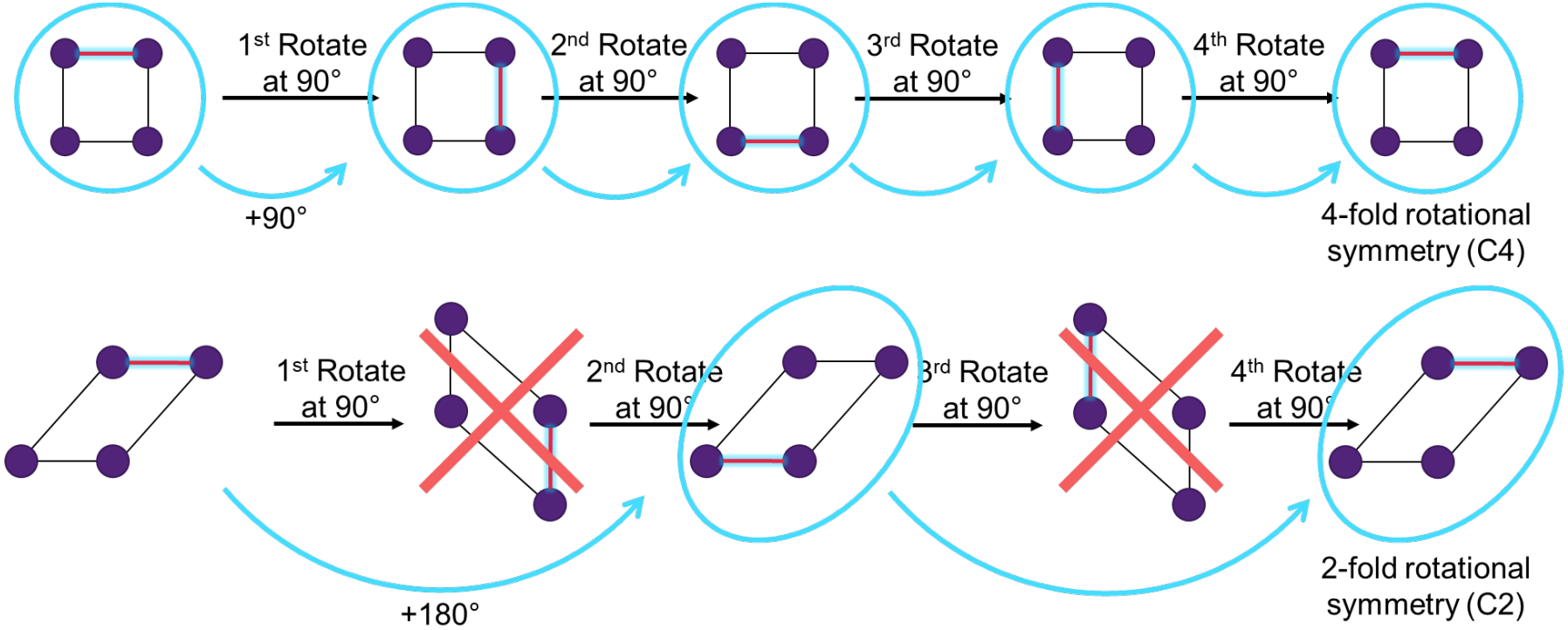
ɔic, tetragonal, orthorhombic,
They differ in the arrangement
ɔonding to the above seven,
nown in the figure.

Therefore, the basic rules of a primitive unit cells of 2D lattices,

Rule #1: **Unit cells should cover up the entire lattice by simple translation (*i.e.,* translational symmetry).**

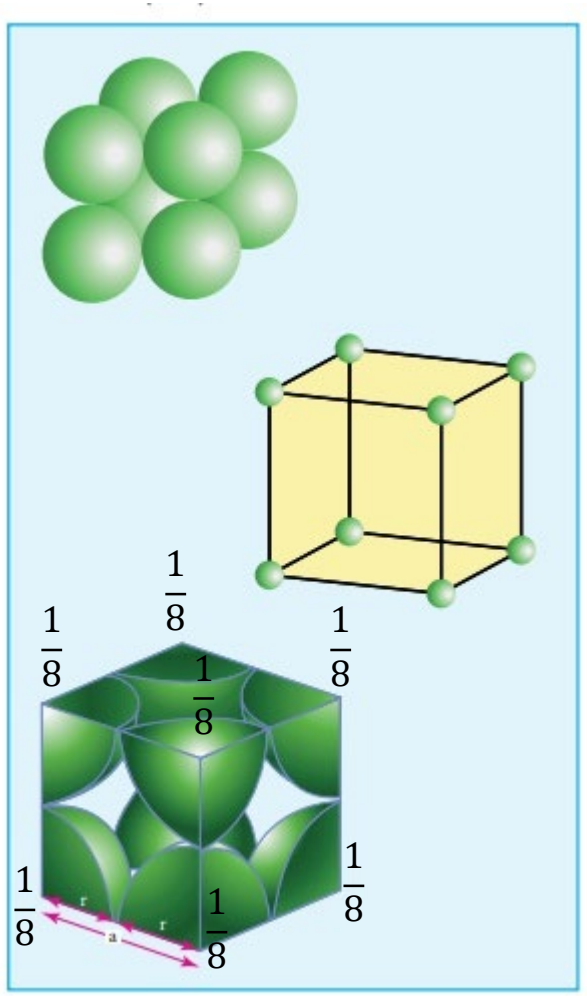
Rule #2: **Primitive unit cells should have lattice points only at the corners**

Rule #3: **Primitive unit cells best describe the symmetry of overall lattice (i.e., smallest rotational angle for symmetry).**



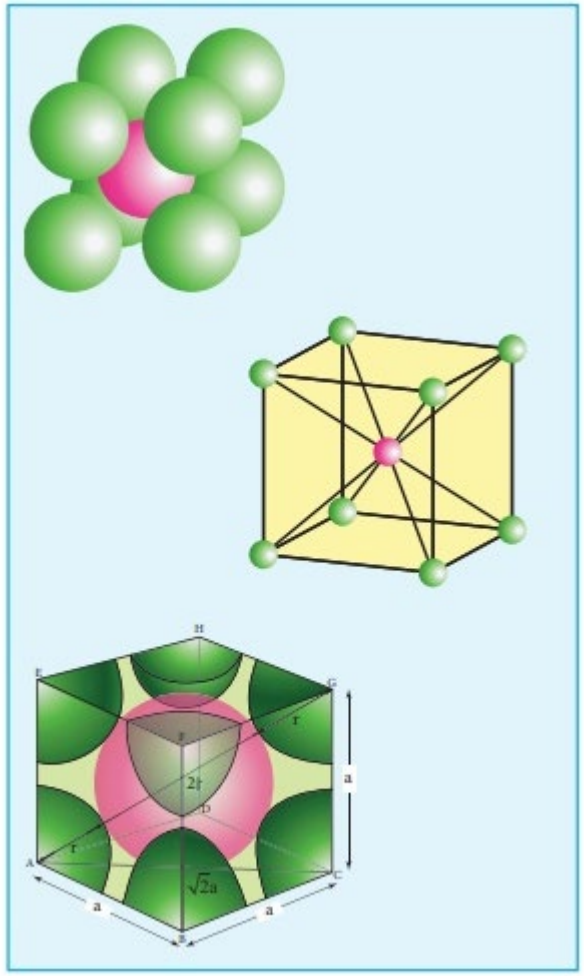
Number of atoms in a unit cell:

Simple Cubic (SC)



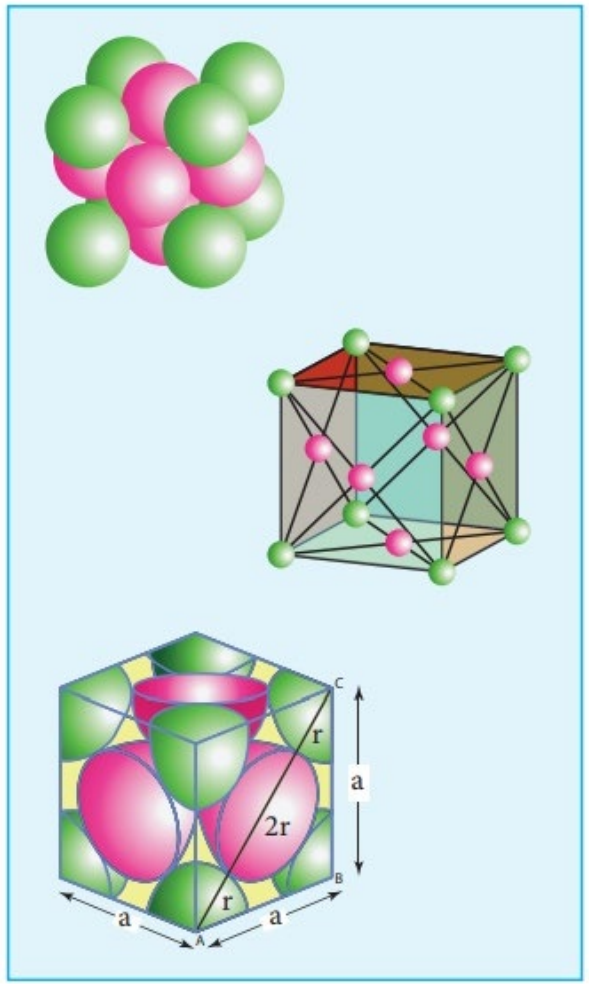
$N_a = 1$

Body Centred Cubic (BCC)



QUESTION: $N_a = ?$

Face Centred Cubic (FCC)



$N_a = ?$

Crystal directions

Directions in lattices are determined relative to the crystal axes defined by the unit vectors of the unit cell.

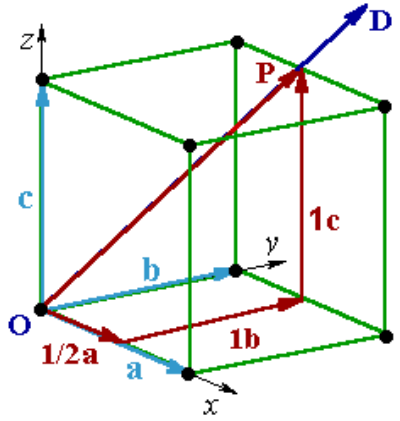
Indexing of crystal directions

A particular direction index is expressed using square brackets [ ]
Directions in crystals are vectors which may be expressed in terms of the unit vectors **a**, **b**, and **c**:

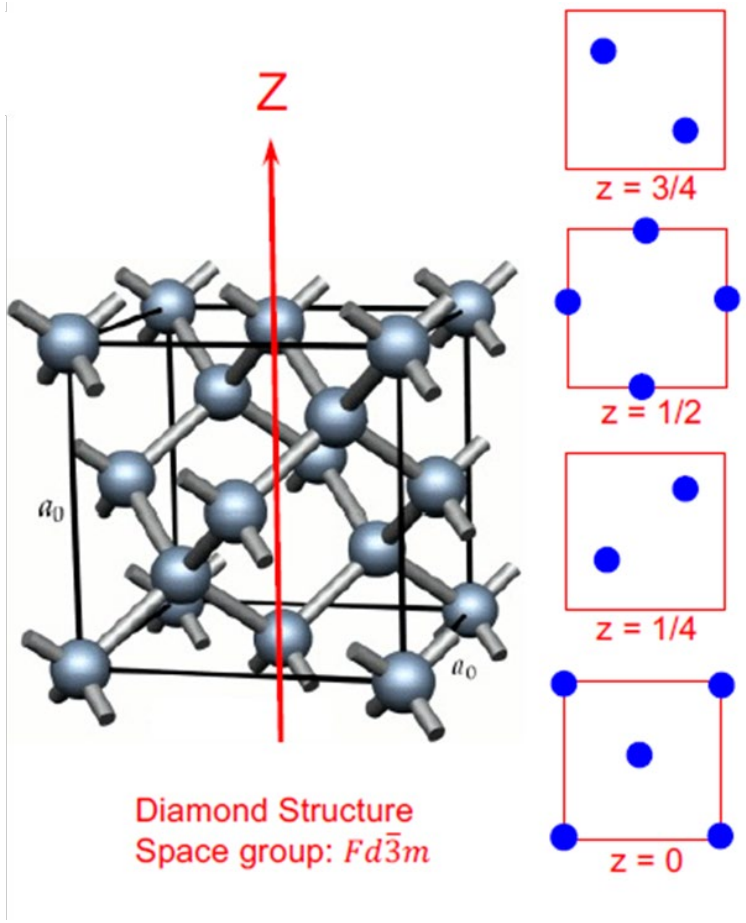
$$\mathbf{r}_{uvw} = u\,\mathbf{a} + v\,\mathbf{b} + w\,\mathbf{c}$$

In general, a **direction index** (or direction symbol, or **zone axis**) is given as **[u v w]** where u, v and w are *integers*.

If negative integers are required the notation is [-u -v -w], read as bar u, bar v, and bar w.



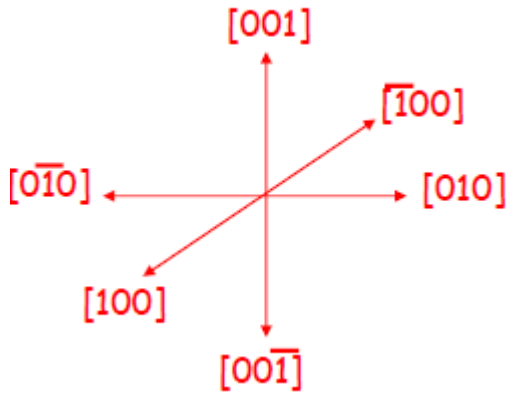
In vector notation:	Direction of
OP = 1/2a + 1b + 1c	OD is [122]

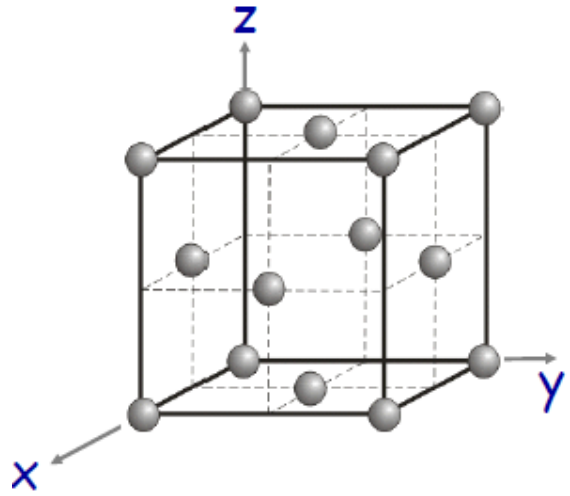


Equivalent crystal directions

- Crystal directions [ ] and their families < >,

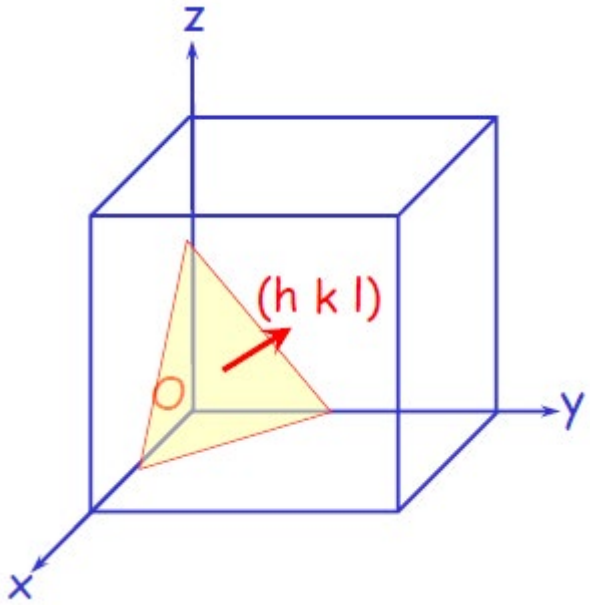
  as well as their determination.





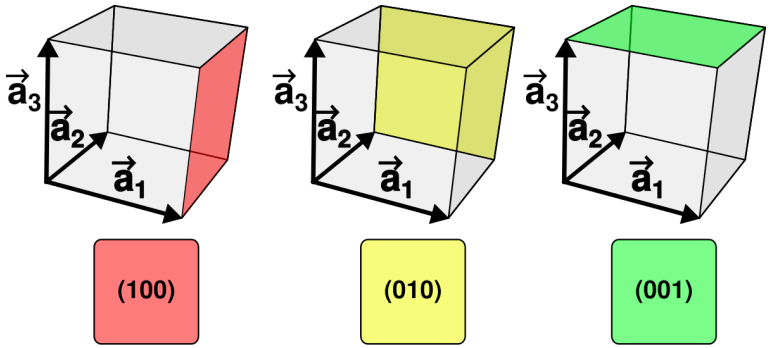
Crystal planes via Miller indices

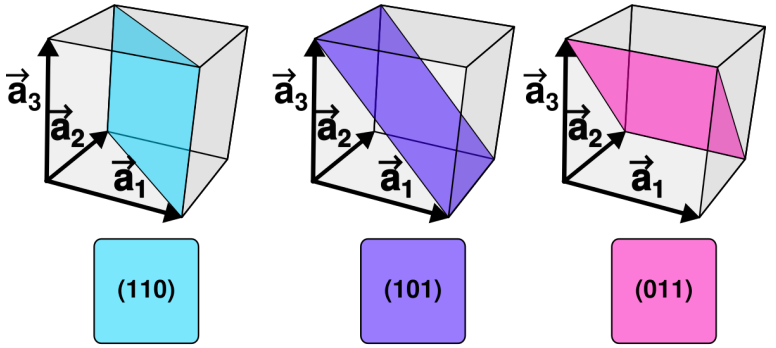
- Since crystal planes are two dimensional objects, the normal to a plane is used to define that plane.
- Miller indices are used to defined the crystal planes.
- A set of crystal planes form an infinite regularly spaced atomic planes, which collectively pass through all the points in a lattice.
- A set of crystal planes (parallel and equally spaced) has the same Miller index.
- Miller index ( ) of crystal planes and their families { }, as well as their determination.

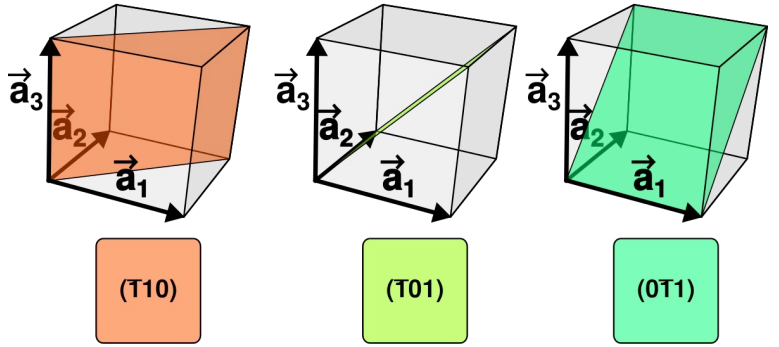


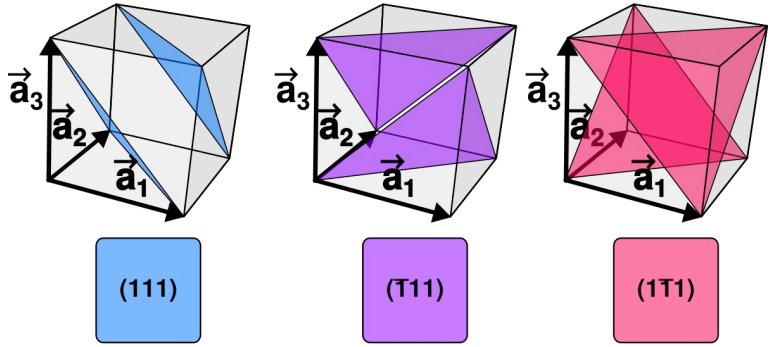
Miller indexing of crystal planes:

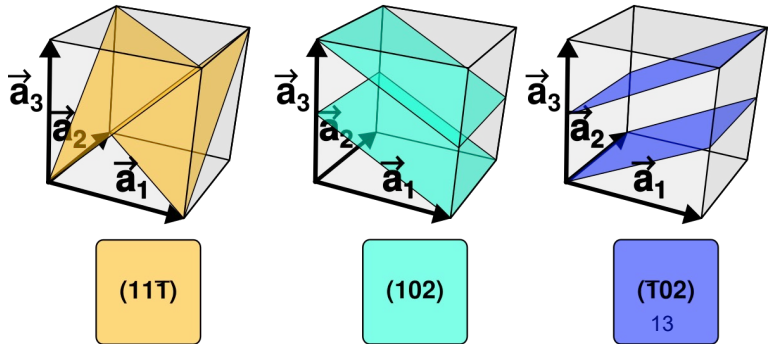
- A particular Miller index is expressed using round brackets ( ).
- A Miller index for a set of planes is expressed as (h k l), where h, k,& l are integers.
- If a crystal plane is parallel to any axis, its intercept on that axis is at infinity, the index for this axis is zero.
- If negative integers are required the notation is $(\bar{h}\ \bar{k}\ \bar{l})$, read as bar h, bar k, & bar l.







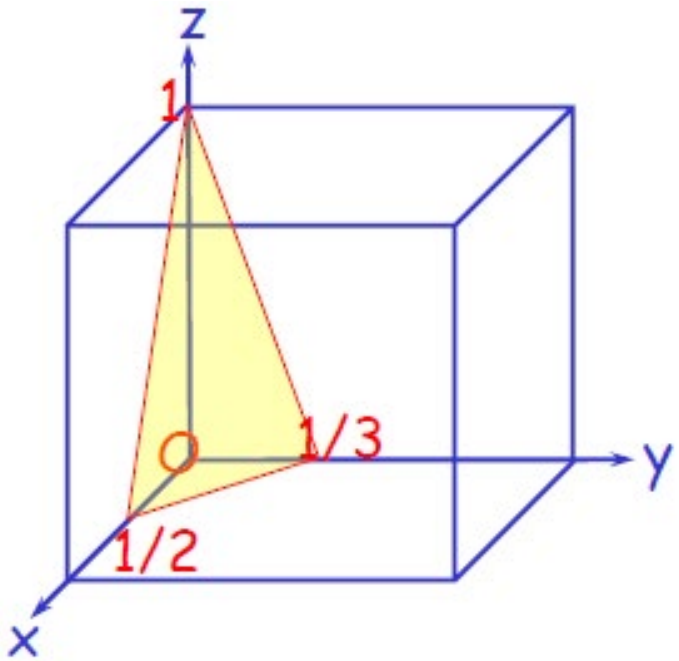


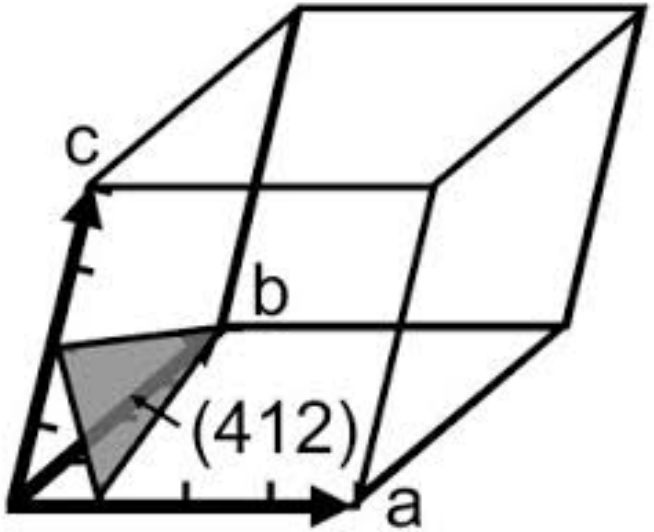


http://calistry.org/calculate/latticePlanesMillerIndices

Determination of Miller index

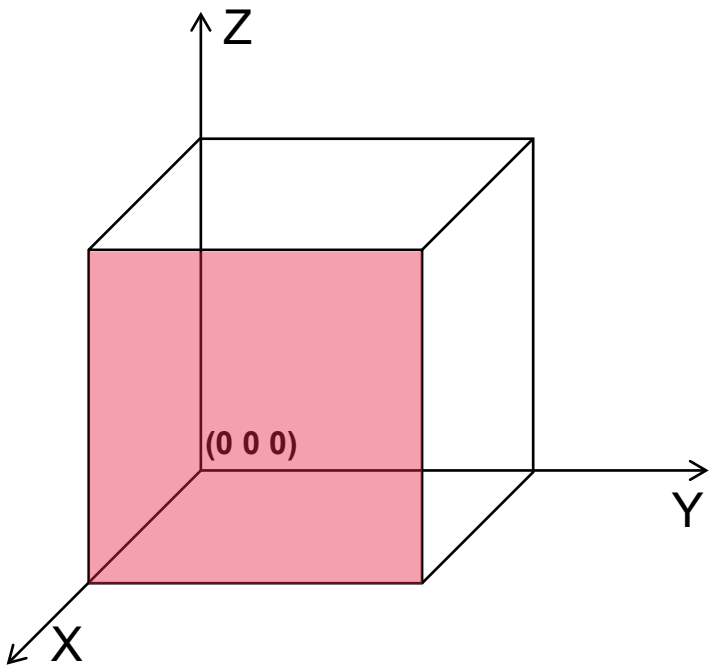
- Determine a set of parallel and equally spaced crystal planes to be indexed.
- Randomly choice the origin in one of these planes;
- Determine the intercepts (1/h)a, (1/k)b, (1/l)c on the axes for the first plane away from the origin.
- For example, (1/2)a, (1/3)b, (1/1)c. So, the plane is (2 3 1).
- If the plane is parallel to an axis, the intercept is ∞, the index for this axis is zero.





The plane which passes through crystal lattice and intersects lattice points is called “Lattice plane”.

Miller indices are set of three numbers used to specify lattice plane.



Miller indices of this plane is (1 0 0)

Step 1: Identify intercepts made by the plane in X, Y, Z axis, respectively.

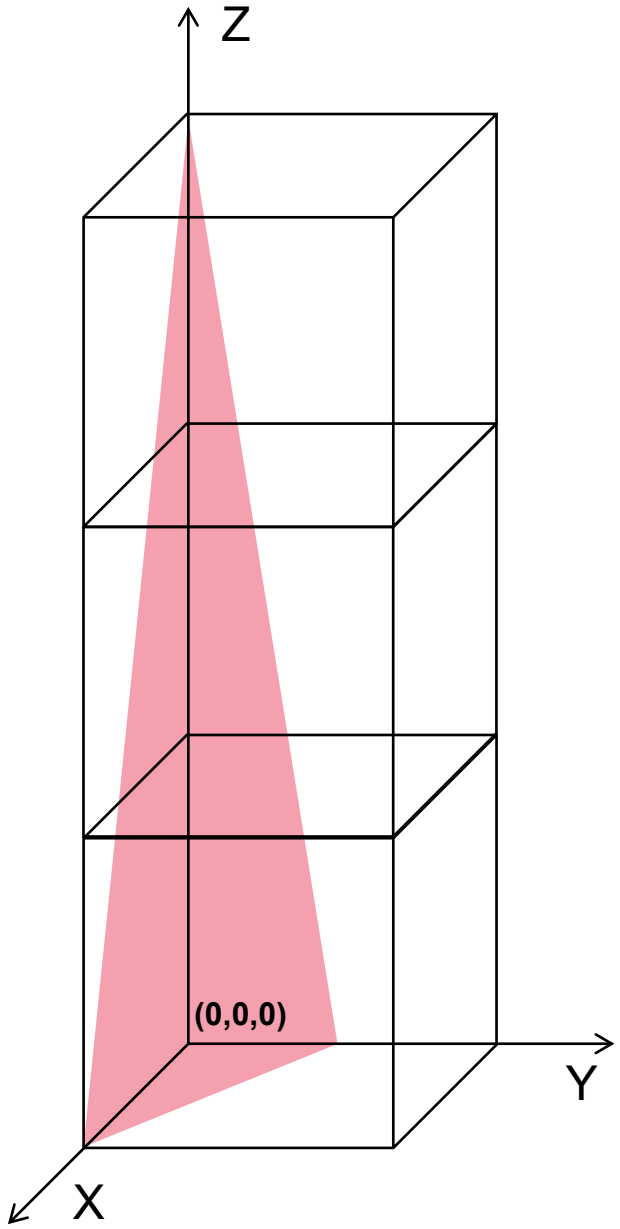
X intercept = 1
Y intercept = ∞
Z intercept = ∞

Step 2: Take reciprocal of each intercept.

X intercept = 1/1 = 1
Y intercept = 1/∞ = 0
Z intercept = 1/∞ = 0

Rules to denote Miller indices:

1. There is **no commas between numbers**.
2. The number of **Miller indices cannot be in the form of fraction**.
   e.g., (0.66, 1, 0) or (2/3 1 0) the correct way is (2 3 0).



Step 1: Identify intercepts made by the plane in X, Y, Z axis, respectively.

X intercept = 1
Y intercept = 1/2
Z intercept = 3

Step 2: Take reciprocal of each intercept.

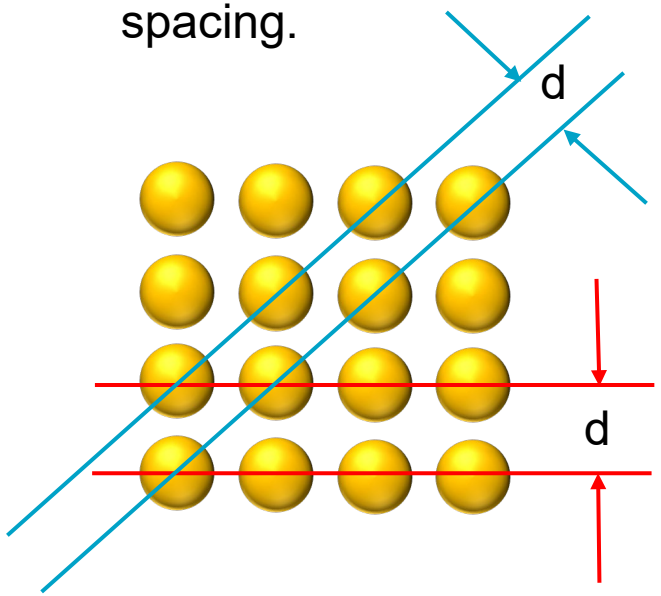
X intercept = 1/1 = 1
Y intercept = 1/0.5 = 2
Z intercept = 1/3

Miller indices of this plane is **(1 2 $^1/_3$)??**

No, the number cannot be in the form of fraction.
So by simply multiplying all numbers by 3, we can get Miller indices of this plane as (3 6 1).

d-spacing of crystal planes

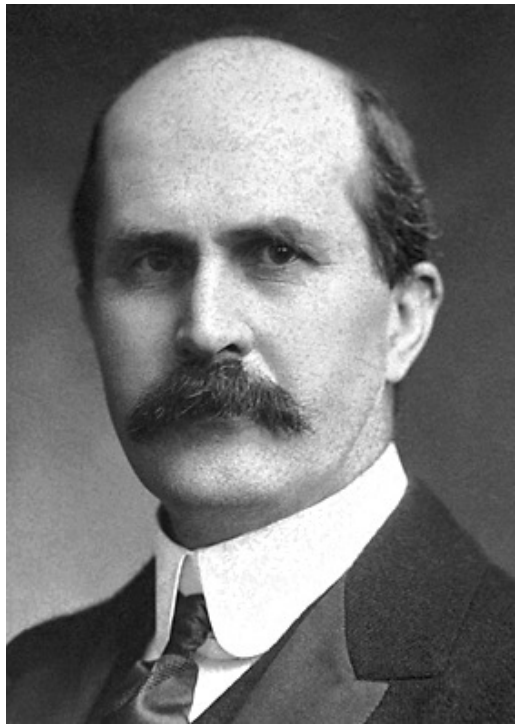
- The **spacing** of a set of planes (hkl), is the perpendicular distance between adjacent planes.
- The spacing is a function of (hkl).
- Generally, the higher the indices the smaller the spacing.

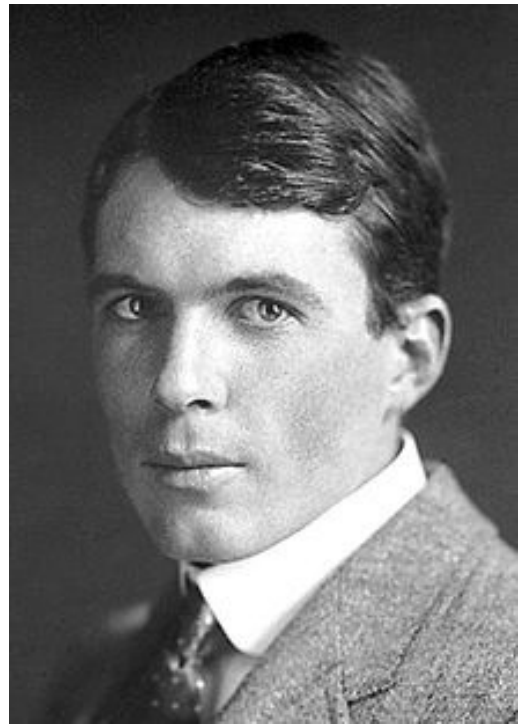


Crystal System	Axial Translations	Axial Angles	Spacing, expressed as: $\frac{1}{d^2}$
Cubic P, F, I	a = b = c	$\alpha = \beta = \gamma = 90°$	$\frac{h^2 + k^2 + \ell^2}{a^2}$
Tetragonal P, I	a = b ≠ c	$\alpha = \beta = \gamma = 90°$	$\frac{h^2 + k^2}{a^2} + \frac{\ell^2}{c^2}$
Hexagonal P	a = b ≠ c	$\alpha = \beta = 90°, \gamma = 120°$	$\frac{4}{3} \cdot \left(\frac{h^2 + hk + k^2}{a^2}\right) + \frac{\ell^2}{c^2}$
Orthorhombic P, I, C, F	a ≠ b ≠ c	$\alpha = \beta = \gamma = 90°$	$\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{\ell^2}{c^2}$
Rhombohedral P	a = b = c	$\alpha = \beta = \gamma \neq 90°$	$\frac{\left(h^2 + k^2 + \ell^2\right)\sin^2\alpha + 2 \cdot \left(hk + k\ell + h\ell\right)\cos^2\alpha - \cos\alpha}{a^2 \cdot \left(1 - 3\cos^2\alpha + 2\cos^3\alpha\right)}$
Monoclinic P, C	a ≠ b ≠ c	$\alpha = \gamma = 90°, \beta > 90°$	$\frac{1}{\sin^2\beta} \cdot \left(\frac{h^2}{a^2} + \frac{k^2 \sin^2\beta}{b^2} + \frac{\ell^2}{c^2} - \frac{2h\ell \cdot \cos\beta}{ac}\right)$
Triclinic P	a ≠ b ≠ c	$\alpha \neq \beta \neq \gamma \neq 90°$	$\frac{1}{V^2} \cdot \left(S_{11} \cdot h^2 + S_{22} \cdot k^2 + S_{33} \cdot \ell^2 + 2S_{12} \cdot hk + 2S_{23} \cdot k\ell + 2S_{13} \cdot h\ell\right)$ V = volume of unit cell $S_{11} = b^2c^2 \sin^2\alpha$ $S_{22} = a^2c^2 \sin^2\beta$ $S_{33} = a^2b^2 \sin^2\gamma$ $S_{12} = abc^2 \cdot \left(\cos\alpha\cos\beta - \cos\gamma\right)$ $S_{23} = a^2bc \cdot \left(\cos\beta\cos\gamma - \cos\alpha\right)$ $S_{13} = ab^2c \cdot \left(\cos\gamma\cos\alpha - \cos\beta\right)$

Pioneers of diffractions

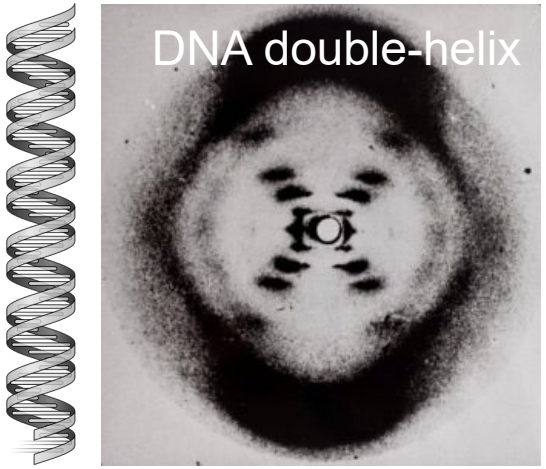
Max von Laue (1879-1960) won the Noble Prize in Physics, 1914 *for his discovery of the diffraction of X-ray by crystals*!

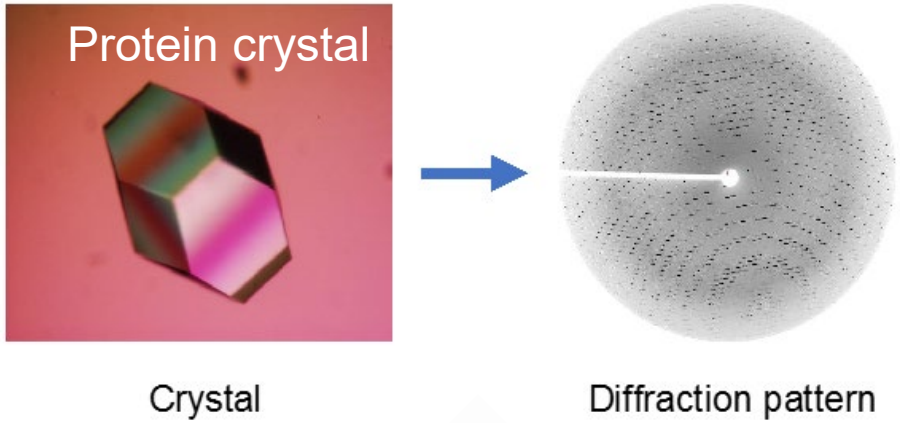


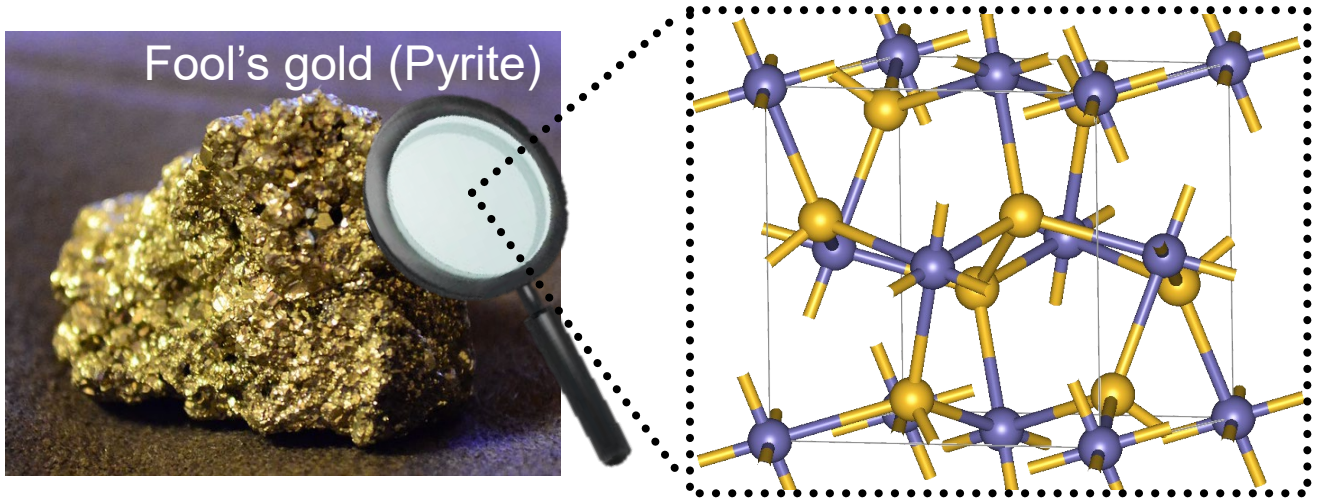


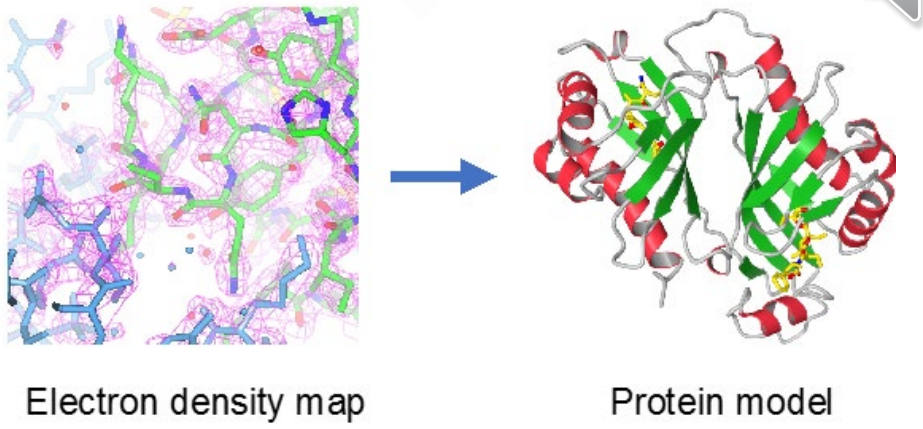
William Henry Bragg (1862-1942) and his Australian born son William Lawrence Bragg (1890-1971) won the Noble Prize in Physics, 1915 *for their services in the analysis of crystal structure by means of X-ray*.

The finding of X-ray revolutionized various fields including
Medical imaging, genetics, proteomics, materials science, manufacturing…



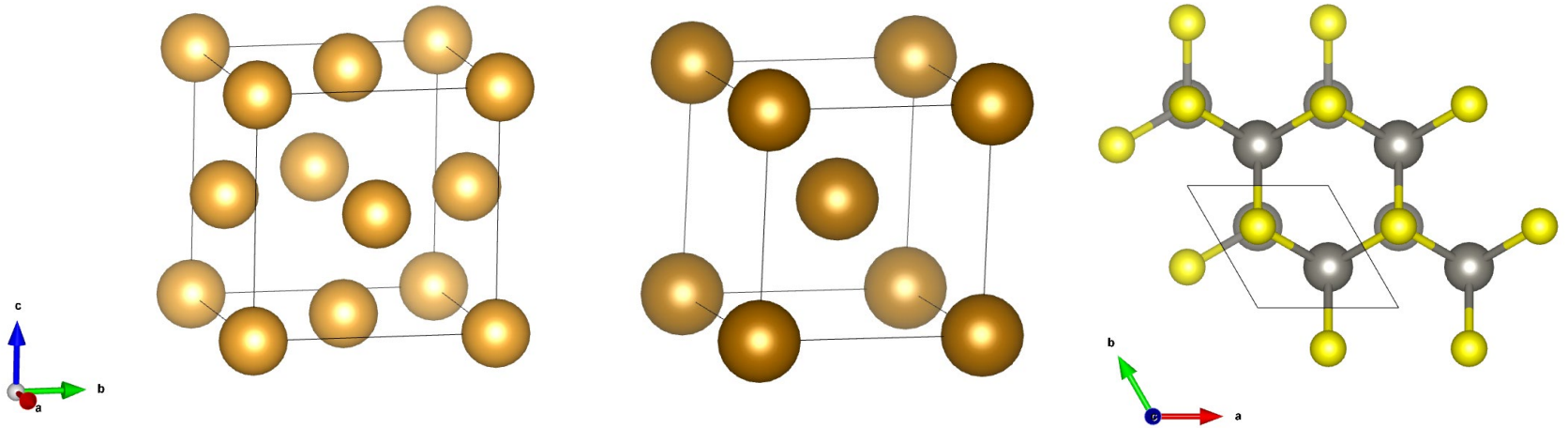






What is XRD?

X-ray diffraction (XRD) is a technique by irradiating a material with incident X-rays and measuring the intensities and scattering angles of the X-rays that leave the material.



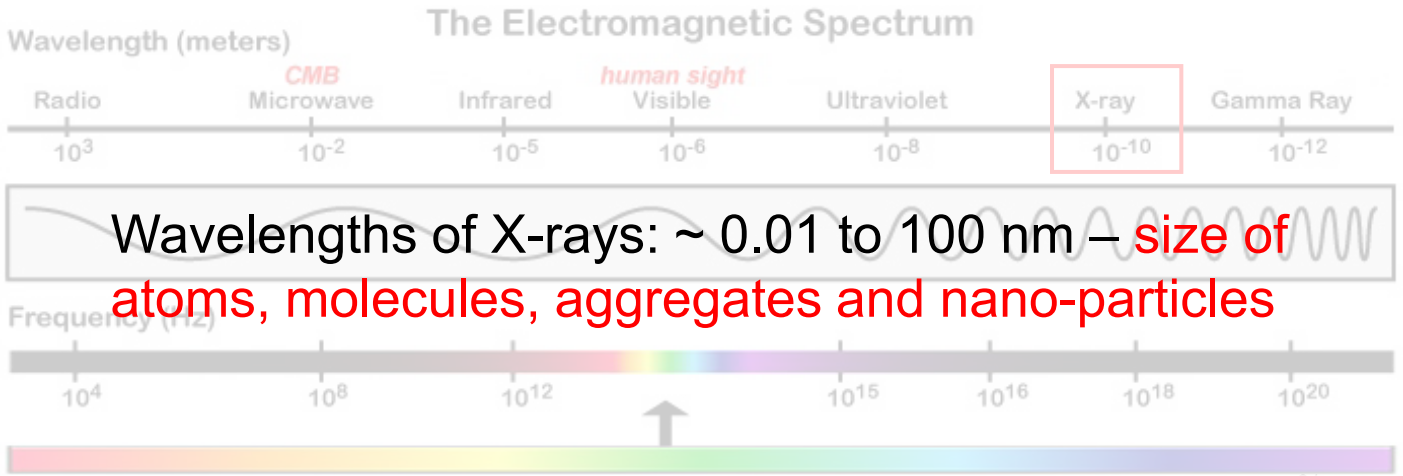
Risk Involved:

Radiation Safety

What material information can be obtained?

It is used to identify

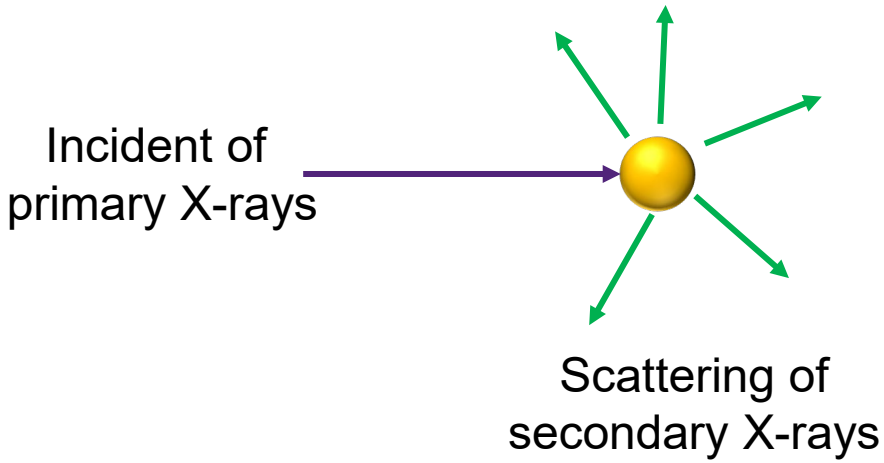
- Crystalline phases and orientation
- Determine structural properties:
  - Lattice parameters
  - Crystallite size
  - Phase composition
  - Strain



Wavelengths of X-rays: ~ 0.01 to 100 nm – size of atoms, molecules, aggregates and nano-particles

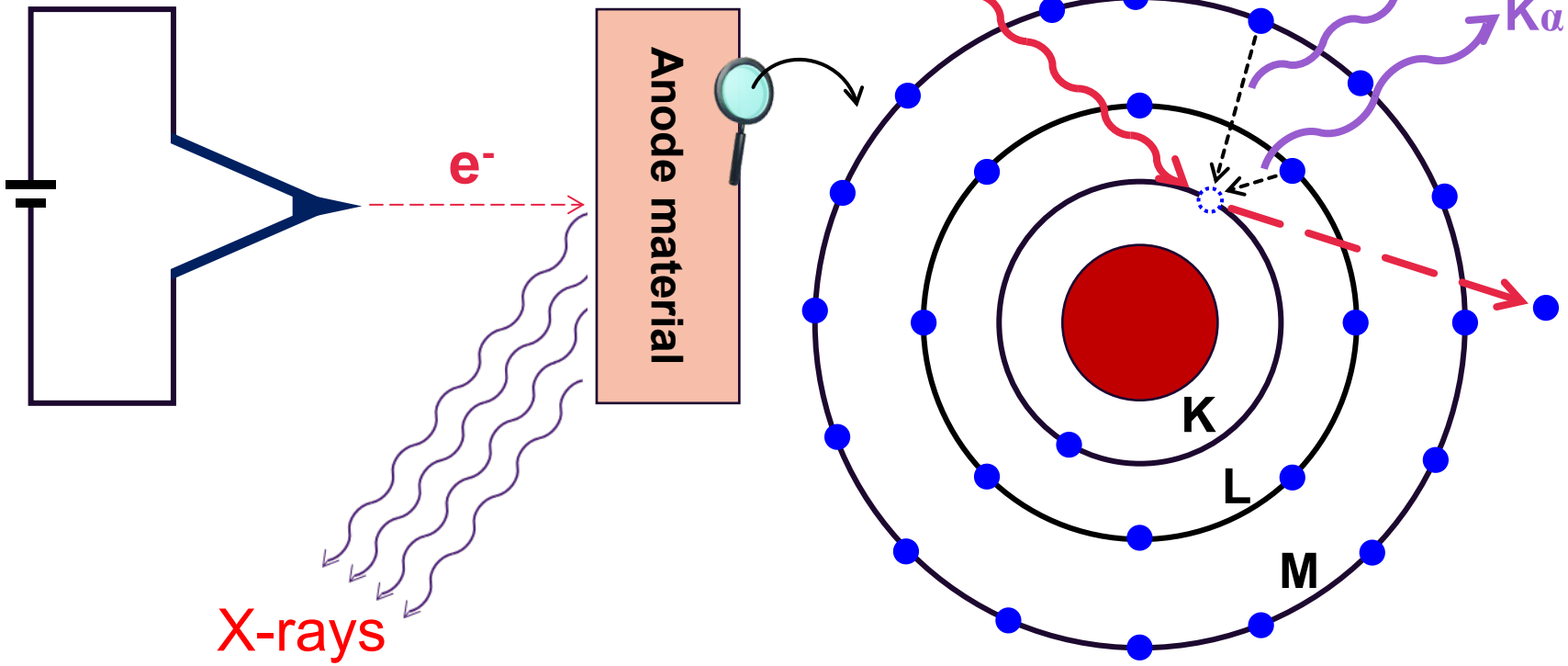
Scattering and diffraction

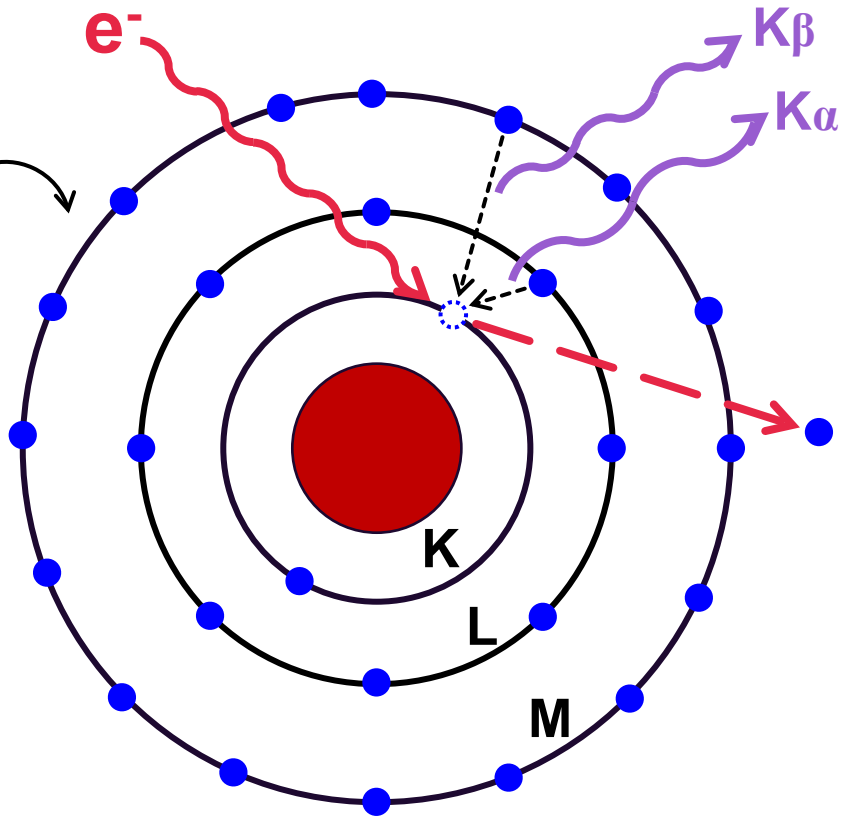
Diffraction is the **sum of scattering** events! When an incident beam (radiation) hits a particle, the wave interacts with a single particle, the particle scatters the incident beam uniformly in all directions.



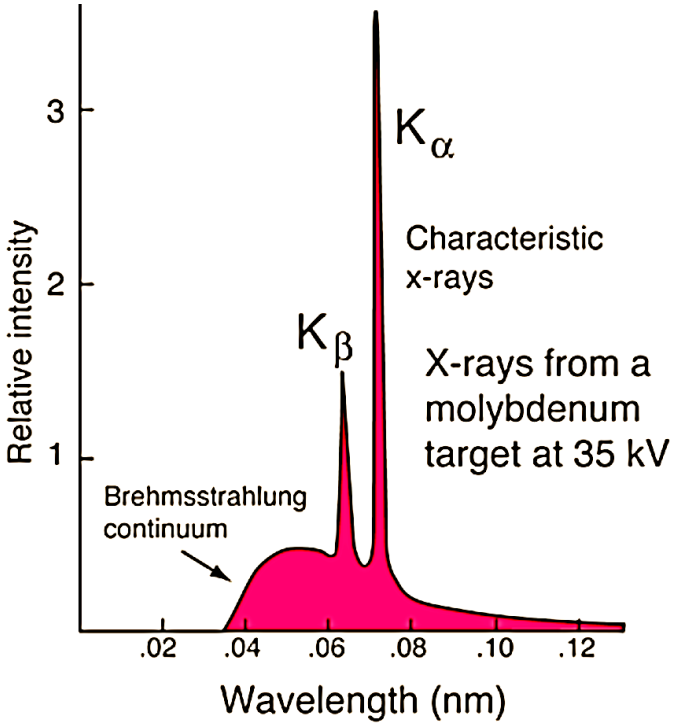
- Diffraction techniques exploit the scattering of radiation from large numbers of sites.
  - Scattering from atoms
  - Scattering from groups of atoms in crystals
- Diffraction is an interference effect which leads to scattering of strong
- beams of radiation in specific directions.
- There are various diffraction techniques: X-ray diffraction, Electron diffraction, Neutron diffraction

Generation of X-rays





Characteristic X-rays (Incident X-ray for XRD)



Continuous X-ray / Characteristic X-rays

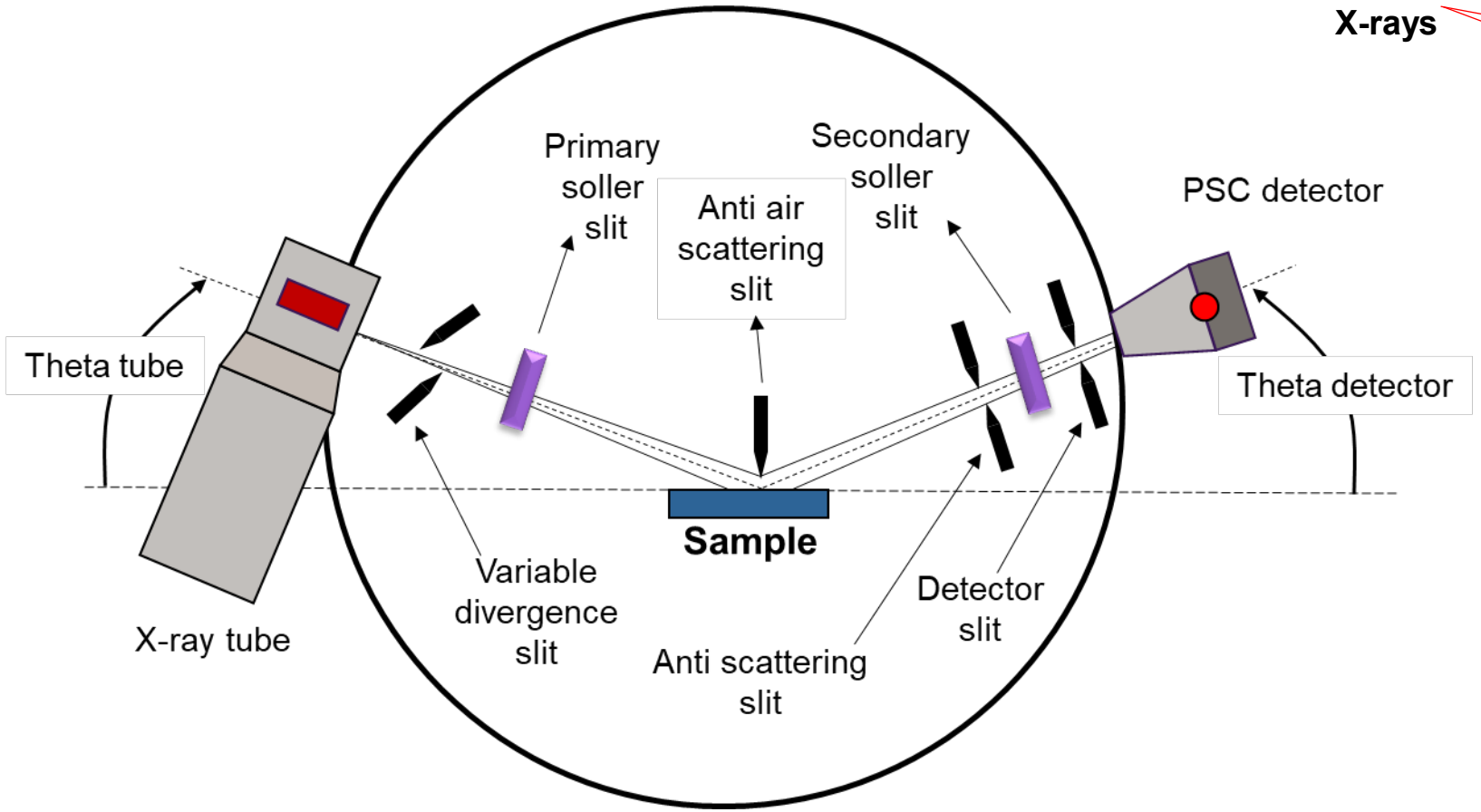
Continuous X-rays

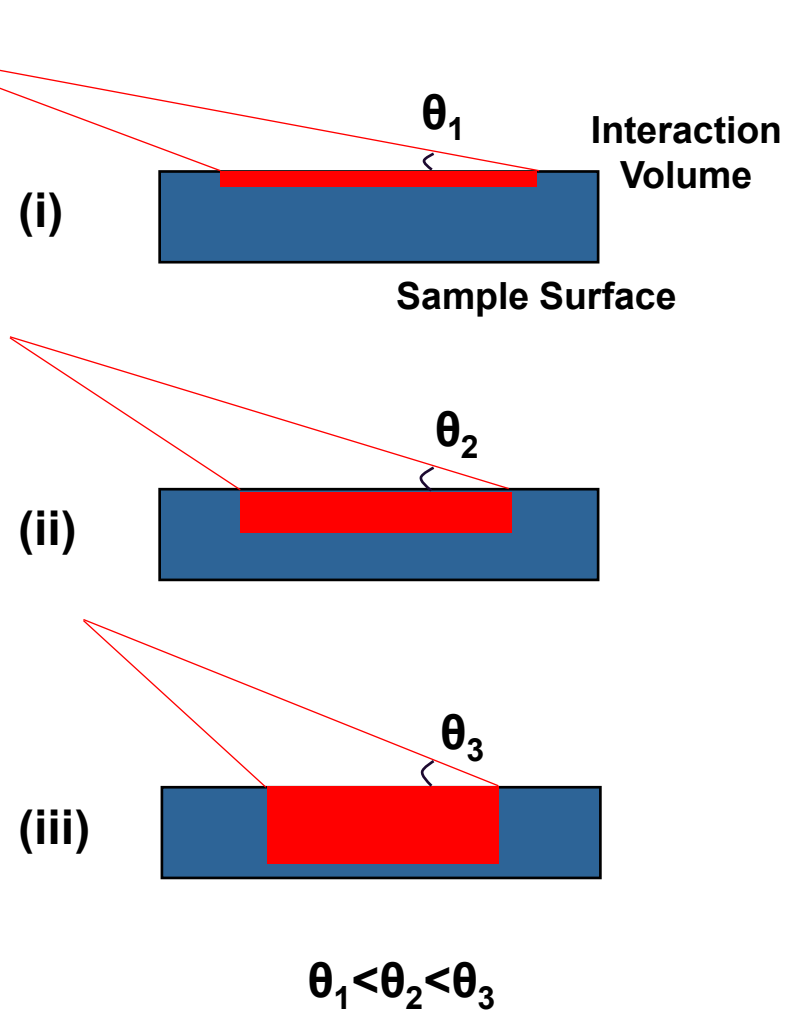
- radiation from braking

$$\lambda_{\min} = \frac{hc}{eV} = \frac{1.24}{V(kV)} nm$$

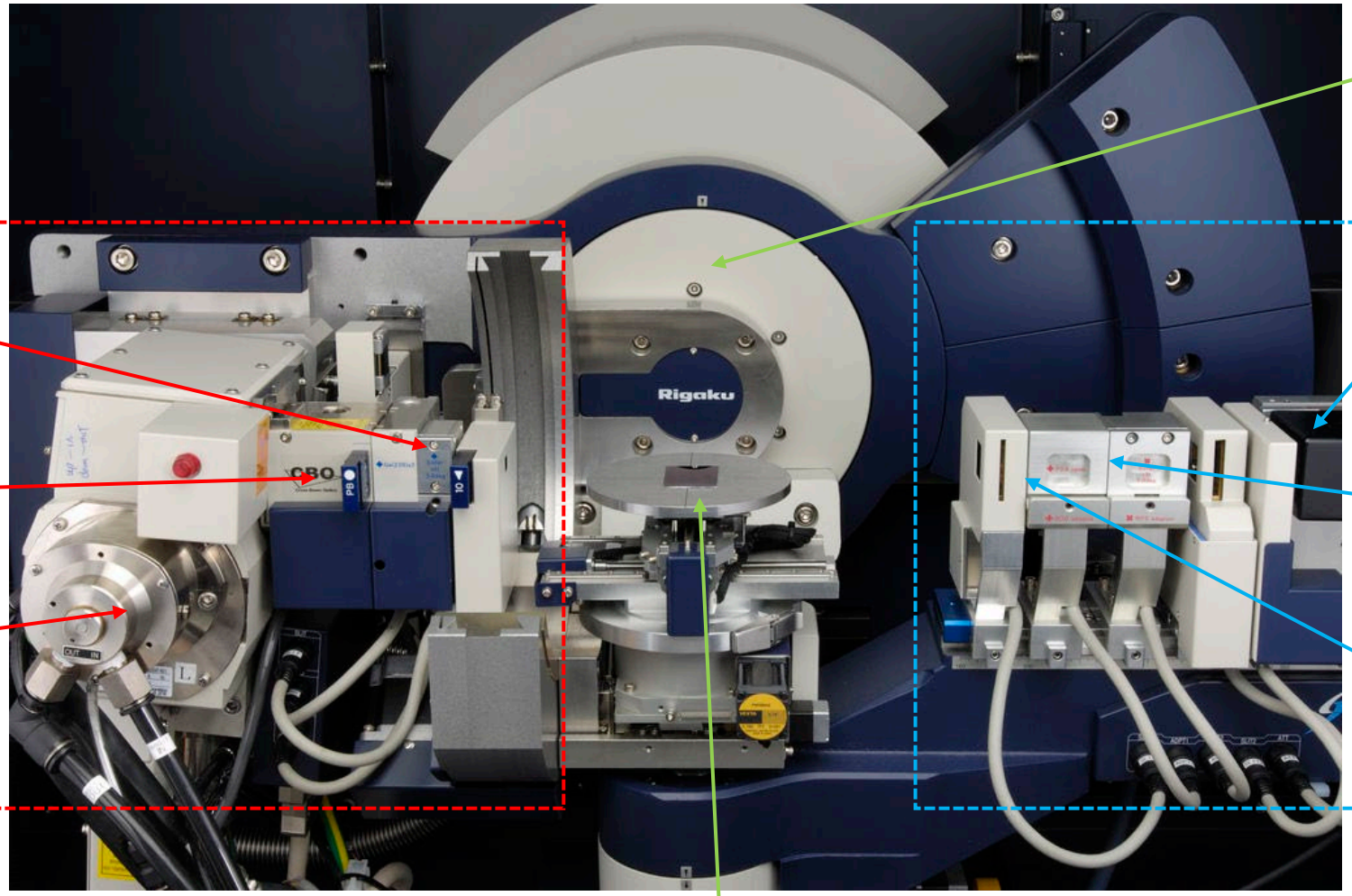
Target	Cr	Fe	Co	Ni	Cu	Mo	Ag
Atom #	24	26	27	28	29	42	47
Kα (Å)	2.291	2.937	1.790	1.659	1.541	0.711	0.561

Fundamentals of XRD

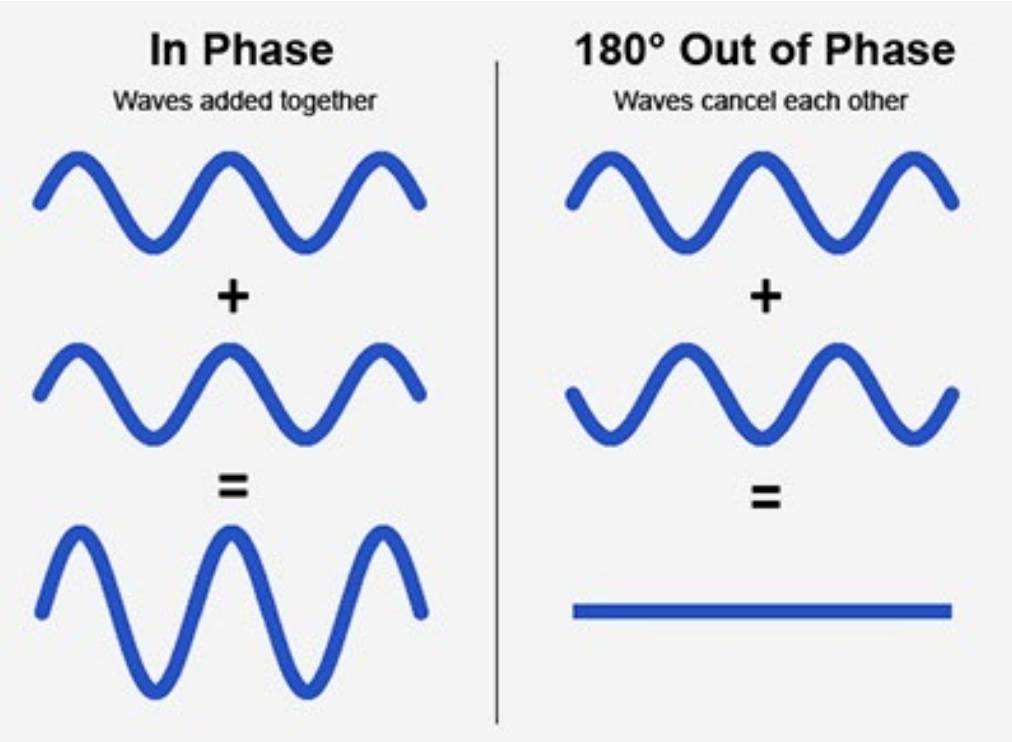


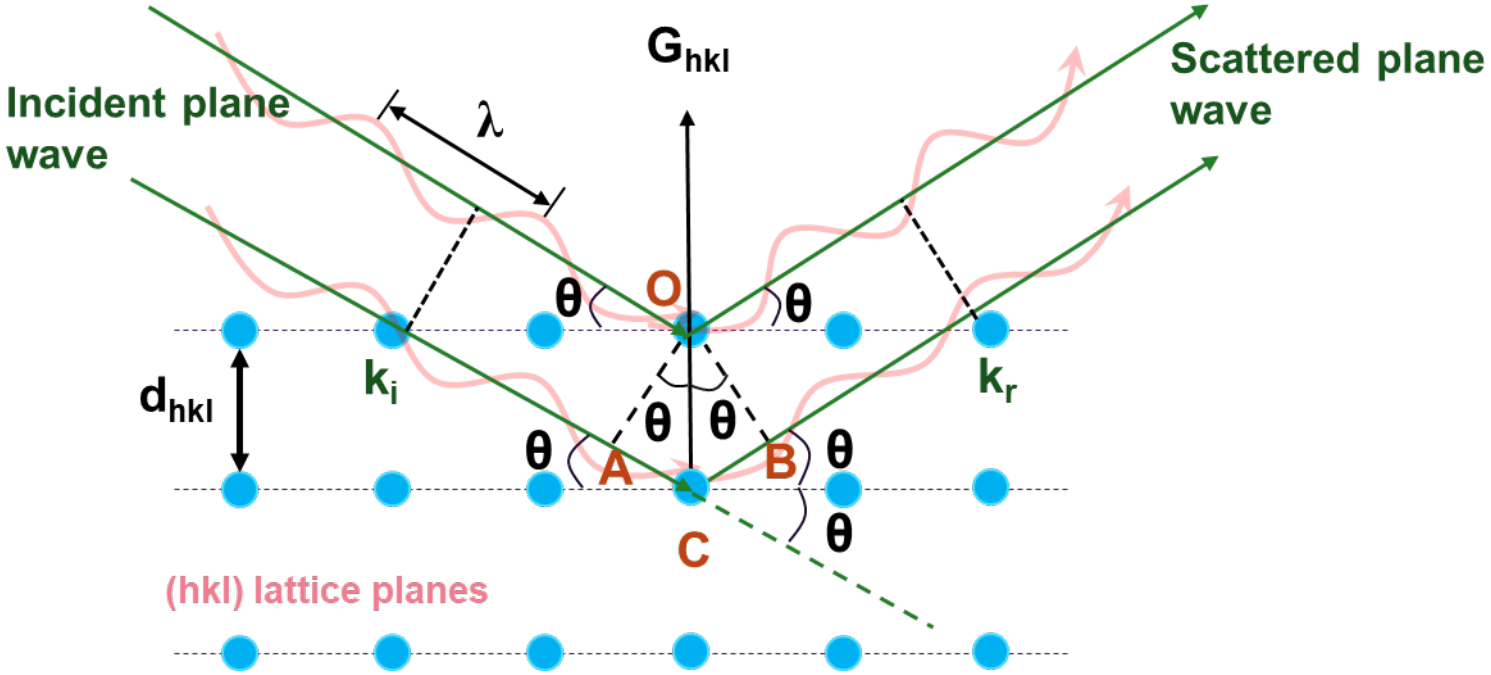


- Fundamental components in an X-ray diffractometer and its reflected angle (θ) dependent change in the X-ray interaction volume on sample surface.


Goniometer
SOURCE
DETECTOR
Divergence slit
Primary x-ray optics
X-ray tube
Detector
Soller slits
Diffraction beam collimator
Sample Stage
Rigaku

Operating Principle





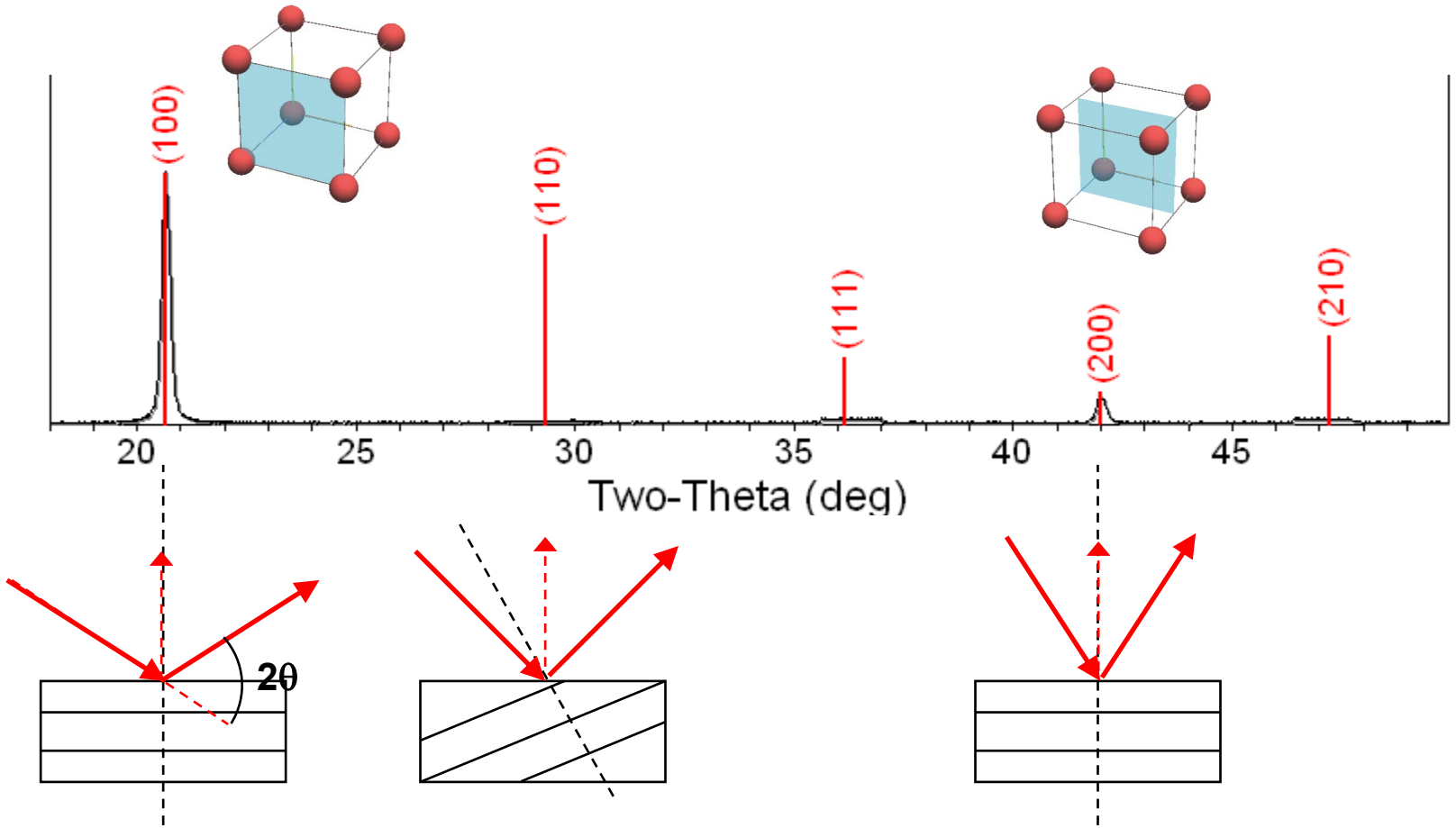
From ΔOCB,

BC = d sinθ ; where, d = OC (interplanar distance) and BC=AC

The total path difference between two x-rays that escape from the crystals is given by: *AC+BC = 2d sinθ*

A constructive interference that results in the bright band (also called Bragg's peak), is observed when the path difference between diffracted X-rays is equal to, *nλ = 2d sinθ.*

A single crystal specimen in a Bragg-Brentano diffractometer would produce only one family of peaks in the diffraction pattern.

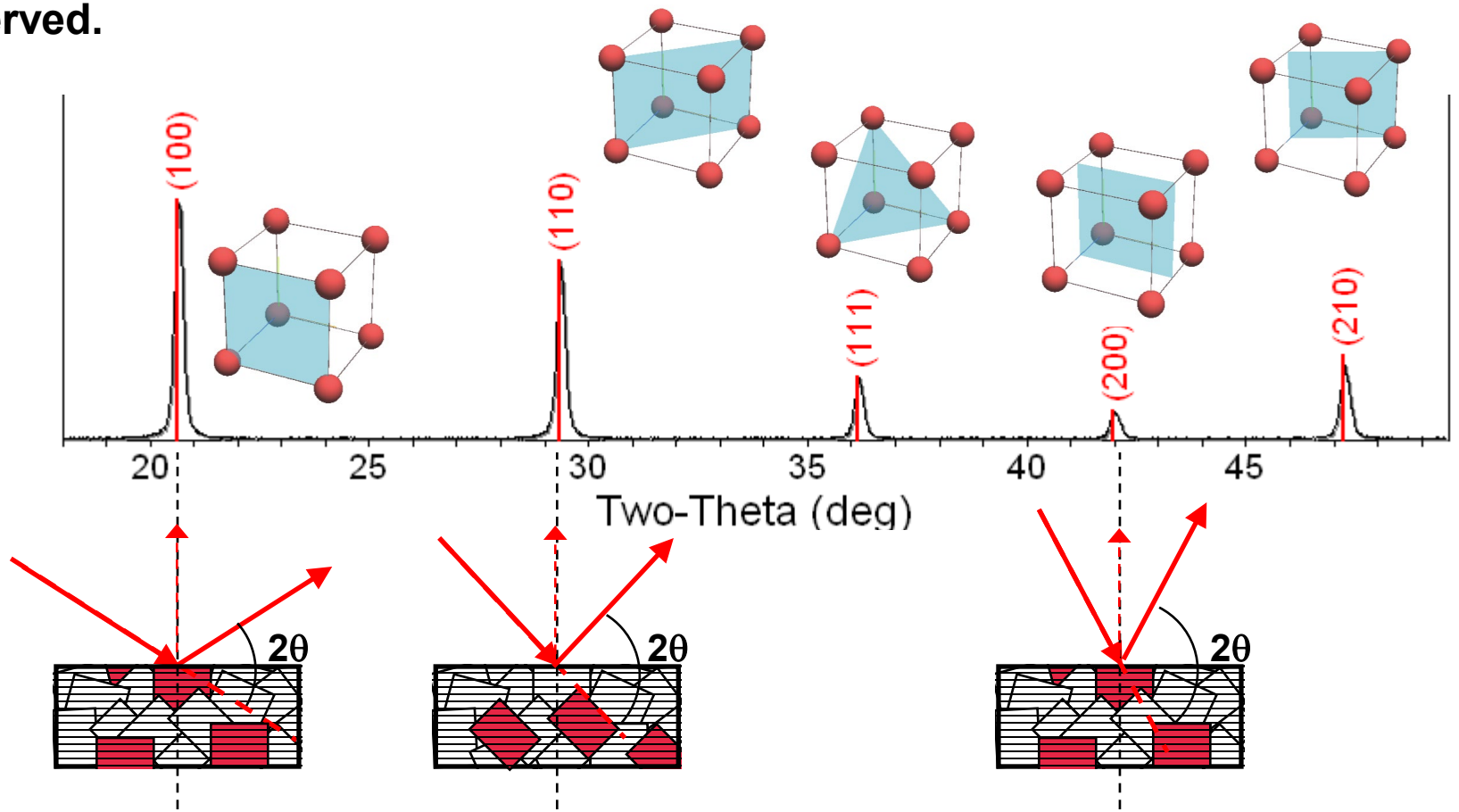


At 20.6 °2θ, Bragg's law fulfilled for the (100) planes, producing a diffraction peak.

The (110) planes would diffract at 29.3 °2θ; however, they are not properly aligned to produce a diffraction peak (the perpendicular to those planes does not bisect the incident and diffracted beams). Only background is observed.

The (200) planes are parallel to the (100) planes. Therefore, they also diffract for this crystal. Since d_{200} is ½ d_{100}, they appear at 42 °2θ.

A polycrystalline sample should contain thousands of crystallites. Therefore, all possible diffraction peaks should be observed.

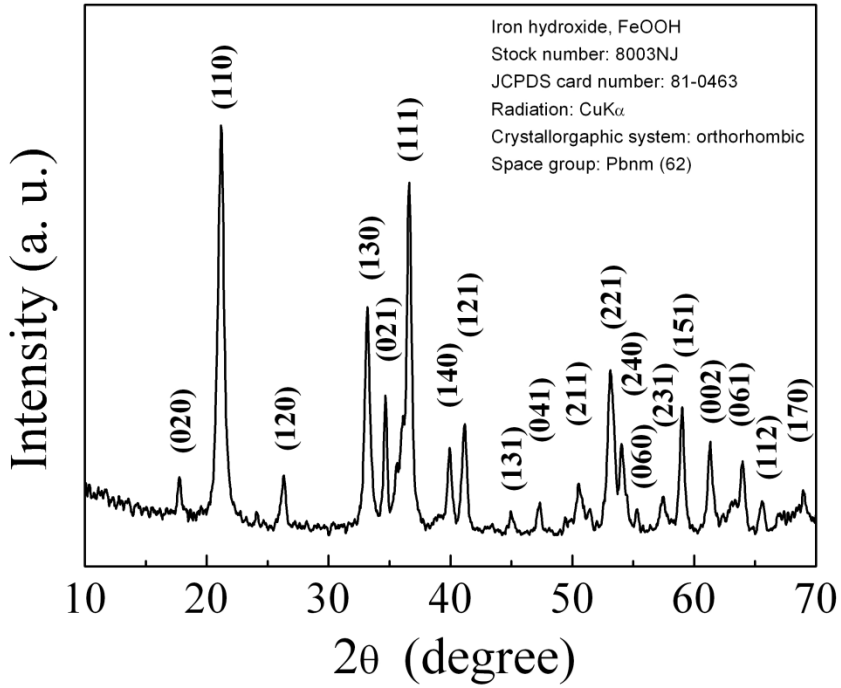


- For every set of planes, there will be a small percentage of crystallites that are properly oriented to diffract (the plane perpendicular bisects the incident and diffracted beams).
- Basic assumptions of powder diffraction are that for every set of planes there is an equal number of crystallites that will diffract and that there is a statistically relevant number of crystallites, not just one or two.

Example: Diffraction patterns from XRD

Each peak represents the solution to Bragg's law for known (*i.e.*, Cu kα) radiation wavelength (λ = 0.15406nm).

The unique relationship between such patterns and crystal structures provide a powerful tool for identification of the phase composition of powders and polycrystalline materials.



Relationship between diffraction peaks, miller indices and lattice spacings

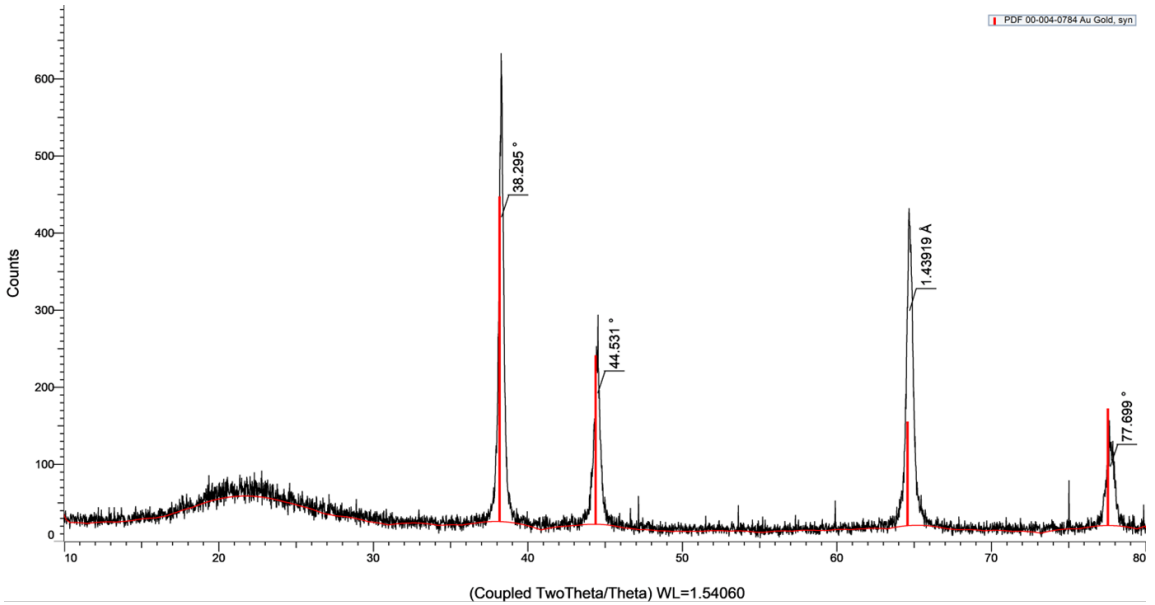
Simple cubic material **a** = 5.0 Å

hkl	d(Å)	2Θ
100	5.00	17.72
110	3.54	25.15
111	2.89	30.94

$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2}$$

Bragg Equation: $\mathbf{n\lambda = 2d\sin\theta}$

How many lattice planes are possible?
How many d-spacings? The number is large but finite.
$\mathbf{n\lambda = 2d\sin\theta}$ so if theta = 180, then d = λ/2. For Cu radiation that means that we can only see d-spacings down to 0.77 Å for Mo radiation, down to about 0.35 Å

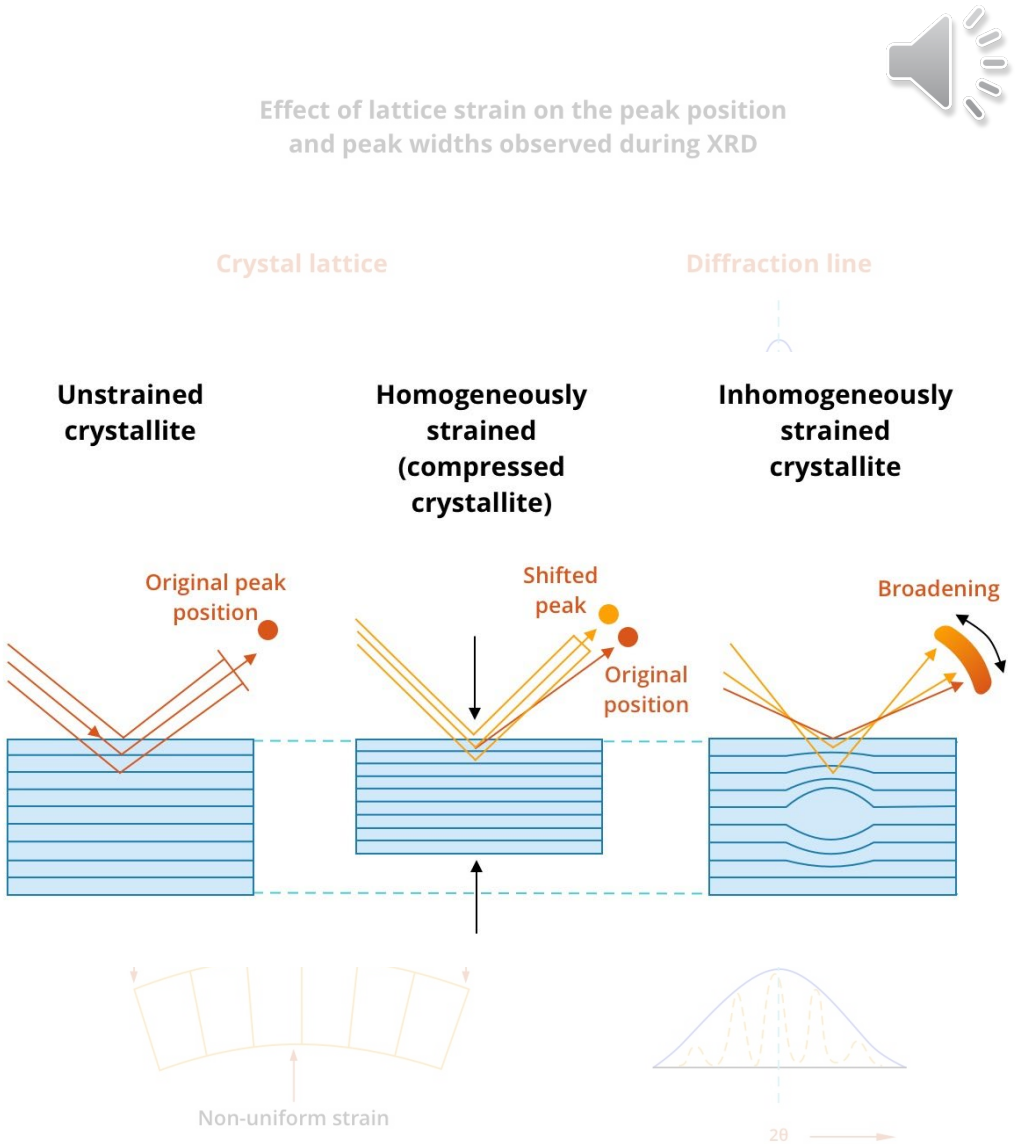


Phase identified (*PDF 00-004-0784, Au*)

$$\lambda = 2d_{hkl} \sin\theta_{hkl}$$

Note: Different anode gives different λ, so 2θ varies with the anode material.

Basis and Bravais Structure Factor Terms

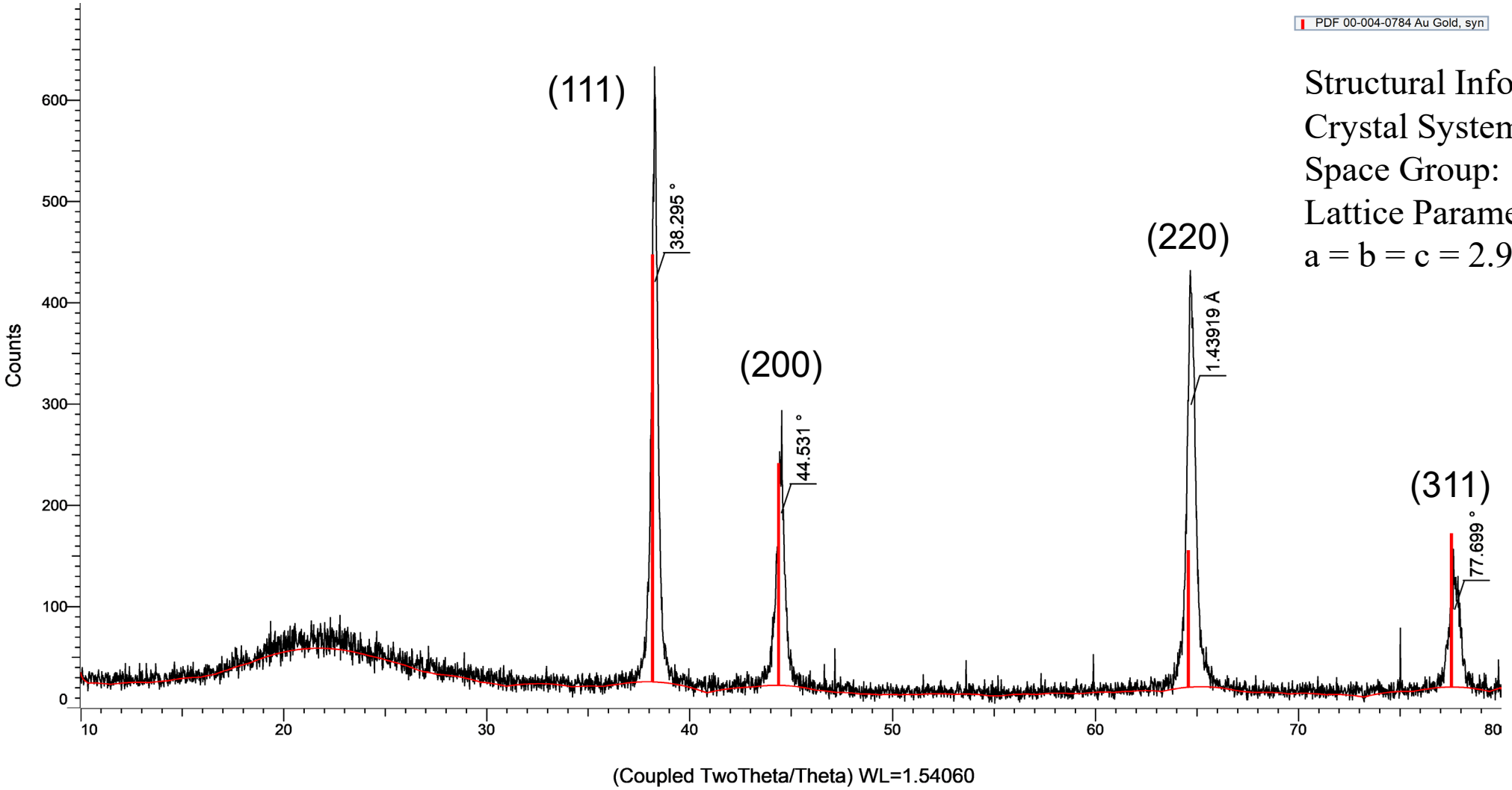


What information we can get from an XRD analysis?

- Lattice parameters
- Phase identity
- Phase purity
- Crystallinity
- Crystal structure
- Percent phase composition

What information we can not get from an XRD analysis?

- Elemental analysis -
  - How much lithium is in this sample?
  - Is there iron in this sample
  - What elements are in this sample
- Tell me what this sample is ????
  - Unless you know something about this sample, powder XRD won't have answers !!!

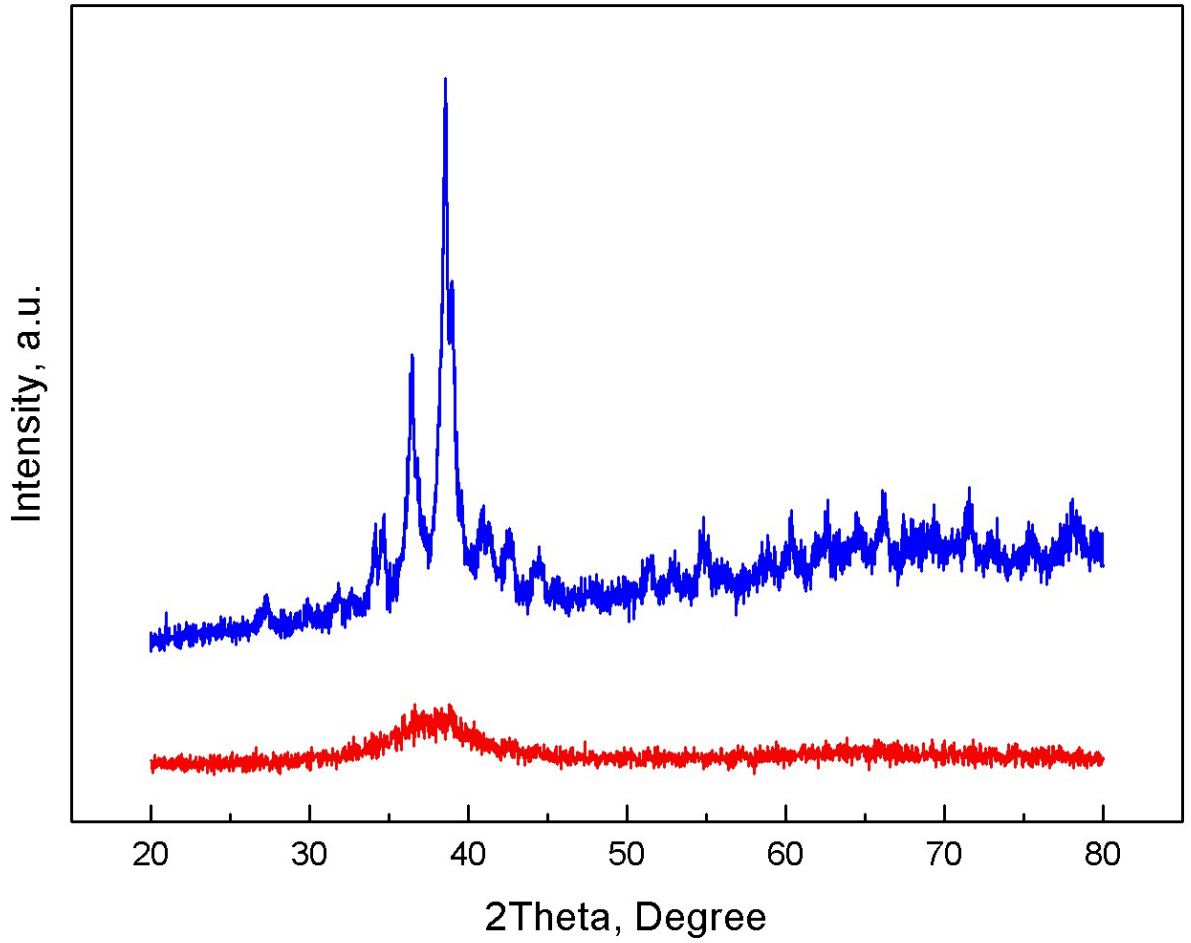


Structural Information: **Au**
Crystal System: Cubic
Space Group: $Fm\bar{3}m$
Lattice Parameters:
$a = b = c = 2.95Å$

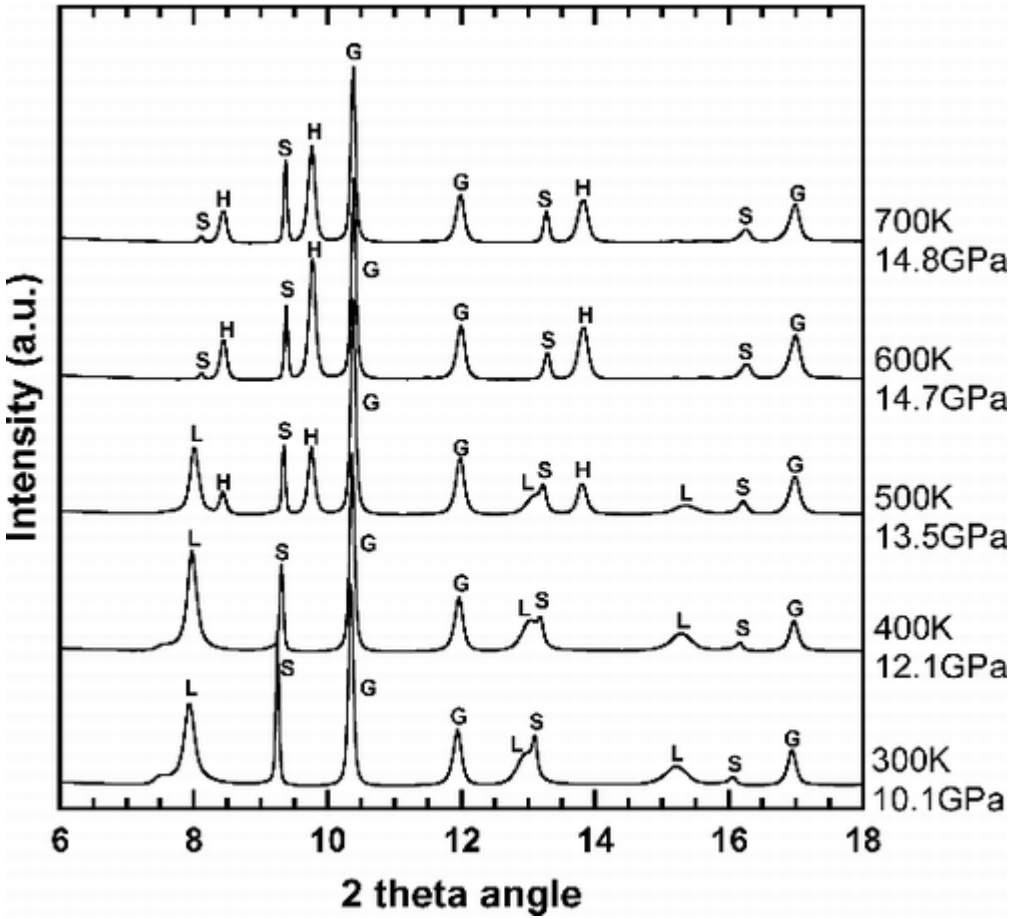
Phase identified (*PDF 00-004-0784, Au*)

Examples of X-ray diffraction

Monitoring crystallization from amorphous



Monitoring phase change during materials processing



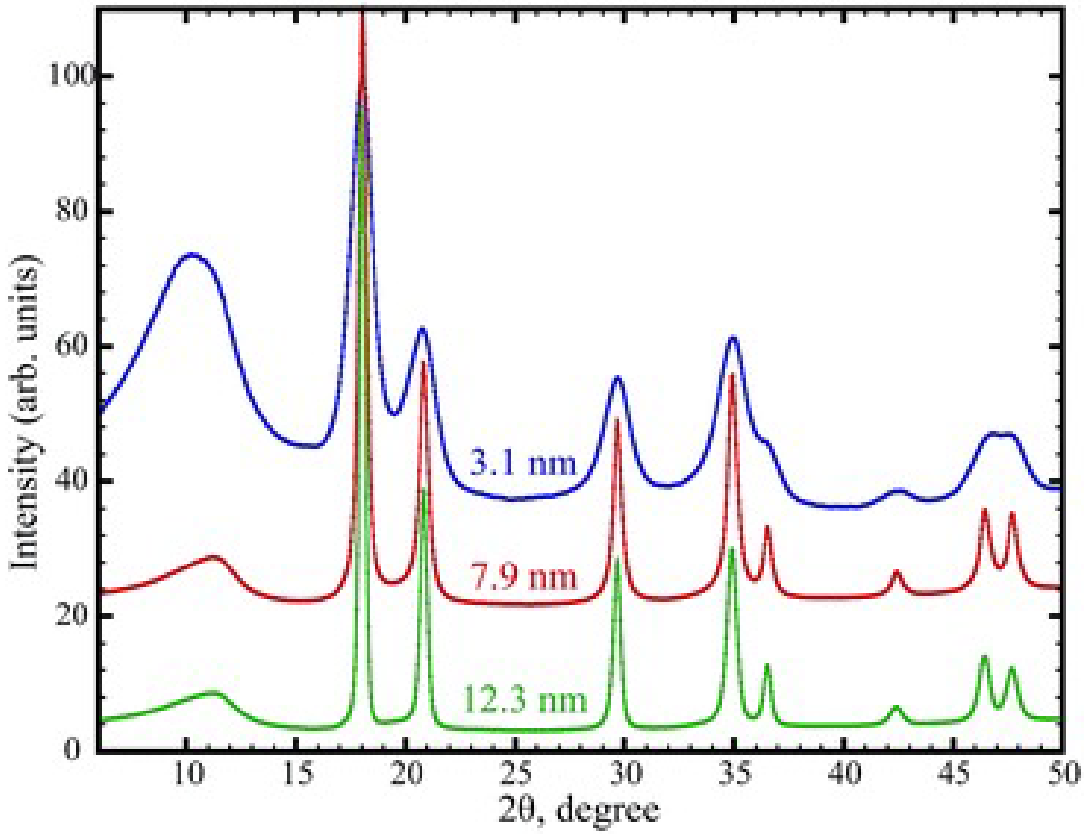
X-ray diffraction of nanocrystals

- XRD patterns of Pt nanoparticles with different sizes.
- The **diffraction peaks become broadening with decreasing particle size**!
- The width of the diffraction peaks provides information on the persistence of long-range order and hence the crystal size.
- Practically, the particle size(D_p) can be estimated by,

Debye-Scherrer equation:

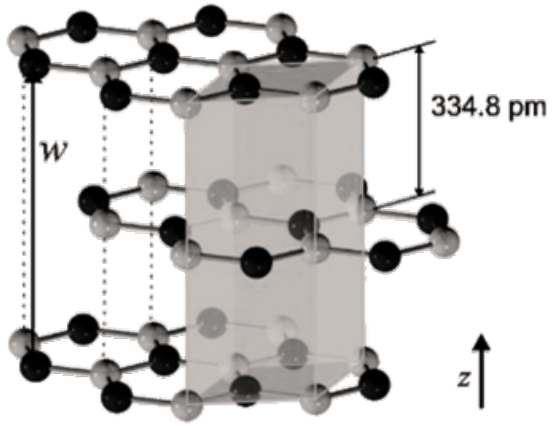
$$D_p = \frac{K\,\lambda}{\beta\,cos\theta}$$

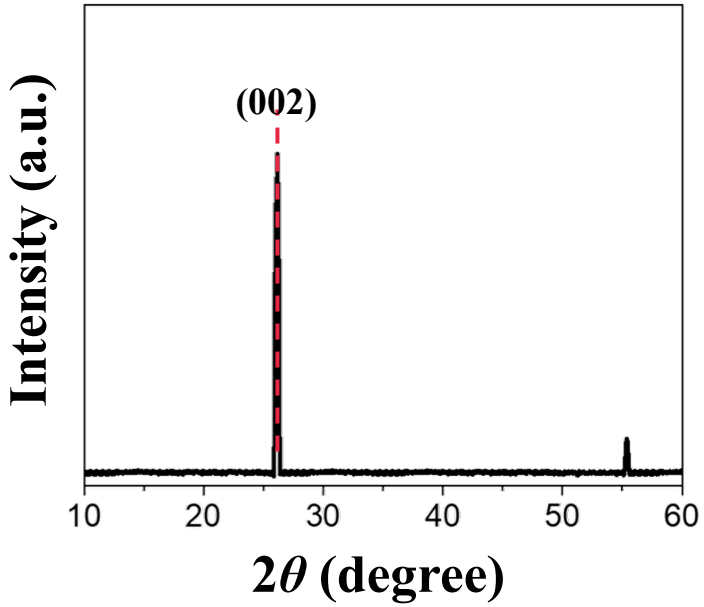
Where,
K = Scherrer's Constant (K=0.94)
β = Line broadening (in radians),
θ = Bragg angle (in radians),
λ = X-Ray wavelength (*i.e.,* for Cu kα 0.15406 nm)



THE UNIVERSITY OF QUEENSLAND AUSTRALIA

Graphitised carbon (graphite)

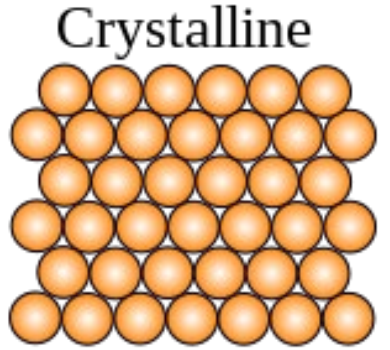


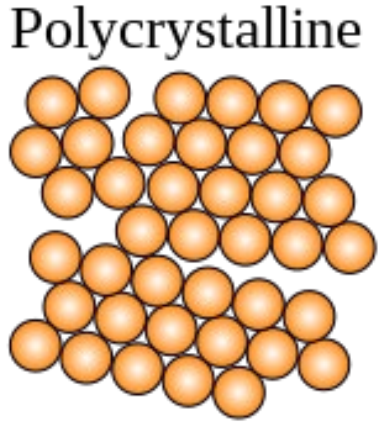


Crystalline material

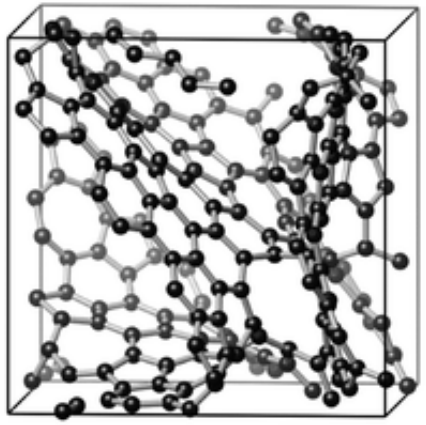
Uniform d-spacing of specific lattice planes.

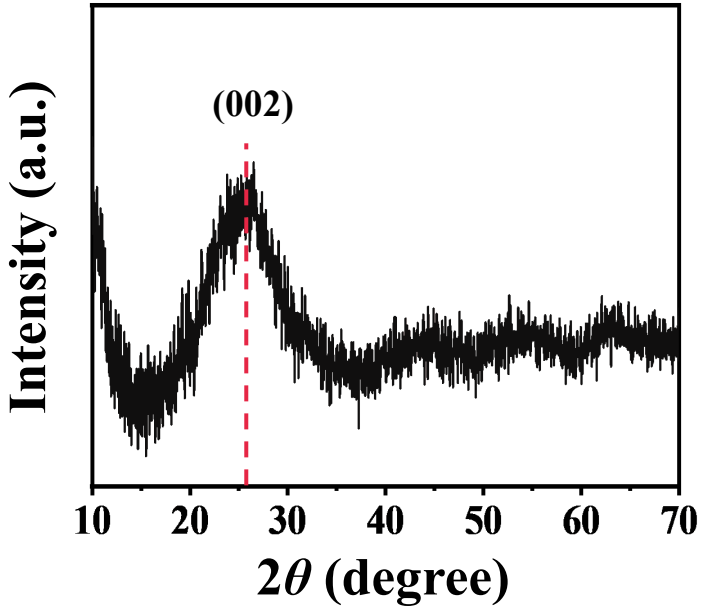
Sharp diffraction peaks at respective angles.





Amorphous carbon

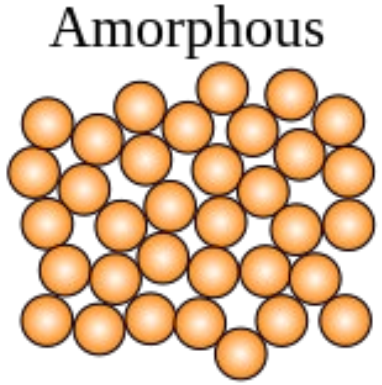




Amorphous material

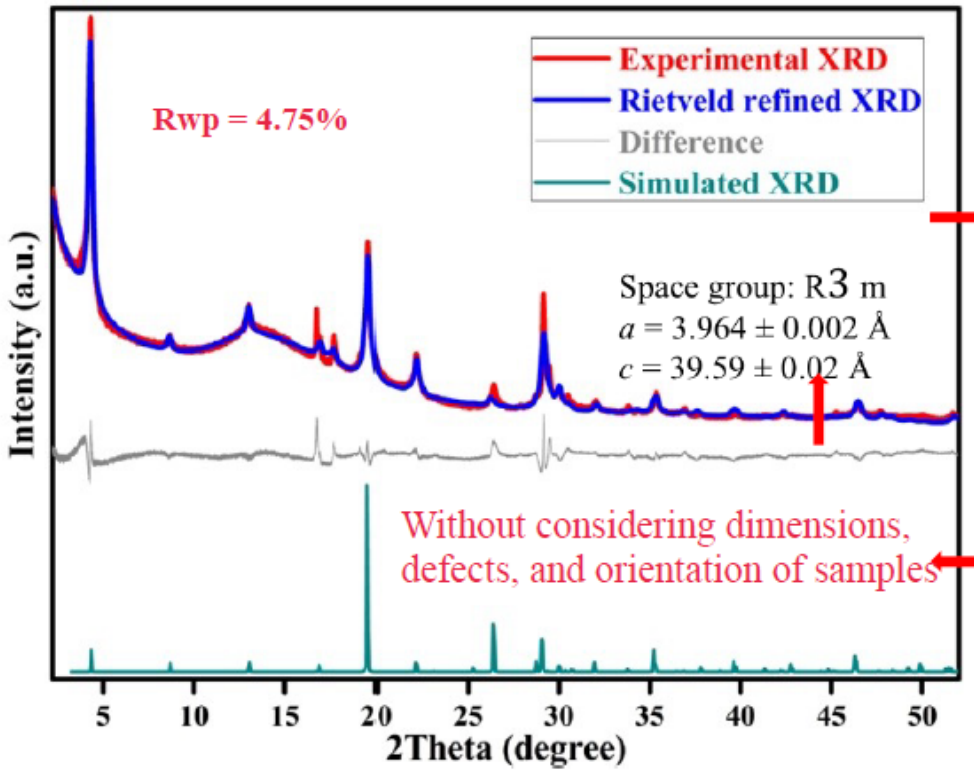
Random d-spacing of specific lattice planes.

Broad diffraction peaks at respective angles.



Synchrotrons X-ray diffraction (XRD)

Structure determination of newly discovered In_3Se_4 phase using synchrotron XRD

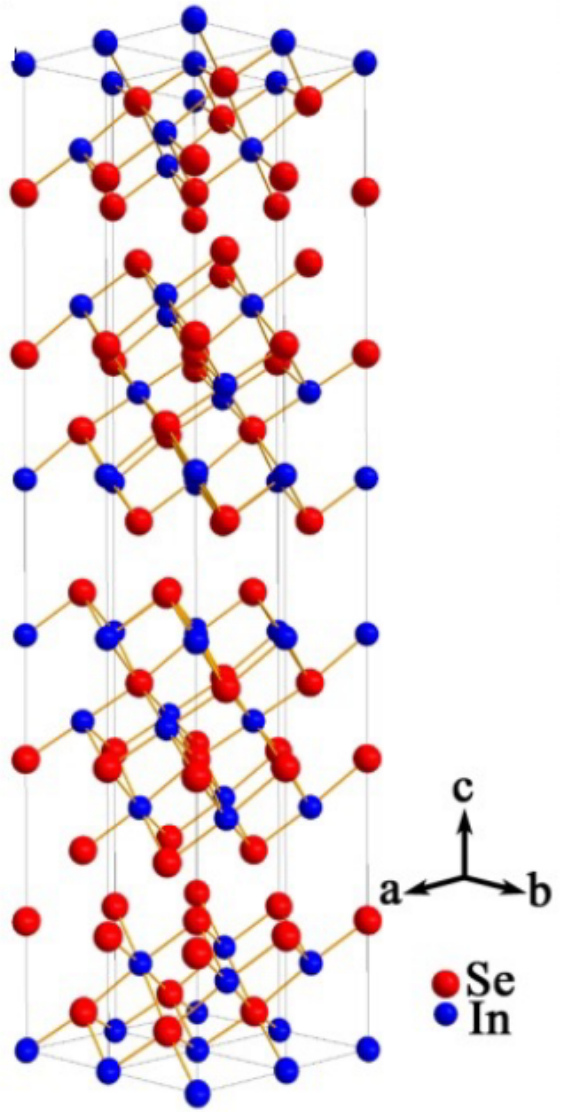


Refined experimental In_3Se_4

Site	Element	Wyckoff Symbol	Symmetry	x	y	z	Occupation
In1	In	3a	-3m	0.00000	0.00000	0.0000	1
In2	In	6c	3m	0.00000	0.00000	0.4150(1)	1
Se1	Se	6c	3m	0.00000	0.00000	0.1242(1)	1
Se2	Se	6c	3m	0.00000	0.00000	0.3012(1)	1

Perfect In_3Se_4

Site	Element	Wyckoff Symbol	Symmetry	x	y	z	Occupation
In1	In	3a	-3m	0.00000	0.00000	0.0000	1
In2	In	6c	3m	0.00000	0.00000	0.4271	1
Se1	Se	6c	3m	0.00000	0.00000	0.1278	1
Se2	Se	6c	3m	0.00000	0.00000	0.2917	1



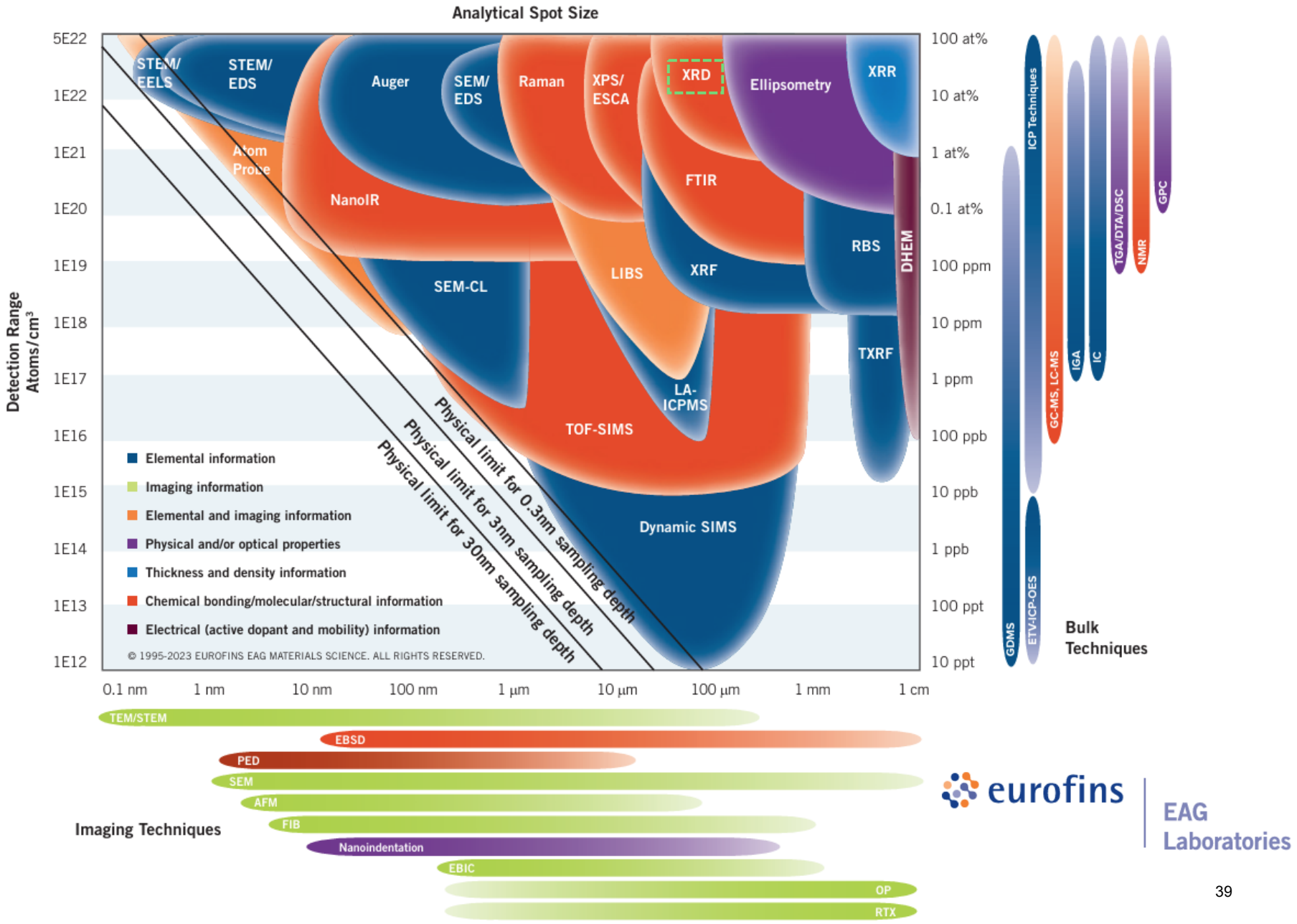
Specifics of X-ray diffraction

- Powders, single/poly-crystalline bulks, thin films of few nm
- Sample surface must be flat, which is achievable by polishing for bulk samples.
- Surface deformation/stressing introduced by polishing should be removed by thermal annealing.
- Small crystal dimensions (*e.g.,* 100 nm or less) lead to peak broadening.
- Large crystals lead to less diffraction occurrence(10-5 nm).

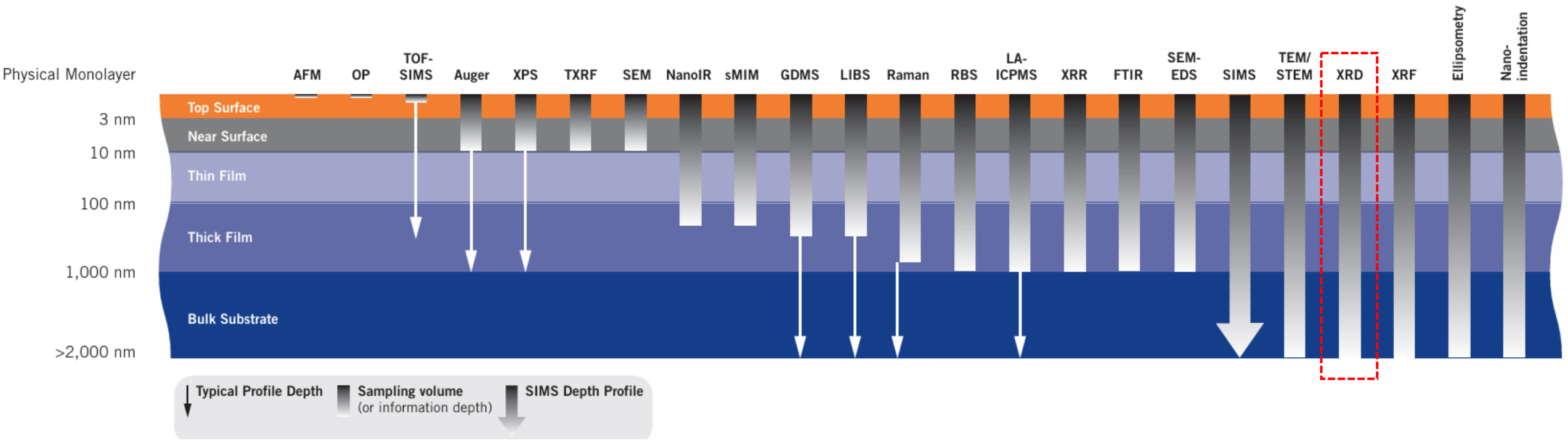
Applications of X-ray diffraction

- Identification of crystalline phases and their percentage (%).
- Determination of crystal structures of new materials/structures (once it is done, it can be added to ICDD database).
- Determination of the preferred orientations of specimens (likely microstructures).
- Measurement of texture, grain size, and internal stress.
- Determination of thermal expansion and compactness of materials due to its high accuracy (10-5 nm).

Analytical resolution *vs.* Detection limit



Typical analysis depth for techniques



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Reference & useful links:

1. Crystallography, W. Borchardt-Ott
2. Crystallography and Crystal Defects, A. Kelly, G.W. Groves, and P. Kidd
3. The Basics of Crystallography and Diffraction, C. Hammond
4. The Structure of Materials, S.M. Allen and E.L. Thomas
5. http://www.matter.org.uk/matscicdrom/manual/xl.html
6. http://www.msm.cam.ac.uk/phasetrans/2001/intro.cryst.pdfhttps://www.bbc.co.uk/bitesize/guides/z463rwx/revision/1
7. https://www.satnanoparticles.com/news-show-2323.html
8. https://chem.libretexts.org/Bookshelves/Introductory_Chemistry/Introductory_Chemistry_%28CK-12%29/13%3A_States_of_Matter/13.14%3A_Unit_Cells
9. https://www.brainkart.com/article/Primitive-and-non-primitive-unit-cell_38660/
10. https://www.xtal.iqf.csic.es/Cristalografia/parte_04-en.html
11. SmartLab | Rigaku Global Website
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Thank you

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CRICOS code 00025B

WEEK 3 · 知识点总结 · Dr Aditya Ashok

第 3 周　扫描电子显微镜

Scanning Electron Microscopy (SEM)

主线：电子波长决定分辨率 → SEM 结构与像差 → 电子–样品相互作用（SE / BSE / X 射线）→ 参数选择与制样

1. 为什么用电子

- 分辨率（Abbe / Rayleigh）：$d = 0.61\lambda / (n \sin\alpha)$；可见光显微镜极限约 **200 nm**。
- de Broglie：$\lambda = h / \sqrt{(2meV)}$；5 kV → 1.72 Å，10 kV → 1.22 Å，**20 kV → 0.86 Å**。
- SEM 相比光镜：形貌信息多、放大倍数高（可达约 10^6）、**景深大**；图像为灰度。

2. SEM 的组成

- **电子枪**：W 灯丝（热发射，寿命约 10^2 h）、LaB_6（约 10^3 h）、**场发射枪 FEG**（约 10^4 h，亮度最高）；Wehnelt 帽 + **阳极**（加速电压 0.5–30 kV）。
- **聚光镜**：决定束斑大小（"塑形"）；**物镜**：把束聚焦到样品表面；**扫描线圈**：光栅扫描，改变扫描区域 = 改变放大倍数。
- **像差**：
  - 色差：不同能量的电子焦点不同；
  - 球差：离轴电子焦点不同；
  - 像散：x、y 方向聚焦不同（操作者可用消像散器校正）。

3. 电子–样品相互作用（★）

信号	来源	能量 / 深度	提供信息
SE 二次电子	非弹性散射	< 50 eV，表层几 nm	**表面形貌**（最常用）
BSE 背散射电子	弹性散射	50 eV – E_0，较深	**原子序数 Z 衬度**（重元素更亮）
特征 X 射线	非弹性（内层电离）	整个作用体积	**元素成分（EDX/EDS）**

- **ETD 探测器**：SE 被约 +10 kV 吸向闪烁体 → 转为光 → 光导 → 光电倍增管。
- **作用体积**随加速电压 ↑ 而增大，随 Z ↑ 而减小。Si 中穿透深度：30 kV 约 8 µm，15 kV 约 2.5 µm，5 kV 约 0.4 µm。

4. 参数选择（★ 取舍）

参数	增大时
加速电压	信号强、可激发 EDX；但作用体积大、表面细节丢失、荷电和损伤增加（低 kV：表面细节好、荷电少）
束流 / 束斑	信号多，但束斑变大 → 分辨率下降（选能给出足够信号的最小束流）
工作距离 WD	WD 长 → 景深大；WD 短 → 分辨率高
扫描速度	越慢 → 每像素信号越多 → 图像越好

5. 制样与局限

- 导电碳胶固定在样品台 → 真空烘箱除气（70 °C, 8 h）→ 绝缘样品**喷镀 Au 或 C**；样品须**干燥、导电**（防止荷电），必须在**真空**中观察（电子在空气中自由程很短）。
- 局限：需导电或镀膜、真空、**只看表面**（除非做截面）、需样品耐电子束。
- EDX 结果：ZAF 无标样定量，给出 mass% 与 atom%（例：介孔 Au 膜 Au 约 95 wt%）。

考试重点：电子波长计算；SE vs BSE vs X 射线的来源与用途；kV、束流、WD 的取舍；绝缘样品为何要镀膜；SEM 的局限。

MATE 6301

Scanning Electron Microscopy (SEM)

MATE 6301 | Light Microscope vs Electron Microscope

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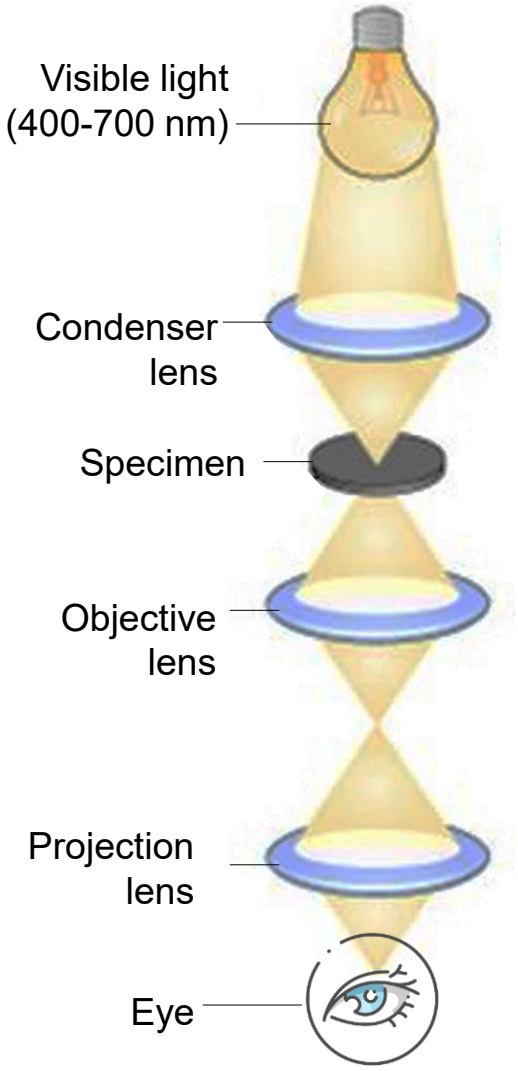
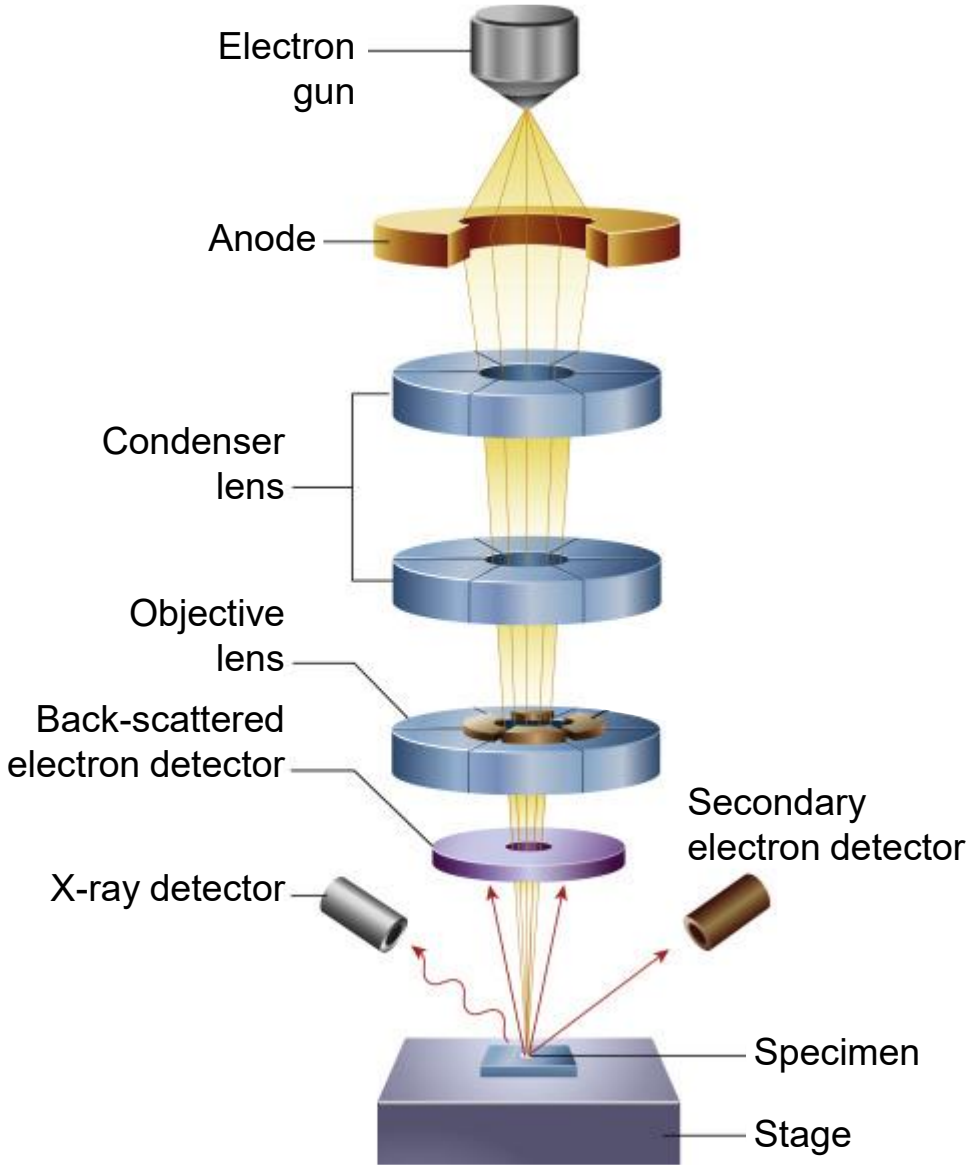
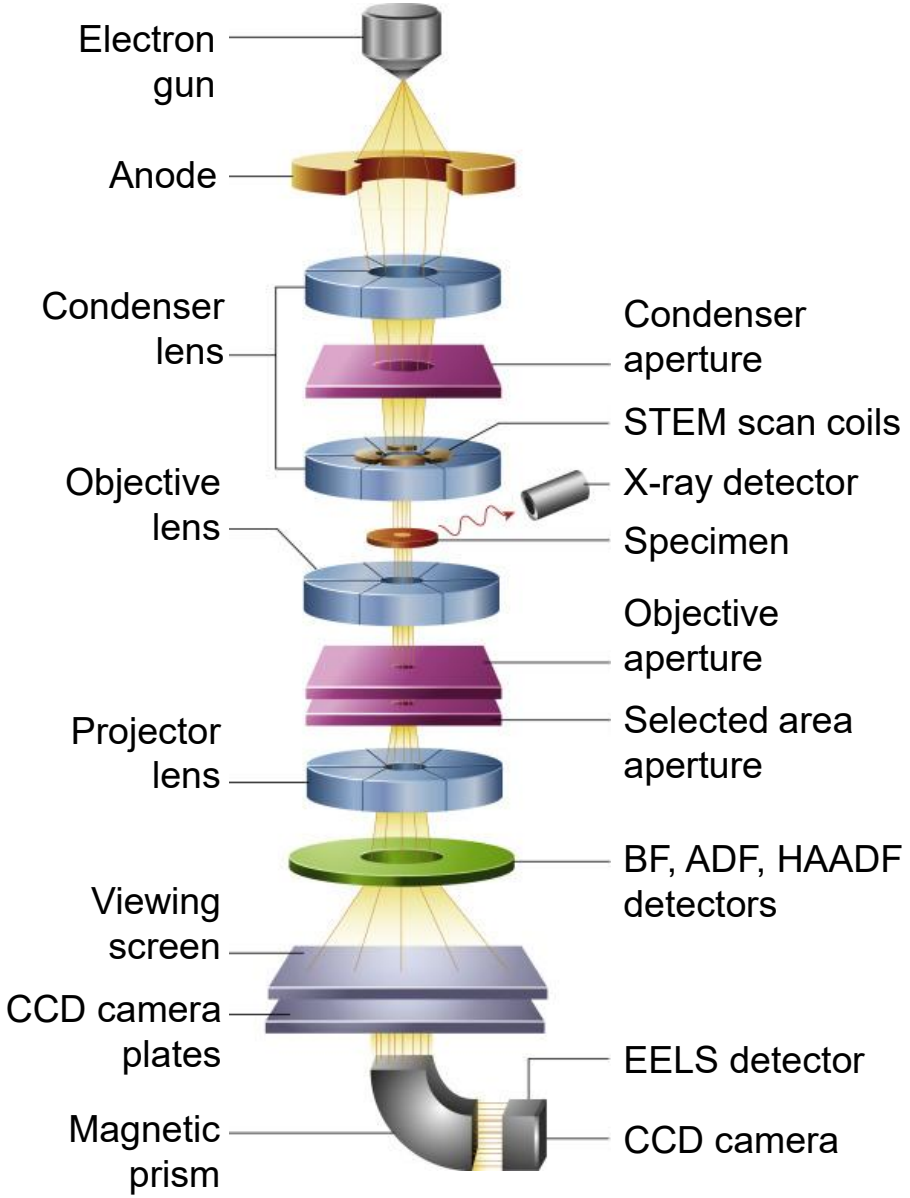
Light microscope



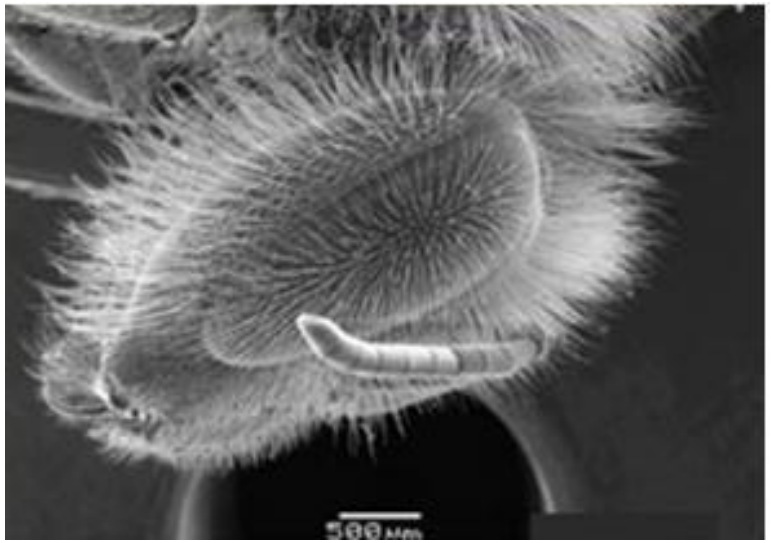
Scanning EM



Transmission EM



Light microscope (Colored)



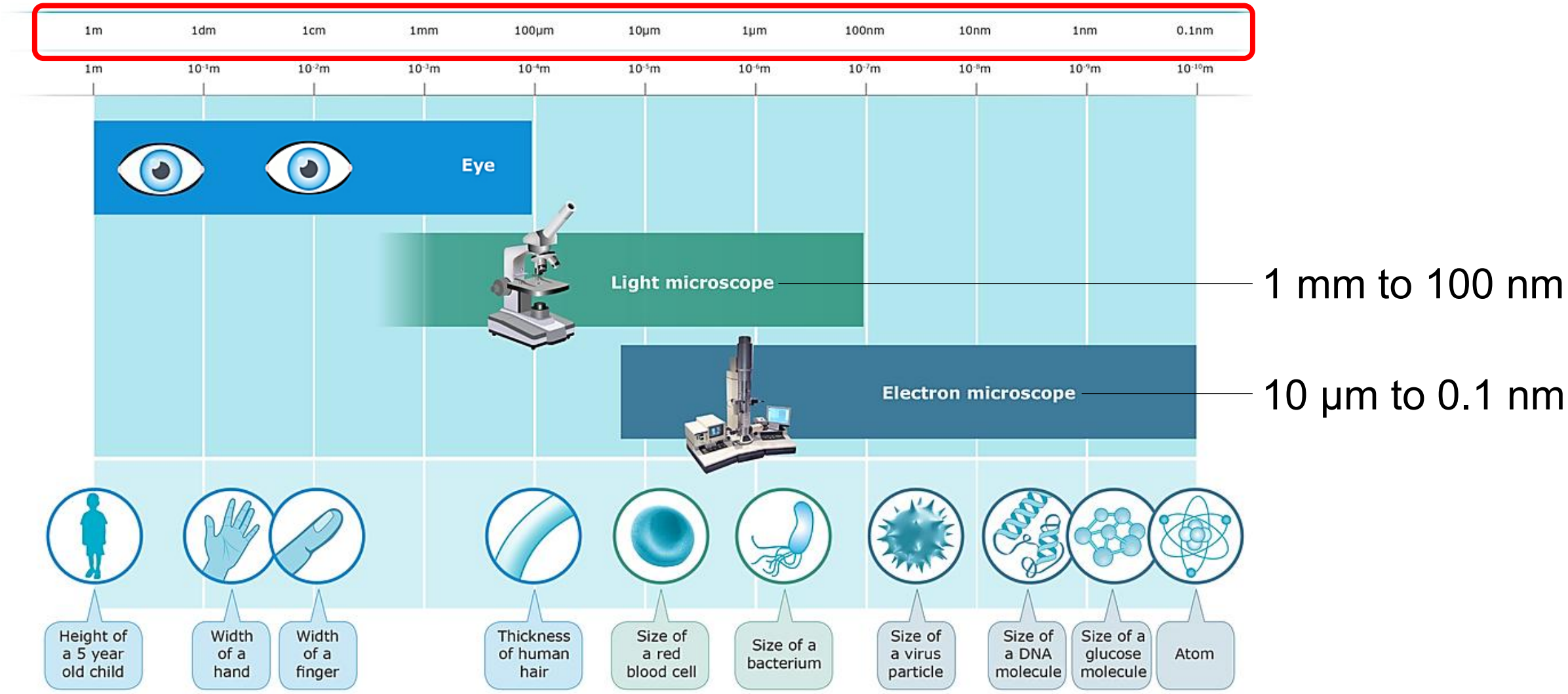
SEM image (Grayscale)

SEM offers…

- Much more topographical information.
- Much greater magnification.
- Much higher depth of field.

Why?

Limit of resolution (resolving power):
The limit or ability of a microscope to distinguish details of a specimen or sample.



1 mm to 100 nm

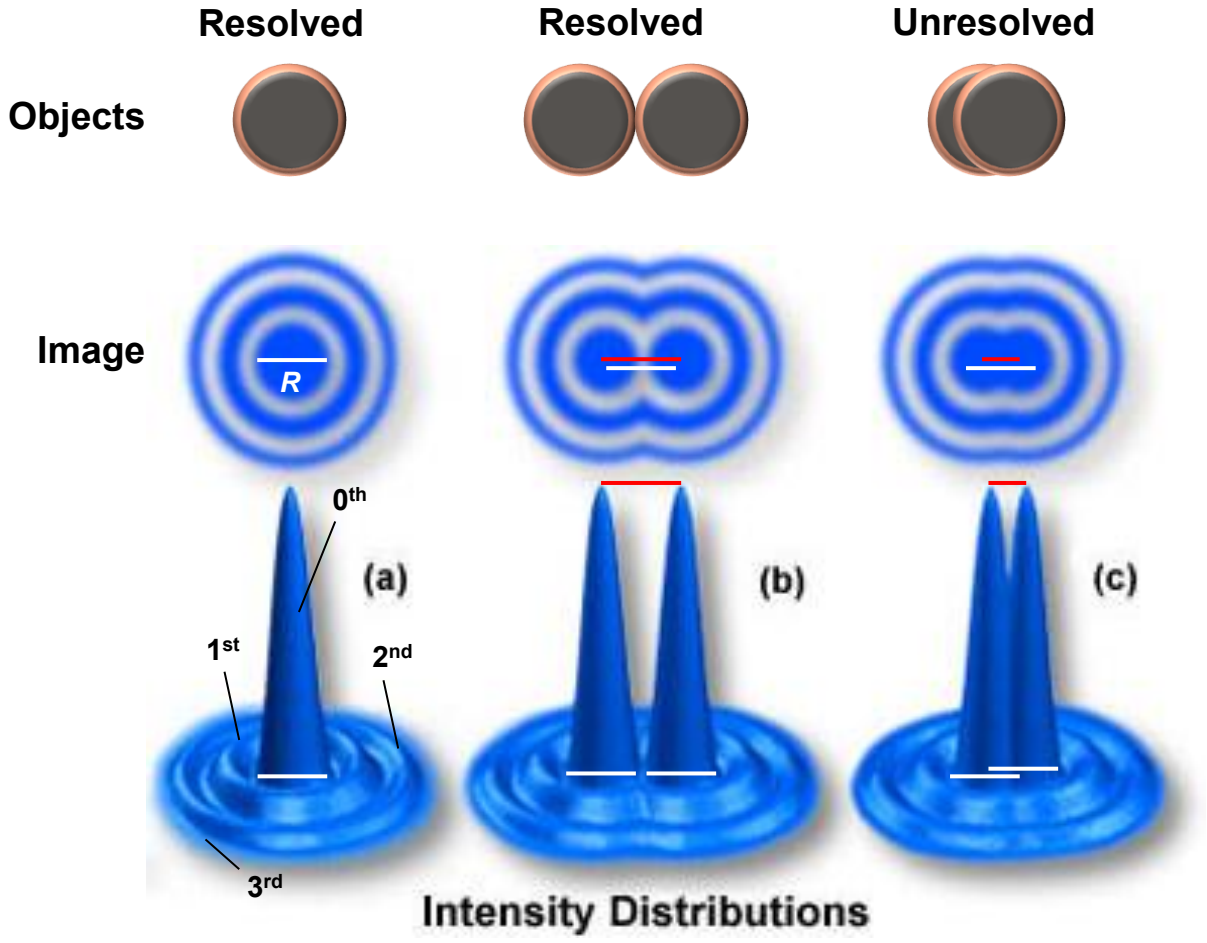
10 µm to 0.1 nm

Light Microscope theoretical resolution:

$$R = \frac{0.61\lambda}{n\,sin(\theta/2)}$$

R = limit of resolution or radius of Airy disk, λ = incident wavelength,
n = refractive index of viewing medium, θ = angle of microscope's aperture.

The **Rayleigh criterion** specifies the minimum separation between two light sources that may be resolved into distinct objects. It is only possible to resolve a pair of sources if the central peaks of the two diffraction patterns are no closer than the radius of the Airy disk.



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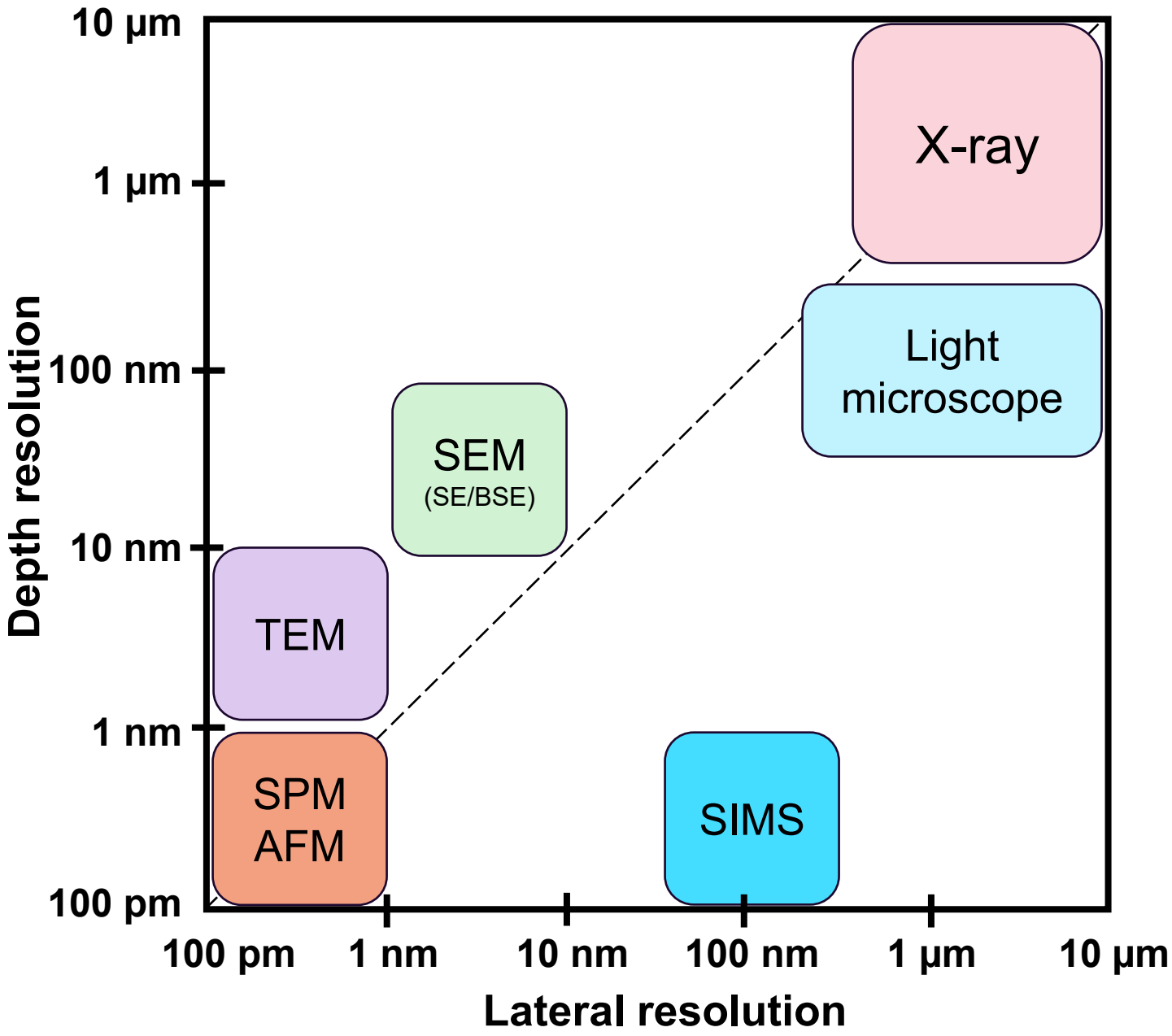
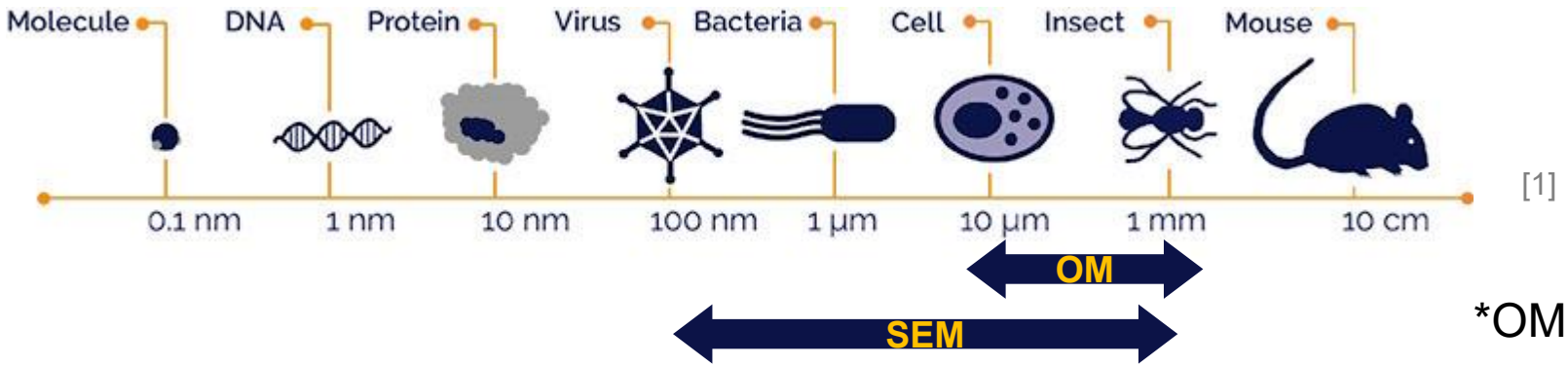


What is SEM?

Scanning Electron Microscope (SEM) is a type of microscope that **uses a fine beam of focused electrons to scan a sample's surface**. The microscope records information about the **interaction between the electrons and the sample**, creating a magnified image. SEM has the **potential to magnify an image up to 1 million times** (100 nanometres).



[1]

*OM: Optical Microscope

What material information can be obtained?

It is used to observe,

- Surface structure information. (SE)
- Chemical composition of the specimen. (EDX)

- Electron Microscopes (SEM & TEM) use electrons to visualise the specimen.
- *Limit of resolution* of electron microscopes is related to the *wavelength of electrons*.
- How can we know the wavelength of electrons?

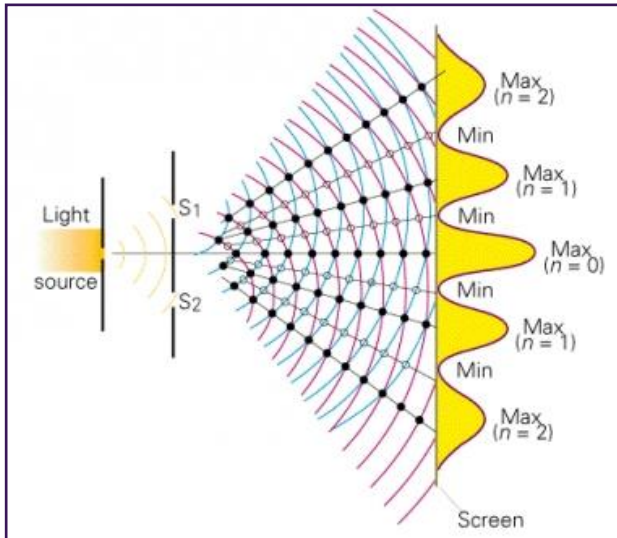
Louise de Broglie

He hypothesised that all matters exhibit wave-particle duality under specific experimental conditions and came up with de Broglie's wavelength equation (1924).

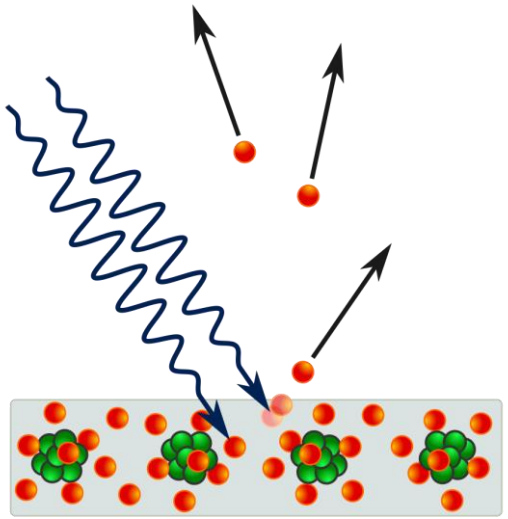
de Broglie's wavelength, $\lambda = \frac{h}{p}$

λ = wavelength of matter, h = Plank's constant, p = momentum of the matter.

In de Broglie's time, wave-particle duality of light was already proven experimentally.



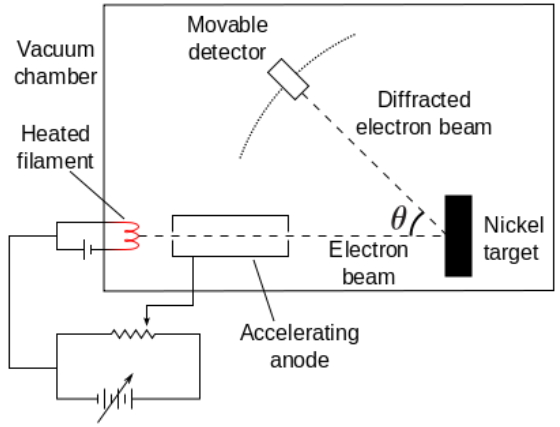
Young's double-slit experiment

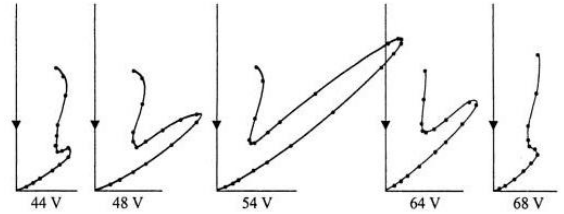


Einstein's Photoelectric effect in a solid

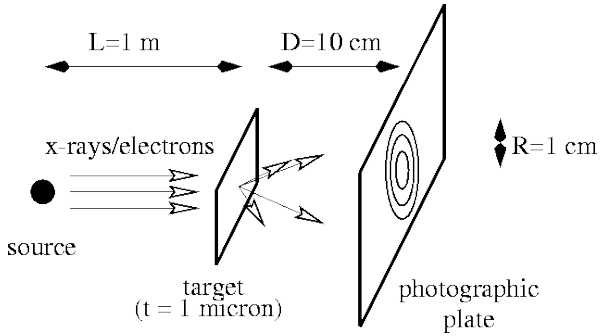
de Broglie's hypothesis that matters exhibit wave-particle duality like light was experimentally proven later using electrons as an example of matter.

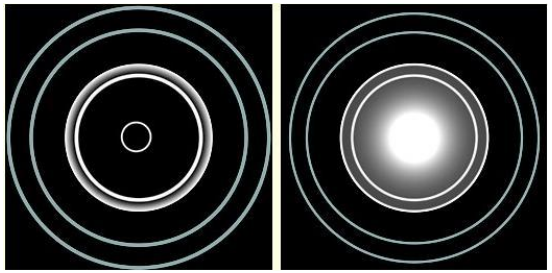
Davisson Germer experiment (1927)





G.P. Thomson Experiment (1926/1927)





Later experiments confirmed that electrons also exhibit wave-like behaviour.

de Broglie wavelength (λ) $\lambda = \frac{h}{p}$ $(p = mv)$

$$\lambda = \frac{h}{mv} \quad \cdots \text{Eqn. (1)}$$

Since the velocity of the electrons (v) is determined by the accelerating voltage, or electron potential (V) where

$$eV = \frac{1}{2}mv^2$$

The velocity of the electrons (v) can be calculated by

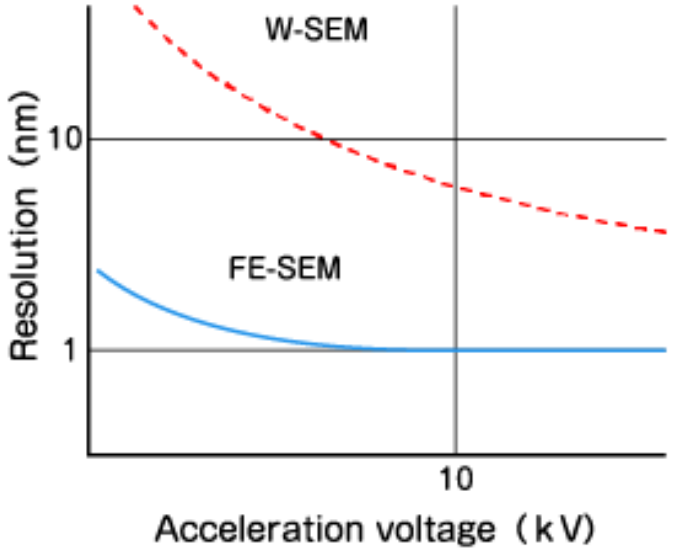
$$v = \sqrt{\frac{2eV}{m}} \quad \cdots \text{Eqn. (2)}$$

By substituting Eqn. (2) to Eqn. (1), $\lambda = \frac{h}{\sqrt{2meV}}$

E.g., at V = 5 kV (5,000 V), $\lambda_{electron}$ = 1.72 Å.

at V = 10 kV (10,000 V), $\lambda_{electron}$ = 1.22 Å

at V = 20 kV (20,000 V), $\lambda_{electron}$ = 0.86 Å



Note: λ is wavelength of electron, h is Plank's constant (6.626 x 10^{-34} Js), p is momentum of electron, m is mass of electron (9.1 x 10^{-31} kg), v is the velocity of electron, e is the charge of electron (1.6 x 10^{-19} C), V is the voltage of electron.

MATE 6301 | SEM Image

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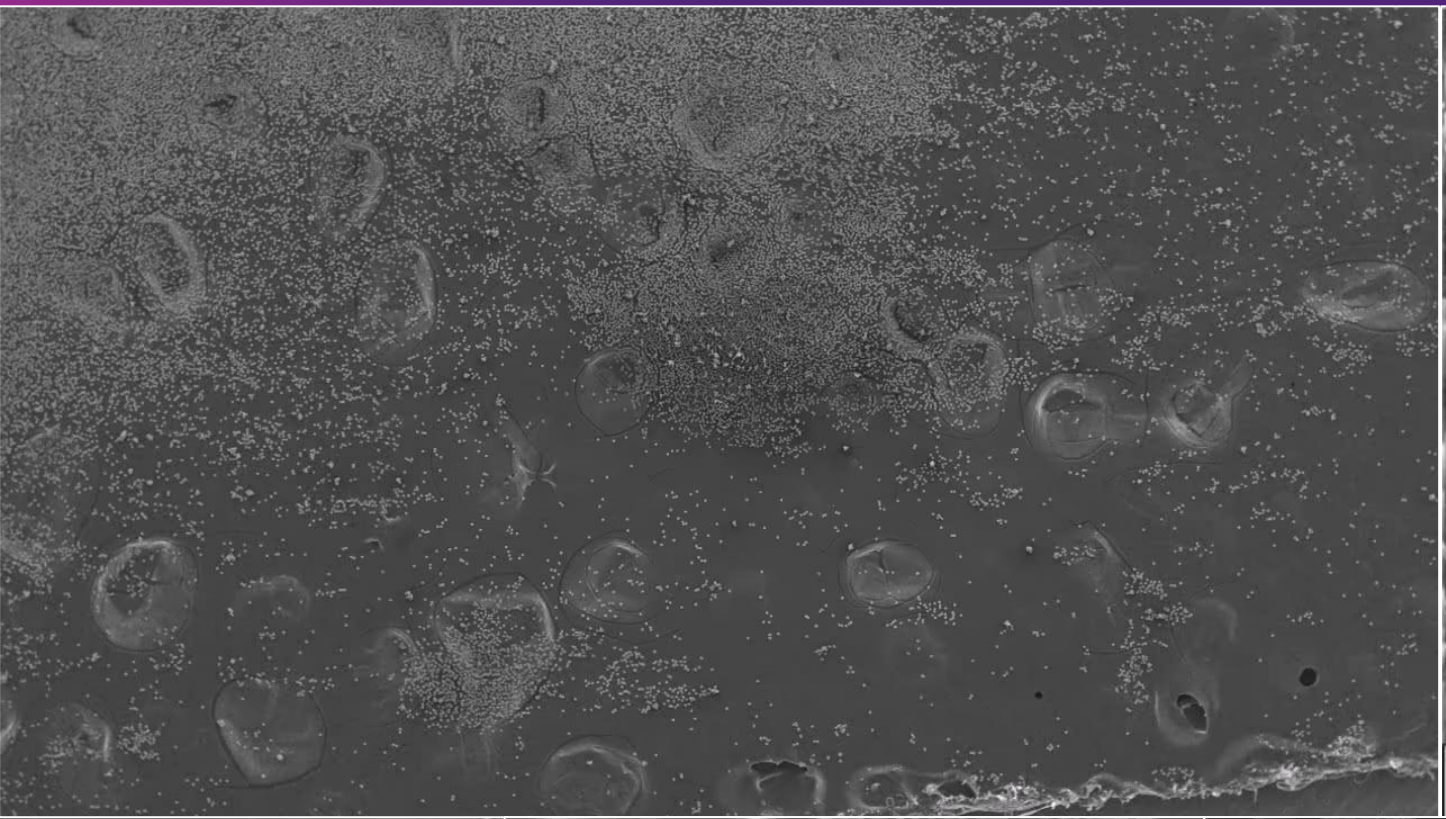

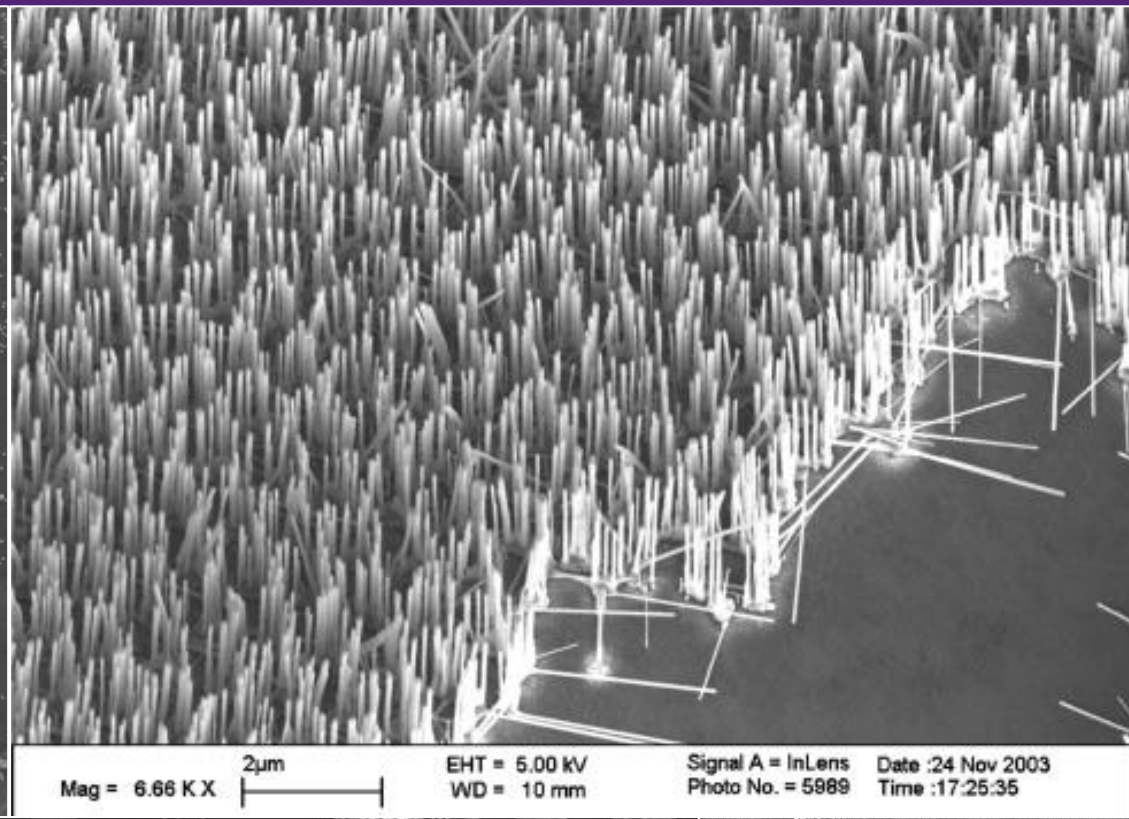

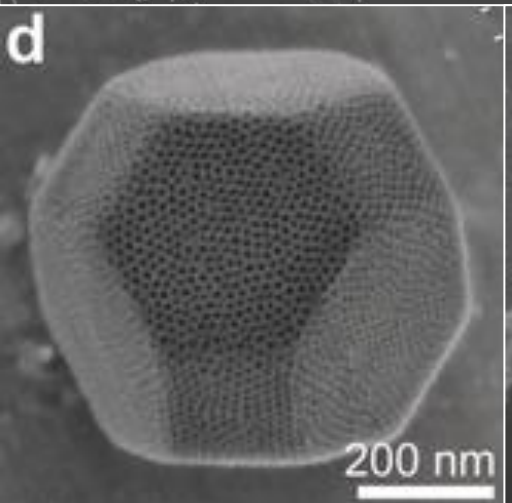

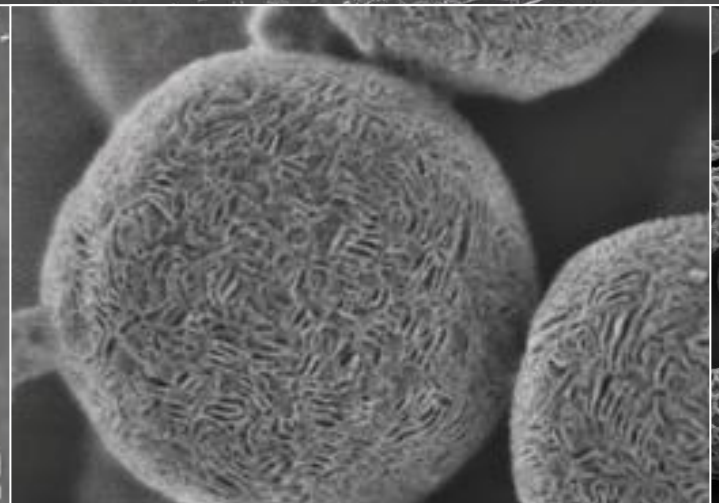

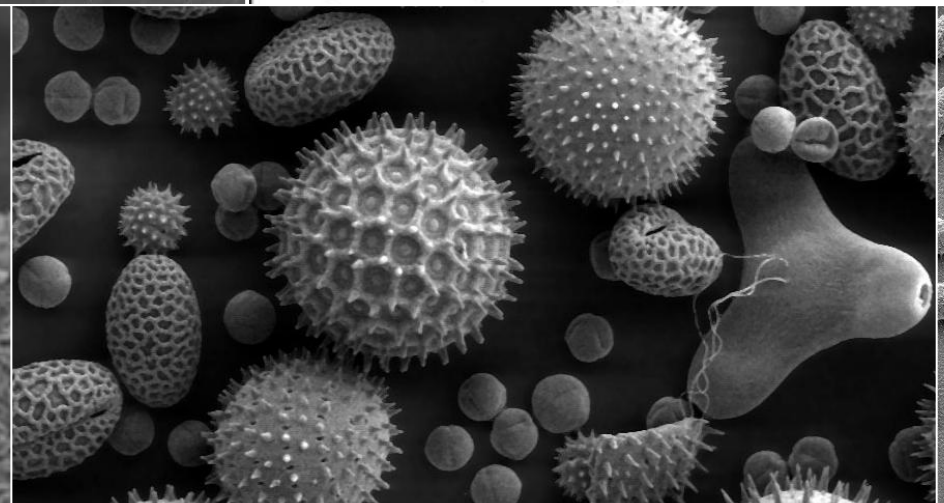

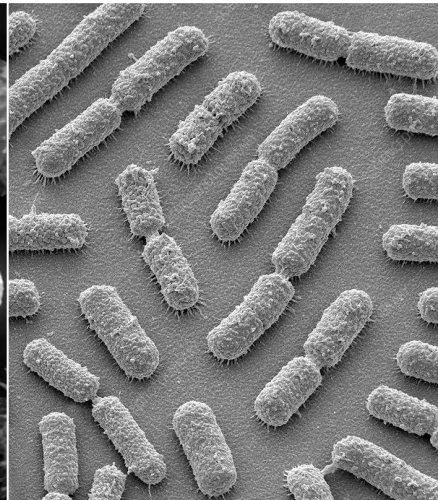

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Why SEM?

- Typical *magnification* for a light microscope is 50x to 2000x, set because of the nature of the resolution (d, key parameter).

- According to Abbe's equation, resolution (for a perfect lens)

$$d = \frac{0.61\lambda}{n \sin \alpha}$$

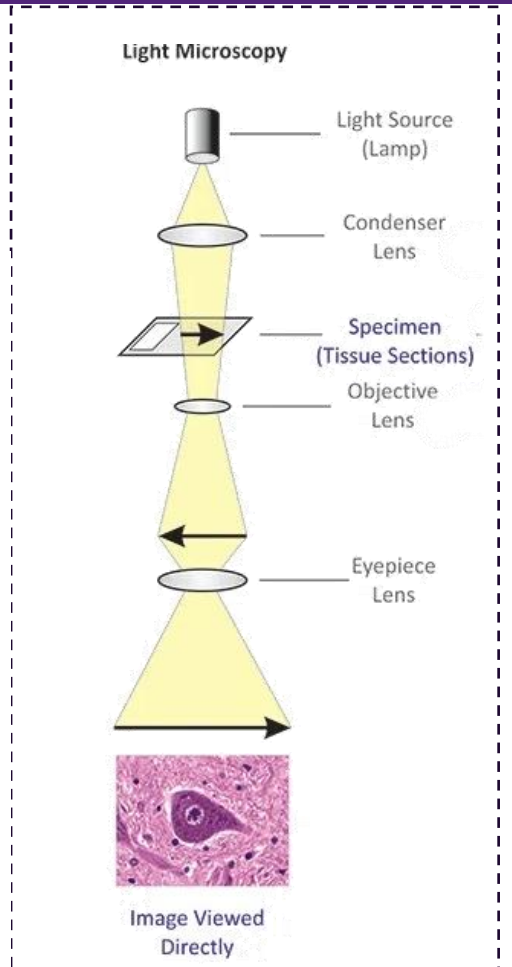
d = resolution (minimum resolvable distance)
λ = wavelength of energy source
n = refractive index of the medium
α = aperture angle

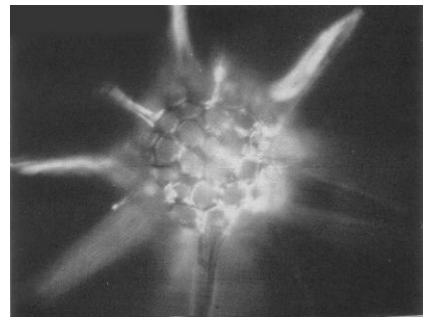
- For light in the visible spectrum, the resolution of a light microscope is approximately 200nm.

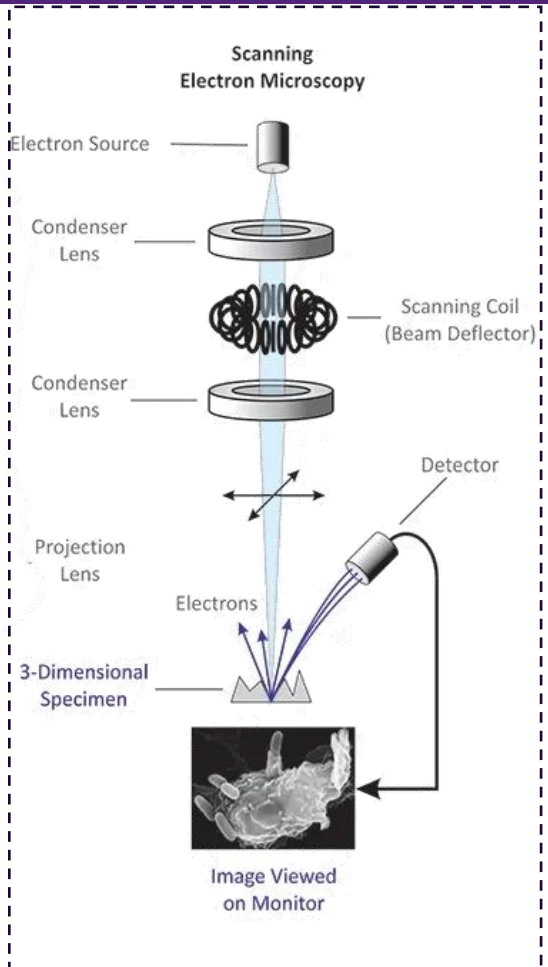
- Electrons have wave-like properties (de Broglie) with wavelength dependent upon their energy.

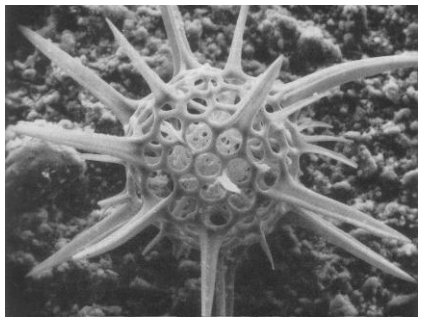
$$\lambda \text{ (Å or } nm) = \frac{h}{\sqrt{2meV}}$$

h = Planck's constant (6.626 x 10^{-34} Js)
m= electron mass (9.109 x 10^{-31} kg)
e = electric charge (1.6 x 10^{-19} C)
V = accelerating voltage (in volts V)



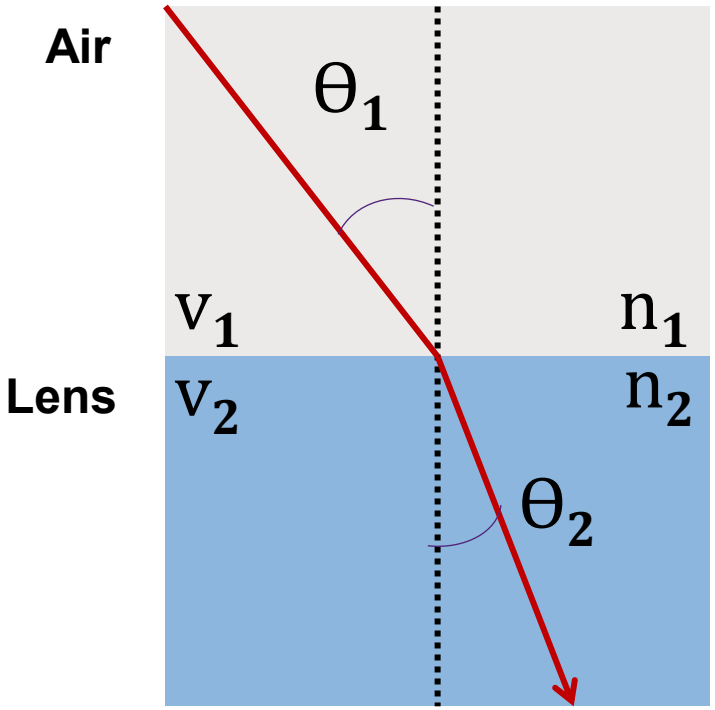






Bending of Light – Snell's Law

Refraction is the bending of light (or any wave) as it passes from one medium to another with a different density, causing a change in its speed. This change in direction occurs at the boundary between the two media and is described by **Snell's Law.**



$$\frac{n_1}{n_2} = \frac{Sin\Theta_1}{Sin\,\Theta_2}$$

Where,

$$n = \frac{c}{v}$$

$$n_1 Sin\Theta_2 = n_2 Sin\Theta_1$$

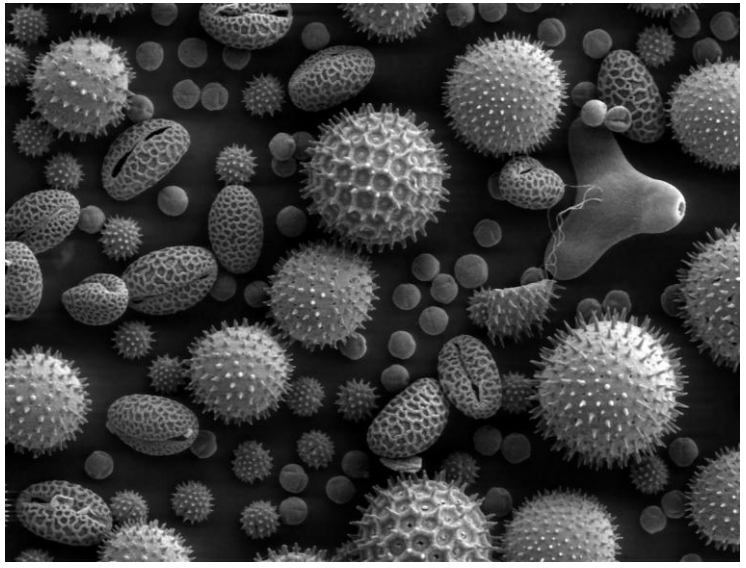
n: refractive index
c: speed of light in a vacuum
v: speed of light through the medium
ϴ: angle of incidence and refraction

e.g., light will travel faster through a vacuum than through air or glass.

Characteristic Information

Morphology

Shape and size of the particle

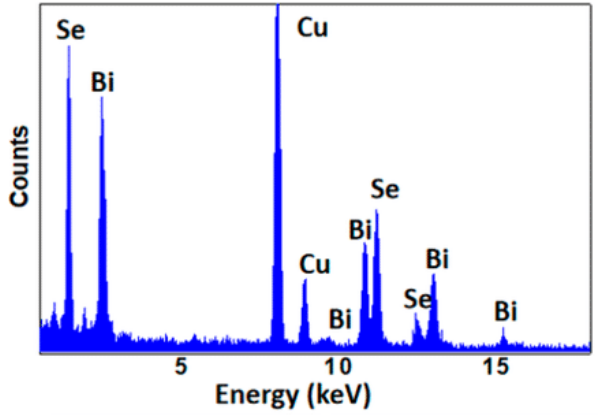


[2]

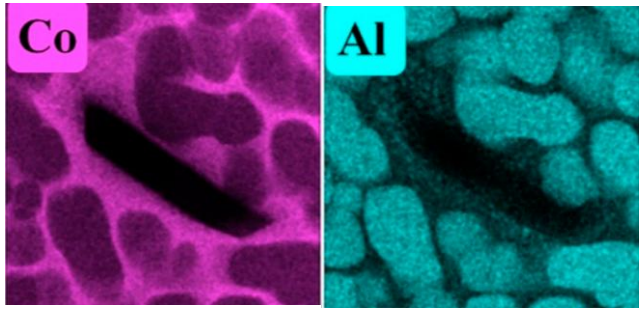
[3]

Chemical Topography

Elemental Spectrum

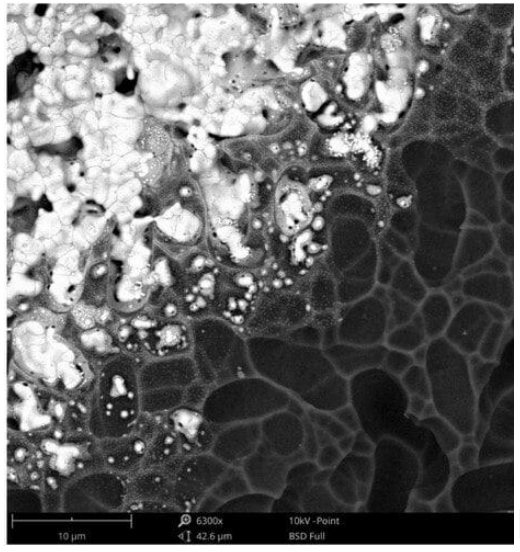


Elemental Mapping



[3]

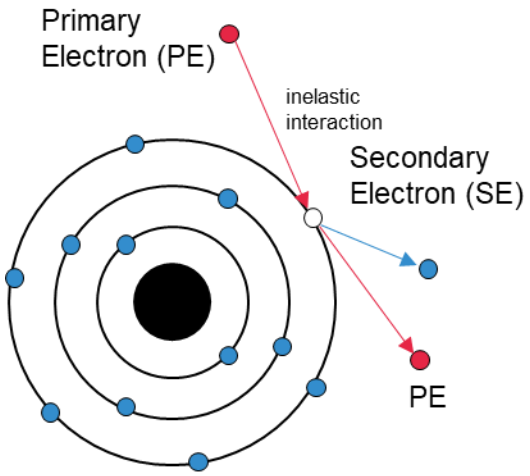
Backscattered (z-contract images)



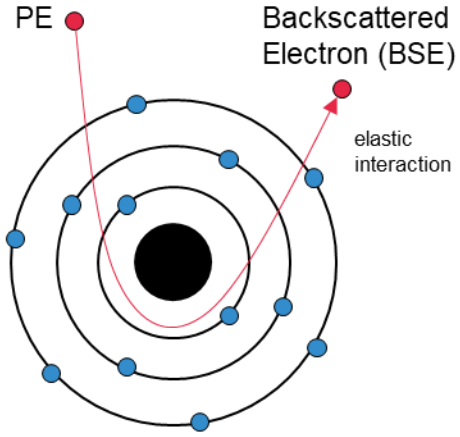
[4]

Operating Principle:

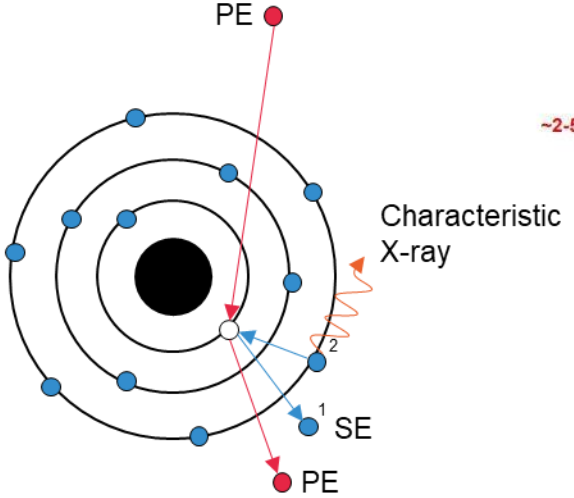
A scanning electron microscope (SEM) projects and scans a focused stream of electrons over a surface to create an image. The electrons in the beam interact with the sample, thereby producing various signals that can be used to obtain information about the surface's topography and composition.



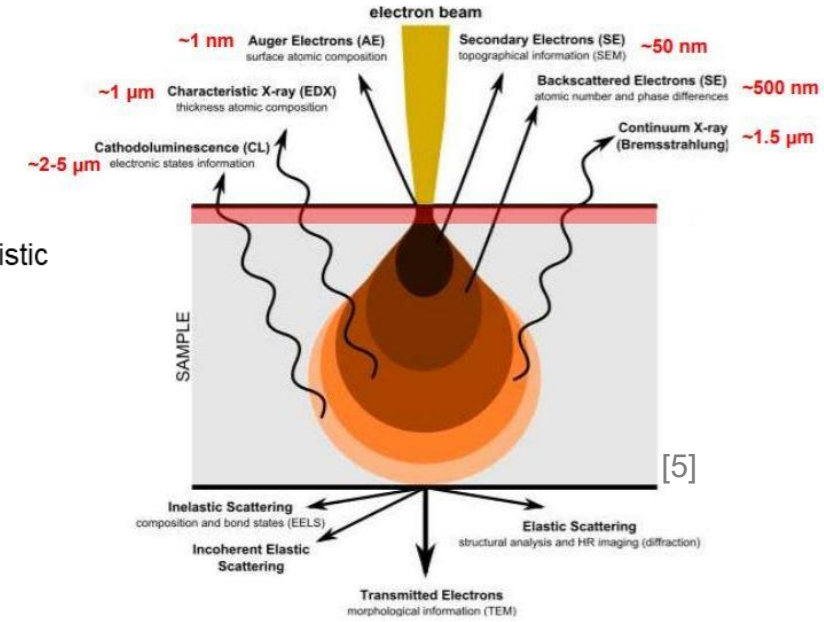
Contrast: **Topography**



Contrast: **Atomic Number (Z)**

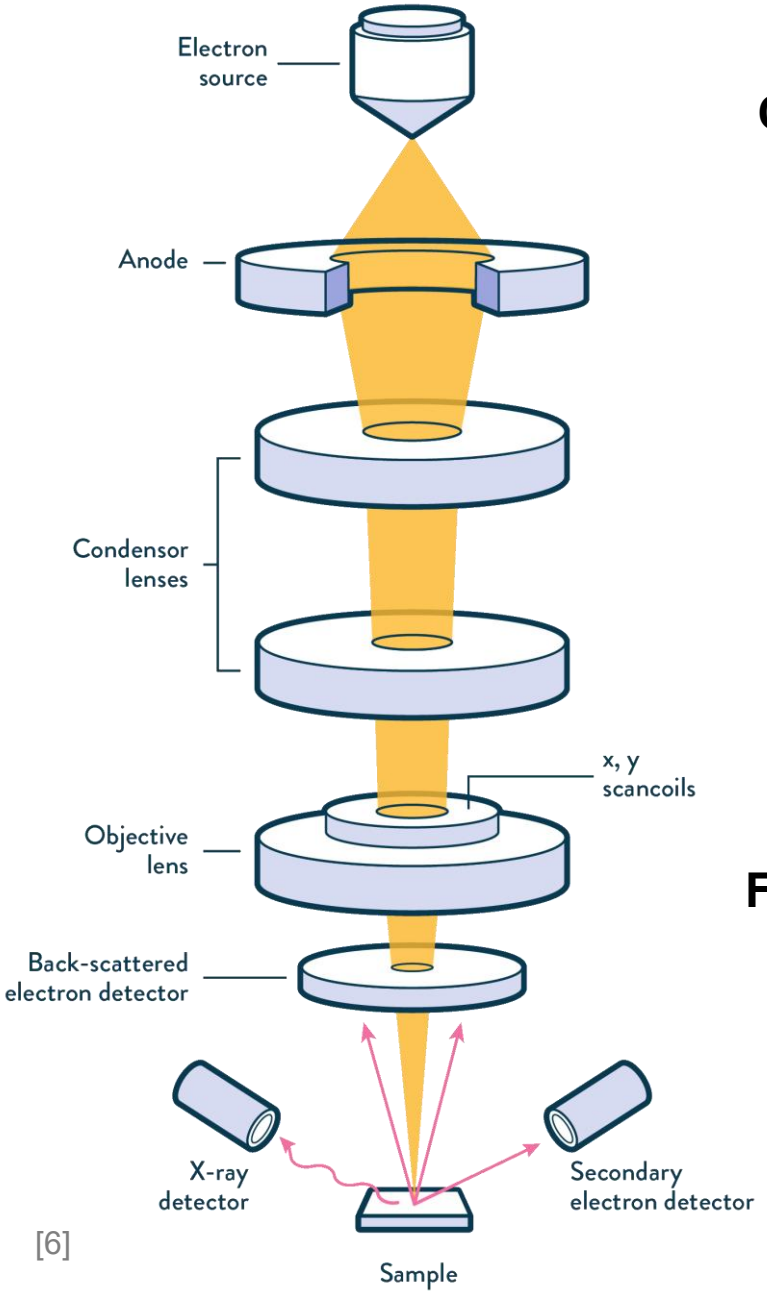


Contrast: **Elemental Composition**



The interaction between the electron beam and the sample creates secondary electrons, backscattered electrons, and X-rays. These interactions are captured to create a magnified image.

Components of SEM



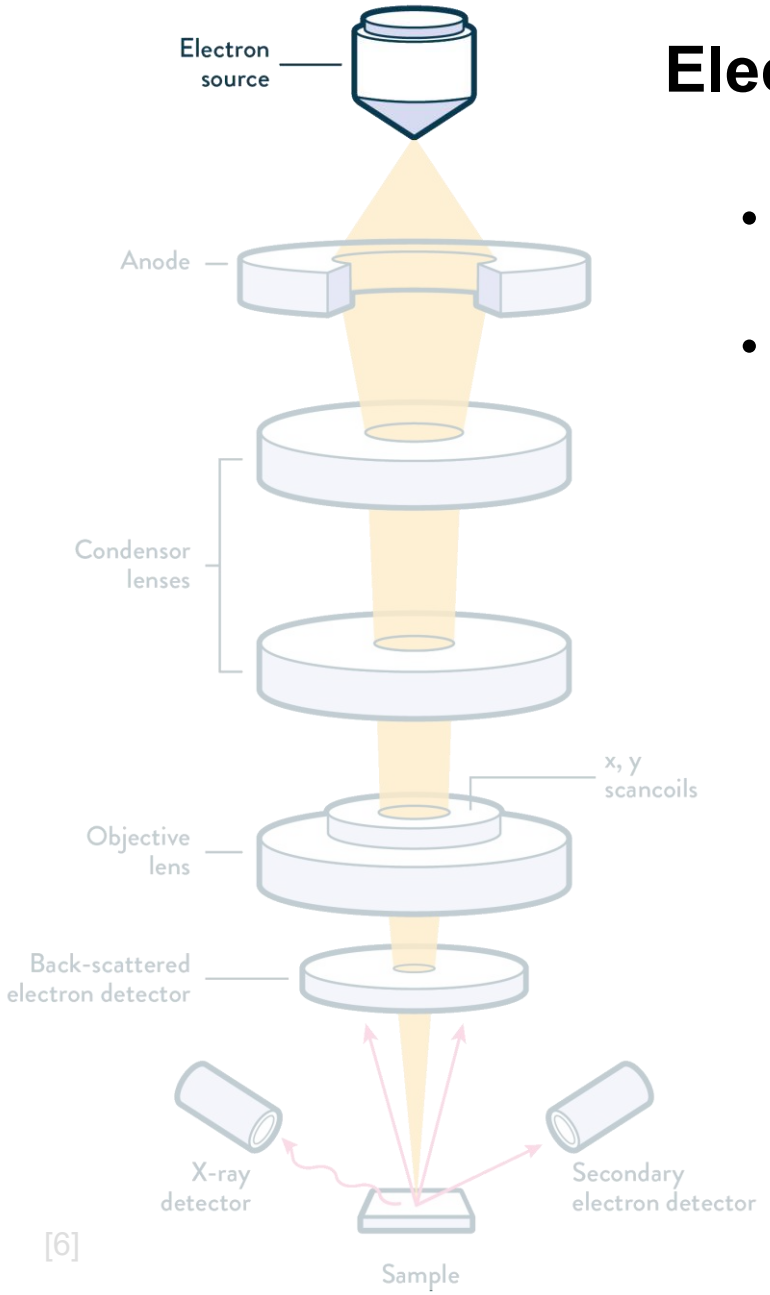
[6]

Components

- Electron gun is the source of electrons with required parameters
- Condenser lens converge the electron beam emitted from electron gun into finer beam
- Deflection coils are used to scan the electron beam in X and Y directions and change size of the area to be scanned (magnification)
- Objective lens converges electron beam into a fine beam and focus it on the sample surface.
- Detectors for capturing signal for each pixel
- Synchronization of data acquisition and scanning system to receive an 2D images.

Fundamentals of SEM

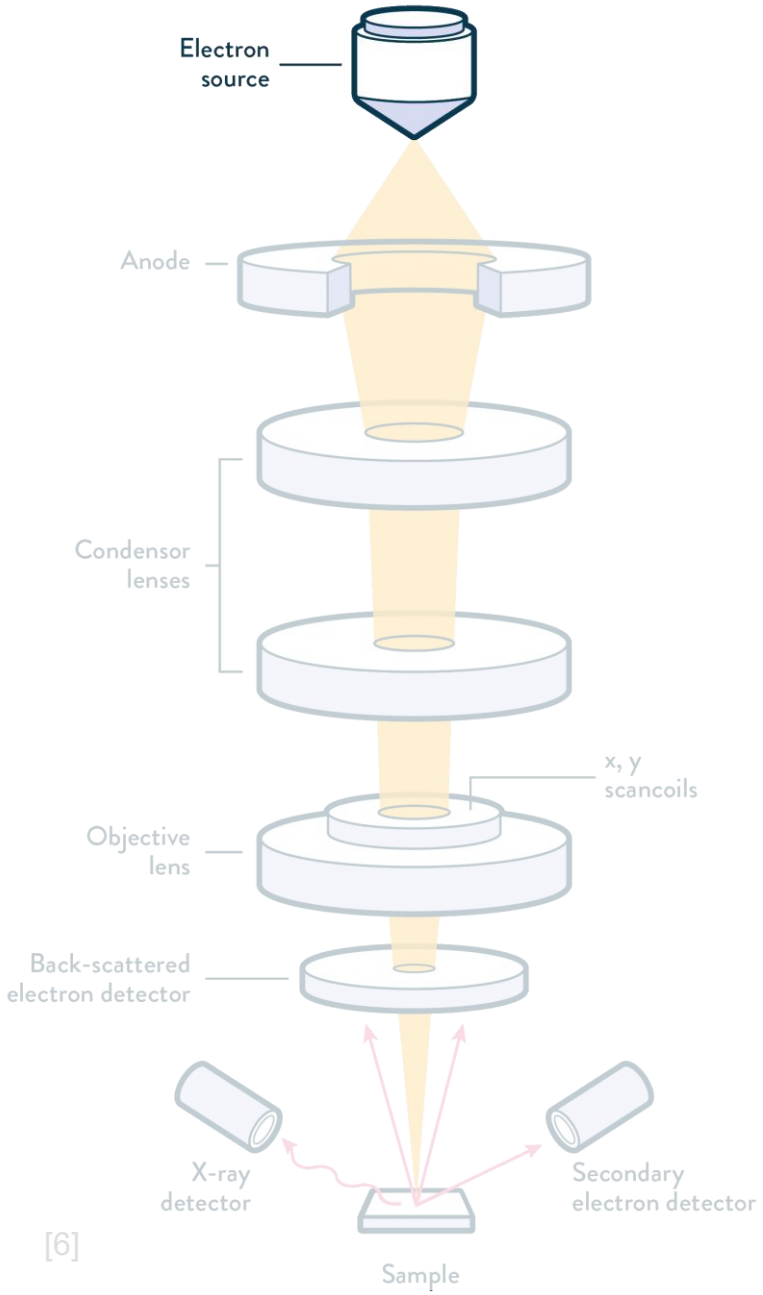
1. Accelerating Voltage (0.5 to 30 kV)
2. Probe Current (2 nA to 10 pA)
3. Working Distance



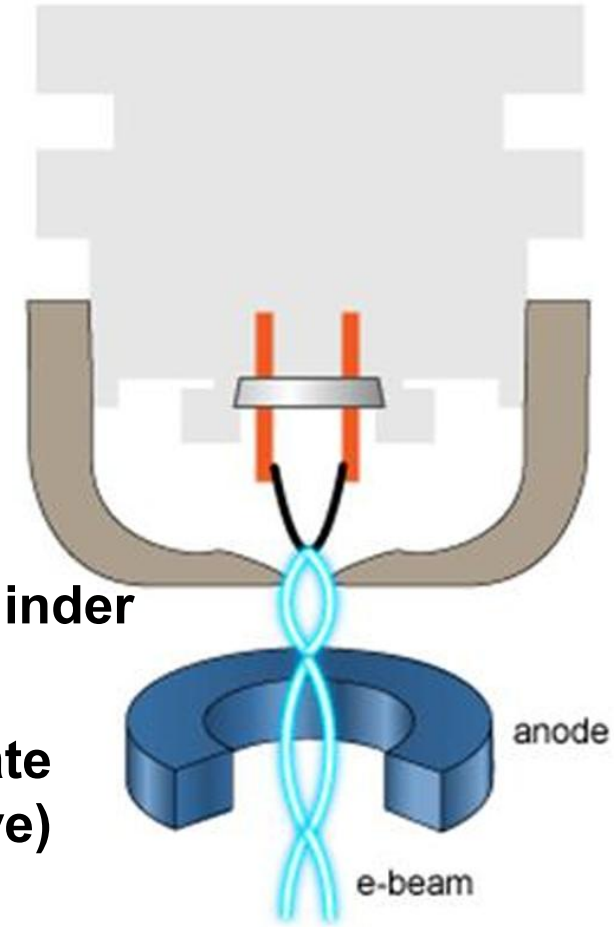
[6]

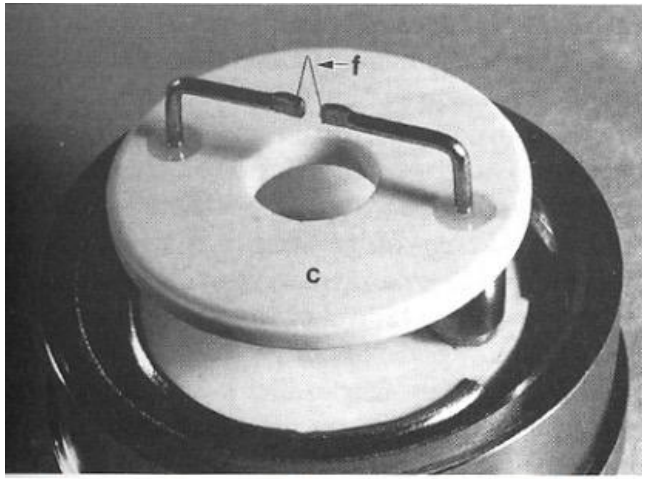
Electron Source/Gun

- **Filament (E-beam source)**
- **Wehnelt cylinder (Cover)**

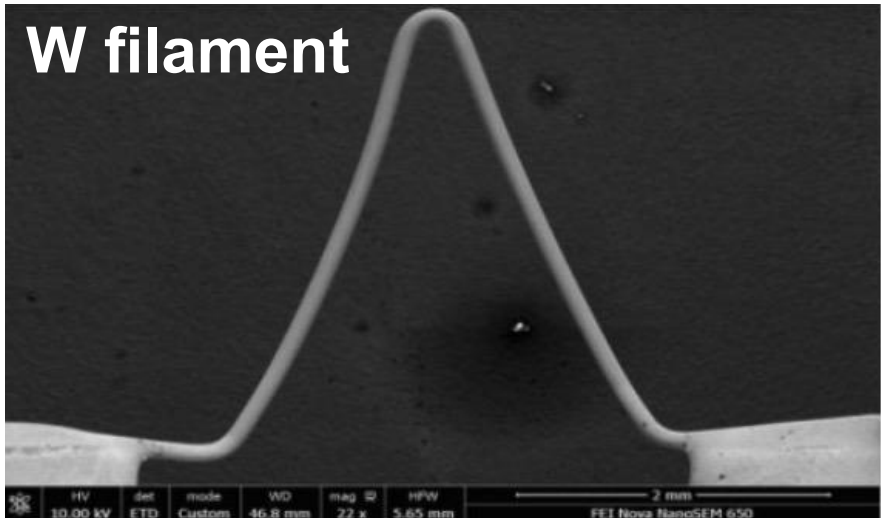

Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample

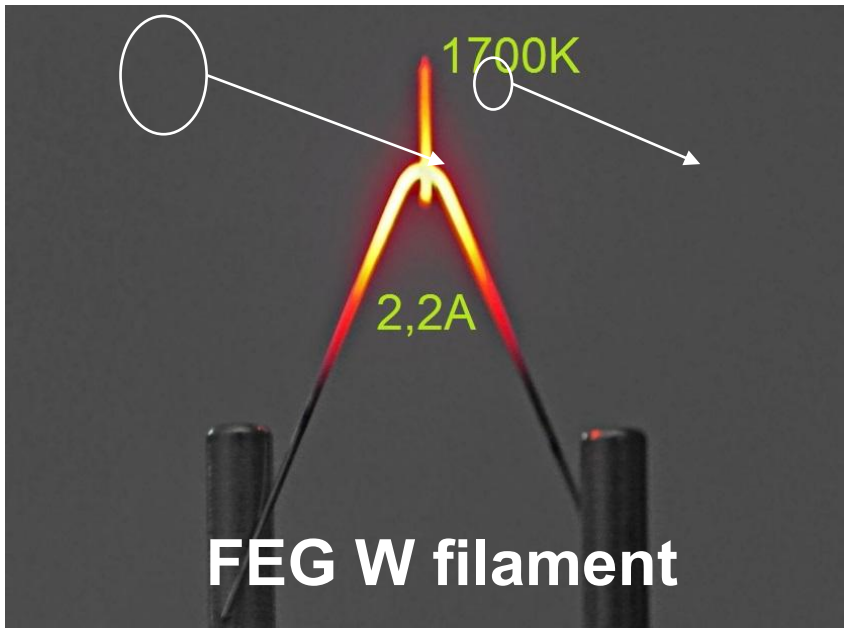
[6]


Wehnelt cylinder
anode plate (+ve)
anode
e-beam

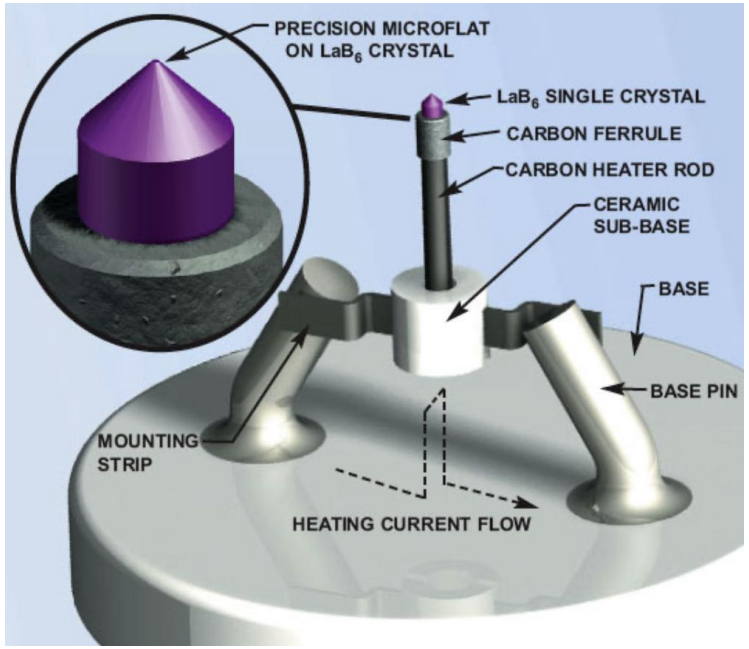


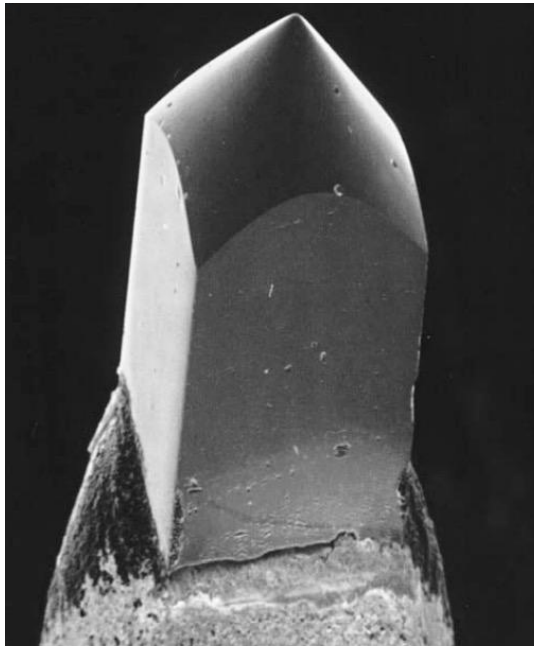
Bozzola and Russell, Fig. 6.22

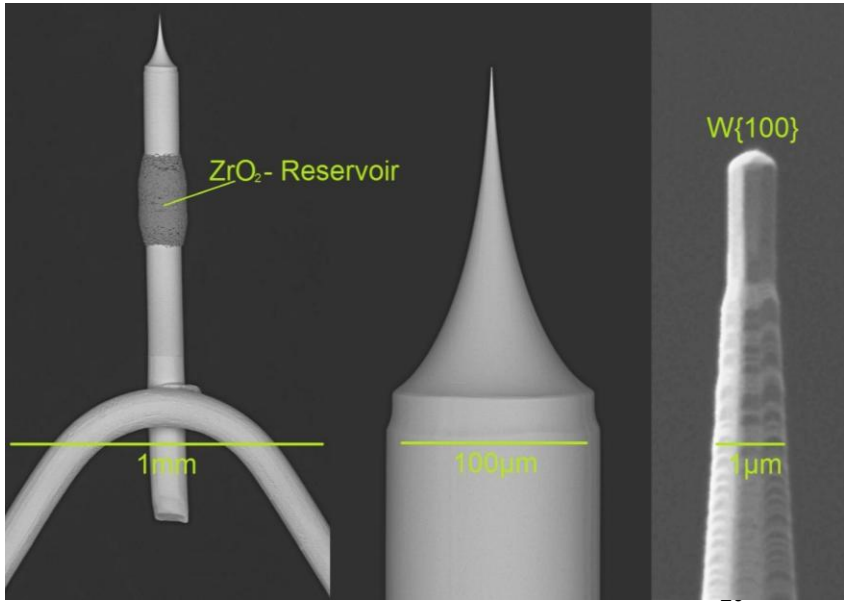




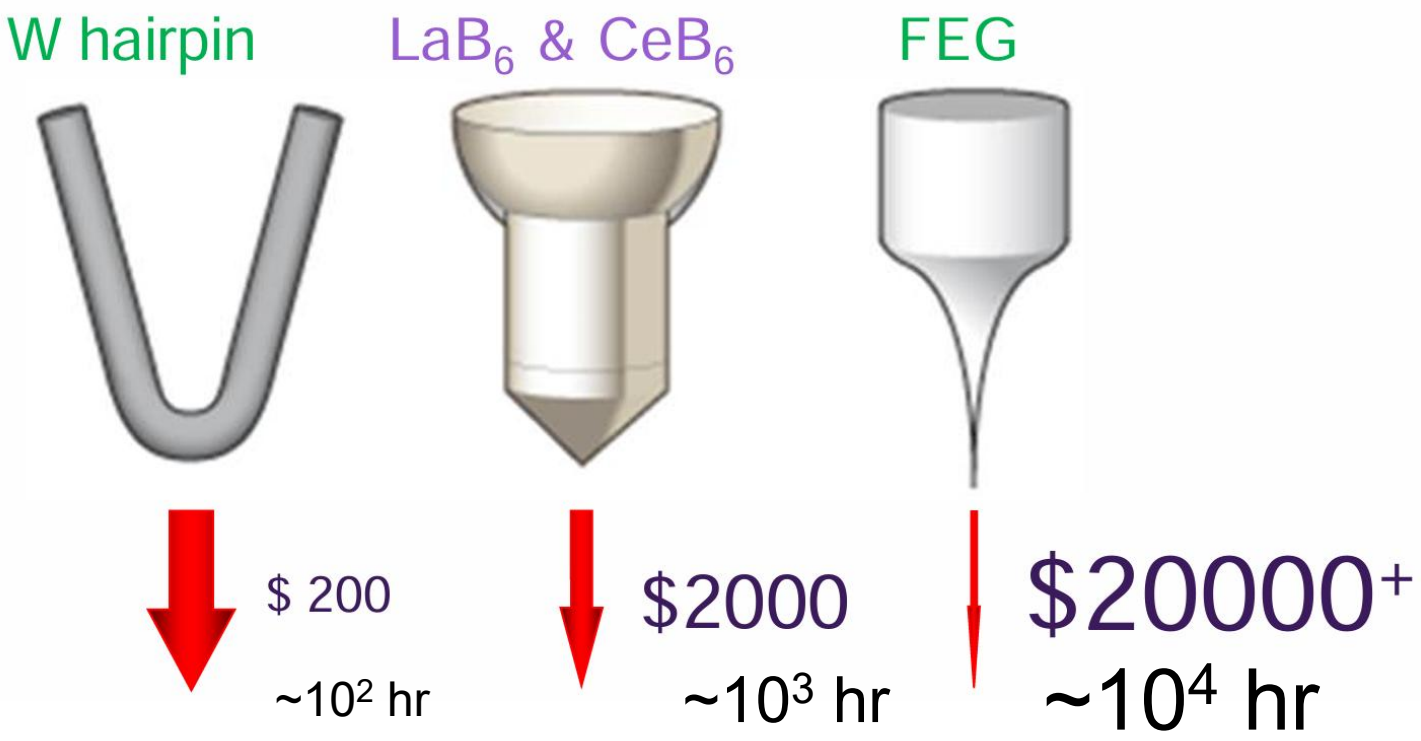
LaB_6 filament

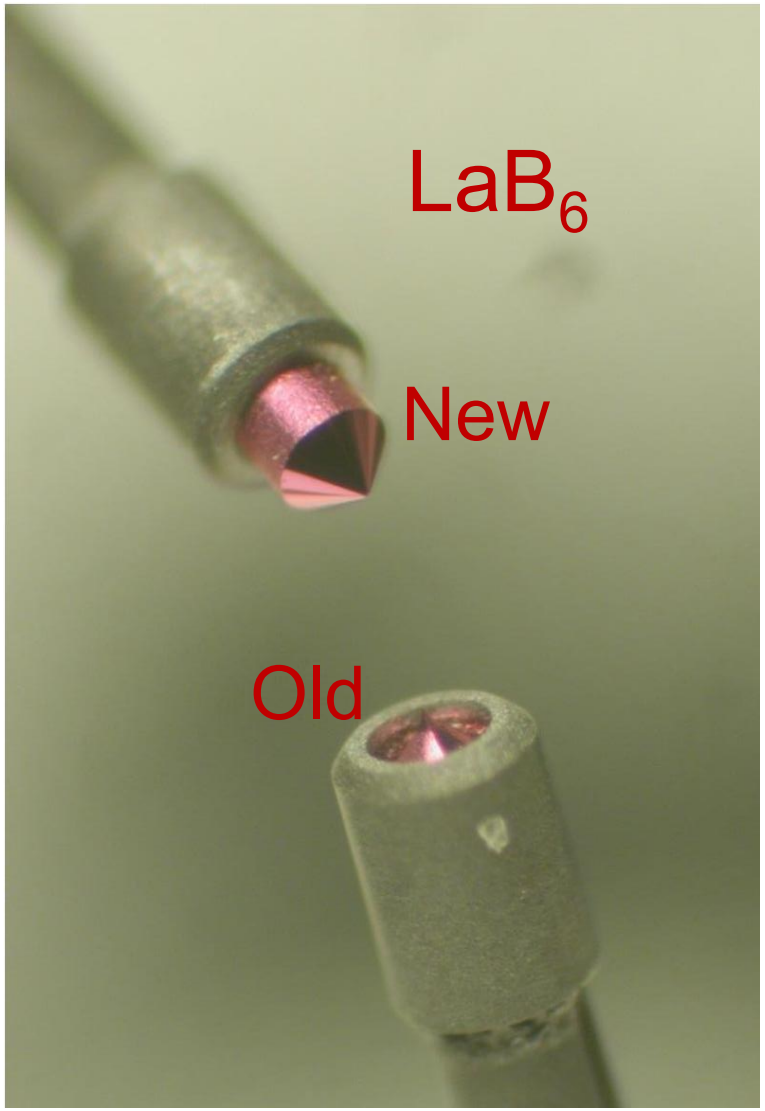


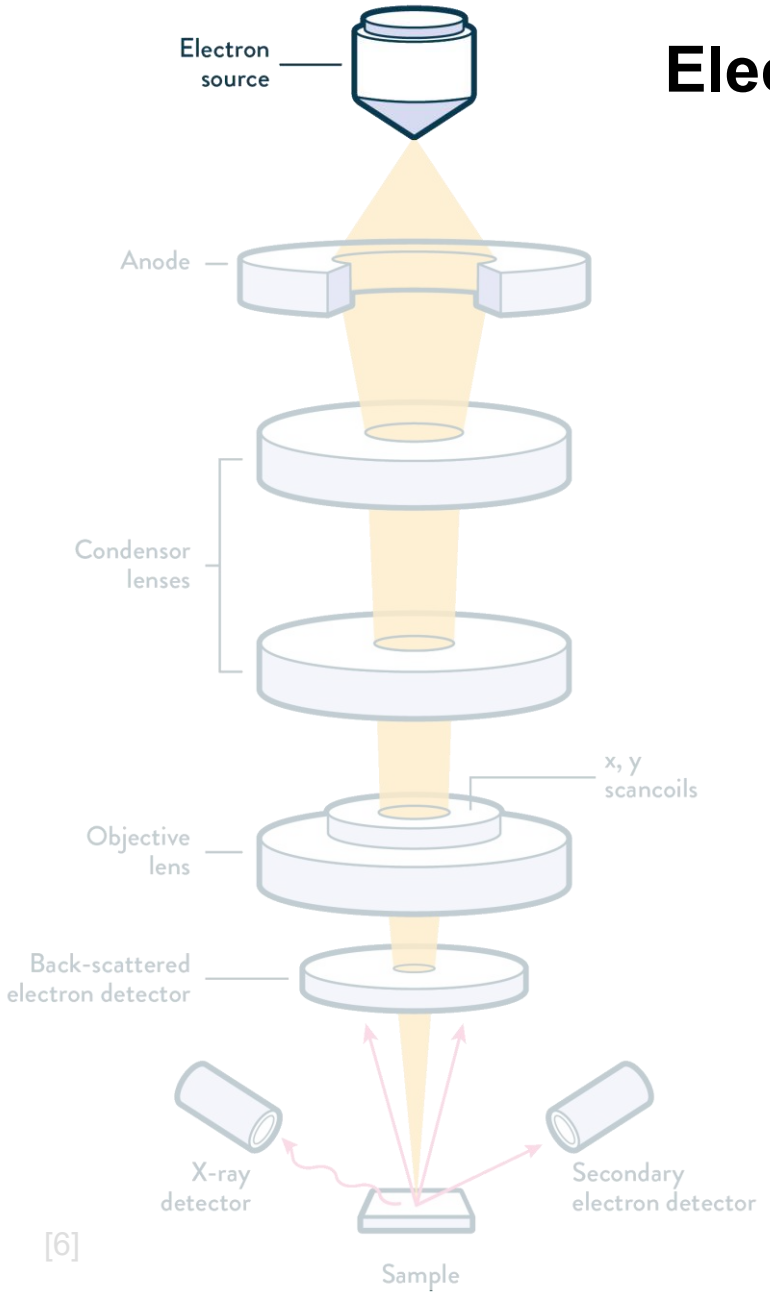




Filament types



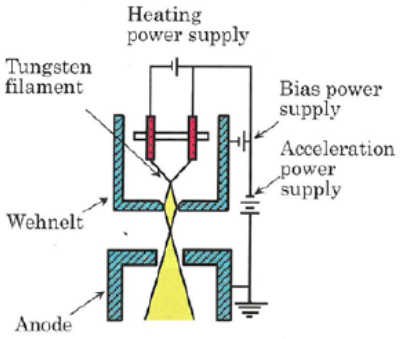




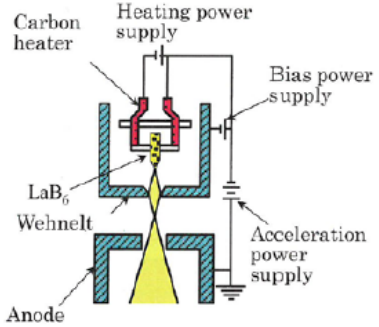
[6]

Electron Source

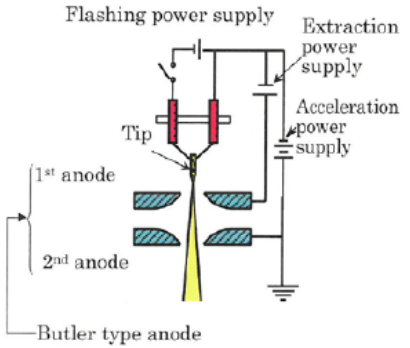
Tungsten



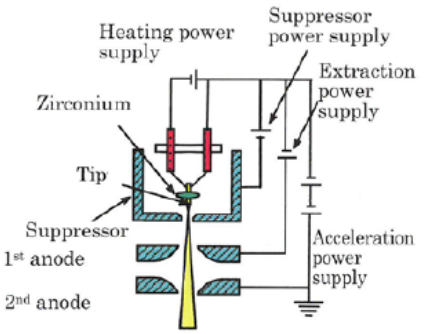
LaB_6



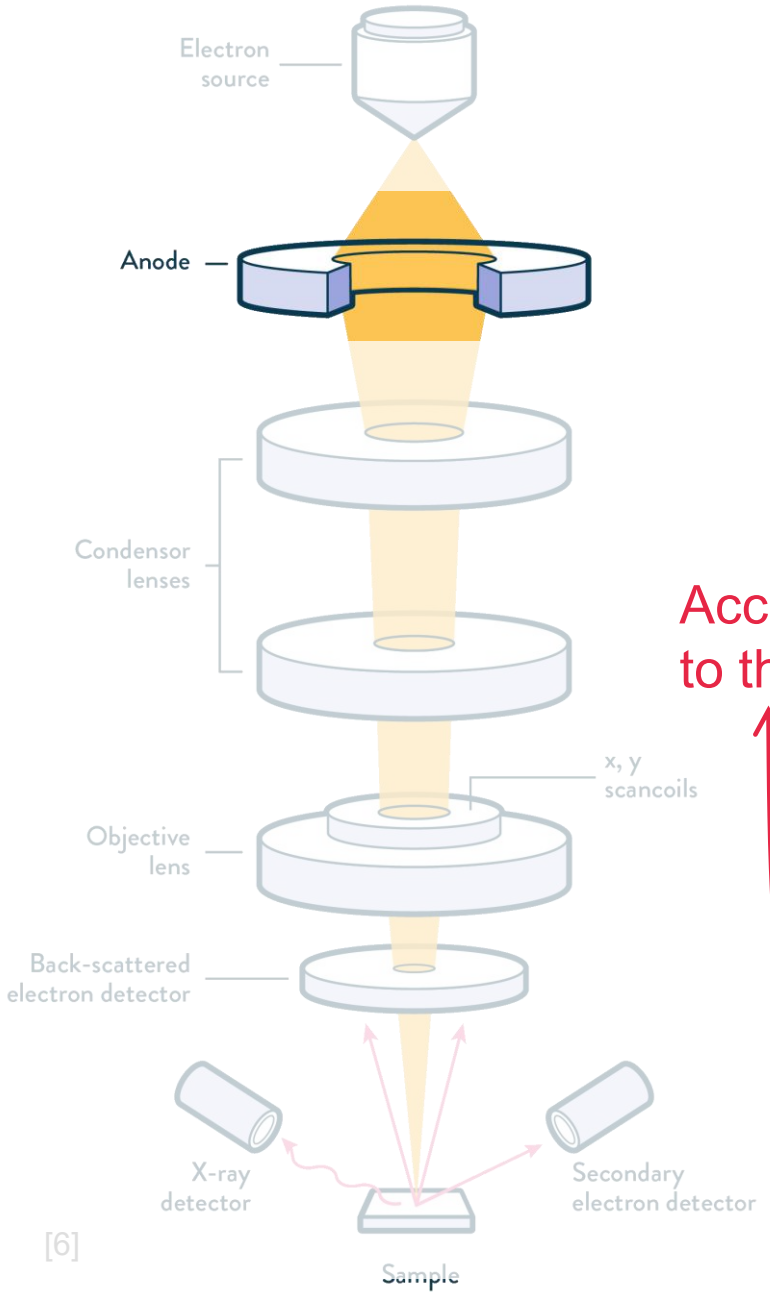
Cold field-emission



Schottky



	Tungsten	LaB_6	CFE	Schottky
Material	Tungsten hairpin	Lanthanum hexaboride single crystal	Tungsten single crystal	Tungsten single crystal/ zirc. oxide
Cathode temp	~2300 °C	~1500 °C	RT °C	~1500 °C
Brightness	10^6 A/cm^2·sr	10^7 A/cm^2·sr	10^9 A/cm^2·sr	10^8 A/cm^2·sr
Energy width	~2 eV	~1.5 eV	0.2-0.3 eV	0.3-1 eV
Max. probe current	~100 nA	~100 nA	~10 µA	~100 nA
Working pressure	10^{-4} Pa	10^{-5} Pa	10^{-8} Pa	10^{-7} Pa
Lifetime	50-100 h	200-1000 h	2-4 y	2-3 y



[6]

Anode Plate (+ve charge)

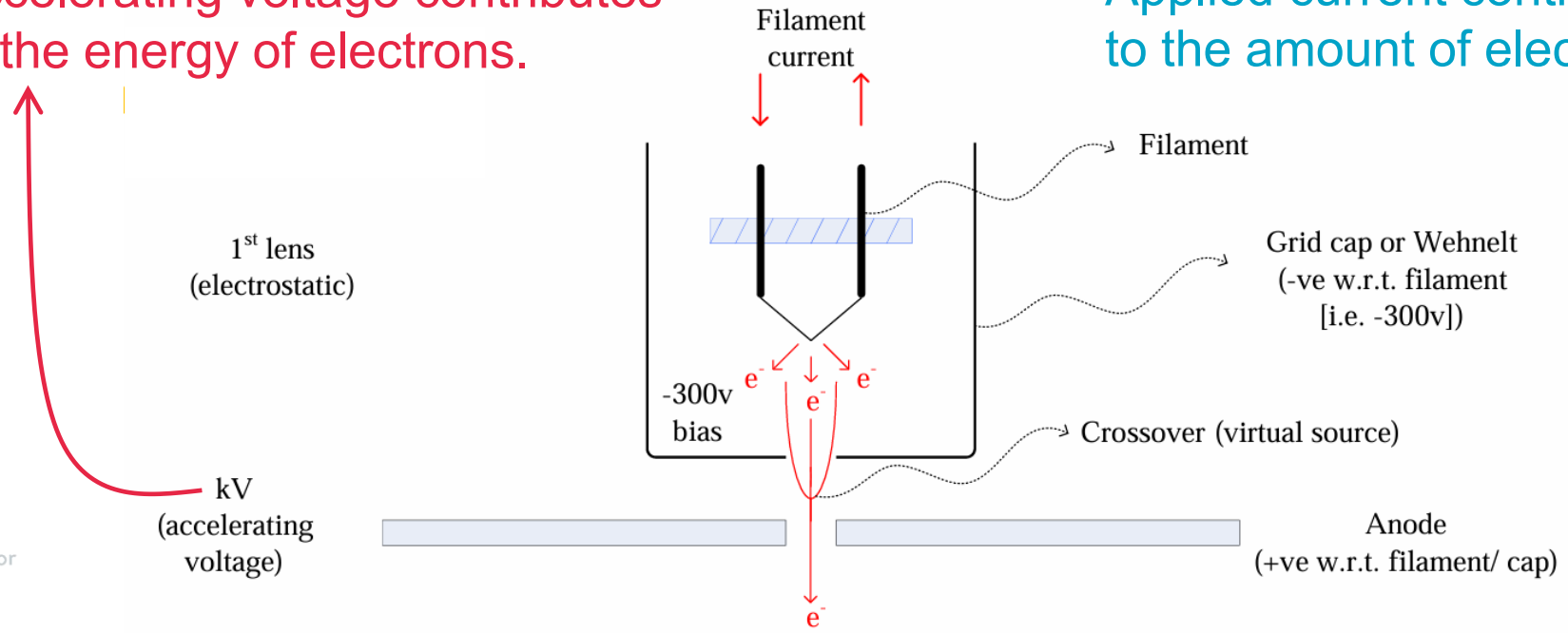
- The electrons are accelerated away from the filament by an applied voltage.
- This voltage can go up to around 30 kV in a typical SEM.

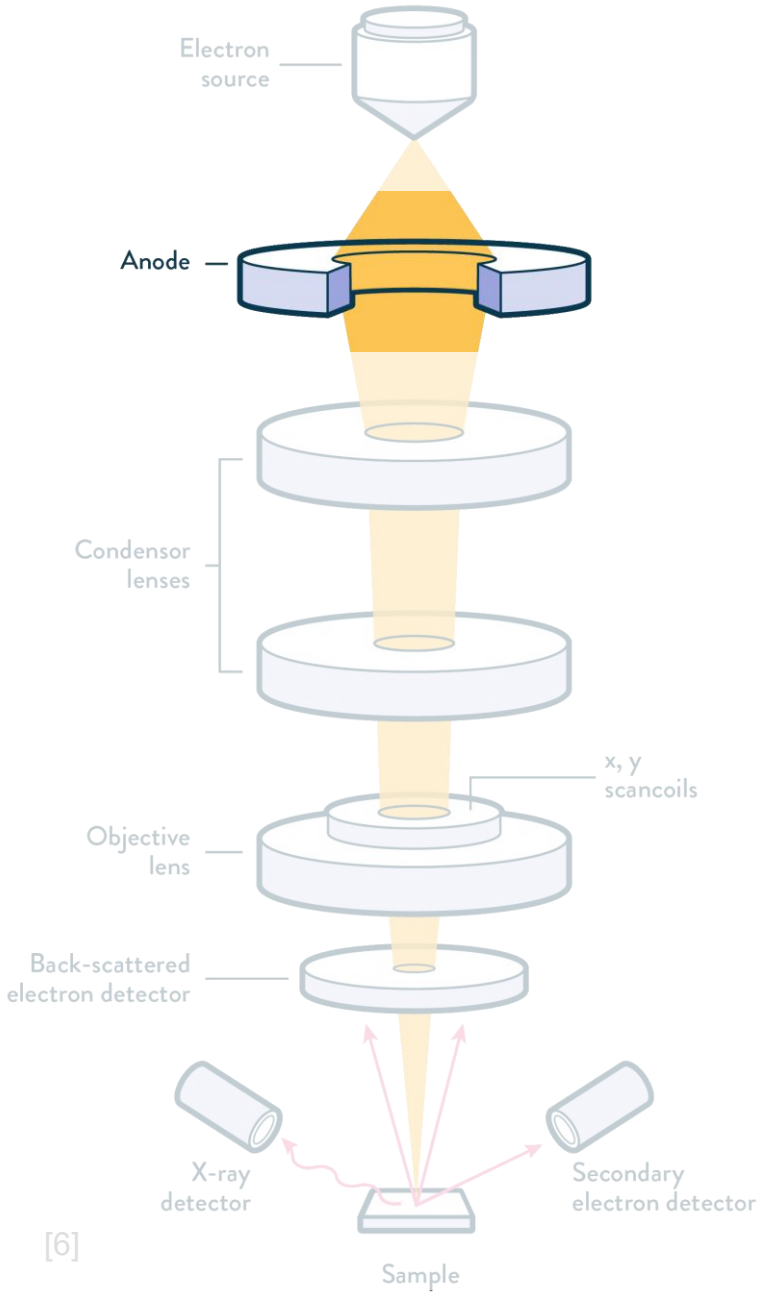
The Electron Gun

= Filament + Grid cap + Anode

Accelerating voltage contributes to the energy of electrons.

Applied current contributes to the amount of electrons.

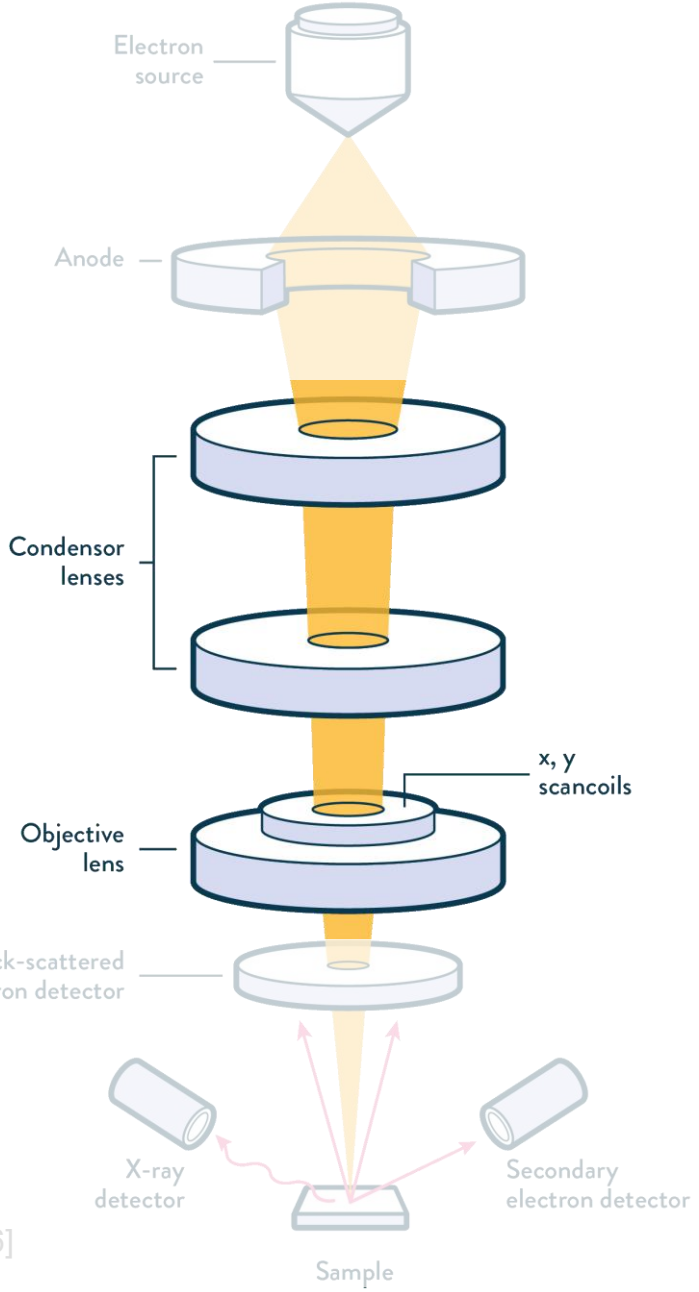




[6]

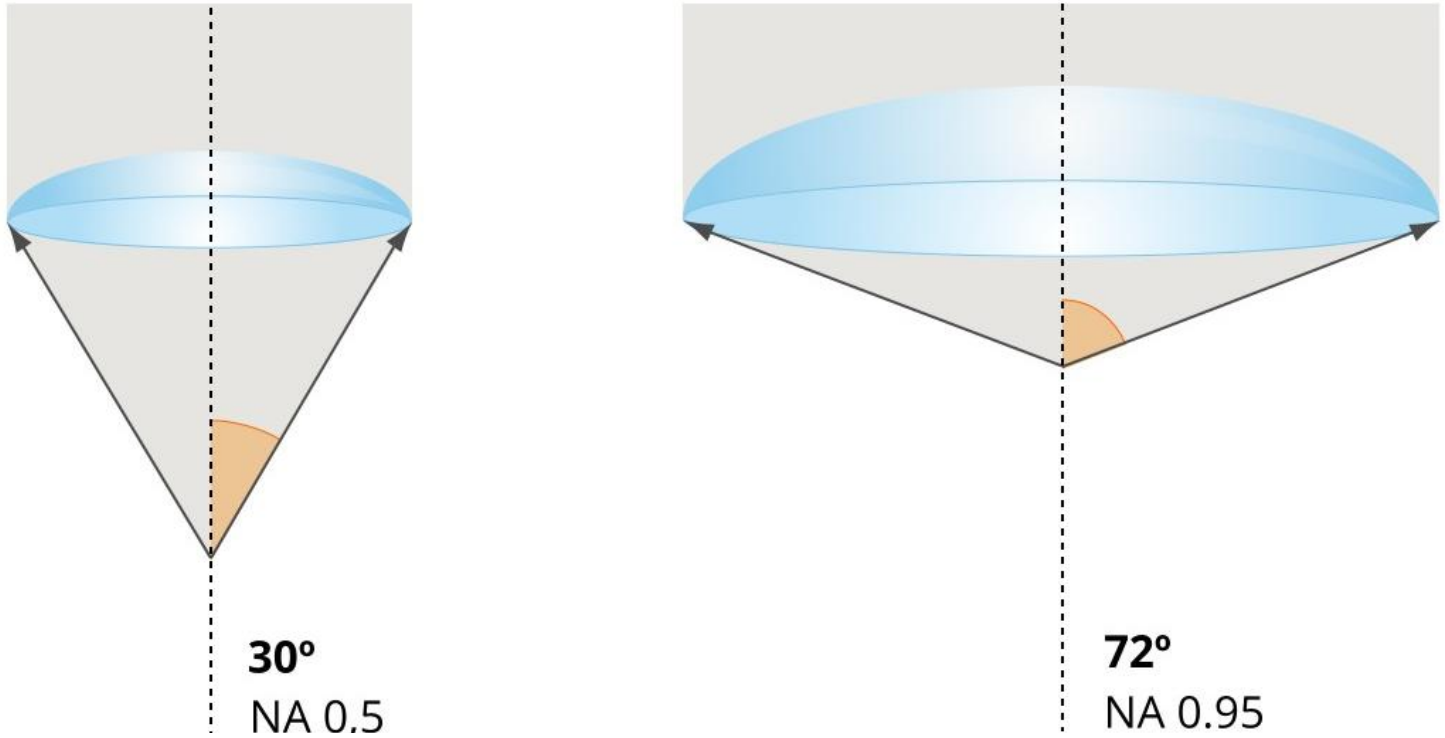
Anode Plate (+ve charge)

- The anode is **positively charged** and **accelerates electrons** emitted from the electron gun toward the specimen.
- **Controls Beam Energy:** It helps define the energy (accelerating voltage) of the electron beam, which influences resolution and penetration depth.
- **Guides Electron Trajectory:** Positioned before the condenser lenses to ensure the beam enters the lens system in a controlled, focused path.
- **Reduces Beam Divergence:** Helps narrow the electron beam for better focusing and improved image resolution.
- **Protects Gun Components:** Keeps high voltage away from sensitive components of the electron source.
- **Enables Stable Beam Generation:** Provides a stable potential difference between the cathode and the anode, essential for consistent beam current.

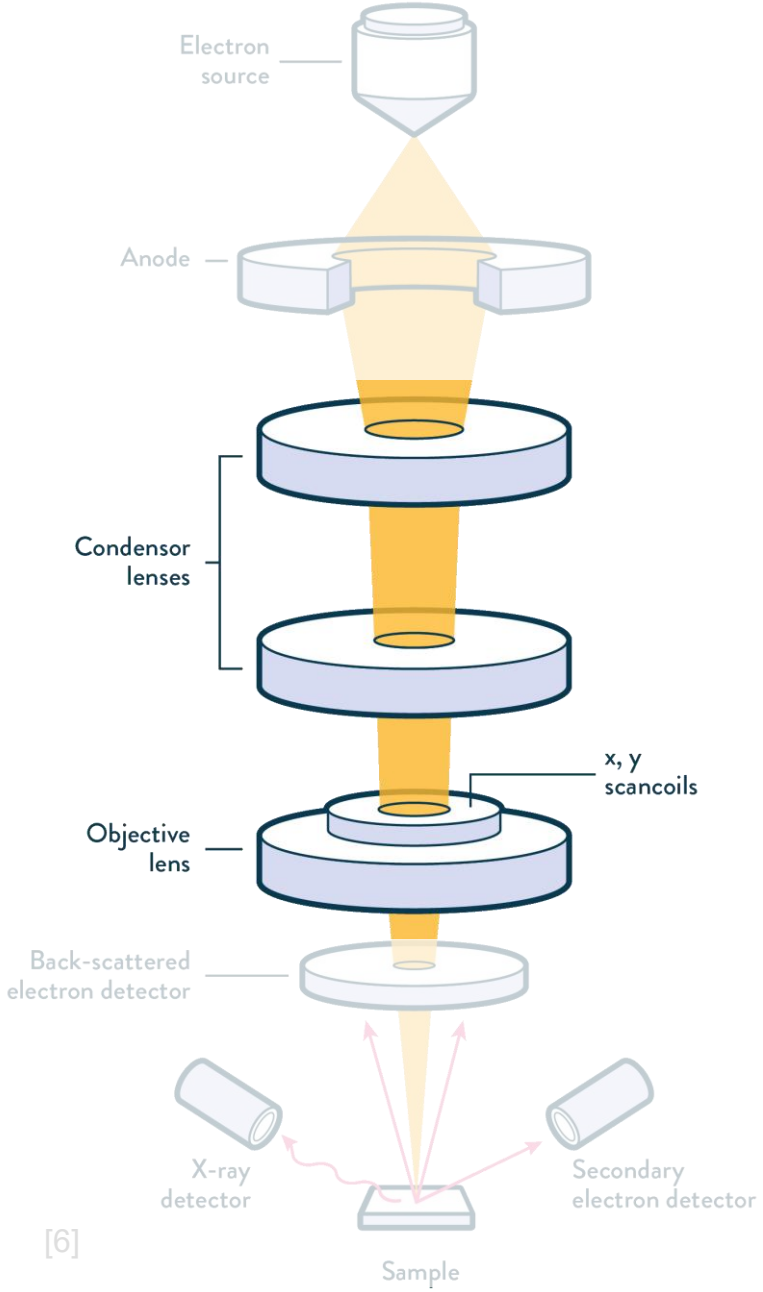


[6]

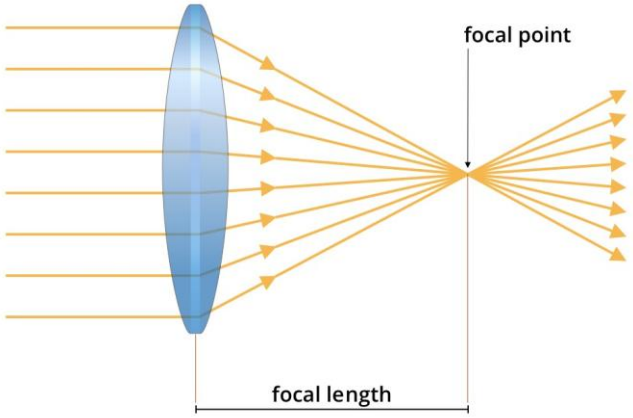
Lens

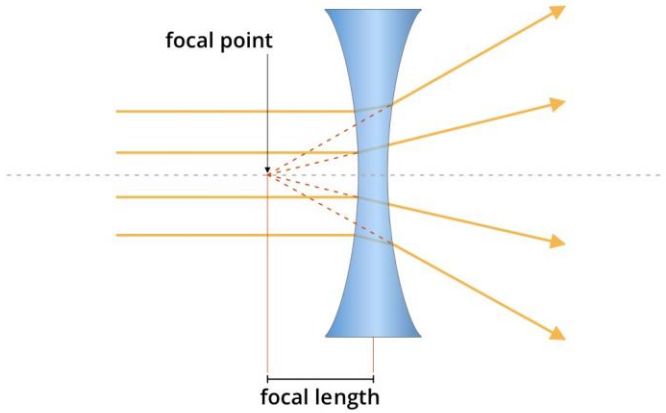


- For visible light microscopy, lenses are manufactured so they can gather more or less light.
- The relative amount of light gathered by a lens is known as its numerical aperture (NA).
- The greater the NA number, the more light it gathers and the greater the resolution it gives.

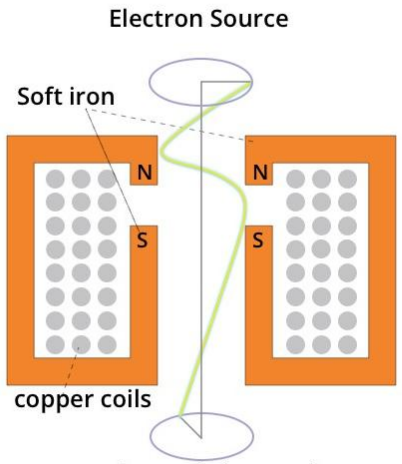


[6]

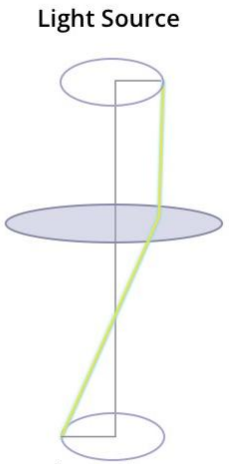


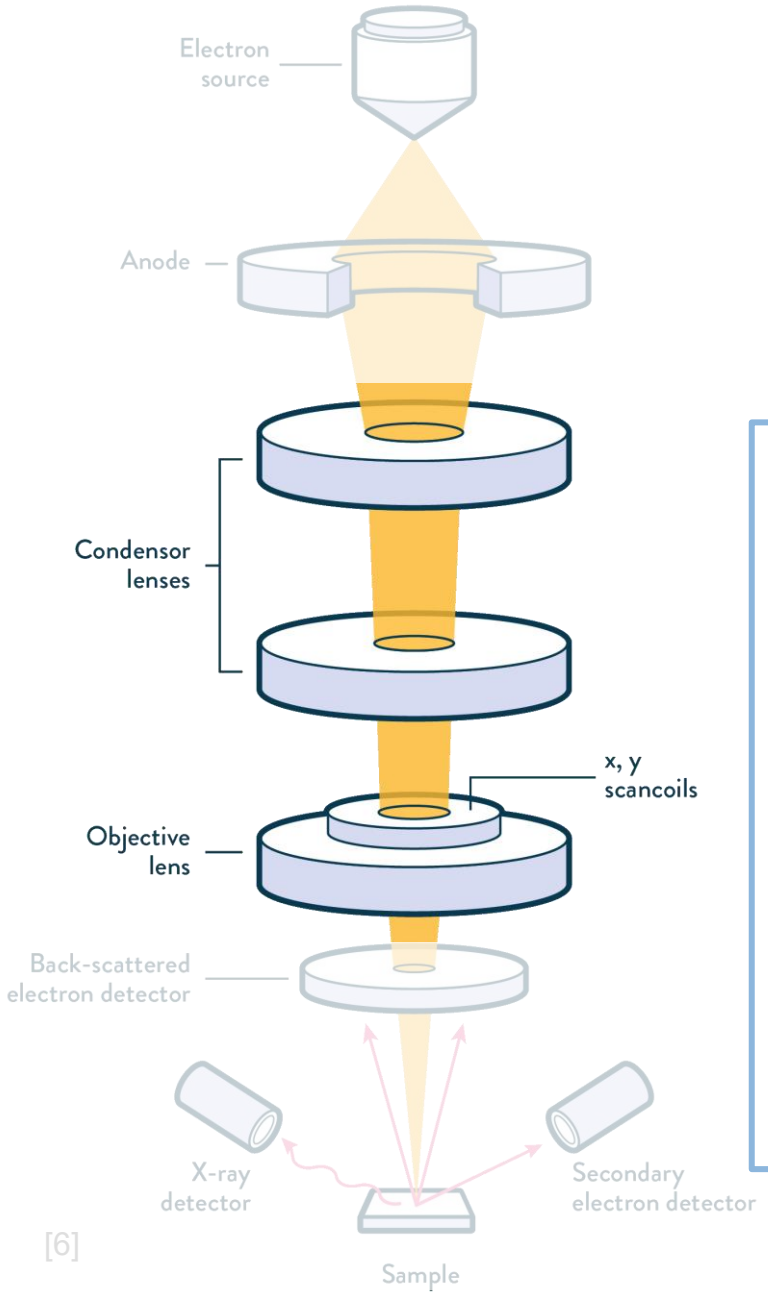


Magnetic lens



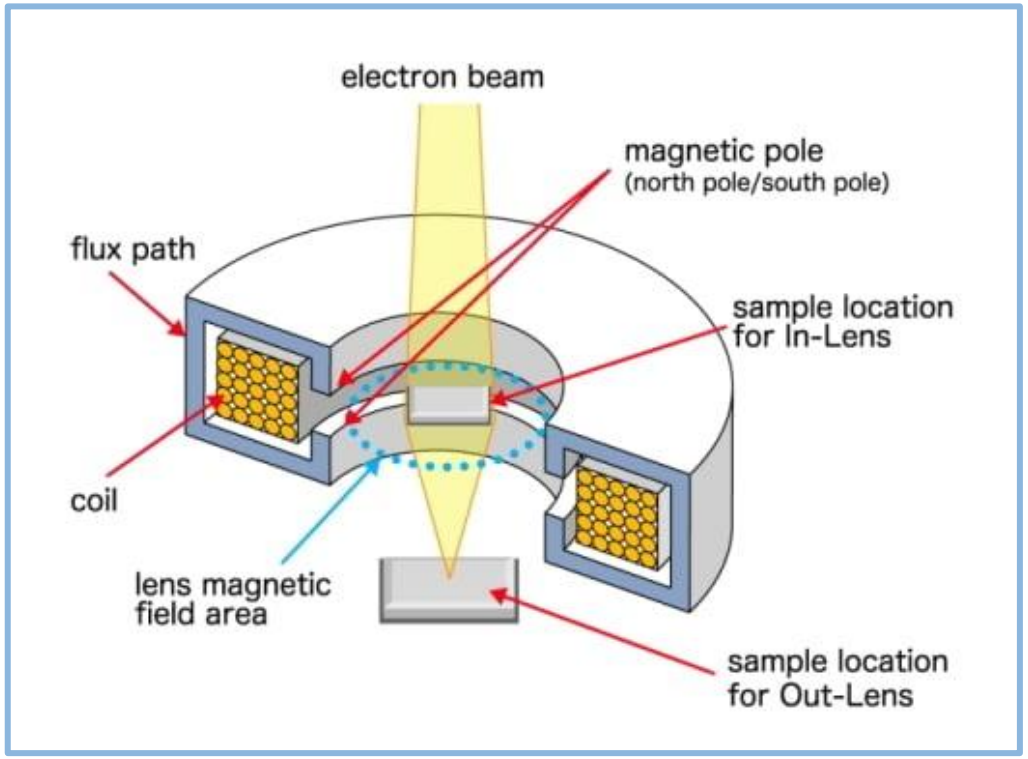
Optical lens

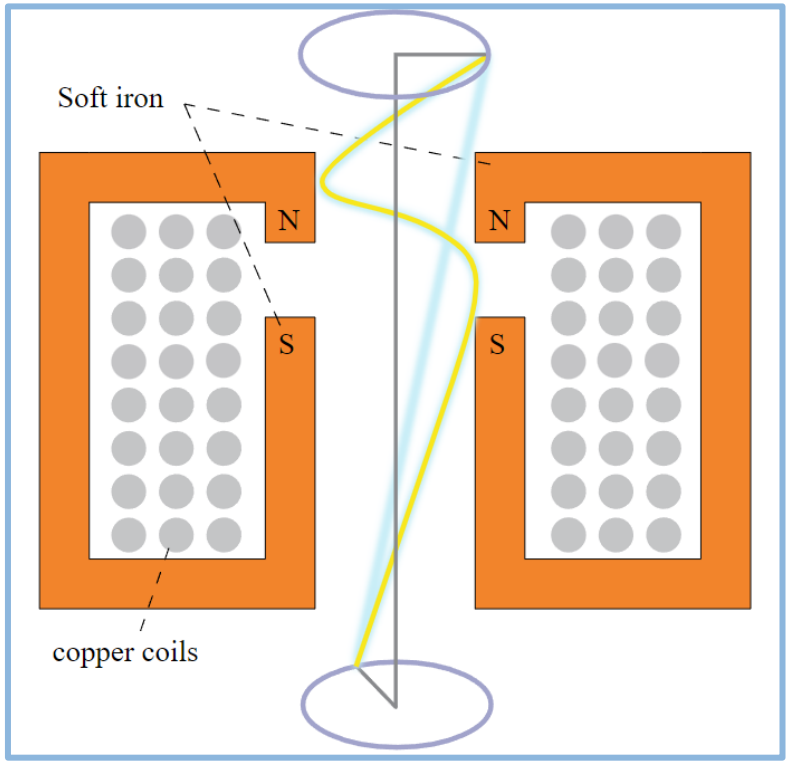


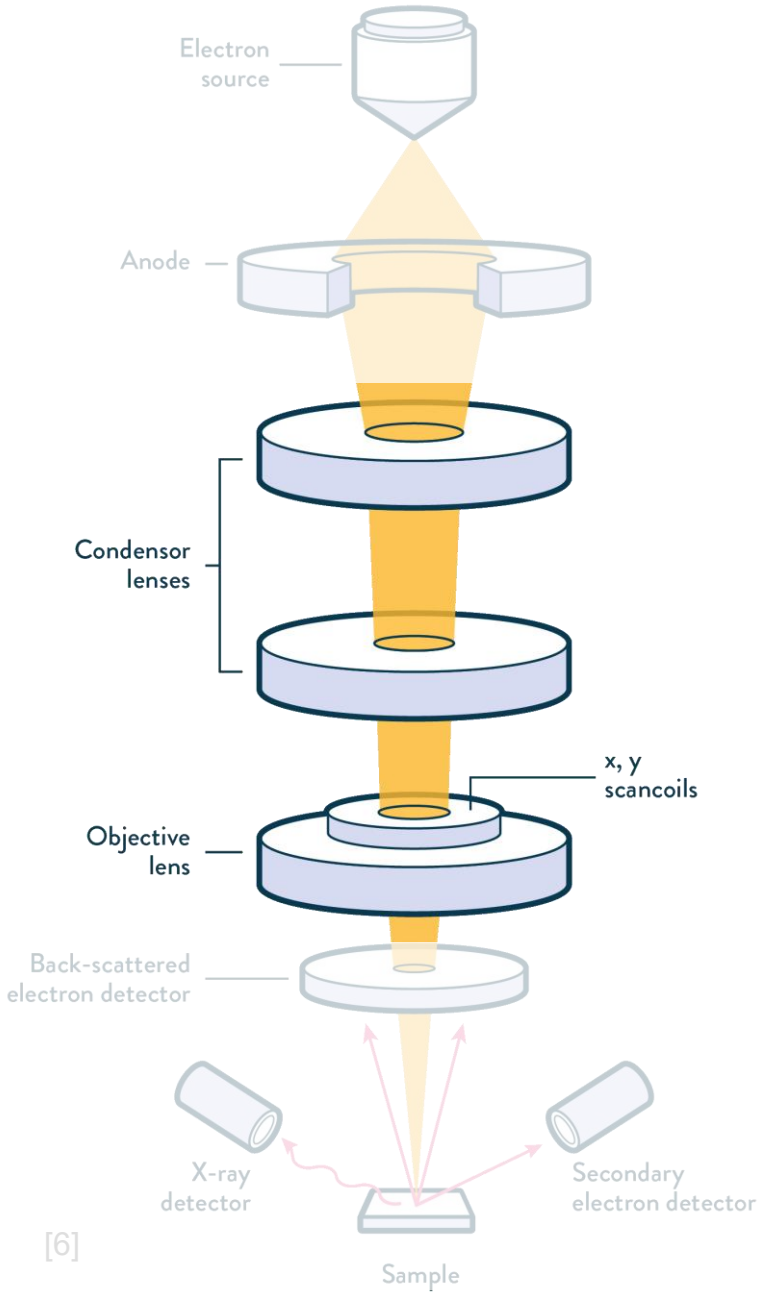


[6]

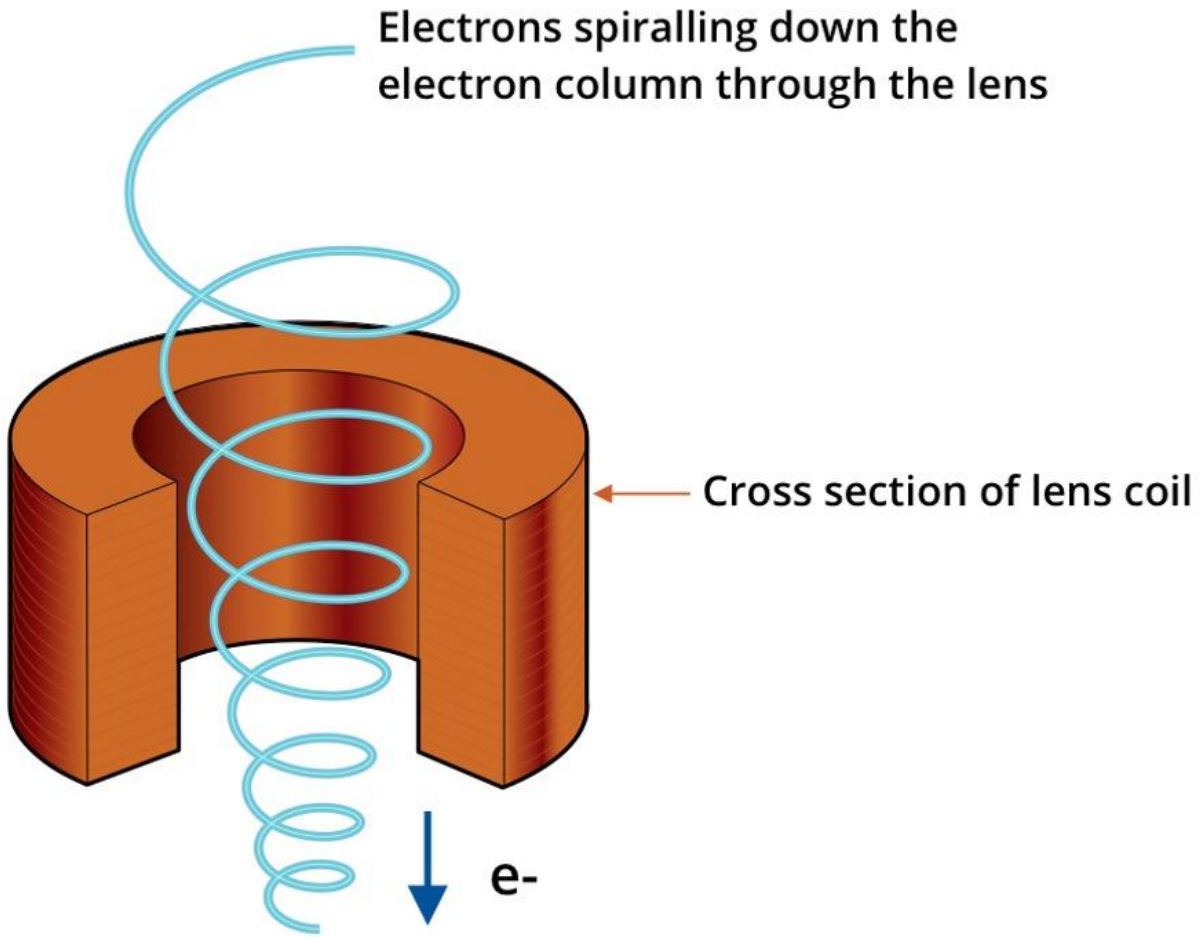
- Electrons pass through a condenser lens, and then through an objective lens.
- By varying the current through the coils, the position of the beam can be rastered.

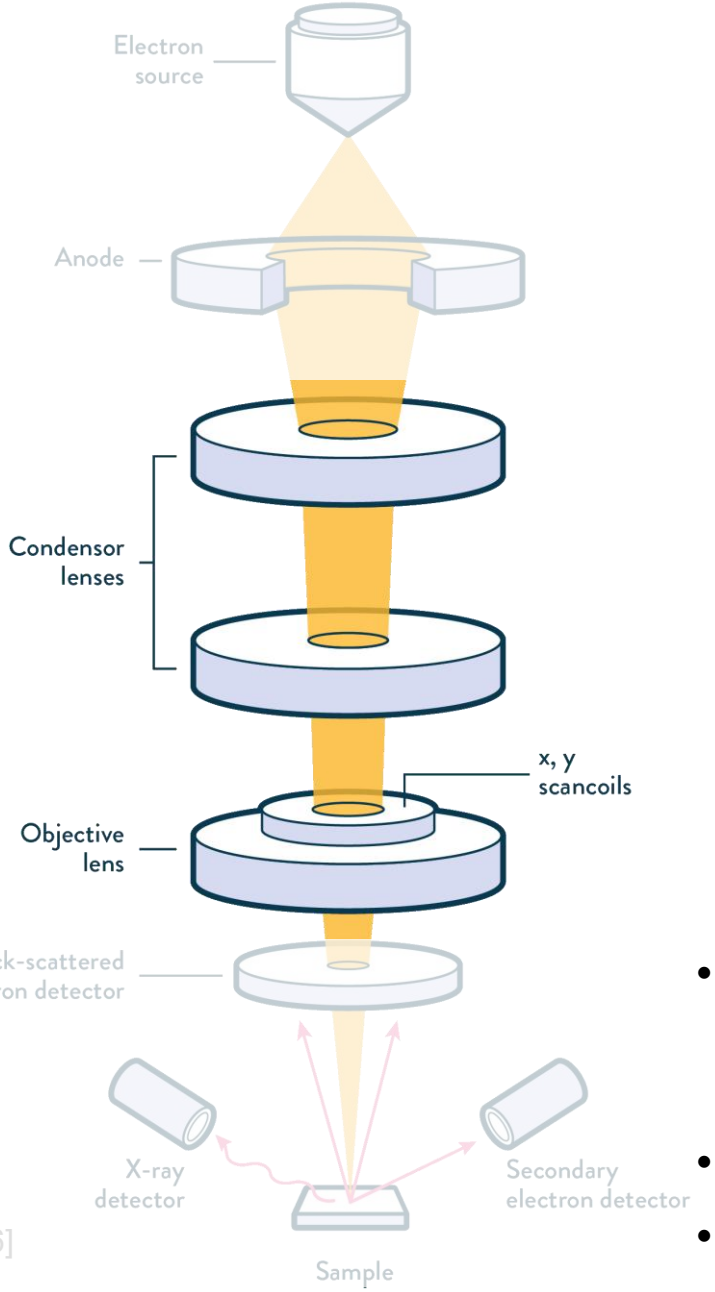





Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample

[6]

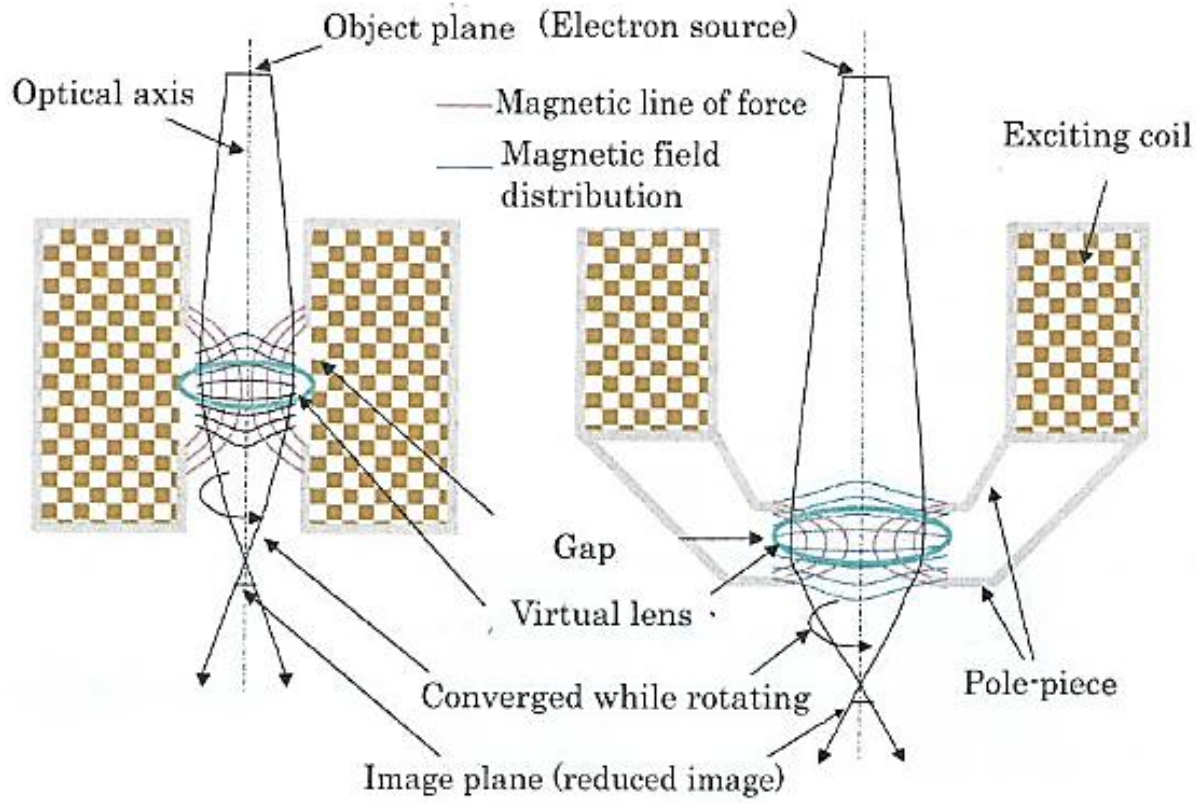

Electrons spiralling down the electron column through the lens
Cross section of lens coil
e-



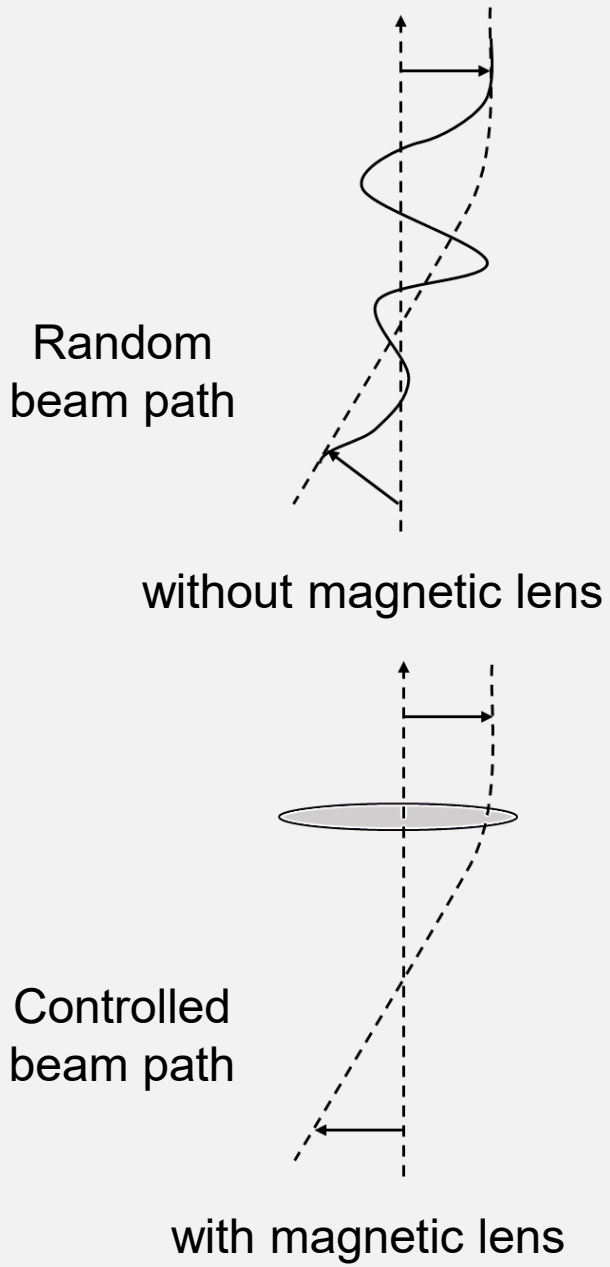
[6]

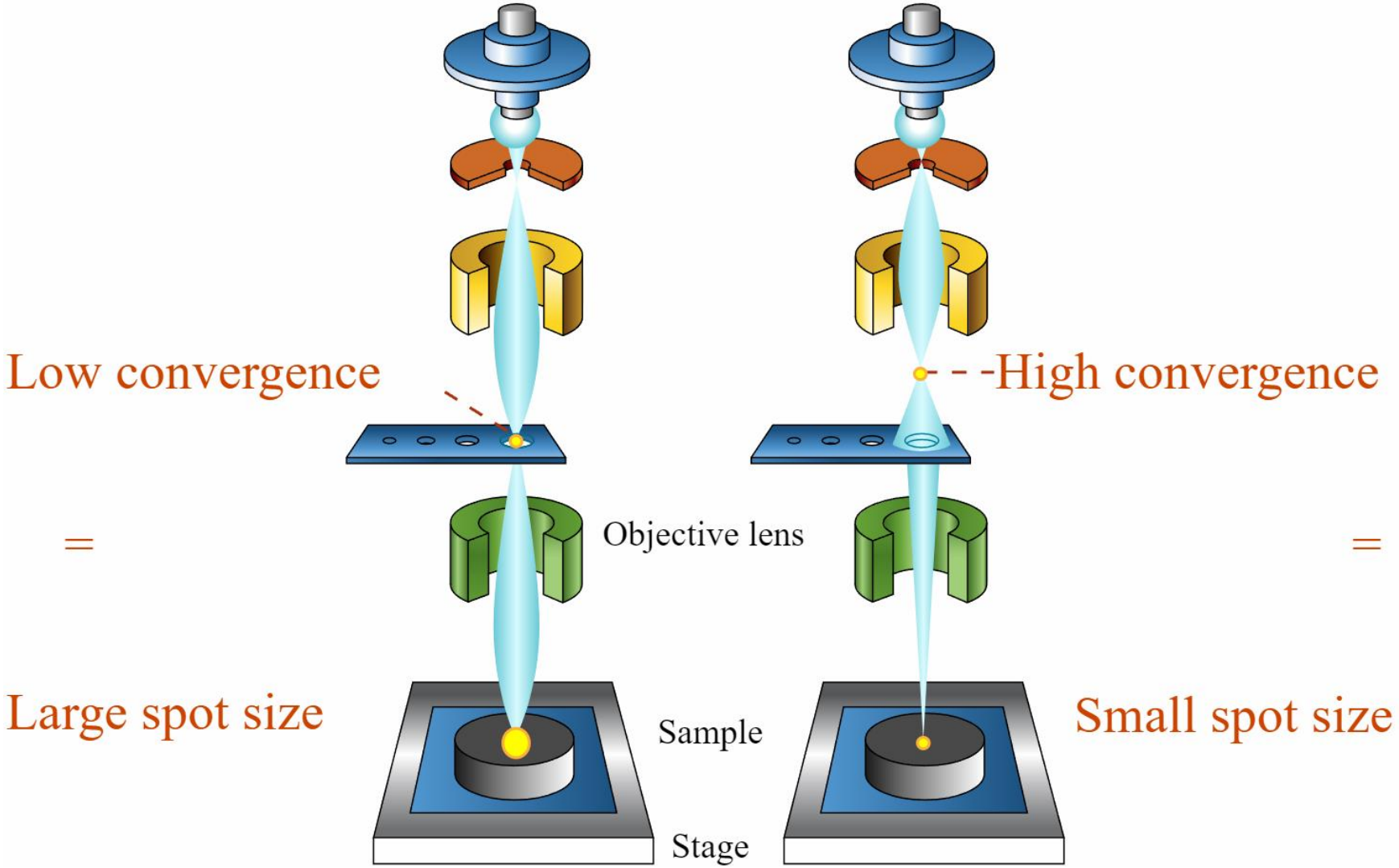
Condenser Lens

Objective Lens

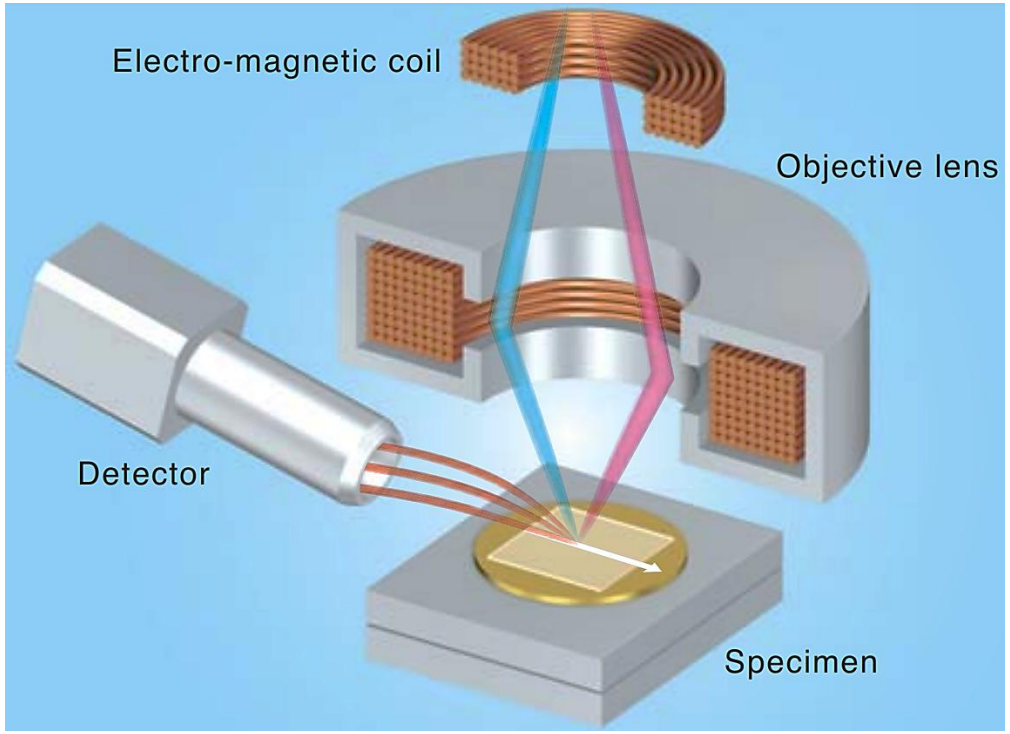


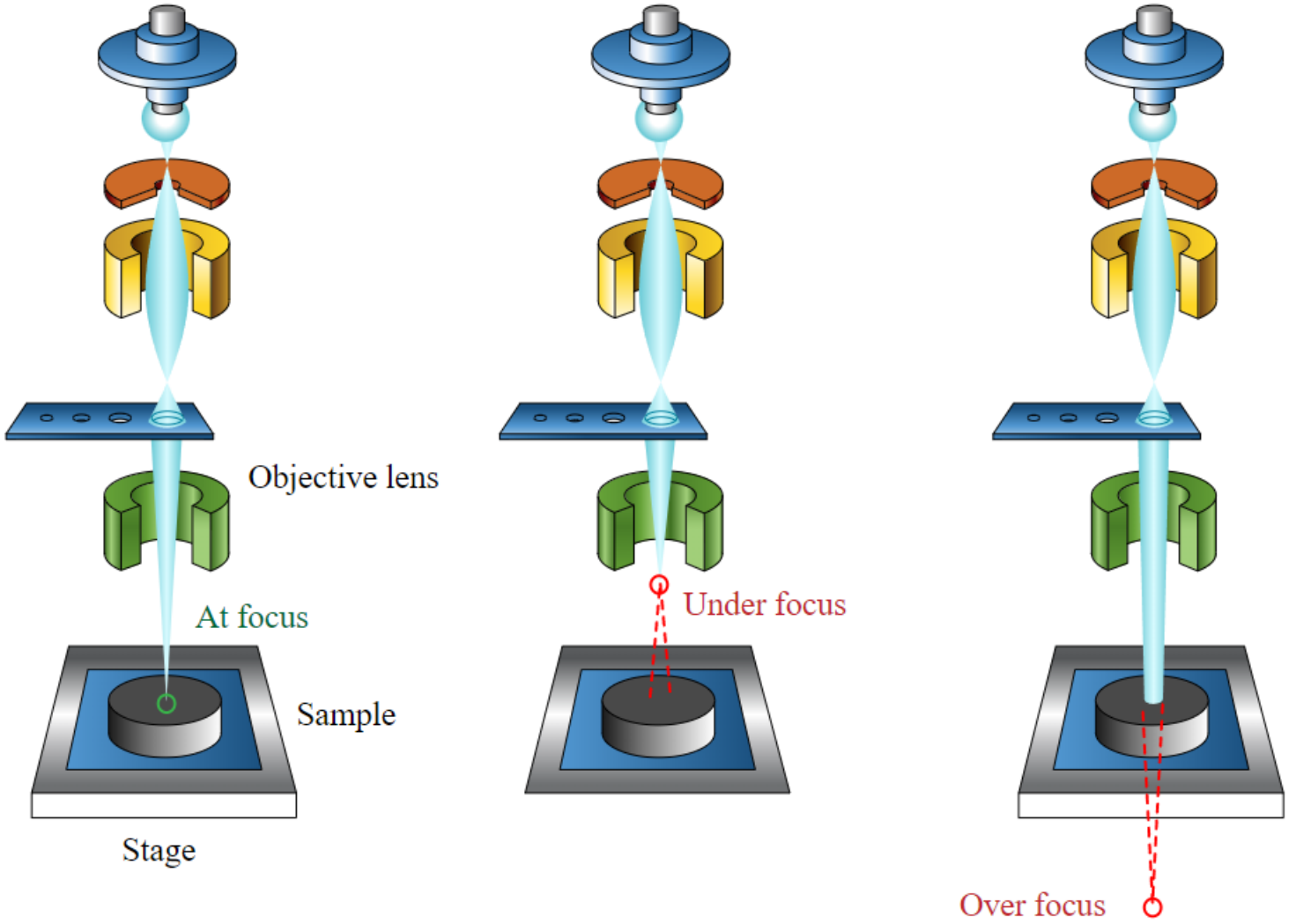
- Electron lenses in SEM: **magnetic/electrostatic lenses** to focus the accelerated virtual electron source from the electron gun into a fine probe onto the sample surface.
- Condenser Lens: **"Shaping"** the electron beam (responsible for its intensity).
- Objective Lens: "**Focusing**" the electron beam into a fine probe.





Condenser lens defines the size of the electron beam (i.e., spot size or probe size).





Objective lens converges the beam again before hitting the sample to focus the beam onto the sample.

CALIBRATION

i. **Beam Alignment**

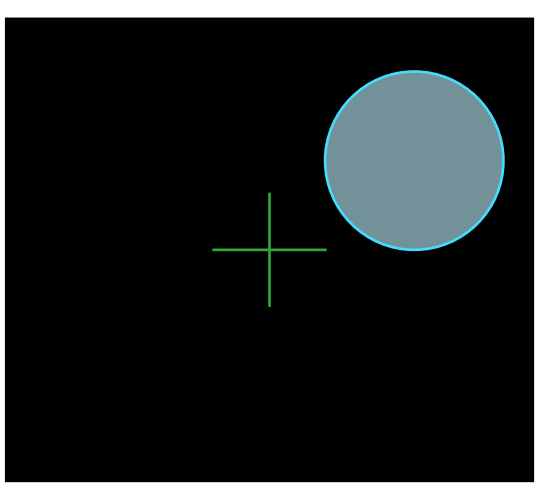
ii. Source Calibration *or* **Chromatic Aberration**

iii. Electromagnetic Lens Calibration *or* **Spherical Aberration**

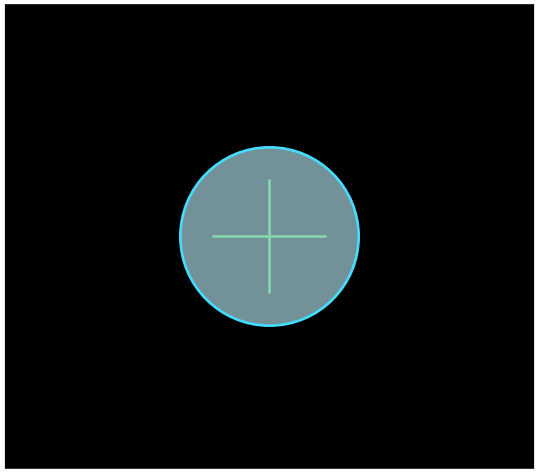
iv. Focal Point Calibration *or* **Astigmatism**

The University Of Queensland AUSTRALIA

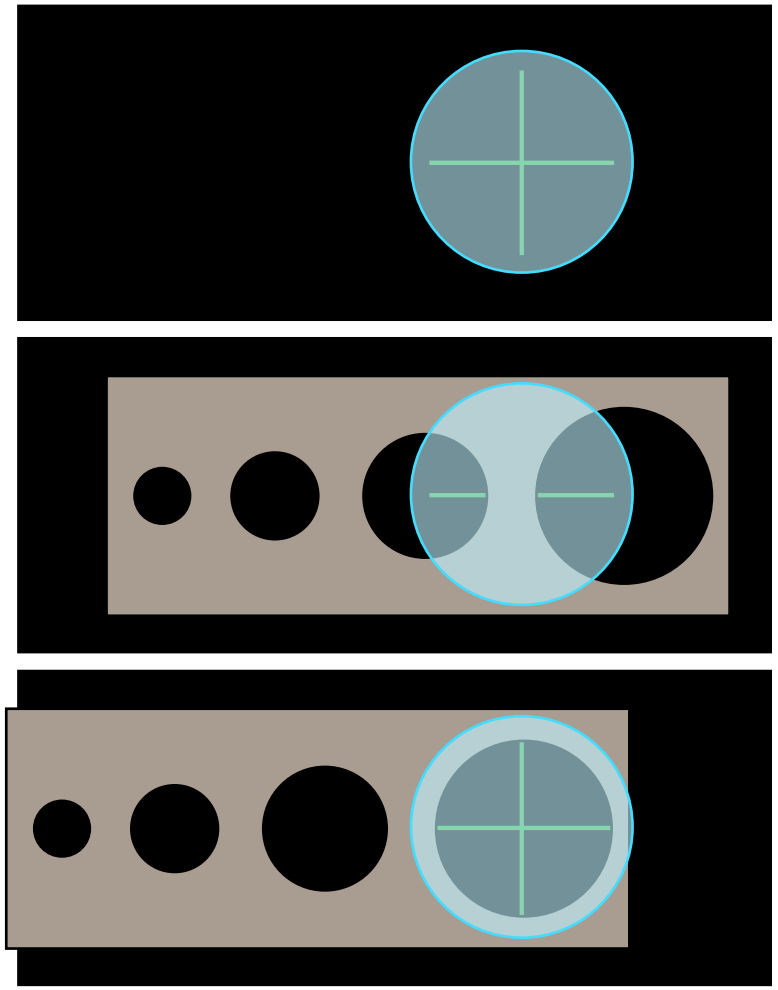
- An electron beam must be aligned first to obtain in-focus images of the sample at high resolution.



Beam off centered



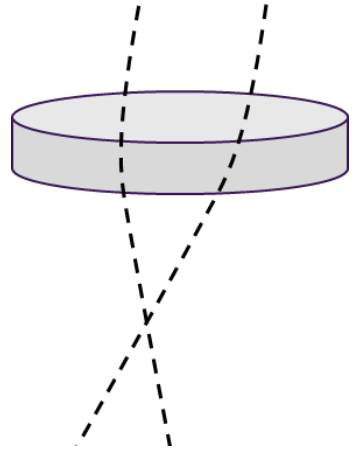
Beam centered



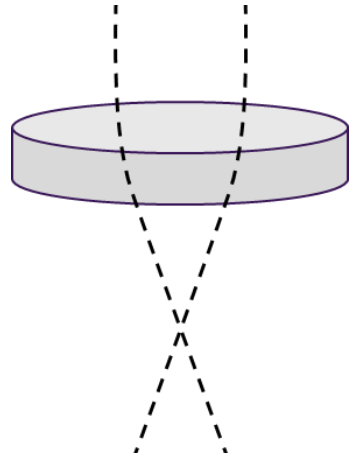
Insert aperture

Aperture not aligned

Aperture aligned

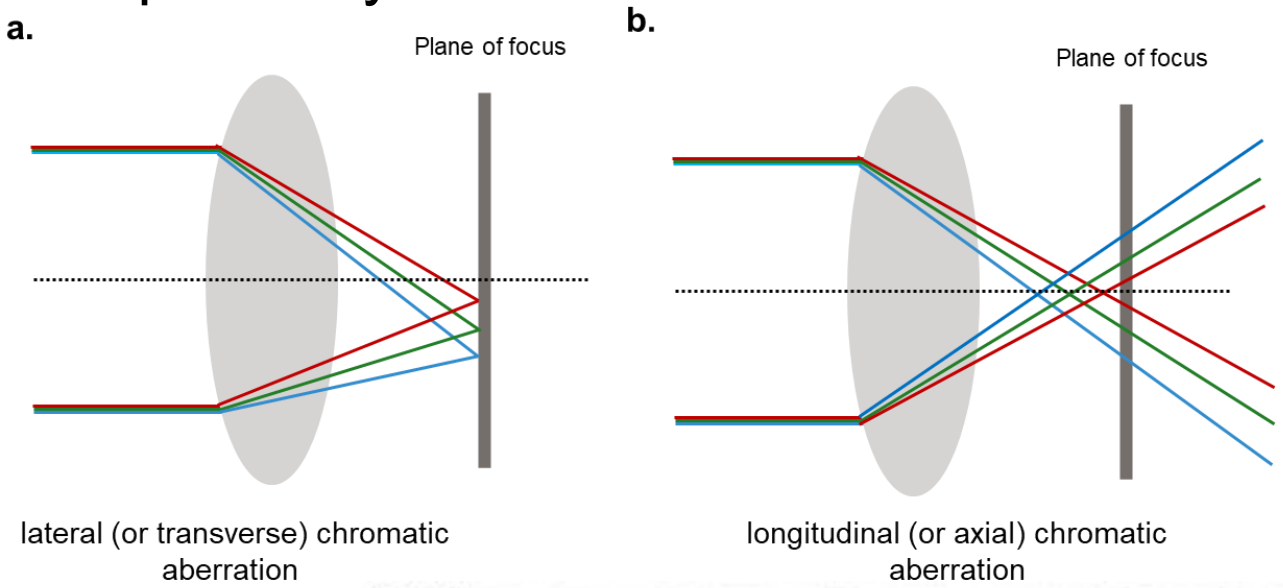


Lens not aligned



Lens aligned

Chromatic Aberration: Arises when electrons with different energies (wavelengths) are focused at different points by the lens.

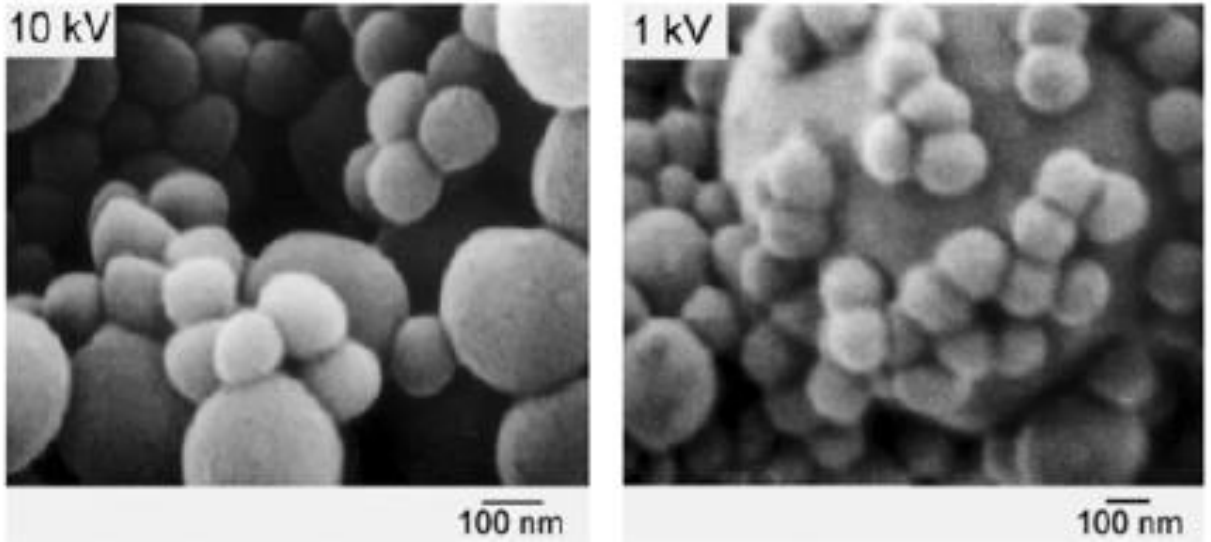


lateral (or transverse) chromatic aberration

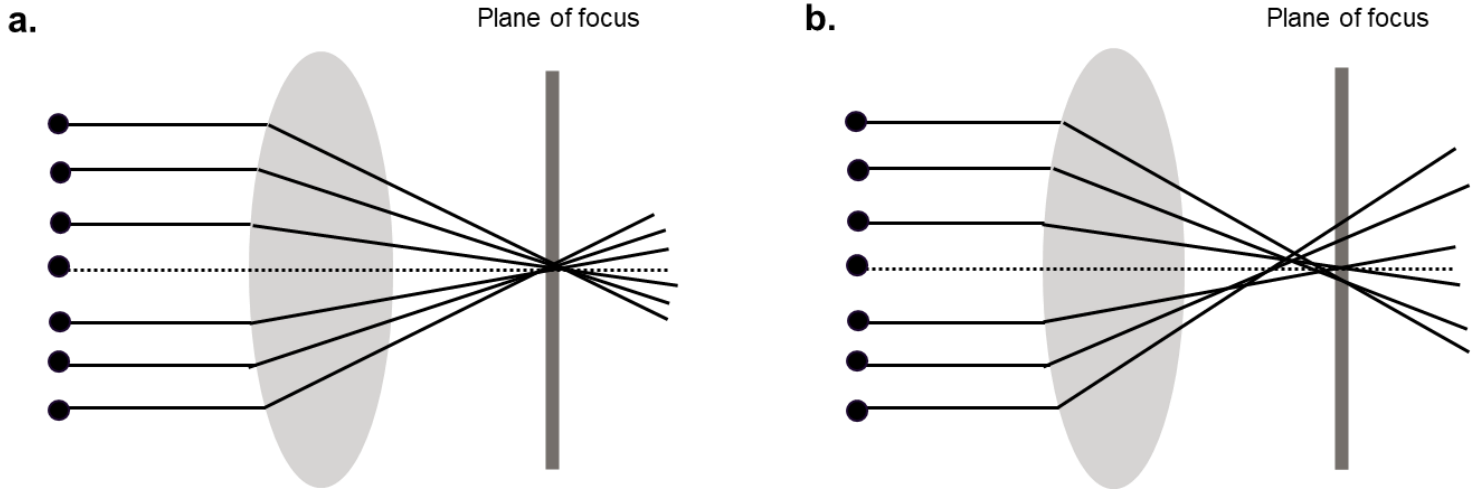
longitudinal (or axial) chromatic aberration

Effect on Image:

- Beam broadening
- Image blurring
- Reduced contrast and resolution

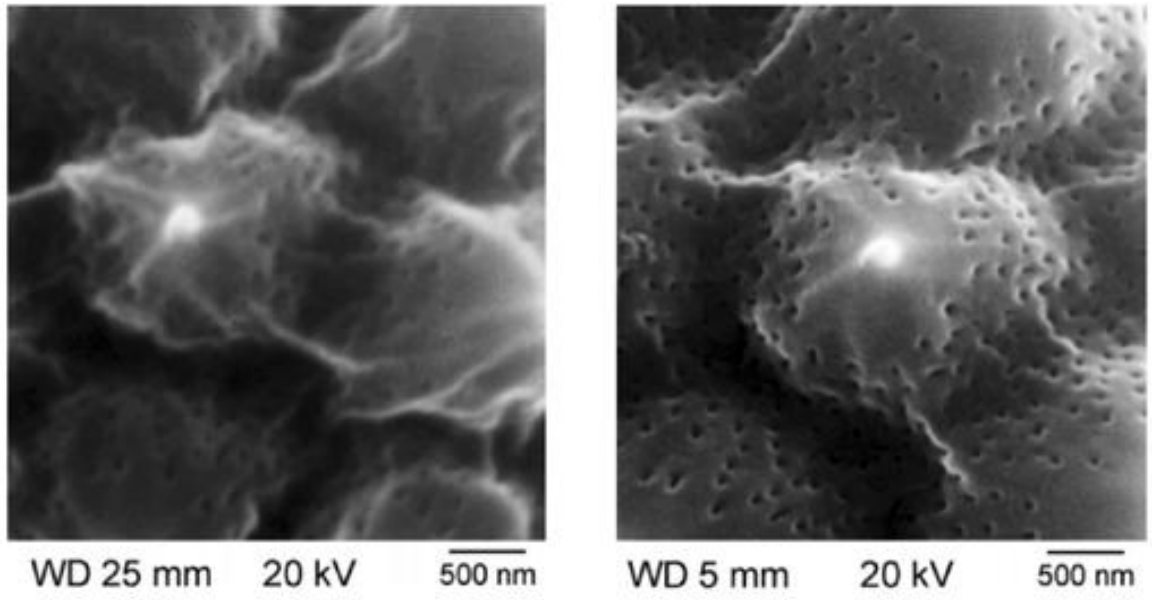


Spherical Aberration: Occurs when electrons passing through the outer regions of an electromagnetic lens are focused on different point than those passing near the lens axis.

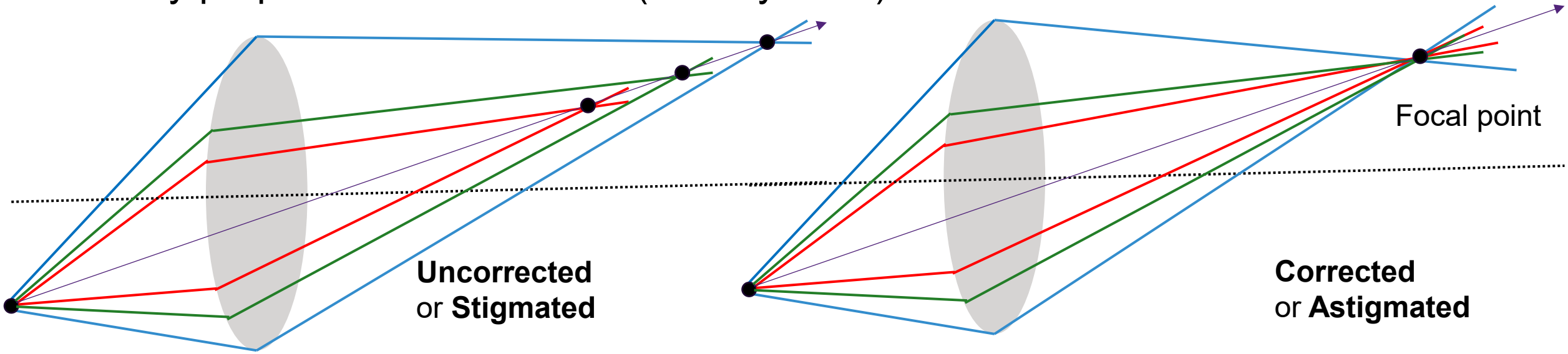


Effect on Images:

- Leading to a loss of sharpness
- Blurred or distorted images
- Reduced resolution

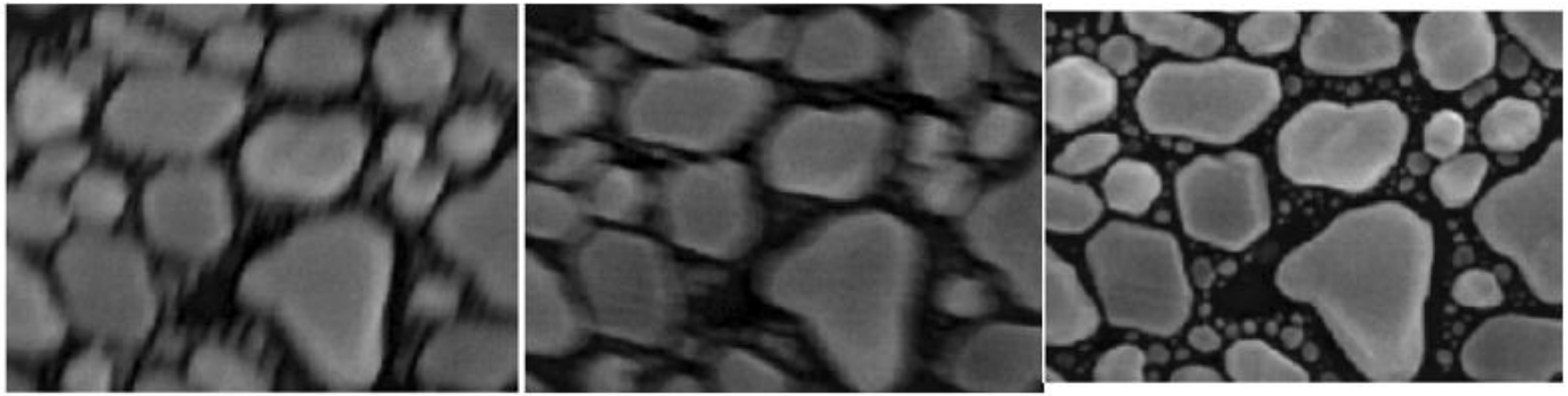


Astigmatism: Refers to a lens imperfection that causes the electron beam to focus differently perpendicular directions (x and y-axes).

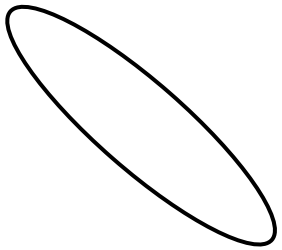


Effect on Image:

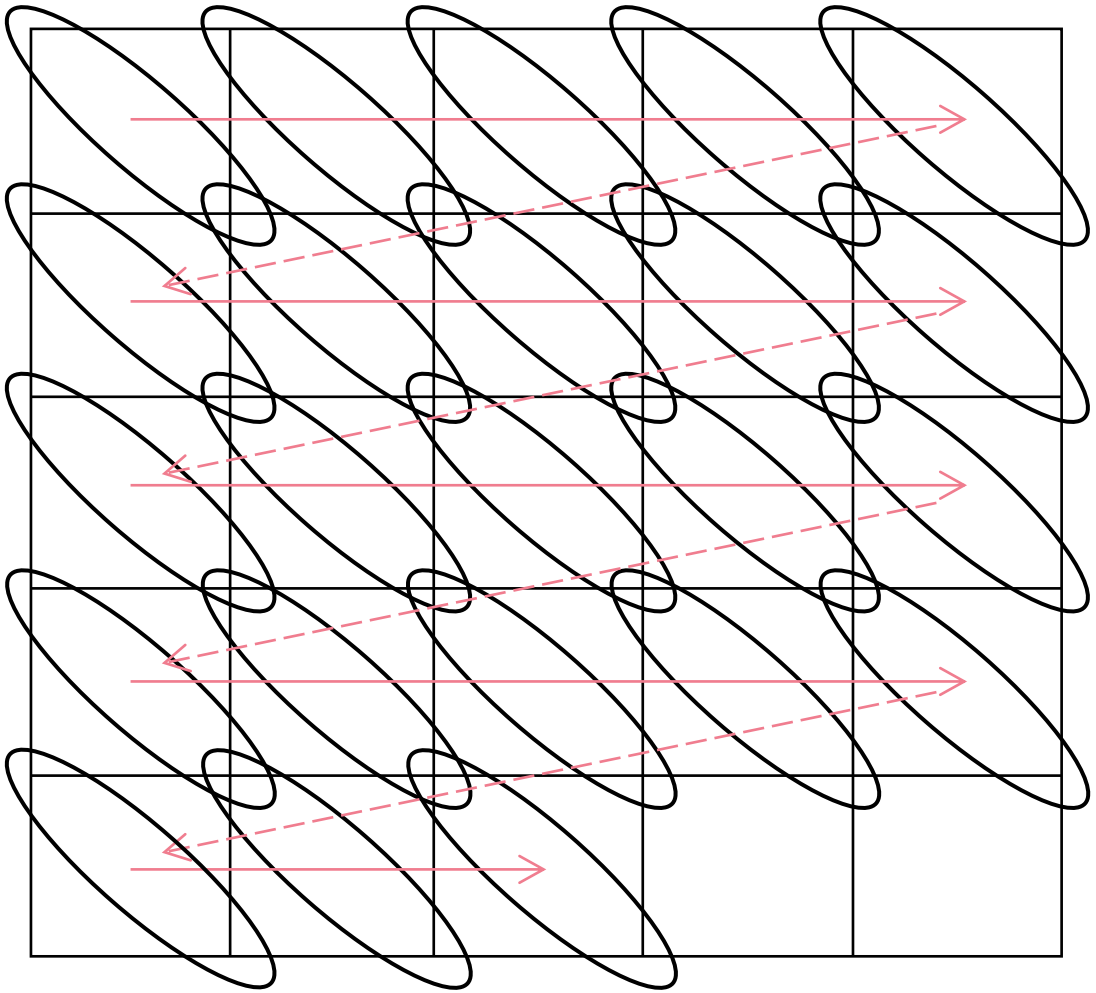
- Image blurring
- One direction appears sharper than the other
- Reduced contrast and resolution

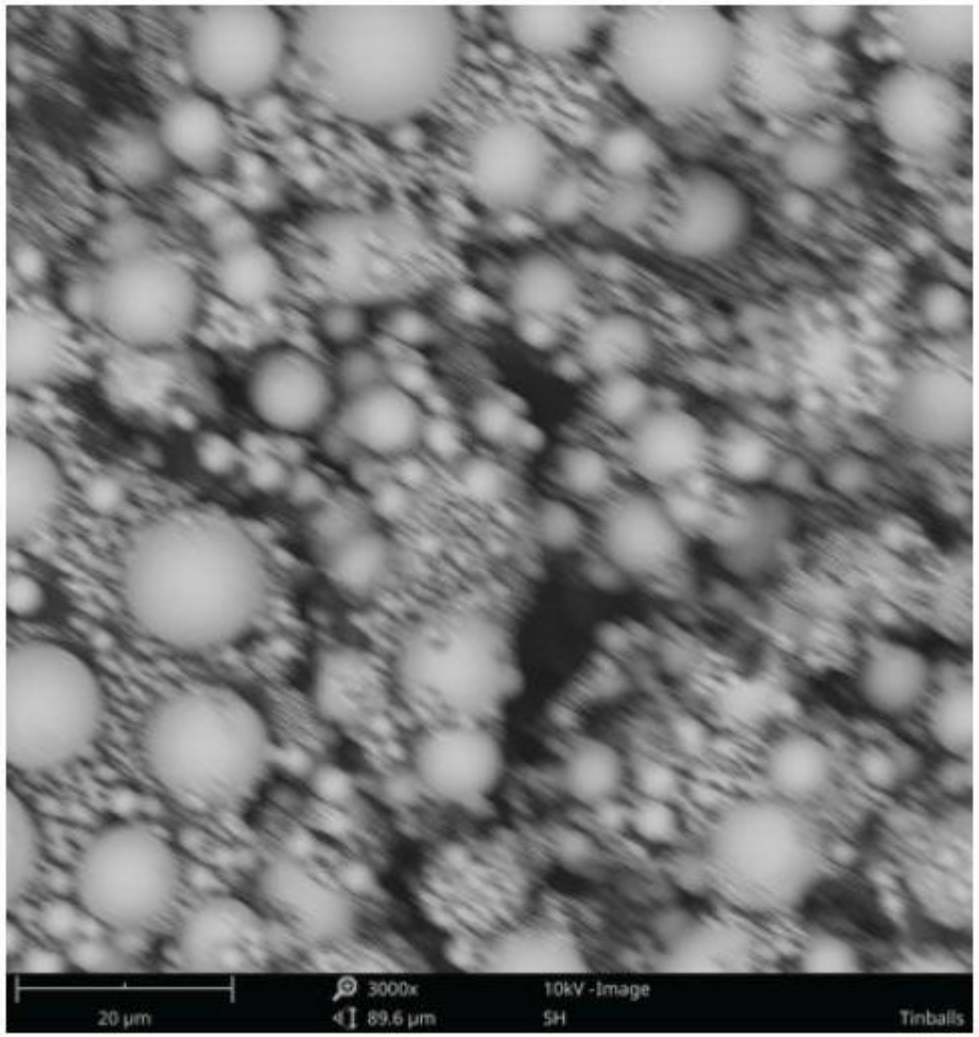


- With stigmatism, the image appears blurry due to the overlapping of probes.

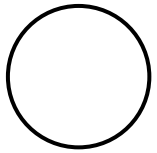


Aspherical probe

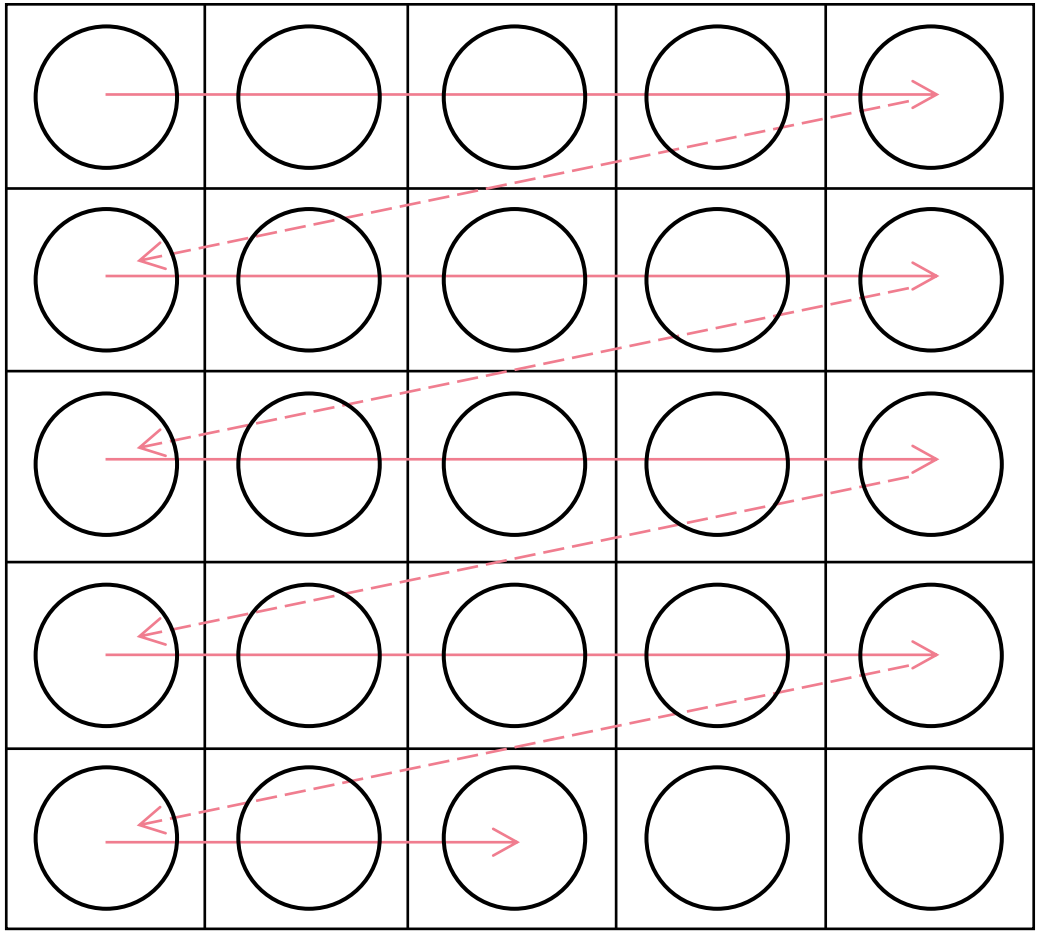


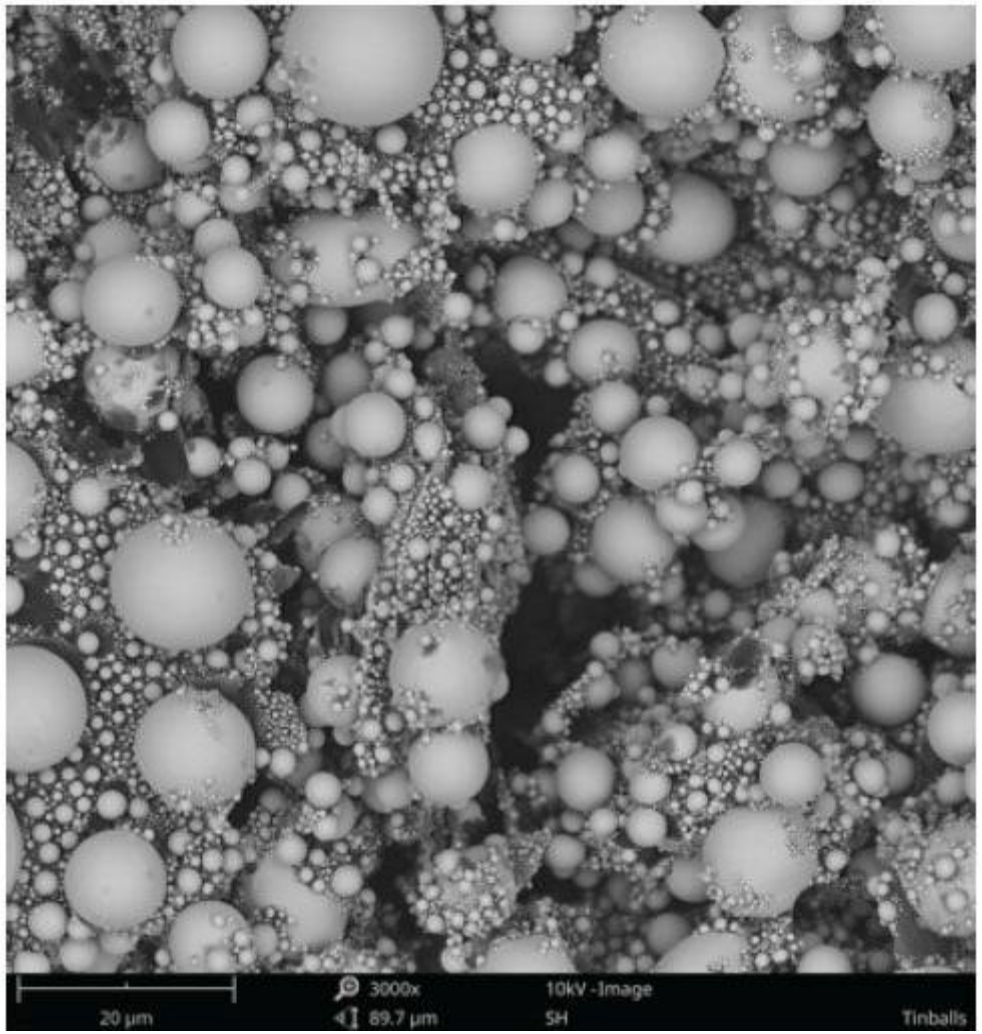


- To get focused image, the probe shape must be spherical.



Spherical probe



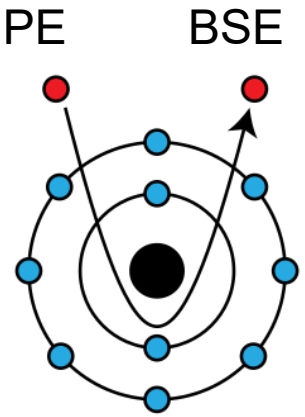


Electron-matter Interactions – Interaction Volume

Electron-matter interactions can be divided into two classes:

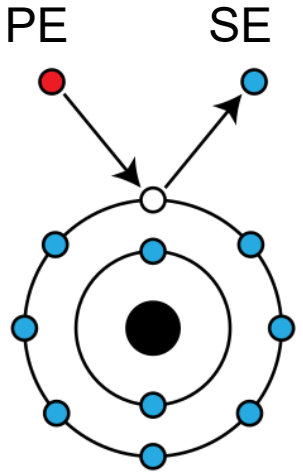
Elastic Scattering (Conserving Energy)

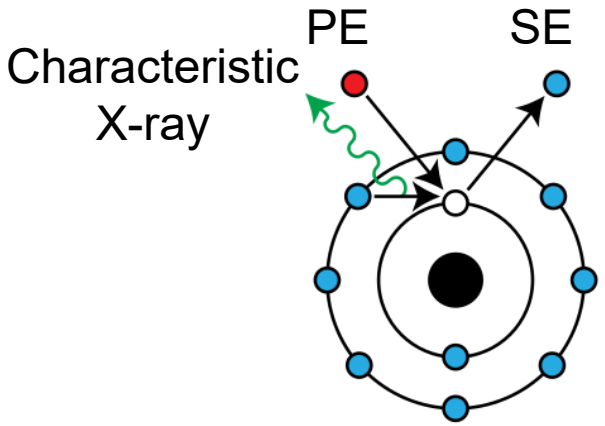
The electron trajectory within the specimen changes but its kinetic energy and velocity remains essentially constant.

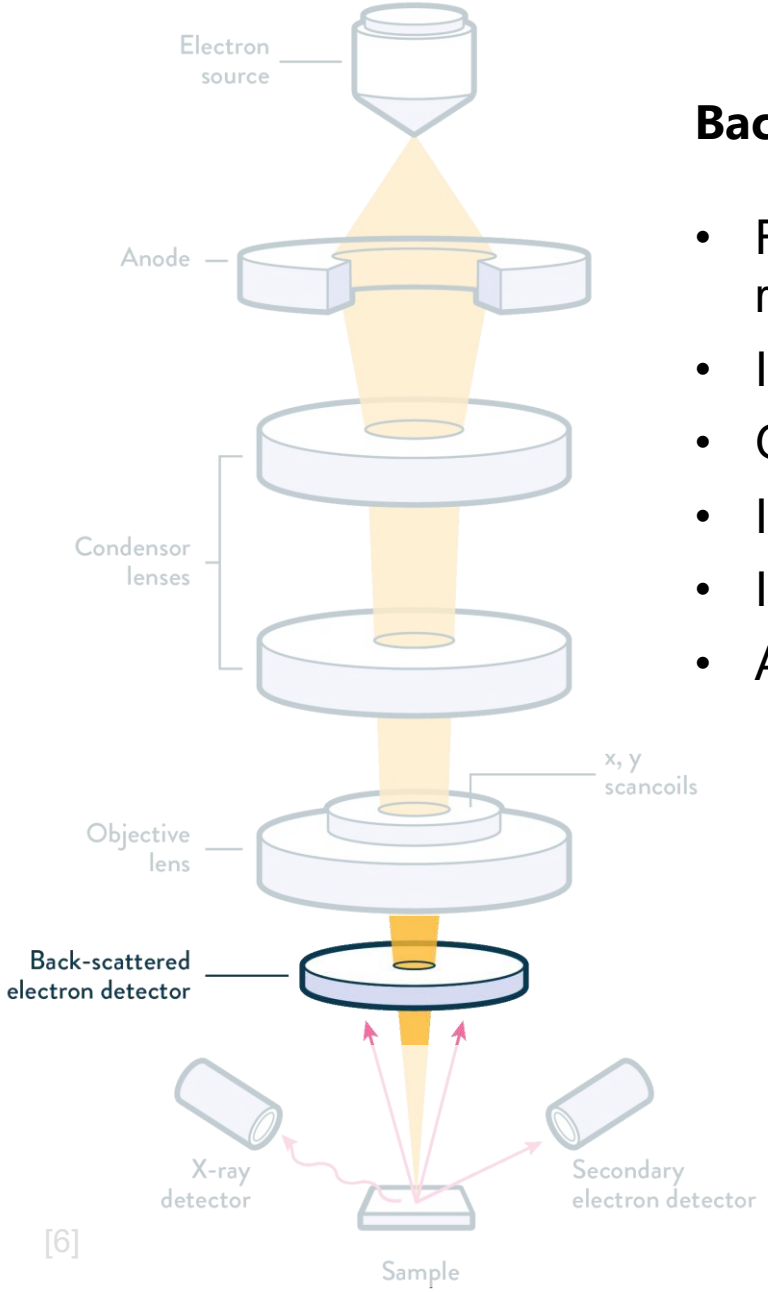


Inelastic Scattering (Losing Energy)

This occurs when the incident electron trajectory is only slightly perturbed, but energy is lost through the transfer of energy to the specimen.



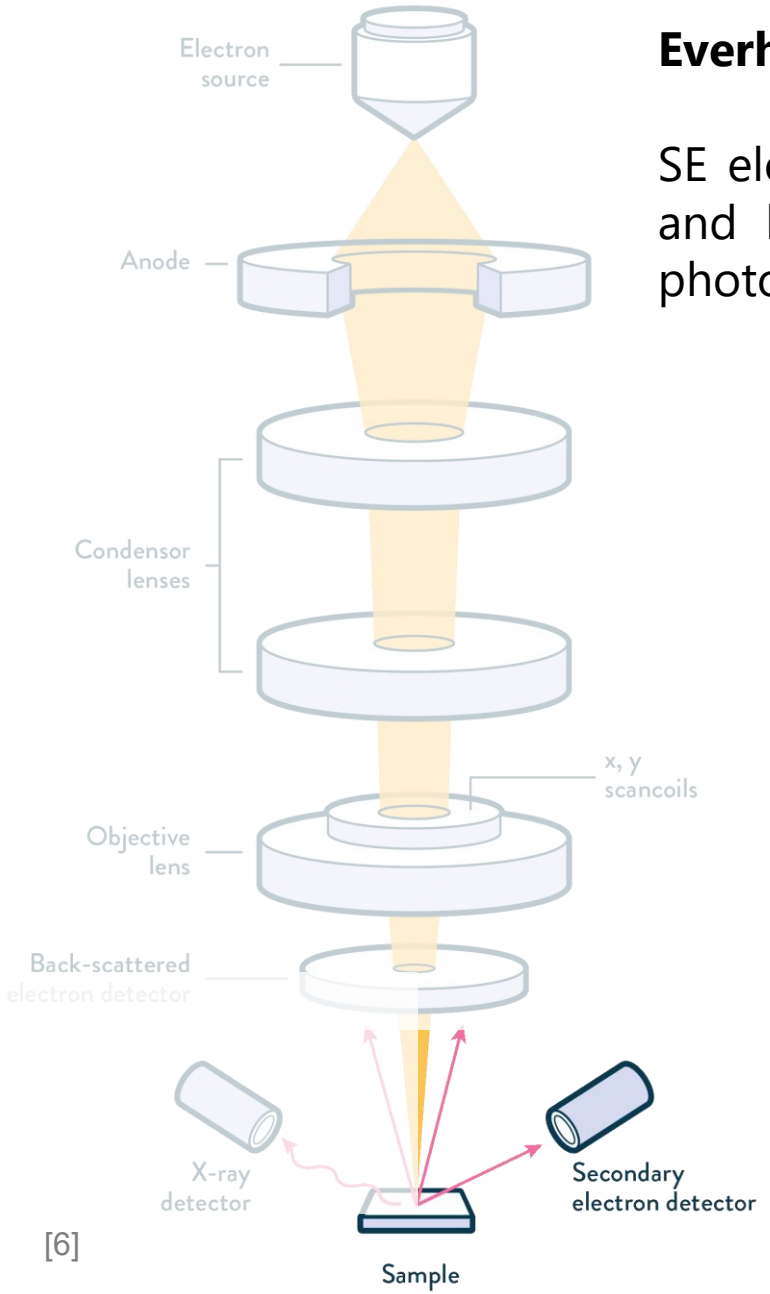




[6]

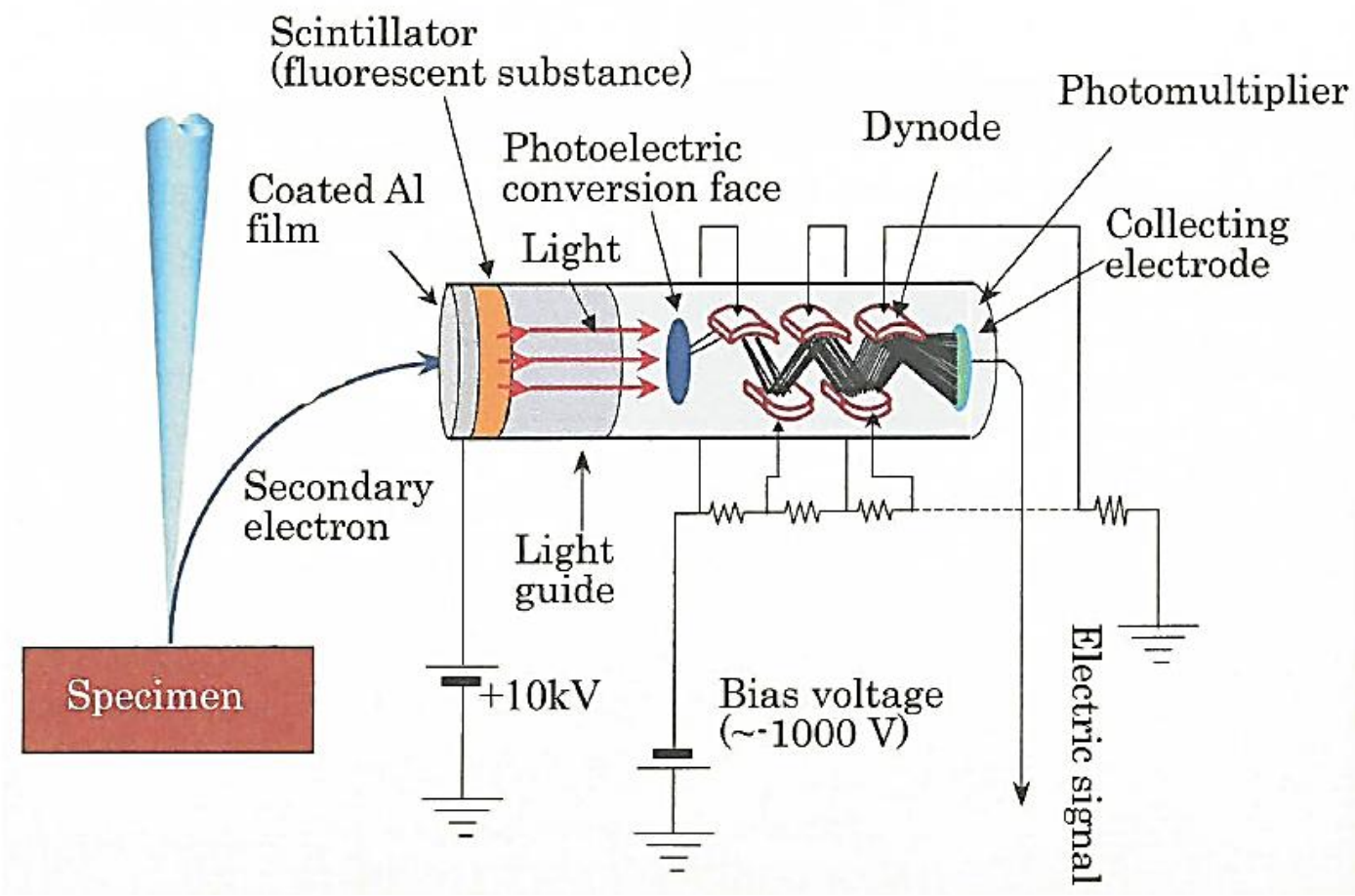
Backscattered electrons (BSE)

- Formed by primary electrons, which are backscattered (all electrons in the energy range from 50 eV to E_0).
- In general number of BSEs is much lower than SEs
- Gives mainly information about material (material contrast), less about topography.
- Increases with increasing atomic number Zup to E0~30 keV.
- Increases with increasing tilt of the sample.
- Almost independent on energy for E_0 >10 keV.

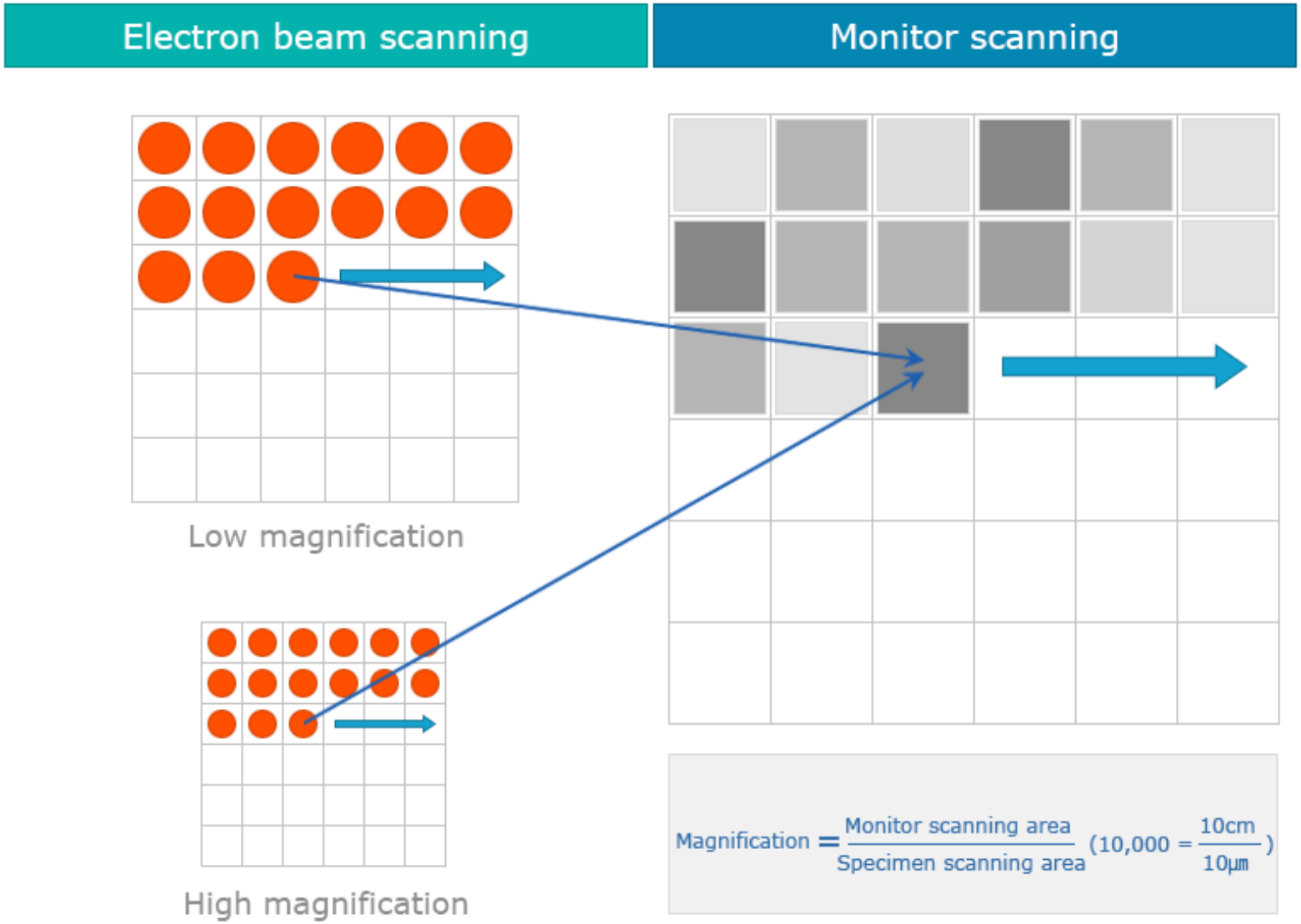


Everhart-Thornley (ET) detector

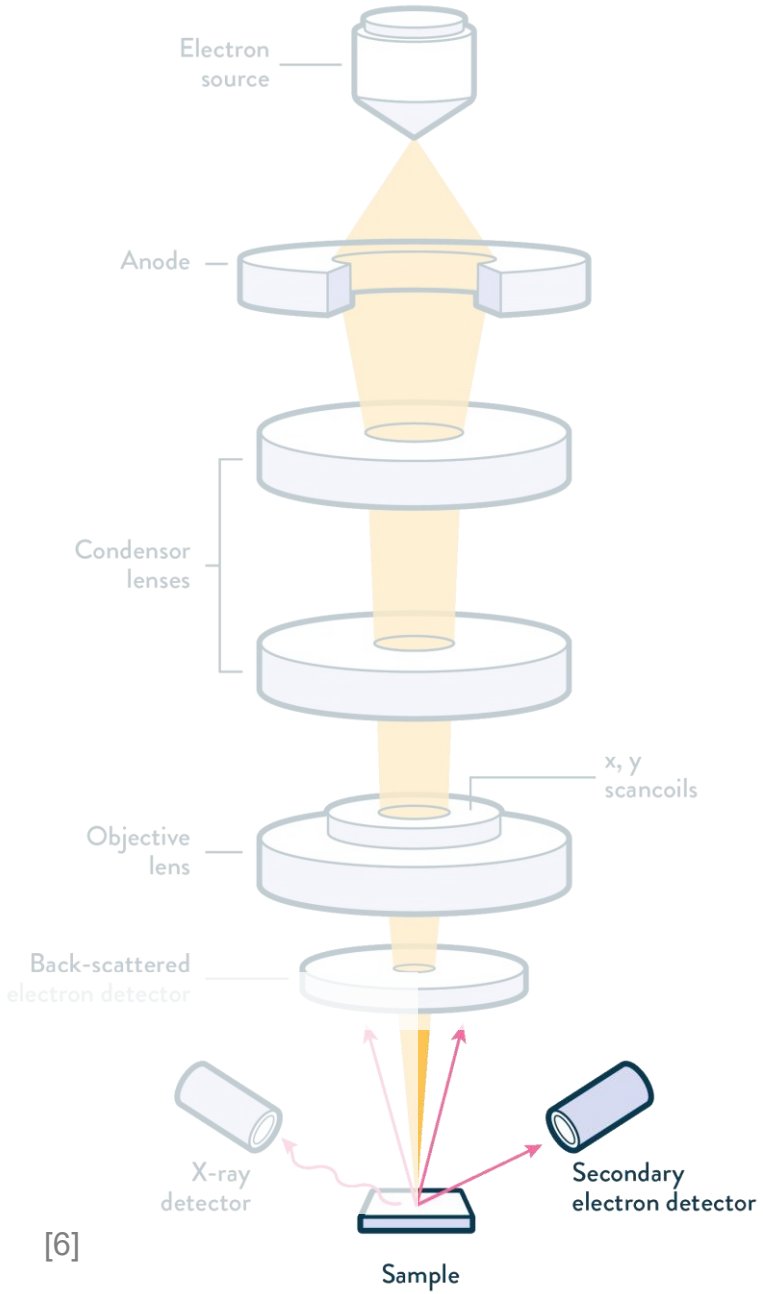
SE electrons (very low energy <50 eV) are accelerated towards 10 kV electric field and hit the scintillator for conversion into the light signal, which leads to the photomultiplier tube (PMT) through a light pipe



[6]

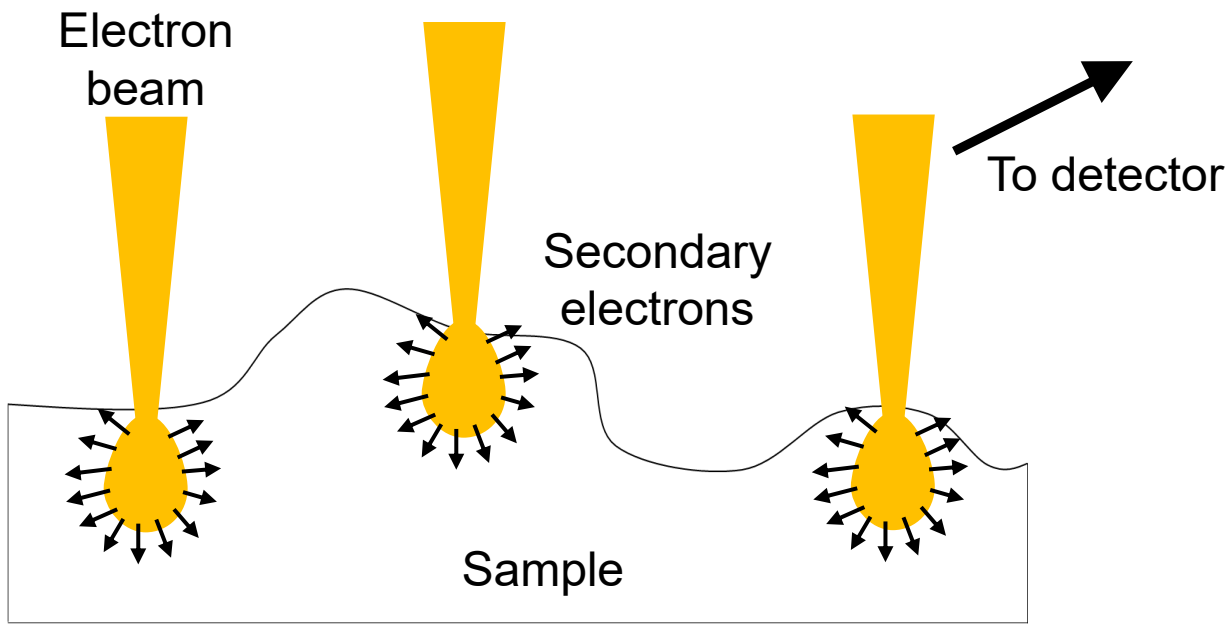


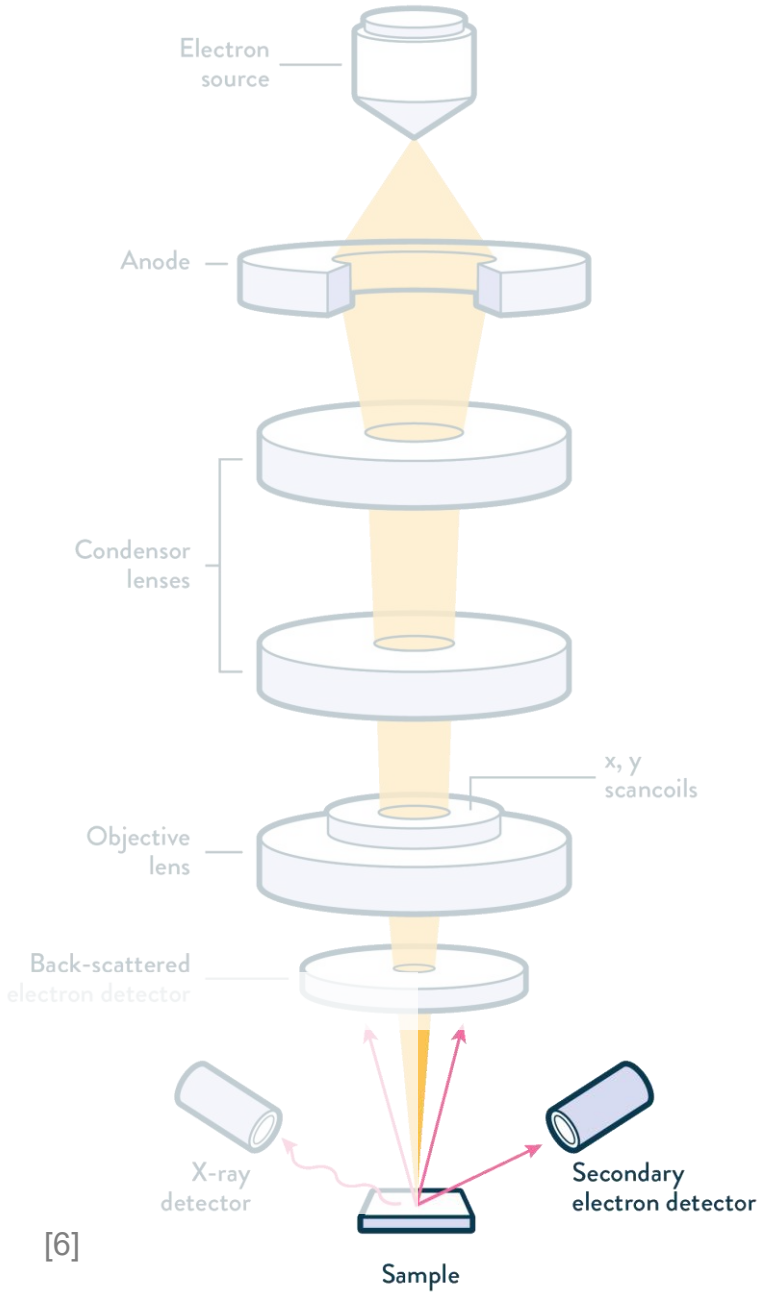
- The electron beam is raster-scanned across the sample.
- For each pixel, the signals come from the detectors. The “brightness” of each pixel is dependent on the strength of the signal from the detector.



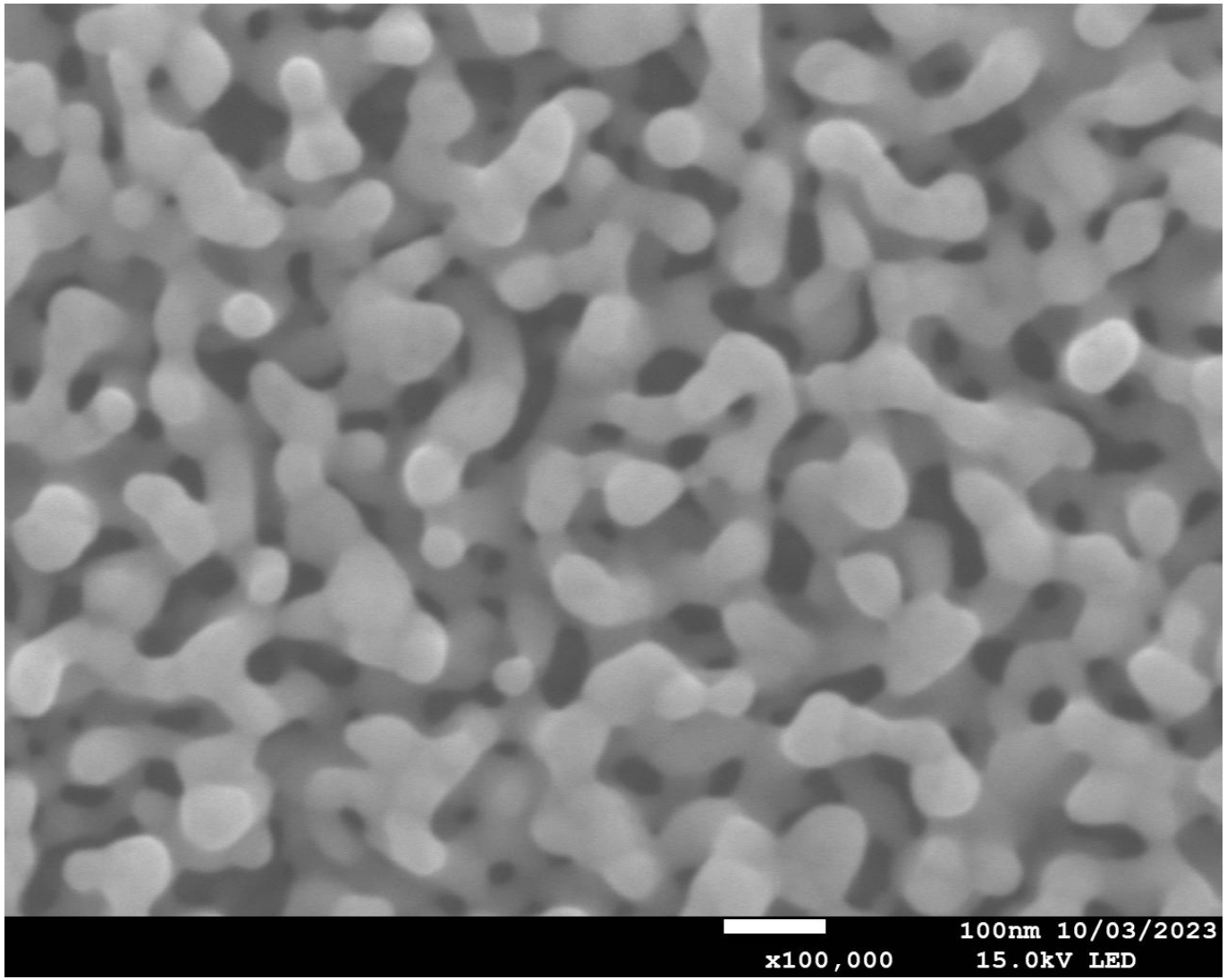
[6]

- Most SEM images you see are secondary electron (SE) images.
- The changing slope of the sample alters the number of secondary electrons that come out.
- This is why you get topographical information from them.



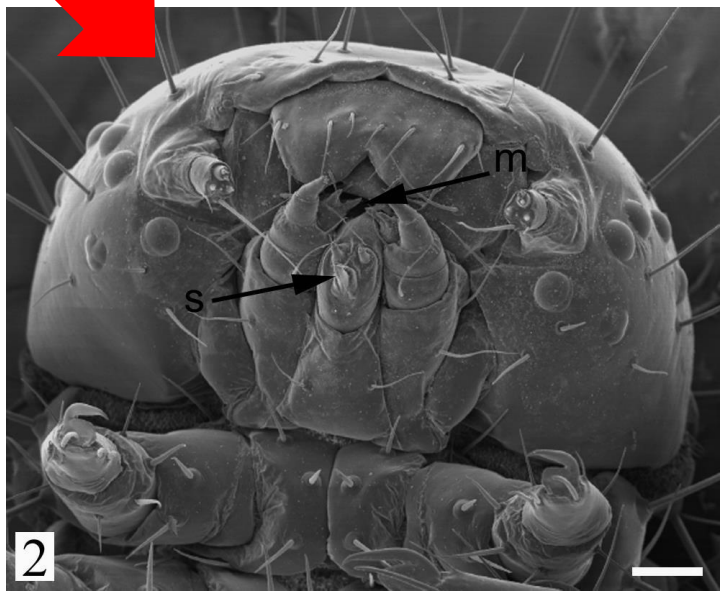

Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample

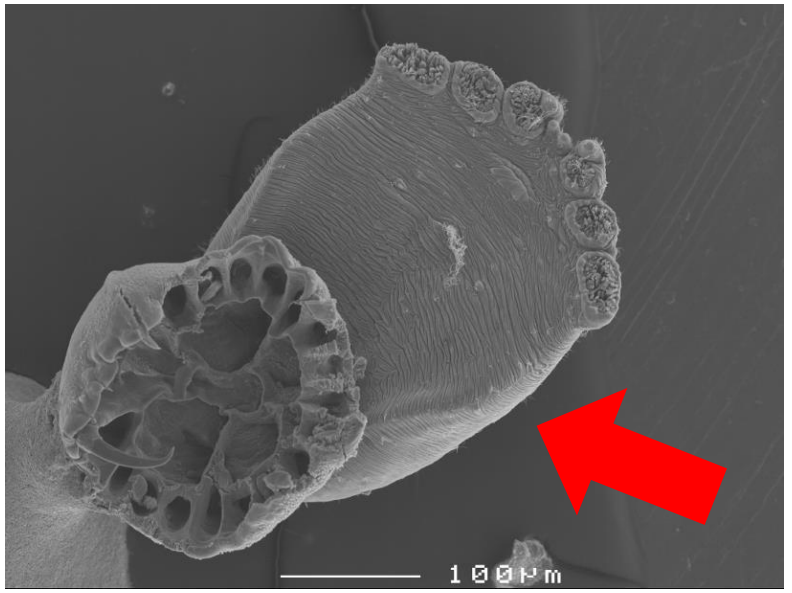
[6]


x100,000
100nm 10/03/2023
15.0kV LED

- The placement of the detector can give rise to “shading” similar to what you would see from a light source.
- The location of the SE detector in the chamber can be guessed.

SE was pointed in this direction

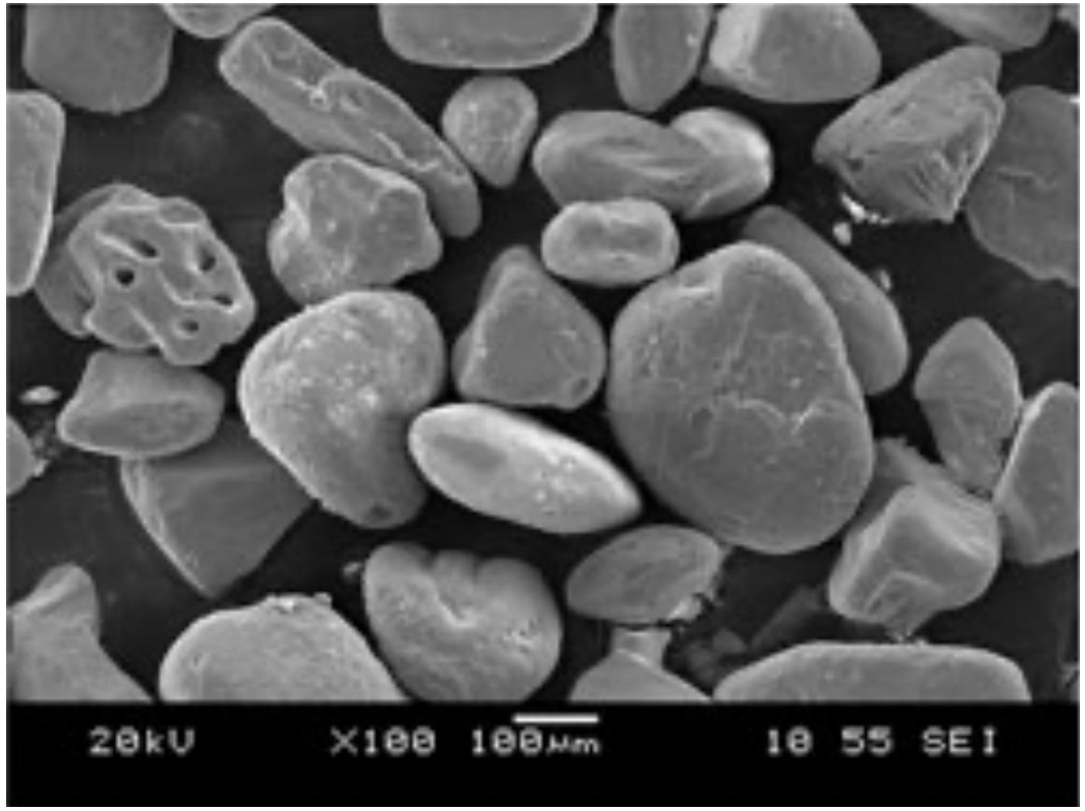




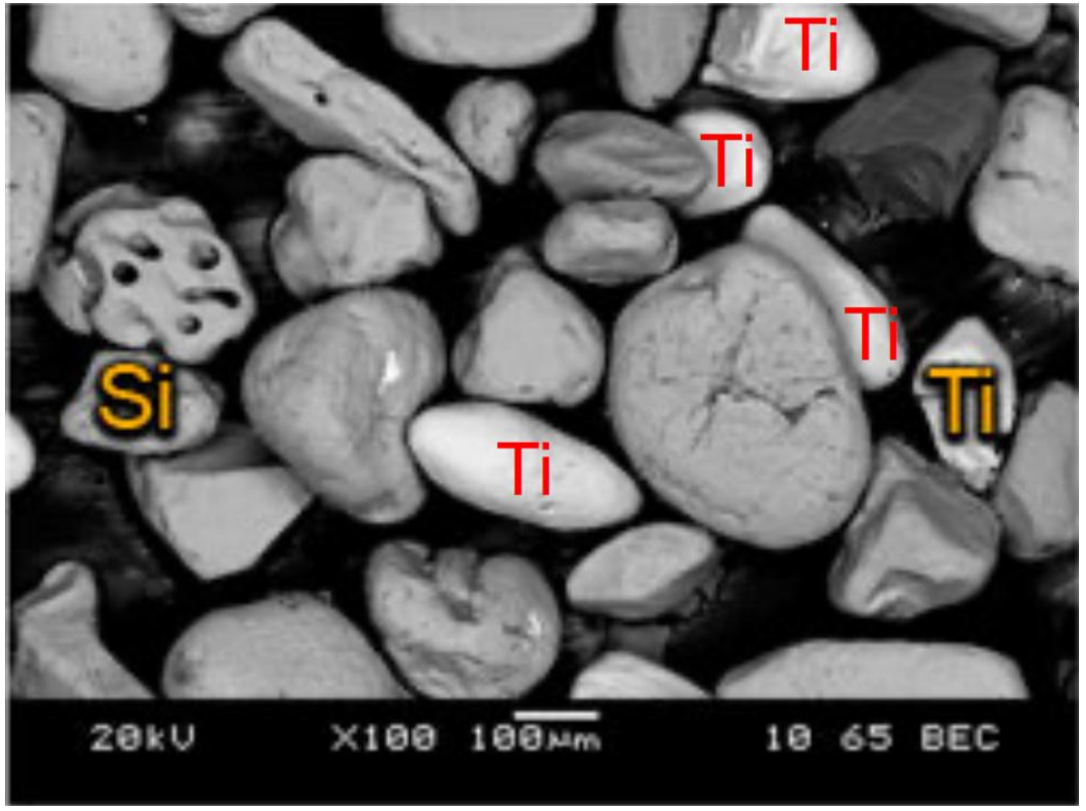
SE was pointed in this direction

It is where the “light source” in the image is.

Secondary electrons



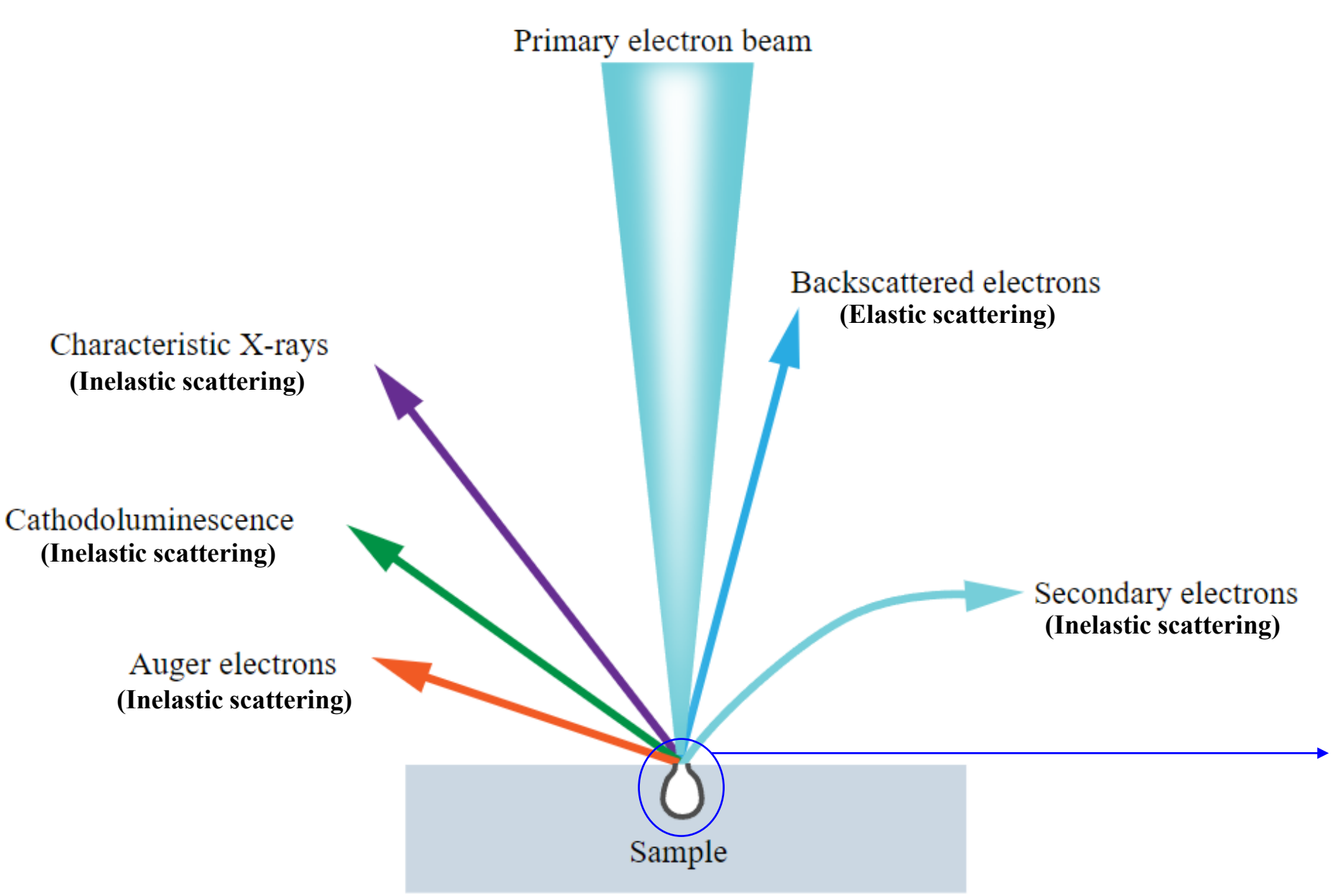
Backscattered electrons

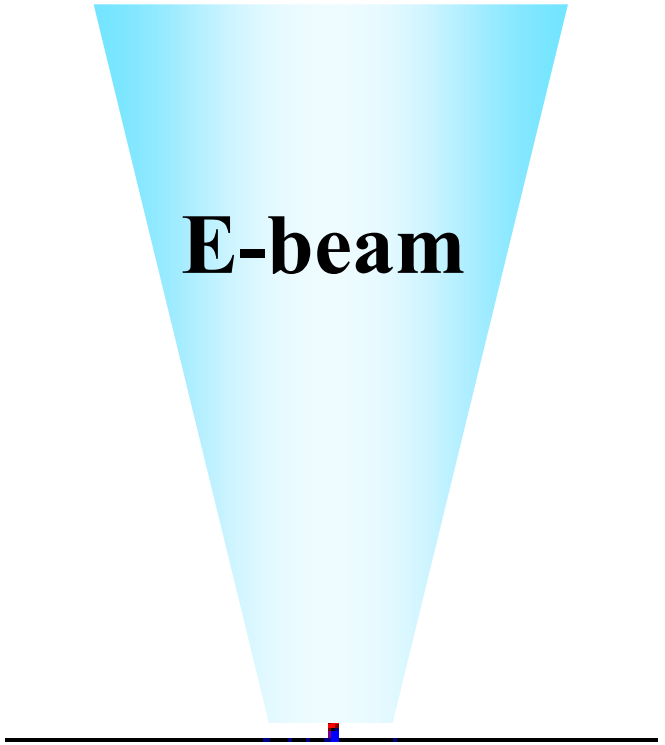


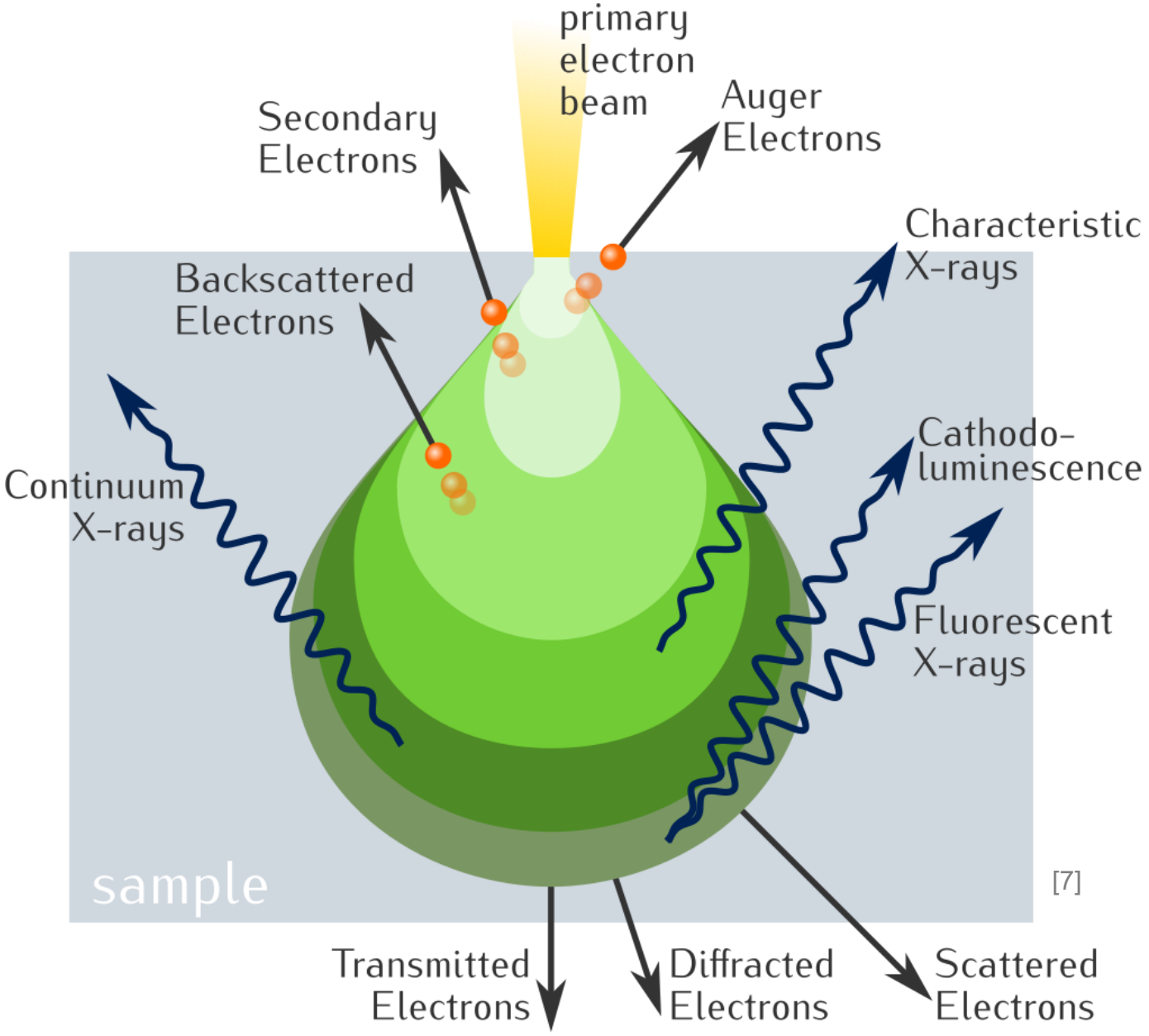
[6]

SE: Sample surface information (topography).

BSE: Due to targeted immunolabeling(gold nanoparticles bound to immunoglobulin) the possibility to identify specific parts on the sample (bright Au nanoparticles).

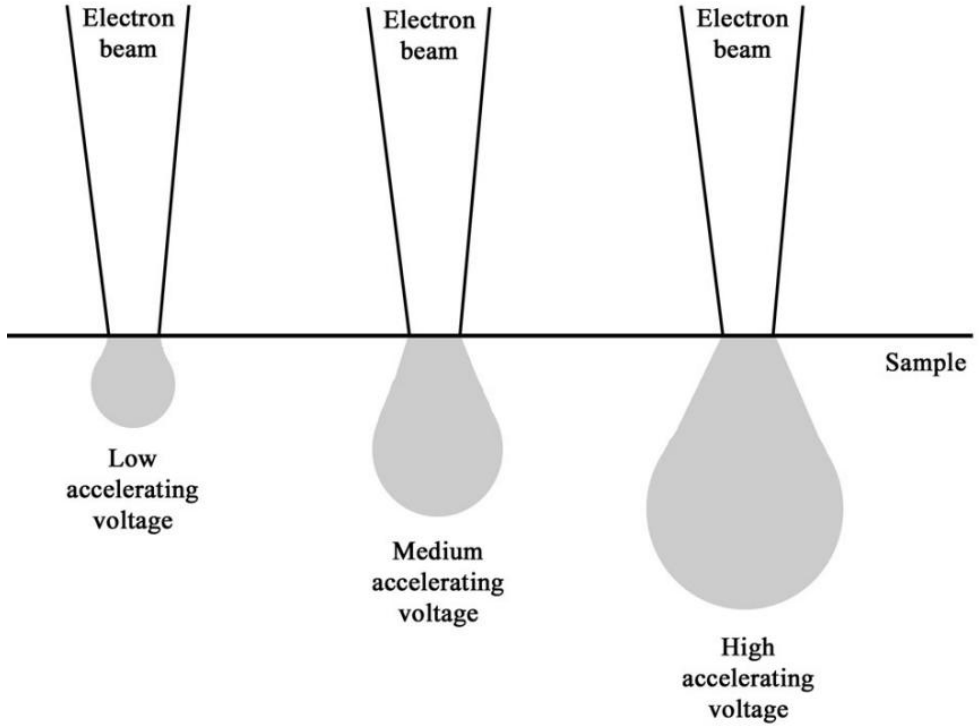

Primary electron beam
Backscattered electrons
(Elastic scattering)
Characteristic X-rays
(Inelastic scattering)
Cathodoluminescence
(Inelastic scattering)
Auger electrons
(Inelastic scattering)
Secondary electrons
(Inelastic scattering)
Sample

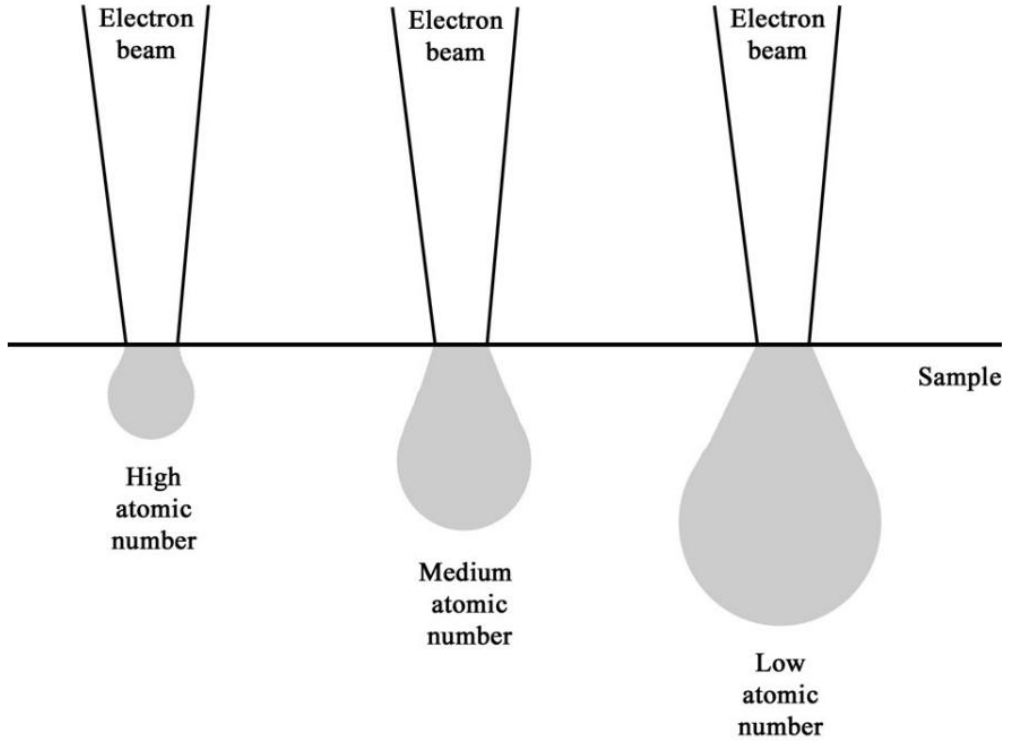

E-beam

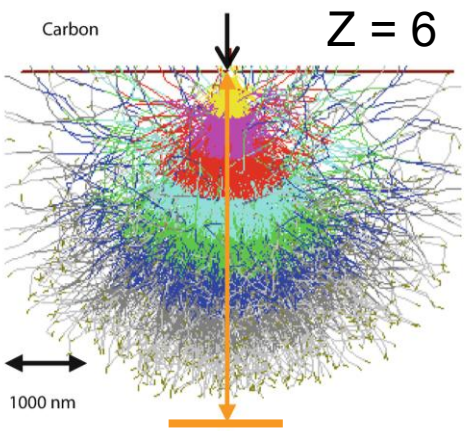

primary
electron
beam
Secondary
Electrons
Auger
Electrons
Characteristic
X-rays
Backscattered
Electrons
Cathodo-
luminescence
Continuum
X-rays
Fluorescent
X-rays
sample
Transmitted
Electrons
Diffracted
Electrons
Scattered
Electrons

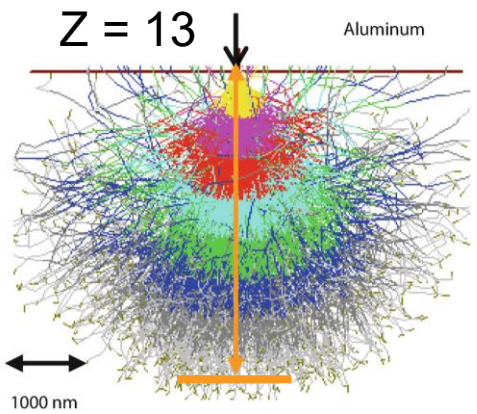
[7]

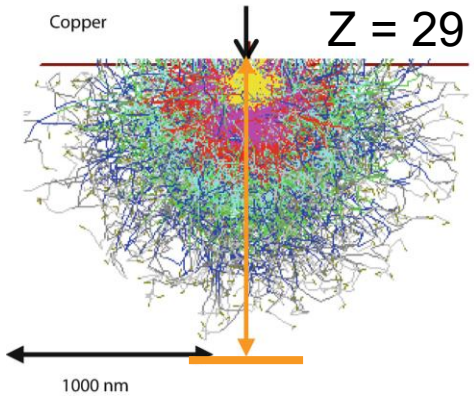
- The interaction volume is dependent on the accelerating voltage and the materials.

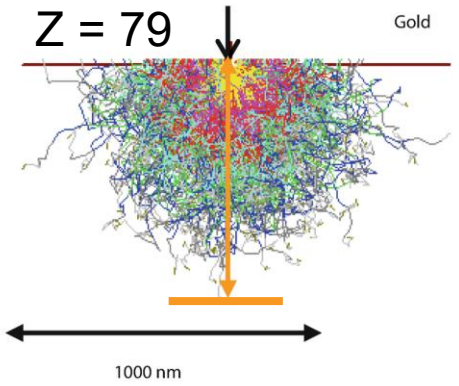




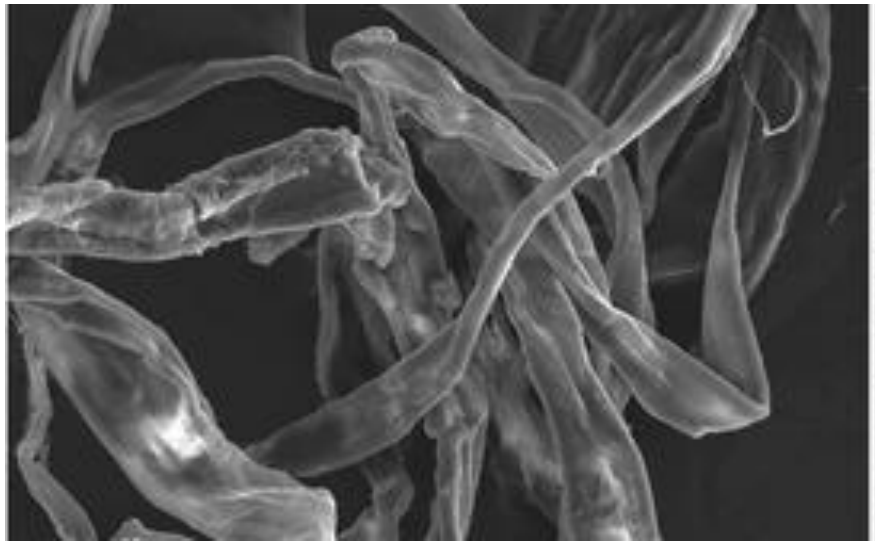


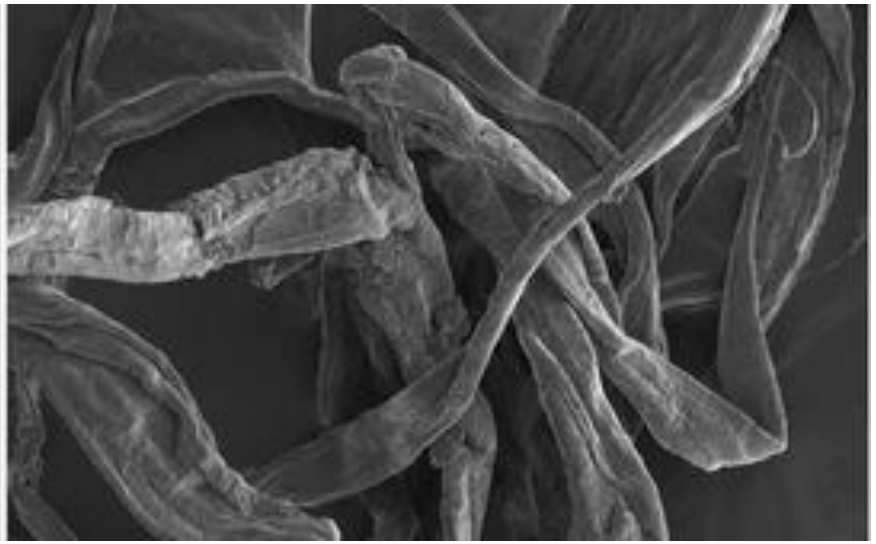




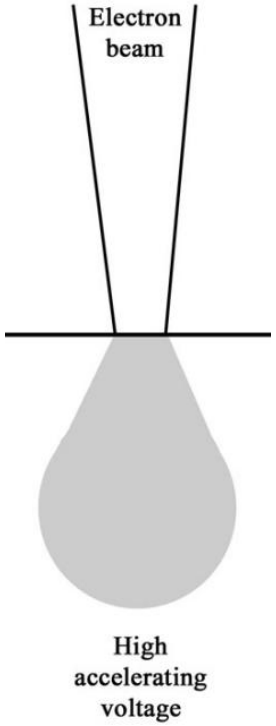


Carbon coated cellulose fiber at 20 kV.



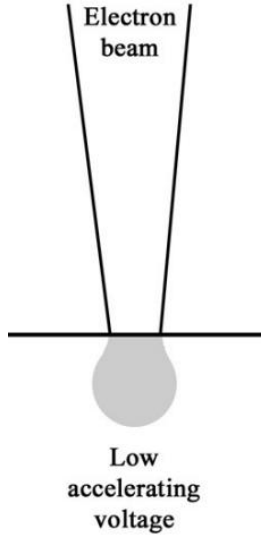


Carbon coated cellulose fiber at 1 kV.

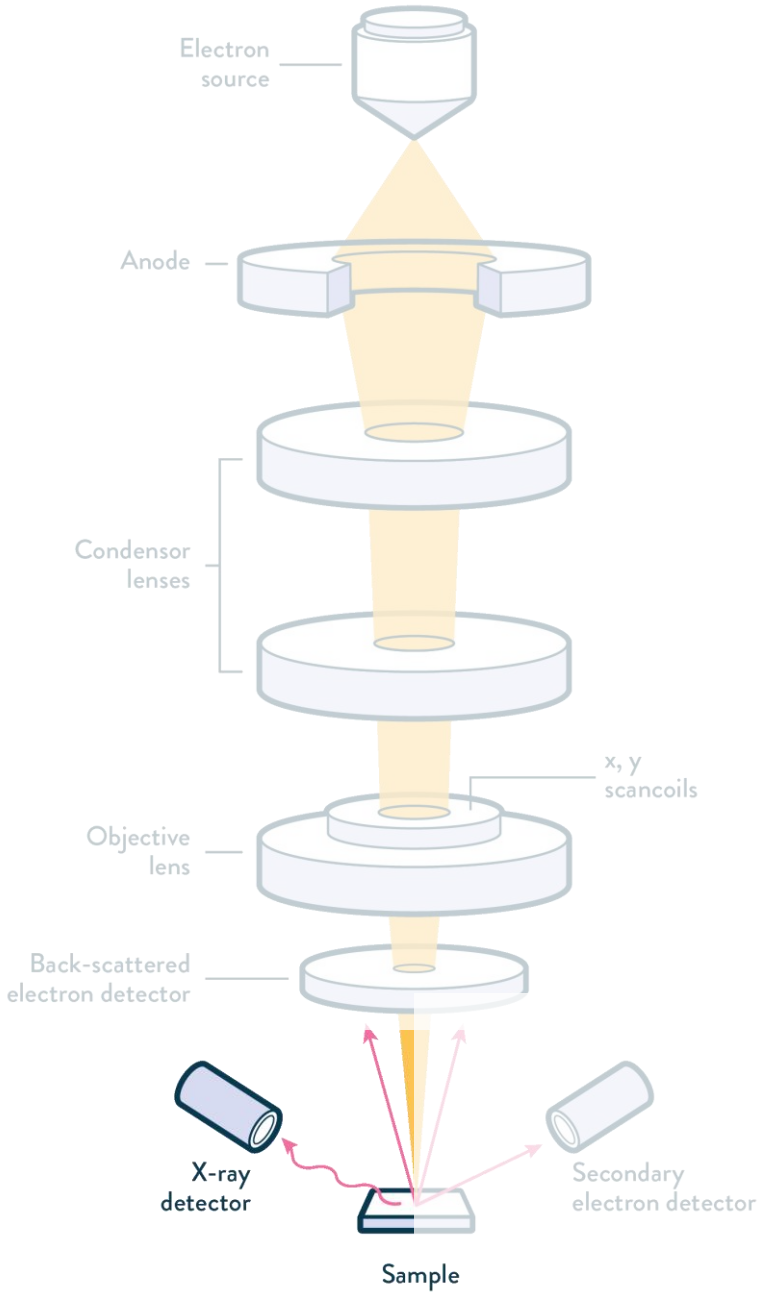


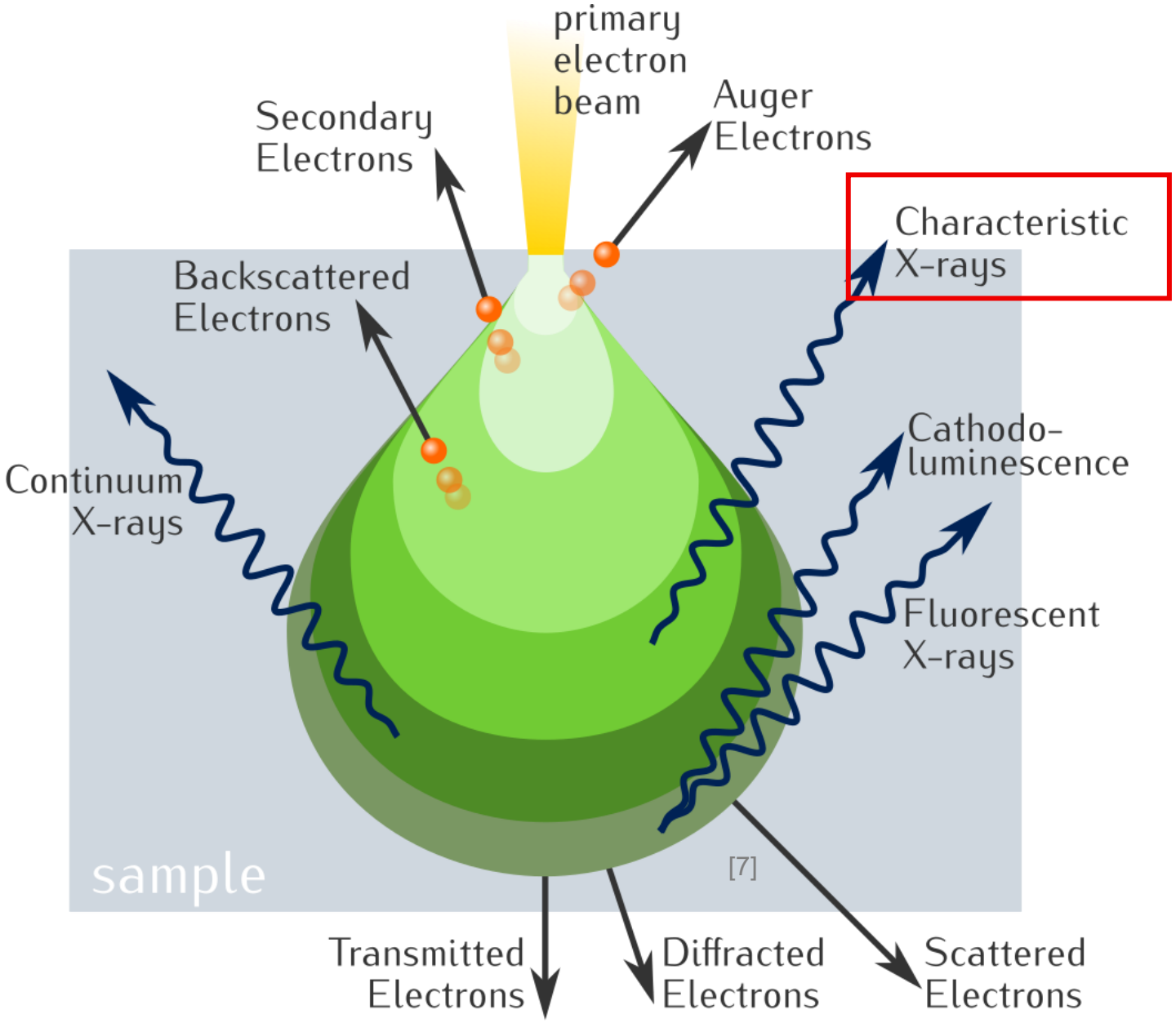
- Larger interaction volume
- Lower resolution
- More edge effect
- More charging
- More surface damage

- Smaller interaction volume
- Higher resolution
- Less edge effect
- Less charging
- Less surface damage

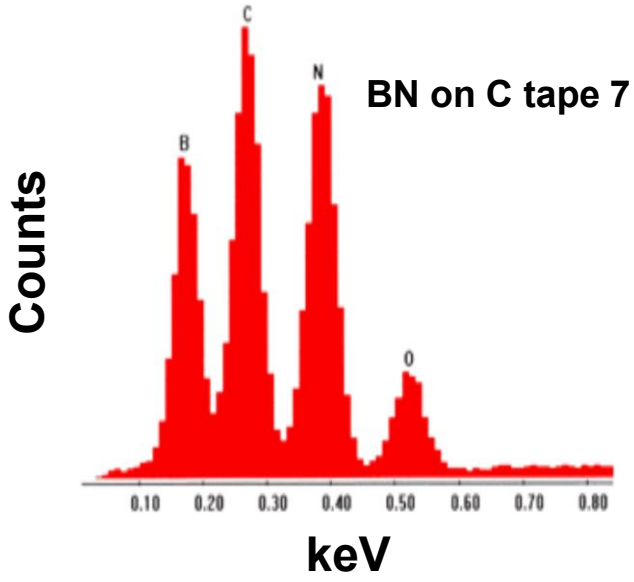


Energy dispersive X-ray spectroscopy (EDX or EDS)

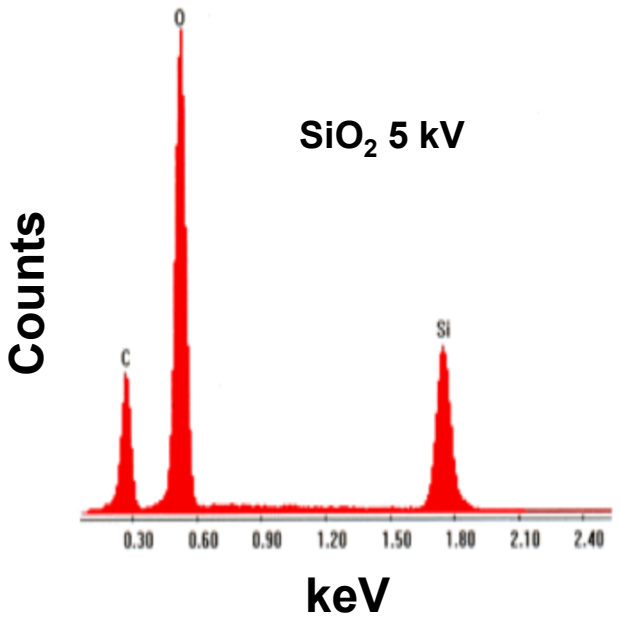

Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample


primary electron beam
Secondary Electrons
Auger Electrons
Characteristic X-rays
Backscattered Electrons
Continuum X-rays
Cathodo-luminescence
Fluorescent X-rays
sample
[7]
Transmitted Electrons
Diffracted Electrons
Scattered Electrons

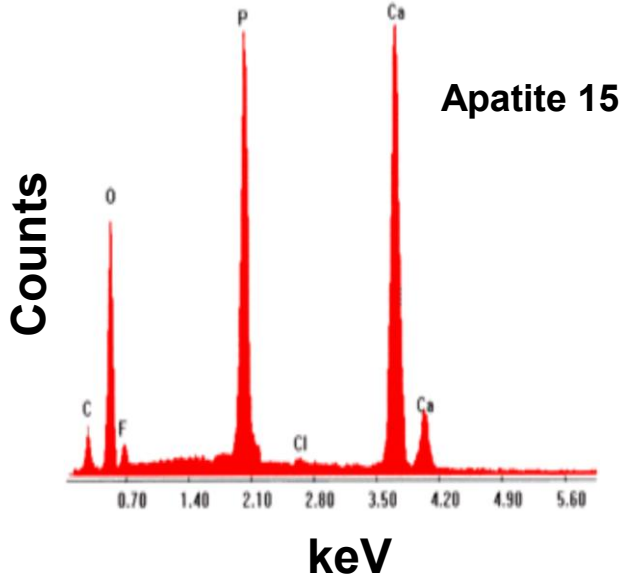
BN on C tape 7 kV



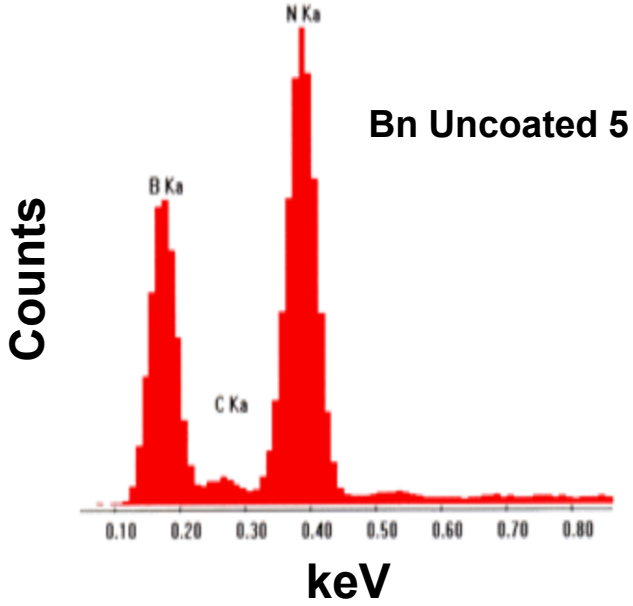
SiO_2 5 kV



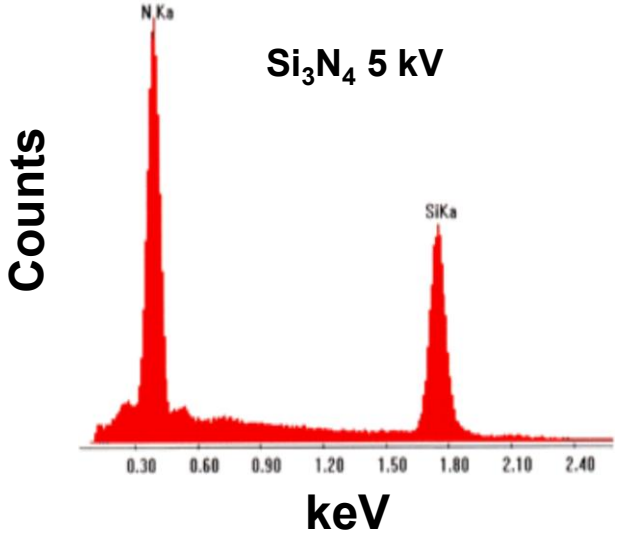
Apatite 15 kV



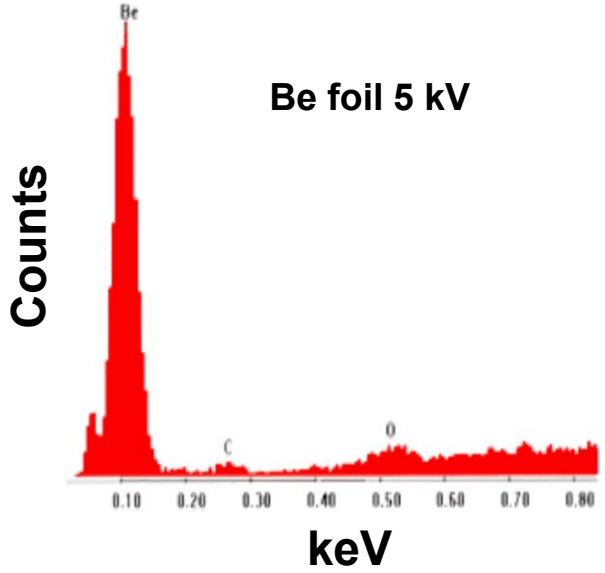
Bn Uncoated 5 kV

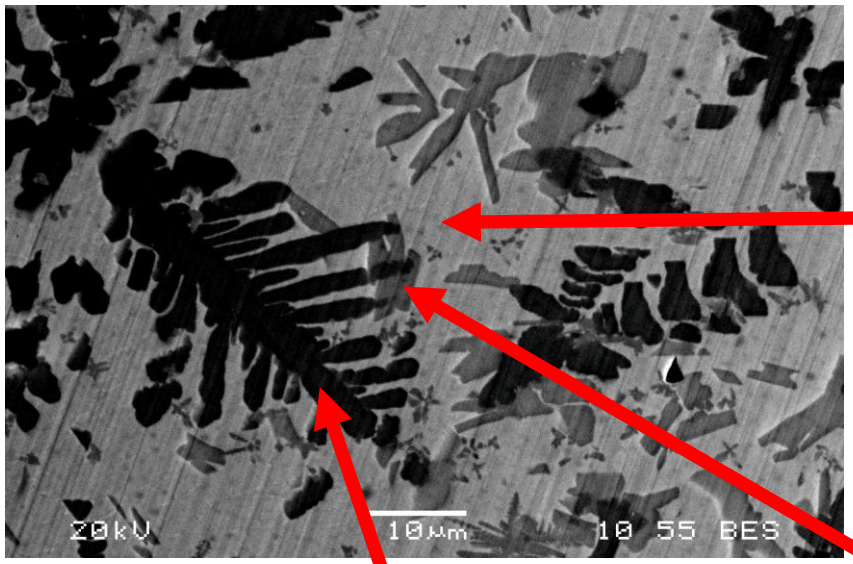


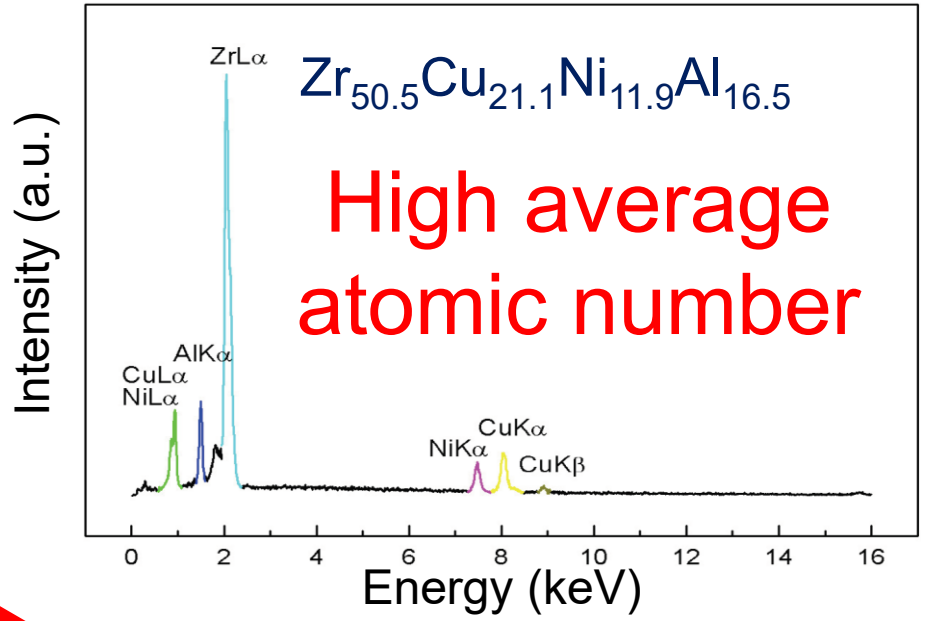
Si_3N_4 5 kV

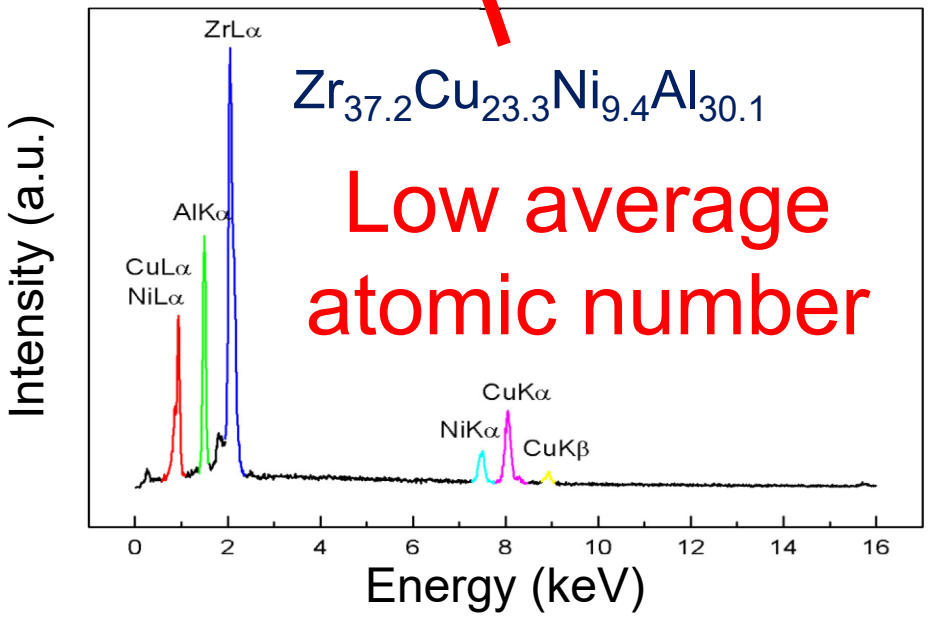


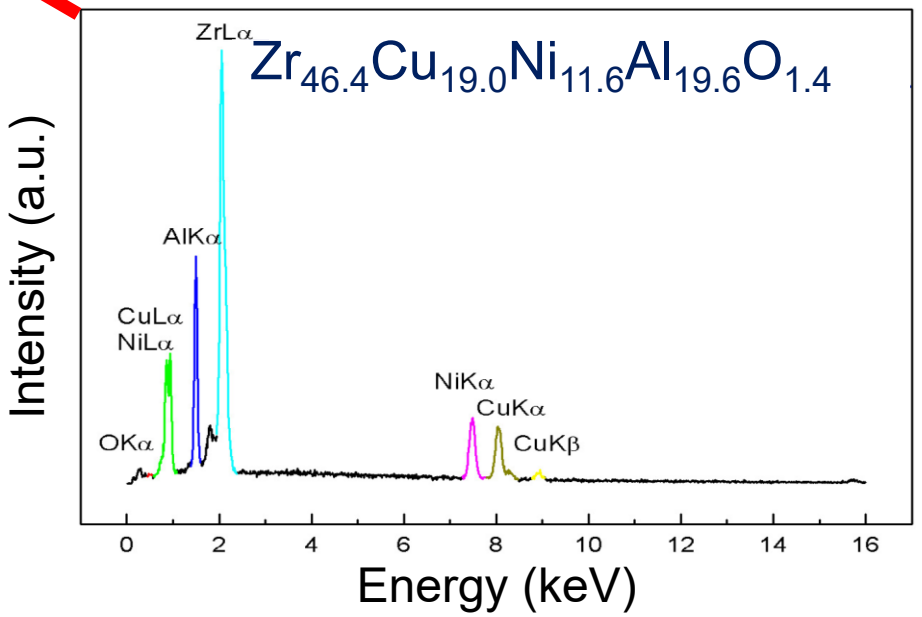
Be foil 5 kV



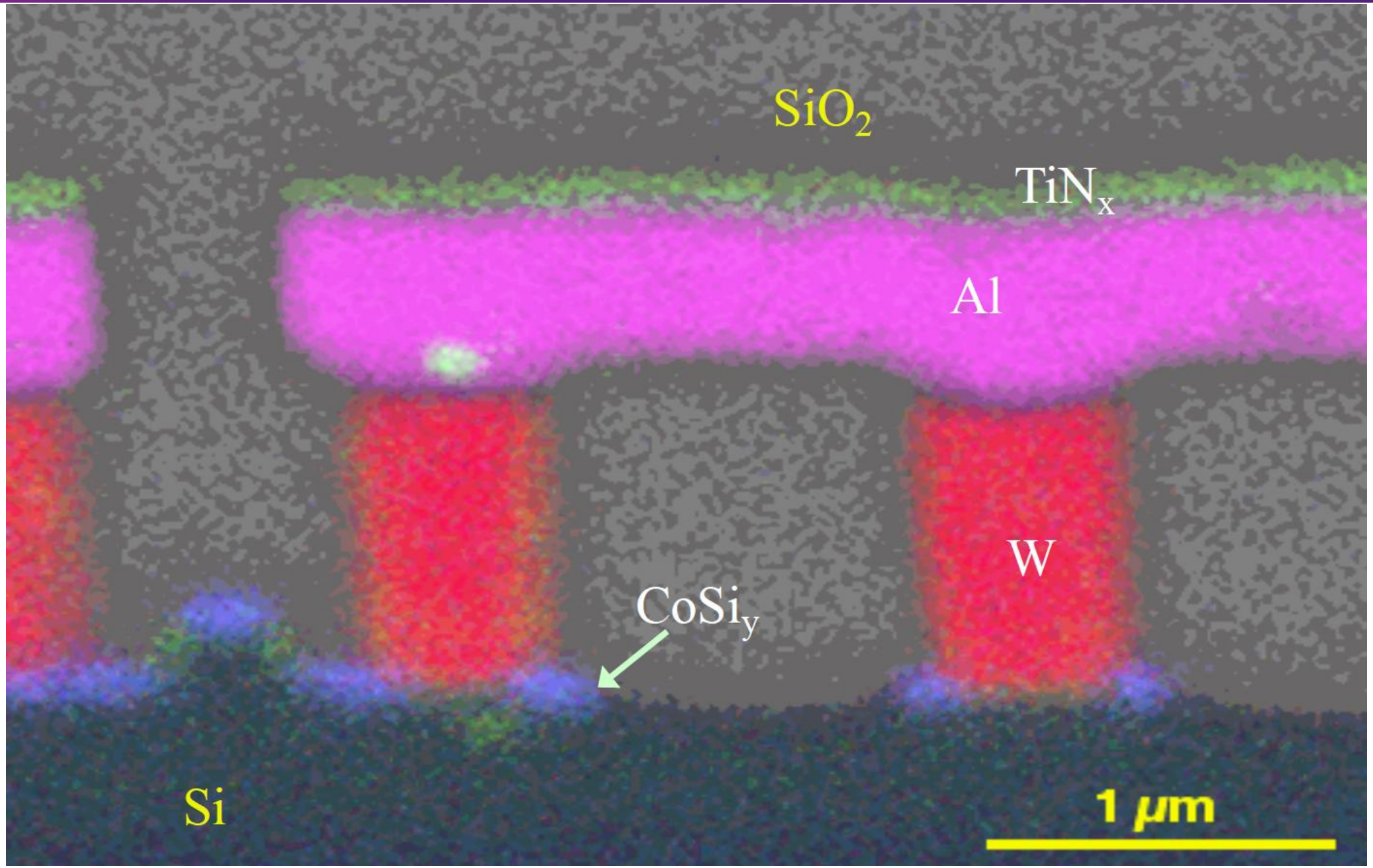

20kV
10µm
10 55 BES


ZrLα
Zr50.5Cu21.1Ni11.9Al16.5
High average atomic number
AlKα
CuLα
NiLα
CuKα
NiKα
CuKβ
Intensity (a.u.)
0 2 4 6 8 10 12 14 16
Energy (keV)

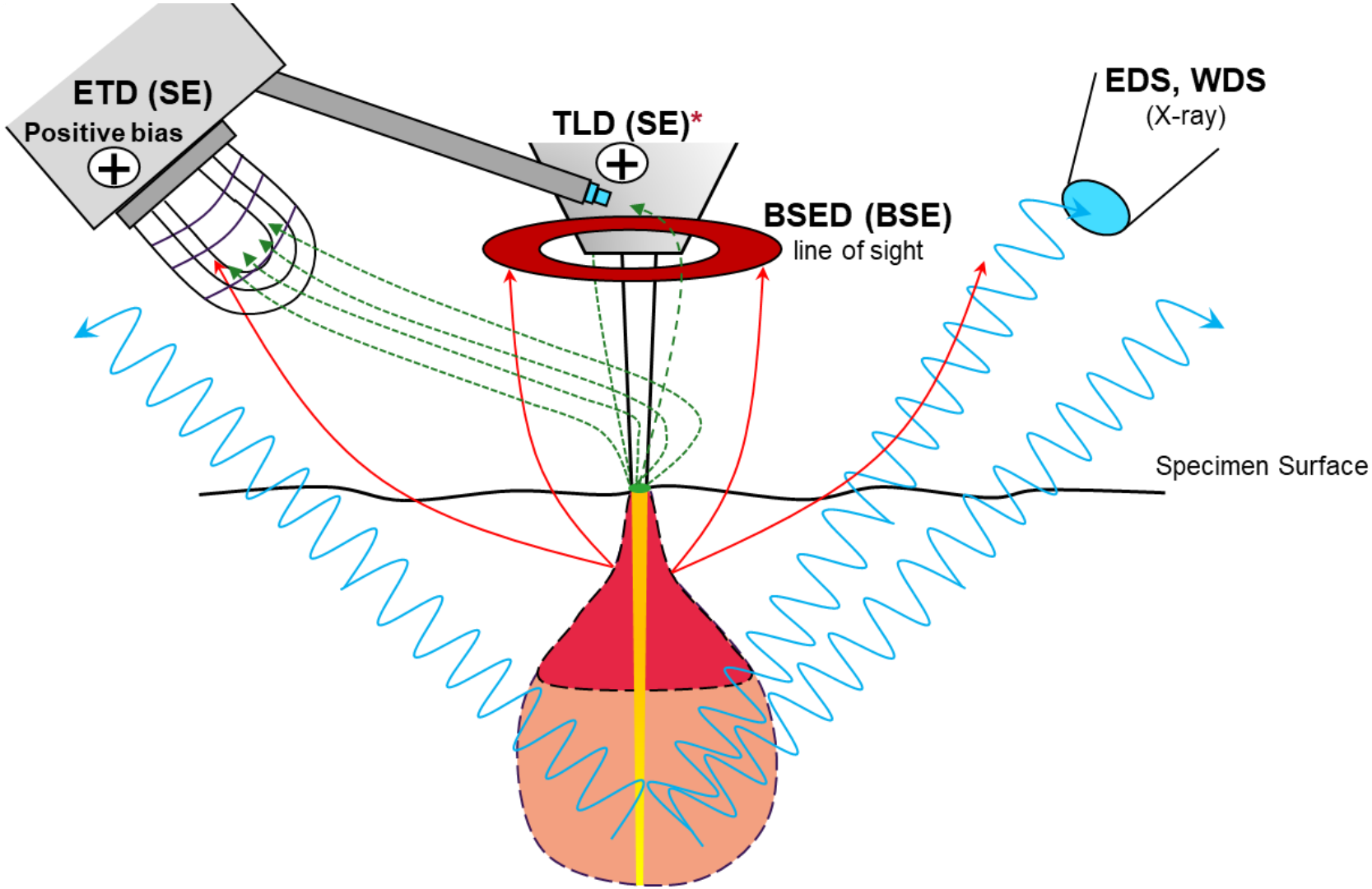

ZrLα
Zr37.2Cu23.3Ni9.4Al30.1
Low average atomic number
AlKα
CuLα
NiLα
CuKα
NiKα
CuKβ
Intensity (a.u.)
0 2 4 6 8 10 12 14 16
Energy (keV)


ZrLα
Zr46.4Cu19.0Ni11.6Al19.6O1.4
AlKα
CuLα
NiLα
OKα
NiKα
CuKα
CuKβ
Intensity (a.u.)
0 2 4 6 8 10 12 14 16
Energy (keV)

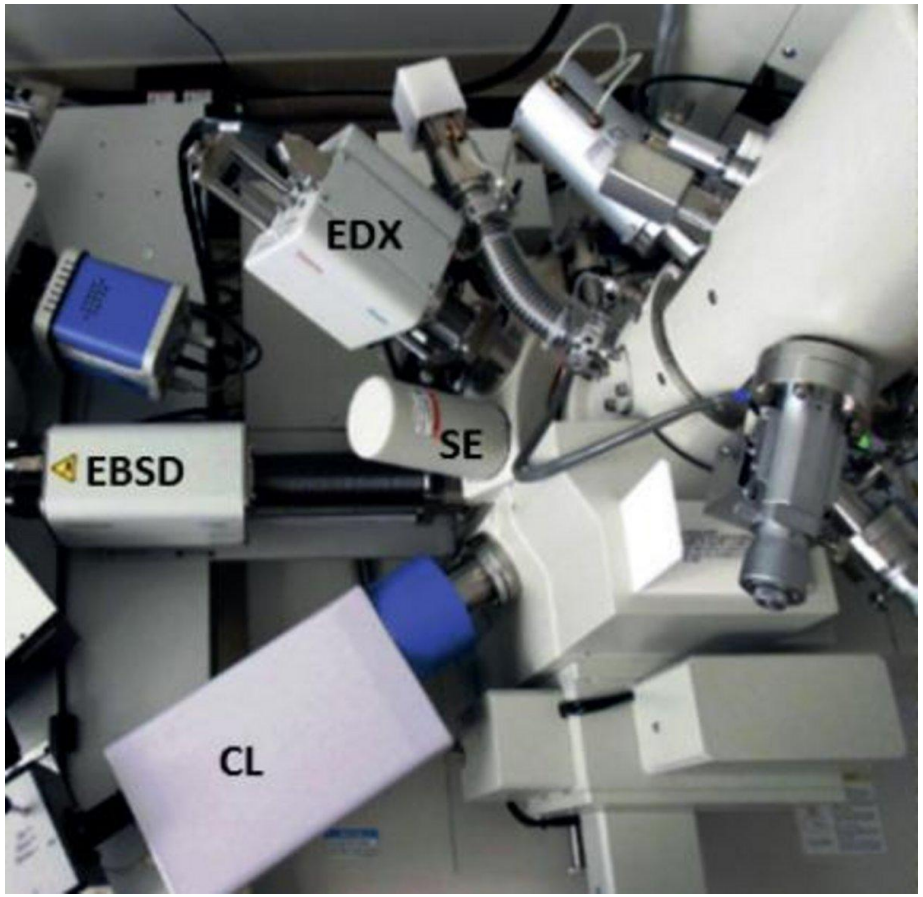
MATE 6301 | Example of X-ray mapping

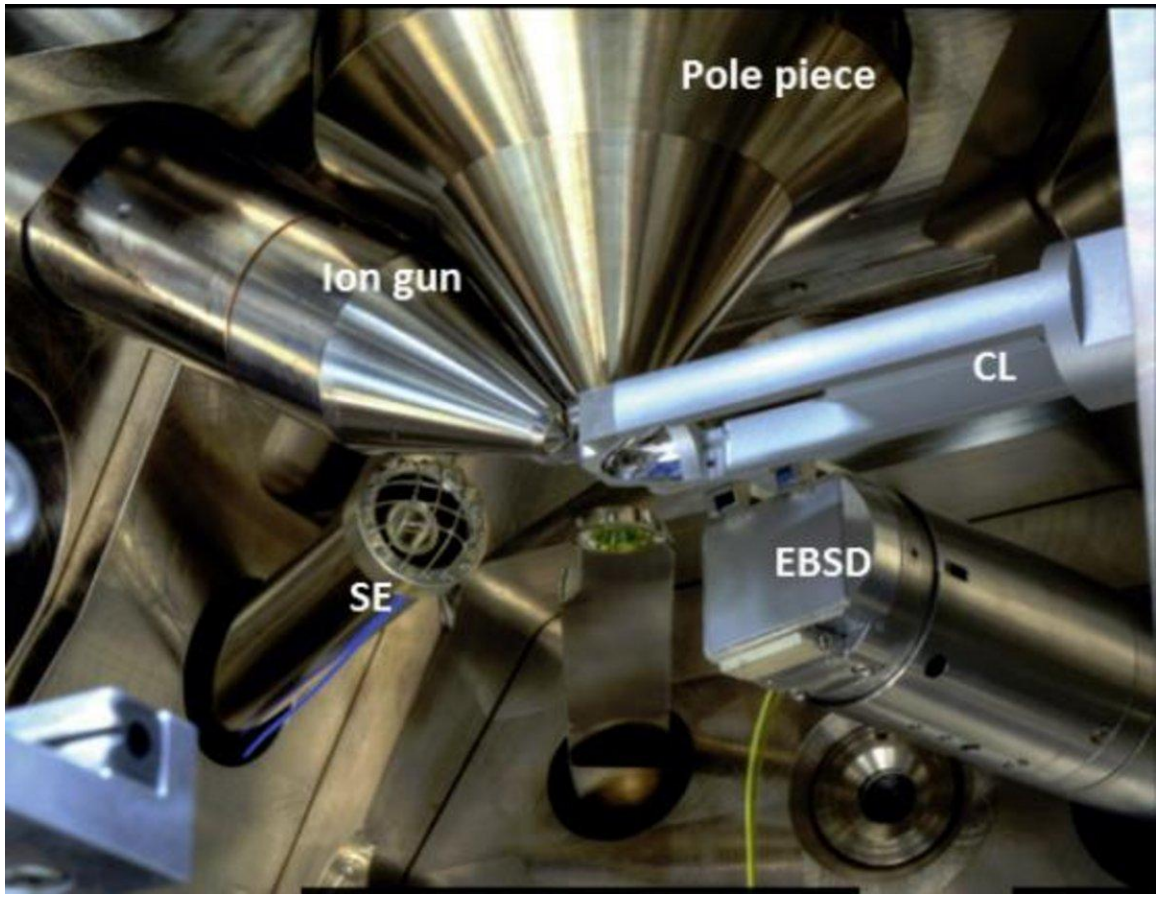


Different Signals & Detectors:



- Detectors inside the microscope collect various signals. These signals are amplified and sent to the computer, and an image is created.



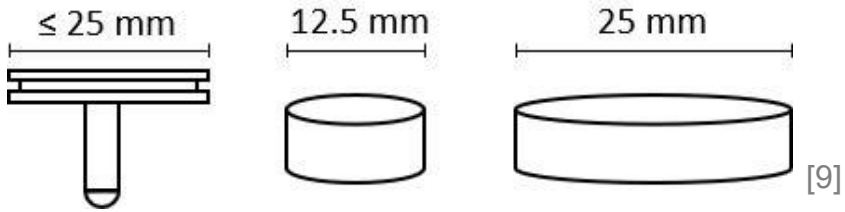


Detector	Signal	Recommended Current	Information Detected
Everhart-Thornley Detector (ETD)	Secondary Electrons (SE)	1 – 200 pA (Spot 1-3)	Surface Topography
Through-the-Lens Detector (TLD)	Secondary Electrons* (SE*)	1 – 200 pA (Spot 1-3)	Surface Topography
Backscattered Electron Detector (BSED)	Backscattered Electron (BSE)	50 pA - 2 nA (Spot 2-5)	Atomic # (Z) Contrast
Energy Dispersive Spectroscopy (EDS)	X-ray	100 pA - 10 nA (Spot 3-7)	Elemental Composition
Wavelength Dispersive Spectroscopy (WDS)	X-ray	100 pA - 10 nA (Spot 3-7)	Elemental Composition

SEM Sample Preparation

Sample Preparation Procedure

1. Mount sample on the sample mount/stub:
   - Sample should be attached to a standard pin stub (≤ 25 mm) or solid cylinder stub (12.5 mm or 25mm) (Fig. 1) or cast in a resin block (12.5 mm or 25 mm diameter, *as shown below*). When using solid cylinder stubs, it is recommended to use thicker (> 8 mm) stubs.



[9]

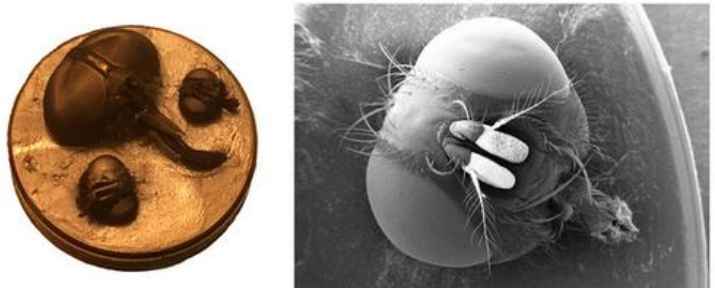
   - Samples are then mounted onto the stub using a conductive carbon tape.

2. Outgas/clean sample:
   - Outgassing is done, by placing the sample in a vacuum oven at 70 °C for 8 hrs or equivalent, followed by ozone plasma cleaning (optional, based on samples reactivity).

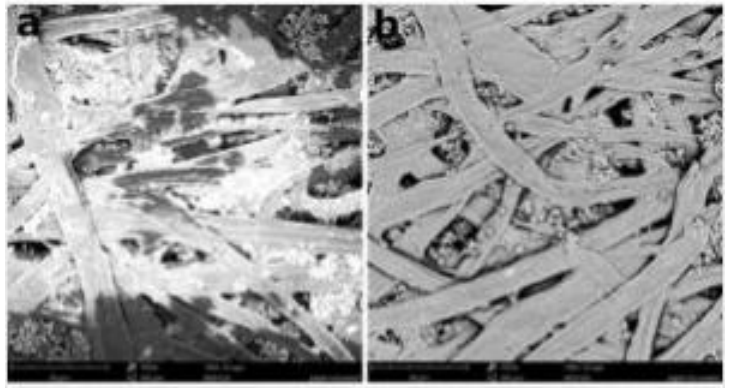
3. Conductive coating:
   - A thin uniform layer of conductive coating (*i.e.,* Au or Carbon) are deposited under vacuum in case of an electrically insulating samples before loading the sample for imaging.

Sample Preparation Tips:

- Dry and conductive samples are required.
- Water containing samples affect vacuuming the chamber and energetic electrons incidence on the sample may evaporate the residual water and affect image quality.
- The conductivity of the sample surface is vital to prevent charge build-up on its surface.
- For non-conductive samples, create a connection to the ground using metallic tape. And sputter coat the sample with gold.



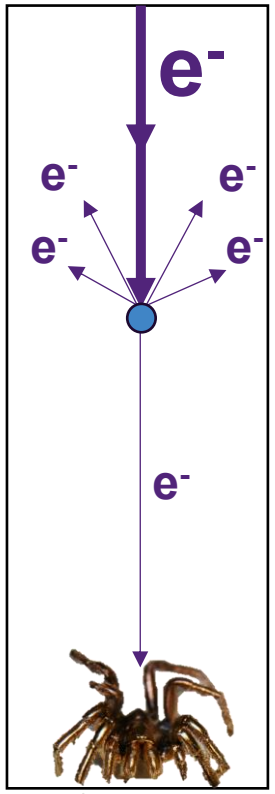
Three insect heads coated with gold on a stub (left) and SEM Image of one of the heads (right)



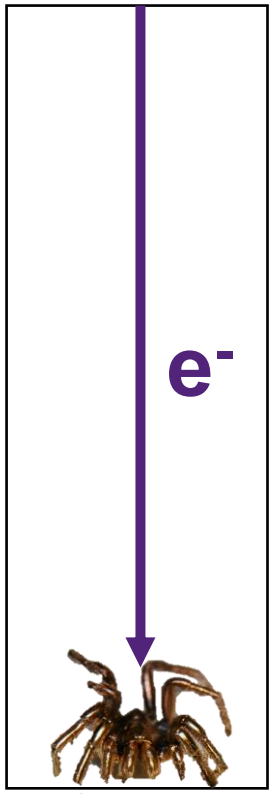
Non-conductive **Conductive**

[10]

- An electron beam wouldn't have a very long mean free path in air. This is why the SEM is under vacuum, so the electrons can make it from the source to the sample.



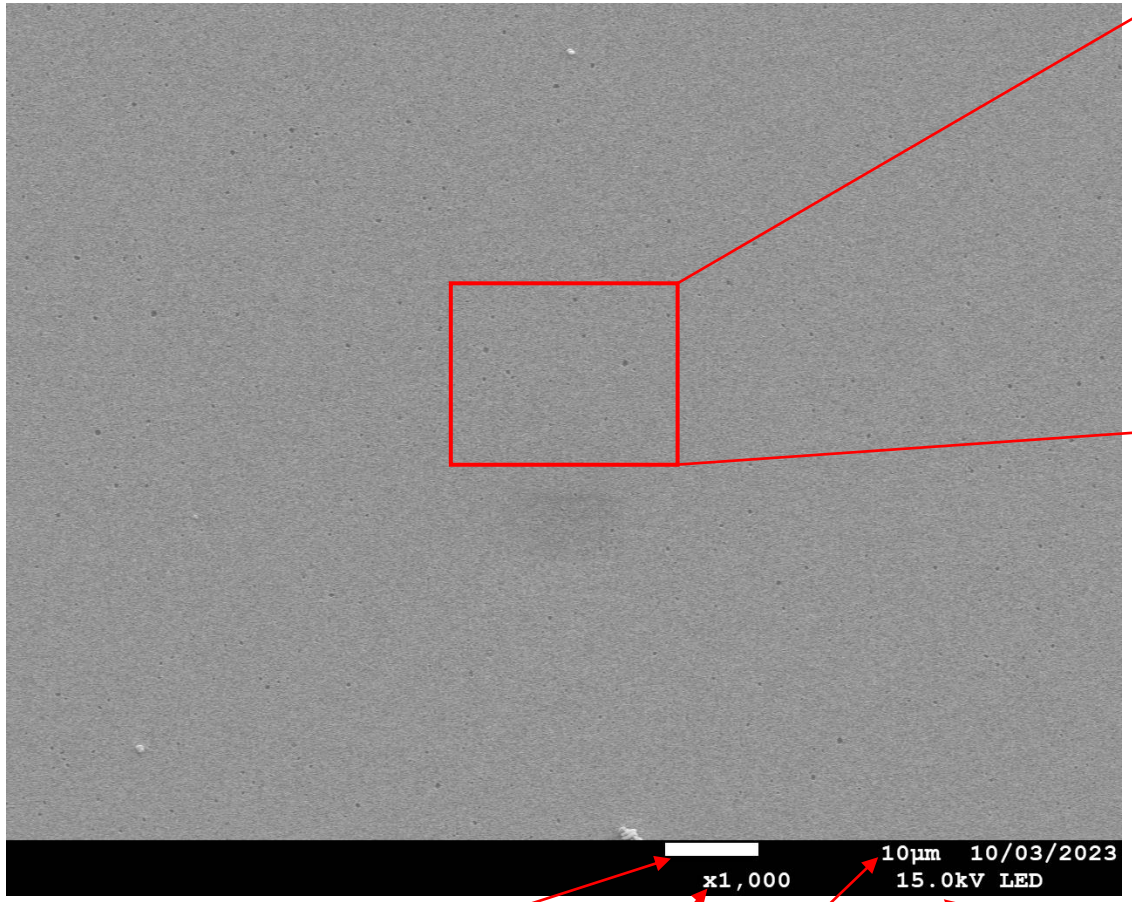
In air



In vacuum

Powder-free gloves

Results: SEM surface morphology



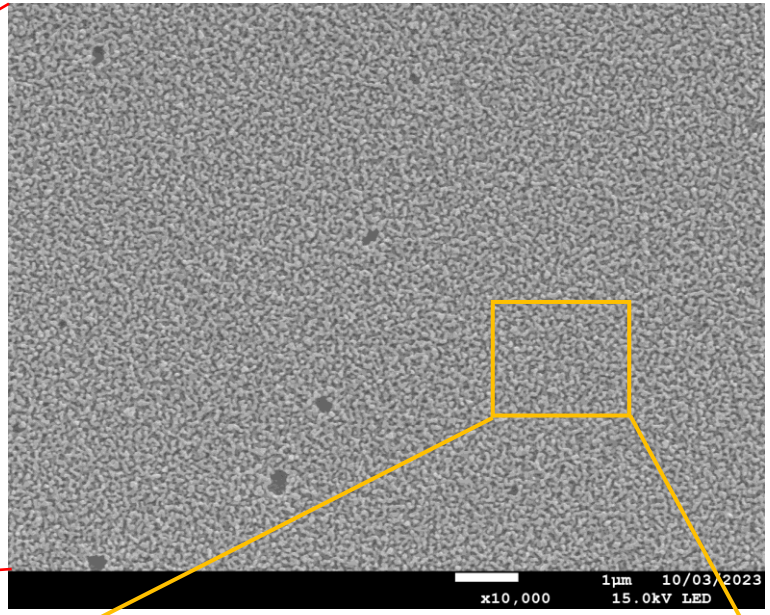
Scale bar

Magnification

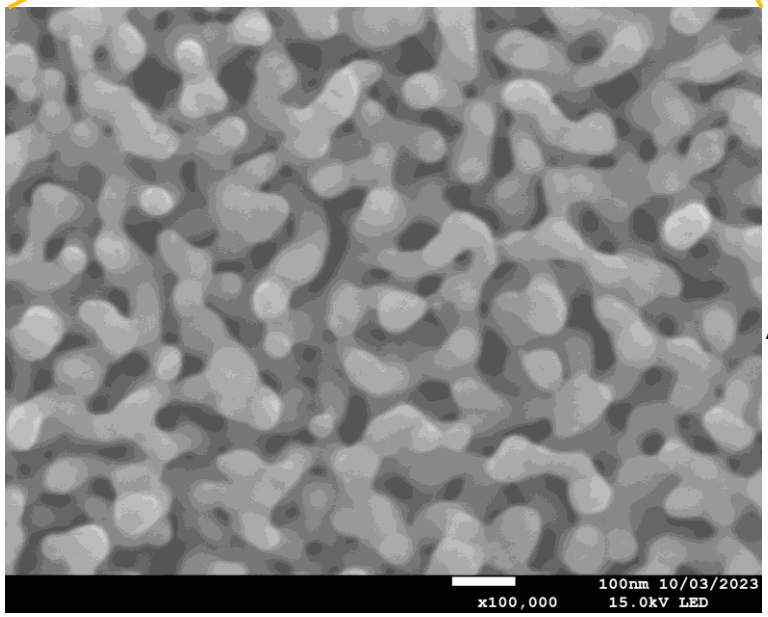
Scale

Acceleration voltage

x10

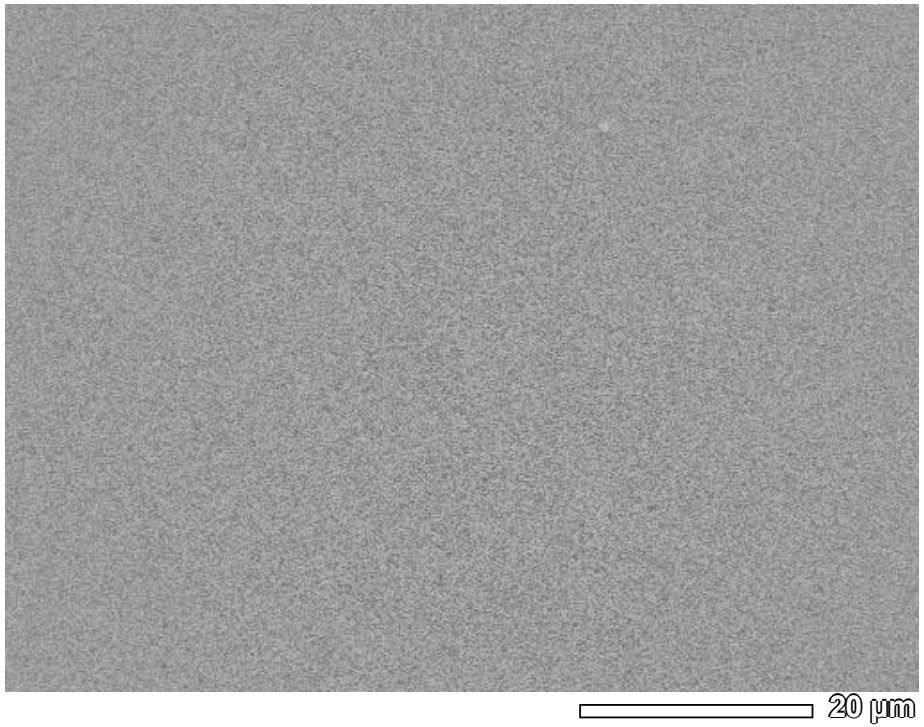


x10

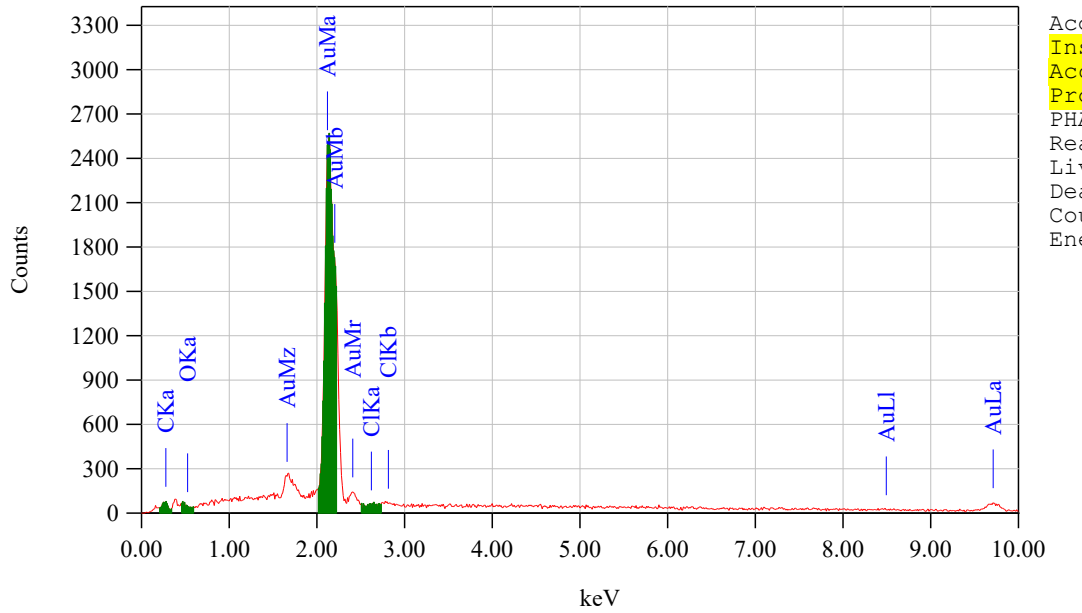


Mesoporous Au surface

Results: EDX data




```
Title        : IMG1
---------------------------
Instrument   : 7100F
Volt         : 15.00 kV
Mag.         : x 1,500
Date         : 2023/03/10
Pixel        : 512 x 384
```




```
Acquisition Parameter
Instrument    : 7100F
Acc. Voltage  : 15.0 kV
Probe Current : 7.47500 nA
PHA mode      : T3
Real Time     : 50.74 sec
Live Time     : 50.00 sec
Dead Time     : 1 %
Counting Rate : 1764 cps
Energy Range  :  0 - 20 keV
```

ZAF Method Standardless Quantitative Analysis
Fitting Coefficient : 0.0763

Element	(keV)	Mass%	Sigma	Atom%	Compound	Mass%	Cation	K
C K*	0.277	3.95	0.17	39.05				1.8573
O K*	0.525	0.37	0.09	2.77				0.3666
Cl K*	2.621	0.21	0.06	0.70				0.2151
Au M*	2.121	95.46	1.00	57.48				97.5611
Total		100.00		100.00				

Key-Points

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1. Sample preparation

- Conductive sample
- Non-conductive sample

2. Beam alignment

- Focal plane (or working distance) adjustment
- Column (aperture & lens) alignment
- Astigmatism correction

3. Aberration correction

- Spherical aberration
- Chromatic aberration

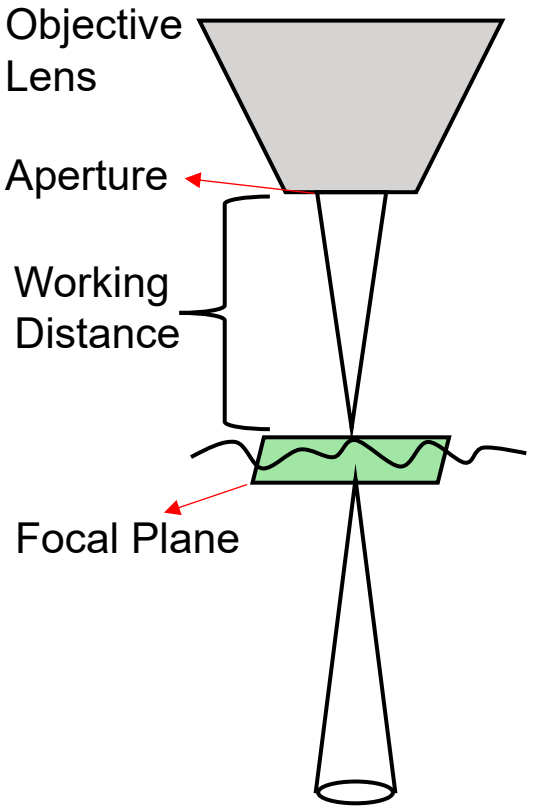
4. Imaging parameters

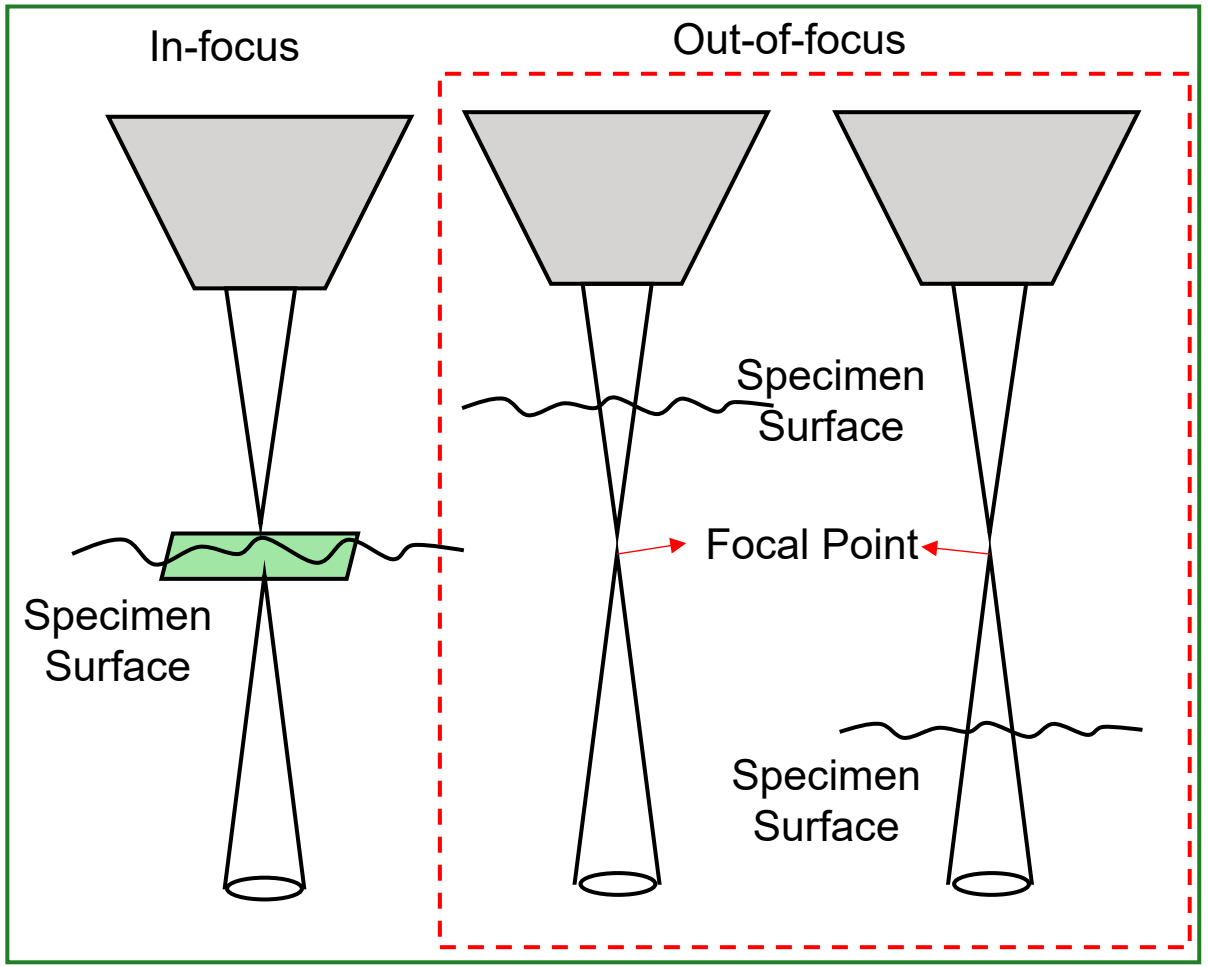
- Accelerating voltage
- Beam current
- Working distance
- Aperture

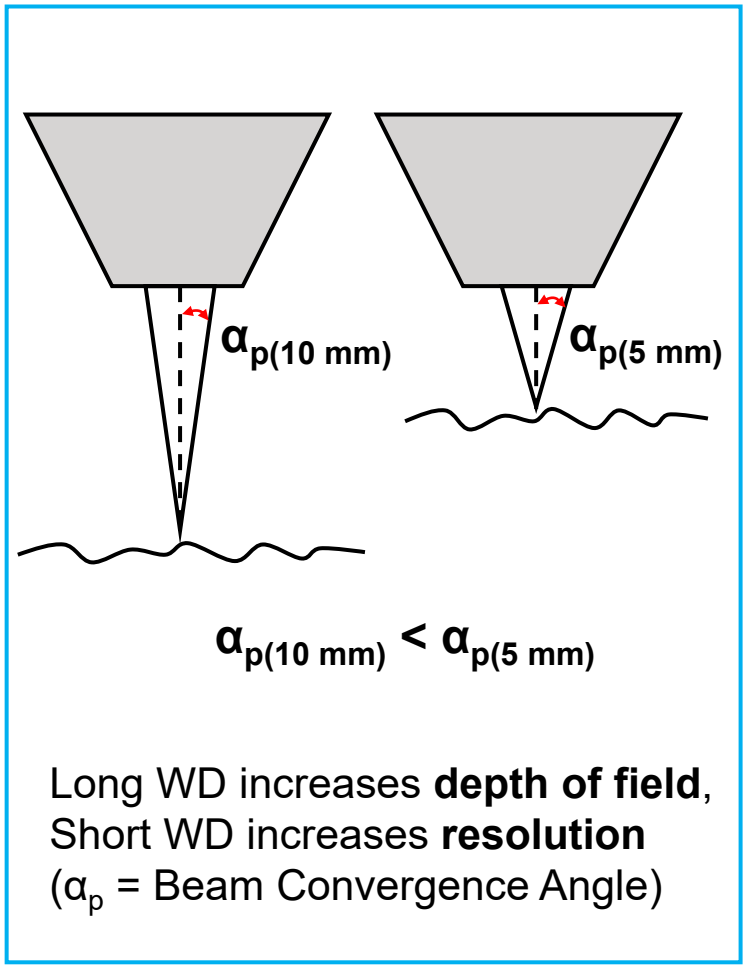
Non-controllable

Controllable

Working Distance (WD):







↓ **WD** ↑ **α_p** ↓ **Depth of Field** ↑ **Resolution**

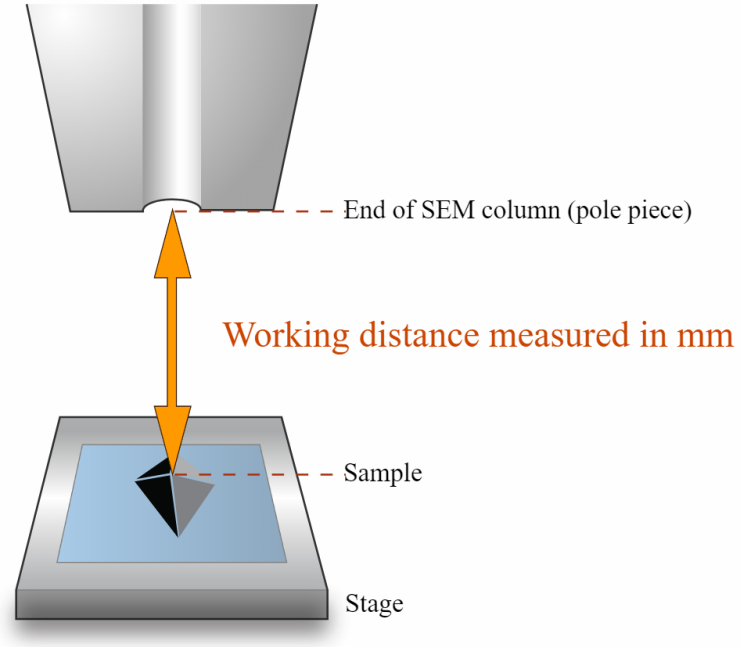
Working distance (WD)

- The distance between the sample and the end of SEM column.

- Controls focal plane height, therefore, focus.

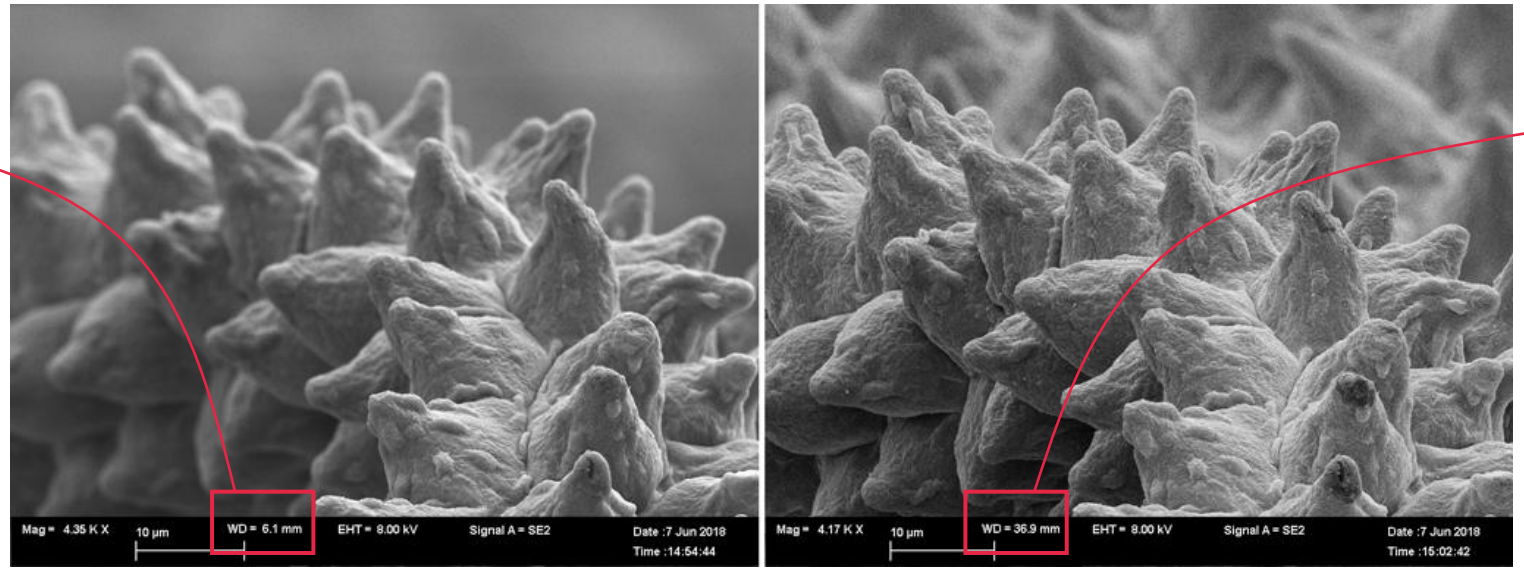
- Controls beam convergence angle (α_p), therefore, depth of field (DoF).

- WD $\propto \alpha_p \propto \frac{1}{\text{DoF}} \propto$ Resolution



WD = 6.1 mm

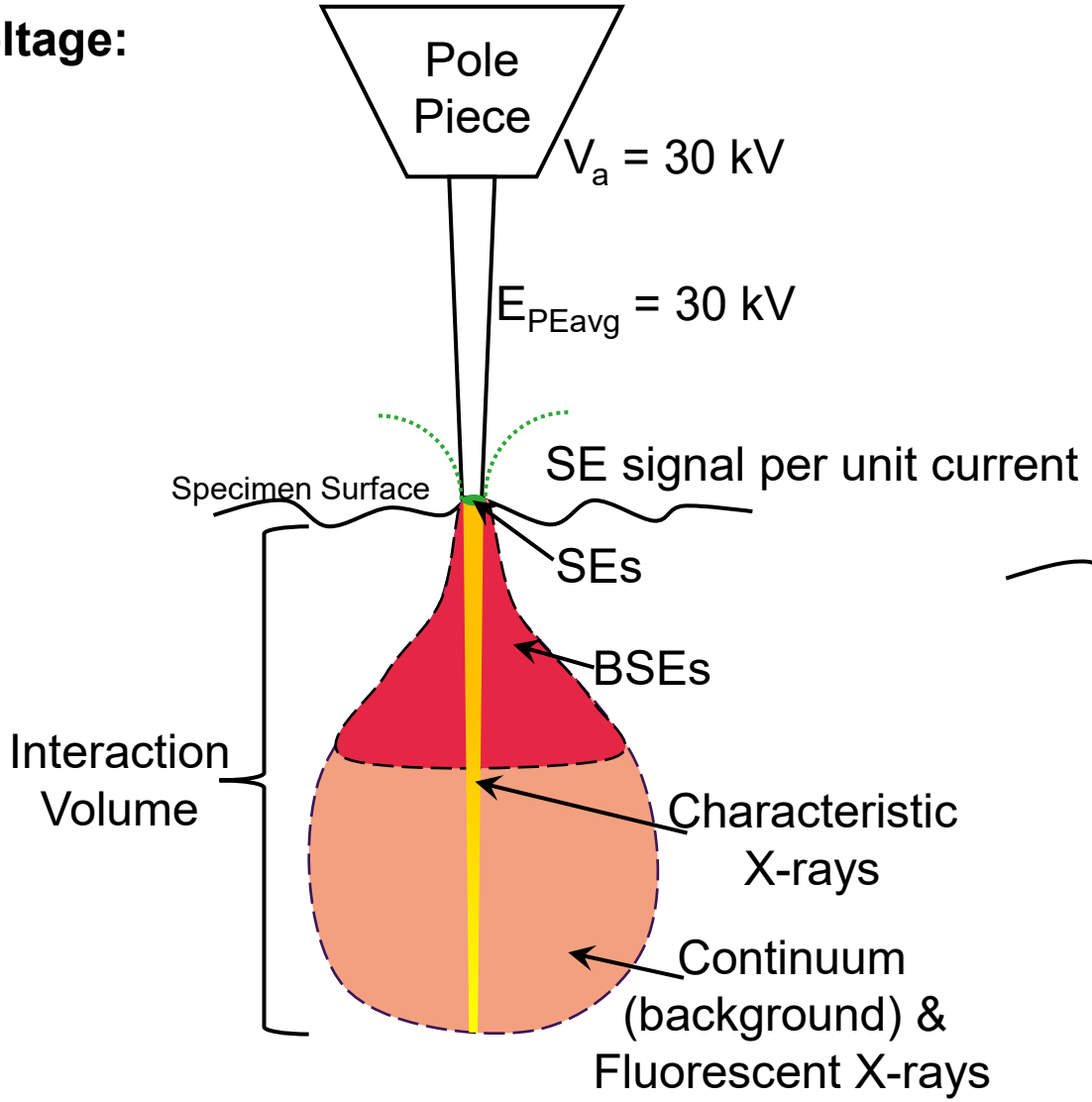
Shallower DoF
Higher resolution

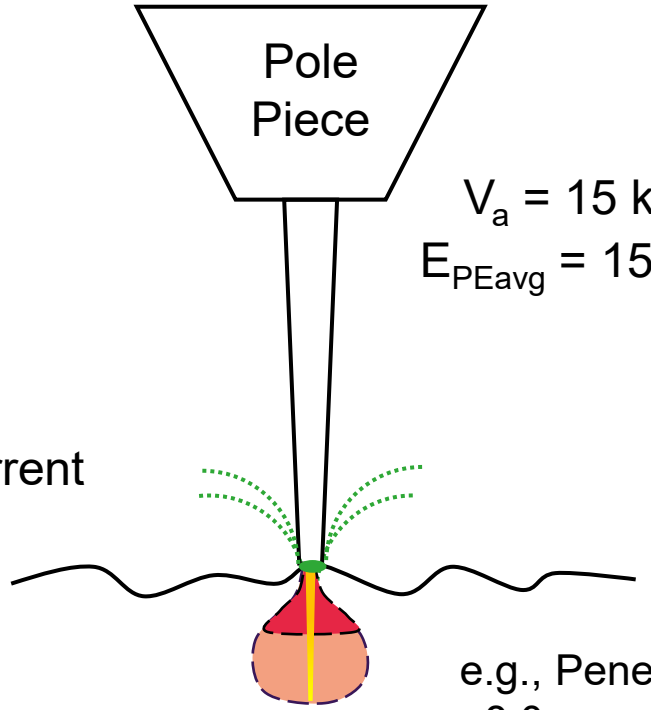


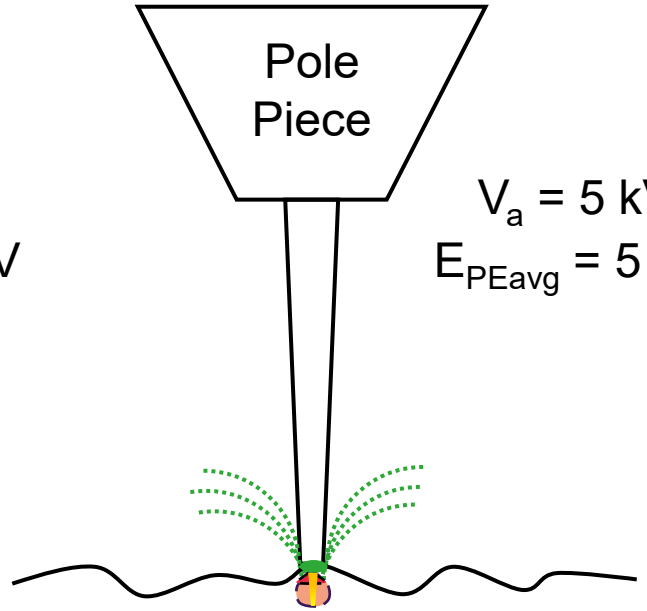
WD = 36.9 mm

Deeper DoF
Lower resolution

Voltage:





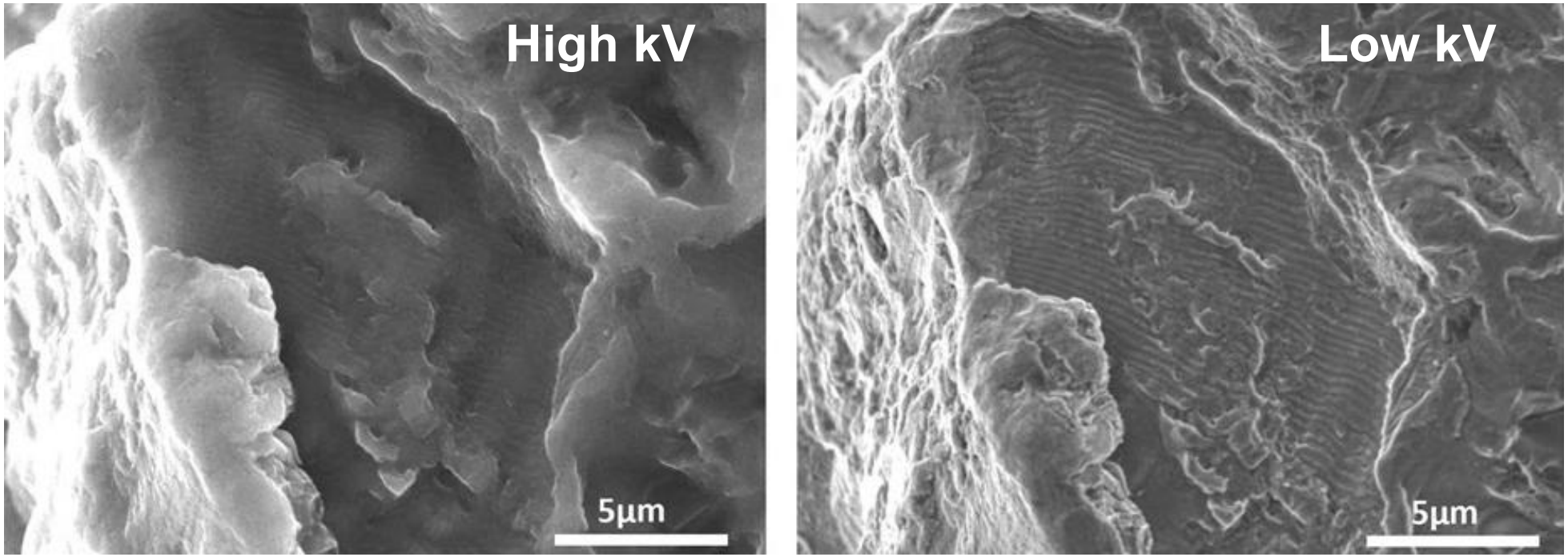


e.g., Penetration depth into Si:
~8.0 μm at 30 kV, 2.5 μm at 15 kV,
0.4 μm at 5 kV (volumes drawn to scale)

Note:

- For non-conductive specimens, deeper penetration at high V_o builds charge decrease image quality.
- For non-conductive specimens, smaller λ – and less severe aberrations – at high V_o increases resolution.

Volume ∝ Voltage ∝ E_{PEavg} $1/\lambda_{PE}$



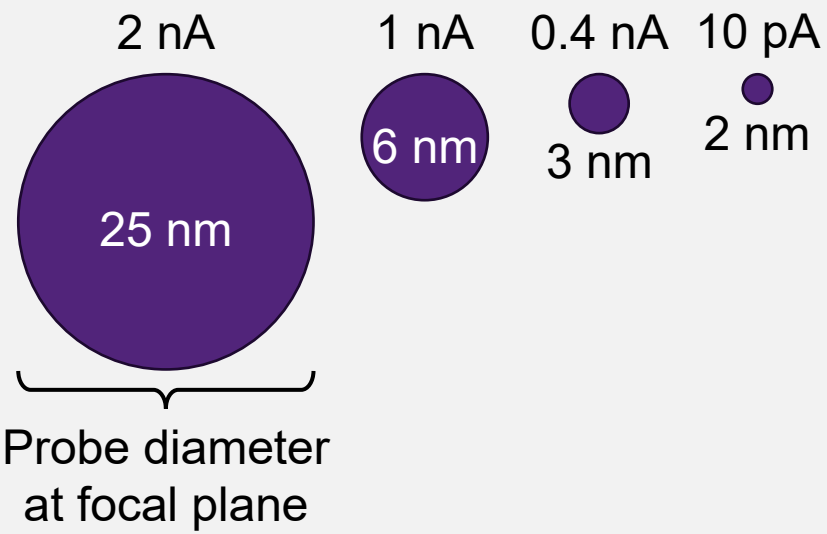
An aluminum fatigue failure imaged at high and low kV.
Because Al is a light (low atomic No.) metal, and self oxidizes, using a high kV penetrates deep into the metal and the signal that comes from below the surface masks the fatigue striations on the fracture surface.

Probe Current

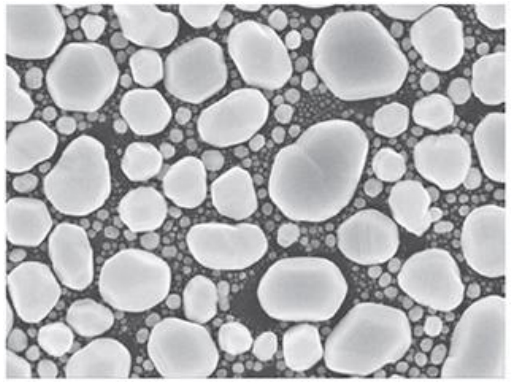
- Controls the amount of electrons.

- Therefore, controls the amount of interactions and signals.

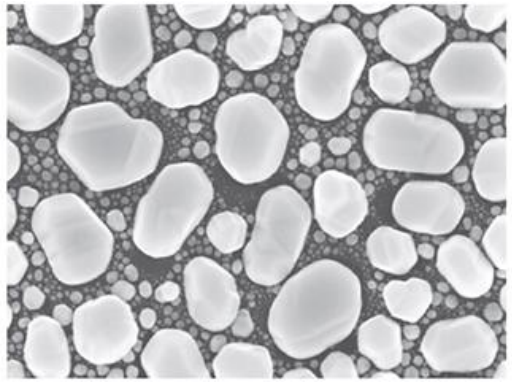
- Probe current $\propto$ Probe diameter $\propto$ Amount of signal $\propto \frac{1}{\text{Resolution}}$



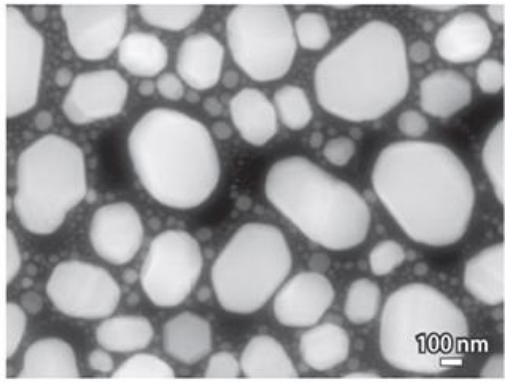
Specimen: Gold coated carbon, 15 kV



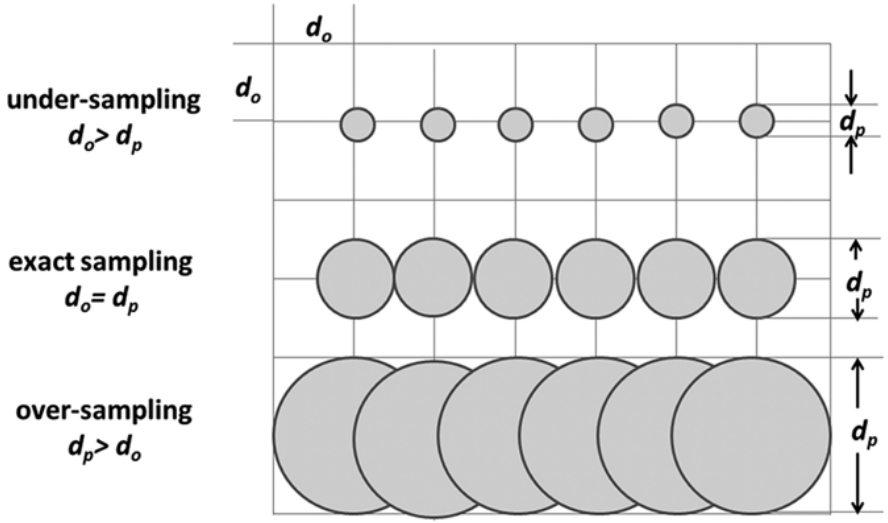
Probe current = 50 pA



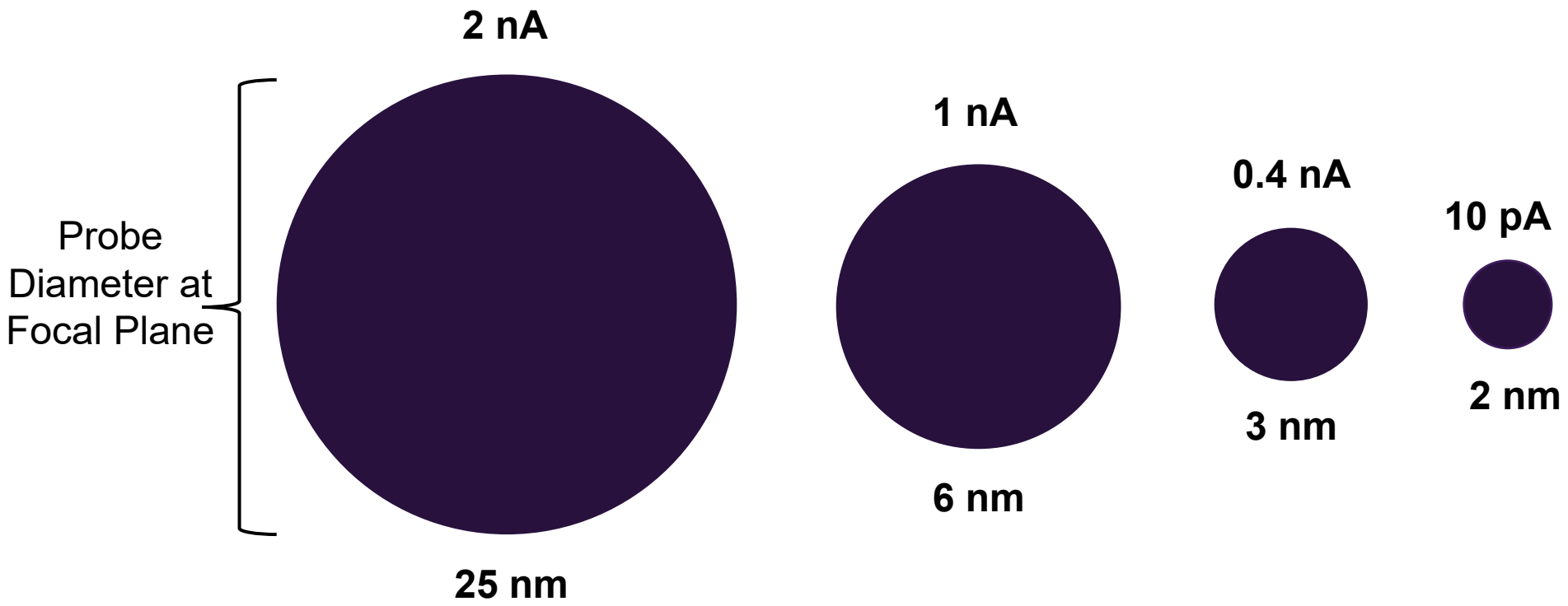
Probe current = 500 pA



Probe current = 100 nA



Probe Current: Controls the amount of electrons set to interact with the specimen surface (amount of interactions ↑, amount of signal ↑)

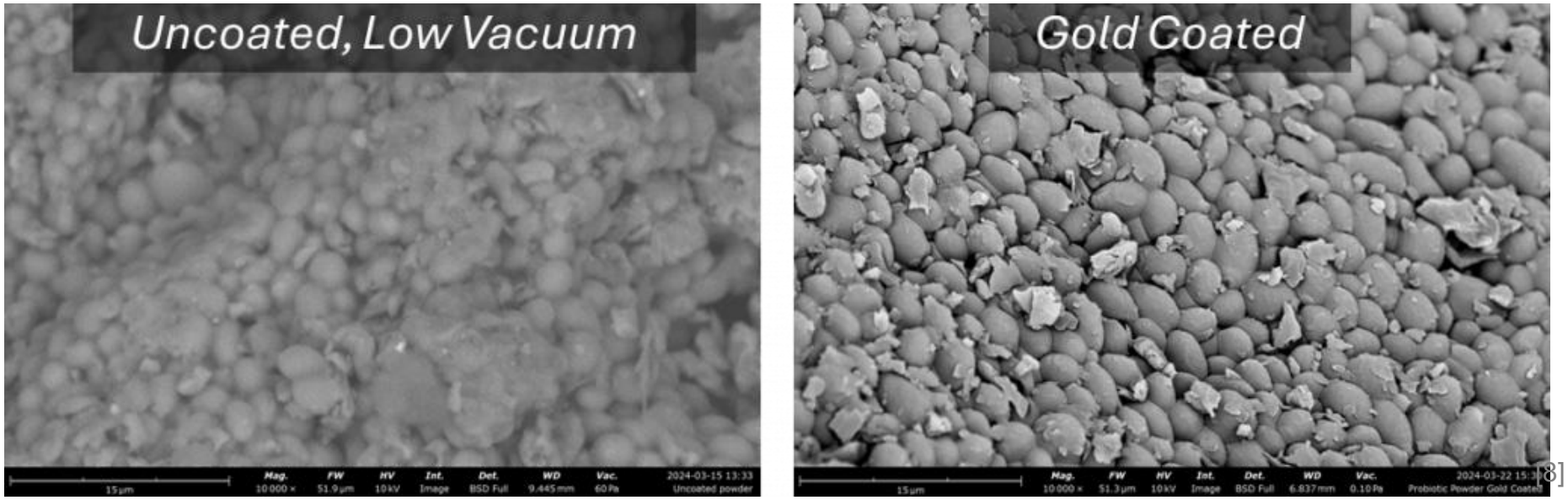


Note:

Choose the lowest current value that still generates an adequate signal for the detector/application. If the contrast in image is less than 75%, dropping the current to improve resolution is something to think about.

↓ **Probe Current** ↓ **Probe Diameter** ↓ **Signal** ↑ **Resolution**

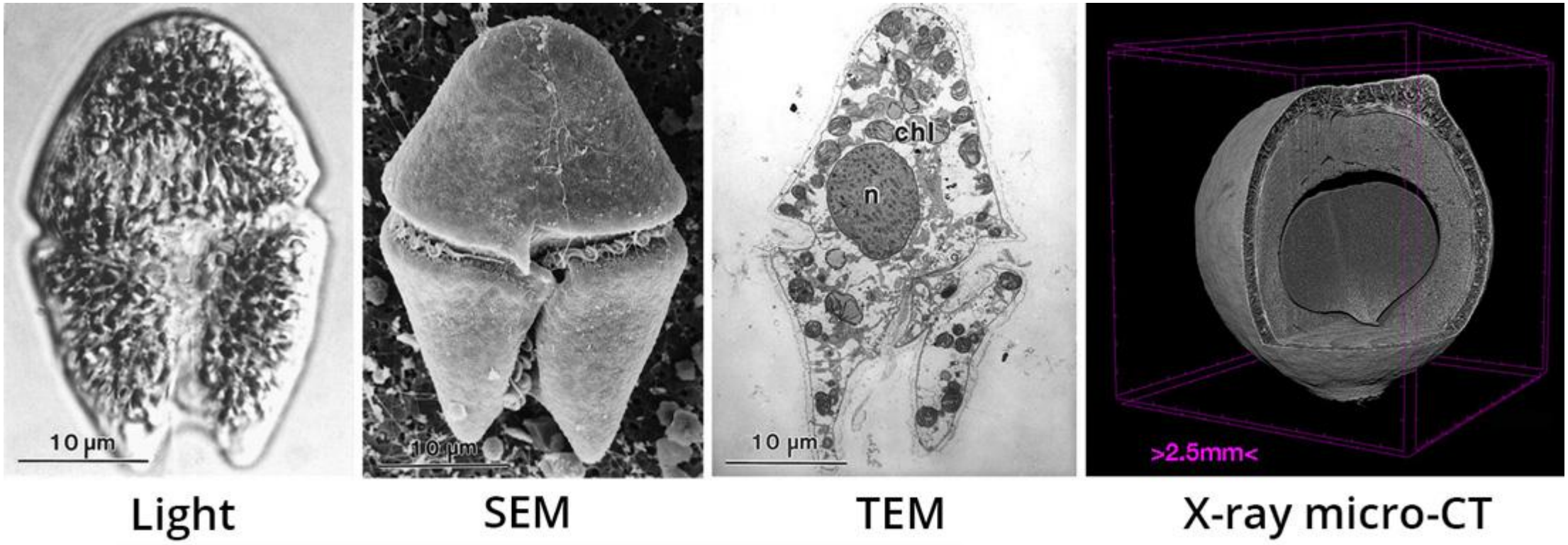
- SEM images are not created all at once. The beam is rastered in a grid you specify, and data is collected for each pixel in the grid.
- Depending on the scan rate, the amount of data per pixel varies.
- Better quality image can be acquired at slower scan rate.



[8]

- Samples to be conductive or coated with a conductive layer (*e.g.*, gold or carbon).
- Operates in a vacuum environment, which can limit the types of samples that can be observed.
- The technique provides mainly surface information, not internal structure, unless cross-sections are prepared.
- The sample surface should be relatively planar.
- The sample should be stable to the electron beam.
  - Beam voltage - sample charring
  - Sample shape & moisture (can damage the pole piece and affect the vacuum)
  - Proper grounding
  - Thin Au or carbon coating for better imaging

Difference:



References:

1. https://andor.oxinst.com/learning/view/article/most-popular-microscopy-techniques
2. File:Misc pollen.jpg – Wikipedia
3. https://myscope.training/Concepts_Electromagnetic_spectrum_and_microscopy
4. metals-13-00416-g001-550.jpg (550×423) (mdpi.com)Backscattered Electron - BSE- SEM Imaging - Advancing Materials (thermofisher.com)
5. Backscattered Electron - BSE- SEM Imaging - Advancing Materials (thermofisher.com))
6. SEM Signal - Electron Imaging - Advancing Materials (thermofisher.com)
7. SEM vs TEM | Technology Networks
8. Theory of Electron Microscopy — IXRF Systems
9. https://www.nanoscience.com/blogs/gold-vs-platinum-choosing-the-right-coating-for-sem-imaging/
10. JEOL 7100F-users manual, UQ-CMM
11. Sample Preparation | 4 SEM Sample Preparation Techniques (vaccoat.com)

CMM's MyScope (Visual Training)

1. https://myscope.training/SEM_simulator.html

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CREATE CHANGE

Acknowledgement

- Centre for Microscopy and Microanalysis (CMM), Hawken (bld. 50).

Thank you

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Yamauchi Group
Australian Institute for Bioengineering and Nanotechnology (AIBN)
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CRICOS code 00025B

WEEK 4 · 知识点总结 · Dr Aditya Ashok

第 4 周　透射电子显微镜

Transmission Electron Microscopy (TEM)

主线：高电压 + 超薄样品 → 三种衬度机制 → 电子衍射、STEM、EDS / EELS → 像差校正

1. SEM vs TEM

	SEM	TEM
电子	散射电子（扫描）	透射电子
图像	表面，类 3D	内部结构的 **2D 投影**
放大倍数	约 1–2 × 10^6	> 5 × 10^7
加速电压	1–30 kV	**60–300 kV**
样品厚度	不限	**< 100–150 nm**
操作 / 成本 / 速度	简单、便宜、快	复杂、昂贵、慢

- 电子波长：100 kV → 0.037 Å，200 kV → 0.025–0.027 Å，300 kV → 0.020 Å → 可实现**原子分辨**。

2. TEM 五大部件

- 电子枪（W / LaB_6 / FEG，100–300 kV）；真空系统（低至约 10^{-8} Pa）；电磁透镜（**聚光镜**形成束，**物镜**成像，**投影镜**放大到屏幕）；样品台（约 3 mm Cu 网，可倾转得到 tilt series）；成像装置（荧光屏 / CCD）。

- TEM 主要用**前向散射**信号（SEM 只用背散射）；弹性散射 → 成像与衍射；非弹性散射 → EDS、EELS。

3. 三种衬度（★）

衬度	形成方式	用途 / 要求
衍射衬度	物镜后焦面放光阑，只选**一个**束成像 → 明场 BF / 暗场 DF	晶体：缺陷、析出相、晶粒尺寸
相位衬度（HRTEM）	**≥ 2 个**衍射束干涉	晶格条纹、界面；样品 **< 20 nm**，晶带轴平行电子束；常需模拟解释
质厚衬度	原子核的非相干弹性散射（Rutherford）	Z 大或厚 → 暗；适合**非晶**、聚合物、生物样品、介孔 SiO_2

4. 衍射与分析

- **SAED**：单晶 → 斑点；多晶 → 衍射环；非晶 → 弥散晕。
- **STEM**：会聚束扫描；**HAADF** 衬度 ∝ Z（Z 衬度，重原子更亮）；可做原子级 EDS 元素分布图。
- **EDS**：元素成分；**EELS**：成分 + **化学键 / 氧化态**。

5. 分辨率与像差

- $\delta_{HRTEM} = 0.64\,(C_s\,\lambda^3)^{1/4}$，$\delta_{STEM} = 0.43\,(C_s\,\lambda^3)^{1/4}$。
- λ 不能无限减小（电子束损伤）→ 只能减小 C_s → **球差校正器**，分辨率达约 **0.05 nm**。
- 球差 $r_{sph} = C_s\,\beta^3$；色差 $r_{chr} = C_c\,\beta\,(\Delta E / E_0)$；除像散外，像差不能由操作者校正，需要硬件。

6. 制样与局限

- 粉末：分散在乙醇或异丙醇中，**短时间超声**（约 10 min，保护孔结构）→ 滴在碳膜 Cu 网上；薄膜：FIB 切片。
- 局限：观察区域极小（**代表性？**）；制样繁琐；**电子束损伤**（轻元素、生物、软材料）；2D 投影解释复杂；需真空。

考试重点：SEM 与 TEM 对比表；三种衬度的成像原理与用途；SAED 斑点 / 环 / 晕；为什么需要球差校正；TEM 结果需要与 XRD 统计结果互相印证（W6 案例 3、W12）。

MATE 6301

Transition Electron Microscopy (TEM)

MATE 6301 | Light Microscope *vs* Electron Microscope

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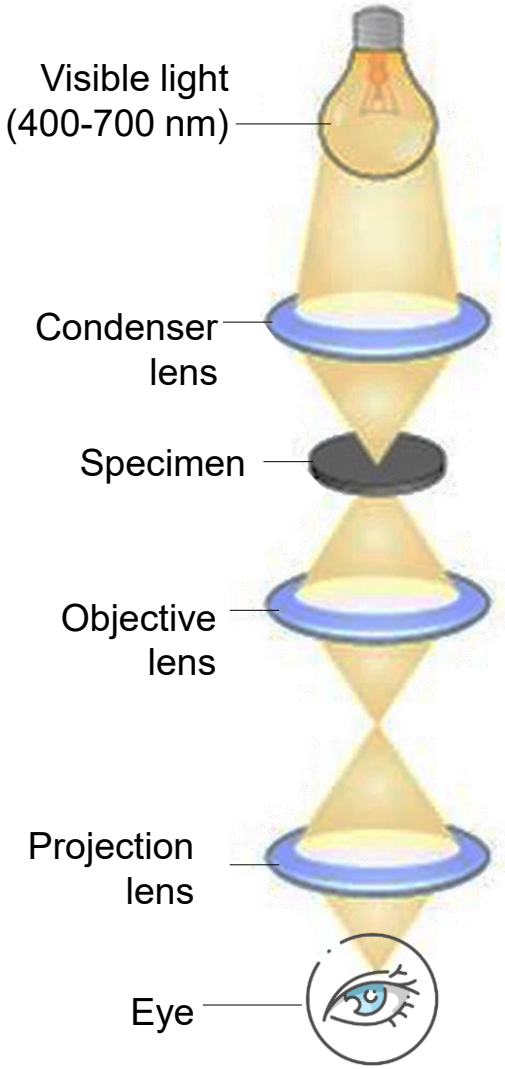
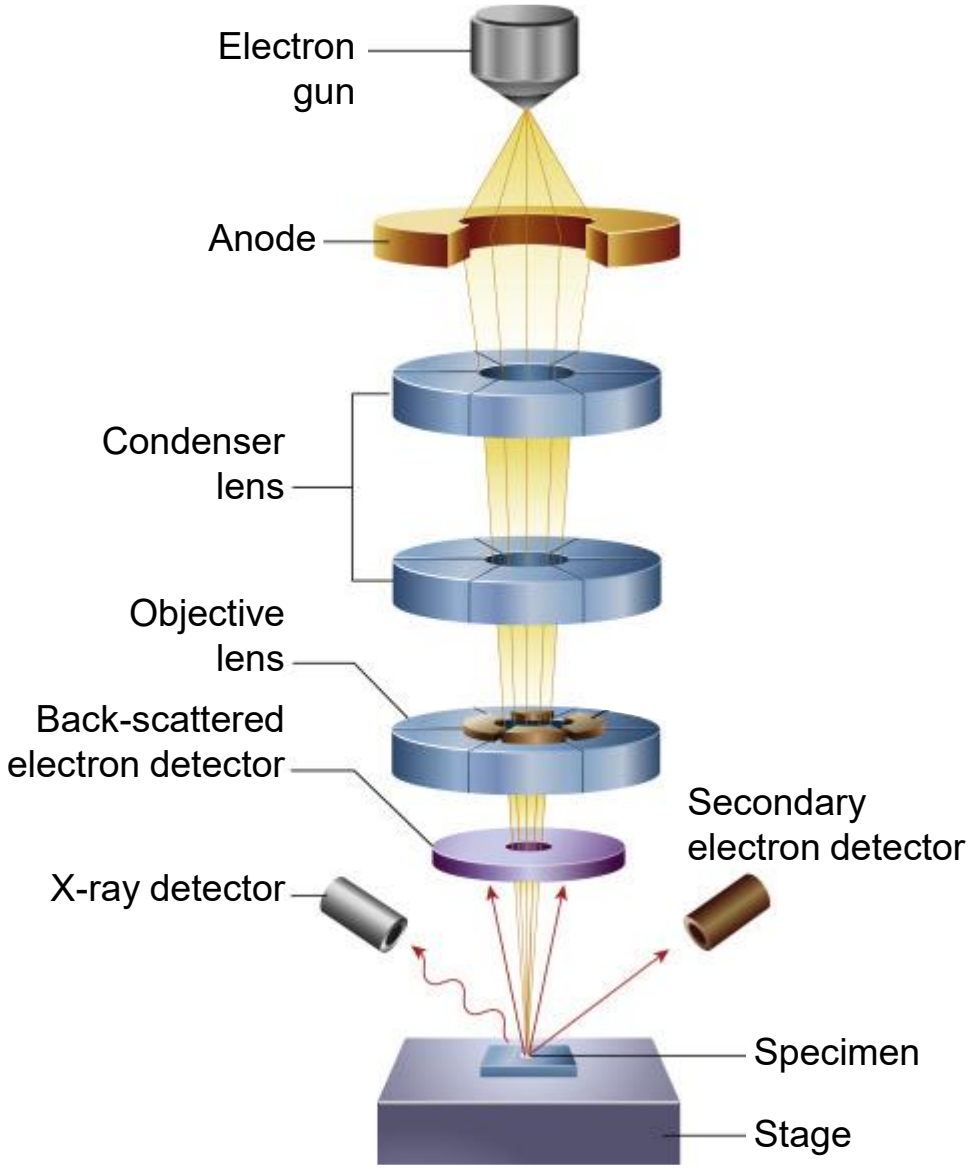
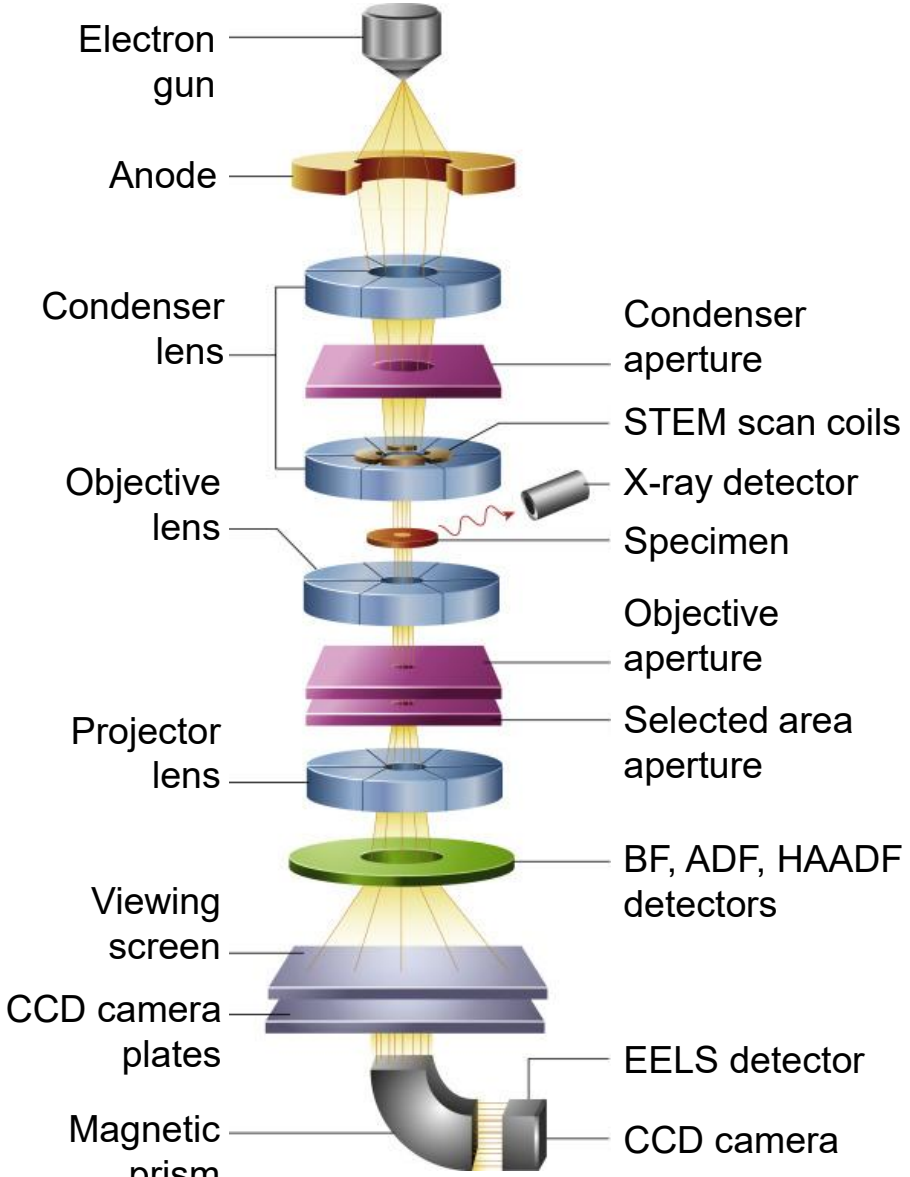
Light microscope



Scanning EM



Transmission EM



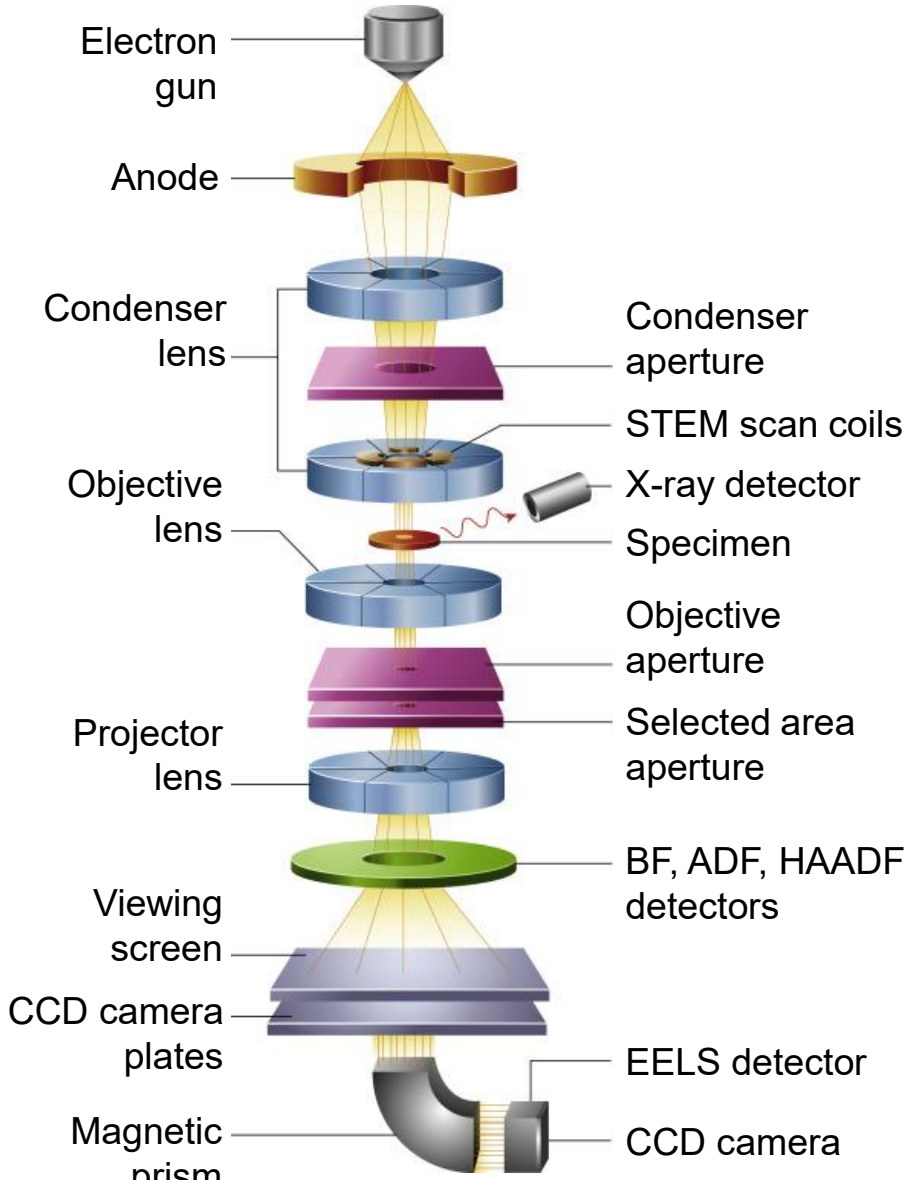
MATE 6301 | Transmission Electron Microscope (TEM)

Ernst Ruska

The father of transmission electron microscopy (TEM).

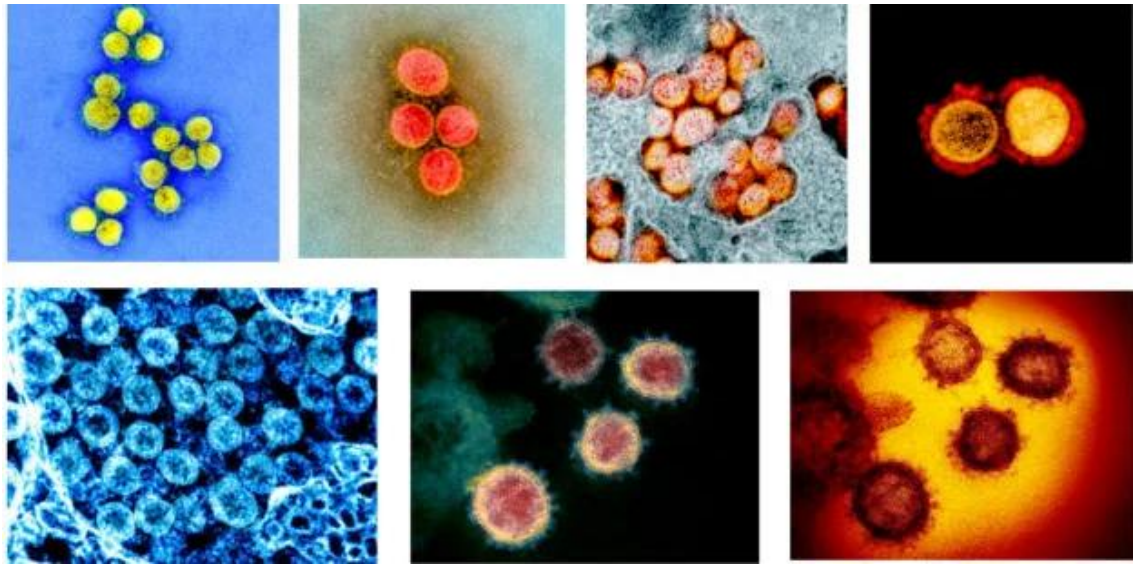
In 1932-33, he came up with the concept and developed a prototype TEM for his Ph.D. He worked further and commercialised the first TEM on the market in 1939.

Transmission EM



What is TEM?

Transmission Electron Microscope (TEM) is a powerful electron microscope that uses a beam of electrons to focus on a specimen. TEM has the potential to magnify an image up to 10-50 million times.



[1]

What material information can be obtained?

Applying TEM to the real world has fuelled many discoveries and advancements. Scientists can acquire a basic understanding of the internal structures of materials and biological systems down to the atomic level.

Information such as the morphological features to high-resolution lattice images, compositions and crystal information of the specimen.

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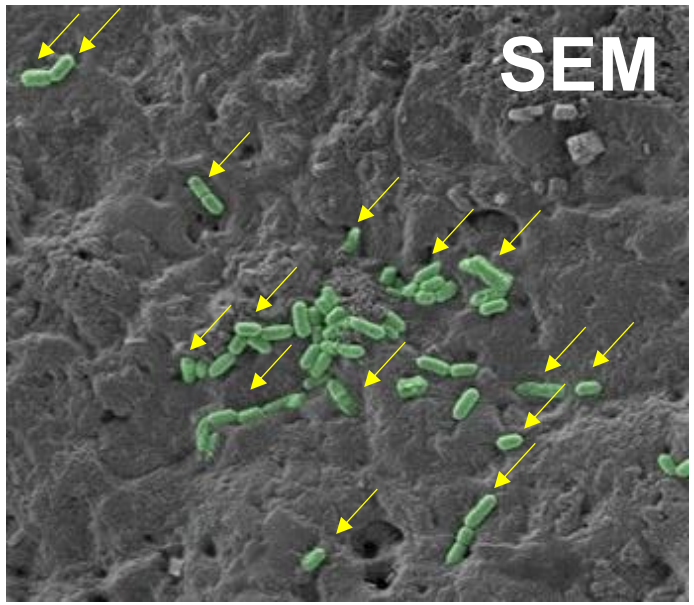
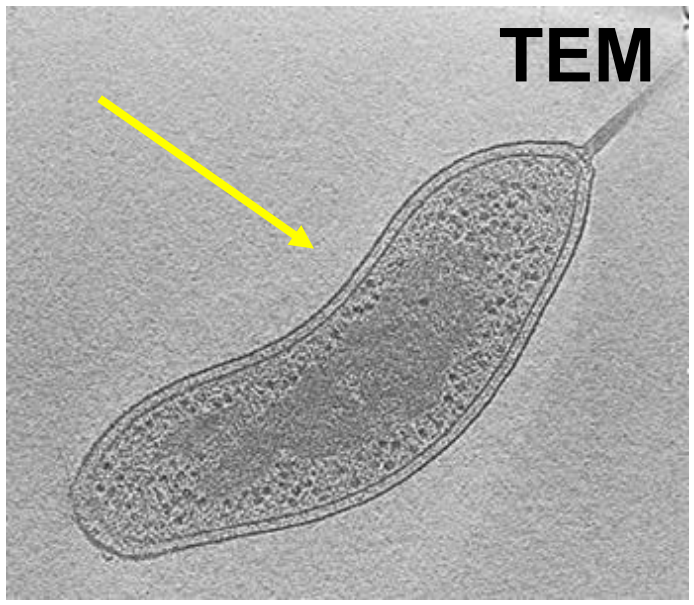
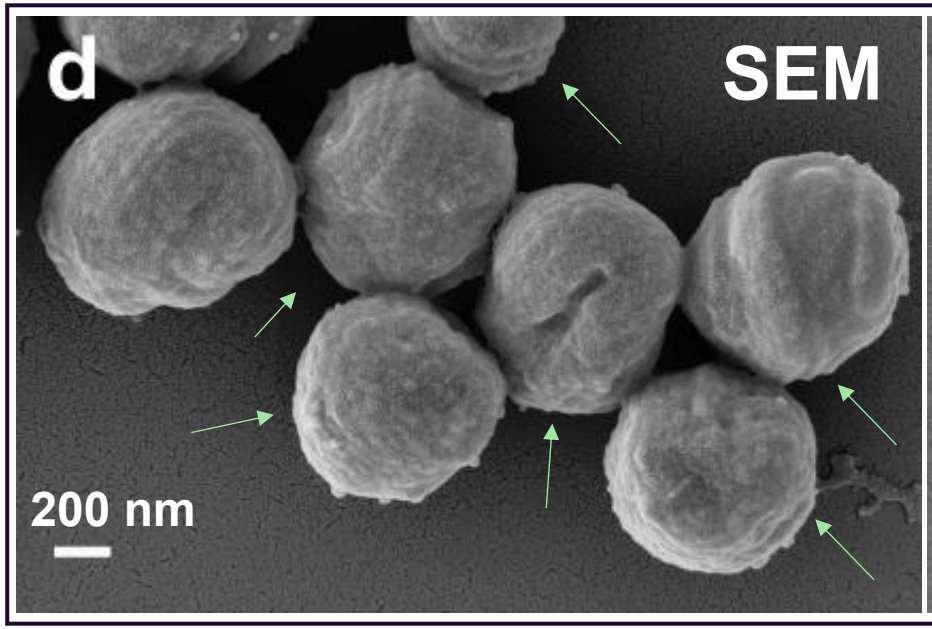


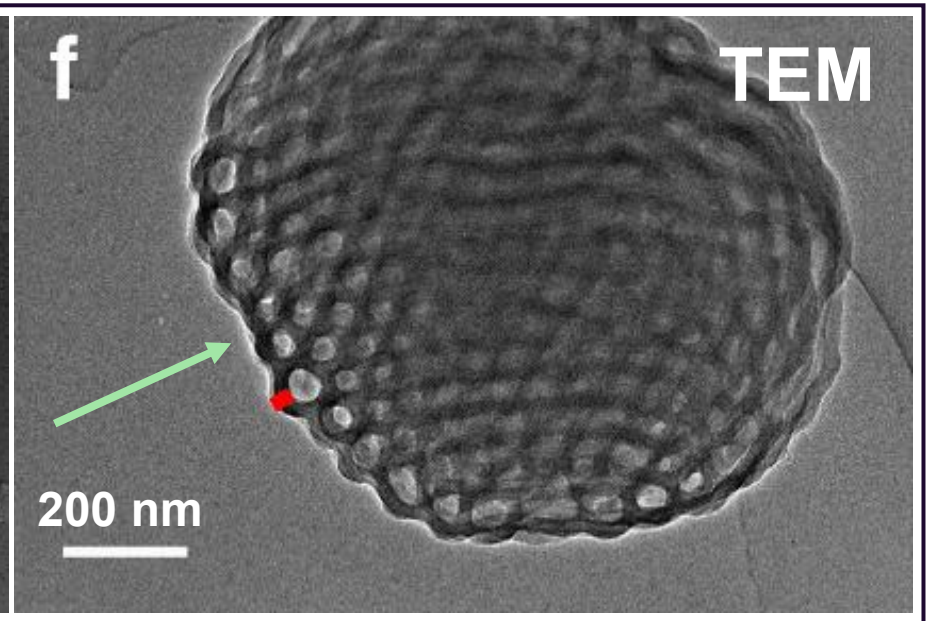


TEM offers…

- 2D projection of 3D objects.
- Finer details of internal structures.
- Much higher resolution & magnification.

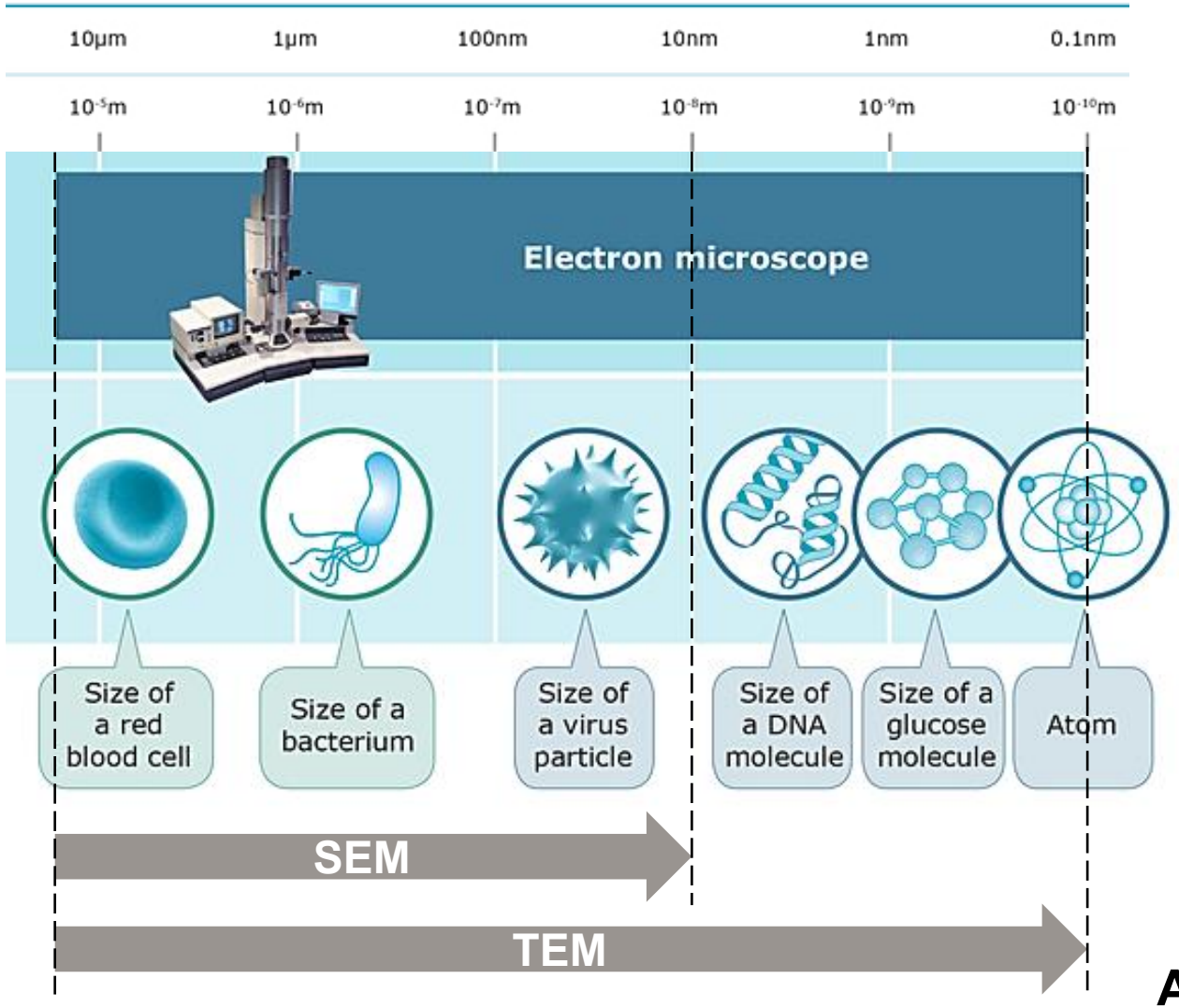
Why?





	SEM	TEM
Electronic type	Scattered scanned electrons	Transmission electrons
Image formation	Electrons are captured and counted by the detector Display on PC computer screen	Imaging directly on a fluorescent screen on a PC computer screen using a charge-coupled device (CCD)
Image information	Surface 3D images	2D internal structure projection image
Maximum multiplier	About 1 – 2 million times	More than 50 million times
Maximum field of view	Large	Limited
Sample thickness	Unrestricted	Usually, < 150 nm.
Acceleration Voltage	1 – 30 kV	60 – 300 kV
Difficulty of operation	Easy to use, little or no sample preparation required	Not easy to use, require sample preparation and training prior to use
Cost	Cheaper	More expensive
Speed	Quick	Slow

Limit of resolution (resolving power): The limit or ability of a microscope to distinguish details of a specimen.



$$\lambda = \frac{h}{p} \longrightarrow \lambda = \frac{h}{\sqrt{2meV}}$$

TEM operates at much higher accelerating voltage (80 – 200 kV) than SEM (1 – 30 kV).

at V = 100 kV (100,000 V), $\lambda_{electron}$ = 0.038 Å

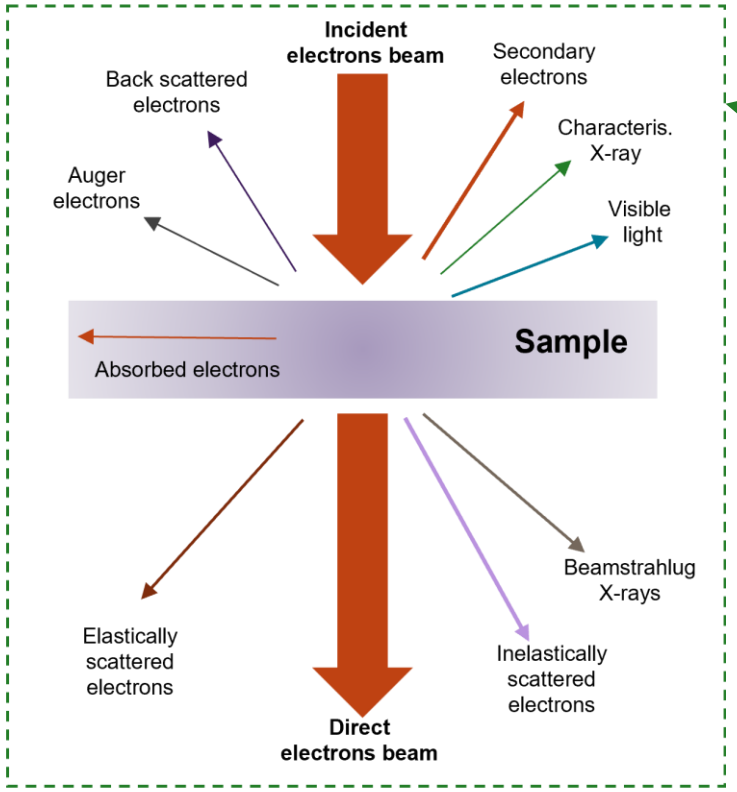
at V = 200 kV (200,000 V), $\lambda_{electron}$ = 0.027 Å

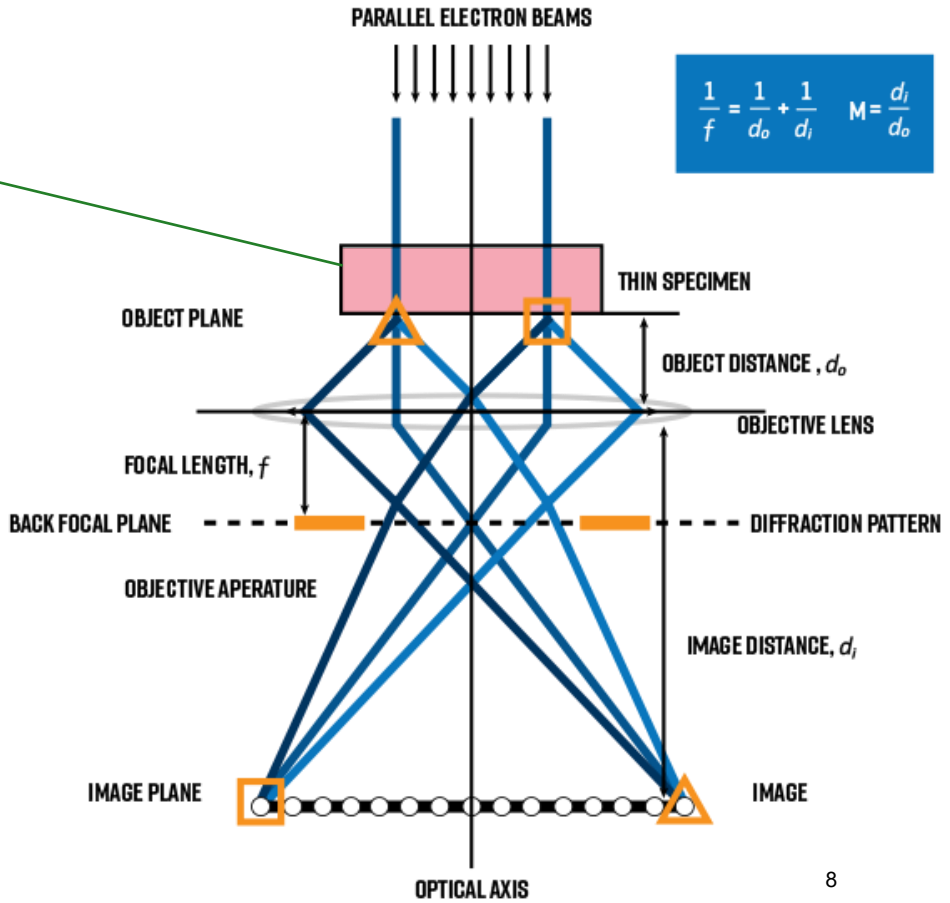
at V = 300 kV (300,000 V), $\lambda_{electron}$ = 0.022 Å

Atomically resolved image collection is possible !

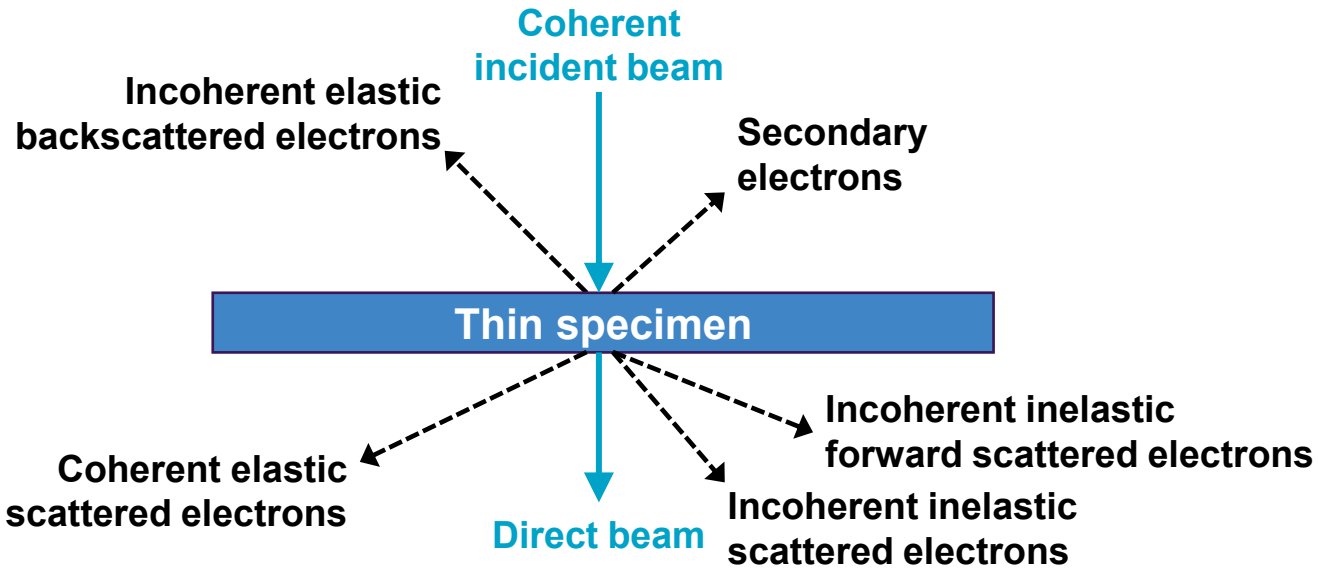
Operating Principle:

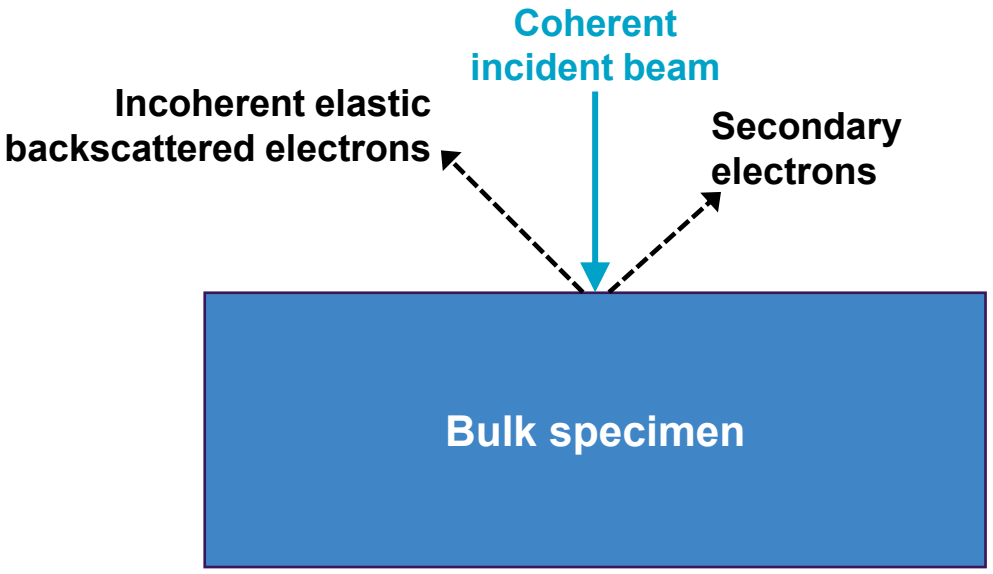
- de-Broglie's equation: $\lambda = h/mv$, When velocity (v) increases, wavelength (λ) decreases.
- Wavelength of electrons is based on the transmission of electrons through the object.
- Bright field imaging relies on incident electrons being scattered and disappearing from the beam depending upon the compositional density and crystal orientation of the sample.





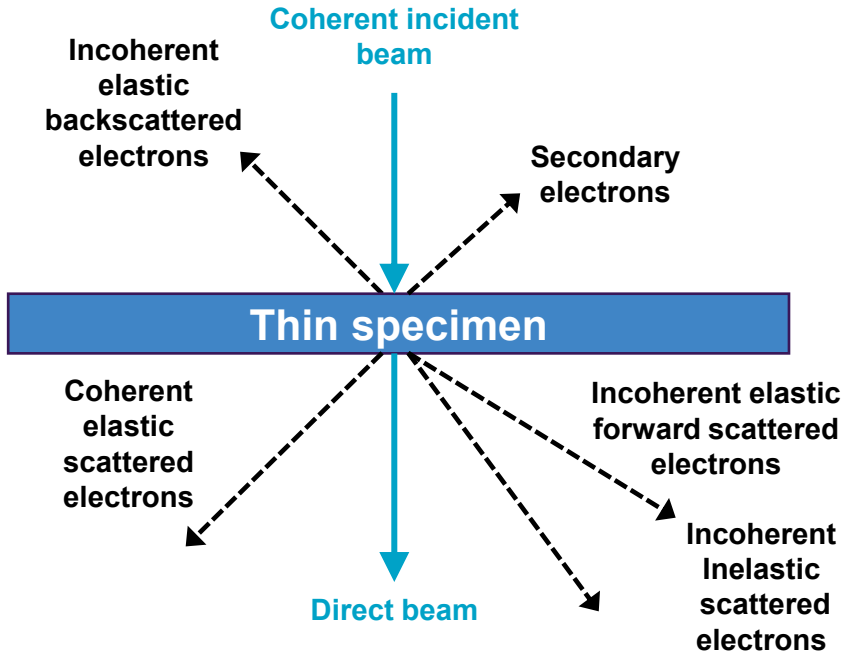
- High accelerating voltage is used in TEM to allow E-beam to penetrate through the specimen.
- This fact restricts the TEM observation to sufficiently thin samples.
- If the sample is too thick, effective penetration of the sample by E-beam is hard to achieve, thus generating dark object in the image without useful information on the internal structure.





Scattering in TEM occurs as both forward and back scattering, though we almost always use the forward scattered signals (Scattering in the SEM is always backscattering).

Coherent scattering has the strong relationship between phase of waves.



Incoherent scattering has the weak or non-existing phase relationship.

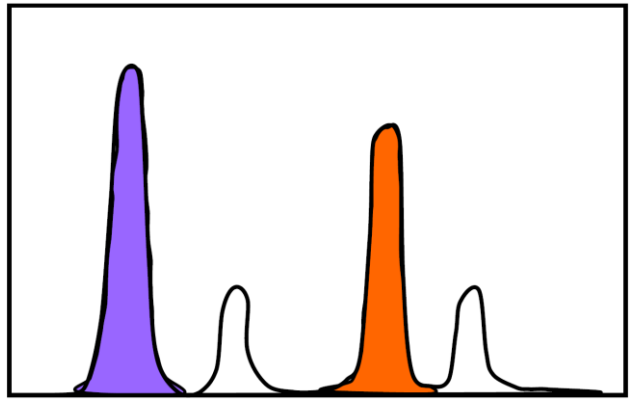
- Elastic scattered electrons conserve the energy from the incident electron beam.
- Inelastic scattered electrons lose the energy form the indecent electron beam.

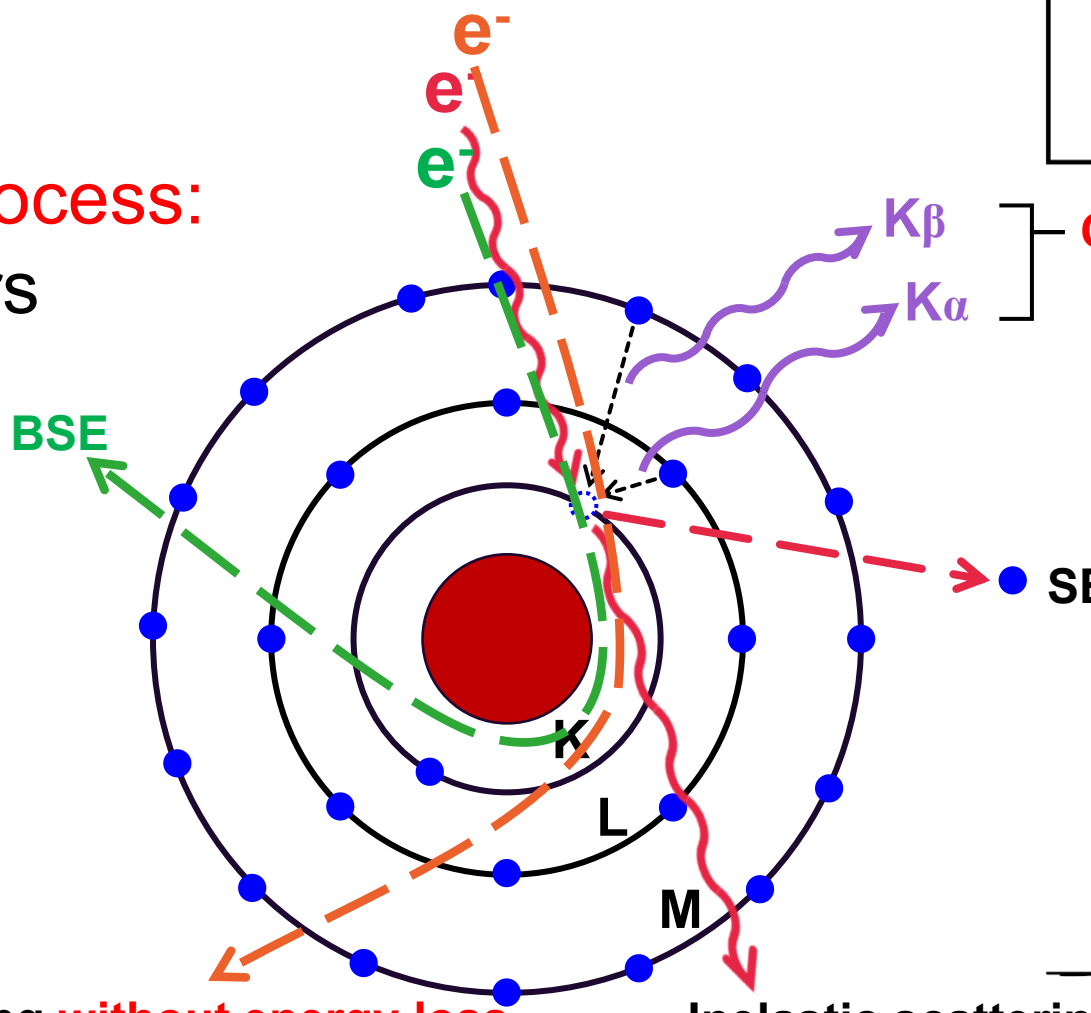
ELASTIC scattering process:

- Incidental beam does not suffer energy loss

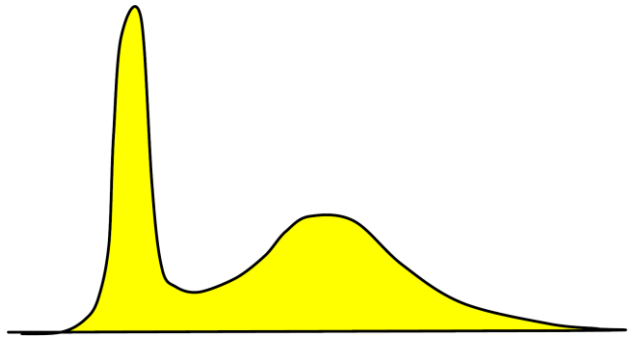
INELASTIC scattering process:

- Incidental beam suffers energy loss





Characteristic X-rays
E = hv



Elastic scattering without energy loss

Inelastic scattering with energy loss

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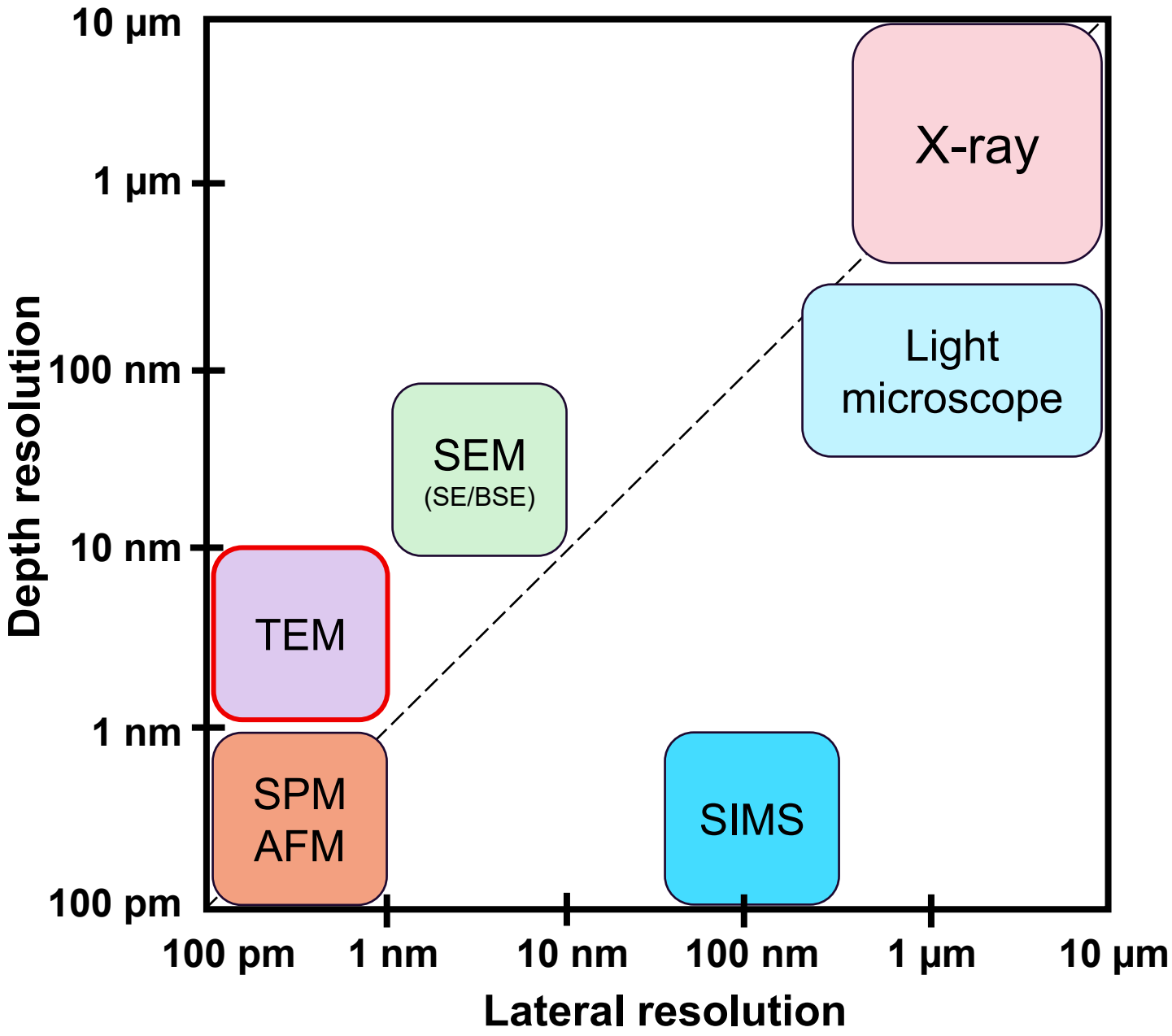
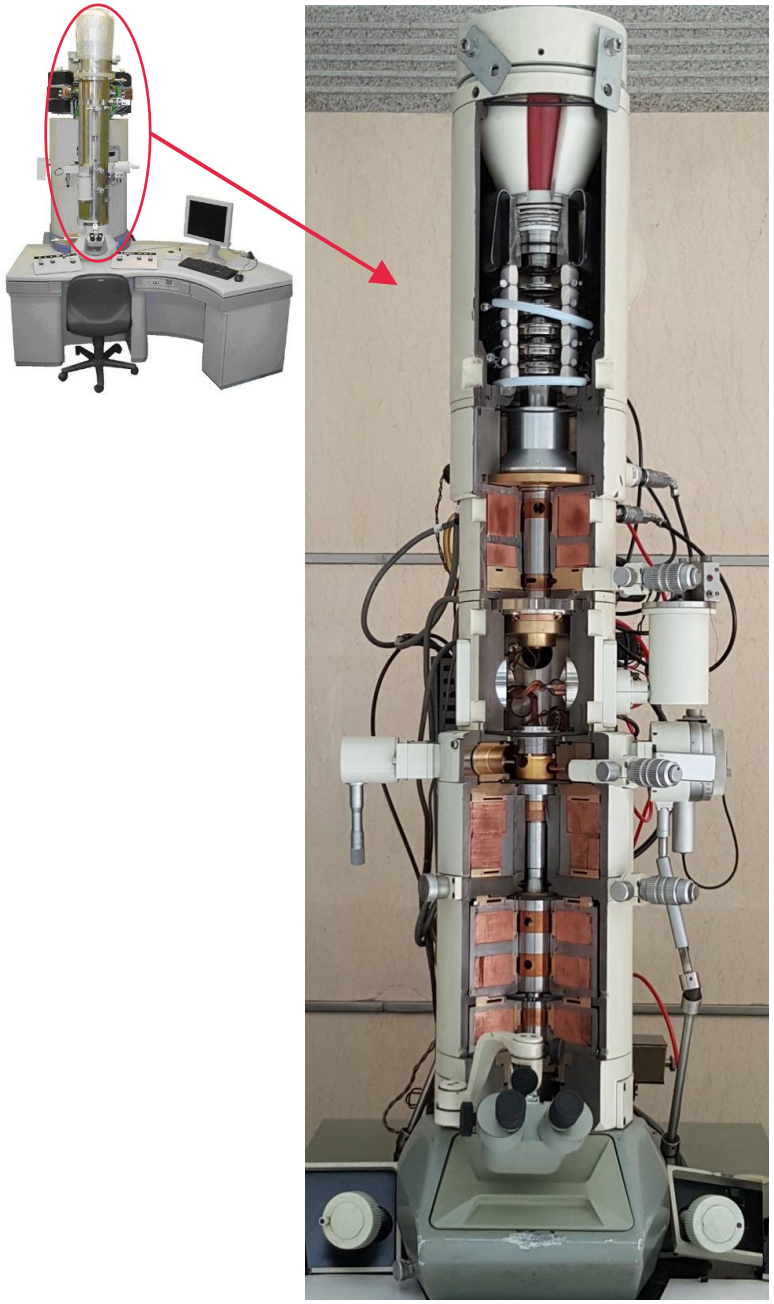
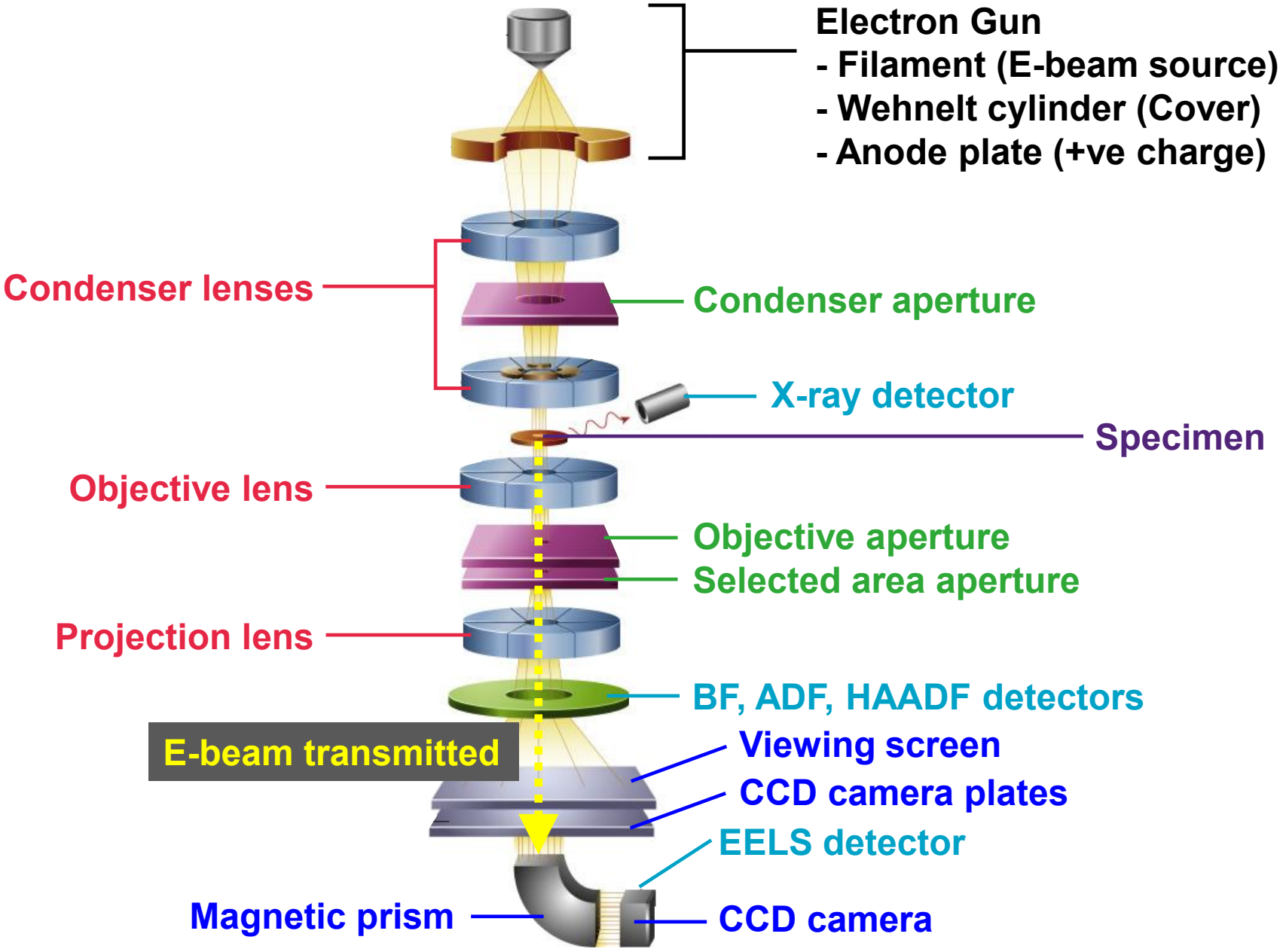
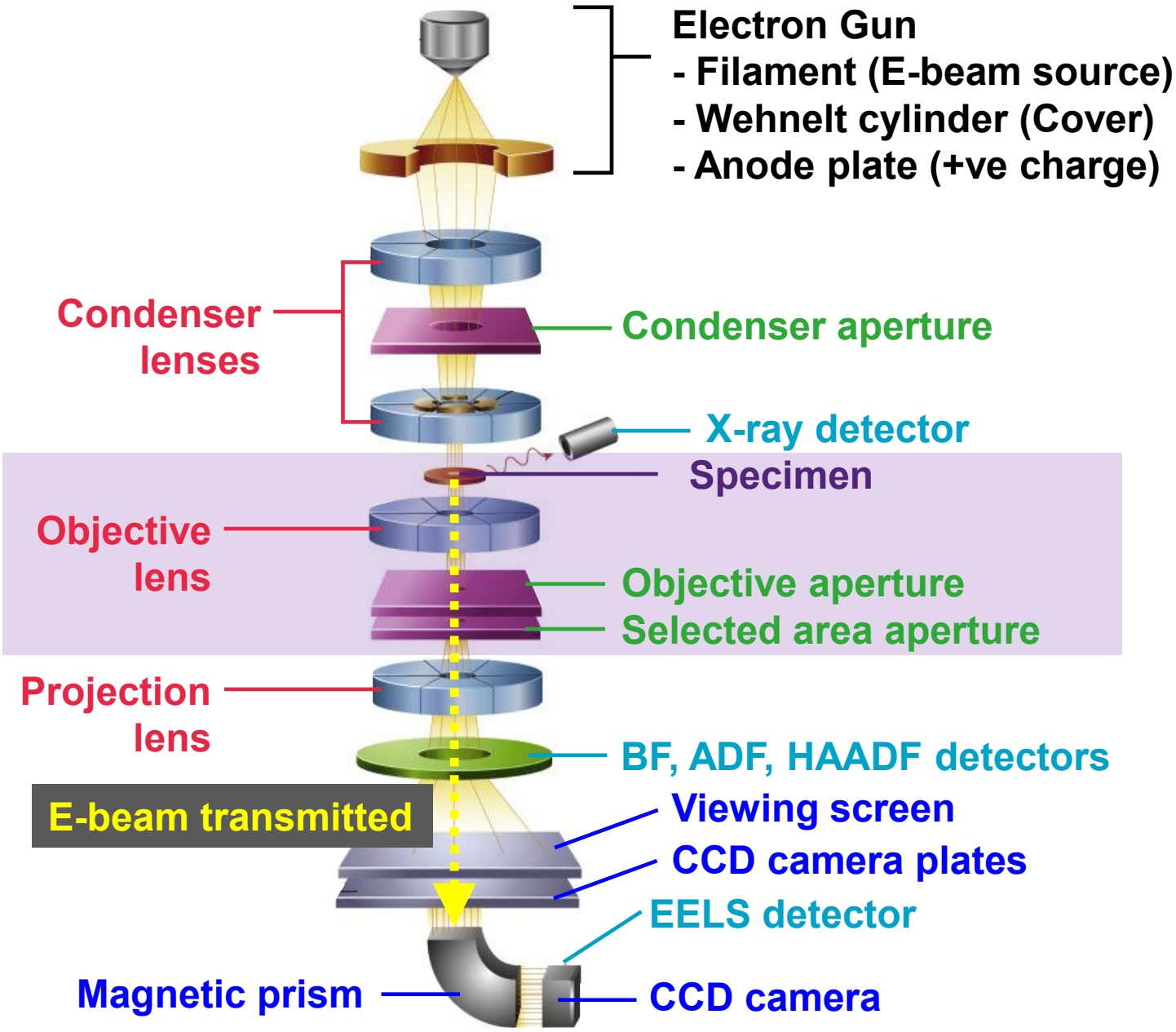


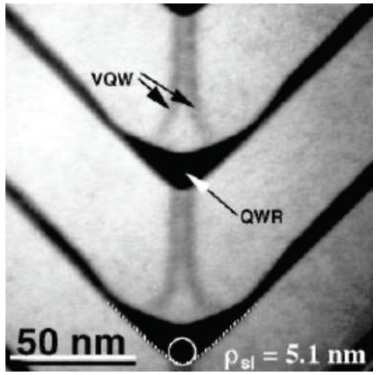
MATE 6301 | How does TEM work?



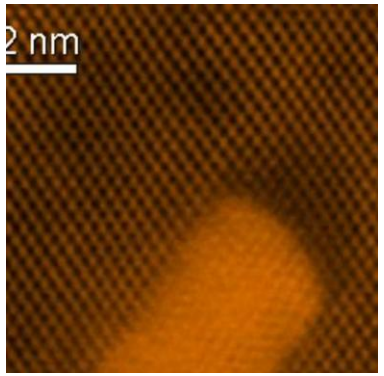


MATE 6301 | How does TEM work?

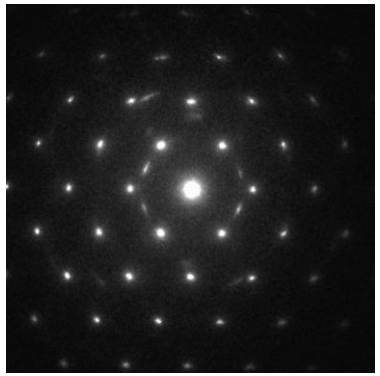


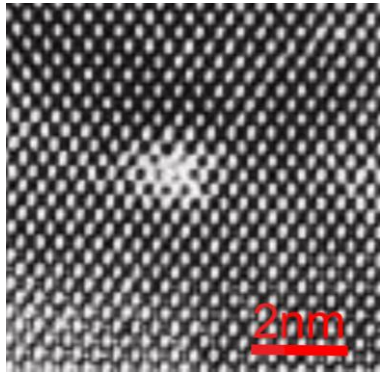


Diffraction contrast

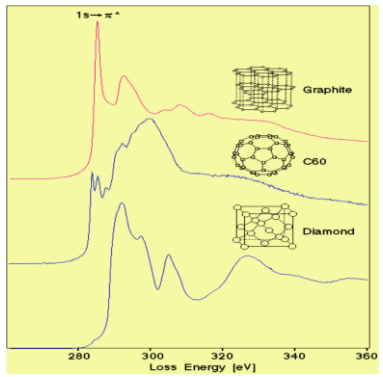


STEM imaging

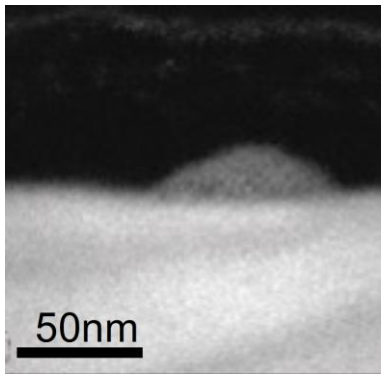

Electron diffraction



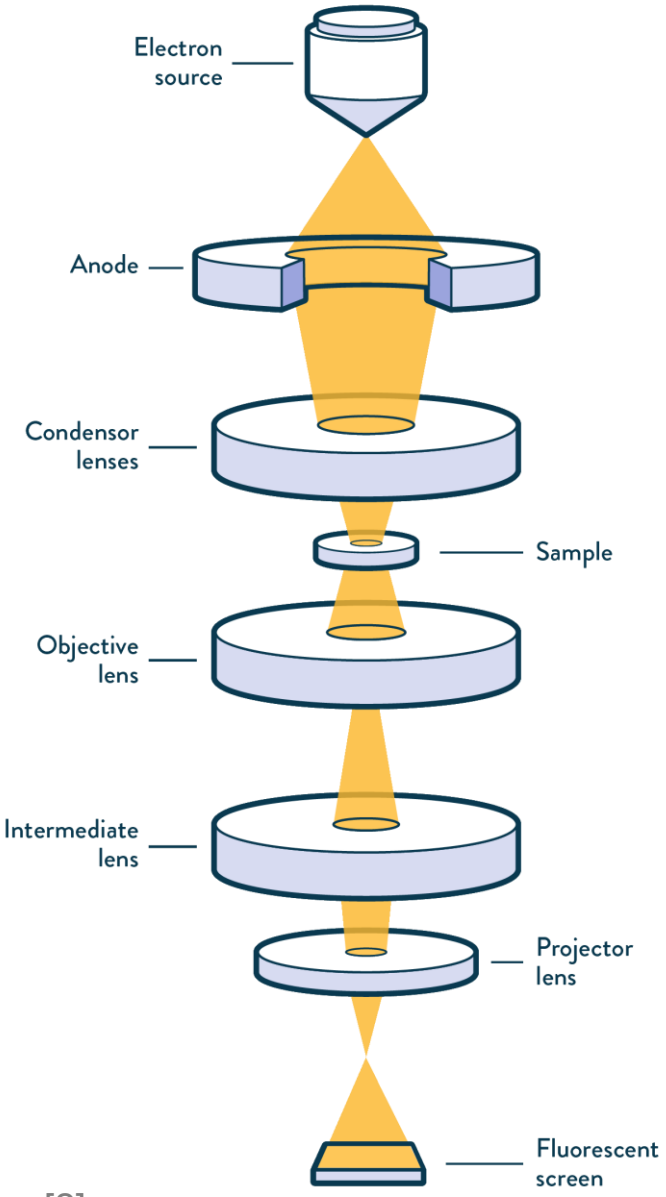
HRTEM imaging



EELS spectroscopy



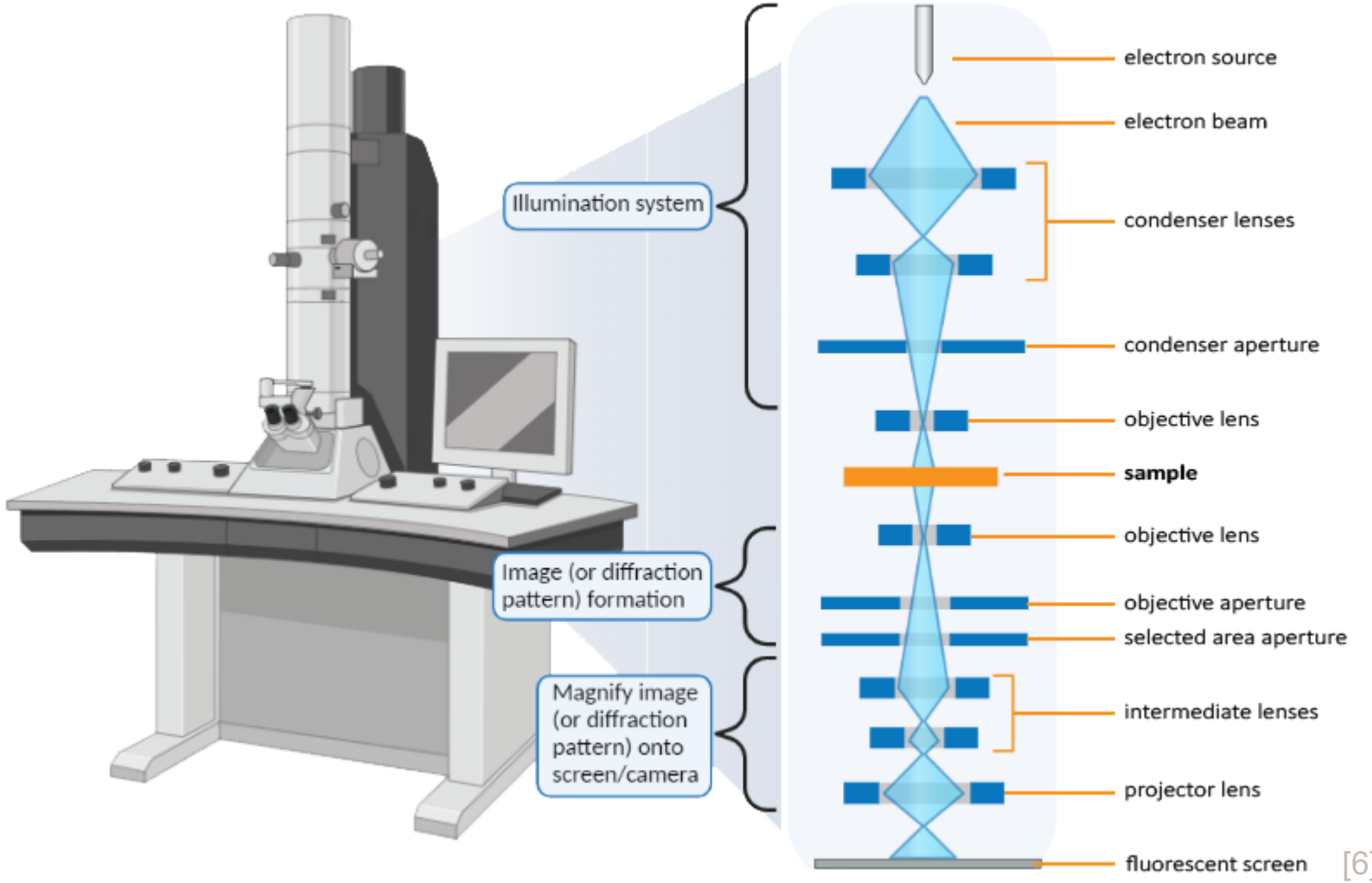
Energy filtered TEM



[2]

Fundamentals of TEM

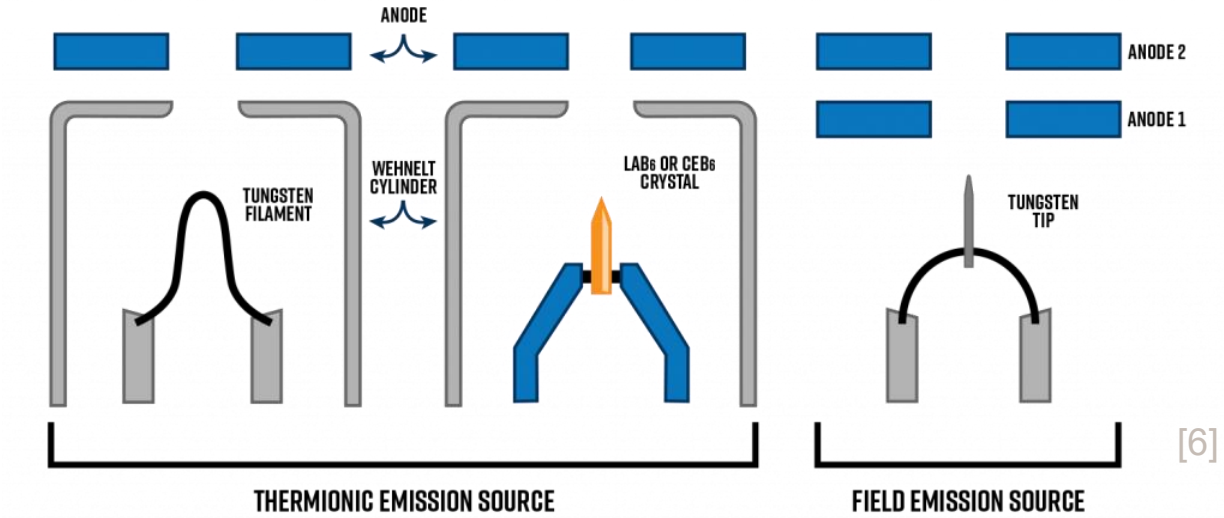
The five major components in TEM are the electron gun, vacuum system, electromagnetic lenses, specimen stage and imaging devices.



[6]

1. Electron Gun:

The TEM consists of an electron emission source, which may be a **tungsten filament or a lanthanum hexaboride (LaB_6) source**, known as an electron gun. By connecting this gun to a high-voltage source of about **100 – 300 kV**, the gun begins to emit electrons by either **thermionic or field electron emission** into the vacuum.



[6]

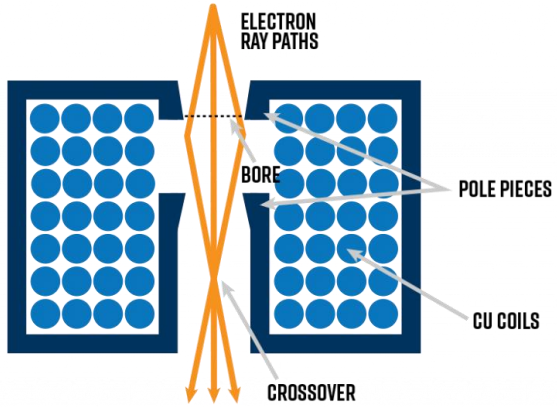
2. Vacuum System:

Electrons behave like light only when they are manipulated in a vacuum. Different vacuum pumps are used to obtain and maintain **high to ultra-high vacuum pressure as low as 10^{-8} Pa**. A number of airlocks and separation valves are used to avoid the need to evacuate the whole column every time a specimen or photographic material or filament is exchanged.

[6]

3. Electromagnetic Lenses:

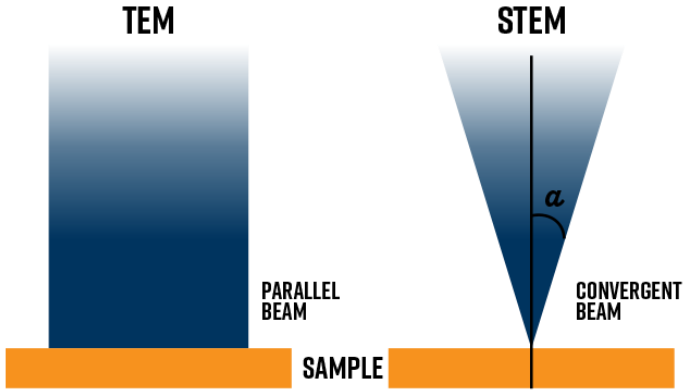
The **electromagnetic lenses typically consist of Cu coils**, and when an electric current is passed through these coils, an **electromagnetic field is created between the pole pieces creating a gap in the magnetic circuit**. By varying the current through the coils, the strength of the field, and thereby the **power of the lens, can be varied**.



Typically, a TEM consists of three stages of lensing – **condenser lenses, objective lenses,** and **projector lenses**.

- The **condenser lenses** are responsible for **primary beam formation**.
- The **objective lenses focus on the beam that comes through the sample itself**.
- The **projector lenses** are used **to expand the beam onto the phosphor screen** or **other imaging devices such as film**.

The magnification of the TEM is due to the ratio of the distances between the specimen and the objective lens' image plane.



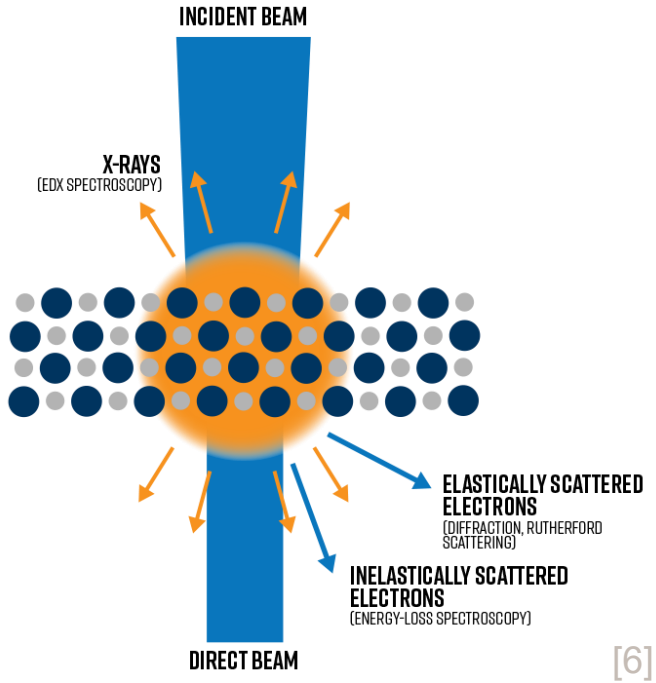
[6]

Comparison of TEM and STEM illumination which are used to form a parallel beam or convergent probe, respectively, at the sample entrance surface.

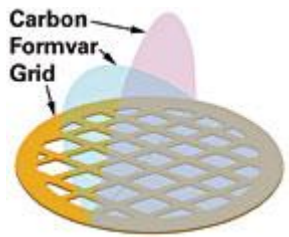
4. Specimen Stage:

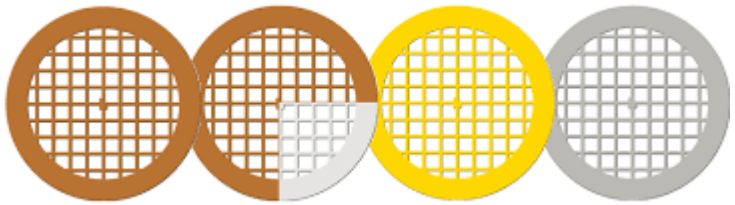
The specimen is placed on a mesh having a diameter of about 2.5 mm and is held by a specimen holder. The **mesh or grid is made of copper, molybdenum, gold,** or **platinum**. The specimen stage also includes airlocks to allow for the insertion of the specimen holder into the vacuum with a minimal increase in pressure in other areas of the microscope.

Sample preparation includes the preparation of thin sections of a maximum thickness of a few tens or hundreds of nanometers. As TEM specimen holders permit the rotation of a sample by a desired angle, multiple views of the same specimen can be obtained by rotating the angle of the sample along an axis perpendicular to the beam. By taking multiple images of a single TEM sample at differing angles, typically in 1° increments, a set of images known as a "**tilt series**" can be collected.



[6]

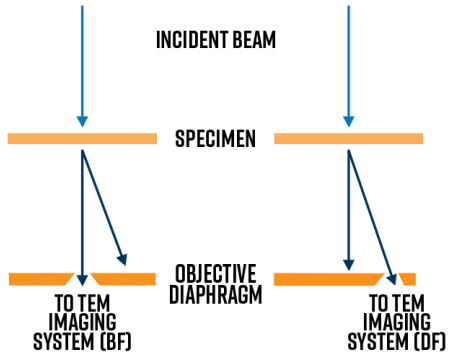




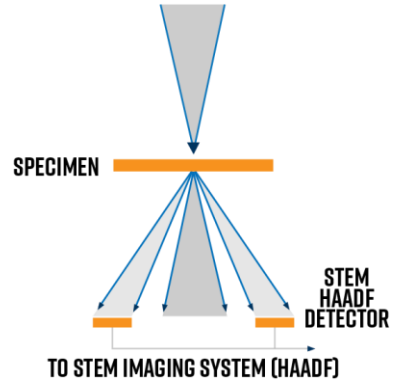
[7]

5. Imaging Devices:

Originally, TEMs used a fluorescent screen that emitted light when impacted by the transmitted electrons for real-time imaging and adjustments; and a film camera to record permanent, high-resolution images. Modern instruments rely primarily on solid-state imaging devices, such as a **CCD (charge-coupled device) camera**, for image capture. They may still include a fluorescent screen, but it may be observed by a video camera. [3]



[6]



[6]

Note:

Samples must be thin (less than 150 nm) to allow an electron beam to pass through the sample, to form an image.

Risk Involved:

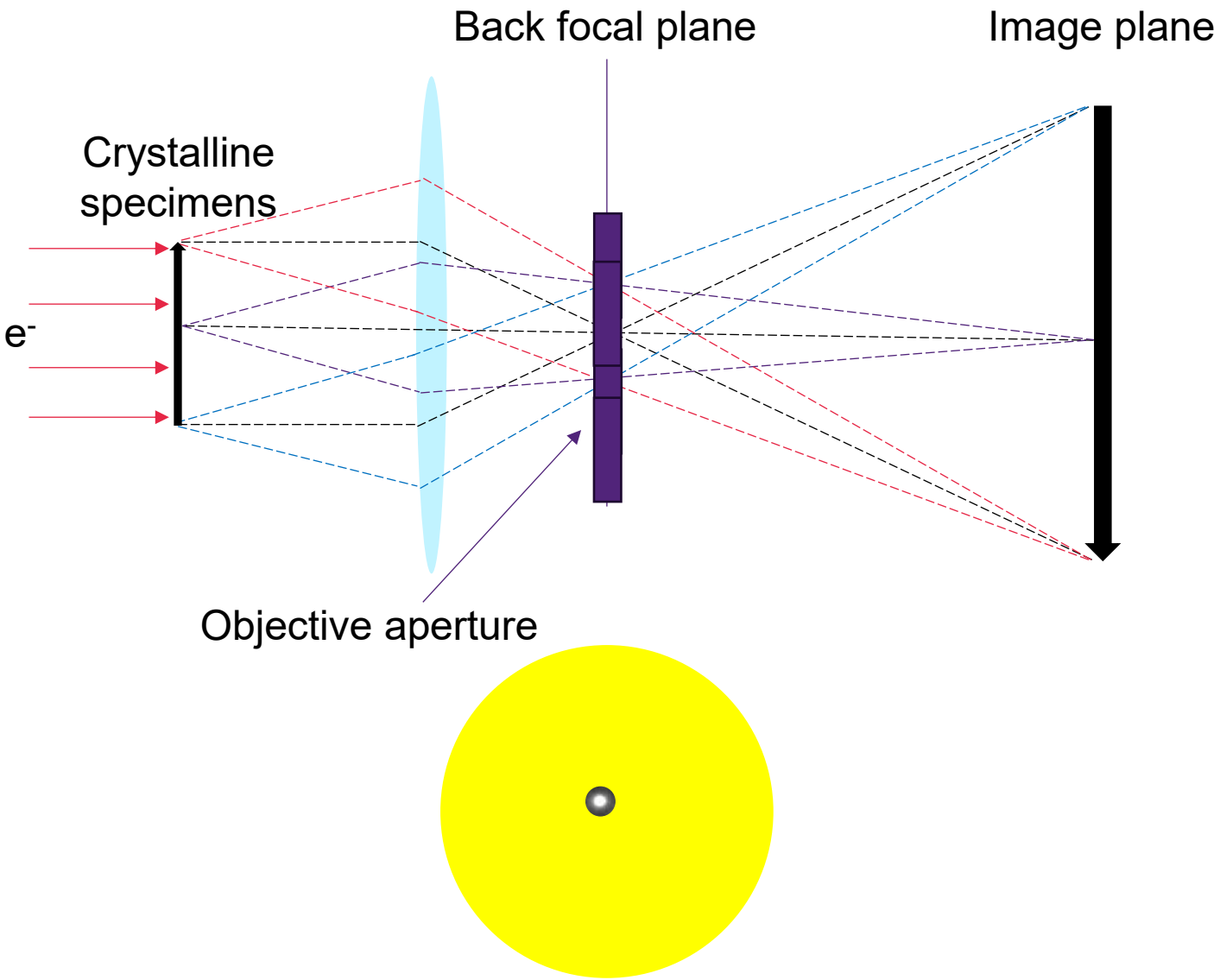
- Longer exposure of organic or biological samples to the electron beam can affect the sample.
- Free from solvents & moisture (can react to the electron beam and affect the vacuum).

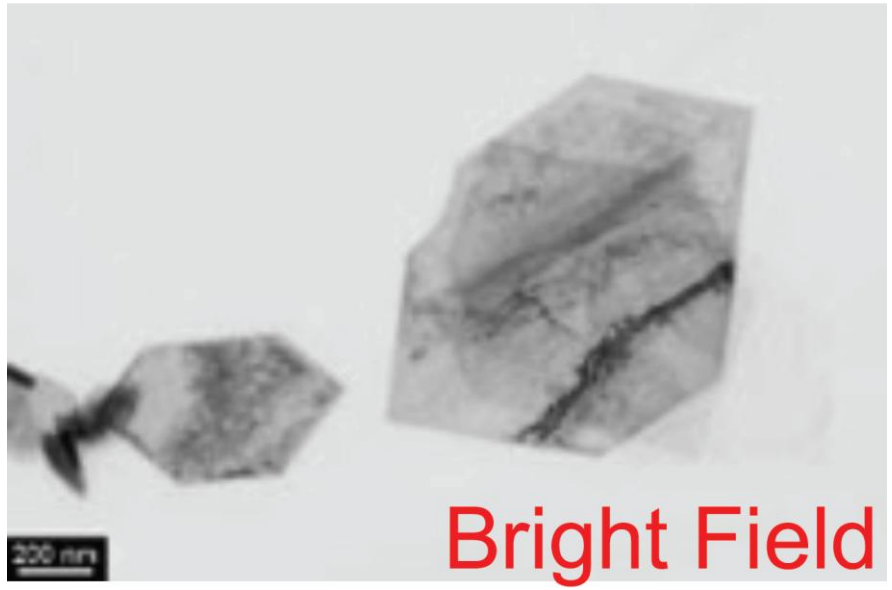
Imaging Techniques

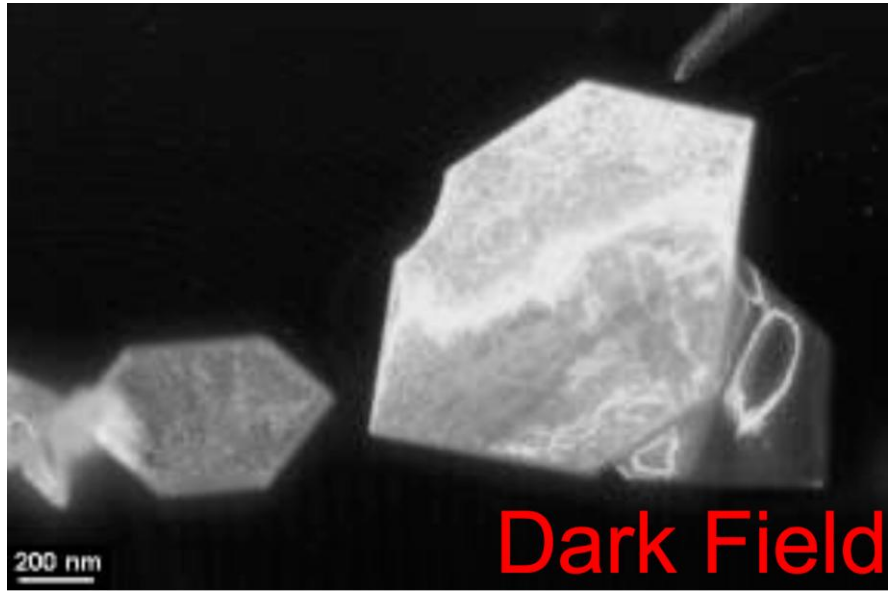
- **Conventional Imaging Modes:**
  - Diffraction Contrast
  - Phase Contrast
  - Mass/Thickness Contrast
- **Diffraction Modes:**
  - Electron Diffraction
- **Inelastic Signals:**
  - Characteristic X-ray (also called Energy Dispersive Spectroscopy – EDS)
  - Electron Energy Loss Spectroscopy (EELS + EFTEM)
- **Scanning Transmission Electron Microscopy (STEM)**

Diffraction contrast

- Elastic scattered electrons are typically used to form images and diffraction patterns







- Diffraction contrast is used for crystalline specimens
- Diffraction contrast is formed **with a single diffracted (can be transmitted) electron beam** and realised with an aperture at the back focal plane of the objective lens
- Diffraction contrast is mainly used for study of
  - Defects
  - Precipitates
  - Grain sizes and distributions
  - ……

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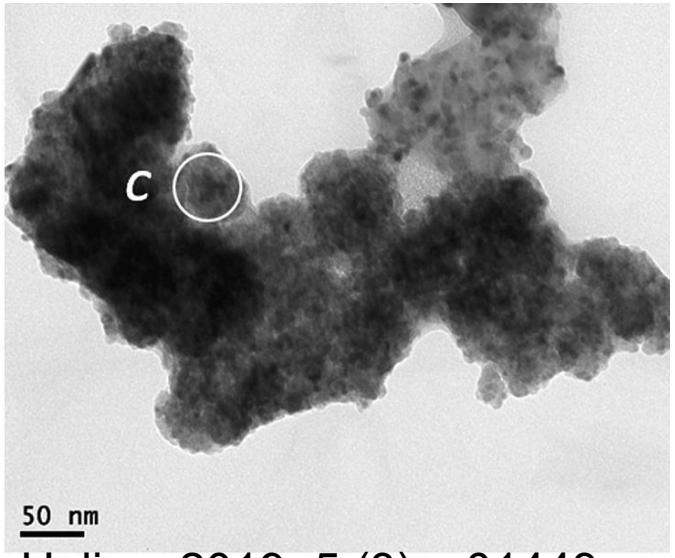
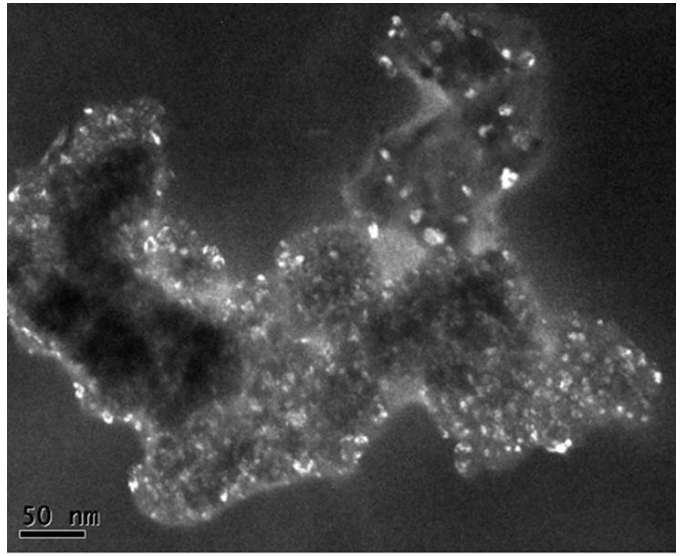
Bright-field TEM image

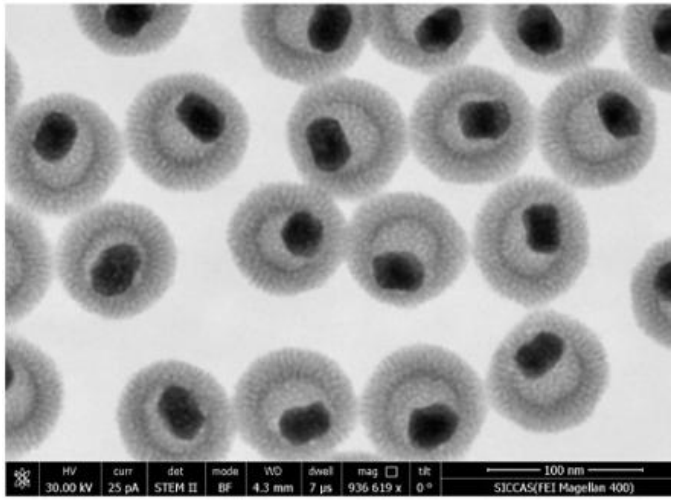


Heliyon2019, 5 (3), e01443

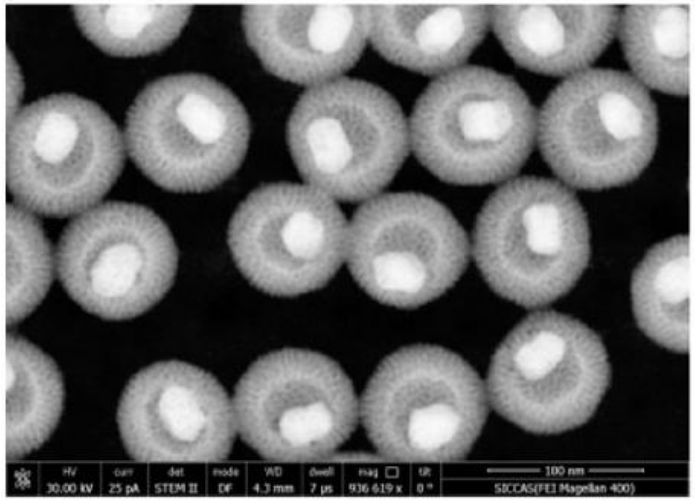
Dark-field TEM image



Fe_2O_3 on SiO_2

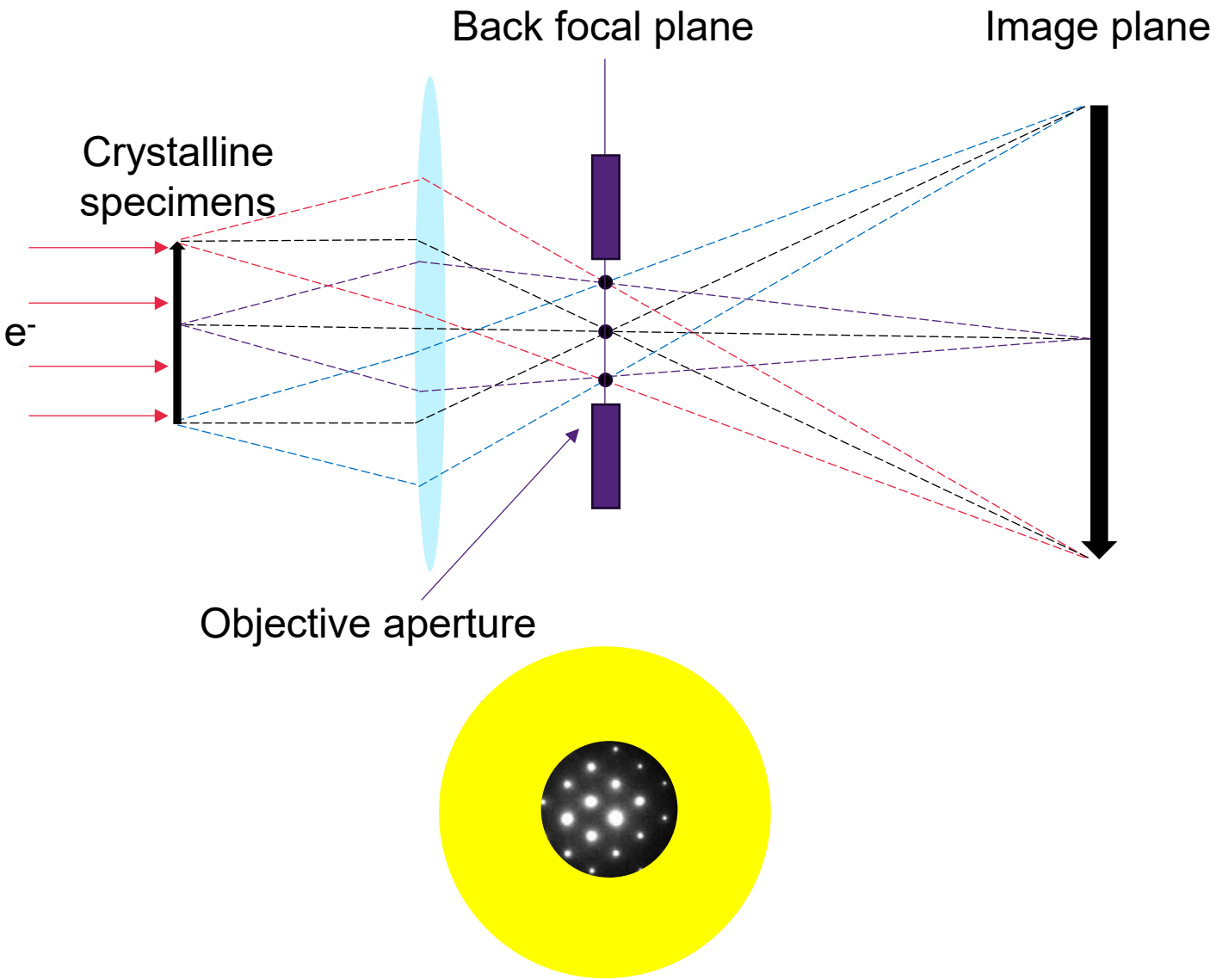


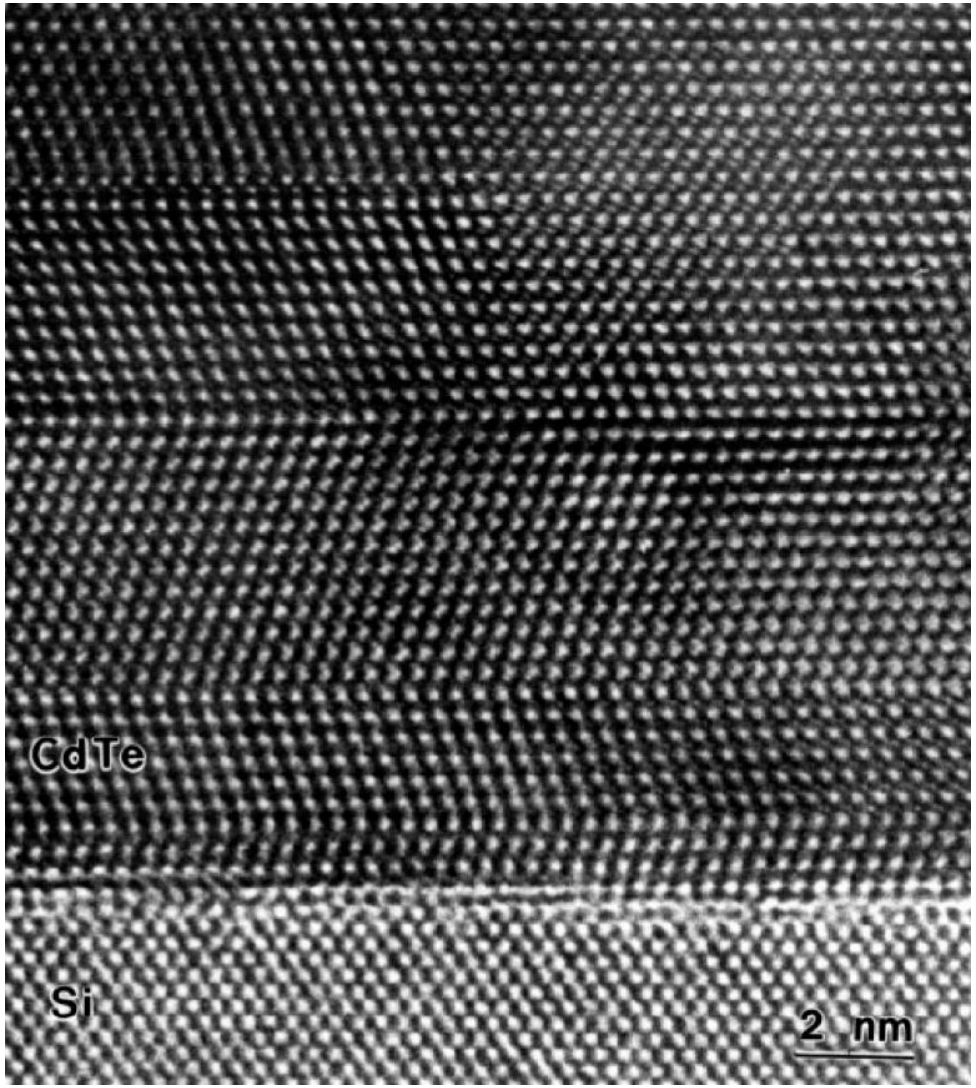
Ruthenium nanoparticle core in SiO_2 shell



J. Am. Chem. Soc. 2014, 136, 27, 9701–9709

- Elastic scattered electrons are typically used to form images and diffraction patterns



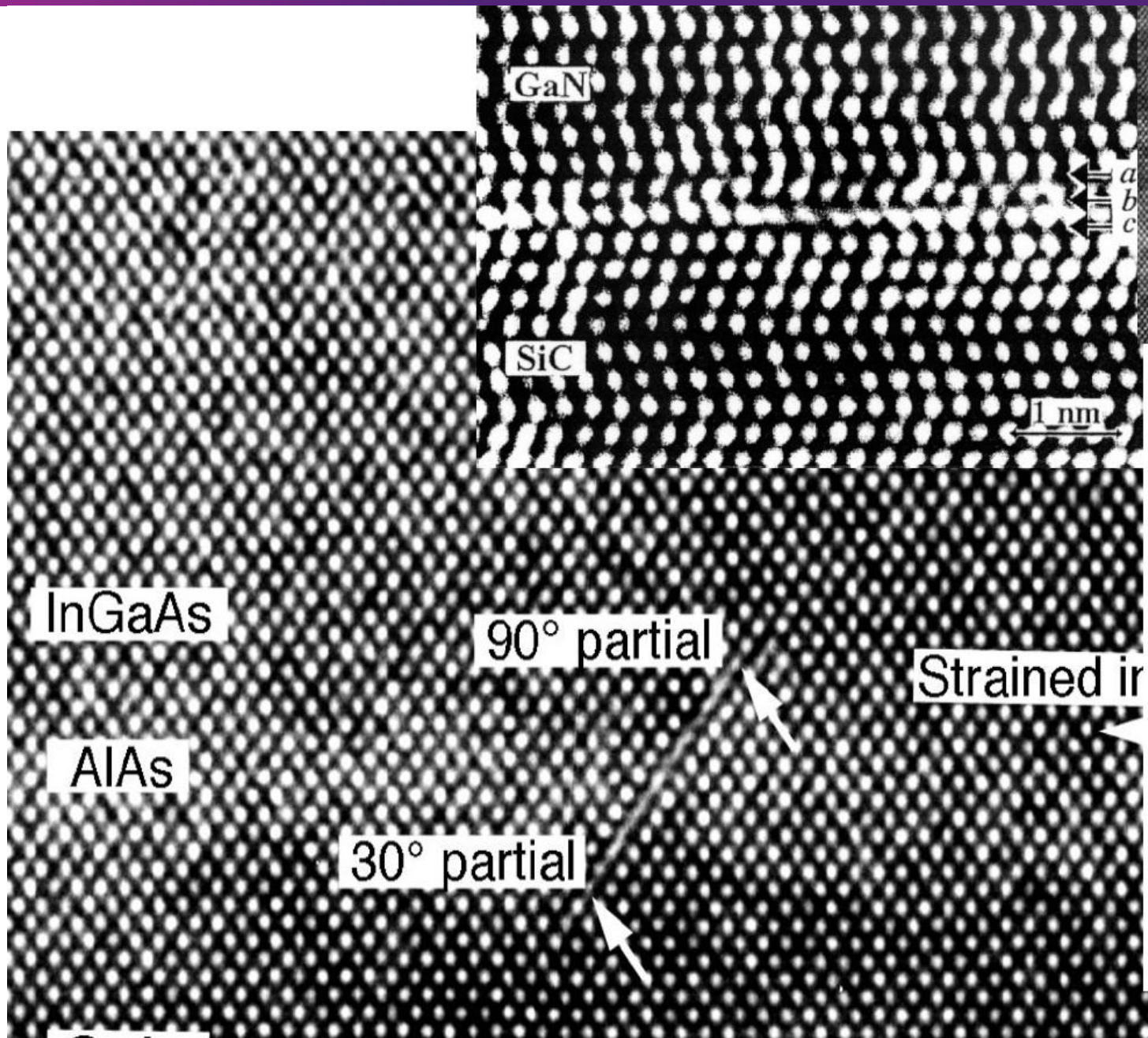


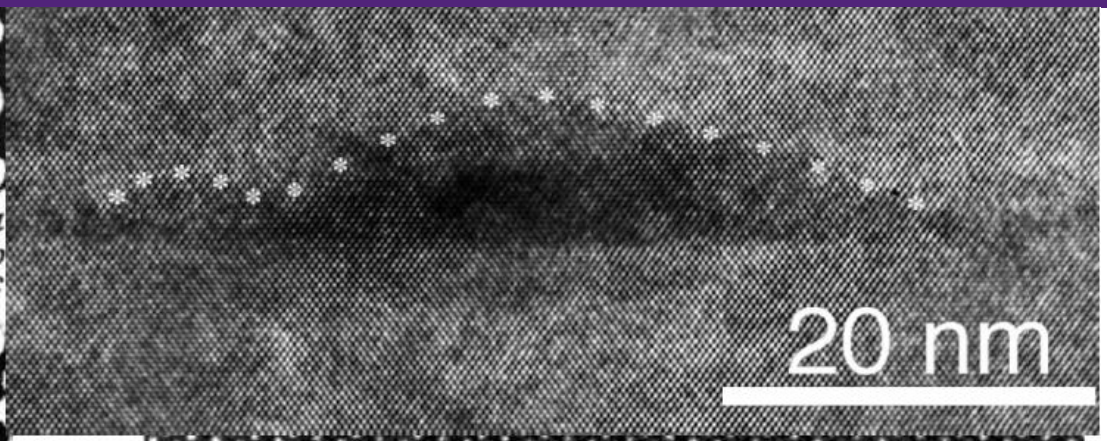
- Phase contrast deals also with crystalline specimens
- Phase contrast is formed **with at least two diffracted electron beams through their interference**.
- Phase contrast is realised with (or without) an aperture at the back focal plane of the objective lens
- Phase contrast is mainly used for study of
  - Defects
  - Precipitates
  - Interfacial structures
  - Crystal growth
  - ……

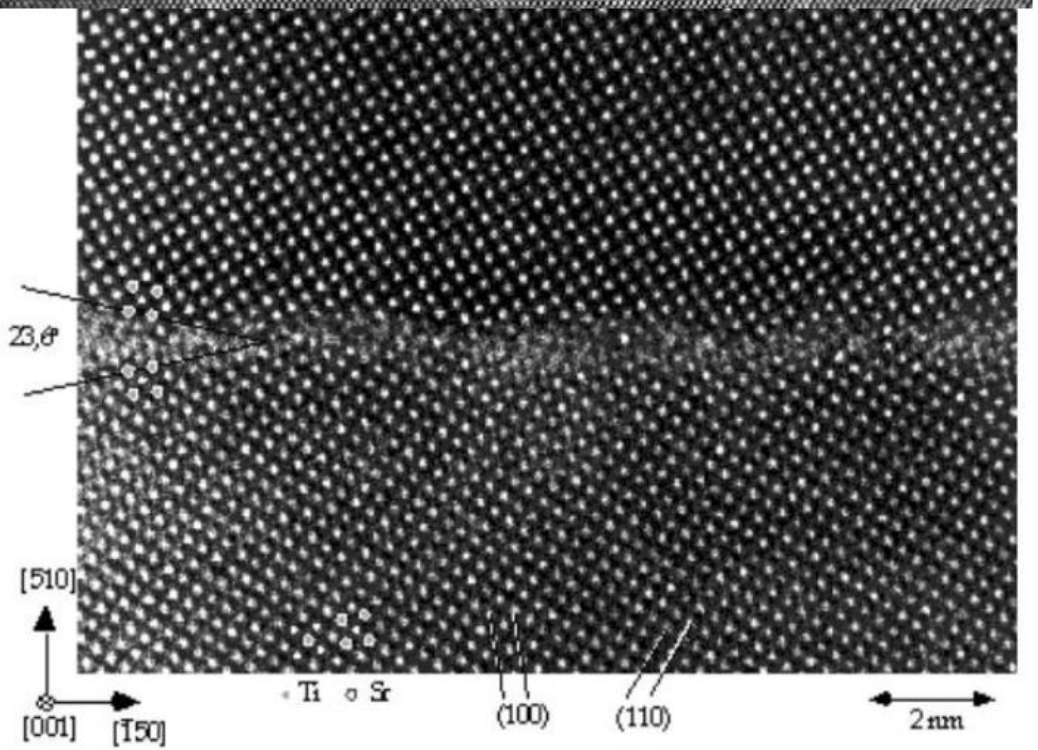
- Phase contrast requires **very thin specimens** (< 20 nm typically) for the Interference

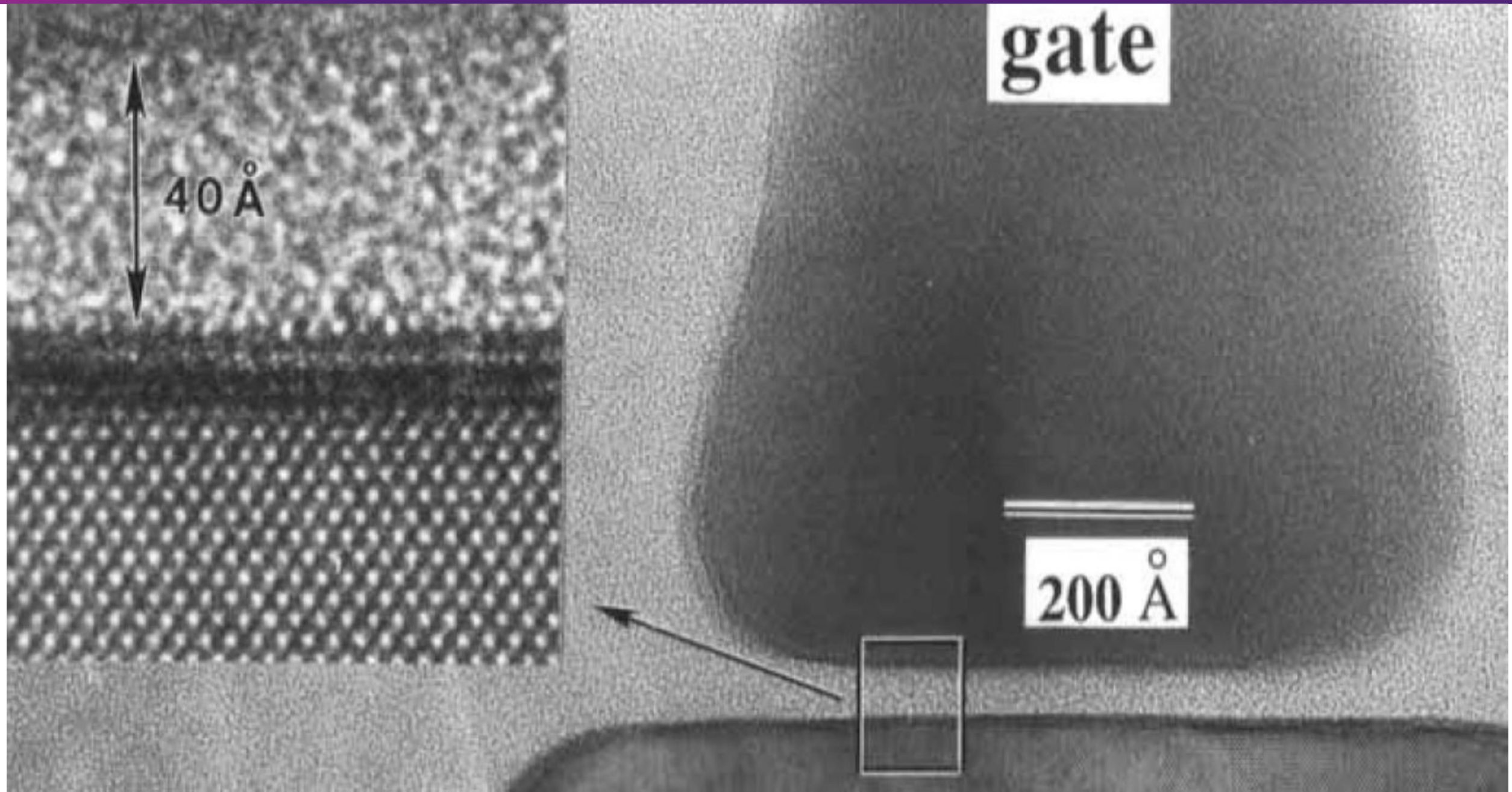
- Crystals should be **orientated** to the directions where the **zone axes** should be parallel to the incidental electron beam

- Sometimes, image simulations are needed for gain correct interpretation as atoms can be black or white in the image


GaN
SiC
1 nm
InGaAs
AlAs
90° partial
30° partial
Strained ir


20 nm

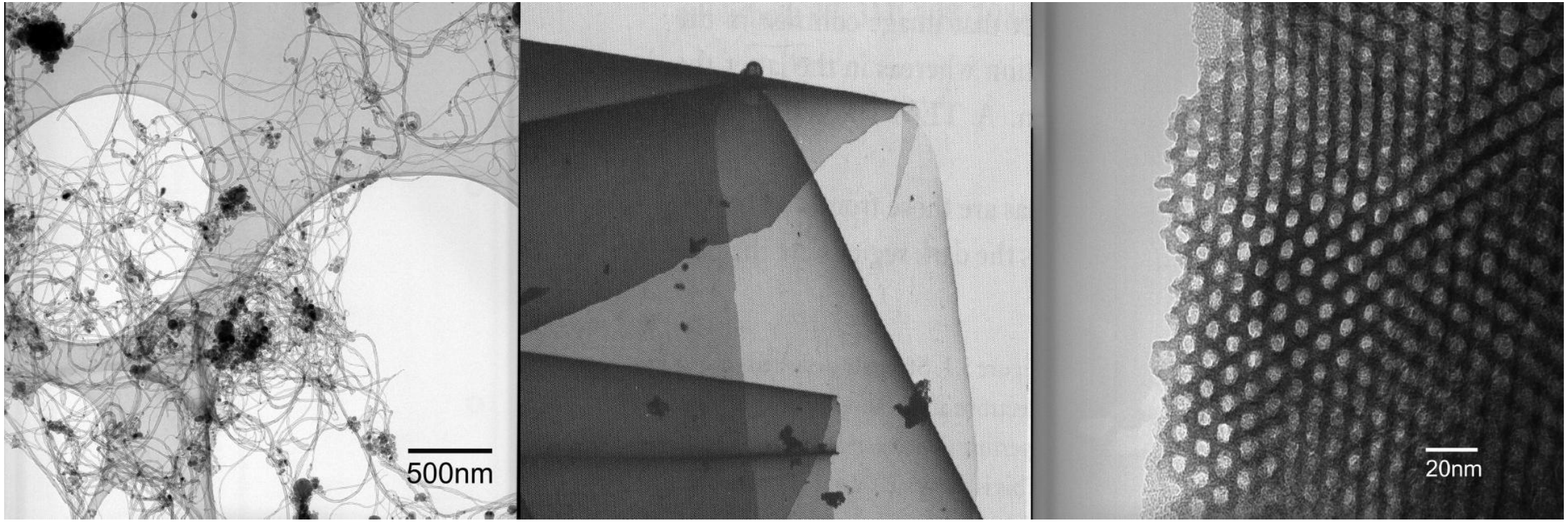

Z3,e
[510]
[001]
[1̄50]
Ti
Sr
(100)
(110)
2 nm

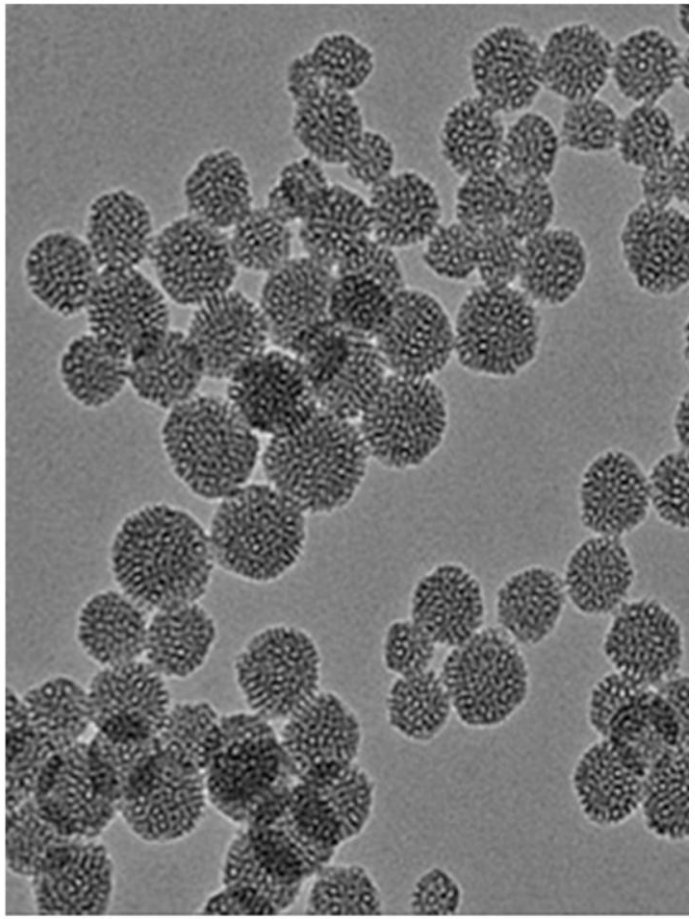

40 Å
gate
200 Å

- Mass/thickness contrast arises from incoherent elastic scattering at the nucleus (Rutherford cross section)
- The interaction between electrons and heavy atoms is stronger than that between electrons and light atoms . If the thickness is homogeneous, areas in which heavy atoms are concentrated appear with darker contrast than such with light atoms (mass contrast). Of course, more electrons are scattered in thick than in thin areas; thus, thick areas appear dark (thickness contrast). However, a thick area with light elements might have the same contrast as a thinner area with heavier atoms.

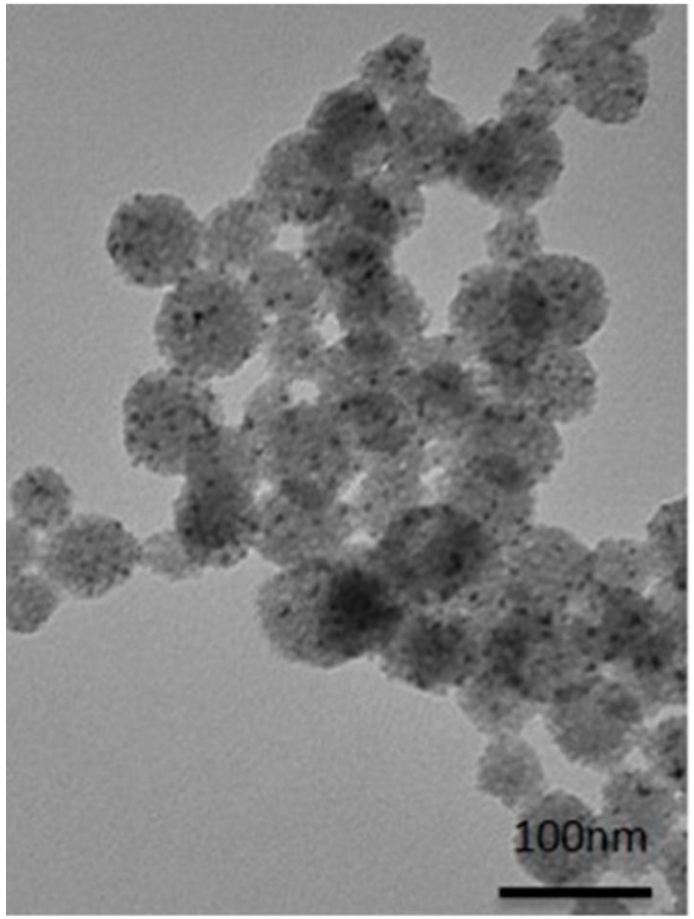
- The contrast depends upon atomic number Z and sample thickness t.

- Amorphous materials, such as polymers and amorphous porous
- Crystalline materials, but with all electron beam participating the formation of the images.
- Biological materials

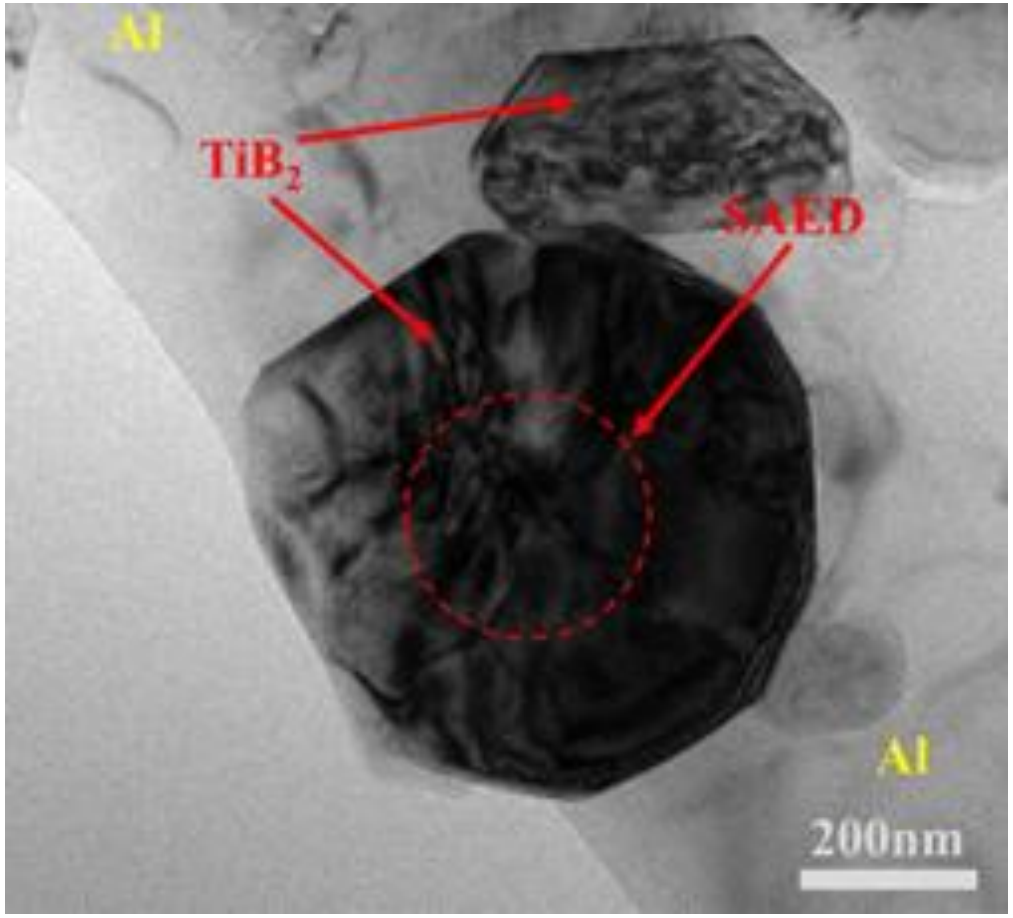




Mesoporous silica



Pt-loaded mesoporous silica

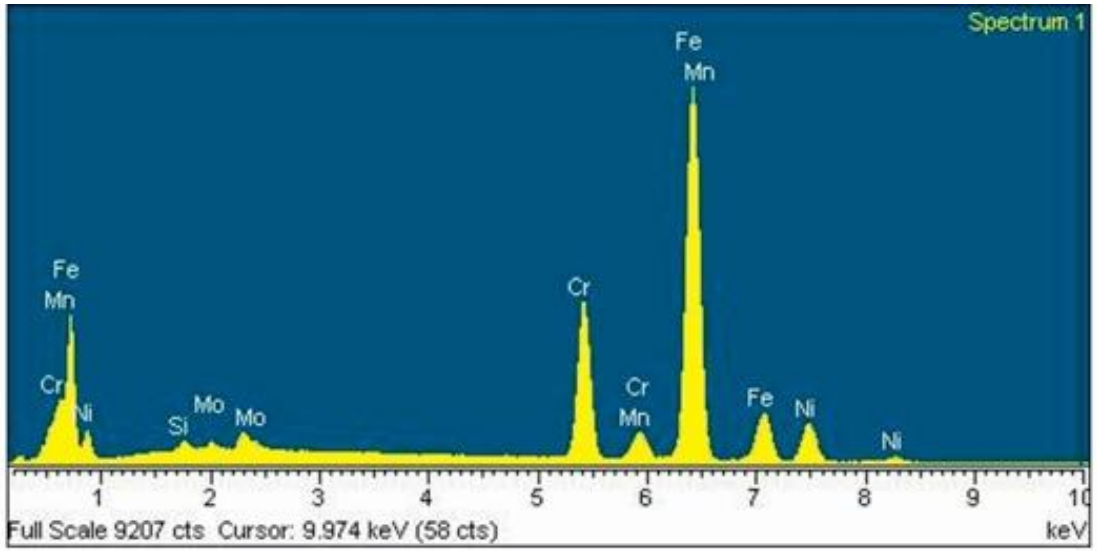


Int. J. Mol. Sci. **20**(7), 1560 (2019).

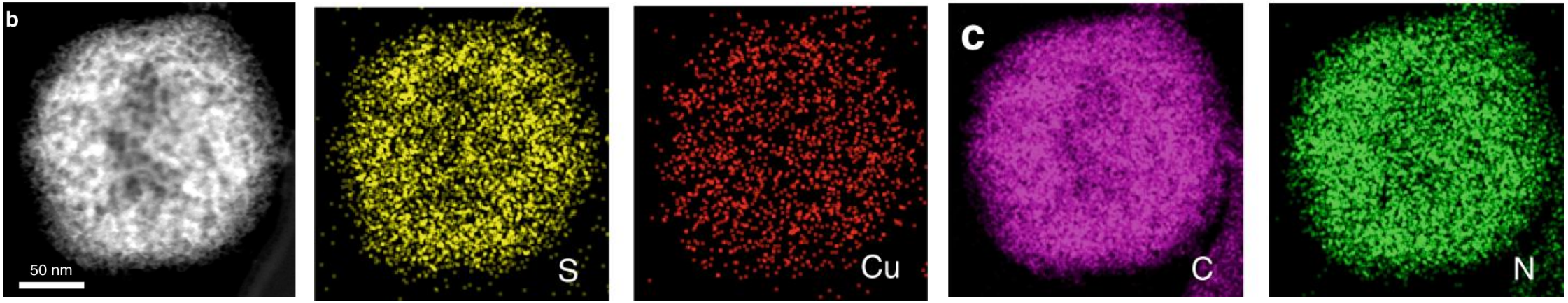
Mater. Res. Express **8** 036514 (2021).

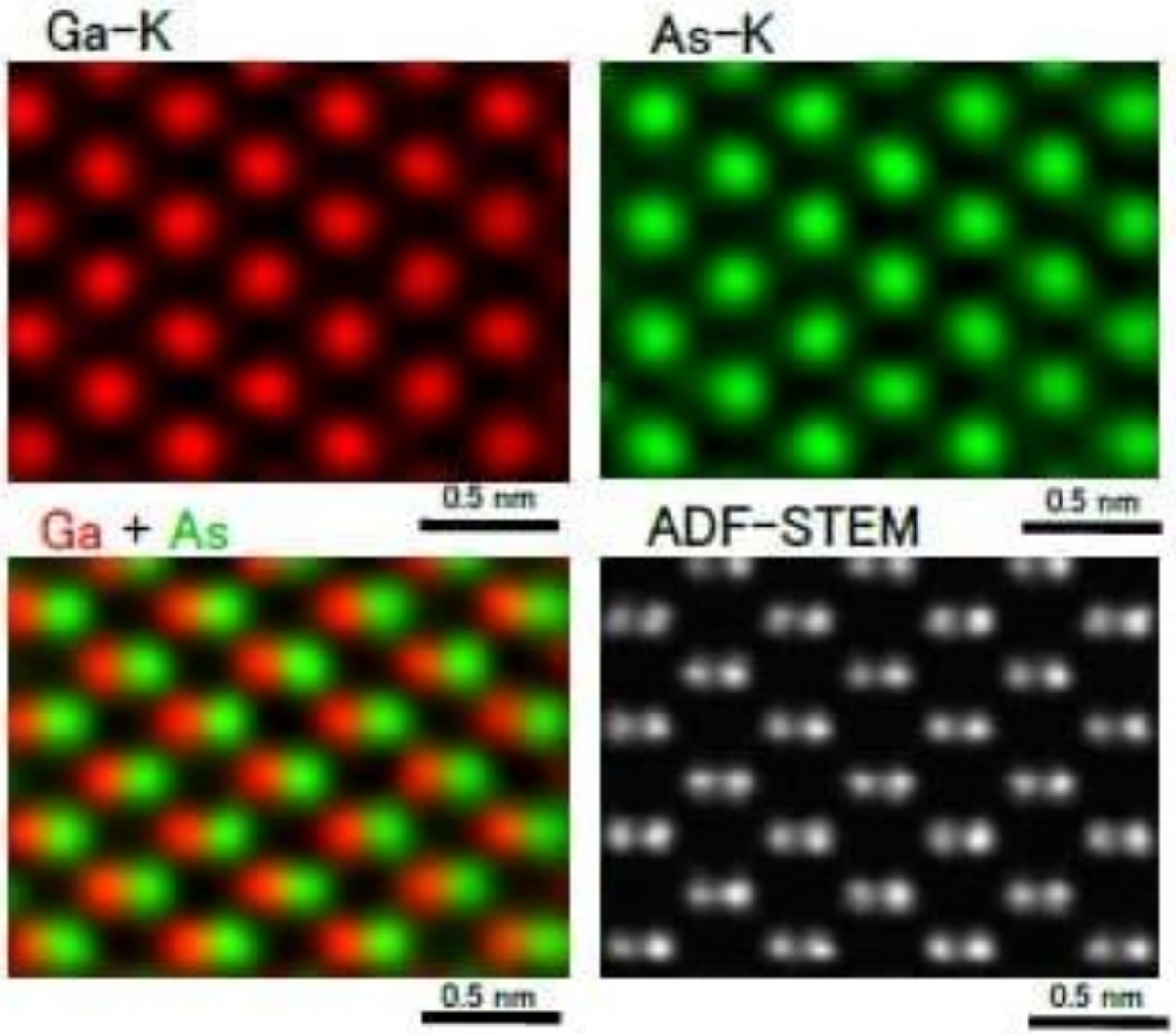
- Inelastic scattered electrons are used to form the basis of spectroscopy inside the TEM.

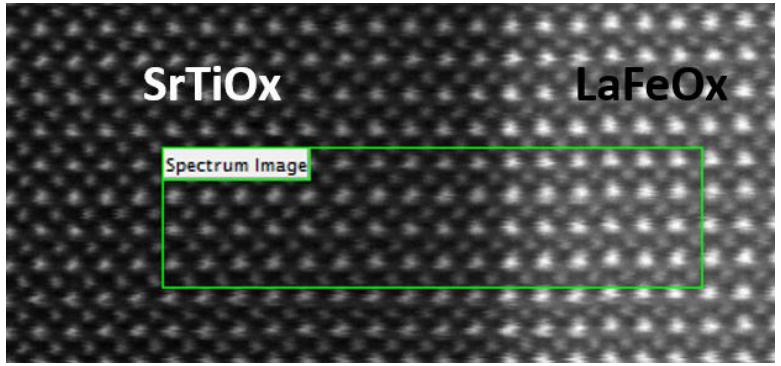
  - X-ray spectroscopy / Energy dispersive spectroscopy

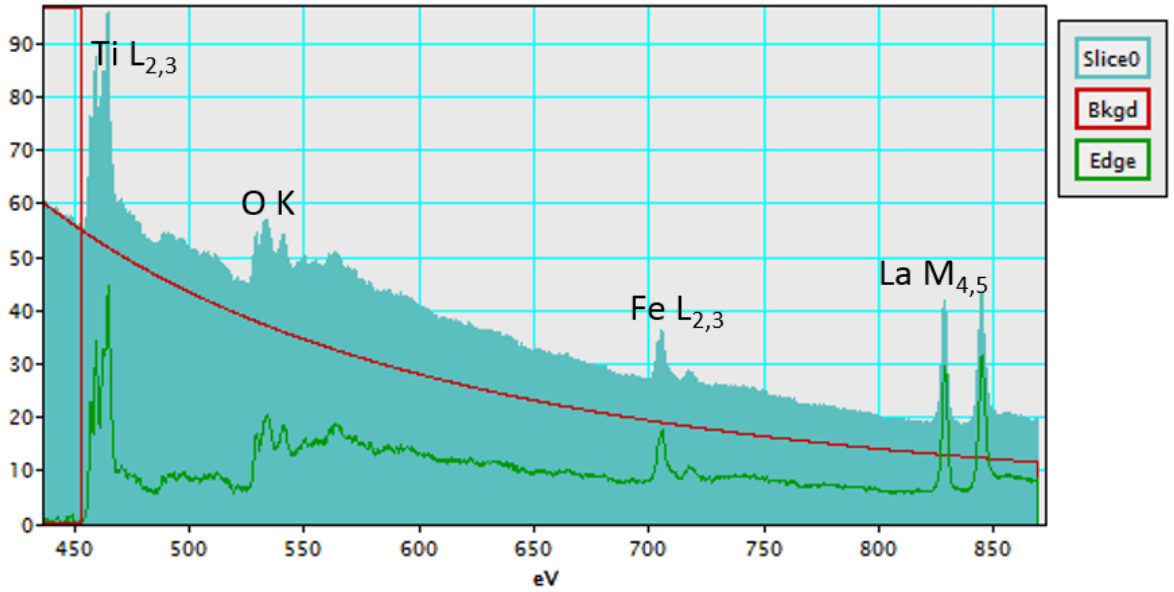


Element	Weight %	Atomic %
Si	0.54	1.08
Cr	17.53	18.80
Mn	1.45	1.47
Fe	66.61	66.50
Ni	11.11	10.55
Mo	2.76	1.60
Totals	100	100









Atomic-level-resolution EDX elemental map of a gallium arsenide monocrystal <011>.

https://www.hitachi-hightech.com/global/en/sinews/topics/0672/

Scanning Transmission Electron Microscope (STEM)

Conventional TEM mode (**T**ransmission **E**lectron **M**icroscope) mode

- Static/fixed beam
- Incident beam is relatively broad, unfocused
- Most relevant subtype: high-resolution (lattice, HRTEM) imaging

STEM mode (**S**canning **T**ransmission **E**lectron **M**icroscope) mode

- Rastered/scanning beam
- Incident beam is focused
- Most relevant subtype: HAADF-STEM imaging

S/TEM is instrument that can operate in TEM or STEM mode.

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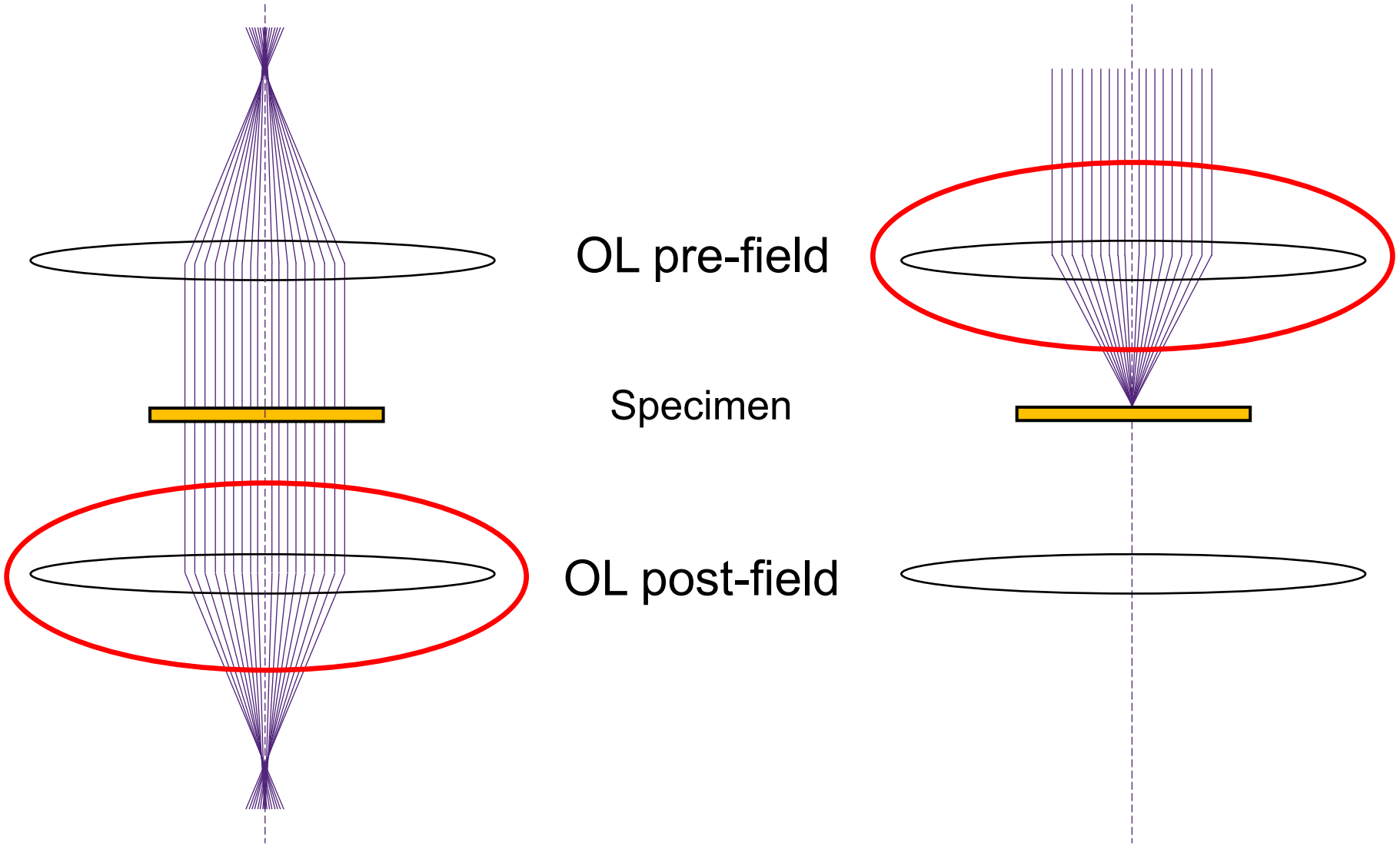
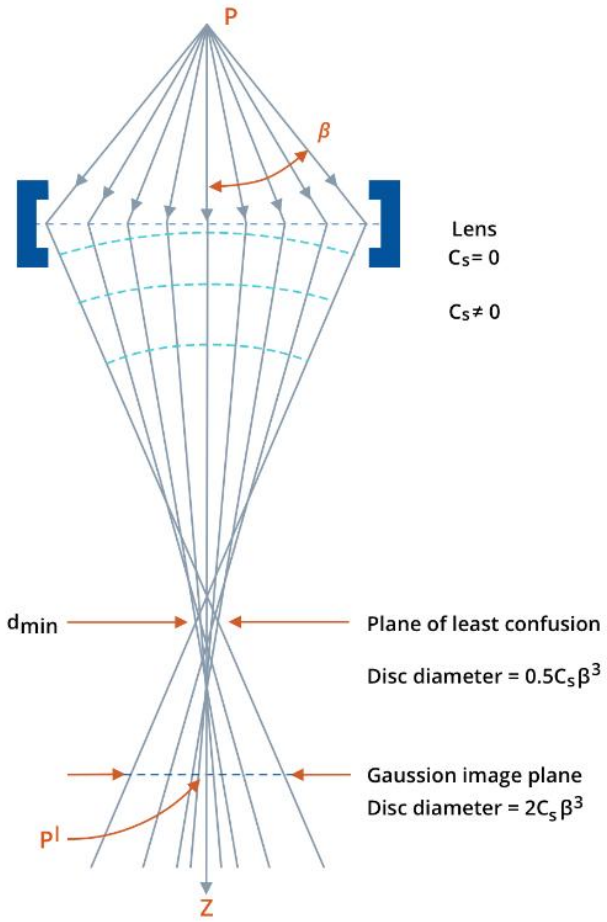


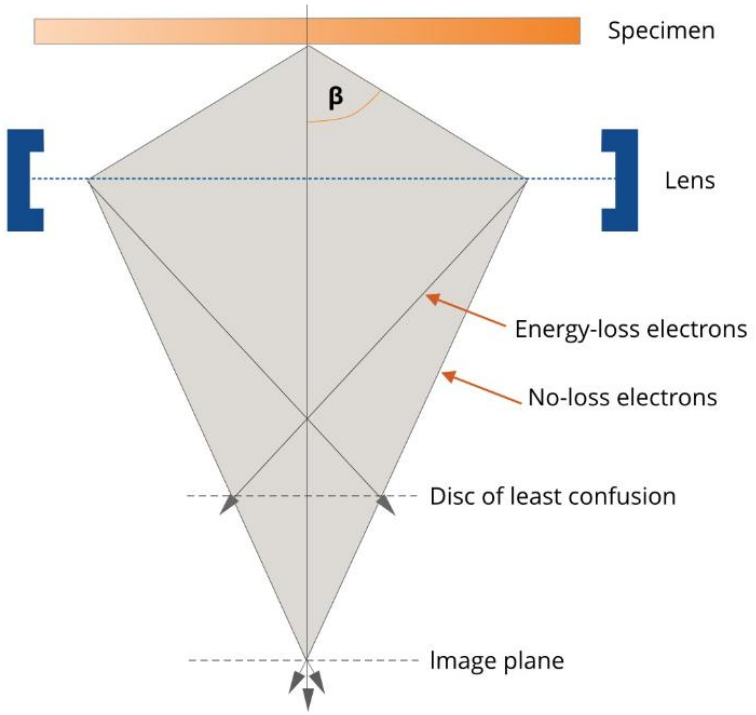
0.2 nm the limit of resolution for typical TEM.

To further improve on the limit of resolution, imperfection in the lenses (aberrations) must be corrected.

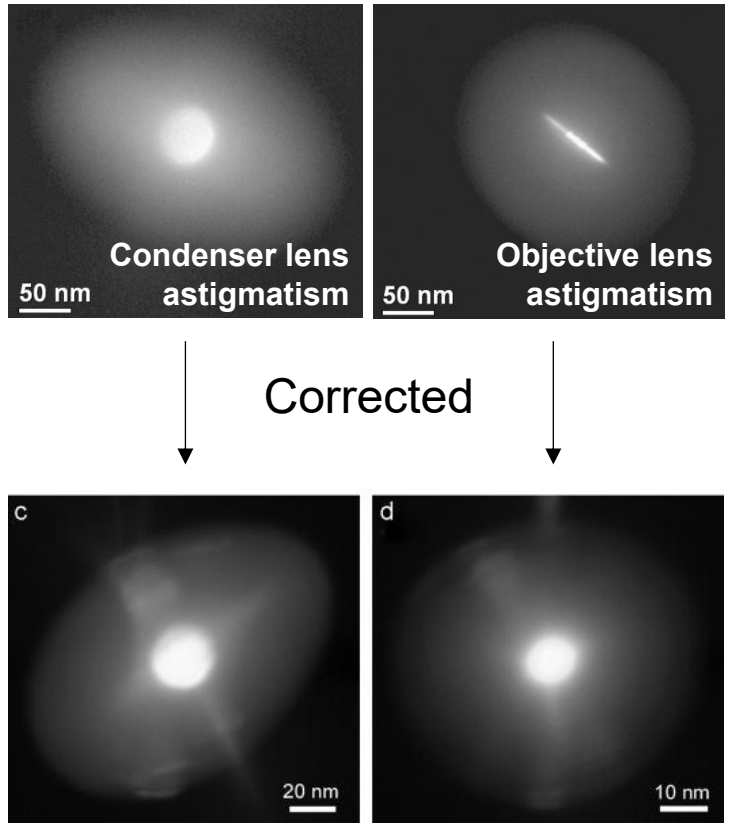
1. **Spherical aberration**



2. **Chromatic aberration**



3. **Astigmatism**

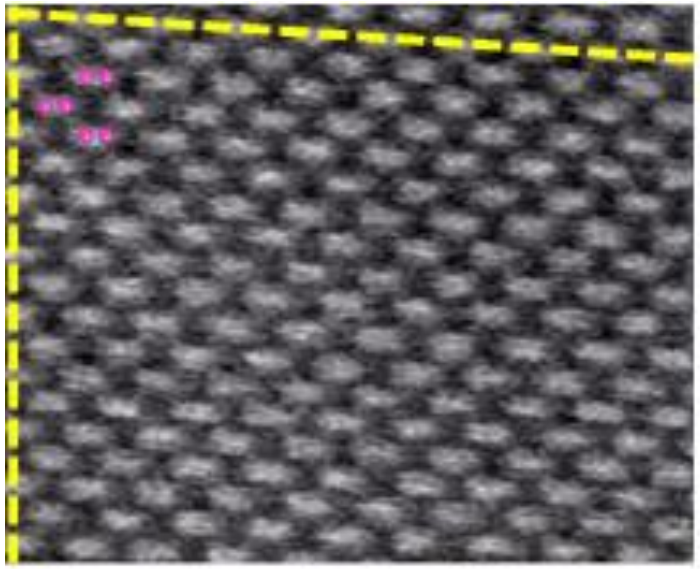


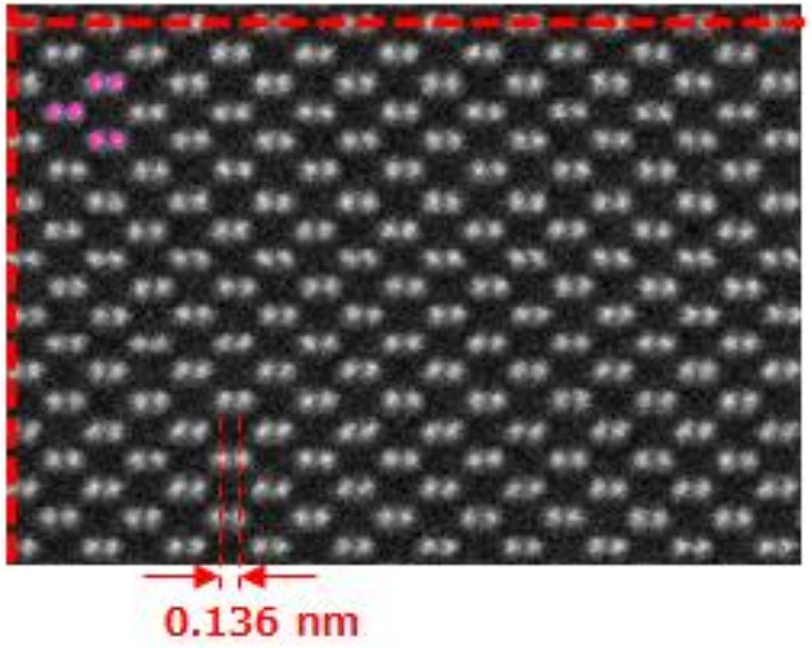
Aberration correction

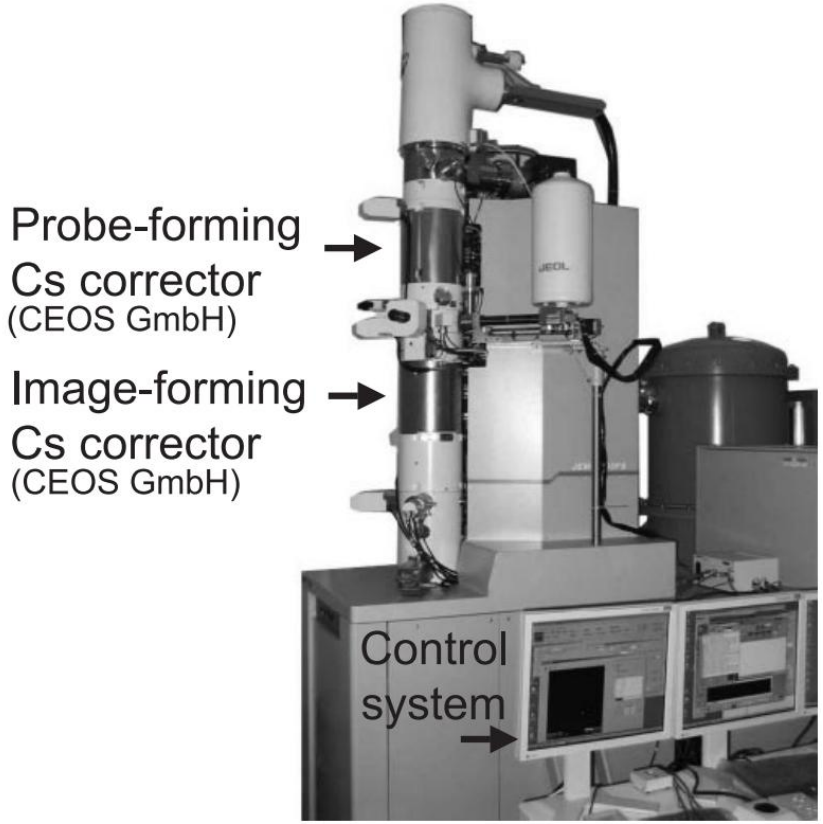
- Spherical aberration
- Chromatic aberration

Aberrations (other than astigmatism) cannot be corrected by TEM operators because it is inherently caused by the imperfection in magnetic lenses.

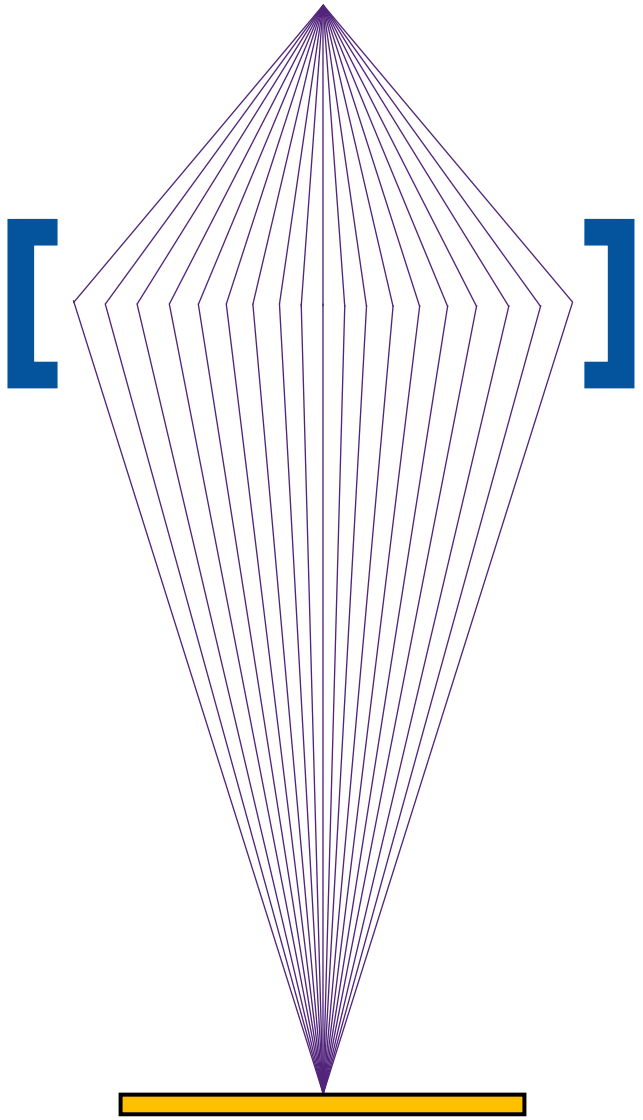
Therefore, they need to be corrected by additional hardware.



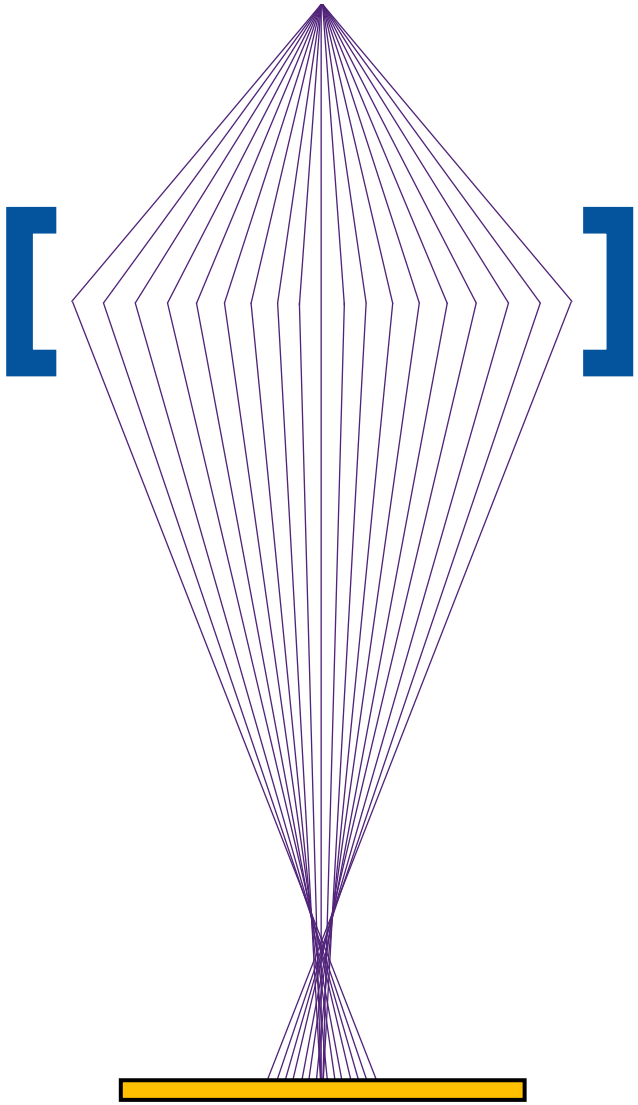




Expectation



Reality



HRTEM (lattice) imaging resolution:

$$\delta_{HRTEM} = 0.64(C_s\lambda^3)^{1/4}$$

STEM imaging resolution:

$$\delta_{STEM} = 0.43(C_s\lambda^3)^{1/4}$$

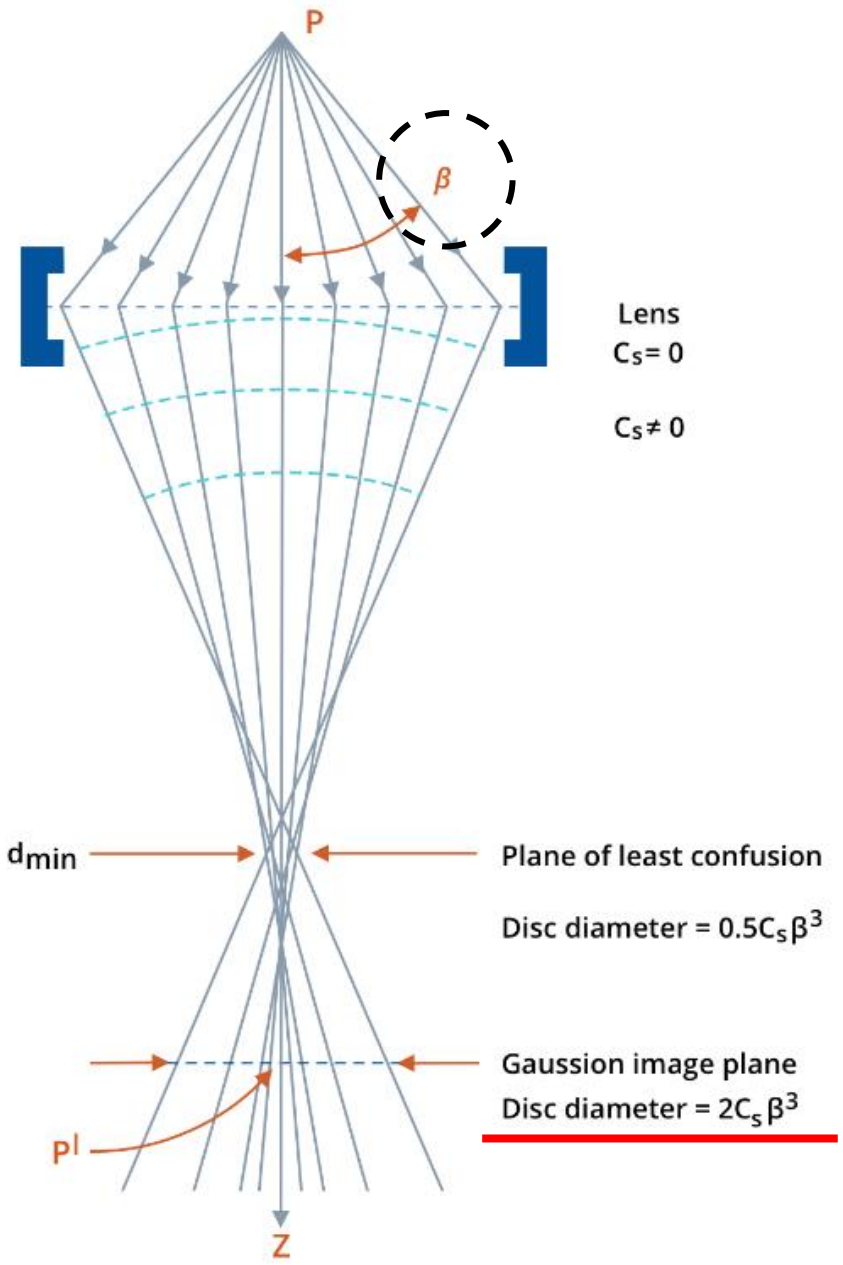
To get high resolution image, δ value needs to be made smaller.
Based on the equations, lower δ can be obtained by

1. *Decreasing C_s.*
2. *Decreasing λ.*

In practice, there is a limit on λ to avoid beam damage.

- λ = 1.97 pm (300 kV) for physical sciences (roughly).

→ Remaining option: Make C_s *smaller*.



The spherical aberration (C_s) is caused by the imperfection of objective lens.

Electron rays entering at large angles or further from optic axis ***bent excessively*** upon exiting.

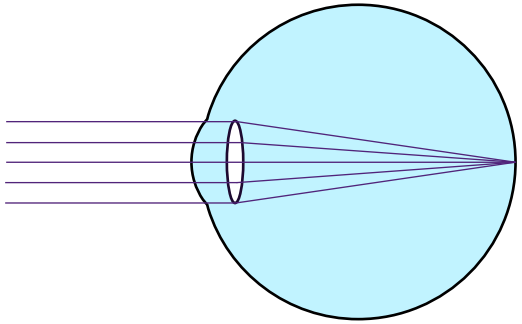
With C_s, off-axis rays are inhomogeneously generated.

The radius of the spherical aberration disk (r_{sph}) in the image plane can be calculated by the below equation.

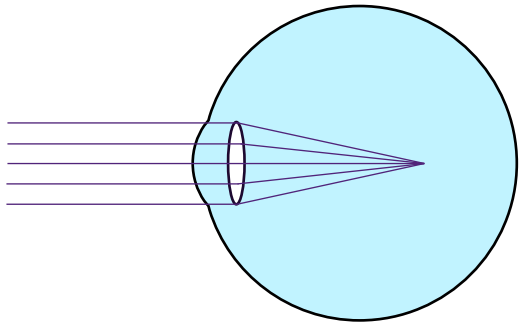
$$r_{sph} = C_s \beta^3$$

C_s = the spherical aberration coefficient (a constant for a particular lens, units of mm).
β = the maximum semi-angle of collection of the lens.

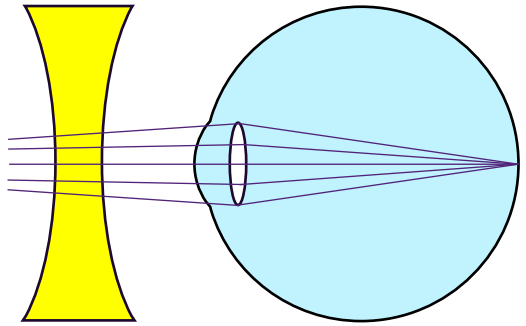
Normal



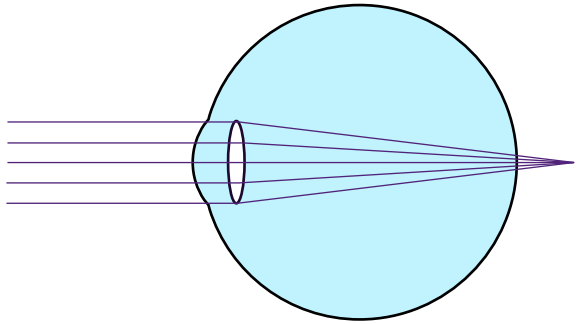
Near-sighted



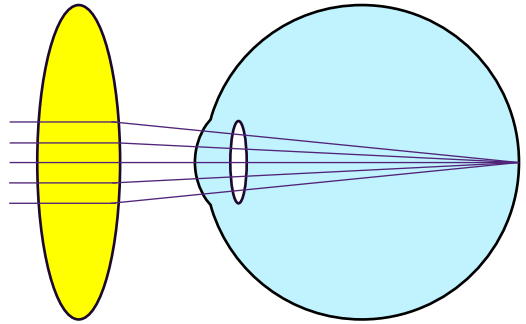
Corrected Near-sighted



Far-sighted



Corrected Far-sighted



Glass lenses do not make you correct your eye but alters the path of light rays to get to the right focal plane.

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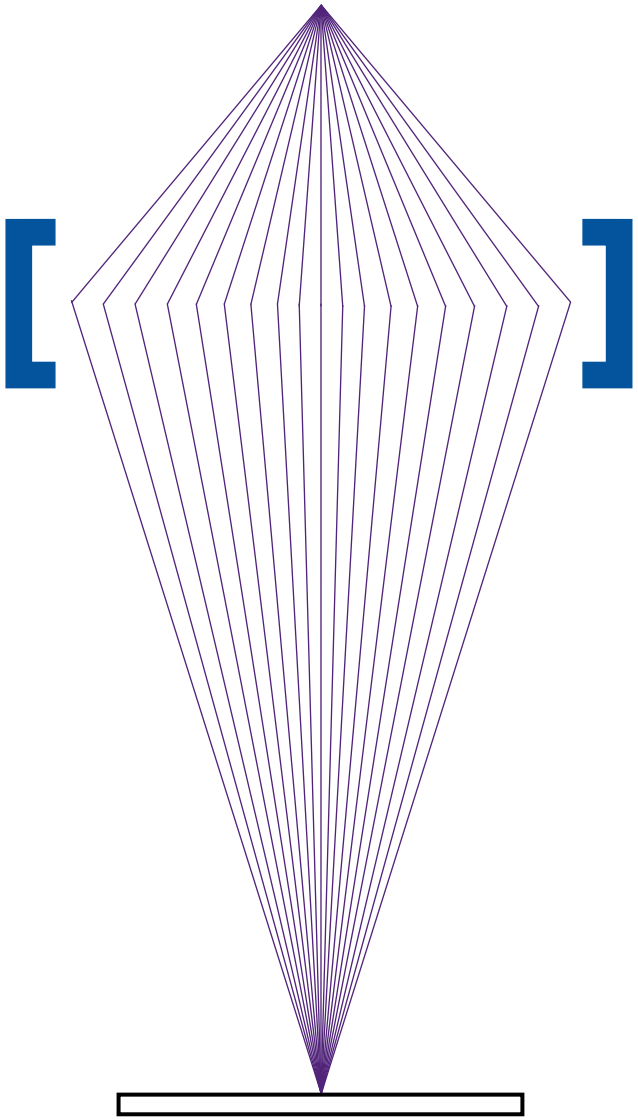
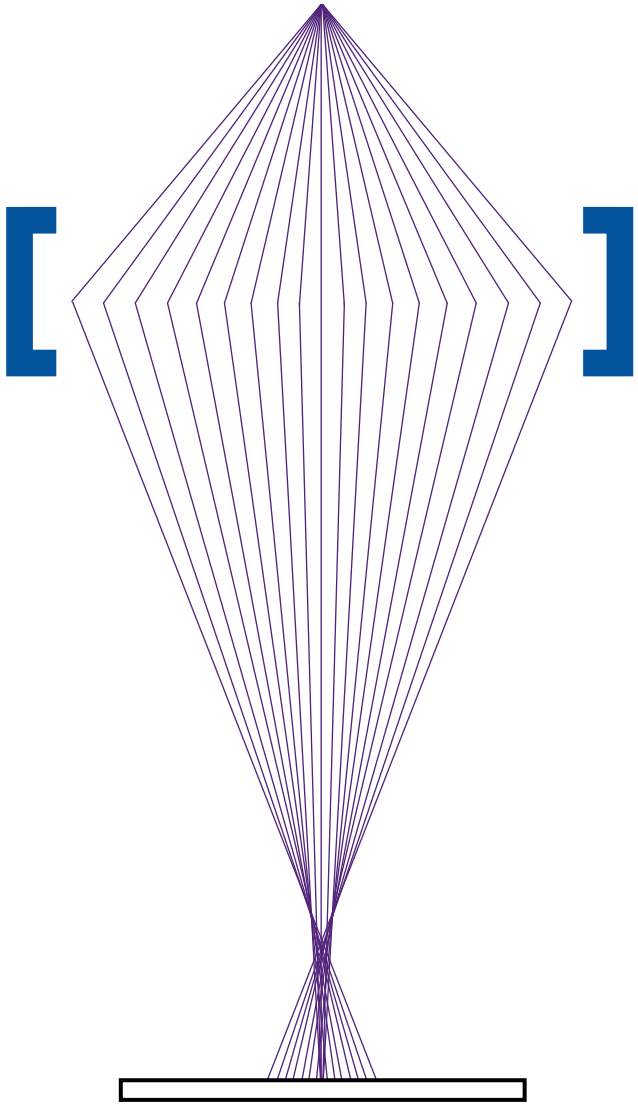
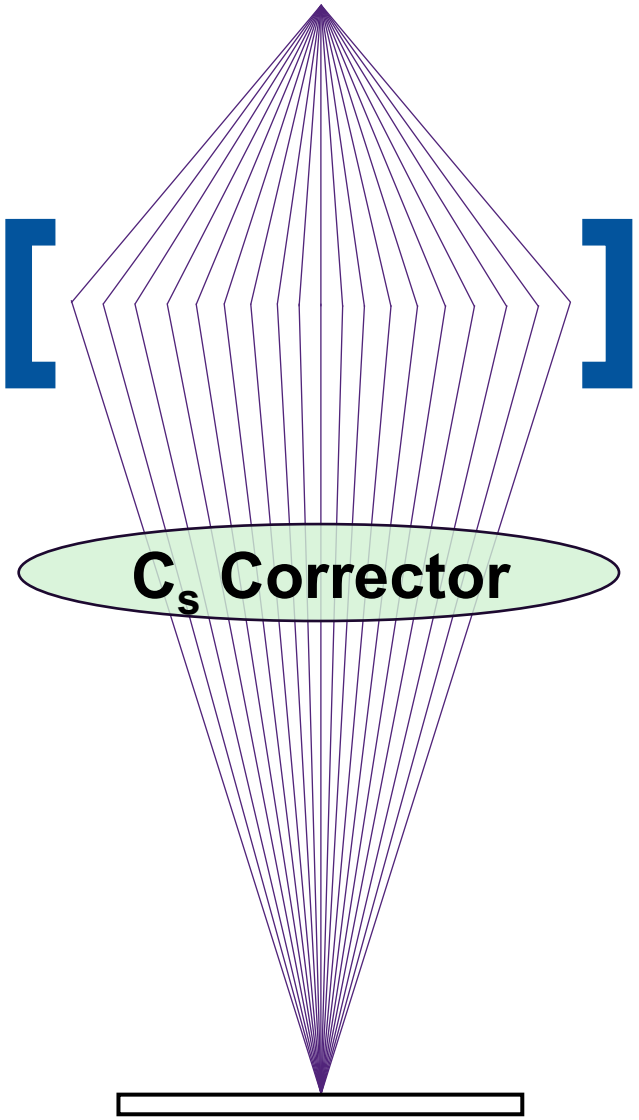
Expectation

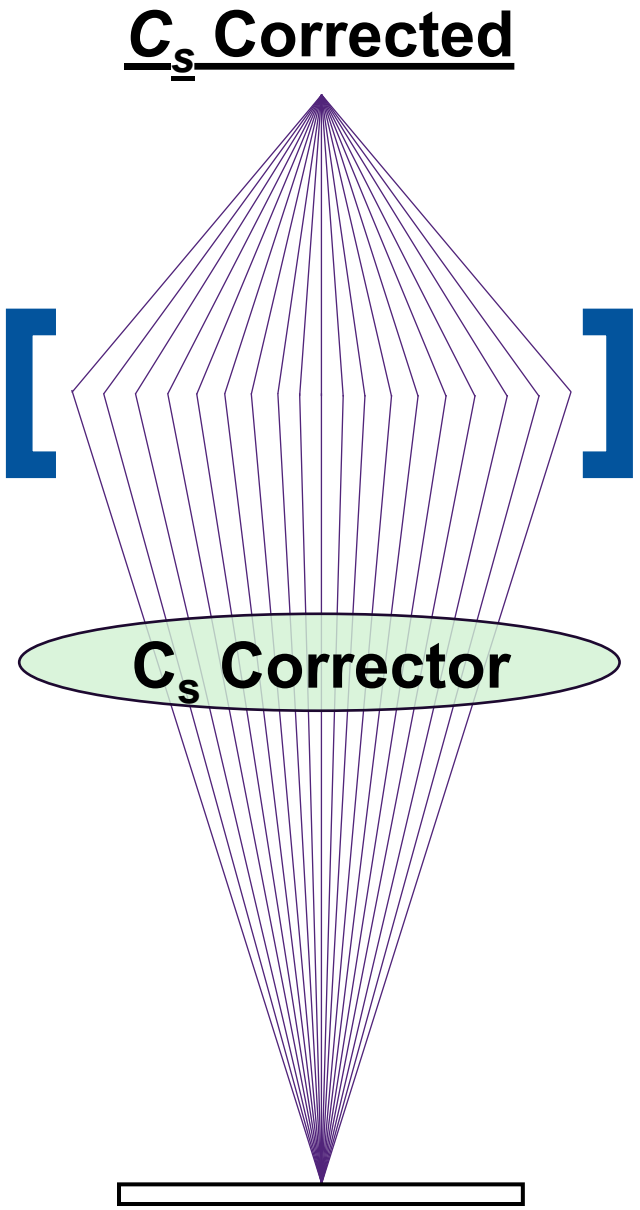


Reality



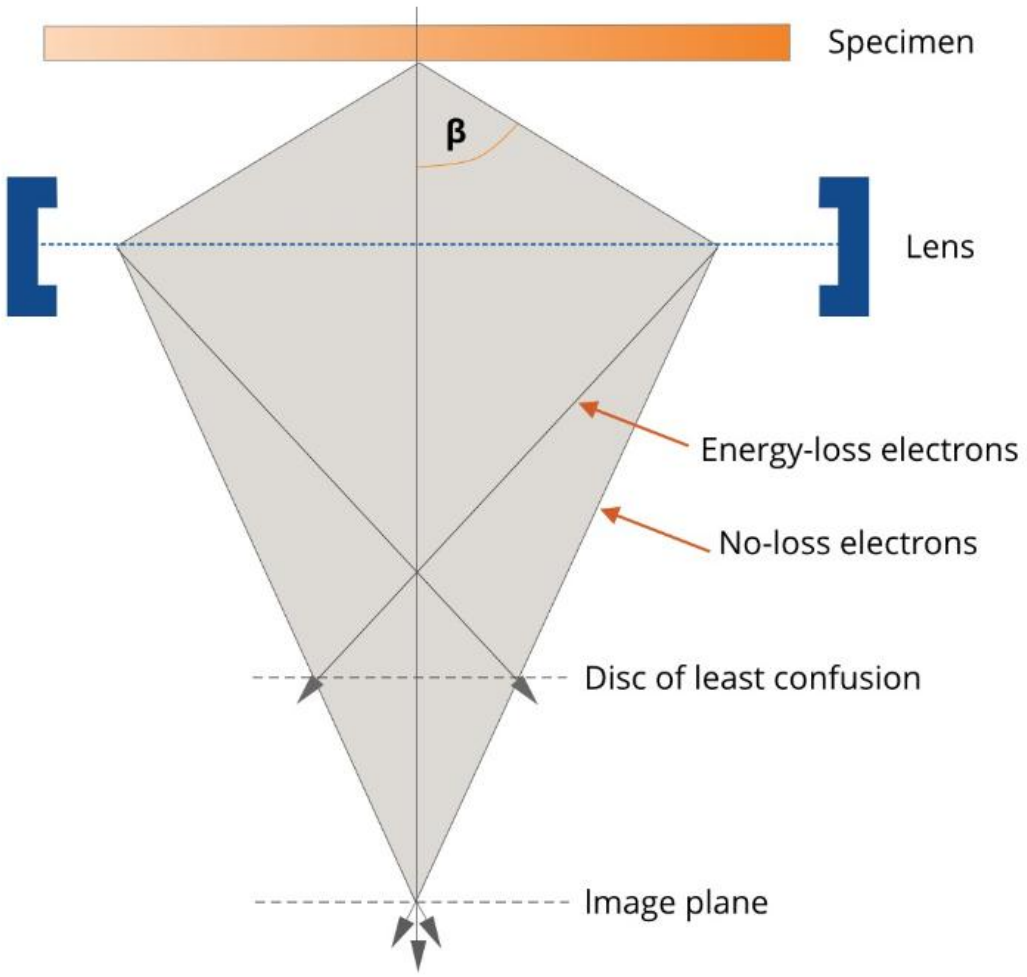
C_s Corrected





- Imaging
  - ➢ Ability of resolve ≤ 0.1 nm features at 30 – 300 kV
    - What is currently commercially available:
      - ➢ < 0.06 nm (300 kV)
      - ➢ ~ 0.1 nm (30 kV)

- Analytical
  - ➢ < 0.1 nm probes with high current (up to ~1 nA)
    Ability of resolve ≤ 0.1 nm features at 30 – 300 kV
    - Fast(er) atomic-resolution chemical analysis
      - ➢ Energy dispersive spectroscopy (EDS)
        - ○ Elemental mapping
      - ➢ Electron energy loss spectroscopy (EELS)
        - ○ Elemental mapping
        - ○ Oxidation state mapping



The term chromatic aberration is related to the energy of the electrons.

Electrons are not monochromatic. Electrons emerge from the gun at a whole range of energies and are bent by the objective lens to different degrees (electrons that have lost energy are bent more strongly). Thus, once again, electrons from a point on the specimen form a disk image, as for spherical aberration. The radius (r_{chr}) of the disk is given by:

$$r_{chr} = C_c \frac{\Delta E}{E_0} \beta$$

C_c = the chromatic aberration coefficient of the lens (length).
ΔE = the energy loss of the electron.
E_o = the initial beam energy.
β = the maximum semi-angle of collection of the lens.

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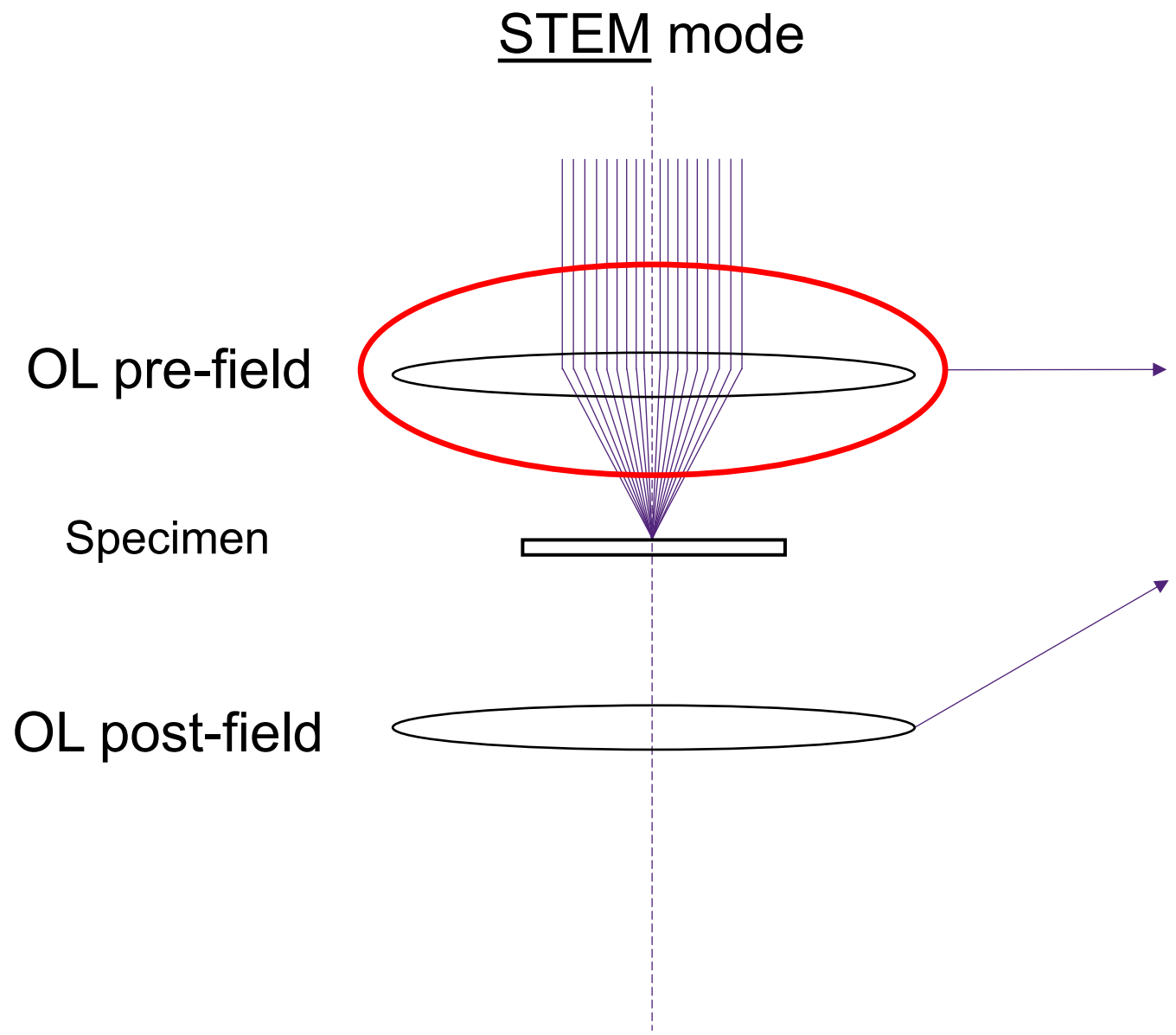
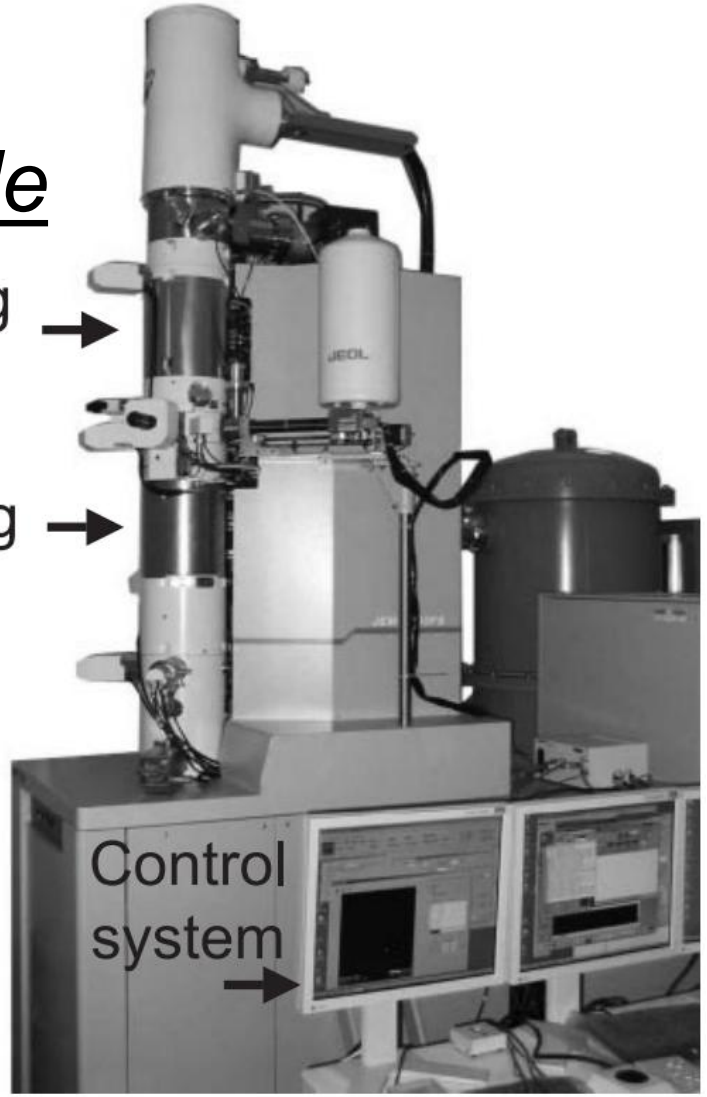


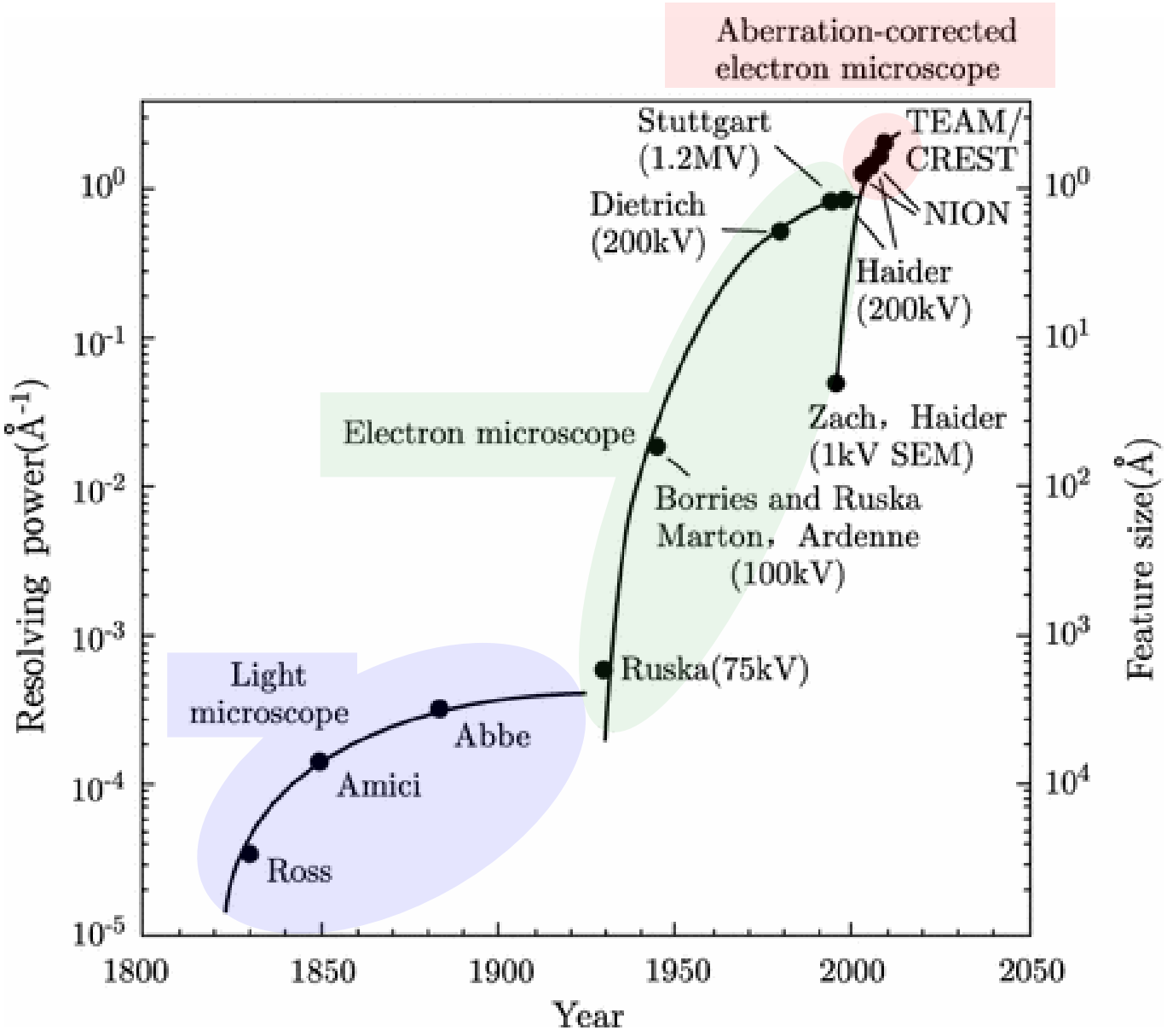
STEM mode

Probe-forming
Cs corrector
(CEOS GmbH)

Image-forming
Cs corrector
(CEOS GmbH)

TEM mode





With aberration correction, the limit of resolution of 0.05 nm can be achieved.

MATE 6301 | Cs-corrected STEM image

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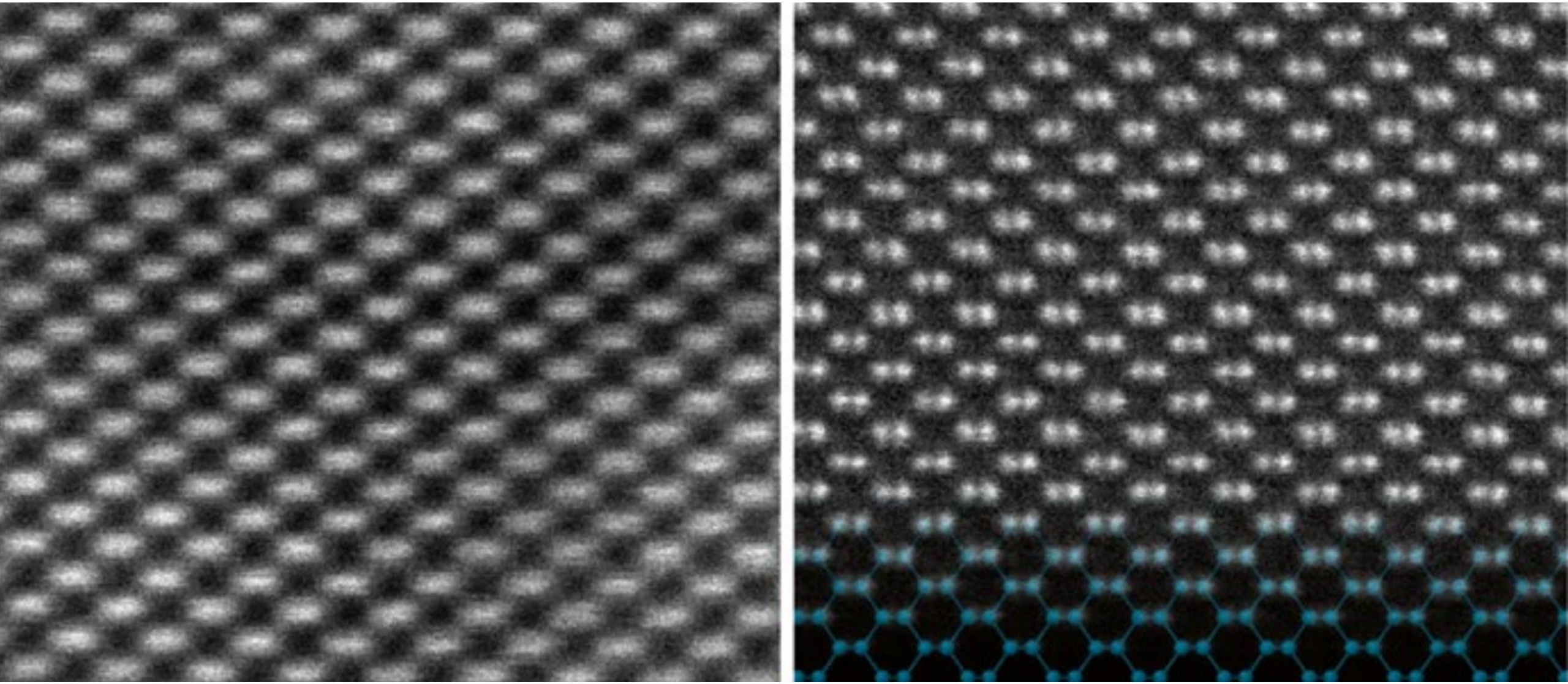
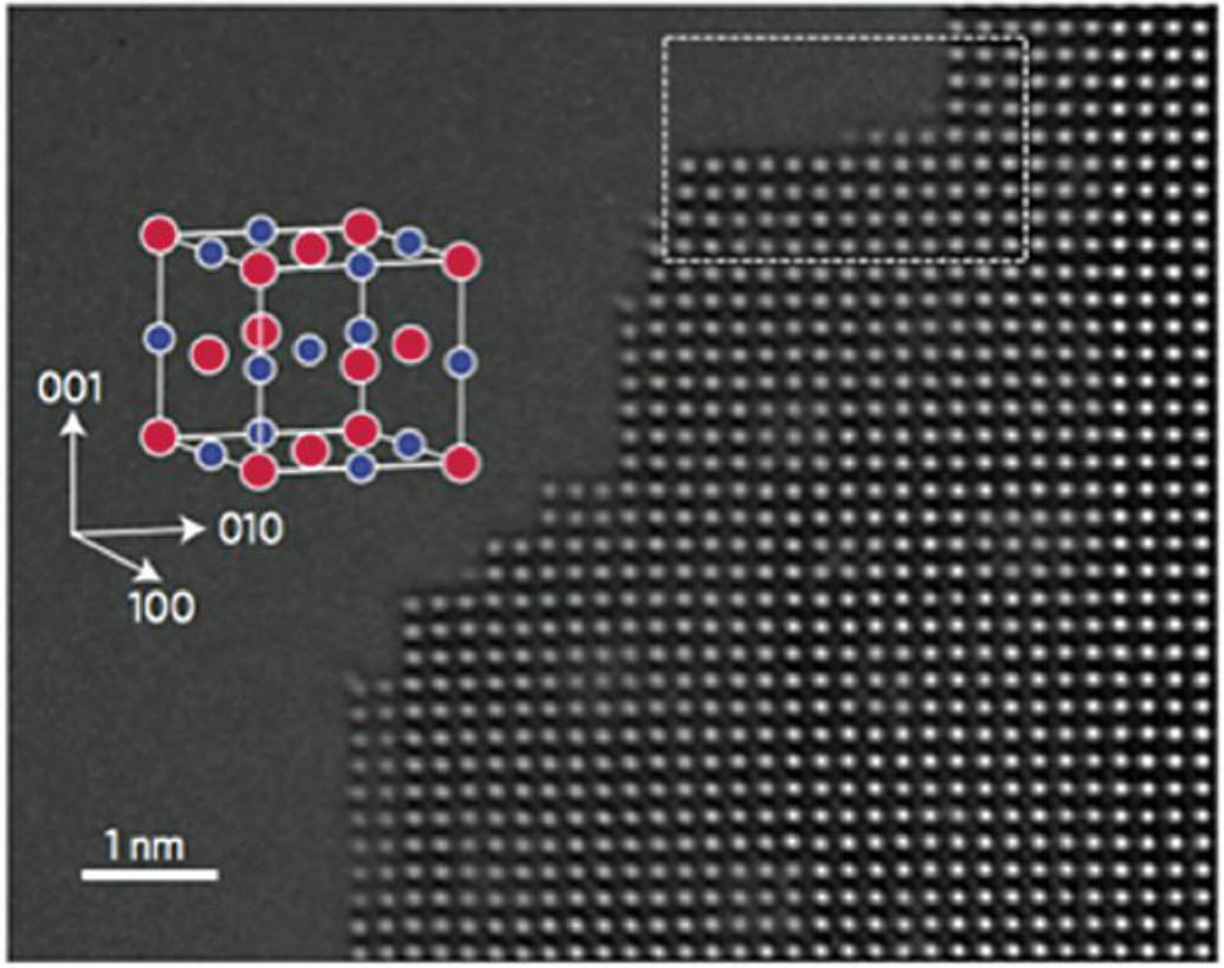
Uncorrected

C_s-corrected

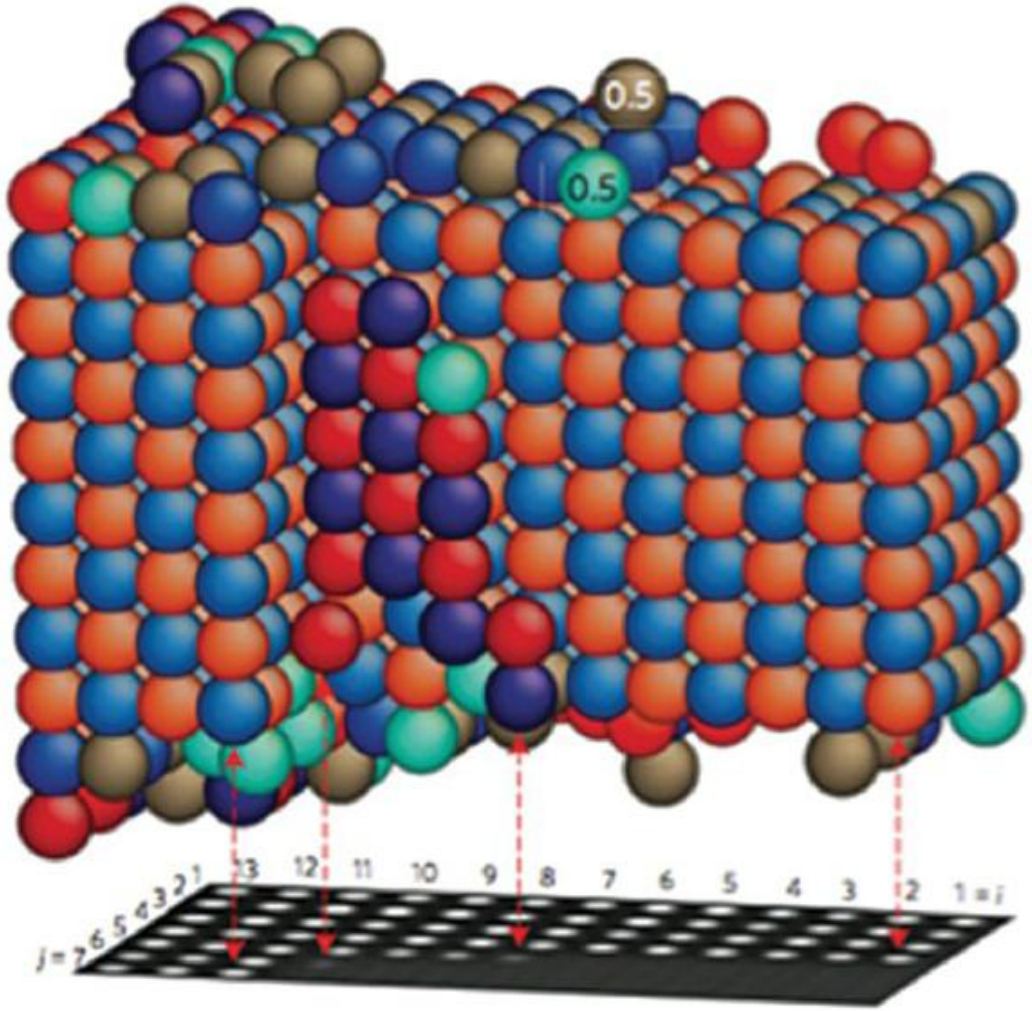


It’s like swtiching from standard definition to 4k UHD TV!

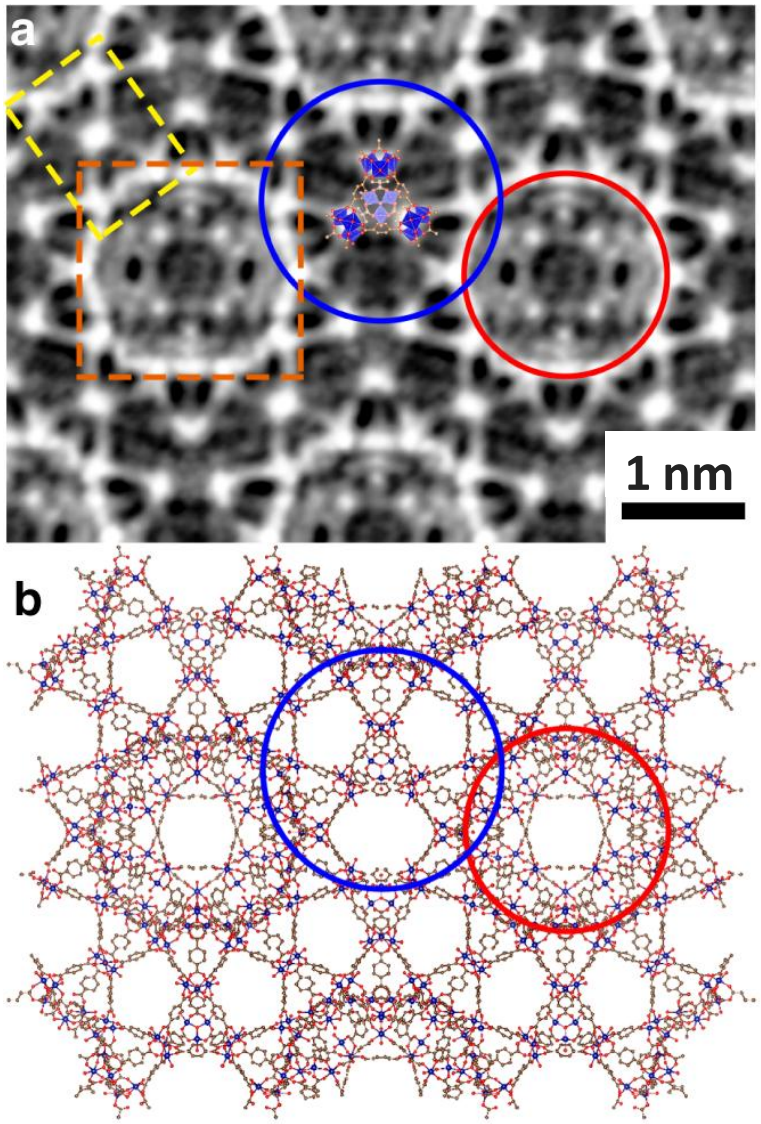
(a)

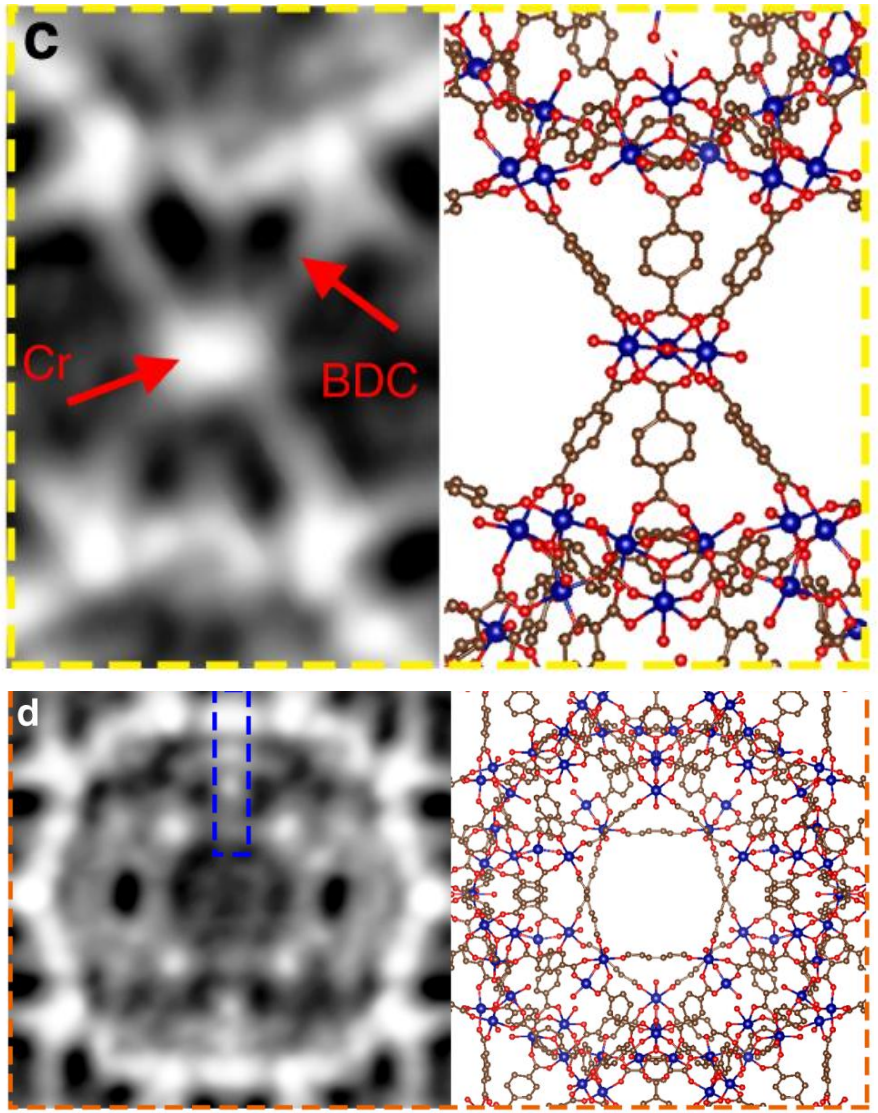


(b)

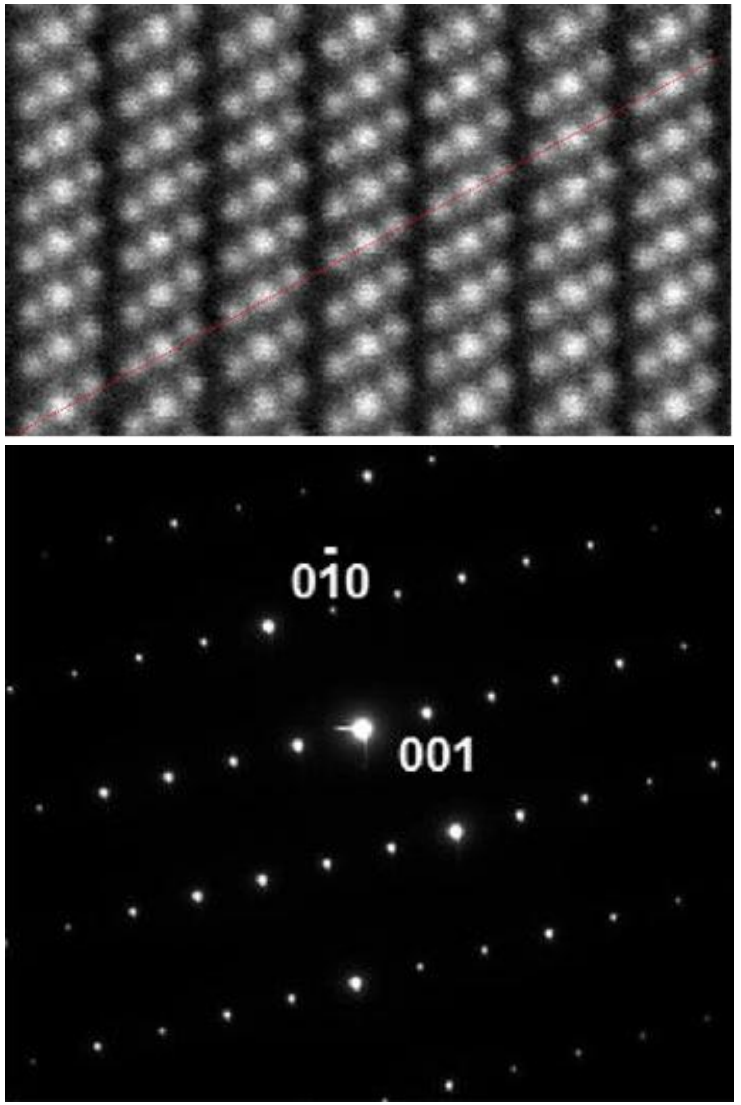


MATE 6301 | Cs-corrected STEM image





Nat Commun **11**, 2692 (2020).



CrystEngComm **16**, 8678-8683 (2014).

- Microstructural analysis
- Defects identification
- Interfacial analysis
- Crystal structure
- Seeing atomic arrangement of crystals
- Elemental analysis at nanometre scale

- **Advantages**
  - ✓ Superior resolution
  - ✓ Ability to obtain compositional and crystallographic information from specimens with high spatial resolution

- **Limitations**
  - ➢ Very small volume of material studied – is it representative?
  - ➢ Limitations on samples
  - ➢ Sample preparation can be quite lengthy and involved
  - ➢ Electrons can damage sample and introduce artifacts

- **Structure of TEM**
  - Principles of TEM contrasts
  - TEM imaging modes include:
  - Diffraction contrast (bright field/dark field)
  - Phase contrast
  - Mass/thickness contrast

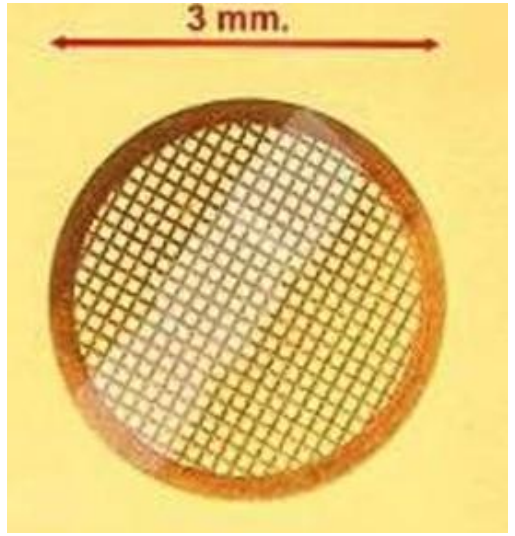
- **Compositional information obtained through:**
  - Characteristic X-rays
  - STEM

Sample Preparation for TEM

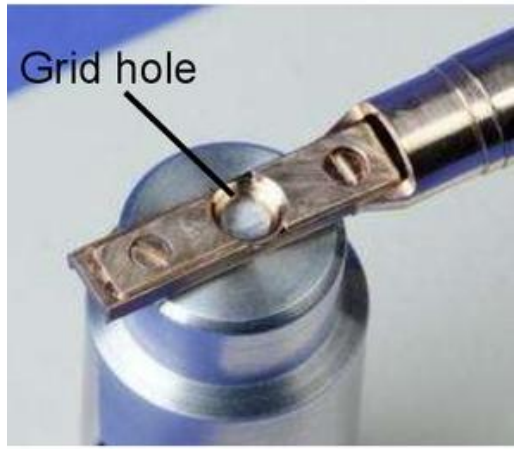
Sample Preparation

Powdered samples/Nanoparticles:

- Small amount of powdered samples or in the case of a thin film sample, we scrape the film deposited on the substrate and transfer it to a volatile solution (*i.e.*, isopropyl alcohol, ethanol).
- Then the solution containing the nanoparticle is allowed to sonicate for 10 minutes (this short duration is just to maintain the pore structure without being collapsed and to avoid heating of samples).
- This solution is then drop-cast on the carbon-coated side of the TEM grid, and the solvent is allowed to evaporate completely before the analysis.

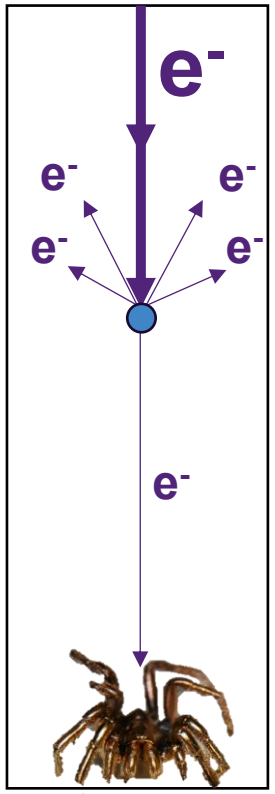


[4]

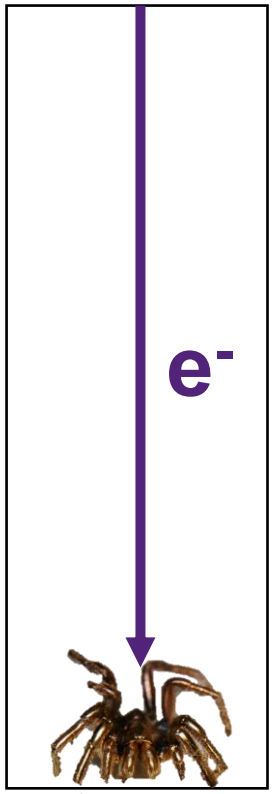


[5]

- An electron beam wouldn't have a very long mean free path in air. This is why the TEM is under vacuum, so the electrons can make it from the source to the sample.



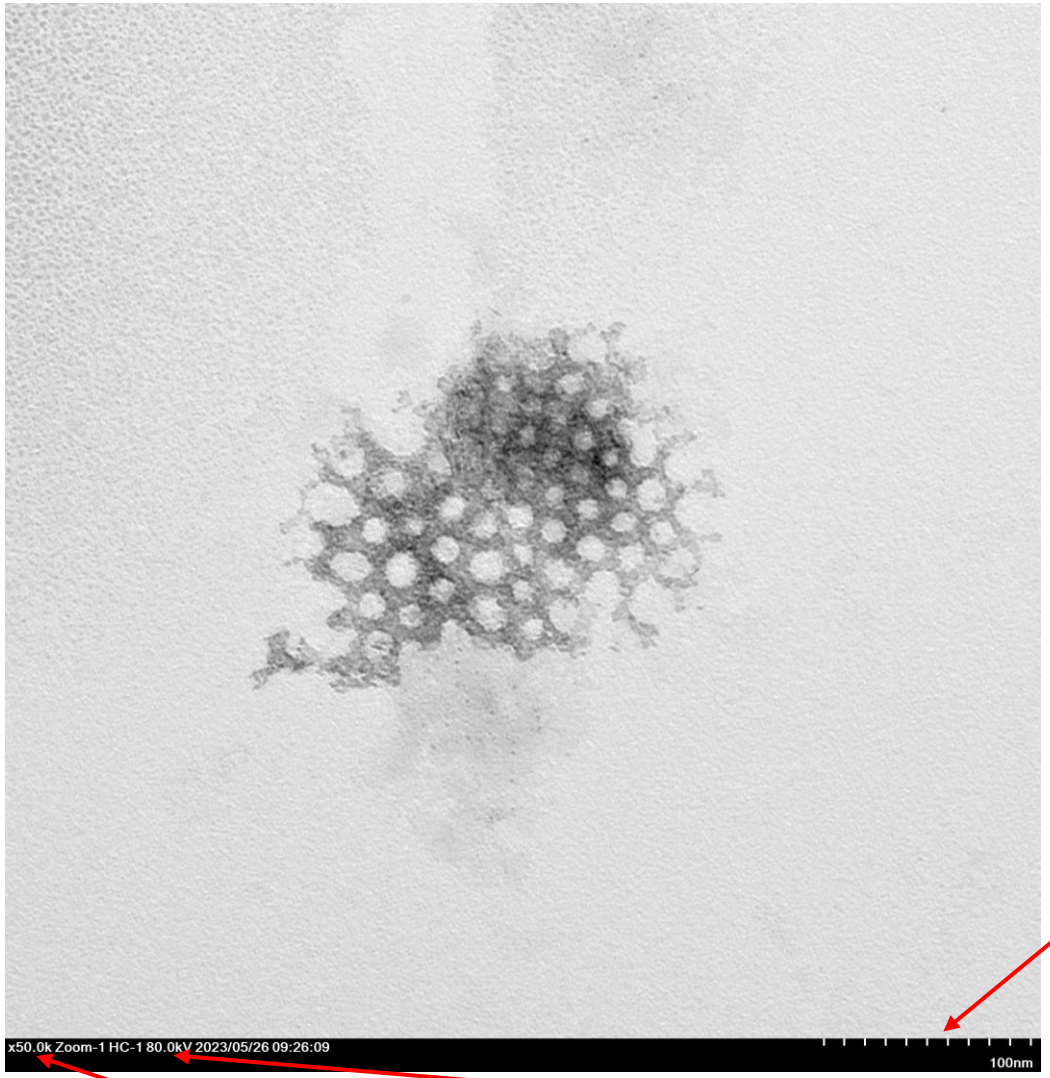
In air

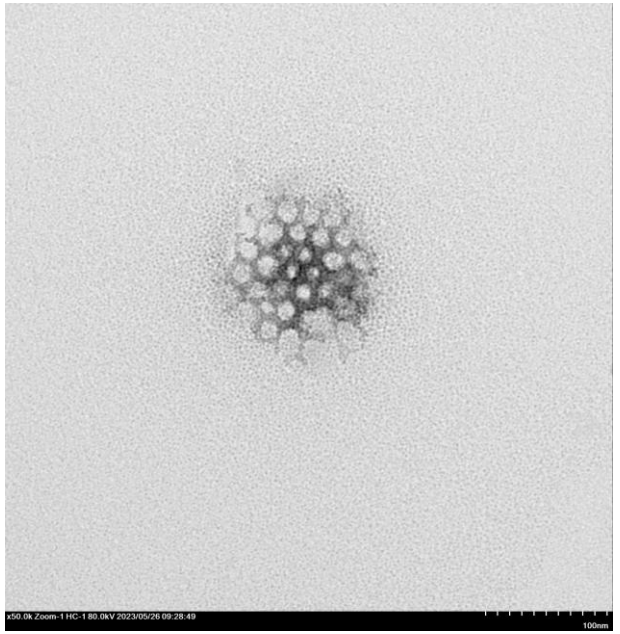


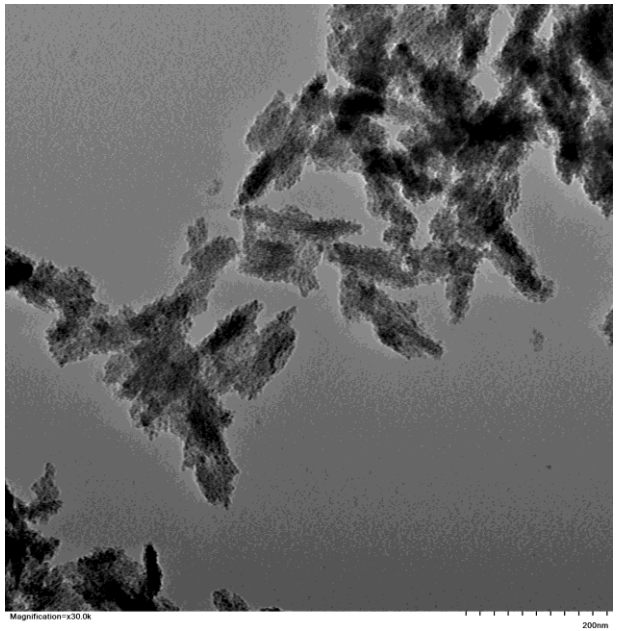
In vacuum

Powder-free gloves

Results:







Scale bar

Magnification

Accelerating Voltage

Key-Points

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1. Sample preparation
   - Conductive sample
   - Non-conductive sample

2. Beam alignment
   - Focal plane (or working distance) adjustment
   - Column (aperture & lens) alignment
   - Astigmatism correction

3. Aberration correction
   - Spherical aberration
   - Chromatic aberration

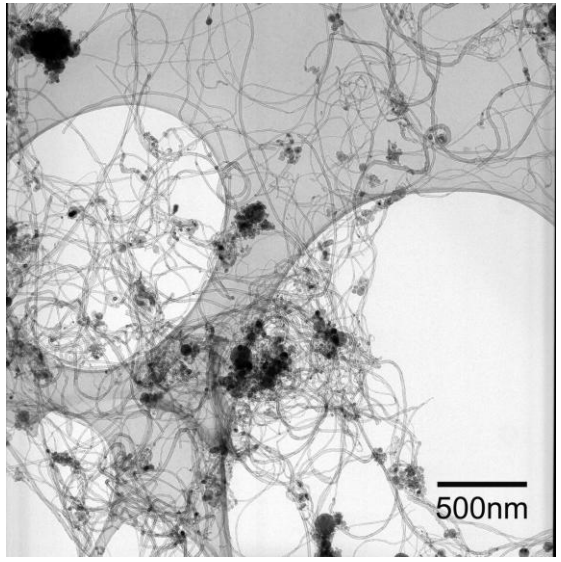
4. Imaging parameters
   - Accelerating voltage
   - Beam current
   - Working distance
   - Aperture

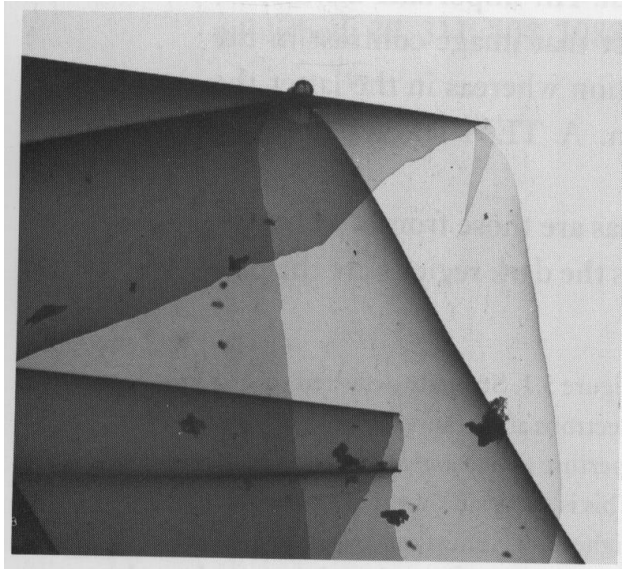
Non-controllable by Operator

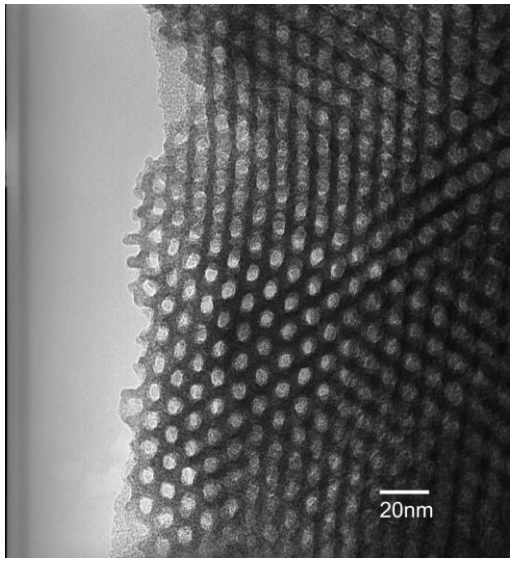
Controllable by Operator

Mass/thickness contrast

- Mass/thickness contrast arises from incoherent elastic scattering at the nucleus (Rutherford cross section).
- The interaction between electrons and heavy atoms is stronger than that between electrons and light atoms . If the thickness is homogeneous, areas in which heavy atoms are concentrated appear with darker contrast than such with light atoms (mass contrast). Of course, more electrons are scattered in thick than in thin areas; thus, thick areas appear dark (thickness contrast). However, a thick area with light elements might have the same contrast as a thinner area with heavier atoms.
- The contrast depends upon atomic number Z and sample thickness t.



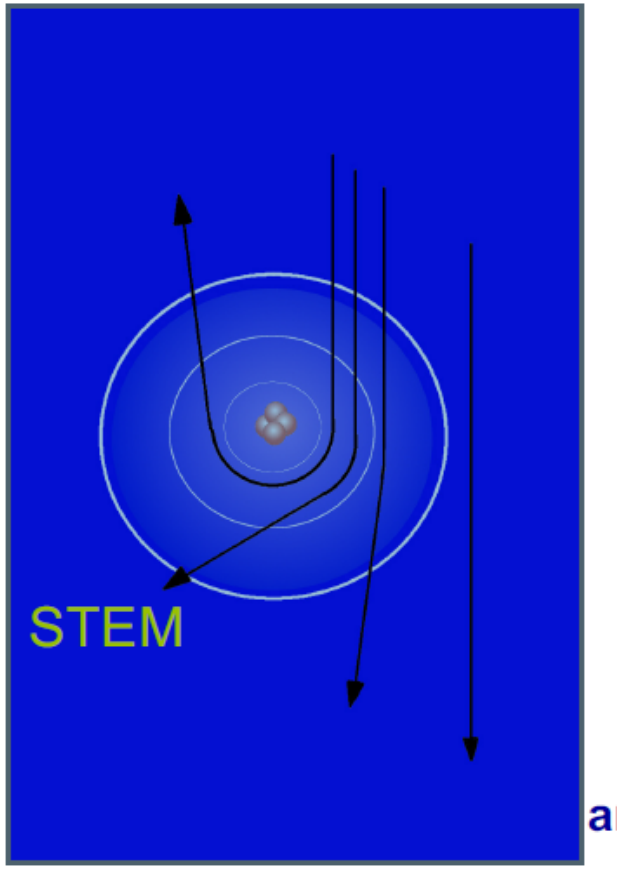




Phase contrast & applications

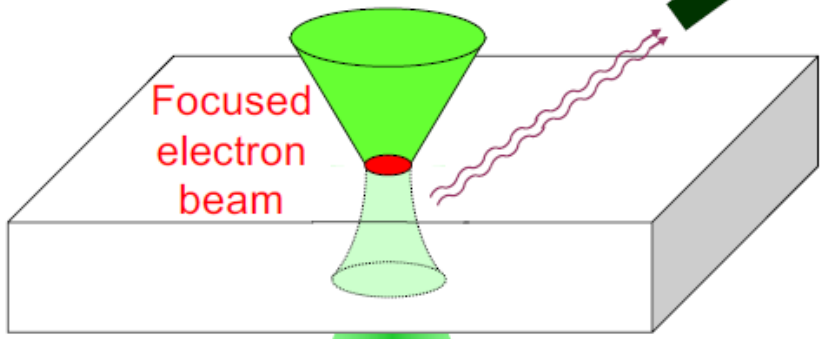
- Phase contrast deals also with crystalline specimens
- Phase contrast is formed **with at least two diffracted electron beams through their interference**.
- Phase contrast is realized with (or without) an aperture at the back focal
- plane of the objective lens
- Phase contrast is mainly used for study of,
    i. Defects
    ii. Precipitates
    iii. Interfacial structures
    iv. Crystal growth
- Phase contrast requires very thin specimens (< 20 nm typically) for the interference.
- Crystals should be orientated to the directions where the zone axes should be parallel to the incidental electron beam.
- Sometimes, image simulations are needed for gain correct interpretation as atoms can be black or white in the image.

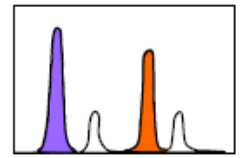
Scanning Transmission Electron Microscopy (STEM)



Small (< 1 nm) electron probe across sample

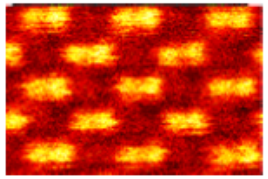
X-ray detector

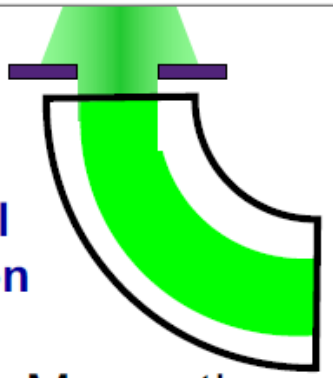


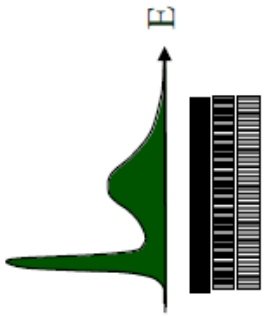


EDS:
Quantitative compositional distributions

STEM Imaging:
Atomic resolution structural and compositional information



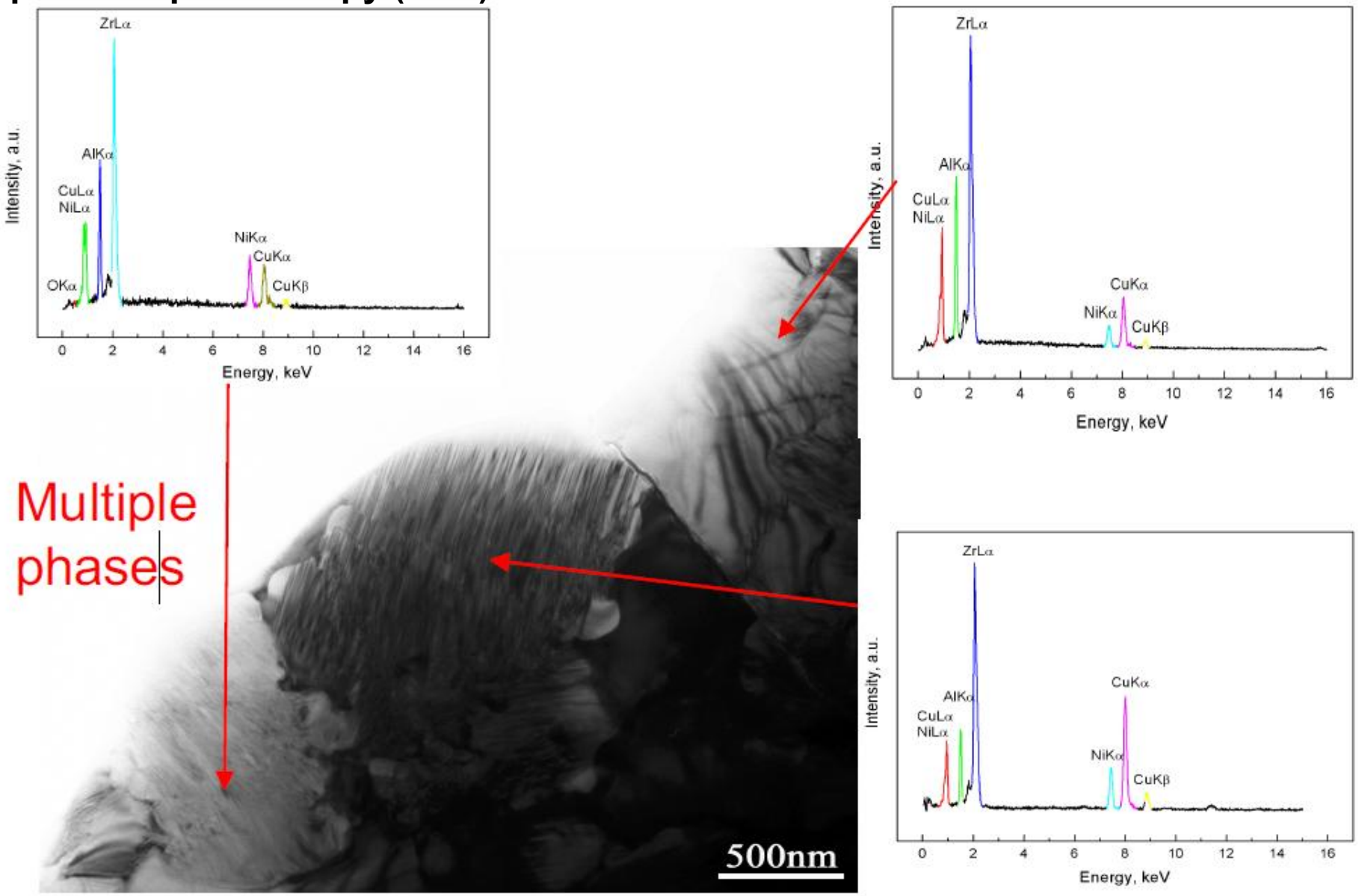




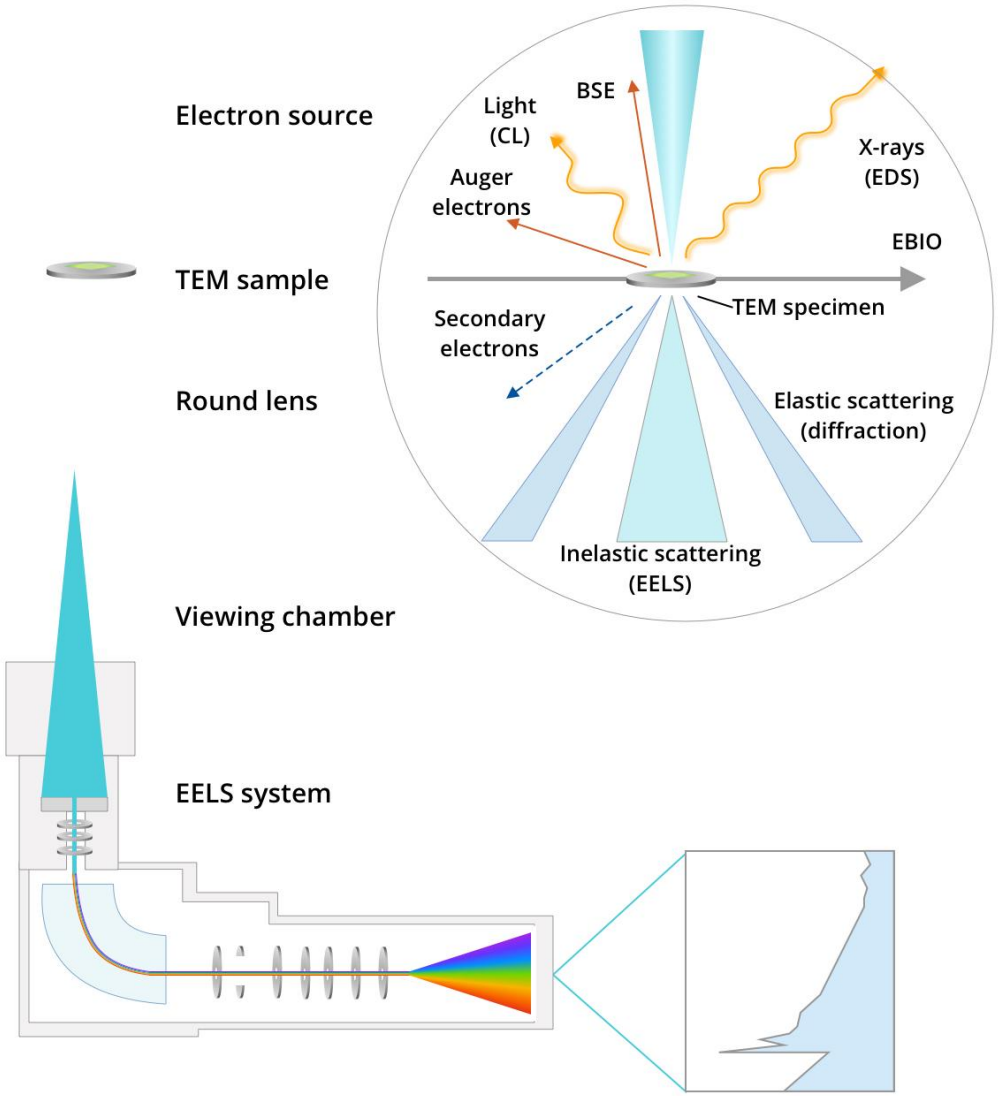
EELS:
Compositional information + electronic structure

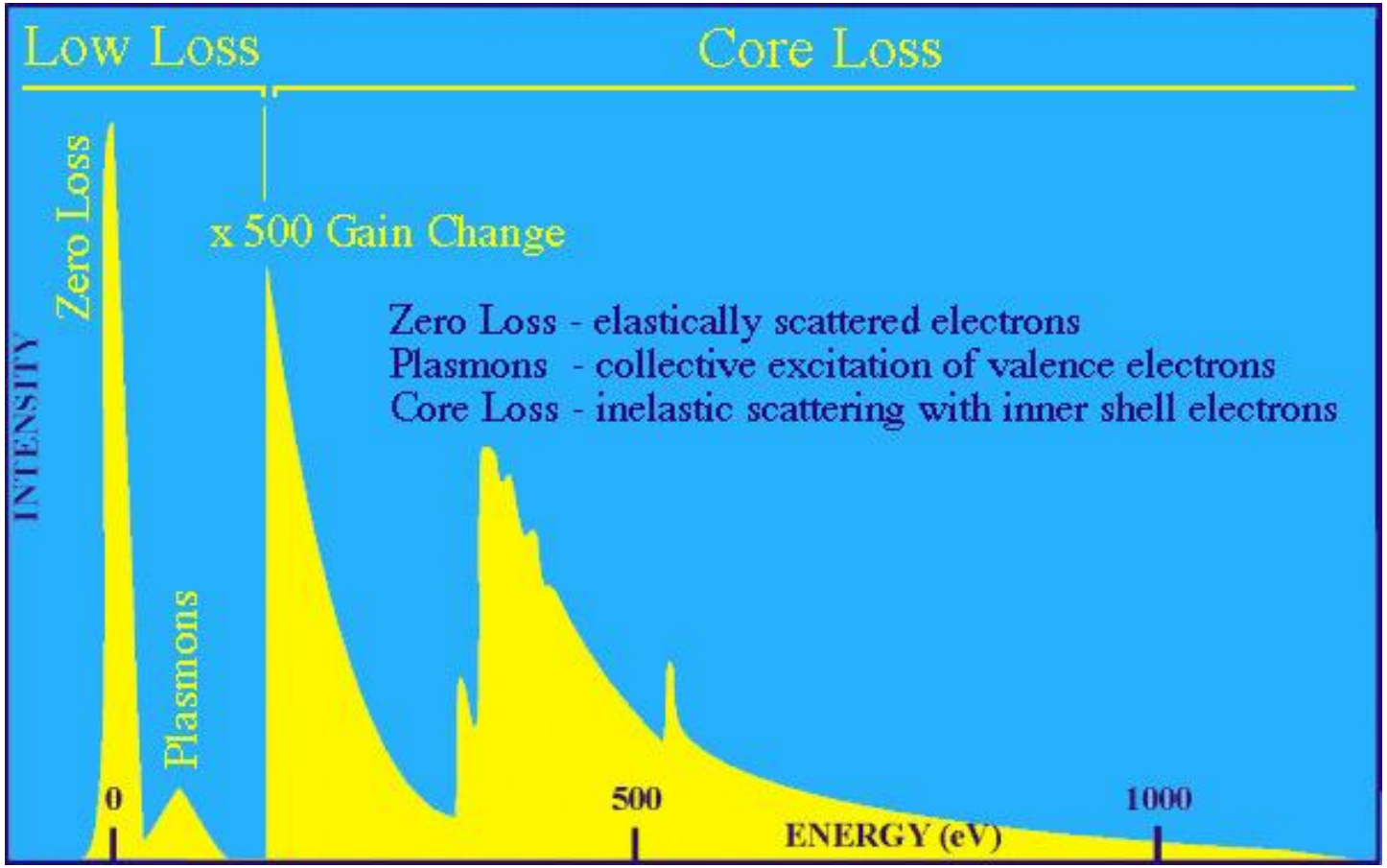
parallel electron detector

Energy Dispersive Spectroscopy (EDS)

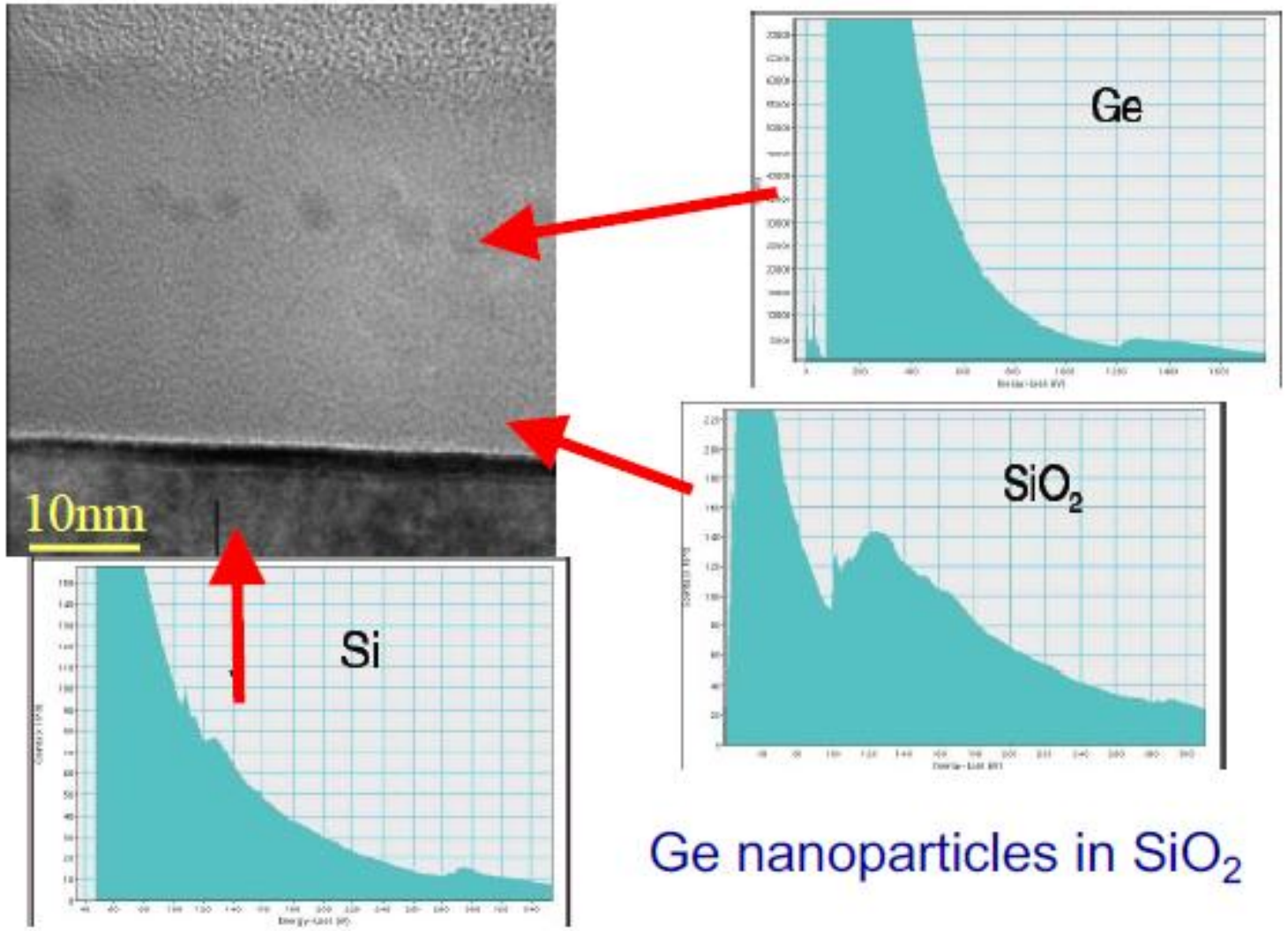


Electron Energy Loss Spectroscopy (EELS)

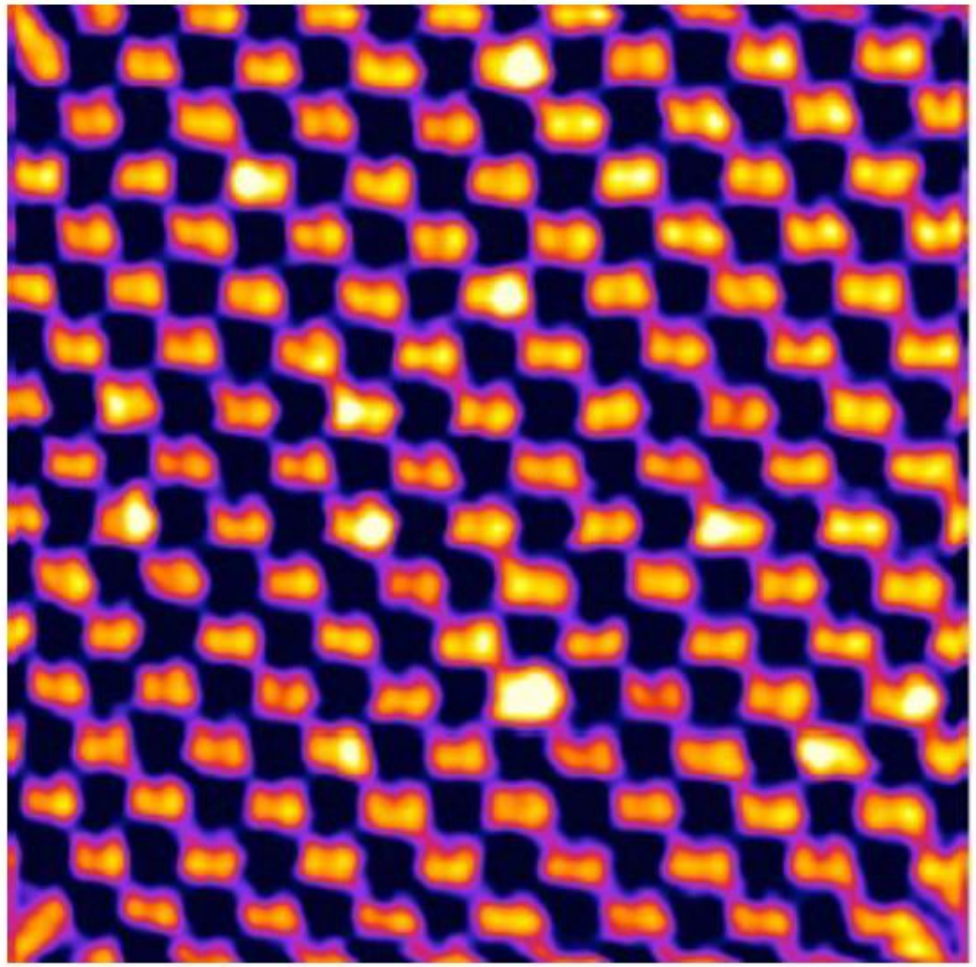




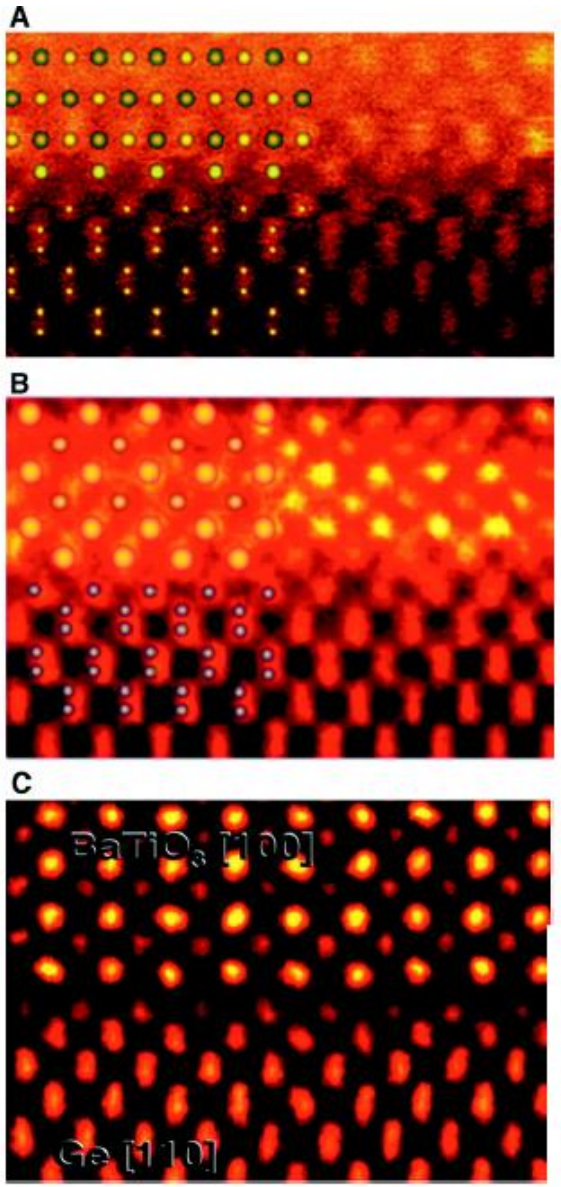
Phase identification using EELS

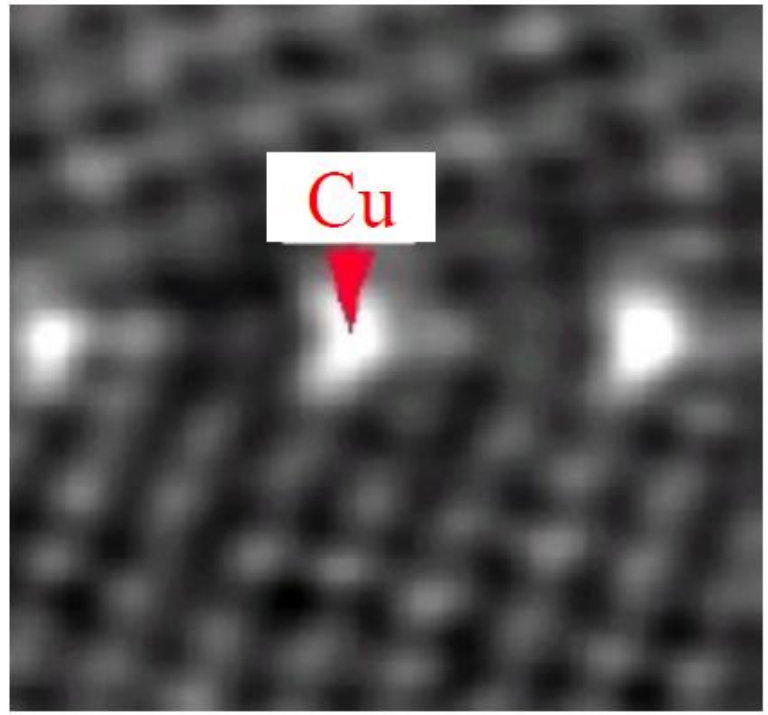


Images from STEM



Bi doped Si





Cu segregated in
Al grain boundary

Perovskite Oxides
grown epitaxially
on Si and Ge

Major applications of TEM

- Microstructural analysis
- Defects identification
- Interfacial analysis
- Crystal structure
- Seeing atomic arrangement of crystals
- Elemental analysis at nanometre scale

Advantages

- Superior resolution.
- Ability to obtain compositional and crystallographic information from specimens with high spatial resolution.

Limitations

- Very small volume of material studied – is it representative?
- Limitations on samples.
- Sample preparation can be quite lengthy and involved.
- Electrons can damage sample and introduce artifacts.

Advantages

- Offers the highest and most powerful magnification of any microscopy technique.
- Versatile imaging modes: dark/bright field and phase contrast (TEM); high-angle annular dark field (STEM).
- Provides the ability to collect electron diffraction patterns (crystallographic information) from nanometer-sized regions by using selected-area electron diffraction (SAED).
- Enables nano-analysis: the ability to collect local information about composition and bonding which can be correlated to high-resolution images.

Disadvantages

- Limited sampling: a typical field of view for an HR-TEM image is no more than 100 nm^2.
- Complex image interpretation: all TEM images (and diffraction patterns) are 2D projections of 3D structures.
- Possibility of electron beam damage, especially for light elements, biological samples, and soft materials. In many cases, this makes it challenging to collect meaningful data before the sample is amorphized.
  - Specialized techniques that utilize advanced equipment like cryo-TEM, low-dose TEM, low-kV TEM can be leveraged to analyze beam-sensitive materials.
- Vacuum environment required – limiting the ability to observe functional materials under realistic “working” conditions.
- Difficult sample preparation to create extremely thin specimens; thin samples often result in imaging artifacts. [6]

The difference between an SEM & TEM:

	Scanning Electron Microscopes (SEM)	Transmission Electron Microscopes (TEM)
Electron stream	Fine, focused beam	Broad beam
Image taken	Topographical/surface	Internal & lattice structures
Resolution	Lower resolution	Higher resolution
Magnification	Up to 2,000,000 times	Up to 50,000,000 times
Image dimension	3D	2D
Sample thickness	Thin and thick samples	Ultrathin samples only
Penetrates sample	No	Yes
Sample restriction	Less restrictive	More restrictive
Sample preparation	Shorter preparation time	Longer preparation time
Cost	Less expensive	More expensive
Speed	Faster	Slower
Operation	Easy to use	More complicated

[1]

Reference

1. Transmission Electron Microscope (TEM)- Definition, Principle, Images (microbenotes.com)
2. SEM vs TEM | Technology Networks
3. Transmission Electron Microscopy: Principle and Components | Soil Mineralogy (soilmanagementindia.com)
4. TEM sample support mesh "grid", with ultramicrotomy sections.@copyright... | Download Scientific Diagram (researchgate.net)
5. Dual-Axis Tomography Holder - Model 2040 - EDEN Instruments | In Situ NanoCharacterization Solutions (eden-instruments.com)
6. Transmission Electron Microscopy - Nanoscience Instruments
7. https://myscope.training/TEM_Electron_energy_loss_spectroscopy__EELS
8. carbon-formvar.jpg (143×115) (tedpella.com)

Other References:

- What if you just keep zooming in? (YouTube)

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CREATE CHANGE

Acknowledgement

- Centre for Microscopy and Microanalysis (CMM), AIBN (bld. 75).

Thank you

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CRICOS code 00025B

WEEK 5 · 知识点总结 · A/Prof. Shahriar Hossain

第 5 周　磁性起源、类型与纳米磁学基础

Magnetism: Origin, Types & Fundamentals of Nano-magnetism

主线：电子磁矩 + 交换作用 → 五种磁性 → 磁畴与磁滞回线 → 尺寸减小到纳米 → 超顺磁性

1. 磁性的起源

- 电子 = 小磁铁：**轨道磁矩**（电流环）+ **自旋磁矩**（更大）。成对电子 ↑ ↓ 抵消。
- 强磁性取决于相邻原子之间的**交换耦合**：平行 → 铁磁；反平行 → 反铁磁；无耦合 → 顺磁。**未成对电子多 ≠ 铁磁**（如 Mn）。

2. 基本物理量

- H（外加场，A/m，线圈 H = NI/l）；B（磁感应强度，T，B = μH，真空 $B_0 = \mu_0 H$）；M（磁化强度）；**χ = M/H**（磁化率）；**μ = B/H**（磁导率）；$\mu_0 = 4\pi \times 10^{-7}$ H/m；$B = \mu_0(H + M)$，$\mu_r = 1 + \chi$。

3. 五种磁性（★）

类型	自旋排列	χ	例子
抗磁	全成对；外场感生相反磁矩	负，约 -10^{-5}	Au, Cu, Zn, Bi, Hg
顺磁	永久磁矩随机，**相邻无相互作用**	正，约 10^{-5}–10^{-3}	Al, Pt, Mn, Ti
铁磁	平行，自发磁化	≫ 1	Fe, Co, Ni, Gd
反铁磁	反平行、**等量** → 抵消	小	MnO, NiO
亚铁磁	反平行、**不等量** → 净磁矩	≫ 1	**Fe_3O_4**, MFe_2O_4

- Fe_3O_4：A、B 位的 Fe^{3+} 反平行抵消，净磁矩来自 Fe^{2+} → **亚铁磁**（不是铁磁！）。

4. 磁畴与磁滞回线（★）

- 磁化：有利取向的磁畴通过**畴壁移动**长大 → 全部对齐 → **Ms**。
- H 降为 0 时残留 **Mr（剩磁）**；使 M = 0 所需反向场为 **Hc（矫顽力）**。
- **回线面积 = 每循环单位体积的热损耗**；功率 = 面积 × 体积 × 频率（W6 磁热疗的基础）。
- **软磁**：回线窄、Hc 低、损耗低（变压器铁芯）；**硬磁**：回线宽、Hc 高（永磁、存储）。

5. 温度效应

- Ms 在 0 K 最大，到**居里温度 T_C** 降为 0 → 顺磁（Fe 768 °C、Co 1120 °C、Ni 335 °C、Fe_3O_4 585 °C）；反铁磁对应**奈尔温度 T_N**。

6. 超顺磁性（★ 计算题）

- 颗粒小到**单畴**，热能使磁矩随机翻转 → **Ms 高、Hc = 0、Mr = 0**；是铁磁性的**尺寸效应**。
- 判据：能垒 $E_B = KV$ 与热能 $25k_BT$（$25 = \ln(\tau_m/\tau_0)$）比较：$KV \ll 25k_BT$ → 超顺磁；$KV \gtrsim 25k_BT$ → 被阻挡。
- 阻挡温度 $T_B = KV / 25k_B$；临界直径 $D = (150k_BT / \pi K)^{1/3}$。
- 例：Fe_3O_4，$K = 10^5$，300 K：8 nm 超顺磁，20 nm 被阻挡；SPION $K = 1.3 \times 10^4$，310 K → 临界约 **25 nm**，选 15 nm。

7. 影响因素与案例

- 影响 Ms、Mr、Hc、T_B：形状、尺寸、成分、合成方法、烧结温度、孔隙、密度、核壳结构。
- **案例 1**：$CoFe_2O_4$ 种子介导生长；**尺寸**由种子 / 前驱体比控制，**形状**由升温速率与温度控制（慢升至 210 °C → 立方；260 °C → 球形）；ZFC 峰 = T_B。
- **案例 2**：共沉淀 Fe_3O_4 退火：→ 250 °C γ-Fe_2O_3（超顺磁）→ 550–650 °C α-Fe_2O_3（硬磁）；Scherrer 晶粒 6.6 → 37.6 nm；XRD 难分磁铁矿 / 磁赤铁矿，需 FTIR；Ms 41.7 → 53.6 → 0.49 emu/g，Hc 4.2 → 1459 Oe。

考试重点：五种磁性对比（反铁磁 vs 亚铁磁）；画并解释磁滞回线；软 / 硬磁；KV vs $25k_BT$ 计算；T_C 与 T_B 的区别。

MATE6301
Nanomaterials

Week 5: Magnetism – Origin, Types and Fundamentals of Nano-magnetism

Assoc./Prof. M Shahriar Hossain
Email: md.hossain@uq.edu.au

Learning Objective

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After successfully completing this module you should be able to:

FABRICATE/CHARACTERISE MAGNETIC and PLASMONIC NANOCOMPOSITES

- LO4.

Design, fabricate, and characterise functional nanomaterials, including porous gold nanostructures and magnetic nanocomposites, and analyse their optical and magnetic properties for practical applications

- LO5.

Select appropriate magnetic and optical nanostructures to meet the requirements of designed applications.

L-6 - Outline

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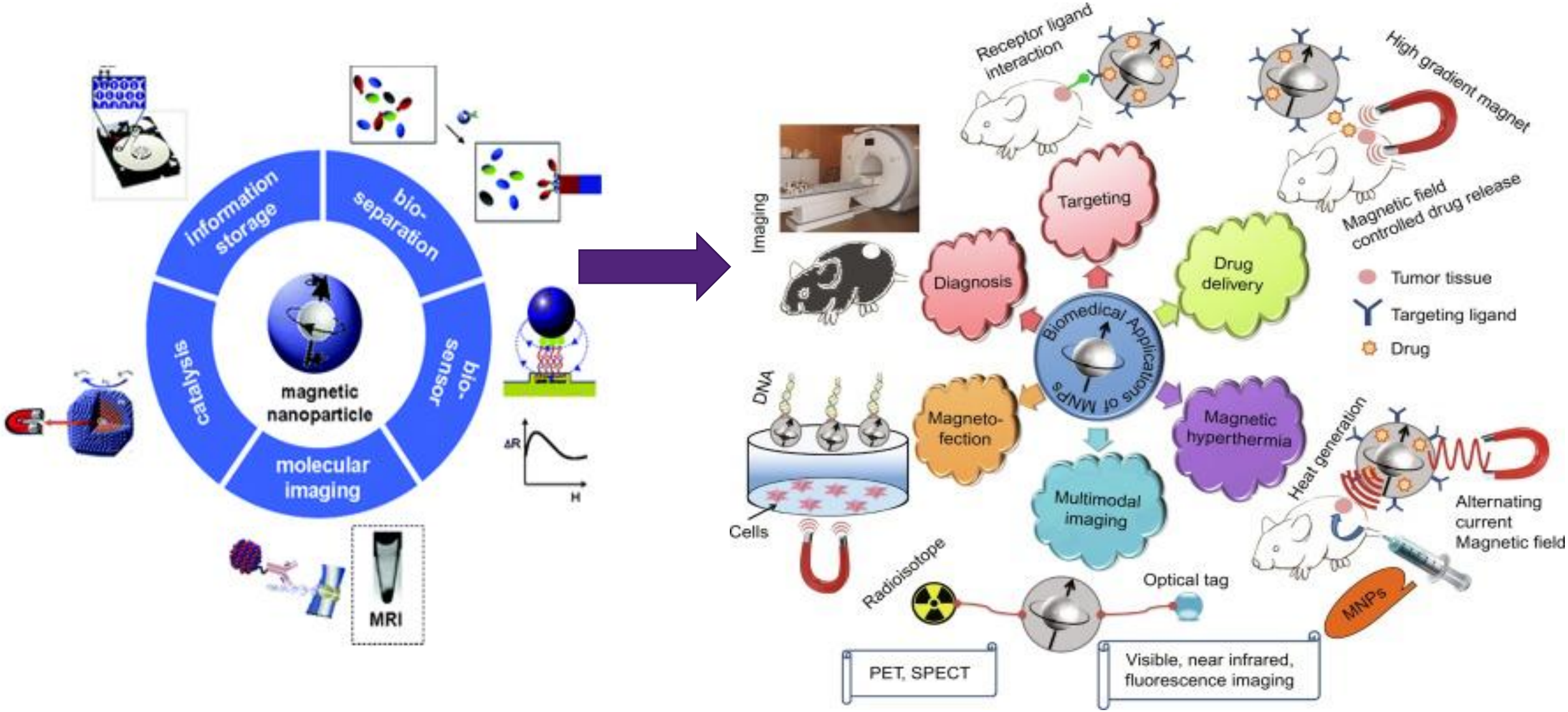
- Origin and fundamentals of magnetism
- Magnetic materials and their types
- Magnetic hysteresis
- Bulk (macro) magnetism (Ferromagnetism) and Nano magnetism (superparamagnetism)
- Synthesis of Magnetic Nanoparticles.
- Parameters influence magnetic properties of nanoparticles/nanocomposites
- Effect of annealing temperatures on magnetic phase transition and magnetic properties of nanoparticles
- Information about the Lab practical.

Why we need to learn Nanomagnetism



https://www.cd-bioparticles.com/t/Properties-and-Applications-of-Magnetic-Nanoparticles_55.html

Understanding the facts of Magnetism

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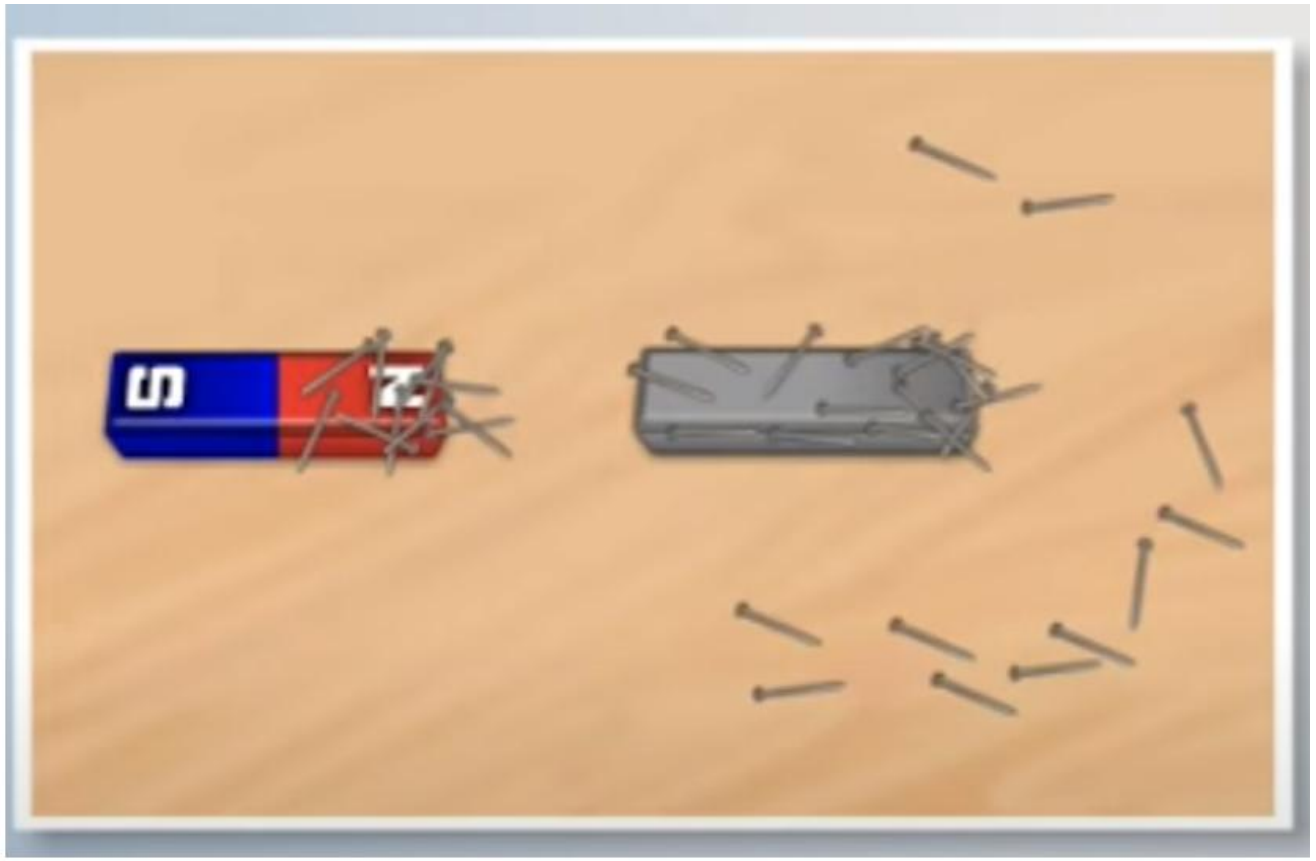
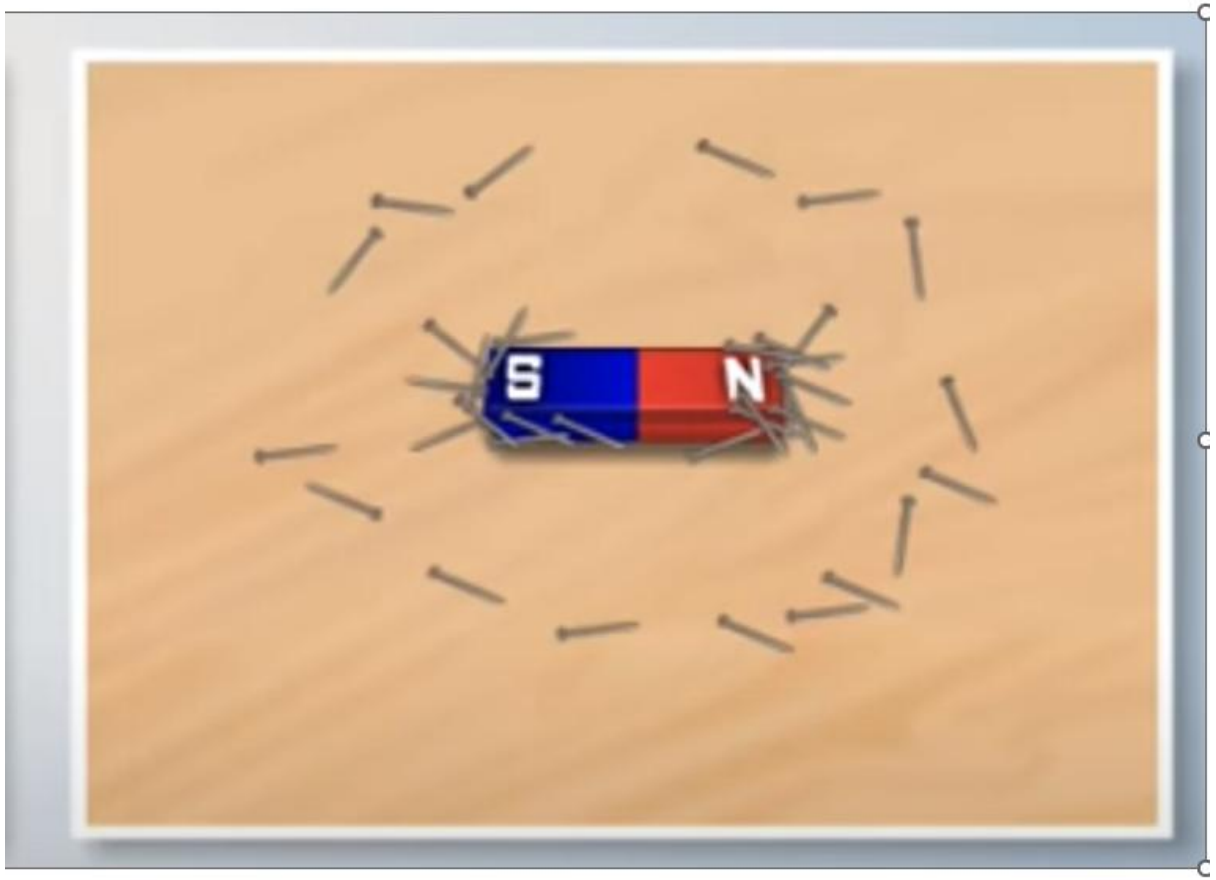
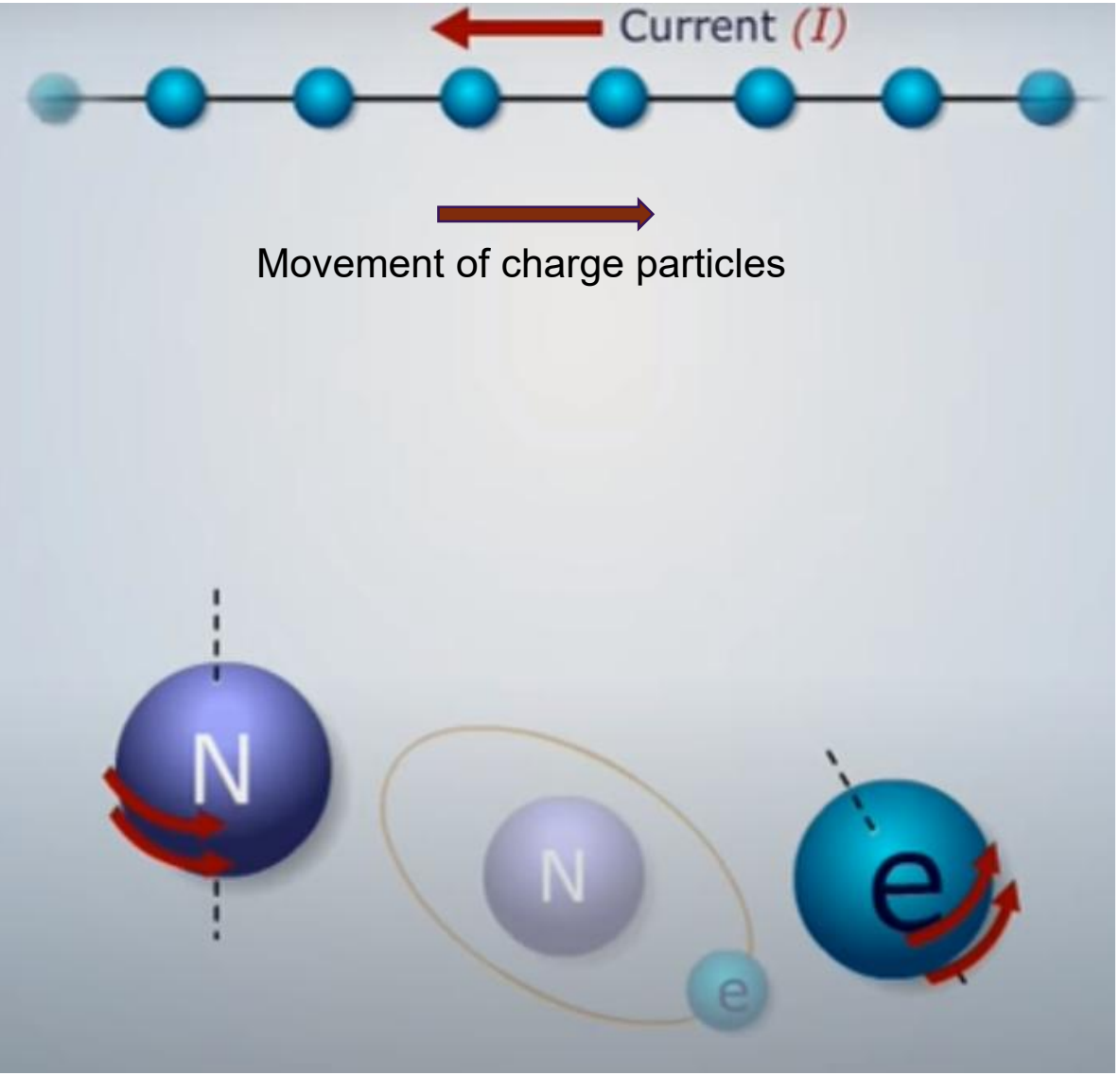
Why such behaviour?

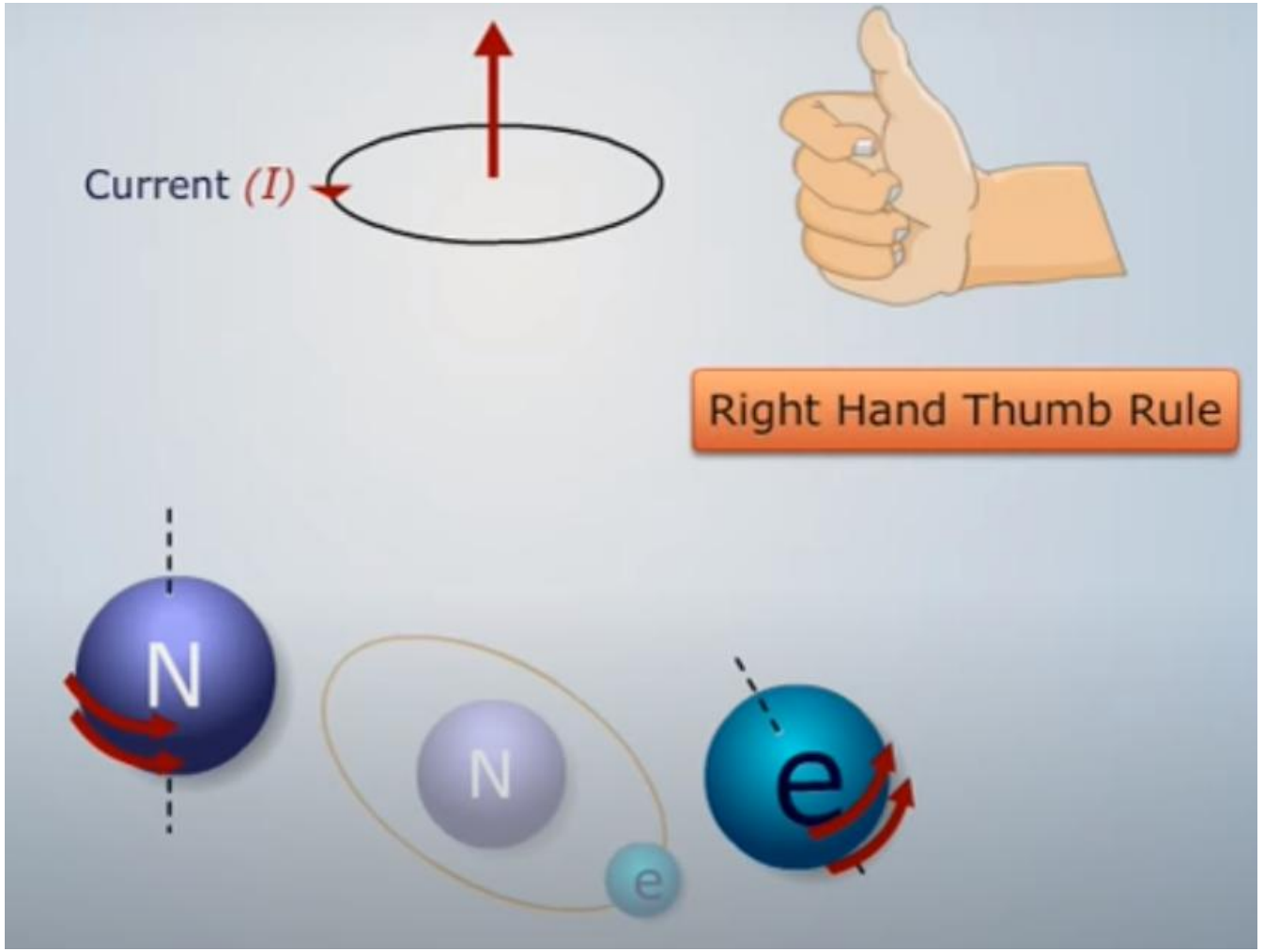




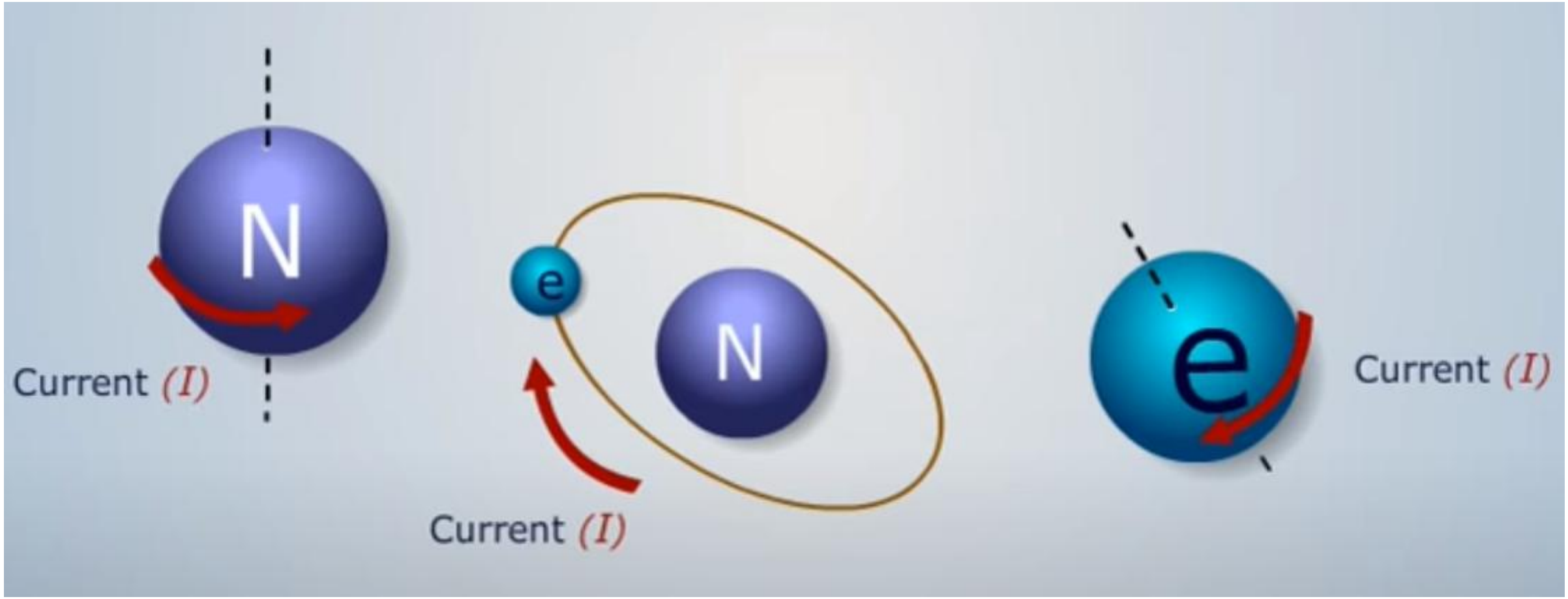
Understanding the facts of Magnetism

Why such behaviour?

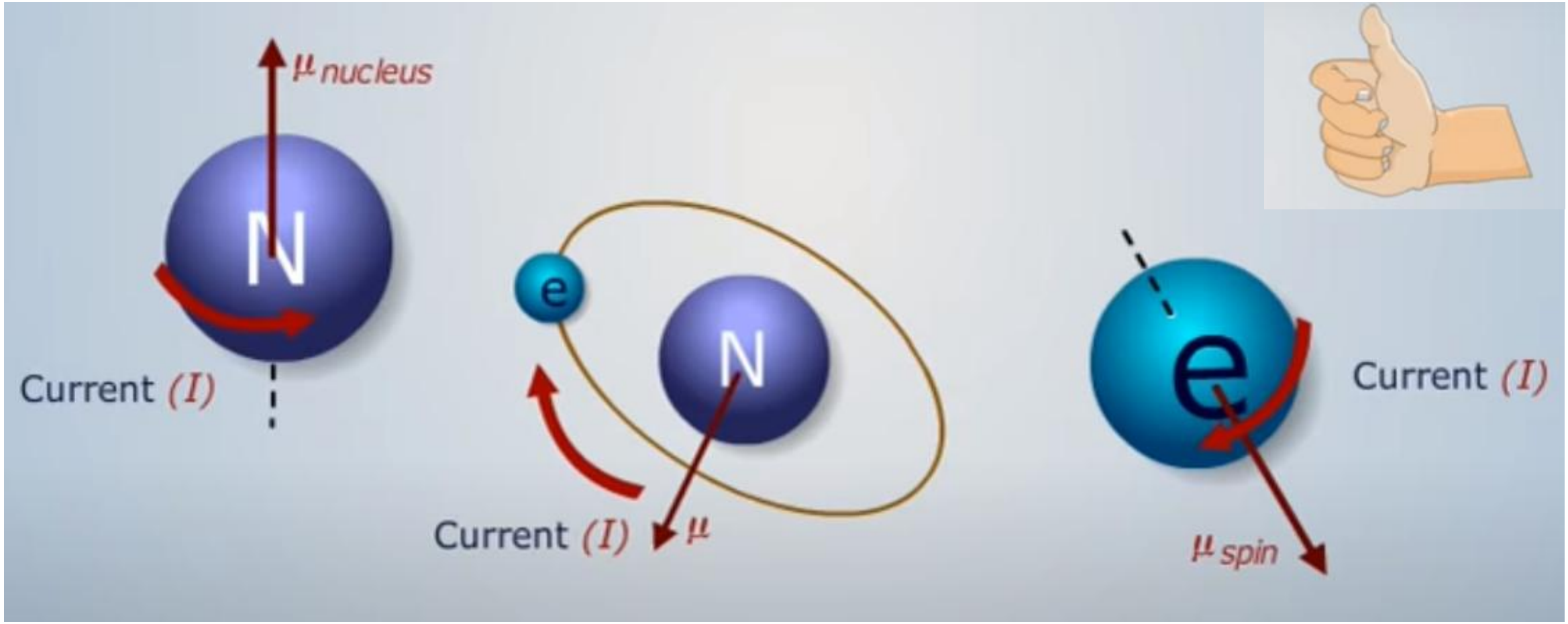




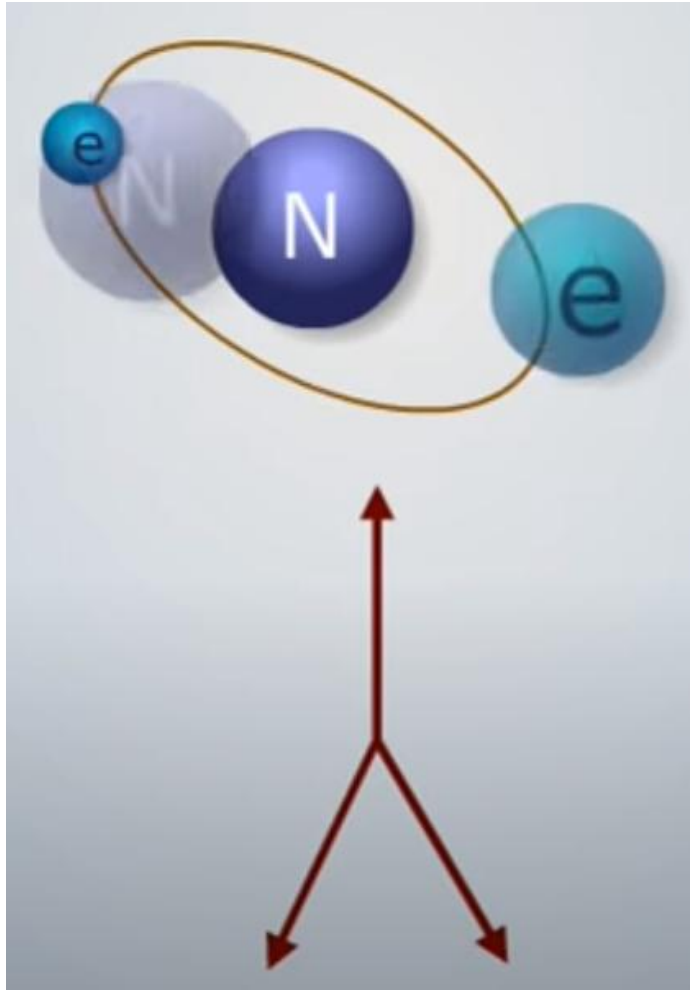
Why such behaviour?

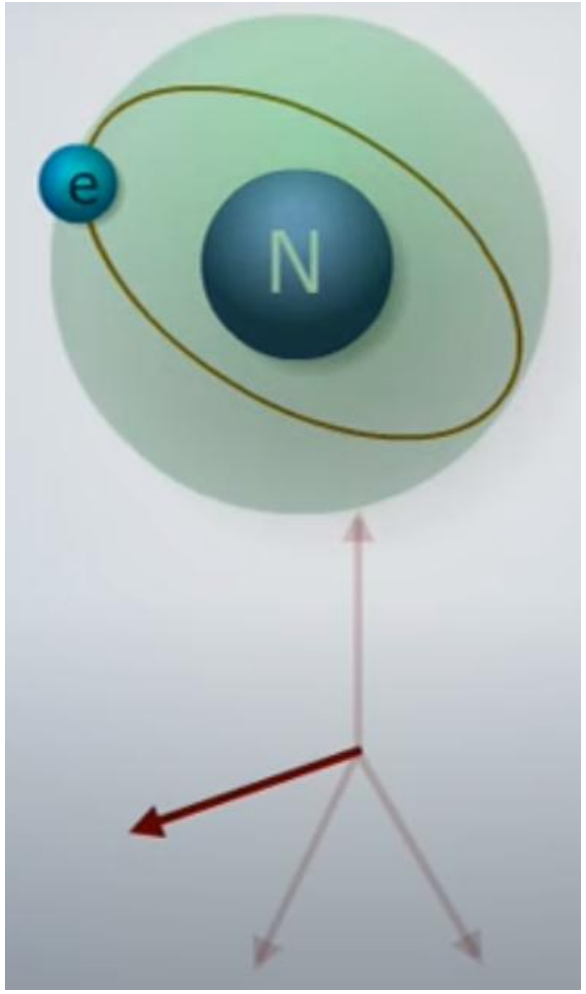


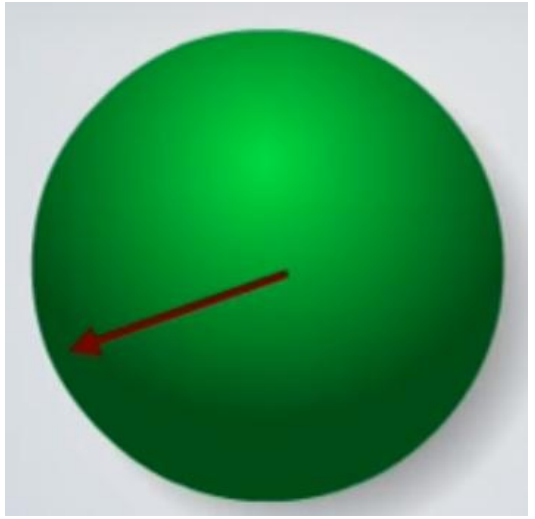
Why such behaviour?



Why such behaviour?







Understanding the facts of Magnetism

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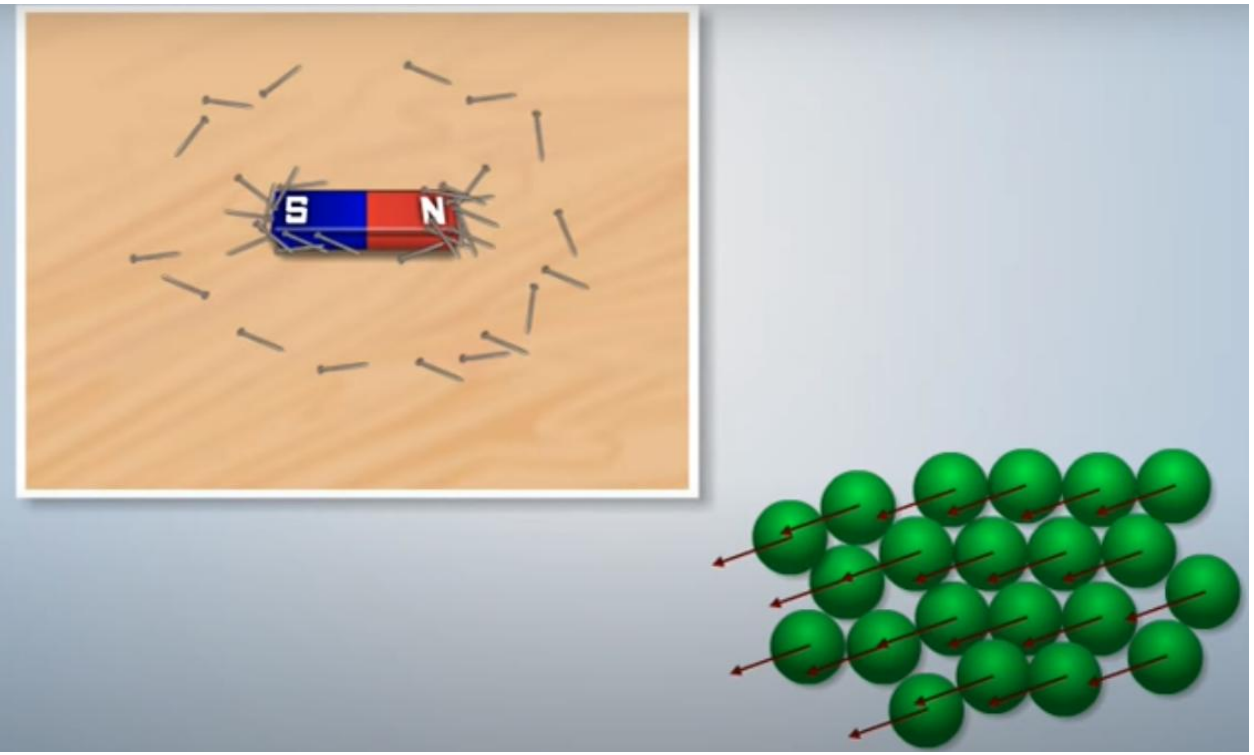
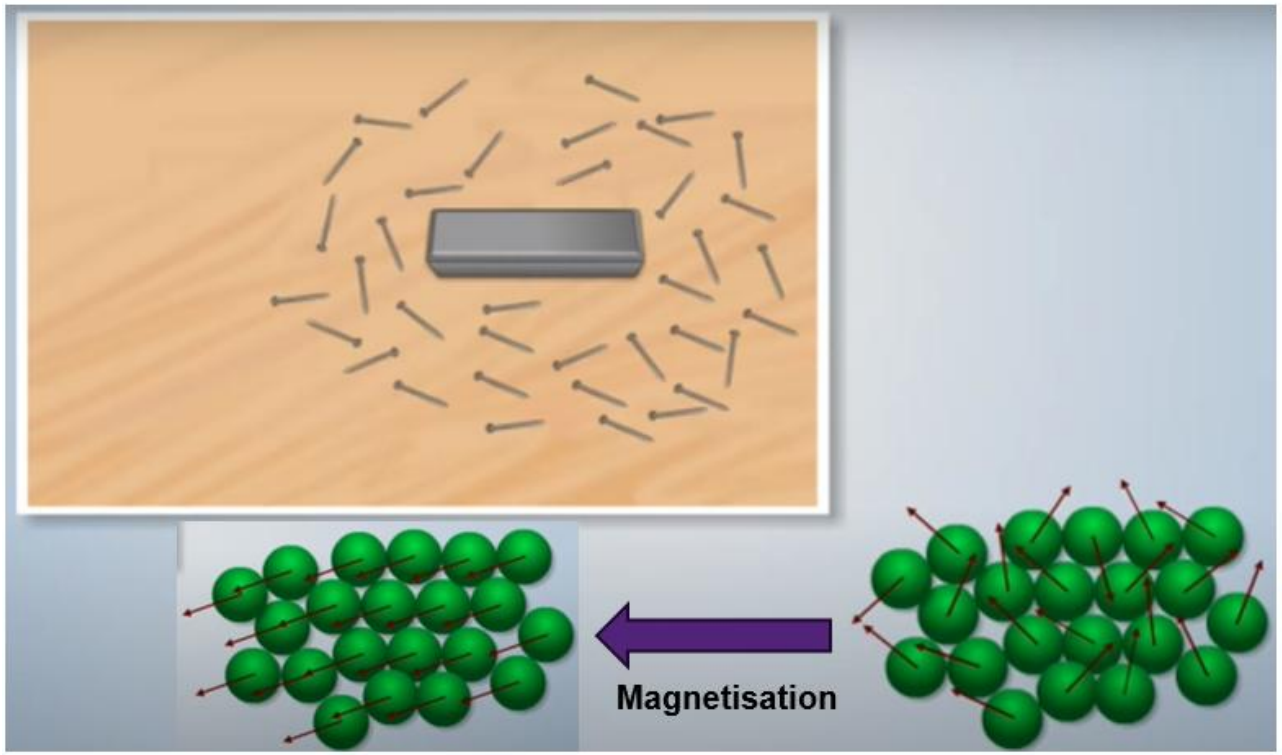
Why such behaviour?





Understanding the facts of Magnetism

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Why such behaviour?

Mathematically

Magnetic Moment due to Revolving Electrons

$$\mu = -\frac{ehl}{4\pi mc}$$

Magnetic Moment due to Spinning Electrons:

$$\mu_{spin} = \frac{eh}{4\pi mc}$$

Magnetic Moment due to Spinning Nucleus:

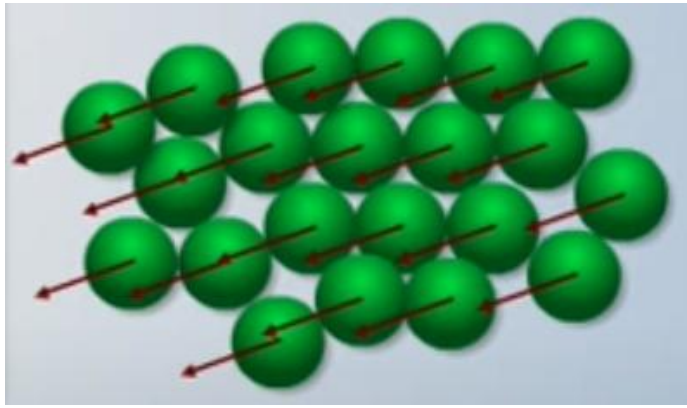
$$\mu_{nucleus} = \frac{eh}{4\pi m_p c}$$

Where,

e = Charge on an Electron
h = Planck's Constant
l = Orbital Quantum Number(= 0,1,2,....)
m = Mass of an Electron
c = Velocity of Light (= 3 X 10^8 m/s)
m_p = Mass of Proton

Keywords

- *Magnetic Moments*
- *Spinning and Revolving Electrons*
- *Spinning Nucleus*



ATOMIC ORIGIN OF MAGNETISM in any **Magnetic Material** can be explained by taking into consideration the *Magnetic Moments* of the *Spinning* and *Revolving Electrons* and also that of the *Spinning Nucleus.*

Basic Magnetism

We are all familiar with some basic properties of electron and atoms

Magnetic dipole of electrons → macroscopic behavior of matter

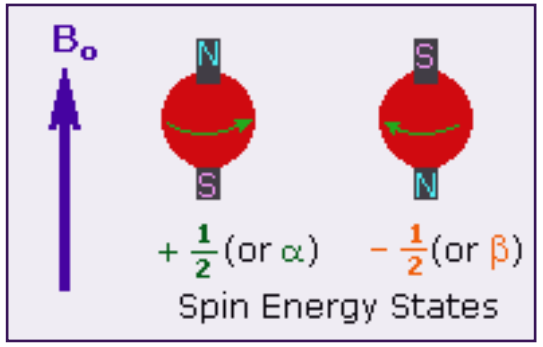
Atom → Electron —Spin→ Magnetic moment → Magnetism

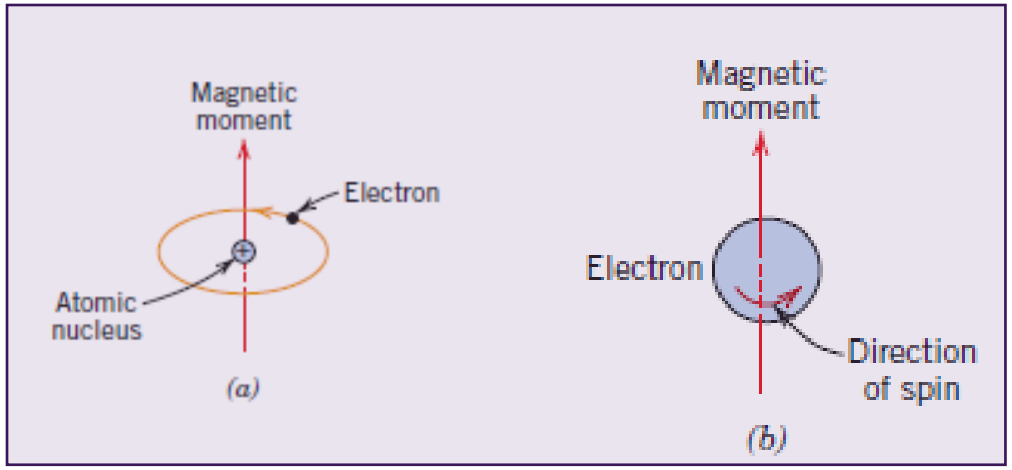
Each electron in an atom may be thought of as being a small magnet having permanent orbital and spin magnetic moments

Net atomic magnetic moment = some of moments from all electrons

Orbital motion around the nucleus; being a moving charge, an electron may be considered to be a small current loop, generating a very small magnetic field having a magnetic moment along its axis of rotation.

- Nucleus at the center, electrons rotating on orbits
- Electron is negatively charged
- Electron has an intrinsic angular momentum (spin)





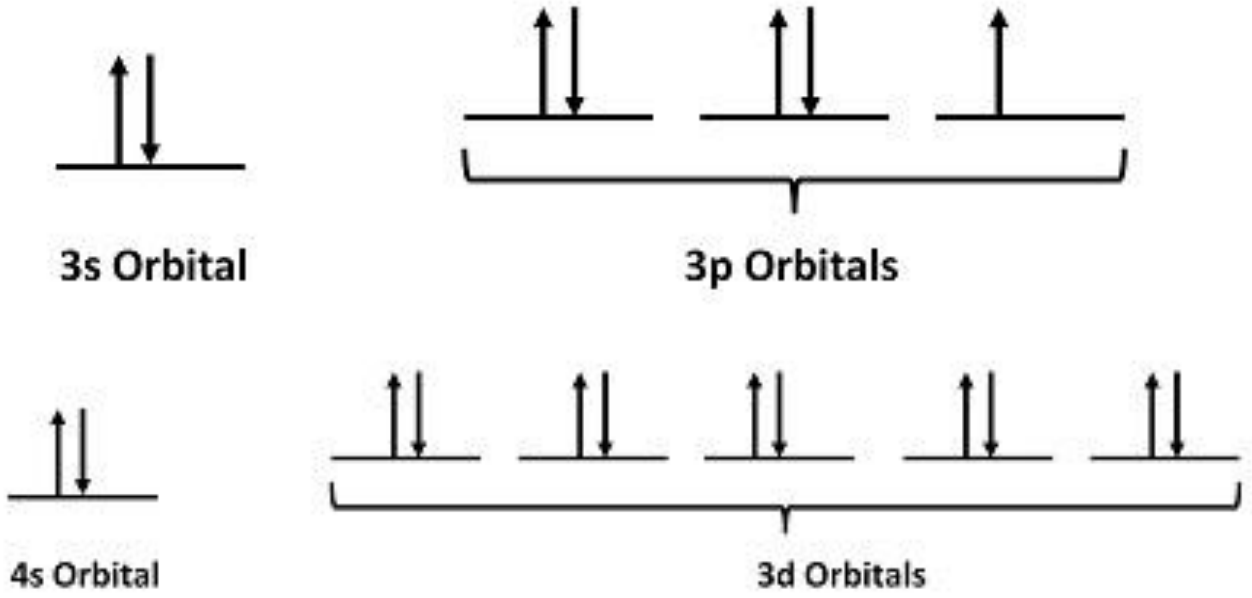
Reference: Callister 8th Ed.; G. Sciolla – MIT

A magnetic moment is a vector quantity, with a magnitude and a direction.

Origins of Magnetism: From Atomic Model

Every materials has orbital and spinning electron, in that case, all materials should be magnetic, right?

Of course : No!
Why?



The arrangement and interaction of these electrons differ significantly.

Paramagnetic

[↑] [↓] [↑] [↓] [↑] ← unpaired electrons, random directions
Random alignment → weak response to magnetic field

Ferromagnetic

[↑] [↑] [↑] [↑] ← unpaired electrons, spins aligned
Spontaneous alignment → strong permanent magnetization

- Number of unpaired electrons alone doesn't guarantee ferromagnetism.
- It's the interaction between spins (exchange coupling) that creates ferromagnetism.
- Net magnetization is important for the consideration of higher or lower magnetisation

Exchange energy interactions

- Exchange energy is the energy associated with the alignment of spins between neighboring atoms.
- It determines whether the material will exhibit strong magnetism or not.
- The exchange interactions between neighboring atoms are strong and lead to a parallel alignment of spins over large regions, resulting in strong magnetism.

↑ ↑ ↑ ↑ ↑

↑ ↑ ↑ ↑ ↑

All arrows aligned

Strong magnetism

↑ ↓ ↑ ↓ ↑ ↓

↑ ↓ ↑ ↓ ↑ ↓

Arrows cancel out

Weak or no magnetism

- Despite having more unpaired electrons, the exchange interactions are not not strong. The spins in atoms tend to align in an antiferromagnetic arrangement (neighboring spins point in opposite directions), which cancels out the overall magnetism.

Origins of Magnetism: From Atomic Model

Martin S. Silberberg, *Chemistry: The Molecular Nature of Matter and Change*, 2nd Edition. Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Table 8.4 Partial Orbital Diagrams and Electron Configurations*

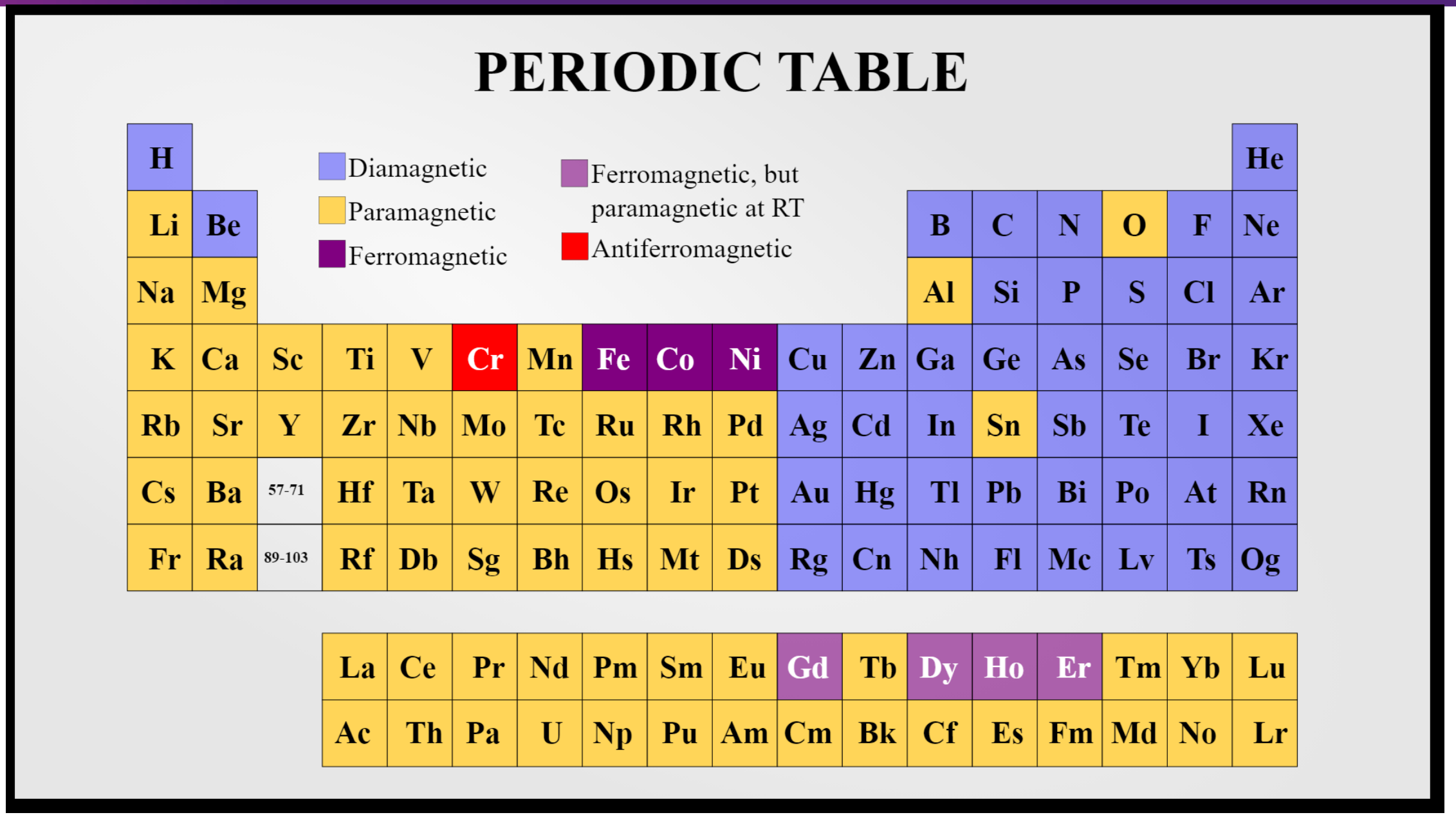
Atomic Number	Element	Partial Orbital Diagram (4s, 3d, and 4p Sublevels Only) 4s	3d	4p	Full Electron Configuration	Condensed Electron Configuration
19	K	↑↓			$[1s^2 2s^2 2p^6 3s^2 3p^6] 4s^1$	[Ar] $4s^1$
20	Ca	↑↓			$[1s^2 2s^2 2p^6 3s^2 3p^6] 4s^2$	[Ar] $4s^2$
21	Sc	↑↓	↑		$[1s^2 2s^2 2p^6 3s^2 3p^6] 4s^2 3d^1$	[Ar] $4s^2 3d^1$
22	Ti	↑↓	↑ ↑		$[1s^2 2s^2 2p^6 3s^2 3p^6] 4s^2 3d^2$	[Ar] $4s^2 3d^2$
23	V	↑↓	↑ ↑ ↑		$[1s^2 2s^2 2p^6 3s^2 3p^6] 4s^2 3d^3$	[Ar] $4s^2 3d^3$
24	Cr	↑	↑ ↑ ↑ ↑ ↑		$[1s^2 2s^2 2p^6 3s^2 3p^6] 4s^1 3d^5$	[Ar] $4s^1 3d^5$
25	Mn	↑↓	↑ ↑ ↑ ↑ ↑		$[1s^2 2s^2 2p^6 3s^2 3p^6] 4s^2 3d^5$	[Ar] $4s^2 3d^5$
26	Fe	↑↓	↑↓ ↑ ↑ ↑ ↑		$[1s^2 2s^2 2p^6 3s^2 3p^6] 4s^2 3d^6$	[Ar] $4s^2 3d^6$
27	Co	↑↓	↑↓ ↑↓ ↑ ↑ ↑		$[1s^2 2s^2 2p^6 3s^2 3p^6] 4s^2 3d^7$	[Ar] $4s^2 3d^7$
28	Ni	↑↓	↑↓ ↑↓ ↑↓ ↑ ↑		$[1s^2 2s^2 2p^6 3s^2 3p^6] 4s^2 3d^8$	[Ar] $4s^2 3d^8$
29	Cu	↑	↑↓ ↑↓ ↑↓ ↑↓ ↑↓		$[1s^2 2s^2 2p^6 3s^2 3p^6] 4s^1 3d^{10}$	[Ar] $4s^1 3d^{10}$
30	Zn	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓		$[1s^2 2s^2 2p^6 3s^2 3p^6] 4s^2 3d^{10}$	[Ar] $4s^2 3d^{10}$
31	Ga	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑	$[1s^2 2s^2 2p^6 3s^2 3p^6] 4s^2 3d^{10} 4p^1$	[Ar] $4s^2 3d^{10} 4p^1$
32	Ge	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑ ↑	$[1s^2 2s^2 2p^6 3s^2 3p^6] 4s^2 3d^{10} 4p^2$	[Ar] $4s^2 3d^{10} 4p^2$
33	As	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑ ↑ ↑	$[1s^2 2s^2 2p^6 3s^2 3p^6] 4s^2 3d^{10} 4p^3$	[Ar] $4s^2 3d^{10} 4p^3$
34	Se	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑↓ ↑ ↑	$[1s^2 2s^2 2p^6 3s^2 3p^6] 4s^2 3d^{10} 4p^4$	[Ar] $4s^2 3d^{10} 4p^4$
35	Br	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑↓ ↑↓ ↑	$[1s^2 2s^2 2p^6 3s^2 3p^6] 4s^2 3d^{10} 4p^5$	[Ar] $4s^2 3d^{10} 4p^5$
36	Kr	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑↓ ↑↓ ↑↓	$[1s^2 2s^2 2p^6 3s^2 3p^6] 4s^2 3d^{10} 4p^6$	[Ar] $4s^2 3d^{10} 4p^6$

*Colored type indicates sublevel(s) whose occupancy changes when last electron is added.

In general, materials that have all paired electrons in the atoms and thus have no net magnetic moment are called **diamagnetic materials.** Diamagnetic materials include zinc, gold, mercury, and bismuth.

Substances with unpaired electrons are called **"paramagnetic"** as they are attracted towards a magnetic field. Most materials with one or more unpaired electrons are at least slightly magnetic. Materials with a small attraction to a magnet are called **paramagnetic materials**, and those with a strong attraction are called **ferromagnetic materials**. Aluminum, platinum, and manganese are some paramagnetic materials. Iron, cobalt, and nickel are examples of ferromagnetic materials.

Types of Magnetism



http://periodic-table-1.blogspot.com/2014/12/periodic-table-magnetic-metals.html

Basic Terminology

The **externally applied magnetic field**, sometimes called the magnetic field strength**,** is designated by *H*. If the magnetic field is generated by means of a cylindrical coil (or solenoid) consisting of *N* closely spaced turns, having a length *l*, and carrying a current of magnitude *I*,:

$$H = \frac{NI}{l}$$

The units of *H* are ampere-turns per meter, or just amperes per meter. H is produced by an electric charge in motion by i) current flowing in a conductor ii) orbital movement and spin of electron

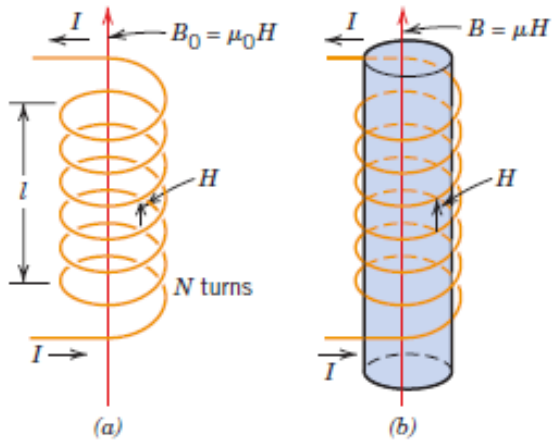
The **magnetic induction, or magnetic flux density,** denoted by *B*, represents the magnitude of the internal field strength within a substance that is subjected to an *H* field. The units for *B* are teslas [or webers per square meter (Wb/m^2)].

Both *B* and *H* are field vectors, being characterized not only by magnitude, but also by direction in space.

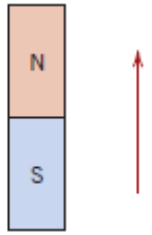
Magnetisation (M): process of being magnetized (iron gets magnetized under external magnetic field)

Magnetic susceptibility is a dimensionless proportionality constant that indicates the degree of magnetization of a material in response to an applied **magnetic** field. ... A closely related parameter is the **permeability**, which expresses the total magnetization of material and volume. (ratio of M/H)

Permeability **(ability to create internal magnetic field)** is the measure of the ability of a material to support the formation of a **magnetic** field within itself. Iron gets magnetized (B) if it is kept in external magnetic field (H). (Ratio of B/H)



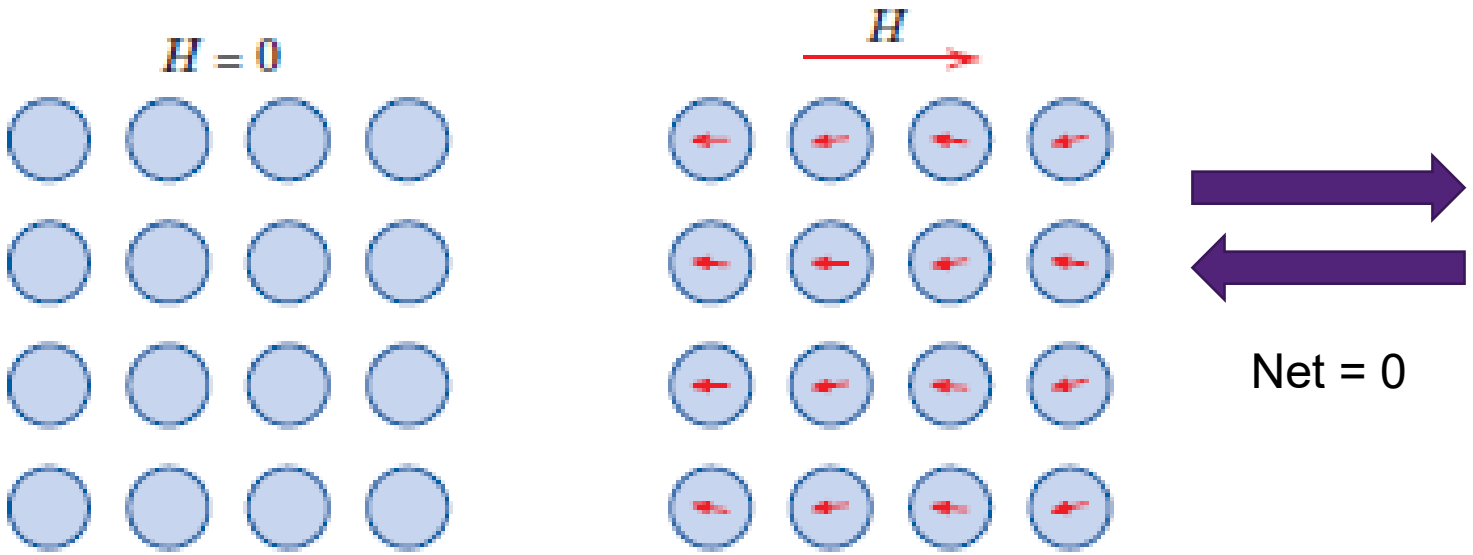
(a) The magnetic field *H* as generated by a cylindrical coil is dependent on the current *I*, the number of turns *N*, and the coil length *l*. The magnetic flux density B_0 in the presence of a vacuum is equal to $\mu_0 H$, where μ_0 is the permeability of a vacuum, $4\pi \times 10^{-7}$ H/m (*b*) The magnetic flux density *B* within a solid material is equal to μH, where μ is the permeability of the solid material.



The magnetic moment as designated by an arrow

Reference: Callister 8th Ed.;

Diamagnetism

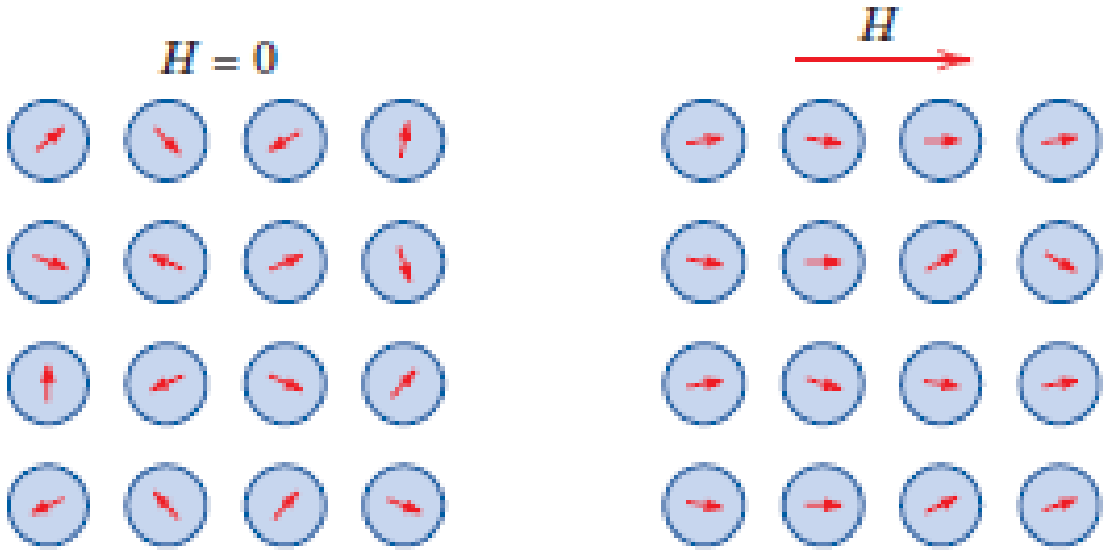


- Completely filled shells or subshells.
- Total cancellation of orbital and spin moments
- a very weak form of magnetism that is nonpermanent and persists only while an external field is being applied.
- It is induced by a change in the orbital motion of electrons due to an applied magnetic field because external magnetic field acting on the atoms slightly unbalances their orbiting electrons and creates small magnetic dipoles within the atoms, which oppose the applied field, and this action produces a negative magnetic effect to the applied field.

- The magnitude of the induced magnetic moment is extremely small, and in a direction opposite to that of the applied field.
- The relative permeability *r is less than unity (however, only very* slightly), and the magnetic susceptibility is negative; that is, the magnitude of the *B* field within a diamagnetic solid is less than that in a vacuum.

A weak, negative, repulsive reaction of a material to an applied magnetic field.

Paramagnetism



Both diamagnetic and paramagnetic materials are considered nonmagnetic because they exhibit magnetization only when in the presence of an external field.

- For some solid materials, each atom possesses a permanent dipole moment by virtue of incomplete cancellation of electron spin and/or orbital magnetic moments
- Permanent Dipoles. Randomly oriented when no field is present. Permanent dipoles align with an external field. No interaction between adjacent dipoles.
- In the absence of an external magnetic field, the orientations of these atomic magnetic moments are random
- as the dipoles align with the external field, they enhance it, giving rise to a relative permeability *r that is greater than* unity, and to a relatively small but positive magnetic susceptibility.

A weak, positive, attractive reaction of a material to an applied field

Diamagnetic and Paramagnetic materials

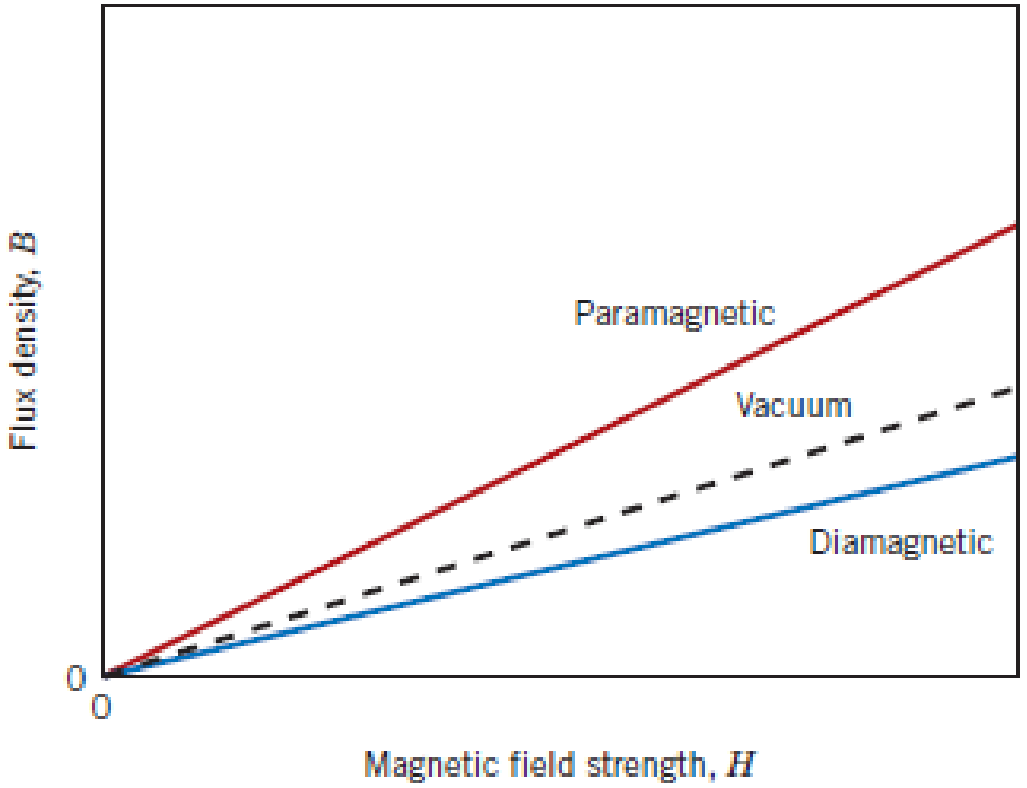
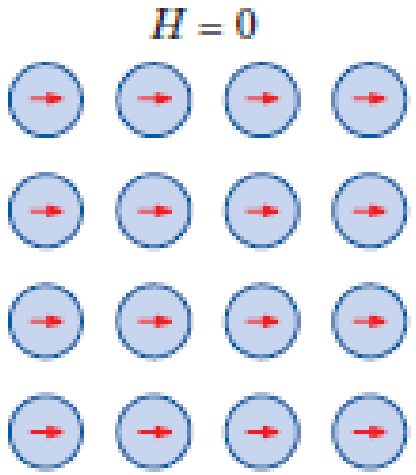


Table 20.2 Room-Temperature Magnetic Susceptibilities for Diamagnetic and Paramagnetic Materials

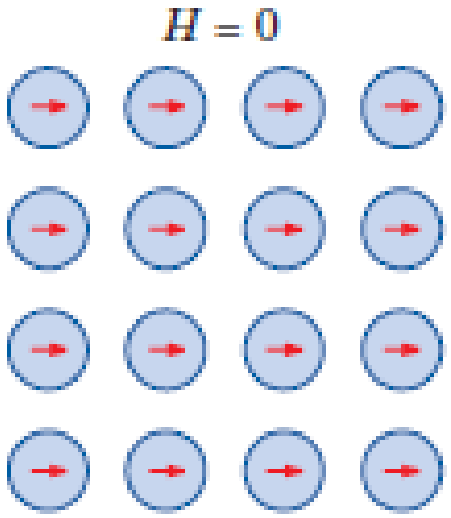
Diamagnetics		*Paramagnetics*	
Material	*Susceptibility χ_m (volume) (SI units)*	*Material*	*Susceptibility χ_m (volume) (SI units)*
Aluminum oxide	-1.81×10^{-5}	Aluminum	2.07×10^{-5}
Copper	-0.96×10^{-5}	Chromium	3.13×10^{-4}
Gold	-3.44×10^{-5}	Chromium chloride	1.51×10^{-3}
Mercury	-2.85×10^{-5}	Manganese sulfate	3.70×10^{-3}
Silicon	-0.41×10^{-5}	Molybdenum	1.19×10^{-4}
Silver	-2.38×10^{-5}	Sodium	8.48×10^{-6}
Sodium chloride	-1.41×10^{-5}	Titanium	1.81×10^{-4}
Zinc	-1.56×10^{-5}	Zirconium	1.09×10^{-4}

Ferromagnetism



- Certain metallic materials possess a permanent magnetic moment in the absence of an external field, and manifest very large and permanent magnetizations
- Permanent magnetic moments in ferromagnetic materials result from atomic magnetic moments due to *uncancelled electron* spins as a consequence of the electron structure
- There is also an *orbital magnetic moment* contribution that is small in comparison to the spin moment.
- In a ferromagnetic material, *coupling interactions of e- spin* cause net spin magnetic moments of adjacent atoms to align with one another, even in the absence of an external field.
- These are the characteristics of ferromagnetism, and they are displayed by the transition metals iron (as BCC -ferrite), cobalt, nickel, and some of the rare earth metals such as gadolinium (Gd).

Ferromagnetism



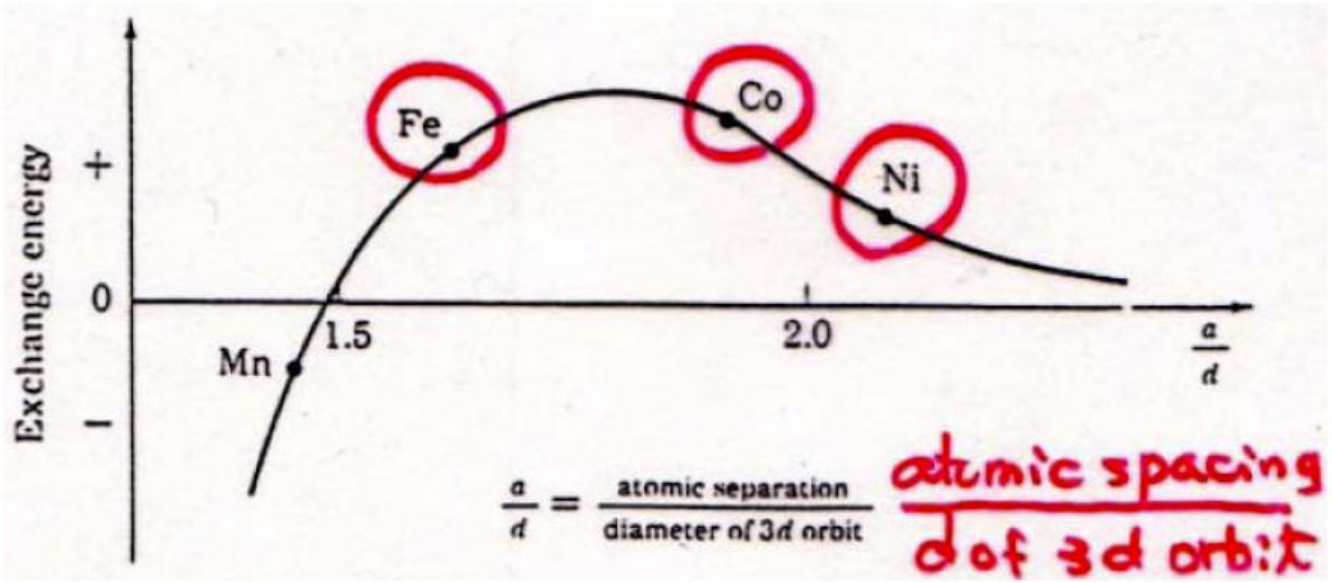
Fe, Co and Ni →ferromagnetic materials

Cr and Mn→not ferromagnetic materials, Why?

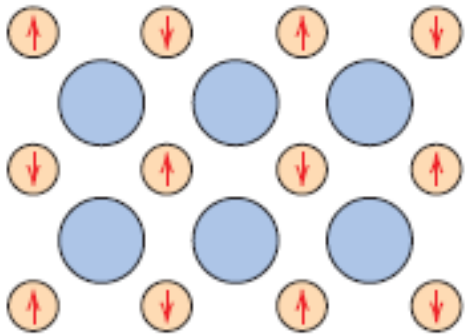
(they all have unpaired 3d electrons)

Magnetic exchange interaction energy → Energy associated with the coupling of individual magnetic dipoles into a single magnetic domain. Only when this exchange energy is **positive** → material be ferromagnetic.

Magnetic exchange energy is related to the *ratio of atomic spacing to 3d orbit*, must be in the range of 1.4 to 2.7.



Antiferromagnetism



- Magnetic moment coupling between adjacent atoms or ions occurs in materials which are not ferromagnetic
- If the coupling of electron spins results in antiparallel alignment then spins will cancel each other and no net magnetic moment will arise. This is known as antiferromagnetism
- Manganese oxide is **a ceramic oxide material** that **is ionic in character**, having both Mn2+ and O2- ions. No net magnetic moment is associated with the O2- ions (O2 ion is not magnetic), because there is a total cancellation of both spin and orbital moments.
- However, the Mn2+ ions possess a net magnetic moment that is predominantly of spin origin. These Mn2+ ions are arrayed in the crystal structure such that the moments of adjacent ions are antiparallel
- Obviously, the opposing magnetic moments cancel one another, and, as a consequence, the solid as a whole possesses no net magnetic moment.

Ferrimagnetism

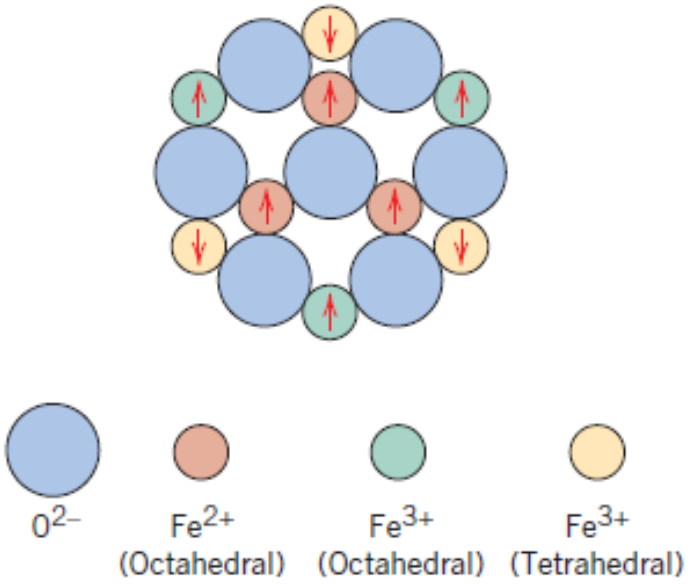


Table 20.3 The Distribution of Spin Magnetic Moments for Fe^{2+} and Fe^{3+} Ions in a Unit Cell Fe_3O_4[a]

Cation	*Octahedral Lattice Site*	*Tetrahedral Lattice Site*	*Net Magnetic Moment*
Fe^{3+}	↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑	↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓	Complete cancellation
Fe^{2+}	↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑	—	↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑

- Some ceramics also exhibit a permanent magnetization, termed **ferrimagnetism**
- The macroscopic magnetic characteristics of ferromagnets and ferrimagnets are similar; the distinction lies in the source of the net magnetic moments
- Certain ionic solids having a general formula MFe2O4 where M is any metal show permanent magnetism , termed ferrimagnetism due to partial cancellation of spin moments.
- The antiparallel coupling between Fe3+ (half in A site and half in B site) moments cancels each others but Fe2+ moments are alighned in the same direction and result in net magnetic moment.

Antiferromagnetism vs Ferrimagnetism

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Antiferromagnets:

Near-neighbor spins are antialigned. No hysteresis.
Examples: MF_2, MO (M = Mn, Fe, Co, Ni, Cu), IrMn …

Ferrimagnets:

Near-neighbor spins are antialigned, and are also not compensated (the spins don't cancel). Hysteresis and magnetization like in a ferromagnet.
Examples: Fe_3O_4, $CoFe_2O_4$, $Y_3Fe_5O_{12}$, $BaFe_6O_{19}$ …

Magnetic Hysteresis

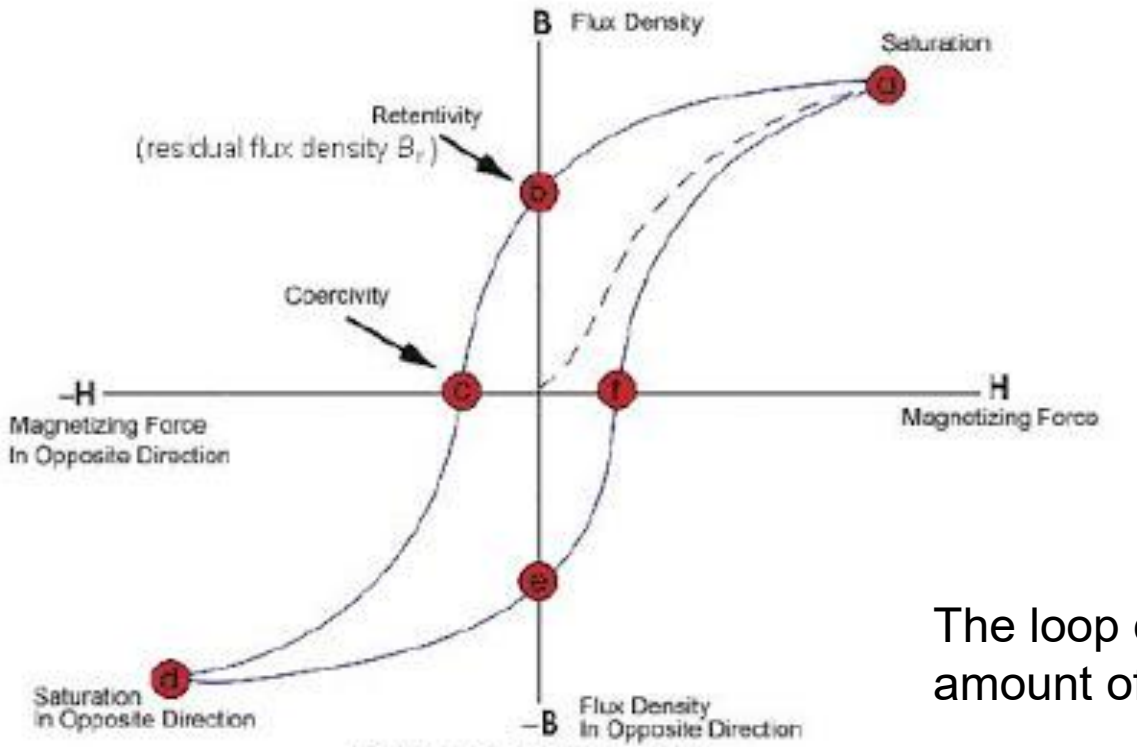


Figure 4.6: Hyteresis loop.

A magnetic hysteresis (hysteresis loop):

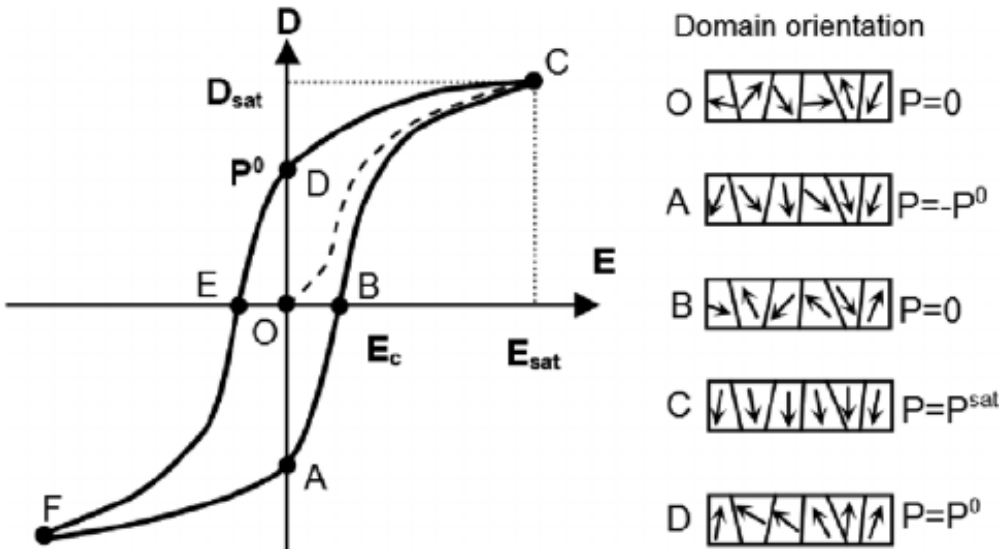
- magnetizing force (H) versus the magnetic flux density (B) of a ferromagnetic material.
- The curvature of the hysteresis is characteristic of the type of material being observed
- Can vary in size and shape (i.e. narrow or wide).

The loop can be generated by using a Hall Effect sensor to measure the amount of magnetic field at various points –

- when in the presence of a magnetic field,
- when it is removed from the magnetic field, and
- when a force is applied to bring the magnetic flux back to zero.

These loops are important in the memory capacity of devices for magnetic storage of data on computer disks.

Magnetic Hysteresis



What will happen if you reduce the magnetic field from saturation magnetization?

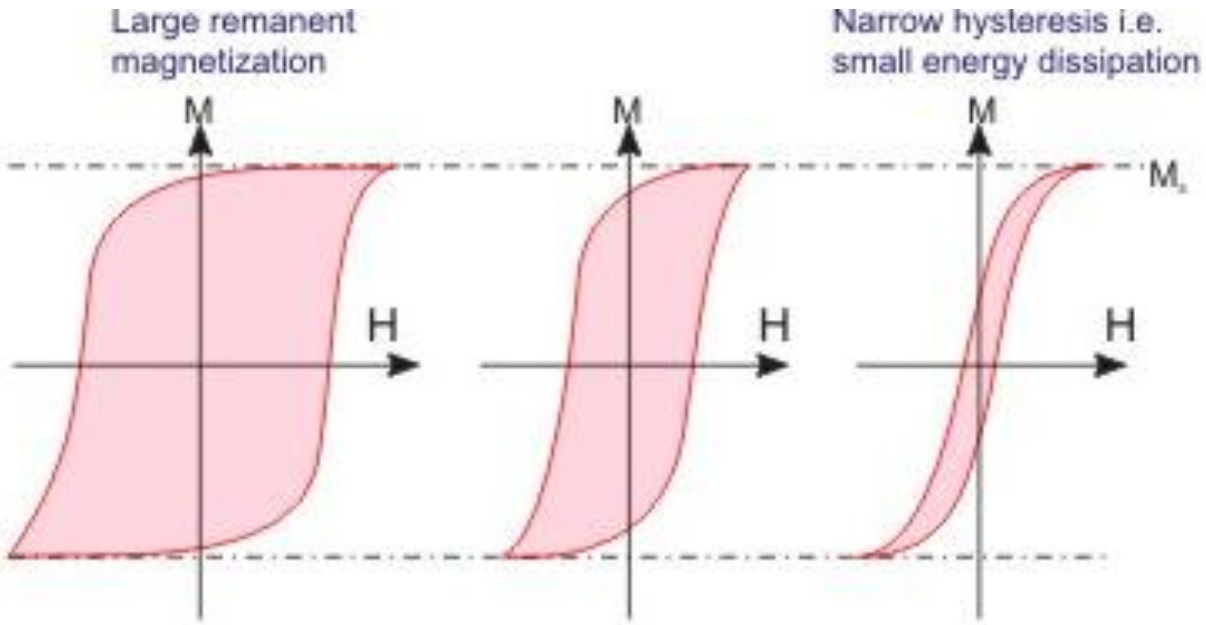
- If the field is reduced from the saturation by magnetic reversal a hysteresis look develops.
- As the field is reversed, the favorably oriented domains tend to align in the new direction.
- When H reaches zero at point P some of the domains still remain aligned in the previous direction giving rise to a residual magnetisation called Remanence,
- The reverse field strength (-Hc) at which magnetization is zero (E), is called coercivity.

The area within a loop represents a magnetic energy loss per unit volume of material per magnetization-demagnetization cycle in the form of heat

https://www.doitpoms.ac.uk/tlplib/ferromagnetic/hysteresis.php

Significance of Hysteresis loop

$$\text{Power Dissipated} = \text{Area of } B-H \text{ Curve} \times \text{Volume of Material} \times \text{Frequency}$$



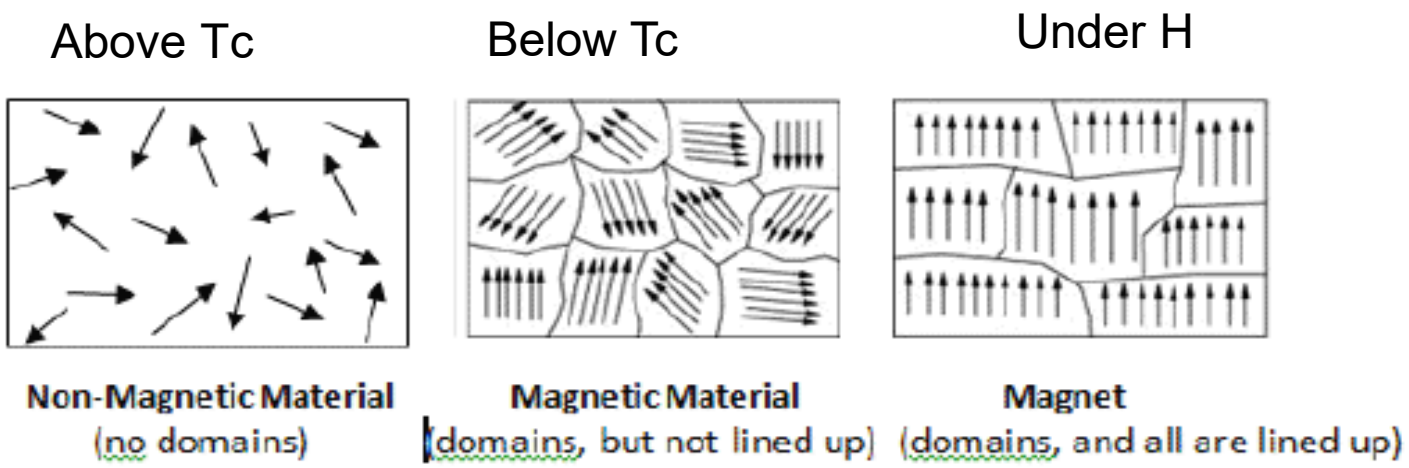
In soft magnets:

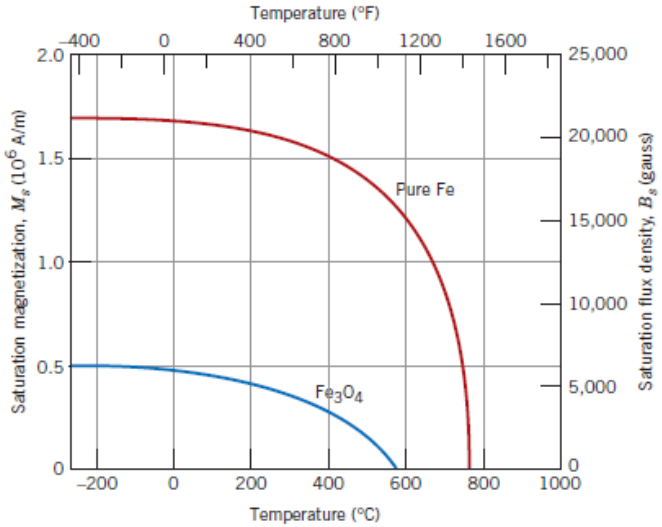
- Narrow hysteresis->>> low loss
- High permeability and low coercivity due to easy movement in domain wall
- Easy to magnetize and demagnetize

Another feature which is used to explain hard and soft magnets is the area of B-H or M-H curve which represents the power dissipated as heat and is expressed as So, since a hysteresis curve implies loss of a certain amount of energy, typically soft magnets show lower energy losses as compared to hard magnets. Soft magnets with reasonably large M_r and smaller H_c are useful.

https://nptel.ac.in/courses/113104005/lecture30a/30_5.htm

Temperature Effect on magnetic properties





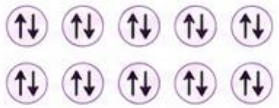
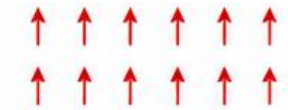
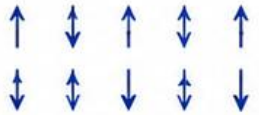
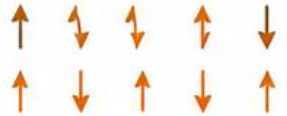
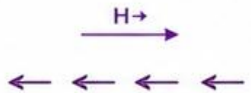
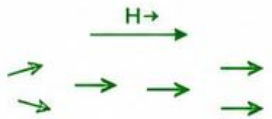
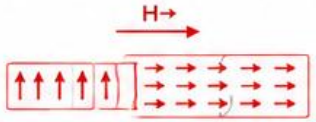
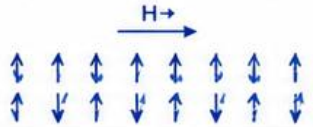
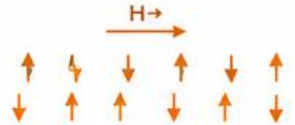
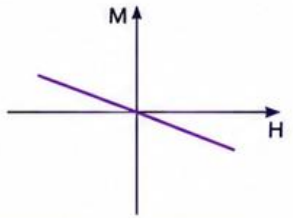
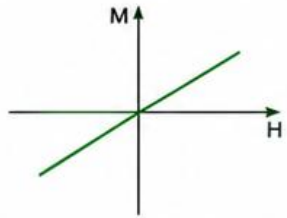
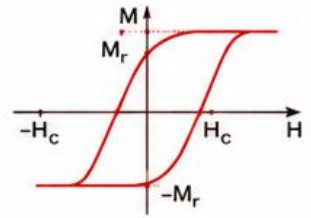
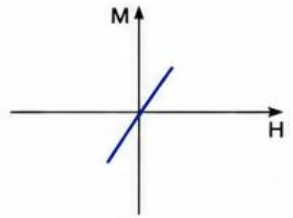
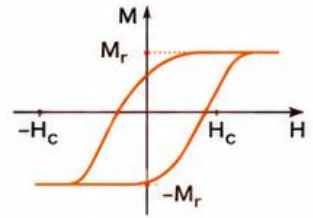
- Raising the temperature of a solid results in an increase in the magnitude of the *thermal vibrations* of atoms. The atomic magnetic moments are free to rotate; hence, with rising temperature, the increased thermal motion of the atoms tends to randomize the directions of any moments that may be aligned

- The saturation magnetization is a maximum at 0 K, at which temperature the thermal vibrations are a minimum. With increasing temperature, the saturation magnetization diminishes gradually and then abruptly drops to zero at what is called the ***Curie temperature, Tc*****.** The magnitude of the Curie temperature varies from material to material; for example, for iron, cobalt, nickel, and Fe3O4, the respective values are 768, 1120, 335, and 585C.

- Antiferromagnetism is also affected by temperature; this behavior vanishes at what is called the ***Néel temperature***. *At temperatures above this point, antiferromagnetic* materials also become paramagnetic

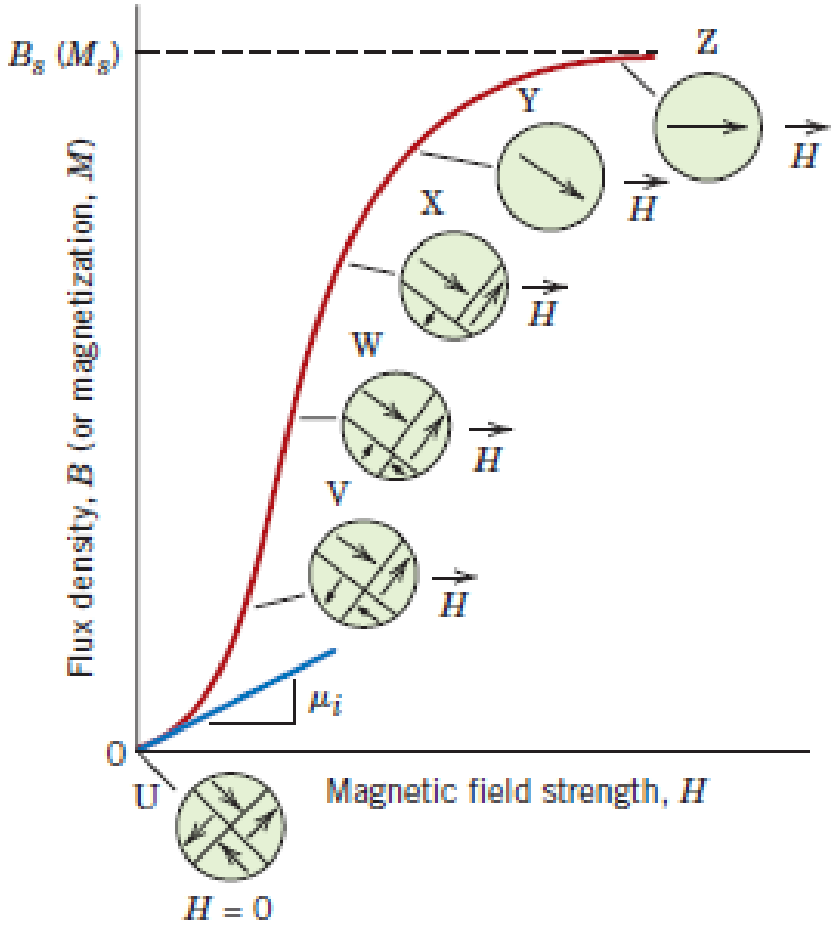
Five Basic Types of Magnetism

The University Of Queensland
AUSTRALIA

	DIAMAGNETISM	PARAMAGNETISM	FERROMAGNETISM	ANTIFERROMAGNETISM	FERRIMAGNETISM
Atomic / Spin Arrangement (H = 0)	No permanent magnetic moment (all electrons paired) 	Permanent moments randomly oriented	Spins align parallel (spontaneous magnetization) 	Neighboring spins align antiparallel and cancel 	Neighboring spins antiparallel but unequal → net magnetization 
Response to Applied Field (H →)	Induced moment opposite to applied field 	Moments tend to align with the field 	Domains grow and align with the field 	Slight canting of spins; small net moment 	Unequal moments lead to net alignment 
M–H Curve (at 300 K)					
Key Comments	• No permanent dipole moment • Induced moment opposite to H • Weak repulsion from magnets • Present in all materials	• Permanent moments exist • Random due to thermal motion • Weak attraction to magnets • No hysteresis	• Strong parallel exchange interaction • Spontaneous magnetization below Curie temperature (T_C) • Hysteresis with large M_r and H_c	• Strong antiparallel exchange • Net moment ≈ 0 (if equal) • Below Néel temperature (T_N) • No hysteresis (or very small)	• Antiparallel but unequal moments • Nonzero net magnetization • Hysteresis similar to ferromagnets • Below Curie temperature (T_C)
Examples	Bi, Cu, Ag, Au, Water, Graphite	Al, Pt, Mg, Cr, O_2, rare earth salts	Fe, Co, Ni, Gd, Dy, Sm	MnO, NiO, FeO, Cr_2O_3, MnF_2	Fe_3O_4 (Magnetite), $CoFe_2O_4$, $NiFe_2O_4$
Magnetic Susceptibility (χ)	$\chi < 0$ (very small, ~10^{-6} to 10^{-5})	$\chi > 0$ (small, ~10^{-6} to 10^{-3})	$\chi >> 1$ (very large, nonlinear)	$\chi \approx 0$ (very small) (positive above T_N, negative below T_N)	$\chi >> 1$ (large, nonlinear)

Magnetism arises from electron spin and orbital motion and their exchange interactions. Temperature can destroy magnetic order: Ferromagnetism and ferrimagnetism disappear above Curie temperature (T_C); Antiferromagnetism disappears above Néel temperature (T_N).

Domain boundary movement and Saturation Magnetisation



- When a magnetic field is applied to a ferromagnetic materials, domain tends to align (similar to a charge particle deforms after applying electric field) in the direction of the field by domain boundary movement and hence the flux density or magnetization increases .
- As the field strength increases domains which are favorably oriented to field direction grow at the expense of the small and un-oriented ones.
- All the domains are aligned to the field direction at high field strength and the material reaches at saturation magnetization Ms
- The initial slope of the curve at H=0 is called initial permeability μ which is a material property

https://www.doitpoms.ac.uk/tlplib/ferromagnetic/hysteresis.php

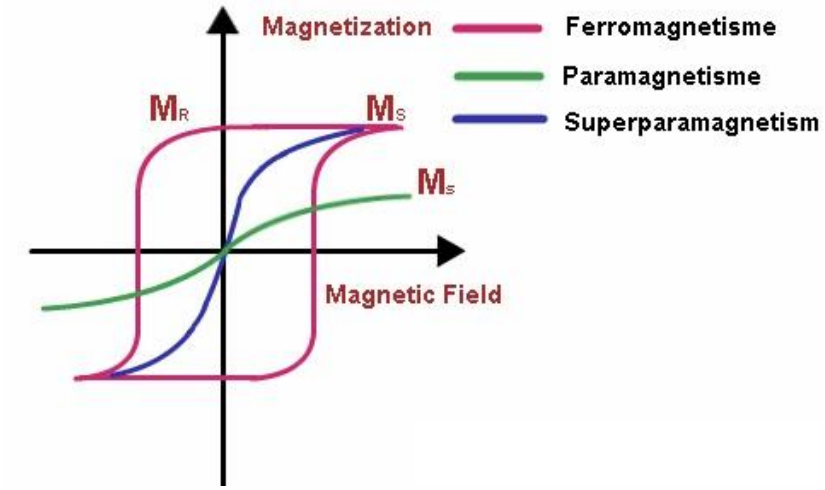
Ferromagnetism to superparamagnetism (transition to nanomagnetism)

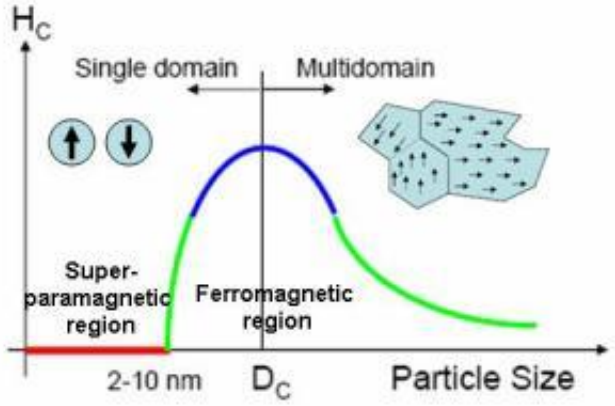
What is ‘Super’ about Superparamagnetism
Very fine ferromagnetic particles have very short relaxation times, even at room temperature, and behave superparamagnetic.

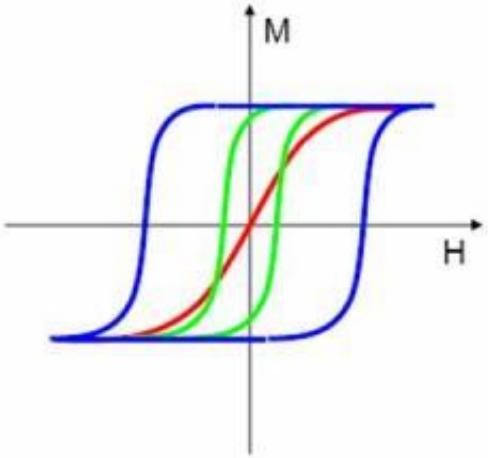
Superparmagnetic materials have:

- A high saturation magnetiszation (Ms)
- Zero coercivity (Hc) and
- Zero remanence (Mr),
- Single domain

Making it to be distinguished from ferromagnetism and paramagnetism - Superparamagnetism is a size effect of ferromagnetism.







The influence of magnetic particle size on magnetic properties. The coercivity changes with the particle size.

https://www.science20.com/mei/blog/blocking_temperatu

https://mappingignorance.org/2014/03/05/superparamagnetic-nanoparticles-and-the-separation-problem/

Ferromagnetism to superparamagnetism (transition to nanomagnetism)

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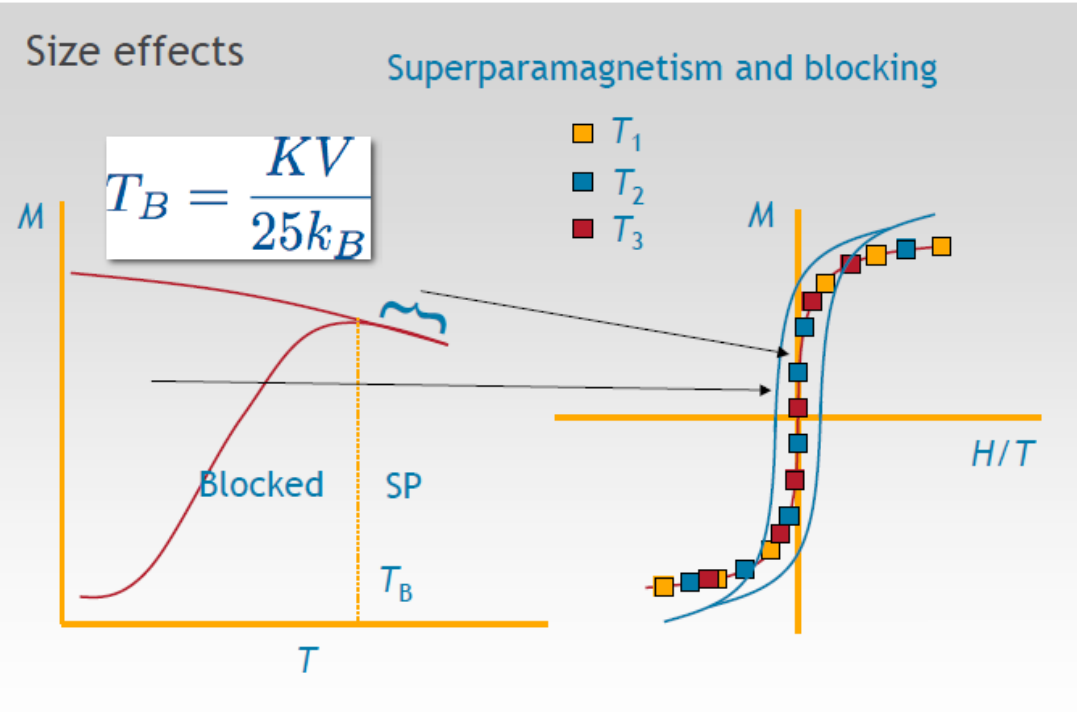
Some important features about Superparamagnetism

- Their behavior is paramagnetic (lower Ms) but their magnetization values are typical of ferromagnetic substances (higher Ms and zero Hc and Mr).

- The individual particles have normal ferromagnetic movements but very short relaxation times, enabling them to rapidly follow directional changes in an applied field.



- Magnetization curves measured at different temperatures superimpose when magnetization (*M*) is plotted as a function of field (*H*)/temperature (*T*).

- This also demonstrates that superparamagnetism can be destroyed by cooling.

- The temperature at which this occurs is called the blocking temperature (T_B) and is dependent linearly on volume and the magnitude of the crystal field anisotropy.

K is the effective magnetic anisotropy constant, V is the magnetic particle volume, kB is Boltzmann's constant, and TB is the **blocking temperature**.

Problem solving

You synthesize two batches of Fe_3O_4 nanoparticles:
Batch A: diameter DA=8 nm
Batch B: diameter DB=20 nm

Assume spherical particles, room temperature T=300K, Cu Kα XRD confirms single-crystal particles, and the effective anisotropy energy density is $K=1.0 \times 10^5$ J m^{-3}
Use the "blocking criterion" based on the Néel–Arrhenius law with a typical measurement window ($\ln(\tau m/\tau 0)\approx 25$), i.e.
Tb = KV/25kb
KV=25kbTb

Energy barrier, Eb = KV
Thermal fluctuation energy, Et = 25kbTb
Blocked/ferromagnetic if $EB=KV \gtrsim 25$ kBT, superparamagnetic if $EB \ll 25$ kBT.

Tasks
1. Compute the particle volumes VA and VB.
2. Compute the energy barriers EB,A and EB,B.
3. Compare each to 25kBT and decide whether each batch is superparamagnetic or blocked at 300 K.

Given Constants:
$kB=1.380649\times10^{-23}$ J/K (Boltzmann constant)
$25kBT=25\times kB\times 300\approx 1.035\times10^{-19}$ J
Sphere volume $V=4/3\pi r^3$

A biomedical company is developing **superparamagnetic iron oxide nanoparticles (SPIONs)** for magnetically targeted drug delivery. The nanoparticles should respond strongly to an external magnetic field so that they can be guided toward a target site. However, once the external magnetic field is removed, the particles should have negligible remanence and coercivity to minimise magnetic aggregation.

The nanoparticles have an effective magnetic anisotropy constant, $K=1.3\times104$ $Jm-3$
and can be approximated as spherical, non-interacting, single-domain particles.

For the measurement timescale being considered, assume the blocking condition is
$KV=25kBTB$

where
$kB=1.38\times10-23$ $JK-1$.

The nanoparticles will operate at body temperature:
$T=310$ K.

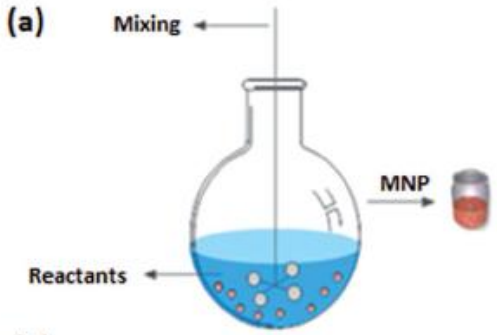
(a) Calculate the maximum approximate particle diameter corresponding to a blocking temperature of 310 K.

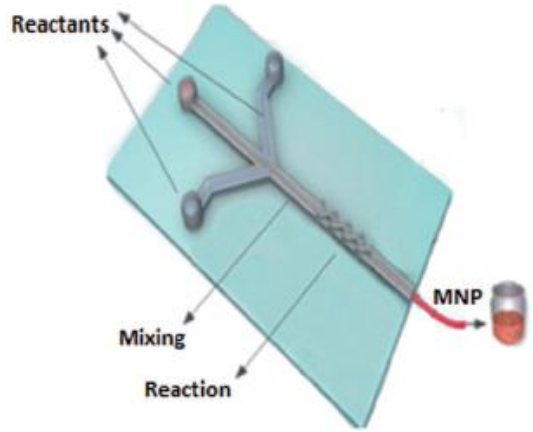
(b) The manufacturing team can produce particles with average diameters of 15 nm, 25 nm, and 35 nm. Which would you recommend for superparamagnetic operation at body temperature? Justify your answer using the energy-barrier concept.

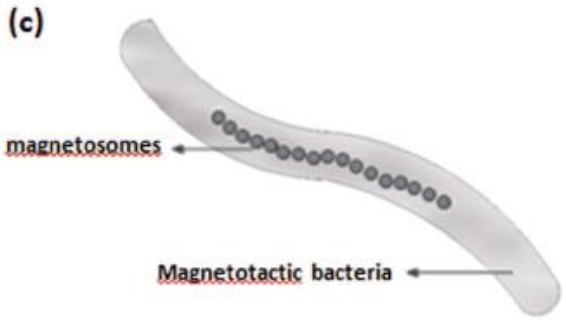
(c) A colleague says:
“The 35 nm particles are better because larger particles contain more magnetic material and therefore will respond more strongly to the applied magnetic field.”

Do you agree? Explain the engineering trade-off between magnetic response and magnetic stability.

List all the synthesis techniques and comparison







- Wet chemistry routes
  - Hydrolytic
    - Coprecipitation
    - Hydrothermal
    - Sonochemistry
    - Microemulsion
  - Non-hydrolytic
    - Thermal decomposition
    - Microwave assisted
- Microfluidic reactors
  - Single-phase flow
    - Capillary tube microreactors
    - Co-axial flow microreactors
  - Multiphase flow
    - Gas/liquid or liquid/liquid microreactors
    - Droplet-based microreactors
  - Micromixing-based microreactors
    - Passive mixing
    - Active mixing
- Biogenic synthesis
  - Bacteria mediated
  - Protein mediated
  - Fungi mediated

Adv. Healthcare Mater. ***2017, 6, 1700306***

List all the synthesis techniques and comparison

Synthesis route	Coprecipitation	Hydrothermal
Reaction temperature	20–90	180–220
Reaction period	Minutes	hours
Solvent	Water	Water, ethanol
Production	High	Medium
Advantages	Mild reaction conditions Synthesis in H_2O Easy surface modification Ease conversion to maghemite Ease scale up	Improved size control Narrow size distribution Synthesis in H_2O Tunable magnetic properties Applicability for continuous synthesis
Disadvantages	Broad size distribution Polydispersity Low reproducibility Uncontrolled oxidation Difficulty in adjusting the pH of the reaction	High temperature High pressure Need for special reactors or autoclaves

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List all the synthesis techniques and comparison

Synthesis route	Sonochemistry	Microemulsion
Reaction temperature	20–90	20–50
Reaction period	Minutes	hours
Solvent	Water	Water
Production	Medium	Low
Advantages	Accelerated reaction time Enhanced mixing by non-invasive source of energy Relatively narrow size distribution Good dispersivity High crystallinity	Improved size control Relatively narrow size distribution Ease of size tunability Uniform magnetic properties
Disadvantages	Need for ultrasound wave irradiation equipment	Low reaction yield Poor crystallinity Difficulty to remove surfactants Need for large amounts of organic solvents

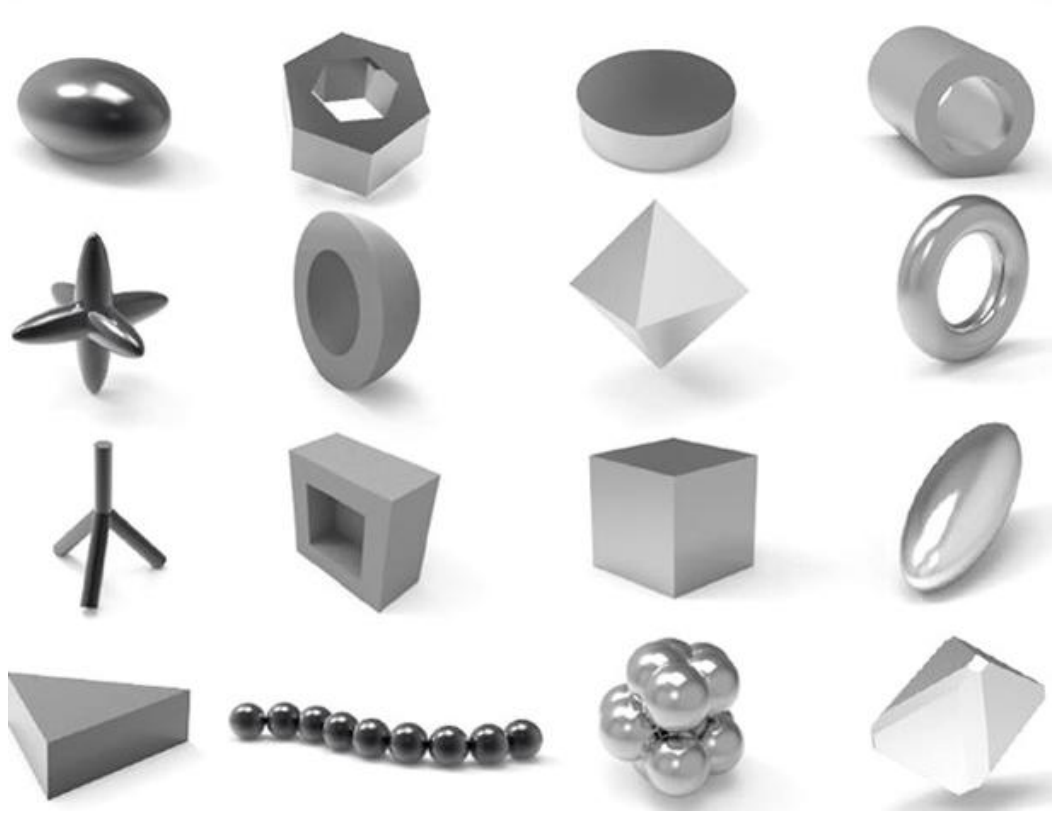
List all the synthesis techniques and comparison

Synthesis route	Thermal decomposition	Microwave assisted
Reaction temperature	100–320	160–210
Reaction period	hours	Minutes
Solvent	Organic compound	Organic compound
Production	High	High
Advantages	Narrow size distribution High crystallinity High size control Tunable magnetic properties	Narrow size distribution, high crystallinity, water dispersible, Rapid processing, high reaction rate, reduced reaction time, high yield of product along with simplicity and high energy efficiency
Disadvantages	Toxic organic solvents High temperature Need for phase transfer	Need for organic solvent Limited penetration depth of the microwave

Parameters influence magnetic properties:

Ms (Saturation magnetisation), Mr (Remanance), Hc (Coercivity), Tb (Blocking temperature)

- Shape
- Size
- Composition
- Synthesis processes
- Sintering temperature
- Induced Porosity
- Density
- Shell-core structure



https://doi.org/10.1016/j.addr.2018.12.008

Case study 1:

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Shape and Size effect on magnetic properties of nanocomposites

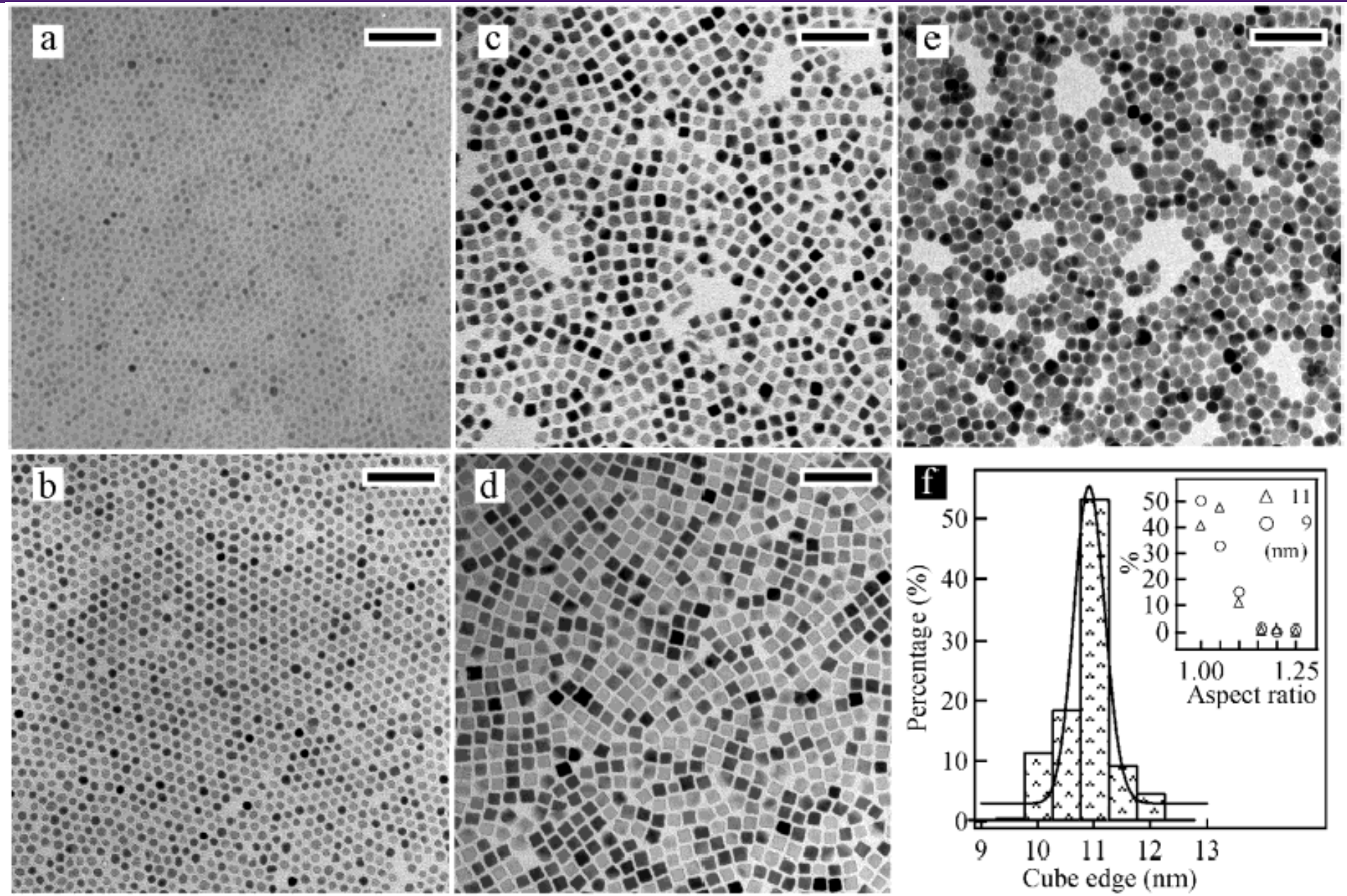
https://pubs-acs-org.ezproxy.library.uq.edu.au/doi/full/10.1021/ja049931r

Case study 1:

Monodisperse CoFe2O4 nanocrystals are synthesized by using a combination of a nonhydrolytic process and seed-mediated growth.

TEM micrographs with the scale bar as 50 nm. The size in panel (a) and (b) is 5.2 +/-1.1 and 7.9 +/-0.5 nm spherical CoFe2O4 nanocrystals, respectively. Panels (c) and (d) show 9.1 +/- 0.5 and 10.9 +/- 0.6 nm cubic CoFe2O4 nanocrystals, respectively. Panel (e) is 11.8 (1.3 nm spherical CoFe2O4 nanocrystals. Panel (f), as a representative histogram, shows the size distribution of cubic nanocrystals in panel (d) after measuring over 400 nanocrystals. The inset in panel (f) displays the aspect ratios of cubic nanocrystals in panels (c) and (d).

Growth reaction parameters such as heating rate, temperature, reaction time, ratio of seed to precursors, and ratio of oleic acid to oleylamine have been systematically studied for the control of size and shape of nanocrystals.



https://pubs-acs-org.ezproxy.library.uq.edu.au/doi/full/10.1021/ja049931r

Case study 1:

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- **Precursors**: coordination compounds of iron(III) and cobalt(II) acetylacetonate, $Fe(acac)_3$ and $Co(acac)_2$

- **5nm Seed preparation**: 2 mmol of $Co(acac)_2$ + 40 mL of phenyl ether + 20 mmol of 1,2-hexadecanediol + 10 mL of oleic acid, + 10 mL of oleylamine --→ 140°C → droplet addition of 4 mmol $Fe(acac)_3$ + 20 mL of a phenyl ether solution → 260 °C → reflux for 30 min before being cooled down to room temperature-→ addition of ethanol and centrifuging-→spherical $CoFe_2O_4$ 5nm seed

- **8 nm spherical nanocrystals**: 5 nm nanocrystals were used as seeds to grow 8 or 9 nm spherical nanocrystals in the solution of $Co(acac)_2$ and $Fe(acac)_3$ precursors. 100 mg of seeds was used in a particle growth solution consisting of 1 mmol of $Co(acac)_2$, 2 mmol of $Fe(acac)_3$, 10 mmol of 1-octadecanol, 5 mL of oleic acid, and 5 mL of oleylamine. Then, the solution temperature was raised to 260 °C at a rate of 10−15°/min and kept at reflux at 260 °C for 30 min. The nanocrystals precipitated out after ethanol was added.

Note: Monodisperse nanocrystals with a 9 nm diameter can be obtained using the same procedures but with increased concentration of precursors in the particle growth solution. Although nanocrystals with a larger diameter can also be prepared with 5 nm seeds, the quality of nanocrystals in term of size distribution usually deteriorated. monodisperse nanocrystals with larger diameters were typically produced by incrementally increasing the size of nanoparticulate seeds.

Growth reaction parameters such as heating rate, temperature, reaction time, ratio of seed to precursors, and ratio of oleic acid to oleylamine have been systematically studied for the control of size and shape of nanocrystals. The results indicated that heating rate and growth temperature control the shape of $CoFe_2O_4$ nanocrystals.

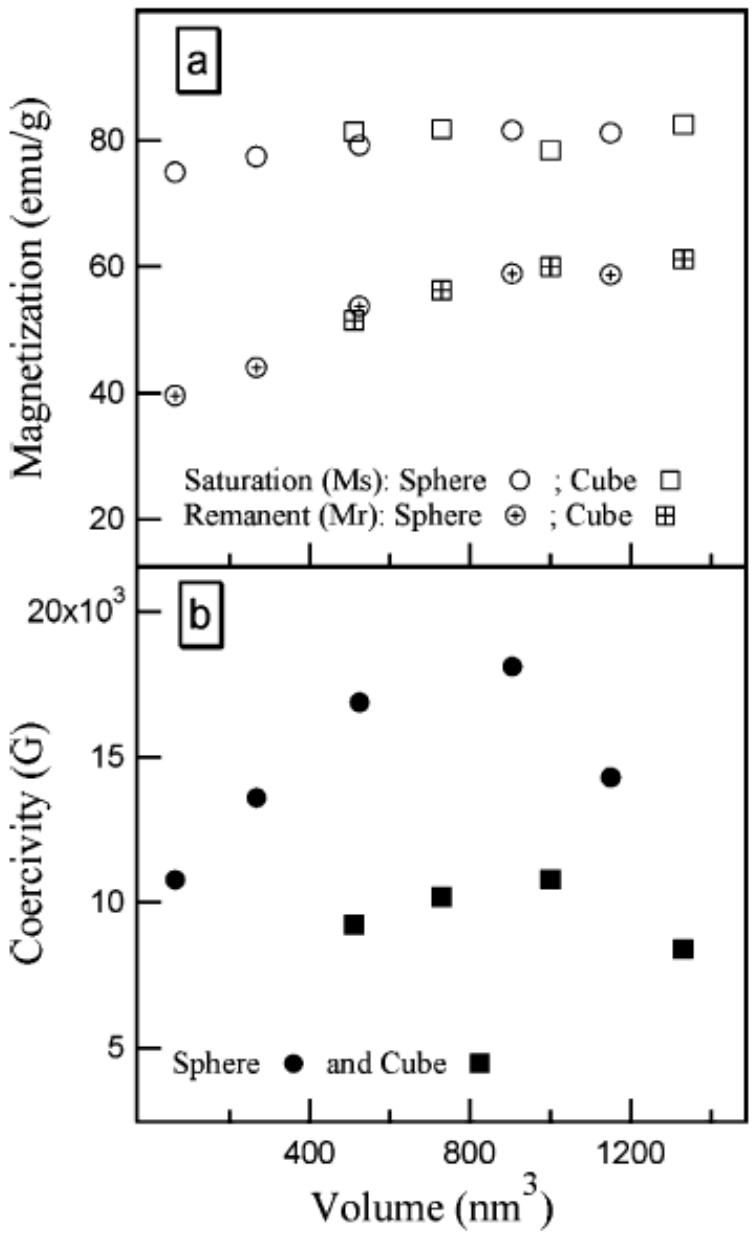
Due to the decisive effect of heating rate and growth temperature on shape control, Figure 1e shows almost spherical monodisperse $CoFe_2O_4$ nanocrystals with a 12 nm diameter, which were prepared by using nanocube seeds with an 8 nm edge length. Such a reversible shape formation clearly demonstrated that the shape of $CoFe_2O_4$ nanocrystals can be precisely controlled through the temperature and heating rate.

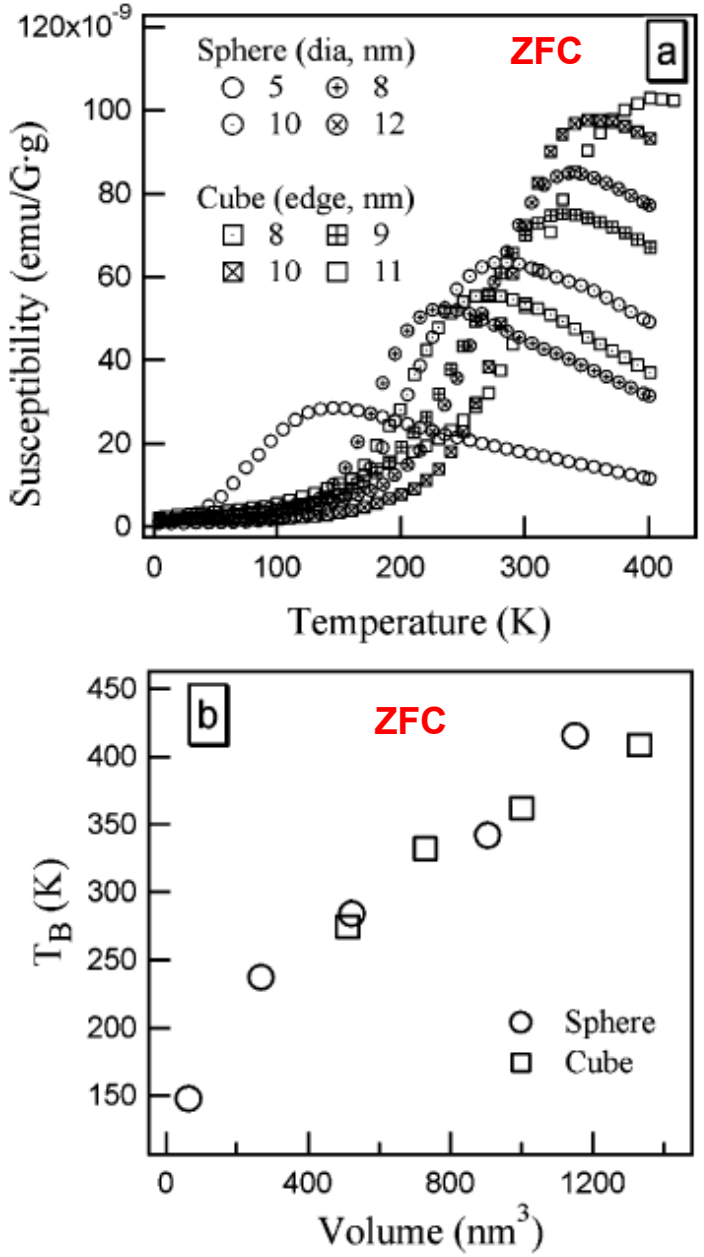
Size control → depends on synthesis parameter → The size of the nanocrystals was controlled through the ratio between the quantity of nanocrystalline seeds and the amount of precursors in the solution for seed-mediated growth.

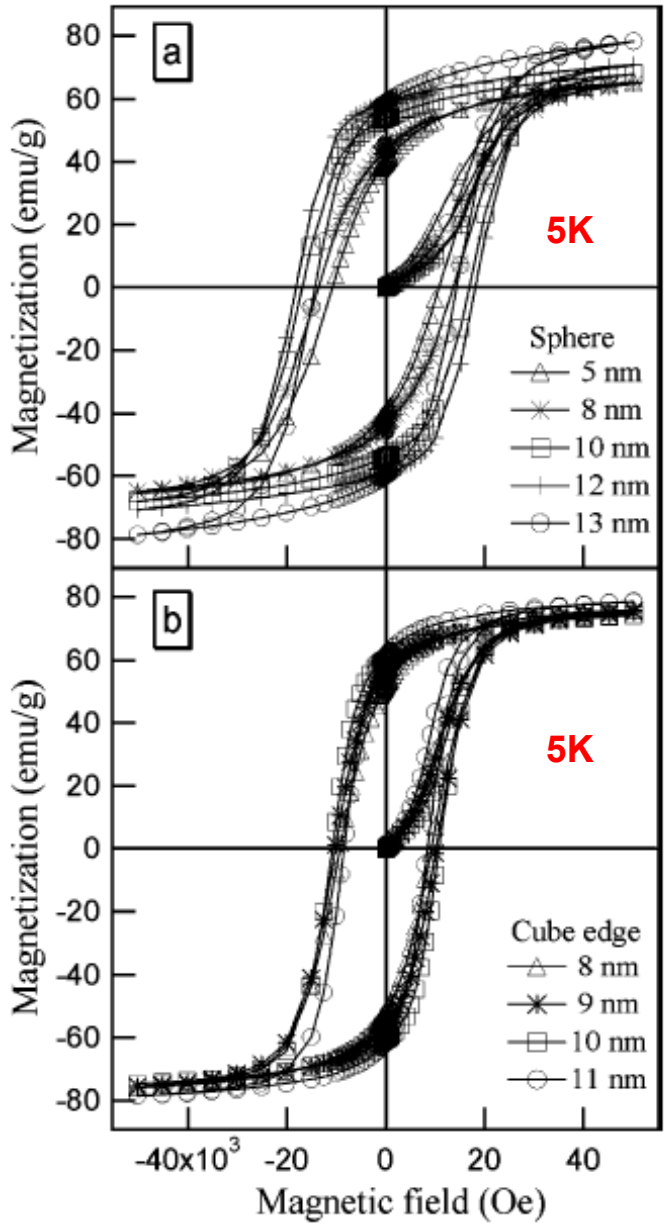
Shape control → depends on processing parameters → cubic $CoFe_2O_4$ nanocrystals were produced with an edge length of 8 or 9 nm when the temperature was raised only to 210 °C at a rate of 2−3°/min.

https://pubs-acs-org.ezproxy.library.uq.edu.au/doi/full/10.1021/ja049931r

Case study 1:


a
Magnetization (emu/g)
Saturation (Ms): Sphere ○ ; Cube □
Remanent (Mr): Sphere ⊕ ; Cube ⊞
b
20x10^3
Coercivity (G)
Sphere ● and Cube ■
Volume (nm^3)


120x10^-9
Susceptibility (emu/G·g)
Sphere (dia, nm)
5 8
10 12
Cube (edge, nm)
8 9
10 11
ZFC
a
Temperature (K)
b
ZFC
T_B (K)
Sphere
Cube
Volume (nm^3)


a
Magnetization (emu/g)
5K
Sphere
5 nm
8 nm
10 nm
12 nm
13 nm
b
5K
Cube edge
8 nm
9 nm
10 nm
11 nm
-40x10^3
Magnetic field (Oe)

Case study 2:

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Effect of annealing temperature on magnetic phase transition in magnetic nanoparticles

https://www.sciencedirect.com/science/article/abs/pii/S0304885314012633

Case study 2

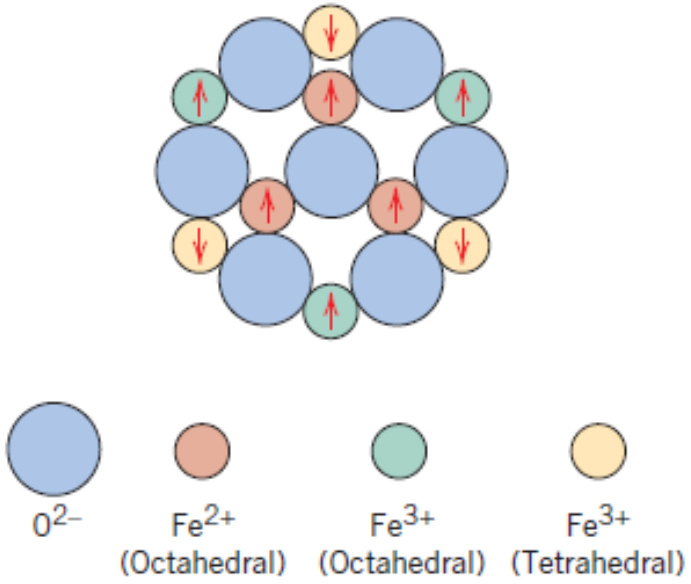


Table 20.3 **The Distribution of Spin Magnetic Moments for Fe^{2+} and Fe^{3+} Ions in a Unit Cell Fe_3O_4[a]**

Cation	*Octahedral Lattice Site*	*Tetrahedral Lattice Site*	*Net Magnetic Moment*
Fe^{3+}	↑↑↑↑ ↑↑↑↑	↓↓↓↓ ↓↓↓↓	Complete cancellation
Fe^{2+}	↑↑↑↑ ↑↑↑↑	—	↑↑↑↑ ↑↑↑↑

26	Fe	↑↓	↑↓ ↑ ↑ ↑ ↑		$[1s^2\,2s^2\,2p^6\,3s^2\,3p^6]\,4s^2\,3d^6$	$[Ar]\,4s^2\,3d^6$

Case study 2:

- Iron oxides, especially magnetite, are widely used in various fields, i.e, medical use in, for example, as contrast agents in MRI, targeting drug delivery, hyperthermia etc.

- Magnetite is thermodynamically unstable and can be oxidized to hematite, maghemite and other type of iron oxides in the presence of oxygen and depending on temperature conditions.

- This feature causes limitations in its applications under different temperature conditions.

- The investigation of the Fe_3O_4 NPs behavior at various temperatures or under different oxidizing conditions is necessary

- Magnetic properties are affected by phase transition processes

- Factors influencing the phase transitions:

Particle size, reaction medium (atmosphere, vacuum or inert gas such as Ar, N2, H2), preparation method, annealing time and heating rate in annealing.

Case study 2:

Synthesis Process:

$FeCl_2 \cdot 4H_2O$ and $FeCl_3 \cdot 6H_2O$ → 1:2 molar ratio dissolve in 25 ml DI water → Fe^{+2} and Fe^{+3}

Dropwise adding with magnetic stirring

200 ml of 1.5 mol/L NaOH (precipitator agent)

magnetic separation

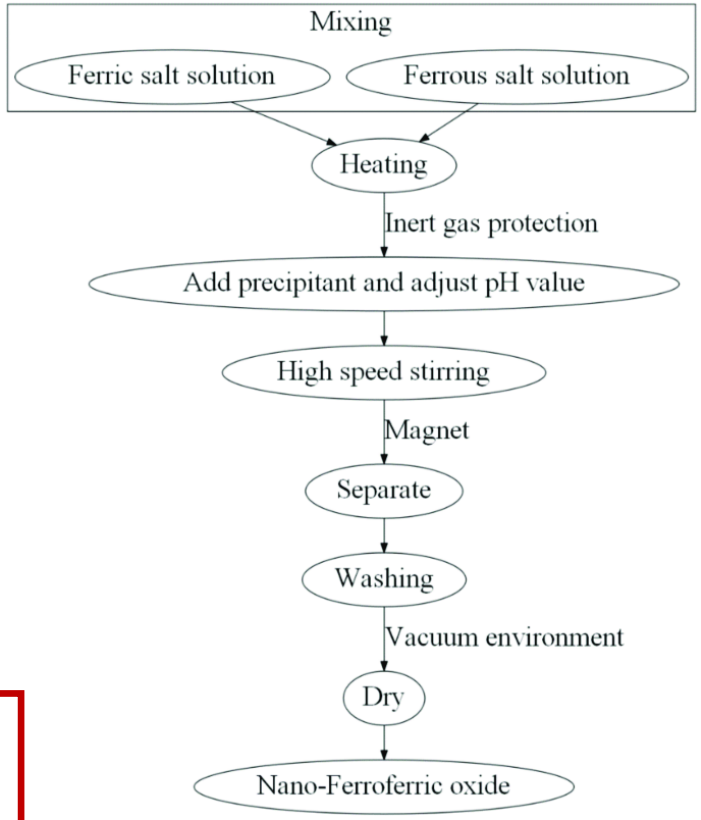
3 times wash with DI and ethanol

Black precipitations

Dry oven

50 C/24hrs

Fe3O4



$$Fe^{+2}(aq) + 2Fe^{+3}(aq) + 8OH^- \rightarrow Fe_3O_4 \downarrow + 4H_2O$$

annealing time (1 h) with heating rate of 5 °C/min

Phase transformation:

Fe3O4 —(250 C, 1 hr)→ Y-Fe2O3 —(550 C-650 C 1 hr)→ α- Fe2O3 —(750 C-800 C 1hr)→ α- Fe2O3

Fe3O4	Y-Fe2O3	α- Fe2O3	α- Fe2O3
Magnetite (Soft magnet)	Maghemite (Soft magnet)	hematite (Soft magnet)	hematite (hard magnet)
Superparamagnetic	Superparamagnetic	Superparamagnetic	Ferromagnetic

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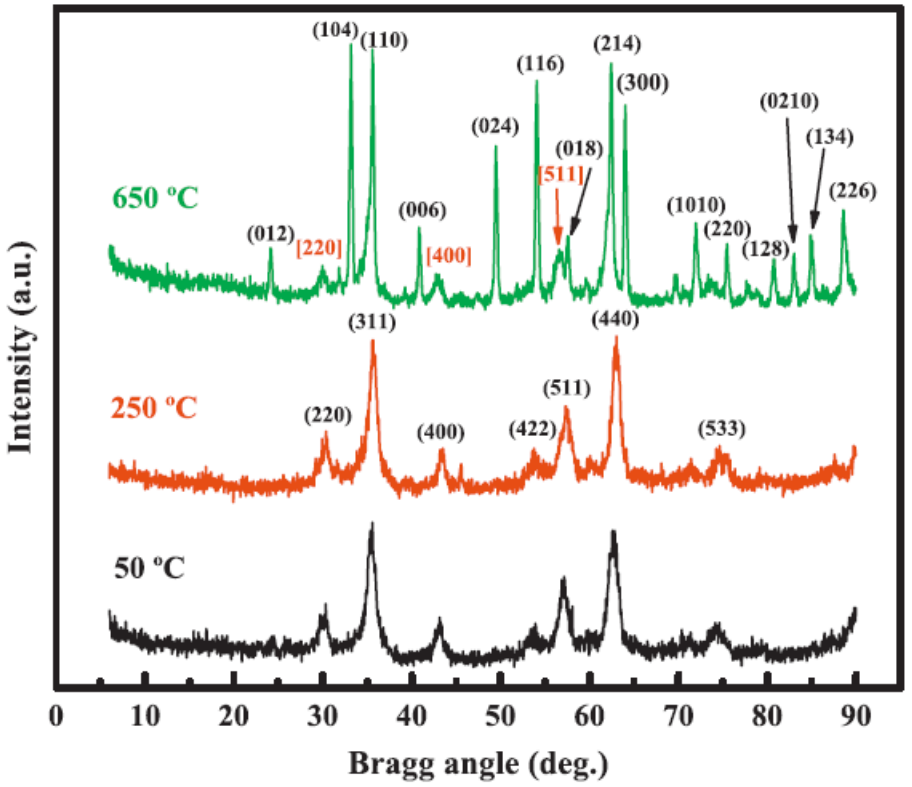
DOI: 10.3390/pr6040033

Case study 2:



Summary of Debye–Scherrer size of the samples.

Annealing temperature (°C)	Debye–Scherrer size (nm)
50	6.6
250	7.5
350	8.8
450	9.8
550	12.7
650	23.9
750	31.2
850	37.6

No difference between the diffraction peaks of Fe3O4 NPs annealed at 50 (magnetite) and 250 °C (maghemite)

Difficult to distinguish magnetite from maghemite except precise calculation of d-value (lattice spacing) of (311) peak is varied from 2.531 to 2.518 Å (Raman spectroscopy)

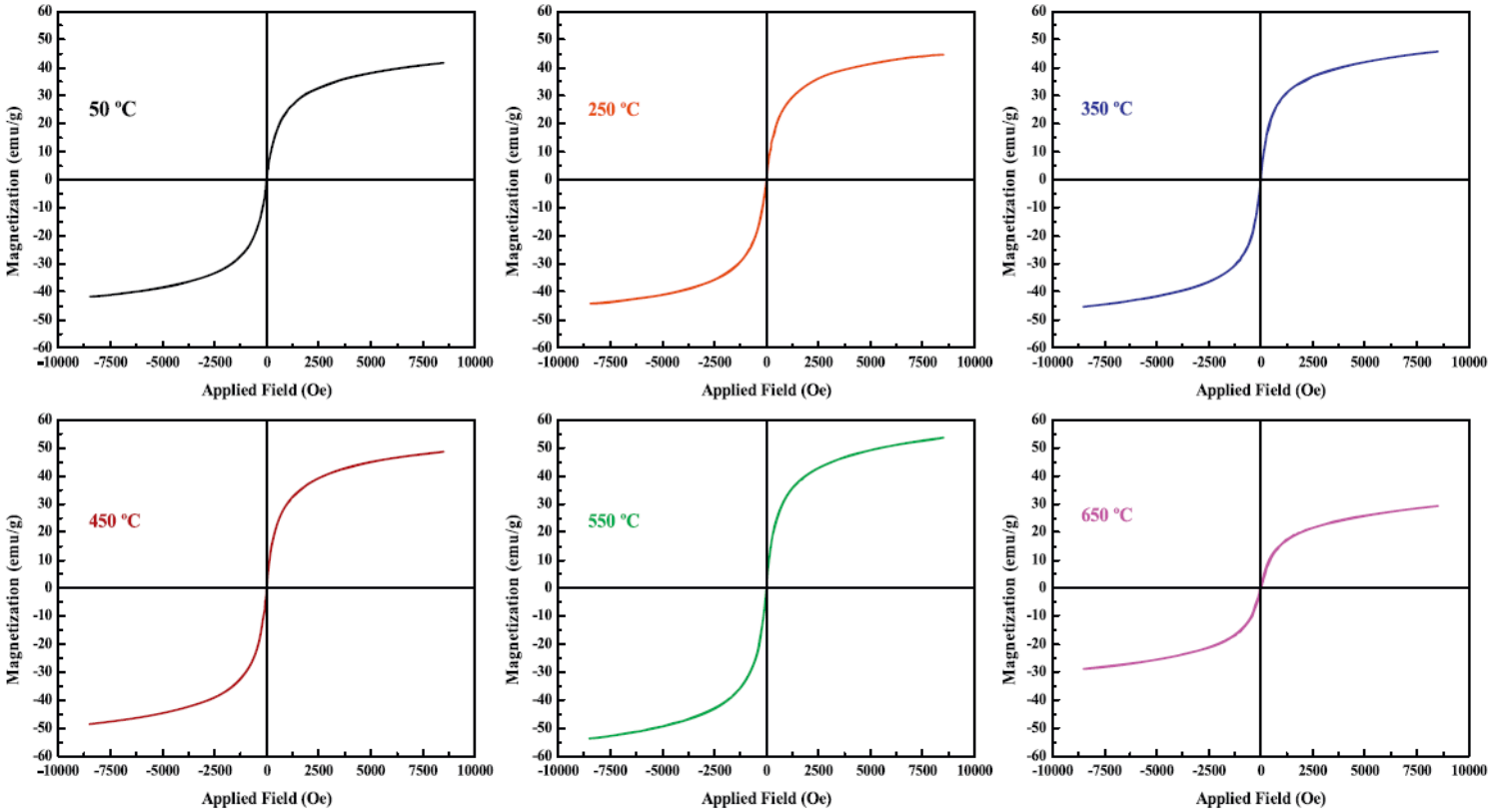
The magnetite (Fe3O4)-to-hematite (α-Fe2O3) conversion started at 650 °C

The average crystallite size of the prepared Fe3O4 NPs after annealing at different temperatures was calculated by the Debye– Scherrer equation:

$$D = \frac{k\lambda}{\beta \cos\theta}$$

Crystallite size of the NPs increased on increasing the annealing temperature due to the grain growth which is attributed to the coalescence of the particles by solid-state diffusion

Case study 2:



Room-temperature hysteresis loops of as-synthesized Fe3O4 NPs after annealing at 50, 250, 350, 450, 550, 650, 750 and 850 °C.

Langevin equation

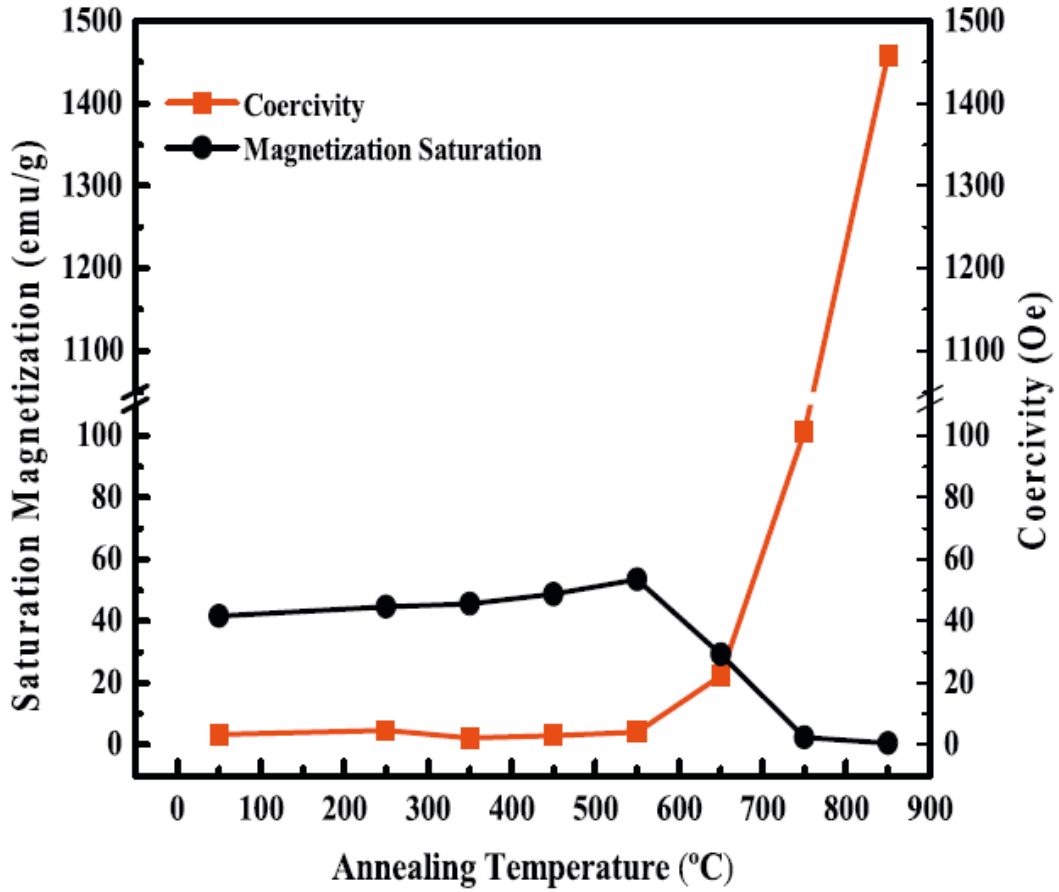
$$M = M_s \left[\coth\left(\frac{\mu B}{k_B T}\right) - \left(\frac{k_B T}{\mu B}\right) \right]$$

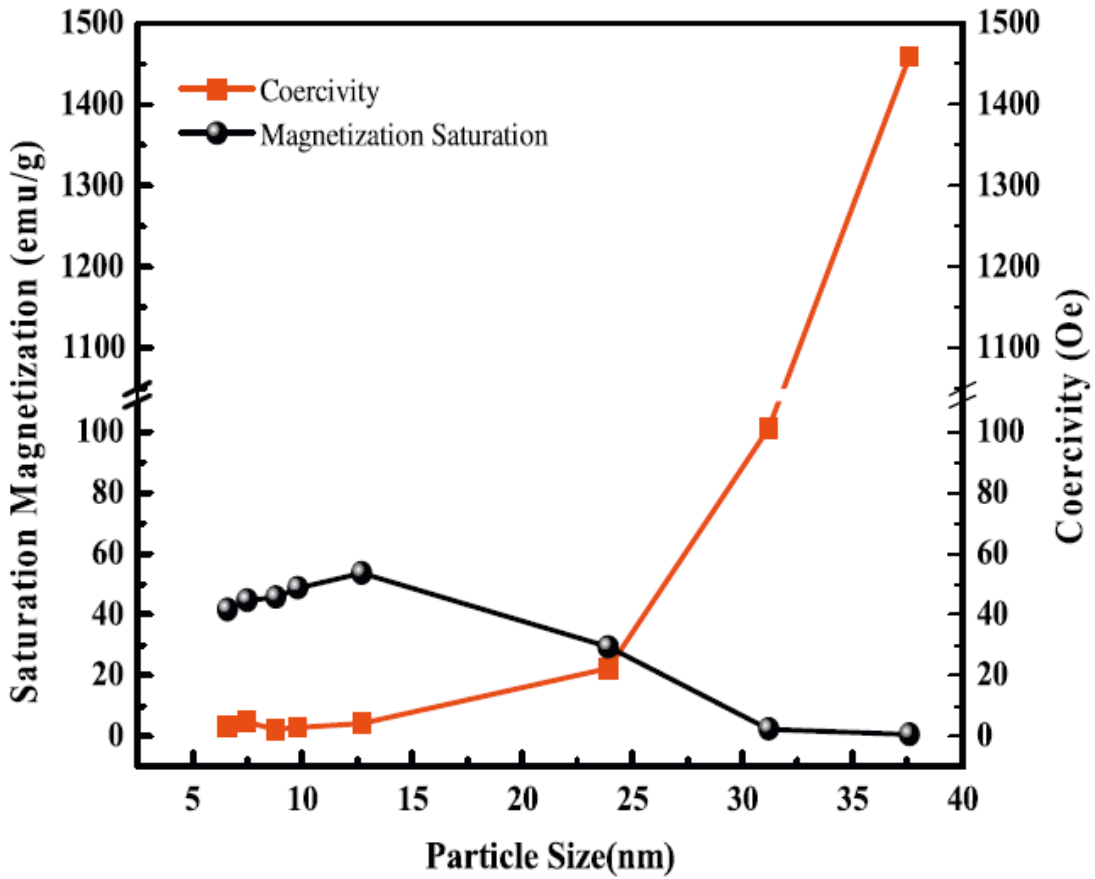
$$M_s = \mu N$$

μ is the magnetic moment of each particle, T the absolute temperature, kB the Boltzmann constant, and Ms the saturation magnetization of the sample, N is the number of particles/weight of sample).

Case study 2:

Variation of saturation magnetization and coercivity with annealing temperature and particle size.





Can you recall last week's lecture about the hysteresis loop?

Transformation from superparamagnetic to ferromagnetic!!!

$$H_c = \frac{0.96K}{M_s}$$

The relationship between Hc, Ms and K at a certain temperatures [46]

Case study 2: Summary

- Fe_3O_4 (magnetite) nanoparticles(NPs) were synthesized using a co-precipitation method.
- Annealed at various temperatures between 50 and 850 °C for 1h in air (heating rate 5°C/min).
- The NPs were characterized by X-ray diffraction (XRD), Fourier transform infrared (FT-IR) spectroscopy and vibrating sample magnetometer (VSM).
- Fe_3O_4 NPs were converted to γ-Fe_2O_3 (maghemite) by annealing at 250 °C for 1h (XRD and FTIR)
- Then to α-Fe_2O_3 (hematite) on annealing in the range of 550–650 °C
- The average crystallite size of the NPs estimated by the Debye–Scherrer equation increased from 6.6 to 37.6nm by increasing annealing temperature from 50 to 850 °C
- The magnetite NPs were superparamagnetic (VSM)
- Converted to the maghemite with super- paramagnetic phase by annealing up to 550 °C
- A phase transition from soft (maghemite) to hard (hematite) ferromagnetic was occurred at annealing temperature 650 and 850 °C, respectively.
- The VSM analysis confirmed the XRD and FTIR results. Magnetite and maghemite have extreme similarity in crystal structure, and XRD is not sensitive enough to differentiate between them, FTIR needed.
- The saturation magnetization(Ms) of Fe_3O_4 NPs was increased from 41.69 to 53.61 emu/g by increasing annealing temperature from 50 to 550 °C, and then decreased intensively to 0.49emu/g after annealing at 850 °C.
- The coercive force (Hc) was reached to 4.20 Oe after annealing at 550 °C and then intensively increased to 1459.02 Oe for any further increasing of particle size up to 850 °C.

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Summary

- Understand the facts fundamental and basic terminology of magnetism
- Understand the origin of magnetism from atomic model
- Determination the magnetism in materials from their electronic configuration
- Understand the types of magnetism and their mechanism to distinguish
- Understand the effect of temperatures on magnetism
- Understand the magnetic hysteresis and their significance in practical application
- Understand the domain structure transformation (multi to single) and magnetic saturation from hysteresis loop
- Understand the ferromagnetism to superparamagnetism transformation (nanomagnetism)
- Understand the effect of size and shape effect on magnetic properties
- Understand the effect of annealing temperatures on magnetic properties of nanoparticles and mechanism of their phase transformation (magnetite to hematite)

Synthesis of magnetic nanoparticles for week 7's lab prac

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1. Nanoparticle synthesis and their biofunctionalisation

2. Ni ferrite nanoparticle using hydrothermal process

3. All the equipment required such as incubator, hydrothermal chamber, vacuum oven, calcination, autoclave etc

4. Characterisation: SEM/magnetic properties

WEEK 6 · 知识点总结 · A/Prof. Shahriar Hossain

第 6 周　纳米磁学的应用：纳米机械

Application of Nano-magnetism – Nano-machineries

主线：临界尺寸的确定（XRD + TEM）→ VSM / SQUID → 磁热疗与 Néel / Brownian 弛豫 → AC/DC 磁场联合给药

1. Hc–颗粒尺寸曲线（★）

- 多畴区：畴壁移动反转，尺寸越大 Hc 越低 → **单畴临界尺寸 D_c**：Hc 最大 → 单畴区：Hc 下降 → **D_{sp}** 以下超顺磁（Hc = Mr = 0）。
- 超顺磁四特征：高 Ms、Hc = 0、Mr = 0、单畴；可被冷却破坏（$T < T_B$）；M 对 H/T 作图重合。

2. 案例 3：确定 Fe_3O_4 单畴临界尺寸（Sci. Rep. 2017）

- 液相沉淀法，控制 Fe^{2+} 浓度、pH、温度；两步氧化 → 立方（C1–C9，9.6–287 nm），一步氧化 → 球形（S1 93 nm、S2 121 nm）。
- **d_p**（SEM/TEM 颗粒尺寸）vs **d_c**（XRD (311) Scherrer 晶粒尺寸）：$d_c \approx d_p$ → 单晶单畴（约 80 nm 以下）；更大时 d_c 进入平台 → 多晶多畴。
- HRTEM：单方向晶格条纹 = 单晶（至 64.7 nm），多方向 = 多晶。
- 磁性：亚铁磁；Ms 随尺寸单调上升；**Hc、Mr 在约 76 nm 达峰**；只有 C1 超顺磁；球形（多晶、取向不一）Hc、Mr 更低。

3. 磁性测量

- **VSM**：法拉第电磁感应；样品在恒定场中被磁化 → **振动**使杂散场随时间变化 → 拾取线圈感应信号 $\propto$ M → 锁相放大 → M–H 回线。
- **SQUID**：两个并联约瑟夫森结的超导环；灵敏度极高（可测生物磁场）；需液氦；用于 ZFC/FC 求 T_B。

4. 磁热疗（★）

- 两大物理基础：可被外场操控（分离、免疫检测、靶向）；交变场中有损耗（热疗）。DC 场 → 成像（MRI），AC 场 → 升温 / 驱动。
- 目标：肿瘤 **> 43 °C**，不伤正常组织；磁场穿透组织，颗粒局部发热；可增强放化疗敏感性（GdIO + DOX 协同）。
- 四种发热机制：**磁滞损耗**（多畴）、**Néel 弛豫**、**Brownian 弛豫**、涡流损耗。

	Néel	Brownian
什么在转	磁矩在颗粒内部	整个颗粒在液体中
公式	$\tau_N = \tau_0 \exp(KV/k_BT)$	$\tau_B = 3\eta V_h/k_BT$
取决于	核体积、各向异性 K	水合体积、黏度 η
颗粒被固定后	仍有效	失效

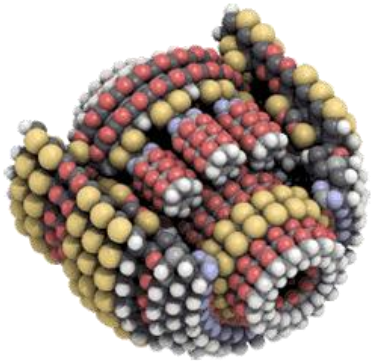
- $\tau_{eff} = \tau_N\tau_B / (\tau_N + \tau_B)$，τ 小者主导。例（$K = 10^4$，300 K）：8/12 nm → $\tau_N \approx 2$ ns，**Néel 主导**；20/22 nm → $\tau_N \approx 25$ μs，$\tau_B \approx 3$–4 μs，**Brownian 主导**。

5. 案例 5：AC/DC 磁场联合给药（Hossain 组）

- **DC 场定位**（把颗粒吸在靶点），**AC 场**（2–16 Hz，磁–机械振动）**释放**；只用 AC 不定点，只用 DC 几乎不释放。
- 载体：多孔 Fe_3O_4（SEM 10–30 nm 团聚体；TEM 内有 2–5 nm 孔、SAED 衍射环 = 多晶）；亚甲基蓝通过**静电作用**吸附（FTIR 证实）；A549 存活率 > 90%；310 K 时 S 形回线、Hc 可忽略、Ms ≈ 77 emu/g → 超顺磁。
- UV-vis 约 580 nm 定量释放；类黄酮药物：AC/DC **约 2 h 释放 100%**，被动释放约 9 天 → **快约 100 倍**。
- 优化因素（载药：孔隙、尺寸、比表面积、静电、吸附…；释放：毒性、场频率 / 振幅、线圈、黏度…）；高浓度（1 mg/mL）有毒；**磁注射器**（坡莫合金软磁针）：注射 → AC 释放 → DC 取回。

考试重点：两类计算（超顺磁判据 / 临界尺寸；Néel vs Brownian）；d_c–d_p 图与 Hc 峰的解读；VSM 原理；为什么需要 DC + AC；为什么热疗用超顺磁颗粒。

MATE6301 – Nanomaterials



Week 6: Application of Nano-magnetism – Nano-machineries

Assoc. Prof. M Shahriar Hossain
Email: md.hossain@uq.edu.au

Outline

The particle size effect on magnetic properties and determination of critical size.

Magnetic properties characterization techniques.

Nano-machineries.

Case Study 4. Magnetic hyperthermia for cancer treatment.

Case Study 5. Magnetically triggered drug delivery system by using combined AC and DC magnetic field.

Lab prac :
Magnetic nanoparticles characterisation for biosensors.

Five Basic Types of Magnetism

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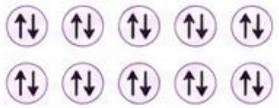
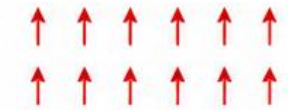
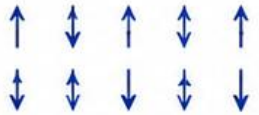
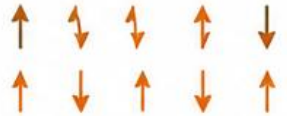
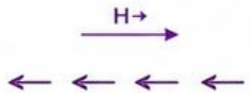
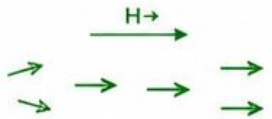
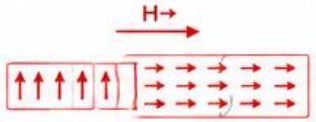
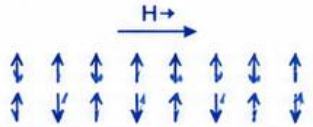
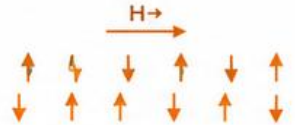
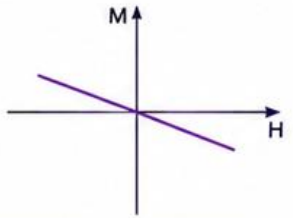
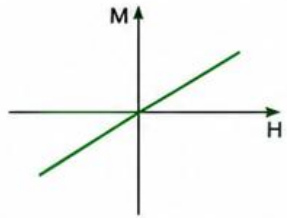
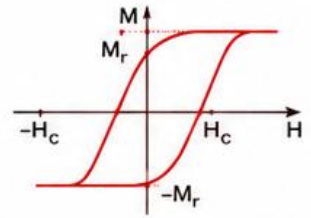
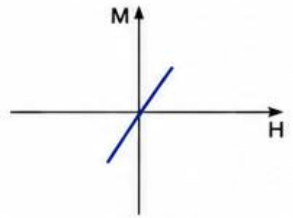
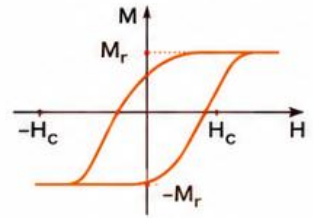
	DIAMAGNETISM	PARAMAGNETISM	FERROMAGNETISM	ANTIFERROMAGNETISM	FERRIMAGNETISM
Atomic / Spin Arrangement (H = 0)	No permanent magnetic moment (all electrons paired) 	Permanent moments randomly oriented	Spins align parallel (spontaneous magnetization) 	Neighboring spins align antiparallel and cancel 	Neighboring spins antiparallel but unequal → net magnetization 
Response to Applied Field (H →)	Induced moment opposite to applied field 	Moments tend to align with the field 	Domains grow and align with the field 	Slight canting of spins; small net moment 	Unequal moments lead to net alignment 
M–H Curve (at 300 K)					
Key Comments	• No permanent dipole moment • Induced moment opposite to H • Weak repulsion from magnets • Present in all materials	• Permanent moments exist • Random due to thermal motion • Weak attraction to magnets • No hysteresis	• Strong parallel exchange interaction • Spontaneous magnetization below Curie temperature (T_C) • Hysteresis with large M_r and H_c	• Strong antiparallel exchange • Net moment ≈ 0 (if equal) • Below Néel temperature (T_N) • No hysteresis (or very small)	• Antiparallel but unequal moments • Nonzero net magnetization • Hysteresis similar to ferromagnets • Below Curie temperature (T_C)
Examples	Bi, Cu, Ag, Au, Water, Graphite	Al, Pt, Mg, Cr, O_2, rare earth salts	Fe, Co, Ni, Gd, Dy, Sm	MnO, NiO, FeO, Cr_2O_3, MnF_2	Fe_3O_4 (Magnetite), $CoFe_2O_4$, $NiFe_2O_4$
Magnetic Susceptibility (χ)	$\chi < 0$ (very small, ~10^{-6} to 10^{-5})	$\chi > 0$ (small, ~10^{-6} to 10^{-3})	$\chi >> 1$ (very large, nonlinear)	$\chi \approx 0$ (very small) (positive above T_N, negative below T_N)	$\chi >> 1$ (large, nonlinear)

Magnetism arises from electron spin and orbital motion and their exchange interactions. Temperature can destroy magnetic order: Ferromagnetism and ferrimagnetism disappear above Curie temperature (T_C); Antiferromagnetism disappears above Néel temperature (T_N).

Ferromagnetism to superparamagnetism (transition to nanomagnetism)

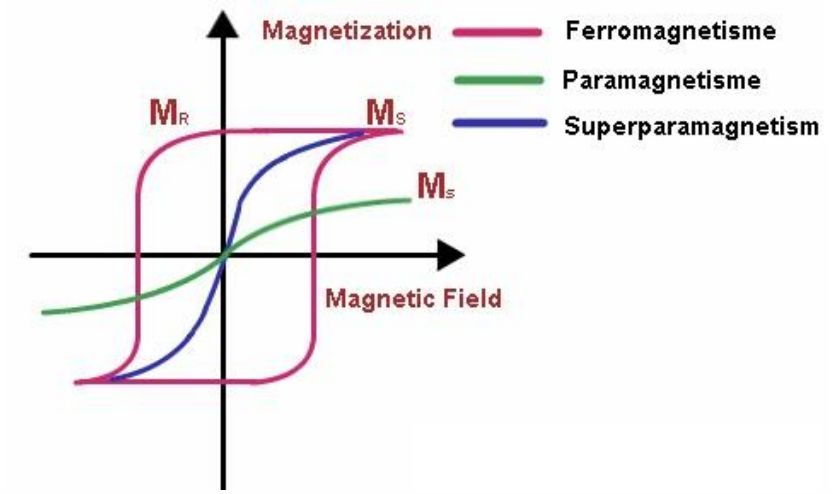
The University Of Queensland AUSTRALIA

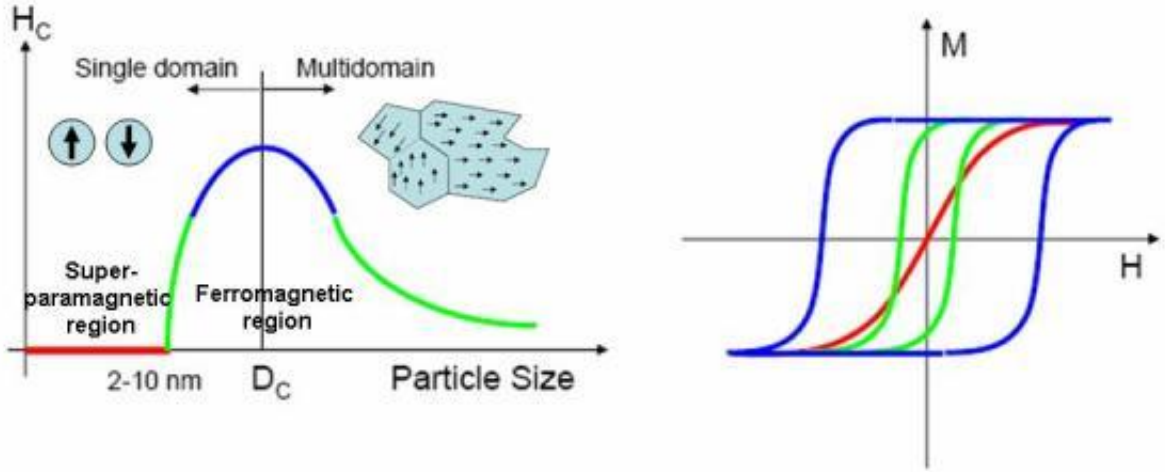
What is 'Super' about Superparamagnetism
Very fine ferromagnetic particles have very short relaxation times, even at room temperature, and behave superparamagnetic.

Superparmagnetic materials have:

- A high saturation magnetiszation (Ms)
- Zero coercivity (Hc) and
- Zero remanence (Mr),
- Single domain

Making it to be distinguished from ferromagnetism and paramagnetism - Superparamagnetism is a size effect of ferromagnetism.





The influence of magnetic particle size on magnetic properties. The coercivity changes with the particle size.

https://www.science20.com/mei/blog/blocking_temperatu

https://mappingignorance.org/2014/03/05/superparamagnetic-nanoparticles-and-the-separation-problem/

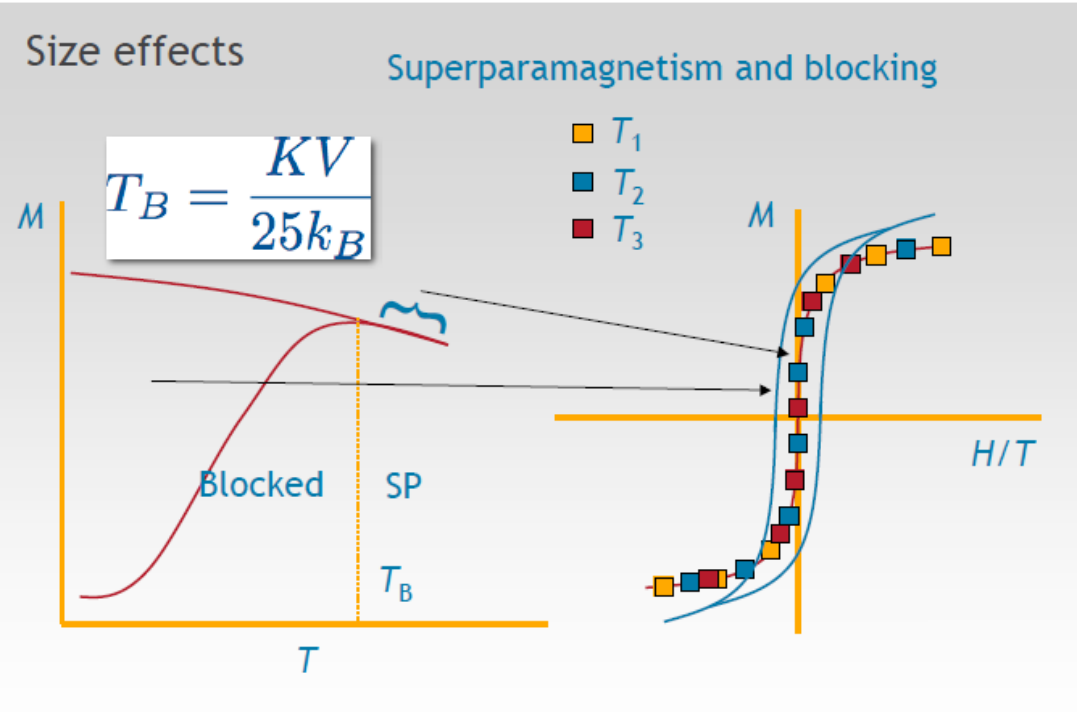
Ferromagnetism to superparamagnetism (transition to nanomagnetism)

THE UNIVERSITY OF QUEENSLAND AUSTRALIA

Some important features about Superparamagnetism

- Their behavior is paramagnetic (lower Ms) but their magnetization values are typical of ferromagnetic substances (higher Ms and zero Hc and Mr).

- The individual particles have normal ferromagnetic movements but very short relaxation times, enabling them to rapidly follow directional changes in an applied field.



- Magnetization curves measured at different temperatures superimpose when magnetization (*M*) is plotted as a function of field (*H*)/temperature (*T*).

- This also demonstrates that superparamagnetism can be destroyed by cooling.

- The temperature at which this occurs is called the blocking temperature (T_B) and is dependent linearly on volume and the magnitude of the crystal field anisotropy.

K is the effective magnetic anisotropy constant, V is the magnetic particle volume, kB is Boltzmann's constant, and TB is the **blocking temperature**.

Problem solving

You synthesize two batches of Fe_3O_4 nanoparticles:
Batch A: diameter DA=8 nm
Batch B: diameter DB=20 nm

Assume spherical particles, room temperature T=300K, Cu Kα XRD confirms single-crystal particles, and the effective anisotropy energy density is $K=1.0 \times 10^5$ J m^{-3}
Use the "blocking criterion" based on the Néel–Arrhenius law with a typical measurement window ($\ln(\tau m/\tau 0)\approx 25$), i.e.
Tb = KV/25kb
KV=25kbTb

Energy barrier, Eb = KV
Thermal fluctuation energy, Et = 25kbTb
Blocked/ferromagnetic if $EB=KV \gtrsim 25$ kBT, superparamagnetic if $EB \ll 25$ kBT.

Tasks
1. Compute the particle volumes VA and VB.
2. Compute the energy barriers EB,A and EB,B.
3. Compare each to 25kBT and decide whether each batch is superparamagnetic or blocked at 300 K.

Given Constants:
$kB=1.380649\times10^{-23}$ J/K (Boltzmann constant)
$25kBT=25\times kB\times 300\approx 1.035\times10^{-19}$ J
Sphere volume $V=4/3\pi r^3$

Solution

Thermal energy, $25k_bT=25\times kB\times 300\approx 1.035\times 10^{-19}$ J
Energy Barrier, $E_b = KV$

1) Volumes
Batch A:
$D_A=8$ nm$\Rightarrow rA=4\times 10^{-9}$ m
$V_A=4/3\pi(4\times 10^{-9})^3=4.18879\times 64\times 10^{-27}\approx 2.681\times 10^{-25}$ m3

Batch B:
$D_B=20$ nm$\Rightarrow rB= 20/2$ nm $= 1.0\times 10^{-8}$ m
$V_B=4/3\pi(1.0\times 10^{-8})^3=4.18879\times 10^{-24}$ m3

2) Energy barrier, $E_b=KV$
$E_b,A=1.0\times 10^5\times 2.681\times 10^{-25}\approx 2.681\times 10^{-20}$ J
$E_b,B=1.0\times 10^5\times 4.18879\times 10^{-24}\approx 4.189\times 10^{-19}$ J

3) Compare to thermal energy, $25k_bT\approx 1.035\times 10^{-19}$ J
Batch A (8 nm): $E_b,A\approx 2.681\times 10^{-20}$ J
thermal energy, $25k_bT > E_b,A \rightarrow$ superparamagnetic at 300 K.

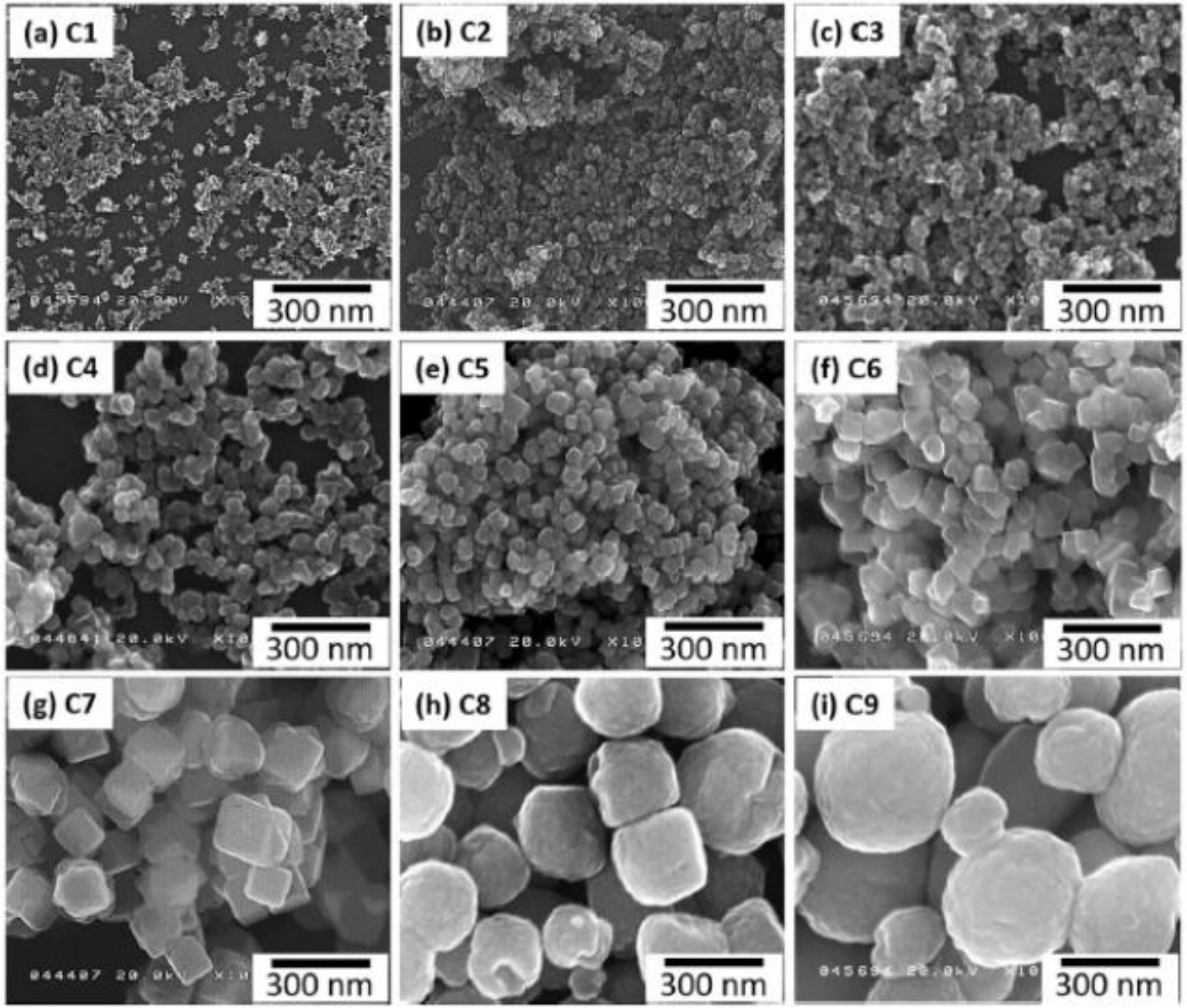
Batch B (20 nm): E_b, $B\approx 4.189\times 10^{-19}$ J
$25kBT < E_b$, $B \rightarrow$ blocked (ferromagnetic-like) at 300 K.
8 nm Fe_3O_4: superparamagnetic at room temperature.
20 nm Fe_3O_4: blocked at room temperature ferromagnetic (lost superparamagnetism).

Case study 3:

- **Highly crystalline single-domain 10-300nm magnetite Fe_3O_4 nanoparticles have been synthesized**
- **Large-scale liquid precipitation method was used**
- **Well-controlled crystal growth (by precise control of Fe^{2+} concentration, PH and temperature)**
- **Cubed and spherical shaped (By 2-stage and 1 stage oxidation reaction, respectively)**
- **Critical size was determined in terms of transforming from single and multi-domain structure**
- **Domain structures were determined by TEM using the lattice fringes**
- **Compared with cube-like NPs, sphere-like NPs have lower magnetic coercivity and remanence values, which is caused by the different orientations of their polycrystalline structure**

https://www.nature.com/articles/s41598-017-09897-5

Case study 3:



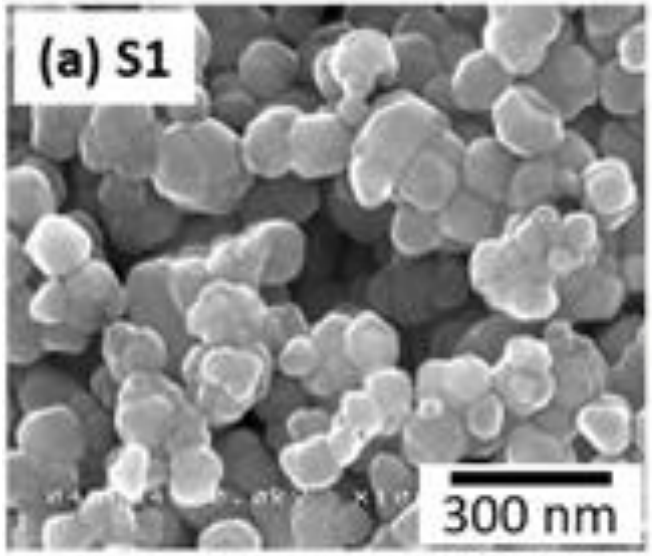
- Highly crystalline single-domain 10-300nm magnetite Fe3O4 nanoparticles have been synthesized using large-scale liquid precipitation method

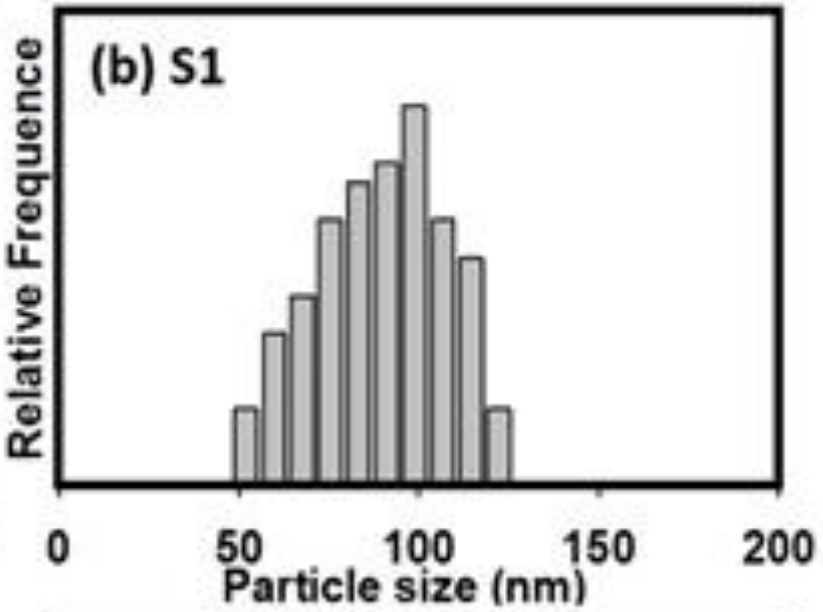
Scanning electron microscopy images of cube-like Fe3O4 nanoparticles with various particle sizes.

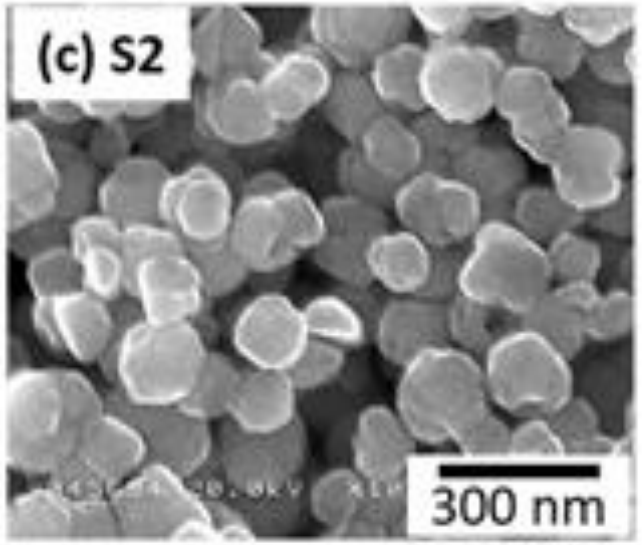
(a)9.6,
(b)19.6,
(c)24.4,
(d)31.9,
(e)45.3,
(f)64.7,
(g)130,
(h)243, and
(i)287 nm,

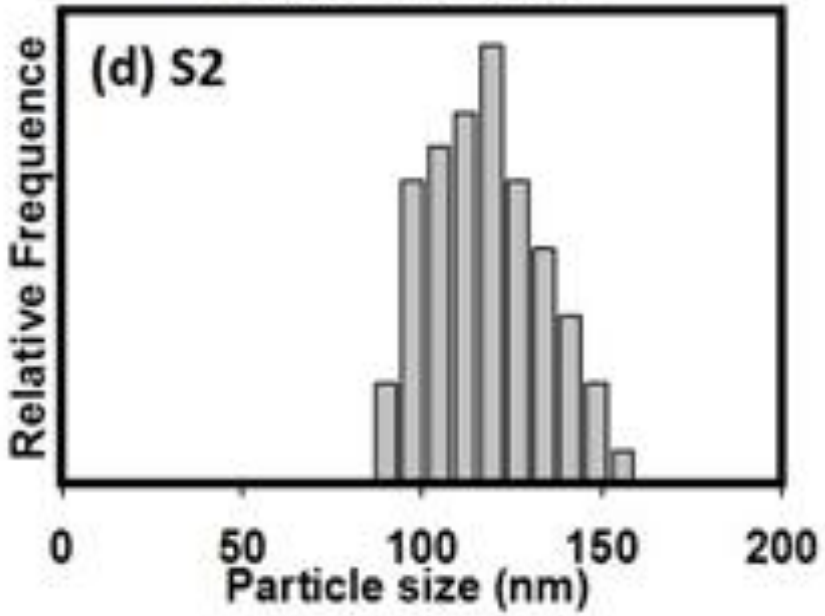
which are named as C1–C9 respectively.

Case study 3:









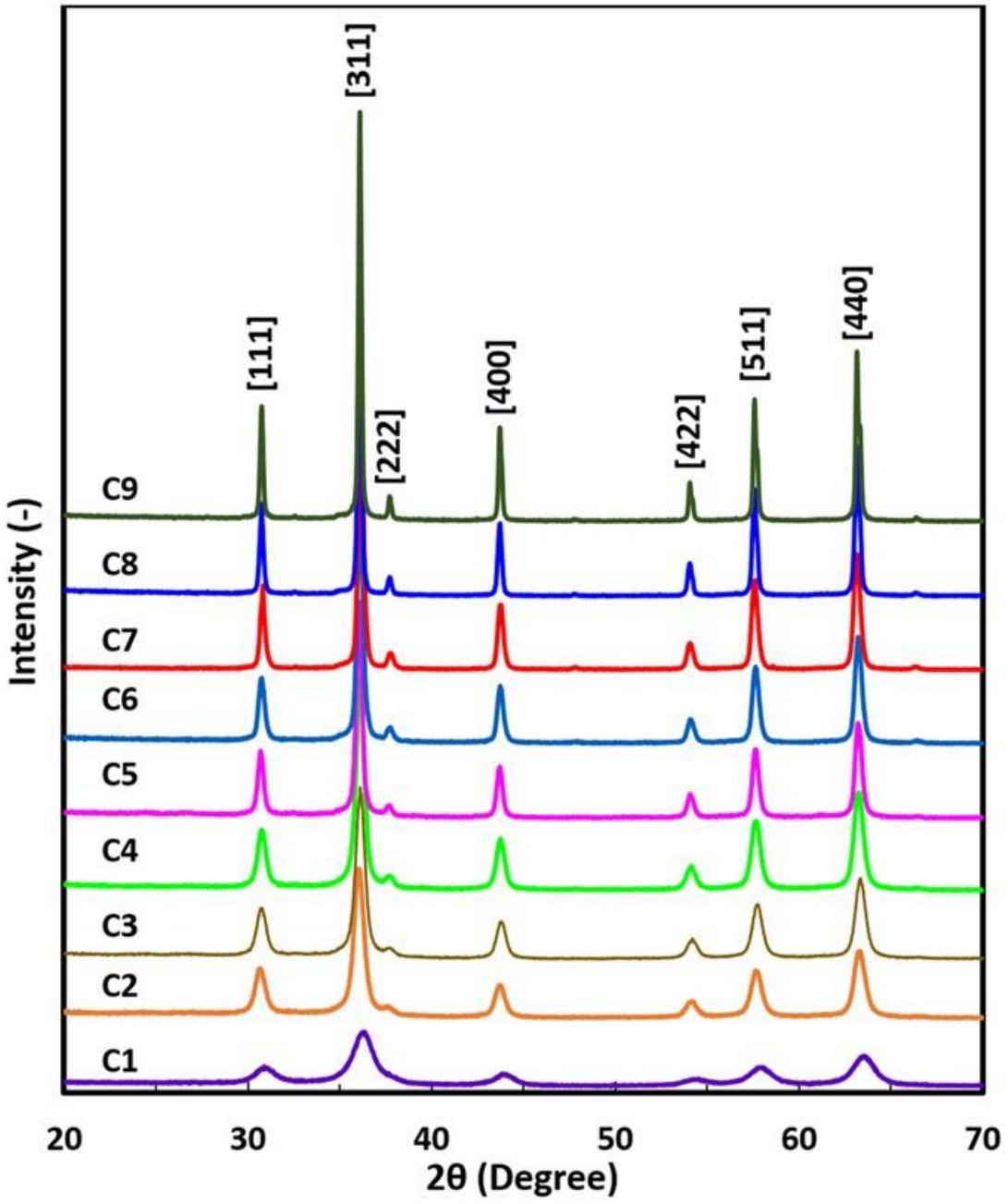
- Highly crystalline single-domain 10-300nm magnetite Fe3O4 nanoparticles have been synthesized using large-scale liquid precipitation method

Particle properties of sphere-like Fe_3O_4 nanoparticles:

S1 (93.3 nm) and
S2 (121 nm):

(a) and (c) scanning electron microscopy images,
(b) and (d) particle size distributions

Case study 3:



Compared to our previous case study:

- The sharp peaks indicate the highly crystalline nature of the Fe_3O_4 NPs.

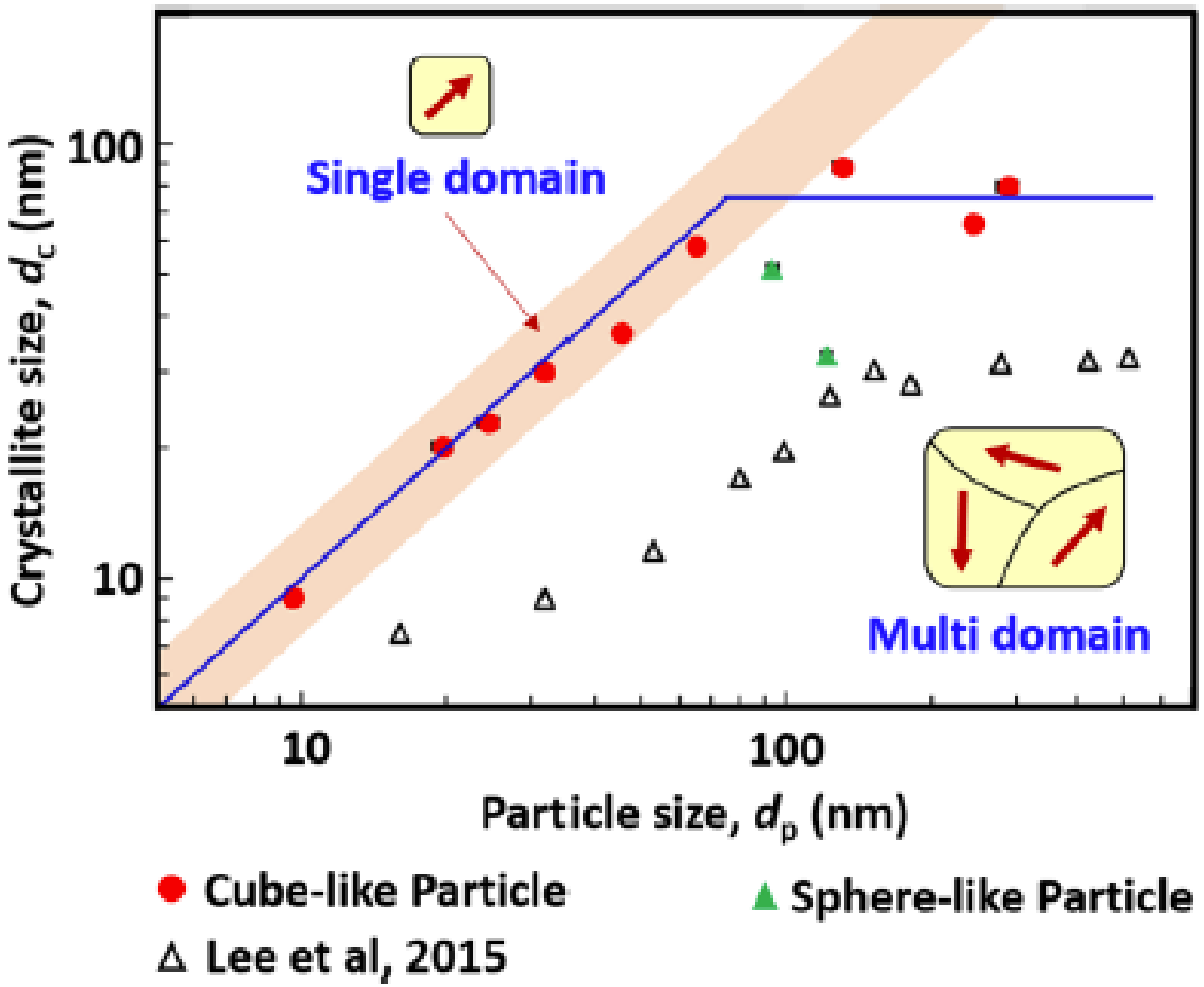
- Impurity peaks and transition phases were not observed, which indicates that the particles prepared by the liquid precipitation method were pure, both chemically and in their crystalline phase.

The crystallite sizes (d_c) of all the Fe_3O_4 NPs were estimated by the Scherrer formula using the highest intensity XRD peak, namely [311], and compared with the d_p obtained from the TEM analysis

Case study 3:

- Highly crystalline single-domain 10-300nm magnetite Fe3O4 nanoparticles have been synthesized using large-scale liquid precipitation method

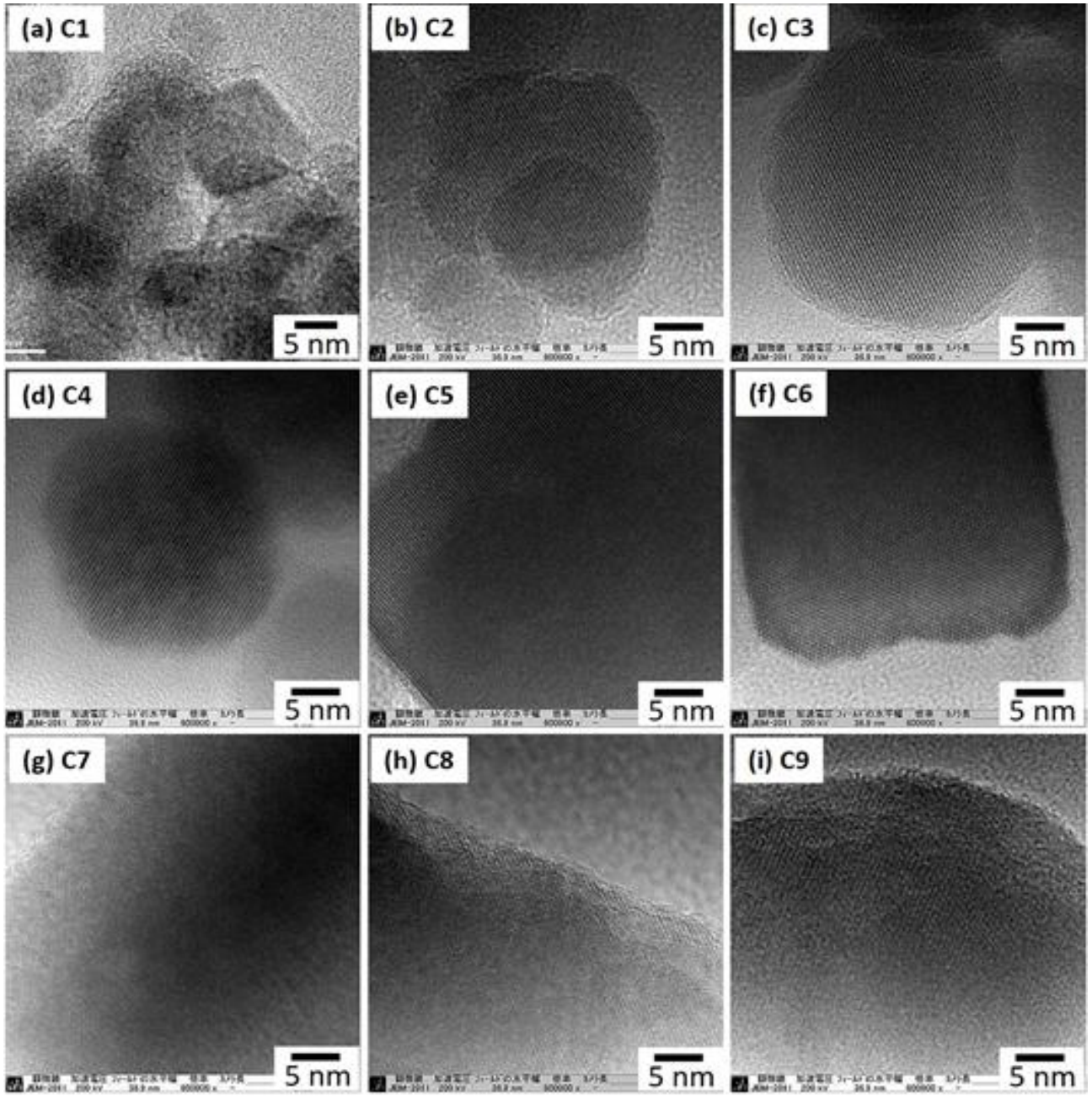
Sample	Particle size d_p (nm)	Crystal size d_c(nm)
C1	9.6	9.1
C2	19.6	20.2
C3	24.4	22.9
C4	31.9	29.9
C5	45.3	36.9
C6	64.7	58.2
C7	130	88.7
C8	243	65.8
C9	287	79.6
S1	93.3	51.7
S2	121	32.4



Particle size was calculated from SEM/TEM/Particle size analyser and crystallite size was calculated by Debye– Scherrer equation from XRD.

The straight line indicates a single crystalline particle, whereas the particles corresponding to points that fall below this line are polycrystalline. the dc and corresponding dp have almost the same value for particles with a diameter of 10 to around 80 nm, which indicates a single-domain structure. The particles with diameters of larger than 80 nm have a constant dc, which indicates a multi-domain structure. This can be explained the by the single and multi directional lattice fringes by TEM

Case study 3:

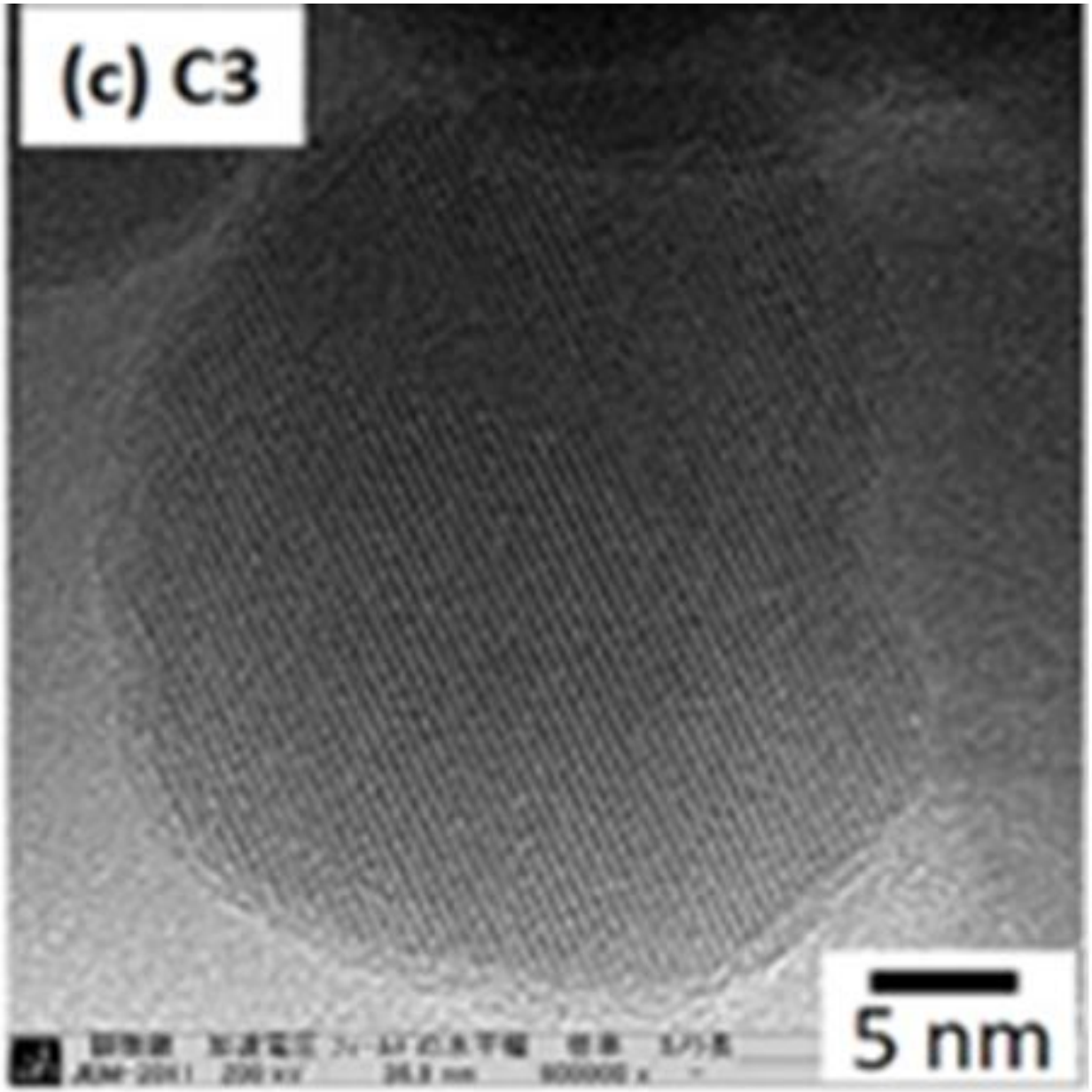


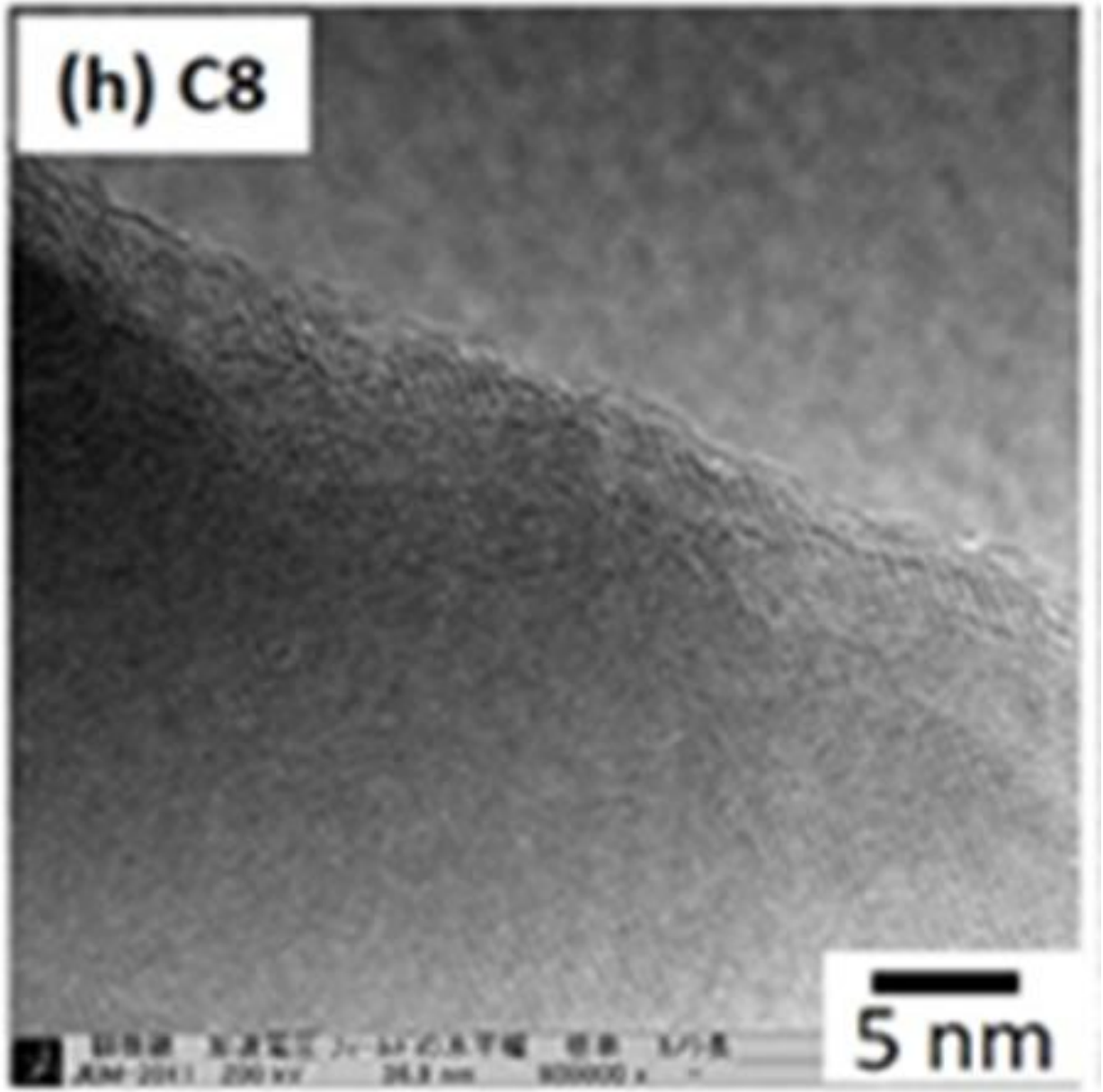
- Highly crystalline single-domain 10-300nm magnetite Fe3O4 nanoparticles have been synthesized using large-scale liquid precipitation method

High resolution transmission electron microscopy images of cube-like Fe_3O_4 nanoparticles with various sizes. (a) 9.6, (b) 19.6, (c) 24.4, (d) 31.9, (e) 45.3, (f) 64.7, (g) 130, (h) 243, and (i) 287 nm, which are named C1–C9, respectively.

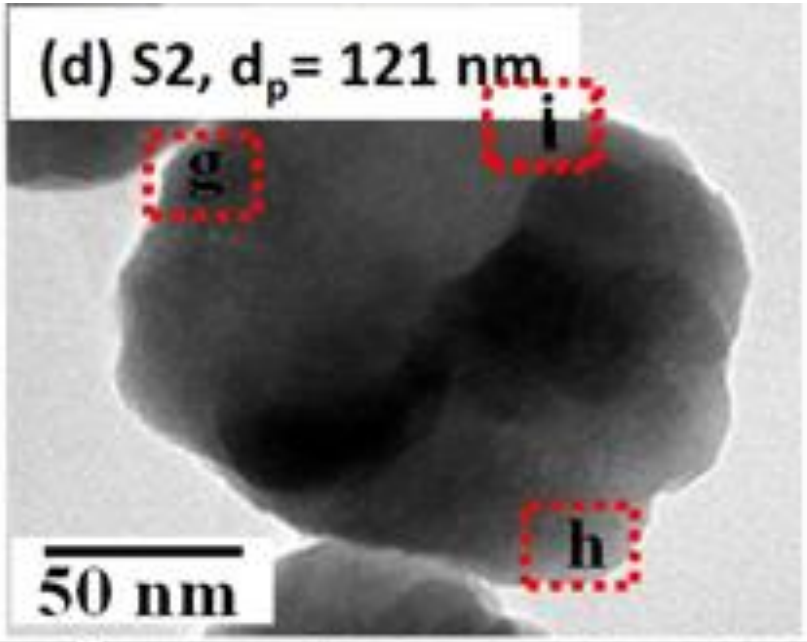
TEM shows the single direction lattice fringes of the NPs, and indicate that the NPs have a single crystalline structure up to a *dp* of 64.7 nm. In contrast, the C7–C9 NPs, with diameters larger than 100 nm, show a polycrystalline structure multi directional fringes.

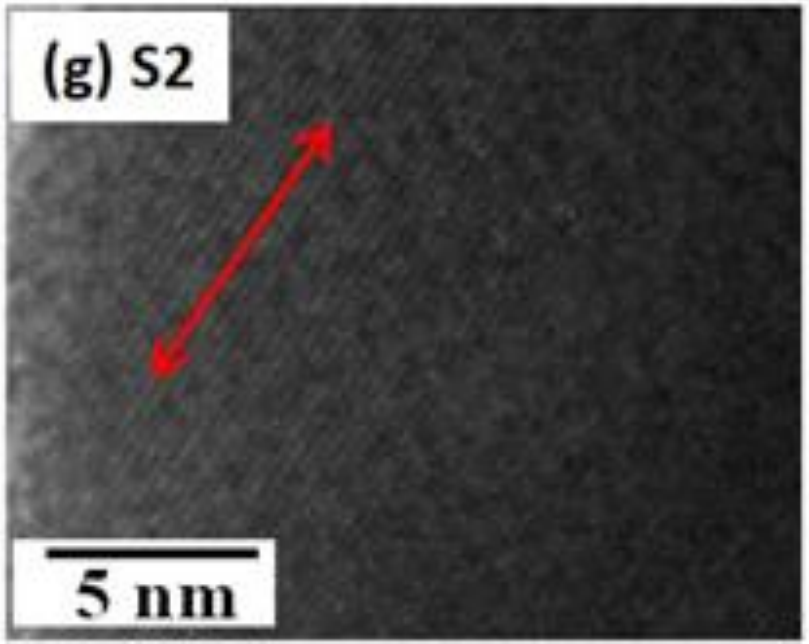
Case study 3:

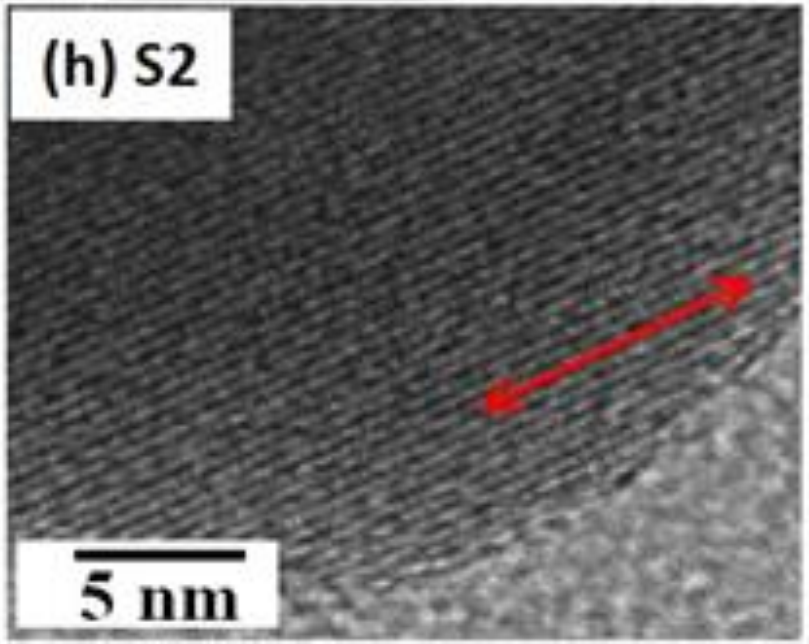


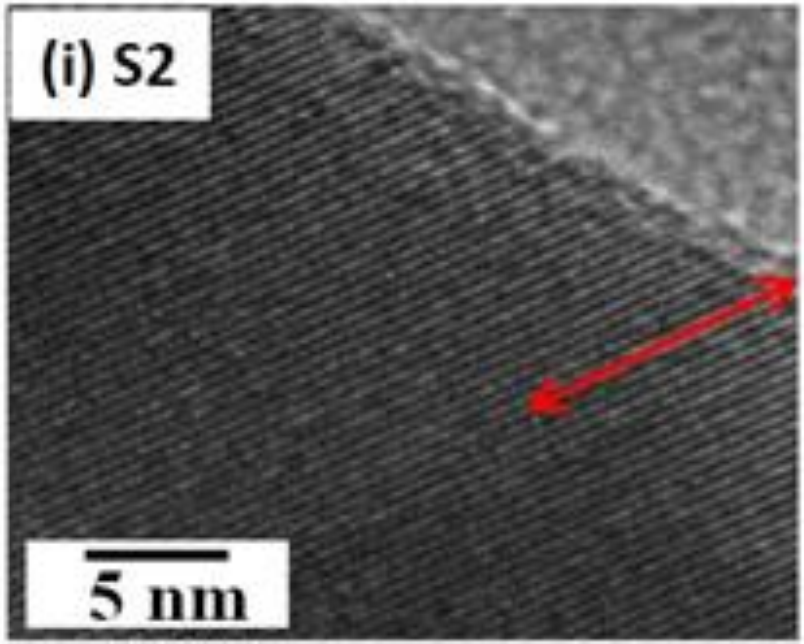


Case study 3:



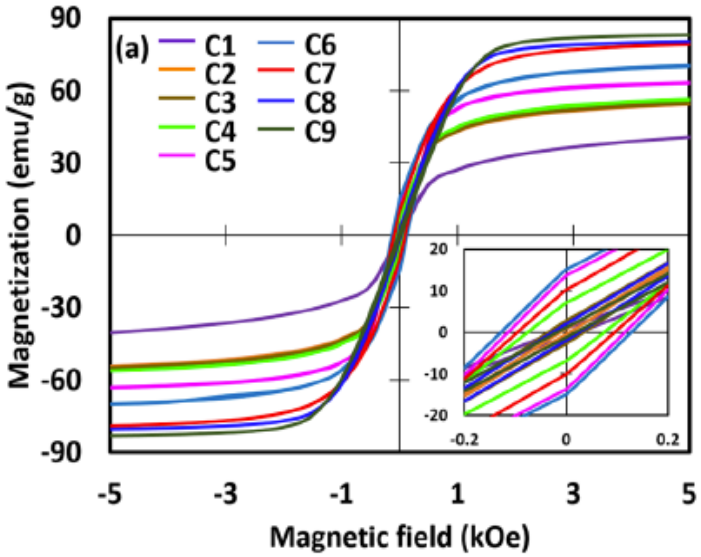


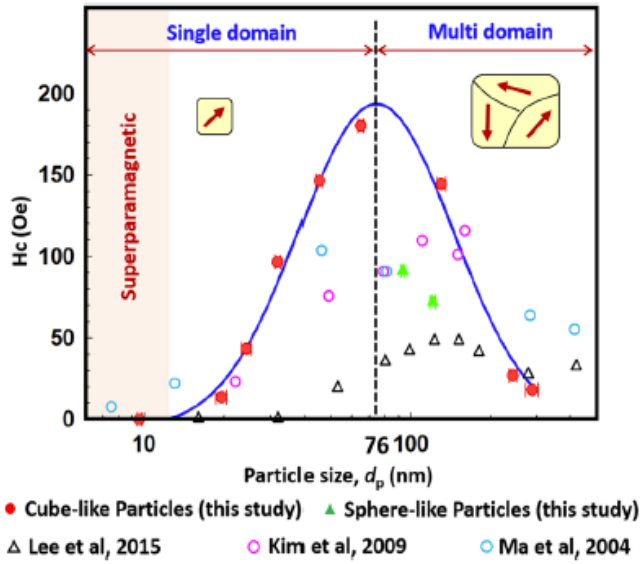


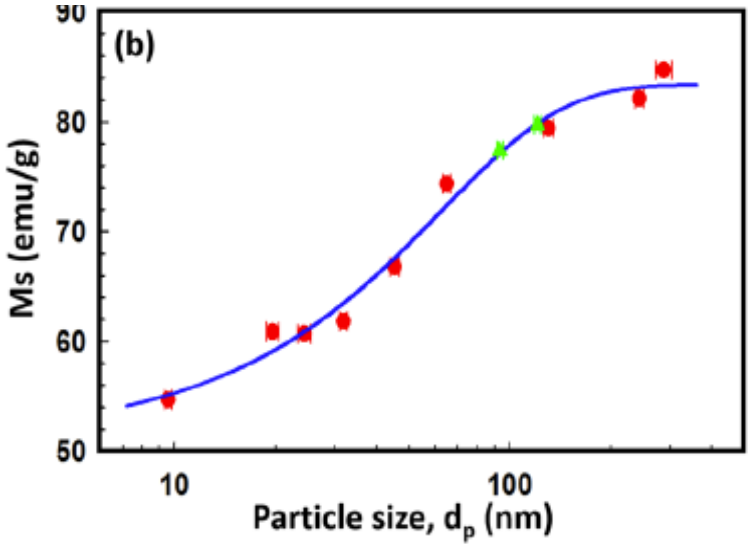


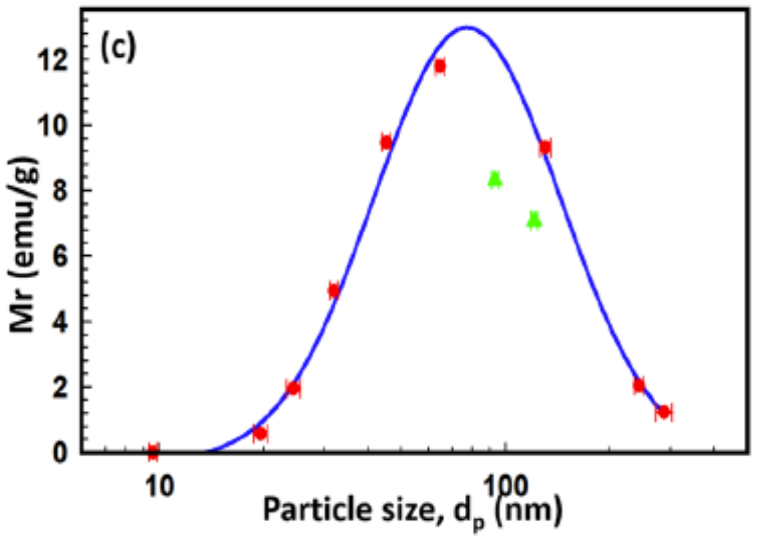
Case study 3:

- Highly crystalline single-domain 10-300nm magnetite Fe3O4 nanoparticles have been synthesized using large-scale liquid precipitation method









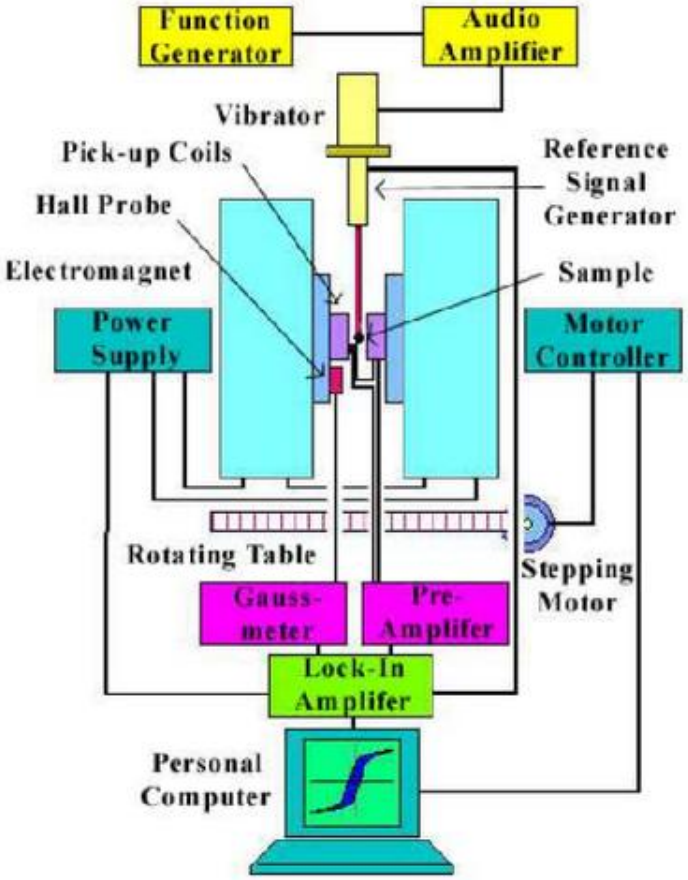
Cube-like Particles Sphere-like Particles

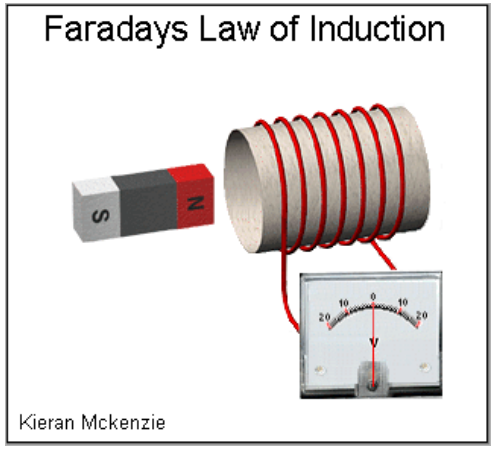
1. The hysteresis loops for these particles, show the ferrimagnetic nature.
2. The *Ms* increases with increasing particle size, regardless of the crystal structure and particle shape.
3. The *Hc* and remanent magnetisation (*Mr*), which are also affected by *d*p
4. Superparamagnetic when Mr = 0 and Hc =0, C1 samples is only superparamagnetic
5. 76nm (single domain) sample has highest Mr and Hc. After that Mr and Hc decreases even further increase of dp (multi domain)
6. Multiple orientation of the polycrystalline samples crystallites within the multi-domain-structured NPs is considered to be the main reason for their lower Mr and Hc.

These findings suggest that considerable attention should be given to the particle size and crystalline properties of Fe3O4 NPs, which have potential biomedical and industrial applications. These applications require that magnetic particles are sized appropriately to achieve a good balance between effective surface area and satisfactory magnetic performance.

Magnetic properties characterization techniques

Vibrating Sample Magnetometer (VSM)





https://www.daenotes.com/electronics/basic-electronics/faraday-laws-of-electromagnetic-induction

1. VSM works as Faraday's electromagnetic induction
2. A VSM is used to measure the magnetic behaviour of magnetic materials.
3. Placing the sample to be studied in a constant magnetic field (by electromagnet)
4. If the sample is magnetic, this constant magnetic field will magnetize the sample
5. The magnetic dipole moment of the sample will create a magnetic field around the sample, sometimes called the magnetic stray field

6. As the sample is moved up and down, this magnetic stray field is changing as a function of time and can be sensed by a set of pick-up coils.

7. The alternating magnetic field will cause an electric field in the pick-up coils according to Faraday's Law of Induction. This current will be proportional to the magnetization of the sample.

8. The induction current is amplified by a trans impedance amplifier and lock-in amplifier

9. system can tell you how much the sample is magnetized and how its magnetization depends on the strength of the constant magnetic field.

http://gato-docs.its.txstate.edu/jcr:7543b8ec-f619-49c3-888f-01c3ad13c7da/vsm.pdf

Magnetic properties characterization techniques

THE UNIVERSITY OF QUEENSLAND AUSTRALIA

Superconducting Quantum Interface Device (SQUID)

The superconducting quantum interference device (SQUID) are very sensitive magnetometers based on superconducting loops consists of two superconductors separated by thin insulating layers to form two parallel **Josephson junctions**.

SQUID magnetometer is one of the most effective and sensitive ways of measuring magnetic properties.

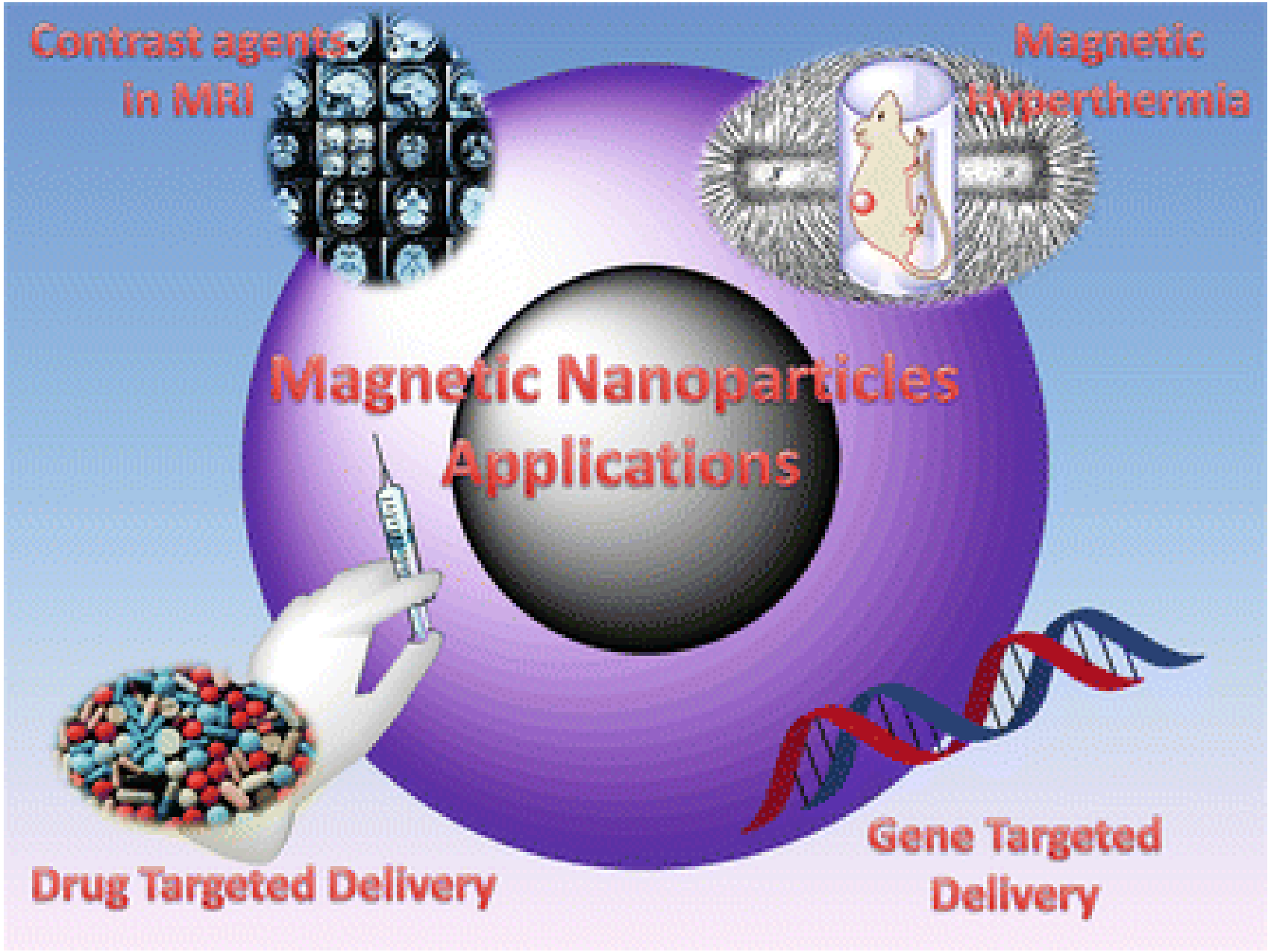
The device may be configured as a magnetometer to detect incredibly small magnetic fields small enough to measure the magnetic fields in living organisms.

Squids have been used to measure the magnetic fields in mouse brains to test whether there might be enough magnetism to attribute their navigational ability to an internal compass.

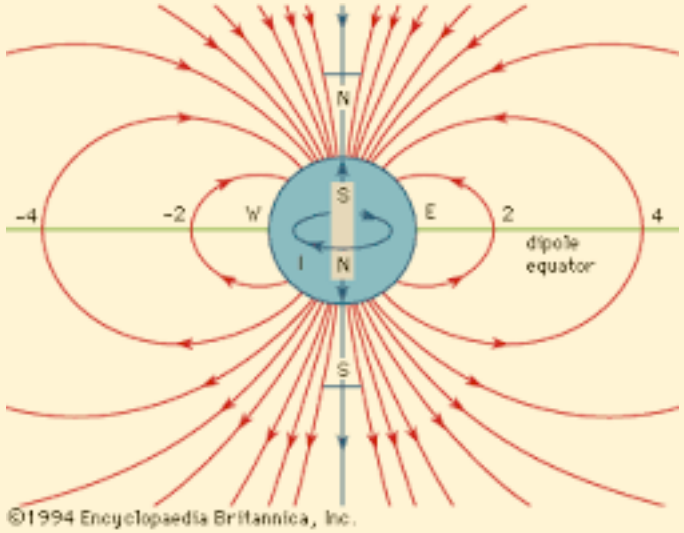
Cryocooler Compressor: The MPMS EverCool system features an integrated cryocooler–Dewar System that recondenses the He directly with in the EverCool Dewar.

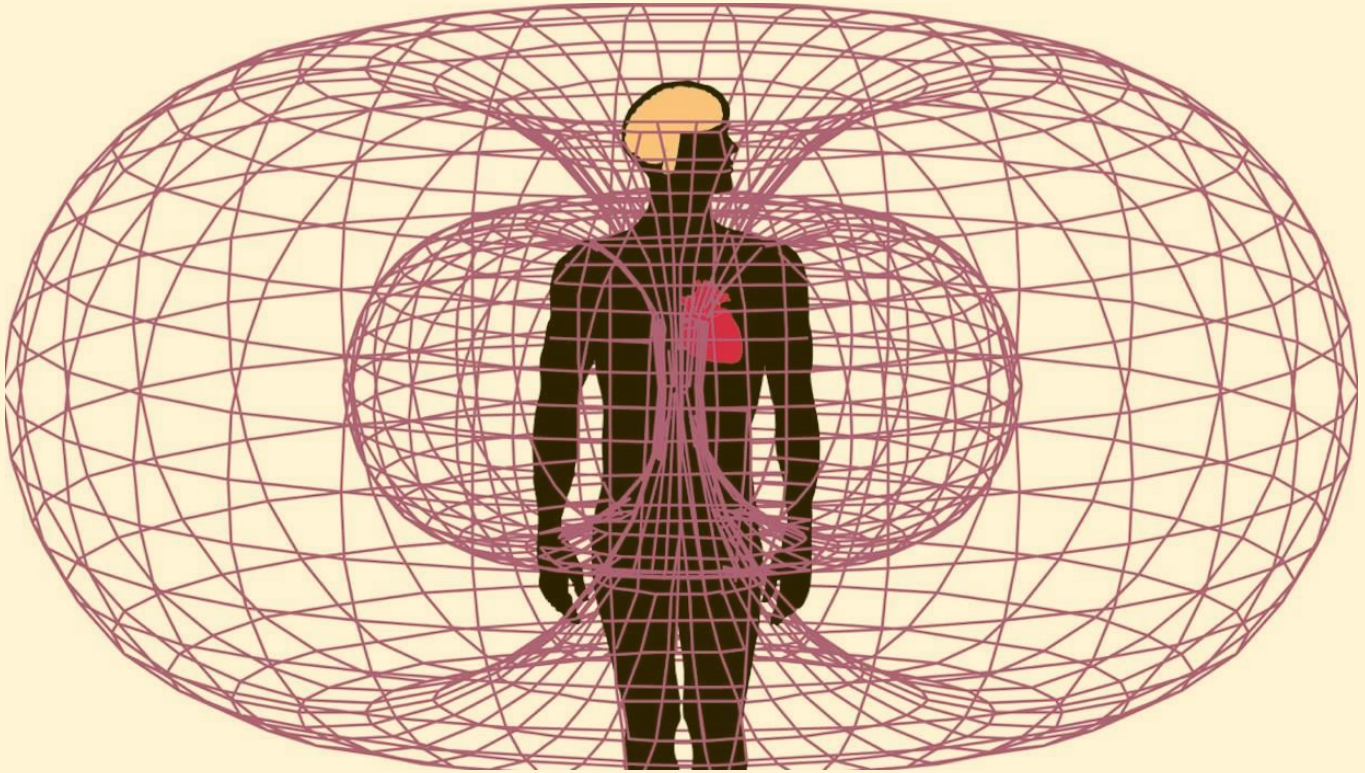
http://squid.iitd.ernet.in/Basic_Literature.htm

Applications – magnetic nanoparticles



Bringing the Magnetic field into Biomedical Applications





Applications – Nano-machineries

DARLINGHURST

Forces fight for a cure

A NEW method of targeting tumours using magnetic forces could improve the treatment and wellbeing of cancer patients.

Researchers at Darlinghurst's Victor Chang Cardiac Research Institute and the University of Wollongong are developing the method that will enable doctors to deliver cancer medications directly to a tumour.

The proposed treatment involves attaching tiny magnetic particles to drug carriers that can be guided through the body to the cancer using a magnetic field. A magnetic field will also be used to release the drugs directly to the cancer cells.

Medications ingested for conventional cancer treatments disperse through the bloodstream and attach to the cancer cells where they interfere with the molecules responsible for its growth.

Dr Shahriar Hossain and Prof Boris Martinac are developing tumours targeting using a magnetic method.

MAGNETIC CANCER TREATMENT

Large doses of highly toxic cancer medications are required to kill a tumour, often leading to serious side effects and damaging healthy tissues and organs.

Prof Boris Martinac of the Victor Chang Institute, said magnetically targeted drug delivery would help to reduce the amount of cancer-killing medication needed to kill the tumour and would reduce side effects.

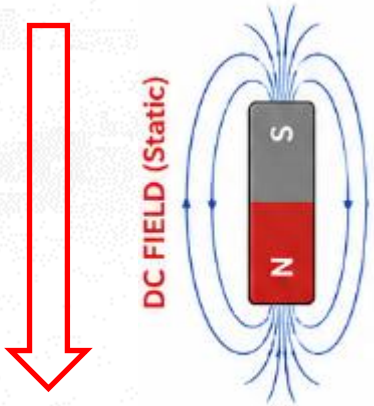
"We want to attach tiny magnetic particles to cancer drugs. The doctor can then direct the drugs through the patient's bloodstream to the tumour using a handheld magnetic device that emits a magnetic field, but is not harmful for the patient," he said.

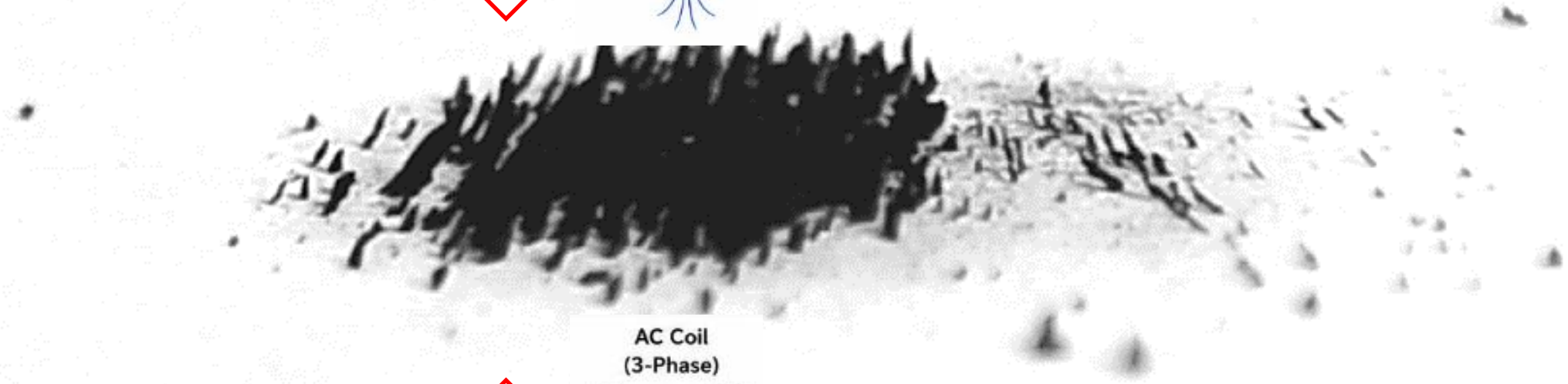
"When the drugs reach the tumour different alternating magnetic forces can be applied to the particles, causing the protein molecules to open, much like a valve or turning on a tap, releasing the drug that will then attack the cancer cells.

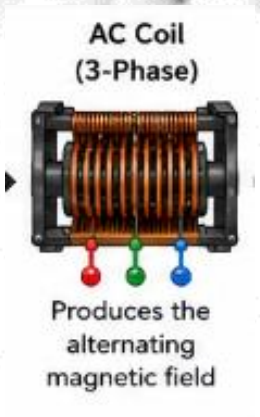
He said it was exciting because it could potentially reduce the need for current treatments such as chemotherapy and surgery.

UOW materials scientist and Australian Research Council Discovery Early Career Researcher Award Fellow, Dr Shahriar Hossain and his team, will build a device that doctors could use to "drive" the drugs through the body.

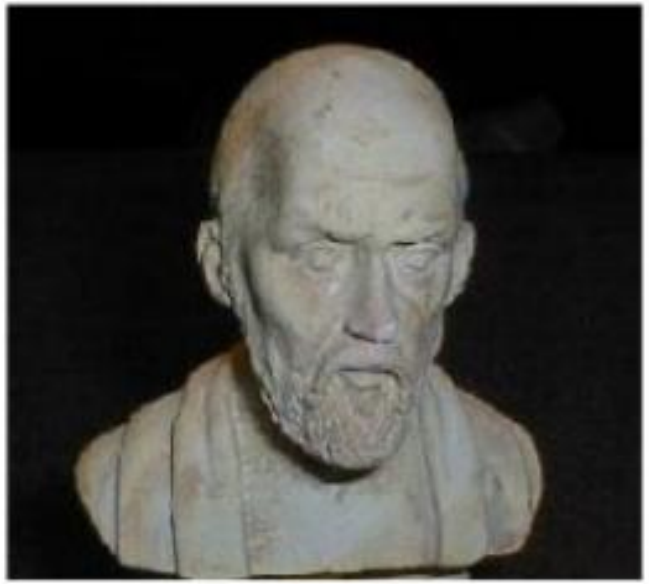
Magneto-Mechanical Nano-machineries for Precision Drug Release







Applications – Magnetic Hyperthermia – Nano-heaters



"Those diseases which medicines do not cure,
the knife cures;
those which the knife cannot cure, fire cures;
and those which fire cannot cure,
are to be reckoned wholly incurable."

—Hippocrates (460-370 BC)

- Physical properties which make magnetic materials attractive for biomedical applications are,
  - First, that they can be manipulated by an external magnetic field – this feature is useful for separating, immunoassay and drug targeting, and
  - Second, hysteresis and other losses occur in alternating magnetic fields – this is useful in hyperthermia applications.

Applications – Hyperthermia – Nano-heaters

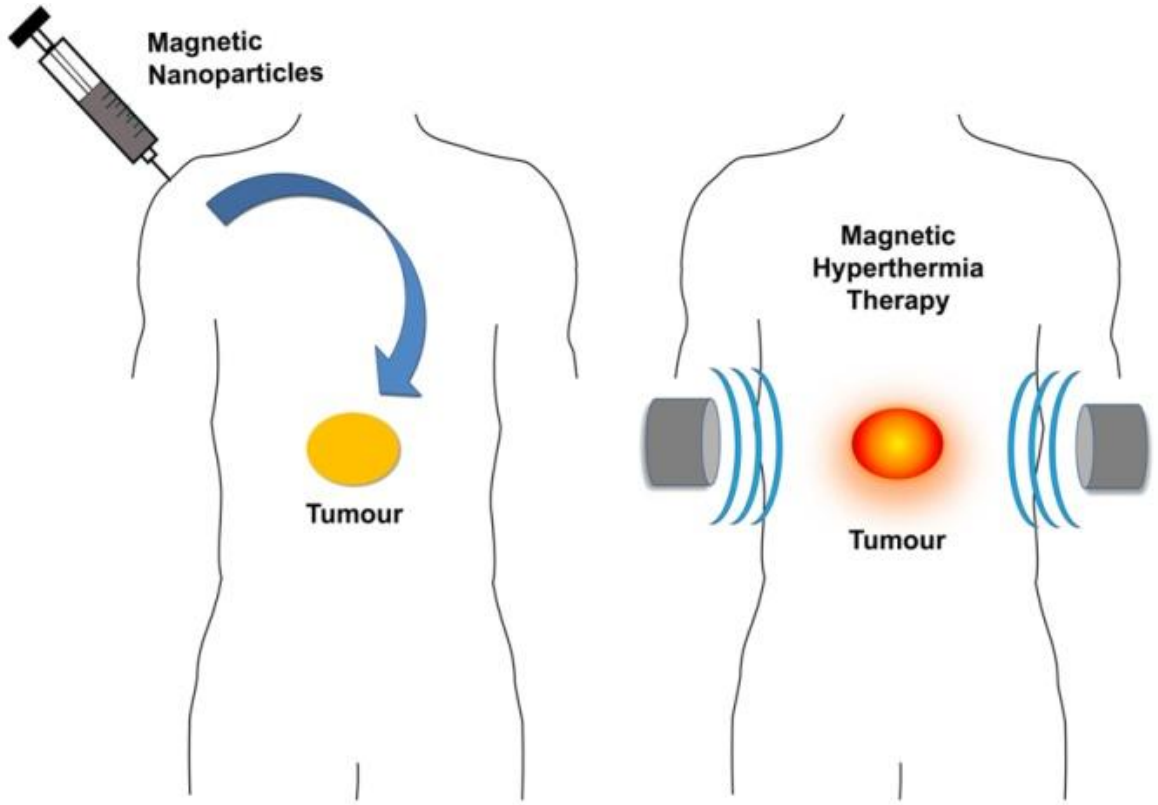
- **Hyperthermia** is an elevated body temperature due to failed thermoregulation. Hyperthermia occurs when the body produces or absorbs more heat than it can dissipate.
- Hyperthermia can be created artificially by drugs or medical devices. Hyperthermia therapy may be used to treat some kinds of cancer and other conditions, most commonly in conjunction with radiotherapy. It is known that cancer cells show more weakness to high temperatures compared with ordinary cells.
- Hyperthermia system that is able to heat tumor cells above 43°C without damaging normal tissues is required.

Applications – Hyperthermia – Nano-heaters

Applications – Hyperthermia – Nano-heaters

The magnetic moments on nanoparticles will align with an external magnetic field. As the external field changes direction, the magnetic moment will also change direction. This produces dissipation leading to heating. One of the major advantages of using magnetic fields to produce heating is that they readily penetrate biological tissues

However, heating the surrounding tissue can also produce unwanted side-effects so there are advantages strictly controlling the region under treatment by using magnetic nanoparticles as the heating element. Superparamagnetic nanoparticles absorb power from AC magnetic fields and heat surrounding tissues.



https://www.intechopen.com/books/biomedical-engineering-frontiers-and-challenges/coating-nanomagnetic-particles-for-biomedical-applications

Applications – Hyperthermia – Nano-heaters

The University Of Queensland AUSTRALIA

This thermotherapy is based on the fact that magnetic nanoparticles can transform electromagnetic energy from an external high-frequency field to heat.

As a consequence, if magnetic nanoparticles are put inside a tumor and the whole patient is placed in an alternating magnetic field of well-chosen amplitude and frequency, only the tumor temperature would rise. The elevation of temperature may enhance tumor oxygenation , radio- and chemosensitivity, thus shrinking tumors.

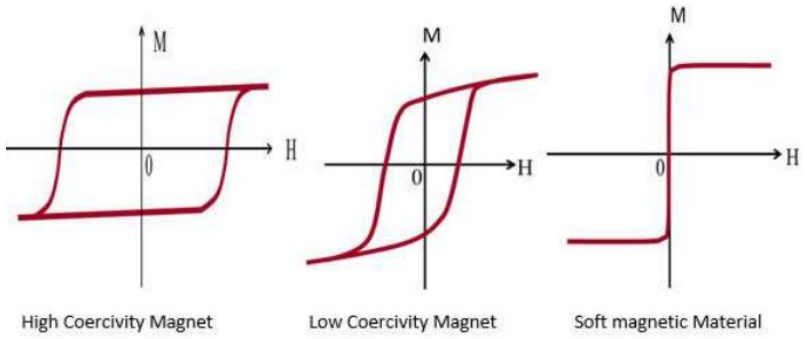
Difficulty in heating the local tumor region WITHOUT DAMAGING normal tissue

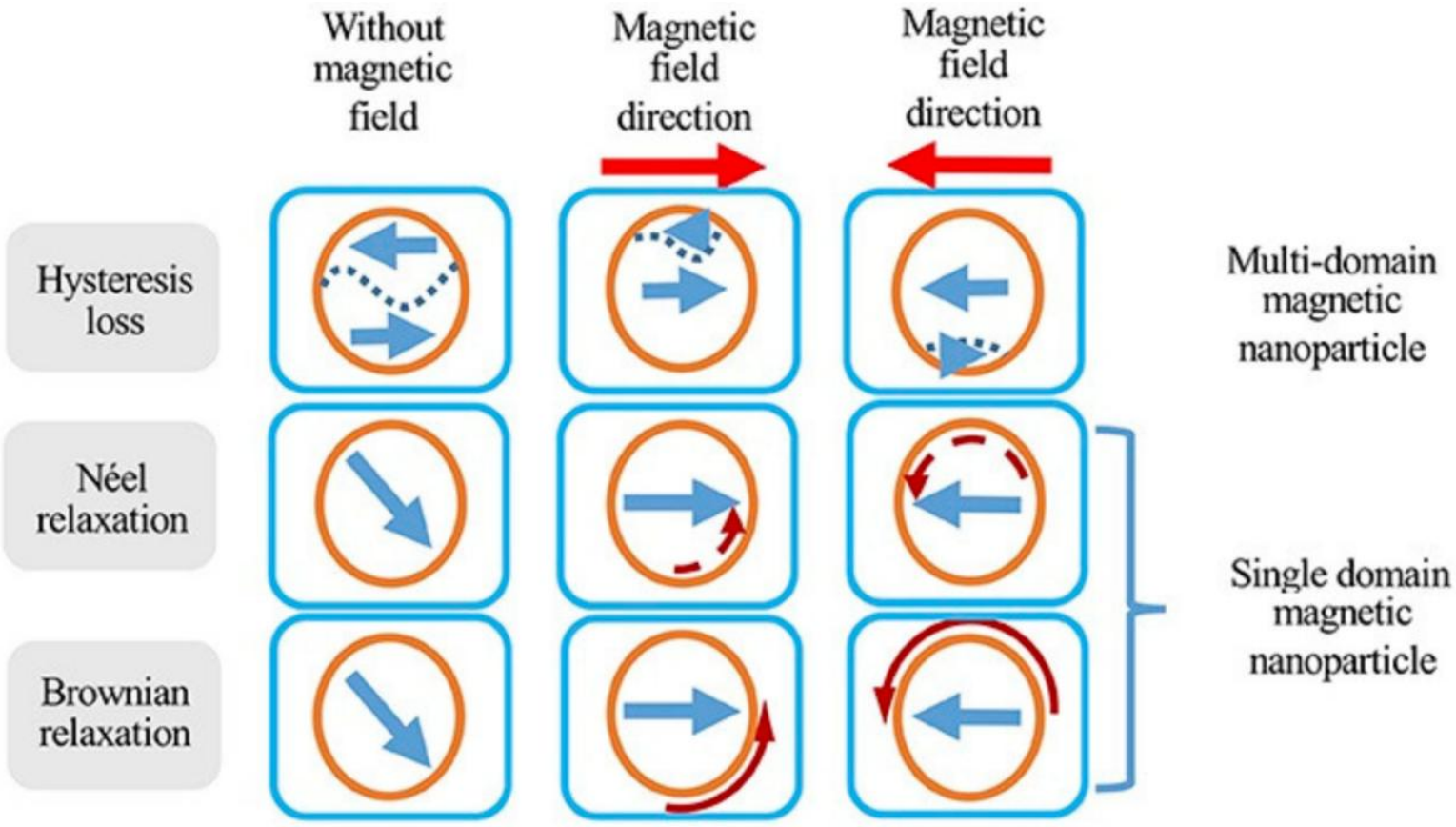
Applications – Hyperthermia – Nano-heaters

What are the mechanism?

Heat dissipation from the magnetic nanoparticles under an AC field is caused by

1)Brownian relaxation loss
2)Neel's spin relaxation
3)Hysteresis loss and
4)Eddy current loss.

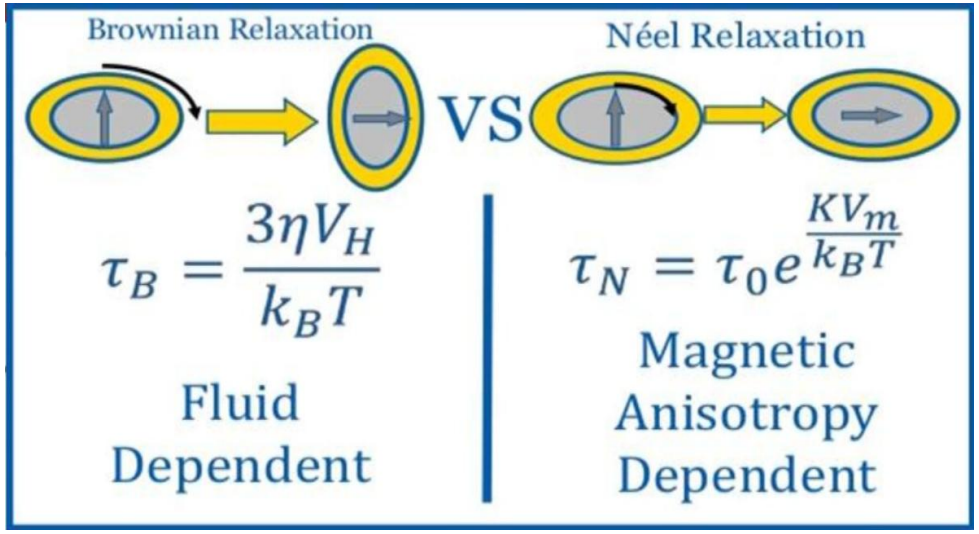




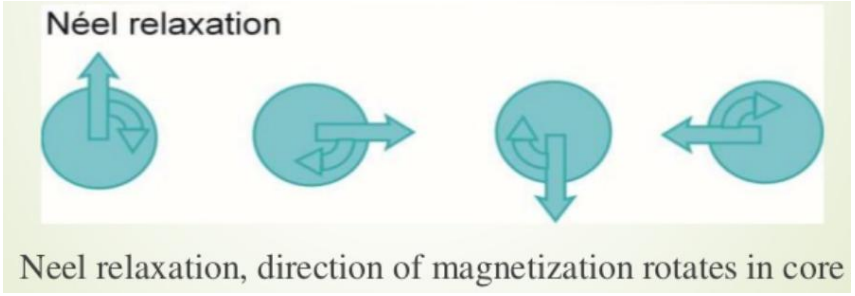
https://www.mdpi.com/1424-8220/20/9/2554/htm

Magnetic Particle Relaxation

The well dispersed magnetic nanoparticles in a fluid would show two types of relaxations, ***Brownian relaxation*** (particle rotation, occurs in all types of nanoparticles) and ***Neel's spin relaxation*** (magnetic moment rotation, occurs only in superparamagnetic nanoparticles where no physical rotation of the particle is possible).



η is the dynamic viscosity of carrier liquid and
V is the hydrodynamic radius of the particle,
k_B is Boltzmann constant,
$\tau 0 = 10^{-9}$ s



Case study problem solving

Two magnetite nanoparticles are being considered for biomedical sensing:

Particle S (small core, coated): core diameter $D_c = 8\ \mathrm{nm}$; hydrodynamic diameter $D_h = 12\ \mathrm{nm}$.
Particle L (large core, lightly coated): core diameter $D_c = 20\ \mathrm{nm}$; hydrodynamic diameter $D_h = 22\ \mathrm{nm}$.

Use:

- $K = 1.0 \times 10^4\ \mathrm{J\,m^{-3}}$
- $k_B = 1.38 \times 10^{-23}\ \mathrm{J\,K^{-1}}$
- $T = 300\ \mathrm{K}$
- $\tau_0 = 1 \times 10^{-9}\ \mathrm{s}$
- viscosity $\eta = 1.0 \times 10^{-3}\ \mathrm{Pa \cdot s}$
- sphere volume $V = \frac{4}{3}\pi\left(D_c/2\right)^3$,hydrodynamic volume $V_h = \frac{4}{3}\pi\left(D_h/2\right)^3$
- Néel relaxation with $E_B = KV$
- Brownian relaxation $\tau_B = 3\eta V_h/(k_B T)$
- Effective relaxation $\tau_{eff} = \frac{\tau_N \tau_B}{\tau_N + \tau_B}$.

Decide for each particle: (a) is it superparamagnetic (use $E_B$vs $25k_BT$ as the blocking criterion)? (b) which relaxation dominates (shorter time)?

Case study problem solving - answer

Common numbers

- $k_B T = 1.38 \times 10^{-23} \times 300 = 4.14 \times 10^{-21}$ J
- $25 k_B T = 1.035 \times 10^{-19}$ J(used as blocking threshold)

Particle S (8 nm core, 12 nm hydrodynamic)

1.Core volume

$r = 4\ \text{nm} = 4.0 \times 10^{-9}\ \text{m}\rightarrow$

$$V = \frac{4}{3}\pi r^3 = 4.18879 \times \left(4.0 \times 10^{-9}\right)^3 = 4.18879 \times 64 \times 10^{-27} = 2.6808 \times 10^{-25}\ \text{m}^3.$$

2.Energy barrier

$$E_B = KV = 1.0 \times 10^4 \times 2.6808 \times 10^{-25} = 2.68 \times 10^{-21}\ \text{J}.$$

3.Superparamagnetic check

$E_B = 2.68 \times 10^{-21} \ll 25 k_B T = 1.035 \times 10^{-19} \rightarrow$**superparamagnetic** (thermally unstable / flips easily).

4.Néel relaxation time

exponent $= E_B/(k_B T) = 2.68 \times 10^{-21}/4.14 \times 10^{-21} \approx 0.647.$

$$\tau_N = 10^{-9}\, e^{0.647} \approx 1.91 \times 10^{-9}\ \text{s}.$$

5.Hydrodynamic volume (Dh = 12 nm → rh = 6 nm)

$$V_h = 4.18879 \times \left(6.0 \times 10^{-9}\right)^3 = 6.556 \times 10^{-23}\ \text{m}^3.$$

6.Brownian relaxation time

$$\tau_B = \frac{3\eta V_h}{k_B T} = \frac{3 \times 10^{-3} \times 6.556 \times 10^{-23}}{4.14 \times 10^{-21}} \approx 6.0 \times 10^{-7}\ \text{s}.$$

7.Effective relaxation & dominant mechanism

$\tau_N \approx 1.9 \times 10^{-9}$ svs $\tau_B \approx 6.0 \times 10^{-7}$ s.

→ **Néel much faster** → **Néel-dominated** (internal flips dominate). $\tau_{eff} \approx \tau_N \approx 2 \times 10^{-9}$s.

Case study problem solving

THE UNIVERSITY OF QUEENSLAND AUSTRALIA

Particle L (20 nm core, 22 nm hydrodynamic)

1.Core volume

$r = 10\ \text{nm} = 1.0 \times 10^{-8}\ \text{m}\rightarrow$

$$V = 4.18879 \times \left(1.0 \times 10^{-8}\right)^3 = 4.18879 \times 10^{-24}\ \text{m}^3.$$

2.Energy barrier

$$E_B = KV = 1.0 \times 10^4 \times 4.18879 \times 10^{-24} = 4.19 \times 10^{-20}\ \text{J}.$$

3.Superparamagnetic check

$E_B = 4.19 \times 10^{-20} < 25k_BT = 1.035 \times 10^{-19}$ →still **technically superparamagnetic** by the $25k_BT$ rule (i.e., thermally able to flip), but the energy barrier is substantially larger than for Particle S.

4.Néel relaxation time

exponent $= E_B/(k_BT) = 4.19 \times 10^{-20}/4.14 \times 10^{-21} \approx 10.12.$

$\tau_N = 10^{-9}\, e^{10.12} \approx 2.48 \times 10^{-5}$ s(≈ 25 μs).

1.Hydrodynamic volume (Dh = 22 nm → rh = 11 nm)

$$V_h = 4.18879 \times \left(11.0 \times 10^{-9}\right)^3 \approx 4.04 \times 10^{-24}\ \text{m}^3.$$

2.Brownian relaxation time

$\tau_B = \frac{3\eta V_h}{k_BT} = \frac{3\times10^{-3}\times4.04\times10^{-24}}{4.14\times10^{-21}} \approx 2.93 \times 10^{-6}$ s (≈ 3 μs).

(Depending on rounding, values ≈ 3–4 μs are typical.)

3.Effective relaxation & dominant mechanism

$\tau_N \approx 25 \times 10^{-6}$ svs $\tau_B \approx 3 \times 10^{-6}$ s.

→ **Brownian is faster** here → **Brownian-dominated**. τ_{eff}≈a few μs (roughly the smaller of the two, slightly reduced by combination).

Case study problem solving

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Interpretation & practical meaning (summary)

τ_0 vs τ_n:

- τ_0 (here 10^{-9}s) is the intrinsic attempt time — same for both particles in the example.
- τ_n (Néel relaxation) depends strongly on particle **core volume**. Particle S: τ_n ~ 2 ns. Particle L: τ_n ~ 25 µs. So even with same τ_0, τ_n can differ by many orders of magnitude.

Which relaxation dominates?

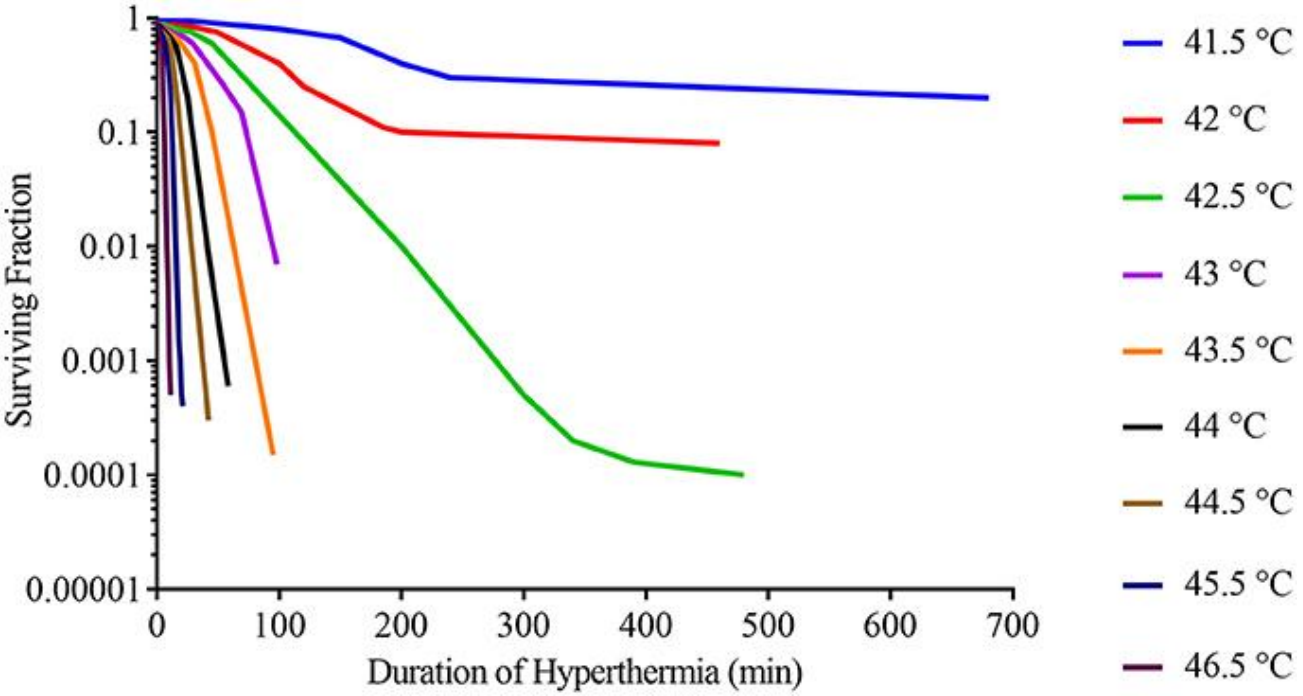
- **Particle S (8 nm core, Dh12):** Néel dominates. Magnetic response comes from internal moment flips — **good** if particles are immobilised (e.g., bound to cells), or if very fast magnetic response is needed.

- **Particle L (20 nm core, Dh22):** Brownian dominates. Magnetic response depends on physical rotation in fluid — **sensitive to viscosity and binding** (if it binds/immobilises, Brownian will be suppressed and the magnetic response will drop).

Superparamagnetism (blocking):

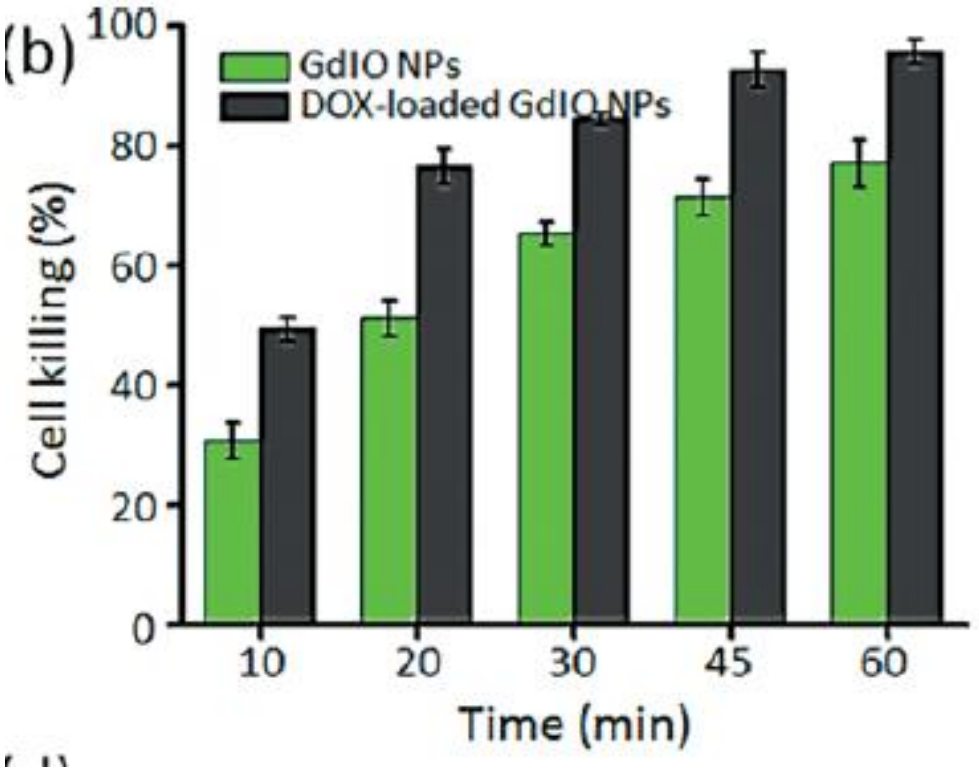
- Both particles have $E_B < 25k_BT$in this example → both are in the superparamagnetic regime by that rule. But their τ_n values differ: Particle S flips on nanosecond timescale, Particle L on microsecond timescale. Whether a particle appears "blocked" depends also on the measurement timescale (e.g., an experimental probe operating at 1 kHz sees dynamics on ~10^{-3} s — both here would appear superparamagnetic; a probe at MHz frequencies might respond differently).

Applications – Hyperthermia – Nano-heaters



https://www.frontiersin.org/articles/10.3389/fphar.2018.00831/full

Survival curves for asynchronous Chinese hamster ovary (CHO) cells heated at different temperatures for varying lengths of time



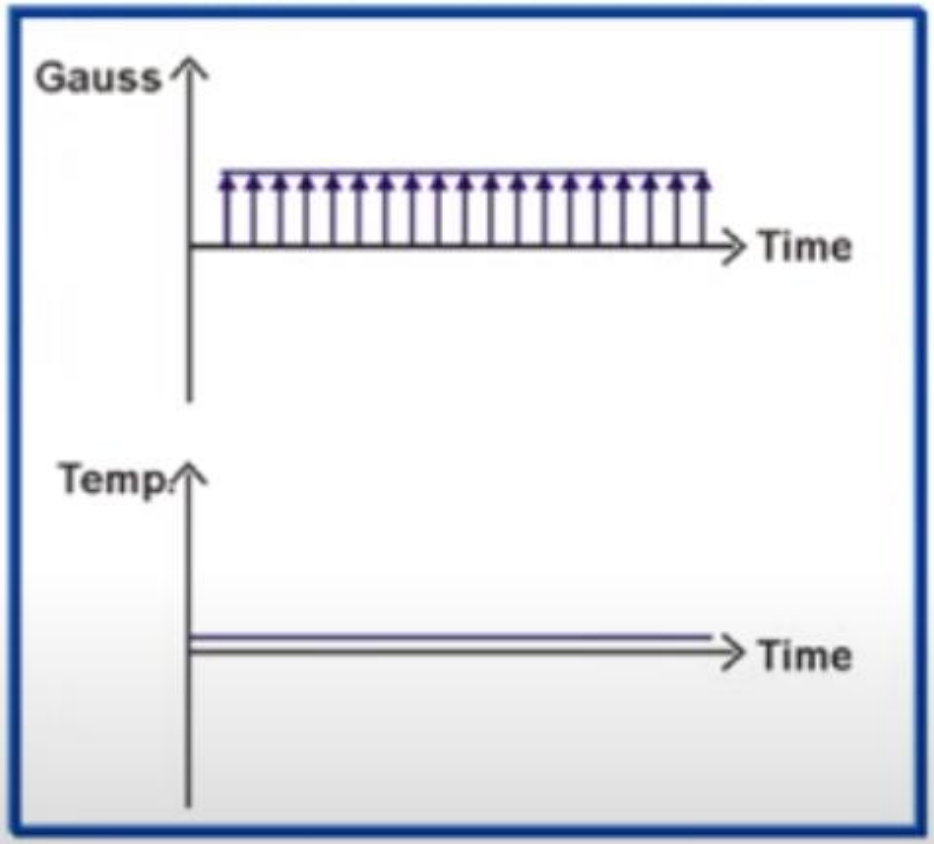
DOI: 10.1002/ejic.201600706

Cancercell killing by magnetic hyperthermia generated by bare GdIO NPs and DOX-GdIO NPs. The cells were treated with bare GdIO NPs or DOX-loaded GdIO NPs for 10–60 min, followed by the application of an alternating magnetic field (300 Oe, 267 kHz). After treatment, the cells were incubated under culture conditions for 24 h.

Applications – Magnetic Drug Delivery

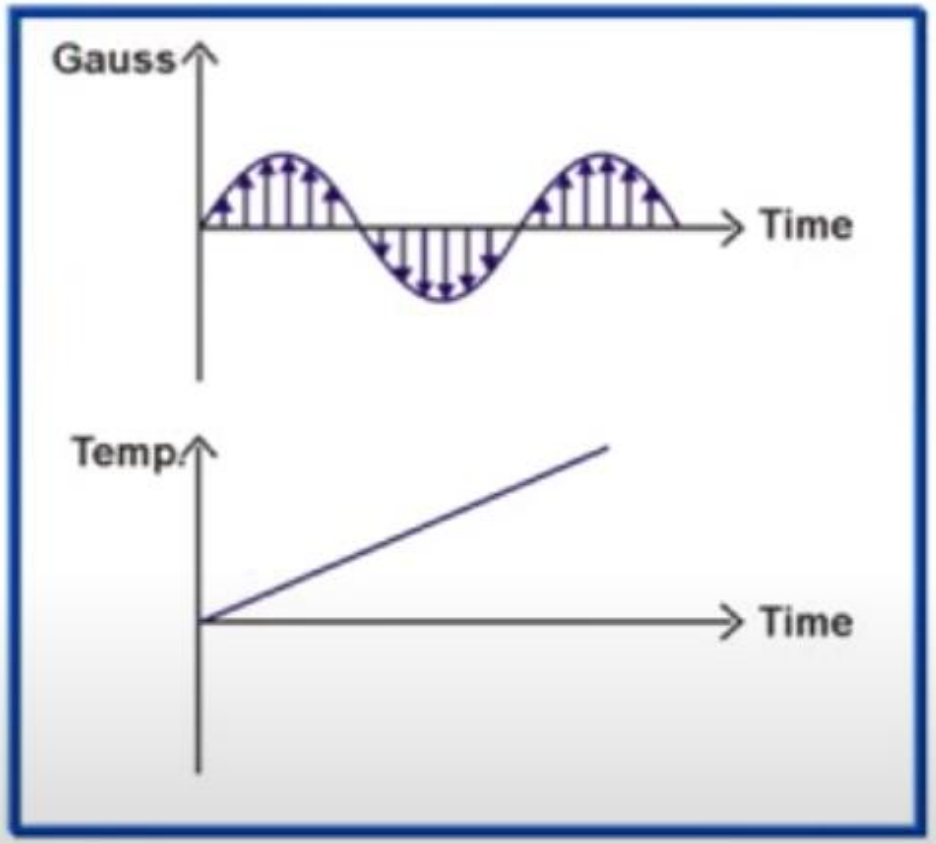
The University Of Queensland AUSTRALIA

Static – time-invariant



Very useful for imaging
Magnetic resonance imaging
(MRI)

Time-varying - alternating



Useful for therapy
Heat or activating devices

Applications – Magnetic Drug Delivery

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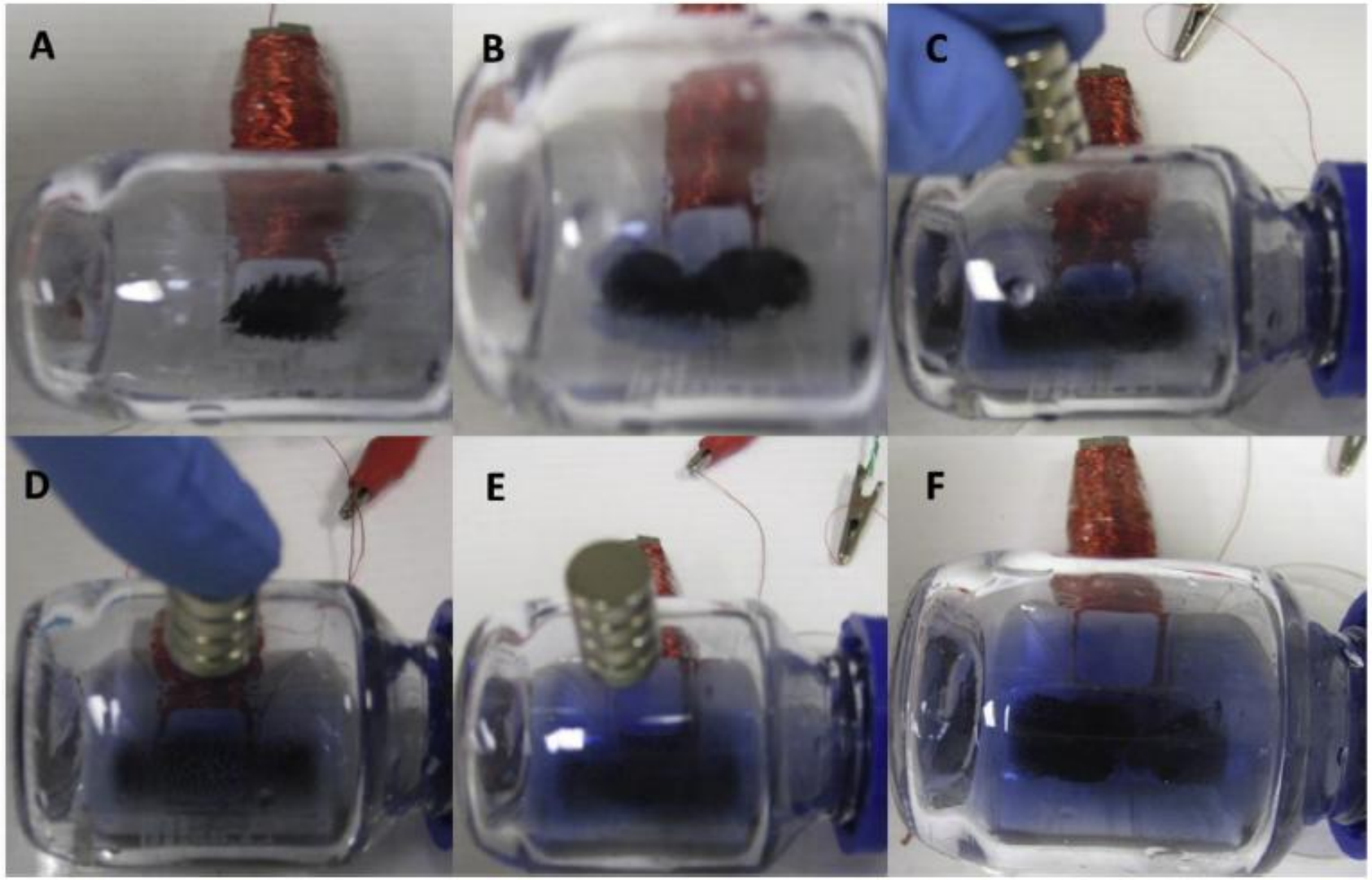
Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as drug carrier

AC field only
(Is it localized or site specific?)

Applications – Magnetic Drug Delivery

Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as drug carrier

Site specific drug delivery system through combined AC and DC magnetic field



Ref. http://dx.doi.org/10.1016/j.micromeso.2015.12.032

Applications – Magnetic Drug Delivery

Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as

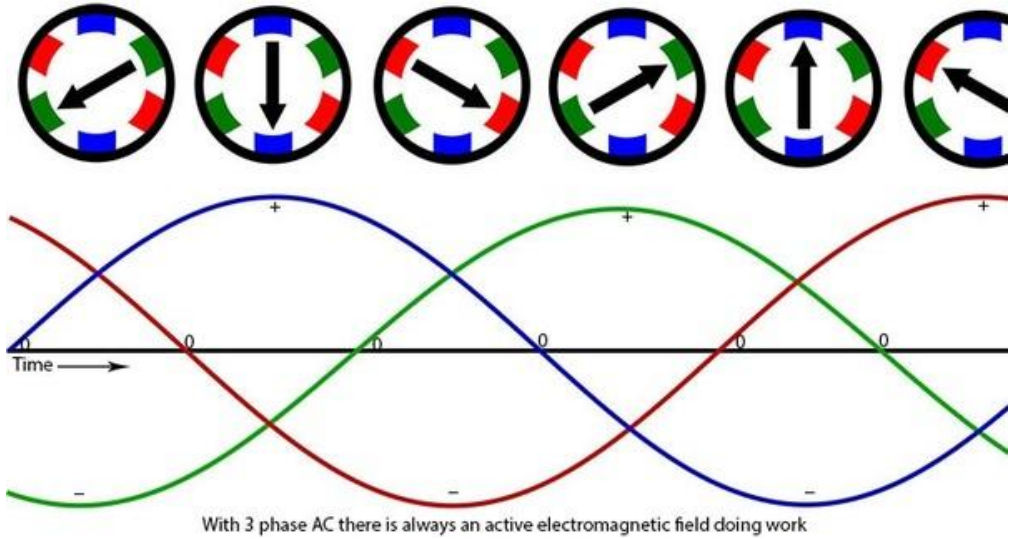
Ref. http://dx.doi.org/10.1016/j.micromeso.2015.12.032

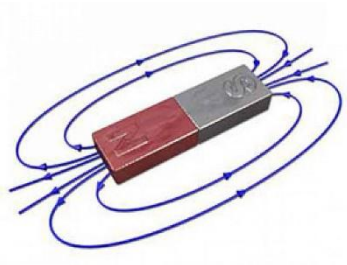
Applications – Magnetic Drug Delivery

THE UNIVERSITY OF QUEENSLAND AUSTRALIA

Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as drug carrier

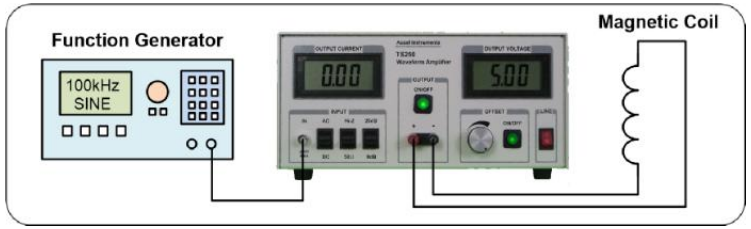
DC and AC magnetic field: Why do you need these two types?





DC field

Site specific targeted zone



AC field

Drug Release

AC field: **magnetic fields** produced by an **alternating** current is a changing **magnetic field** (Its direction and magnitude changes with time

DC the **field** is constant both in magnitude and direction.

Ref. http://dx.doi.org/10.1016/j.micromeso.2015.12.032

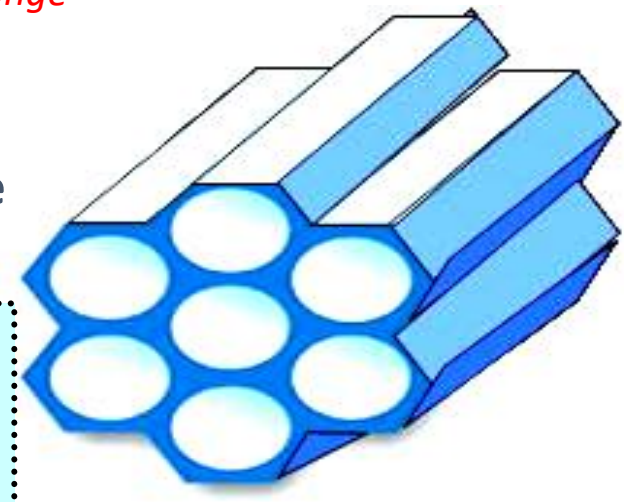
Applications – Magnetic Drug Delivery

Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as drug carrier

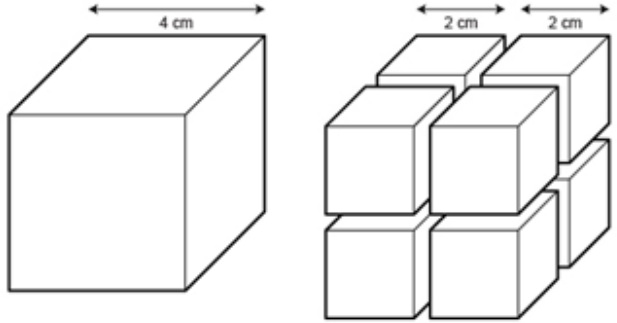
Importance of porous metals

Sponge

Nanoporous structure



Advantage:
(i) Increase of surface area
(ii) Accessible mesopores



Surface area
= (4 cm x 4 cm x 6 faces) = 96 cm^2

Surface area of one cube
= (2 cm x 2 cm) x 6 faces = 24 cm^2

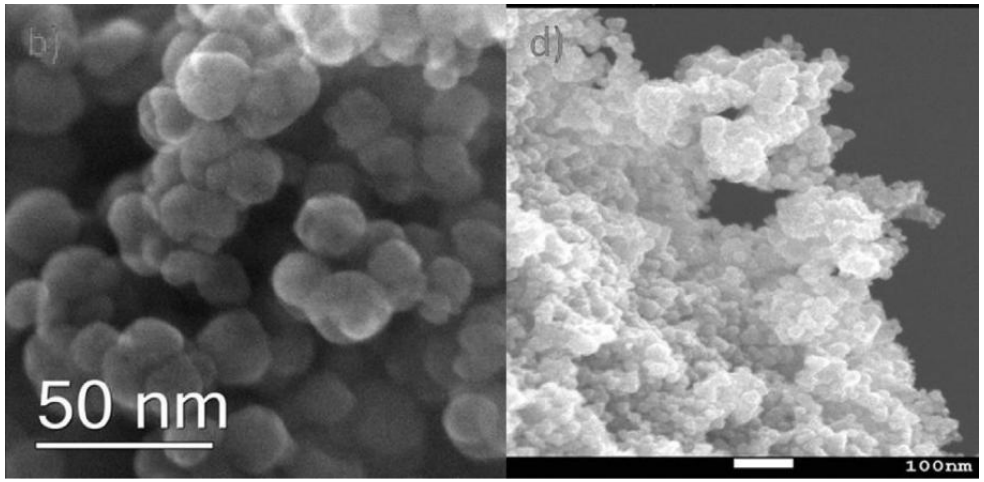
Total surface area
= 24 cm^2 x 8 cubes = 192 cm^2

Ref. http://dx.doi.org/10.1016/j.micromeso.2015.12.032

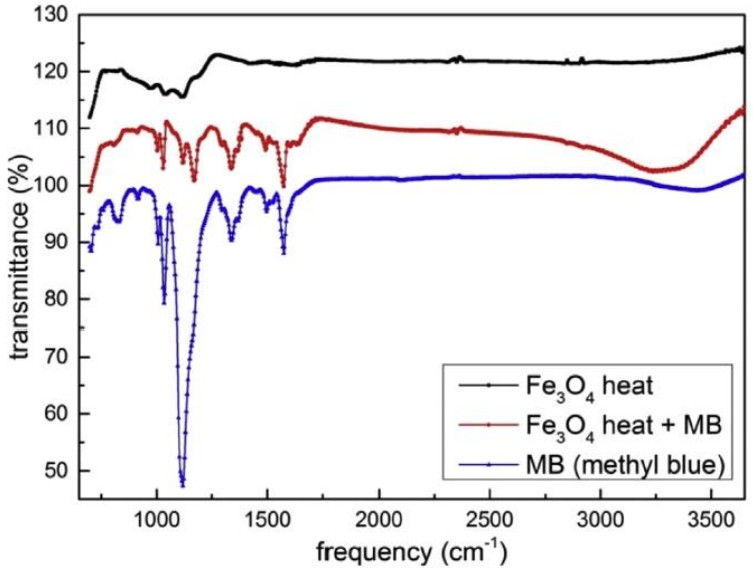
http://www.bbc.co.uk/schools/gcsebitesize/science/21c/materials_choices/nanotechnologyre shtml

Applications – Magnetic Drug Delivery

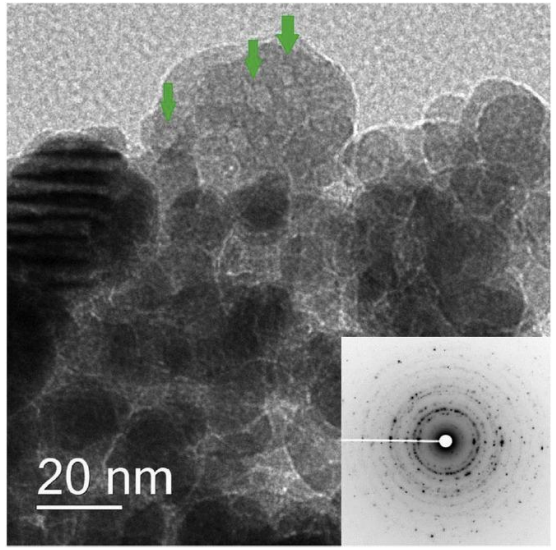
Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as drug carrier



Secondary electron (SE) images of Fe3O4 nanoparticles: STEM SE image of the surface of an agglomerate composed of 10-30 nm diameter particles; Detail of a) showing the smooth, rounded particles with extensive open porosity;



FTIR spectra of as-prepared porous Fe3O4, porous Fe3O4 after exposure to methyl blue, and methyl blue.



Bright field TEM image showing crystalline particles 10-50 nm in diameter. Extensive 2-5 nm diameter porosity is evident within the larger particles (green arrows)

Porous Fe3O4 structures have been used as a vehicle for the transport of drugs (Methylene blue as model drug)

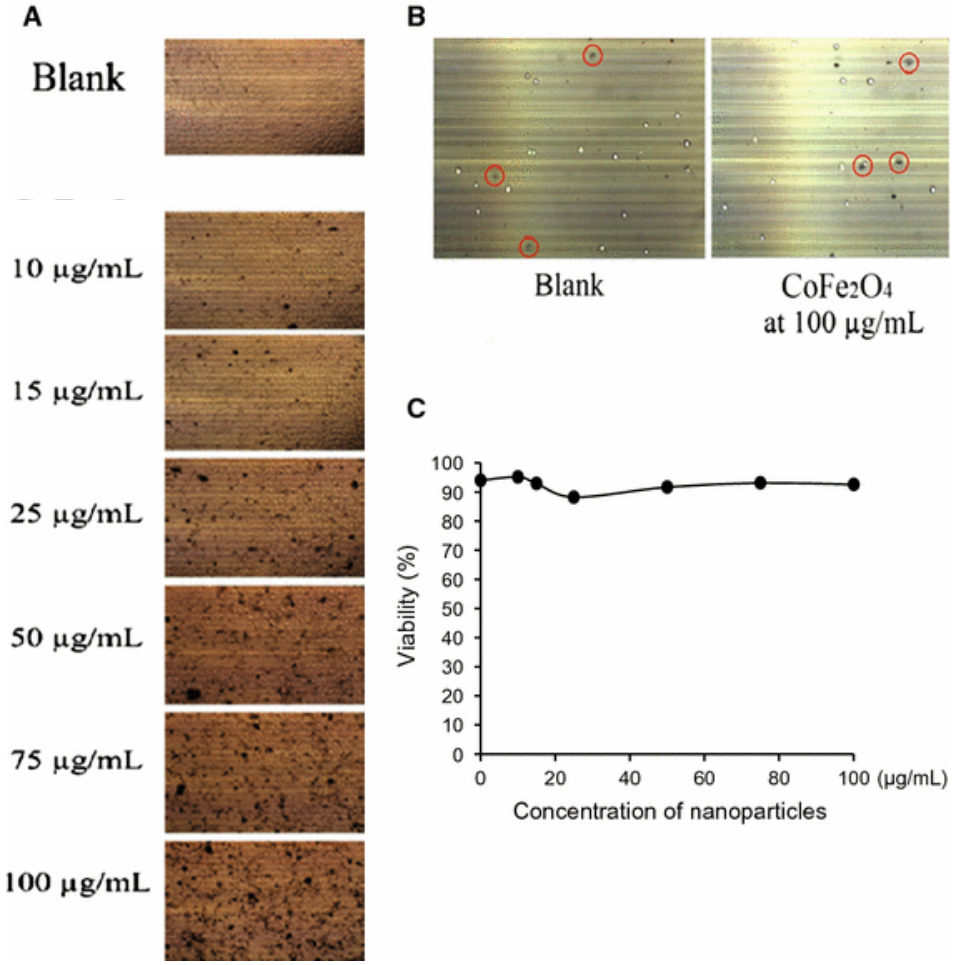
1. Crystalline Fe3O4 nanoparticle
2. Pores form from individual crystallite and between crystallites
3. Metheylene Blue as model drug
4. Adsorbed onto the porous surface through electrostatic interaction.

Ref. http://dx.doi.org/10.1016/j.micromeso.2015.12.032

Applications – Magnetic Drug Delivery

Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as drug carrier

Cytotoxicity Test of the Fe_3O_4 nanoparticles

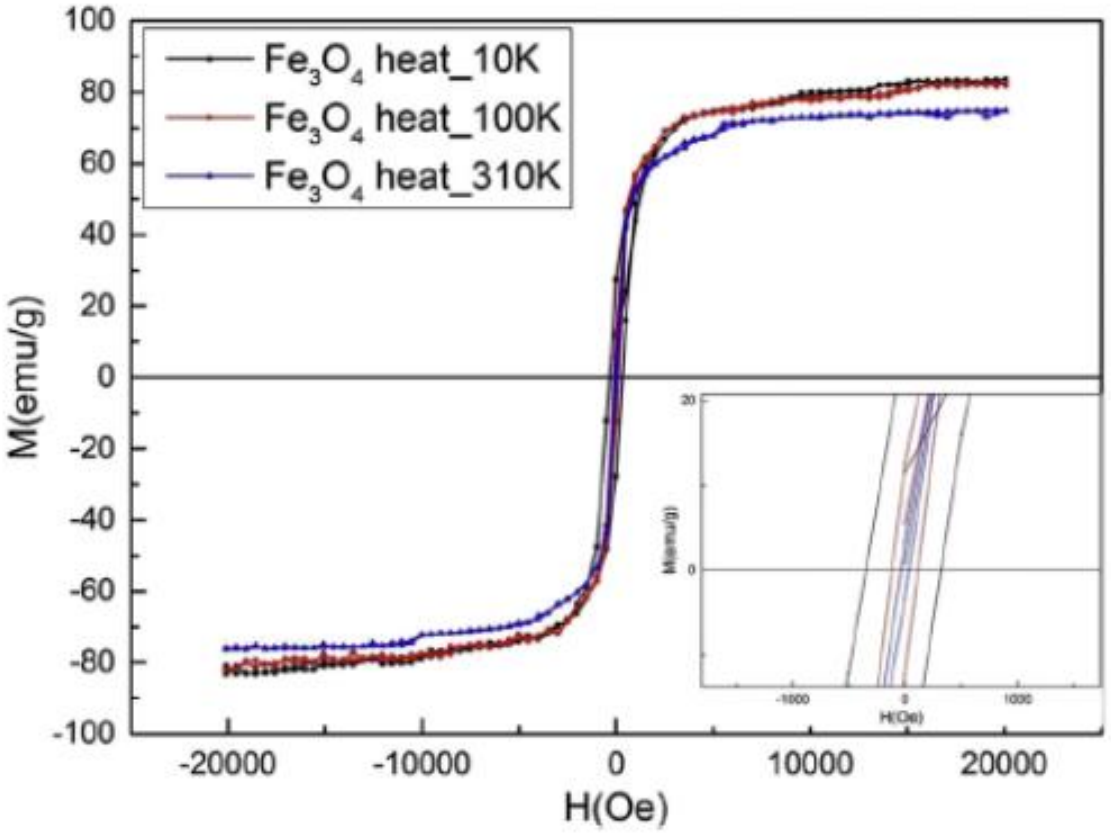


1. Cultures of A549 human cancer cells
2. Different concentrations of nano-conjugates ranging from 10 to 100 μg/ml
3. Cell viability is more than 90%

Ref. http://dx.doi.org/10.1016/j.micromeso.2015.12.032

Applications – Magnetic Drug Delivery

Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as drug carrier



The magnetic hysteresis loops of porous Fe3O4 at temperatures of 10, 100, and 310 K. The zoomed low-field region (inset) shows values of the coercive field.

Ref. http://dx.doi.org/10.1016/j.micromeso.2015.12.032

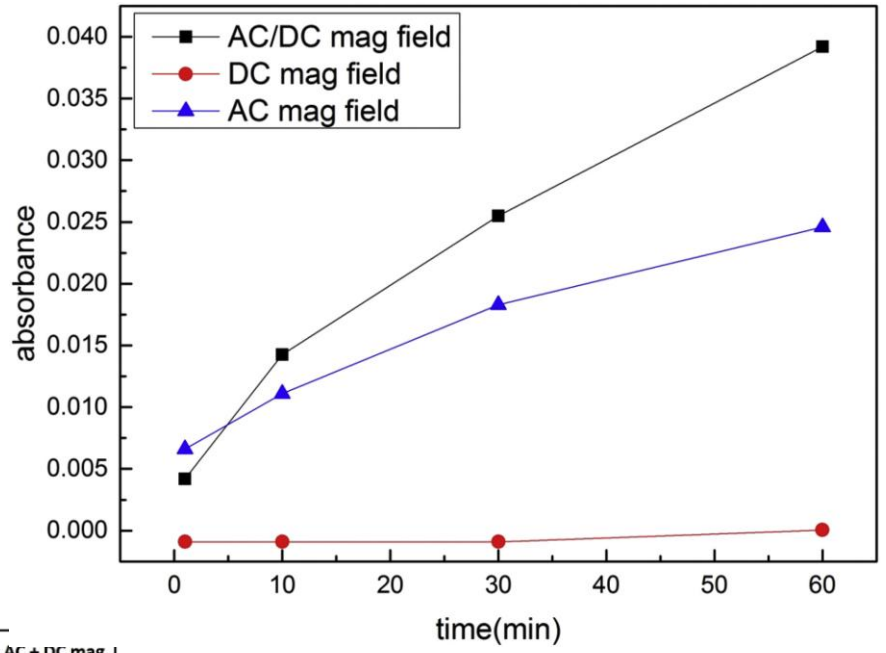
Magnetic properties of the Fe3O4 nanoparticles

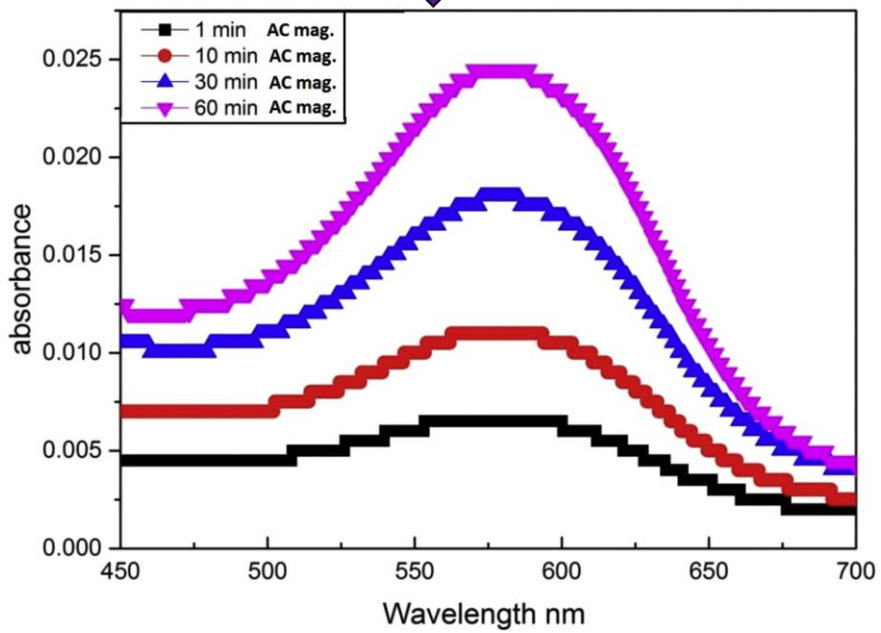
The S-shaped hysteresis loops with negligible coercive field (HC) are a typical feature of superparamagnetic Nanoparticles

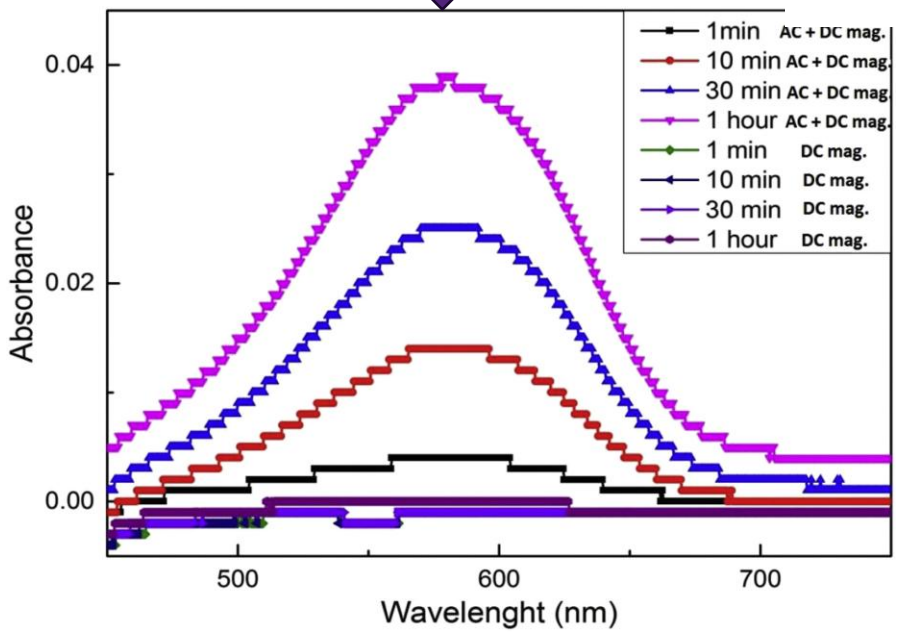
The saturation moment (Ms) measured at 10K is quite high and has a value of 85.5 emu/gm, while at 310 K the saturation moment was slightly smaller at around 77 emu/gm.

Small Hc values were obtained for the nanoparticulate material, because the magnetic moment of each nanocrystallite making up this material was small enough to become unsettled by thermal excitation

Magneto-Mechanical Nano-machineries for Precision Drug Release





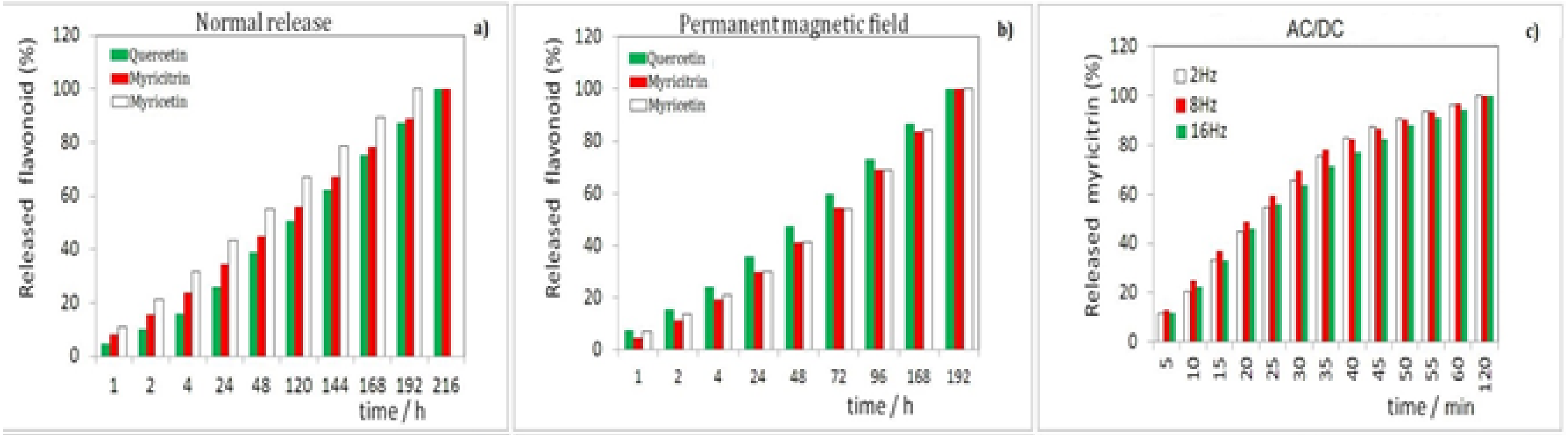


The released model drug has a strong absorption band around **580 nm**. This wavelength acts like the drug's **optical fingerprint**.

Applications – Magnetic Drug Delivery

Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as drug carrier

Same mechanism using real drugs:



Cumulative released profiles of quercetin, myricetin and myricitrin in (a) normal condition, (b) permanent magnetic field and (c) and AC/DC magnetic fields, 2Hz, 8 Hz and 16 Hz.

100% drug can be released within 2 hrs if combined AC/DC technique is used where 100 times faster compared to normal release and only DC field is used.

Ref. http://dx.doi.org/10.1016/j.micromeso.2015.12.032

Practical contribution of Brownian and Neel relaxation in site-specific delivery

1.Guidance with external magnetic fields

1. When nanoparticles are circulating in blood, both Brownian and Néel relaxation matter.
2. Brownian relaxation affects how quickly particles reorient with the guiding field → helps efficient steering.
3. Néel relaxation ensures fast internal magnetic moment flips → maintains strong response even if Brownian rotation is slowed by viscous environments (like blood plasma).

2.Hyperthermia for therapy

1. In magnetic hyperthermia (using AC fields to heat particles at the tumour site), energy dissipation comes from relaxation.
2. Small particles (Néel-dominated) flip internally to generate heat.
3. Larger or coated particles (Brownian-dominated) rotate physically in fluid to dissipate heat.
4. Which dominates depends on the core size and hydrodynamic size.

3.Drug release mechanism

1. If nanoparticles are designed with a drug-loaded coating, Brownian rotation can induce *mechanical agitation* to help drug release.
2. If particles are immobilised at the target site, only Néel relaxation can continue to dissipate energy → useful for triggering local heating and drug release without particle rotation.

AC/DC magnetic syringe device for stimulated drug release, injection and ejection of nanocarriers

The University Of Queensland AUSTRALIA

What factors need to be considered here to optimise for successful magneto-mechanical site-specific drug release using Fe_3O_4?

Drug loading into carrier:

1. Porosity
2. Size
3. High surface to volume ratio
4. Electrostatic interaction
5. Adsorption
6. Self vibration
4. Collision between the particles
5. Collision between particles and media.

Challenges:

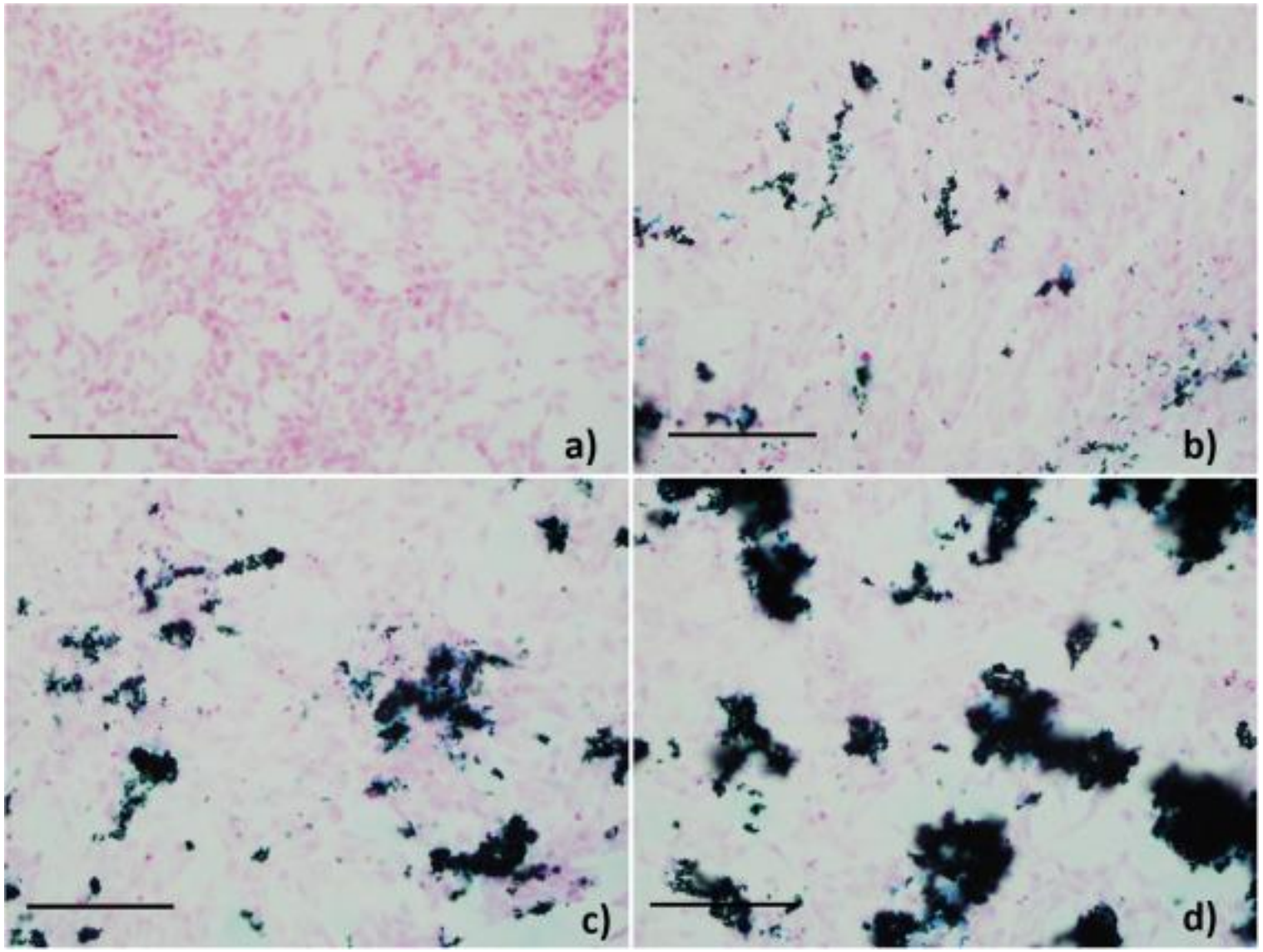
Removing the nano-carrier from body and re-use?

Site specific drug release:

1. Cytotoxicity
2. DC magnetic field
3. AC magnetic field
4. Light absorption (release evidence
5. magnetic moment of the particle.
6. Frequency
7. Amplitude
8. Applied current
9. DC Magnetic field
10. AC magnetic field
11. Coil design
12. Viscosity of the media

https://www.sciencedirect.com/science/article/pii/S0304389420329095
https://www.sciencedirect.com/science/article/pii/S0304885319310236

Magnetic drug delivery based on superparamagnetic porous nanostructures

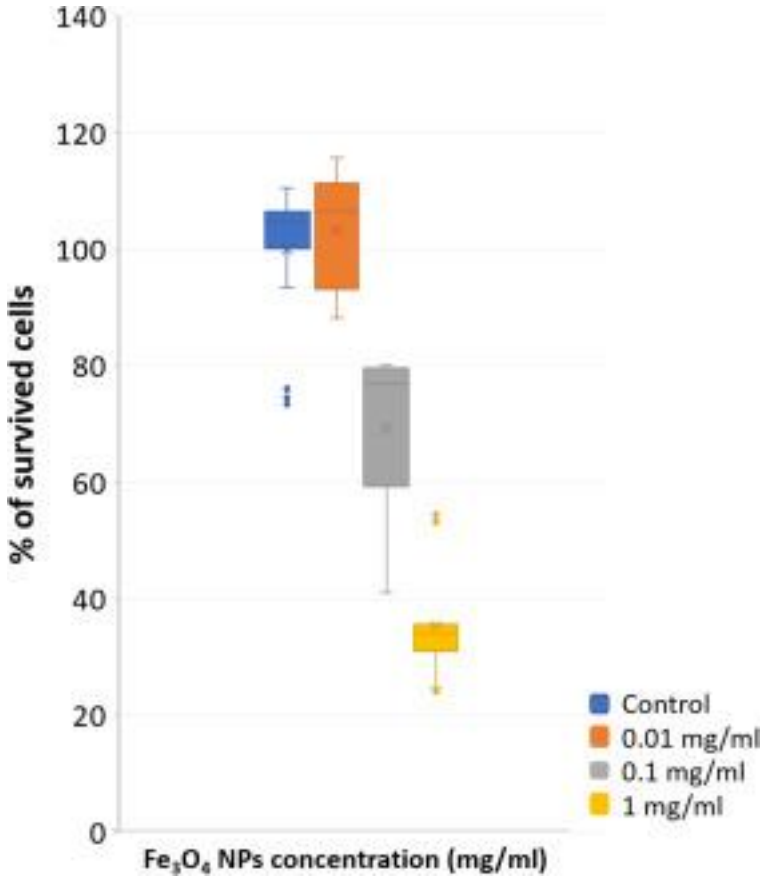


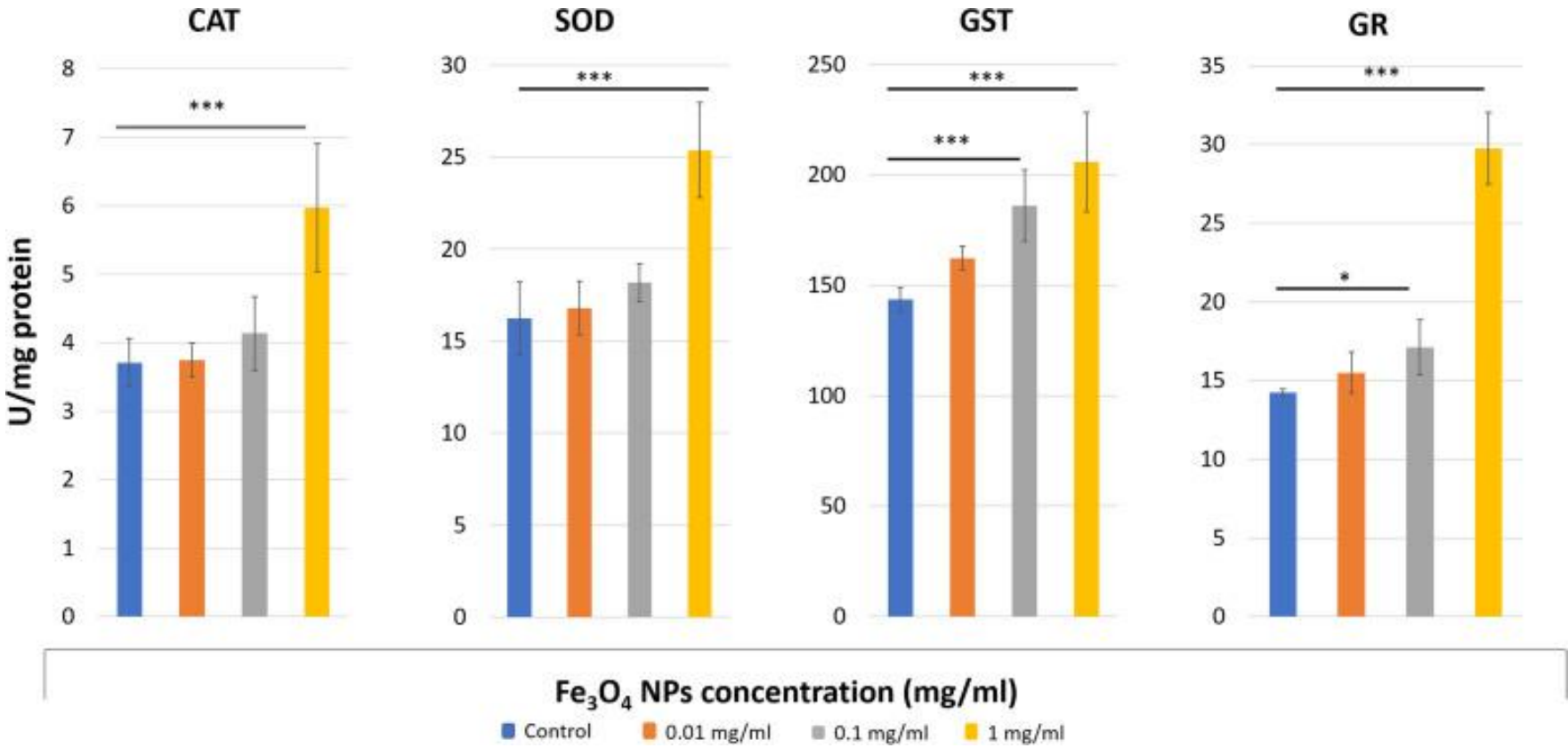
Prussian blue staining against iron on differentiated SH-SY5Y human neuroblastoma cell line at 24 h post exposure to different concentrations of Fe_3O_4 nanoparticles; A – 0 mg/ml (control), B – 0.01 mg/ml, C – 0.1 mg/ml, D – 1 mg/ml; Blue – metabolised iron positive stain, Red – nuclei and cytoplasm, Black – non metabolised nanoparticles; magnification 400 ×; Scale 100 μm. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

J. of Hazardous Materials 2021, 409, 124918 -https://www.sciencedirect.com/science/article/pii/S0304389420329095#fig0055

CRICOS code 00025B

Higher concentration of particles (Fe3O4) inside biological cell is toxic





J. of Hazardous Materials 2021, 409, 124918 -https://www.sciencedirect.com/science/article/pii/S0304389420329095#fig0055

Applications – Magnetic Syringe

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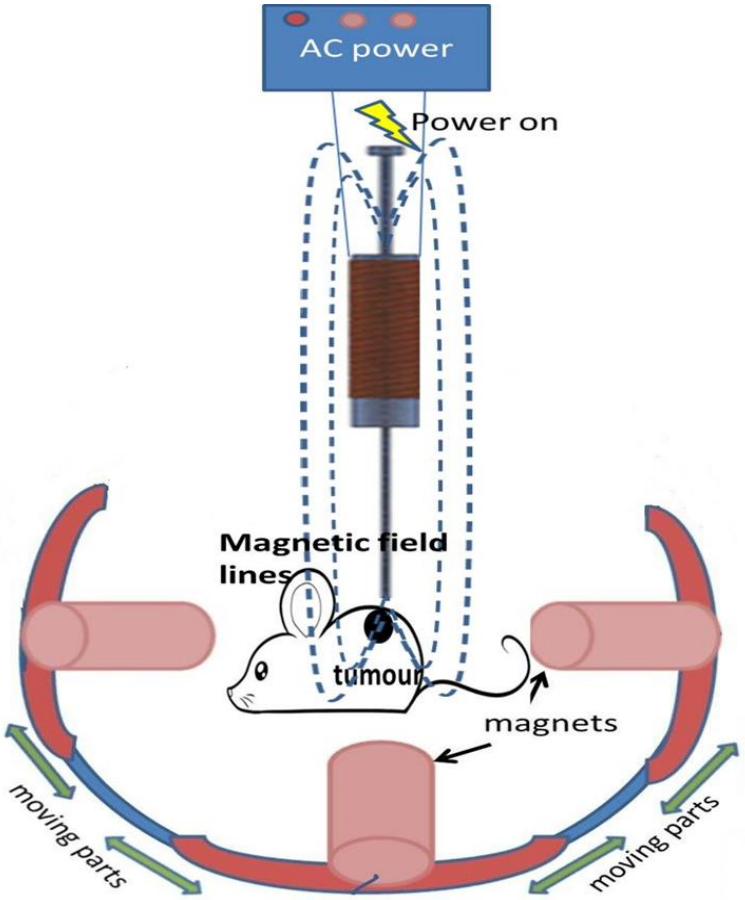
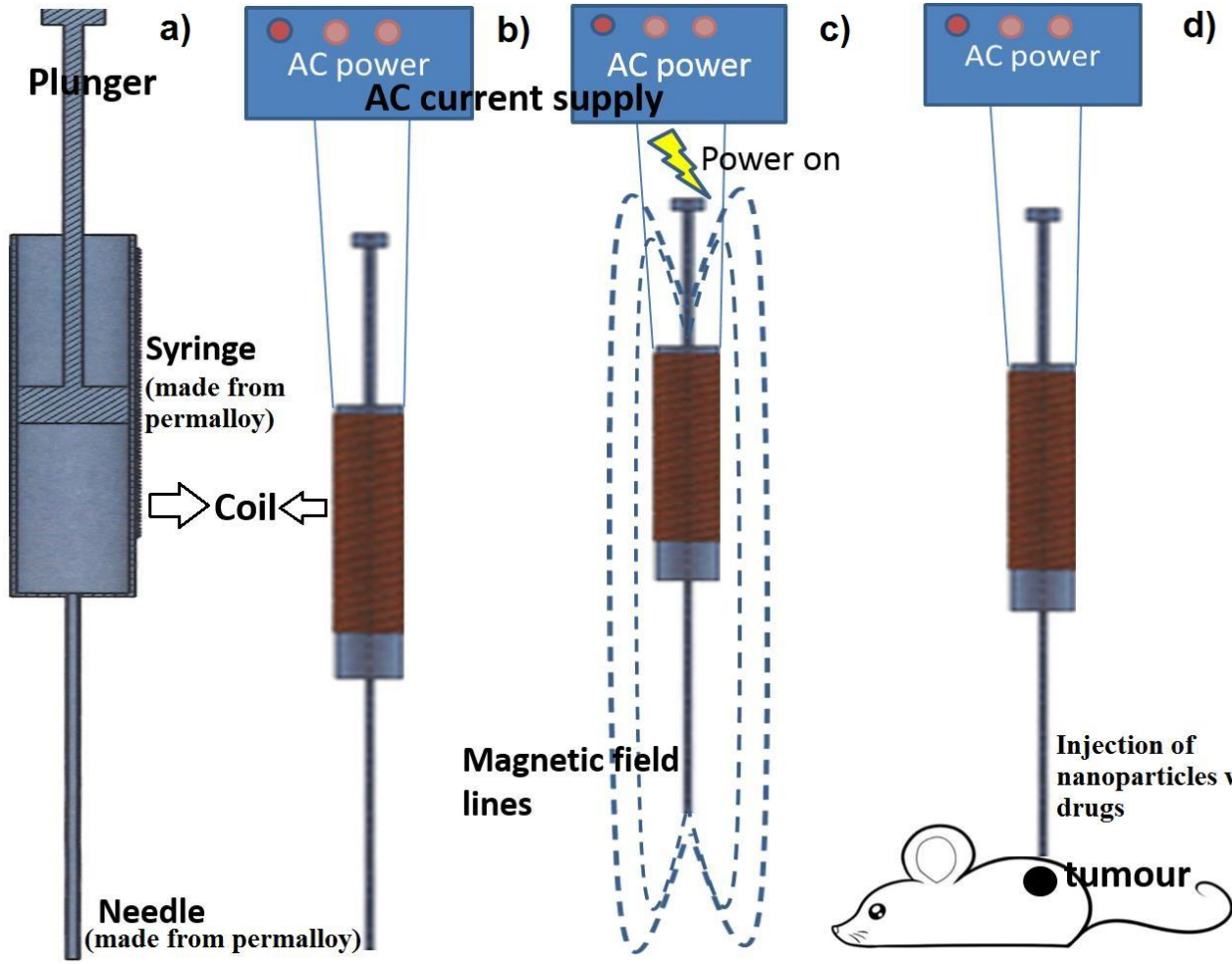
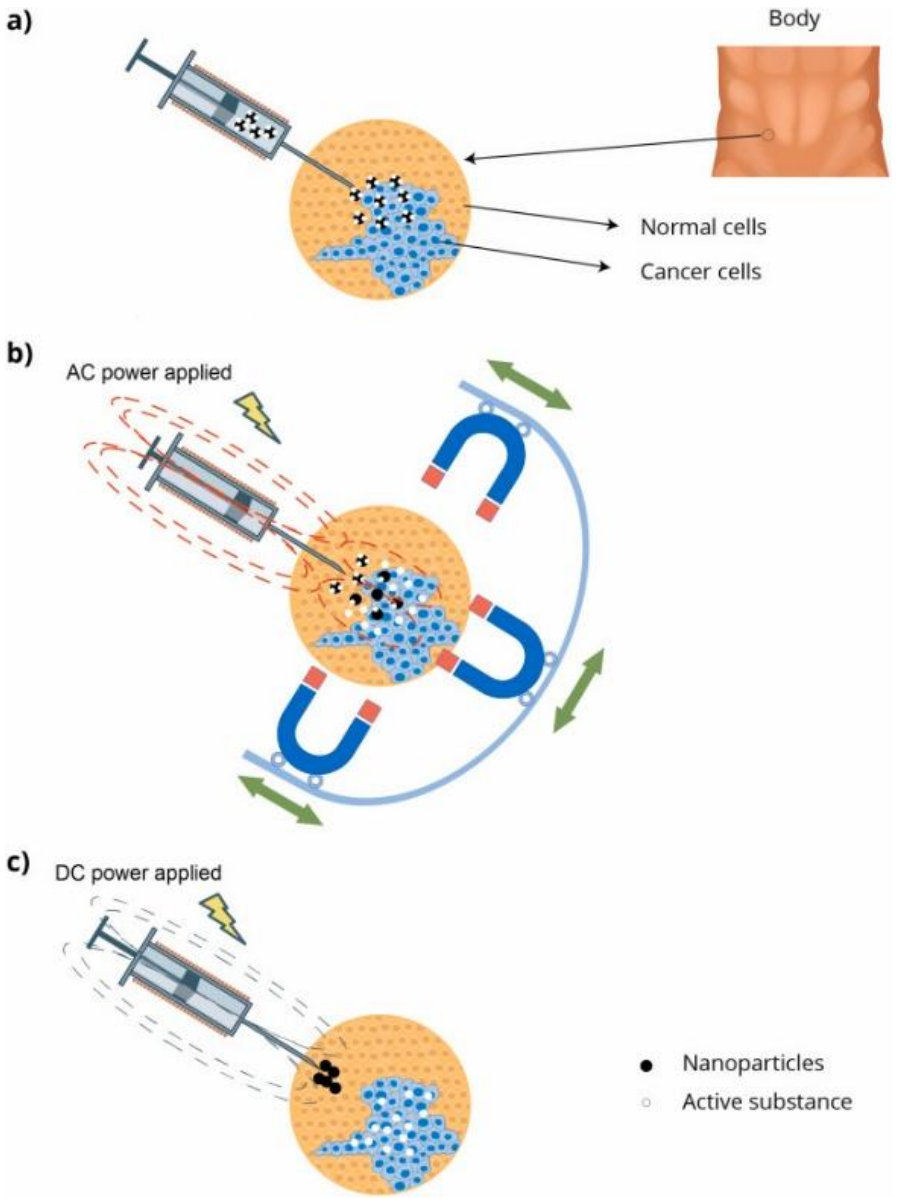
Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as drug carrier

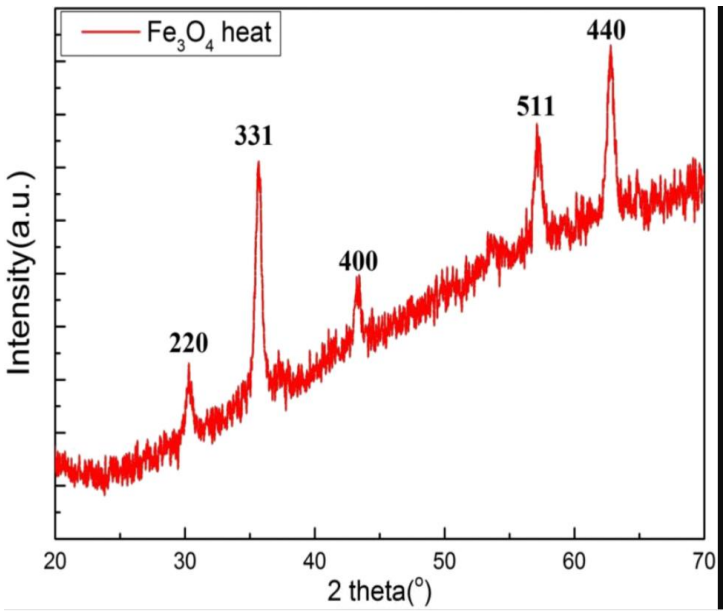


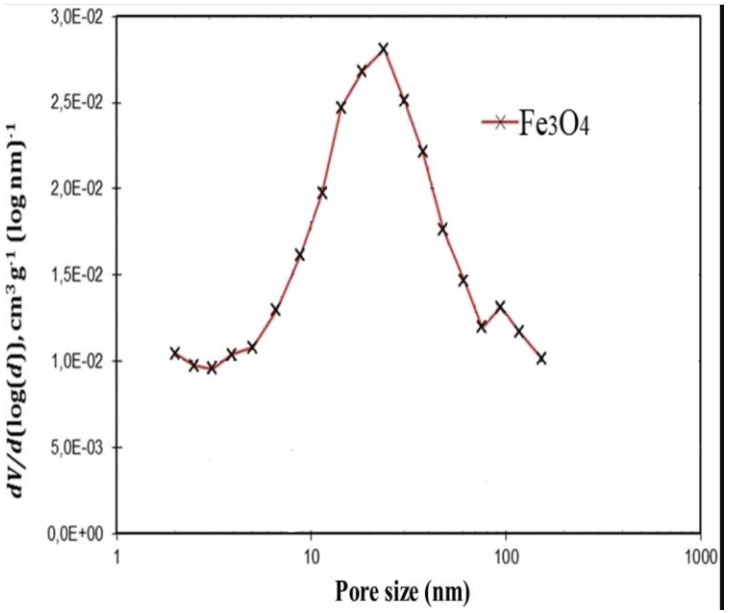


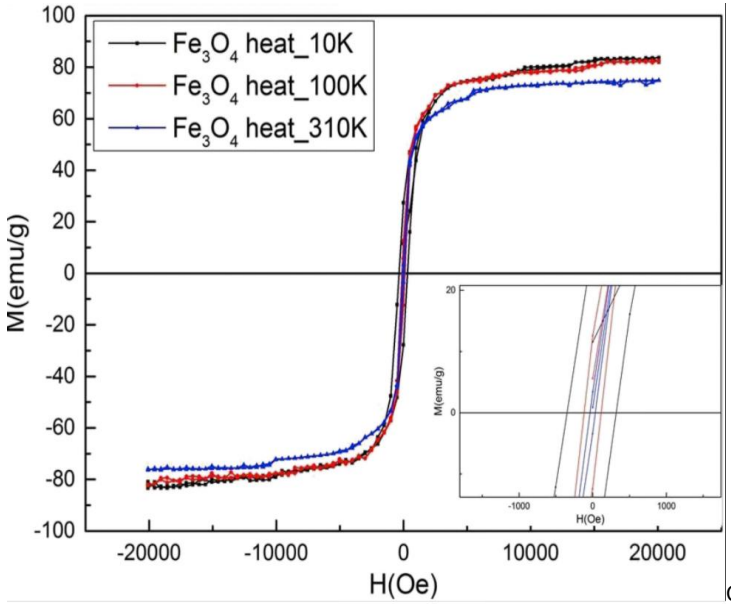
Magnetic syringe

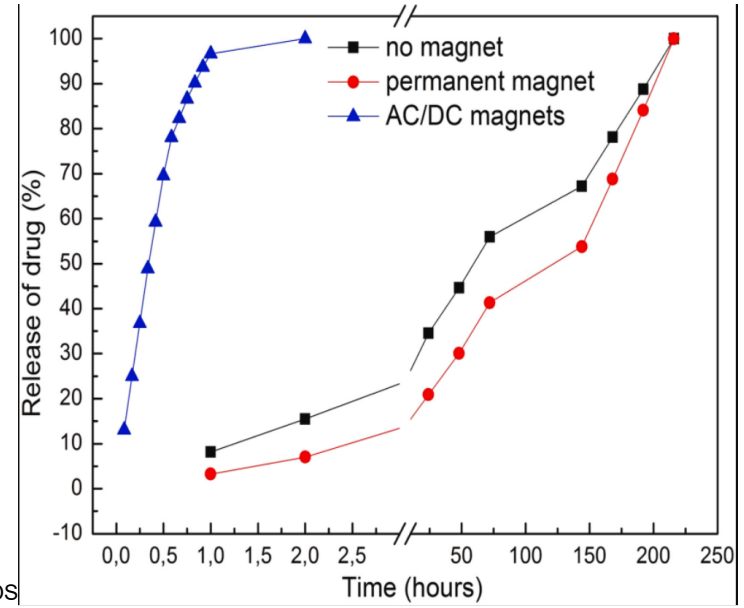


J. of Hazardous Materials 2021, 409, 124918









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Applications – Magnetic Syringe

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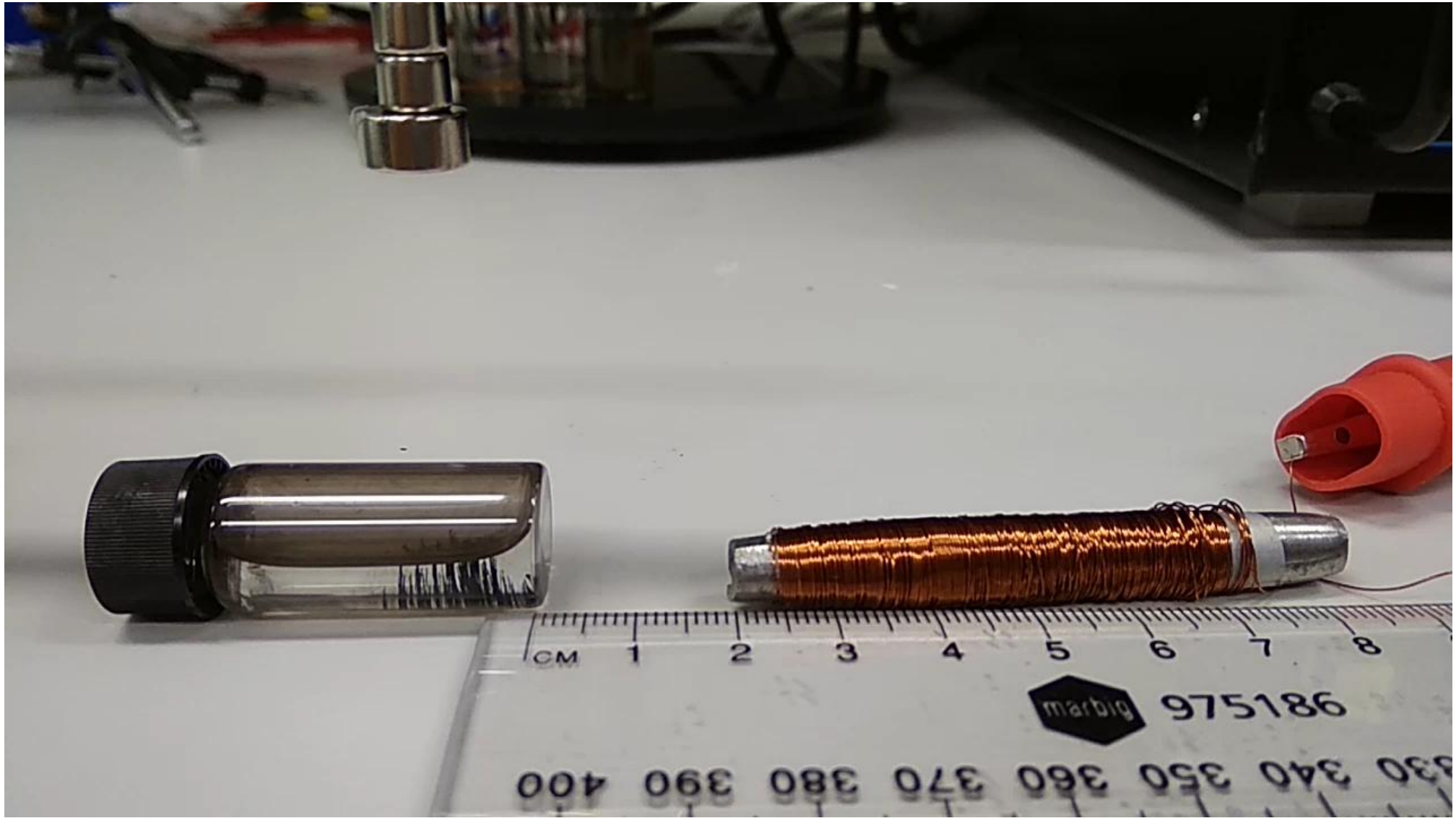
Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as drug carrier



Lab Practical on Electrochemical Characterisation of Magnetic Nanoparticles:

WEEK 7 · 知识点总结 · Dr Mostafa Kamal Masud

第 7 周　等离激元多孔金属纳米复合材料

Plasmonic Porous Metal Nanocomposite

主线：纳米结构 → LSPR 与电磁热点 → Raman 到 SERS → SERS 体系组成与三种传感策略 → 智能传感器

1. 为什么要造孔（★ 7 参数表）

结构参数	工程后果
孔径	传质
孔隙率	比表面积
孔连通性	扩散 / 可达性
韧带尺寸	力学 + 光学性能
表面粗糙度	电磁热点
金属成分	LSPR 波长
结构均一性	传感器重现性

- IUPAC：微孔 < 2 nm，介孔 2–50 nm，大孔 > 50 nm；纳米复合材料：比表面积大、力学更好、光学更好。

2. Raman 与 SERS（★）

- Raman：激光被化学键散射；绝大部分为 **Rayleigh**（λ 不变）；约 10^{-9} 为 **Raman**（λ 变长），位移是分子"指纹"。无损、对水不敏感，但信号弱。
- **SERS**：分子吸附在**粗糙贵金属（Au/Ag）**表面 → 信号放大多个数量级；适合低浓度、荧光干扰场合；谱峰**更少、更宽**。
- 两种增强机制：**电磁增强**（主要；入射光激发 LSPR → 热点处局域电场放大，信号约 $\propto |E|^4$）；**化学增强**（次要；化学吸附 → 金属–分子电荷转移）。

3. 结构如何影响等离激元

- **LSPR**：纳米金属中自由电子在光电场驱动下集体共振，表面附近电场被放大。
- **等离激元杂化**：实心球（ω_{sp}）+ 空腔（ω_c）→ 成键模式 ω_-（低能、红移）+ 反键模式 ω_+；壳越薄杂化越强。
- Ag 立方 → AuAg 纳米盒 → 纳米框架：空腔增大，共振**红移到低能**（溶液黄 → 蓝）；原因：① 合金化（成分）② 空腔形成（形貌）。
- 热点：**球**（均匀，热点少）< **立方 / 尖角**（角上集中）< **团聚**（颗粒间隙）< **介孔金属**（大量连通热点）。多孔 Au 增强最强，但信号波动大（重现性差）。

4. SERS 体系五要素

- **基底**（Au/Ag，LSPR；颗粒太小 LSPR 丧失；立方优于球；团聚和多孔都增强）；**报告分子**（Raman 染料 R6G、CV、MB；巯基分子 MBA、DTNB，形成 SAM，谱带少适合多重检测）；**稳定剂**（PEG、SiO_2、石墨烯：防团聚、防脱落、防非特异吸附）；**偶联材料**（EDC/NHS 酰胺键、巯基 Au–S 最牢、静电、生物素–链霉亲和素）；**制备方式**。

5. 三种传感策略（★）

策略	简便	选择性	灵敏度	复杂度	局限 / 优点
无标记 / 未修饰	★★★	★	★★	★	选择性低、胶体批次不一、团聚不可控
功能化表面	★★	★★★	★★	★★	大受体（抗体）使分子远离热点
SERS 纳米标签	★	★★★	★★★	★★★	特异、灵敏、假阳性少

6. 应用

- 智能传感器 8 步：目标物 → 多孔传感器 → 激光 → 收集 → 光谱仪 → 信号处理 → 机器学习 → 数字输出。
- 多孔 Au 功能化基底检测水中痕量污染物；病原体：DEP 微流控**先富集** + 多重 SERS；Cu^{2+}：Cys 使 Ag 团聚（信号强）→ Cu^{2+} 氧化 Cys → 散开（turn-off）；**SERS 侧流试纸条**（Au@Ag 纳米棒 + DTNB + 抗体，T 点夹心、C 点对照，可定量）。
- 总结：结构控制等离激元；孔隙增强传感；界面控制选择性；需平衡**灵敏度、选择性、稳定性、重现性、可制造性**。

考试重点：Raman vs SERS 与两种增强机制；为什么多孔 Au 比球形 Au 强；7 参数表；三种策略对比与便携传感器的选择；设计题（材料 + 架构 + 识别 + 如何兼顾灵敏度 / 选择性 / 重现性）。

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CREATE CHANGE

Plasmonic Porous Metal Nanocomposite

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[MATE6301] Nanomaterials

School of Mechanical and Mining Engineering

The University of Queensland

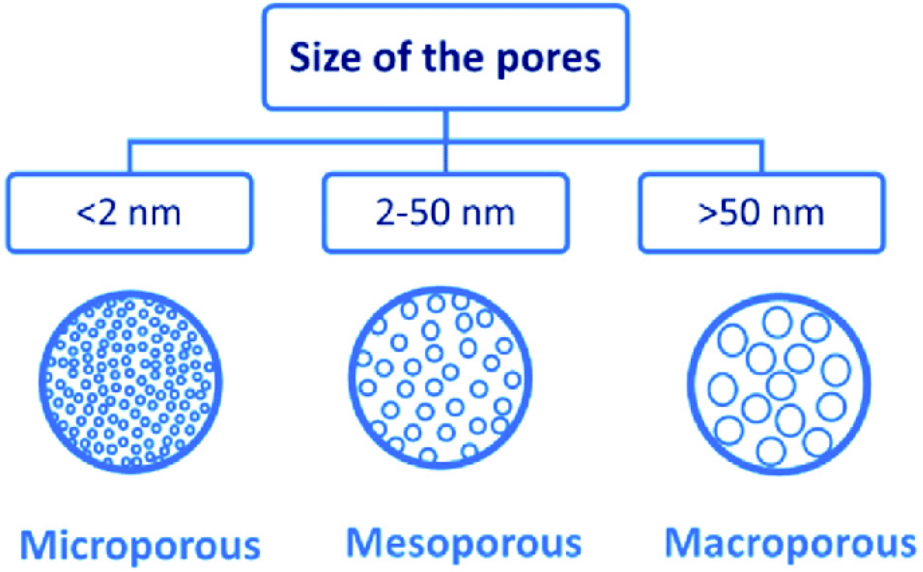
Plasmonic Porous Metal Nanocomposite

Learning outcomes

- ❑ By the end of this lecture, students should be able to:
- ❑ Explain how porosity, size, shape and composition influence plasmonic behaviour.
- ❑ Explain the origin of SERS enhancement and electromagnetic hotspots.
- ❑ Relate nanostructure design to sensor sensitivity and reproducibility.
- ❑ Compare different SERS sensor architectures.
- ❑ Design a simple plasmonic sensor for an engineering or biomedical application.

ACS Nano 2021, 15(4), 6038–6060

Nanocomposite and porous nanocomposite



The nanocomposite material, which is made of non-metallic, metallic and polymeric materials through a certain method, offers extra benefits!

- High surface-to-volume ratio, which results in small nanofiller sizes and tight filler particle spacing
- Superior mechanical properties (such as high plasticity without sacrificing stability, scratch resistance) and
- Improved optical characteristics.

Materials Today Bio, 2021, *10*, 100100.

Why engineer porosity at the nanoscale?

Porous nanocomposites consist of a regular organic or inorganic bulk phase in which a porous structure is present.

- Nanoporous materials exhibit pore diameters that are most appropriately quantified using units of nanometers.
- The diameter of pores in nanoporous materials is thus typically 100 nanometers or smaller.
- Pores may be open or closed, and pore connectivity and void fraction vary considerably, as with other porous materials.

Structural parameter	**Engineering consequence**
➢ Pore size	✓ Mass transport
➢ Porosity	✓ Surface area
➢ Pore connectivity	✓ Diffusion/accessibility
➢ Ligament size	✓ Mechanical + optical properties
➢ Surface roughness	✓ Electromagnetic hotspots
➢ Metal composition	✓ LSPR wavelength
➢ Structural uniformity	✓ Sensor reproducibility

Materials Today Bio, 2021, *10*, 100100.

CRICOS code 00025B

Raman Spectroscopy

Raman Spectroscopy provides detailed information about chemical structure, phase and polymorphy, crystallinity and molecular interactions.

- It is based upon the interaction of light with the chemical bonds within a material. Whereby a molecule scatters incident light from a high-intensity laser light source.
- Most of the scattered light is at the same wavelength (or color) as the laser source and does not provide useful information – this is called Rayleigh Scatter.
- However, a small amount of light (typically 0.0000001%) is scattered at different wavelengths (or colors), which depend on the chemical structure of the analyte – this is called Raman Scatter.

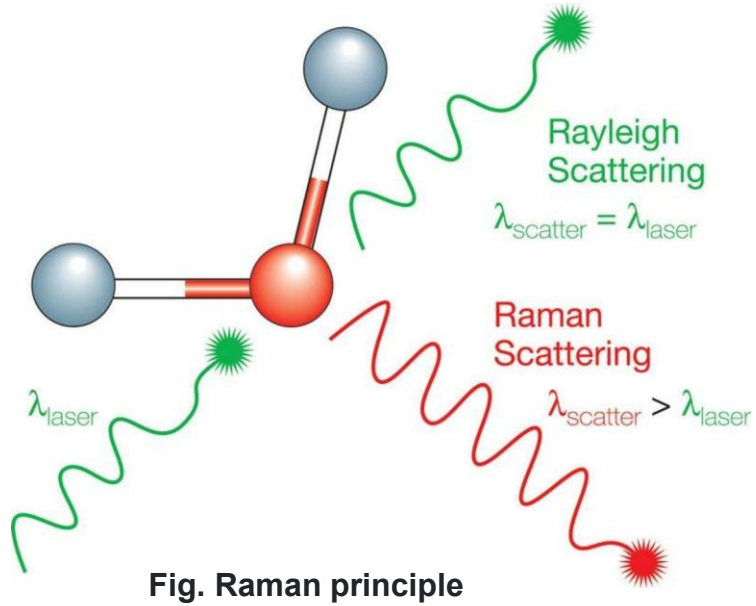


Fig. Raman principle

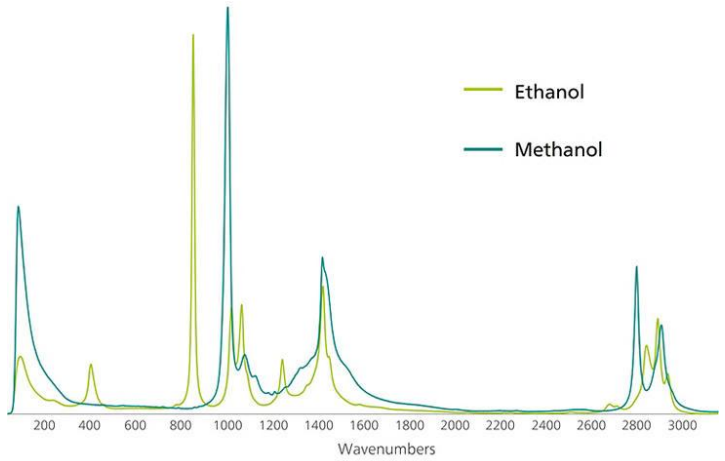


Fig. Raman spectra of ethanol and methanol

CRICOS code 00025B

Plasmonic Nanocompisites for Biosensor-Raman versus SERS

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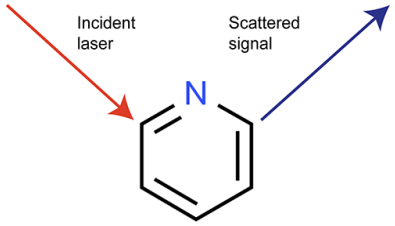
Raman

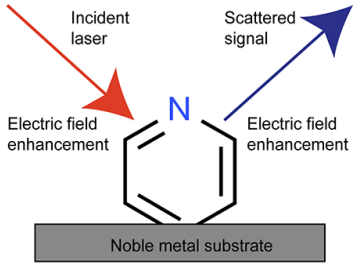
- A laser source is incident directly on a sample.
- The laser light is scattered by the bonds of the analyte, and the inelastically scattered light is collected and processed into a Raman spectrum.
- The non-destructive nature of the technique, the selectivity of Raman bands, and the insensitivity to water make Raman a useful analytical tool for both qualitative and quantitative studies of both organic and inorganic systems.
- Raman spectrum contain more peaks than corresponding SERS spectrum.

SERS

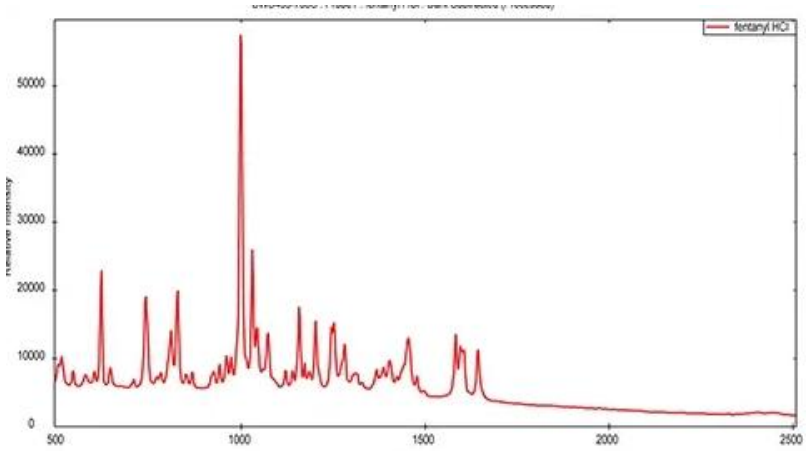
- A incident light falls on the surface of precious metals with a certain amount of roughness.
- SERS enhancement can primarily occur via two mechanisms: electromagnetic enhancement and chemical enhancement.
- In low-concentration sensing where fluorescence overwhelms Raman signal, SERS is an invaluable techniques.
- The SERS bands are noticeably broader than normal Raman bands.



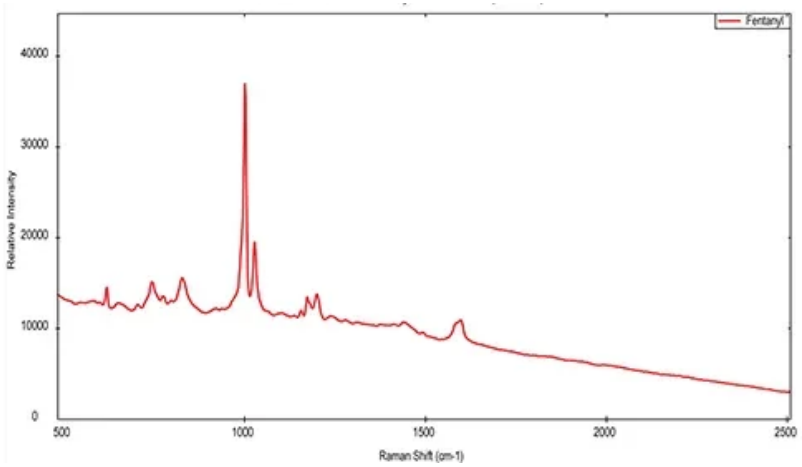
Raman Scattering



Surface-enhanced Raman scattering



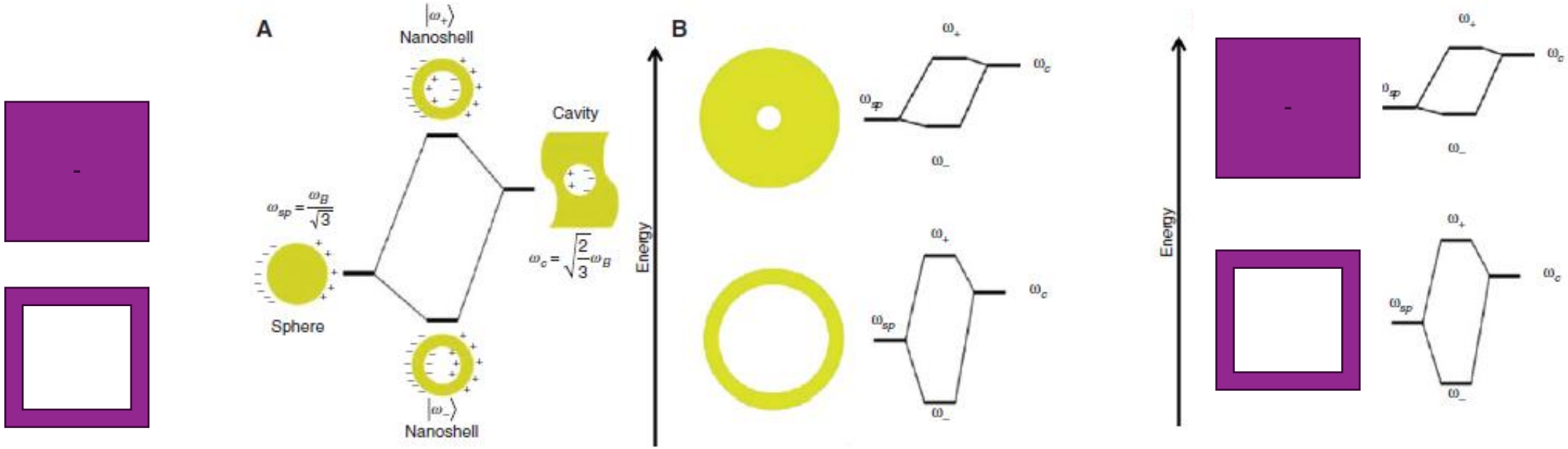
Raman



SERS

Plasmonic properties of composites: Why porous?

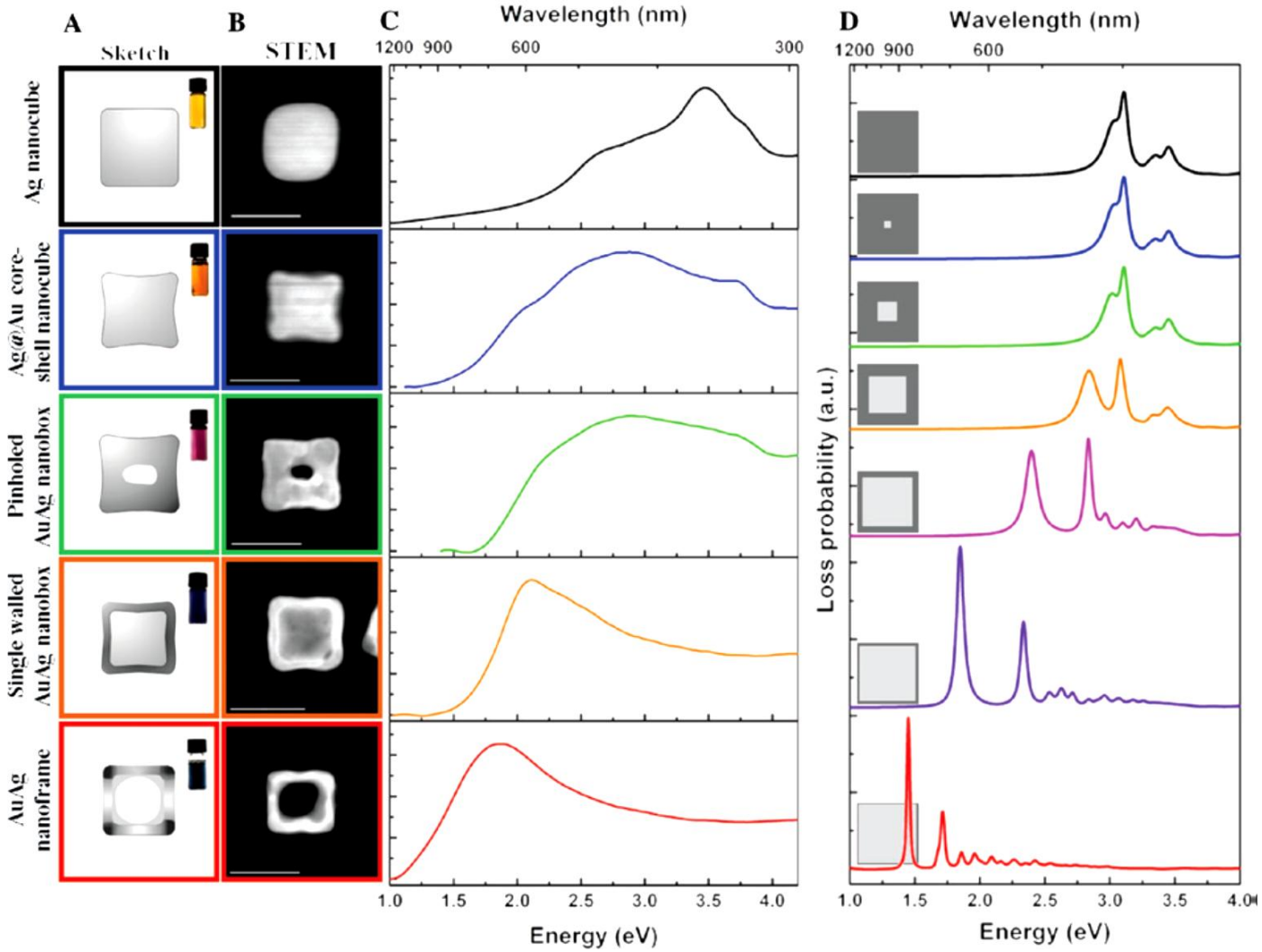
Plasmon Hybridization Mechanism



(A) Schematic energy-level diagram illustrating the plasmon hybridization mechanism in metal nanoshells as a result of the interaction between the sphere (ω_{sp}) and cavity (ω_c) plasmons. Anti-symmetrically coupled, anti-bonding, mode (ω_+) and symmetrically coupled, bonding, mode (ω_-).
(B) Extent of the hybridization for thick (upper) and thin (lower) metallic shells revealing that the hybridization is much more significant for the thin metallic shell.

Nanophotonics 2017; 6(1): 193–213

Porous effect on Plasmonic



Plasmon resonances shift to lower energies with increasing void size. Generally, such an energy shift is the result of two main effects:

(i) compositional effects due to alloying and

(ii) morphological effects due to void formation.

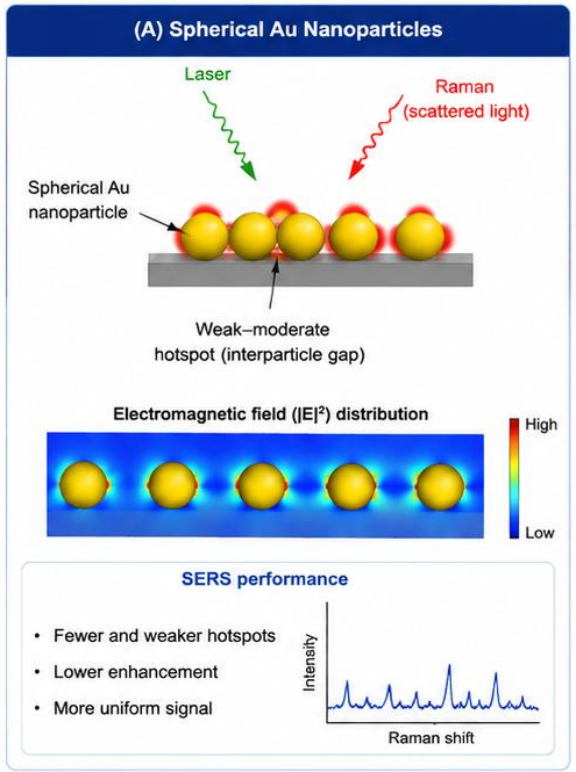
A) Structural sketches and corresponding solution colors from solid Ag nanocubes to single-walled AuAg nanobox and AuAg nanoframe. (B) HAADF STEM images (C) evolution of localized SPRs with structural changes. (D) BEM-simulated EEL spectra revealing the effect of void size for Ag nanostructures.

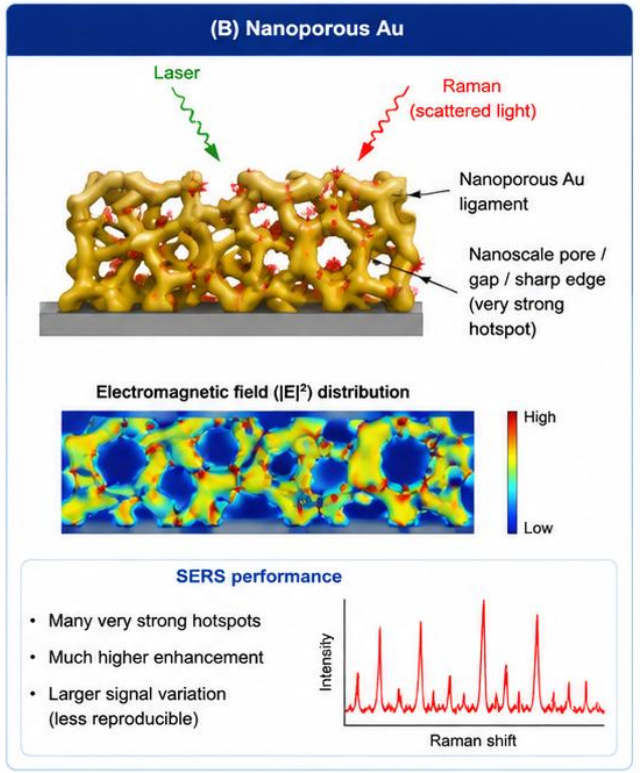
CRICOS code 00025B

ACS Photonics 2016;3:770–9.

Porous effect on Plasmonic

- ✓ **Sphere → relatively uniform field → fewer intense hotspots**
- ✓ **Cube / sharp edges → field concentration at corners**
- ✓ **Aggregated particles → interparticle gaps**
- ✓ **Mesoporous metal → many interconnected nanoscale hotspots**





Why nanoporous Au (B) gives stronger SERS

- ✓ Many nanoscale **pores**, gaps and sharp edges
- ✓ Very strong electromagnetic hotspots (**"hot spots"**)
- ✓ Much larger active surface area
- ✓ More analyte adsorption sites
- ✓ More molecules located in hotspots → stronger SERS

ACS Photonics 2016;3:770–9.

CRICOS code 00025B

Activity (15 min)

(A) Why can a nanoporous Au structure produce stronger SERS enhancement than spherical Au nanoparticles?

(B) How do pore size, pore connectivity, surface roughness and structural uniformity influence sensing performance?

(C) Design challenge: You need to detect a trace contaminant in water. Which plasmonic structure would you choose—spherical nanoparticles, aggregated nanoparticles or a porous metal substrate? Justify your choice.

CRICOS code 00025B

Components of SERS

- Substrate
- Reporter molecule
- Stabilizer molecule
- Conjugating material, and
- Type of fabrication

Substrate

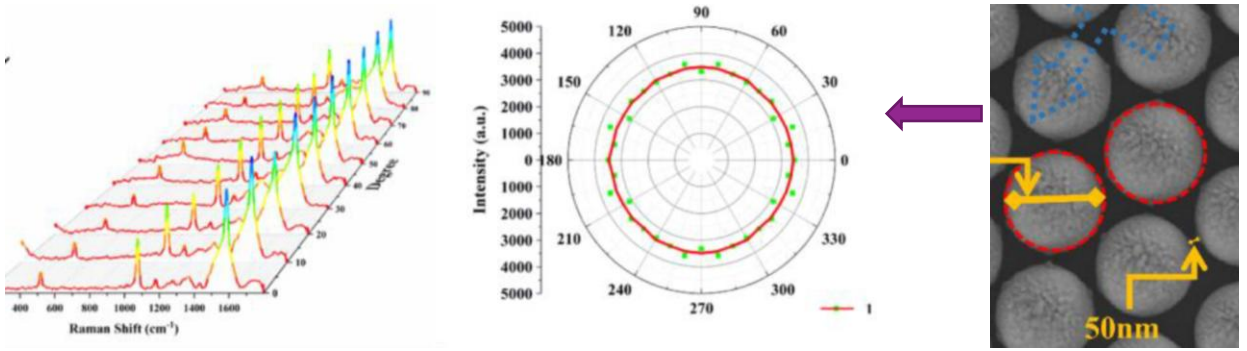
- Substrate is an important component of SERS-based assays.
- It acts as a solid surface that enables the capturing of biomarkers after appropriate functionalization.
- The performance of these substrates strongly depends on the composition of SERS-active materials.
- Ag and Au are widely applied in SERS substrates because of their localized surface plasmon resonance (SPR)
- SPR occurs when the photons of incident light strike the metallic surface at a certain angle of incidence and excite the surface electrons. The exposure of a metal surface to a laser beam scatters conduction electrons into frequency electrons, also known as plasmons, that generate an additional electric field corresponding to the electrons of the laser.
- In nanostructures, these electrons oscillate all over the surface and, thus, are commonly defined as localized surface plasmons, thereby increasing the intensity of the local electromagnetic field, a phenomenon known as localized SPR (LSPR).

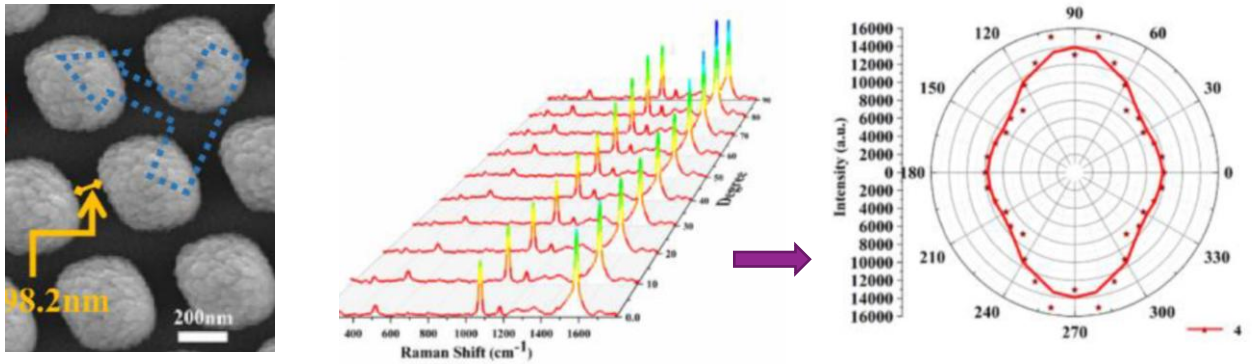
Components of SERS: Substrate

- LSPR depends on the NP size, smaller NPs display poor polarization and result in loss of LSPR properties. The reduced LSPR causes smaller SERS, thereby making these NPs impractical for various SERS-based applications.

- The shape of the NPs also influences the overall SERS efficiency. Spherical NPs have lower efficiency because of the uniform spread of electric field density (hotspots).

- Changing the shape from spherical to cubical NPs improves SERS efficiency, mainly because of the focused non-uniform field close to the sharp ends.

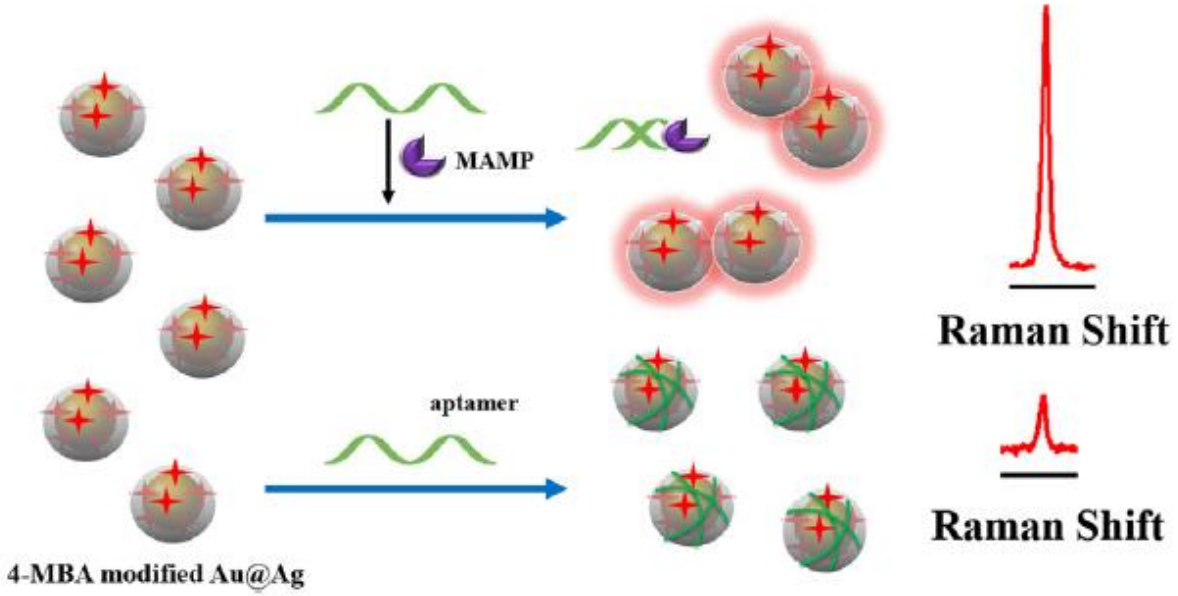




Molecules, 2021, 26, 187

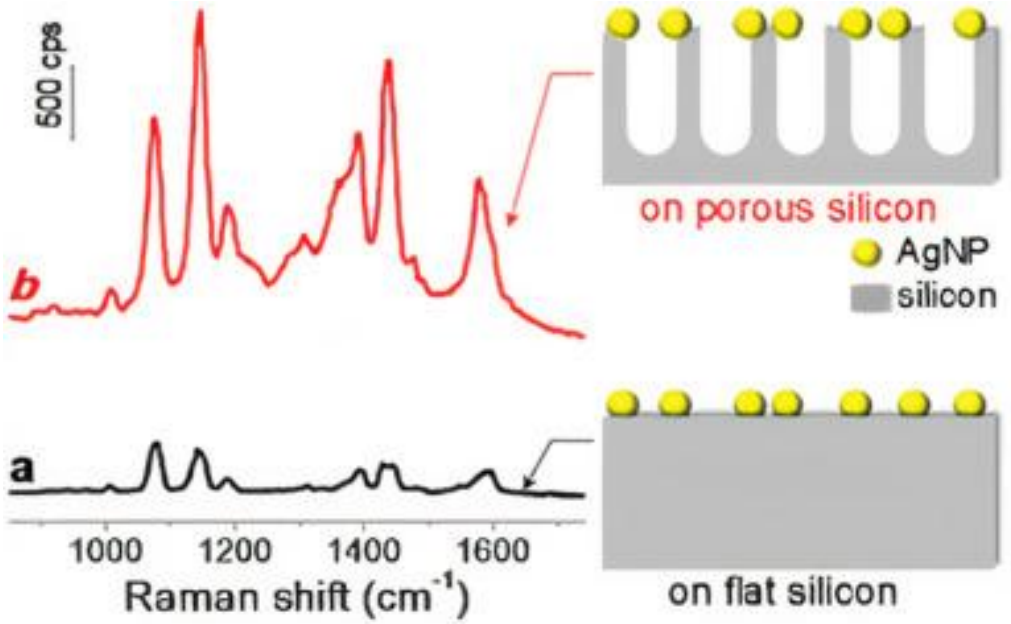
Components of SERS: Aggregation and Porous Substrate

- More hotspots are created with NP aggregation compared with single NPs.



Talanta 190 (2018) 263–268

- Incorporating pores also induces the SERS signal enhancement



Macroporus and Mesoporous Materials, 2021, 111204

Components of SERS: Reporter molecule

The most commonly used reporter molecules are Raman dyes, thiolated reporters, and other small molecules. Usually, Raman active molecules are immobilized on the surface of either AuNPs or AgNPs to form SERS nanotags.

- The functional abilities of Raman reporter molecules (RRMs) include properties such as the formation of self-assembled monolayers (SAMs), elevated Raman signal intensity, and in vivo imaging applications.

- SAM benefits include variability in their molecular orientation and the relative distance of the vibrating group to the metal surface. Thiophenol molecules are particularly relevant for the formation of SAMs. In addition, because of the small number of detectable Raman bands, these thiophenols are preferred when designing multiplex assays.

- Raman dyes [e.g., carboxyfluorescein (FAM), malachite green (MG), rhodamine 6G (R6G), malachite green isothiocyanate (MGITC), methylene blue (MB), and crystal violet (CV)]

- Thiolated reporters [e.g., 5,50-dithiobis(2-nitrobenzoic) (DTNB), 4-nitrothiophenol (NTP4), and mercaptobenzoic acid (MBA), 4-methoxythiophenol. These reporters have the ability to directly conjugate to the surface of metal nanostructures.

CRICOS code 00025B

Components of SERS

Stabilizer

To ensure the stability of SERS-encoded particles in biological fluids and prevent the release of the Raman reporter or nonspecific protein adsorption, several polymeric and nonpolymeric media, such as poly(ethylene glycol) (PEG), silica, and graphene, have been used as protective coatings. After the addition of the RRM, the stabilizers are adsorbed onto the surface of the NPs and prevent aggregation, thereby assisting bioconjugation.

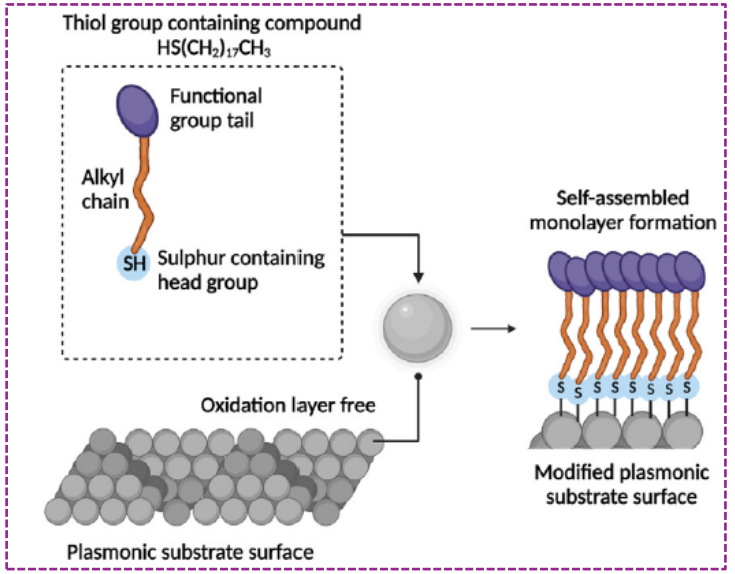
Conjugating Materials

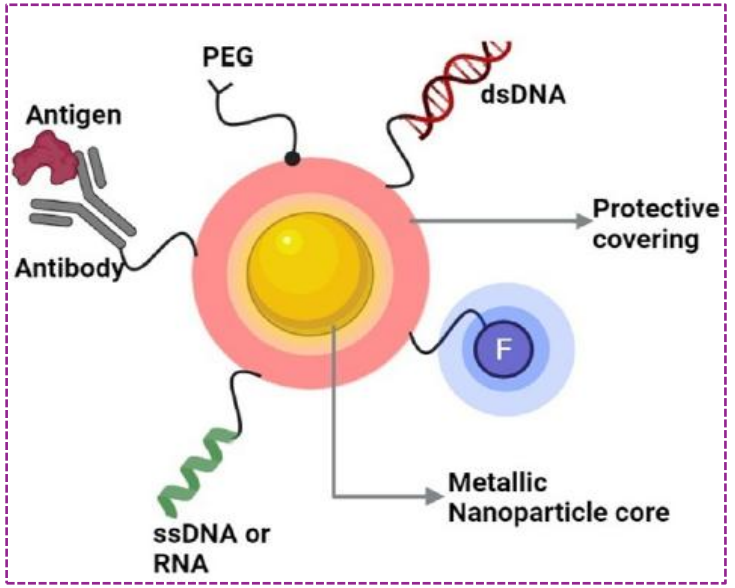
The tag or reporter molecules often have carboxyl groups, thiol groups, or primary amines.

The creation of an amide bond by carbodiimide activation facilitates bio-conjugation of the ligands on the surface of NPs.

The bio-conjugation formed by thiol-group attachment is considered a strong and effective conjugate as it provides better stability.

Other electrostatic interactions and attachment of biotin-streptavidin conjugates to NPs are also used.

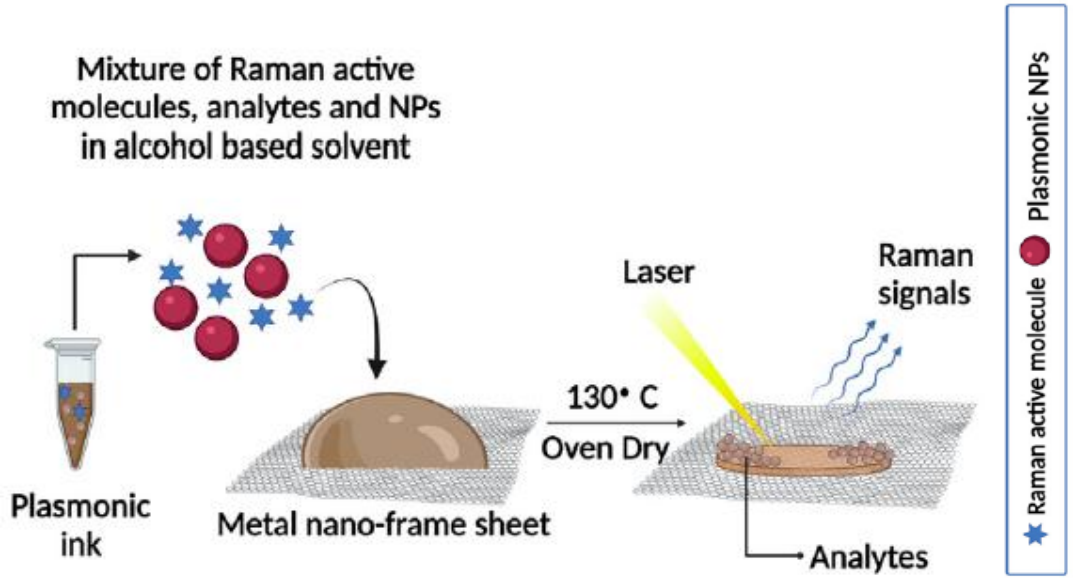




Drug Discovery Today, 2020, 27
(https://doi.org/10.1016/j.drudis.2022.04.016)

Sensors based on unmodified surfaces

The colloid substrates/metal substrate is coated with the standard materials to create a SAM. These SAMs form steady layers around the metal substrate with high coverage and enhance the chemisorption of an analyte to the coated surface.



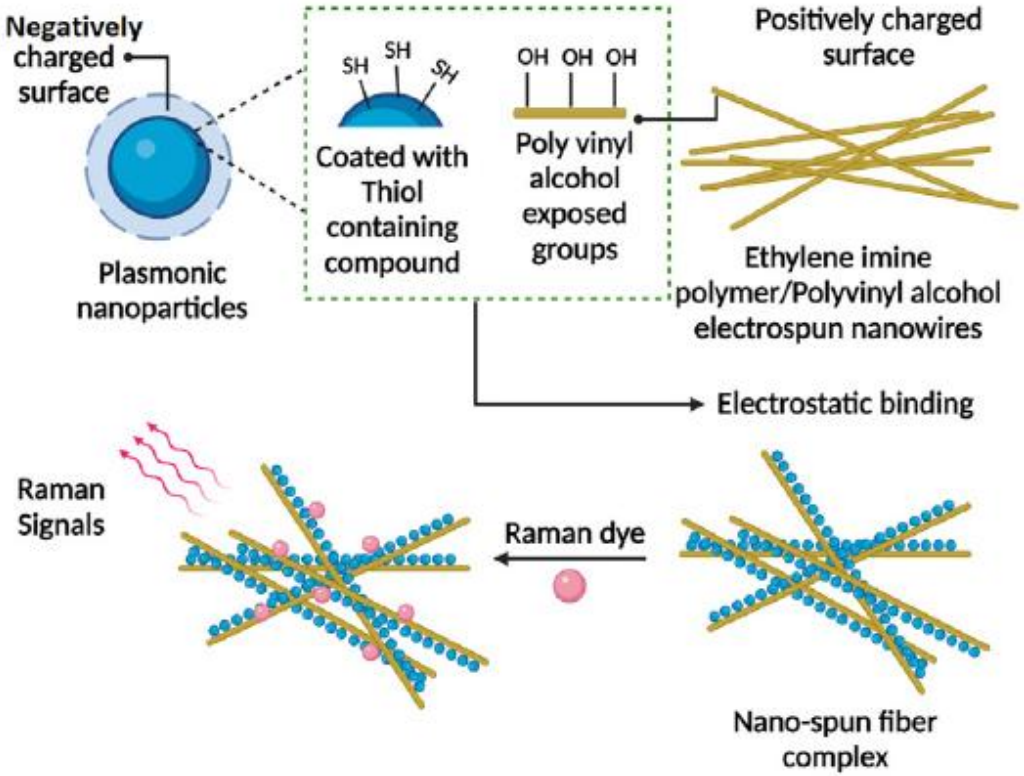
Drug Discovery Today, 2020, 27 (https://doi.org/10.1016/j.drudis.2022.04.016)

Limitation

- Low selectivity- because of imprecise binding and an absence of analyte affinity.
- It is difficult to prepare colloids with uniform sizes and shapes. Thus, the distribution of size and shapes in each batch might differ, which can significantly influence
- the Raman spectrum.
- Uncontrolled and irreproducible colloid aggregation might also make analyte absorption difficult.

Sensors based on modified surfaces

To enhance the selectivity and specificity, functionalized nanostructures with a huge variety of receptor molecules can be used.



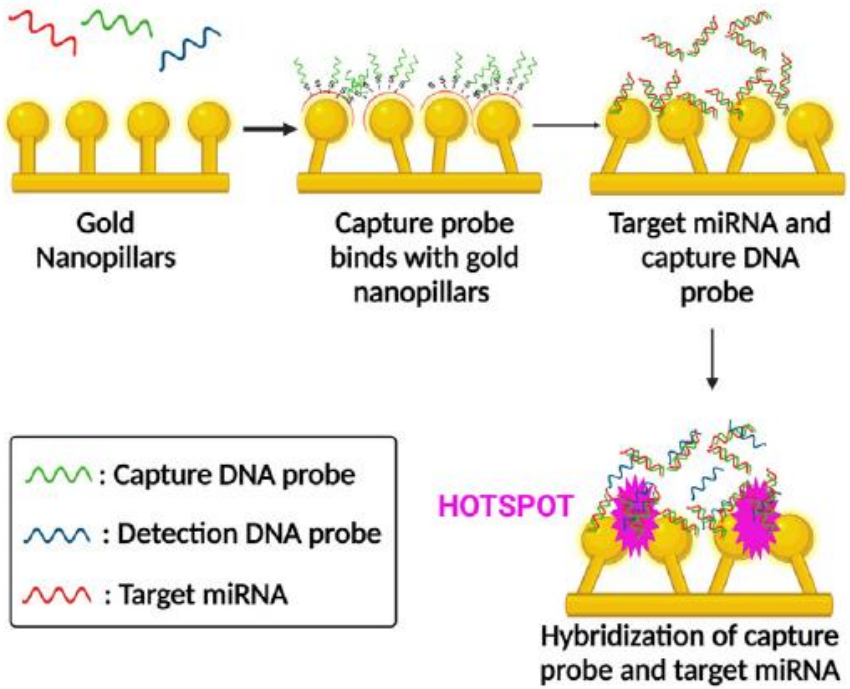
Drug Discovery Today, 2020, 27
(https://doi.org/10.1016/j.drudis.2022.04.016)

Limitation

- Selection of the receptor molecules is limited by the size and the Raman cross-section of the receptor molecules in proportion to the analytes.
- For bulky receptor molecules (e.g., antibodies), the captured molecules might be too distant from the surface to feel the effective increase.

Sensors based on SERS labelling

- SERS label or tags have specific properties, such as photostability, fluorescent potential, and enhanced Raman signals resulting from the in-built hotspots.
- Targeting ligands, including antibodies, aptamers, small molecules, or molecular imprinted polymers, are commonly used in these strategies.
- Enhancement of the Raman signal is directly proportional to the specific binding between an analyte and ligand.



Drug Discovery Today, 2020, 27 (https://doi.org/10.1016/j.drudis.2022.04.016)

Advantage

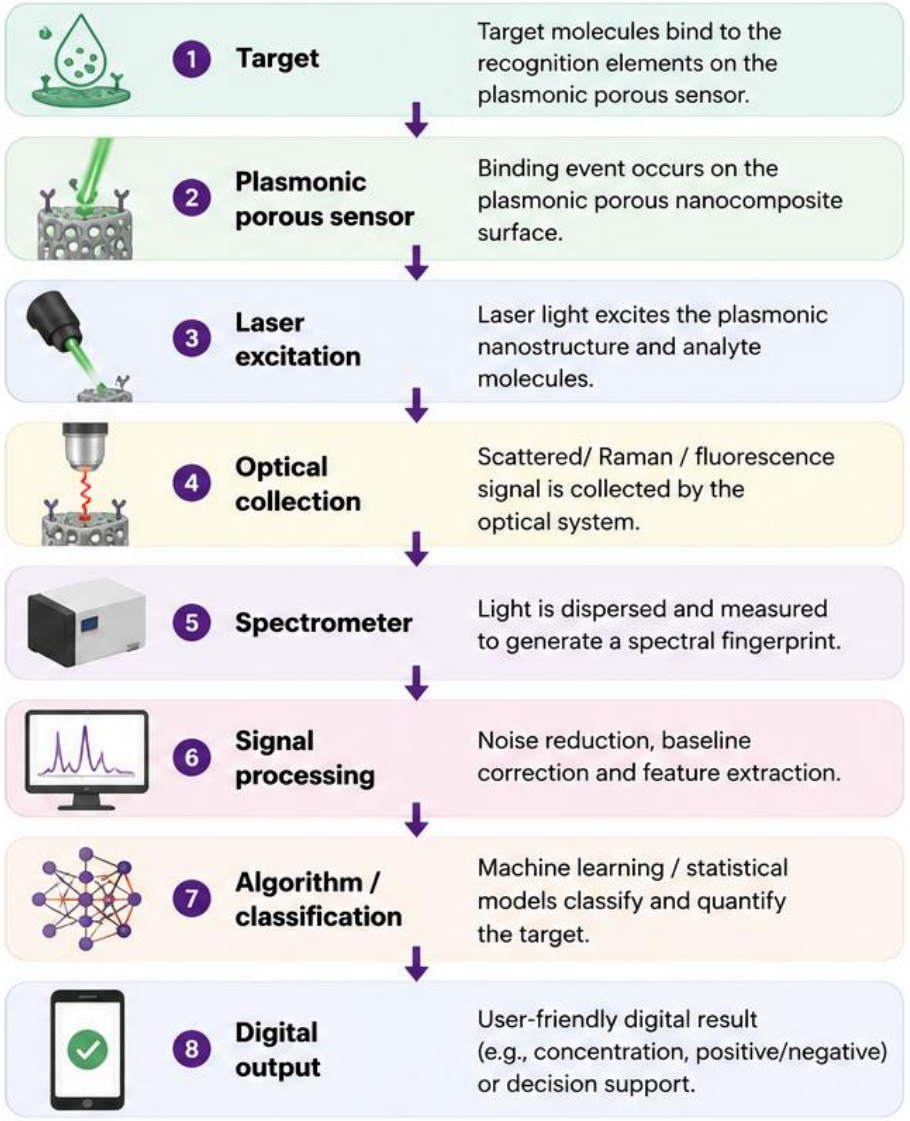
- When the analyte is exposed to the tag, changes such as immobilization of the SERS-tags at specific locations, generation or removal of hotspots, and change in distance between the Raman reporter molecule and the SERS-active substrate are observed.
- The formation of the SERS substrate– ligand complex not only enhances the specificity and sensitivity for trace analysis, but also reduces the chance of false spectra.

SERS-biosensor strategies

Approach	Simplicity	Selectivity	Sensitivity	Complexity
✓ Label-free/unmodified	★★★	★	★★	★
✓ Functionalised surface	★★	★★★	★★	★★
✓ SERS nanotag	★	★★★	★★★	★★★

Which architecture would you choose for a portable sensor and why?

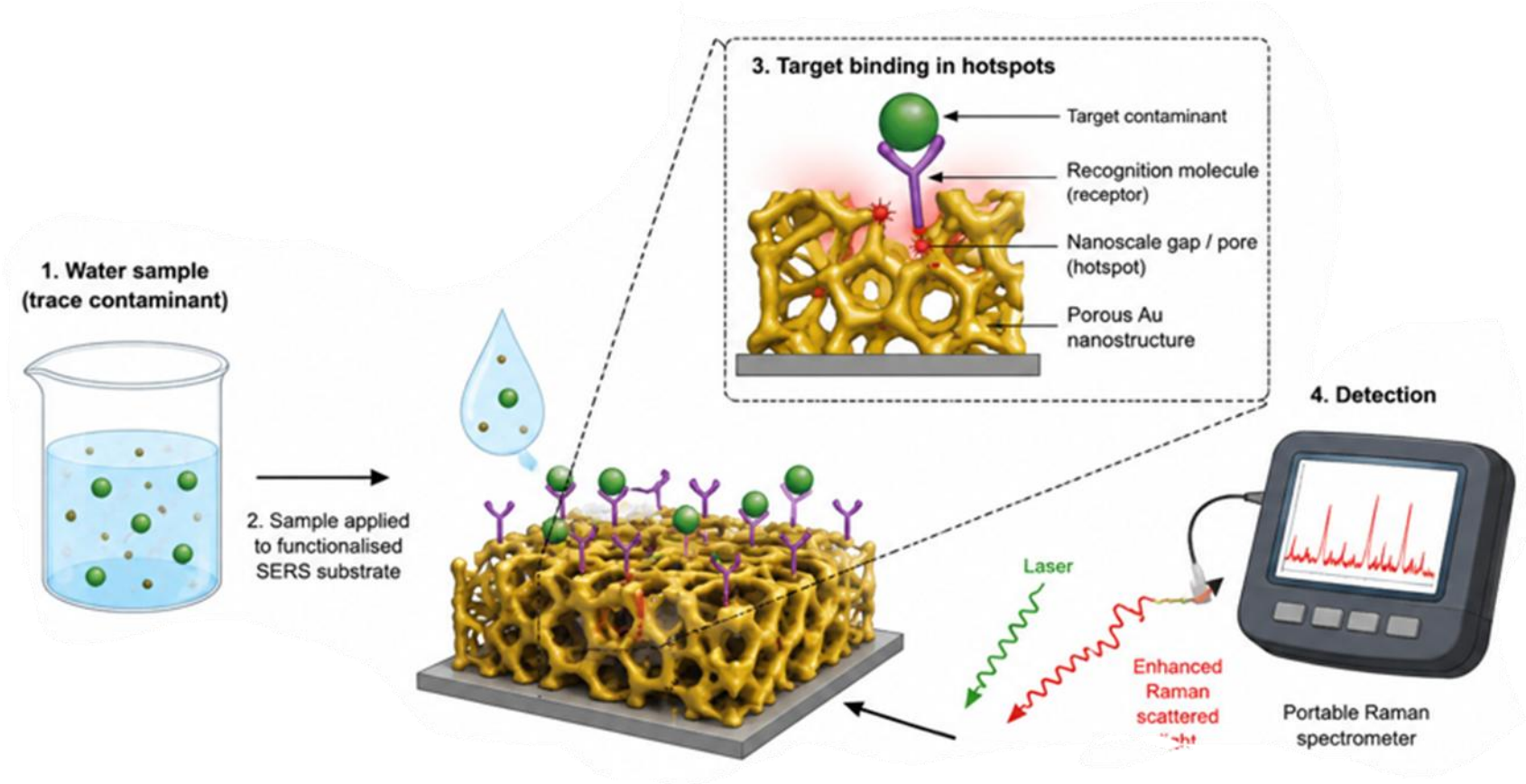
From nanomaterial to smart sensor Target



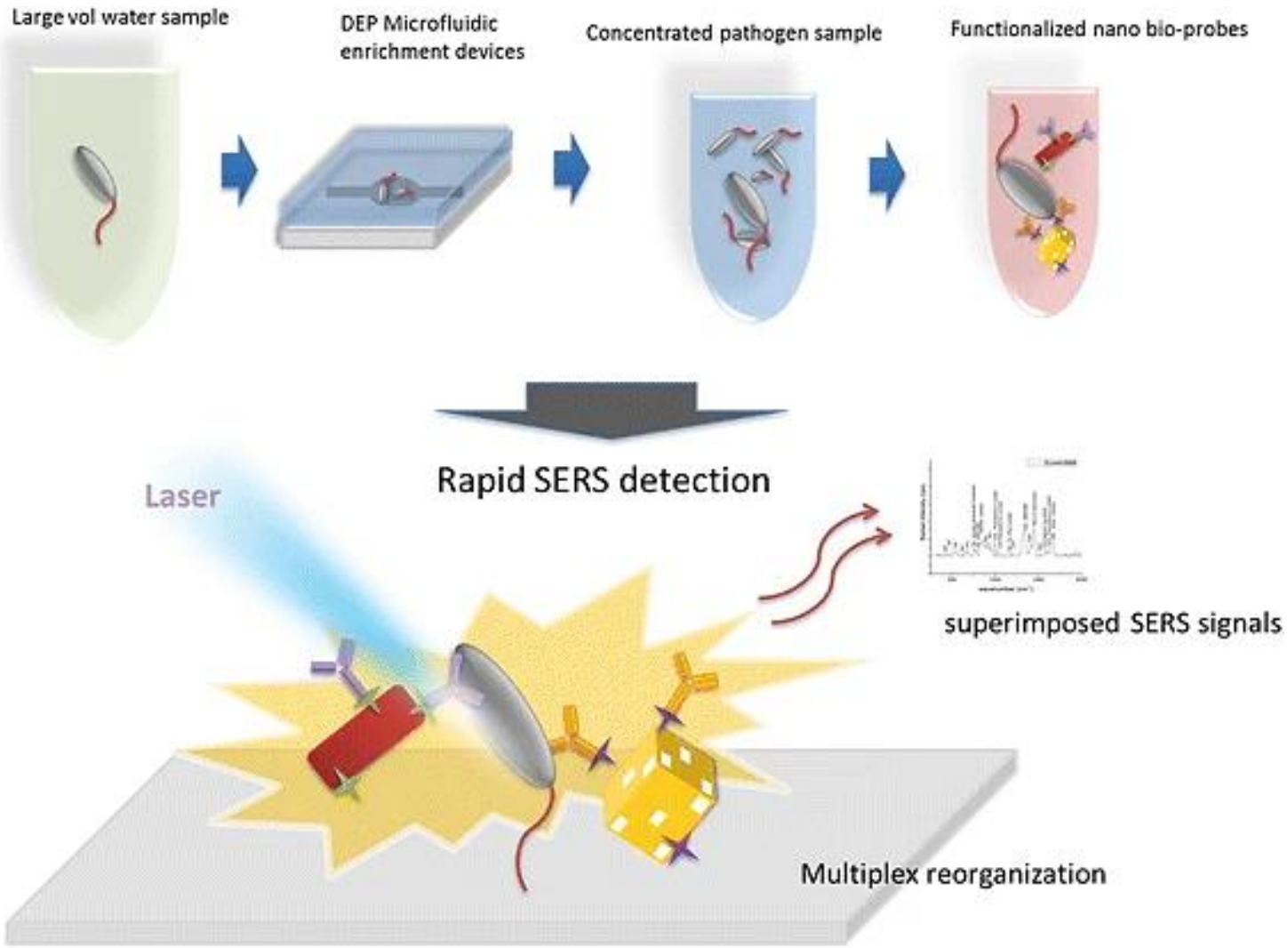
Identify your scope here!

Q. How can you progress this smart sensor?

Porous substrate for water contamination detection

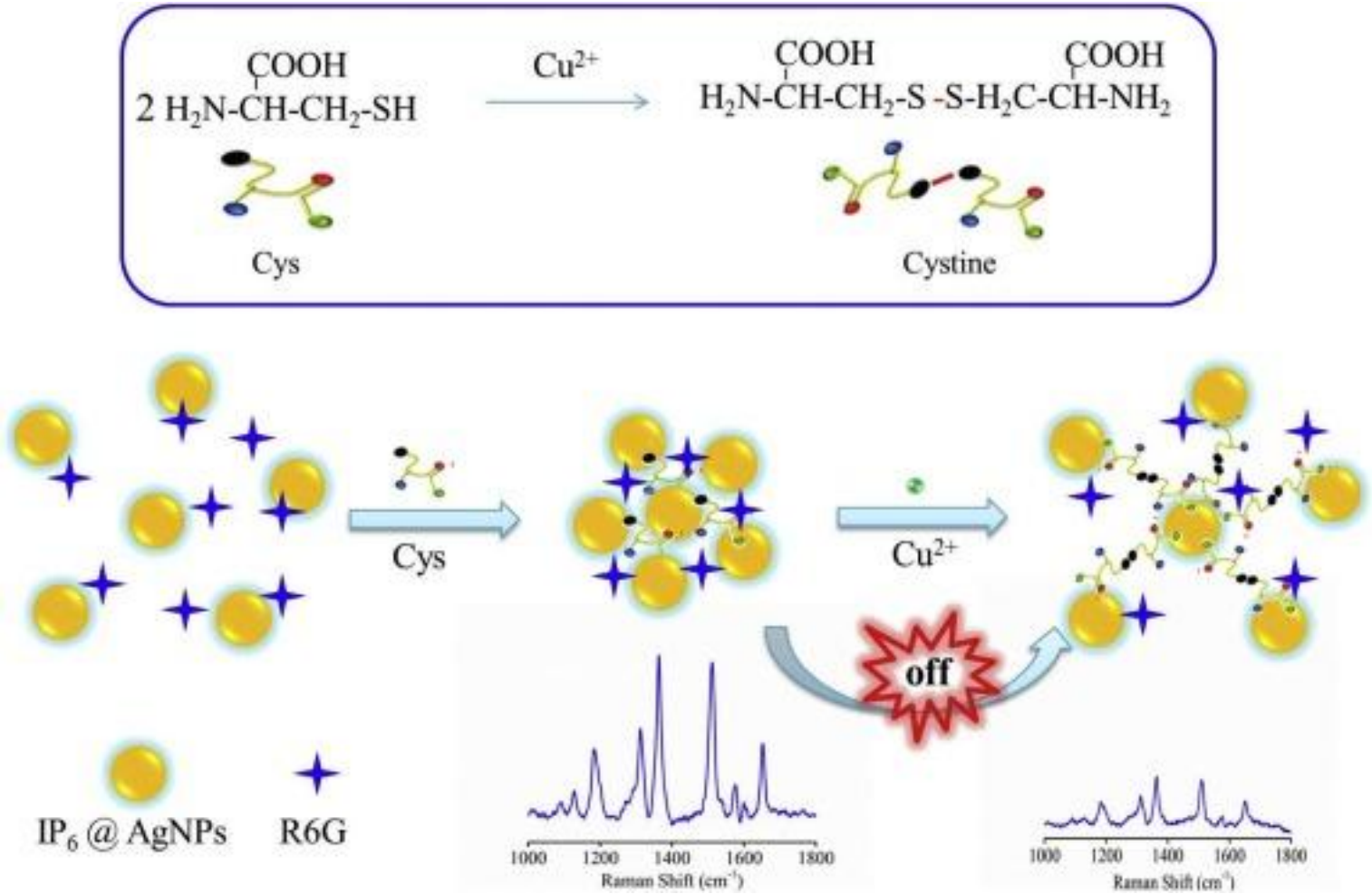


Porous substrate for water borne pathogen detection



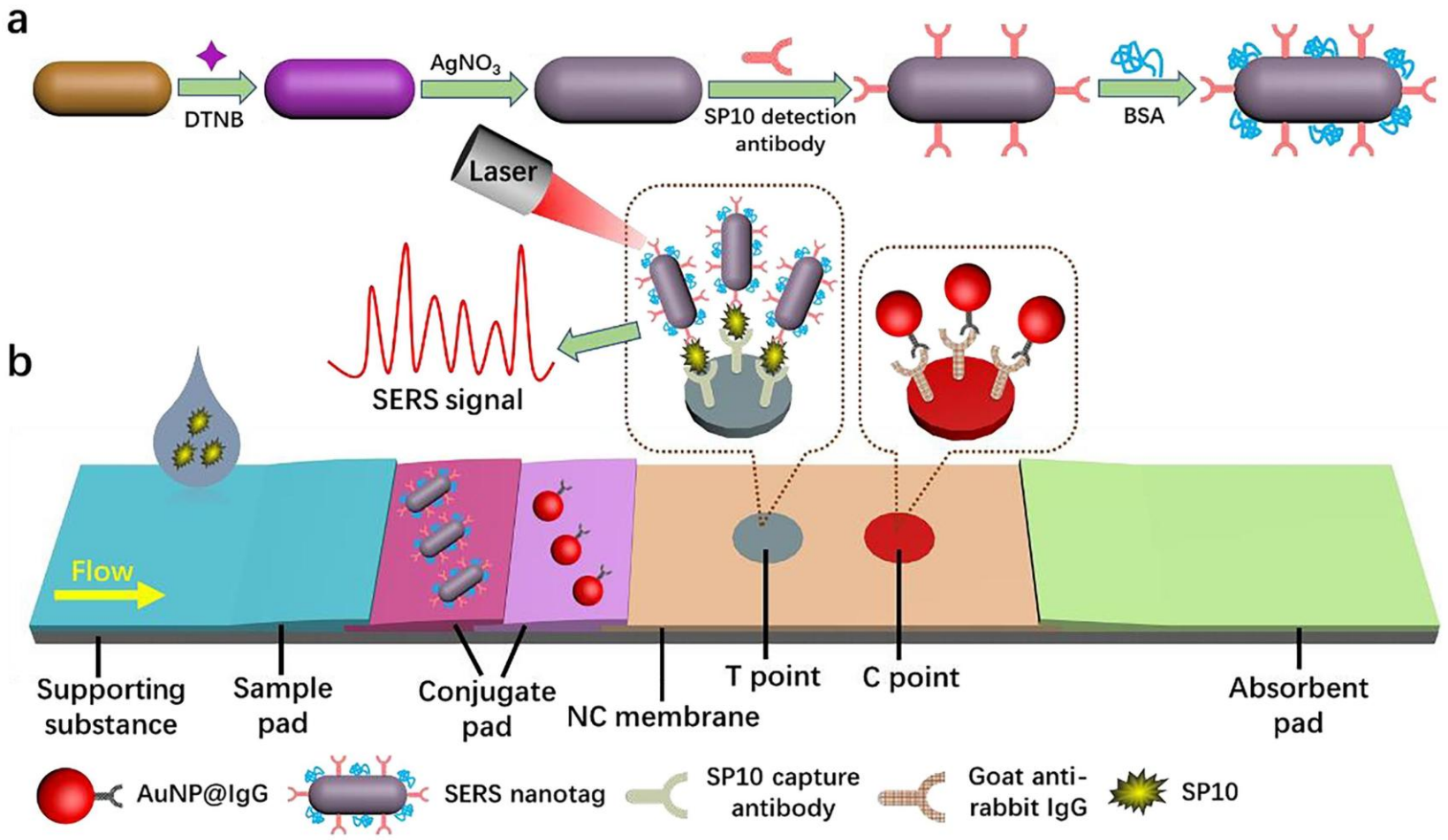
Journal of Biological Engineering, 2017, 11, 9

SERS for heavy metal detection



Sensors and Actuators B: Chemical, 2019, 283, 278-283

SERS-based lateral flow immunoassay strip



Microchemical Journal, **2022**, 175, 107191

CRICOS code 00025B

Activity (10 min)

1. A SERS sensor gives a strong signal but poor reproducibility. What could cause this problem, and how would you improve it?

2. Design challenge: Design a portable SERS sensor for detecting a low-concentration target in a real sample. Explain your choice of plasmonic material, sensing architecture and recognition strategy, and how you would achieve good sensitivity, selectivity and reproducibility.

CRICOS code 00025B

Recap and Questions

1. **Structure controls plasmonics: Size • Shape • Porosity • Roughness • Composition**

-determine LSPR electromagnetic hotspots and SERS enhancement.

2. Porosity enhances sensing: High surface area + interconnected pores + nanoscale hotspots

-enable efficient target interaction and strong signal enhancement.

3. Interface controls selectivity: Surface functionalisation + recognition molecules

-bring the target to the plasmonic surface and enable selective detection.

4. Engineering determines real-world performance:

-A successful sensor must balance **Sensitivity • Selectivity • Stability • Reproducibility • Manufacturability**

Nanostructure → Plasmonic Properties → Target Recognition → Signal → Smart Sensor → Real-World Application

WEEK 8 · 知识点总结 · Dr Yusuf Valentino Kaneti

第 8 周　金属有机框架（MOF）用于水处理

Metal-Organic Frameworks for Water Treatment

主线：纳米多孔材料 → 沸石 vs MOF → MOF 合成、尺寸控制与气凝胶 → 表征 → 吸附等温线与动力学（计算）

1. 纳米多孔材料与沸石

- 纳米多孔：孔径 < 100 nm；IUPAC：**微孔 < 2 nm**（沸石、MOF）、**介孔 2–50 nm**（介孔碳、介孔硅）、**大孔 > 50 nm**。
- **沸石**：水合铝硅酸盐，AlO_4 与 SiO_4 四面体相连，$M_{x/n}Si_{1-x}Al_xO_2{\cdot}yH_2O$（M = Na^+, K^+, Ca^{2+}…）。天然（> 40 种，火山成因，孔很小）；合成（> 100 种，0.2–1 nm，石油裂化如 ZSM-5，重金属去除）。

2. MOF

- 定义：**金属离子（节点）+ 有机配体（连接体）配位形成的晶态多孔材料**；Yaghi（UC Berkeley）与 Kitagawa（京都大学）于 90 年代末开创；已合成 > 90,000 种。
- 金属：二价 Zn、Cu、Co、Ni、Mn；三价 Al、Cr、In；四价 Ti、Zr。配体：二羧酸 **BDC**、三羧酸 **BTC**、四羧酸、咪唑类 **2-MeIM**。
- **ZIF-8（Zn）/ ZIF-67（Co）** + 2-甲基咪唑：结构类似沸石（Si/Al → 过渡金属，桥连 O → 咪唑）。
- 性质：组成 / 结构 / 孔可控；通常微孔；比表面积**可达 8000 m^2/g**；热稳定 300–600 °C；热导和电导低。

MOF	沸石
比表面积高（≤ 8000 m^2/g）	较低（500–800 m^2/g）
组成高度可控，孔径可调（微 / 介 / 多级）	组成控制有限，孔径 0.2–1 nm
热稳定性较低（300–600 °C），化学稳定性较低	更高（至 800 °C），化学稳定性更高
成本较高	成本低，大量商业化

3. 合成方法（6 种）

- **沉淀法**：室温至 < 100 °C、常压（ZIF-8、ZIF-67、HKUST-1）。**水热 / 溶剂热**：高温高压不锈钢釜（MIL-125）。**微波**：更均匀、快、产率高（UiO-66）。**电化学**：阳极溶解供应金属离子，快、低温。**超声**：空化，局部约 4727 °C、1000 bar，得细小晶体。**机械化学**：无溶剂球磨，可放大，得纳米 MOF。

4. 表征与尺寸控制

- **XRD**（与 CCDC 模拟谱比对，晶态）；**FTIR**（有机键，与配体比较确认成键）；**BET**（N_2 吸附：比表面积与孔径分布；MOF 多为 **I 型等温线** = 微孔）；**SEM/TEM/EDS**（形貌、空心结构、成分）。
- **成核快 → 晶核多 → 生长时间短 → 颗粒小**：阴离子活性 NO_3^- > CH_3COO^- > Cl^-（ZIF-8：88 / 240 / 533 nm）；配体 : 金属比高 → 小；温度高 → 小；时间长 → 大。减小尺寸 → 比表面积和活性位点增加、扩散路径缩短。
- **MOF 气凝胶**（解决粉末难加工、难回收、二次污染）：**混合法**（ZIF-8 + 琼脂糖 → 冷冻干燥）；**原位生长**（ZIF-67 长在棉浆纤维素气凝胶上，分布更均匀，更优）。

5. 水处理应用

- 吸油：ZIF-8@rGO@PU 海绵，吸附量 1400–2900 wt%，循环 100 次（介孔 + 超疏水）。
- 染料吸附：rGO/ZIF-67 对阳离子 CV 吸附 > 阴离子 MO（pH 6 表面带负电）。三要素：**表面电荷**（异电相吸）、**孔径 > 染料分子**、**π–π 作用**。
- 商业：NuMat ION-X 气瓶，1 atm 储存剧毒掺杂气体。

6. 定量分析（★ 计算）

- **Beer–Lambert**：$A = \varepsilon l C$，用标准曲线把吸光度换算成浓度。
- **Langmuir**（均匀表面、单层）：$C_e/q_e = 1/(q_m K_L) + C_e/q_m$；**Freundlich**（非均匀、多层）：$\log q_e = \log K_F + (1/n) \log C_e$。
- **准一级**（物理吸附限速，适合初期）：$\log(q_e - q_t) = \log q_e - k_1 t/2.303$；**准二级**（化学吸附限速，全程）：$t/q_t = 1/(k_2 q_e^2) + t/q_e$。**$R^2$ 越接近 1 越合适**。
- 例：t/q_t–t 斜率 1.1314 → q_e = 1/1.1314 = **0.88 mg/g**；截距 6.5421 → $k_2 = 1/(6.5421 \times 0.88^2)$ = **0.20 g mg^{-1} min^{-1}**。

考试重点：IUPAC 孔分类；MOF vs 沸石表；合成方法特点；尺寸控制规律；表征手段各测什么；染料吸附三要素；Langmuir / Freundlich、准一级 / 准二级的假设与从线性图求参数。

MATE6301
METAL-ORGANIC FRAMEWORKS FOR WATER TREATMENT

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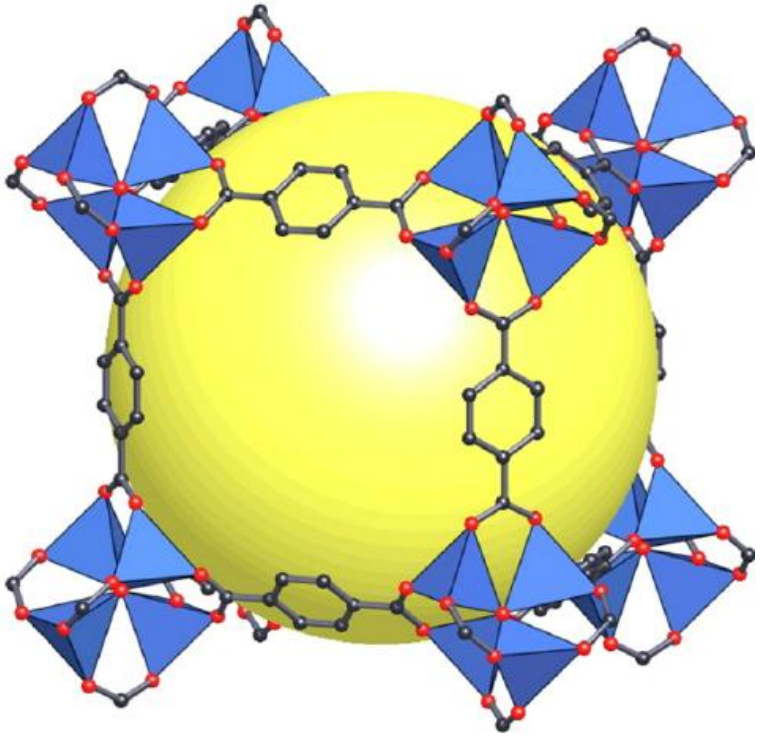
Australian Institute for Bioengineering and Nanotechnology
The University of Queensland, St Lucia Campus

LECTURE OUTLINE

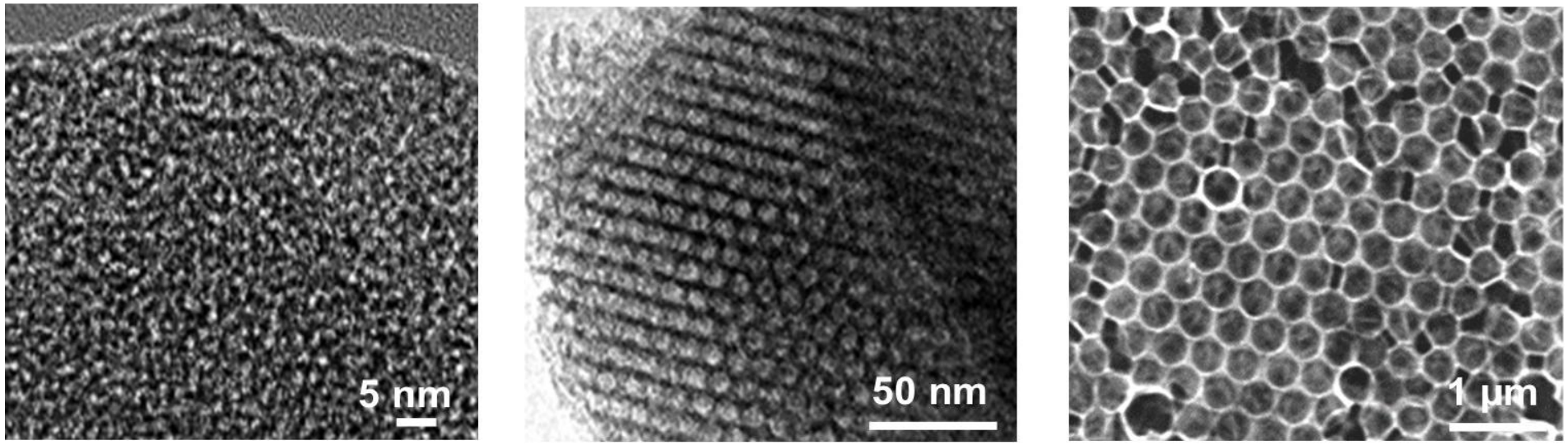
The University Of Queensland
AUSTRALIA

1. Nanoporous Materials
2. Zeolites
3. Metal-organic frameworks (MOFs)
   - Synthesis of MOFs
   - Fabrication of MOF aerogels
   - Characterisation of MOFs
   - Applications of MOFs in water treatment



1. NANOPOROUS MATERIALS

- **Definition:** Porous materials with pore size less than 100 nm.
- Classification of nanoporous materials:



Microporous
(pore size < 2 nm)

Mesoporous
(pore size: 2-50 nm)

Macroporous
(pore size >50 nm)

Figure 1. IUPAC classification of nanoporous materials.

- **Microporous materials:** zeolites, metal-organic frameworks (MOFs)
- **Mesoporous materials:** mesoporous carbon, mesoporous silica, etc.

2. ZEOLITES

- Zeolites are hydrated aluminosilicate minerals made from interlinked tetrahedra of alumina (AlO_4) and silica (SiO_4).

- The general chemical composition of a zeolite is:

$$M^n_x/_n Si_{1-x} Al_x O_2 \cdot yH_2O$$

where M = Na^+, K^+, Li^+, Ca^{2+}, Ba^{2+}…

- Two types of zeolites:

(1) Natural zeolites

- Aluminosilicate minerals
- Occur in nature → volcanic origin
- Over 40 types of natural zeolites
- Commonly used for adsorption, water treatment → very small pore size

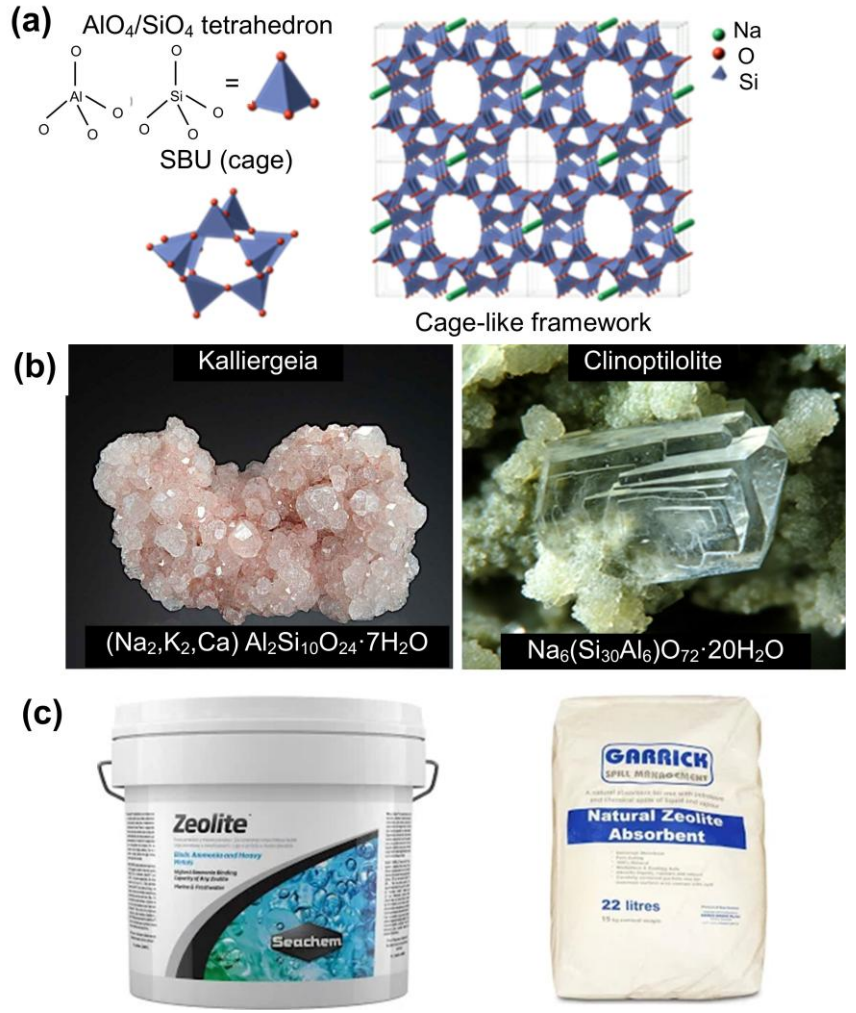


Figure 2. (a) Basic zeolite structure. (b) Examples of natural zeolites, and (c) applications of zeolites.

2. ZEOLITES

THE UNIVERSITY OF QUEENSLAND AUSTRALIA

(2) Synthetic zeolites

- Over 100 types of synthetic zeolites.
- Manufactured synthetically using a thermal process.
- Microporous but pore sizes are larger than natural zeolites (~0.2-1 nm).
- Used as catalysts for petroleum cracking, wastewater treatment (heavy metal ion removal) and adsorption.

Figure 3. Examples of synthetic zeolite catalysts (ZSM-5 & 13A).

3. METAL-ORGANIC FRAMEWORKS (MOFs)

- **Definition**: Crystalline, porous materials formed by the simple coordination of metal ions and organic linkers (ligands).
- Pioneered in late 90's by Professor Omar Yaghi from UC Berkeley (USA) and Professor Susumu Kitagawa from Kyoto University (Japan)
- Since late 90's, >90,000 MOFs have been synthesized and over 500,000 MOFs have been predicted by simulations.

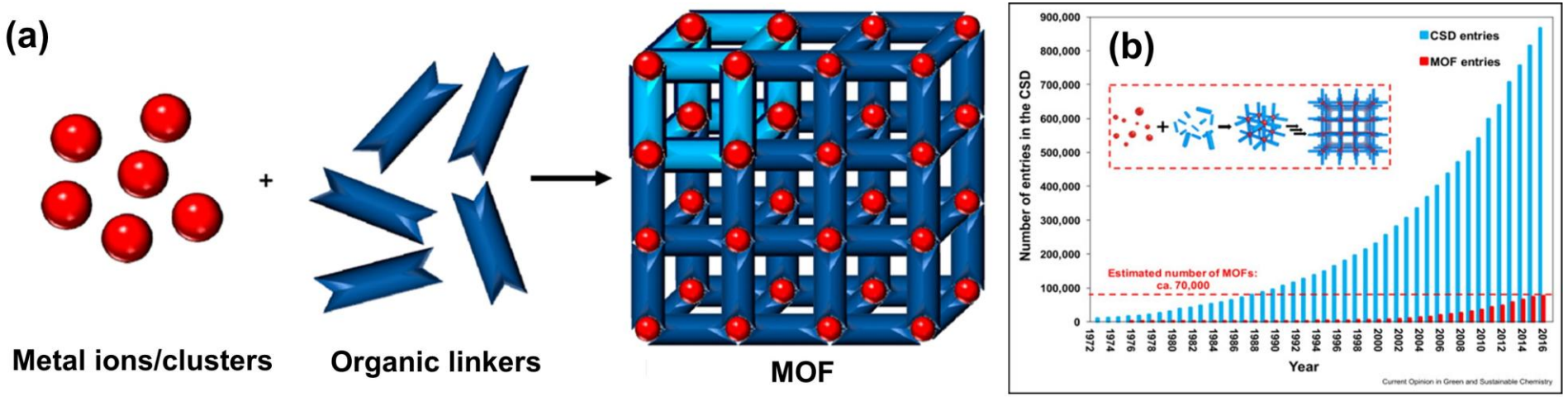


Figure 4. (a) Schematic illustration showing the general assembly process of MOF. (b) A plot showing the number of MOFs listed in Cambridge Crystal Database from 1972 to 2016.

3.1. COMPOSITIONS OF MOFs

- MOFs have controllable composition. They can be prepared by combining different metal ions (centers) and organic linkers.

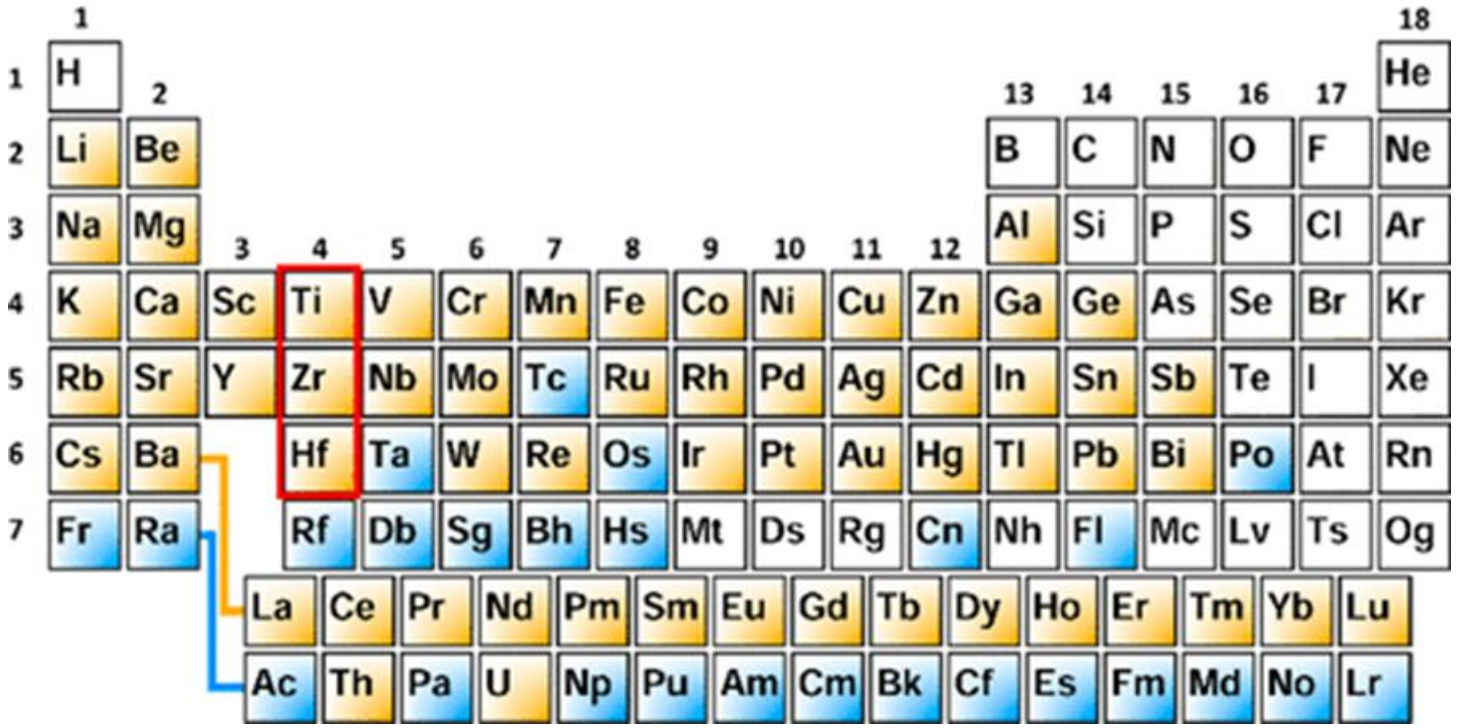


Figure 5. Various metal centres used for the preparation of MOFs.

- Metal centres in MOFs:

Divalent metals: Ni^{2+}, Co^{2+}, Mn^{2+}, Zn^{2+}, Cu^{2+}

Trivalent metals: Al^{3+}, Cr^{3+}, In^{3+}

Tetravalent metals: Ti^{4+}, Zr^{4+}, Sn^{4+}

Higher valency metals: W^{6+}

3.2. ORGANIC LINKERS FOR MOFs

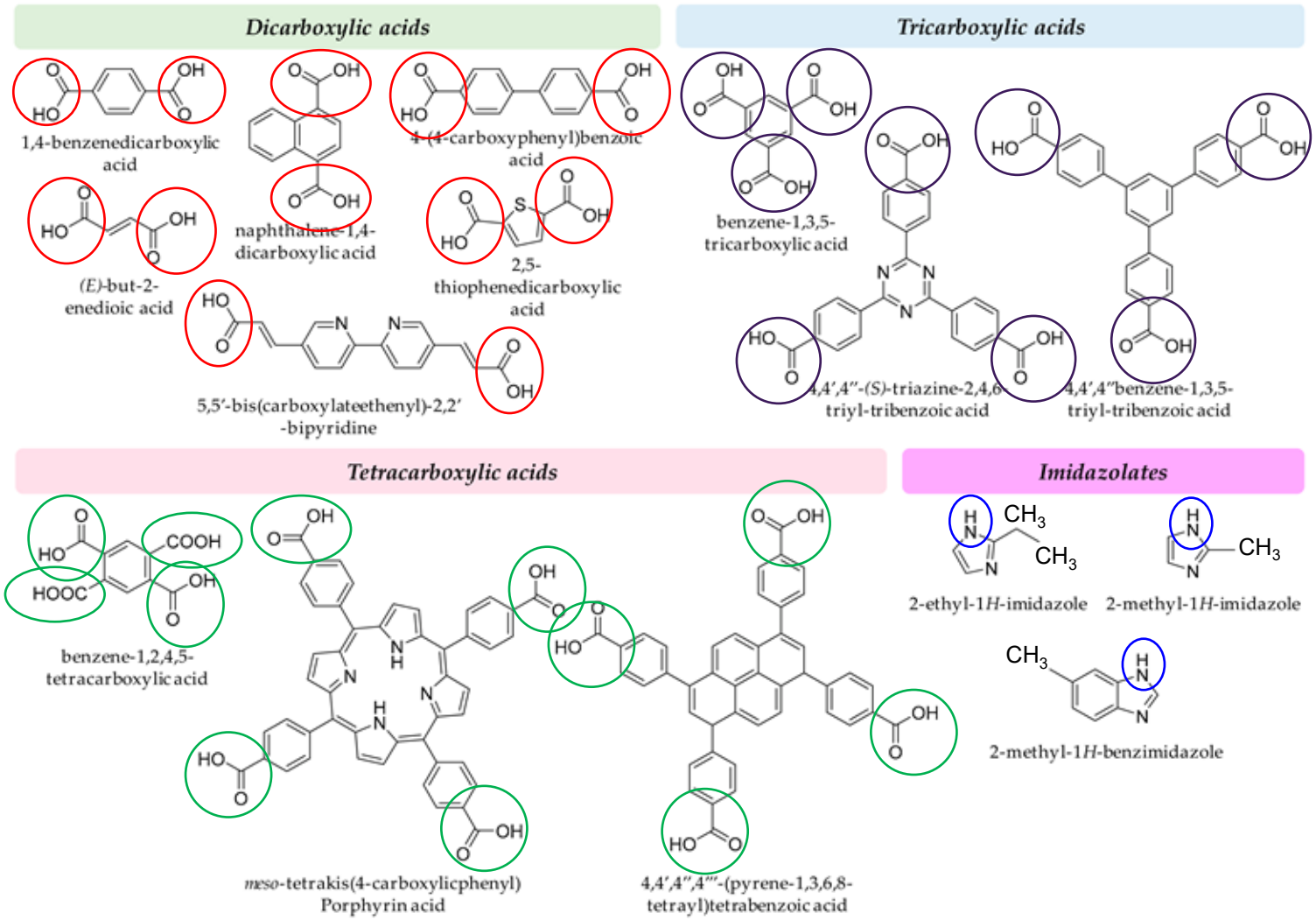


Figure 6. Common linkers used for the preparation of metal-organic frameworks (MOFs).

Typical linkers in MOFs:

- **Dicarboxylic acids:** contain two carboxylic acid (-COOH) groups, e.g., 1,4-benzenedicarboxylic acid (often shortened as “BDC”)
- **Tricarboxylic acids:** contain three carboxylic acid groups, e.g., benzene-1,3,5-tricarboxylic acid (often shortened as “BTC”)
- **Tetracarboxylic acids:** contain four carboxylic acid groups, e.g., benzene-1,2,4,5-tetracarboxylic acid
- **Imidazolates:** contain two non-adjacent nitrogen atoms, e.g., 2-methylimidazole (2-MeIM)

3.3. EXAMPLES OF MOFs

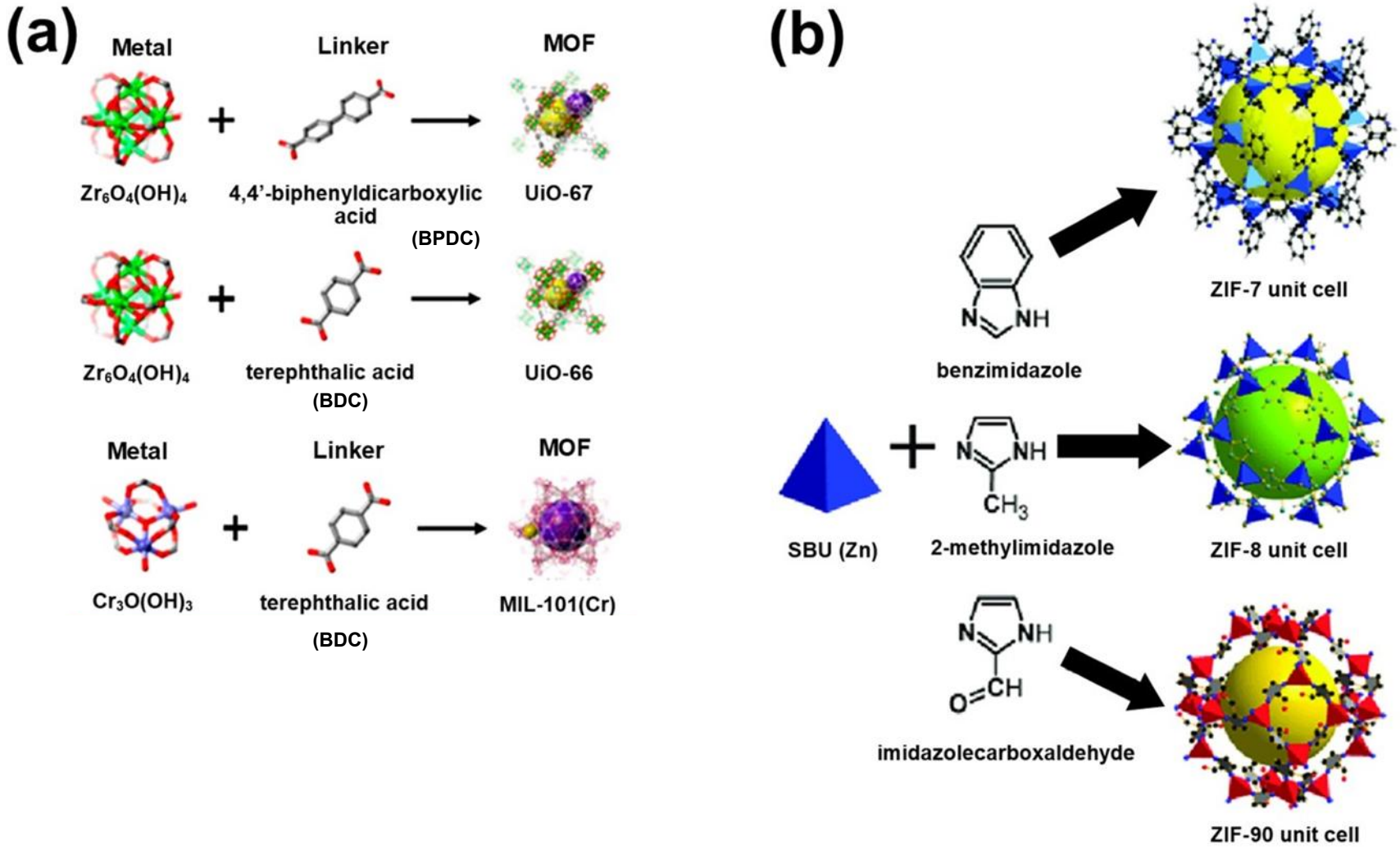
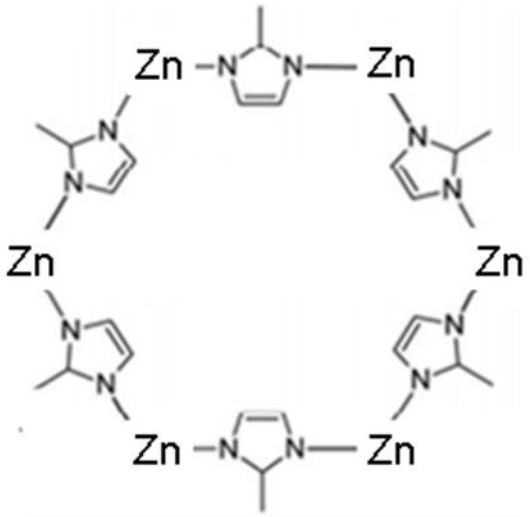


Figure 7. (a) Examples of different MOFs obtained by combining different metal centres and organic linkers. (b) Examples of different ZIF-based MOFs which can be obtained using Zn metal centre and different imidazolate-based linkers.

Zeolitic imidazolate frameworks (ZIFs)

- ZIF-8 (ZIF-67) is formed by coordinating zinc (cobalt) ions with 2-methylimidazole linker.
- Exhibit similar structu conventional alumino
- The tetrahedral Si or O in zeolites are repla metals (e.g., Zn or Co linkers, respectively.
- Properties: microporous, high surface area, good thermal stability, decent chemical stability, high crystallinity, etc.



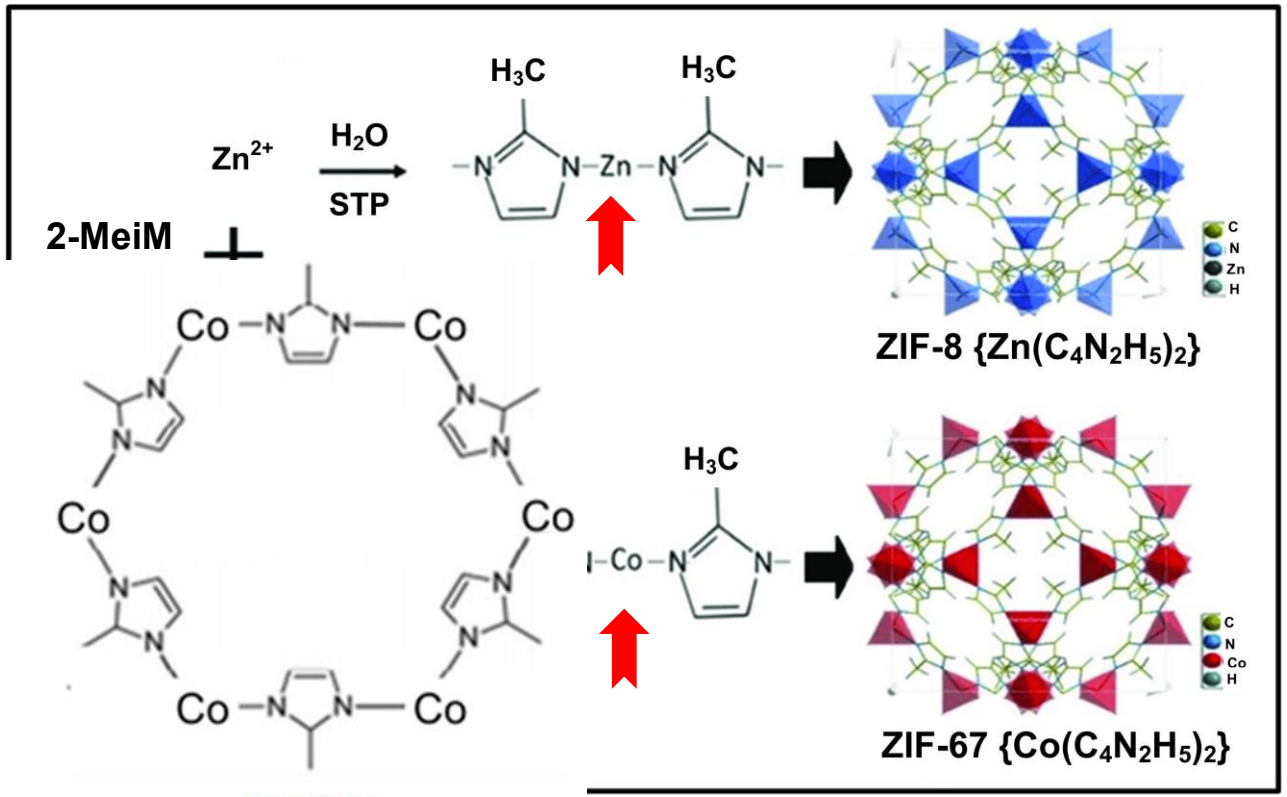


Figure 8. Schematic illustration showing the formation of ZIF-8 and ZIF-67 from zinc (cobalt) ions and 2-methylimidazole. Inset shows the typical morphology of ZIF-8 or ZIF-67 particles.

3.5. PROPERTIES OF MOFs

- Controllable composition, structure, and porosity.
- Generally microporous.
- High internal surface area (up to 8,000 m^2/g).
- High thermal stability (300-600 ºC in air).
- Low thermal conductivity. (<2 $W\ m^{-1}\ K^{-1}$ at RT ≈ concrete)
- Generally low electrical conductivity.

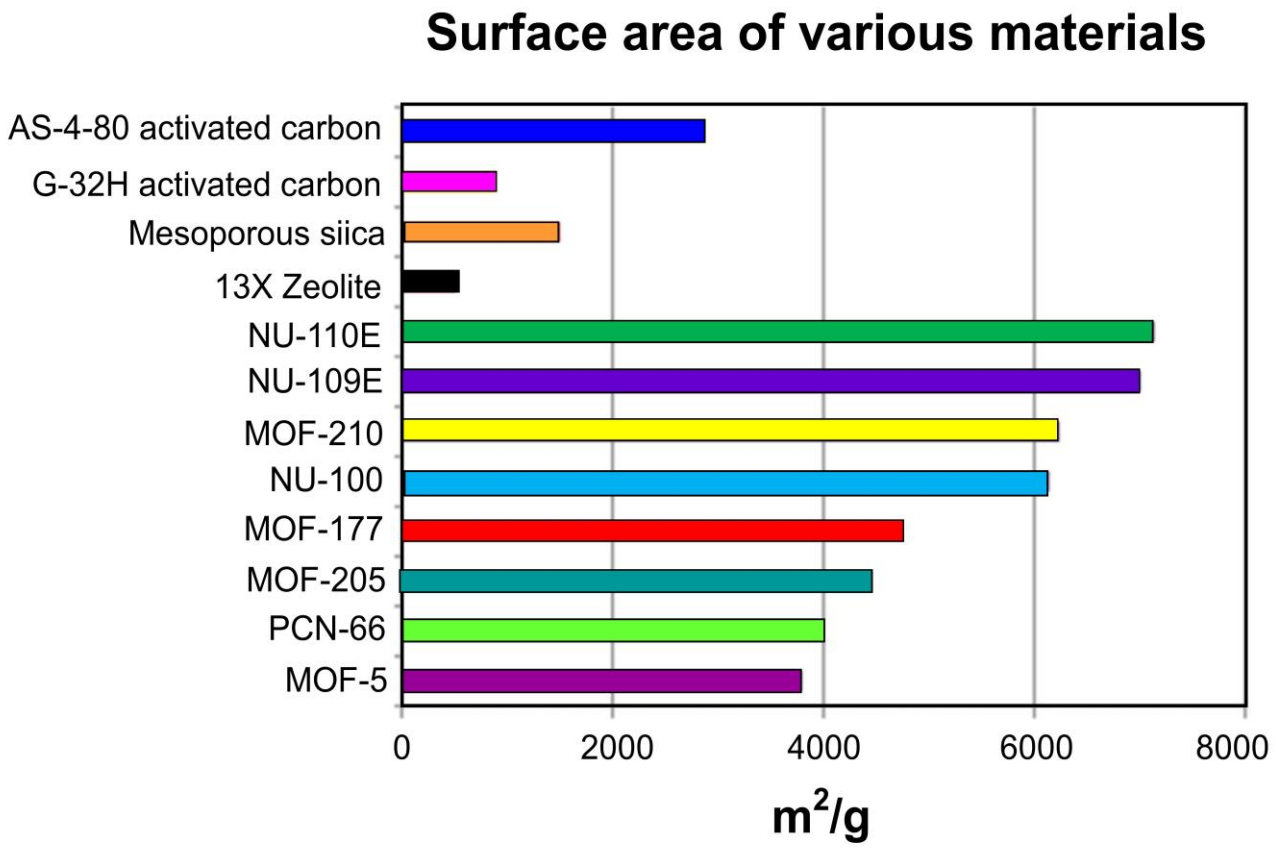


Figure 9. Comparison of surface areas of MOFs and other porous materials.

3.5. PROPERTIES OF MOFs vs. ZEOLITES

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Metal-organic frameworks	Zeolites
Higher surface area (up to 8,000 m^2/g)	Lower surface area (500-800 m^2/g)
Highly controllable composition	Limited control of composition
Controllable pore size (micro, meso, hierarchical pores)	Limited pore size (microporous, 0.2-1 nm)
Lower thermal stability (300-600 ºC)	Higher thermal stability (up to 800 ºC)
Lower chemical stability	Higher chemical stability
Higher cost (some commercialised)	Lower cost (many commercialised)

- Introduction to MOFs: https://www.youtube.com/watch?v=m91P-R3kxOs

3.6. SYNTHESIS OF MOFs

(1) Precipitation

- Aging at low temperature (RT to <100 ºC).
- Performed at ambient pressure.
- Used to synthesize MOFs that readily precipitate at low temperatures
  e.g., ZIF-8, ZIF-67, HKUST-1 (Cu-BTC)

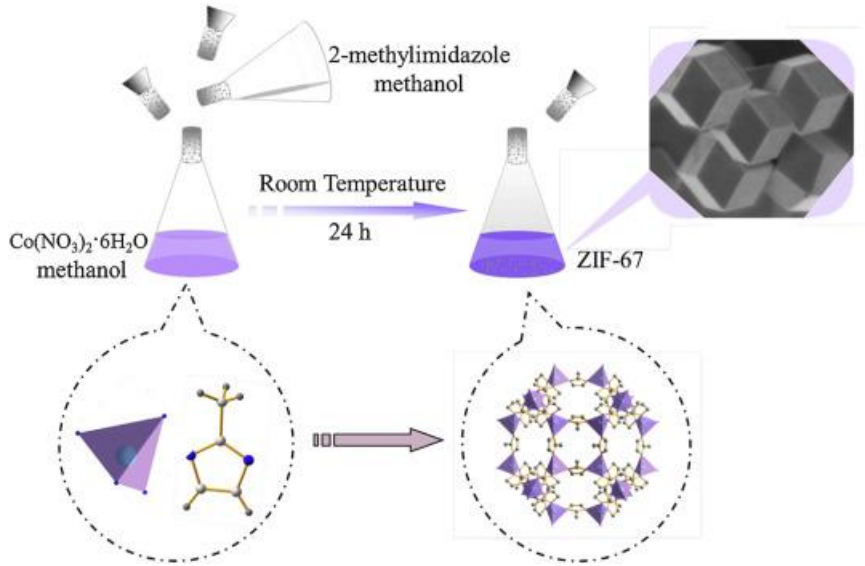

Figure 10. Synthesis of ZIF-67 particles by precipitation method.

(2) Hydrothermal/solvothermal method

- *Hydrothermal*: uses aqueous media
- *Solvothermal*: uses non-aqueous media
- Conducted at high temperature and high pressure.
- Performed in stainless-steel autoclaves.
- Examples: ZIF-8, ZIF-67, MIL-125 (Ti-BDC)

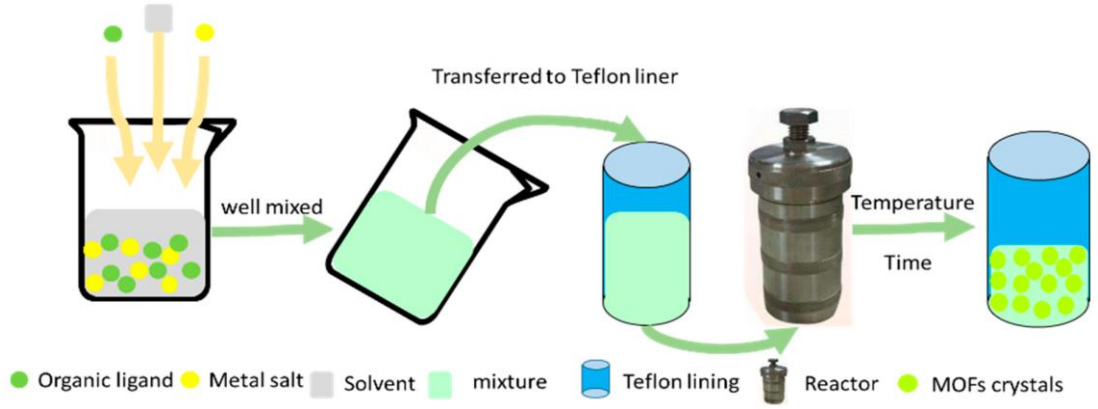

Figure 11. General synthesis of MOF particles by hydro-/solvothermal method.

3.6. SYNTHESIS OF MOFs

(3) Microwave method

- More homogenous than hydro-/solvothermal method.
- Fast reaction kinetics.
- High product yield.
- Examples: ZIF-8, UiO-66 (Zr-BDC)

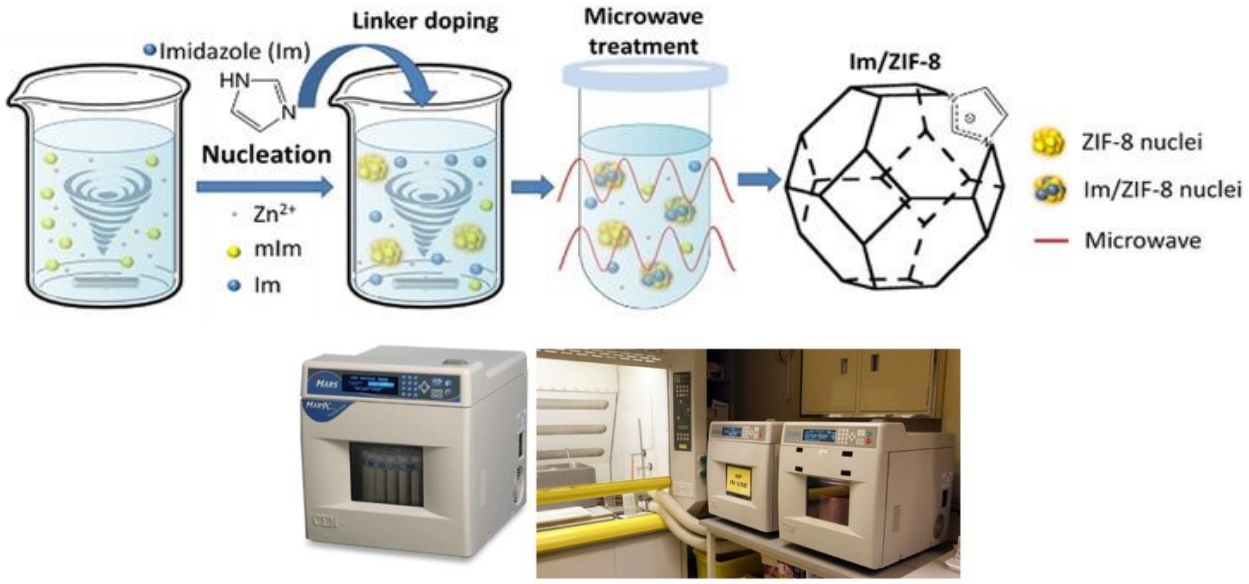

Figure 12. Microwave synthesis of ZIF-8 particles.

(4) Electrochemical method

- Metal ions are supplied from anodic dissolution (instead of metal salts) to react with dissolved linker molecules.
- Cu cathode is used & metal deposition is avoided by using protic solvents.
- Rapid, low temperature, and separation of metal ions are not needed.
- Examples: ZIF-8, ZIF-67, HKUST-1

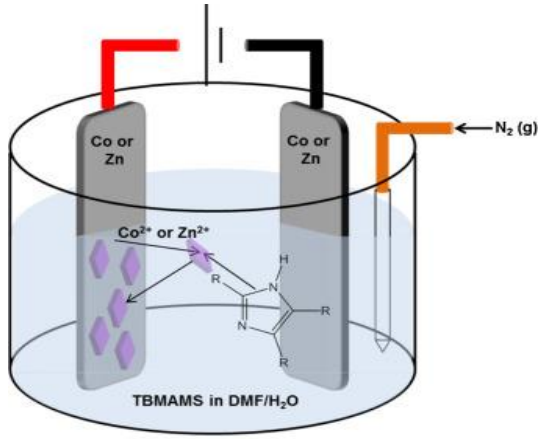

Figure 13. Electrochemical synthesis of ZIF particles.

3.6. SYNTHESIS OF MOFs

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(5) Sonochemical method

- Involves formation and collapse of bubbles ("acoustic cavitation").
- Very high local T (~4727 ºC) & P (~1,000 bar).
- Very fast heating & cooling rates (>734 ºC/s).
- Produce fine crystallites.

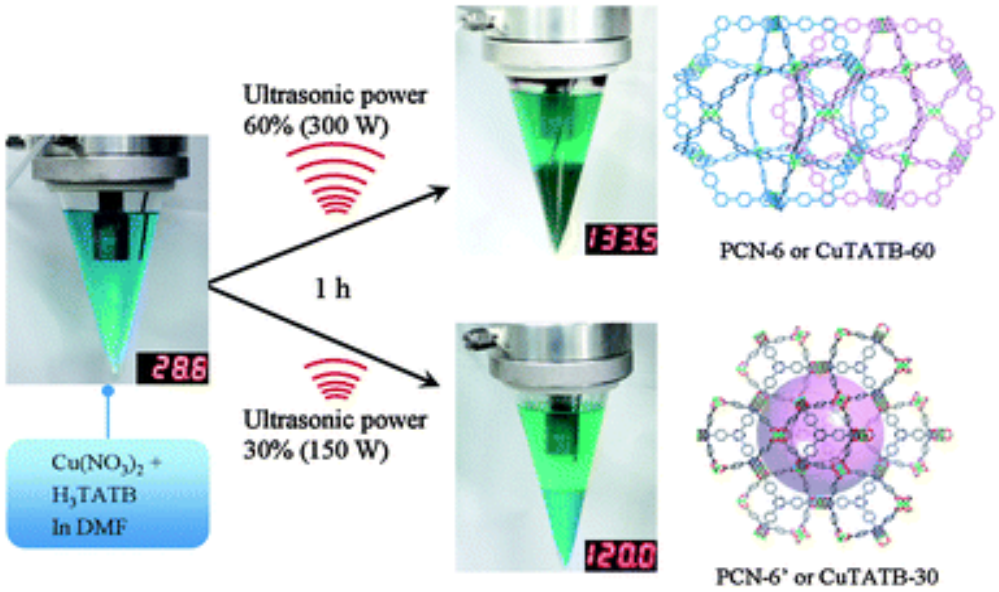

Figure 14. Synthesis of MOF particles by sonochemical method.

(6) Mechanochemical method

- Does not require solvents.
- Good for scalable production.
- Generates nanosized MOFs.
- Performed at room temperature using ball milling or mortar and pestle.
- Structural control can be challenging.

Figure 15. Synthesis of ZIF-8 and ZIF-67 particles by ball-milling method.

3.7. MOF CHARACTERISATION

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XRD

FTIR

BET

EM

Figure 16. Different techniques used for characterizing the structural and compositional features of MOFs.

- Composition
- Crystal structure

- Composition (organic compounds, bonds: C−C, C−H, C=O, −COOH, etc.)

- Surface area
- Pore size distribution

- Exterior morphology
- Pores (mesopores and macropores)
- Composition (EDS)

3.7. MOF CHARACTERISATION –XRD & EM

XRD

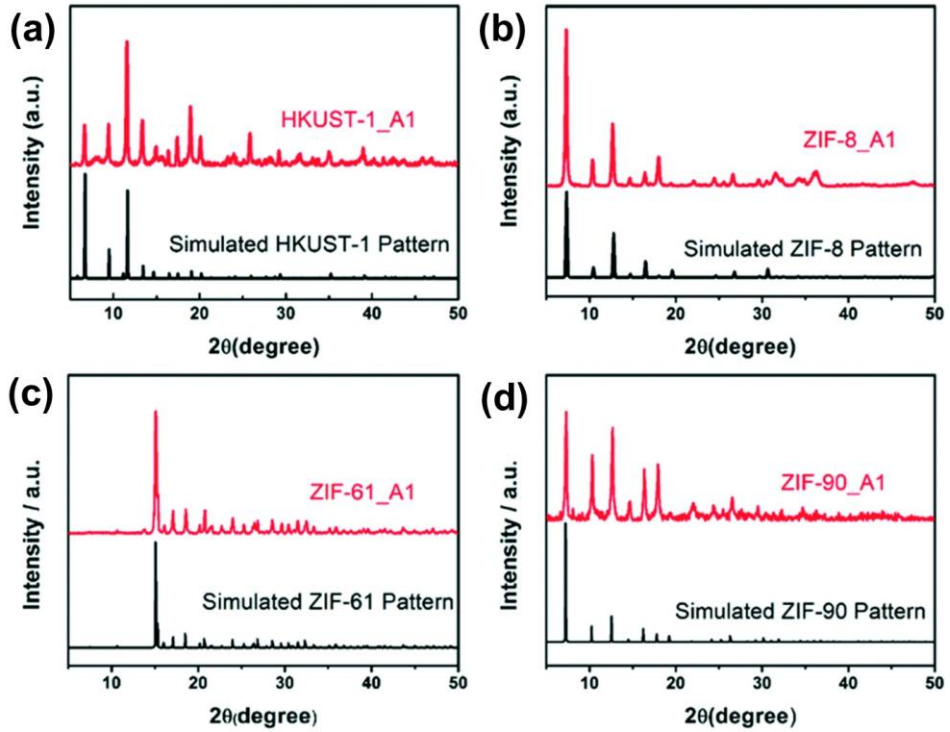

Figure 17. XRD patterns of (a) HKUST-1, (b) ZIF-8, (c) ZIF-61, and (d) ZIF-90

- Highly crystalline.
- Match experimental pattern with simulated pattern (CCDC Database).

SEM & TEM – ZIF-67

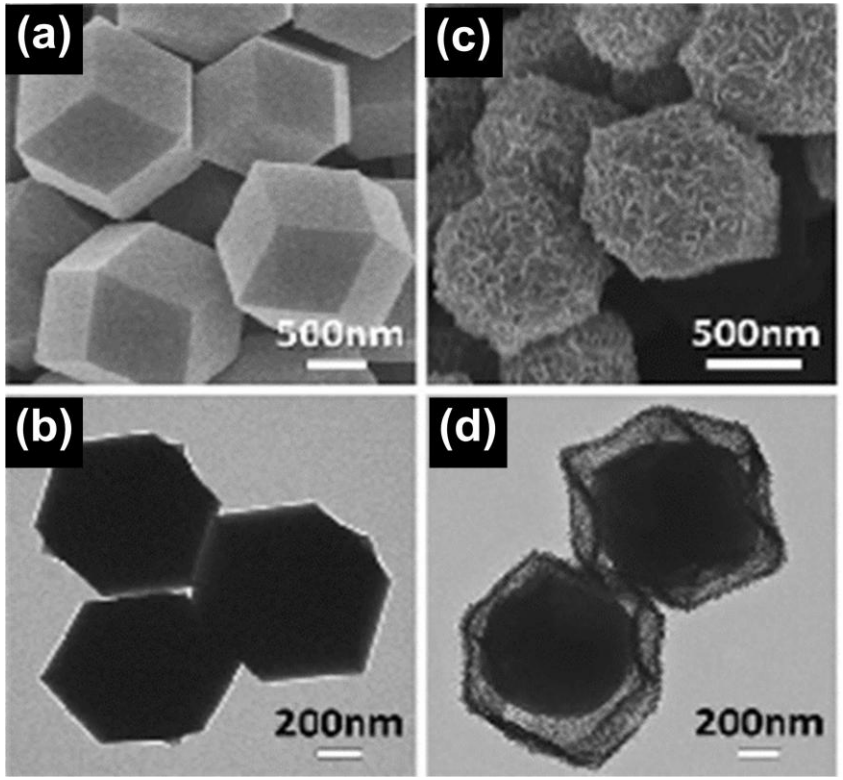

Figure 18. SEM and TEM images of (a, b) solid and (c, d) hollow ZIF-67 particles.

- **SEM:**
- **TEM:**

3.7. MOF CHARACTERISATION-FTIR & BET

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FTIR

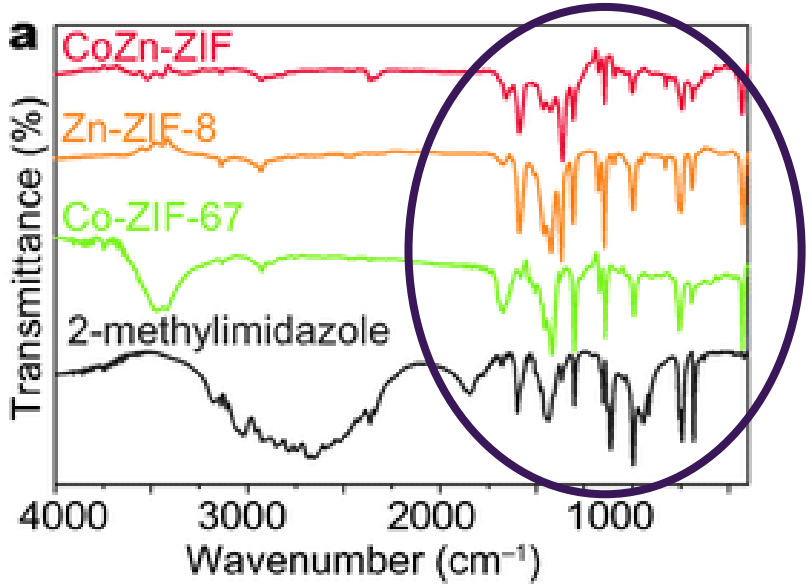

Figure 19. FTIR spectra of 2-MeIM and ZIF particles.

- **FTIR:** information about organic bonds present in the samples.
- Check similarities of MOF peaks with the linker molecules to confirm successful formation of MOFs.

BET

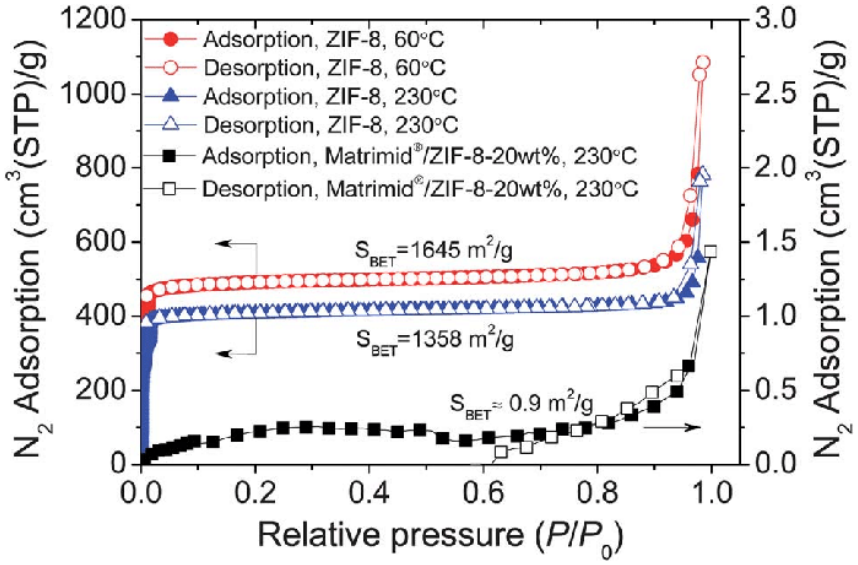

Figure 20. N_2 sorption isotherms of ZIF particles.

- **BET** (Brunauer-Emmett-Teller) surface area: information about surface area and pore size distribution.
- Measures volume of N_2 adsorbed/desorbed.

3.7. MOF CHARACTERISATION-BET S.A.

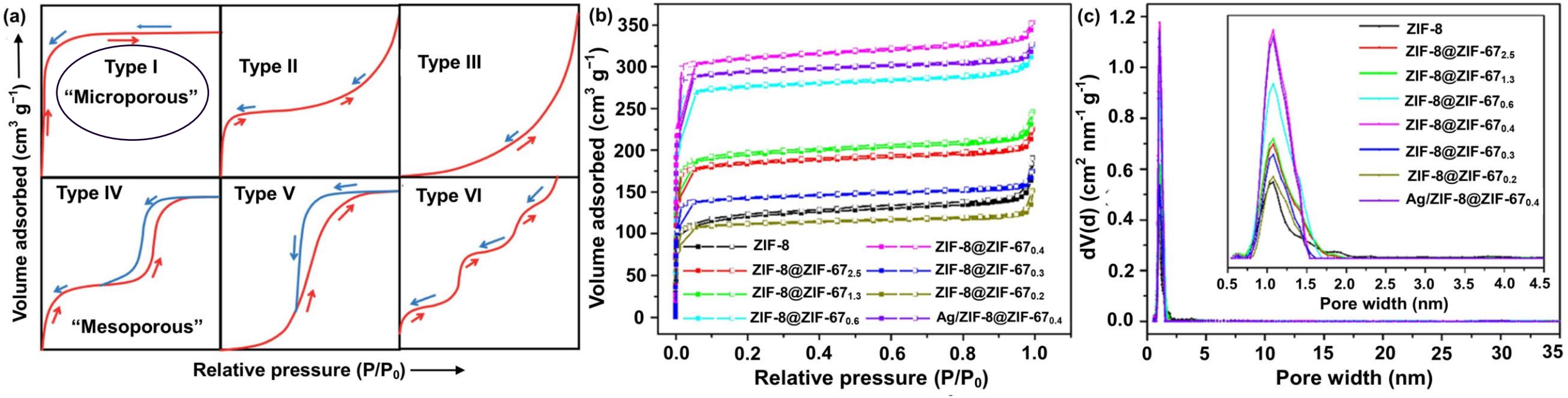

Figure 21. (a) Types of sorption isotherms. (b) N_2 sorption isotherms and (c) pore size distribution plots of ZIF-8 and ZIF-8@ZIF-67 composites.

- The N_2 sorption isotherms provide information about the type of pores present in the material, while the pore size distribution data gives information about the size distribution of pores inside the MOFs.

- MOFs typically exhibit type I isotherm due to their microporosity (pore size< 2.0 nm).

3.8. SIZE CONTROL OF MOFs

- The formation of MOFs can be divided into several stages:

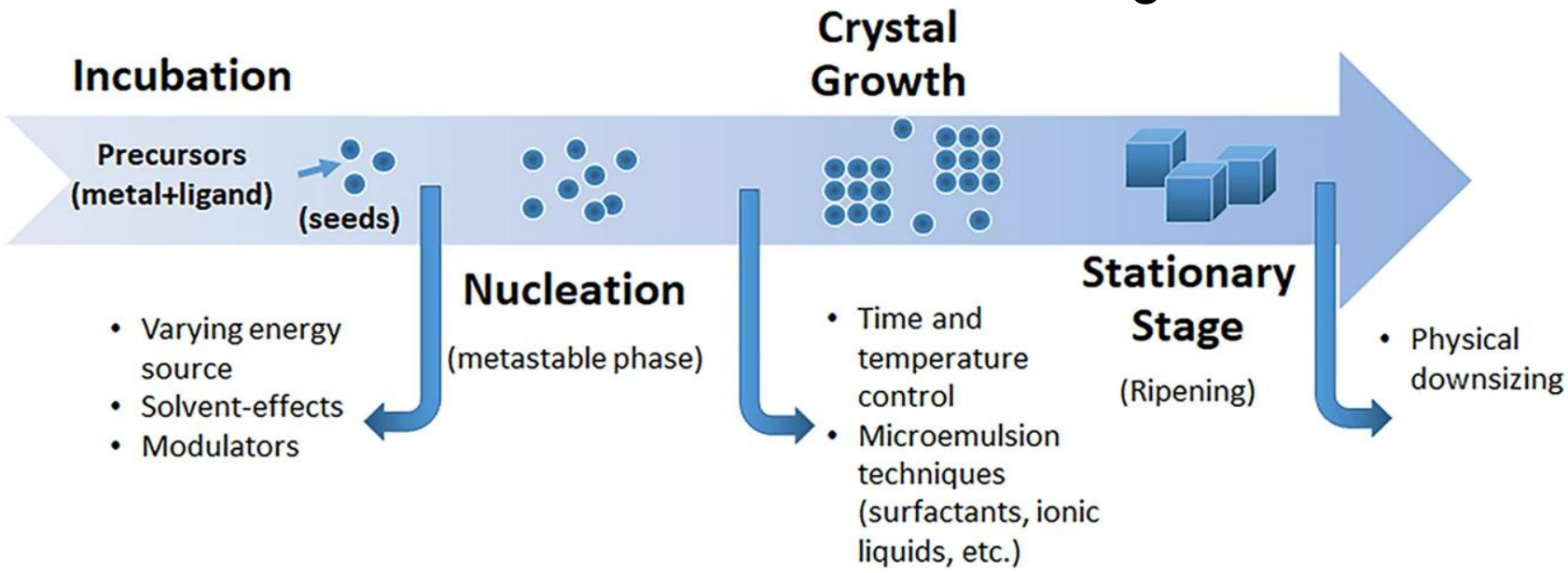

Figure 22. Major stages of MOF crystallization.

- MOF particle size is determined by the nucleation and growth rates.
- Faster nucleation → more nuclei → shorter crystal growth → smaller particles
- Slower nucleation → less nuclei → longer crystal growth → larger particles
- **Why control the size of MOFs?**
  (1) Increase surface area → increase the number of active sites
  (2) Improve ion/electron transfer and reduce ion diffusion length

3.8. SIZE CONTROL OF MOFs: ZIF-8

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- Size control of ZIF-8 particles can be achieved by modifying:

(1) The anion reactivity of metal salts: more reactive anions → faster nucleation → more nuclei → shorter crystal growth → smaller particles.

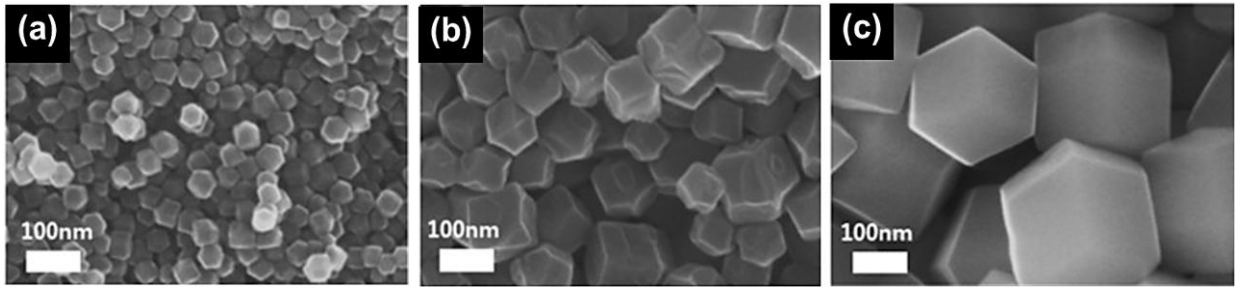

Figure 23. SEM images of ZIF-8 particles obtained by mixing (a) zinc nitrate (88 nm), (b) zinc acetate (240 nm), and (c) zinc chloride (533 nm) and 2-methylimidazole at 70 ºC for 10 mins.

$$NO_3^- > CH_3COO^- > Cl^-$$

(2) Metal to linker ratio: Higher L:M ratio (i.e., excess linker) → faster nucleation → more nuclei → shorter crystal growth → smaller particles.

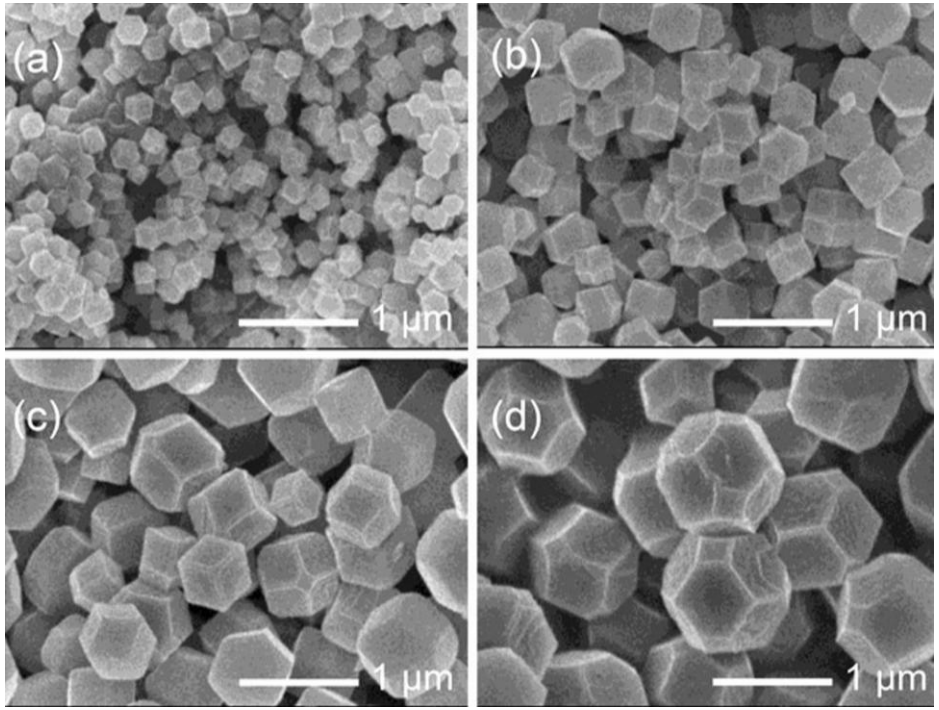

Figure 24. SEM images of ZIF-8 particles obtained with 2-MeIM: Zn^{2+} ratios of (a) 40:1, (b) 30:1, (c) 20:1, and (d) 10:1.

3.8. SIZE CONTROL OF MOFs: ZIF-8

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(3) Reaction temperature: higher temperature → faster nucleation → more nuclei → shorter crystal growth → smaller particles.

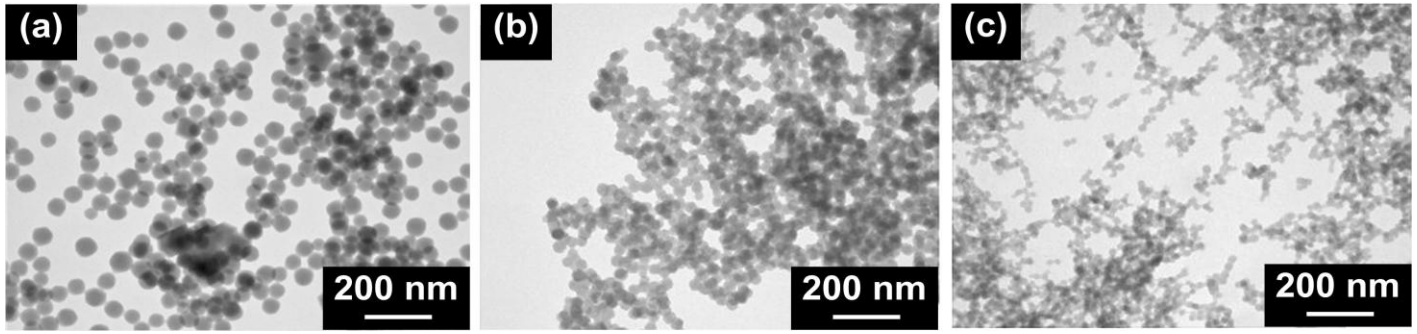

Figure 25. SEM images of ZIF-8 particles obtained at reaction temperatures of (a) -15 ºC, (b) 25 ºC, and (c) 60 ºC in methanol (Zn^{2+}:2-MeIM = 1:8)

(4) Reaction time: longer time → longer crystal growth → larger particles

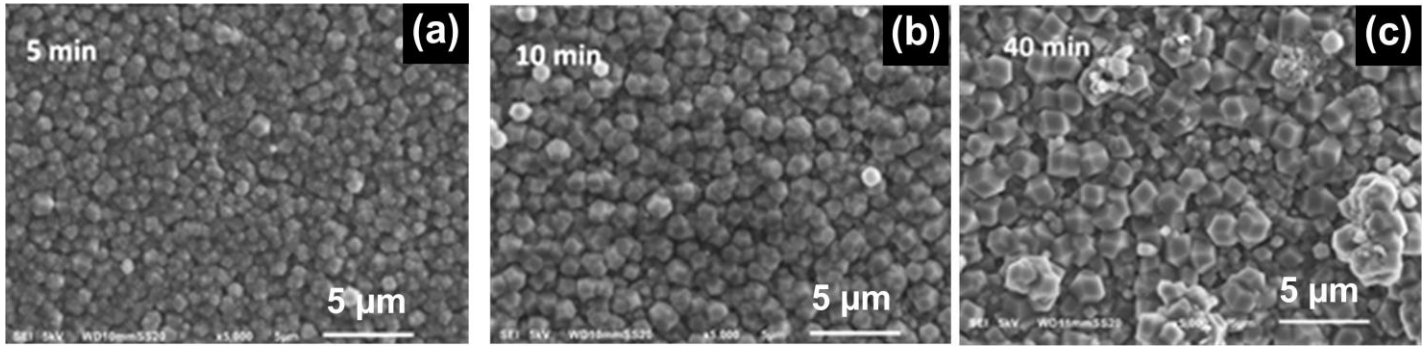

Figure 26. SEM images of ZIF-8 particles obtained with reaction time of (a) 5 mins, (b) 10 mins, and (c) 40 mins using zinc oxide rods and 2 M 2-MeIM.

3.10. MOF AEROGELS

- MOFs are typically produced in powder forms.

- Limitations of MOF powders for industrial applications:
  - Low processibility
  - Difficult recovery
  - Potential of causing secondary/additional pollution

- Enhancing processibility of MOFs → by forming aerogels.

- What is an aerogel?
  "**Aerogel** is a synthetic porous ultralight material obtained by combining a polymer with a solvent to form a gel, and then removing the liquid from the gel."

- **Characteristics of aerogels**: very light, highly porous, high surface area, low density, low thermal conductivity.

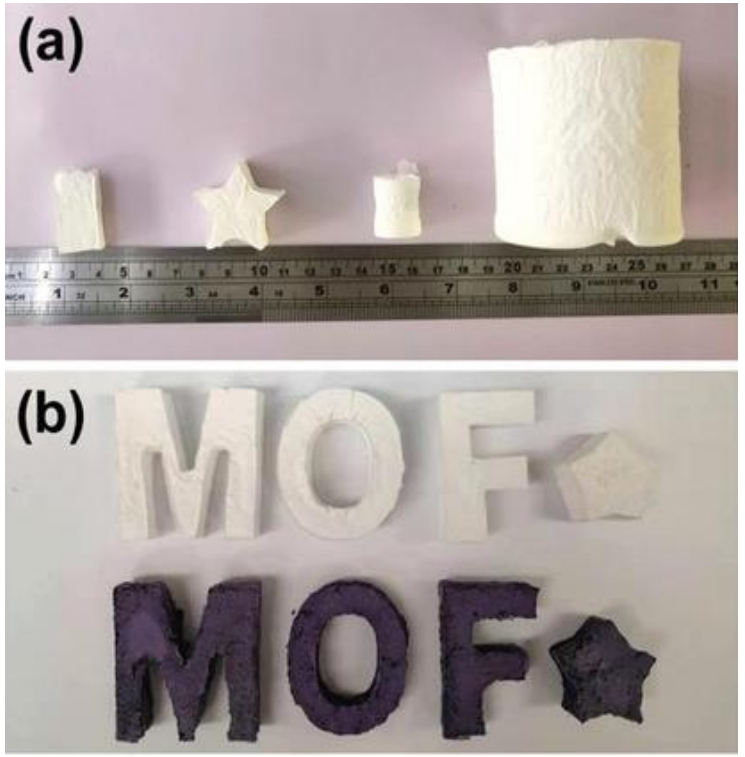

Figure 31. (a, b) Digital photographs of ZIF-8 and ZIF-67 aerogels.

3.10. MOF AEROGELS

Two general methods for fabricating MOF aerogels:

(1) **Mixing method**: Pre-synthesized MOF particles are mixed directly with the supporting matrix with heating to form an aerogel.

(2) **In situ growth:** MOF particles are grown directly on the supporting matrix to form an aerogel. This method is superior as it allows for more uniform distribution of the MOF particles on the supporting matrix.

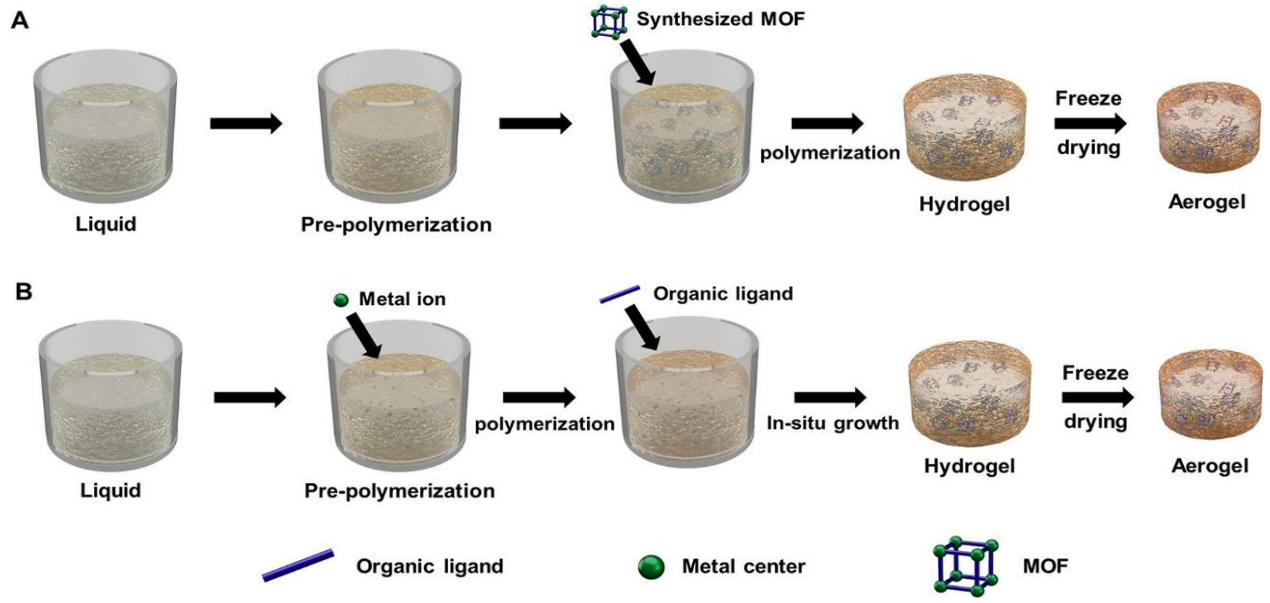

Figure 32. Schematic illustrations showing the preparation of aerogel by (a) mixing method and (b) in-situ growth method.

3.10. MOF AEROGELS- MIXING METHOD

- ZIF-8 aerogel can be prepared by mixing ZIF-8 powder with agarose (as a supporting matrix to promote crosslinking) and subsequent heating at mild temperatures to prepare the ZIF-8 hydrogel.

- Freeze drying is applied to remove water from the hydrogel and generate ultralight ZIF-8 aerogel.

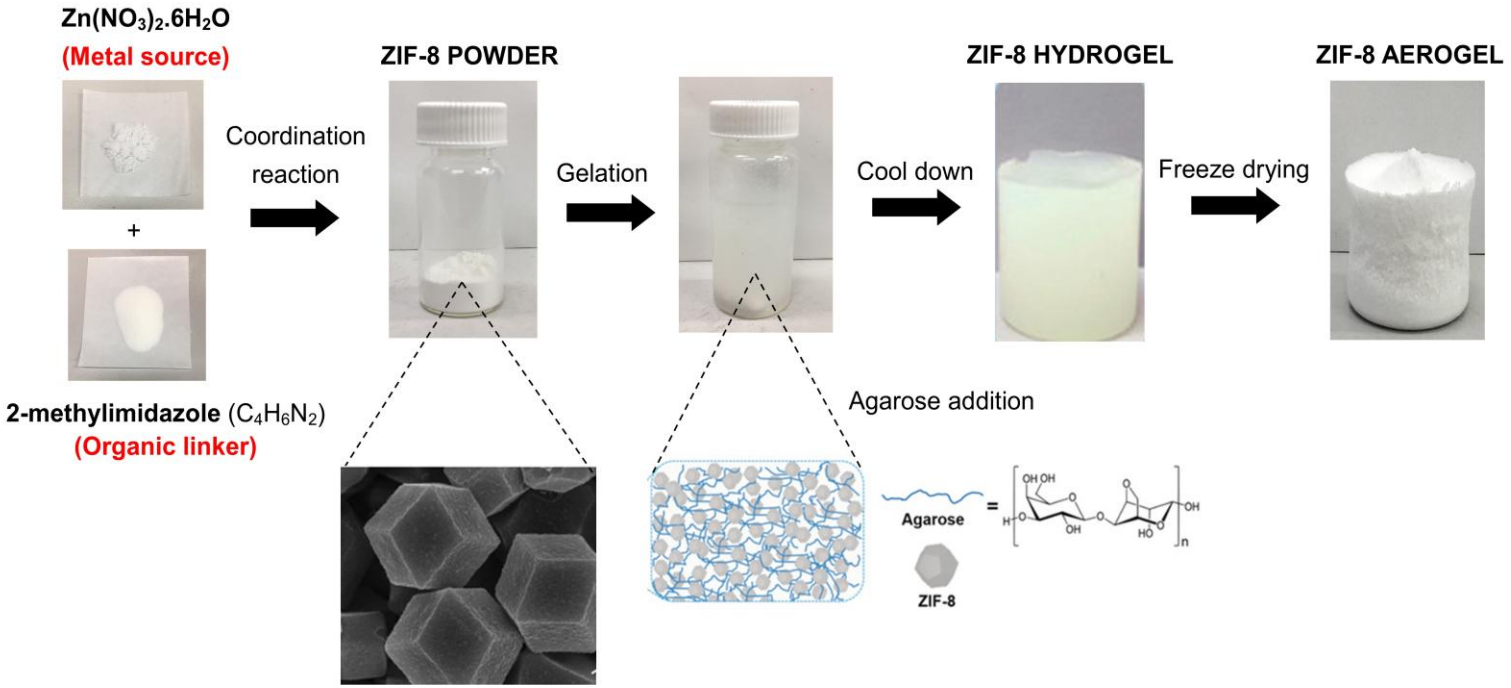

Figure 33. Preparation process of ZIF-8 aerogel by mixing with agarose.

3.10. MOF AEROGELS- IN-SITU METHOD

- Cellulose aerogel (CA) derived from cotton pulp using epichlorohydrin as a crosslinking agent.
- ZIF-67 particles can be directly grown ("in-situ" growth) on the CA aerogel by dispersing it in a methanolic solution containing cobalt nitrate and 2-methylimidazole (ZIF-67 growth solution) followed by aging for 24 h.
- Other supporting matrices used to make aerogel by in-situ method: reduced graphene oxide, wood, and carbon nitride (C_3N_4), etc.

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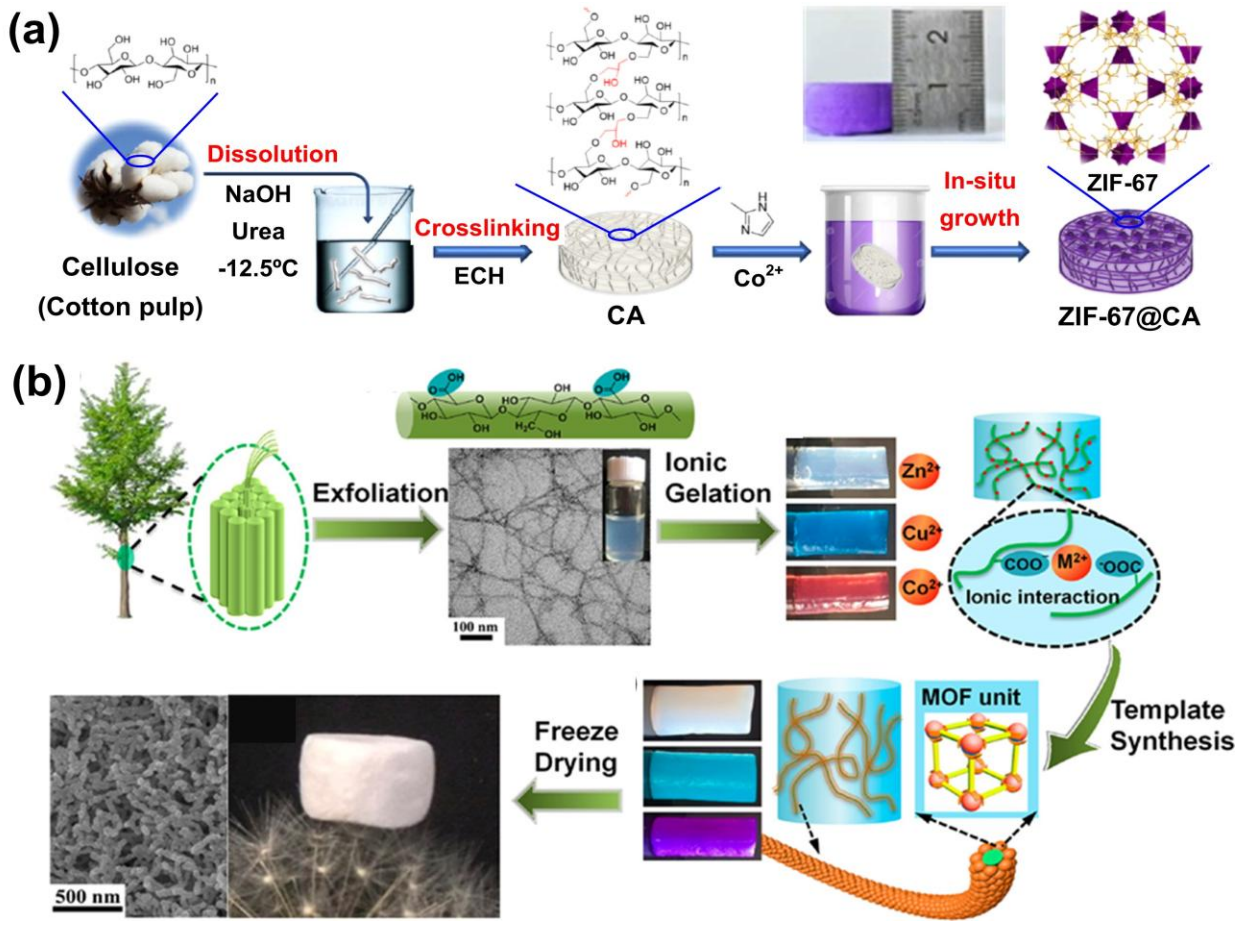
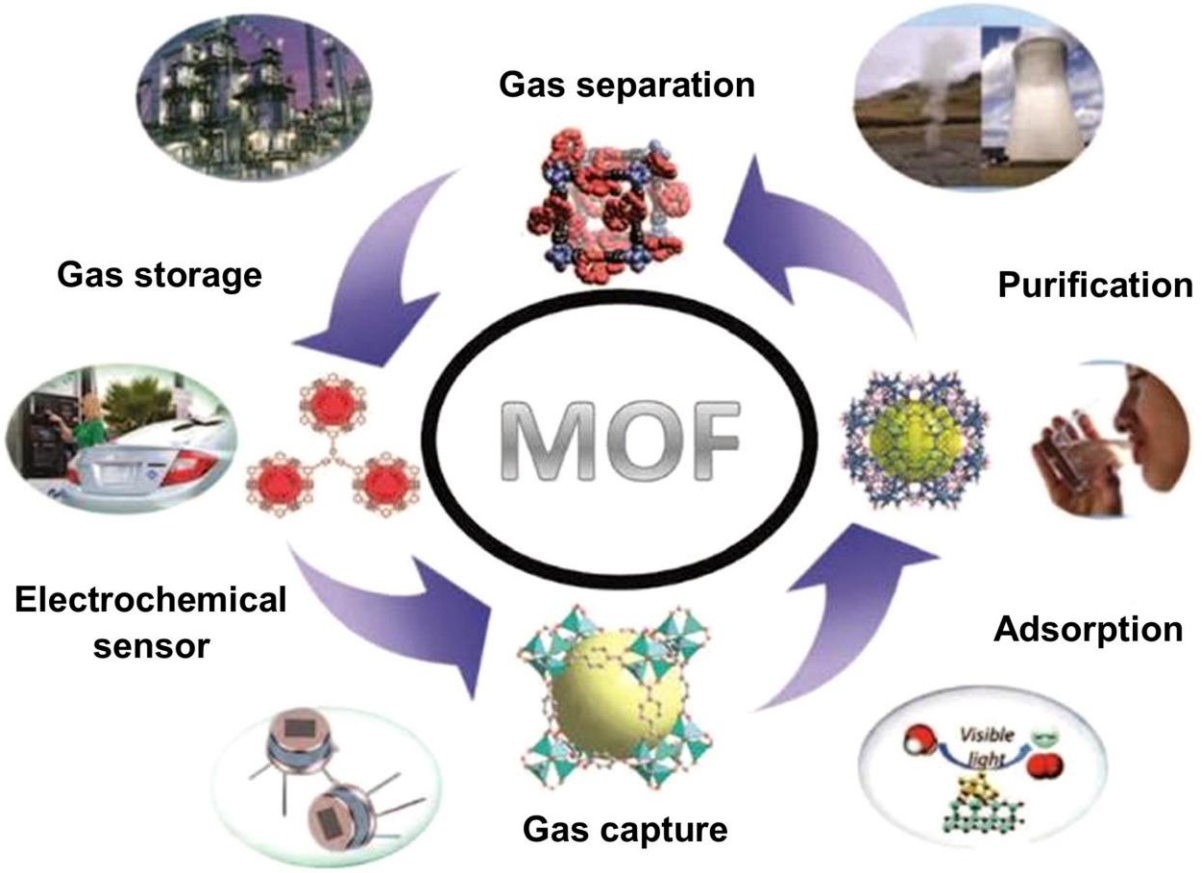


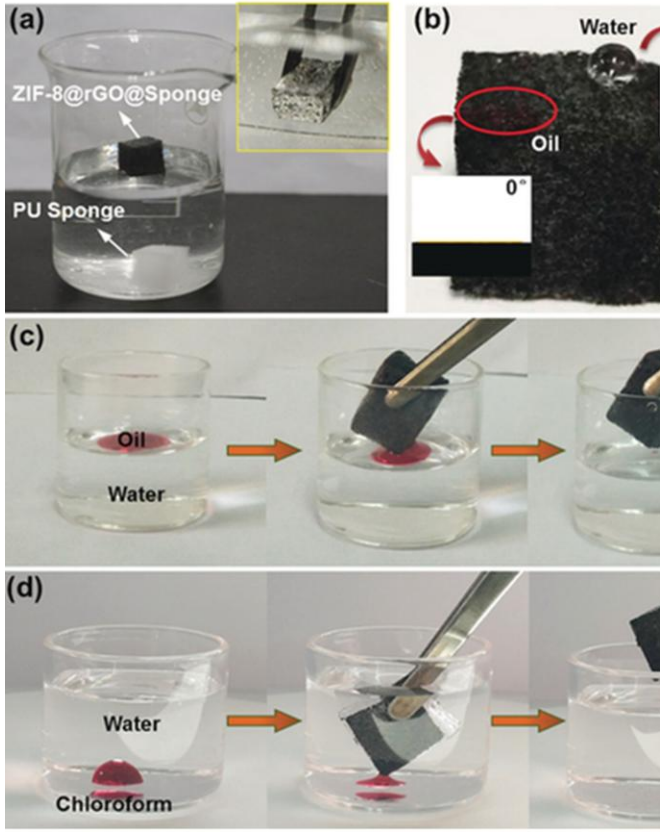
Figure 34. (a) Fabrication of ZIF-67@CA aerogel by in-situ growth of ZIF-67 on cotton pulp-derived cellulose. (b) Fabrication of different MOF aerogels by in-situ growth of different MOFs on cellulose nanofibrils derived from dry kraft pulp.

3.11. APPLICATIONS OF MOFs

- Antibacterial agent
- Oil adsorption
- Dye degradation (adsorption)
- Electrochemical sensor
- Gas capture/separation
- Gas storage



3.12. MOF FOR OIL ADSORPTION



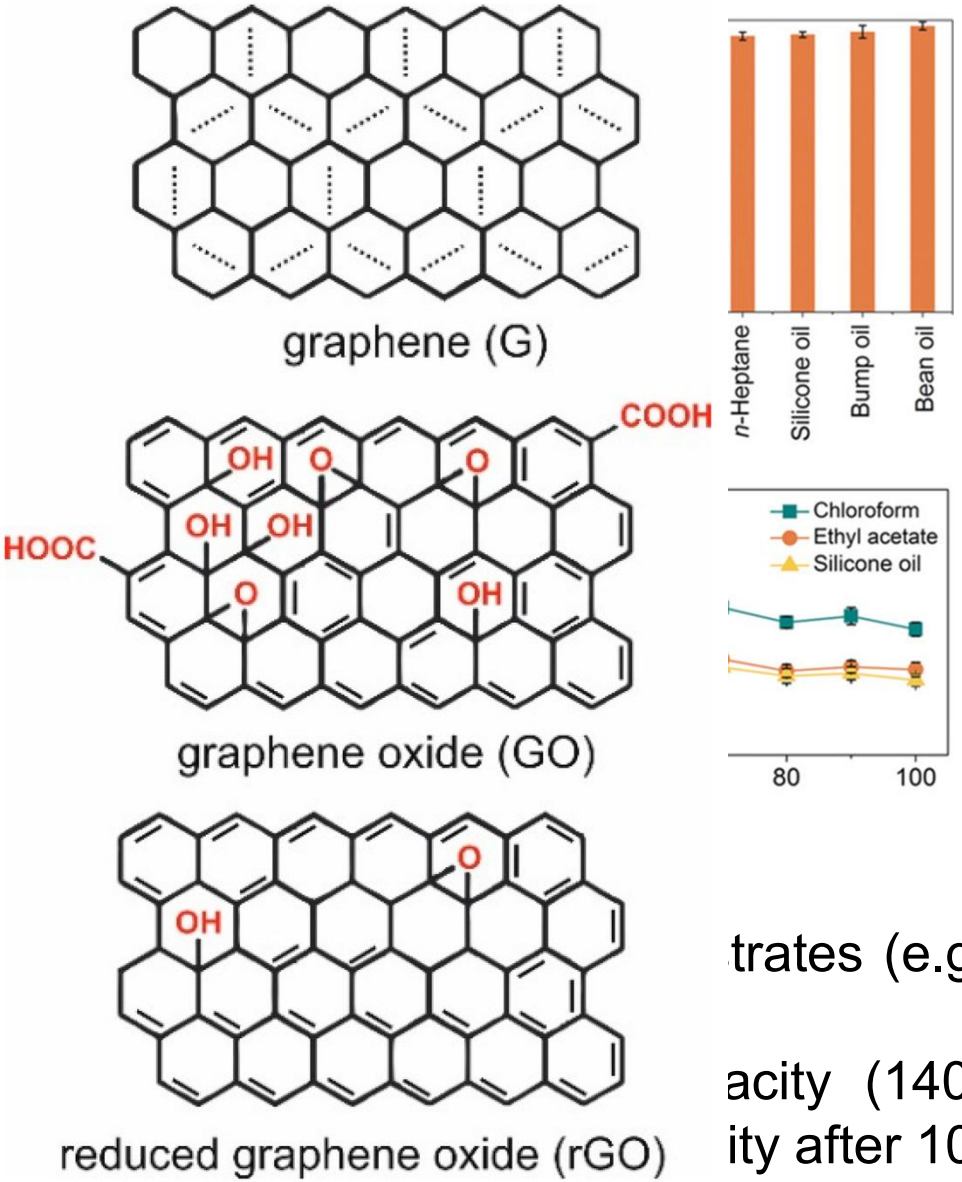


Figure 35. (a) Photo of pure PU and ZIF-8@rGO@PU sponge in water. (b) Photo depicting oil and water droplets on ZIF-8@rGO@PU sponge. Photos showing the adsorption of dyed silicone oil (c) and chloroform (d) from water. (e) Absorption capacities toward oils and organic solvents. (f) Water-oil separation efficiency. (g) Absorption recyclability test of the ZIF-8@rGO@sponge.

Angew. Chem

- MOFs, such as ZIF-8 cai trates (e.g., foams, papers, etc.) and used to absorb oil and other orgai
- ZIF-8@rGO@PU sponge acity (1400-2900 wt.%) for oil and organic solvents, efficient oil-wate ity after 100 cycles.
- High absorption capacity → mesoporosity and superhydrophobicity.

3.13. MOF FOR DYE ADSORPTION

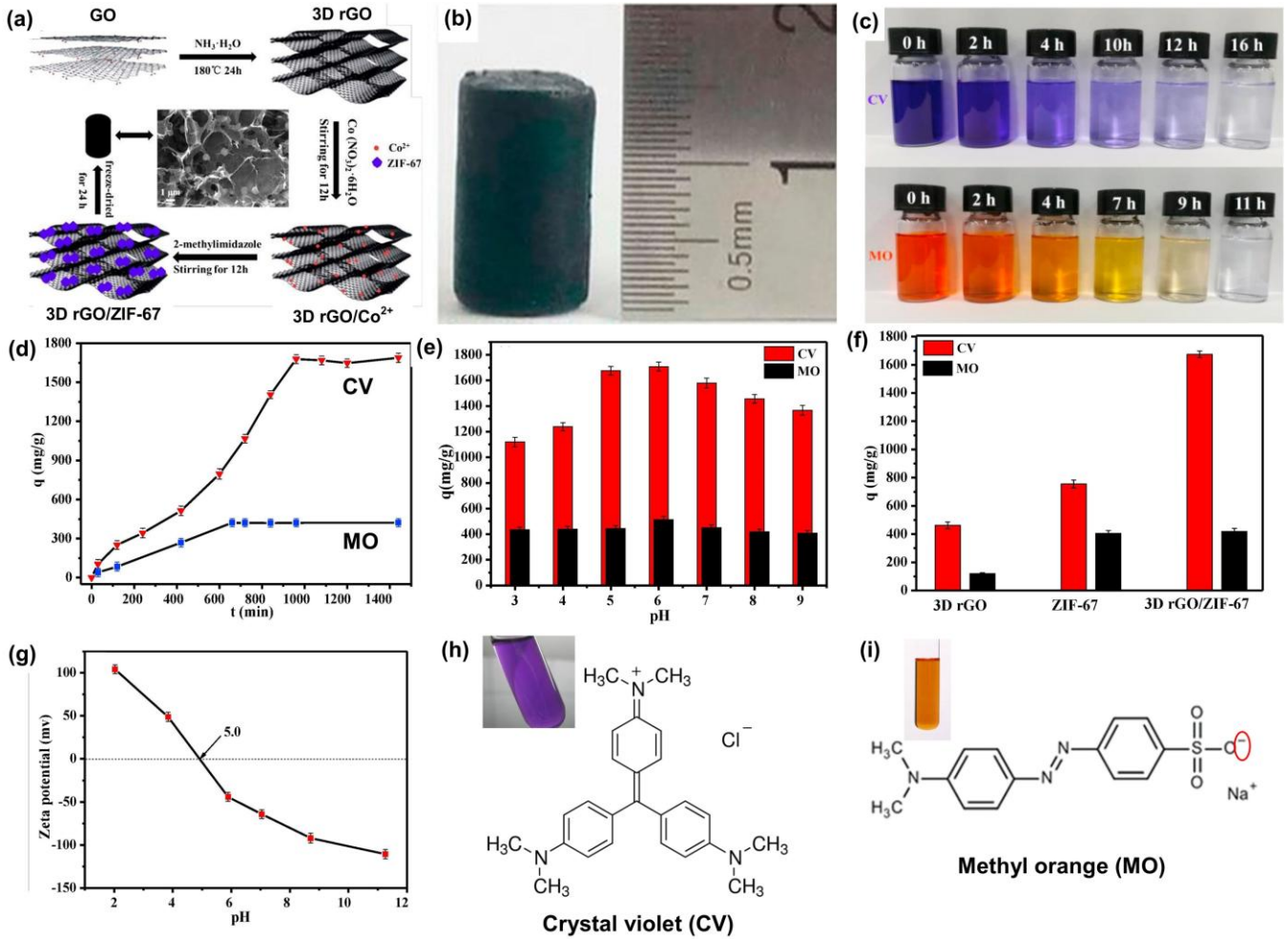


Figure 36. (a) Preparation process and (b) digital photo of reduced graphene oxide (rGO)/ZIF-67 aerogel. (c) Digital photo of crystal violet (CV) and methyl orange (MO) solution at different adsorption time. Effects of (d) contact time and (e) pH on CV and MO adsorption capacity. (f) Adsorption capacities of 3D rGO, ZIF-67 and 3D rGO/ZIF-67 aerogel for CV and MO. (g) Zeta potential of 3D rGO/ZIF-67 aerogel. Chemical structures of (h) CV and (i) MO.

- Chemical interactions between benzene rings in rGO/ZIF-67 and CV & MO by π-π interactions.
- rGO/ZIF-67 shows higher adsorption capacity for CV than MO.

- CV is a positively charged (“cationic”) dye and MO is a negatively charged (“anionic”) dye.
- At pH 6, the surface charge of rGO/ZIF-67 aerogel is negative → promotes <u>electrostatic adsorption with CV but not MO</u>.

3.14. DYES AND MOF ADSORBENTS

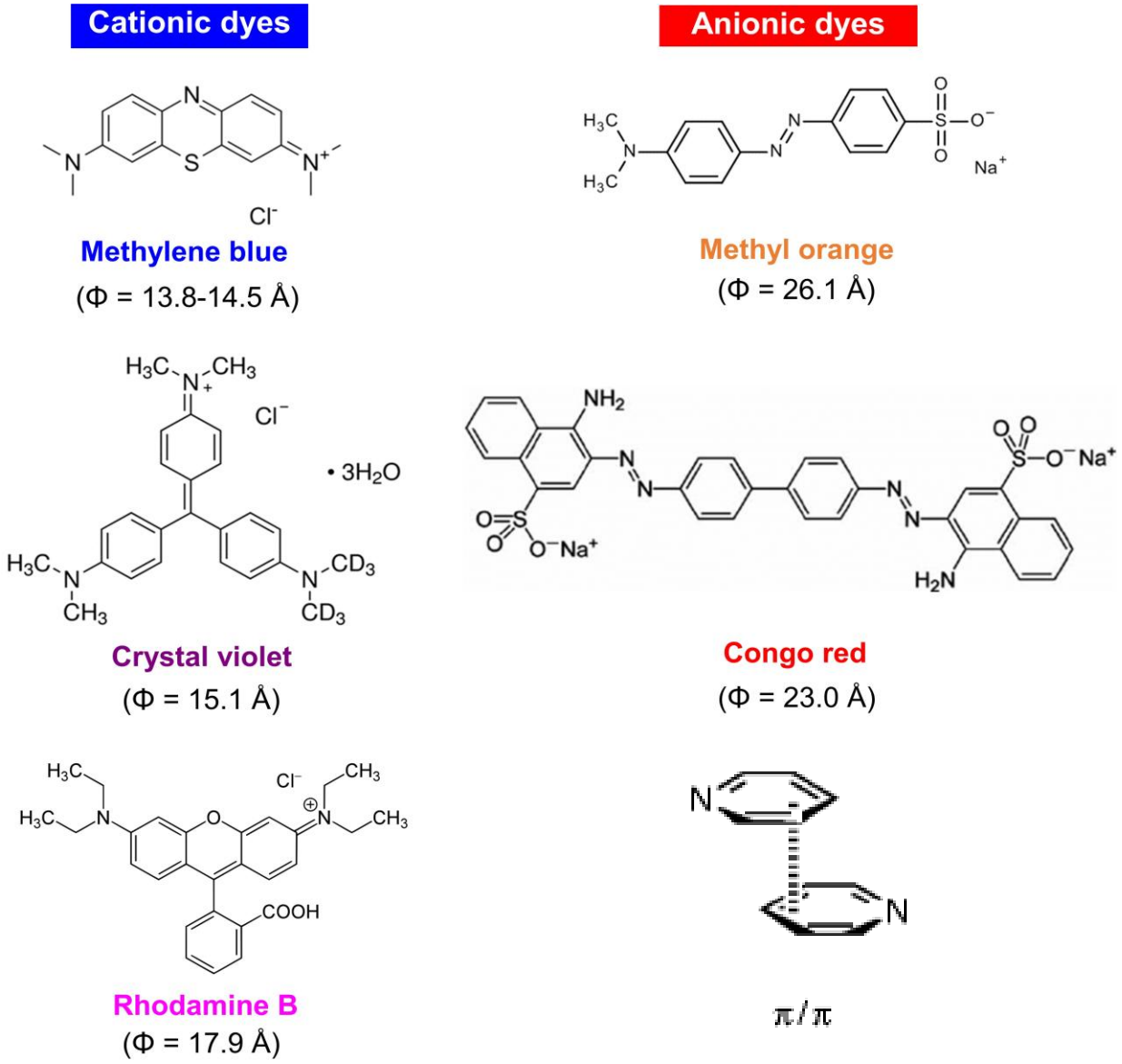


Figure 37. Examples of cationic and anionic dyes.

- Three important factors to consider: surface charge, pore size, and π-π interactions.
- **Surface charge:** anionic dye → MOF with positively charged surface & cationic dye → MOF with negatively charged surface.
- **Pore size:** pore diameter > dye molecule size → to enable effective adsorption of dye inside the pores of MOF.
- **π-π interactions:** desirable to promote chemical (non-covalent) interactions between the aromatic groups in dye and MOF structure.

3.15. OPTICAL PROPERTIES BY UV-VIS

- **UV-visible spectroscopy:** to obtain the absorbance spectra of a compound in solution (e.g., dye solution) or as a solid (visible range: 200-800 nm)

- **Components in UV-vis spectrophotometer:** Light source, monochromator, sample, and detector.

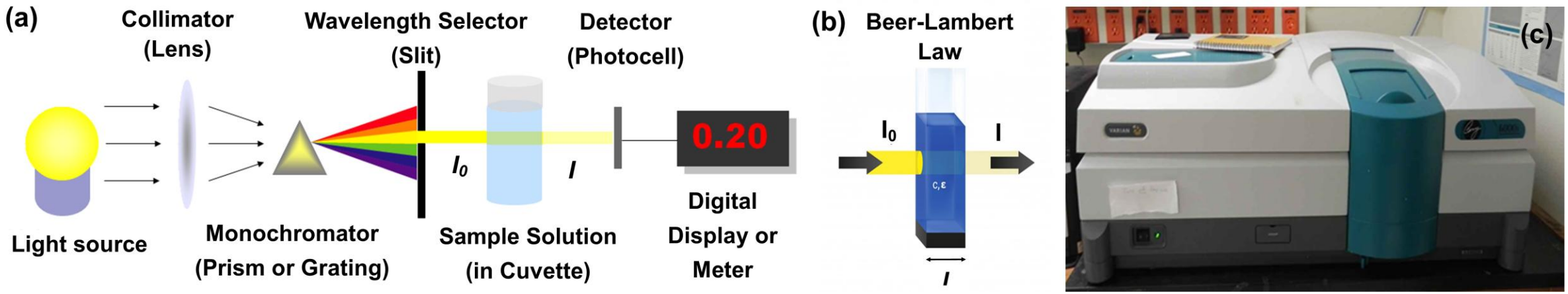


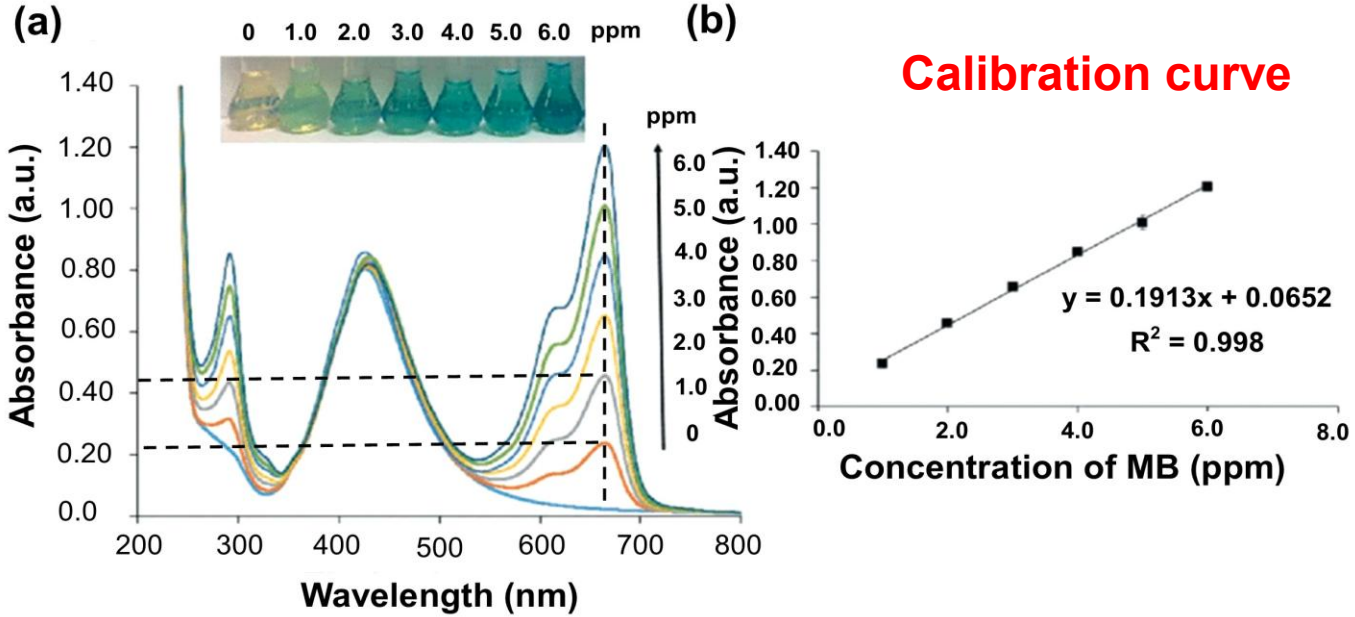
Figure 38. (a) Typical components of a UV-vis spectrophotometer. (b) Beer-Lambert Law. (c) Digital photograph of a UV-vis spectrophotometer.

3.16. INTERPRETING UV-VIS RESULTS

- **Beer-Lambert Law:** *"The amount of absorbed light (absorbance) is proportional to the solution concentration."*

$$A = \varepsilon l C$$

where A is absorbance (unitless or arbitrary units), ε is the molar absorption coefficient ($M^{-1}\,cm^{-1}$), l is the path length of the cuvette (usually 1 cm), and C is the solution concentration (M).



$$A = 0.1913C + 0.0652$$

$$C = A - 0.0652/0.1913$$

Figure 39. (a) UV-vis spectra of methylene blue (MB) solutions with different concentrations and (b) the corresponding concentration *versus* absorbance plot, showing a linear correlation.

3.17. ADSORPTION ISOTHERMS

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❑ **Adsorption isotherms:**

- Describe the amount of substance adsorbed on the adsorbent surface as a function of the concentration of the substance in gas phase or solution.
- Give information about **the type of adsorption (monolayer or multilayer)**

❑ Two common adsorption isotherms:

(1) Langmuir isotherm: assumes that the adsorbent surface is homogeneous, and all binding sites have the same affinity for the adsorbate, leading to monolayer adsorption.

$$\frac{C_e}{q_e} = \frac{1}{q_m K_l} + \left(\frac{1}{q_m}\right) C_e$$

y = b + ax

where C_e is adsorbate concentration at equilibrium. q_e and q_m (mg/g) are the adsorption capacity at equilibrium and maximum adsorption capacity, respectively; K_l is the Langmuir constant.

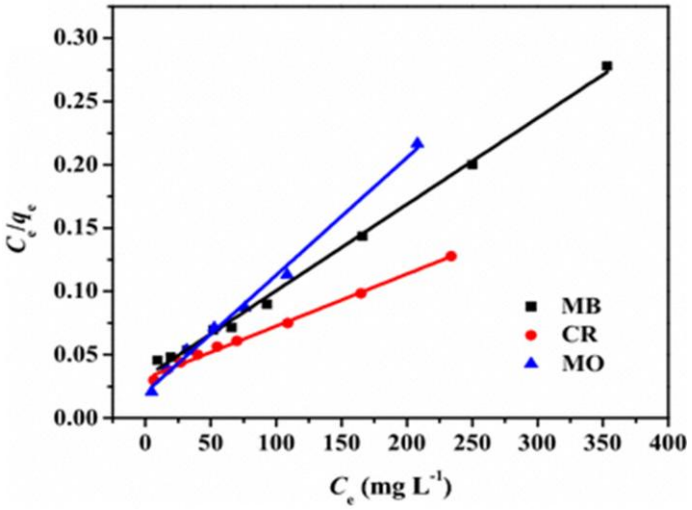

Figure 40. Langmuir isotherms of dye adsorption.

3.17. ADSORPTION ISOTHERMS

(2) Freundlich isotherm: assumes that the adsorbent surface is heterogeneous, and different binding sites have varying affinity for the adsorbate, leading to multilayer adsorption.

$$\log (q_e) = \log (K_f) + (1/n) \log (C_e)$$

$$y \quad = \quad b \quad + \quad ax$$

where C_e is adsorbate concentration at equilibrium. q_e is the adsorption capacity at equilibrium, and K_f and 1/n are Freundlich constants.

$$q_e = \frac{(C_0 - C_e) \times V}{m}$$

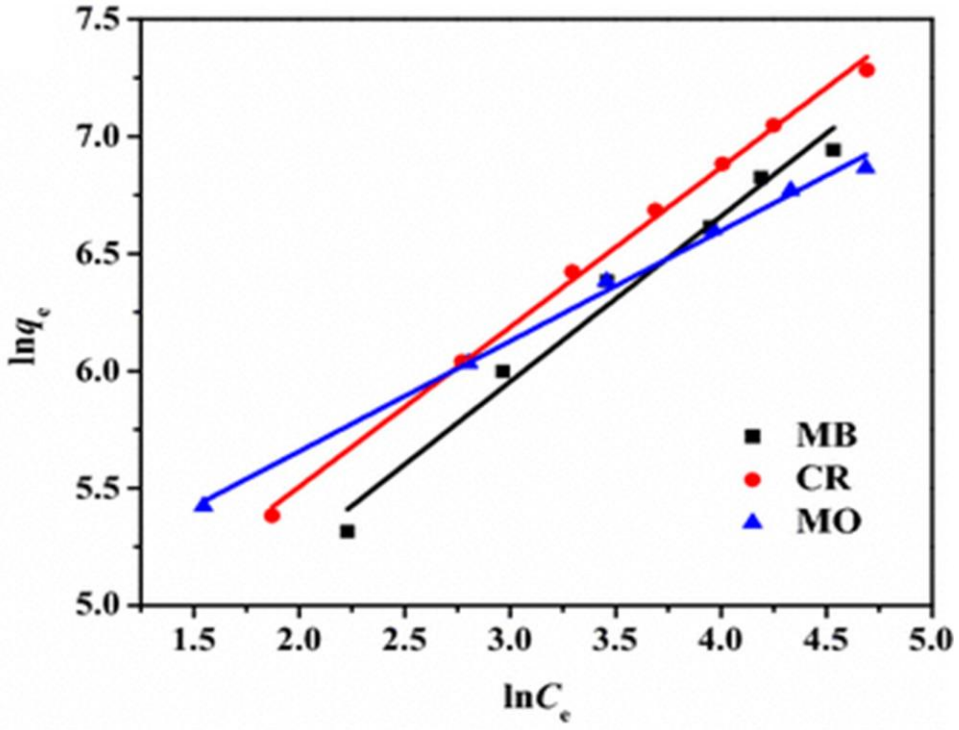

Figure 41. Freundlich isotherms of dye adsorption.

3.17. ADSORPTION ISOTHERMS

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Criteria	Langmuir	Freundlich
Adsorbent surface		
Binding sites		
Type of adsorption		

3.18. ADSORPTION KINETICS

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- **Adsorption kinetics:** useful for predicting the adsorption mechanism and rate-limiting step.

- Two common models to describe the adsorption kinetics for dye adsorption:

(1) Pseudo first-order model: assumes that *physisorption is the rate-limiting step*. Hence, the *adsorption rate depends on adsorbate concentration*. More suitable for describing initial adsorption.

$$\log(q_e - q_t) = \log(q_e) - \frac{k_1}{2.303}t$$

$$\downarrow \quad\quad \downarrow \quad\quad \downarrow$$

$$\mathbf{y} \quad = \quad \mathbf{b} \quad + \quad \mathbf{ax}$$

where q_e and q_t (mg/g) are the adsorption capacity at equilibrium and time t, respectively; k_1 is the equilibrium rate constant (L/min) in the pseudo first-order model.

3.18. ADSORPTION KINETICS

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(2) Pseudo second-order model: based on the assumption that *the rate-limiting step is chemical adsorption.* In this case, *the adsorption rate depends on the adsorption capacity rather than the concentration of the adsorbate*. Predicts the behavior in the entire adsorption range.

$$\frac{t}{q_t} = \frac{1}{k_2 q_e^2} + \left(\frac{1}{q_e}\right)t$$

$$y = b + ax$$

where q_e (unitless) and q_t (mg/g) are the adsorption capacity at equilibrium and time t, respectively, and k_2 is the equilibrium rate constant of the pseudo second-order model (g mg^{-1} min^{-1}).

- Rearranging the above equation to get an equation describing q_t as a function of t and then plot q_t versus t

$$q_t = \frac{k_2 q_e^2 t}{1 + k_2 q_e t}$$

3.18. ADSORPTION KINETICS

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Criteria	Pseudo 1st-order	Pseudo 2nd-order
Rate-limiting step		
Adsorption rate		
Behaviour prediction		

3.18. ADSORPTION KINETICS

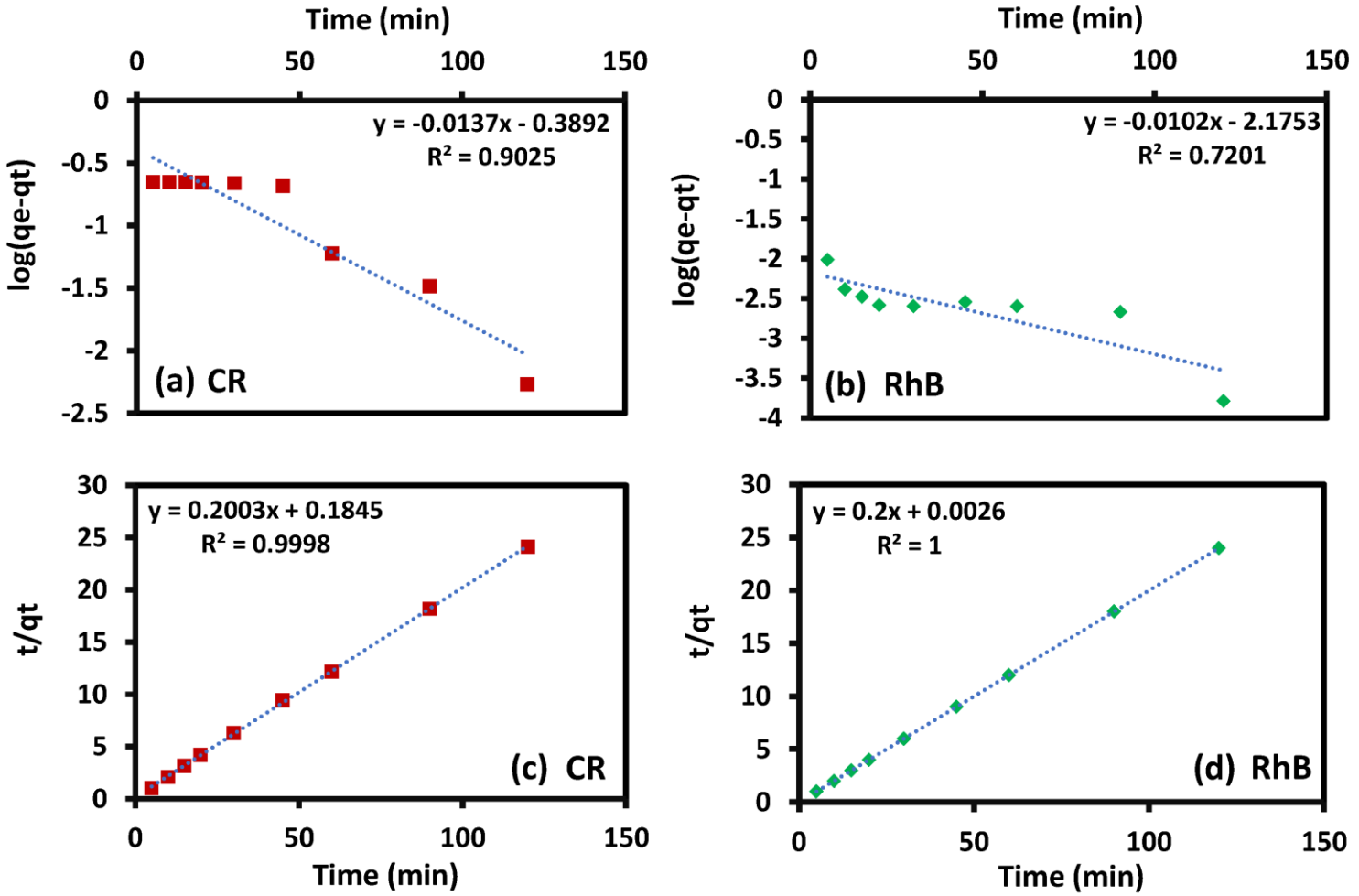


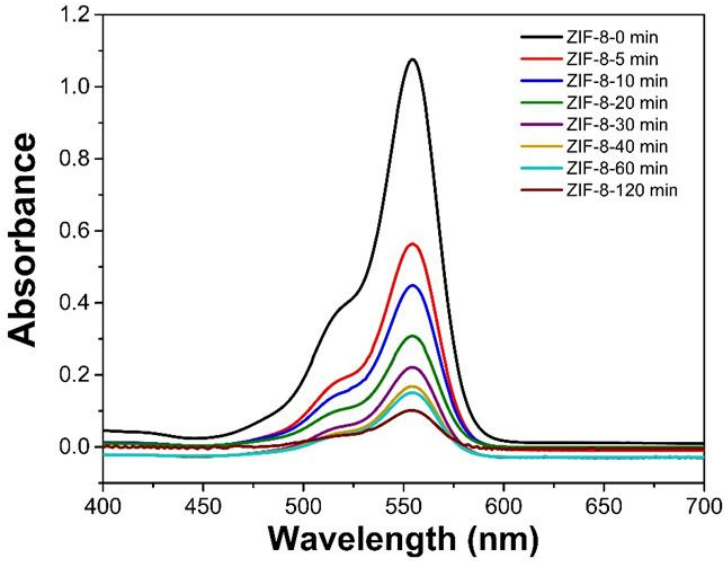
Figure 42. Adsorption kinetics: pseudo-first-order model of (a) CR dye; (b) RhB dye and pseudo-second-order model for the adsorption of (c) CR dye; (d) RhB dye (T= 298 K, pH = 2.0, Co = 25 mg L^{-1}, mass = 0.15 g, V = 20 mL).

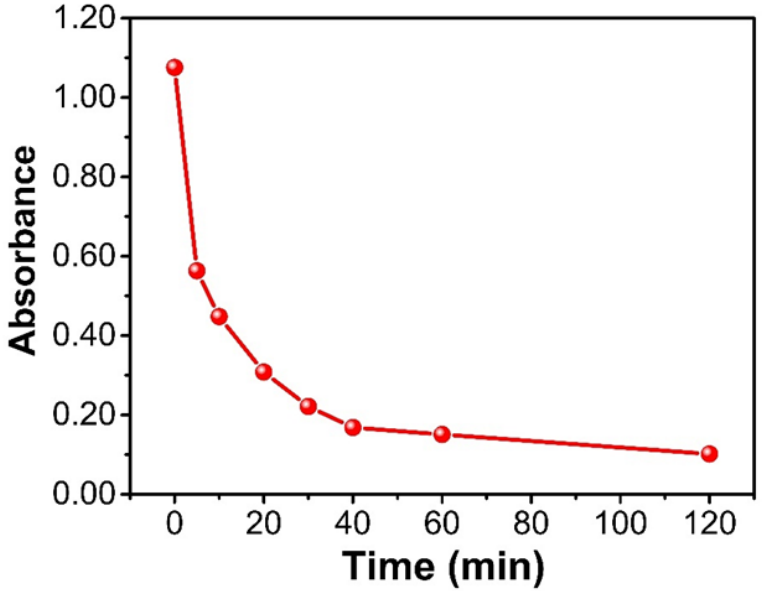
- **Pseudo first-order:** Plot t *versus* log (q_e-q_t).
- **Pseudo second-order:** Plot t/qt *versus* t.
- Then compare the **R^2 value** to determine which model is more suitable to describe the adsorption kinetics of a particular dye by the adsorbent.
- **Closer R^2 value to 1** indicates higher suitability.

QUESTIONS?

QUESTION 1

Tom employed a ZIF-8 aerogel to absorb or remove Rhodamine B dye contaminant from water. He measured the absorbance of the Rhodamine B solution with increasing contact time with the ZIF-8 aerogel from 0 minute to 120 minutes using the UV-vis spectroscopy technique and obtained the following result:

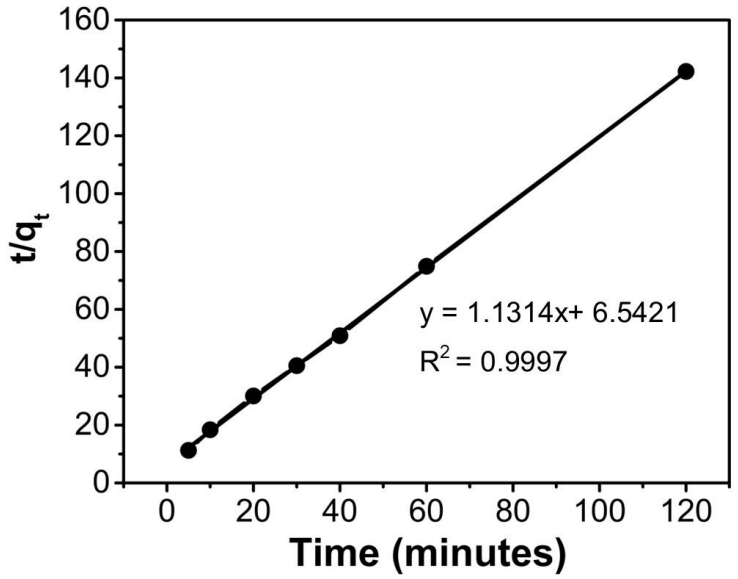




Using the above data, plot the absorbance *versus* time plot for the Rhodamine B dye degradation by the ZIF-8 aerogel.

QUESTION 2

(b) From the above data, Tom plotted the pseudo second-order model to analyse the adsorption kinetics of Rhodamine B dye by the ZIF-8 aerogel and found the model to be highly suitable to describe the adsorption kinetics due to the high R^2 value (almost 1), as shown below:



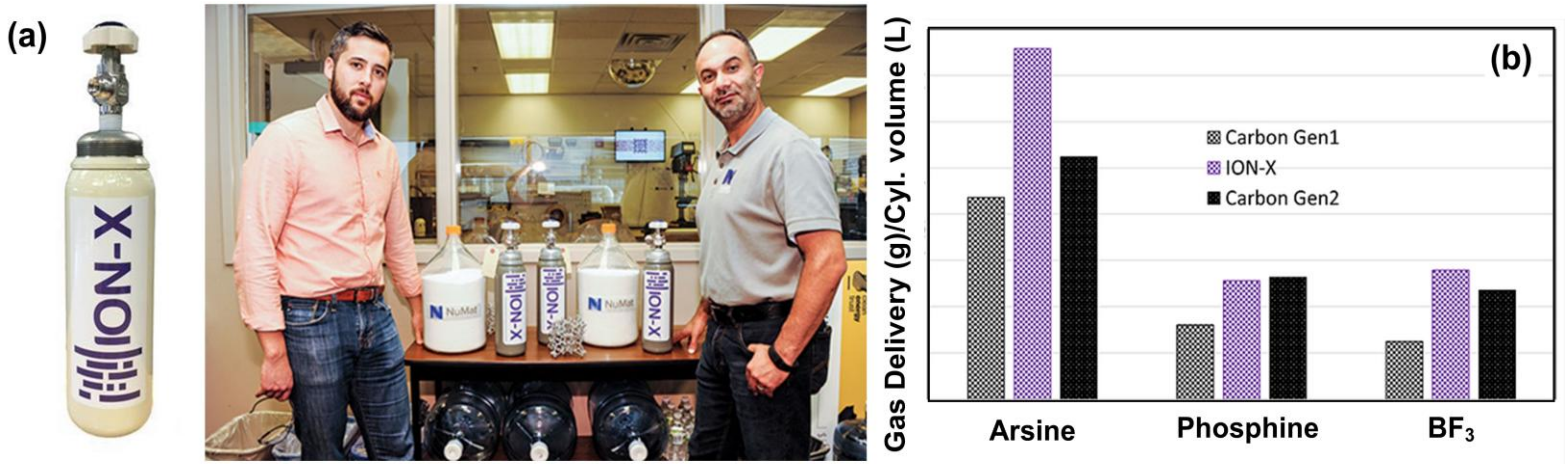
Using the above linear plot, determine the pseudo-second-order rate constant (k_2) and the adsorption capacity at equilibrium (q_e) values (round both values to two decimal places).

Answer: From the linear plot: slope = 1.1314 and intercept = 6.5421

Slope = $\frac{1}{q_e}$ = 1.1314, so $q_e = \frac{1}{1.1314}$ = 0.88 mg/g

Intercept = $\frac{1}{k_2 q_e^2} = \frac{1}{k_2 * (0.88)^2}$ = 6.5421; so $k_2 = \frac{1}{6.5421 * (0.88)^2}$ = 0.20 g/mg/min

COMMERCIAL MOF-BASED STORAGE



Sources:

https://www.chemistryworld.com/news/mofs-offer-safer-toxic-gas-storage-/1017610.article

https://www.numat.tech/ion-x-oem-data

https://www.youtube.com/watch?v=kxoeD8vBmZY

Figure. (a) Digital photograph of ION-X gas cylinder and (b) dopant gas delivery comparison between carbon-based incumbent products and ION-X.

- Commercial ION-X gas cylinder (2.4 L) was developed by a start-up company in U.S., NuMat Technologies.
- Used to store arsine, phosphine & boron trifluoride dopant gases → semiconductor industries.
- ION-X gas cylinder shows higher gas delivery than conventional carbon-based adsorbents → ultrahigh surface area and tailored pore size. Also, safer than conventional gas cylinders → low internal pressure (1 atm).

WEEK 9 · 知识点总结 · Dr Mostafa Kamal Masud

第 9 周　双金属 / 三金属纳米复合材料用于生物传感

Bi- and Trimetallic Nanocomposite for Biosensing

主线：成分（Au / Ag / Cu）→ 结构（合金 / 核壳 / 多孔）→ 合成路线 → 表面功能化 → 生物传感平台

1. 为什么用多金属

- **协同效应**：合金化产生单金属没有的电子、催化、力学性质。
- **性能可调**：调节金属种类、比例、结构。
- **性能增强**：三金属可在活性、稳定性、选择性上优于双金属和单金属。
- **同成分、不同结构 → 不同性质**：合金、核壳、多孔。

2. 三种金属各自的角色

金属	优点	缺点
Au	化学稳定、耐腐蚀；LSPR 在可见–近红外；**Au–S** 易功能化；导电性好；生物相容性好	贵
Ag	贵金属中 **LSPR 激发效率最高**、波长范围宽；消光系数高、峰尖锐	易氧化、稳定性差；抗菌但有毒性
Cu	必需微量元素；地球储量丰富、便宜；催化与光致发光；**点击化学**（Cu(I) 催化叠氮–炔环加成）	易氧化

- **Au–Cu–Ag 三金属纳米棒**：结合 Ag 的高 SERS 活性、Au 的表面稳定性与生物相容性、Cu 的结构稳定性。
- 合成取决于**还原电位**（Au 最易被还原，Cu 最慢）、加料顺序、时间间隔、电极电位与**电偶置换**。Au/Ag 比例不同，光学响应不同。

3. 介孔多金属的合成路线

方法	做法	优缺点
硬模板（纳米浇铸）	以介孔 SiO_2 等无机模板为模具，得到反相复制品	直观可靠；但去模板慢、需反复使用危险化学品、形貌控制差
软模板	用溶致液晶（LLC）、嵌段共聚物等有机表面活性剂引导介孔形成	有前景的替代方案（W1 实验、W12 CuTe 即此类）
无模板（去合金化）	化学处理优先溶解一种元素，留下多孔金属（如 Cu–Al）	简单；但多孔层与基体性质差异大，易应力腐蚀开裂、脆性断裂

4. 表面功能化（★）

- 目的：① 防止在水相、酸性、血液等环境中**降解**；② 抑制相互作用、**防止团聚**；③ 提供**官能团**以连接蛋白质、核酸等生物分子。
- 四种机制：**巯基–纳米材料表面**、**生物素–链霉亲和素**、**π–π 堆积**、**EDC/NHS 碳二亚胺反应**。
- 功能化提高稳定性、特异性和重现性，但可能影响生物相容性 → 需平衡。

5. 生物传感平台案例

- **病毒检测**：Au/Ag 平台（思考：选了什么材料、为什么、如何制备、信号如何转换、产业化的瓶颈）。
- **细菌检测**：引入 Fe_2O_3（磁性，可磁分离富集）。
- **癌细胞检测**：Au@Ag（Ag 壳增强 SERS、Au 核稳定）+ **微流控** + **多重 SERS** 区分不同生物标志物。
- **人 IgM 检测**。

6. 练习题方向

- 双金属 SERS 基底灵敏度低于预期的三个原因（成分比例、合金 vs 核壳结构、氧化、热点不足、制备不均）。
- 为成像 / 传感 / 治疗设计 Au–Ag–Cu 复合材料并说明各金属的作用；机器学习 / AI 如何辅助优化成分与合成。

考试重点：多金属的三点优势；Au、Ag、Cu 各自的优缺点；合金 / 核壳 / 多孔对比；三种介孔合成路线的优缺点；表面功能化的目的与方法。主线：**Composition → Structure → Properties → Sensing performance**（与 W7 互补）。

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CREATE CHANGE

Bi- and Trimetallic Nanocomposite for Biosensing

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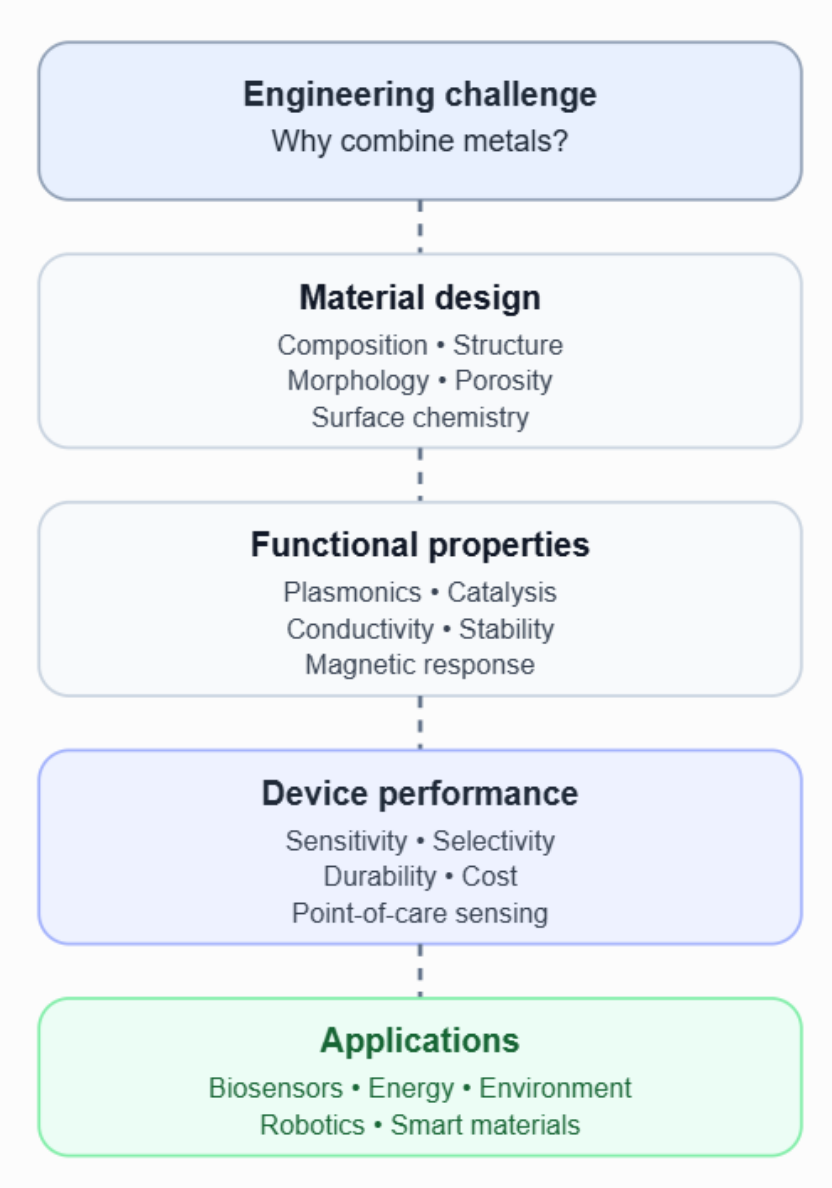
[MATE6301] Nanomaterials
School of Mechanical and Mining Engineering
The University of Queensland

Why do engineers design multimetallic nanomaterials?



What will we Learn?

- ❑ Why bimetallic and trimetallic nanomaterials exhibit synergistic properties.
- ❑ Compare alloy, core–shell and porous multimetallic structures.
- ❑ Describe how composition and morphology influence optical sensing performance.
- ❑ Evaluate fabrication strategies for mesoporous multimetallic materials.
- ❑ Propose a nanocomposite design for an engineering sensing application.

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Why they are important?

Synergistic Effects: Alloying metals produces new electronic, catalytic, and mechanical properties that outperform single-metal systems.

Tunable Properties: Engineers can adjust metal type, ratio, and structure to fine-tune performance for targeted applications (e.g., sensors, catalysts, coatings).

Enhanced Performance: Trimetallic systems can improve catalytic activity, stability, and selectivity compared to bi-metallic and monometallic nanoparticles, enabling more efficient and durable designs.

Design parameter	Engineering consequence
Metal composition	Electronic band structure changes
Atomic arrangement	Catalytic activity changes
Nanostructure geometry	Optical field enhancement
Porosity	Surface area and mass transport
Surface chemistry	Biomolecule attachment
Fabrication route	Reproducibility and scalability

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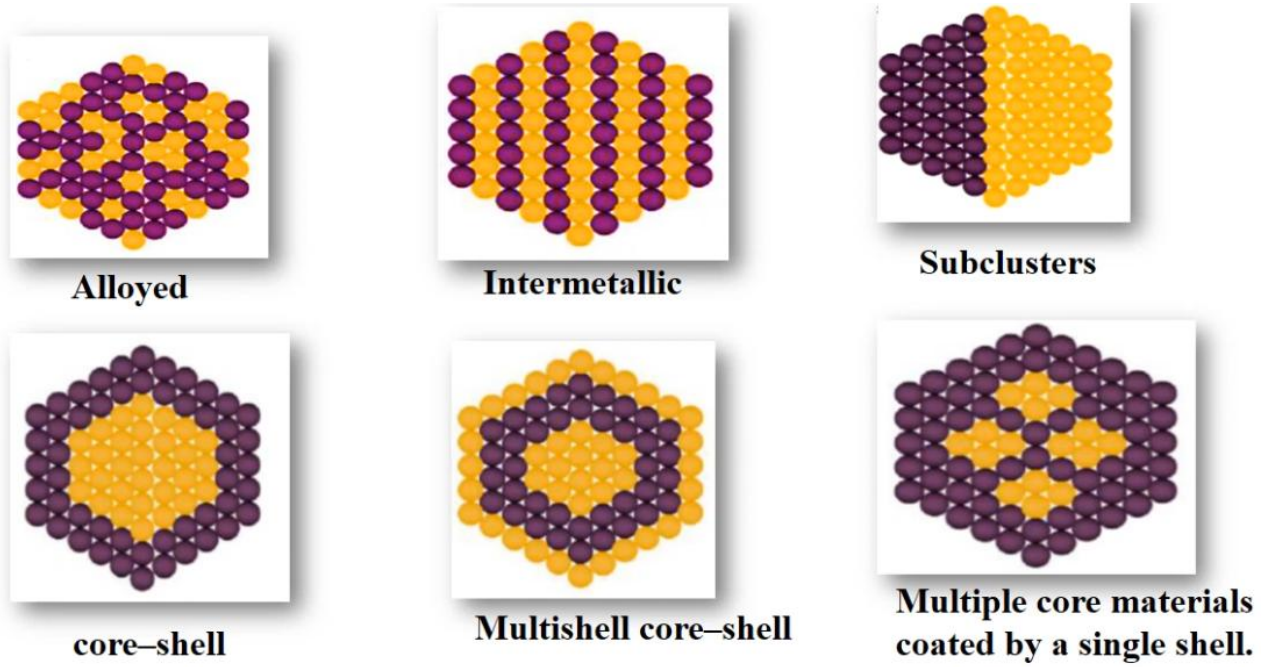
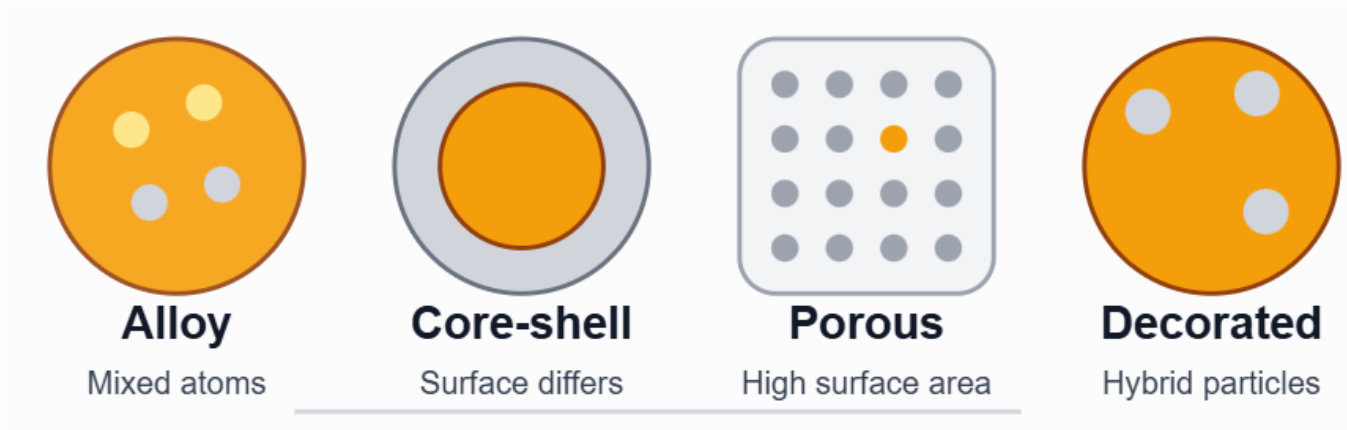
Types of Metal Composites





Same Composition + Different Architecture → Different Properties

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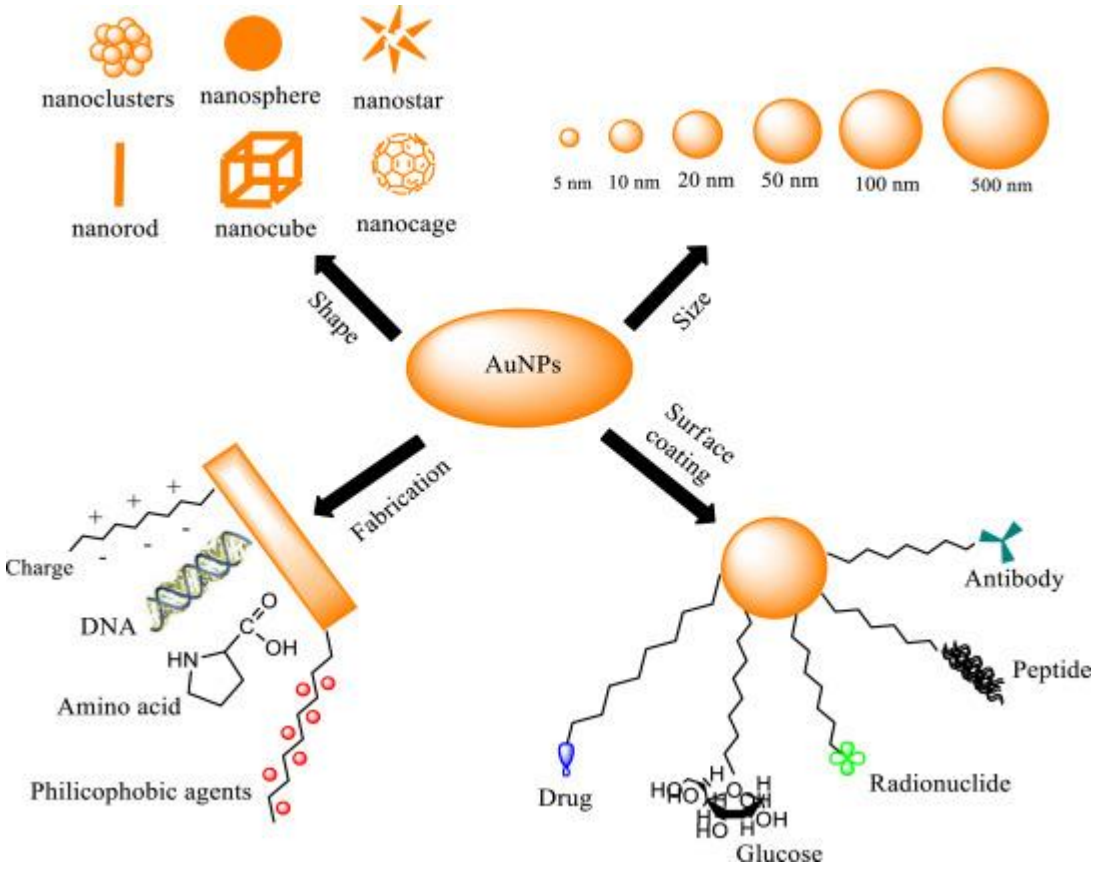
Why Au?

- ✓ Excellent chemical stability and corrosion resistance
- ✓ Strong localized surface plasmon resonance in the visible–NIR region
- ✓ Easy surface functionalisation through Au–thiol chemistry
- ✓ High electrical conductivity
- ✓ Excellent biocompatibility for biomedical devices
- ✓ Expensive



Measurement: Sensors 2021, 16, 100050

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Why Ag?

- Among the noble metals that exhibit plasmon resonance, silver (Ag) displays the highest efficiency of localized surface plasmon resonances (LSPR) excitation with a wider wavelength range. The LSPR band varies according to size, shape, composition and dielectric constant variations.
- The electrical, optical, catalytic and colour variations are tuned by inducing a change in morphology, size and aggregation behaviour.
- Ag NPs are more reactive and can easily undergo electrochemical oxidation due to their high extinction coefficients, larger scattering-to-extinction ratio and clear sharp bands.
- Ag possesses antimicrobial activity (bactericide properties, which can cause undesired effects such as DNA denaturation), vulnerable stability and toxicity.

Advantages	Limitations
• Strongest plasmonic enhancement • High scattering efficiency • Excellent SERS performance • Lower cost than Au	• Oxidation in ambient conditions • Lower chemical stability • Potential cytotoxicity • Requires protective coating in biomedical devices

Sensors 2021, 21(15), 5114

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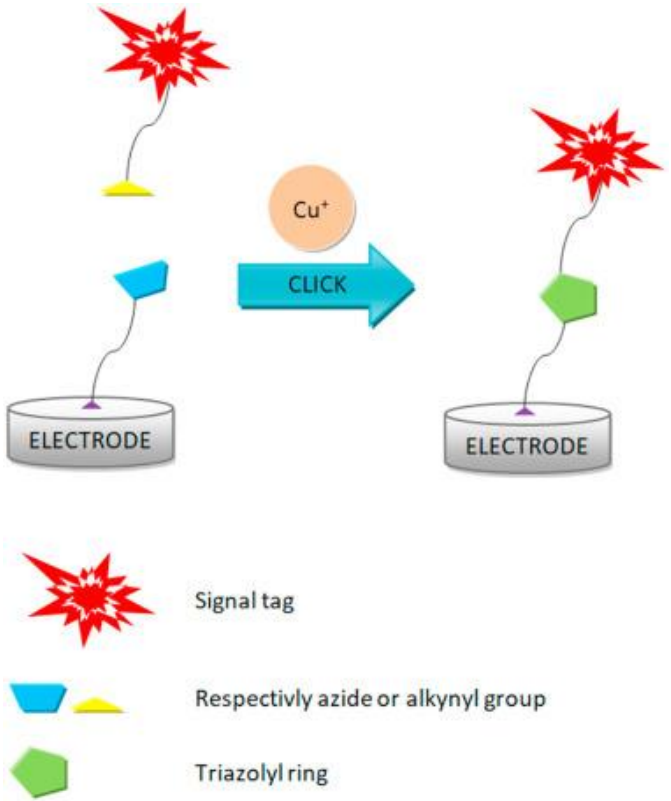
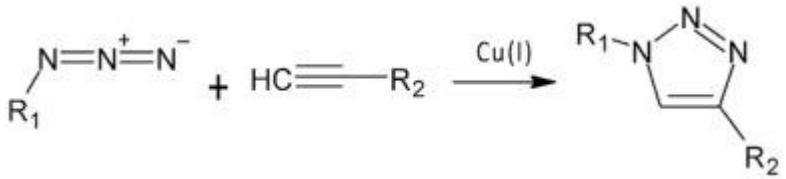
Why Cu?





General scheme for Cu^{I}-catalyzed azide/alkyne cycloaddition.

- ✓ **Copper is one of the most essential trace elements in organisms as it is involved in a wide range of metabolic processes.**
- ✓ **Non-precious metal Cu is earth-abundant, significantly cheaper and widely used in industries**
- ✓ **Cu possess unique catalytic and photoluminescence (PL) properties.**
- ✓ **Cu is popular for click Chemistry.**
- ✓ **Inexpensive**

Biosensors and Bioelectronics, 169, 2020, 112614

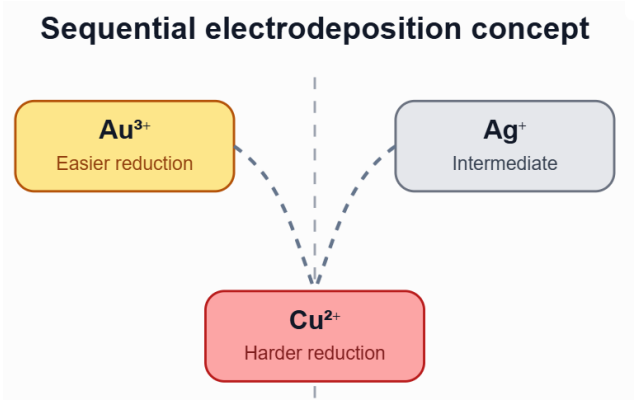
Why bimetallic or trimetallic over monometallic?

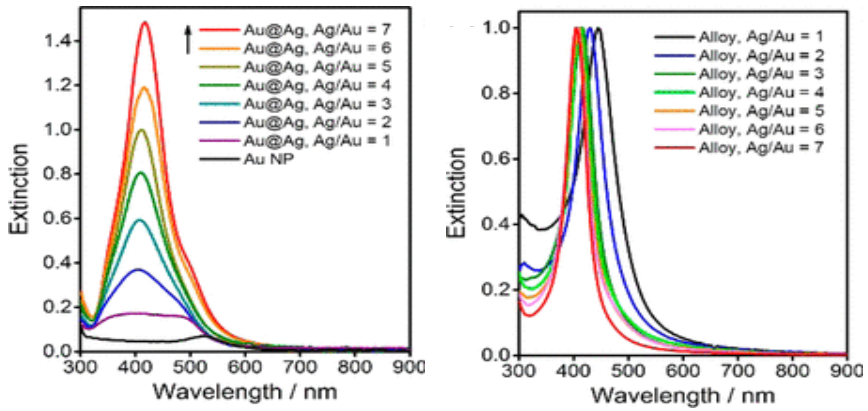
$Au^0 = +1.50$ eV

$Ag^0 = +0.80$ eV

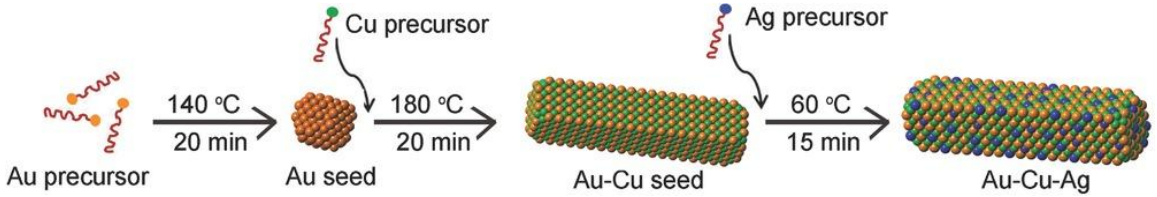
$Cu^0 = +0.34$ eV

Reduction potential of Au, Ag and Cu





Responses of Au/Ag system with their different ration



Schematic illustration of the synthetic strategy of Au–Cu–Ag trimetallic nanorods.

- Ag, Au, and Cu in a single unit to integrate the individual merits of each metal, the higher SERS activity of Ag, better surface stability and biocompatibility of Au, and structural stability of Cu.

- The reduction of Cu will be slower than Ag. Au will be more reductive than Ag. Therefore, the synthesis depends upon the sequence of metal, time gap, electrode potential and galvanic replacement of metal ion.

J. Am. Chem. Soc. 2014, 136, 20, 7474–7479
Plasmonic Particles, 34, 2017, 1600384

Exercise!

Q1. Q1. A bimetallic SERS substrate shows lower sensitivity than expected. Suggest three possible reasons related to its composition, structure or fabrication.

Q2. Suggest an Au–Ag–Cu tri-metallic nanocomposite design for biomedical applications (e.g., imaging, sensing, or therapy). Justify your choice based on the unique roles of each metal.

Q3. How could Material Informatics machine learning and AI contribute to optimizing the composition, synthesis, and performance prediction of multi-metallic nanocomposites?

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Synthesis of bi and tr-metallic nanocomposite (mesoporous)

- **Hard-templating approach** (*i.e.*, nanocasting)
  - Hard templates made of inorganic materials such as mesoporous silica are commonly used to guide and control the building of mesoporous metals.
  - The generated replicas reverse the template's morphology; however, the dissolution process is typically slow and requires repeated exposure to dangerous chemicals.
  - Although hard templates provide a consistent and intuitive method for producing mesoporous metals, the complexity of the technique, lack of morphology control, and use of harsh chemicals have confined their use in wider applications
- **Soft-templating approach** (*e.g.*, lyotropic liquid crystalline (LLC) and di-block polymer-based templating)
  - Soft-templating methods are promising alternatives that use organic surfactants (*e.g.*, block polymers, lyotropic liquid crystals, *etc.*) as molecular templates to direct the formation of mesopores in metallic structures.
- **Template-free approach** (*e.g.*, post-synthetic dealloying)
  - The chemical treatment preferentially dissolves a non-targeted element, leaving a porous metal or alloy surface behind.
  - This technique is commonly seen in technologically significant alloy systems, such as stainless steel, brass, and different alloys especially Cu–Al alloys.
  - The mechanical and physical properties of a porous overlayer differ greatly from those of the bulk alloy on which it sits, resulting in stress corrosion cracking, brittle crack propagation as well as other unwanted material failure.

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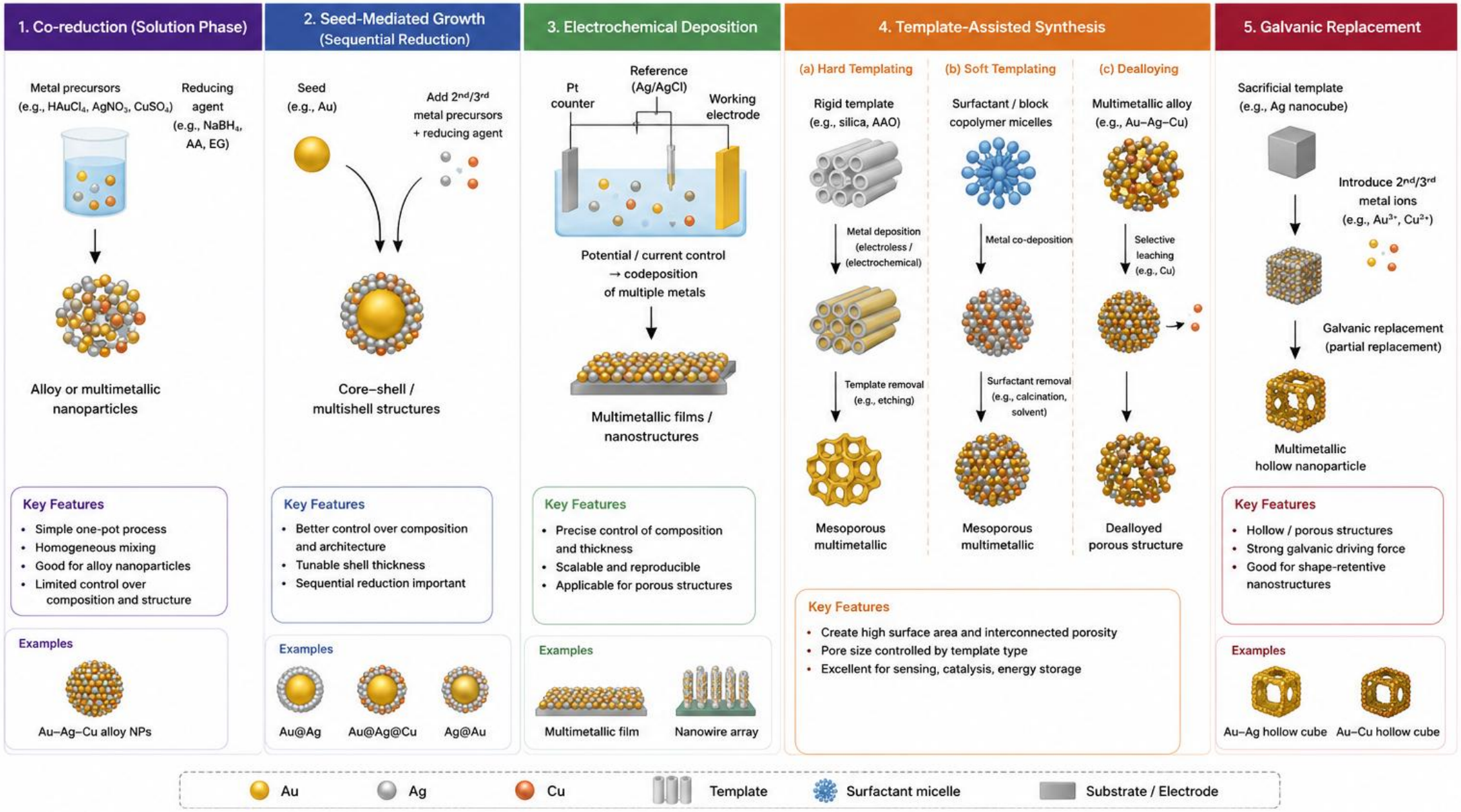
Synthesis of bi and tr-metallic nanocomposite (mesoporous)



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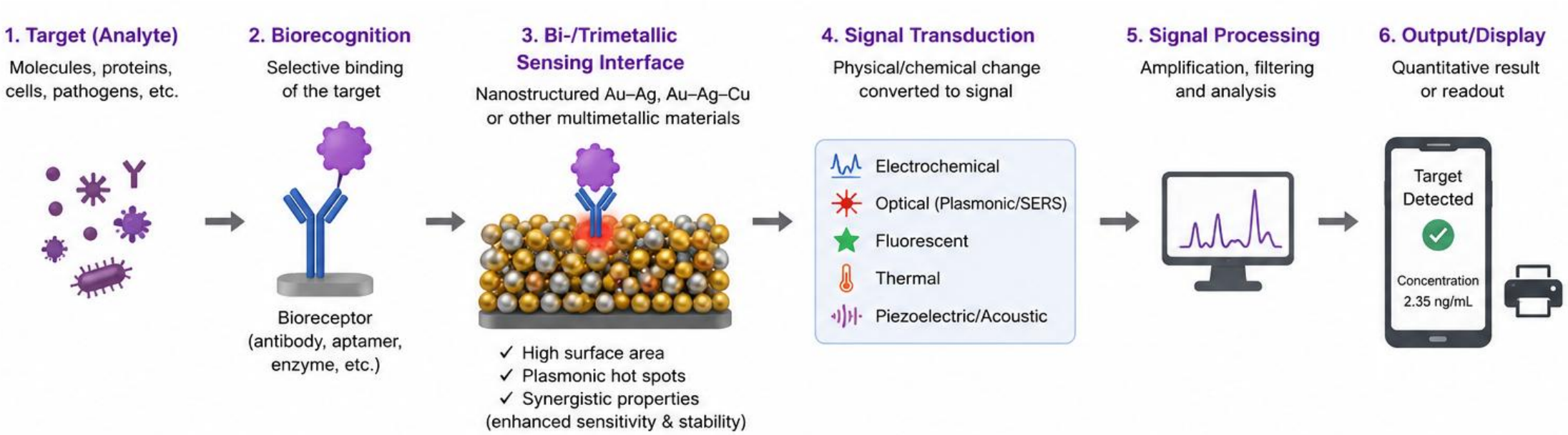
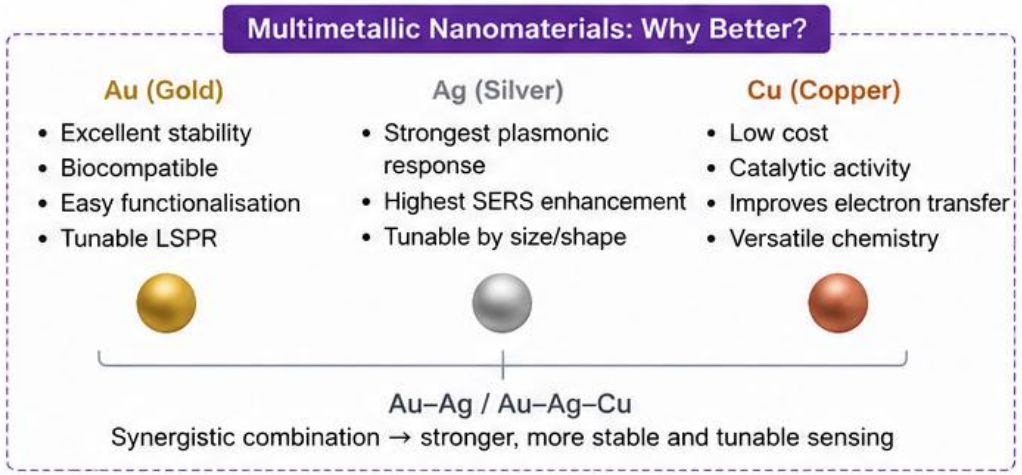
Synthesis of bi and tr-metallic nanocomposite (mesoporous)

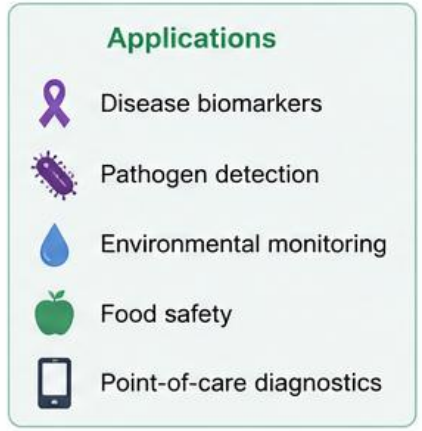
Method	Advantage	Limitation	Engineering use
Hard templating	Excellent structural fidelity	Slow, harsh chemistry	Ordered porous metals
Soft templating	Tunable pores	Sensitive to processing	Mesoporous electrodes
Dealloying	Simple porous formation	Limited composition control	Porous alloys
Electrodeposition	Scalable and controllable	Requires conductive substrate	Sensor fabrication

CRICOS code 00025B

Biosensor







Biosensors **2018**, *8*(4), 104

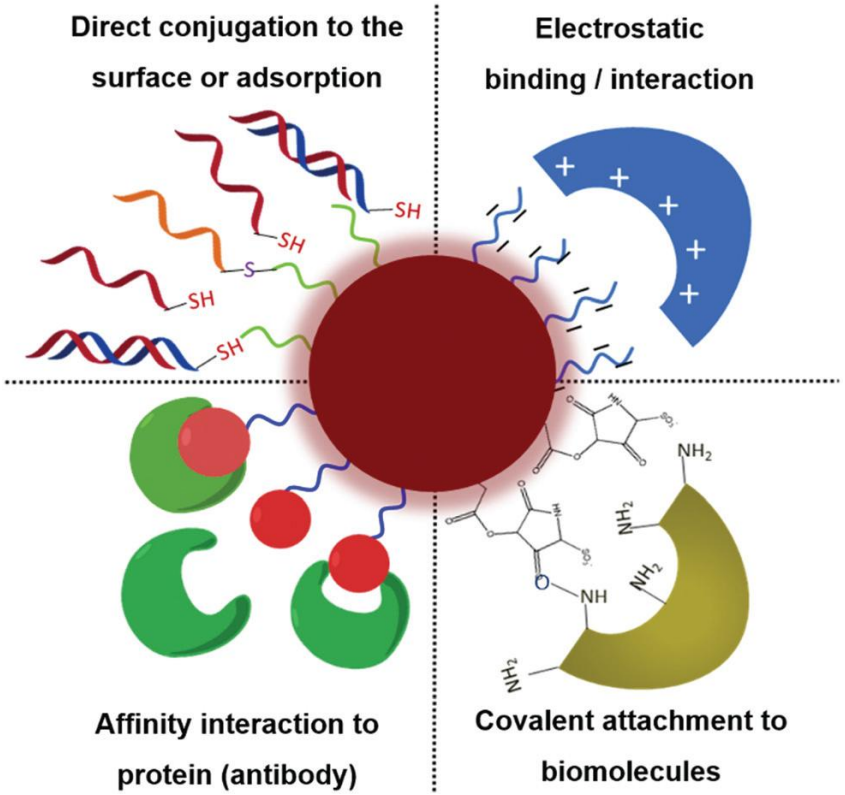
CRICOS code 00025B

Au, Ag and Cu in Biosensing : Surface Functionalization

- Their use in biosensing is inevitably about their high SA conferred greater functionalization abilities, making it possible to track any stimulus or material response in biological and non-biological configurations.

- The surface functionalization or covering of the surface with biocompatible compounds is crucial to (i) prevent degradation in an aqueous, acidic, or aggressive (blood) environment; (ii) suppress the magnetic interactions to avoid aggregation; and (iii) provide a functional group to bind or attach a wide range of biomolecules (e.g., proteins and nucleic acids).

- Notable functionalization mechanisms for tagging biological moieties to nanomaterial surfaces include;
  (i) thiol-NM surface interaction
  (ii) streptavidin-biotin interaction
  (iii) π-π stacking interactions, and
  (iv) NHS-EDC carbodiimide reaction.



Chem. Soc. Rev., 2019, 48, 5717

Surface Functionalization

Thiol–Au

Strong anchoring
SAM formation

EDC–NHS

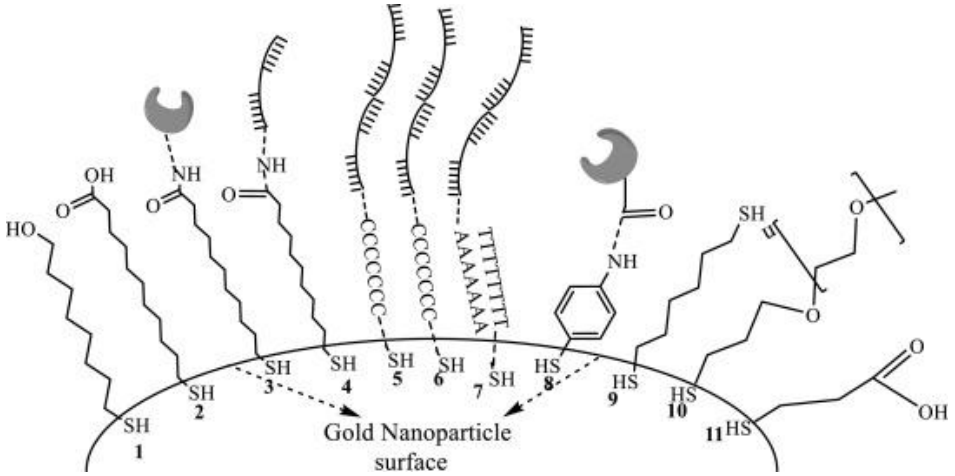
COOH + NH_2
Covalent amide bond

Streptavidin

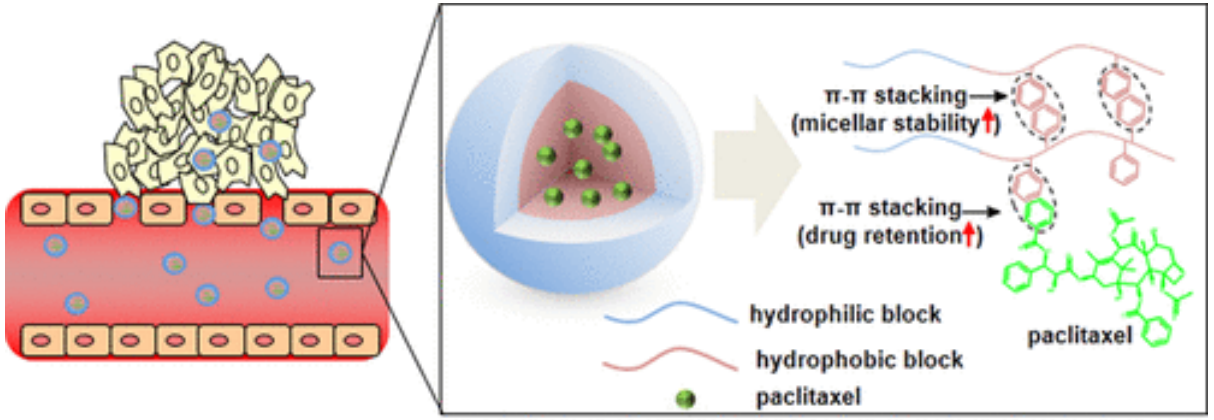
Biotin recognition
High affinity

π–π stacking

Aromatic surfaces
Non-covalent assembly



Modification with thiol compounds



Modification through π-π stacking interactions

Chem. Soc. Rev., 2019, 48, 5717

Exercise!

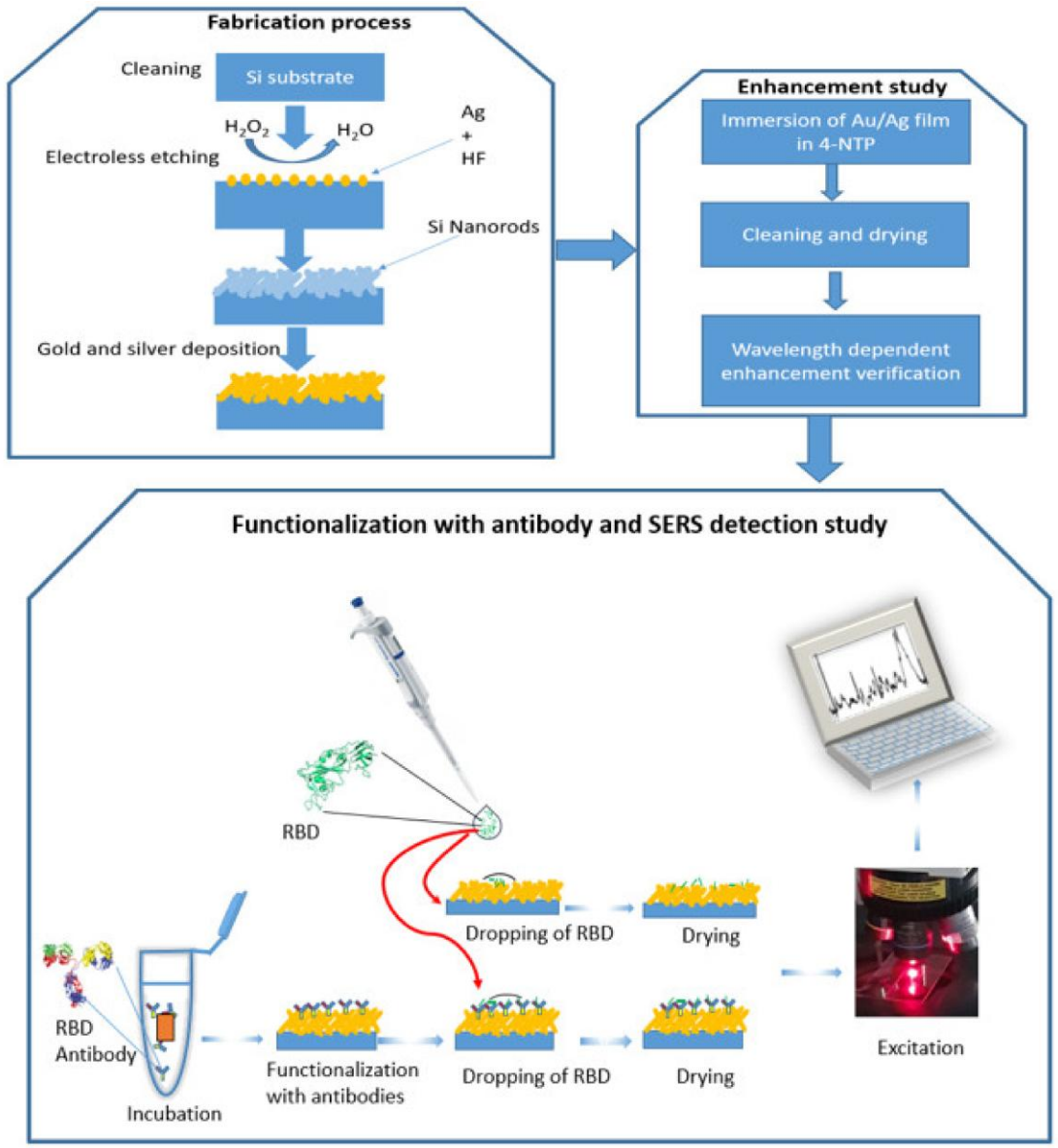
Q1. Functionalization improves stability, but may affect biocompatibility. How would you balance both for biomedical applications?

Q2. How can surface functionalization improve specificity of Au-Ag-Cu nanocomposites for detecting biomolecules?

Q3. A SERS sensor has excellent sensitivity but poor stability and reproducibility. How could surface functionalisation improve its performance?

CRICOS code 00025B

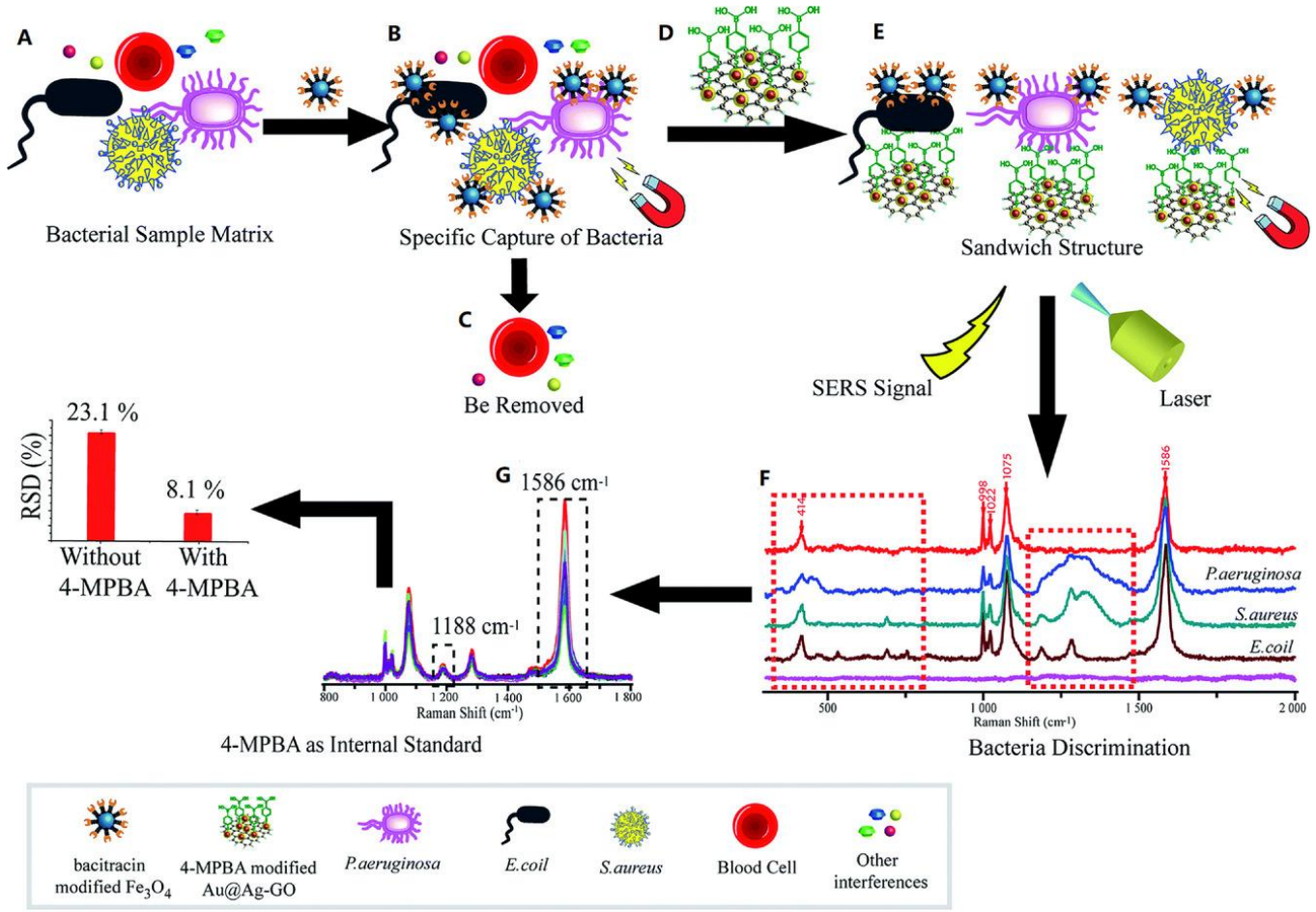
Au/Ag based Biosensor Platform: Viral detection



- ❑ What material was engineered?
- ❑ Why was Au/Ag chosen?
- ❑ What fabrication technique was used?
- ❑ What signal transduction occurred?
- ❑ What limits translation into manufacturing?

Sensors **2021**, *21*(13), 4617

Au/Ag based Biosensor Platform: Bacterial detection

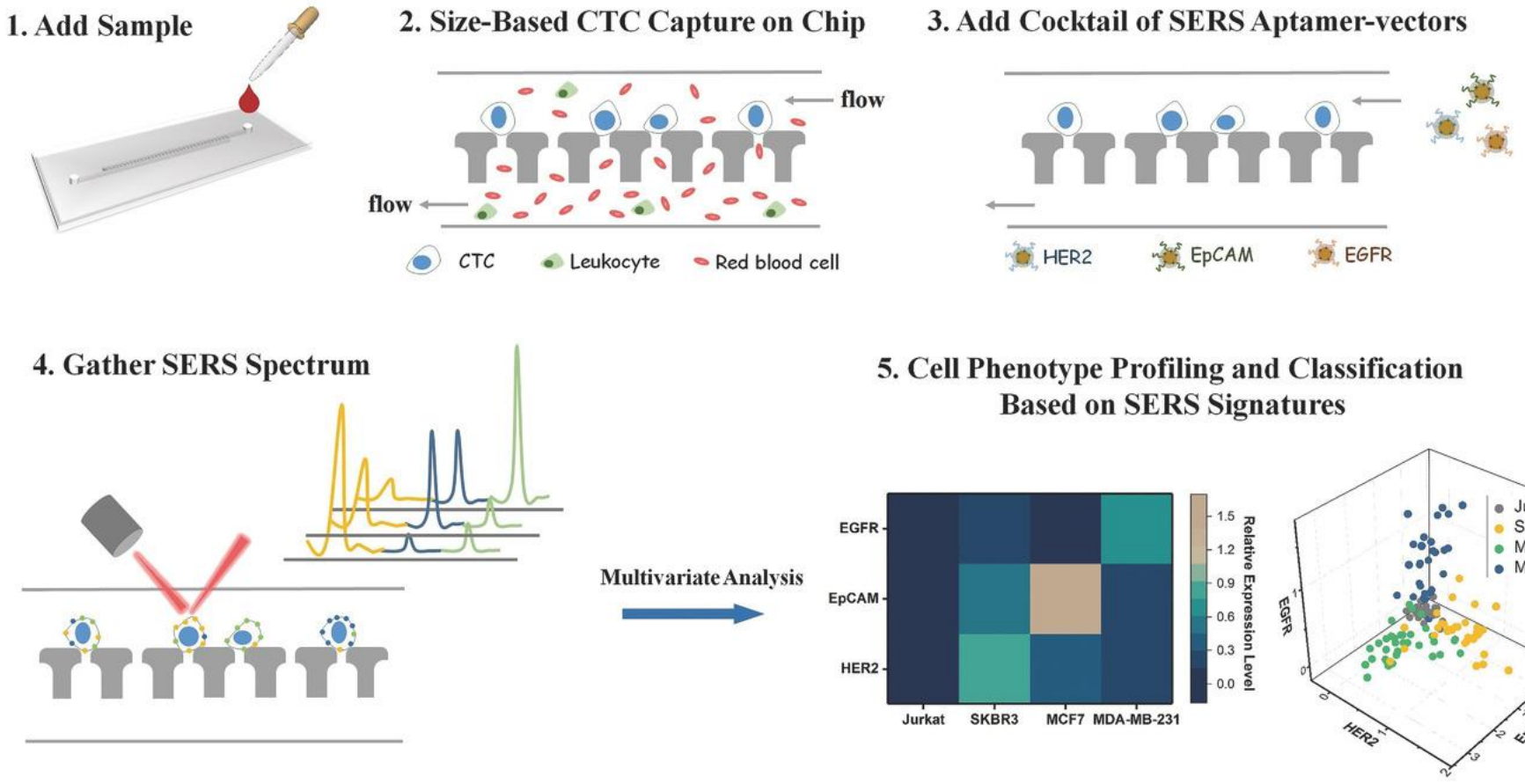


- ❑ What material was engineered?
- ❑ Why was Fe_2O_3 chosen?
- ❑ What fabrication technique was used?
- ❑ What signal transduction occurred?
- ❑ What limits translation into manufacturing?

Chem. Sci., 2018,**9**, 8781-8795

CRICOS code 00025B

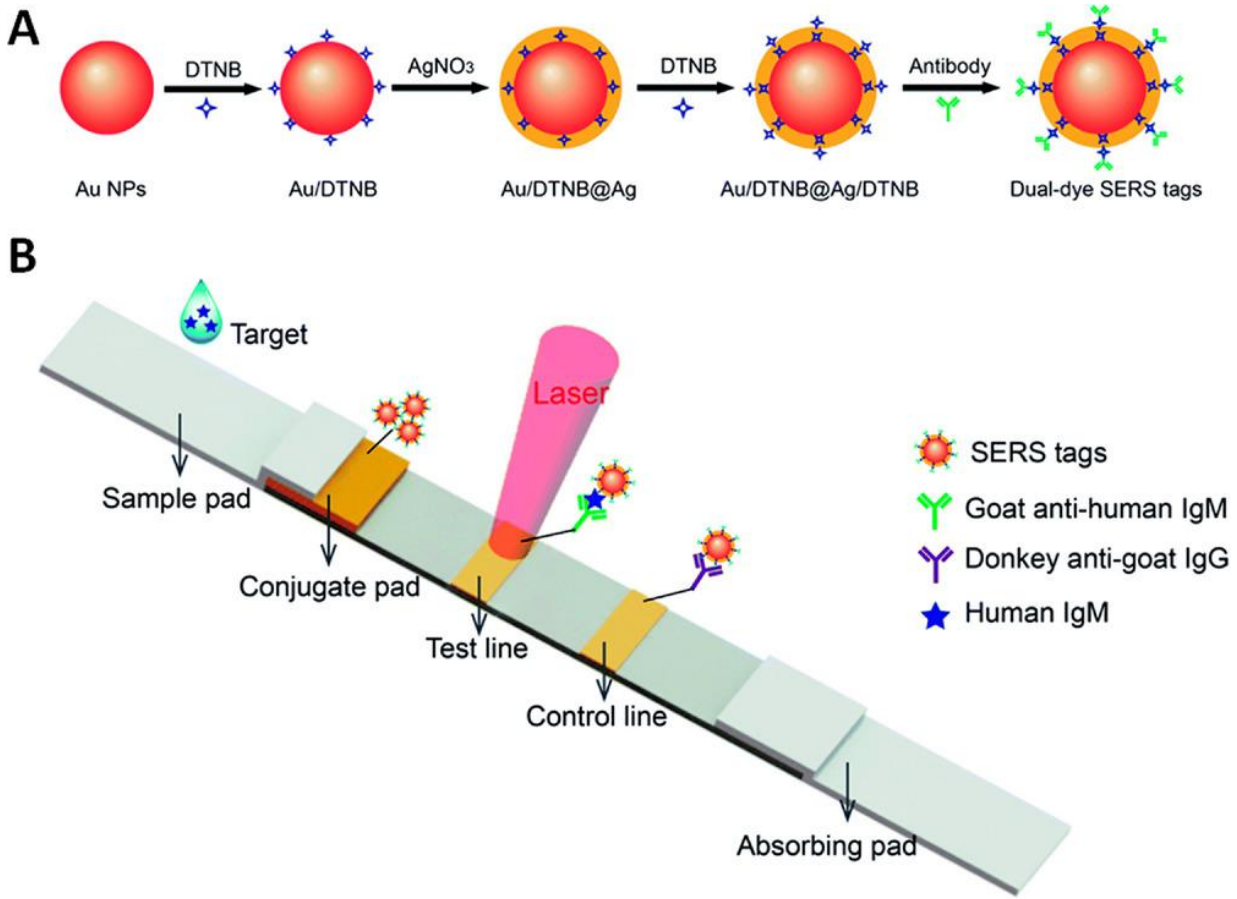
Au/Ag based Biosensor Platform: cancer cells detection



- ❑ Why use Au@Ag rather than Au alone?
- ❑ What role does the microfluidic structure play?
- ❑ How does multiplex SERS distinguish different biomarkers?

Small 2018, *17*, *1704433*

Au/Ag based Biosensor Platform: human IgM detection



- ❑ What material was engineered?
- ❑ Why was Au/Ag chosen?
- ❑ What fabrication technique was used?
- ❑ What signal transduction occurred?
- ❑ What limits translation into manufacturing?

RSC Adv. 2018, **8**, 21243.

Take-Home Messages

- ✓ **Composition matters: Combining metals can generate synergistic optical, catalytic and electronic properties.**
- ✓ **Architecture matters: Alloy, core–shell and porous structures can behave differently even with similar compositions.**
- ✓ **Processing matters: Synthesis controls composition, morphology, porosity and ultimately sensing performance.**
- ✓ **Engineering matters: The best sensor balances sensitivity, stability, reproducibility and manufacturability.**

Composition → Structure → Properties → Sensing Performance

CRICOS code 00025B

WEEK 10 • 知识点总结 • A/Prof. Nasim Amiralian

第 10 周　纳米纤维素

Nanocellulose: from plant fibre to functional, sustainable material

主线：可持续性背景 → 纤维素的来源与结构 → CNF / CNC 的制备 → 表征与挑战 → 两个 UQ 案例 → 规模化与负责任制造

1. 背景：为什么需要替代化石基塑料

- 自 1950 年代以来累计生产超过 **83 亿吨**塑料；仅约 **9% 被回收**，12% 焚烧，约 79% 进入填埋或环境；微塑料广泛存在于水与人体。
- 问题在于"化石原料 → 制造 → 短期使用 → 废弃"的**线性模式**；纳米纤维素的目标：以可再生纳米结构材料实现同等功能、降低全生命周期负担。

2. 纤维素

- 最丰富的天然聚合物（约 10^{12} t/年）；**D-葡萄糖以 β-1,4 糖苷键连接**的线性均聚物；形成**半结晶微纤丝**；不溶于水。
- 来源：**植物**（草本：亚麻、棉、黄麻、竹、剑麻…；木本）；**细菌纤维素**（木醋杆菌胞外合成，**唯一纯净形式**，微纤丝较大）；**被囊动物**（唯一能合成纤维素的动物，结晶度高）。
- 植物细胞壁 = 纤维素 + **半纤维素**（杂多糖，DP < 200，无定形）+ **木质素**（非线性多酚，"胶水"，疏水，棕色）。

3. 制备

- **制浆**：洗涤、干燥、粉碎 → **脱木质素** → **漂白**（去残余木质素与半纤维素）→ 纯化纤维。
- **CNF（纤维素纳米纤维）**：预处理（化学 / 机械 / 酶）+ 机械解纤：高压均质（直径约 8 ± 2 nm）、挤出（约 11 ± 3 nm）、研磨、超声、Silverson 剪切；在能耗、直径、可放大性之间取舍。
- **CNC（纤维素纳米晶）**：**酸水解**（H_2SO_4、HCl、H_3PO_4、HNO_3）去除无定形区 → 离心、透析 → 超声分散 → 棒状晶体。

CNF	CNC
低浓度（0.5–1.5 wt%）即成凝胶	低浓度为流体分散液
长径比极高（约 1000），柔软、"意面状"	长径比较低，刚性棒状
适合复合材料增强相	适合填料、黏度调节、高端应用

4. 表征、优点与挑战

- 表征：形貌 TEM / SEM / DLS；表面电荷 **Zeta 电位**；比表面积 BET；**结晶度 XRD**；热稳定性 TGA / DSC；力学：流变、拉伸、单根纤维 AFM；表面官能团 FTIR / XPS。
- 优点：高长径比 → 增强与网络；**富含羟基** → 易功能化、氢键；成膜性与氧阻隔；流变调节；生物基。
- **四大挑战**：高比表面能 → **团聚**；与疏水聚合物和非极性溶剂**不相容**；**吸湿**；**干燥与运输成本** → 解决：**表面改性**（吸附、硅烷化、乙酰化 / 酯化、羰基化、聚合物接枝）。
- 商业产品：CelluForce（CNC）、Borregaard Exilva（MFC）、FiberLean、UPM GrowDex（3D 细胞培养）、Nippon Paper Cellenpia；多为"使能材料"而非终端产品。

5. 案例 1：Spinifex 草纳米纤维素

- Triodia（spinifex）覆盖澳洲 > 200 万 km^2；与原住民社区合作（Camooweal，QLD）；澳洲首个纳米纤维素中试厂。
- **伦理合作五原则**：社区主导、双向知识、**自由事先知情同意（FPIC）**、利益共享、材料转移协议（参考 Nagoya Protocol）。从"抽取式研究"转向"共同设计的价值创造"。
- 应用：增强天然乳胶**医用手套**——普通填料提高强度但变硬、断裂伸长率下降 → 用**软而韧**的纳米纤维 → 薄而触感好的膜。

6. 案例 2：自消毒纤维素涂层

- 背景：**抗菌药物耐药（AMR）**，2019 年约 130 万人死亡；材料型抗菌表面不依赖抗生素。
- 纳米纤维素优势：表面 OH 易化学修饰、比表面积大、生物相容、可降解、可制成膜 / 凝胶 / 气凝胶 / 涂层。
- 方法：**高碘酸盐氧化 → 双醛纤维素**（DACNF / DACMF）；或在纳米纤维素上模板合成金属（CuO）纳米颗粒。杀灭 S. aureus、MRSA、E. coli、H1N1 流感病毒；细胞毒性按 **ISO 10993-5**（L929 成纤维细胞）。

7. 规模化

- 实验室（mg–g，证明概念）→ 中试（kg，证明工艺、批次一致性、能耗）→ 商业（吨，证明商业模式、供应链、法规）。
- 瓶颈：原料波动、预处理强度、产率与固含量低、重现性、**干燥与再分散**、市场接受；需控制：输入、过程、质量、风险、文档。技术可放大 ≠ 负责任的生物制造。

考试重点：纤维素结构与来源；CNF vs CNC 的制备与性质对比；表征手段对应关系；四大挑战与表面改性；伦理合作原则；抗菌机理与规模化瓶颈（简答题素材）。

Acknowledgment of **Country**

The University of Queensland (UQ) acknowledges the Traditional Owners and their custodianship of the lands on which we meet.

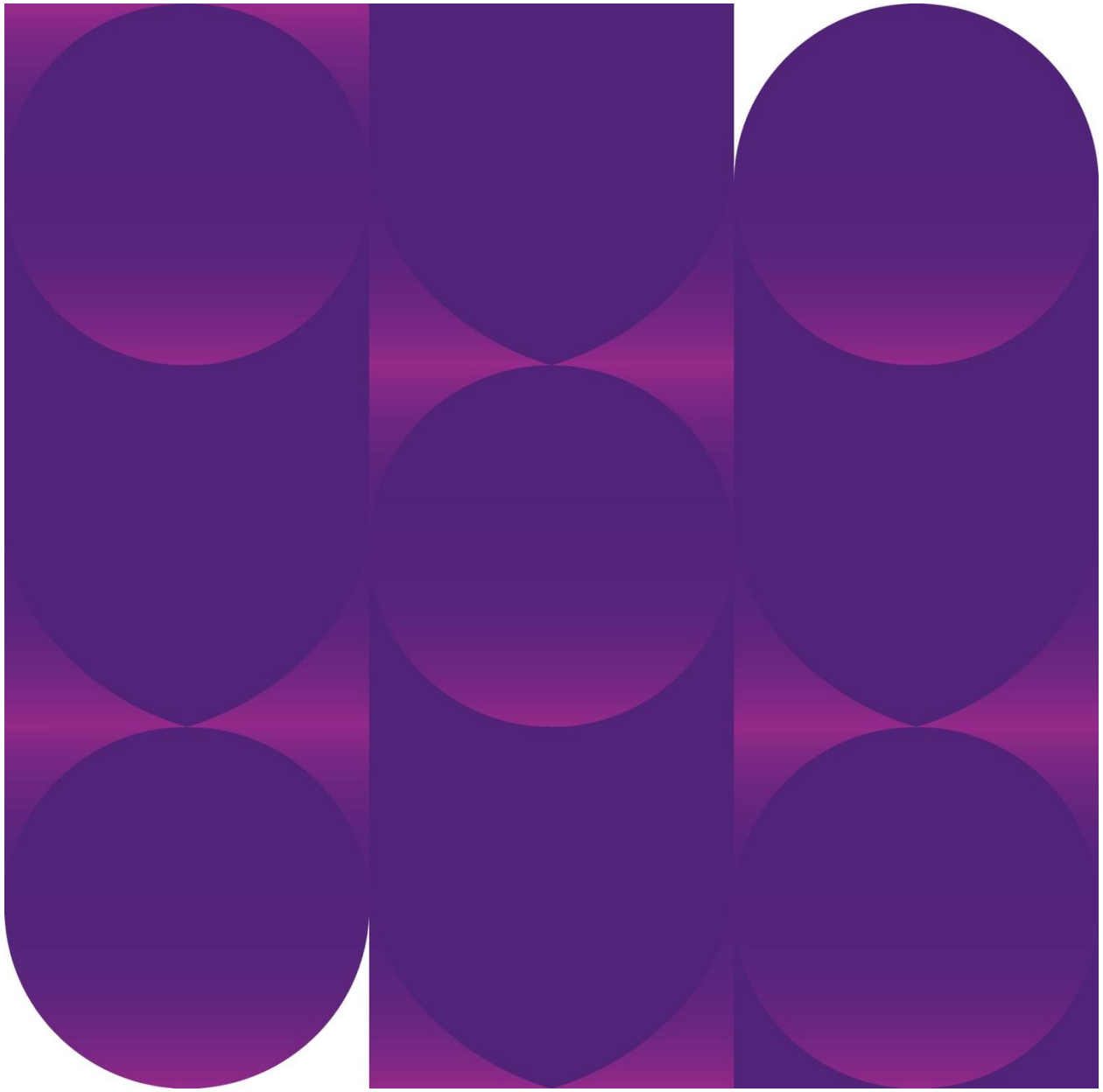
We pay our respects to their Ancestors and their descendants, who continue cultural and spiritual connections to Country.

We recognise their valuable contributions to Australian and global society.

THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

Nanocellulose

From plant fiber to functional, sustainable material



MATE6301 · Nanomaterials

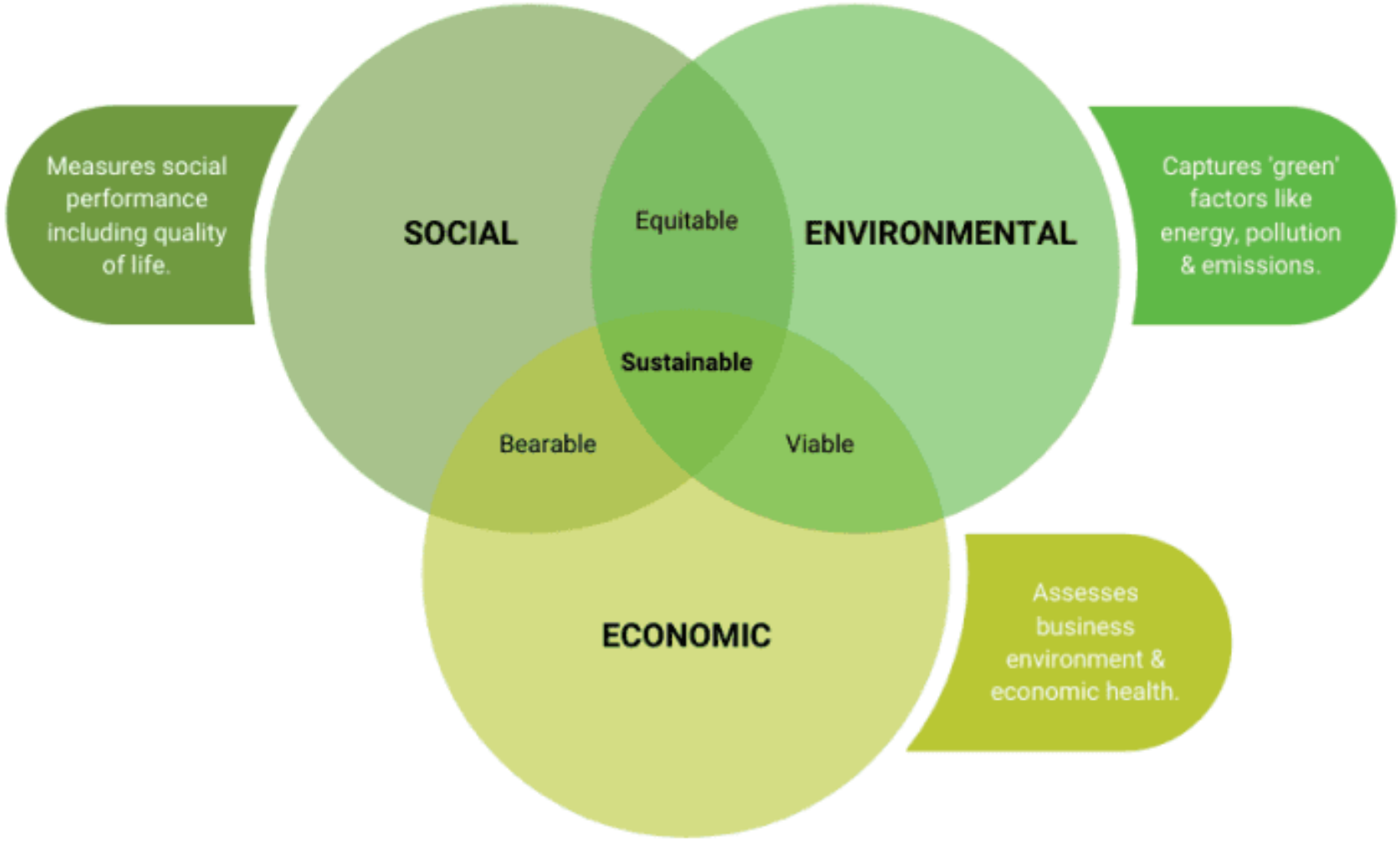
A/Prof Nasim Amiralian

What we'll cover today

- Why materials science needs an alternative to fossil-based plastics
- What cellulose is, and the different natural sources it comes from
- How nanocellulose is made — from plant fibre to nanoscale material
- How we characterise nanocellulose, and the challenges it still faces
- Where nanocellulose is already being used, commercially and in research
- Two UQ case studies: spinifex nanocellulose and antimicrobial cellulose coatings

Sustainability

Sustainability means **meeting our own needs without compromising the ability of future generations to meet their own needs**.



https://www.mcgill.ca/sustainability/files/sustainability/what-is-sustainability.pdf

https://www.sustain.ucla.edu/what-is-sustainability/

Excessive consumption

Climate Change
Greenhouse Gas Emissions

Plastic pollution
Microplastics

Economic Costs

Environmental pollution
Resource depletion

Social impact
Declining ecosystem health

Impact of plastic pollution: key statistics

Global Production: Over 8.3 billion metric tons of plastic have been produced since the 1950s.

Recycling Rates: Only about 9% of plastic waste has been recycled globally.

Landfills and Environment: Approximately 79% of plastic waste accumulates in landfills or the natural environment.

Incineration: About 12% of plastic waste is incinerated.

Economic Cost: Plastic pollution costs the global economy an estimated $13 billion annually.

Microplastics: Microplastics are found in 83% of tap water samples worldwide.

Air Pollution: Incineration of plastics releases harmful chemicals like dioxins, furans, and PCBs.

Human Health: Over 90% of humans have plastic particles in their bodies.

Wildlife Impact: 100,000 marine mammals and 1 million seabirds die annually due to plastic pollution.

Soil Contamination: Up to 23% of plastic waste can end up in agricultural soils.

Legislation: More than 60 countries have implemented bans or levies on single-use plastics.

https://ourworldindata.org/plastic-pollution | *Nature* 619, 222 (2023) | https://www.oecd-ilibrary.org/environment/data/global-plastic-outlook_c0821f81-en

Did you know?

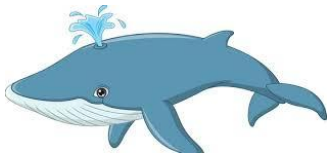
Single use plastic production per year = 300 million tons

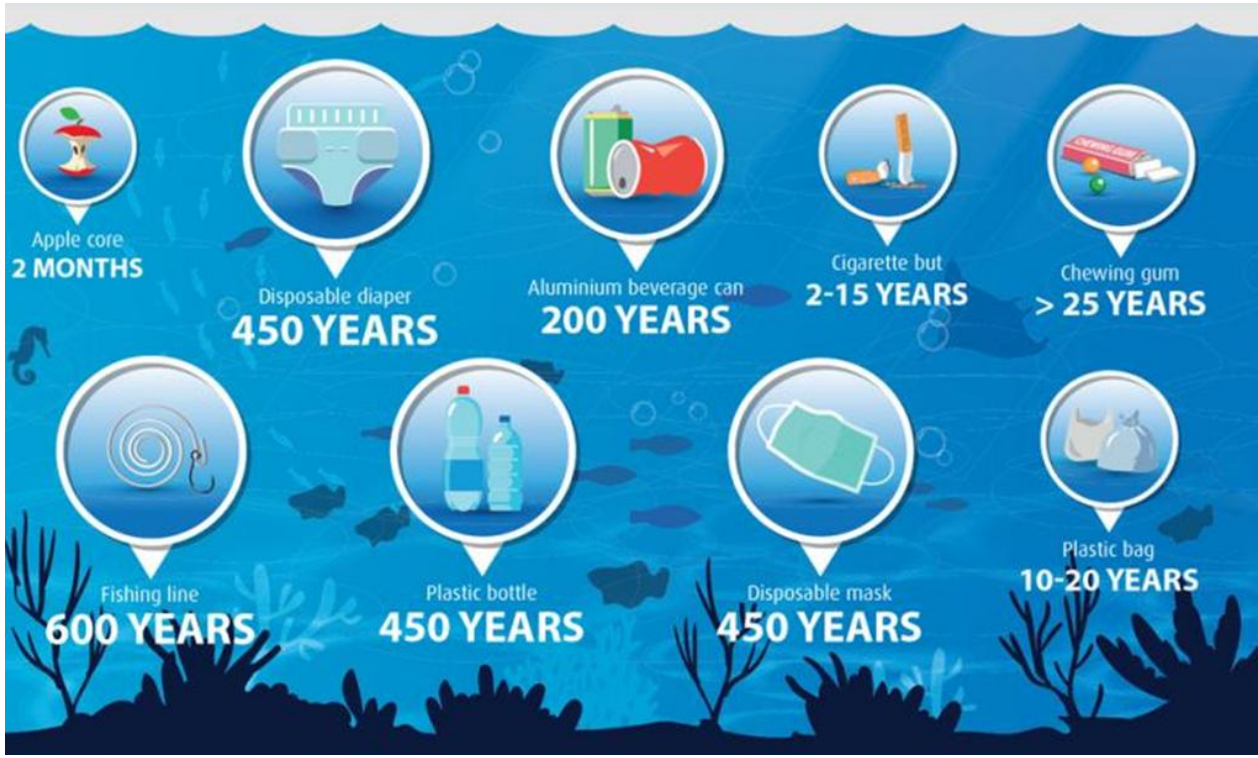
=

Entire human population

2,700,000 blue whales

681 million football pitches





Did you know?

10M tons of plastic are dumped in the ocean/pers = more than a garbage truck every minute!

Did you know?

There are more particles of microplastics in the ocean than the stars in the milky way.

Did you know?

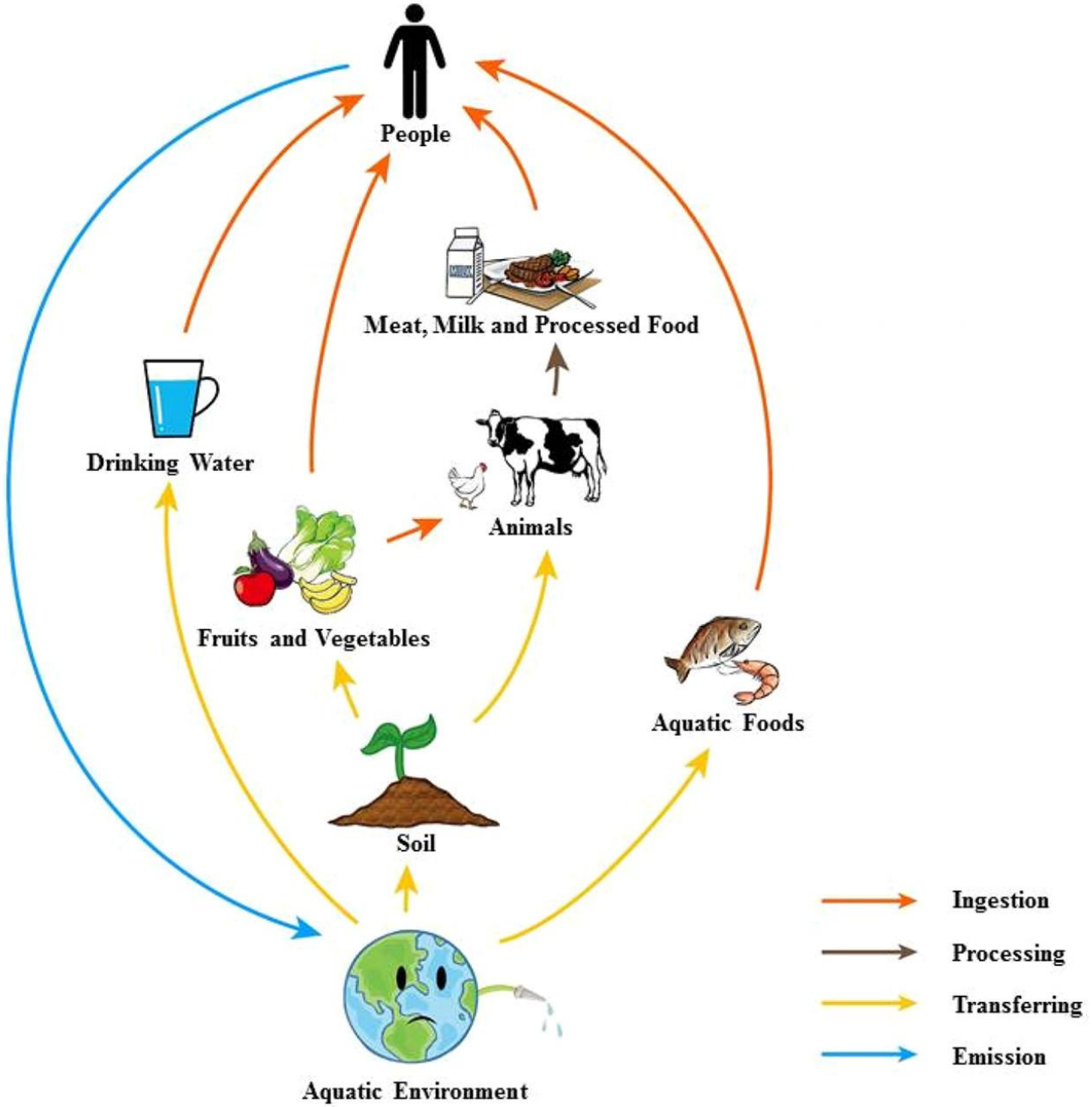
By 2050, there will be more plastic in the ocean than fish.

Did you know?

100% of mussels tested have microplastics!

Microplastics in the food chain

Humans eat over 18 kg of plastic in their lifetime.



The problem: linear, fossil-based materials

Fossil feedstock

Oil and gas are the primary carbon input for most plastics, coatings and packaging

→

Manufacture

High-throughput production optimised for cost and performance, not end-of-life

→

Short use phase

A large share of plastic is single-use, discarded within months or even minutes

→

Waste & pollution

Recycling alone cannot solve design, contamination and collection limits

Nanocellulose asks: can a renewable, nanostructured material deliver equivalent function with a lower lifecycle burden?

Cellulose

Cellulose is the building block of plants.

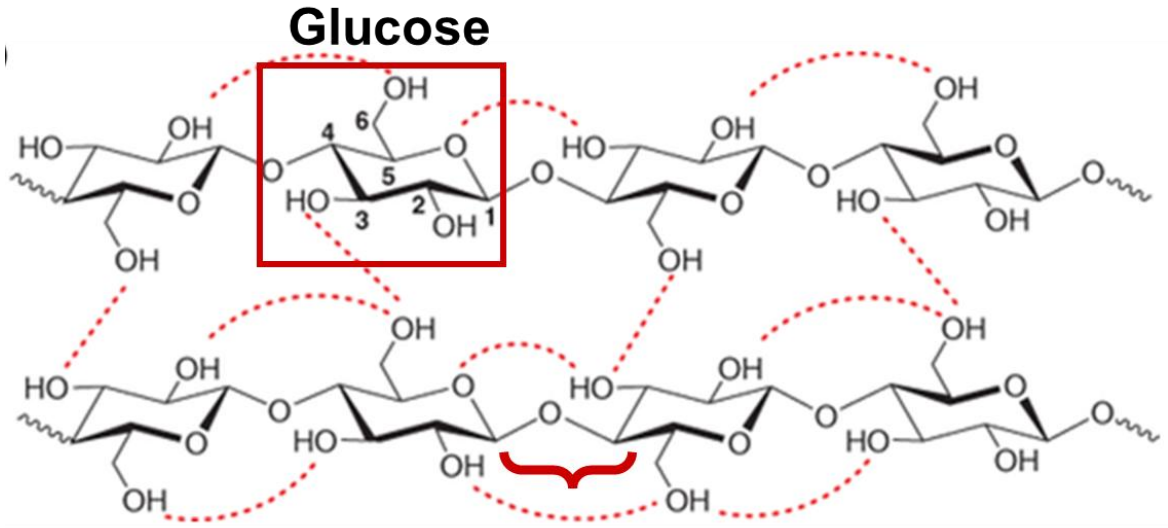
The most abundant polymer is available worldwide(10^{12} tons produced per year).

Cellulose is a polysaccharide made up of thousands of D-glucose subunits linked via beta 1-4 glycosidic bonds.

Linear homopolymer

Forms semi-crystalline microfibrils

Insoluble

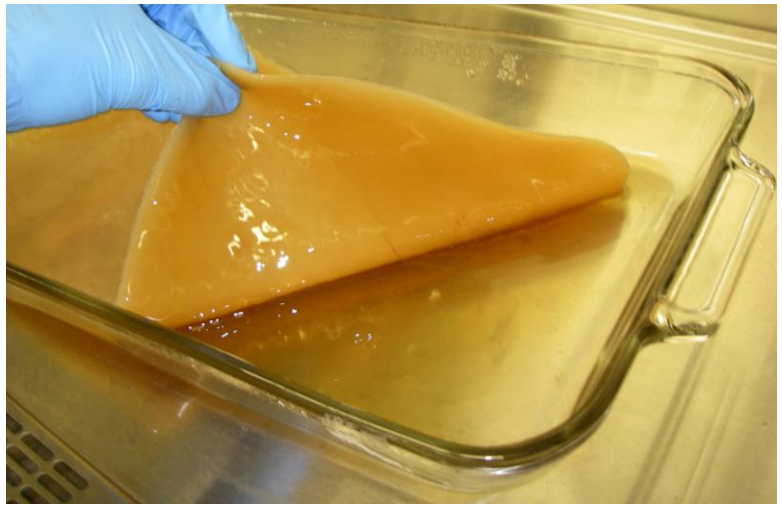


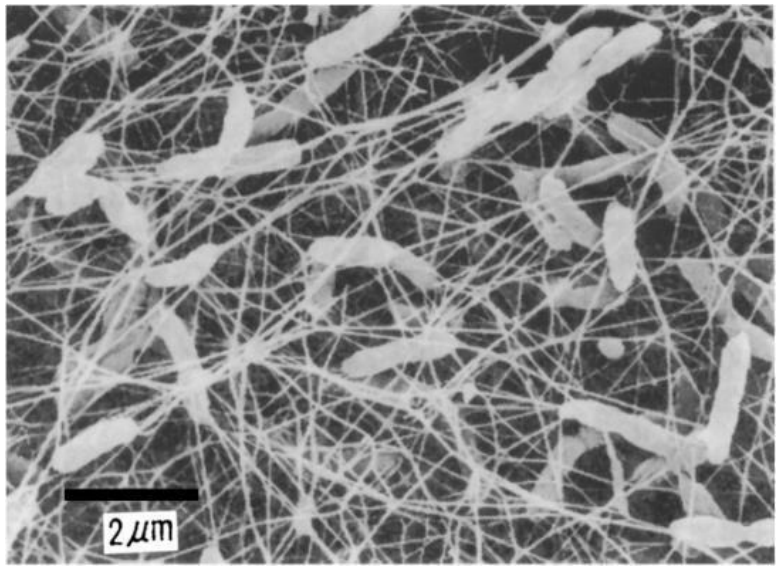
1-4 beta glycosidic bonds

Source of cellulose

Industrial Crops and Products, 93(25),2016, 26-38

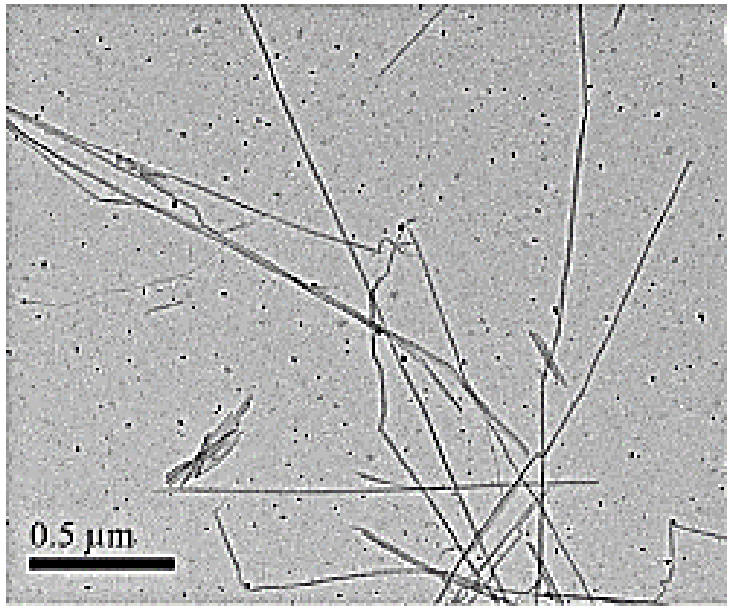
Bacterial cellulose





- Some species of bacteria can biosynthesise cellulose (notably *acetobacteria xylinum*)
- The cellulose is extracellular, it is not in the bacteria but extruded by the bacteria
- Bacterial cellulose is the only pure form of cellulose in nature, no other components are synthesised with cellulose
- Microfibrils in bacterial cellulose are generally larger than in other cellulose grades

Tunicate cellulose



Tunicates are small marine animals

- They are the *only species of animals* that biosynthesize cellulose

- The body of a tunicate is surrounded by a test or a tunic where the cellulose is Produced

- Tunicate cellulose is in the form of microfibrils that are highly crystalline compared with most plant celluloses

Plant cellulose

Herbaceous plants

- Agro fibres
- At least the part above ground dies after the growing season

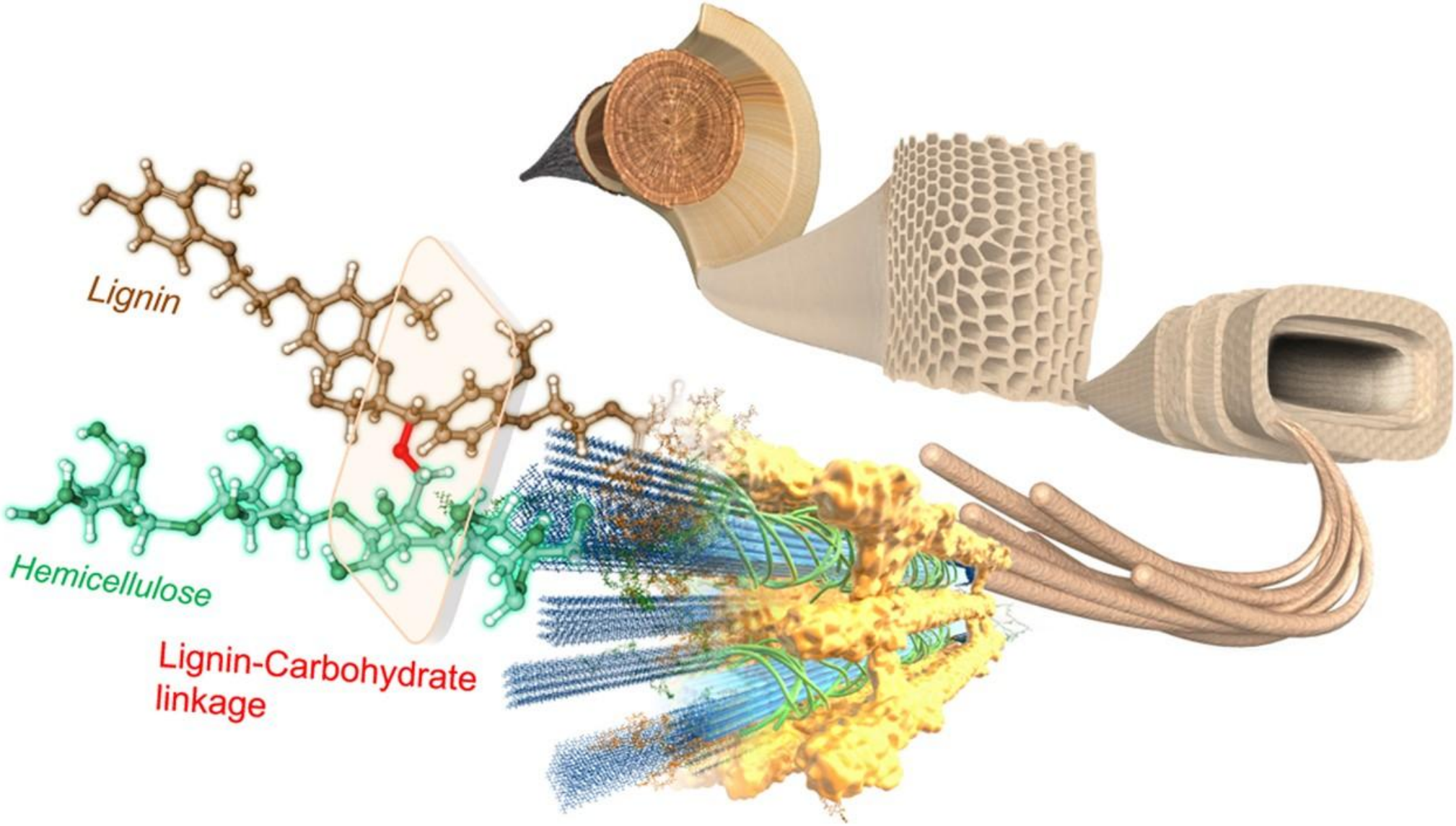
- Flax
- Cotton
- Jute
- Kenaf
- Bamboo
- Ramie
- Sisal

Woody plants

- Wood fibres
- Remain alive during dormant season; reinforced by secondary xylem

- Trees
- Shrubs

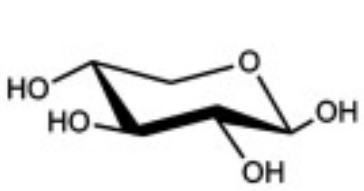
Plant structure



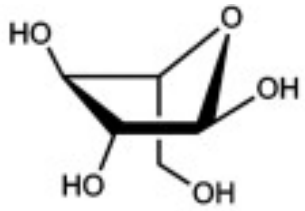
To get to nanocellulose, this natural composite has to be taken apart.

Hemicellulose

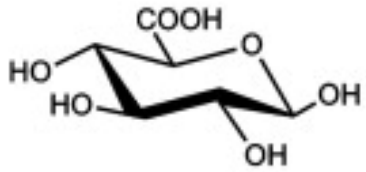
Monomer sugars in hemicellulose



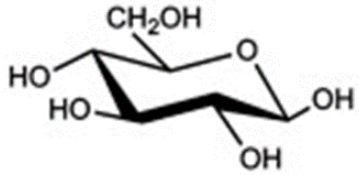
Xylose



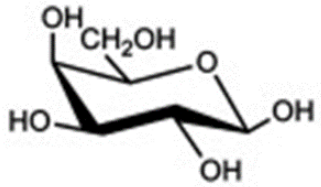
Arabinose



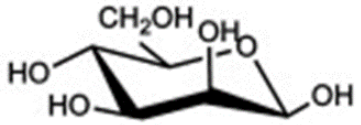
Glucuronic acid



Glucose

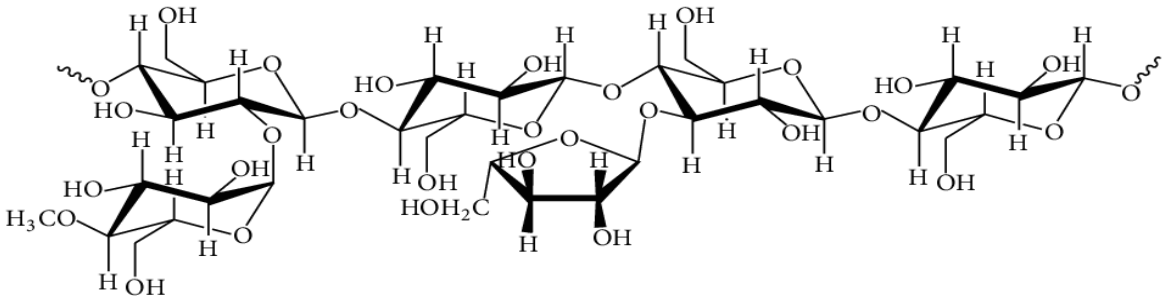


Galactose



Mannose

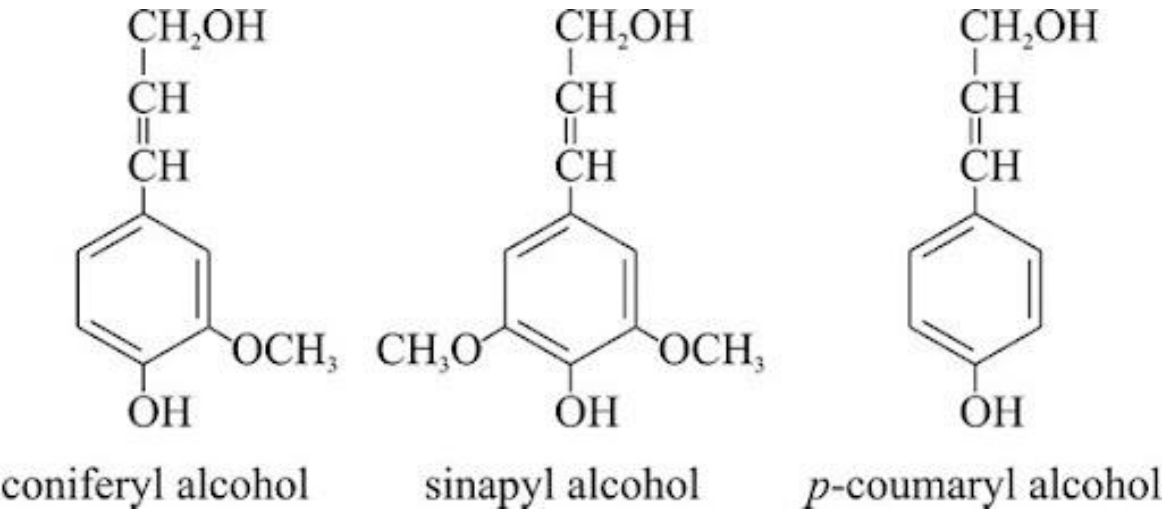
Xylan



Hemicelluloses are heteropolysaccharides with low DP (<200) and they form amorphous structures in the cell wall

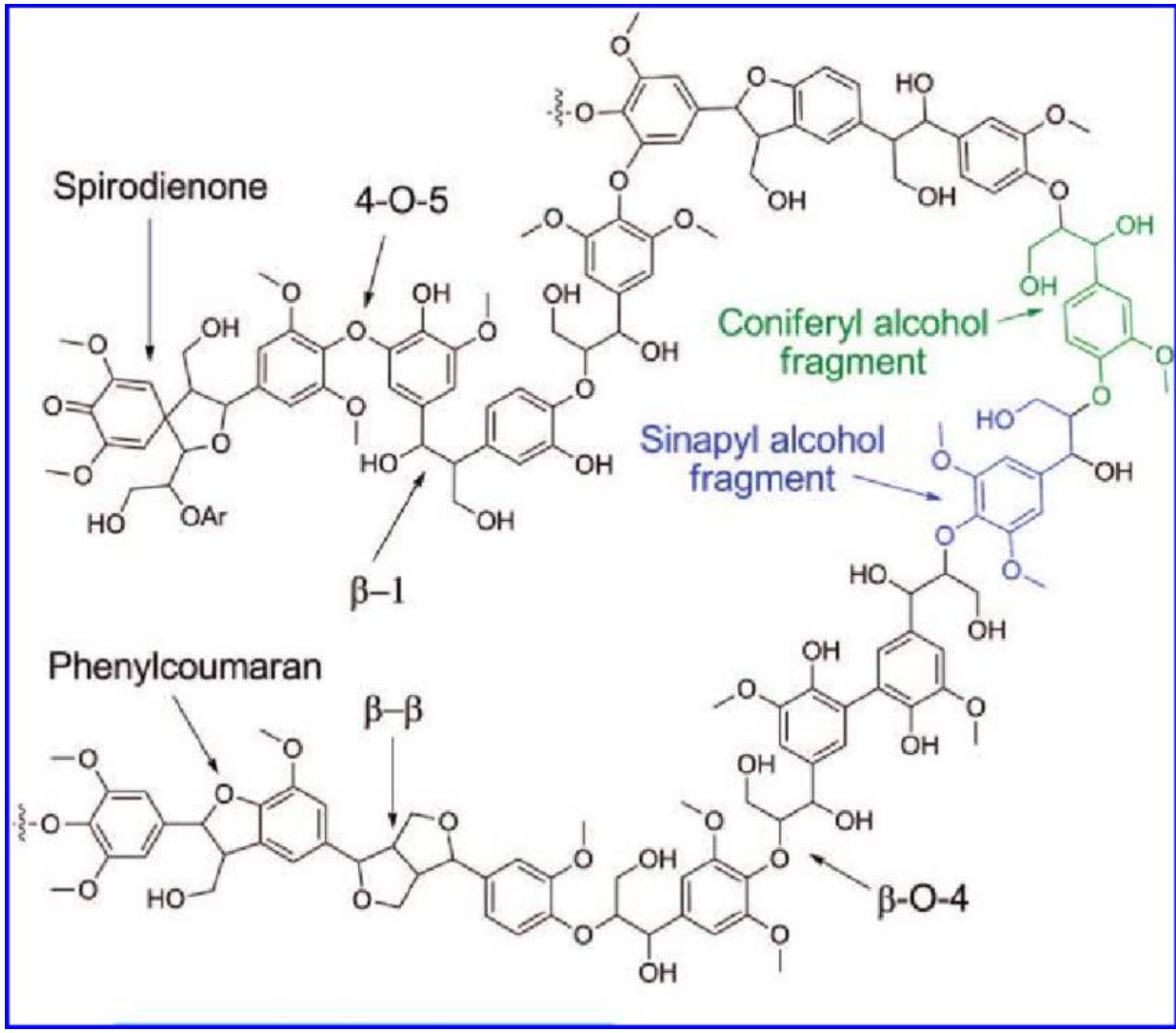
Lignin

Monolignols



- Non-linear polyphenol
- Structurally extremely diverse
- Glues fibres together as the main component in the middle lamella
- Hydrophobic: controls the amount of water inside the cell wall
- Responsible for the brown colour of the wood

Hardwood lignin structure



Applied Microbiology and Biotechnology. 2020, 104, 5201–5212

Step 1: pulping — isolating the fibers

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Wood chips or plant fibre

Raw lignocellulosic material, straight from harvest

→

Washing, drying & grinding

Removes surface debris and reduces material to a workable size

→

Delignification

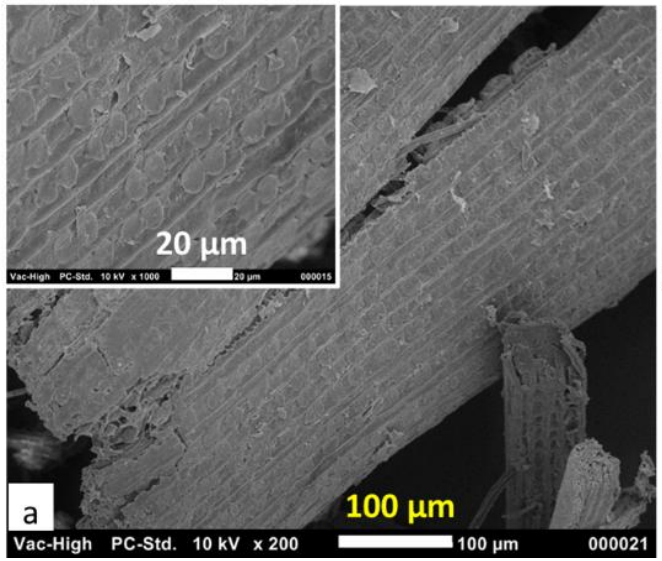
Chemically removes lignin, the rigid “glue” between fibres

→

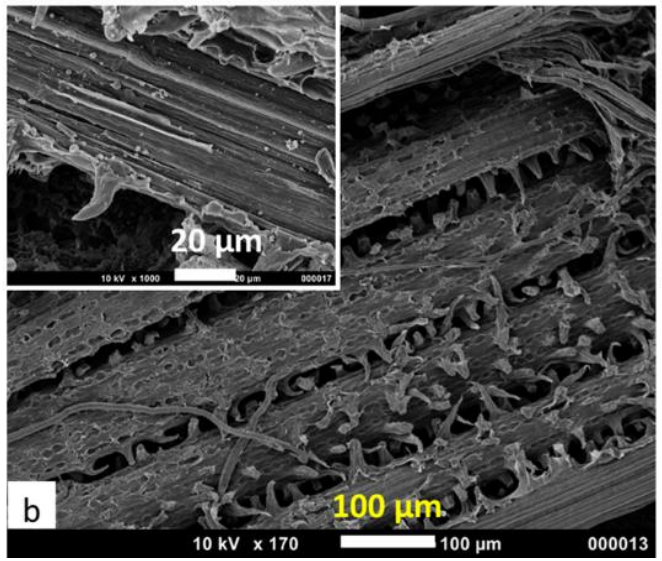
Bleaching

Removes residual lignin and hemicellulose, leaving purified fibre

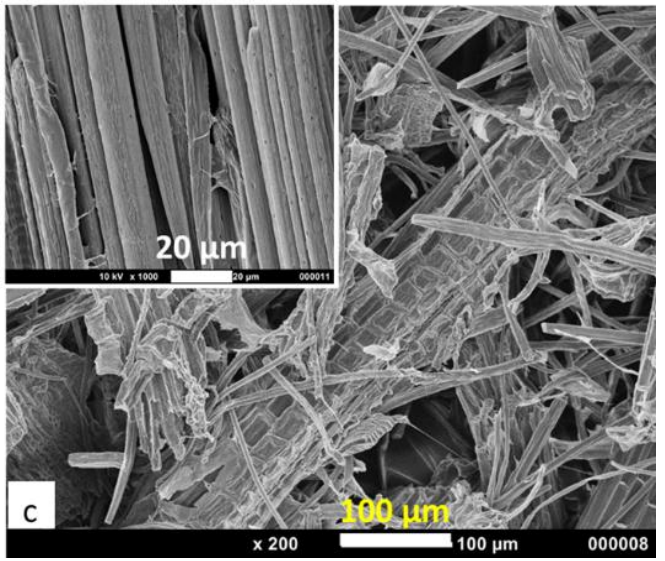
Pulping removes lignin and hemicellulose, leaving purified cellulose fibre ready for nanoscale disintegration.



Water washed grass



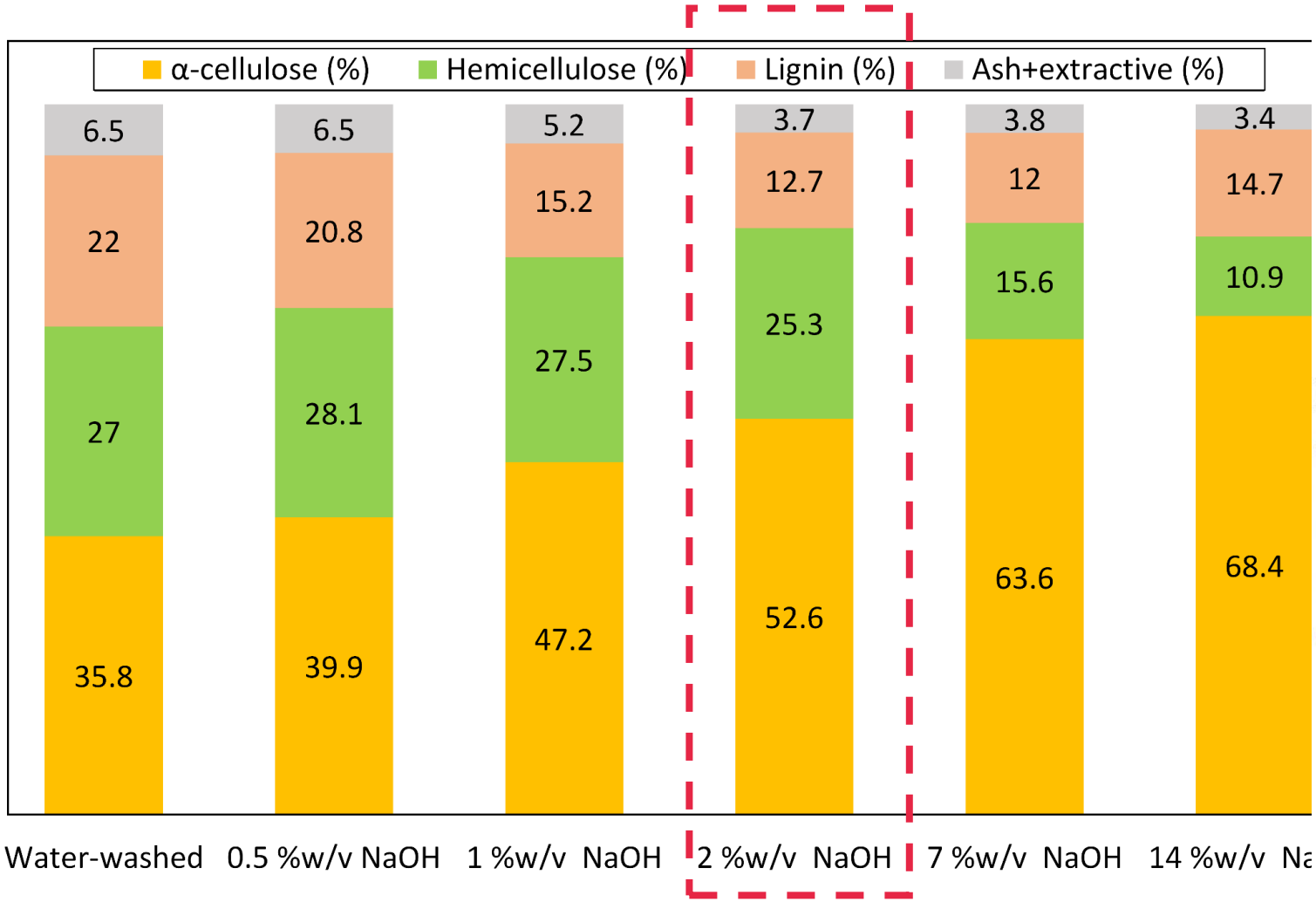
Delignified fibre

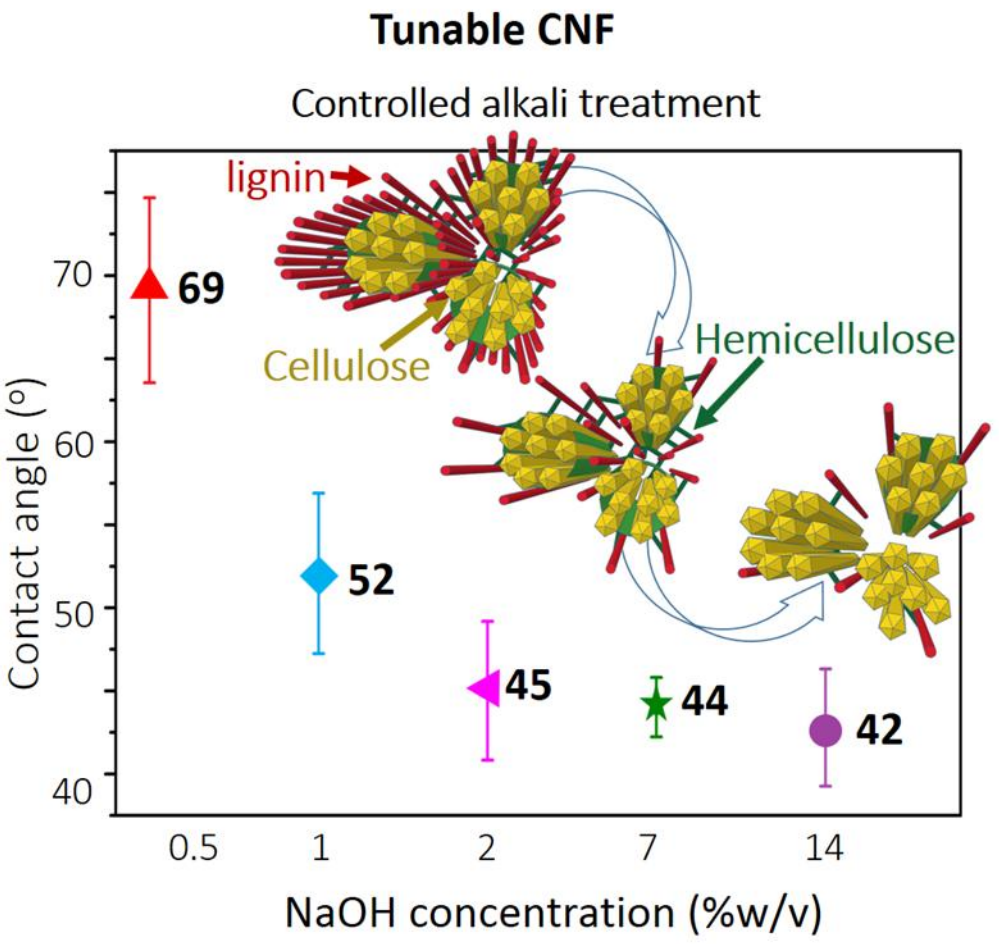


Bleached fibre

RSC Advances, 2015, 5 (41), 32124-32132

Effect of pulping conditions on fibres composition





ACS omega 2019 3 (11), 15933-15942

Making CNF: from pulp to nanofibre

THE UNIVERSITY OF QUEENSLAND
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Purified pulp

From the pulping stage — fibres ready for disintegration

→

Pretreatment

Chemical, mechanical or enzymatic — loosens the fibre structure

→

Mechanical disintegration

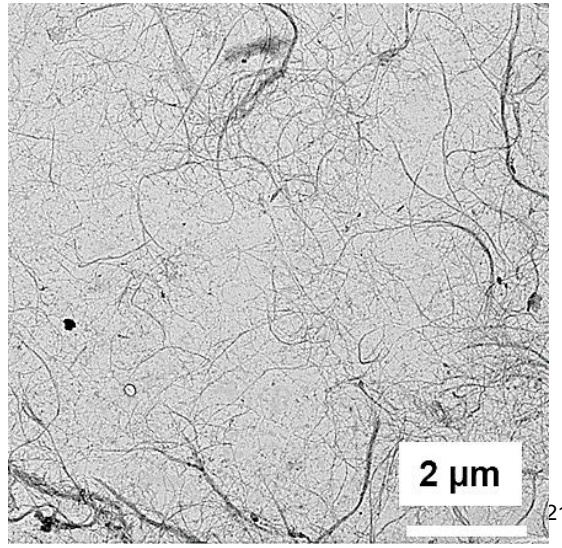
Homogeniser, milling, ultrasonication, Silverson or extrusion

→

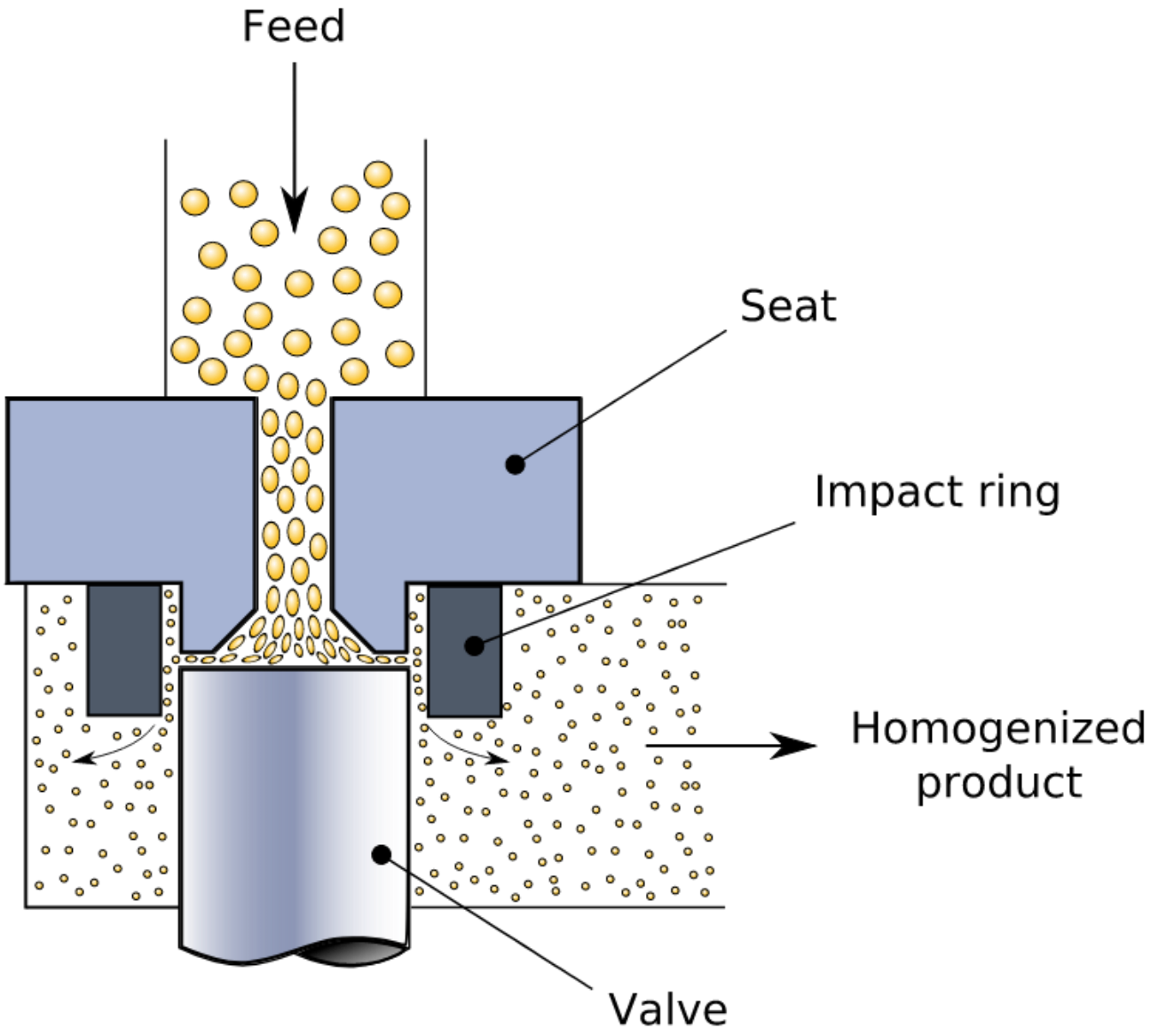
CNF

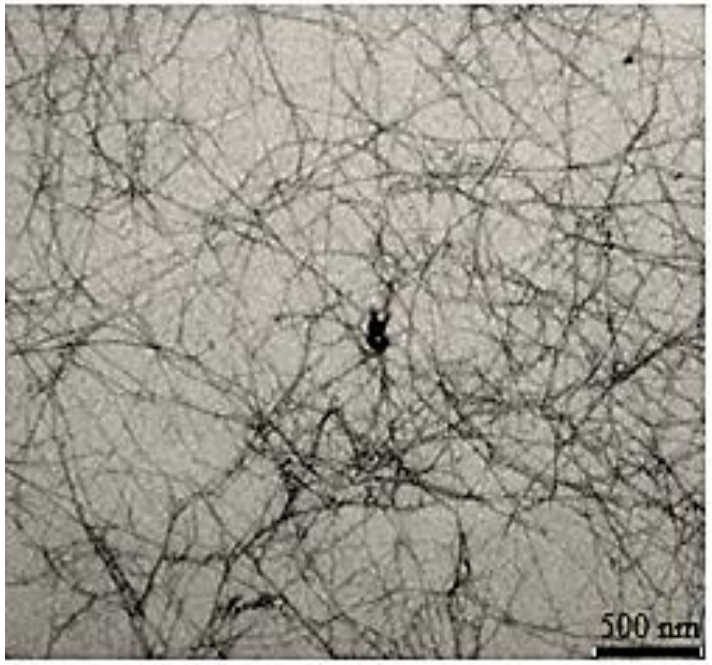
Flexible, high-aspect-ratio cellulose nanofibres

Several mechanical routes exist — they trade off fibre diameter, energy cost and scalability.



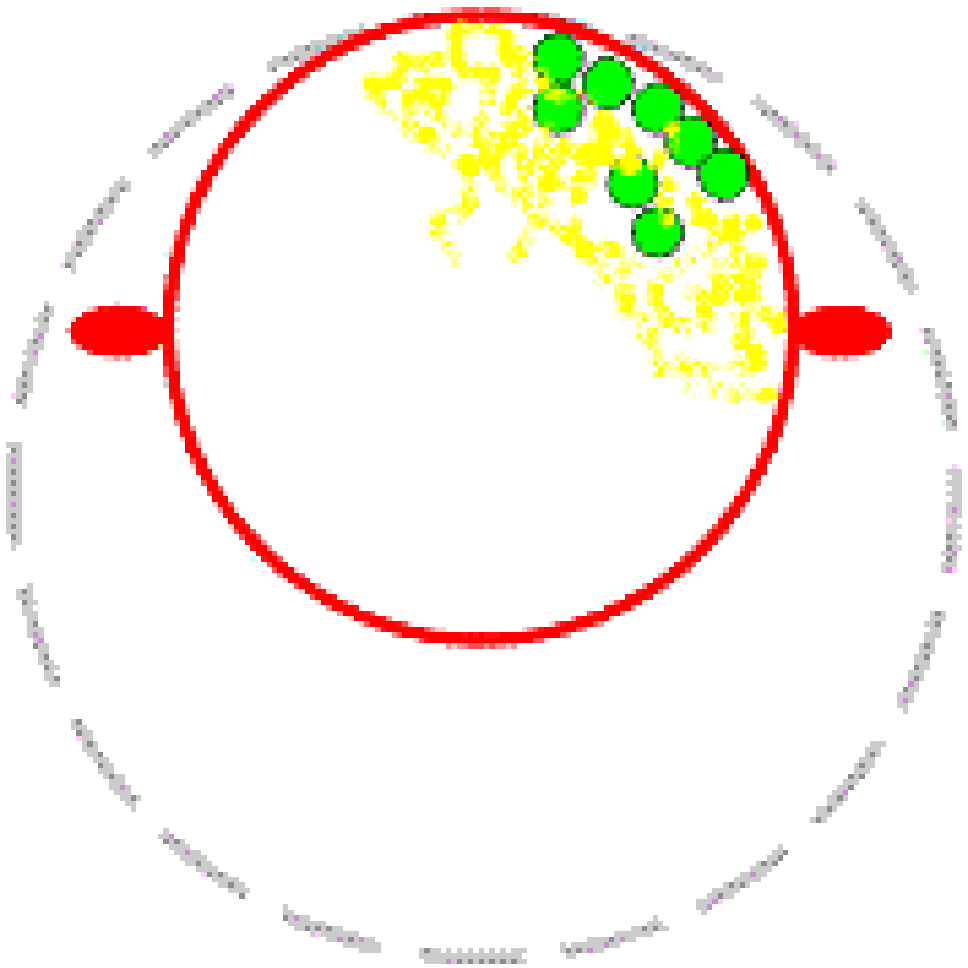
High pressure homogeniser

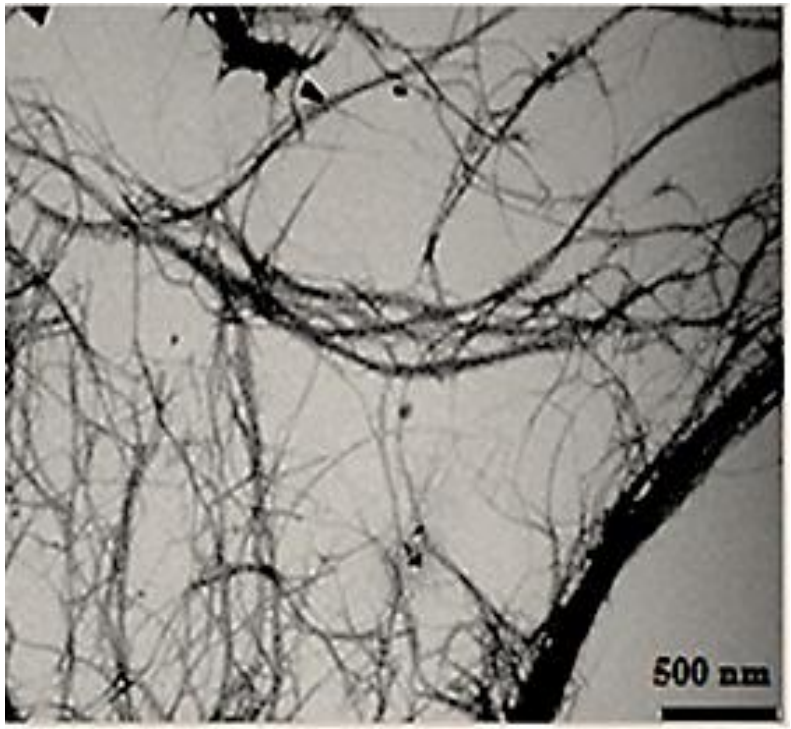




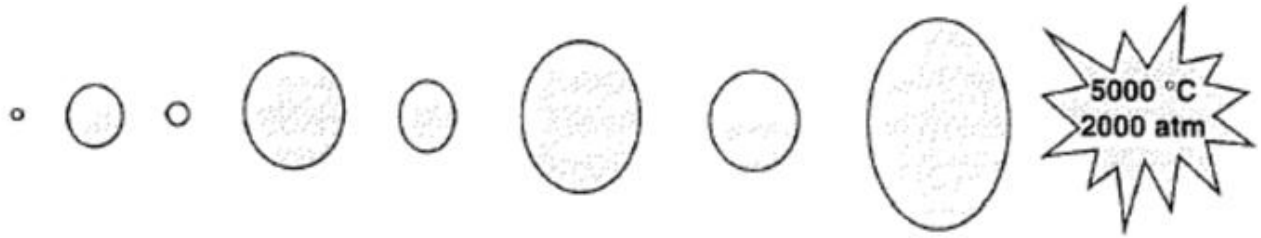
https://www.youtube.com/watch?v=kbyBgLzSSs4

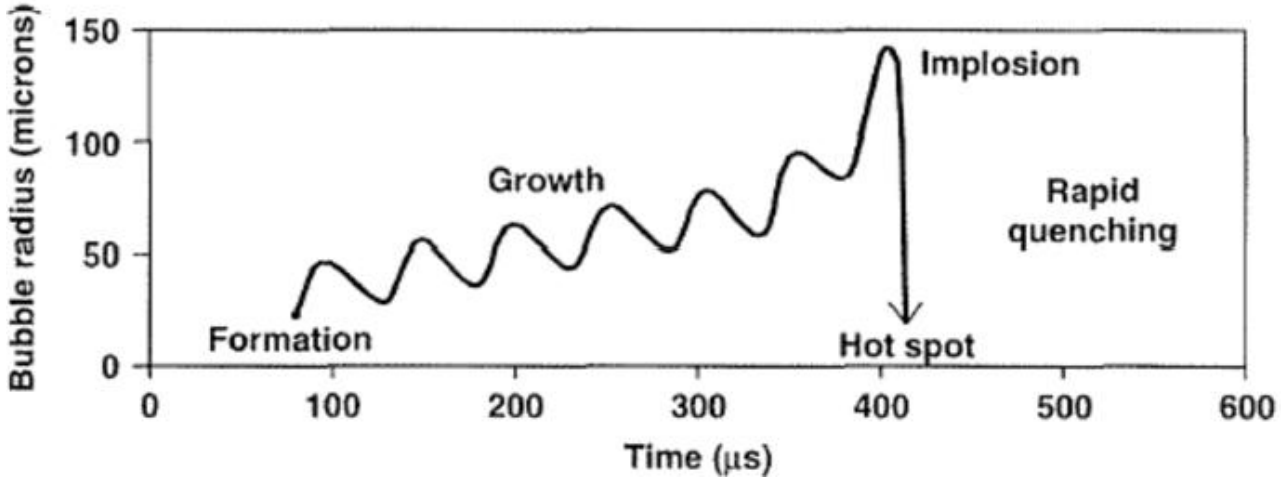
High energy milling

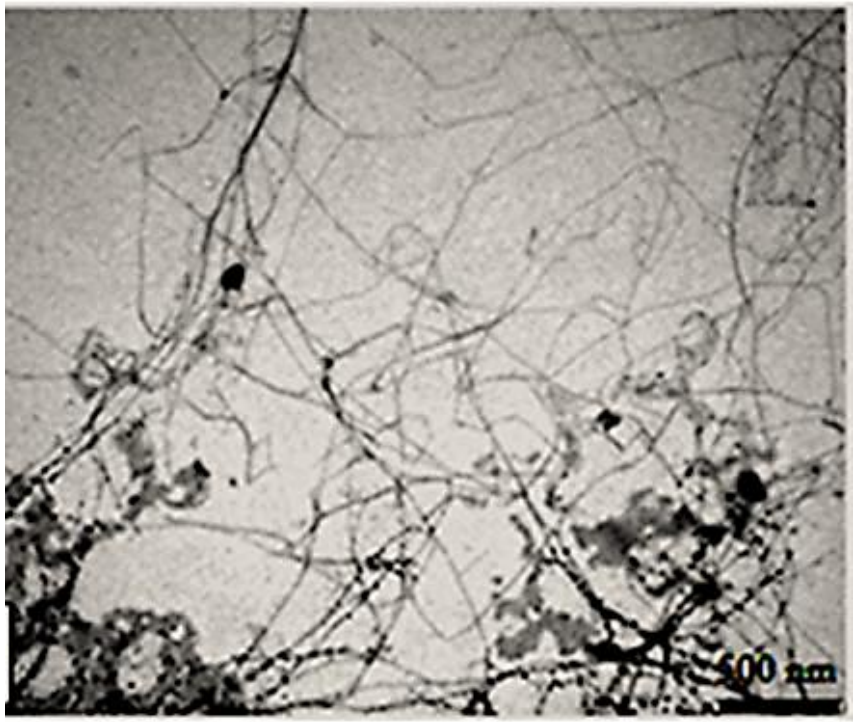


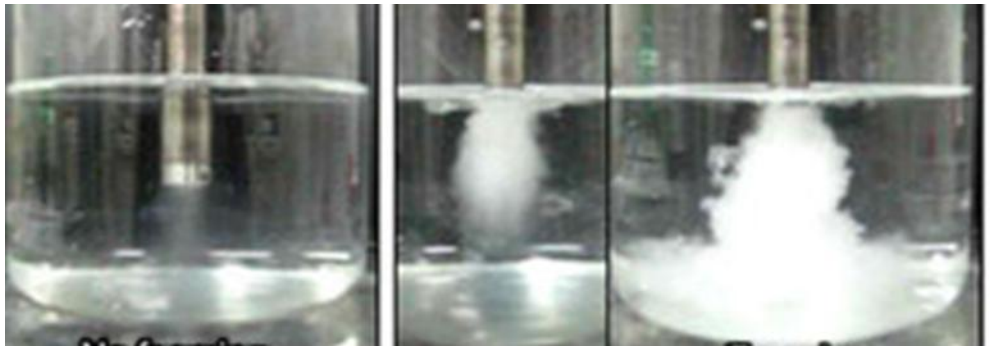


Ultrasonication









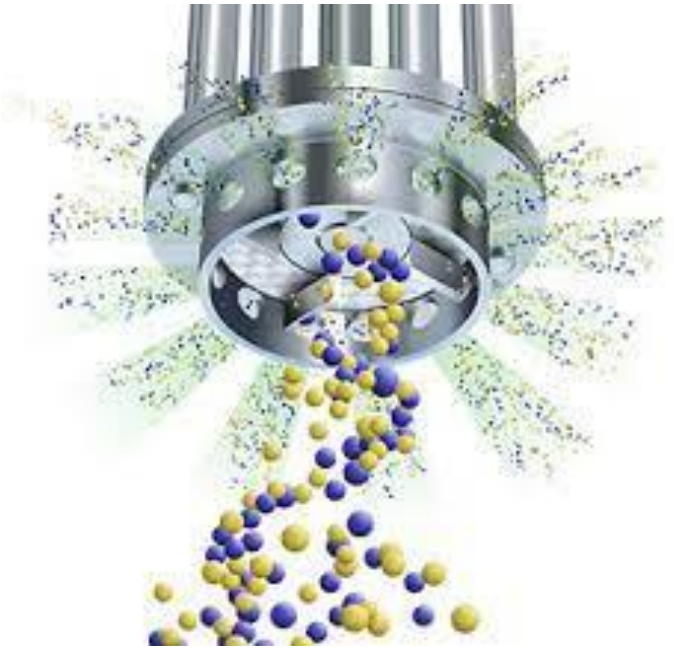
Silverson

Stage 1: The rotor blades inside the special mixing workhead spin really fast and create a strong pull. This pulls liquids and solids from the bottom of the container into the centre of the mixing workhead.

Stage 2: Because of the spinning, the materials move to the edges of the mixing workhead. There, they get milled and crushed between the rotor blades and the wall of the mixing workhead.

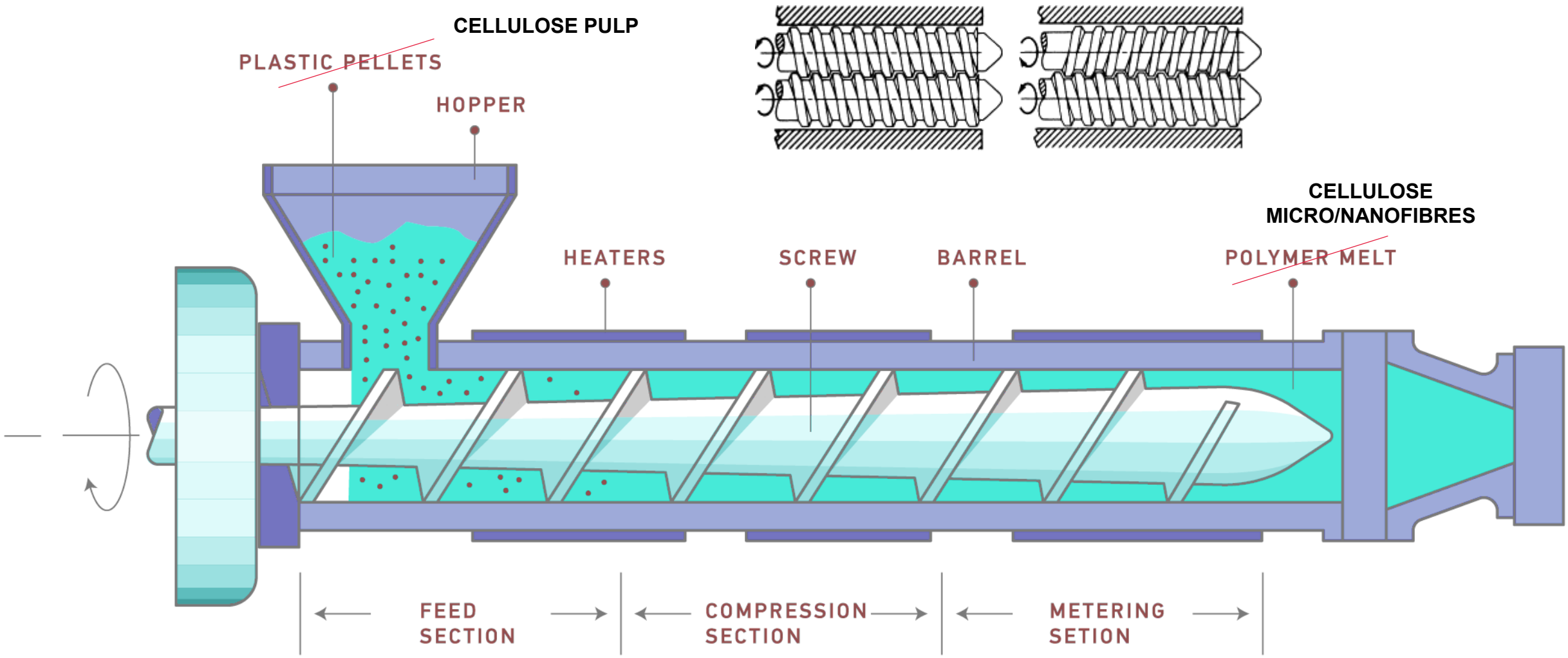
Stage 3: Then, the materials are pushed through tiny holes in the mixing workhead, creating a strong cutting force.

Stage 4: The materials that come out of the mixing workhead move outward quickly and hit the sides of the container. At the same time, more material gets pulled into the mixing workhead, so the process keeps going. This movement pattern helps avoid creating bubbles in the liquid.



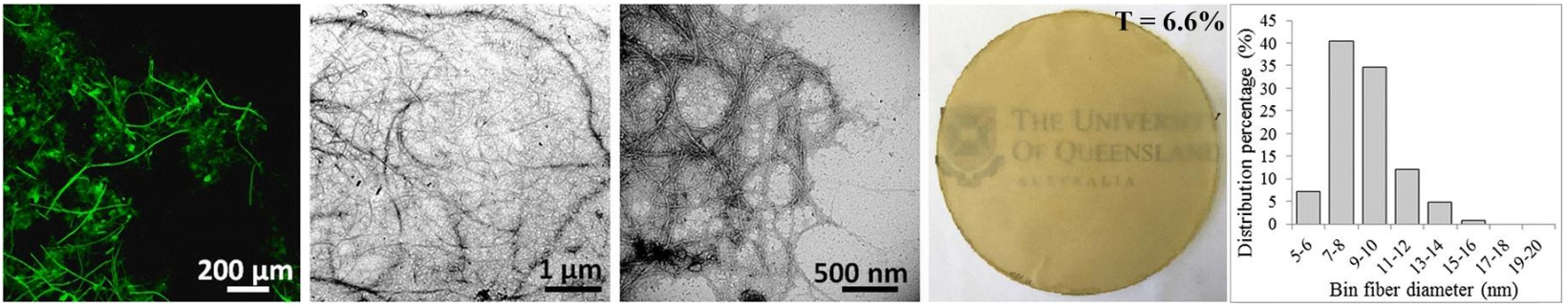
https://www.youtube.com/watch?v=cep81yJlgDk

Extrusion

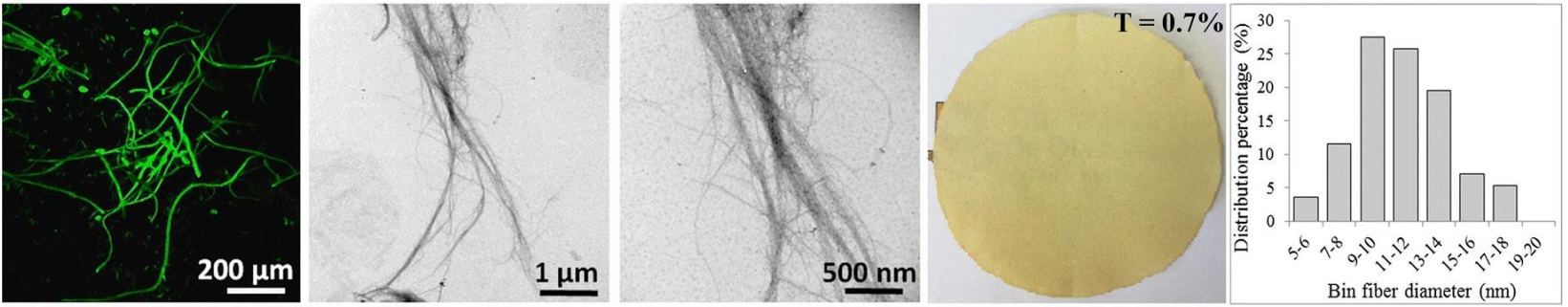


https://www.youtube.com/watch?v=uj-CaGmgsgA

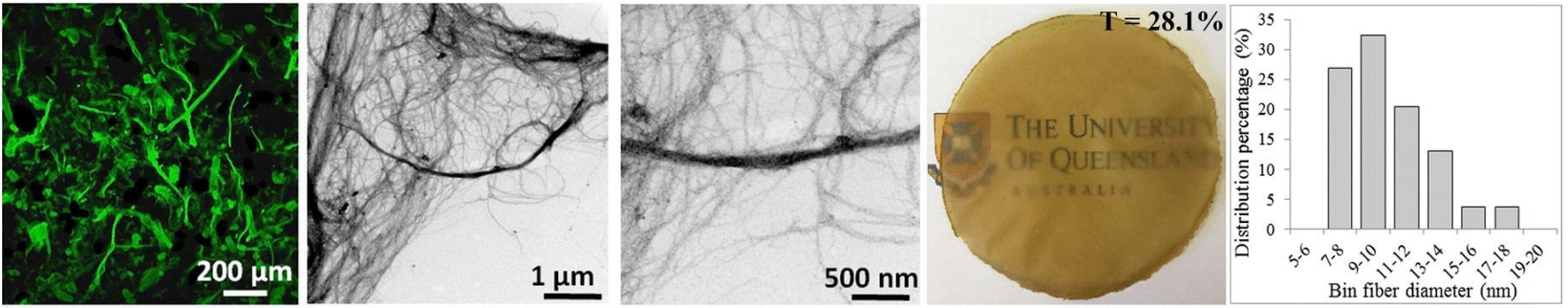
Nanofibres produced using different methods



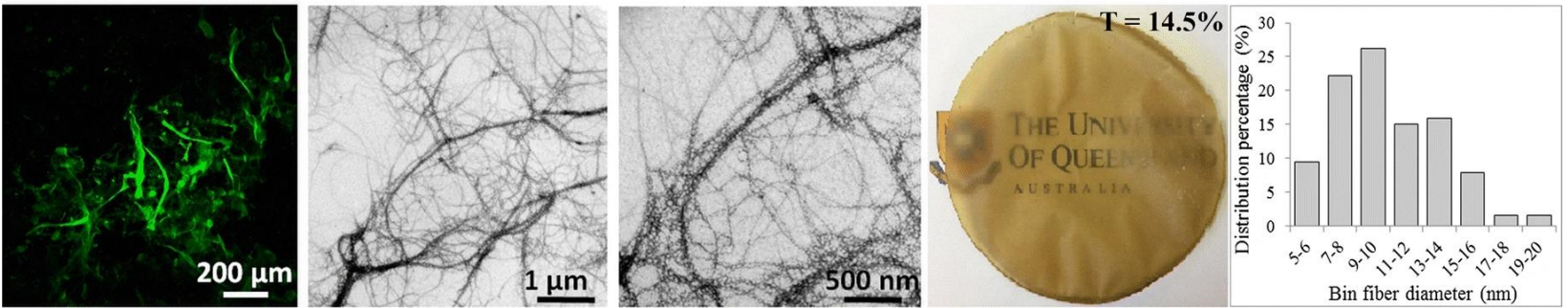
High pressure homogeniser (400,700,3x1100)
Diameter 8 ± 2 nm



Extrusion (3x)
Diameter 11 ± 3 nm



Extrusion (3x) and milling (20 min)
Diameter 10 ± 3 nm



Milling (20 min)
Diameter 10 ± 3 nm

Cellulose, 26, 6555–6569 (2019)

Making CNC: acid hydrolysis

Purified pulp

From the pulping stage — fibres ready for hydrolysis

Acid hydrolysis

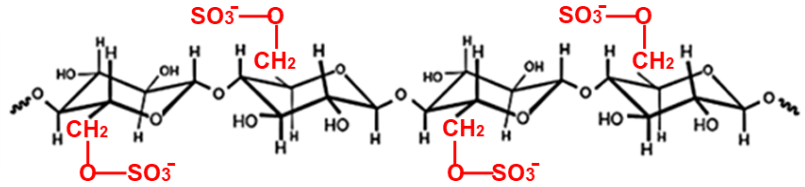
Sulphuric, hydrochloric, phosphoric, or nitric acid + sodium nitrate

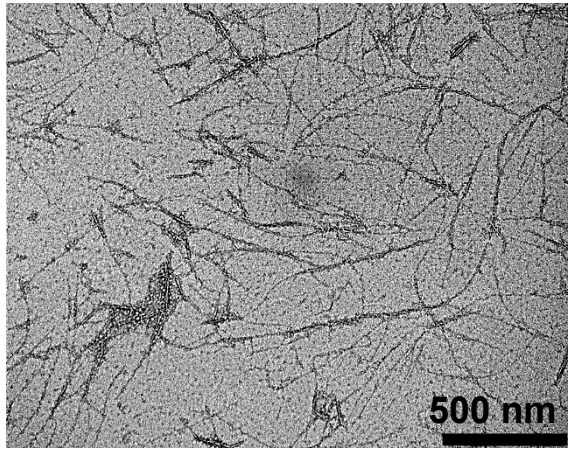
Centrifuge & dialysis

Removes excess acid and reaction by-products

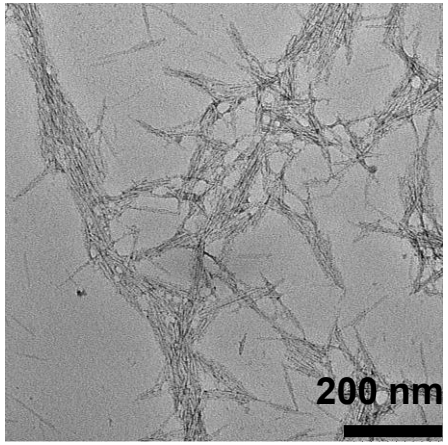
Ultrasonication → CNC

Disperses the crystals into rod-like cellulose nanocrystals

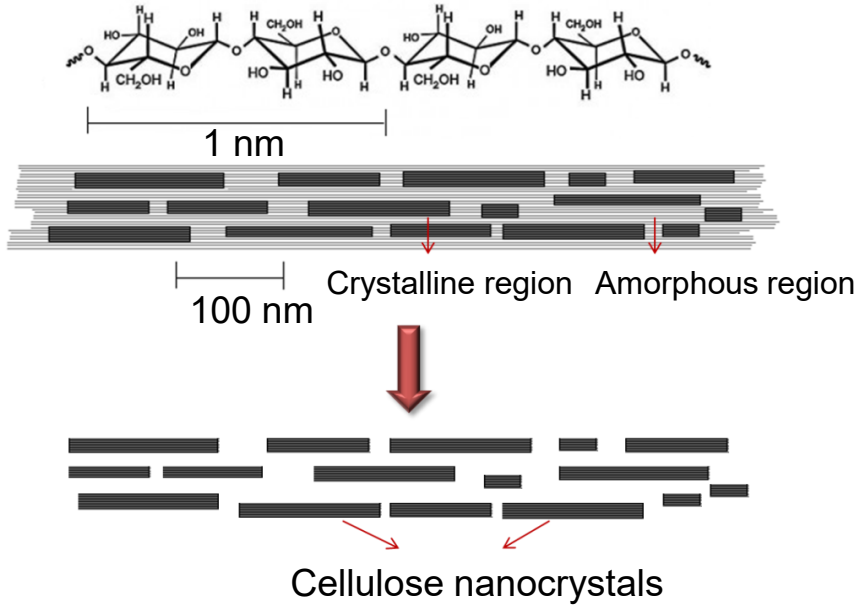




Mild acid treatment of grass



Acid treatment of wood



Cellulose, 2017 24 (9), 3753-3766

Different types of nanocellulose – a comparison

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Cellulose nanofibers

- Gels at low concentrations (0.5-1.5 w%)
- Very high aspect ratio (l/d can be ~1000)
- Flexible, spaghetti-like shape
- Behaviour is not always well defined
- Intensive research from 2007 onwards
- Promising in especially composites applications (reinforcing phase)

Cellulose nanocrystals

- Fluid dispersion at low concentrations
- Not very high aspect ratio
- Rod-like: always straight shape
- Intensive research from 1992 onwards (although a surge after 2005 occurred)
- Promising as filler material, viscosity controller etc.
- Many high-end applications have been proposed

Nanocellulose characterisation

Morphology: Transmission electron microscopy (TEM)
Scanning electron microscopy (SEM)
Dynamic Light Scattering (DLS)

Surface charge: Zeta potential

Surface area: BET

Degree of crystallinity: X-ray diffraction (XRD)

Thermal stability: Thermogravimetric analysis (TGA)
Differential scanning calorimetry (DSC)

Mechanical properties: Bulk nanofibers: Rheology, Instron
Individual nanofiber: AFM

Surface functionality: Fourier transform infrared (FTIR)
X-ray photoelectron spectroscopy (XPS)

Lignocellulose composition

Why nanocellulose is useful for sustainable materials

High aspect ratio

reinforcement and network formation

Hydroxyl-rich surface

functionalisation and hydrogen bonding

Film-forming ability

dense oxygen-barrier layers when dry

Rheology modification

stabilisation of coatings, inks and emulsions

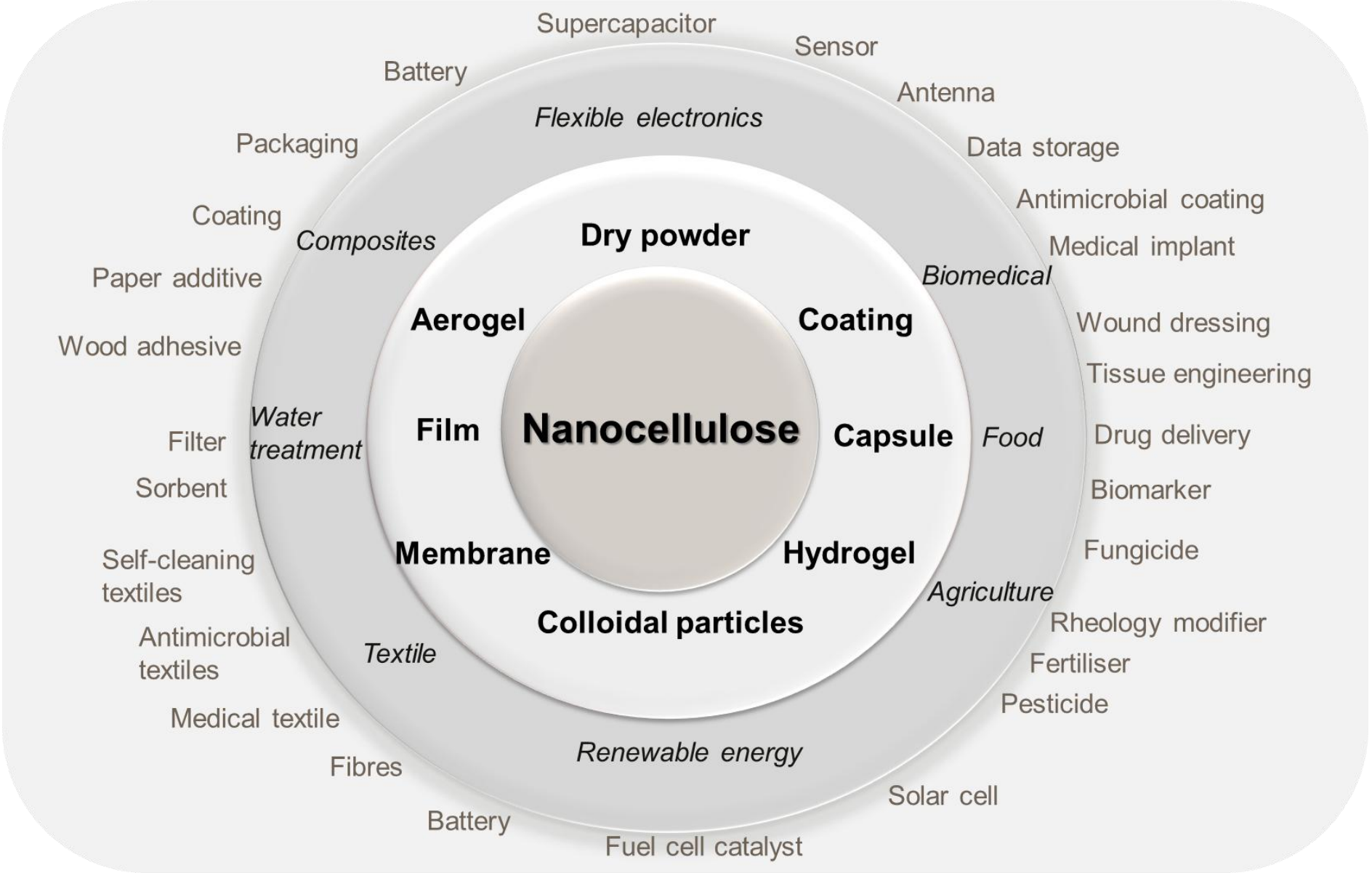
Bio-based origin

renewable carbon input — not automatically sustainable

Surface tunability

antimicrobial, hydrophobic or low-fouling functions

Why industry cares about nanocellulose



Challenges

High surface area and high surface energy lead to agglomeration

Not compatible with hydrophobic polymers and non-polar solvent

Moisture sensitivity

Drying and shipping cost

Most of today's research effort goes into solving exactly these four problems.

Surface modification: the engineering answer

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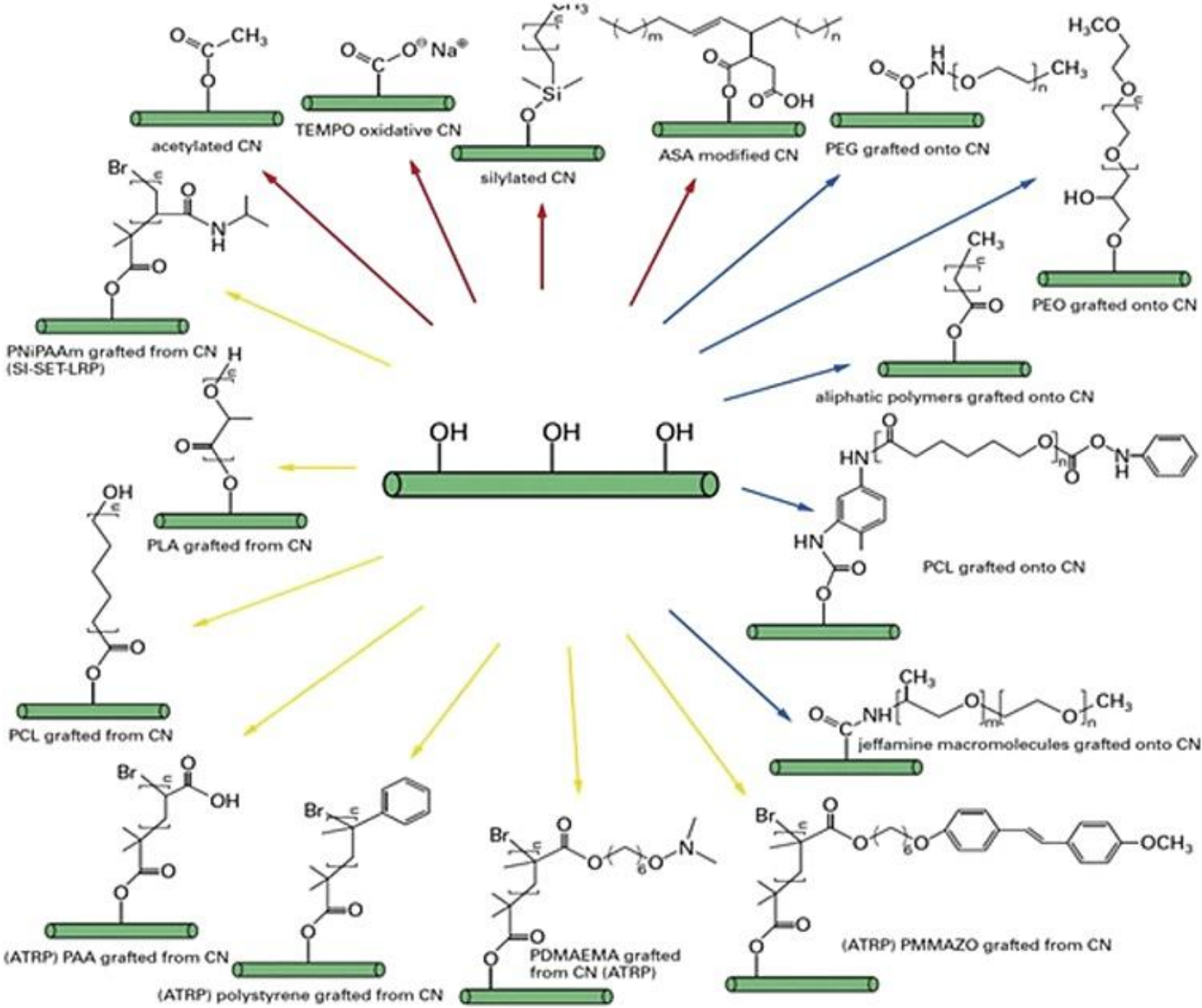
Adsorption methods

Chemical methods

- Silylation
- Acetylation/Esterification
- Carbonylation
- Functionalised CNF reactions

Modification by polymer grafting

Each approach trades off cost, scalability, and how much of the native fibre properties survive.



Biotechnology Reports, 29, 2021, e00582

Nanocellulose-based commercial products

Company / product	Type	Commercial application
CelluForce NCC® / CelluShield™	CNC	Barrier coatings and performance additives for packaging and other formulated products.
Borregaard Exilva®	MFC	Rheology modifier and stabiliser in paints, coatings, adhesives, detergents, and personal care.
FiberLean MFC	MFC	Paper, packaging, tissue, and speciality fibre applications.
UPM GrowDex® / FibDex® / FibGel™	NFC	3D cell culture hydrogel, wound care, and medical-device hydrogel applications.
Nippon Paper Cellenpia®	CNF	Food, cosmetics, rubber, and industrial material applications.

Most nanocellulose products are not sold directly to consumers; they are enabling materials used to improve function, sustainability, and processability.

Existing commercial applications of nanocellulose

Adult diaper
(CNF, Japan)

Viscosity control in
oil drilling
(CNC, Canada)

Ink dispersing in
ball point pen
(CNF, Japan)

CelluShield™-
barrier coating

Multi-tonne pilot and production facilities are already running across North America, Japan and Europe.

Case study: from spiky grass to nanocellulose

From material extraction to community-led biomass valorisation

Years of knowledge – the world's longest-running R&D program

Country and knowledge systems

- Aboriginal and Torres Strait Islander peoples have developed and refined knowledge of Australia's unique flora, fauna, and materials for over 100,000+ years.
- This knowledge is not 'folklore' – it is a sophisticated empirical system developed through rigorous observation, intergenerational transmission, and adaptive management.
- Plant-derived materials, natural antimicrobials (e.g. tea tree oil, eucalyptus), and fibre processing techniques are part of this knowledge base.
- This knowledge has high relevance to biomanufacturing – but engaging with it ethically is non-negotiable.

Why this matters for biomanufacturing

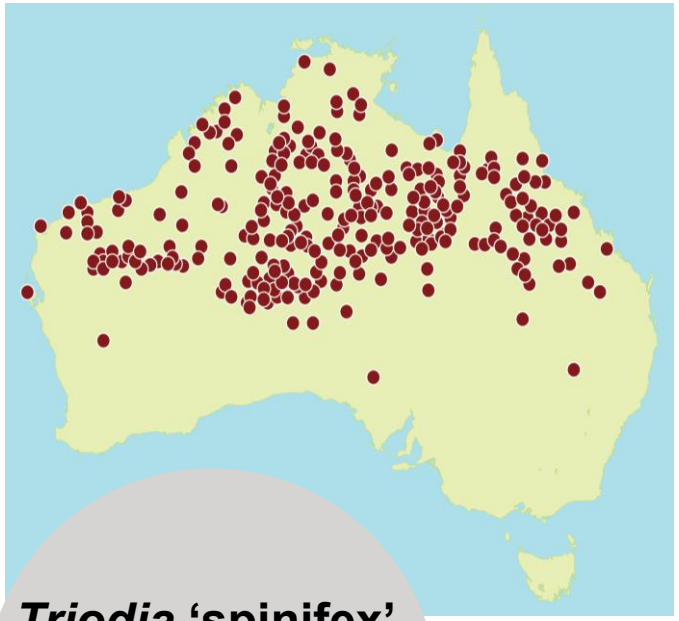
- Traditional knowledge can identify promising feedstocks and applications faster than systematic screening.
- Community partnerships enable access to land and materials under ethical frameworks – not extraction.
- Indigenous communities are increasingly asserting sovereignty over biological resources and associated knowledge.
- A biomanufacturing researcher who does not understand this context will make costly ethical and legal errors.

Principles of ethical research collaboration

Five principles are drawn from participatory action research frameworks used in Australian Indigenous health and environmental research, and from the Nagoya Protocol on access and benefit-sharing:

Community-led	Research questions emerge from community priorities, not only from lab priorities. The community defines what is useful, not just what is scientifically interesting.
Two-way knowledge	Both knowledge systems – Western scientific and Indigenous – contribute, and both are respected. Results are shared back with the community in accessible, plain-language formats.
Free, Prior & Informed Consent	No material collection, no knowledge documentation, no publication without explicit consent from the appropriate knowledge holders. This is an ongoing relationship, not a one-time agreement.
Benefit sharing	Economic and non-economic benefits flow back to the community – capacity building, employment, IP co-ownership where relevant, community-identified outcomes.
Material Transfer Agreements	Formal legal agreements govern what plant or biological material can be taken, for what purpose, and with what restrictions on further use or commercialisation.

From spiky grass…



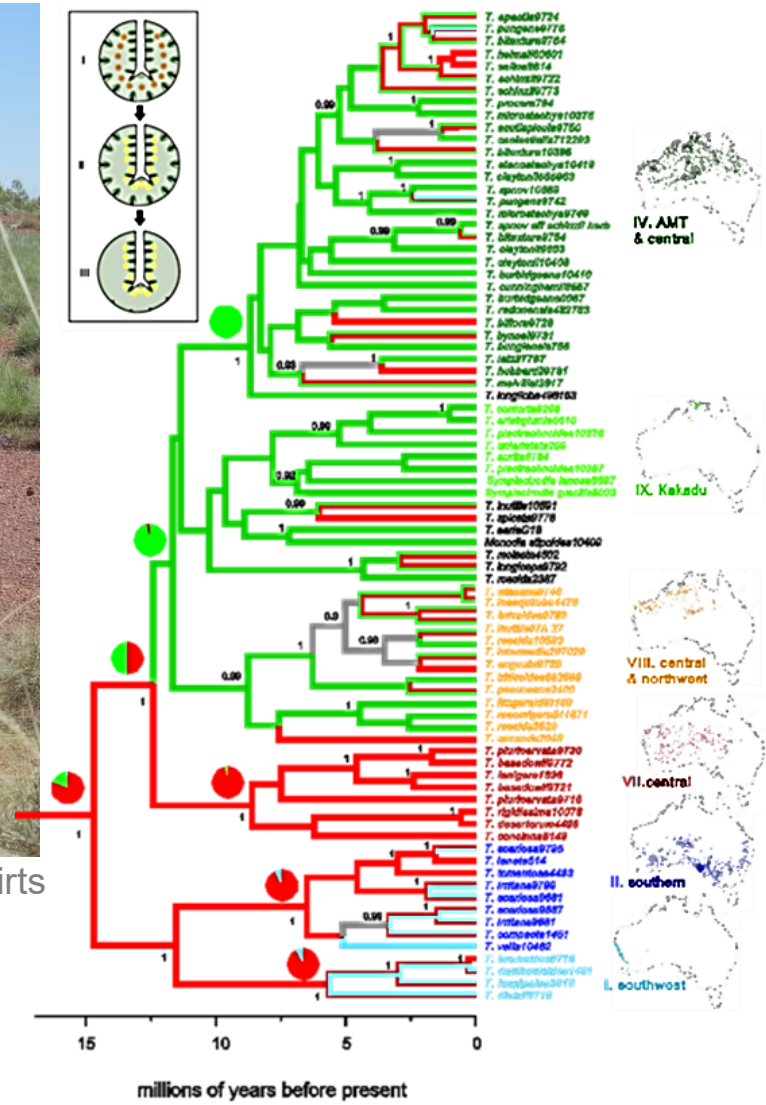
***Triodia* 'spinifex'**

Covers >2 mln km^2

~ 69 species

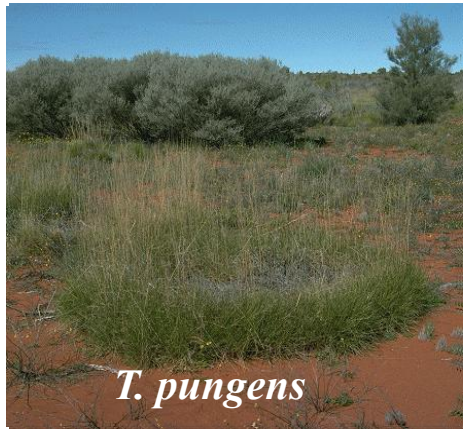
Spinifex harvesting in the outskirts of Camooweal, QLD

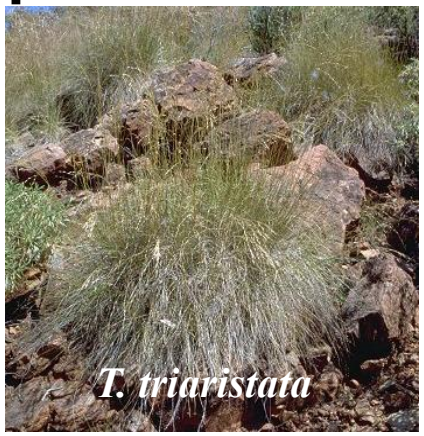
Extremophile, abundant in remote, arid zones of Australia

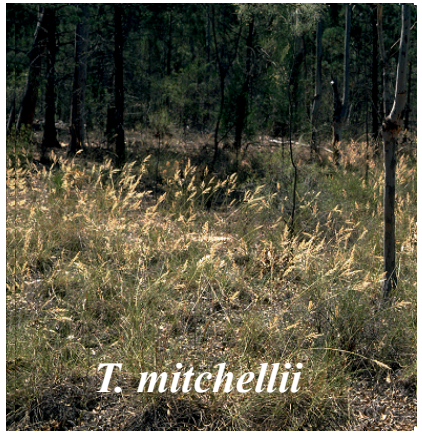


Some species of spinifex

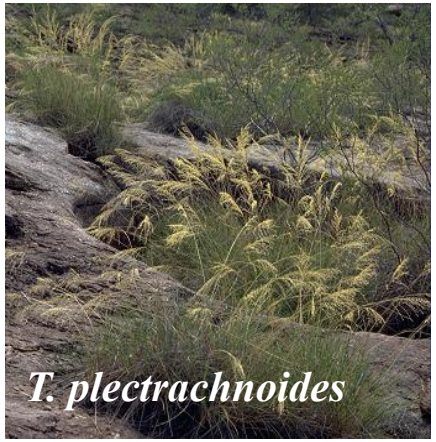
Soft spinifex

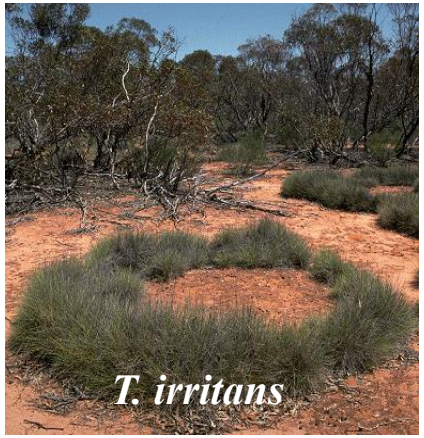


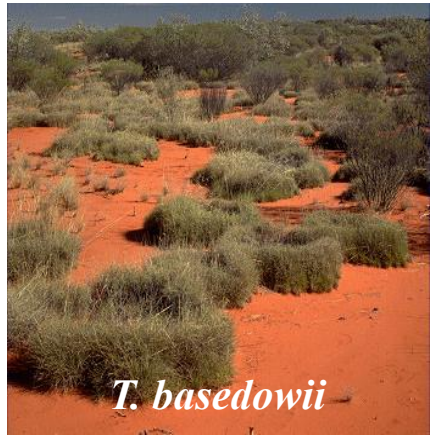




Hard spinifex

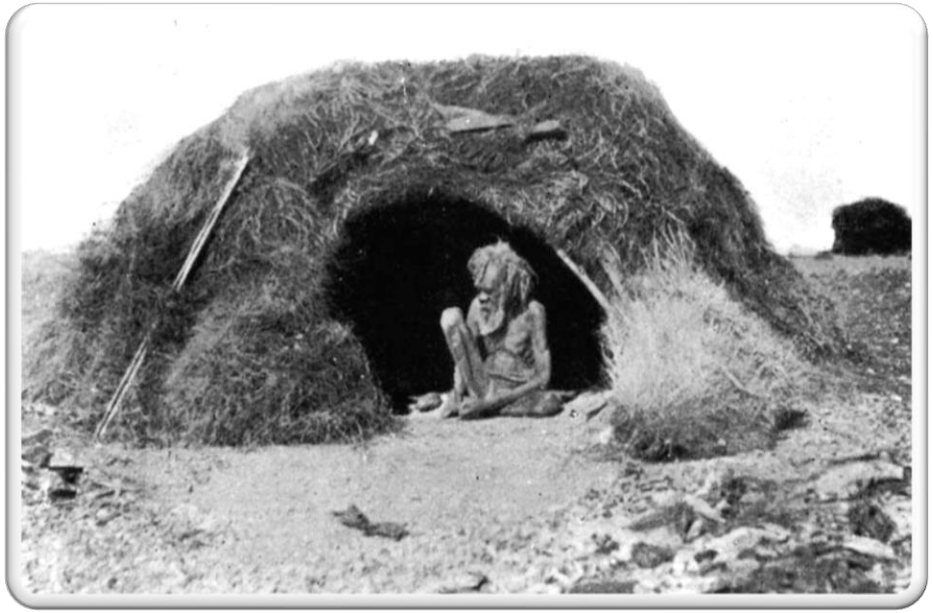




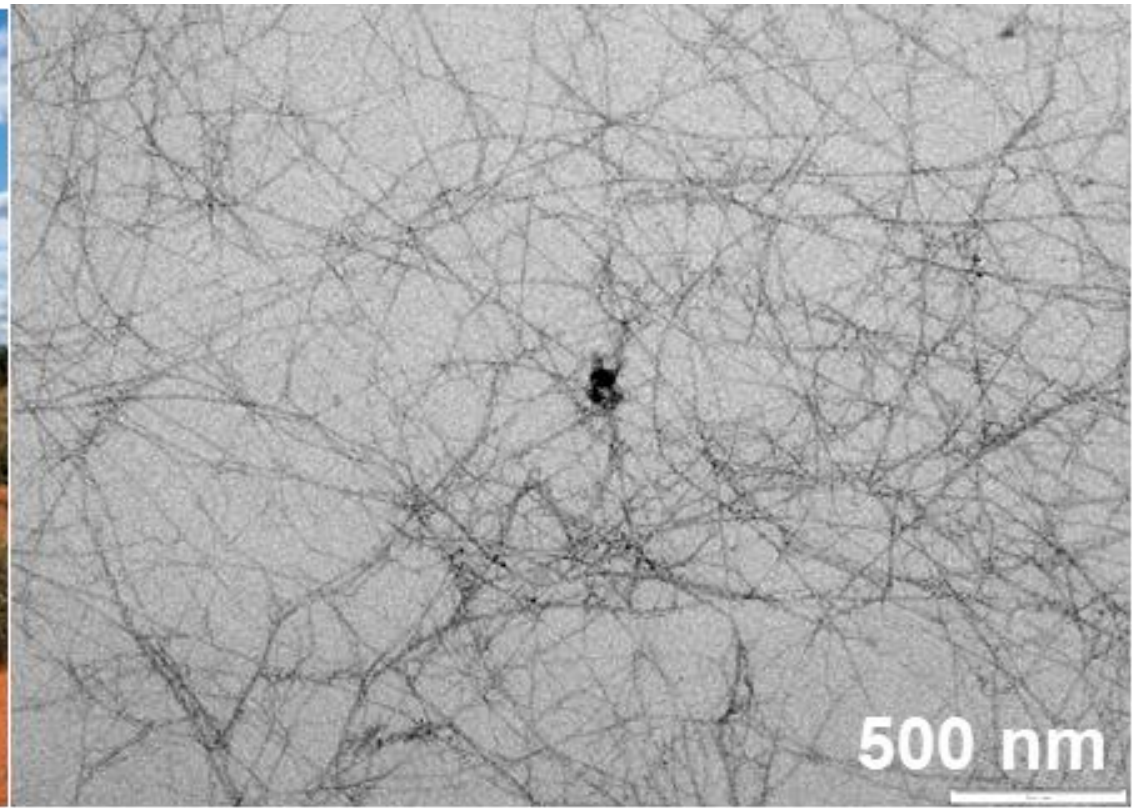


Photos: Aus Grass CD

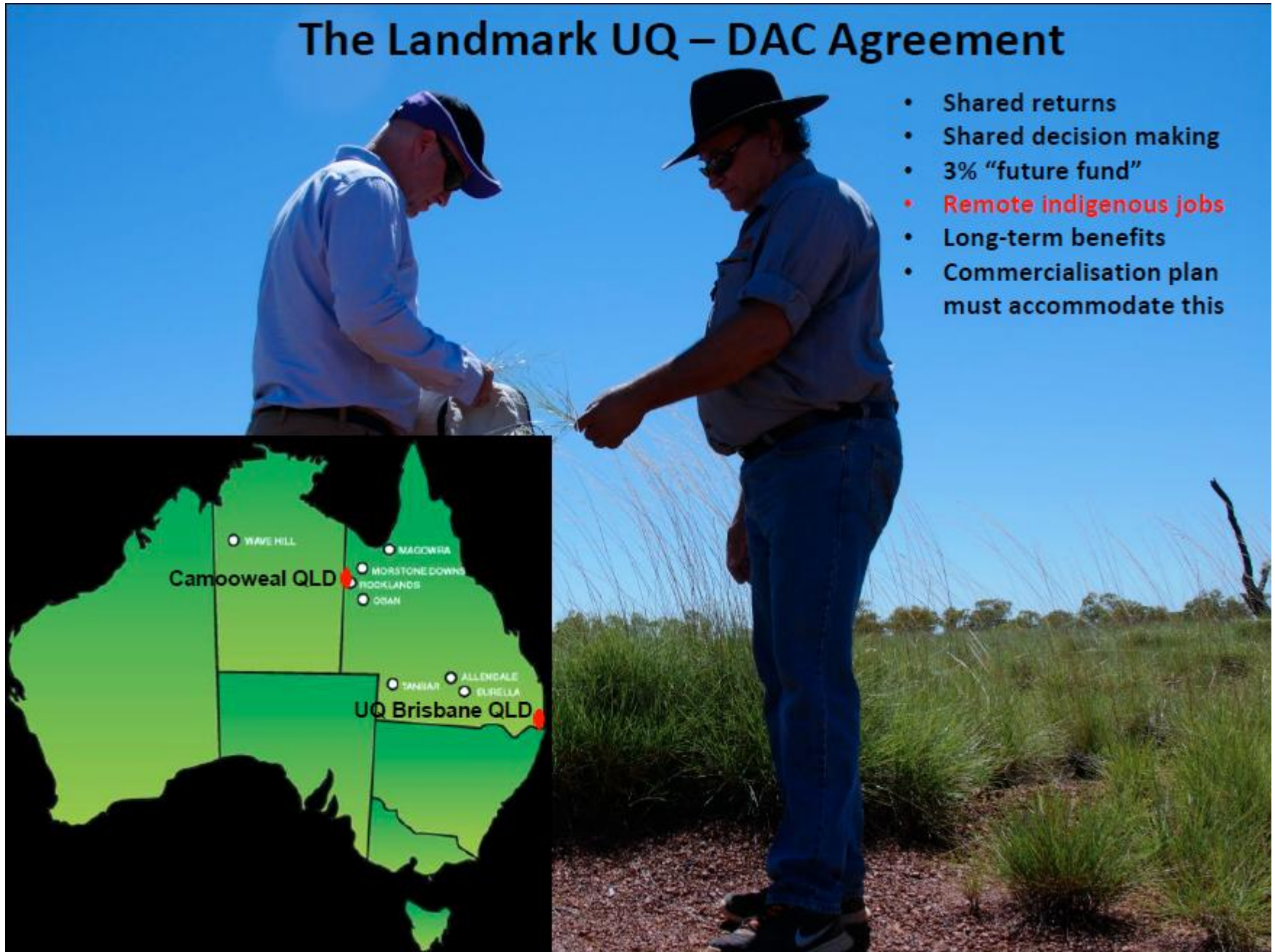
Traditional applications of spinifex



Spinifex cellulose nanofibers



Spinifex nanotechnology



Spinifex nanocellulose technology

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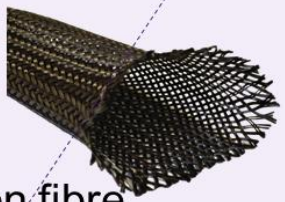
- ✓ Innovate solutions to solve society's problems
- ✓ Sustainable materials
- ✓ Translational progress

Australia's first Nanocellulose pilot plant facility

Cement and concrete

Water filtration

Bitumen



Carbon fibre

Dry compou

ard

Triodia Wilingi

AU Patent WO2015074120 A1
AU Patent WO2016090425

What changed because it was a partnership?

The partnership influenced not only who participated, but how the technology was designed, governed and translated.

Research design

The research questions, materials focus and intended applications were shaped through partnership rather than defined only by the laboratory.

Governance

Access to biological resources, knowledge use, consent and decision-making required explicit agreement rather than being treated as informal background issues.

Capability building

The work was linked to skills, participation and enterprise opportunities, not only publication outputs.

Value creation

The pathway was considered in terms of who benefits, how value is shared and whether local manufacturing or community outcomes are possible.

Long-term accountability

Success was judged not only by the invention itself, but by whether relationships, responsibilities and benefits could be sustained over time.

Partnership changed the translation pathway, not just the source of the biomass.

Shift the model: from extractive research to co-designed value creation

Extractive model

Researcher defines problem

Community provides material or access

Benefits captured mainly by institution

Limited long-term accountability

Co-designed model

Community defines priorities with researchers

Governance, IP and benefit are agreed early

Capability and local value are considered

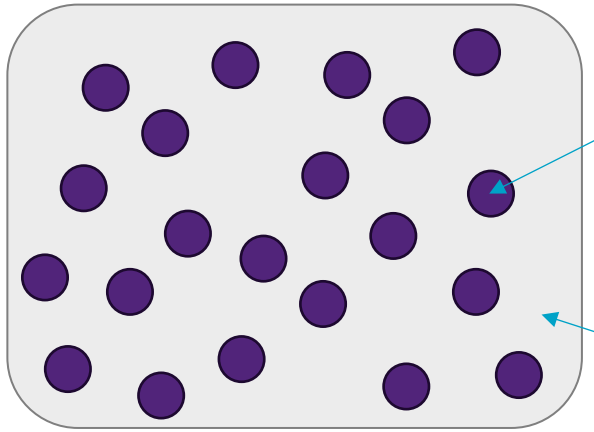
Long-term accountability is explicit

Case study: from spiky grass to nanocellulose

Reinforcing natural rubber latex for medical gloves

Nanocomposites from nanocellulose

Schematic cross section
of a nanocomposite



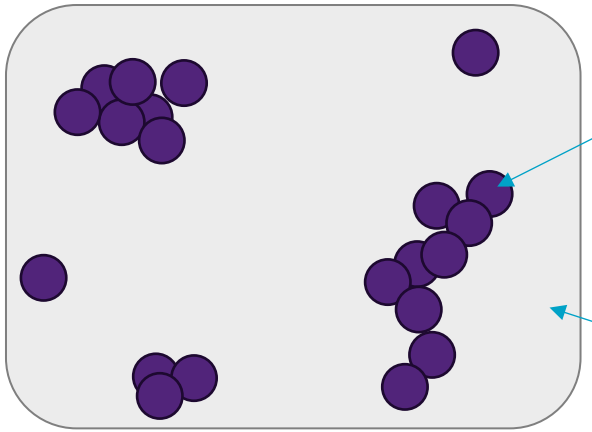
Reinforcing phase: Nanofibril or nanocrystal cross section (cellulose)

Continuous polymer matrix
(preferably from a biodegradable polymer)

WHY NANOCOMPOSITES?
Nanocellulose has a very large surface area, that is, many contact points with the continuous polymer matrix, plus it is very strong.

Nanocomposites from nanocellulose

Schematic cross section
of a nanocomposite



Reinforcing phase: Nanofibril or nanocrystal cross section (cellulose)

Continuous polymer matrix
(preferably from a biodegradable polymer)

WHAT IS THE PROBLEM?
Cellulose aggregates easily with itself. It is incompatible with nearly anything other than the lignin/hemicellulose matrix in the plant cell wall.
→ Loss in surface area → No nanocomposite anymore

Nanocomposite

Natural Rubber latex membrane for glove applications

Market	Revenue	Growth (CAGR)
Medical gloves	$100 M	9.2%

Rubber product with good tensile strength and toughness, but very high flexibility

↓ Desired properties

Product that is resistant to breakage but also allows for a close fit

Solution → Reinforce latex with fillers

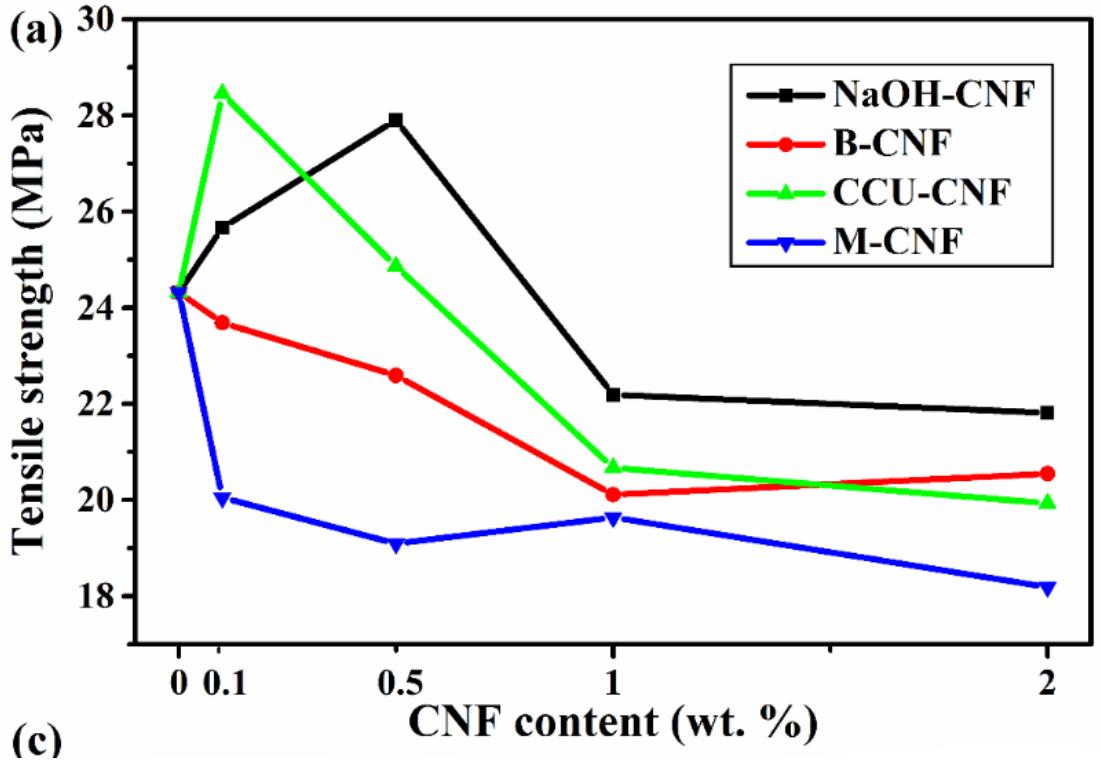
↓ Results

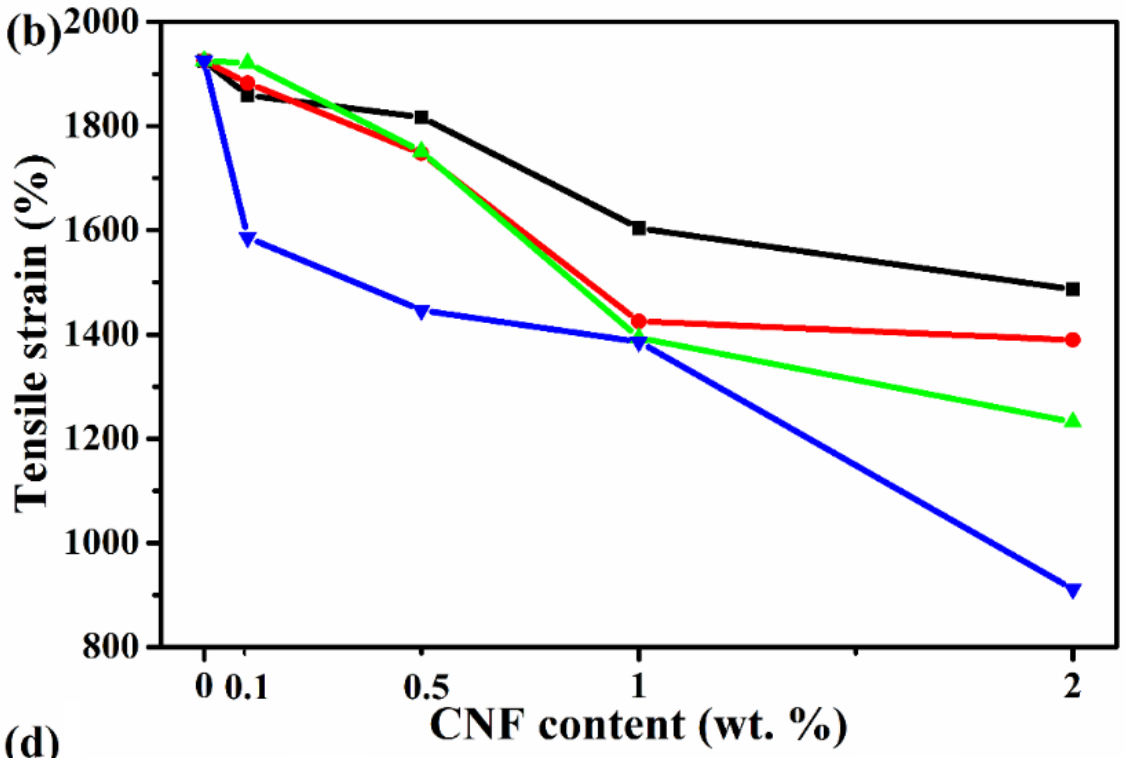
- Increase in tensile strength
- Undesirable increase in stiffness
- Reduction in elongation at break

A solution to this is using SOFT + TOUGH nanofillers = THIN + TACTILE membranes

Polymer reinforcement

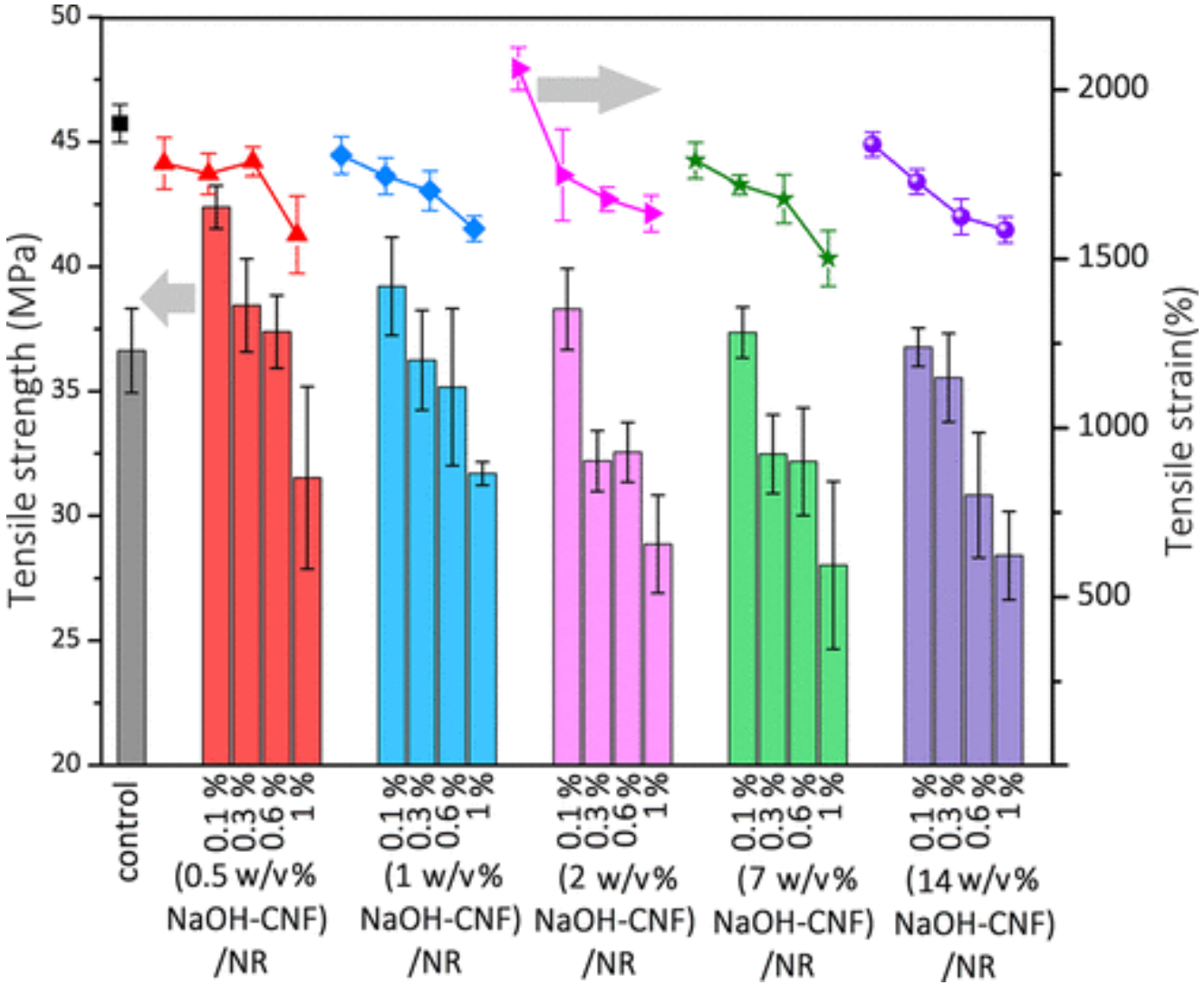
Sample name	NaOH-CNF	B-CNF	CCU-CNF	M-CNF	Control sample NR
Contact angle (θ)	45° ± 4.6°	30° ± 3.5°	62.7° ± 5.1°	8.6° ± 1.4°	82.7° ± 3.1°

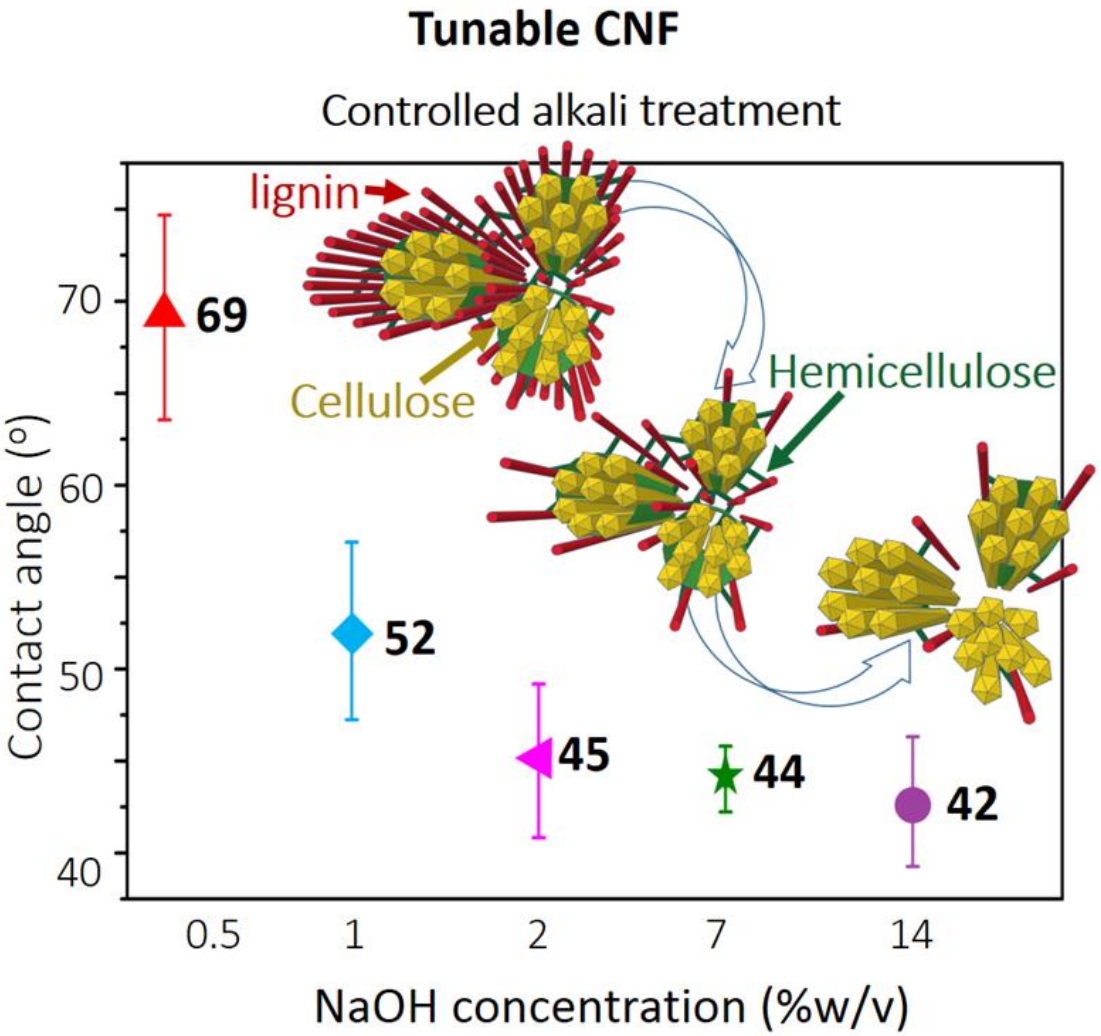




Nanoscale, 2017, 9, 9510–9519

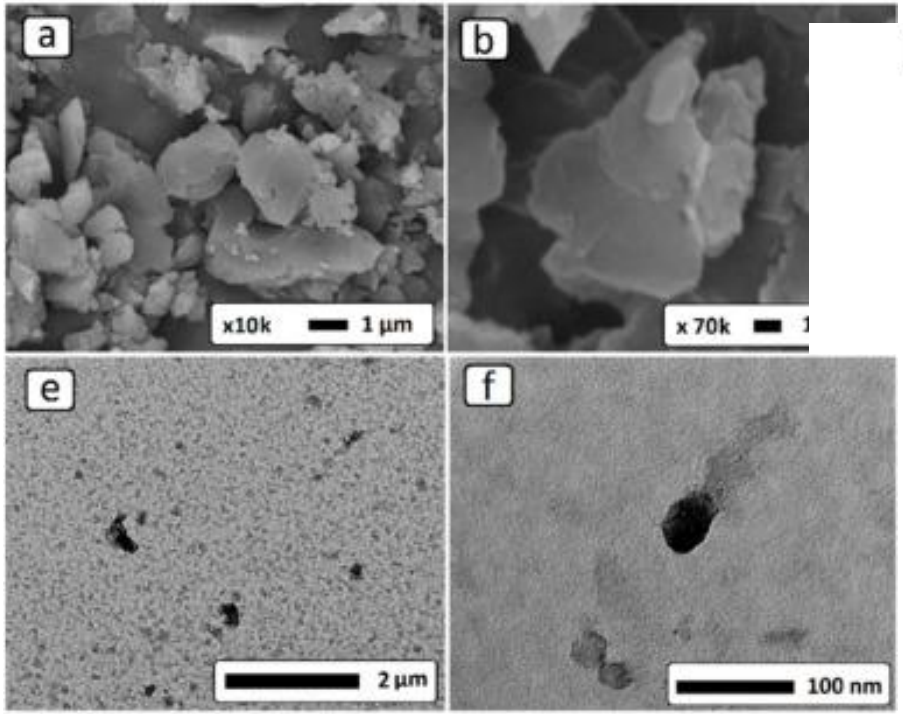
Polymer reinforcement

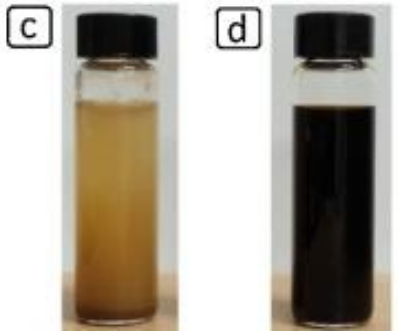


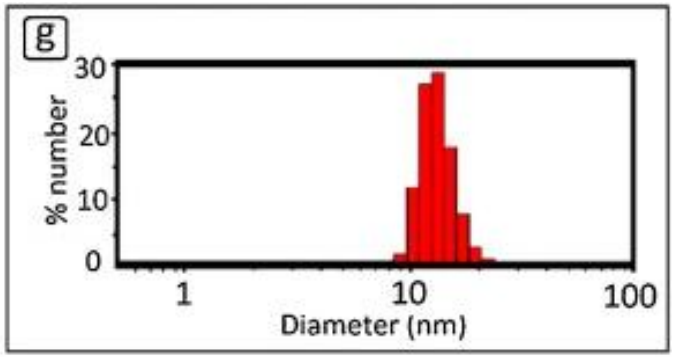


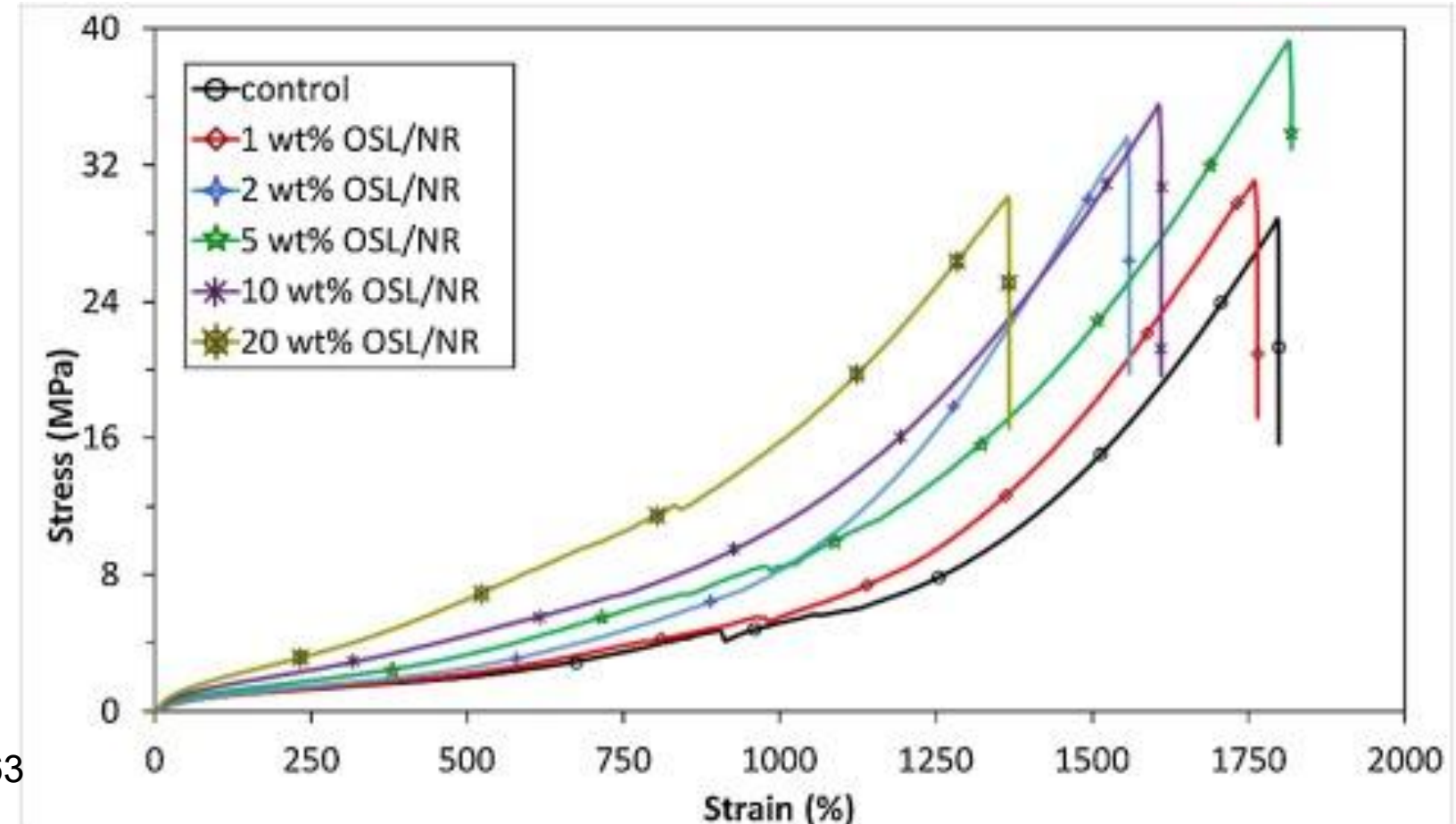
ACS omega 2019 3 (11), 15933-15942

Polymer reinforcement









Industrial Crops and Products, 2021 159, 113063

Case study: a self-disinfecting cellulose coating

Antimicrobial and low-fouling materials

Antimicrobial and low-fouling materials

Surface access

active groups must be exposed at the interface

Killing mechanism

contact-killing, ion release, ROS, low-fouling

Durability

adhesion, abrasion, leaching, cleaning cycles

Safety

cytotoxicity, biocompatibility, migration, regulation

Antimicrobial resistance why new materials matter

The AMR crisis

- ~1.3 million deaths directly attributed to AMR infections in 2019 (Lancet 2022) — projected to exceed cancer mortality by 2050.
- Conventional antibiotics: slow pipeline, rapid resistance development.
- Materials-based solutions: antimicrobial surfaces and coatings can kill or inhibit bacteria without relying on antibiotics that drive resistance.
- Key targets: hospital-acquired infections, food packaging (extending shelf life without preservatives), wound dressings.

Why nanocellulose?

- Surface OH groups → easy, tuneable chemistry to attach antimicrobial agents.
- High surface area → more contact surface per gram than conventional materials.
- Biocompatible → safe for wound contact, food contact.
- Biodegradable → avoids the long-term persistence problem of plastic antimicrobial films.
- Can be formed into films, hydrogels, aerogels, coatings - multiple delivery formats.

Our platform solutions

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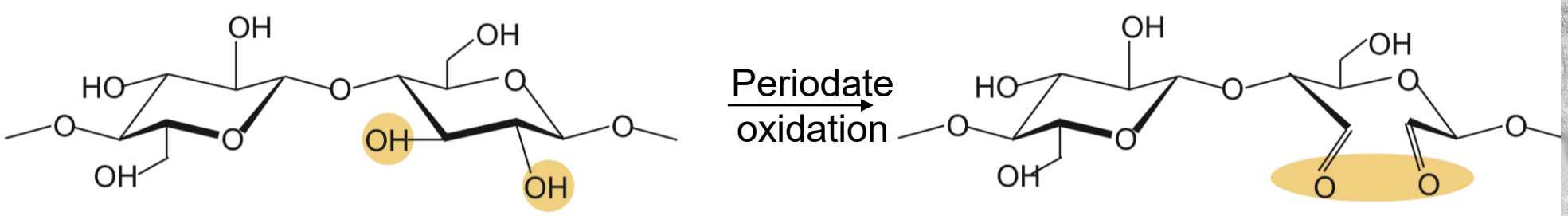
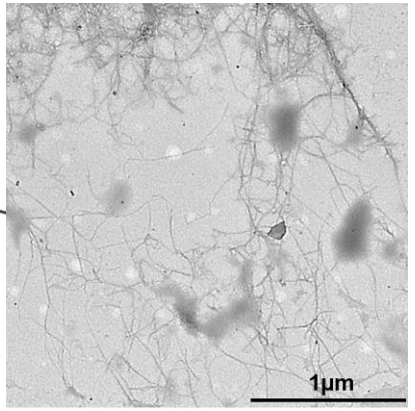
Alteration of cellulose backbone to antimicrobial functionalities

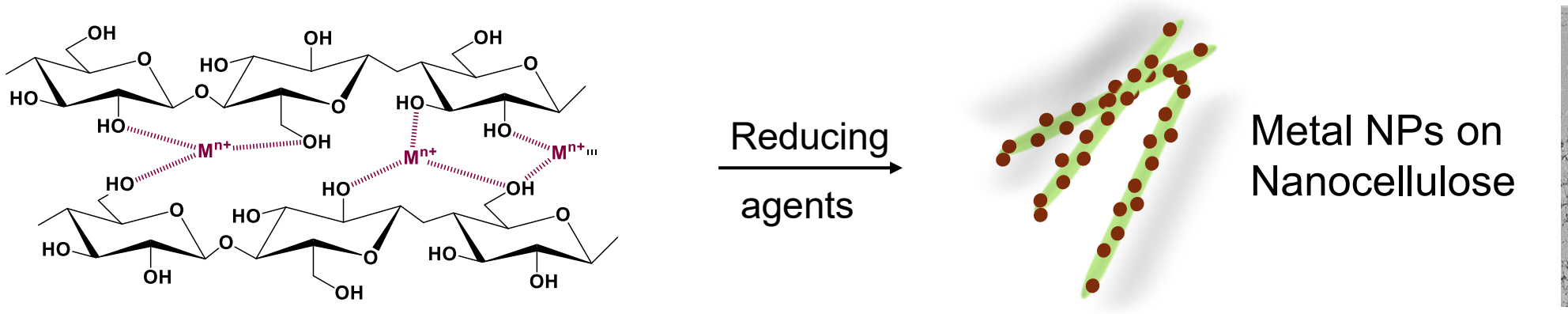


Cellulose

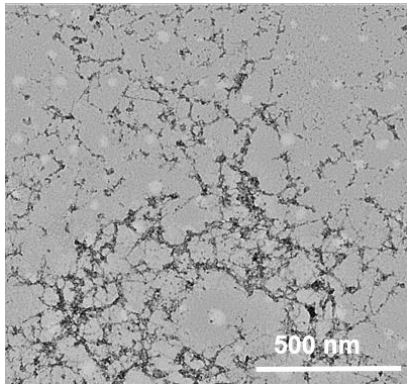
Dialdehyde cellulose



Metal nanoparticles templating on nanocellulose surface



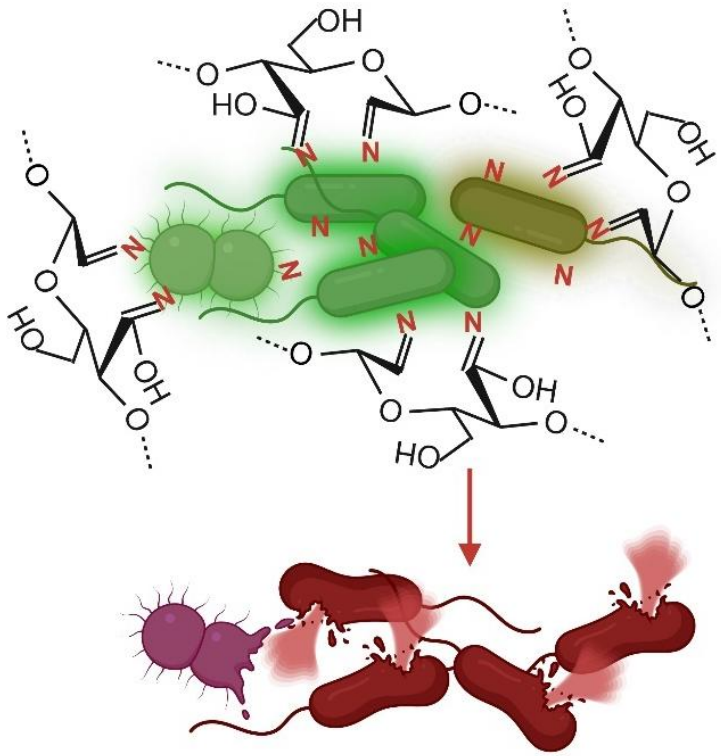
Metal NPs on Nanocellulose



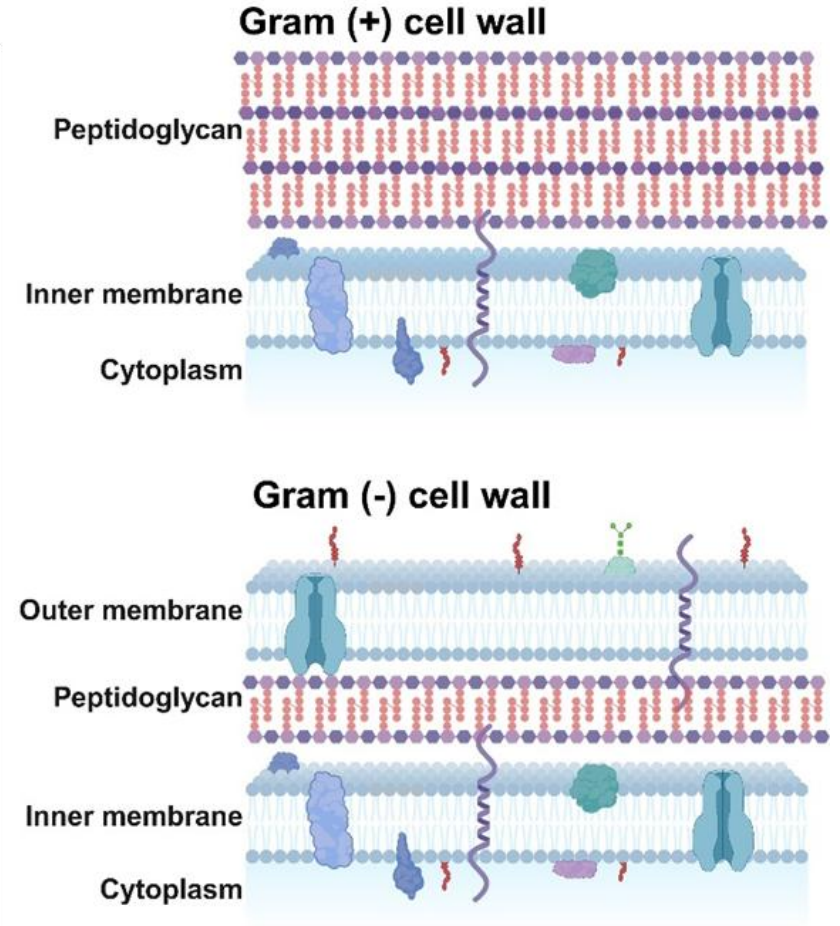
ACS Appl. Mater. Interfaces, 2025, 17 (10), 16027-16039

Antibacterial mechanism of dialdehyde cellulose

Antibacterial mechanism of dialdehyde cellulose



Structural comparison between Gram (+) & Gram (-) bacterial cell walls



ACS Appl. Mater. Interfaces, 2025, 17 (10), 16027-16039

(created with Biorender.com)

Antibacterial efficacy of dialdehyde cellulose

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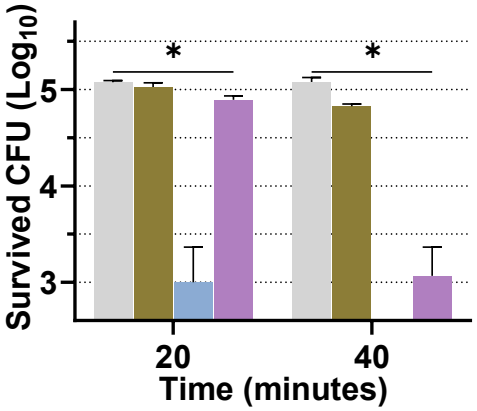
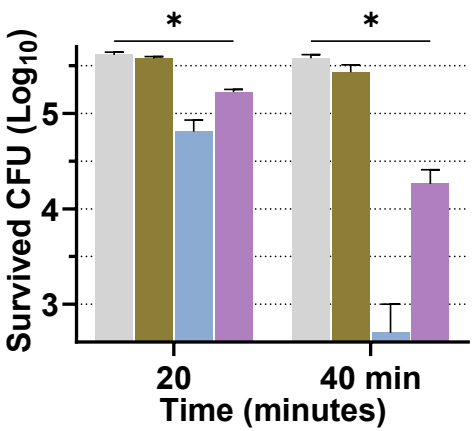
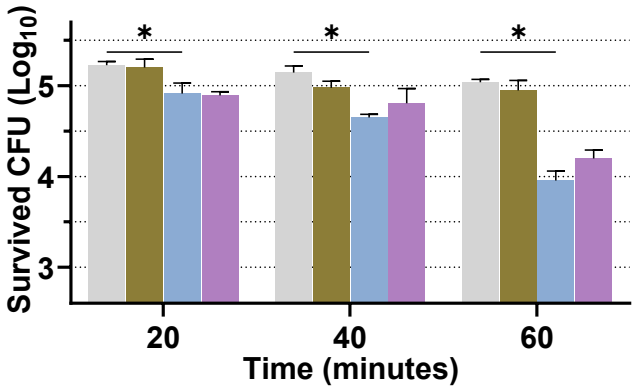
S. Aureus (Gram +)



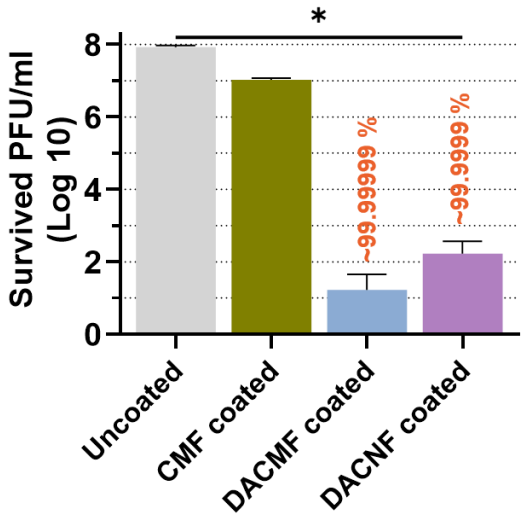
MRSA (Gram +)



E. Coli (Gram -)

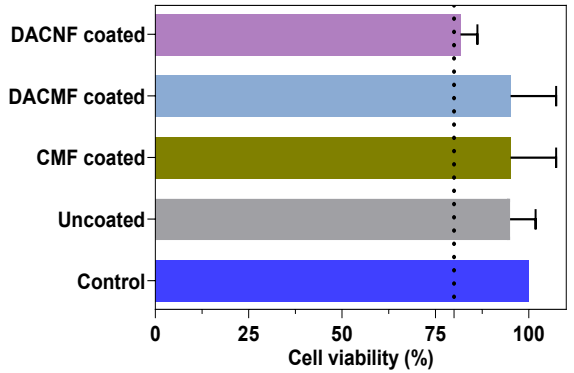


A/H1N1 (2009) influenza virus



Uncoated CMF DACMF DACNF

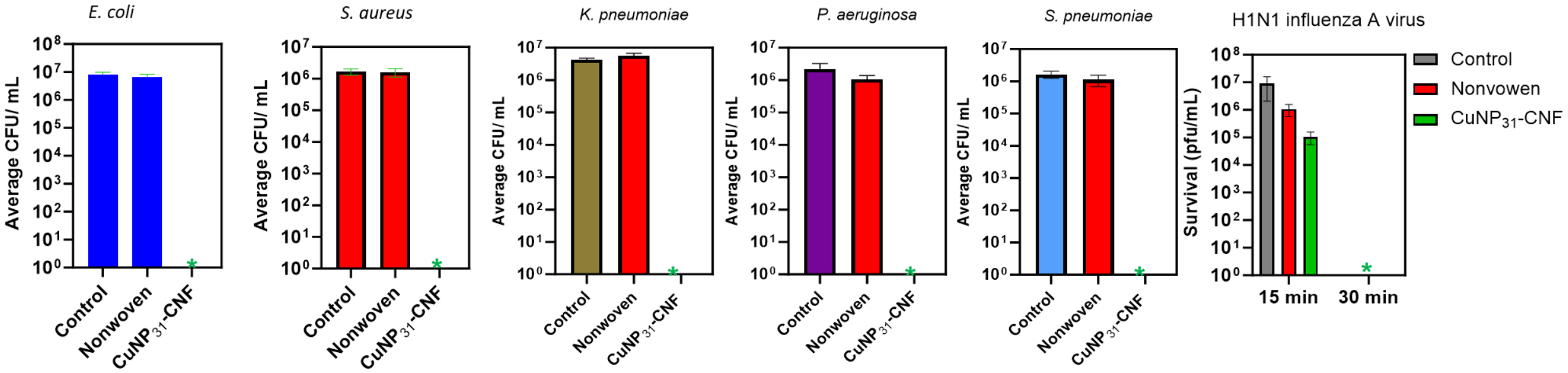
Cytotoxicity

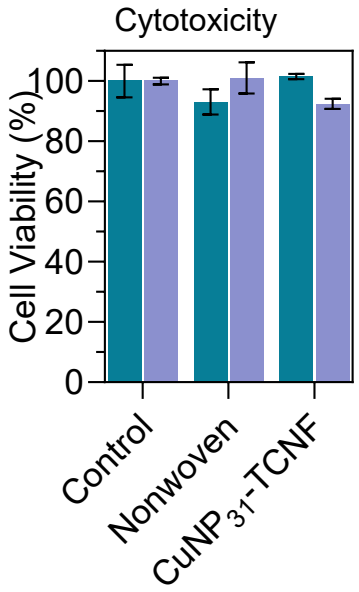


ISO 10993-5, using L929 cell line- mouse fibroblast

ACS Appl. Mater. Interfaces, 2025, 17 (10), 16027-16039

Antibacterial properties of copper oxide nanocellulose





ISO 10993-5, using L929 cell line- mouse fibroblast

Nanoscale 18 (35), 19165-19177

What changes when you leave the lab?

Lab scale (mg–g)

Prove the concept. Optimise chemistry. Characterise the material. Cost and yield don't matter yet. This is where most academic research lives.

Pilot scale (kg–100 kg)

Prove the process. Batch consistency becomes critical. Energy consumption is now measurable and expensive. This is where most promising materials hit their first wall.

Commercial scale (tonnes)

Prove the business. Feedstock supply chain, regulatory compliance, customer qualification, and unit economics all must work simultaneously.

As you move from the laboratory to manufacturing, the dominant questions shift from material performance to process reproducibility, cost, supply reliability and product qualification.

Scale-up bottlenecks for biomass-derived materials

Feedstock variability

season, geography, moisture, ash, contaminants

Pretreatment intensity

chemicals, water, energy, waste streams

Yield and concentration

low solids increase downstream cost

Product reproducibility

batch-to-batch quality and specifications

Drying and redispersion

often hidden cost and performance bottlenecks

Customer adoption

incumbent materials are cheap, optimised and qualified

Manufacturing readiness: what must be controlled?

Input control

feedstock specification, supplier, storage, contamination

Process control

unit operations, critical parameters, mass balance

Quality control

identity, purity, functional performance, stability

Risk control

toxicity, migration, exposure, worker safety, waste streams

Documentation

traceability, SOPs, change control, claims evidence

From manufacturing readiness to responsible manufacturing

Scaling a biomass-derived product is not only a technical challenge. Governance, ownership and benefit-sharing also shape whether a value chain can translate responsibly.

Technical readiness	Partnership readiness	Translation readiness
Can the feedstock, process and product be controlled reproducibly at scale?	Who has access to the resource, who helps define value, and how are decisions made?	Can the pathway scale in a way that is ethical, acceptable and beneficial to all parties?

A technically scalable process is not automatically a responsible biomanufacturing system.

Question to keep in mind

If a biomass resource is connected to Indigenous knowledge, what must be agreed before the technology is developed further?

Biomass manufacturing opportunities for Australia

The University Of Queensland AUSTRALIA

Local feedstocks

Sugarcane residues, forestry waste, cotton and grain waste, algae and seaweed

High-value products

Nanocellulose, biopolymers, functional materials, biofuel, composites and specialty chemicals

Regional opportunity

Manufacturing close to biomass sources, skilled jobs and new revenue for producers

Where Australia can compete

- Build on local biomass rather than imported feedstocks: sugarcane bagasse and trash, forestry residues, cotton and grain waste, algae and seaweed.
- Target high-value markets where Australia has research and industry strengths: sustainable packaging, biomedical materials, composites, mining sorbents and specialty chemicals.
- Close the scale-up gap through pilot manufacturing, feedstock standardisation, product qualification, techno-economic and life-cycle assessment, and regulatory pathways.
- Create regional and Indigenous-led enterprises through equitable partnerships, consent, benefit-sharing and protection of Indigenous knowledge.

CSIRO; Queensland Government biofutures resources; National Reconstruction Fund priority areas

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Thank you

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CRICOS code 00025B

WEEK 12 · 知识点总结 · Dr Aditya Ashok

第 12 周　光电探测器

Photodetectors: from photons to signals

主线：光吸收与带隙 → p–n 结与性能指标 → 用 SEM / EDX / TEM / XRD 看结构 → 多孔半导体 → 等离子体缺陷工程案例

1. 光与半导体

- 转换链：光子 → 吸收 → 电子–空穴对 → 分离 → 收集 → 电流。
- $E = hc/\lambda$，$E(eV) \approx 1240/\lambda(nm)$；只有 $h\nu \geq E_g$ 才吸收；截止波长 $\lambda_c(\mu m) = 1.24/E_g(eV)$（Si 1.12 eV → 1.11 μm，所以在 1550 nm 不响应）。
- 反射 $R = [(n-1)^2 + \kappa^2]/[(n+1)^2 + \kappa^2]$（裸 Si 约 32%）；吸收 $I(x) = I_0(1-R)e^{-\alpha x}$，$\eta_{abs} = (1-R)(1-e^{-\alpha d})$。
- 直接带隙 $\alpha \propto (h\nu - E_g)^{1/2}$（强吸收：GaAs、钙钛矿）；间接带隙 $\alpha \propto (h\nu - E_g \mp E_p)^2$（需声子：Si）；无序导致 **Urbach 带尾** $\alpha = \alpha_0 \exp(h\nu/E_U)$。
- 选材原则：选**能覆盖目标波长的最小带隙**（带隙越小，暗电流越大、越需冷却）。

2. 器件物理与性能指标

- p–n 结：内建电势 $V_{bi} = (kT/q)\ln(N_A N_D/n_i^2)$；耗尽区内电子–空穴对被快速分离；区外需扩散 $L = \sqrt{(D\tau)}$。
- 光照 I–V：$I = I_0[\exp(qV/nkT) - 1] - I_{ph}$；第 III 象限 = 光导模式（反偏，线性、快）；第 IV 象限 = 光伏模式（自供能）。
- 光电导增益 $G = \tau/\tau_{tr}$，$\tau_{tr} = L^2/(\mu V)$；**增益 × 带宽 ≈ 常数**：陷阱带来增益但牺牲速度（例：$G = 10^5$，$f_{3dB} \approx 160$ Hz）。
- $EQE = (I_{ph}/q)/(P/h\nu)$；响应度 $R = \eta\lambda(\mu m)/1.24$ (A/W)（Si @ 850 nm，η = 0.8 → 0.55 A/W）。
- 噪声：散粒 $2qI\Delta f$，热噪声 $4kT\Delta f/R$；$NEP = i_n/R$；$D^* = \sqrt{(A\Delta f)}/NEP$（Jones）；只用暗电流算 D* 会高估。
- 器件对比：光电导（增益高、慢）、p–i–n（快）、APD（雪崩增益）、PEC、测辐射热计；纳米多孔膜通常属于光电导 / PEC 类。

3. 表征技术综合复习（W2–W4）

- SEM 看形貌；EDX 看成分；TEM 看局部结构；XRD 看物相与平均结晶度；XPS 看表面 3–10 nm 的化学态。
- Kanaya–Okayama 作用范围：20 kV 下 Si 约 4.7 μm，Au 约 0.86 μm。EDX 过电压 $U = E_0/E_{edge} \geq 1.5$–2（Cu K 边 8.98 keV → 需 ≳ 15 kV）；精度 ±1–2 at%，轻元素不可靠，薄膜要注意衬底信号。
- TEM：200 kV λ ≈ 2.51 pm；量 10 条条纹取平均求 d；SAED $d = \lambda L/R$；单晶点、多晶环、非晶晕。
- XRD：峰位 → 物相、应变；强度 → 织构；峰宽 → 尺寸（例：2θ = 25°，β = 0.40° → D ≈ 20 nm）；应变也会展宽 → 用 Williamson–Hall；薄膜用**掠入射 GIXD**。
- **没有单一技术足够**：看到（SEM/TEM）+ 鉴定（EDX/XPS）+ 量化有序度（XRD/SAED）。

4. 多孔半导体

- 造孔：硬模板、软模板（嵌段共聚物胶束 PS-b-PEO + 电沉积 + THF 去模板）、刻蚀 / 去合金化 / 阳极氧化。
- 四个优点：比表面积大（约 90 倍）、载流子路径短（10–20 nm 孔壁）、光管理、传质快。
- 代价：悬挂键与**表面态**（陷阱、复合、响应慢）、小而有应变的畴、带尾（E_U 增大）、表面氧化；常规**退火在 ≥ 140 °C 时孔塌陷**。

5. 案例：等离子体畴工程（Ashok et al., Angew. Chem. 2022）

- 介孔 CuTe 薄膜，低功率 Ar 等离子体 **30 W、180 s**，接近室温。
- SEM：孔 9.4 / 16.2 / 20.9 nm 保持；EDX Cu:Te ≈ 48:52（XPS ≈ 49:51）；UV-vis：E_U 599 → 713 / 842 / 900 meV（壁越薄越无序），等离子体后降低。
- GIXD：宽包 → 尖峰，**结晶度约 70% → 85%**；XPS：表面氧化物去除；TEM：5–20 nm 杂乱畴 → 更大、取向一致（d(001) = 6.98 Å）。
- 器件：响应快约 60%，恢复快约 86%，光电流 +30%；10 W 无效，30 W 超过 240 s 开始刻蚀。

考试重点：$\lambda_c = 1.24/E_g$、响应度、Scherrer 等计算；增益与带宽的权衡；各表征技术回答什么问题、局限是什么；EDX vs XPS；多孔的优点与代价；如何用多种技术交叉证明"结晶度提高、孔不变、缺陷减少"。

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LECTURE · MATE6301 NANOMATERIALS

Photodetectors

From photons to signals, and how we see the nanostructure behind them

Light–matter interaction · junctions · figures of merit · SEM, EDX, TEM, XRD · porous semiconductors · A case study in plasma defect engineering

Dr Aditya Ashok
Australian Institute for Bioengineering and Nanotechnology
The University of Queensland

Roadmap: from photon to signal to structure

ROADMAP · 90 MINUTES · 5 BLOCKS

A 0–35 MIN

Photodetector essentials

B 35–60 MIN

Seeing the nanostructure: SEM, EDX, TEM, XRD

C 60–68 MIN

Porous semiconductors

D 68–84 MIN

Case study: plasma domain engineering

E 84–90 MIN

Summary and discussion

THROUGHOUT

18 live simulations · 4 checkpoints · figures from the case-study paper

make the material → **see** its structure (SEM · EDX · TEM · XRD) → link structure to **performance**

A Photodetector essentials

0–35 MINUTES

Photon energy and absorption, why semiconductors, the p–n junction, and the figures of merit that decide whether a detector is any good.

Why light–matter interaction matters

A · THE CONVERSION CHAIN

Sensing

UV index, flame and gas detection, ambient light, pulse oximetry.

Si, GaN, Ga_2O_3, organic photodiodes

Imaging

CMOS cameras, thermal imaging, X-ray and medical imaging, astronomy.

Si pinned photodiodes, InGaAs, HgCdTe, microbolometers

Communication

Fibre links at 1310/1550 nm, LiFi, free-space optics, LiDAR.

InGaAs p–i–n and APDs, Ge-on-Si, Si SPADs

Energy & catalysis

Photovoltaics and photo(electro)catalysis share the same first steps.

absorb → separate → collect (or react)

photon → absorption → e^-/h^+ pair → separation → collection → current

A photodetector makes each step as efficient, fast and quiet as the application needs. Every step depends on the **material's structure**: band gap, crystallinity, defects and surface area. That is why the second half of this lecture is about seeing that structure.

Photon energy decides who can absorb it

A · SIMULATION 1

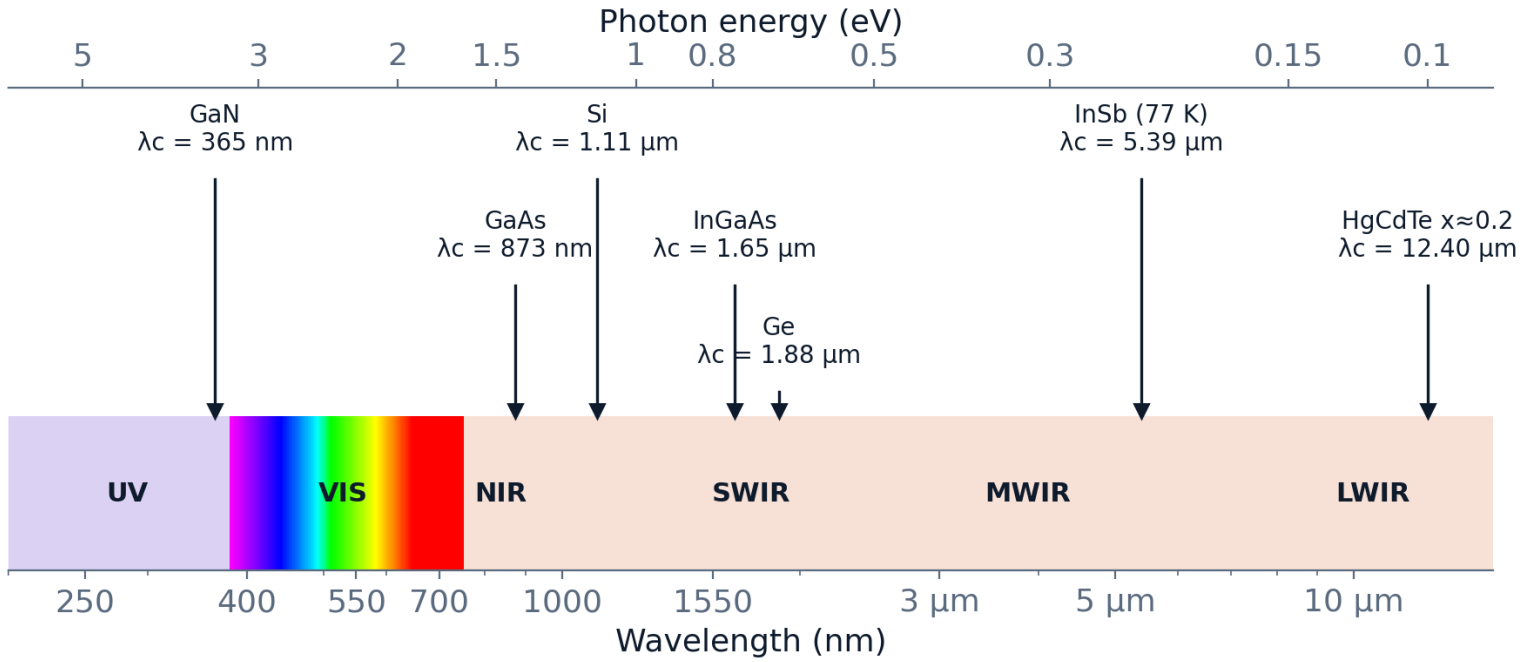
$$E = h\nu = hc/\lambda$$

$$E\ [\text{eV}] \approx \mathbf{1240} / \lambda\ [\text{nm}]$$

$$\lambda_c\ [\mu\text{m}] = 1.24 / E_g\ [\text{eV}]$$

A semiconductor absorbs a photon across its gap only if **$h\nu \geq E_g$**. That one inequality explains why silicon cameras are blind at 1550 nm.

Anchors: 400 nm = 3.10 eV · 850 nm = 1.46 eV · 1550 nm = 0.80 eV · 10 µm = 0.124 eV



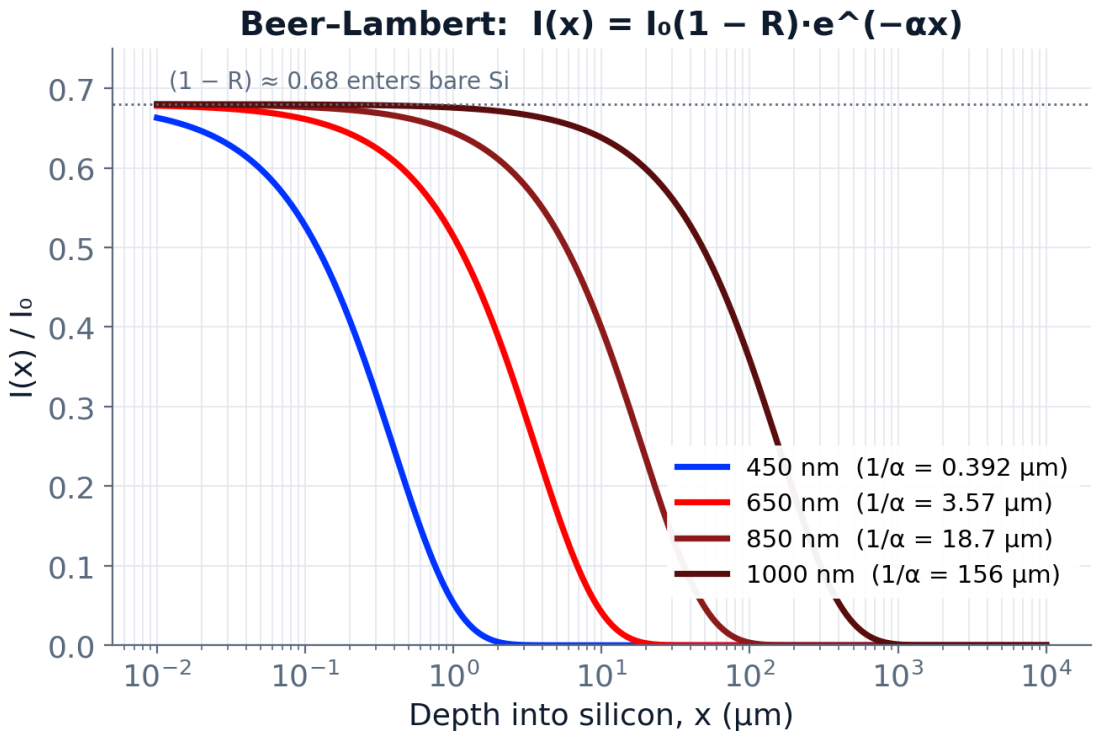
Reflection and Beer–Lambert: how deep does light go?

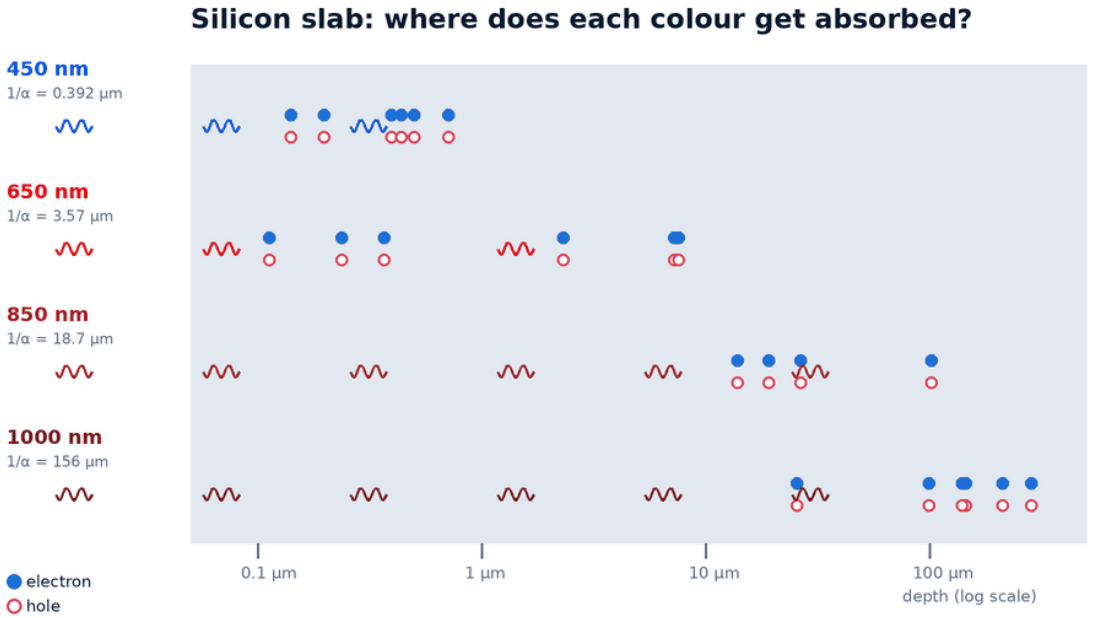
A · SIMULATION 2

$R = [(n - 1)^2 + \kappa^2] / [(n + 1)^2 + \kappa^2]$

$I(x) = I_0(1 - R)\cdot e^{-\alpha x}$

$\eta_{abs} = (1 - R)(1 - e^{-\alpha d})$

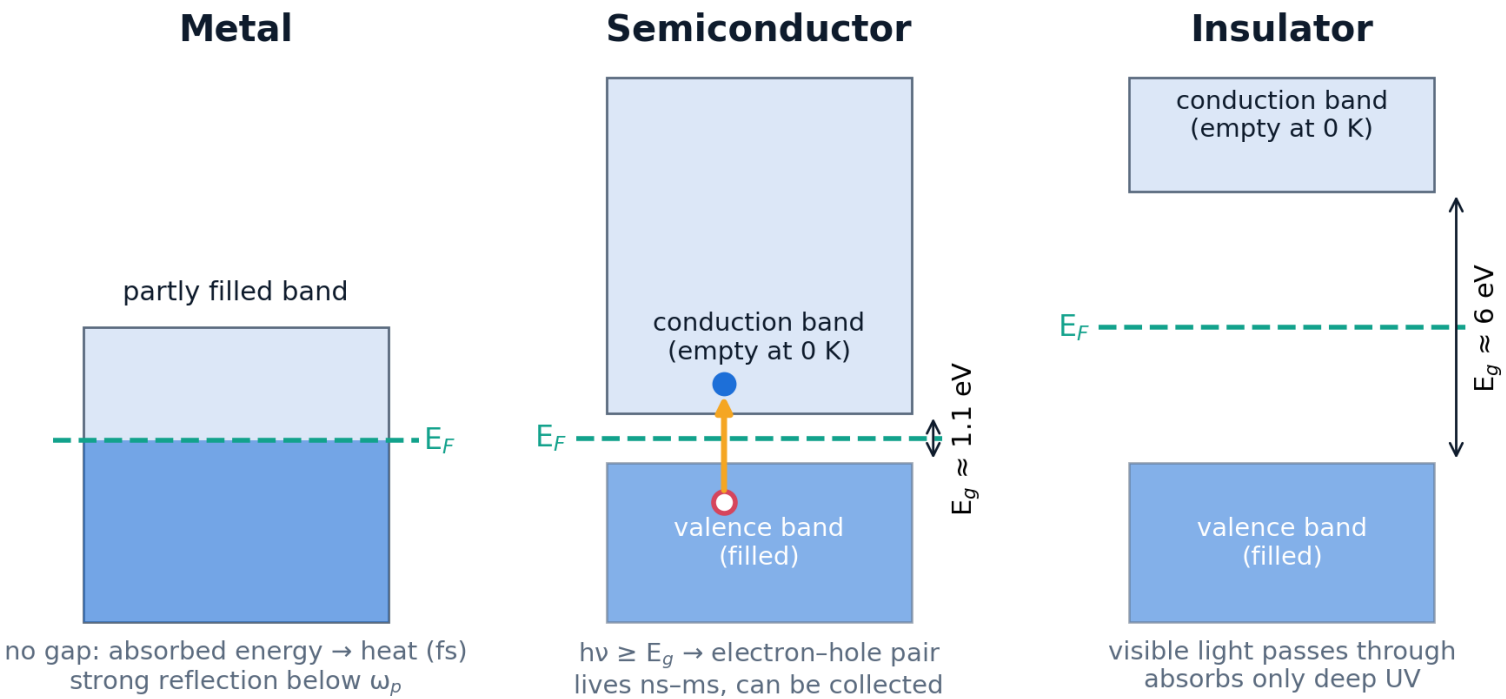




Bare Si reflects ≈ 32 % (n ≈ 3.6) before any absorption. Animation: blue light stops within 1 μm; 1000 nm light needs hundreds of μm.

Metal vs semiconductor vs insulator under light

A · SIMULATION 3



Drude: $\varepsilon(\omega) = \varepsilon_\infty - \omega_p^2 / (\omega^2 + i\gamma\omega)$

Copper chalcogenides such as the CuTe in our case study are semiconductors that absorb across the visible. Ultrawide-gap materials (diamond, AlN, Ga_2O_3) are used for solar-blind UV detection.

Metal

E_F inside a band: reflects below ω_p; absorbed energy → heat in fs.

Semiconductor

Gap 0.1–3.5 eV, controllable doping, ns–ms lifetimes: the sweet spot.

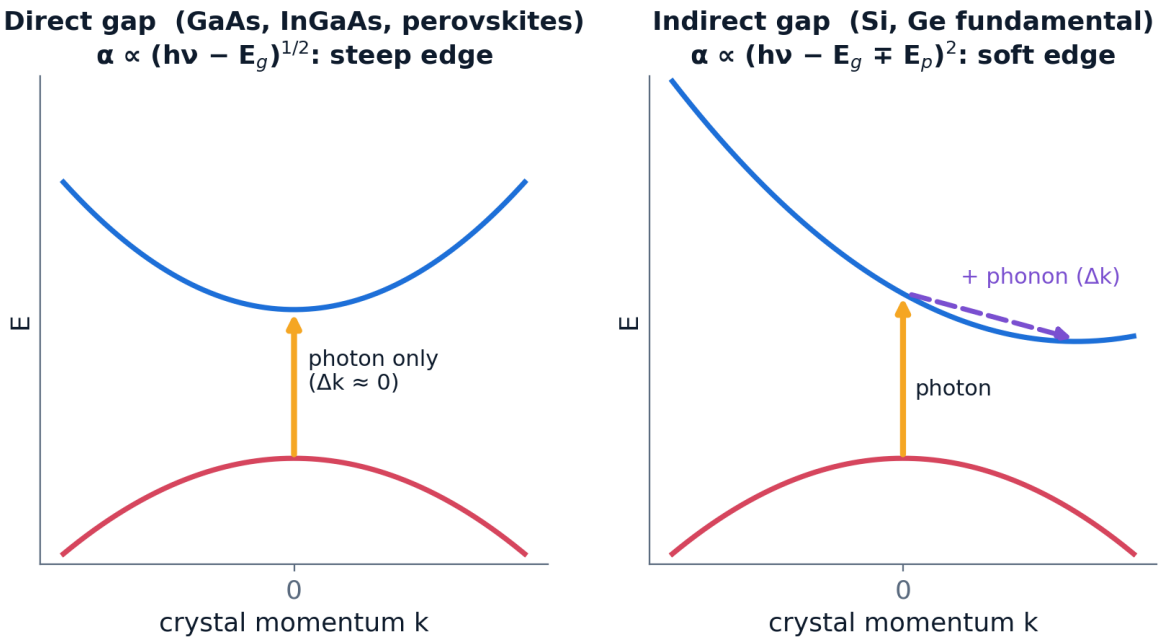
Insulator

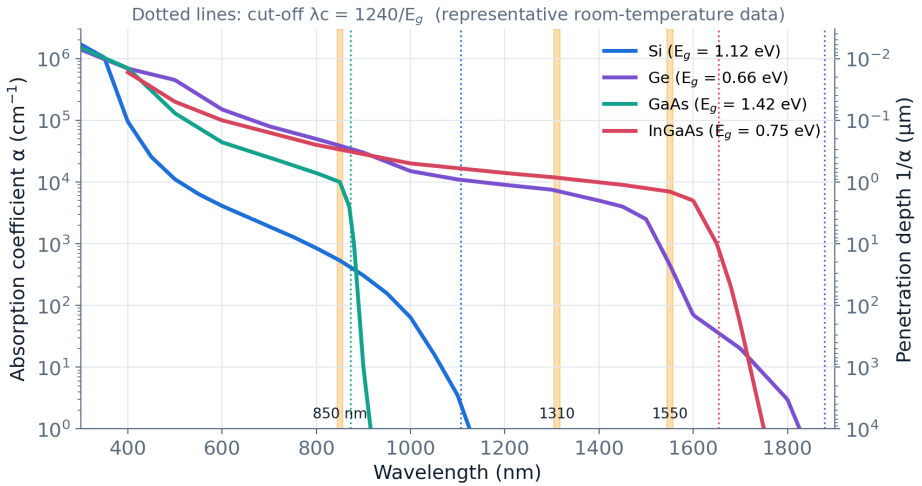
Gap > 5 eV ($SiO_2 \approx 9$ eV): transparent until deep UV.

Direct vs indirect gaps and the absorption edge

A · SIMULATION 4

direct: $\alpha \propto (h\nu - E_g)^{1/2}$ · indirect: $\alpha \propto (h\nu - E_g \mp E_p)^2$





Direct (GaAs, InGaAs, perovskites)

Photon alone does it: strong absorption just above E_g.

Indirect (Si)

A phonon supplies Δk: weak edge, thick absorbers.

Edges are blurred by **Urbach tails** from disorder. The Urbach energy becomes a key number in the case study: it tracks how disordered the pore walls are.

Choosing a material is choosing a spectral window

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A · MATERIAL TABLE

Material	E_g (eV)	Type	λ_c	Typical band	Notes
AlN	~6.0	direct	207 nm	deep UV	AlGaN alloys for solar-blind
β-Ga_2O_3	~4.9 (4.4–4.9)	direct/near-direct	253 nm	solar-blind UV-C	intrinsically solar-blind
GaN	3.4	direct	365 nm	UV-A	visible-blind
4H-SiC	3.26	indirect	380 nm	UV	robust, low dark current
ZnO	3.37	direct	368 nm	UV	persistent photoconductivity
$MAPbBr_3$	~2.3	direct	540 nm	visible	narrowband single-crystal demos
MoS_2 (1L)	~1.85	direct	670 nm	visible	bulk ~1.2 eV, indirect
$MAPbI_3$	~1.55	direct	800 nm	vis–NIR	tunable by halide and A-site
GaAs	1.42	direct	873 nm	vis–NIR	high α, fast
Si	1.12	indirect	1.11 µm	0.2–1.1 µm	CMOS workhorse
$In_{0.53}Ga_{0.47}As$	0.75	direct	1.65 µm	SWIR 0.9–1.7 µm	telecom; lattice-matched to InP
Ge	0.66 (direct 0.80)	indirect	1.88 µm	NIR/SWIR	higher dark current
PbS CQD	~0.7–1.5	direct	tunable	NIR/SWIR	size-tunable; CQD-on-CMOS
Black phosphorus	~0.3 (bulk)	direct	4.1 µm	MIR–vis	polarisation-sensitive
InSb (77 K)	0.23	direct	5.4 µm	MWIR 3–5 µm	cooled
HgCdTe x≈0.2 (77 K)	~0.1	direct	~12 µm	LWIR 8–14 µm	alloy-tunable
Graphene	0	semimetal	none	UV–THz	2.3 % absorption per layer

Rule of thumb: pick the smallest-gap material that still covers your wavelength. A smaller gap means more intrinsic carriers, more dark current and more cooling.

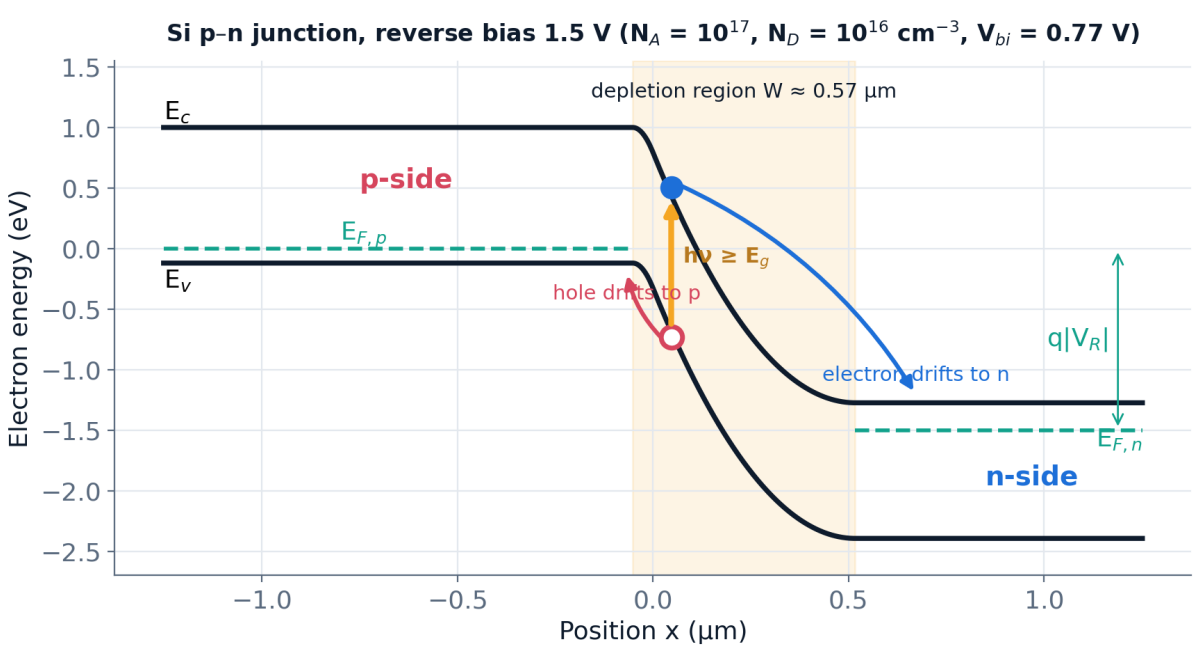
The p–n junction: a built-in field separates the pair

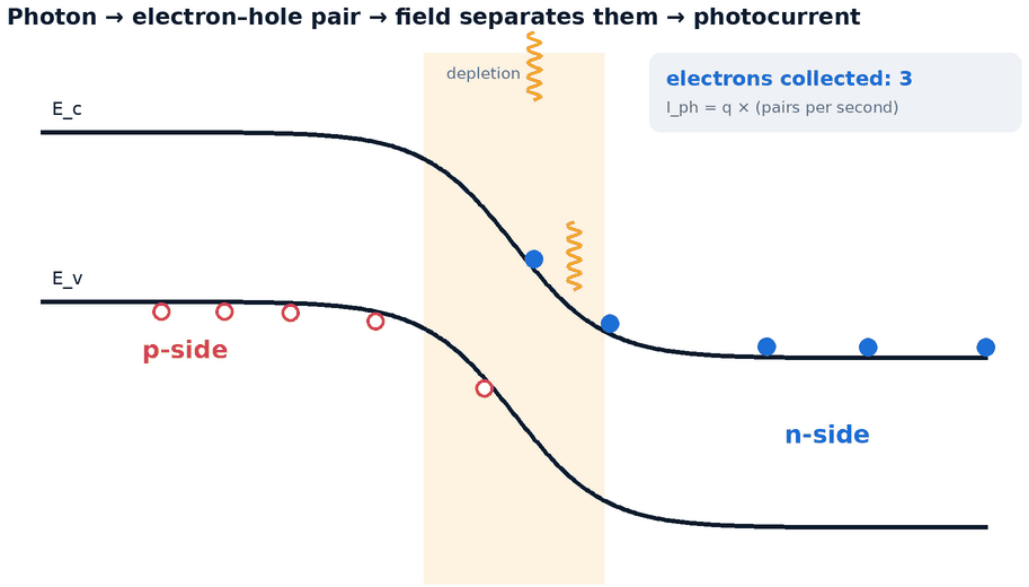
A · SIMULATION 5

$$V_{bi} = (kT/q)\cdot\ln(N_A N_D/n_i^2)$$

$$W = \sqrt{[2\varepsilon_s(V_{bi} - V)(N_A + N_D)/(qN_A N_D)]}$$

$$L = \sqrt{(D\tau)}$$



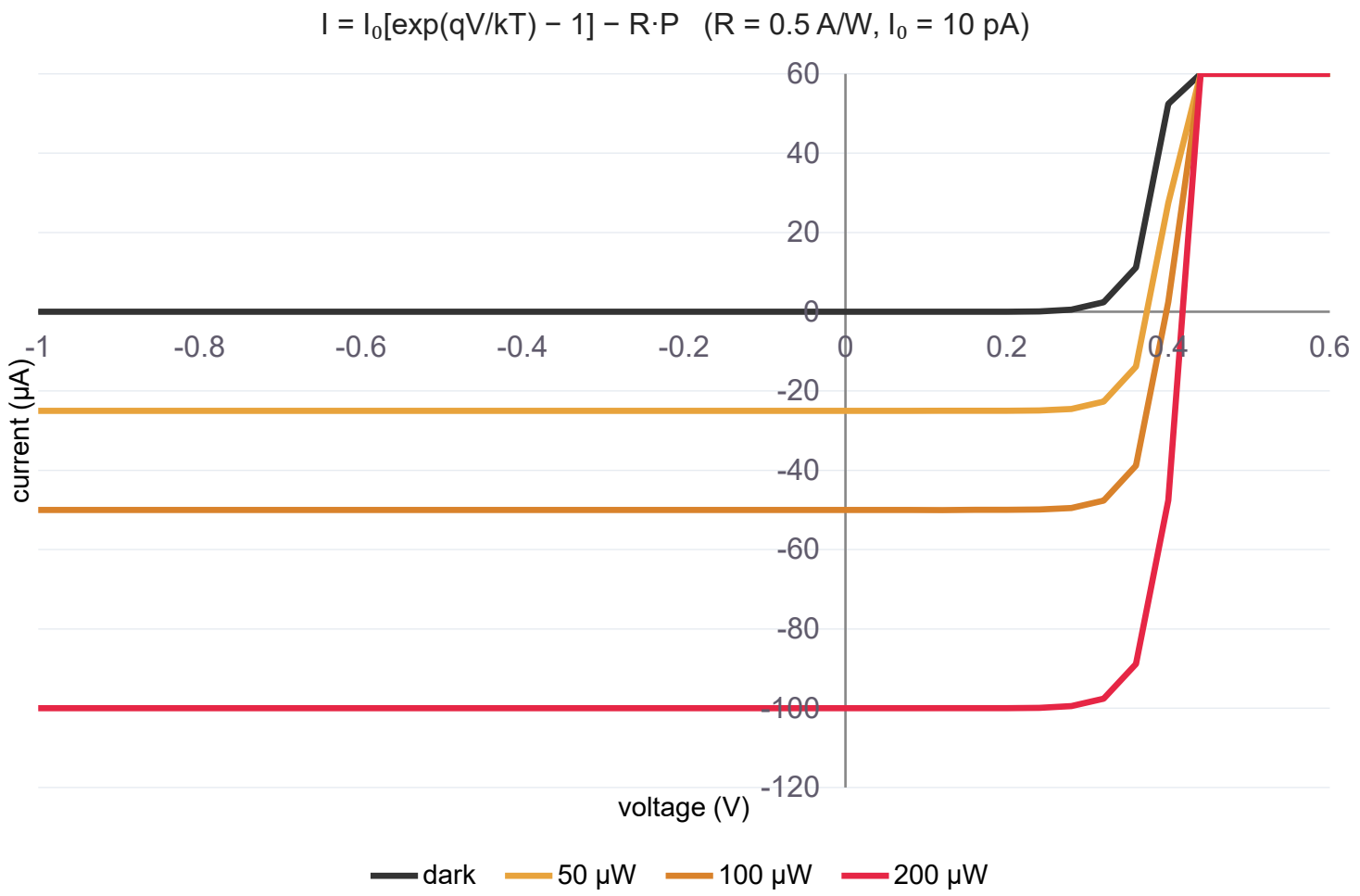


Pairs generated in the depletion region are swept apart in ps–ns; pairs generated further away must diffuse a distance L before recombining.

In a mesoporous film every wall is only 10–20 nm thick: carriers are always close to a surface.

The illuminated diode: dark and light I–V in four quadrants

A · SIMULATION 6



$$I = I_0[e^{qV/nkT} - 1] - I_{ph}$$

$I_{sc} = I_{ph}$ (linear) · $V_{oc} \propto \ln(I_{ph}/I_0)$ (log)

III · Photoconductive mode

Reverse bias: linear, fast, more dark current.

IV · Photovoltaic mode

Power out: solar cells, self-powered detectors.

The case-study films were tested by recording current with light chopped on and off.

Photoconductors: gain from trapped carriers

A · SIMULATION 7

$$\Delta\sigma = q(\Delta n \cdot \mu_n + \Delta p \cdot \mu_p)$$

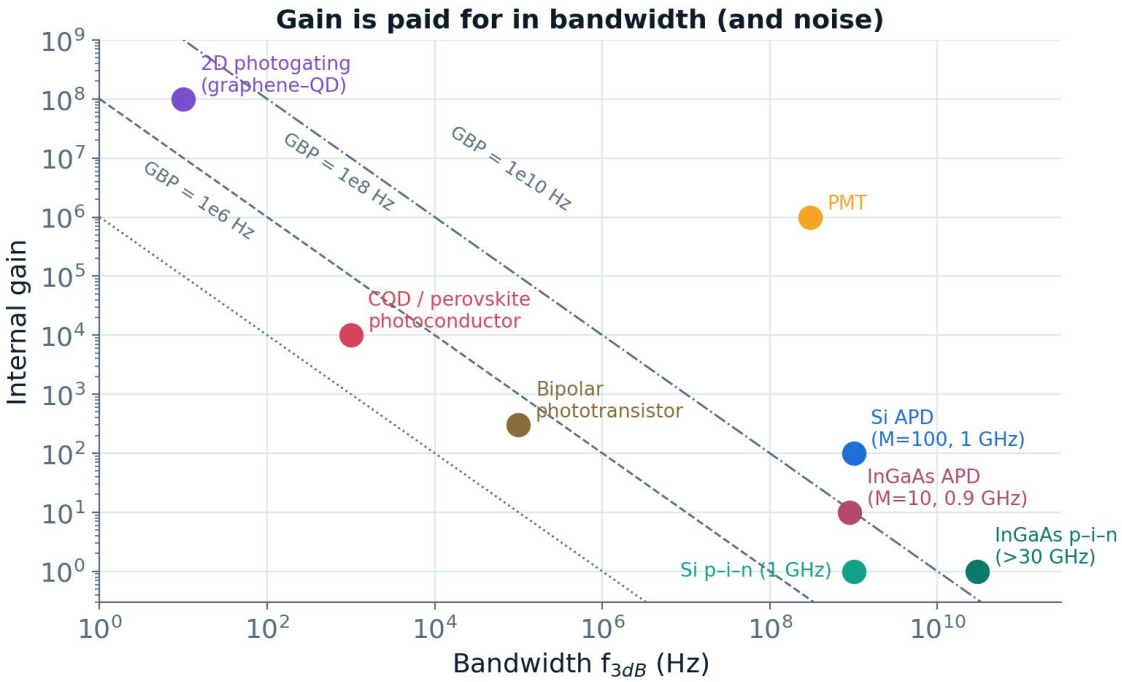
$$G = \tau / \tau_{tr}, \quad \tau_{tr} = L^2 / (\mu V)$$

If one carrier is trapped for τ while the other crosses in $\tau_{tr} \ll \tau$, the contacts re-inject carriers: each photon moves many electrons.

Bandwidth ≈ $1/(2\pi\tau)$, so G × bandwidth ≈ $1/(2\pi\tau_{tr})$ is fixed. **Traps give gain and cost speed.**

Surface traps in nanostructures (pores, grain boundaries, dangling bonds) are exactly these traps. Passivating them speeds up the response, as the case study shows.

Worked: L = 10 µm, µ = 100 cm²/Vs, V = 1 V, τ = 1 ms → τ_{tr} = 10 ns, G = 10^5, f_{3dB} ≈ 160 Hz



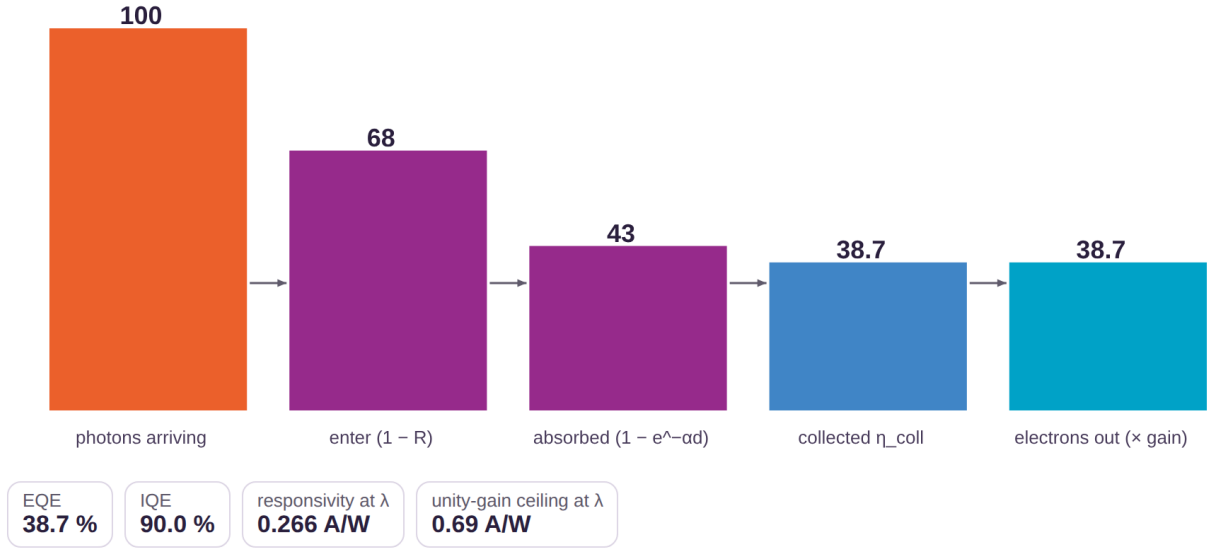
Figures of merit 1: quantum efficiency and responsivity

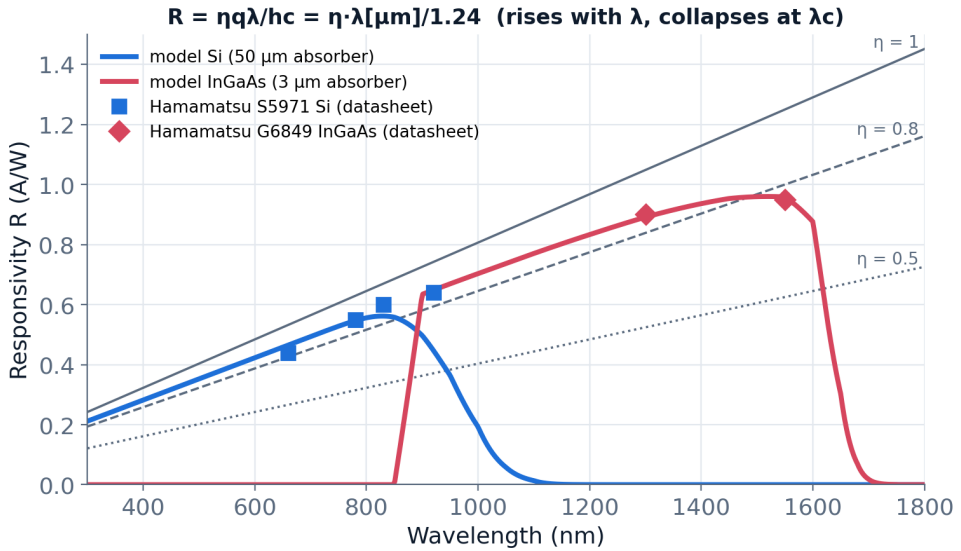
A · SIMULATION 8

$EQE = (I_{ph}/q) / (P/hv)$

$EQE = (1 - R)(1 - e^{-\alpha d})\cdot\eta_{coll}\cdot(G)$

$R = I_{ph}/P = \eta\cdot\lambda[\mu m]/1.24$ (A/W)





Worked: Si at 850 nm, η = 0.8 → R = 0.8 × 0.85 / 1.24 = **0.55 A/W**. Commercial unity-gain photodiodes sit at η ≈ 0.76–0.86: a sanity anchor when a paper claims thousands of A/W.

Figures of merit 2: noise, NEP and detectivity

A · SIMULATION 9

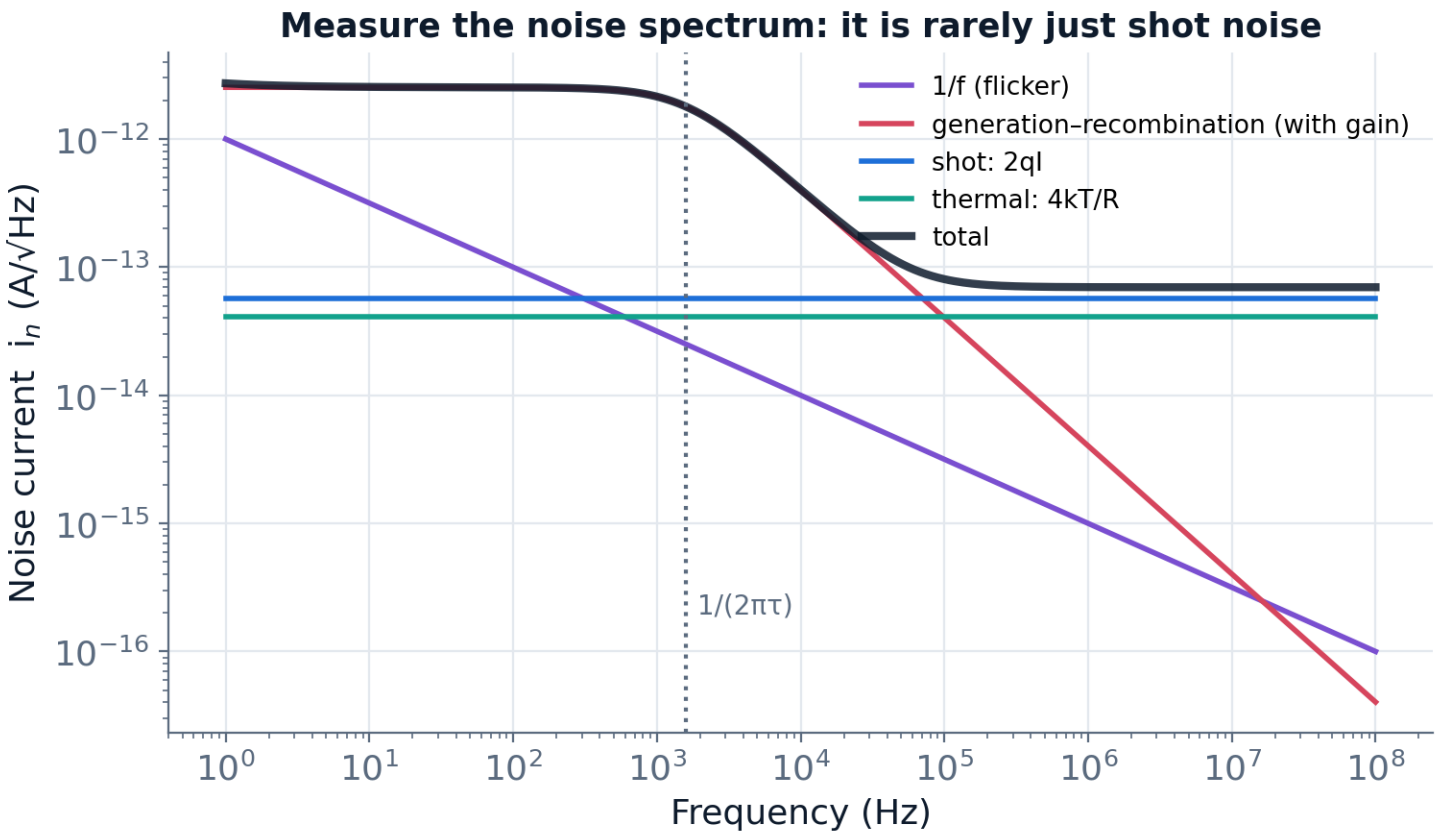
Shot: $\langle i^2 \rangle = 2qI\Delta f$ · Thermal: $4kT\Delta f/R$

$NEP = i_n / R$ (W/√Hz)

$D^* = \sqrt{(A\Delta f)} / NEP$ (Jones)

Trap states add generation–recombination and 1/f noise. A "D*" calculated only from dark current ignores them and can overestimate detectivity by orders of magnitude. **Measure the noise.**

Hamamatsu S5971 Si PIN: I_d = 0.07 nA, R = 0.64 A/W → NEP ≈ 7.4×10^{-15} W/√Hz, $D^* \approx 1.4 \times 10^{13}$ Jones.



Device types and trade-offs

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A · FILTER BY NEED

Device	Internal gain	Typical bandwidth	Noise	Bias	Typical applications
Photoconductor	1–10^8 (traps)	Hz–MHz	g–r + 1/f, high dark current	required	cheap IR (PbS), UV (ZnO), nanostructured films
p–n photodiode	1	kHz–100 MHz	shot + thermal; lowest at 0 V	0 V or reverse	light meters, PPG
p–i–n photodiode	1	100 MHz to >50 GHz	shot + thermal	reverse, few V	fibre receivers, LiFi
Schottky / MSM	1	GHz to 100s GHz	higher dark current	reverse / bias	UV (GaN, Ga_2O_3), ultrafast
APD	10–1000	100 MHz to 25+ GHz	shot × M^2F	tens–hundreds V	LiDAR, telecom
Phototransistor	10^2–10^8	Hz–MHz	gain-amplified; 1/f	required	ambient sensing, 2D research
Photoelectrochemical (PEC)	1	Hz–kHz	electrolyte, surface states	0 V or small	self-powered sensing, photocatalysis
Thermal (bolometer)	none	ms	thermal fluctuation	bias / none	thermal imaging, THz

Every gain mechanism has a bandwidth bill and a noise bill. Nanostructured and porous films usually sit in the photoconductor / PEC rows: their surfaces and grain boundaries decide gain, speed and noise.

B Seeing the nanostructure

35–60 MINUTES

Once a sample is made: SEM for morphology, EDX for composition, TEM for nanoscale and atomic structure, XRD for phase and crystallinity.

Every technique answers a different question

SEM

What does it look like?

Morphology, pore size, film thickness, particle shape.

EDX

What is it made of?

Elements present, atomic ratios, elemental maps.

TEM

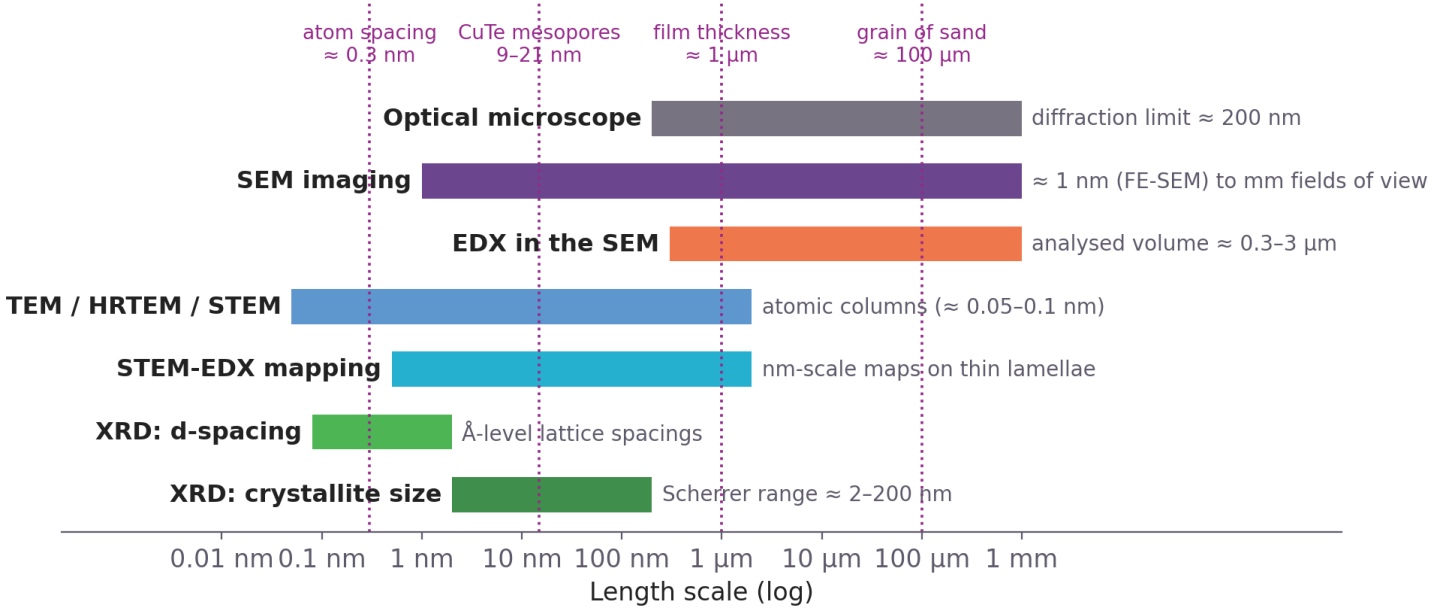
How is it arranged inside?

Grains, lattice fringes, d-spacings, defects, cross-sections.

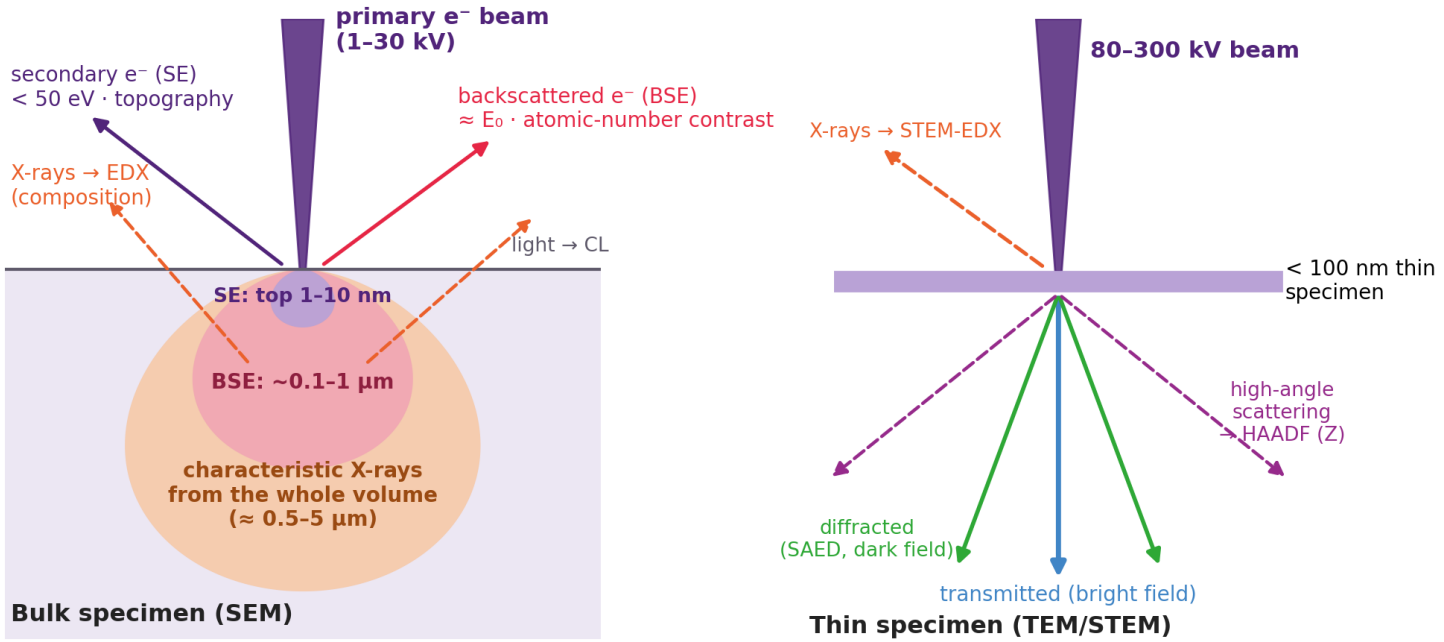
XRD

Which crystal, how crystalline?

Phase ID, crystallite size, strain, texture.



Electron–matter interaction: one beam, many signals



Secondary electrons (SE)

< 50 eV, escape from the top 1–10 nm → surface topography (SEM).

Backscattered electrons (BSE)

Energy ≈ E_0, from ~0.1–1 µm → atomic-number (Z) contrast.

Characteristic X-rays

From the whole interaction volume → composition (EDX).

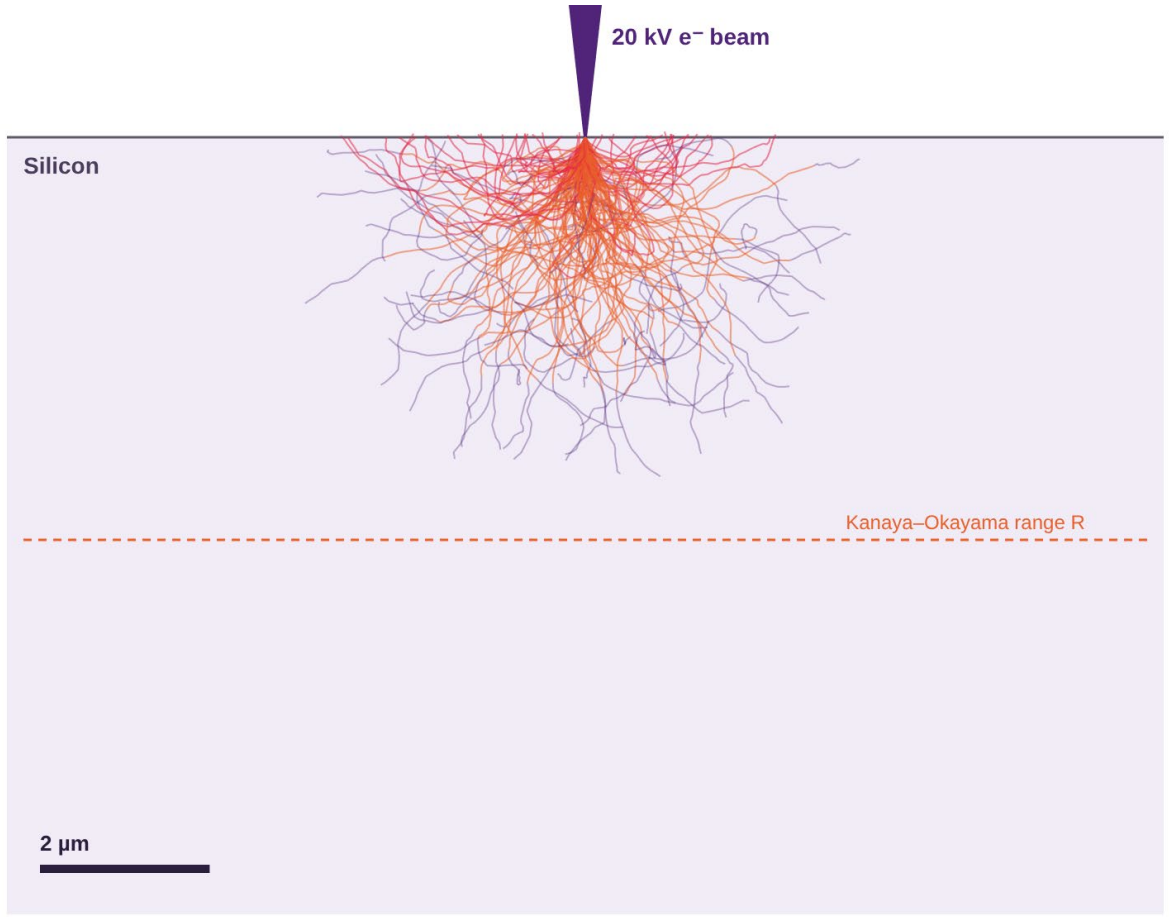
Thin specimen (TEM)

Transmitted, diffracted and high-angle scattered electrons → images, SAED, HAADF.

Kanaya–Okayama range: $R\ [\mu m] = 0.0276 \cdot A \cdot E_0^{1.67} / (Z^{0.889} \cdot \rho)$ → at 20 kV: Si ≈ 4.7 µm · Cu ≈ 1.5 µm · Au ≈ 0.86 µm

Simulation: how big is the interaction volume?

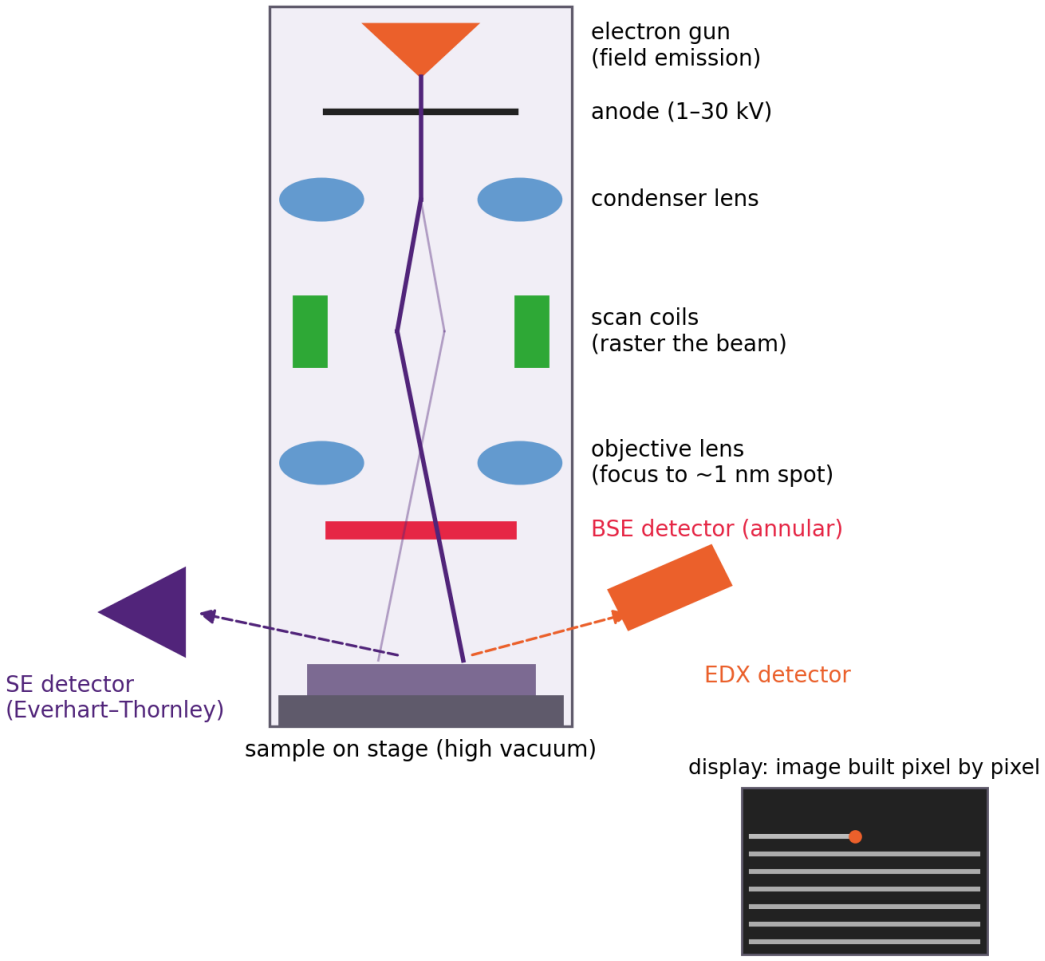
B · MONTE CARLO ELECTRON TRAJECTORIES



- **Si at 20 kV**: a pear-shaped volume ~4–5 µm deep.
- **Au**: under 1 µm, and many more electrons bounce back out (BSE): heavy elements look bright.
- **5 kV**: the volume shrinks dramatically → true surface detail.
- **Cu K edge highlight**: below ~12 kV almost no trajectory stays above 8.98 keV, so Cu Kα is barely excited.

E_0 in keV, A in g/mol, ρ in g/cm³ for the range formula.

SEM: a focused beam, scanned point by point



Electron source

Field-emission gun, accelerating voltage 1–30 kV, high vacuum.

Lenses and scan coils

Condenser and objective lenses focus a ~1 nm spot; scan coils raster it across the sample.

Detectors

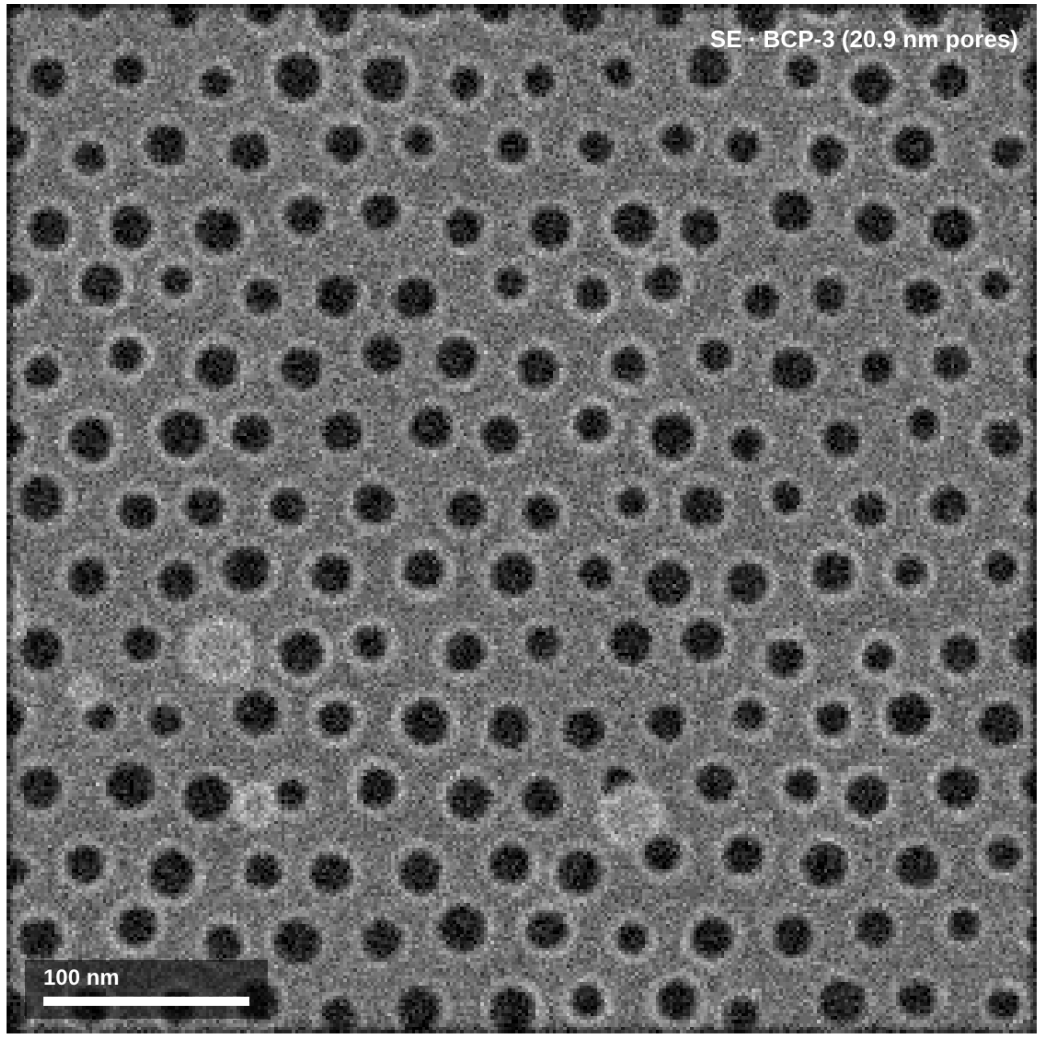
SE (Everhart–Thornley or in-lens), BSE (annular, under the pole piece), EDX (X-rays).

Image

Each pixel = detector signal at that spot. Magnification = display width / scanned width. Resolution is set by spot size and interaction volume, not magnification.

Simulation: build an SEM image of a mesoporous film

B · SIMULATION 11 · RASTER SCAN, SPOT SIZE, DWELL TIME, DETECTOR



- Start with **BCP-3** at 500 nm field of view.
- Enlarge the **spot** past the pore size: pores vanish at the same magnification.
- Shorten the **dwell time**: noise grows.
- Switch to **BSE**: Au particles light up (higher Z); edge contrast weakens.
- Zoom out to **3 µm**: 9 nm pores are lost once pixel > pore.

SEM in practice: getting a trustworthy image

SE vs BSE

SE (< 50 eV, top 1–10 nm): edges and topography. Use for pore shape and size.
BSE (≈ E_0): brighter for heavier elements. Use for phase contrast and particles.

Choosing the voltage

Low kV (1–5 kV): small interaction volume, true surface detail, less charging.
High kV (15–20 kV): better signal on conductive samples; needed to excite K-lines for EDX.

Charging and coating

Insulators accumulate charge: bright streaks, drifting images.
Fixes: thin Pt/Au or C coating, low kV, low-vacuum mode, conductive tape to ground.

Measuring pores

Threshold the image (e.g. ImageJ/Fiji), measure ≥ 100 pores, report mean ± SD and the histogram, not one hand-picked pore.

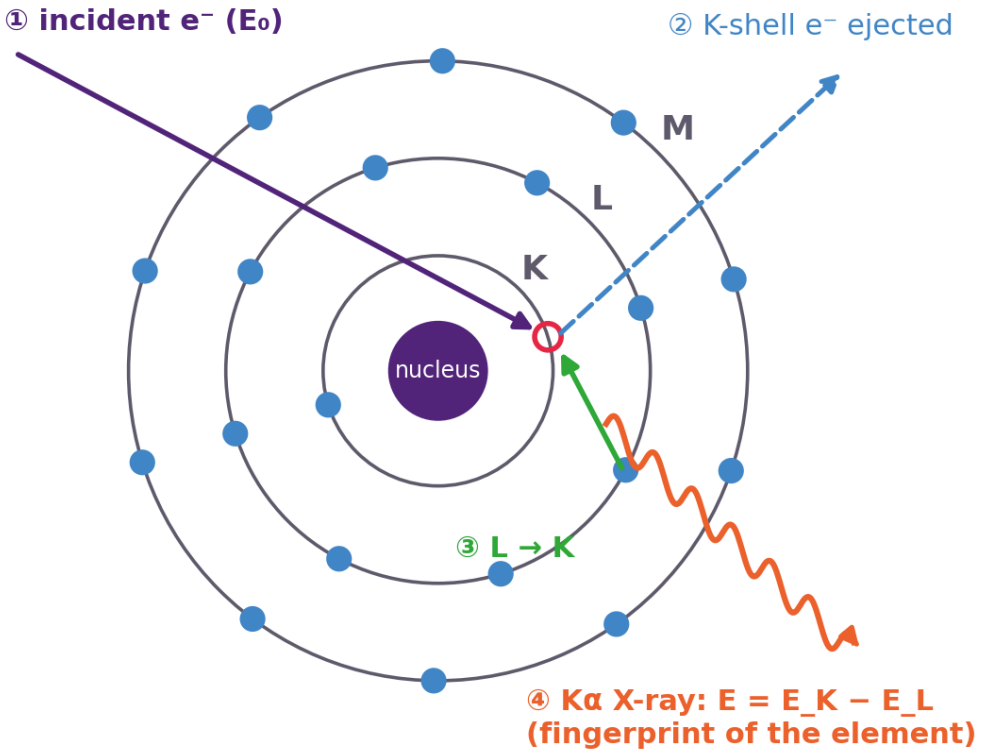
Cross-sections

Tilt or fracture the sample (or FIB-cut it) to measure film thickness and see whether pores run through the film.

Limits

Surface only. Tells you nothing about crystal structure. Beam damage on polymers and soft materials.

EDX: characteristic X-rays fingerprint the elements



① – ④ The process

The beam ejects an inner-shell (K) electron; an outer electron (L) fills the vacancy; the energy difference leaves as an X-ray.

$E(K\alpha) = E_K - E_L$ · Moseley: $\sqrt{\nu} \propto (Z - \sigma)$

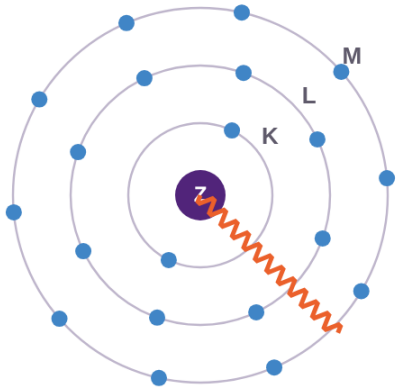
Lines you will see for CuTe on Au

Cu Kα 8.05 keV · Cu Lα 0.93 keV · Te Lα 3.77 keV · Au Mα 2.12 keV (substrate) · O Kα 0.52 keV

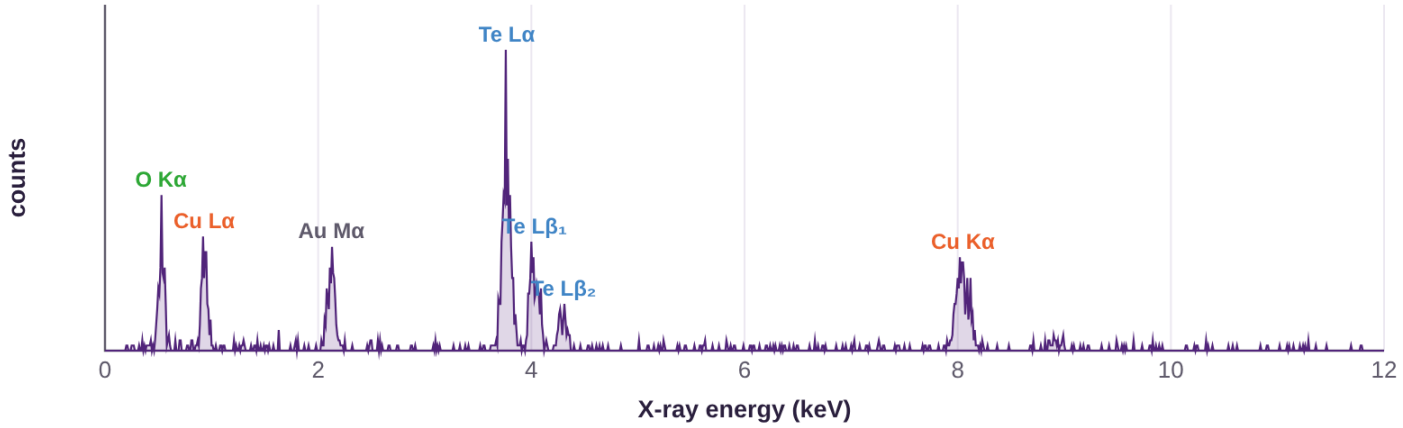
Overvoltage

$U = E_0 / E_{edge}$ should be ≥ 1.5–2. Cu K edge = 8.98 keV, so Cu Kα needs ≳ 15 kV.

Simulation: watch an EDX spectrum build up

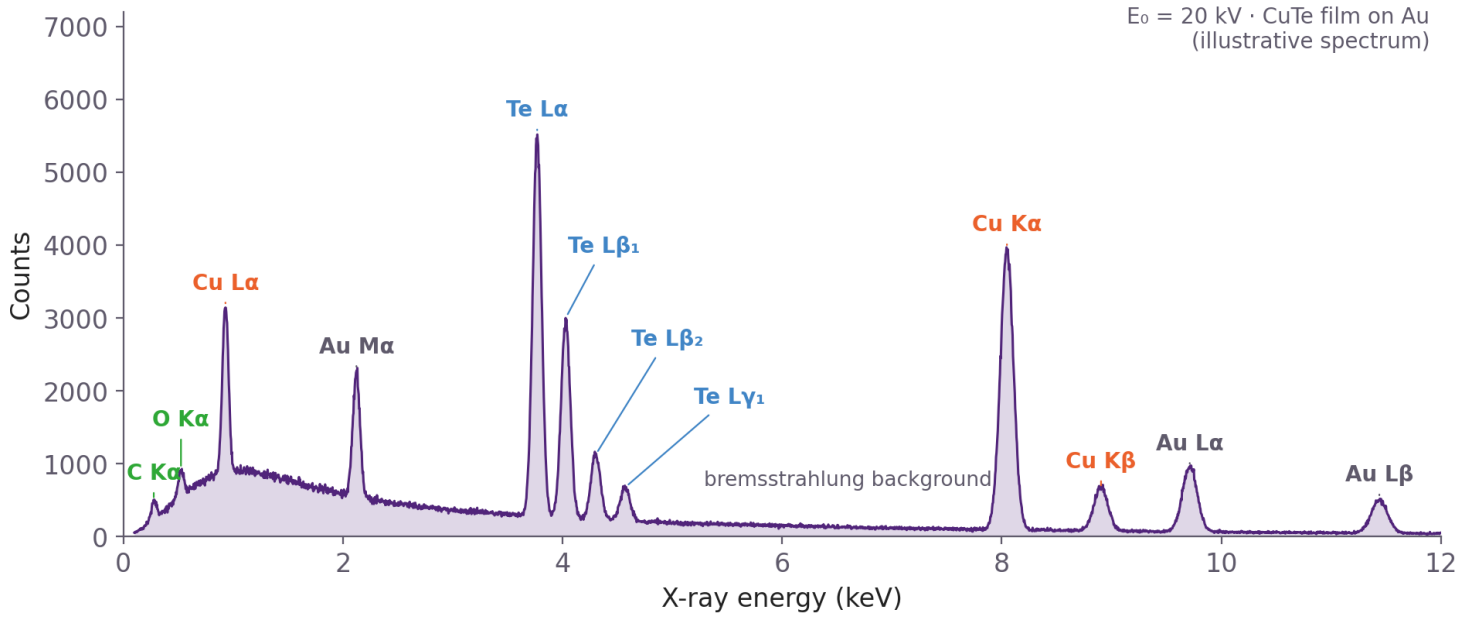


Kα X-ray → detector



- Set Cu : Te = 50 : 50 at 20 kV and let counts accumulate: the measured ratio converges; error falls as 1/√counts.
- Drop to **10 kV**: Cu Kα disappears (10/8.98 too close to 1). Quantify Cu with the L-line instead.
- The **Au** peaks come from the substrate under a thin film: a classic quantification trap.

EDX in practice: point, line, map, and the error bars



Point / area

Composition at a spot or averaged over a frame.

Line scan

Composition along a line, e.g. across a film cross-section.

Map

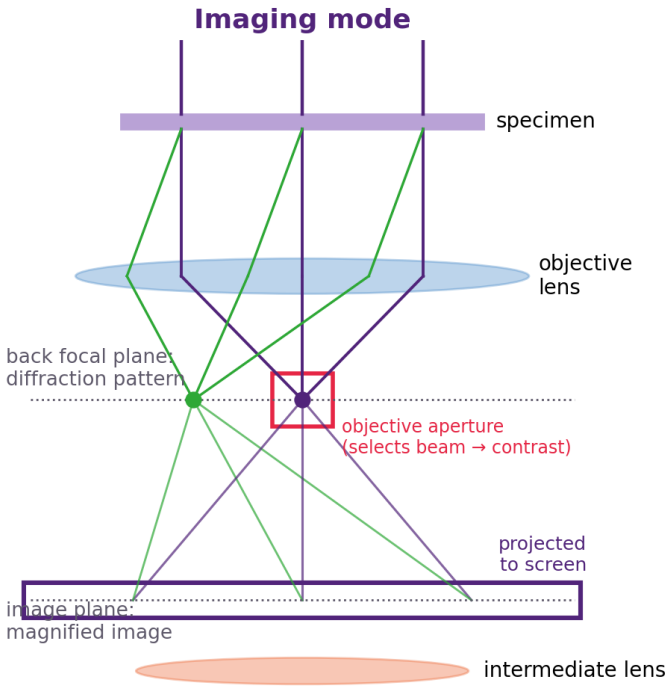
Where each element sits: uniform or segregated?

Limits to quote honestly

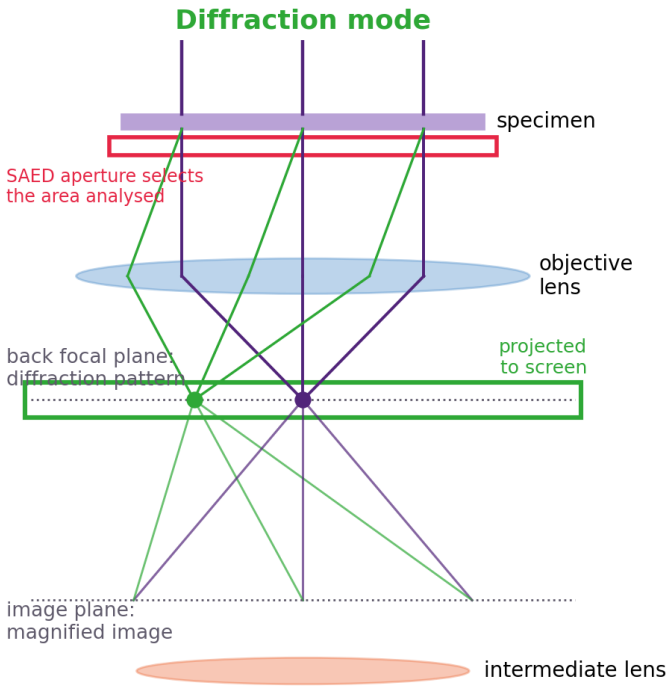
- Spatial resolution ≈ interaction volume: ~1 µm in bulk SEM, ~nm in STEM on thin lamellae
- Typical accuracy ±1–2 at% (standardless); detection limit ~0.1–1 wt%
- Light elements (C, N, O) unreliable
- Peak overlaps; substrate signal from thin films

Cross-check with XPS, which probes only the top few nm. EDX samples the bulk of the film; XPS sees the surface chemistry (oxides, contamination). The case study reports both.

TEM: electrons through a thin slice



BF image / HRTEM lattice fringes



SAED pattern (spots / rings)

$\lambda = h/\sqrt{[2m_0eV(1 + eV/2m_0c^2)]} \rightarrow 2.51$ pm (200 kV)

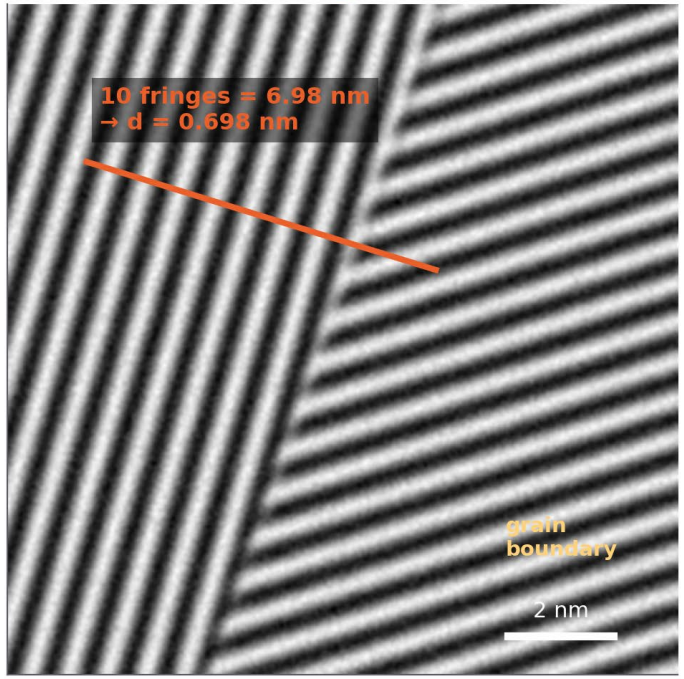
- **Specimen < 100 nm thick**: drop-cast nanoparticles, or a FIB-cut lamella for films
- **Imaging mode**: bright field (mass–thickness and diffraction contrast) and HRTEM lattice fringes
- **Diffraction mode**: SAED from a selected area
- **STEM**: scan a fine probe; HAADF intensity $\propto Z^{\sim 1.7}$, plus STEM-EDX maps at nm resolution

One instrument, two pictures of the same region: the real-space image and its diffraction pattern. Switching modes changes which plane the intermediate lens projects.

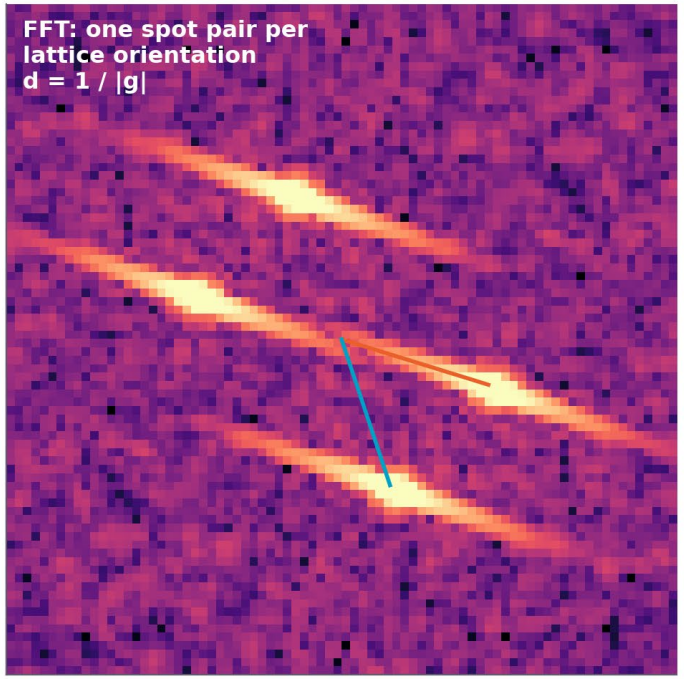
Reading TEM data: lattice fringes and their FFT

B · HRTEM

HRTEM: lattice fringes in two domains



FFT (digital diffractogram)



Measure d from fringes

Measure across 10 fringes and divide by 10: here 6.98 nm / 10 = 0.698 nm.

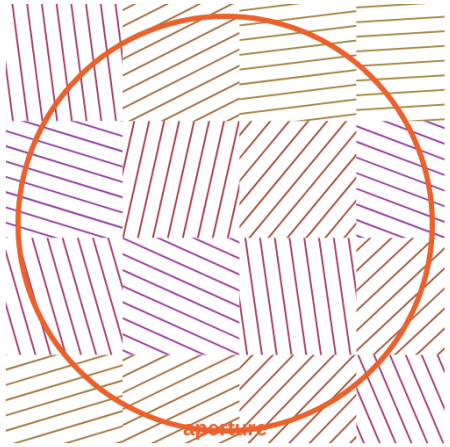
FFT = digital diffraction

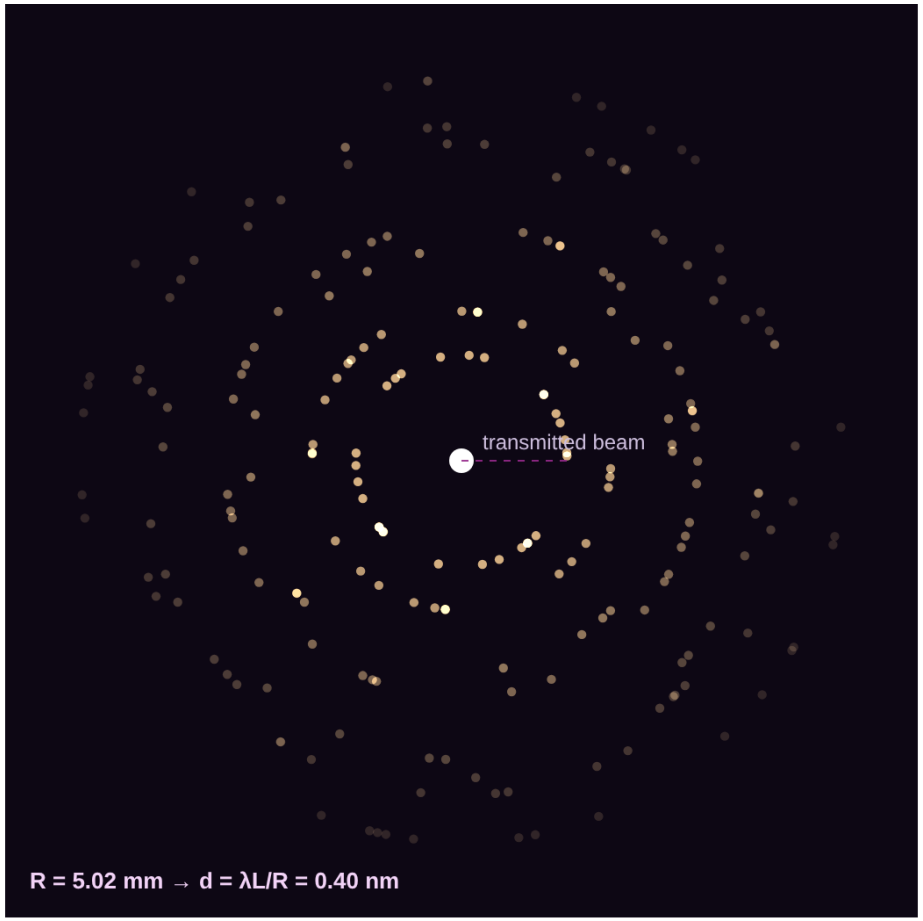
One spot pair per lattice orientation, at |g| = 1/d. Two domains → two spot pairs.

SAED camera constant

R·d = λL: measure the spot or ring radius R on the pattern to get d.

Simulation: spots or rings? Grains inside the aperture





- Small aperture over large grains: **one spot pattern**.
- Widen the aperture or shrink the grains: spots fill in to **rings**.
- Tick amorphous: diffuse **halos**.
- Readout: d = λL/R for the first ring.

Diffraction patterns are a crystallinity fingerprint

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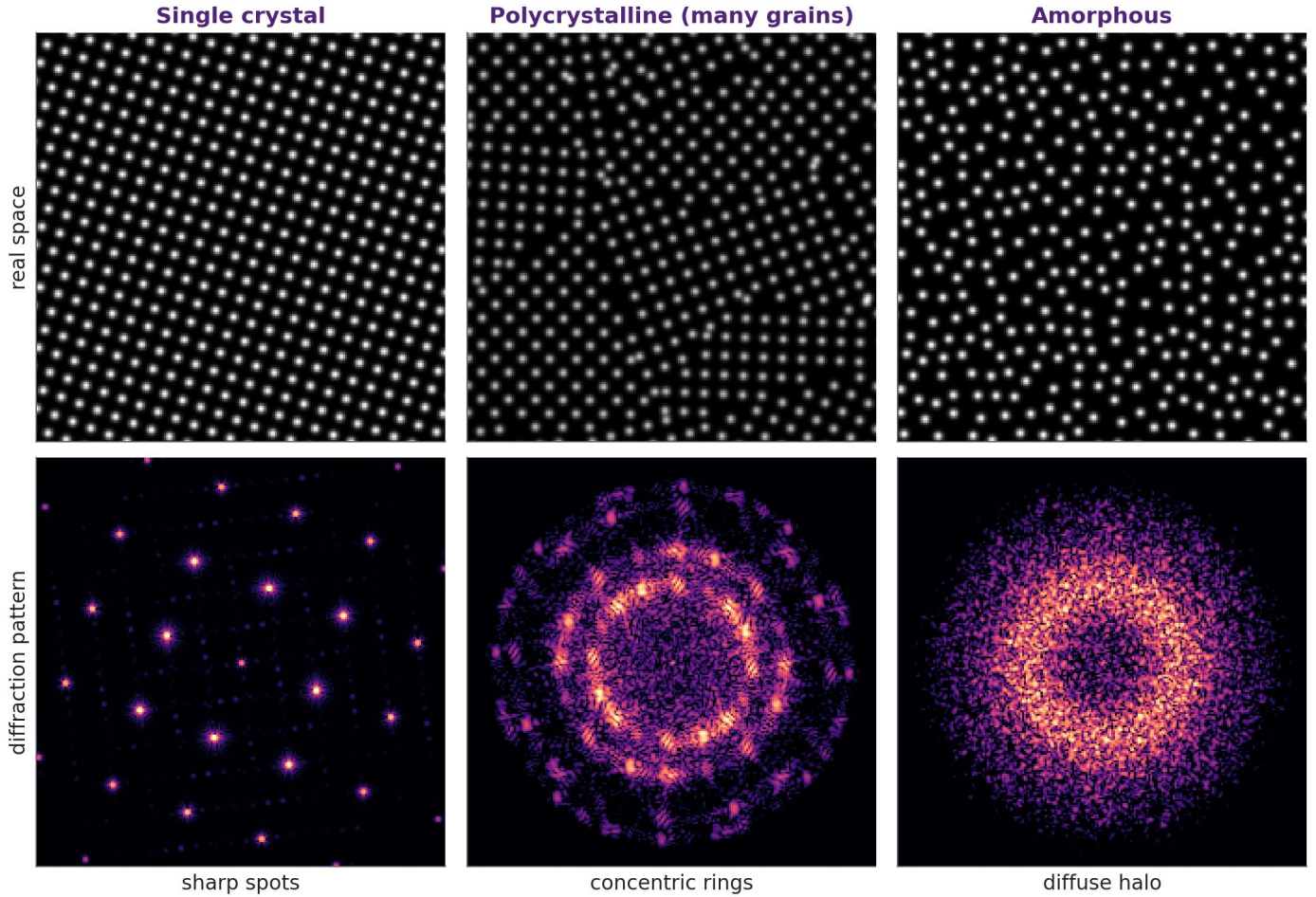
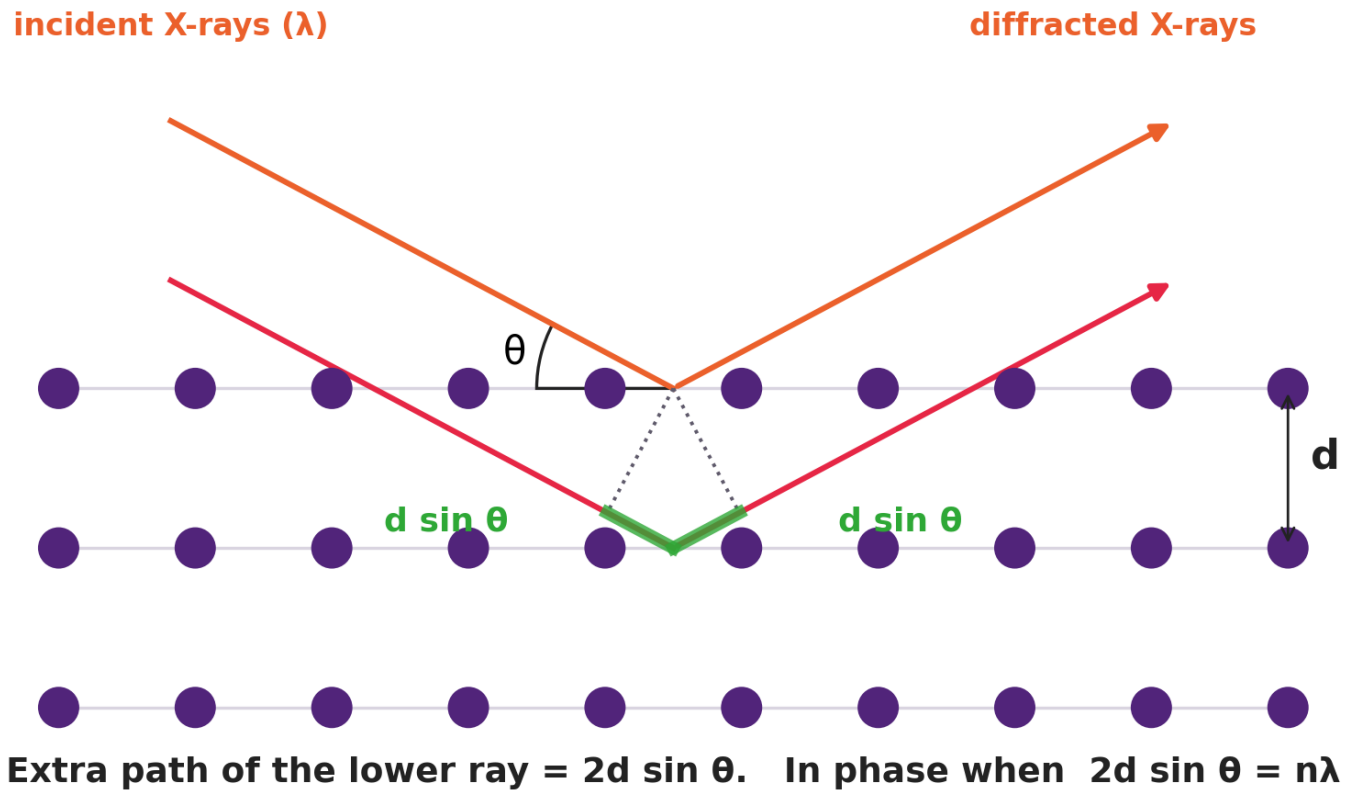
B · COMPUTED PATTERNS (FFT OF MODEL STRUCTURES)



Single crystal → spots · polycrystal → rings · amorphous → diffuse halo. The same three outcomes appear in SAED and in XRD: sharp, ring-like, or a broad hump.

XRD: Bragg’s law



$$2d \sin \theta = n\lambda$$

Source

Cu Kα, λ = 0.15406 nm (Mo Kα 0.07093 nm for heavy absorbers).

Scan

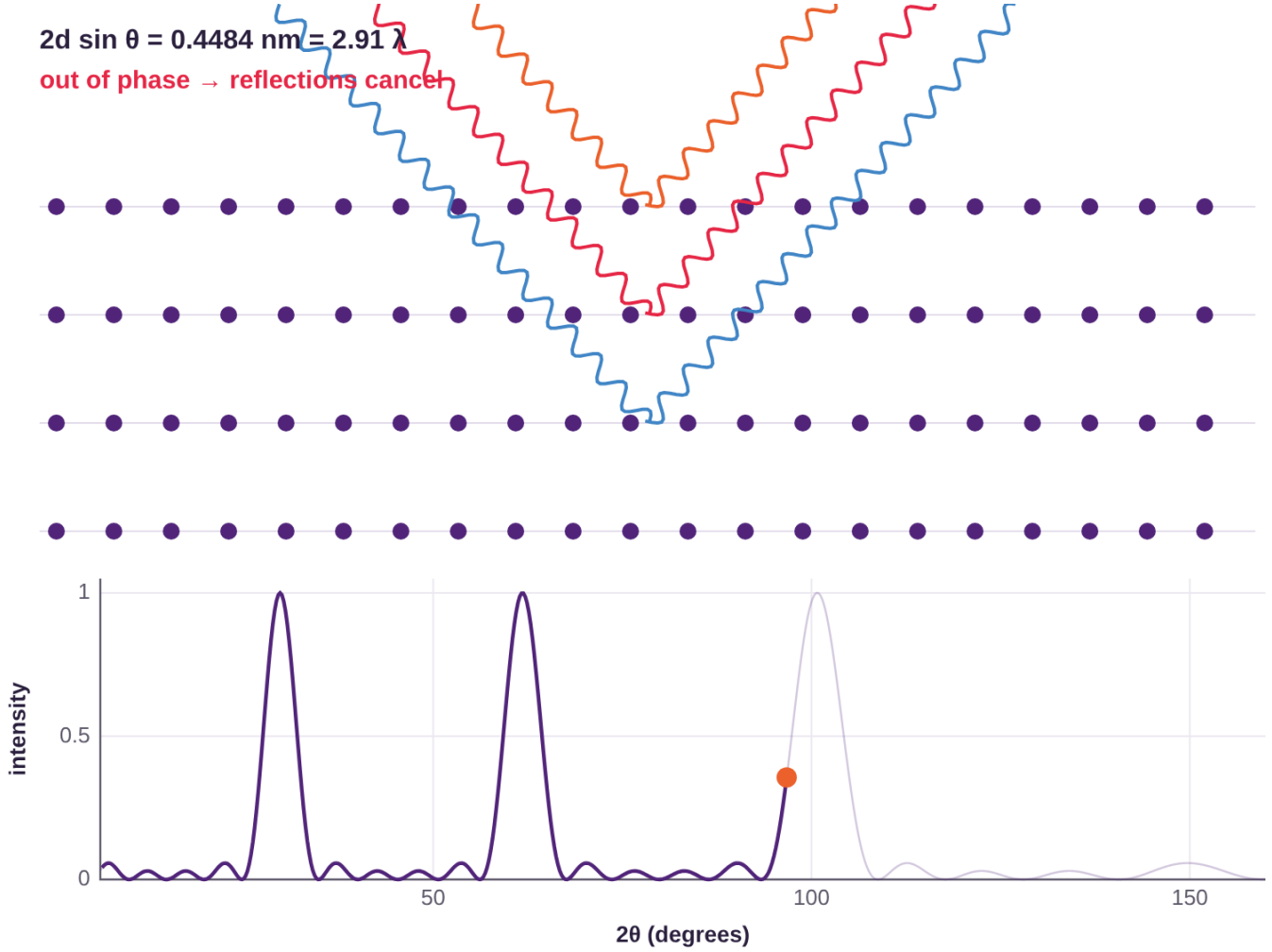
Rotate sample and detector, record intensity versus 2θ. Each peak gives one d-spacing.

Identify

Match the set of d values and intensities to a reference pattern (ICDD PDF, COD) to identify the phase.

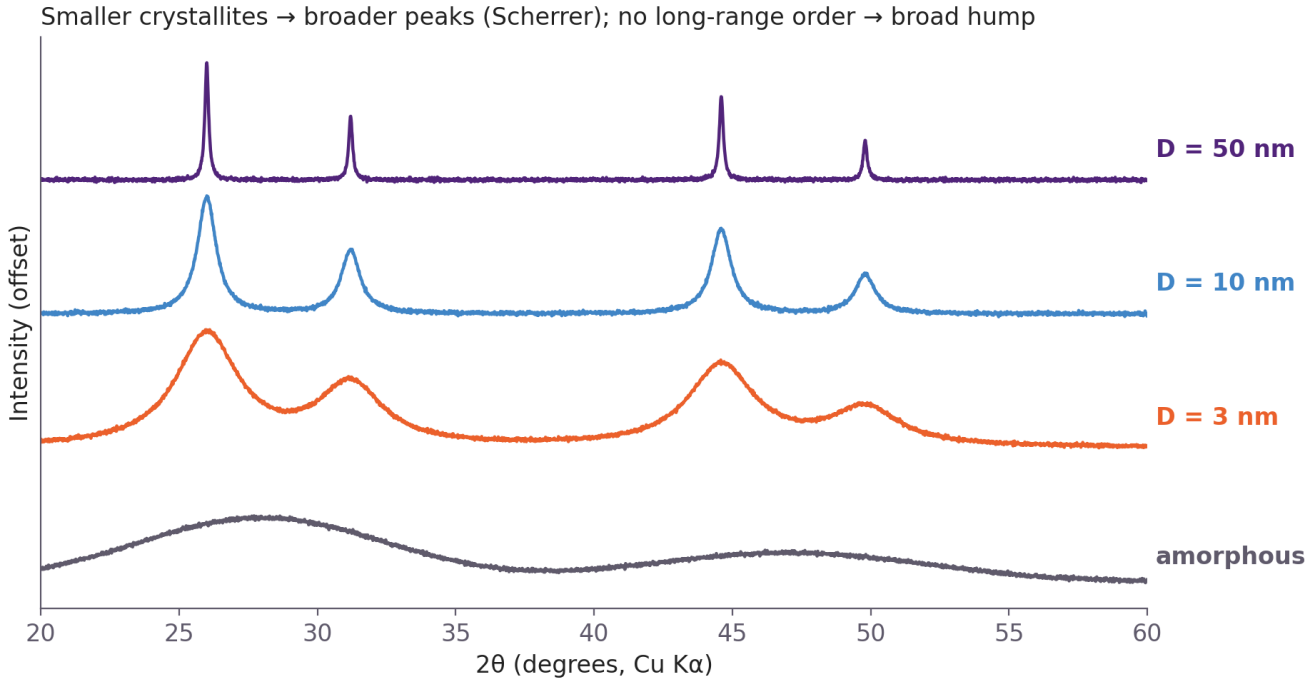
Simulation: sweep θ and watch the Bragg peak appear

B · SIMULATION 14



- **Sweep θ**: waves from successive planes are drawn with their phases.
- When 2d sin θ = nλ the readout turns green.
- Increase the number of planes **N**: the peak sharpens.
- That is the origin of **Scherrer broadening**: few planes, less selective interference, broad peaks.

Reading an XRD pattern: position, intensity, width



Position → phase, d-spacing

Shifts indicate strain or alloying.

Intensity → texture

Ratios that differ from the reference mean preferred orientation.

Width → size and strain

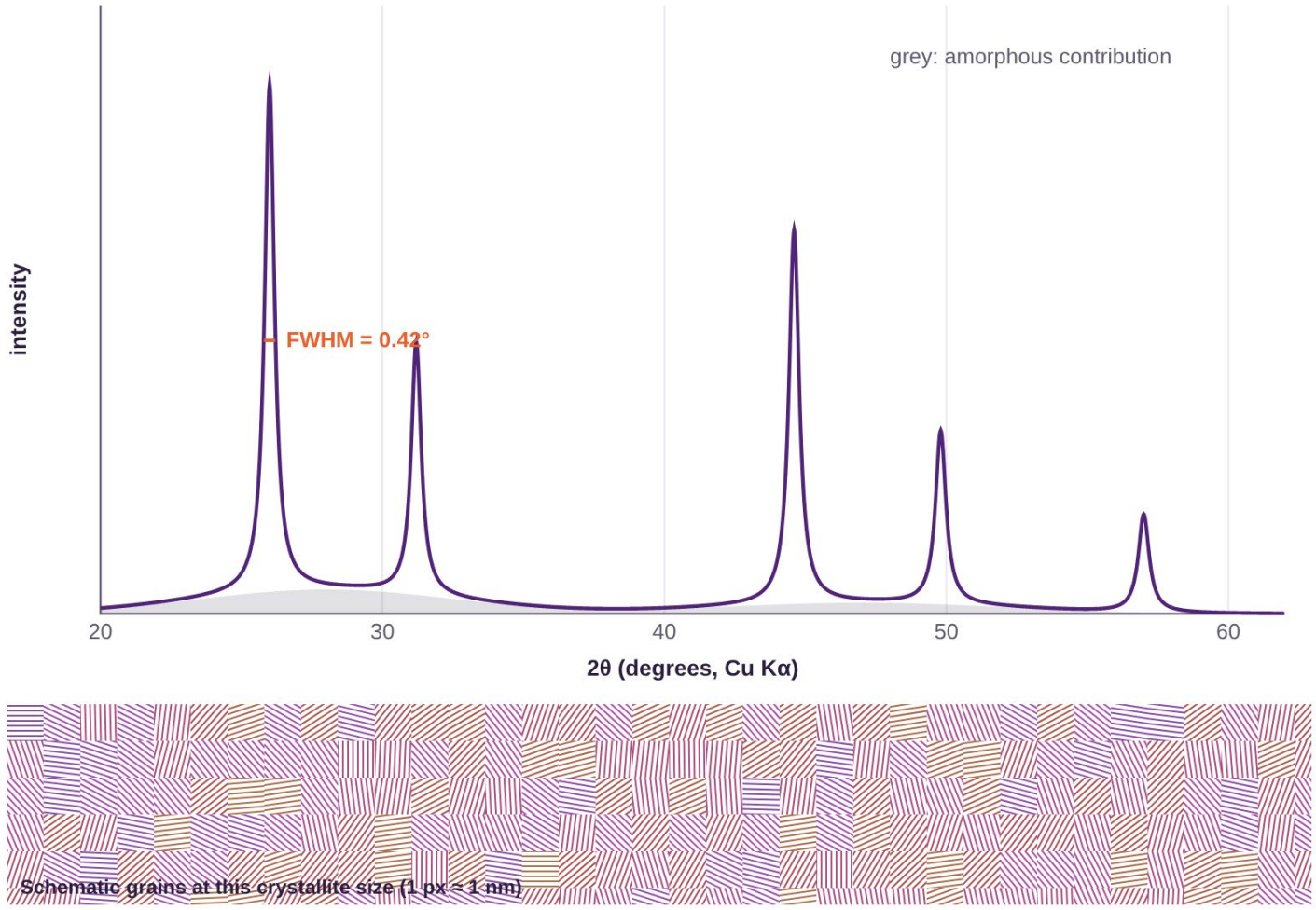
$$D = K\lambda / (\beta \cos \theta)$$

K ≈ 0.9; β = FWHM in radians, instrument-corrected. Valid ≈ 2–200 nm. Crystallite (domain) size, not particle size.

Worked: 2θ = 25.0°, β = 0.40° → D = 0.9 × 0.15406 / (0.00698 × cos 12.5°) ≈ **20 nm**.

Simulation: crystallite size, strain and crystallinity in one pattern

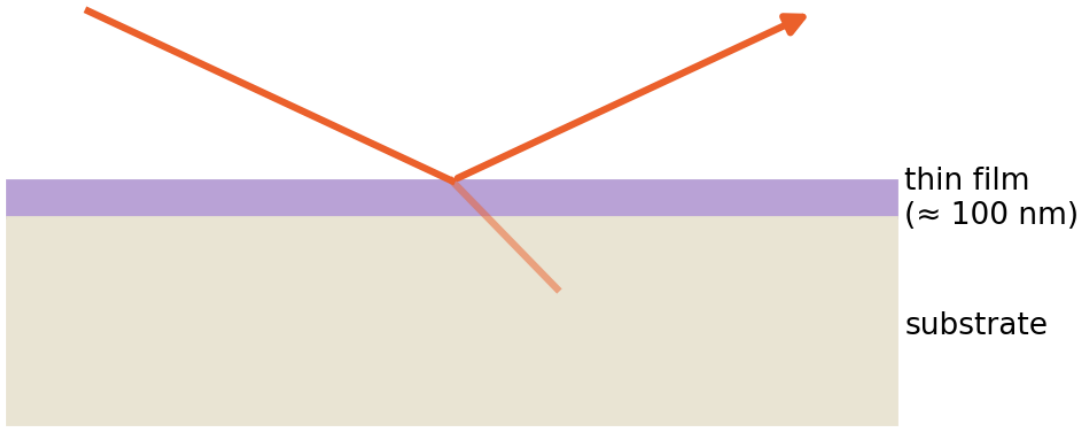
B · SIMULATION 15



- Shrink the crystallite size to **3 nm**: peaks smear into humps (like pristine mesoporous CuTe).
- Add **microstrain**: the Scherrer estimate goes wrong (strain mistaken for small size). Use Williamson–Hall.
- Raise the **amorphous fraction**: the hump grows and crystallinity falls.

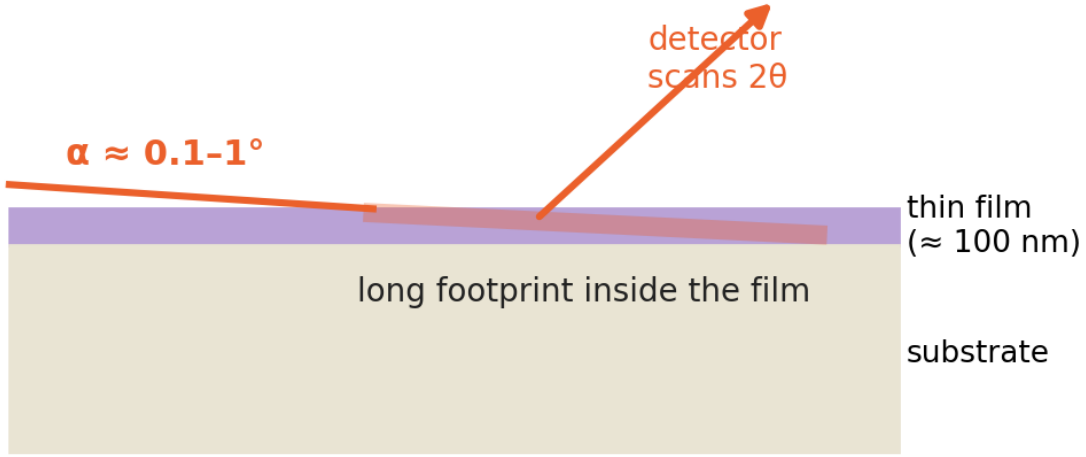
Thin films: grazing-incidence XRD (GIXD)

Symmetric θ–2θ (Bragg–Brentano)



Beam goes straight through the film: substrate peaks dominate

Grazing incidence (GIXD)



Path stays in the film: film peaks enhanced, substrate suppressed

Problem

A 100 nm film diffracts weakly; in θ–2θ the beam penetrates microns and the substrate dominates.

Fix

Keep the incident angle near the critical angle (0.1–1°): a long footprint in the film, little substrate signal.

Out-of-plane vs in-plane

Choose which lattice planes you probe: parallel or perpendicular to the substrate. Useful for oriented films.

Which technique, for which question?

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B · SUMMARY TABLE

	Probe → signal	Tells you	Resolution / sampling	Sample needs	Blind to
SEM	e^- (1–30 kV) → SE, BSE	morphology, pores, thickness	~1–5 nm; local	vacuum-stable, conductive or coated	crystal structure
EDX	e^- → characteristic X-rays	elements, ratios, maps	~1 µm (SEM), ~1 nm (STEM)	flat, enough overvoltage	phase, oxidation state, light elements
TEM	e^- (80–300 kV) → transmitted, diffracted	grains, d-spacing, defects, SAED	~0.1 nm; very local	< 100 nm thick (FIB lamella)	statistics over large areas
XRD	X-rays → diffracted X-rays	phase, crystallite size, strain, texture	Å in d; averages mm²	flat; GIXD for thin films	amorphous detail, local defects
XPS	X-rays → photoelectrons	surface composition, oxidation states	top 3–10 nm	UHV-compatible	bulk below ~10 nm

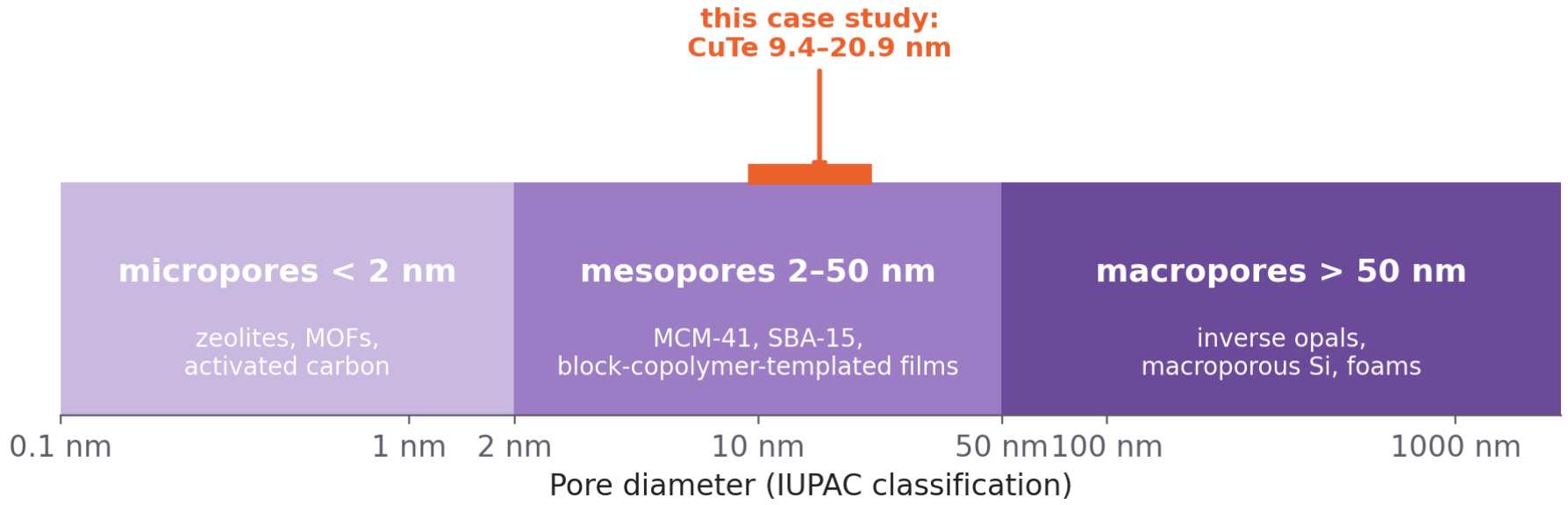
No single technique is enough. A convincing materials claim triangulates: **see it** (SEM/TEM), **identify it** (EDX/XPS), **quantify the order** (XRD/SAED).

C Porous semiconductors

60–68 MINUTES

What a mesoporous semiconductor is, why its pores raise activity, and the price paid in surface states and crystallinity.

What makes a semiconductor porous?



Hard templating

Fill a rigid mould (mesoporous silica, SBA-15/KIT-6) with precursor, then etch the mould away.

Soft templating

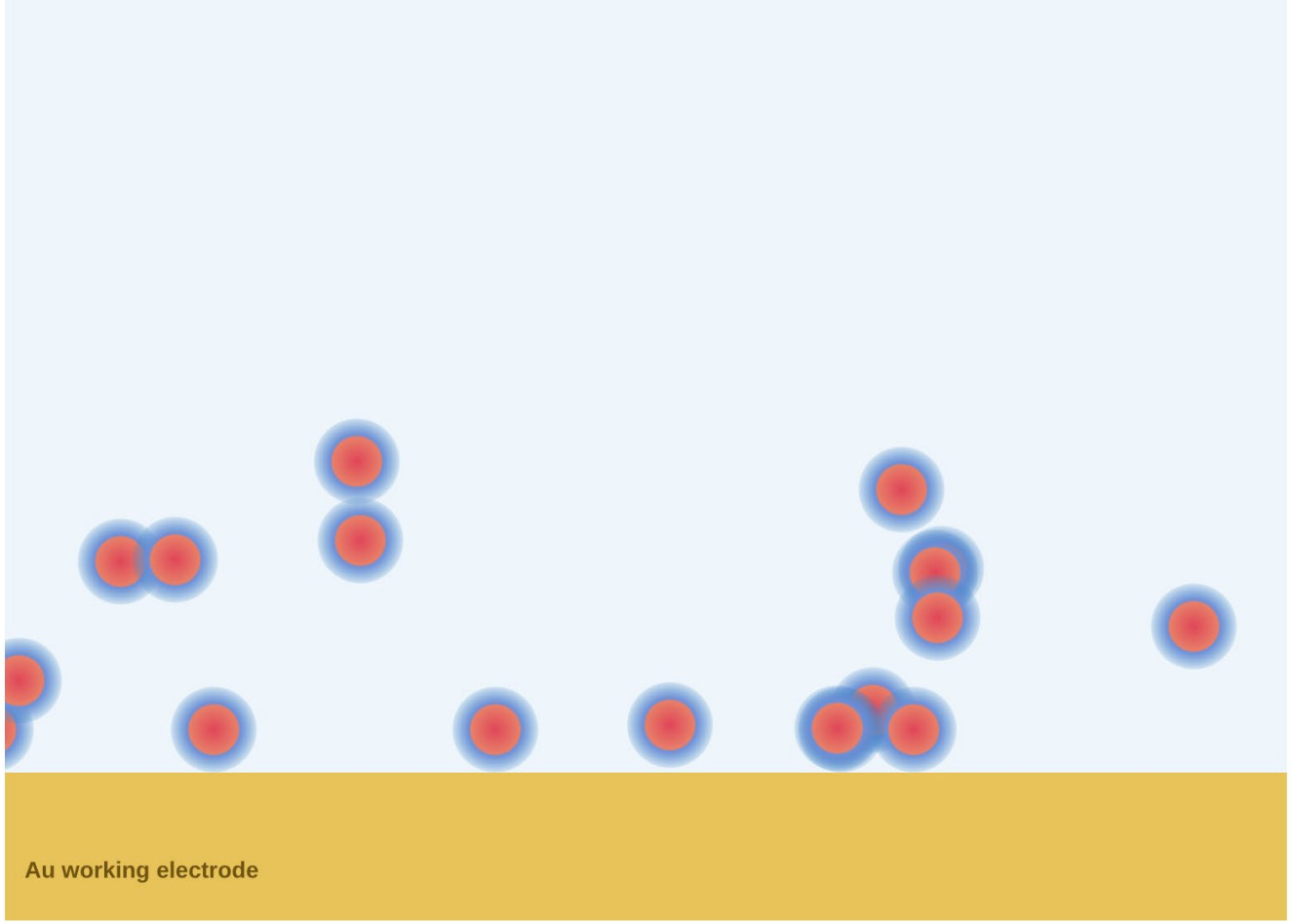
Surfactant or block-copolymer micelles act as removable placeholders during growth (sol–gel, electrodeposition).

Top-down etching

Electrochemical etching (porous Si), dealloying, or anodisation (TiO_2 nanotubes, porous alumina).

Soft templating + electrodeposition, step by step

C · ANIMATION 16



1 Micelles adsorb

PS-*b*-PEO micelles sit on the Au electrode.

2 Electrodeposit

CuTe grows around the micelles.

3 Dissolve

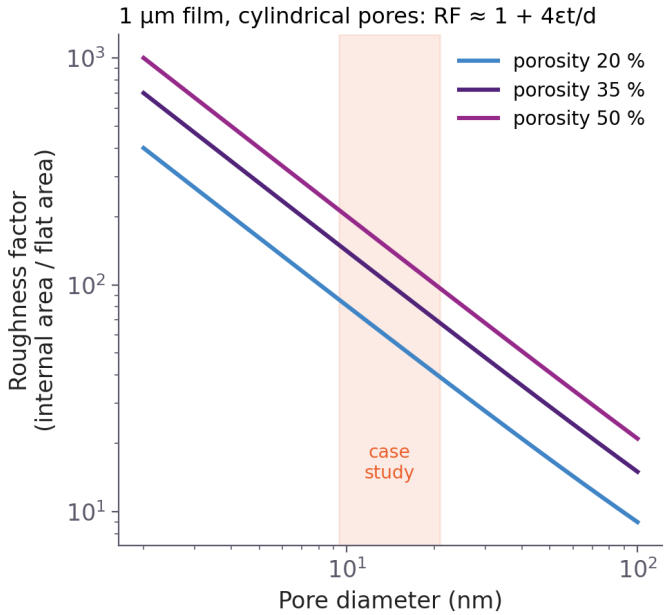
THF removes the polymer.

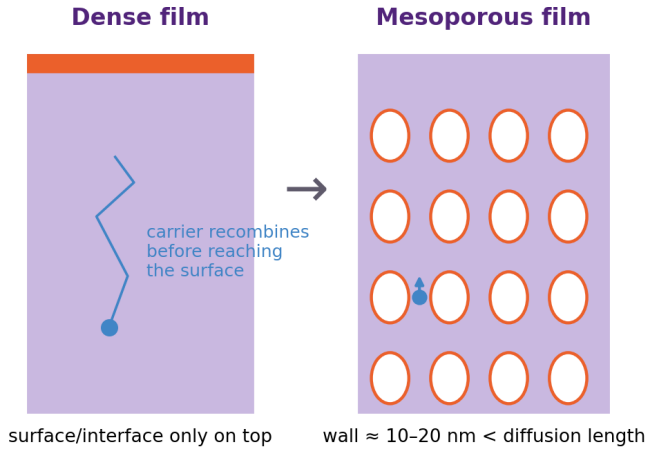
4 Pores remain

Micelle size sets pore size (BCP-1, -2, -3).

Why porosity raises activity

C · FOUR ADVANTAGES





1 · Much more surface

1 µm film, 15 nm pores, 35 % porosity: ~90× the area of a flat film. More active sites.

2 · Short carrier paths

10–20 nm walls are thinner than typical diffusion lengths: carriers reach a surface before recombining.

3 · Light management

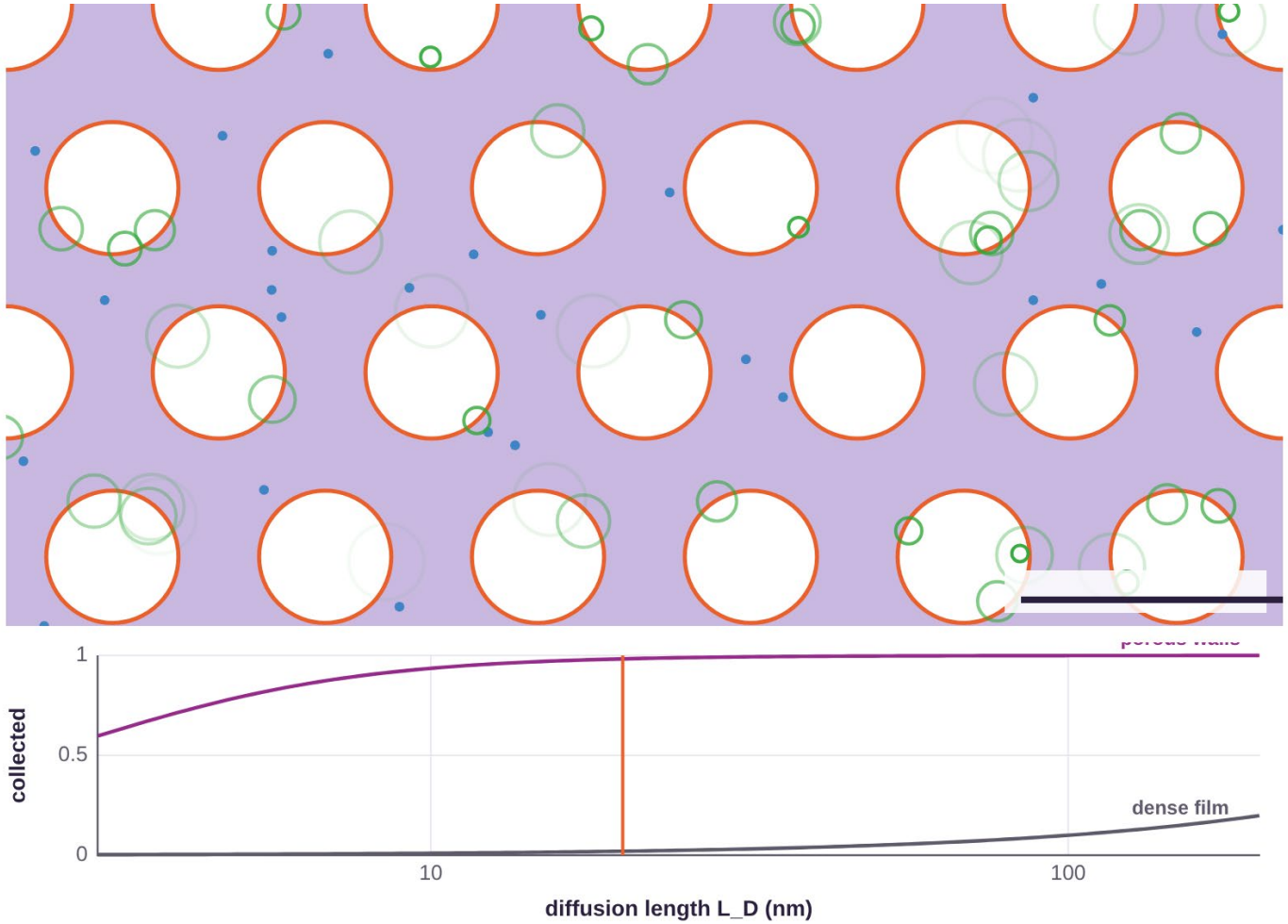
Pores scatter light and lengthen its path; band edges shift with wall size.

4 · Mass transport

Open, interconnected pores let electrolyte and gas molecules in quickly.

Simulation: surface area and carrier collection in a porous film

C · SIMULATION 17



$$RF = 1 + 4\varepsilon t/d$$

- Set L_D = 20 nm: in the porous film almost every carrier reaches a pore surface (green flashes).
- Untick the box: in a dense 1 µm film most carriers recombine before reaching the top surface.
- Slide pore size and porosity: wall thickness and roughness factor change.

The price of porosity: surfaces are also defects

C · WHY DEFECT ENGINEERING IS NEEDED

Dangling bonds and surface states

Unsaturated bonds at pore walls create states in the gap: carrier traps, recombination centres, Fermi-level pinning, slow response.

Small, strained domains

Growth confined around micelles gives nanocrystalline walls with many domain boundaries: lower crystallinity, broad XRD peaks.

Band tailing

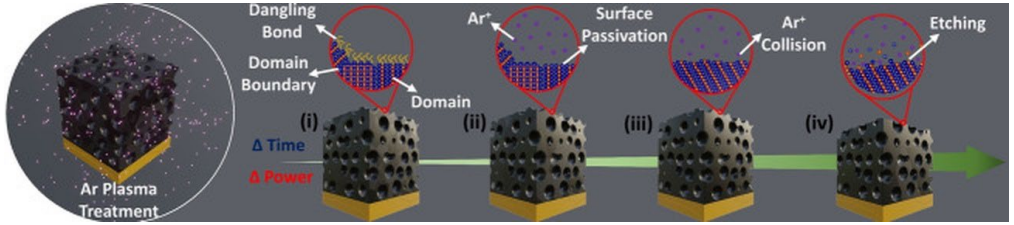
Disorder broadens the absorption edge: a larger Urbach energy E_U ($\alpha = \alpha_0 \cdot \exp(h\nu/E_U)$).

Surface oxidation

More surface means more oxide and contamination.

The usual fix fails. Thermal annealing improves crystallinity, but at a few hundred °C the pores sinter, distort and collapse.

The question for the case study: can we grow the domains and passivate the surfaces **without** destroying the pores?



Graphical abstract: Ashok et al., Angew. Chem. Int. Ed. 2022, 61, e202114729 (CC BY-NC 4.0).

D Case study: plasma domain engineering

68–84 MINUTES

Mesoporous CuTe films, one low-power Ar plasma step, and what SEM, EDX, TEM, XRD and XPS each revealed.

The paper and the question

D · ASHOK ET AL., ANGEW. CHEM. INT. ED. 2022

Plasma-Induced Nanocrystalline Domain Engineering and Surface Passivation in Mesoporous Chalcogenide Semiconductor Thin Films

A. Ashok, A. Vasanth, T. Nagaura, M. Eguchi, N. Motta, H.-P. Phan, N.-T. Nguyen, J. G. Shapter, J. Na, Y. Yamauchi. Angew. Chem. Int. Ed. 2022, 61, e202114729. doi:10.1002/anie.202114729 (open access, CC BY-NC 4.0)

Material

Mesoporous copper telluride (CuTe, orthorhombic vulcanite), electrodeposited on Au with PS-b-PEO micelle templates. Three pore sizes plus a non-porous reference.

Treatment

Low-power Ar plasma (13.56 MHz RF), optimum **30 W for 180 s**, substrate at about 23–35 °C. Benchmarked against annealing at 70, 140 and 210 °C.

Claim

Larger, aligned crystalline domains and passivated surfaces, with the pores intact: crystallinity ~70 % → ~85 %.

85 %

crystallinity after plasma (from ~70 %)

180 s at 30 W

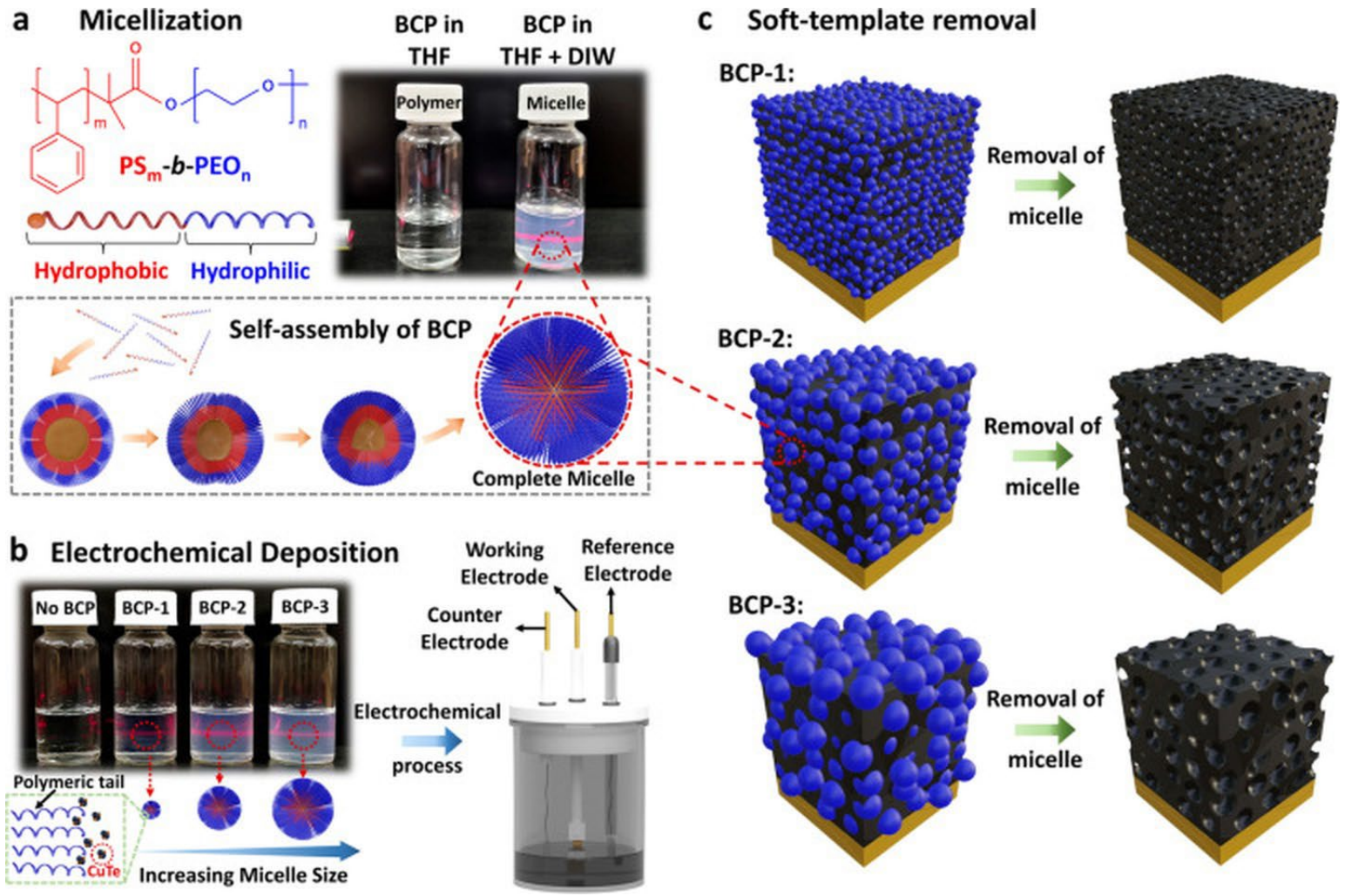
the whole treatment, near room temperature

≥ 140 °C

annealing temperature at which pores distorted and collapsed

Step 1 · Making the films: micelles + electrodeposition

D · FIG. 1



a · Micellisation

PS$_m$-b-PEO$_n$ in THF; adding water forms micelles (hydrophobic PS core, hydrophilic PEO corona).

b · Electrodeposition

Cu and Te precursors with micelles; −0.2 V vs Ag/AgCl; growth ~0.2 nm/s. Polymer molecular weight sets micelle size: BCP-1, -2, -3.

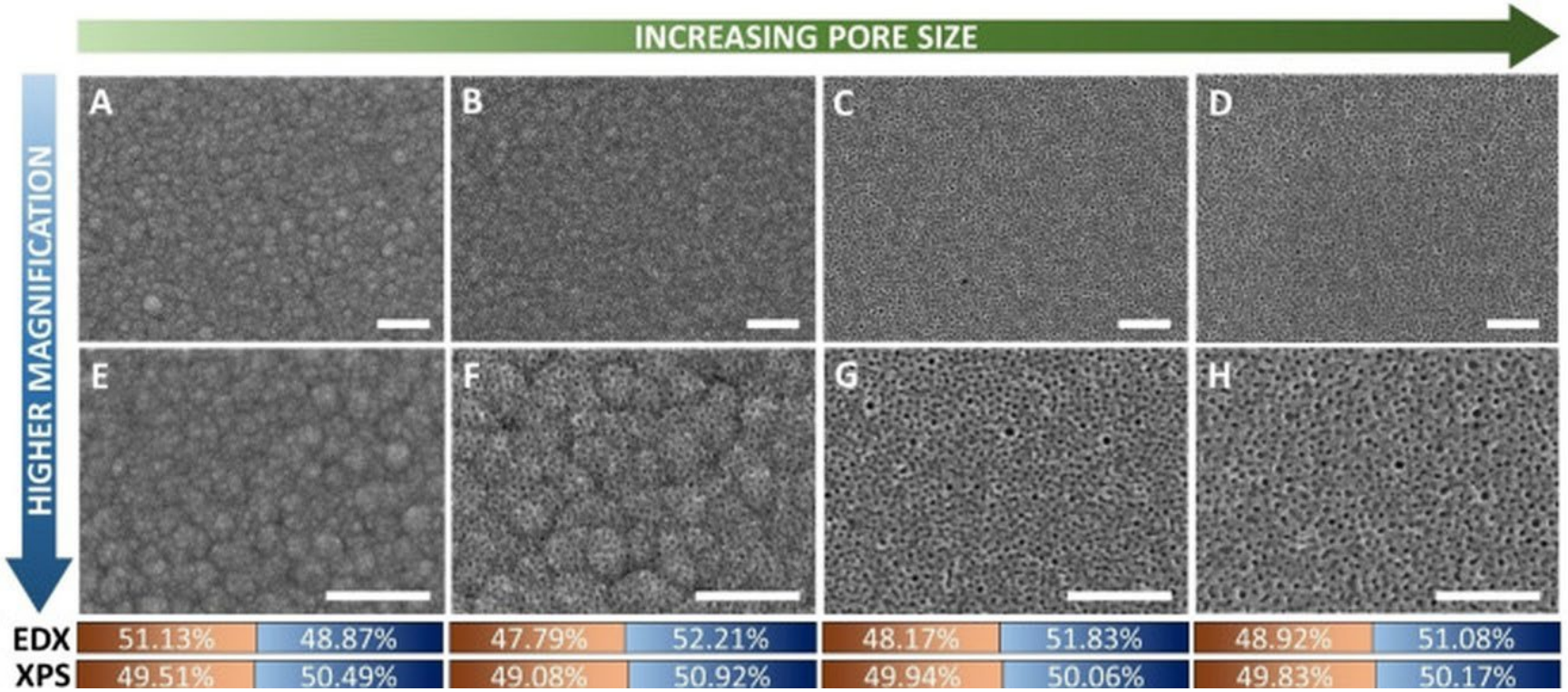
c · Template removal

THF dissolves the micelles, leaving mesopores.

Figure: Ashok et al., Angew. Chem. Int. Ed. 2022, 61, e202114729, Fig. 1 (CC BY-NC 4.0).

Step 2 · SEM and EDX: pores as designed, composition near 1 : 1

D · FIG. 2A–H · SEM · EDX · XPS



9.4 nm
BCP-1 pores · walls 9.2 nm

16.2 nm
BCP-2 pores · walls 14.8 nm

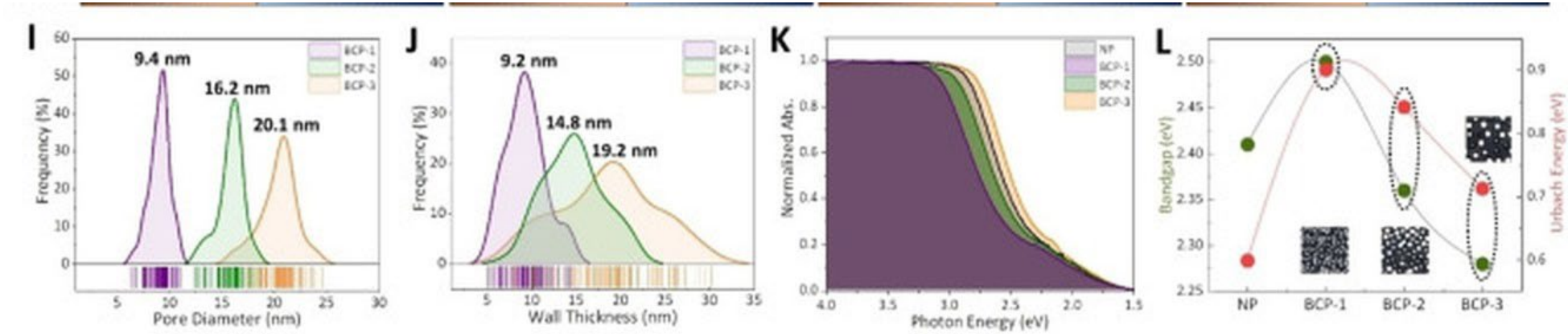
20.9 nm
BCP-3 pores · walls 19.2 nm

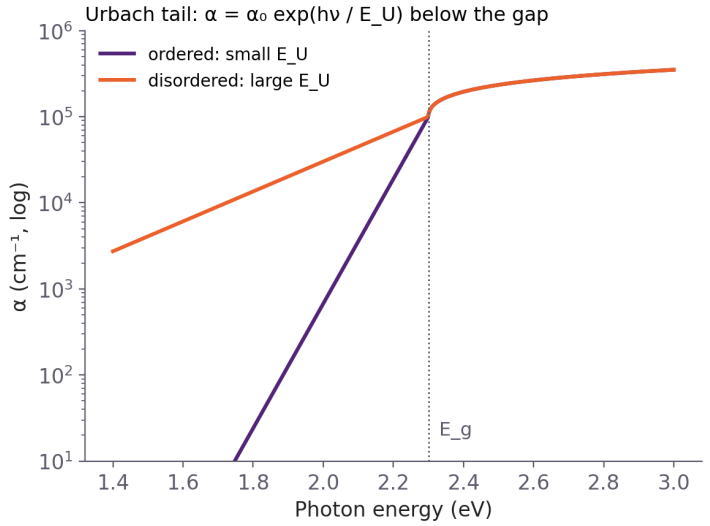
≈ 48 : 52
Cu : Te by EDX (at%); XPS ≈ 49 : 51

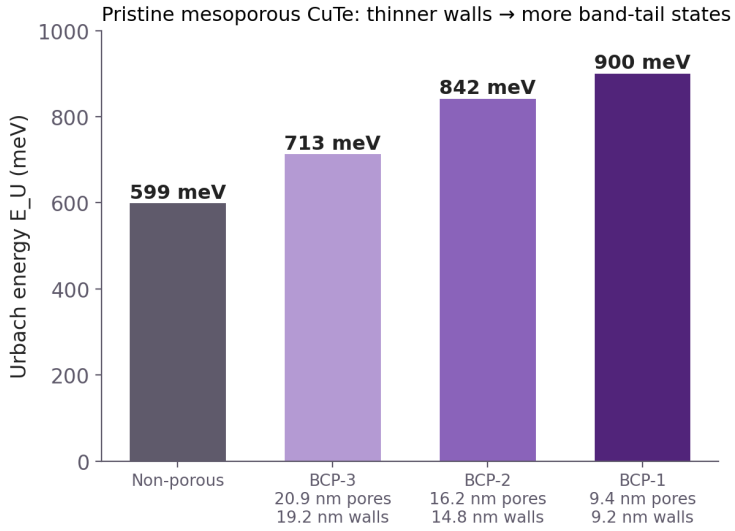
Scale bars 250 nm. Figure: Ashok et al. 2022, Fig. 2 (CC BY-NC 4.0). The text reports 20.9 nm for BCP-3; Fig. 2I labels 20.1 nm.

Step 3 · Thinner walls, more disorder: the Urbach tail

D · FIG. 2I–L · UV–VIS







Reading it

- Absorbance blue-shifts as pores and walls shrink.
- E_U: 599 meV (non-porous) → 713, 842, 900 meV as walls thin from 19 to 9 nm.
- Larger E_U = more tail states (surfaces, strain, small domains).
- After plasma: E_U fell; E_g changed little.

Top: Ashok et al. 2022, Fig. 2I–L (CC BY-NC 4.0). Bottom: Urbach energies replotted from the paper.

Step 4 · Low-power Ar plasma: passivate, crystallise, but do not etch

D · FIG. 3

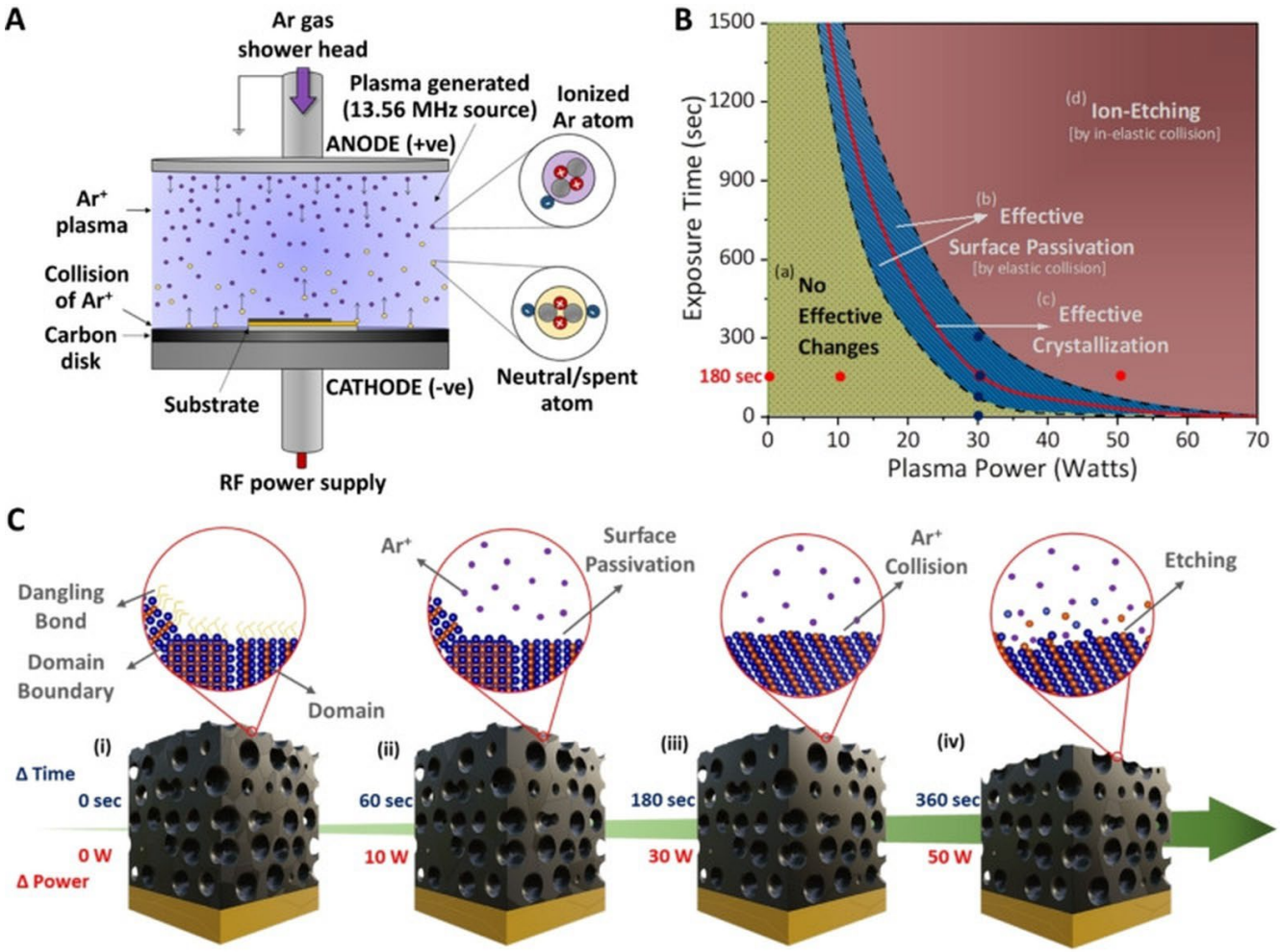


Figure: Ashok et al., Angew. Chem. Int. Ed. 2022, Fig. 3 (CC BY-NC 4.0).

A · Chamber

Parallel-plate RF (13.56 MHz) Ar plasma; Ar^+ accelerated through the sheath to the sample on the cathode.

B · Process map

(a) no change at low power · (b) surface passivation (elastic collisions) · (c) crystallisation · (d) ion etching (inelastic). Optimum **30 W, 180 s**.

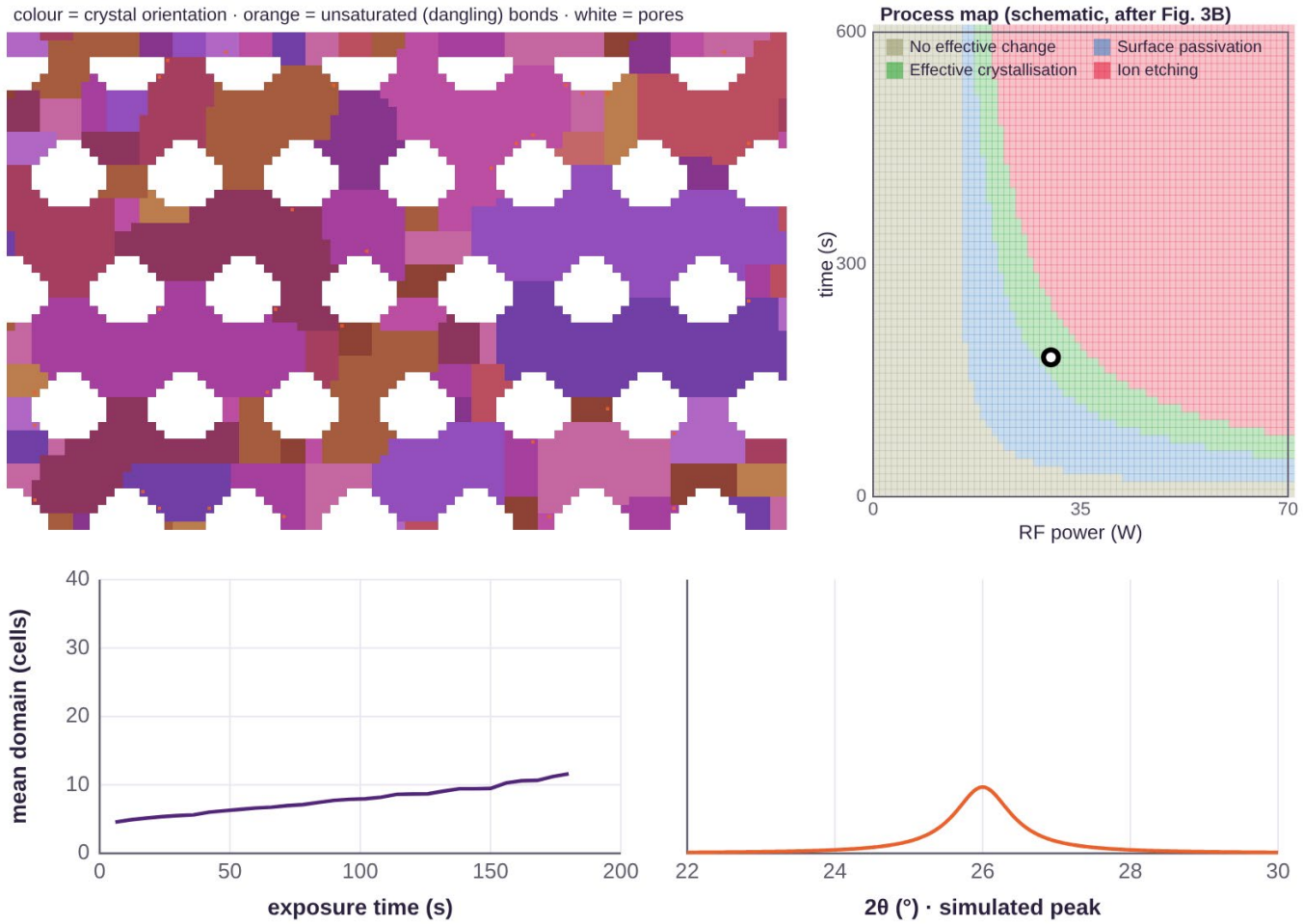
C · What happens

Dangling bonds passivated → domains rearranged and enlarged → over-exposure etches the walls.

At 10 W: no change even for long exposures. At 30 W beyond ~240 s: etching, and E_U rises again.

Simulation: explore the plasma process map

D · ILLUSTRATIVE LATTICE MODEL



10 W, 180 s

nothing happens

30 W, 180 s

bonds passivated, domains coarsen, XRD peak sharpens

50 W, 180 s

domains grow but the surface starts to etch

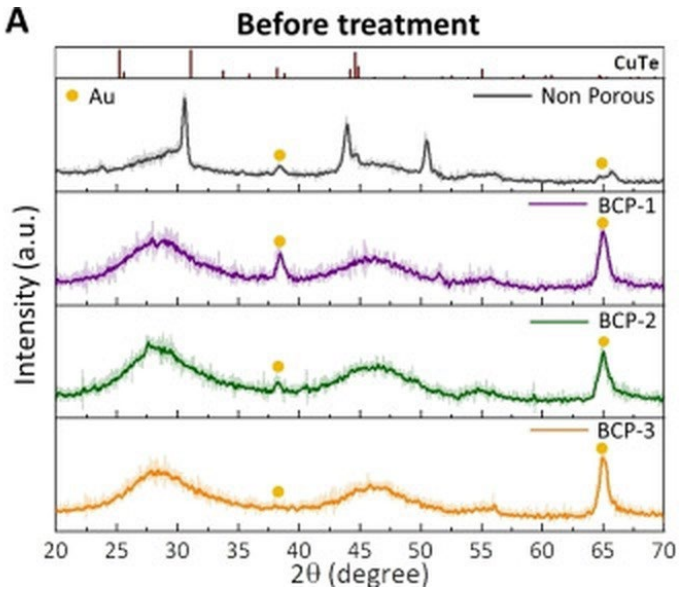
30 W, 360 s

over-exposure etches the walls

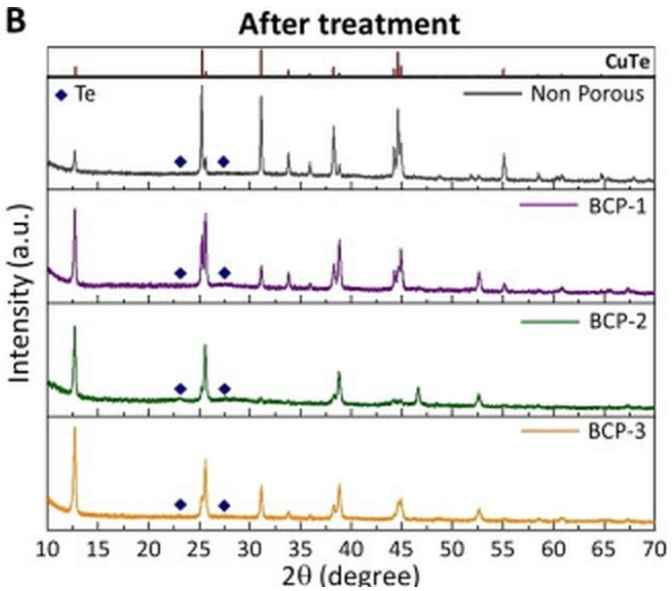
A teaching model tuned to reproduce the regimes of Fig. 3B, not a physical simulation of the plasma.

Step 5 · GIXD: broad humps become sharp peaks

D · FIG. 4A, B · XRD



Before: weak, broad humps in all porous films; sharp Au substrate peaks.



After 30 W, 180 s: many sharp, intense CuTe peaks; small Te peaks (◆) appear.

Width

Broad → sharp: larger coherent domains (Scherrer) and relieved strain. Pristine peaks were too broad to index reliably.

Position

d-spacings changed only slightly and were similar for all pore sizes: same phase, better order.

New peaks

Te reflections: excess Te knocked out of the walls (consistent with the EDX Te excess).

Limit

No further sharpening after 180 s at 30 W; 10 W gave no change.

Figures: Ashok et al. 2022, Fig. 4A,B (CC BY-NC 4.0). Note the different 2θ ranges of the two panels.

Step 6 · XPS: the surface oxides are stripped away

D · FIG. 4C, D · XPS

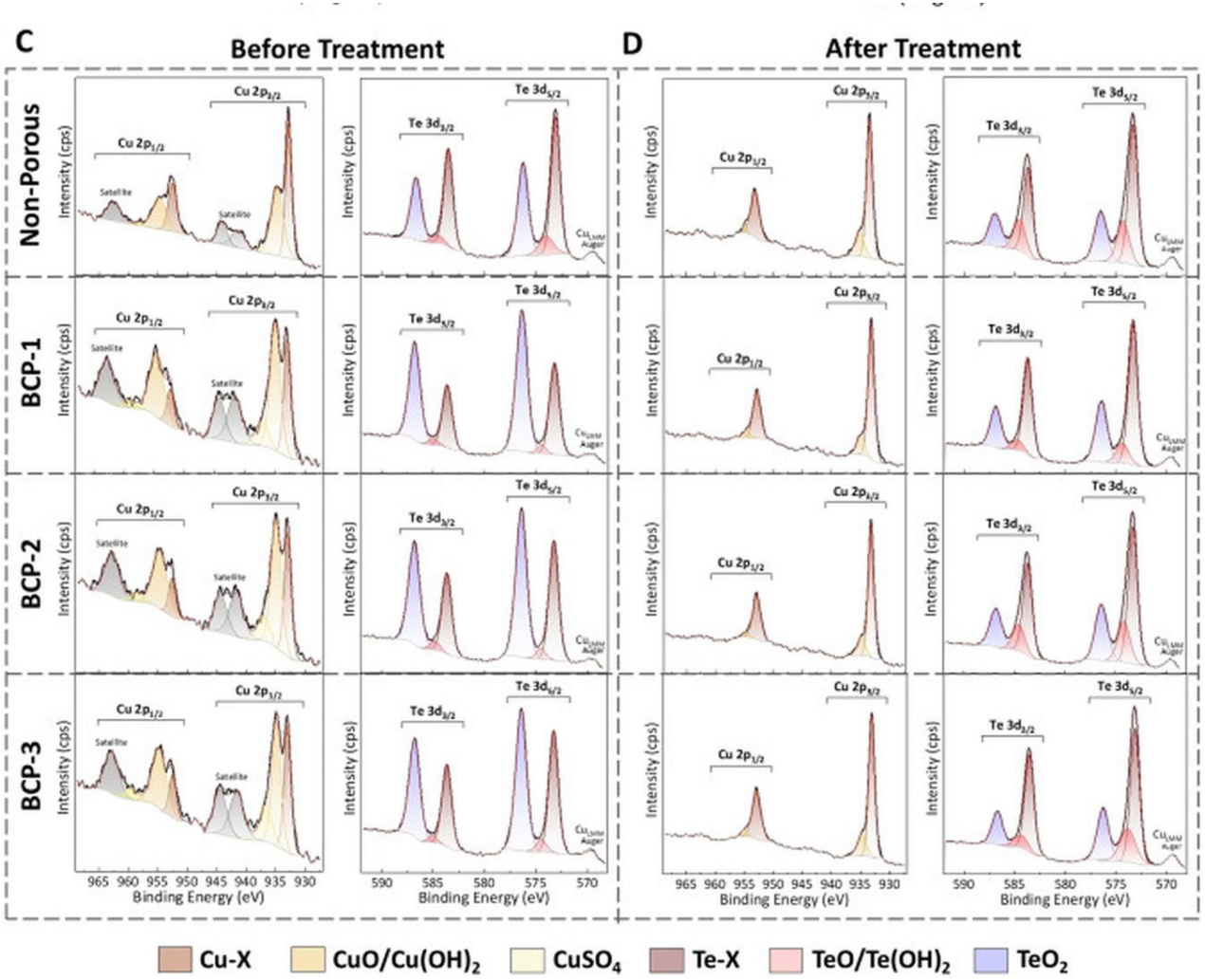


Figure: Ashok et al. 2022, Fig. 4C,D (CC BY-NC 4.0).

Before

Strong Cu 2p satellites and $CuO/Cu(OH)_2$, $CuSO_4$ components; large TeO_2 in Te 3d. Oxide signals are larger in the porous films (more surface).

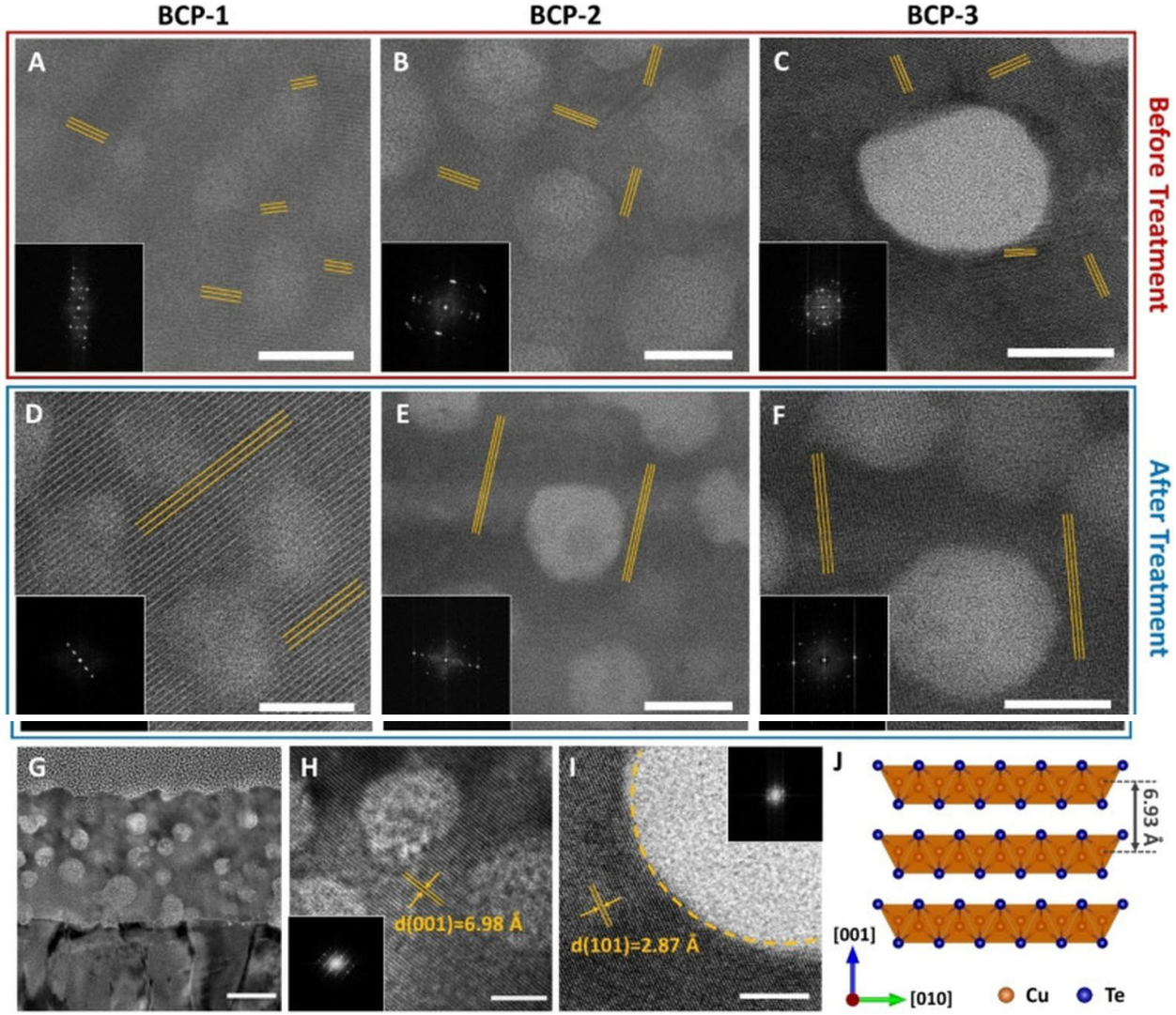
After

Oxide components and satellites largely gone; Cu–Te and Te–Cu dominate.

XPS sees only the top few nm, which is exactly where pores add area. EDX could not show this: it averages through the film and says nothing about oxidation state.

Step 7 · TEM: domains grow and line up

D · FIG. 5 · TEM



Before (STEM-BF, A–C)

Small domains (~5–20 nm) with several orientations (yellow fringe markers); FFT insets show many spots.

After plasma (D–F)

Larger domains with fringes running one way, oriented vertically from the substrate; cleaner FFT.

Lattice (G–J)

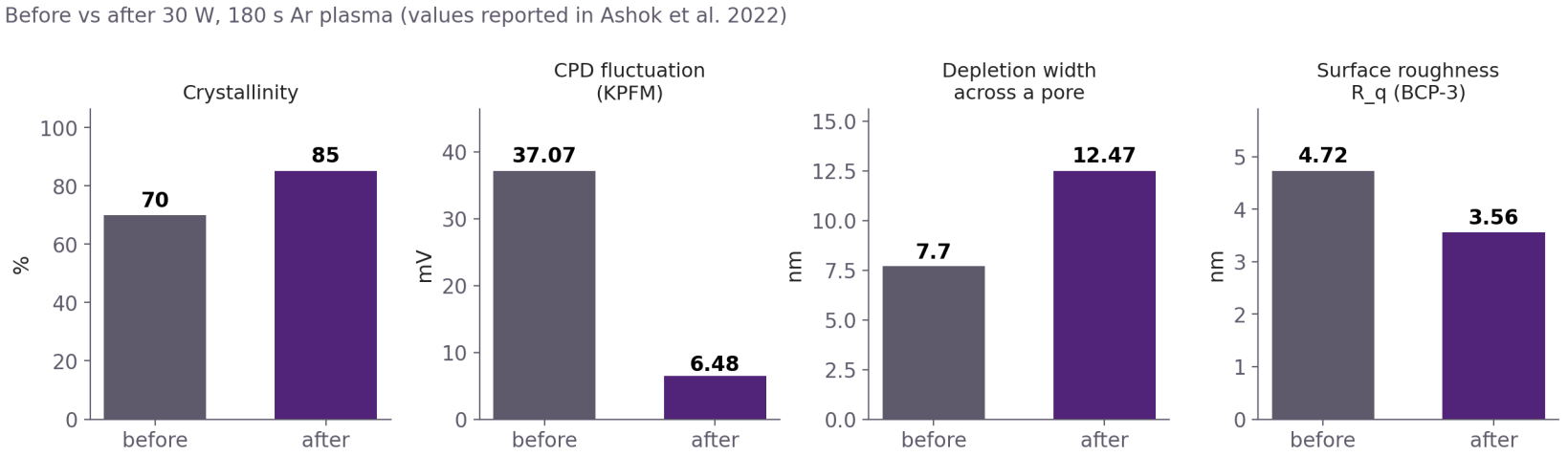
HRTEM: d(001) = 6.98 Å, d(101) = 2.87 Å, vulcanite CuTe.
HAADF: ~2 nm Te clusters on pore surfaces in BCP-3.

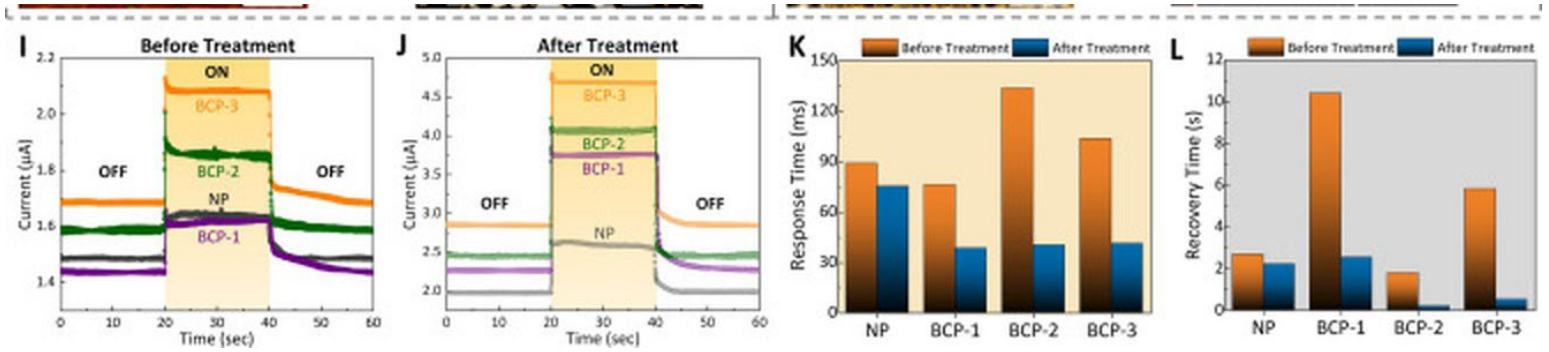
TEM is very local: it must agree with the GIXD average before we believe the claim. Before/after wipe slider: HTML slide 54.

Figures: Ashok et al. 2022, Fig. 5 (CC BY-NC 4.0). Scale bars: A–F 15 nm; G 50 nm; H 10 nm; I 5 nm.

Step 8 · Does it matter? Surface potential and photoresponse

D · FIG. 6 · KPFM / C-AFM + PHOTOELECTROCHEMICAL RESPONSE





~60 % faster response
mesoporous films (non-porous: ~15 %)

~86 % faster recovery
mesoporous films (non-porous: ~18 %)

+30 % light current
porous vs non-porous after plasma (+15 % dark)

Figure: Ashok et al. 2022, Fig. 6 (CC BY-NC 4.0); metrics from the paper text.

What each technique told us

Technique	Before plasma	After 30 W, 180 s	Conclusion
SEM	9.4 / 16.2 / 20.9 nm pores, uniform	pores unchanged (annealing ≥ 140 °C collapsed them)	architecture preserved
EDX	Cu : Te ≈ 1 : 1, slight Te excess	excess Te expelled	stoichiometry kept
UV–vis	E_U 599 → 900 meV as walls thin	E_U reduced; E_g slightly lower	fewer tail states
GIXD	weak, broad humps	sharp, intense peaks; crystallinity ~70 → 85 %	larger domains
XPS	$CuO/Cu(OH)_2$, $CuSO_4$, TeO_2	oxides largely removed	surface passivated
TEM	5–20 nm domains, mixed orientation	larger, aligned domains; d(001) 6.98 Å	local proof of domain growth
KPFM + PEC	CPD fluctuation 37 mV; slow	6.5 mV; response ~60 %, recovery ~86 % faster	better device

Take-home: no single image proves “higher crystallinity, same pores, fewer defects”. SEM + EDX + XRD + TEM + XPS + a device measurement together do.

Summary

E · SIX IDEAS TO KEEP

WHAT IT CAN SEE

$\lambda_c = 1.24 / E_g$

Band gap sets the cut-off.

HOW MUCH SIGNAL

$R = \eta\lambda/1.24 \times G$

Gain costs bandwidth and adds noise.

HOW FAINT

$D^* = \sqrt{(A\Delta f)}/NEP$

With measured noise.

HOW TO SEE STRUCTURE

SEM · EDX · TEM · XRD

Morphology, composition, local order, average order.

WHY PORES

area + short paths

Paid for with surface states.

DEFECT ENGINEERING

30 W · 180 s Ar

Bigger domains, same pores, faster response.

Discussion questions

E · OPEN DISCUSSION · Q&A

1 · Proving “more crystalline”

Which two independent measurements would you insist on before accepting that a treatment increased crystallinity? Why one of each kind (average vs local)?

2 · EDX vs XPS

EDX says Cu : Te ≈ 1 : 1 but XPS shows strong oxides. Can both be right? What does each one sample?

3 · How small a pore?

Smaller pores mean more area but thinner, more defective walls. How would you choose the pore size for a photodetector versus a photocatalyst?

4 · Plasma vs annealing

What other low-temperature routes could crystallise a fragile nanostructure (laser, microwave, chemical passivation)? What would you measure to compare them?

References and resources

E · FURTHER READING

Case study

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Questions and discussion

Thank you

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